Pexip Infinity

Administrator Guide

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The Pexip Infinity platform

Pexip Infinity is a scalable meeting platform that connects virtually any communications tool, such as Microsoft Skype for Business, and traditional video and audio conferencing together for a seamless meeting experience.

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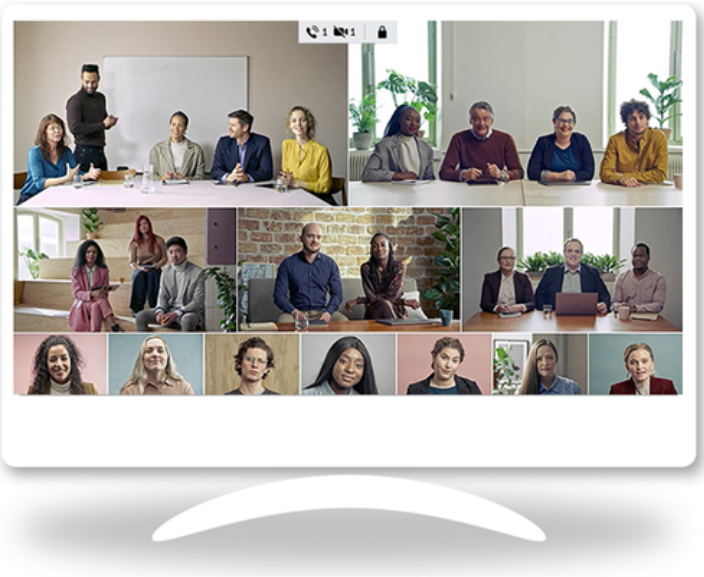
Introduction to Pexip Infinity

Pexip Infinity is a self-hosted, virtualized and distributed multipoint video conferencing platform. It can be deployed in an organization's own datacenter, or in a private or public cloud such as Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP) or Oracle Cloud Infrastructure, as well as in any hybrid combination. It enables scaling of video, voice and data collaboration across organizations, regardless of technology platforms and security requirements, seamlessly allowing everyone to engage in high definition video, web, and audio conferencing.

It provides any number of users with their own personal Virtual Meeting Rooms (VMRs), which they can use to hold conferences, share presentations, and chat. Participants can join over audio or video from any location using the endpoint or client of their choice, including:

- Professional video conferencing room systems (Microsoft Teams Rooms, SIP and H.323 devices)
- Desktop/mobile (with the Pexip app suite of clients)
- Web browsers (WebRTC - no downloads required)
- Traditional audio conferencing (PSTN dialing)



Pexip VMRs maintain the same customized address and are always available for spontaneous 1-to-1 or group meetings.

VMRs can also be accessed through a Virtual Reception IVR service, which allows all participants to dial a single number to access Pexip Infinity, and then use the dial tones on their endpoint or phone to select the conference they want to join.

There is a wide choice of layouts. You can select from a range of classic layouts, Pexip's AI-driven Adaptive Composition layout featuring real-time automatic face detection and framing, or you can even design your own custom layouts.

The platform also includes the Infinity Gateway service, which allows end users to place calls to other endpoints that use different protocols and media formats, or to seamlessly connect into an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

Pexip's Media Playback Service allows you to play prerecorded video content (such as adverts and informational videos) to consumers. When the media finishes playing, the user can be transferred to another service, such as a VMR conference, or they can be disconnected.

It automatically transcodes all the popular video and audio codecs and supports standard protocols including SIP, H.323, and WebRTC. It supports all standards-based devices including those from Cisco, Poly, Lifesize, Sony, Radvision, Yealink, and Avaya. It also supports software clients such as Microsoft Teams Rooms, Skype for Business and Surface Hub.

Secure meetings

The Pexip Infinity secure video collaboration platform offers secure meetings via:

- a range of deployment environments to suit your physical security requirements
- integration with internal and external services to authenticate, authorize and provision access to clients and services — for a full summary see Authentication overview
- the optional Pexip Secure Scheduler for Exchange feature, which integrates with Outlook to allow users to add a secure Pexip VMR to their meeting invitations

- the Pexip Secure Scheduler for Web product, a web-based tool for creating and managing secure virtual meetings
- the Pexip Secure Meetings for Justice product, specifically tailored to the unique requirements of judicial VMRs.

Privacy and security

The Pexip Infinity self-hosted solution supports the industry standards for communication encryption for end-user devices, ensuring that all video calls and shared media content is secure and kept private even if it crosses the internet. The entire deployment and all its data, including call status, diagnostic logs and call history, is completely under the ownership and control of the enterprise, even when deployed in the cloud.

It uses advanced participant authentication to ensure only the right people attend the right meeting at the right time. Access can be granted based on attributes such as role, device type, location, and IP address.

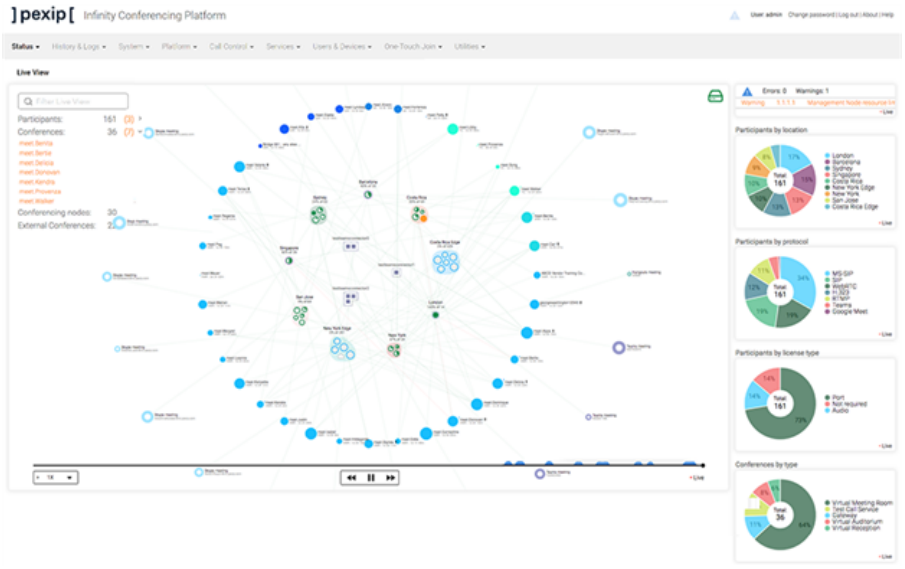
The Pexip Infinity platform has been designed to meet the strictest U.S. Federal & European requirements.

Scheduling and joining meetings

Pexip offers a number of scheduling solutions:

- Pexip's One-Touch Join (OTJ) allows users to schedule a meeting in Microsoft Outlook or Google Calendar and include in the invitation a meeting room with a supported Cisco or Poly videoconferencing endpoint, so that the endpoint in the chosen meeting room displays a **Join** button just before the meeting is scheduled to begin. Participants can then simply walk into the room and select the button, and the endpoint will automatically dial in to the meeting.
- Secure Scheduler for Exchange integrates Pexip Infinity with Microsoft Exchange. It enables Microsoft Outlook desktop and Pexip web app users to schedule meetings using Pexip VMRs as a meeting resource.
- Pexip Secure Scheduler for Web is a web-based scheduling solution for creating and managing secure, virtual meetings. It provides enhanced security with features such as one-time VMRs, secure pin codes, and single sign-on (SSO).

Management and control



The administrator interface provides a single place to manage the entire Pexip Infinity platform, including the ability to:

- Configure your Pexip Infinity conferencing services (Virtual Meeting Rooms, Virtual Receptions and so on).
- Create and manage the Conferencing Nodes that host your conferences.
- Scale on-demand, including dynamic bursting to a cloud service when required, creating capacity only when you need it to save daily running costs.
- Monitor live and historical video usage across the entire platform, as well as call quality and other analytics.

- Perform active conference management tasks such as adding or disconnecting participants, locking a conference, or muting a participant's audio.
- PIN-protect conferences and differentiate between Hosts and Guests to give different users different controls.
- Stream content directly from a VMR to a range of third-party RTMP streaming platforms.

It comes with comprehensive RESTful APIs allowing deep and advanced integrations with numerous services and tools such as PowerBI, plus authentication, authorization and provisioning capabilities against an AD/LDAP database.

You can extend Pexip Infinity's built-in functionality by using external and/or local policy to apply bespoke call policy and routing decisions based on your own specific requirements.

Pexip's unique distributed architecture is purely software-based and virtualized, running on industry-standard servers, meaning it can be deployed quickly and simply with the flexibility to scale as required. Administrators have access to upgrades as soon as they are released.

Customizable to your standards

You can make Pexip Infinity your own with a variety of customization options. You can easily apply your own branding to the Pexip Infinity platform to produce a personalized user experience.

For more information, see:

- Components of the Pexip Infinity platform
- Choosing a deployment environment
- Pexip Infinity features and specifications

Pexip Infinity features and specifications

The Pexip Infinity platform is designed to use industry-standard servers from any vendor to provide high-quality, scalable, efficient and secure video collaboration. The following tables cover the platform, Pexip apps, audio and video (including codecs), host hardware, capacity and hypervisor specifications and requirements.

Pexip Infinity platform

Feature	Description
Application deployment and management	• Software-based, virtualized application architecture, running on industry-standard servers. • Management using industry-standard tools, including VMware vSphere, Microsoft Hyper-V and KVM, and the ability to deploy onto generic hypervisors and orchestration layers. • Ability to deploy on Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP) and Oracle Cloud Infrastructure cloud platforms, including dynamic bursting into Azure, AWS or GCP services when primary conferencing capabilities are reaching their capacity limits. • Support for booting VMs via BIOS or UEFI. • Integration with the Pexip Private Cloud, where your entire deployment can be hosted privately and securely by Pexip on your behalf. • Flexible deployment model allowing customers to deploy the platform in the way that is most appropriate for them without needing to consume additional software licenses or purchase dedicated hardware. • Ability to seamlessly increase capacity by deploying new, updated, or additional hardware resources. • Management API supporting configuration, status reporting and call control. • Support for Russian, Japanese, Korean, Traditional Chinese and Simplified Chinese language in the Pexip Infinity Administrator interface.
Distributed architecture	• Efficient distribution to reduce bandwidth consumption over expensive WANs. • Able to deploy dedicated Proxying Edge Nodes to handle all external connections, and leave the conference media processing to privately-addressed Transcoding Conferencing Nodes. • Keeps media as local to each endpoint as possible, reducing the negative impacts of latency, jitter, and packet loss commonly experienced on centralized deployments. • Able to overflow capacity between nodes and locations, providing support for conferences that span multiple physical boxes. • Industry-leading resilience and redundancy capabilities. • A flexible licensing model that allows you to pool conference resources and quickly increase capacity in response to current local requirements.
Intelligent conference management	• Upscaling all connected participants to provide a seamless experience to all. • Ability to respond dynamically to fluctuating network conditions by downspeeding and upspeeding individual participants, and support for endpoint-based packet loss recovery and adaptation methodologies (such as packet loss concealment and dynamically adapting bandwidth), thereby protecting the user experience in the event of information loss. • Bandwidth-optimized content sharing towards the Pexip apps for crisp image at low bandwidth. • Full support for individual transcoding and transrating of both main stream video and audio, and dual stream content. • Direct media capabilities (end-to-end encrypted calls) between any two WebRTC-based participants. • Simple conference management and interaction for conference participants using the Pexip apps, including the ability for Host participants to add, disconnect, mute and unmute other participants. • Advanced conference management and interaction for administrators (using the web-based Administrator interface or the management API). • Optional tagging of services to allow service providers to track VMR use in CDRs and logs.

Feature	Description

Conferencing services	• Virtual Meeting Rooms providing personal meeting spaces for everyone within the organization.
	• Virtual Auditoriums designed to hold larger lecture-style conferences.
	• Virtual Reception IVR (Interactive Voice Response) service.
	• Media Playback Service that allows you to play prerecorded video content to consumers.
	• Infinity Gateway interoperability enables endpoints to:
	  ◦ Call into an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.
	  ◦ Make point-to-point calls to other endpoints that use different protocols and media formats (e.g. from Microsoft Teams Rooms or WebRTC to H.323 or SIP). Includes DTMF support.
	• Secure Scheduler for Exchange enables Microsoft Outlook desktop and Pexip web app users to schedule meetings using Pexip VMRs as a meeting resource.
	• Pexip Secure Scheduler for Web is a web-based scheduling solution for creating and managing secure, virtual meetings. It provides enhanced security with features such as one-time VMRs, secure pin codes, and single sign-on (SSO).
	• One-Touch Join enables the "click to join" functionality available in VTC endpoints.
	• VMRs, devices and users can be bulk-provisioned from directory information contained in a Windows Active Directory LDAP server, or any other LDAP-accessible database.
	• Pexip VMR self-service portal that allows end-users to manage their personal Virtual Meeting Room without having to send requests to their administrator to change the configuration or branding of their VMR.
	• Choice of layouts: select from a range of classic layouts, Pexip's AI-driven Adaptive Composition layout featuring real-time automatic face detection and framing, or design your own custom layouts.
	• Conference participants can chat and share messaging content.
	• Can output a dedicated multimedia stream to enterprise CDN (Content Delivery Network) streaming and recording services such as Wowza, Quickchannel, Qumu, VideoTool, and Microsoft Stream, and to public streaming services such as YouTube and Facebook.
	• Live captions: the meeting audio can be converted into readable text (live transcription).
	• Can integrate with Epic telehealthcare providers.
	• Ability to manage conferences and participants:
	  ◦ Require participants to authenticate in order to join a conference.
	  ◦ PIN-protect conferences and differentiate between Hosts and Guests.
	  ◦ Lock a conference to prevent any further participants from joining.
	  ◦ Split up a meeting into breakout rooms where participants can talk, present, and chat with each other.
	  ◦ Change the layout during a conference (individual selection or for all participants).
	  ◦ Transfer a participant to another conference.
	  ◦ Limit the number of participants in a conference, on a per-conference basis.
	  ◦ Limit the bandwidth used by each participant, on a per-conference and/or global basis.
	• Ability to re-brand with your own images and voice prompts, on a per-conference basis.
	• Ability to re-brand the Pexip app, and to offer multiple differently-branded web app experiences within your environment.
	• Ability to integrate Pexip app (WebRTC) functionality with third-party applications via our client REST API and with websites via the PexRTC library.
	• Call policy decisions can be taken by an external system or a local policy script.
	• Test call service that allows users to check their connectivity and the quality of their video and audio.

Feature	Description
Broad interoperability and protocol support	• Full support for existing industry-standard protocols (SIP, H.323), as well as other technologies (HTML5, Microsoft Skype for Business, RTMP, WebRTC). • Integration with Microsoft Teams and Teams Rooms. • Integration with Google Meet. • Integration with Microsoft Exchange and Office 365. • Ability to enable and disable support for individual audio and video codecs. • Easy integration with existing SIP and H.323 call control solutions including Cisco UCM, Cisco VCS, Polycom CMA, Polycom DMA, Avaya Aura, Skype for Business and others. • Conferencing Nodes can act as SIP registrars and as H.323 gatekeepers; nodes in the same system location act as alternate gatekeepers for the purposes of H.323 registration. • Support for automatic call escalation (Cisco VCS), call transfer capability (Cisco UCM), and CCCP to a Microsoft Skype for Business meeting. • Support for presence and customizable avatar published to a Microsoft Skype for Business client. • Support for automatic dial-out to audio bridges, including automatically issuing conference aliases and pass codes via DTMF tone generation. • IPv4 and IPv6 support. • Support for Far-End Camera Control (FECC). • Support for Cisco One Button to Push (OBTP) and Poly One Touch Dial (OTD). • Ability to tag management, call signaling, and media packets independently with DSCP QoS support. • Support for Forward Error Correction (FEC), downspeeding, bandwidth throttling, and other packet loss concealment technologies. • Unicode support (SIP, Pexip app, Administrator interface).
Firewall traversal	• Static NAT support. • Support for static routes. • Conferencing Nodes can be deployed with dual network interfaces. • Web proxy support. • Far-end NAT traversal (media latching). • Support for media over a TCP connection to assist with firewall traversal.
Security and monitoring	• Designed to comply with US Federal security requirements. • TLS certificate management, HSTS, certificate signing requests (CSRs). • DTLS support. • Active Directory / LDAP / OIDC / OAuth integration for administrator account authentication and authorization. • SNMPv2c and SNMPv3 support. • Support for multiple roles of access. • Authenticated SIP trunks. • Limit Infinity Gateway calls to registered devices only.

Pexip apps

The Pexip apps are a suite of free client software allowing users to connect to Pexip Infinity services from a web browser, installable desktop client, or mobile device.

Feature	Description
Standard features for all Pexip apps	• Can be used to join conferences as a full audio/video participant, an audio-only participant, or as a presentation and control-only participant. • Can be used to make point-to-point calls in conjunction with the Infinity Gateway. • Provides conference control to Host participants. • Allows participants to share and view content (screen share and full-motion HD presentations), whether or not they are connected with video and/or audio. • Pexip app for Windows and web app users (via Chrome or Firefox) can share their screen. • Chat (Instant Messaging) support. • Supports sending of DTMF tones.
Web app	• Allows participants to join a Virtual Meeting Room or Virtual Auditorium, or make a call via the Infinity Gateway, using a web browser as their video endpoint. The web app is supported in: • Google Chrome version 138 and later (64-bit only) on Windows, Linux, macOS, iOS*, and Android* • Mozilla Firefox version 140 and later on Windows, Linux, macOS, and iOS* • Microsoft Edge version 138 and later (64-bit only) on Windows and iOS* • Apple Safari version 18 and later on macOS and iOS* * For the best experience on mobile devices, we recommend using the Pexip app for mobile. ⓘ We strongly recommend using the latest publicly-released version (i.e. "stable version" or "supported release") of a browser.
Pexip app for Windows	• Allows a participant to join a Virtual Meeting Room or Virtual Auditorium, or make a call via the Infinity Gateway, using a lightweight client on any PC with a Windows operating system. • Allows users to register their clients in order to receive incoming calls and use directory services. Supported on Windows 10 or later and requires **Pexip Infinity version 37** or later.
Pexip app for mobile	• Mobile application optimized for Android and iOS devices, providing calendar integration and a meeting experience similar to that of the Pexip web app. • Allows a participant to join a scheduled meetings directly from their in-app calendar or make a call by dialing a video address. • The Pexip app for iOS app requires: ◦ **Pexip Infinity version 35** or later ◦ **iOS 18** or later • The Pexip app for Android app requires: ◦ **Pexip Infinity version 29** or later ◦ **Android 10** or later

Audio and video specifications and codecs

Feature	Description
Supported protocols	• H.323 • SIP • WebRTC • RTMP • Microsoft Skype for Business • Individual protocols can be administratively enabled and disabled
Audio codecs	• G.711(a/μ) • G.719 (this product is covered by patent rights licensed from Telefonaktiebolaget LM Ericsson) • G.722 • G.722.1, G.722.1 Annex C (SIP only) (licensed from Polycom®) • Siren7™, Siren14™ (licensed from Polycom®) • G.729, G.729A, G.729B • Opus • MPEG-4 AAC-LD (MPEG-4 video technology licensed by Fraunhofer IIS) • Speex • AAC-LC
Video codecs	• H.261 • H.263, H.263+ • H.264 AVC (Constrained Baseline Profile, Baseline Profile and High Profile), H.264 SVC (UCIF Profiles 0, 1) • VP8 • VP9 (for connections to Conferencing Nodes with processors using AVX2 or later)
Content sharing	• H.239 (for H.323) • BFCP (UDP for SIP) • VbSS (for Microsoft Teams and Skype for Business) • RDP (for Microsoft Skype for Business) • PSOM (for presenting PowerPoint files from Microsoft Skype for Business clients) • VP8, VP9 (for WebRTC high frame rate)
Bandwidth	• Connections from 8 kbps per participant (G.729, audio-only), up to 6 Mbps per participant (will vary depending on the deployment environment, video resolutions, etc)
Other audio and video features	• Video resolutions from QCIF to Full HD 1080p (1920 x 1080); 4:3 and 16:9 aspect ratios • Content resolutions up to 1920 x 1200 for transcoded calls (depending on remote side capabilities) • Frame rates up to 30 fps for transcoded calls • Non-transcoded calls (e.g. WebRTC direct media and SIP-SIP) can negotiate higher capabilities such as 720p60 / 1080p60 • Customizable video watermarking and content classification indicators • Pexip StudioSound™ for recording-studio audio quality • Wideband audio mixing • Automatic gain control • Control individual audio via Pexip apps • Support for AES (128-bit and 256-bit key size), DTLS SRTP, and H.235 for H.323 media encryption

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Host hardware requirements

Feature	Description

CPU	**Conferencing Nodes**
	• We recommend 3rd-generation (Ice Lake) or newer Intel Xeon Scalable Series Gold processors for Transcoding Conferencing Nodes.
	  ◦ Earlier Intel Xeon Scalable Series processors and Intel Xeon E5/E7-v3 and -v4 series processors are also supported where newer hardware is not available. Machines based on these architectures will work well for Management and Proxying Edge nodes, we recommend prioritizing the newest hardware for transcoding nodes.
	  ◦ Other x86-64 processors from Intel and AMD that support at least the AVX instruction set can be used but are not recommended. Some features are only available on underlying hardware that supports at least the AVX2 instruction set.
	• 2.6 GHz (or faster) base clock speed if using Hyper-Threading on 3rd-generation Intel Xeon Scalable Series (Ice Lake) processors or newer.
	  ◦ 2.8 GHz+ for older Intel Xeon processors where Hyper-Threading is in use
	  ◦ 2.3 GHz+ where Hyper-Threading is not in use
	• Minimum 4 vCPU per node
	• Maximum 48 vCPU per node, i.e. 24 cores if using Hyper-Threading
	  ◦ Higher core-counts are possible on fast processors: up to 56 vCPU has been tested successfully
	  ◦ Slow (under 2.3GHz) processors may require lower core counts
	Management Node
	• Any processor, 2.0 GHz or faster
	• Minimum 4 vCPUs

RAM	**Conferencing Nodes** 1 GB RAM per vCPU, so either: • 1 GB RAM per physical core (if deploying 1 vCPU per core), or • 2 GB RAM per physical core (if using Hyper-Threading and NUMA affinity to deploy 2 vCPUs per core). **Management Node** • Minimum 4 GB RAM (minimum 1 GB RAM for each Management Node vCPU)
Storage	**Conferencing Nodes** • 500 GB total per server (to allow for snapshots etc.), including: • 50 GB minimum per Conferencing Node • SSD recommended • RAID 1 mirrored storage (recommended) **Management Node** • 100 GB SSD
GPU	• Host servers do not require any specific hardware cards or GPUs.
OS	• The Pexip Infinity VMs are delivered as VM images (.ova etc.) to be run directly on the hypervisor. No OS should be installed.
Network	• Gigabit Ethernet connectivity is strongly recommended. • In general, you can expect 0.5-3 Mbps per call, depending on call control setup.

Feature	Description
Multiple VMs sharing the same hardware	• Pexip Infinity Conferencing Nodes and Management Nodes may share the same physical host. • Pexip nodes may also share the same physical host with other virtual machines. • Pexip virtual machines must be configured with dedicated CPU and memory resources, i.e. Pexip virtual machines do not support oversubscription.
Service provider considerations	A Pexip deployment can manage multiple customers in various ways: • **Single Management Node, multiple domains, shared Conferencing Nodes** A single installation with one Management Node and one or more Conferencing Nodes is used by all customers. Call control or DNS sends calls for all domains to the shared Conferencing Nodes. Does not provide dedicated capacity per customer. • **Single Management Node, multiple domains, dedicated Conferencing Nodes** One or more Conferencing Nodes per customer. Allows for dedicated capacity per customer. • **Dedicated Management Node and dedicated Conferencing Nodes per customer instance** Allows for close customer network integration, using VLANs, hosted on a shared server farm with multiple VLANs. The dedicated Management Node allows for customer self-management.

Capacity

Feature	Description
Call capacity	Capacity is dependent on server specifications. Servers that are older, have slower processors, or have fewer CPUs, will have a lower overall capacity. Newer servers with faster processors will have a greater capacity. The use of NUMA affinity and Hyper-Threading can significantly increase capacity.

Hypervisor requirements

Feature	Description
VMware	• Version 41 of the Pexip Infinity platform supports VMware vSphere ESXi 6.7, 7.0 and 8.0. • We recommend at least the **Standard** edition. • The **Enterprise** and **Enterprise Plus** editions have additional features that can be taken advantage of by Pexip Infinity in larger deployments. • The Pexip Infinity platform will run on the **free edition** of vSphere Hypervisor. However, this edition has a number of limitations that mean we do not recommend its use except in smaller deployments, or test or demo environments.
Microsoft Hyper-V	• The Pexip Infinity platform supports Microsoft Hyper-V Server 2025.
KVM	• You can deploy the Pexip Infinity platform in a KVM environment.
Other hypervisors and orchestration layers	• Conferencing Nodes can be provisioned with a configuration document generated independently of a generic VM image. This permits deployment of Pexip Infinity onto unsupported hypervisors as well as onto supported hypervisors that are managed by an orchestration layer. • Pexip Infinity can be deployed on Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP) or Oracle Cloud Infrastructure, and on the HPE Helion Openstack® Cloud platform.

What's new in version 41?

The new features and enhancements and changes in functionality included in version 41 of the Pexip Infinity platform and associated web app, along with any planned changes, are described below.

For full information about the release, see the Pexip Infinity release notes and the Pexip web app release notes.

For information about new features and changes to earlier versions of Pexip Infinity, see Features added in previous releases.

Pexip Infinity v41 new features and enhancements

Pexip Infinity platform

Feature	Description	More information
Teams Connector enhancements: DSC status information, Azure Automation removed	This release contains the following administrative changes and improvements: • DSC (desired state configuration) status information for each Teams Connector instance is now shown in **Status > Live View** in the Pexip Infinity Administrator interface when hovering over a Teams Connector instance, in the detailed status view, and in **Status > Microsoft Teams Connectors**. • The **Enable PowerPoint Live** option in **Platform > Global Settings** in the Pexip Infinity Administrator interface has moved from the **Tech Preview Features** section to the **Connectivity** section. • The Azure Automation PowerShell module and Automation account resource is no longer required from version 41 (due to Microsoft's deprecation of Azure Automation State Configuration). Note that version 41 of the Teams Connector contains updates that necessitate an upgrade to your Pexip platform to ensure compatibility with the latest updates to the Microsoft Teams APIs and to the Teams Connector's latest features. We strongly recommend that you upgrade your Pexip deployment — both the Pexip Infinity platform and the Pexip Teams Connector — to version 41 as soon as practicable.	
Audio announcement for changes to content classification levels	You now have the option to trigger an audio announcement if the classification level changes during a conference. Content classification levels and announcements are configured via the theme associated with the conference.	Content classifications
Google Meet enhancements: smart note announcements	When Gemini note taking is enabled in a Google Meet conference, a new "smart notes" in-conference indicator is displayed, and appropriate audio announcements are played out to VTC participants.	Introduction
Layout enhancements: participant pinning improvements	When using participant pinning, you can now: • Provide a dynamic pinning configuration while a conference is in progress via the client REST API. Previously a pinning configuration had to be specified within a theme. • When defining and assigning a dynamic pinning configuration via the client REST API, a specific participant can now be assigned to a layout group (via their participant UUID). • Added the ability to exclude a reserved slot from the layout when there are no participants found for that slot.	Pinning participants to the layout during a conference

Feature	Description	More information

Theme updates: new audio files and settings	• There are new theme audio files and settings in support of Gemini note taking in Google Meet conferences:
	  ◦ New audio files: **conf-smart_notes_started_48kHz_mono.wav** and **conf-smart_notes_stopped_48kHz_mono.wav**.
	  ◦ New fields in the **themeconfig.json** file: **disable_smart_notes_indicator** and **smart_notes_indicator_text**.
	• There is a new theme audio file in support of content classification level changes: **conf-classification-changed_48kHz_mono.wav** (note this is an empty/silent file by default).
	• There is a new notification message shown to a direct media participant when the call is about to be terminated because the Host participant has left.
	• The following fields have also been added to the **themeconfig.json** file to allow control of pre-existing features: **disable_recording_indicator** and **disable_transcribing_indicator** (they both default to **false**).
		Rules and requirements for customized themes

OTJ: improvements to regular expressions	A new One-Touch Join regular expression meeting type of **Regex RE2** has been introduced. This offers improved performance over the previous **Regex** meeting type, which is being deprecated.	

Policy enhancements: custom properties fields added to service and participant responses	Local and external policy has the following enhancements:
	• New custom properties fields have been added to the service and participant responses:
	  ◦ Service/gateway responses: new **custom_properties** and **host_custom_properties** fields.
	  ◦ Participant responses: new **custom_properties** and **private_custom_properties** fields.
	These new fields can contain any form of JSON dictionary and would typically be subsequently utilized by client applications and plugins.
	The properties supplied in these fields are provided in the conference and participant update events/callbacks in the client APIs.	

Administrative improvements: new syslog configuration options, security wizard DHE ciphersuites option	This release contains the following administrative improvements: • When configuring a syslog server you can now specify the protocol format (defaults to RFC3164), and optionally whether to send system log entries. • There is a new option in the security wizard that provides the ability to disable DHE ciphersuites.	Using a syslog server

Pexip web app (Webapp3)

Feature	Description	More information
New and improved features		
Background blur and replacement	• Changes to the way in which users select background effects from within their self-view. (See also data-testid changes below.) • Improvements to background blur and image replacement, including smoother edge definition and more natural compositing between foreground and background.	Blurring or replacing your background
Additional "flow" URL parameter	Preconfigured URLs used to launch the web app now support a new "flow" parameter. When set to **flow=express** and used in conjunction with the **role=guest** parameter, the participant will join as a Guest but bypass the step-by-step flow.	flow Support for first-time and infrequent users

Feature	Description	More information
Changes in functionality		
Display name and join flow	When joining a conference that requires SSO authentication, users are no longer asked to enter their display name (because it is provided by the Identity Provider and will override anything provided by the user). When joining a conference that does not require SSO, users who have not already provided their name (either when joining a previous meeting, or via the Settings menu) are now asked to enter their display name at the end of the joining flow rather than at the beginning. As part of this change, the link to the **Terms of Service** now appears when users are asked to enter the video address. When using preconfigured links that include the video address, users will skip this stage. If this applies in your organization but you require users to accept the **Terms of Service** before use, you should instead implement this as part of a Custom Step. This can be configured to reflect your organization's specific requirements for displaying and accepting the terms of service information.	
`data-testid` changes	The `data-testid` (referenced when hiding elements from the web interface) has changed for the following elements: • The button to remove background effects has changed from `select-effect-none` to `button-background-effect-none` • The button to blur the background has changed from `select-effect-blur` to `button-background-effect-blur` • The button to upload a new background image has changed from `button-add-overlay` to `button-background-effect-upload`	
Live captions location	The live captions text overlay has been replaced by a scrollable, expandable live transcript box at the bottom of the video window.	View a live transcription of the meeting
Webapp1 no longer available	Webapp1 was deprecated from Pexip Infinity v32 and is no longer available from v41.	

Pexip Infinity v41 changes in functionality

Feature	Description	More information
Backup file format and checksums	• The format of the backup files no longer includes **.tar**. The format is now **pexip_auto_backup_<hostname>_<version>_<build>_<date>_<time>.pexbak** • SHA-256 Checksums are now available for each backup file. These can be used to verify the file before restoring.	Backing up and restoring configuration
Removal of insecure transport option for LDAP integrations	The option to **Allow insecure transport** for LDAP integrations (available when configuring LDAP sync sources and LDAP-based authentication for the Administrator interface) has been removed. This option relied on Digest-MD5 which was marked as "Historic" in 2011 and is insecure and unsupported. If your existing deployment still uses the option to **Allow insecure transport**, **before** upgrading to v41 you **must**: • migrate to TLS-based LDAP integration • ensure you know the password of your local administrator account, so you can log in as an administrator without LDAP access if necessary. Contact your Pexip authorized support representative if you require any further information or assistance.	Managing administrator access via LDAP Configuring an LDAP data source

Feature	Description	More information

Enhanced SSH security	The security of SSH authentication has been tightened. As a result, if your SSH client has multiple authentication keys available, you may find you are unable to login due to "Too many authentication failures". You can mitigate this by configuring your SSH client to use the specific key for that host. For example:
	`Host *.example.vc`
	`  User admin`
	`  IdentityFile /path/to/example.vc/key`	

Planned changes / actions required

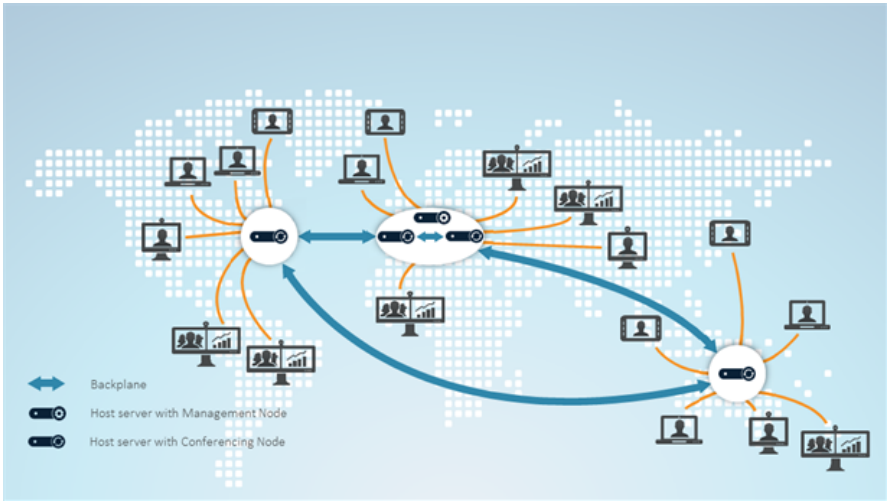
Feature	Description	More information
OTJ: Microsoft Teams domain change	In Q1 2026 Microsoft will start transitioning Teams links from **teams.microsoft.com** to **teams.cloud.microsoft**. One-Touch Join will automatically support invitations using **teams.cloud.microsoft** after upgrading to **v39 or later**, but if you use any custom rules that use **teams.microsoft.com**, you must update them manually to use the new format.	
Remove support for VMware ESXi 6.7	Support for VMware ESXi 6.7 will be removed in a future release. When this occurs, Pexip will support VMware installation on ESXi 7.0 and 8.0.	
Remove support for AVX	In 2027 we intend to cease support for the AVX instruction set on Pexip Infinity Conferencing Nodes. AVX2 and AVX512 will continue to be supported. This means that any Conferencing Nodes based on Intel Sandy Bridge or Ivy Bridge will cease to work. These parts were last sold in 2015 and Intel ended support in 2020. Any newer hardware that has been deployed with a VMware virtual hardware revision older than 11 will need to be updated to a newer virtual hardware revision.	
Use of application impersonation in One-Touch Join and Secure Scheduler for Exchange	In a future release we will remove the ***OAuth (delegate access with a service account)*** Authentication method option when configuring a Secure Scheduler for Exchange Integration. This is because from **February 2025** Microsoft disabled Application Impersonation role assignments to service accounts (for more information, see Microsoft's announcement). Therefore: • **One-Touch Join** customers using O365 with an OTJ Exchange Integration must migrate to using Graph API instead. For full instructions, see Migrating from EWS API to Graph API for One-Touch Join. • **Secure Scheduler for Exchange** customers still using O365 with a service account to access equipment resources and mailboxes must migrate to using app permissions instead. For full instructions, see Configuring Office 365 for scheduling using app permissions.	
Secure Scheduler for Exchange: deprecation of {{start_time}} and {{end_time}} variables	In a future release we will remove the **{{start_time}}** and **{{end_time}}** variables from the following Secure Scheduler for Exchange jinja2 template fields: • Accept new single meeting template • Accept edited single meeting template • Accept new recurring meeting template • Accept edited occurrence template • Accept edited recurring meeting template If you are using these variables, please update your templates to remove them.	
Deprecation of password-based authentication for the Teams Connector CVI application	Certificate-based authentication (CBA) is now the default method to authenticate the Teams Connector CVI application towards MS Graph. You can still use the previous password-based authentication method, but we plan to deprecate it in a future release, thus we recommend using CBA for new installations, or migrating to CBA as soon as practicable when upgrading existing deployments.	

Feature	Description	More information
Deprecation of v1 legacy themes	We plan to deprecate legacy style "version 1" themes in a future release. If you still use v1 themes you should plan to move to using "version 2" themes instead, which have been available since version 18 of Pexip Infinity.	
Deprecation of Webapp2	Webapp2 is being deprecated, and from January 2027 will no longer be included with Pexip Infinity. If this will impact your deployment, please contact your Pexip authorized support representative.	
Deprecation of OTJ "Regex" meeting type	Versions of Pexip Infinity prior to v41 supported a regular expression **Meeting type** of ***Regex***. This is being deprecated and the alternative ***Regex RE2*** should be used because it offers improved performance. Any existing ***Regex*** meeting types should be updated to ***Regex RE2***.	

Components of the Pexip Infinity platform

The Pexip Infinity conferencing platform is a virtual entity that consists of a **Management Node** and one or more securely interconnected **Conferencing Nodes**. Both are software applications that you deploy as Virtual Machines (VMs) on **host servers** distributed around the globe, or via a **cloud service**. You can add, remove or move Conferencing Nodes according to your conferencing capacity requirements.

- Conferences take place in **Virtual Meeting Rooms** and **Virtual Auditoriums**, with each having one or more associated **aliases**. Conference participants access a Virtual Meeting Room or Virtual Auditorium by dialing any one of its aliases directly, or via the **Virtual Reception** IVR service. This connects them to the Virtual Meeting Room or Virtual Auditorium on their nearest Conferencing Node. A single such conference can take place across one, two, or more Conferencing Nodes with no difference in conference experience from the participants' perspective.
- Conference participants can access Virtual Meeting Rooms and Virtual Auditoriums from virtually any endpoint, including any of the **Pexip apps** (which include desktop, mobile and web apps). Pexip apps can also be used to control the conference, view presentations, share content, and chat with other conference participants.
- The **Pexip Infinity Distributed Gateway** allows users to make person-to-person calls between virtually any type of endpoint (including SIP and H.323 devices, Skype for Business, and Pexip's own Pexip apps). It also enables these endpoints to join an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.
- The **Media Playback Service** allows you to play prerecorded video content (such as adverts and informational videos) to consumers.
- **Secure Scheduler for Exchange** enables Microsoft Outlook desktop and Pexip web app users (using Office 365, Exchange 2013 or Exchange 2016) to schedule meetings using Pexip VMRs as a meeting resource.
- **Pexip Secure Scheduler for Web** is a web-based scheduling solution for creating and managing secure, virtual meetings. It provides enhanced security with features such as one-time VMRs, secure pin codes, and single sign-on (SSO).
- Pexip **One-Touch Join** integrates support for videoconferencing endpoints' "click to join" workflows. It can be integrated within an existing or new Pexip Infinity deployment, or installed as a stand-alone deployment with its own dedicated Management Node and Conferencing Nodes.
- The Pexip VMR self-service portal is a separately-installable component that allows end-users to manage their personal Virtual Meeting Room without having to send requests to their administrator to change the configuration or branding of their VMR.
- Pexip's AI Media Server (AIMS) is a self-hosted, standalone virtual machine (VM) that allows you to access Pexip's AI-powered features (such as live captions) in a secure environment.



Pexip Infinity deployment showing Management Node and four Conferencing Nodes with participants connected locally

Management Node

The Management Node is the administrative interface of the Pexip Infinity platform, from which administrators can:

- Create and manage Conferencing Nodes.
- Configure Pexip Infinity services (Virtual Meeting Rooms, Virtual Receptions and so on).
- View platform and conference status across all Conferencing Nodes.
- Perform active conference management functions such as adding and disconnecting participants, enabling streaming or recording services, locking a conference, or muting a participant's audio.

The Management Node does not handle any conference media or signaling.

It is deployed using a virtual machine management application such as VMware's vCenter Server, or Microsoft Hyper-V, or on a cloud service such as Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP) or Oracle Cloud Infrastructure.

Conferencing Nodes

The Conferencing Nodes provide the capacity for conferences.

- They handle all conference signaling and media (except for direct media scenarios).
- A Conferencing Node can have either a transcoding or a proxying role:
  - Transcoding Conferencing Nodes are required in all deployments; they manage all of the media processing required to host a conference, and can also handle direct connections to/from endpoints if required.
  - Proxying Edge Nodes are optional; they handle call signaling and the media connection with the endpoint, but forward the media on to a Transcoding Conferencing Node for processing. For more information, see Distributed Proxying Edge Nodes.
- There is no limit on the number of Conferencing Nodes that you can add to the Pexip Infinity platform.
- All Conferencing Nodes get the same service configuration from the Management Node. This means that participants throughout your organization can access the same Pexip Infinity services (Virtual Meeting Rooms, Virtual Receptions and so on) even though they might be connected to different Conferencing Nodes.
- Conferencing Nodes are deployed via the Management Node. You use the Management Node to configure the new Conferencing Node and generate a configuration file, then complete the deployment using the appropriate hypervisor or cloud-provider tools.
- The Pexip Infinity platform can have Conferencing Nodes that are deployed on one or more host servers, across one or more system locations and managed by one or more types of hypervisor, or it can be a hybrid deployment with nodes running on a combination of on-premises and cloud-hosted servers. A Conferencing Node can co-exist on the same host server as a Management Node.
- Conferencing Nodes can be deployed with dual network interfaces.

Pexip apps

Conference participants do not need to have a traditional video endpoint in order to access Pexip Infinity services.

The complementary Pexip apps allow users to connect to any conference, either:

- directly from a web browser without any special downloads or plugins
- from an installable desktop app
- from a mobile app, available for iOS or Android.

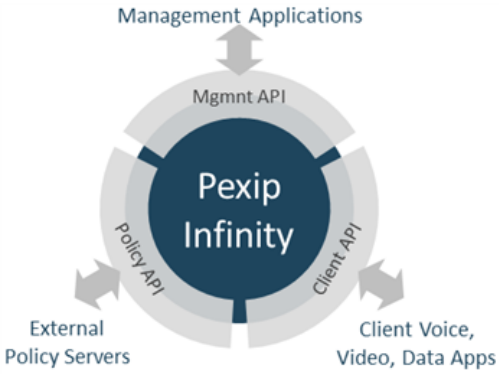
In addition to connecting with video and audio, Pexip app users can control the conference, view presentations, share content and chat. Pexip apps can also be used to make direct calls to other devices or systems when used in conjunction with the Infinity Gateway.

For more information on using and administering Pexip apps, see Introduction to Pexip apps.

Pexip Infinity APIs and SDKs

Pexip Infinity incorporates several powerful and comprehensive APIs:

- **Management API**: a REST API used for configuring the entire Pexip Infinity deployment, viewing history and status, and issuing commands.
- **Client API**: a REST API used for managing calls and participants, such as connect, disconnect, mute and unmute, presentation controls, DTMF, etc. See Pexip client REST API for more information.
- **Policy API**: a REST API used to defer decision-making to external policy servers instead of using the built-in call policies within Pexip Infinity. See Using external and local policy to control Pexip Infinity behavior for more information.



In addition to these REST APIs, a Javascript API is also available for building custom web-based clients. See PexRTC JavaScript client API for more information.

Conference types and services

The Pexip Infinity platform offers a variety of conference types and services:

- Virtual Meeting Rooms and Virtual Auditoriums are used to hold conferences, share presentations, and chat. Participants can join over audio or video from any location using virtually any type of communications tool, such as Skype for Business, a traditional conferencing endpoint, a mobile telephone, or a Pexip app.
- The Virtual Reception IVR service provides a way for conference participants who cannot dial Virtual Meeting Room and Virtual Auditorium aliases directly, to access these services from a central point using DTMF tones. It can also be used to route calls via the Infinity Gateway.
- The Pexip Infinity Distributed Gateway:
  - Enables any type of endpoint, including traditional VTC endpoints, to join externally-hosted meeting services such as Microsoft Teams and Google Meet.
  - Enables endpoints within your deployment to make direct calls to other endpoints. As with calls into VMRs, the gateway can interwork the protocols and media formats used by each type of device (SIP, H.323, WebRTC etc).
  - Can be used with call control systems and other third party services to enable calls from your deployment to external devices including PSTN and mobile phones.
- Pexip's Media Playback Service allows you to play prerecorded video content (such as adverts and informational videos) to consumers. When the media finishes playing, the user can be transferred to another service, such as a VMR conference, or they can be disconnected.
- Secure Scheduler for Exchange integrates Pexip Infinity with Microsoft Exchange. It enables Microsoft Outlook desktop and Pexip web app users (using Office 365, Exchange 2013 or Exchange 2016) to schedule meetings using Pexip VMRs as a meeting resource.
- Pexip's One-Touch Join (OTJ) allows users to schedule a meeting in Microsoft Outlook or Google Calendar and include in the invitation a meeting room with a supported Cisco or Poly videoconferencing endpoint, so that the endpoint in the chosen meeting room displays a **Join** button just before the meeting is scheduled to begin. Participants can then simply walk into the room and select the button, and the endpoint will automatically dial in to the meeting.
- A Test Call Service provides a test loopback service that allows users to check the quality of their video and audio (i.e. that their local camera, microphone and speakers are working properly), and verifies that they can connect to a Conferencing Node.

VMR self-service portal

The Pexip VMR self-service portal is a separately-installable component that allows end-users to manage their personal Virtual Meeting Room without having to send requests to their administrator to change the configuration or branding of their VMR.

Pexip provides the VMR portal appliance via an OVA template suitable for deployment on VMware ESXi. The OVA template is provided "as-is" and provides a reference installation which is suitable for integrating with an existing Pexip Infinity deployment.

Pexip Private AI and AIMS

The Pexip Private AI platform allows you to access Pexip's AI-powered features (such as live captions) in a secure environment. It uses Pexip's AI Media Server (AIMS), a self-hosted standalone virtual machine, which you deploy on your own hardware or private cloud environment, giving you complete control of your data.

The Pexip Private AI platform is deployed alongside, but entirely separately to, your Pexip Infinity platform. You configure Pexip Infinity to integrate with Pexip Private AI where required for supported features.

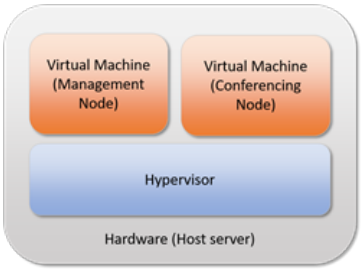
Host servers

The Management Node and Conferencing Nodes are virtual machines (VMs) that run on industry-standard host servers. A Management Node can run on the same host server as a Conferencing Node. Other Conferencing Nodes can run on host servers in the same or different locations, allowing you to create a globally distributed system.

You can have two Conferencing Nodes running on the same host server, for example to ensure service continuity during upgrade of one of the Conferencing Nodes. However, you must ensure that your hardware is not over-committed.

The Pexip Infinity platform can also be deployed as a cloud service via **Amazon Web Services (AWS)**, **Microsoft Azure**, **Google Cloud Platform**, or **Oracle Cloud Infrastructure**, with private, public or hybrid deployment options.

Hypervisors



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ach host server runs a hypervisor, an application which manages virtual machines and the physical hardware on which they are hosted. Pexip Infinity version 41 includes specific support for the following hypervisors:

- VMware vSphere ESXi (6.7, 7.0 and 8.0)
- Microsoft Hyper-V Server 2025
- KVM

Other hypervisors and orchestration layers may be used but are not officially supported. If you wish to deploy Pexip Infinity using a non-supported hypervisor, we recommend that you contact your Pexip authorized support representative for assistance.

Call control

Supported call control solutions

Pexip Infinity can be easily integrated with virtually any existing SIP, H.323 and Skype for Business call control solutions including Cisco UCM, Cisco VCS, Polycom CMA, Polycom DMA, Avaya Aura and others.

Local and external policy

You can extend Pexip Infinity's built-in functionality by using external and/or local policy to apply bespoke call policy and routing decisions based on your own specific requirements.

See Using external and local policy to control Pexip Infinity behavior for more information.

Endpoint registrations

Pexip Infinity can act as a SIP registrar and H.323 gatekeeper, which means that SIP and H.323 endpoints can register directly to Pexip Infinity. This allows Pexip Infinity to route calls to those registered devices without having to go via an external SIP proxy or H.323 gatekeeper, or rely on DNS.

The Pexip app for Windows can also register to Pexip Infinity. Registration is mandatory and allows these apps to receive calls via Pexip Infinity and use directory lookup services.

For more information, see Registering devices to Pexip Infinity and DNS record examples.

Note that the Pexip Infinity platform does not register with external gatekeepers as an MCU.

Using an external gatekeeper to route calls to Pexip Infinity conferences

To ensure that calls can be routed to Pexip Infinity, your gatekeeper or call control system must be configured with appropriate neighbor/zone relationships towards the Pexip Infinity Conferencing Nodes. These zones must be set up so that when an endpoint places a call to a Pexip Infinity alias, the call is routed to the endpoint's local Conferencing Node(s) as a first preference. Other non-local Conferencing Nodes can be used as secondary choices to provide redundancy.

For further information about how to configure your specific call management system to work with Pexip Infinity, see the following documentation:

- Pexip Infinity and Microsoft Skype for Business Deployment Guide
- Pexip Infinity and Cisco VCS Deployment Guide
- Pexip Infinity and Cisco Unified Communications Manager Deployment Guide
- Pexip Infinity and Polycom DMA Deployment Guide

Room management

Enhanced Room Management (ERM) is a separately-installable Pexip product that provides management of your Cisco and Poly devices and room systems. From system monitoring to bulk provisioning of software upgrades, address books, and branding profiles, it provides everything you need to manage your systems from one single management interface.

ERM was End of Sale as of 6 March 2026.

Distributed architecture

Benefits of the Pexip Infinity distributed architecture

Pexip Infinity is built on top of a distributed architecture which provides:

- Centralized management of any number of Conferencing Nodes in any number of locations.
- Ability to deploy conferencing resources where and when required, without service outage.
- Ability to deploy dedicated Proxying Edge Nodes to handle all external connections, and leave the conference media processing to privately-addressed Transcoding Conferencing Nodes.
- Significant WAN bandwidth savings in conferences that span locations.
- Consistent user experience, independent of the number of Conferencing Nodes.
- Increased resilience to temporary network outages.
- Ability to use Conferencing Nodes as gateways for person-to-person calls, thus avoiding hairpinning of media back to a centralized datacenter.
- Allocation of licenses from a central pool.

Distributed architecture components

Pexip Infinity distributed architecture is purely software-based and virtualized, running on industry-standard servers. It consists of:

- A single Management Node. A Pexip Infinity deployment, regardless of size, has just one Management Node. The purpose of the Management Node is to create and manage Conferencing Nodes. The Management Node is in neither the signaling nor the media path of a conference.
- One or more Conferencing Nodes. Conferencing Nodes handle all aspects of the media and signaling connections to endpoints and other devices, and host the associated conferences. A minimal Pexip Infinity deployment has one Management Node and one Conferencing Node. However, most Pexip Infinity deployments will have multiple Conferencing Nodes.

A Management Node can run on the same host server as a Conferencing Node. Other Conferencing Nodes can run on host servers in the same or different locations, allowing you to create a globally distributed system. You can have two Conferencing Nodes running on the same host server, for example to ensure service continuity during upgrade of one of the Conferencing Nodes, and for maximum performance.

The Pexip Infinity platform can also be deployed as a cloud service via **Amazon Web Services (AWS)**, **Microsoft Azure**, **Google Cloud Platform**, or **Oracle Cloud Infrastructure**, with private, public or hybrid deployment options.

Centralized management

Configuration and provisioning data is pushed out from the Management Node to all the Conferencing Nodes, and diagnostics data and event information is sent from the Conferencing Nodes back to the Management Node. Administrators never have to manage a Conferencing Node directly. This centralized management ensures that the entire deployment is configured with a consistent data set – the entire deployment acts as a single application.

Scaling up

You can easily scale a deployment up by creating several Conferencing Nodes in the same location (i.e. the same datacenter). Capacity can even be added “on the fly” – Conferencing Nodes can be added in a couple of minutes if more capacity is needed. Alternatively, each location can be configured to overflow to another location if it reaches its capacity, including bursting to temporary resources on a cloud service.

Scaling out

A typical Pexip Infinity deployment consists of two or more locations. If a customer has three main offices such as New York, London and Tokyo, and a concentration of users in those locations, you would typically deploy Conferencing Nodes in all three locations. There is no limit on the number of locations in a Pexip Infinity deployment. Additional locations can be added “on the fly” while the system is operating, with no impact on service availability.

Application level resiliency

Application level resiliency greatly improves on a conference experience during for instance temporary network outages, as Pexip Infinity will automatically re-establish the conference when the network connection is re-established.

Conference distribution

All Pexip Infinity services (Virtual Meeting Rooms, Virtual Auditoriums, Virtual Receptions and the Infinity Gateway) can be accessed via any Conferencing Node. When a user dials in to a Virtual Meeting Room, or Virtual Auditorium a **conference instance** is created. As more users dial in to the conference, it may be managed across one or more Conferencing Nodes.

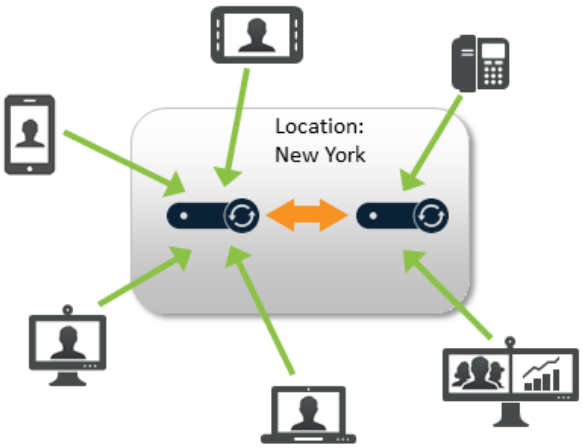
When you deploy a Conferencing Node, you select its system location and role:

- **System location**: this is normally used within Pexip Infinity to group together those nodes that are in the same physical location. Conferences that span more than one Transcoding Conferencing Node can be locally distributed, globally distributed, or both, depending on the system location of each of the nodes involved. These various distribution scenarios are described in more detail below.
- **Role**: determines if the Conferencing Node has a **transcoding** role (i.e. it is a Transcoding Conferencing Node that manages and hosts conferences) or has a **proxying** role (i.e. it is a Proxying Edge Node that forwards the client's media onto a Transcoding Conferencing Node). Proxying Edge Nodes (and their associated locations) do not affect whether a conference is locally or globally distributed, as they simply forward media onto the Transcoding Conferencing Nodes that are responsible for hosting the conference — but the locations of the transcoding nodes that process the media do affect how the conference is distributed. See Conference distribution and Proxying Edge Nodes for an example of calls received on proxying nodes that are forwarded onto transcoding nodes in a locally and geographically distributed conference.

When two or more Transcoding Conferencing Nodes are hosting the same conference, they send the call media between each other over a secure IPsec **backplane**.

Locally distributed conferences

Locally distributed conferences exist across multiple Transcoding Conferencing Nodes in the same **System location**, as shown in the diagram opposite.



Having more than one transcoding node in a single location allows you to increase conferencing capacity and provide redundancy.

To maintain efficiency, no more than four transcoding nodes **per location** will handle the media for a **particular conference instance**. However, you can configure "overflow" locations that will handle the media if a location reaches its capacity for a conference instance. For more information, see Handling of media and signaling.

Globally distributed conferences

Globally distributed conferences exist across several Transcoding Conferencing Nodes, where each node is in a different system location. As system locations are typically used to represent different physical locations, this allows participants in different regions to access the conference from their local Conferencing Node. The nodes send the call media for the conference to each other over a single geo backplane, with each node sending the media on behalf of all the endpoints connected (or proxied) to it, thus minimizing WAN bandwidth usage between locations.

This helps to provide a superior meeting experience as it reduces the distance that the media has to travel between the endpoint / client and the Conferencing Nodes.

Locally and globally distributed conferences

Locally and globally distributed conference with four Conferencing Nodes across four host servers in three locations

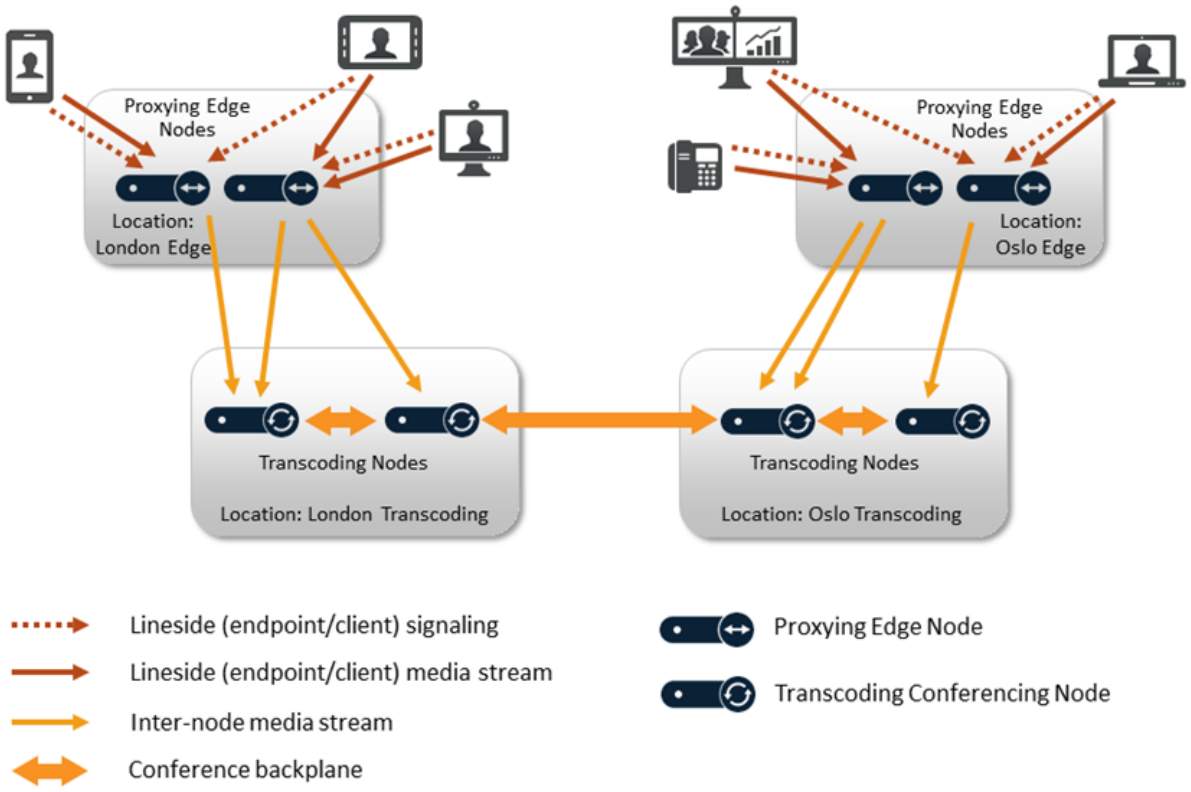
A conference can be **locally and globally distributed** at the same time, if two or more Transcoding Conferencing Nodes in one location and at least one other transcoding node in a different location are involved. In such cases, one transcoding node in each location acts as the intermediary for any other transcoding nodes in the same location that are handling the media for that conference. Call media for each location is sent between the intermediaries only, thus minimizing WAN bandwidth usage between locations.

Conference escalation from locally to globally distributed is handled automatically by Pexip Infinity and is seamless to conference participants.

Conference distribution and Proxying Edge Nodes

Proxying Edge Nodes (and their associated locations) do not affect whether a conference is locally or globally distributed, as they simply forward media onto the Transcoding Conferencing Nodes that are responsible for hosting the conference — but the locations of the transcoding nodes that process the media do affect how the conference is distributed.

A system location should not contain a mixture of proxying nodes and transcoding nodes. Hence, in the example scenario shown here, the Conferencing Nodes in the two locations "London Edge" and "Oslo Edge" are Proxying Edge Nodes and thus those nodes and locations are not involved in the actual hosting of any conferences. They forward the media onto the Transcoding Conferencing Nodes in the "London Transcoding" and "Oslo Transcoding" locations respectively. This conference is locally distributed within both of those "transcoding" locations (as there are multiple nodes hosting the conference within each location, with a local backplane between those nodes), and as the conference is split across two locations, it is also geographically distributed and therefore there is a geo backplane between one of the nodes in the "London Transcoding" location and one of the nodes in the "Oslo Transcoding" location.



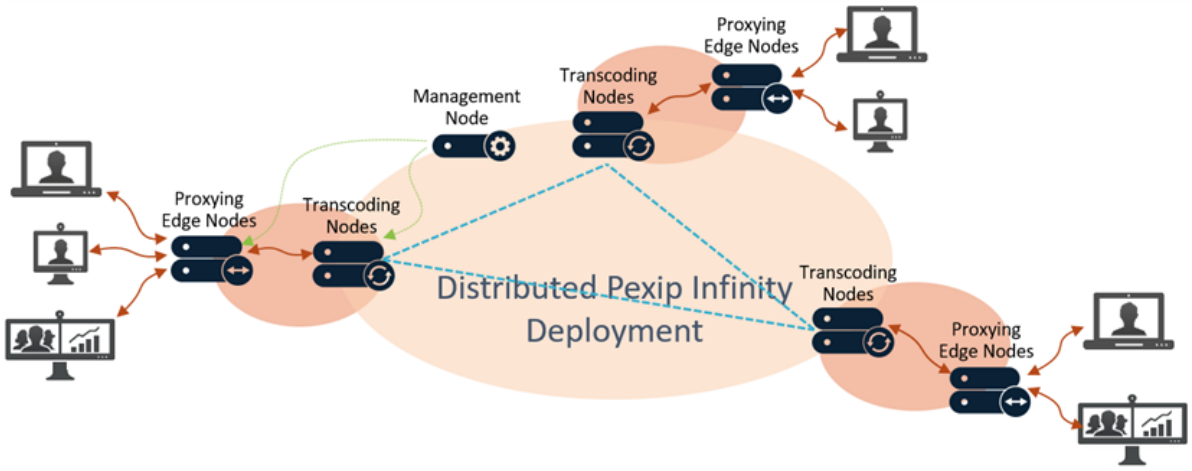
Distributed Proxying Edge Nodes

When designing your Pexip Infinity deployment, one of the main considerations is how and where to deploy your Conferencing Nodes. A Conferencing Node can perform one of two roles:

- **Proxying**: a Proxying Edge Node handles all media and signaling connections with an endpoint or external device, but does not host any conferences — instead it forwards the media on to a Transcoding Conferencing Node for processing.
- **Transcoding**: a Transcoding Conferencing Node handles all the media processing, protocol interworking, mixing and so on that is required in hosting Pexip Infinity calls and conferences. When combined with Proxying Edge Nodes, a transcoding node typically only processes the media forwarded on to it by those proxying nodes and has no direct connection with endpoints or external devices. However, a transcoding node can still receive and process the signaling and media directly from an endpoint or external device if required.

You can deploy your Pexip Infinity platform as either a mix of Proxying Edge Nodes and Transcoding Conferencing Nodes, or as a system that only contains Transcoding Conferencing Nodes.

A typical deployment scenario is to use Proxying Edge Nodes as a front for many privately-addressed Transcoding Conferencing Nodes. Those outward-facing proxying nodes would receive all the signaling and media from endpoints and other external systems, and then forward that media onto other internally-located transcoding nodes to perform the standard Pexip Infinity transcoding, gatewaying and conferencing hosting functions.



How it works

When you deploy a new Conferencing Node, you must decide on its role — either **transcoding** or **proxying** (although you can change its role later).

- If an endpoint connects to a Transcoding Conferencing Node, that node will handle the signaling connection with that endpoint, and (subject to media allocation rules) it may also host the associated conference or gateway call on that node.
- If an endpoint connects to a Proxying Edge Node, that node will still handle the signaling connection to the endpoint, but it (or another proxying node in that location) will then proxy the media on to a Transcoding Conferencing Node for processing. The proxying node will perform any encryption or decryption of the media stream that may be required.

The benefits of using Proxying Edge Nodes to handle all connections from endpoints and other systems — and leaving the media processing to internally-located Transcoding Conferencing Nodes — are:

- You only need to deploy certificates on your Proxying Edge Nodes — only those nodes that handle the signaling connection to an endpoint or other system (such as a Skype for Business Edge Server) need to be configured with the appropriate certificates to allow that endpoint/system to communicate with Pexip Infinity. If you subsequently deploy more Transcoding Conferencing Nodes to increase conferencing capacity, you do not need to add certificates onto those additional nodes.
- You only need to set up and maintain DNS records for the Proxying Edge Nodes. If you subsequently add more Transcoding Conferencing Nodes you do not have to update any call routing logic on third-party systems that connect to Pexip Infinity.
- When an endpoint has established a signaling and a media connection to one or more Proxying Edge Nodes, the ports used for that connection will not change (even if, for example, the call is transferred to a VMR via a Virtual Reception).
- The servers hosting Proxying Edge Nodes do not require as high a specification as those servers hosting Transcoding Conferencing Nodes because they do not process any media. You still need multiple proxying nodes for resilience and capacity. We recommend allocating 4 vCPU and 4 GB RAM (which must both be dedicated resource) to each Proxying Edge Node, with a maximum of 8 vCPU and 8 GB RAM for large or busy deployments.

For more information, see Deployment guidelines for Proxying Edge Nodes.

Bandwidth optimization

The Pexip Infinity distributed architecture provides major bandwidth savings when compared with traditional MCU deployments.

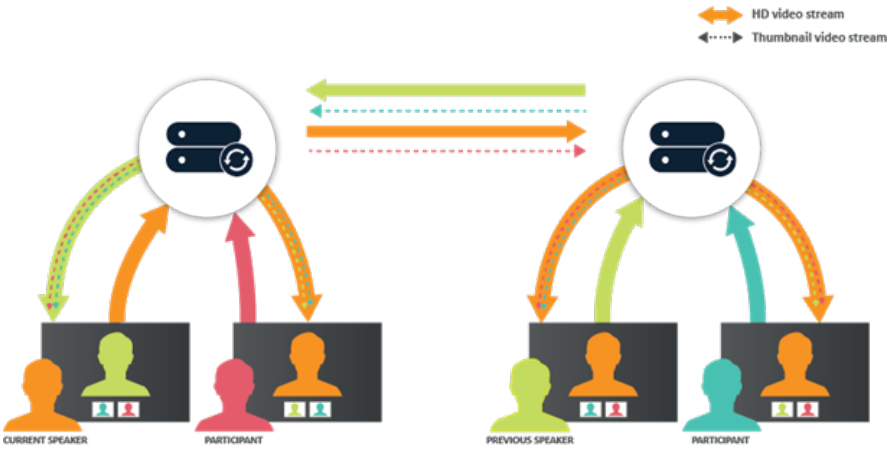
Bandwidth usage (1 + 7 layout)

When Conferencing Nodes that are hosting a conference (in the default 1 + 7 layout) communicate with each other over a backplane:

- The Conferencing Node to which the current speaker is connected sends an HD video stream of the current speaker, plus a lower-resolution thumbnail (a smaller image at the bottom of the screen which shows the participant's video) of each of the other participants connected to it.
- The Conferencing Node to which the previous speaker is connected sends an HD video stream of the previous speaker, plus lower-

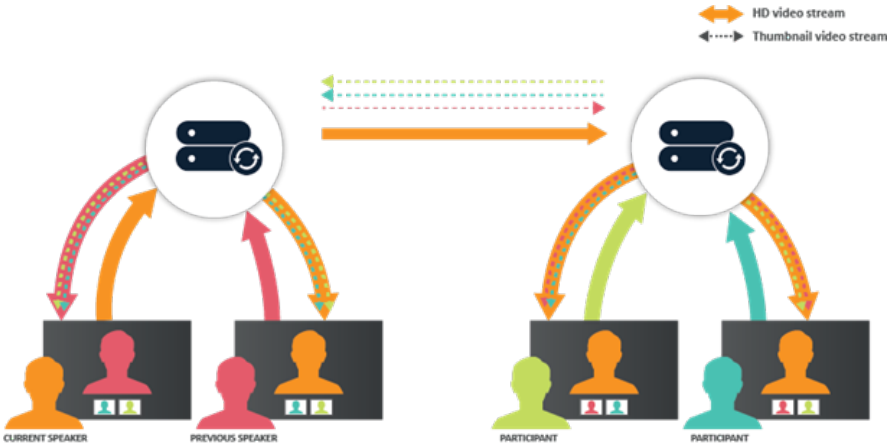
resolution thumbnails of each of the other participants connected to it.

- All other Conferencing Nodes send lower-resolution thumbnails of each of the participants connected to it.



Bandwidth usage when current and previous speakers are connected to different Conferencing Nodes (1 + 7 layout)

As shown in the diagram above, if the current and previous speakers are connected to different Conferencing Nodes, there is one HD stream in each direction in the backplane between the two nodes, plus one smaller stream for every participant being shown in a thumbnail.



Bandwidth usage when current and previous speakers are connected to the same Conferencing Node (1 + 7 layout)

As shown in the diagram above, if the current and previous speakers are connected to the same Conferencing Node, that Conferencing Node sends an HD video stream of the current speaker plus lower-resolution thumbnails of each of the other participants, to all other Conferencing Nodes. The other Conferencing Nodes send lower-resolution thumbnails of the other participants.

This architecture means that there are major bandwidth savings when compared with traditional MCU deployments where all conference participants regardless of location connect to the same MCU and individual HD and thumbnail video streams are sent between the MCU and every endpoint.

Usage guidelines and other conference layouts

The examples above are from the perspective of a conference using a "1 + 7" layout (1 large main speaker and up to 7 other thumbnail participants).

When using other layouts:

- In a 4 + 0 layout, when there are 1 or 2 participants to display, their streams are sent at high resolution over the backplane between nodes; if there are 3 or more participants then up to 5 streams are sent at a medium resolution.

- In a 2 + 21 layout, up to 3 HD streams may be sent between any 2 nodes (up to 2 current speakers and potentially 1 previous speaker).
- When using Adaptive Composition, different resolutions are sent as required for each position in the layout. Up to 13 streams could be sent between nodes: 3 top-row streams (one extra so that current speakers do not see themselves), 3 center-row streams and 7 bottom-row streams, all at their respective call qualities as required for the conference.

You should also note that:

- The bandwidth required by a HD stream on a backplane depends on a combination of many factors. Typically it uses about 1.6 Mbps but could use up to 4 Mbps.
- One thumbnail stream uses about 64-192 kbps.
- A presentation stream uses an additional 1.6 Mbps from the presenter.
- Pexip Infinity supports up to 6 Mbps per participant (this varies depending on the deployment environment, video resolutions, etc).

Downspeeding

Pexip Infinity will automatically downspeed and upspeed individual calls in response to fluctuating network conditions.

Bandwidth restrictions

Administrators can individually restrict the bandwidth available to participants accessing each Virtual Reception, Virtual Meeting Room and Virtual Auditorium, and per Call Routing Rule. Restrictions can also be applied across an entire deployment. Bandwidth limitations cannot be applied to the forwarding connection between a Proxying Edge Node and a Transcoding Conferencing Node. For more information, see Managing and restricting call bandwidth.

Load balancing, redundancy and scalability

This topic summarizes the load balancing, redundancy and scalability aspects of Pexip Infinity.

Load balancing

Pexip Infinity load-balances intelligently across all Conferencing Nodes that are grouped within a system location, and can also utilize media overflow locations when all of the conferencing resource within a location has reached its capacity.

Signaling

- The load balancing of signaling across Conferencing Nodes in a single location is achieved via your call management system and/or DNS SRV records. They should be configured so that calls from your endpoints or other systems are routed to only those Conferencing Nodes that you want to receive signaling.
- Signaling always remains routed to the node that received the call. This could be a Proxying Edge Node or a Transcoding Conferencing Node.

Media

- If the signaling is received on a Proxying Edge Node, media proxying is allocated to the proxying node with the most available capacity in the location that received the signaling. The selected proxying node will always handle the media connection with the endpoint, acting as a proxy between the endpoint and a Transcoding Conferencing Node (which should be in a different location).
- If the signaling is received on a Transcoding Conferencing Node, then whichever transcoding node is selected to process the media (which may be a different node to the signaling node) will also directly handle the media connection with the endpoint.
- You can nominate which location's Transcoding Conferencing Nodes to use to process the call media (either directly or proxied), based on the location of the node that is handling the call signaling. If a transcoding node in the nominated location is already processing media for the conference, and it has spare capacity, then it will also process the media from the new caller, otherwise the transcoding node that currently has the most available capacity is selected to process the media (up to a maximum of four transcoding nodes per location per conference instance). If there is no transcoding resource available in that location, then transcoding nodes in the overflow locations (if configured) are used. The selection of the transcoding node in the media overflow location follows the same balancing rules: if a node is currently processing the conference media and has capacity to take the new call then that node is selected, otherwise the node in the overflow location that currently has the most available capacity is selected.

For full details about how media is routed, load-balanced and directed to overflow locations, see Handling of media and signaling.

For further information about how to configure your specific call management system to work with Pexip Infinity, see the following documentation:

- Pexip Infinity and Microsoft Skype for Business Deployment Guide
- Pexip Infinity and Cisco VCS Deployment Guide
- Pexip Infinity and Cisco Unified Communications Manager Deployment Guide
- Pexip Infinity and Polycom DMA Deployment Guide

Redundancy

Management Node

The Management Node is used to manage Conferencing Nodes, configure conference settings, manage licenses and collate system logs. It does not handle any call processing, so if it were to become unavailable the Pexip Infinity service would be unaffected. However:

- You would not be able to make changes to services (Virtual Meeting Rooms, Virtual Receptions and so on).
- Logging would be affected: the Management Node would not receive and collate logs from the Conferencing Nodes. This can be mitigated by using a syslog server to collate logs.
- After 14 days, licenses would cease to be allocated and the Pexip Infinity service would no longer allow calls.

Conferencing Nodes

You should ensure that your call control system is set up so that, in the event of a Conferencing Node failing, new calls to the conference are routed to another available Conferencing Node.

Basic alternate gatekeeper support is available for H.323 endpoints. When a H.323 device makes a registration request, the Conferencing Node returns a list of any alternate nodes in the same Pexip system location; the device will attempt to use an alternate should the original node be unresponsive.

If multiple Conferencing Nodes are configured in a single location, new calls to a conference can be routed by the call control system to only those nodes that are currently available. To achieve this, you must configure the call control system with all possible Conferencing Nodes (either explicitly or through DNS records), and for liveness checks to be carried out.

Note that there is no live failover; calls in progress on a Conferencing Node that fails will be lost, and participants will need to redial to reconnect to the conference.

Scalability

To increase the capacity of the Pexip Infinity platform, simply deploy one or more new Conferencing Nodes. You may also need to increase the number of call licenses.

Customizing the Pexip Infinity user experience

You can easily apply your own corporate branding to the Pexip Infinity platform, and produce a personalized user experience for all of your Pexip Infinity services.

This topic gives some examples of how themes can be used to customize the voice prompts and images used in your Pexip Infinity services, and how branding customizations can be applied to change the look and feel of the Pexip apps (Pexip's free video apps, which can also be used to access those services).

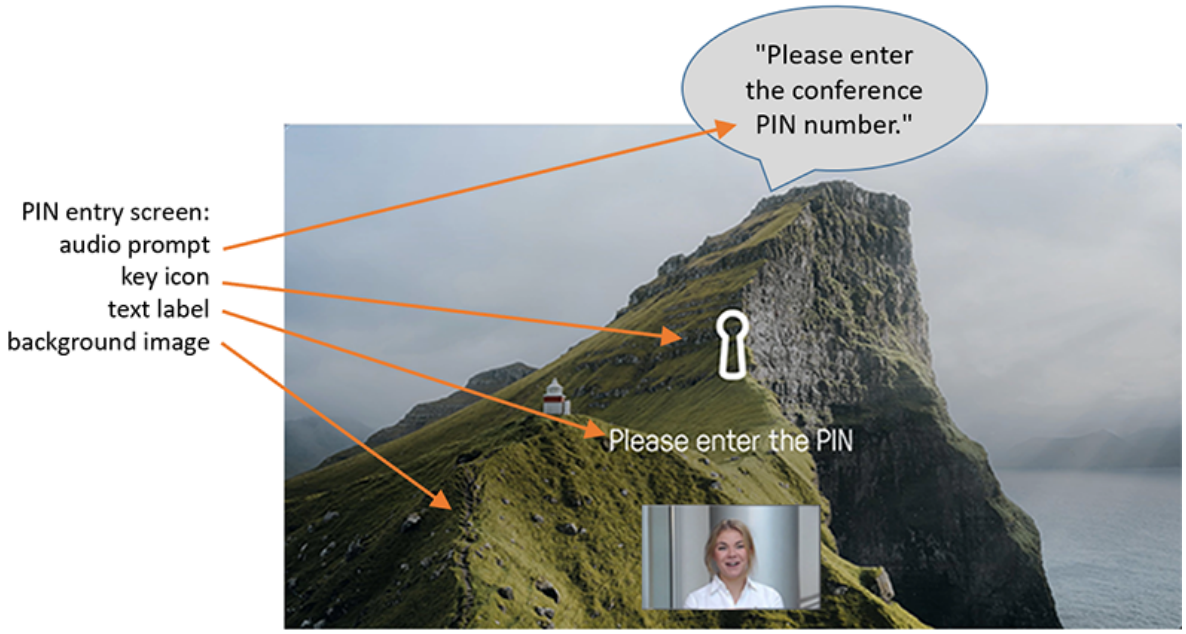
Themes

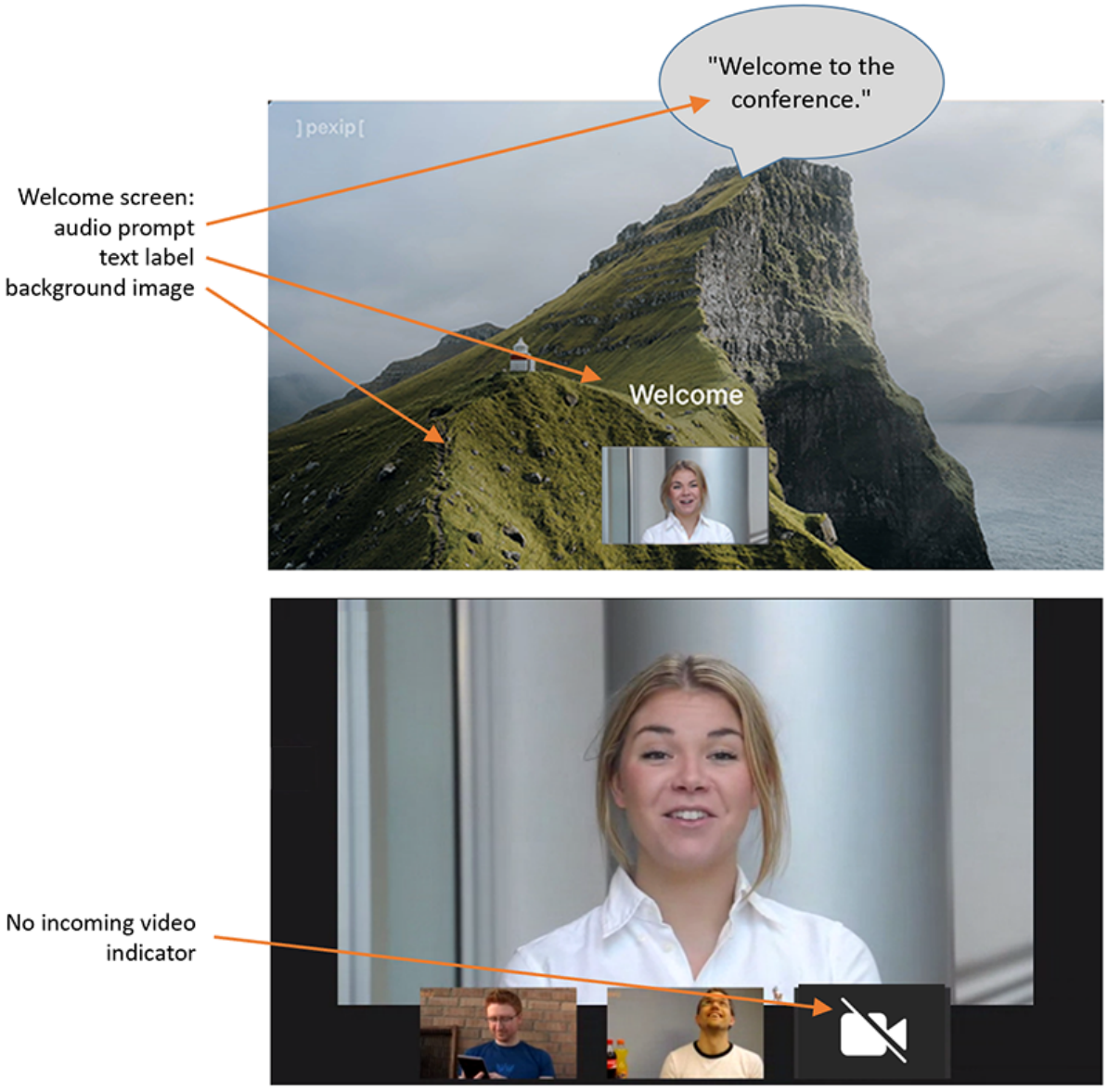
Themes are applied to one or more services (such as a VMR, Virtual Auditorium, Virtual Reception, or Infinity Gateway call routing rule), and affect all participants using that service, regardless of the device or video client that the participant is using to make the call. A different theme can be applied to each service if required.

Themes are used to control:

- all of the voice prompts used when accessing or participating in a conference
- the appearance of PIN entry screens (on devices other than the Pexip apps)
- watermark and content classification indicators
- the appearance of information screens such as waiting for host, PIN entry, capacity exceeded etc.

The following images show some examples of which elements of the user experience can be customized via themes. All of the examples show a video client that is accessing a service that uses the base (default) Pexip theme.





Themes are uploaded and managed via the Pexip Infinity Administrator interface (**Services > Themes**).

For more information, including the full set of theme elements that can be customized, see Customizing conference images and voice prompts using themes.

Pexip app customization

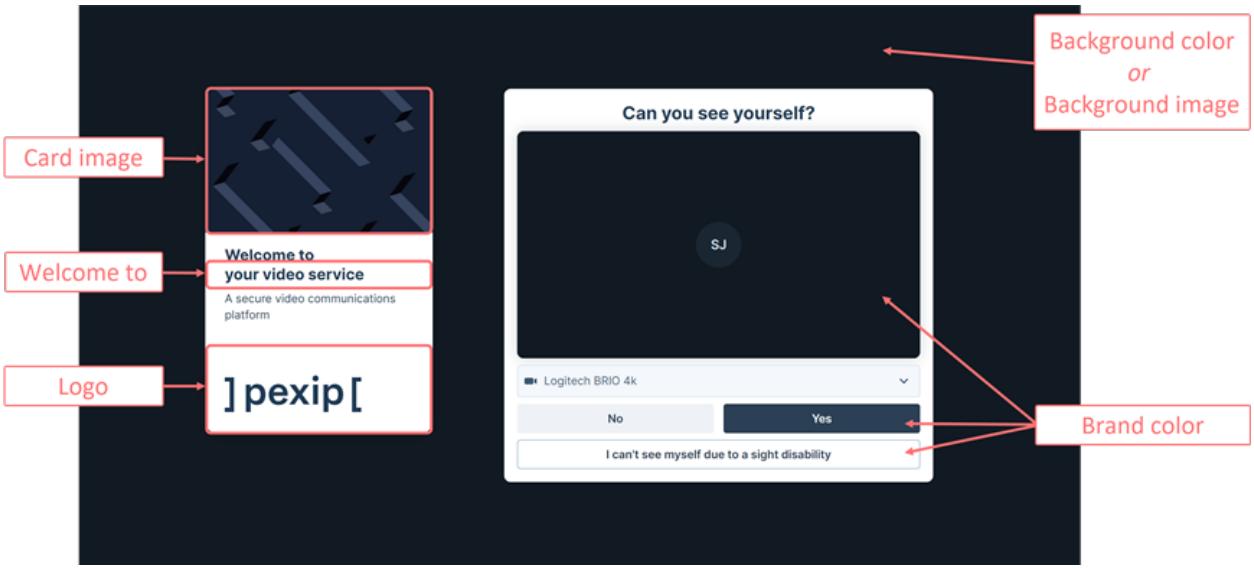
The branding and styling of the web app can be customized. This changes the look and feel of the web app regardless of which service is being accessed. (However, the theme-based elements of each individual service may also have been customized — a theme changes the look and feel of the actual conference you have joined, or are trying to join.)

Pexip web app customization can be used to control:

- default app settings such as bandwidth, default background image, screen sharing frame rate
- default user settings such as whether microphone and camera are muted on join
- the ability to display an image/logo and accompanying welcome text on a landing page, and to use a custom favicon
- language translations and the default language
- the color scheme for buttons, icons and other graphic indicators; elements can be customized individually or a general color scheme can be applied to all similar items.

To customize the **Pexip web app** you typically create a branding package and then upload it to the Management Node. You can use Pexip's branding portal to quickly and easily create branding packages, or you can create them manually for more advanced customizations.

The image below shows some examples of which elements of the web app can be customized via branding.



A beginner's guide to Pexip Infinity

If you're new to video conferencing, or just to Pexip Infinity, this section covers the basics of what you need to know about the Pexip Infinity solution.

What is Pexip Infinity?

Pexip Infinity is a scalable, secure video conferencing platform designed for organizations that need control over their communications. It offers its own native web-based video client, and the ability to create and configure your own meeting rooms hosted directly on the platform. You can also use Pexip Infinity to connect any video device (regardless of protocol) to any video meeting — whether hosted on Pexip Infinity or another provider such as Microsoft Teams, Zoom, or Google Meet.

The Pexip Infinity platform is supplied as software that you install on standard servers, either on-premises or in the cloud. Its distributed architecture is designed to allow as many of these servers as required to be spread around a range of locations, providing videoconferencing resource where and when it is required, reducing bandwidth usage across your organization.

Why shouldn't I use what I have now?

Many free video calling and video conferencing solutions exist, but they often lack the security, flexibility, and scalability required for organizations with strict operational and compliance requirements. There may be limitations to the number of people who can connect into a single meeting; a lack of cross-platform support (for example, for connections from others not using the same solution, or wanting to use devices such as telephones); and concerns over data sovereignty that make these solutions unsuitable.

What to look for in a videoconferencing solution

There are many things you should consider when deciding upon a videoconferencing solution.

Security

Security is a critical consideration when choosing a video conferencing platform. Pexip Infinity is designed for organizations that need full control over meeting access, user authentication, and data handling. Whether you need to comply with government or industry regulations regarding data retention, data sovereignty, or personally identifiable information (for example GDPR, HIPAA,or FedRAMP), or deploy in air-gapped environments, Pexip Infinity ensures your communications stay private and secure.

Pexip Infinity integrates with standard single sign-on (SSO) Identity Providers to give you granular control of who can configure the platform and access meetings. And because Pexip Infinity software can be installed on any compatible server, you decide where and how it is deployed — from completely air-gapped environments, on-premises datacenters, through to sovereign, private, or public clouds, or a hybrid approach.

Customization

Out-of-the box solutions may not meet your organization's specific workflow and security requirements, and often can't be customized to reflect your organization's branding. Pexip Infinity gives you complete control over the configuration and use of your video environment via a web-based Administrator interface and extensive APIs. The native web-based client is also fully customizable and brandable, allowing you to control the look, feel and functionality available to end users.

Lets any device talk to any other

Most organizations have a mix of endpoints that are used to make video calls. Meeting participants might want to call from a dedicated meeting room equipped with a big-screen video endpoint; a desktop client such as Skype for Business; a web browser; or even a telephone. In the past, these systems weren't able to connect to each other. But Pexip Infinity can act as an interpreter for all these systems, allowing participants to use whatever device they prefer to call in to a meeting. We even offer our own desktop, mobile and web-based clients for those users who don't have access to traditional video devices.

Virtual meeting rooms for everyone

Pexip Infinity Virtual Meeting Rooms (VMRs) are always available and ready for use. VMRs that aren't currently being used don't take up any resources, so you can create one for every person in your organization. Whenever someone wants to hold a meeting over video, they can just invite the participants into their VMR.

Making direct calls to other people or other meeting platforms

Pexip Infinity doesn't just provide VMRs – the Infinity Gateway feature also allows users to make direct calls to each other. This is particularly significant if the two people on the call are using different video devices or solutions that would not otherwise be able to connect to each other.

It also lets you join meetings that are running on other systems such as Microsoft Teams, Zoom, or Google Meet.

Registering your device so it can be called

If you have access to a video endpoint, you can usually use it to make outbound calls. However, if you want to be able to receive calls, you need to register your device and the address that can be used to find it, so that others can make calls to it. There are special systems that can handle registrations within an organization, but Pexip Infinity has this functionality built in.

Integrations

Your organization may already be using many standard enterprise products for scheduling meetings, providing authentication, and managing user accounts. Pexip Infinity integrates with many of these products, allowing you to make use of the resources you already have.

The advantages of software

Latest features

Because Pexip Infinity is simply software, you have immediate access to the latest releases and all the new features – for free. Just download the software and install it on your servers – it really is that simple.

Add and remove resources as required

As long as you have access to computing resources – either on-premises servers or cloud-based solutions such as AWS or Azure – you can create additional instances of Pexip Infinity to provide any additional capacity, as and when required, and in a matter of minutes. This can even happen automatically.

Next steps

- Choosing a deployment environment explains your options for deploying Pexip Infinity on your own servers or on a cloud service.
- Our installation guides then provide full information about how to obtain and install the Pexip Infinity software on your chosen platform.

Using your Virtual Meeting Room

Pexip Infinity provides any number of users with their own personal Virtual Meeting Room (VMR) which they can use to hold conferences, share presentations, and chat.

Here's what you need to know for you and your guests to use and make the most of your VMR.

It's always available

Your VMR is a personal space that is **always available** for you to meet with others over video or audio and share content. You can use it whenever you want — for pre-arranged meetings or for ad hoc calls — without making a reservation in advance.

VMR addresses and PINs

Your administrator will let you know the address of your VMR. Often you'll have two addresses — one is typically in the format **name@example.com** (like an email address), and the other may be a number, for example **555678**. Which address is used depends on how you're being called.

In most cases you can dial your VMR directly using the email-style address, but in some circumstances you may need to go through a central Virtual Reception, and then enter your VMR number.

Your VMR may be set up with a PIN code that you as the Host can use, which enables you to control the conference (for example, locking it, and muting guests). Your guests may be able to join your VMR without a PIN, or they may need to enter a different Guest PIN.

Use any device

You and your guests can use almost any video-enabled device – or even a normal telephone – to join a meeting in your VMR, including:

- **A video conferencing endpoint**: on most video endpoints and room systems you can simply enter the address of your VMR.
- **A web browser**: if you or your guests don't have a fancy endpoint or a software client, you can still call into your VMR from almost any browser including Chrome and Edge – no downloads required! Your administrator will need to tell you the link to use in your browser, and then from there you can enter your VMR address to join the meeting.
- **One of the Pexip apps**: these are available for desktop (Windows) and mobile (iOS and Android). If you use one of the Pexip apps, you also get built-in access to a wide selection of meeting information and controls, including a list of attendees. Again, simply enter the VMR address.
- **A telephone**: if you want to join the conference over audio, then your administrator will tell you the number you can dial.

Using your keypad to control the conference

If you aren't using a Pexip app, you can still control some aspects of the conference from your endpoint's keypad (using what are known as DTMF controls).

The default DTMF controls that can be used within a conference are:

DTMF digits	Control
*1	Request (cycle through) a personal layout †
*3	Toggle raise hand (see Conference indicators)
*4	Toggle presentation in the layout mix †
*5	Toggle mute and unmute all Guests
*6	Toggle self mute † (see Video watermarking for information about how to enable mute status indicators via themes)
*7	Toggle conference lock and unlock
*8	Cycle through the set of available layouts (this applies to all participants in the conference)
*9	Toggle multiscreen participant display †
##	Terminate the conference (disconnect all participants including yourself)

† This only applies to the endpoint sending the command, and not to all participants in the conference.

Try out the features in the Pexip apps!

There are many features built in to the Pexip apps, which are available for desktop, mobile and browsers. Most are self-explanatory, but here are some of the most popular:

- **Controlling the meeting room**: when you've joined the meeting room, you can lock it and then only let in the people you want. You can also dial out to people that you want to bring into the meeting.
- **Viewing a list of participants**: you'll see a list of conference attendees on the left of your screen. From here you can mute and disconnect any of them, or change their role from Guest to Host.
- **Sharing content**: you and your Guests can share your screens and full motion HD presentations with others in your VMR.

Pexip Infinity installation guidelines

Pexip Infinity is installed as a Management Node and one or more Conferencing Nodes. All nodes are virtual machines (VMs) that are either deployed onto host hardware using hypervisors, or hosted on a cloud platform service. The Management Node is deployed first, and is then used to configure the Pexip Infinity platform and deploy Conferencing Nodes.

For information about how to install and deploy Pexip Infinity, see:

Installation overview 46
Planning and prerequisites 46
Testing and next steps after initial installation 89

To upgrade your existing Pexip Infinity platform to the latest software version, see Upgrading the Pexip Infinity platform.

Installation overview

Pexip Infinity is installed as a Management Node and one or more Conferencing Nodes. All nodes are virtual machines (VMs) that are either deployed onto host hardware using hypervisors, or hosted on a cloud platform service. The Management Node is deployed first, and is then used to configure the Pexip Infinity platform and deploy Conferencing Nodes.

Full instructions on how to install Pexip Infinity, including how to install the Management Node, are available on our website and to download as individual hypervisor-specific installation guide PDFs.

To upgrade your existing Pexip Infinity platform to the latest software version, see Upgrading the Pexip Infinity platform.

Planning and prerequisites

Choosing a deployment environment

Pexip Infinity supports a wide range of network scenarios, from small, local, private deployments to large, global, cloud-based implementations. How many Conferencing Nodes you deploy, where they are located, and the hypervisors you use will depend on the makeup and requirements of your individual organization.

While your Pexip authorized support representative is the best person to advise you on the most appropriate deployment environment, below are some of the factors to consider.

On-premises, cloud or both?

Your Pexip Infinity platform can be hosted wholly or partially on premises or in the cloud — either by you, or by Pexip on your behalf. There are a number of options, so you can select what's right for your organization:

On-premises deployments use your own network, hardware and hypervisors. You will need to source appropriate servers to host the Management Node and Conferencing Nodes. You will also need to consider whether the majority of users are connecting from within your private network or whether you will have users connecting externally, and therefore if your Conferencing Nodes need to be publicly accessible. If you already have a mature IT network with datacenters in appropriate locations, this may be a suitable option.

Cloud deployments use Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP) or Oracle Cloud Infrastructure to host your Pexip platform nodes. This option does not require initial investment in hardware; instead you must have an agreement with the cloud service provider, and you pay for usage.

Pexip Private Cloud (PPC) deployments are your own private instance of Pexip Infinity hosted on your behalf by Pexip entirely within the Pexip Private Cloud. The Management Node and all Conferencing Nodes are hosted by Pexip, but you still have full, secure access to the Management Node for monitoring, configuration and routine maintenance. You choose the geographical region(s) in which the nodes are deployed to ensure data sovereignty. This option does not require any investment in hardware, or agreements between you and the cloud service provider. Contact your Pexip authorized support representative for further information on this option.

You can also deploy Pexip Infinity using a **hybrid combination of on-premises and cloud**, with the Management Node and some Conferencing Nodes deployed within your own network, and other Conferencing Nodes deployed using a cloud service, for increased flexibility. You can even set up Dynamic bursting to a cloud service to give you temporary increases in capacity whenever you need it.

You don't have to stay committed to one particular deployment environment, and you can always increase or decrease the number of Conferencing Nodes as your capacity requirements evolve. You could, for example, start with an on-premises deployment and then add some cloud-hosted nodes later.

For more details on the on-premises and cloud options, see Network deployment options.

Hypervisor

For on-premises deployments, you need to install a hypervisor on your servers before you can deploy the Management Node and Conferencing Nodes. Pexip Infinity currently supports four hypervisors:

- **VMware** is an independent product with a range of features, dependent on the licensing options you choose.
- **Microsoft Hyper-V** is included with Windows Server and also offers a range of features.
- **KVM** is an open-source option that is not as feature-rich.

For more information, see Supported hypervisors.

Number, location and role of Conferencing Nodes

The capacity of your Pexip Infinity deployment, in terms of how many calls can be handled at any one time, is largely determined by the number of Conferencing Nodes you deploy. To optimize the conferencing experience, we recommend that, where possible, you locate Conferencing Nodes close to each of your main user bases. For example, if the majority of your users are located in three main offices then you should consider concentrating the majority of your Conferencing Nodes among those three locations. If however the majority of your users are spread across many geographic locations, you should deploy Conferencing Nodes evenly among your datacenters.

You must also consider the role of your Conferencing Nodes, in terms of which nodes will be accessed by the client devices and endpoints that need to participate in conferences. In a large Pexip deployment with 5 or more Conferencing Nodes, or where you need to transcode media in multiple locations such as within a DMZ, you should consider deploying Proxying Edge Nodes in addition to your Transcoding Conferencing Nodes.

For more information, see Capacity planning and Distributed Proxying Edge Nodes.

Server and network requirements

This topic summarizes the host server and network requirements when deploying Pexip Infinity.

Host servers

The Management Node and Conferencing Nodes are virtual machines (VMs) that run on industry-standard host servers. A Management Node can run on the same host server as a Conferencing Node. Other Conferencing Nodes can run on host servers in the same or different locations, allowing you to create a globally distributed system.

Our Pexip Infinity Server Design Guide help you choose and configure appropriate servers on which to host your Management Node and Conferencing Nodes.

Network requirements

Depending on your network deployment scenario, you may have to configure your system to operate behind a static NAT and to ensure that traffic can be routed between nodes. See Network deployment options and Network routing and addressing options for Conferencing Nodes for more information about the requirements and implications of your deployment scenario.

Note that in all Pexip Infinity deployment scenarios:

- The Management Node must be able to reach all Conferencing Nodes (Proxying Edge Nodes and Transcoding Conferencing Nodes) and vice versa.
- Each Conferencing Node must be able to reach every other Conferencing Node (Proxying Edge Nodes and Transcoding Conferencing Nodes), except:
  - When a location contains Proxying Edge Nodes, those nodes only require IPsec connectivity with:
    - any other proxying nodes in that location
    - all nodes in the transcoding location, and the primary and secondary overflow locations that are associated with that location
    - the Management Node.

    This means that the proxying nodes in one location do not need to have a direct network connection to other proxying nodes in other locations.
- Any internal firewalls must be configured to allow UDP port 500 and traffic using IP protocol 50 (ESP) in both directions between all Pexip nodes.
- There cannot be a NAT between any Pexip nodes.

NTP servers

Pexip Infinity uses NTP servers to obtain accurate system time. This is necessary to ensure correct operation, including configuration replication and log timestamps.

All host servers **must** be synchronized with accurate time before you install the Management Node or Conferencing Nodes on them.

NTP **must** be enabled on the Management Node VM before you deploy any Conferencing Nodes (this is done during installation of the Management Node).

We strongly recommend that you configure at least three distinct NTP servers or NTP server pools on all your host servers and the Management Node itself. This ensures that log entries from all nodes are properly synchronized.

The VMs hosting the Management Node and Conferencing Nodes use the UTC timezone, and all logs are in UTC. Do not attempt to change the timezone on these systems. Note however that the administrator web interface uses your local time.

DNS servers

Pexip Infinity uses DNS to resolve the hostnames of external system components including NTP servers, syslog servers, SNMP servers and web proxies. It is also used for call routing purposes — SIP proxies, gatekeepers, external call control and conferencing systems and so on. The address of at least one DNS server must be added to your system.

After configuring the DNS servers available to your system, you should assign appropriate DNS servers to each location (**Platform > Locations**). Each Conferencing Node in that location will then use those DNS servers.

Using a VPN to connect to Pexip Infinity

Pexip Infinity is designed to remove the need for Virtual Private Networks (VPNs). We recommend that you deploy Proxying Edge Nodes to enable external connectivity to your enterprise deployment, instead of asking users to connect to internally-located nodes over a VPN. In most cases, connections to Pexip Infinity over a VPN work successfully, but in some cases a lower MTU may be required, or it could lead to UDP media being TCP encapsulated which can cause additional latency.

Call control

If your deployment includes a call control system, it must be configured to route calls to Pexip Infinity appropriately. The exact configuration will depend on your deployment and dial plan, but in general calls placed from an endpoint to a Virtual Meeting Room alias should be routed to the endpoint's local Conferencing Nodes. For more information, see Implementing a dial plan.

Skype for Business server

If your deployment includes on-premises Skype for Business, you must set up static routes to domains used by Pexip Infinity aliases. For more information, see Pexip Infinity and Microsoft Skype for Business Deployment Guide.

Capacity planning

The capacity of each Transcoding Conferencing Node in your Pexip Infinity deployment — in terms of the number of connections* that can be handled simultaneously — depends on a variety of factors including:

- The server capacity and hardware configuration.
- The type of call — Full HD, HD, SD, or audio-only, the codec, and whether there is a presentation stream included.
- The number of unique VMRs being used (and thus the number of backplanes being reserved).
- The type of gateway call — whether the inbound and outbound legs are on the same transcoding node or not, and the types of client involved in the call.

* A connection can be a call or presentation from an endpoint to a Virtual Meeting Room or Virtual Auditorium, a backplane between Transcoding Conferencing Nodes, or a call into or out of the Infinity Gateway. In this context, a **connection** is analogous to a **port**. Note that a Skype for Business client may require two connections (one for the video call, and one for presentation content).

When a connection is proxied via a Proxying Edge Node, the proxying node also consumes connection resources in order to forward the media streams on to a Transcoding Conferencing Node. A transcoding node always consumes the same amount of connection resources regardless of whether it has a direct connection to an endpoint, or it is receiving the media streams via a proxying node.

In all cases, you must also have sufficient concurrent call licenses available.

The following sections explain each of these factors. For some comprehensive examples showing how these different factors can combine to influence capacity, see Resource allocation examples.

Server capacity

The capacity and configuration of the server on which the Conferencing Node is running determines the number of calls that can be handled. This is influenced by a number of factors, including processor generation, number of cores, processor speed, hypervisor and

BIOS settings.

Call types and resource requirements

On startup, each Conferencing Node runs an internal capacity check. This capacity is translated into an estimated maximum number of Full HD, HD, SD or audio-only calls, and can be viewed on the status page (**Status > Conferencing Nodes**) for each Conferencing Node. The status also shows the current media load on each Conferencing Node as a percentage of its total capacity. These figures are an estimate and are for guidance purposes only. Actual capacity will depend on a number of variables, including but not limited to: the number of conferences, the size of each conference, the number of calls into VMRs vs gateways, and the number of participants using audio only vs audio and video.

The type of call (Full HD, HD, SD, or audio-only) affects the amount of resource required by a Transcoding Conferencing Node to handle the call.

In general, when compared to a single high definition **HD** 720p call:

- a **Full HD** 1080p call uses twice the resource
- an **SD** standard definition call uses half the resource
- an **audio-only** call uses one sixteenth of the resource.

However, note that:

- A WebRTC call using the **VP8** codec uses the same amount of resource as H.264, and the **VP9** codec uses around 25% more resource. Therefore:
  - VP9 at 720p uses the equivalent of 1.25 HD resources
  - VP9 at 1080p uses the equivalent of 2.5 HD resources
  - the maximum number of calls a given Conferencing Node can support will be fewer for VP9 calls, for example a node that supports 39 **H.264**/**VP8** SD calls will support 31 **VP9** SD calls.

  Note that within the same conference some participants may use VP9 (if they are connected to a Conferencing Node using the AVX2 or later instruction set) while other participants may use VP8 (if they are connected to a Conferencing Node on older hardware).
- Conferences or gateway calls that use the **Adaptive Composition** layout, and Teams conferences that use the Teams-like layout, consume additional Conferencing Node resources. The actual amount of additional resource depends on many factors, but as a guide, it uses an additional 1 HD of resource per conference, for up to 3 other video participants, plus approximately another 0.5 HD for each additional (4th, 5th etc.) video participant that is on stage. This is regardless of the call quality / resolution of the conference itself and each individual participant's connection (codec, bandwidth and so on).
- H.323 audio-only calls are treated the same as video calls for resource usage purposes.
- Connections to a Media Playback Service use 1.2 times as much resource as a connection to a VMR.
- When transferring a participant, the transferee can temporarily take two sets of resources while in the process of being transferred. This should not normally last more than a few seconds.

If you want to limit video calls to specific resolutions (and limit the transcoding node resources that are reserved for calls), you should use the **Maximum call quality** setting.

Proxying Edge Node resource requirements

When a connection is proxied via a Proxying Edge Node, the proxying node also consumes connection resources in order to forward the media streams on to a Transcoding Conferencing Node.

A proxying node uses approximately the equivalent of 3 audio-only resources to proxy a video call (of any resolution), and 1 audio-only resource to proxy an audio call.

We recommend allocating 4 vCPU and 4 GB RAM (which must both be dedicated resource) to each Proxying Edge Node, with a maximum of 8 vCPU and 8 GB RAM for large or busy deployments.

Backplane reservation

Each conference instance on each Transcoding Conferencing Node reserves a backplane connection at a resource level corresponding to the conference's **Maximum call quality** setting, to allow the conference to become geographically distributed if required. The exceptions to this are:

- Deployments with a single Conferencing Node. In such cases, no backplanes will ever be required, so capacity is not reserved.
- Conferences that are audio-only (in other words, where the conference has its **Conference capabilities** set to ***Audio-only***). In such cases, capacity equivalent to one audio connection is reserved for the backplane.

For some reservation examples, see Resource allocation examples.

Gateway calls

Gateway calls (person-to-person calls, or calls to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet) require sufficient capacity for both the inbound leg and the outbound leg. In general, this means that each gateway call consumes resources equivalent to two connections.

Non-distributed gateway calls (where the inbound and outbound legs are on the same transcoding node) involving only SIP or H.323 clients do not use any additional ports/connections. However, in other scenarios, additional ports may be used as described below (assuming that **Maximum call quality** is HD unless otherwise specified):

- Distributed gateway calls (where the outbound leg is on a different transcoding node to the inbound leg) consume backplane ports — thus 1 HD video + 1 backplane for participant A plus 1 HD video + 1 backplane for participant B. This typically occurs when calling registered endpoints (where the outbound call to the registered endpoint will originate from the node the endpoint is registered to), or when using Call Routing Rules with an ***Outgoing location*** set to something other than ***Automatic***.
- For a gateway call to a **Microsoft Teams** meeting, the connection to Teams uses 1.5 HD of resource if **Maximum call quality** is SD or HD, otherwise it uses 1.5 Full HD resources. The resources required for the VTC leg of the connection depend upon the **Maximum call quality** setting. If any participant presents content, additional resources (typically **0.5 HD**) would be required, either on the Teams backplane (when an endpoint presents) or on the node handling the endpoint's media connection (when a Teams client presents). The exact amount of resource used depends on the codec, resolution and frame rate of the presentation stream.
- For a gateway call to **Google Meet**, the connection to Google Meet always uses 1 HD resource (it uses VP8) for main video. The resources required for the VTC leg of the connection depend upon the type of endpoint and the **Maximum call quality** setting. If the VTC endpoint starts to present content then an extra 1 HD resource is used for the connection from Pexip Infinity to Google Meet. However, no additional resources are required on the Google Meet leg if presentation content is sent from Google Meet, but **0.5 HD of additional resource** would typically be required for each endpoint receiving presentation.

Call protocols and presentation content

The various call protocols have different behavior and limitations for sending and receiving video and presentation content, in addition to what Pexip Infinity will request based on its configured maximum call quality and bandwidth settings. As the endpoints ultimately decide what to send to Pexip Infinity, the following information should be seen as a guide only.

WebRTC

A Pexip app / WebRTC client sends or receives presentation content (files or screen sharing) via the existing call connection used for audio and video media. WebRTC may send resolutions up to 1080p to Pexip Infinity, depending on the camera capabilities and available bandwidth. Incoming presentation content is viewed by Pexip apps in full motion HD video by default.

H.323 and SIP

H.323 and SIP video endpoints may send resolutions up to 1080p to Pexip Infinity at whatever bandwidth and frame rate they decide. Presentation content shares the bandwidth of the main video channel, meaning that when a participant starts presenting, this might decrease resolution of the main video. Most H.323 and SIP endpoints prioritize motion (higher frame rate) for main video and sharpness (higher resolution) for content.

Licenses

The total number of concurrent calls that can be made in your deployment (regardless of whether those calls are HD, SD or audio-only) is limited by your license. For more information, see Pexip Infinity license installation and usage.

Scaling up capacity, media overflow and dynamic bursting

You can easily scale a deployment up by creating several Conferencing Nodes in the same location (i.e. the same datacenter). Capacity can even be added “on the fly” – Conferencing Nodes can be added in a couple of minutes if more capacity is needed. Alternatively,

each location can be configured to overflow to another location if it reaches its capacity, including bursting to temporary resources on a cloud service.

For more information on media overflow and dynamic bursting, see:

- Handling of media and signaling
- Dynamic bursting to a cloud service

Supported hypervisors

Pexip Infinity currently offers specific support for VMware ESXi, Microsoft Hyper-V and KVM hypervisors. You may use a mixture of these hypervisors across your estate. Other hypervisors and orchestration layers, e.g. Citrix XenServer, and those derived from KVM such as Proxmox or Nutanix, may be used but are not officially supported by Pexip.

Note that suspending and resuming Virtual Machines that are running the Pexip Infinity Management Node or Conferencing Nodes is not supported.

The Pexip Infinity platform can also be deployed as a cloud service via **Amazon Web Services (AWS)**, **Microsoft Azure**, **Google Cloud Platform**, or **Oracle Cloud Infrastructure**, with private, public or hybrid deployment options.

VMware vSphere ESXi

VMware versions

Version 41 of the Pexip Infinity platform supports VMware vSphere ESXi 6.7, 7.0 and 8.0.

Standalone ESXi hosts are not supported.

You must have a suitable VMware environment already installed.

VMware editions

The Pexip Infinity platform will run on the **free edition** of vSphere Hypervisor. However, this edition has a number of limitations (limited support from VMware, no access to vCenter or vMotion). For this reason we do not recommend its use except in smaller deployments, or test or demo environments.

The minimum edition of VMware that we recommend is the vSphere **Standard edition**. This does not have the limitations of the free edition. If you do not already use VMware in your enterprise, the vSphere **Essentials Kit** is a simple way to get started and will provide you with Standard edition licenses for 3 servers (with 2 CPUs each) plus a vCenter license.

The **Enterprise Plus edition** includes further additional features relevant to the Pexip Infinity platform that could be of benefit to larger deployments. These include Storage DRS and Distributed Switch.

For a comparison of the VMware editions, see http://www.vmware.com/products/vsphere.html#compare.

Microsoft Hyper-V

Version 41 of Pexip Infinity supports Microsoft Hyper-V 2025. You must have a suitable Hyper-V environment already installed.

KVM

Pexip Infinity supports the following KVM versions:

- Linux: 5.15.0 or later
- QEMU: 6.2.0 or later

For information on configuring a KVM environment specifically for use with Pexip Infinity, see Configuring KVM for Pexip Infinity.

Other hypervisors and orchestration layers

Pexip Infinity Management Nodes and Conferencing Nodes support static IP addressing only. Therefore the hypervisor or orchestration layer must also support static IP addressing.

Cloud service providers

The Pexip Infinity platform can also be deployed as a cloud service via **Amazon Web Services (AWS)**, **Microsoft Azure**, **Google Cloud Platform**, or **Oracle Cloud Infrastructure**, with private, public or hybrid deployment options.

About the Pexip Infinity software files

The Pexip Infinity platform is entirely software-based, and is deployed by installing the files provided by Pexip onto your hypervisor or cloud platform to create Virtual Machines (VMs), which you then configure appropriately for your environment.

No changes should be made to any Pexip VM via the terminal interface (other than as described when running the initial Pexip installation wizard) unless directed to do so by Pexip support. This includes (but is not limited to) changes to the time zone, changes to IP tables, configuration of Ethernet interfaces, or the installation of any third-party code/applications.

The Pexip Infinity version 41 release includes the files described below. Which set of files you require depends on:

- the hypervisor or cloud platform you are deploying on
- whether you are deploying a new instance of Pexip Infinity, upgrading an existing deployment, or deploying a new Conferencing Node
- if you are deploying new generic Conferencing Nodes in unsupported or non-standard environments.

Upgrading an existing deployment (all environments)

The Pexip Infinity upgrade package is required when upgrading an existing deployment. It is used to upgrade all deployment environments.

Description	File name	Location	Hypervisor	Notes
Pexip Infinity upgrade package	Pexip_Infinity_v41_ UPGRADE_<build>.tar	Pexip download page	Any hypervisor or cloud platform	Used when **upgrading** the Pexip Infinity platform to version 41. For more information, see Upgrading the Pexip Infinity platform.

Deploying the Management Node on an on-prem hypervisor

Use one of these files when deploying the Management Node on an on-prem hypervisor.

Description	File name	Location	Hypervisor	Notes
Pexip Infinity OVA	Pexip_Infinity_v41_ generic_pxMgr_ <build>.ova	Pexip download page	Any hypervisor **except** Microsoft Hyper-V	Used to deploy a new generic **Management Node** VM using VMware, KVM or other hypervisors except Microsoft Hyper-V.
Pexip Infinity ZIP	Pexip_Infinity_v41_ HyperV_pxMgr_ <build>.zip	Pexip download page	Microsoft Hyper-V	Used to deploy a generic **Management Node** VM using Hyper-V.

Deploying new Conferencing Nodes on supported hypervisors

There is no requirement to download any files from the Pexip website when deploying new Conferencing Nodes on VMware, Hyper-V, or KVM hypervisors, as the relevant file is generated on request by the Management Node.

- For VMware and KVM, a file in the format **pexip-<hostname>.<domain>.ova** is generated. See Deploying new Conferencing Nodes for full information.
- For Hyper-V, a file in the format **pexip-<hostname>.<domain>.zip** is generated. See Deploying a Conferencing Node for full information.

Deploying the Management Node or Conferencing Nodes on cloud environments (Azure, AWS, GCP or Oracle)

There is no requirement to download any files from the Pexip website when you are deploying the Pexip Infinity Management Node or Conferencing Nodes on an AWS or Azure cloud platform — the relevant Pexip image files are hosted on the AWS/Azure platform instead. However, GCP and Oracle image files are downloadable from the Pexip website.

For specific deployment information about each platform, see:

- Installing Pexip Infinity in a Microsoft Hyper-V environment
- Deploying Pexip Infinity on Amazon Web Services
- Deploying Pexip Infinity on Microsoft Azure
- Deploying Pexip Infinity on Google Cloud Platform
- Deploying Pexip Infinity on Oracle Cloud Infrastructure

Deploying new generic Conferencing Nodes on non-supported hypervisors or orchestration layers

Use one of these files when deploying a Conferencing Node on other non-supported hypervisors or orchestration layers.

Description	File name	Location	Hypervisor	Notes
Generic Pexip Infinity Conferencing Node OVA	Pexip_Infinity_v41_generic_ConfNode_<build>.ova	https://dl.pexip.com/infinity/index.html and then select the appropriate directory for your software version	Other environments that take .ova or .ovf files as input	Used to deploy a generic **Conferencing Node** VM in environments that take .ova or .ovf files as input. For more information, see Deploying a Conferencing Node using a generic VM template and configuration file.
Generic Pexip Infinity Conferencing Node ZIP	Pexip_Infinity_v41_HyperV_ConfNode_<build>.zip	https://dl.pexip.com/infinity/index.html and then select the appropriate directory for your software version	Hyper-V in a cloud-based environment	Used to deploy a generic **Conferencing Node** VM using Hyper-V in cloud-based environments or other orchestration layers where standard deployment is problematic. For more information, see Deploying a Conferencing Node using a generic VM template and configuration file.

Network deployment options

If you need to support business-to-business video calls and provide access to Pexip Infinity resources from external systems and endpoints such as remote or federated Skype for Business clients, remote SIP and H.323 endpoints, and Pexip apps, you need to consider how to deploy your Pexip Infinity Conferencing Nodes.

This section explains how the Pexip Infinity platform fits into typical network deployment scenarios:

- A private "on-premises" deployment of privately-addressed Transcoding Conferencing Nodes, with external connections routed via Proxying Edge Nodes. There are several variations to this option:
  - Proxying Edge Nodes in combination with third-party video call control
  - Routing all external calls via Proxying Edge Nodes
- A public DMZ deployment with publicly-addressable Pexip Infinity nodes (with optional static NAT).
  - For additional flexibility you can deploy Conferencing Nodes with dual network interfaces (one "internal" interface for inter-node communication, and one "external" interface for signaling and media to endpoints and other video devices).

- The Pexip Infinity platform can also be deployed as a cloud service via **Amazon Web Services (AWS)**, **Microsoft Azure**, **Google Cloud Platform**, or **Oracle Cloud Infrastructure**, with private, public or hybrid deployment options.

General network requirements

Note that in all Pexip Infinity deployment scenarios:

- The Management Node must be able to reach all Conferencing Nodes (Proxying Edge Nodes and Transcoding Conferencing Nodes) and vice versa.
- Each Conferencing Node must be able to reach every other Conferencing Node (Proxying Edge Nodes and Transcoding Conferencing Nodes), except:
  - When a location contains Proxying Edge Nodes, those nodes only require IPsec connectivity with:
    - any other proxying nodes in that location
    - all nodes in the transcoding location, and the primary and secondary overflow locations that are associated with that location
    - the Management Node.

    This means that the proxying nodes in one location do not need to have a direct network connection to other proxying nodes in other locations.
- Any internal firewalls must be configured to allow UDP port 500 and traffic using IP protocol 50 (ESP) in both directions between all Pexip nodes.
- There cannot be a NAT between any Pexip nodes.

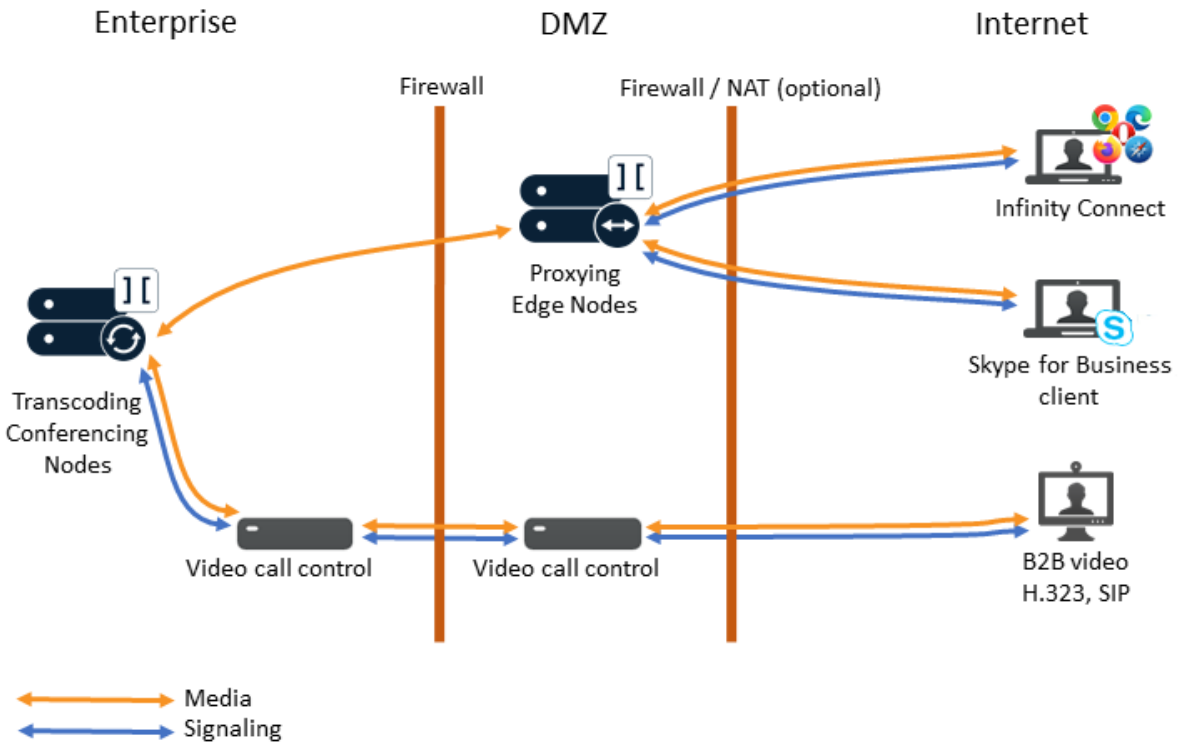
Privately-addressed "on-premises" Transcoding Conferencing Nodes

If you have an on-premises deployment of privately-addressed Pexip Infinity Transcoding Conferencing Nodes, we recommend that you deploy publicly-routable Proxying Edge Nodes to connect external clients and systems to those on-premises transcoding nodes.

The following examples also show how your Pexip Infinity platform may be deployed alongside any existing third-party call control systems and how it can be integrated with on-premises Skype for Business.

Proxying Edge Nodes in combination with third-party video call control

This example deployment scenario uses Proxying Edge Nodes in combination with third-party video call control:



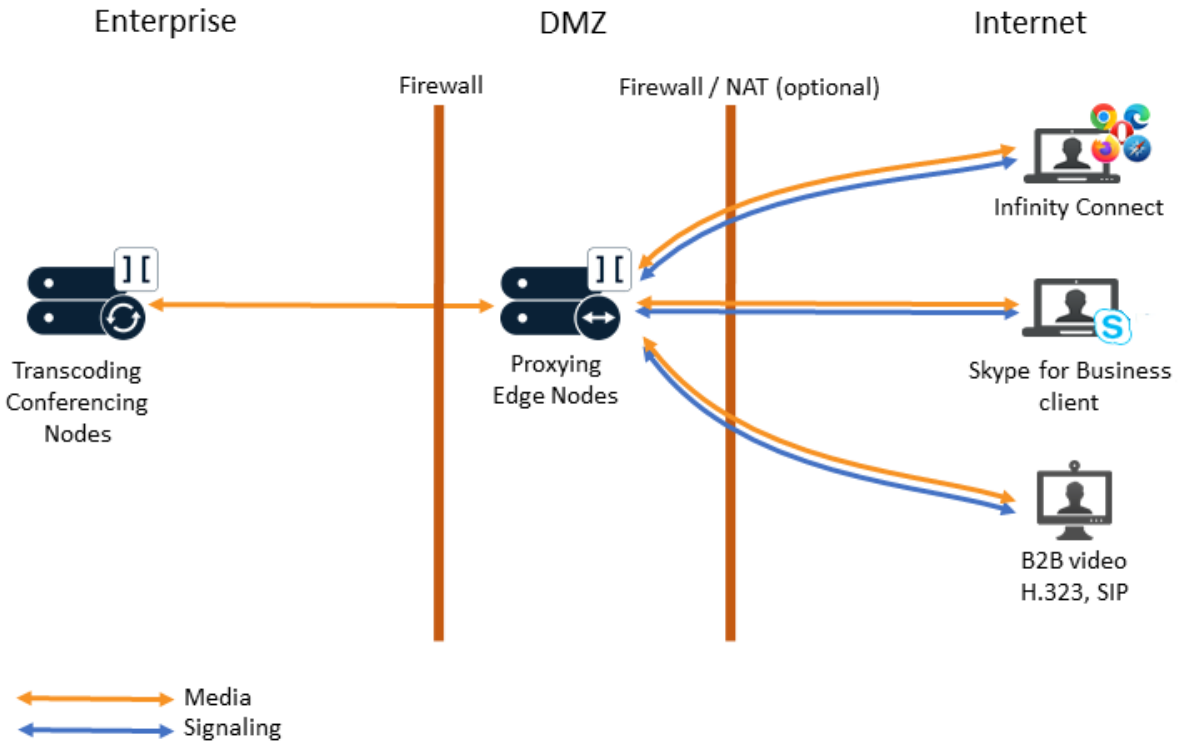
In this type of deployment scenario:

- The Management Node and the Transcoding Conferencing Nodes are deployed with private IP addresses in the local enterprise network.
- External Pexip apps connect to Proxying Edge Nodes. Their signaling is terminated on the proxying node and their media is proxied through to the internal Transcoding Conferencing Nodes. In addition you can optionally deploy a reverse proxy for load balancing and resiliency, or branding.
- Federated Skype for Business clients connect to Proxying Edge Nodes. Their signaling is terminated on the proxying node and their media is proxied through to the internal Transcoding Conferencing Nodes (an on-premises Skype for Business server is not needed). As proxying nodes support ICE, a TURN server is not required.
- External SIP and H.323 endpoints and other forms of business-to-business video calls connect to the Transcoding Conferencing Nodes via a firewall traversal / video call control solution such as a Cisco VCS.
- The external firewall can optionally be a NAT device.

For more information about proxying nodes, see Deployment guidelines for Proxying Edge Nodes.

Routing all external calls via Proxying Edge Nodes

This option extends the previous example by using Proxying Edge Nodes to handle all external calls.



This example deployment scenario is the same as the previous one (with third-party call control), except that:

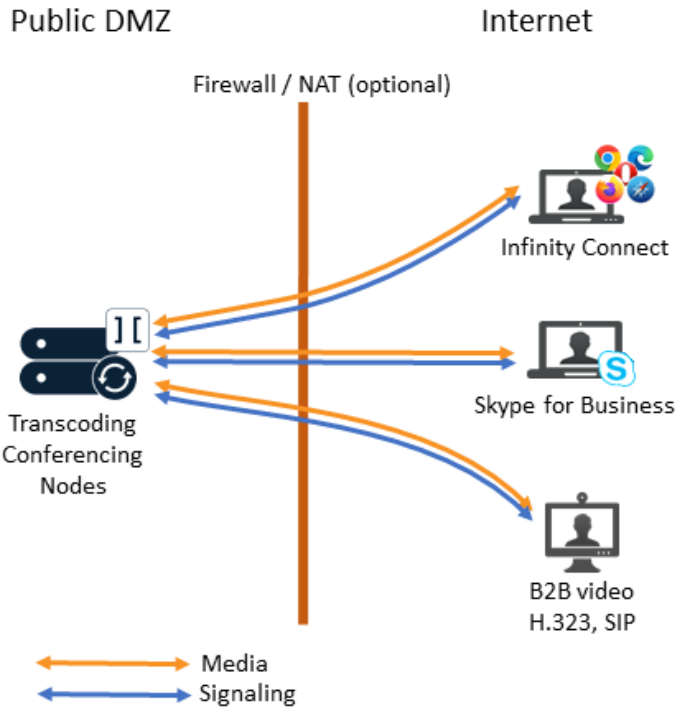
- SIP and H.323 devices connect directly to Proxying Edge Nodes, which terminate the signaling and proxy their media through to the Transcoding Conferencing Nodes. A third-party call control solution is not required.
- Those SIP and H.323 devices can also register to a Proxying Edge Node if required.

Publicly-addressable Transcoding Conferencing Nodes

You can deploy all of your Transcoding Conferencing Nodes in a public DMZ, as shown below.

In this type of deployment:

- One or more Transcoding Conferencing Nodes are deployed in a public DMZ. They have publicly-reachable IP addresses — either directly or via static NAT.
- External SIP and H.323 endpoints and other forms of business-to-business video calls can connect directly to the public-facing IP addresses of the Conferencing Nodes.
- Remote federated Skype for Business clients connect to Pexip Infinity and media can be routed directly through the public-facing IP addresses of the Conferencing Nodes (an on-premises Skype for Business server is not needed). As Conferencing Nodes support ICE, a TURN server is not required.
- Pexip apps can connect to Pexip Infinity and media can be routed directly to the Conferencing Nodes. In addition you can optionally deploy a reverse proxy for load balancing and resiliency, or branding.
- In large deployments you may want to consider whether to deploy some dedicated Proxying Edge Nodes to manage the signaling and media connections between endpoints situated on the public internet and your Transcoding Conferencing Nodes.

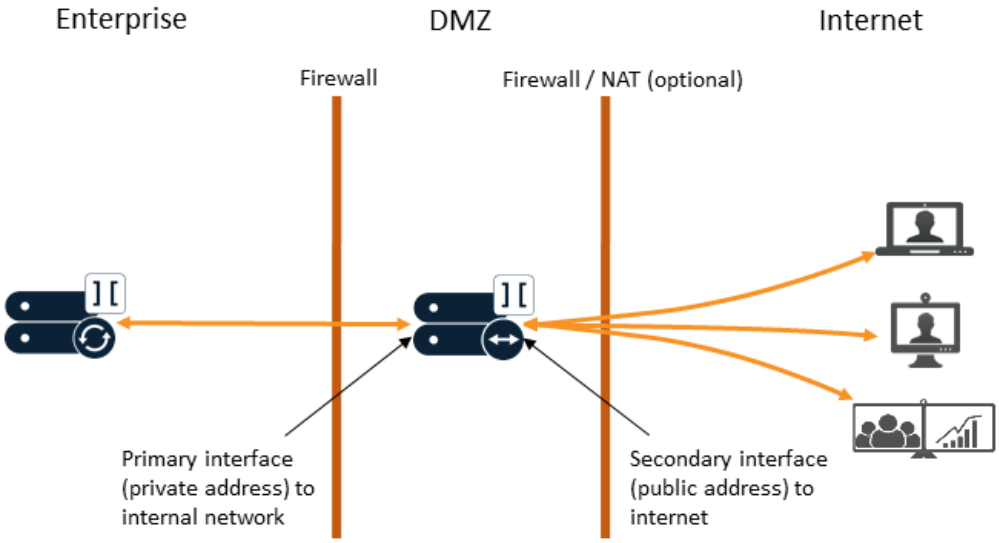


- Note that the Management Node will typically be deployed with a private IP address in the local enterprise network. You must ensure that there is no NAT between the Management Node and the Conferencing Nodes in the DMZ.

Conferencing Nodes with dual network interfaces (NICs)

For additional deployment flexibility, you can configure a secondary network address on a Conferencing Node. Dual NICs are supported on both Transcoding Conferencing Nodes and Proxying Edge Nodes.

You would typically deploy a Conferencing Node with dual network interfaces when it is connected to a dedicated video zone or it is being deployed in a public DMZ where the primary interface would be to an internal, private network segment within the enterprise (where it can connect to the Management Node and other Conferencing Nodes) and the secondary interface would be towards the



video zone or on the publicly-addressable side of the DMZ perimeter network and used for connecting to external endpoints and devices.

When a secondary network address is configured:

- The primary address is always used for inter-node communication to the Management Node and to other Conferencing Nodes.
- SSH connections can be made only to the primary interface.
- The secondary address is always used for signaling and media (to endpoints and other video devices).
- Connections to DNS, SNMP, NTP, syslog and so on, go out from whichever interface is appropriate, based on routing.
- You can have a mixture of any number of single-interfaced and dual-interfaced Conferencing Nodes, providing all nodes can communicate with each other via their primary interfaces.

Note that dual network interfaces are not supported on Conferencing Nodes deployed in public cloud services (Azure, AWS, GCP or Oracle).

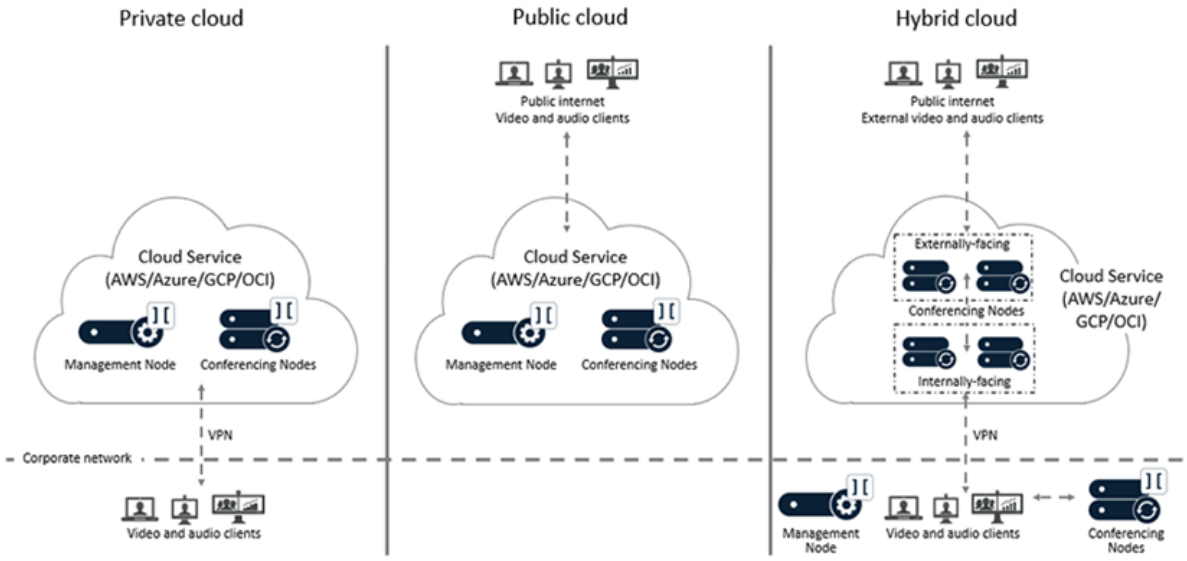
For more information on configuring dual network interfaces, see Deploying Conferencing Nodes with dual network interfaces (NICs) and Firewall/NAT routing and addressing examples.

Deploying as a cloud service via Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP) or Oracle Cloud Infrastructure

The Pexip Infinity platform can be deployed in the Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP) or Oracle Cloud Infrastructure cloud.

There are three main deployment options for your Pexip Infinity platform when using a cloud service:

- **Private cloud**: all nodes are deployed within a private cloud service. Private addressing is used for all nodes and connectivity is achieved by configuring a VPN tunnel between the corporate network and the cloud network. As all nodes are private, this is equivalent to an on-premises deployment which is only available to users internal to the organization.
- **Public cloud**: all nodes are deployed within the cloud network. All nodes have a private address but, in addition, public IP addresses are allocated to each node. The node's private addresses are only used for inter-node communications. Each node's public address is then configured on the relevant node as a static NAT address. Access to the nodes is permitted from the public internet, or a restricted subset of networks, as required. Any systems or endpoints that will send signaling and media traffic to those Pexip Infinity nodes must send that traffic to the public address of those nodes. If you have internal systems or endpoints communicating with those nodes, you must ensure that your local network allows such routing.
- **Hybrid cloud**: the Management Node, and optionally some Conferencing Nodes, are deployed in the corporate network. A VPN tunnel is created between the corporate network and the cloud network. Additional Conferencing Nodes are deployed in the cloud network and are managed from the on-premises Management Node. The cloud-hosted Conferencing Nodes can be either internally-facing, privately-addressed (private cloud) nodes; or externally-facing, publicly-addressed (public cloud) nodes; or a combination of private and public nodes (where the private nodes are in a different Pexip Infinity system location to the public nodes). You may also want to consider dynamic bursting, where the cloud-hosted Conferencing Nodes are only started up and used when you have reached capacity on your on-premises nodes.



All of the Pexip nodes that you deploy in the cloud are completely dedicated to running the Pexip Infinity platform— you maintain full data ownership and control of those nodes.

For specific deployment information about each platform, see:

- Installing Pexip Infinity in a Microsoft Hyper-V environment
- Deploying Pexip Infinity on Amazon Web Services
- Deploying Pexip Infinity on Microsoft Azure
- Deploying Pexip Infinity on Google Cloud Platform
- Deploying Pexip Infinity on Oracle Cloud Infrastructure

Deployment guidelines for Proxying Edge Nodes

A typical deployment scenario is to use Proxying Edge Nodes as a front for many privately-addressed Transcoding Conferencing Nodes. In this scenario, those outward-facing proxying nodes would receive all the signaling and media from endpoints and other external systems, and then forward that media onto other internally-located transcoding nodes to perform the standard Pexip Infinity transcoding, gatewaying and conferencing hosting functions.

The benefit of using only Proxying Edge Nodes to handle all connections from endpoints and other systems is that it can simplify your deployment:

- You only need to set up and maintain DNS records for the proxying nodes.
- As lineside media is terminated at the proxying nodes, you only need to deploy certificates on those proxying nodes.
- You can easily scale up the media processing capacity by adding more transcoding nodes — either in existing locations or by deploying them in new overflow or bursting locations.

In a large Pexip deployment with 5 or more Conferencing Nodes, or where you need to transcode media in multiple locations such as within a DMZ, you should consider deploying Proxying Edge Nodes in addition to your Transcoding Conferencing Nodes. You can easily switch the role of a deployed Conferencing Node from a transcoding node to a proxying node and vice versa.

This topic covers:

- Configuration summary
- Configuring a Conferencing Node's proxying or transcoding role
- Deployment recommendations
- Deployment scenarios
- Additional information when deploying Proxying Edge Nodes

Configuration summary

To enable Proxying Edge Nodes on your Pexip Infinity platform:

1. A system location should not contain a mixture of proxying nodes and transcoding nodes. Hence you should decide if you need to create new locations and assign the Conferencing Nodes that you want to use as proxying nodes to those locations:
   - Go to **Platform > Locations** to configure your locations.
   - Go to **Platform > Conferencing Nodes** to change the location assigned to your existing Conferencing Nodes.

   If you change the system location of a Conferencing Node, all existing calls will be disconnected and the Conferencing Node will be restarted.
2. Assign a **Role** of ***Proxying Edge Node*** to those Conferencing Nodes that you want to deploy as proxying nodes:
   - For existing nodes, do this via **Platform > Conferencing Nodes**.
   - When deploying new Conferencing Nodes, you select the **Role** when providing the name and network settings for the new node.

   If you change the role of a Conferencing Node, all existing calls will be disconnected and the Conferencing Node will be restarted.
3. Ensure that the locations containing your proxying nodes are **not** configured with a **Transcoding location** of ***This location***. Instead, set the **Transcoding location** to a location that contains transcoding nodes.
4. Ensure that your call control systems and DNS are configured to route calls to your proxying nodes only.
5. Ensure that your proxying nodes have appropriate certificates installed (**Certificates > TLS Certificates.**).

Full deployment guidelines and example scenarios are described below.

Configuring a Conferencing Node's proxying or transcoding role

Each Conferencing Node is configured with a role — either as a Proxying Edge Node or as a Transcoding Conferencing Node. You specify the Conferencing Node's role when it is first deployed, but you can change its role later if required.

To change a Conferencing Node's role:

1. Go to **Platform > Conferencing Nodes** and select the name of the Conferencing Node.
2. Select the new **Role** as appropriate and select **Save**.

If you change the role of a Conferencing Node, all existing calls will be disconnected and the Conferencing Node will be restarted.

If a call is received in a location that contains Proxying Edge Nodes, that location must be configured with a **Transcoding location** that contains your Transcoding Conferencing Nodes.

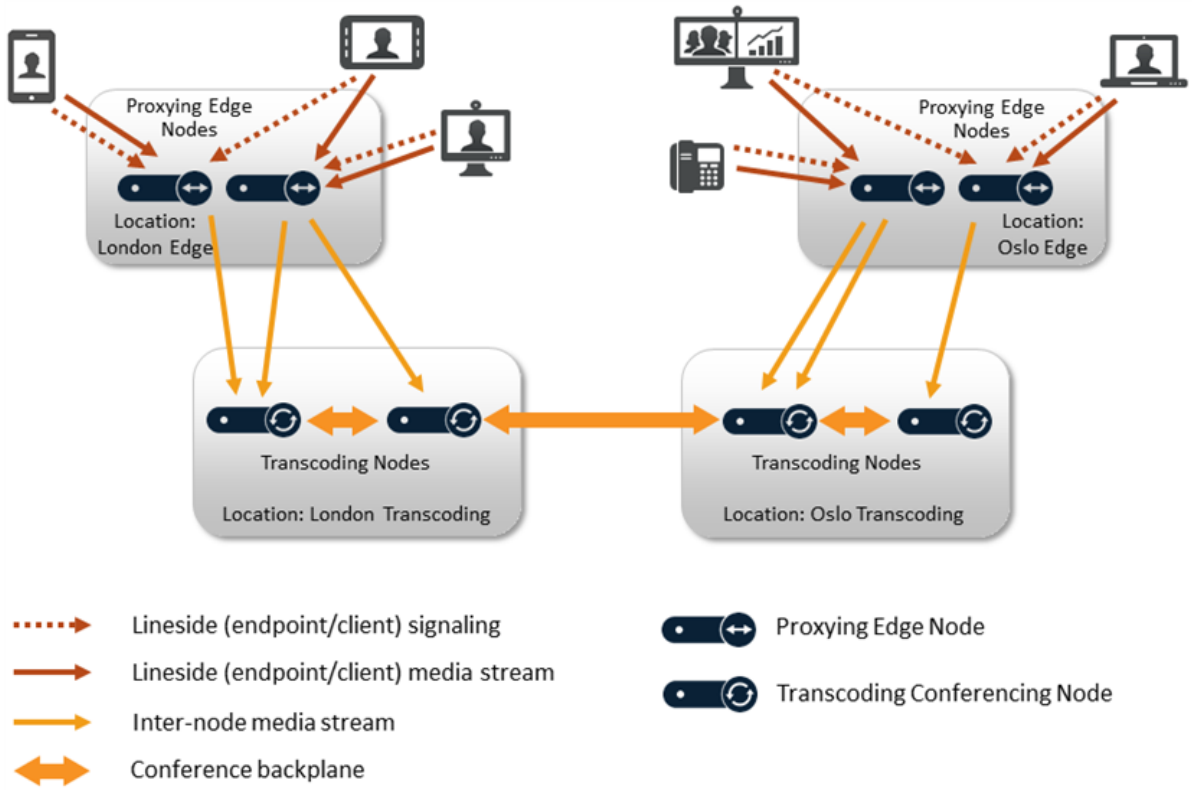
Each role has different compute and host requirements. You should plan for these changes and understand what you might need to alter on the guest VM (such as vCPUs and vRAM within the hypervisor).

Deployment recommendations

If you want to deploy Proxying Edge Nodes in your Pexip Infinity platform, we recommend the following guidelines:

- A system location should not contain a mixture of proxying nodes and transcoding nodes. This separation of roles to locations simplifies load-balancing and conference distribution, and makes it easier for you to manage and monitor your externally-facing Proxying Edge Nodes distinctly from your Transcoding Conferencing Nodes. Hence, in the example scenario shown here, the Conferencing Nodes in the two locations "London Edge" and "Oslo Edge" are Proxying Edge Nodes and thus those nodes and locations are not involved in the actual hosting of any conferences. They forward the media onto the Transcoding Conferencing Nodes in the "London Transcoding" and "Oslo Transcoding" locations respectively.
- The location containing your proxying nodes must set its **Transcoding location** to a location that contains your transcoding nodes. You can also optionally specify a primary overflow location and a secondary overflow location for additional transcoding resources i.e. the proxying nodes will proxy to the **Transcoding location** in the first instance, then proxy to the primary and then the secondary overflow locations if the **Transcoding location** runs out of capacity.
- When a location contains Proxying Edge Nodes, those nodes only require IPsec connectivity with:
  - any other proxying nodes in that location
  - all nodes in the transcoding location, and the primary and secondary overflow locations that are associated with that location
  - the Management Node.

  This means that the proxying nodes in one location do not need to have a direct network connection to other proxying nodes in other locations.



Where you have a geographically distributed platform, each physical region should typically have one location containing 1-3 proxying nodes (for resilience and capacity), and one or more additional locations containing transcoding nodes (typically between 2-25 nodes per location depending on capacity requirements).

- Ensure that your call control systems and DNS are configured to route calls to your proxying nodes only, and ensure that your proxying nodes have appropriate certificates installed. (If a call is routed to a transcoding node it can still accept the signaling and handle the call media to and from the endpoint, but we recommend that you defer these functions to your proxying nodes.)
- Lineside media handling is allocated to the proxying node with the most available capacity in the location that receives the signaling (it will **not** allocate the media to proxying nodes in other locations — there is currently no location overflow for proxying nodes). The node that receives the signaling always continues to handle the signaling.
- The proxying node must then forward the media onto a transcoding node. Pexip Infinity's standard media allocation rules are used to decide which transcoding node will receive the proxied media: in the first case it will use a transcoding node in the **Transcoding location** that is associated with the location of the proxying node that is forwarding the media. If there is no capacity to host the conference in the transcoding location then a transcoding node in the primary overflow location, or the secondary overflow location is used (if overflow locations are configured).
- As with all deployment scenarios, there cannot be NAT between any Pexip nodes, but there can be NAT between external devices/internet and proxying nodes. Proxying nodes support all call protocols, and can be deployed with dual network interfaces and static NAT if required.
- You only need to deploy certificates on your Proxying Edge Nodes — only those nodes that handle the signaling connection to an endpoint or other system (such as a Skype for Business Edge Server) need to be configured with the appropriate certificates to allow that endpoint/system to communicate with Pexip Infinity. If you subsequently deploy more Transcoding Conferencing Nodes to increase conferencing capacity, you do not need to add certificates onto those additional nodes.
- If you use dynamic bursting to a cloud service, the nodes in your cloud overflow locations should all be Transcoding Conferencing Nodes. You only have to ensure that your proxying nodes can route to your cloud-hosted nodes subnet — as, in this scenario, endpoints will never connect directly to a cloud-hosted node. Note that you cannot increase your proxying resources via dynamic bursting — you can only have dynamic bursting of transcoding nodes.
- The servers hosting Proxying Edge Nodes do not require as high a specification as those servers hosting Transcoding Conferencing Nodes because they do not process any media. You still need multiple proxying nodes for resilience and capacity. We recommend allocating 4 vCPU and 4 GB RAM (which must both be dedicated resource) to each Proxying Edge Node, with a maximum of 8 vCPU and 8 GB RAM for large or busy deployments.

Deployment scenarios

Depending on your requirements, there are a range of scenarios where you can make use of a Distributed Edge deployment. These are explained below and range from a basic DMZ Edge scenario to an advanced multi-Edge deployment with dynamic cloud bursting.

- Basic public DMZ deployment with single NIC Proxying Edge Node
- Basic public DMZ deployment with dual NIC Proxying Edge Node
- Basic public DMZ deployment with additional cloud-hosted transcoding resource
- Multi-Edge deployment with public DMZ and video zone routing
- Multi-Edge deployment with public DMZ, video zone and PC zone routing
- Multi-Edge deployment with public DMZ, video zone, PC zone routing and cloud resources

Some of these scenarios require routed connections from DMZ nodes — see Applying static routes to enable routing between externally-facing nodes and local network nodes for more information.

Note that in all Pexip Infinity deployment scenarios:

- The Management Node must be able to reach all Conferencing Nodes (Proxying Edge Nodes and Transcoding Conferencing Nodes) and vice versa.
- Each Conferencing Node must be able to reach every other Conferencing Node (Proxying Edge Nodes and Transcoding Conferencing Nodes), except:
  - When a location contains Proxying Edge Nodes, those nodes only require IPsec connectivity with:
    - any other proxying nodes in that location
    - all nodes in the transcoding location, and the primary and secondary overflow locations that are associated with that location
    - the Management Node.

    This means that the proxying nodes in one location do not need to have a direct network connection to other proxying nodes in other locations.
- Any internal firewalls must be configured to allow UDP port 500 and traffic using IP protocol 50 (ESP) in both directions between all Pexip nodes.
- There cannot be a NAT between any Pexip nodes.

The deployment examples use the following symbols:

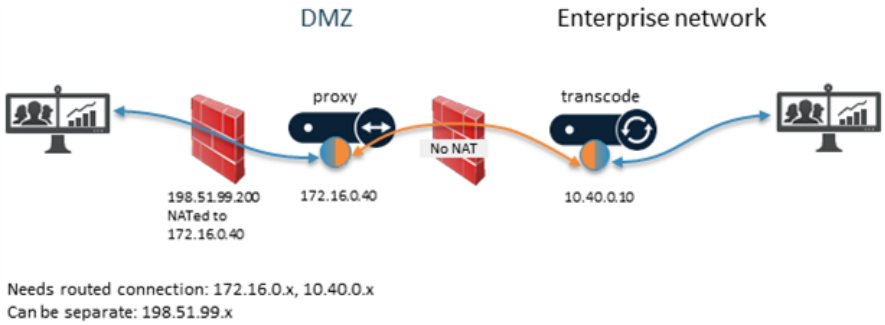
Symbol	Description	Symbol	Description
	Lineside (endpoint/client) media path.		Single NIC Proxying Edge Node with or without lineside NAT.
	Pexip to Pexip inter-node media path.		Dual NIC Proxying Edge Node. Blue = line side interface (clients/endpoints). Orange = Pexip internal interface.
	Pexip to Pexip inter-node signaling.		Single NIC Transcoding Conferencing Node with or without lineside NAT.
	Conferencing Node lineside interface (blue dots). These are the only connection points that endpoints/clients see.		Single NIC Transcoding Conferencing Node. Does not (in these scenarios) need to talk to clients as all media is proxied.
	Conferencing Node internal interface (orange dots).		

Basic public DMZ deployment with single NIC Proxying Edge Node

This deployment scenario shows a Proxying Edge Node with a single NIC in a public DMZ. In this example:

- The Proxying Edge Node in the DMZ has a public IP address which is NATted to a private IP address.
- The Proxying Edge Node's private IP address in the DMZ needs a routed connection to the enterprise network (10.40.0.x).

- The Transcoding Conferencing Node in the enterprise network also supports lineside connections to devices in the enterprise network.

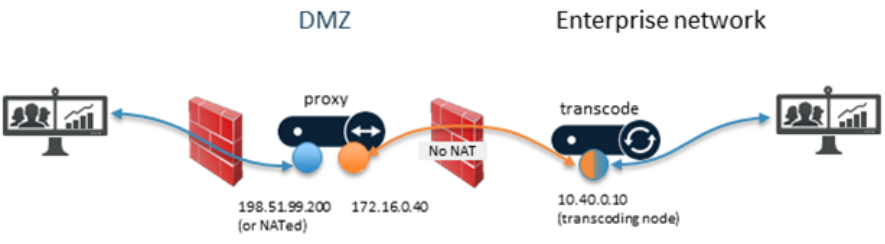


Needs routed connection: 172.16.0.x, 10.40.0.x
Can be separate: 198.51.99.x

Basic public DMZ deployment with dual NIC Proxying Edge Node

This deployment scenario is identical to the previous example, except in this case the Proxying Edge Node has a dual NIC. In this example:

- The secondary NIC on the Proxying Edge Node in the DMZ has a public IP address (which can optionally be NATted).
- The primary NIC on the Proxying Edge Node has a private IP address in the DMZ which needs a routed connection to the enterprise network (10.40.0.x).

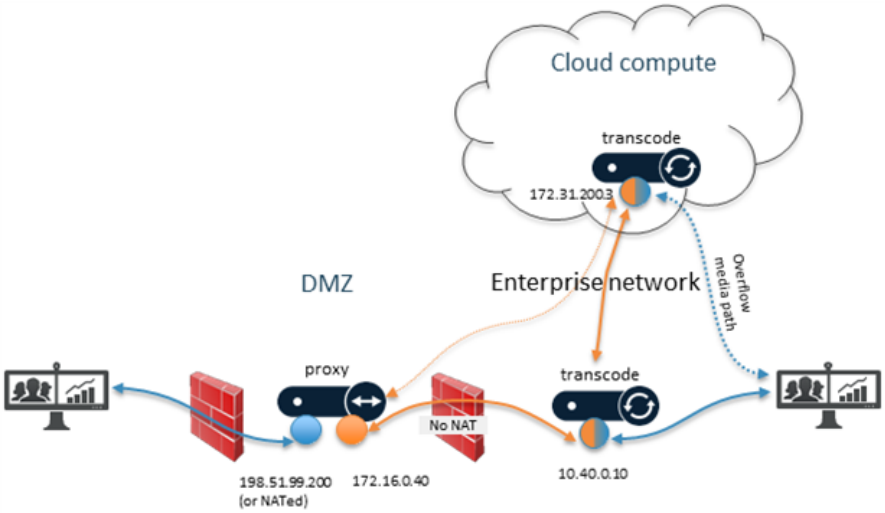


Needs routed connection: 172.16.0.x, 10.40.0.x
Can be separate: 198.51.99.x

Basic public DMZ deployment with additional cloud-hosted transcoding resource

This deployment scenario builds on the previous example by adding additional overflow Transcoding Conferencing Node resources on a cloud-hosted service. In this example:

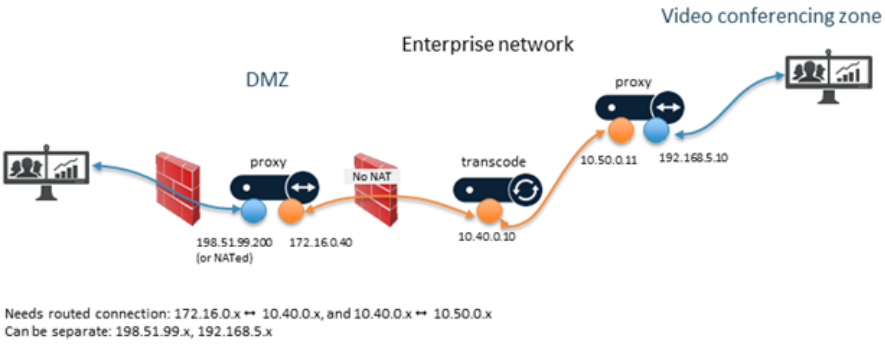
- Enterprise-based devices could get their media assigned to overflow nodes in the cloud service if the on-premises Transcoding Conferencing Nodes are at full capacity.
- The cloud-hosted overflow Transcoding Conferencing Nodes could be "always on" or configured for dynamic bursting.



Multi-Edge deployment with public DMZ and video zone routing

This deployment scenario uses another Proxying Edge Node to handle signaling and media traffic between the enterprise network and the video conferencing zone. In this example we have:

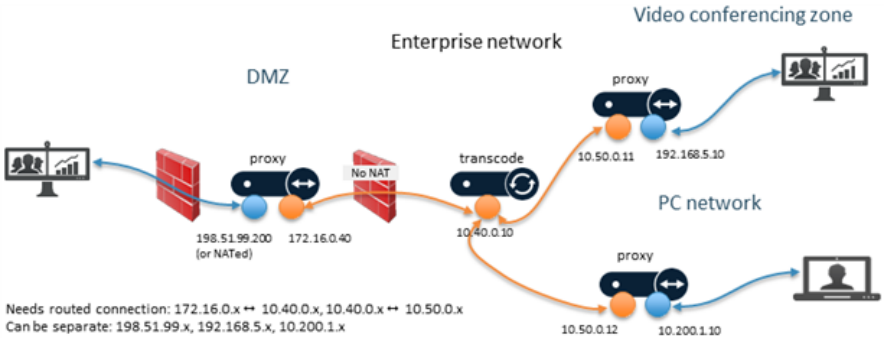
- **Dual NIC Proxying Edge Node in the DMZ**: the secondary lineside NIC has a public IP address (which can optionally be NATted); the primary NIC has a private IP address in the DMZ which needs a routed connection to the enterprise network (10.40.0.x).
- **Dual NIC Proxying Edge Node in the video conferencing network**: the secondary lineside NIC only needs connectivity to the video conferencing zone endpoints (192.168.5.x). The primary internal interface needs a routed connection to the enterprise network.
- A routed connection is not required between proxying nodes in the DMZ and the proxying nodes in the enterprise network.
- No endpoints/clients talk directly to the Transcoding Conferencing Nodes (their media is terminated lineside on Proxying Edge Nodes).



Multi-Edge deployment with public DMZ, video zone and PC zone routing

This deployment scenario adds to the previous example by deploying another Proxying Edge Node to manage connections to a PC network on another internal subnet. Thus, in this example:

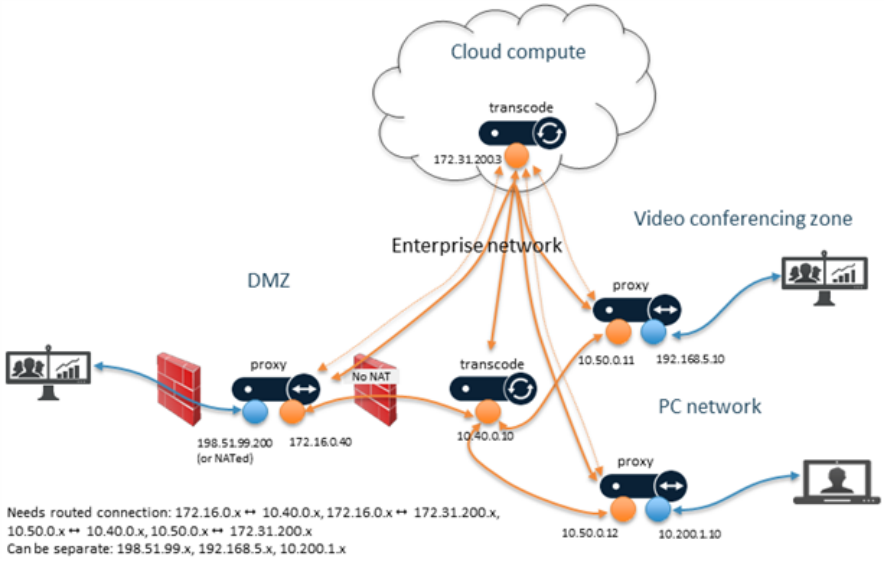
- The private IP addresses in the DMZ (172.16.0.x) and the internal NICs (10.50.0.x) on the proxying nodes handling the video conferencing and PC networks need a routed connection to the enterprise network (10.40.0.x).
- The video conferencing network lineside interface only needs connectivity to video conferencing zone endpoints (192.168.5.x), and the PC network lineside interface only needs connectivity to the PC network (10.200.1.x).
- A routed connection is not required between proxying nodes in the DMZ and the proxying nodes in the enterprise network.



Multi-Edge deployment with public DMZ, video zone, PC zone routing and cloud resources

This deployment scenario brings together all of the elements from the previous examples:

- The private IP addresses in the DMZ (172.16.0.x) and the internal NICs (10.50.0.x) on the proxying nodes handling the video conferencing and PC networks need a routed connection to the enterprise network (10.40.0.x).
- The video conferencing network lineside interface only needs connectivity to video conferencing zone endpoints (192.168.5.x), and the PC network lineside interface only needs connectivity to the PC network (10.200.1.x).
- No endpoints/clients talk directly to the Transcoding Conferencing Nodes in the enterprise network or the cloud-hosted network (their media is terminated lineside on Proxying Edge Nodes). Only the Pexip nodes need to be able to route traffic to the cloud-hosted nodes.
- A routed connection is not required between proxying nodes in the DMZ and the proxying nodes in the enterprise network.



Additional information when deploying Proxying Edge Nodes

- Any Conferencing Node can be a Proxying Edge Node or a Transcoding Conferencing Node, regardless of the hypervisor or cloud platform it is deployed on.
- Even though any media encryption/decryption is performed by a Proxying Edge Node, all subsequent communications with Transcoding Conferencing Nodes is over a secure IPsec connection.
- Proxying Edge Nodes can be deployed anywhere in your network (they do not have to be in a DMZ, for example), but they must be able to forward the media onto at least one Transcoding Conferencing Node.
- Devices can register to Proxying Edge Nodes or to Transcoding Conferencing Nodes.

- A Proxying Edge Node uses less resources per connection than is required by a node that is performing transcoding. A proxying node uses approximately the equivalent of 3 audio-only resources to proxy a video call (of any resolution), and 1 audio-only resource to proxy an audio call.
- Media streams for presentation content to/from an endpoint are forwarded in the same manner as video/audio streams, however they could be handled by different Proxying Edge Nodes (and Transcoding Conferencing Nodes).
- Bandwidth limitations cannot be applied to the forwarding connection between a Proxying Edge Node and a Transcoding Conferencing Node.
- Proxying Edge Nodes do not affect the call licensing requirements for an endpoint connection.

Network routing and addressing options for Conferencing Nodes

This section describes Pexip Infinity's network routing and addressing configuration options that are often required when deploying Conferencing Nodes in a public DMZ, or when routing to a dedicated video zone. It covers the following scenarios:

- Configuring Pexip Infinity nodes to work behind a static NAT device
- Applying static routes to enable routing between externally-facing nodes and local network nodes
- Deploying Conferencing Nodes with dual network interfaces (NICs)
- Remote SIP endpoints behind a remote firewall/NAT

Note that in all Pexip Infinity deployment scenarios:

- The Management Node must be able to reach all Conferencing Nodes (Proxying Edge Nodes and Transcoding Conferencing Nodes) and vice versa.
- Each Conferencing Node must be able to reach every other Conferencing Node (Proxying Edge Nodes and Transcoding Conferencing Nodes), except:
  - When a location contains Proxying Edge Nodes, those nodes only require IPsec connectivity with:
    - any other proxying nodes in that location
    - all nodes in the transcoding location, and the primary and secondary overflow locations that are associated with that location
    - the Management Node.

    This means that the proxying nodes in one location do not need to have a direct network connection to other proxying nodes in other locations.
- Any internal firewalls must be configured to allow UDP port 500 and traffic using IP protocol 50 (ESP) in both directions between all Pexip nodes.
- There cannot be a NAT between any Pexip nodes.

In addition, you must ensure that all appropriate firewall ports have been opened as described in Pexip Infinity port usage and firewall guidance.

Further information is available in Network deployment options, Firewall/NAT routing and addressing examples and Deployment guidelines for Proxying Edge Nodes.

Configuring Pexip Infinity nodes to work behind a static NAT device

To configure your Pexip Infinity deployment to work behind a static NAT device (from the perspective of clients located on the Internet or in a dedicated video zone) you must:

1. Configure the NAT device / firewall with the static, publicly-reachable IP address of each Conferencing Node that you want to be accessible from devices in the internet / video zone, and then map the public address to the node's corresponding internal IP address. Note that it must be a 1:1 NAT.
2. Configure each publicly-reachable Conferencing Node with its **IPv4 static NAT address** (**Platform > Conferencing Nodes**) i.e. the public address of the node that you have configured on the NAT device.

Note that:

- Any Conferencing Nodes that are configured with a static NAT address must not be configured with the same **System location** as nodes that do not have static NAT enabled. This is to ensure that load balancing is not performed across nodes servicing external clients and nodes that can only service private IP addresses.
- Static NAT must be on the secondary interface if the Conferencing Node has dual network interfaces.

- Any internal systems such as Cisco VCSs or endpoints that will send signaling and media traffic to Pexip Infinity nodes that are enabled for static NAT should send that traffic to the public address of those nodes. You must ensure that your local network allows this.
- We do not recommend that you allow the Management Node to be accessible from devices in the public internet. However, if you want to do this, you must assign and configure the Management Node with its static NAT address. You should also configure your firewall to only allow access to the Management Node from the specific IP addresses from where you want to allow management tasks to be performed.
- There cannot be a NAT device between any Pexip Infinity nodes.

Deploying Conferencing Nodes with dual network interfaces (NICs)

For additional deployment flexibility, you can configure a secondary network address on a Conferencing Node.

You would typically deploy a Conferencing Node with dual network interfaces when it is connected to a dedicated video zone or it is being deployed in a public DMZ where the primary interface would be to an internal, private network segment within the enterprise (where it can connect to the Management Node and other Conferencing Nodes) and the secondary interface would be towards the video zone or on the publicly-addressable side of the DMZ perimeter network and used for connecting to external endpoints and devices.

As with specifying the IP address of a Conferencing Node with a single interface, the IP address and network mask of the secondary interface must be specified when the Conferencing Node is initially deployed and cannot be subsequently modified.

When a secondary network address is configured:

- The primary address is always used for inter-node communication to the Management Node and to other Conferencing Nodes.
- SSH connections can be made only to the primary interface.
- The secondary address is always used for signaling and media (to endpoints and other video devices).
- Connections to DNS, SNMP, NTP, syslog and so on, go out from whichever interface is appropriate, based on routing.
- You can have a mixture of any number of single-interfaced and dual-interfaced Conferencing Nodes, providing all nodes can communicate with each other via their primary interfaces.
- Each interface must be in a different subnet.
- The Conferencing Node's default gateway can be in either subnet and is automatically associated with the correct interface. For the typical DMZ deployment scenario where the secondary NIC is the public NIC, the default gateway sits on the secondary NIC.
- Static routes can be in either subnet and are automatically associated with the correct interface.
- Static NAT, if configured, must be on the secondary interface.

Applying static routes to enable routing between externally-facing nodes and local network nodes

In some deployment scenarios you may have the Management Node and some Conferencing Nodes deployed in the local enterprise network, and some Conferencing Nodes deployed in a public DMZ or connected to a dedicated video zone.

If the default gateway on those non-local nodes is configured to route traffic out to the internet / video zone, you will need to configure static routes on those Conferencing Nodes to allow them to communicate with Pexip Infinity nodes or other systems in the local, internal network. See Managing static routes for information about to how to configure and assign static routes to Pexip Infinity nodes.

In situations where the Conferencing Node's default gateway is not towards the Management Node, the static route back to the Pexip Infinity platform must be applied to the Conferencing Node during its initial deployment phase (otherwise it will not be able to communicate with the Management Node and pick up its configuration).

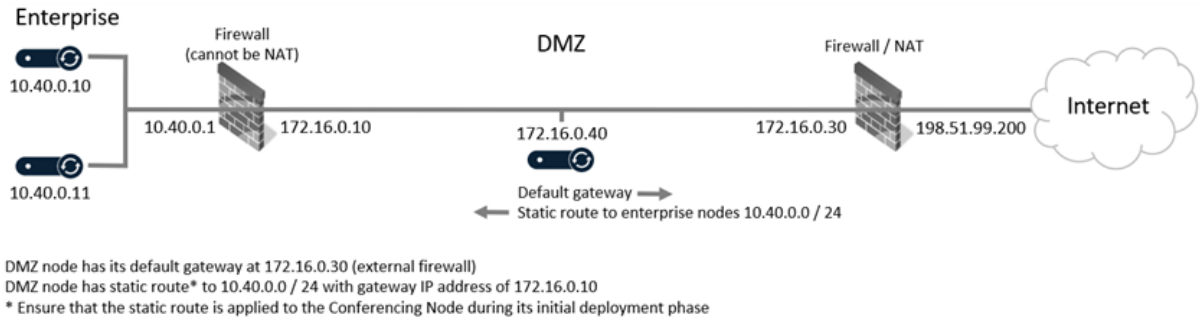
Note that:

- Conferencing Nodes in a DMZ must not be configured with the same **System location** as nodes in a local network. This is to ensure that load balancing is not performed across nodes in the DMZ and nodes in the local network.
- You must configure any internal firewall to allow UDP port 500 and traffic using IP protocol 50 (ESP) in both directions between the Pexip Infinity nodes in the DMZ and the nodes in the local network.
- The firewall between the Pexip Infinity nodes in the DMZ and the nodes in the local network cannot be a NAT device.
- The external firewall between Conferencing Nodes in the DMZ and the internet can be configured with static NAT. In this case you would also need to configure each Conferencing Node in the DMZ with its relevant static NAT address.

- If you deploy the Management Node in the DMZ (although we do not recommend this for security reasons), it must also be assigned with the relevant static route (**Platform > Management Node**).

Example

For example, consider a Pexip Infinity system which is deployed as shown below:



Example network with Pexip Infinity nodes in LAN and DMZ

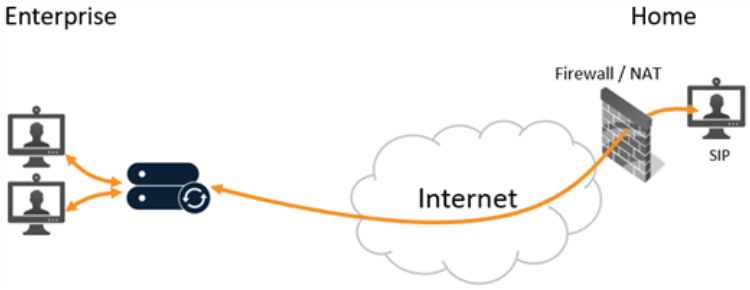
This deployment has:

- Pexip Infinity nodes on the enterprise LAN with addresses in the 10.40.0.0/24 subnet
- an internal firewall (without NAT) with LAN address 10.40.0.1 and DMZ address 172.16.0.10
- an external firewall with DMZ address 172.16.0.30 and public address 198.51.99.200
- a Conferencing Node in the DMZ with a single network interface with address 172.16.0.40 and which is configured with a default gateway address of 172.16.0.30 (the external firewall).

In this situation the Conferencing Node in the DMZ will, by default, send all of its traffic out through its default gateway — the external firewall at 172.16.0.30. To ensure that traffic from the Conferencing Node in the DMZ that is destined for Pexip Infinity nodes on the enterprise LAN can be routed to those nodes, you must deploy the Conferencing Node with a suitable static route applied to it. In this example the **Destination network address** of the static route would be **10.40.0.0** with a **Network prefix** of **24** (to route addresses in the range **10.40.0.0** to **10.40.0.255**) and the **Gateway IP address** would be 172.16.0.10 (the DMZ address of the internal firewall).

Remote SIP endpoints behind a remote firewall/NAT

Remote SIP endpoints that are behind remote firewalls/NATs can join Pexip Infinity conferences.



Media latching with remote endpoints behind a remote firewall/NAT

You do not have to apply any explicit configuration to Conferencing Nodes in order to allow remote SIP endpoints behind remote firewalls/NATs to join a conference. Pexip Infinity automatically uses signaling and media port latching to establish routable paths.

(Port latching involves Pexip Infinity waiting until it receives signaling and media traffic from the remote endpoint, and then it uses — or "latches" on to — the source address and port of that traffic as a destination for all traffic bound in the opposite direction. Typically these source addresses/ports will belong to the public interface of the NAT in front of the remote endpoint, and thus anything sent by Pexip Infinity to that address/port should ultimately reach the endpoint.)

Firewall/NAT routing and addressing examples

Here are some example Pexip Infinity routing and addressing scenarios that demonstrate the various network deployment options that are available when deploying the Pexip Infinity platform:

- Public DMZ node with single NIC and externally-facing default gateway (with/without NAT)
- Public DMZ node with single NIC and three-legged firewall (with/without NAT)
- Public DMZ node with dual NICs and externally-facing default gateway (with/without NAT)
- Conferencing Node with dual NICs and routing to a video zone (with/without NAT)

For more information about how to configure static NAT, dual network interfaces and static routes, see Network routing and addressing options for Conferencing Nodes.

For specific examples that show how to deploy Proxying Edge Nodes in combination with Transcoding Conferencing Nodes, see Deployment guidelines for Proxying Edge Nodes.

Public DMZ node with single NIC and externally-facing default gateway (with/without NAT)

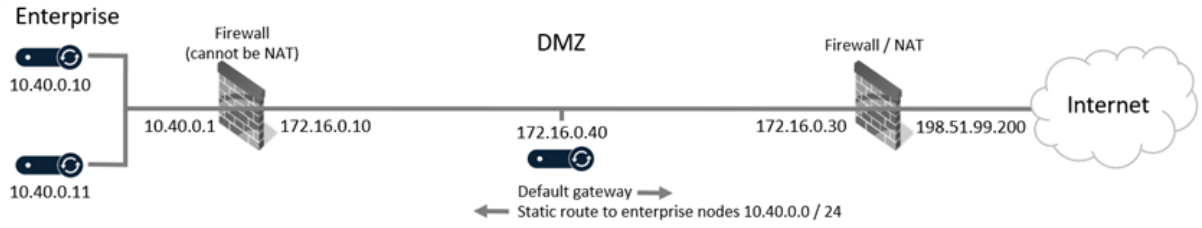
These types of deployment have a Conferencing Node in the public DMZ with:

- a single externally-facing interface (default gateway is out to the public internet) — this can be with or without NAT
- a static route back into the enterprise (to communicate with any private Conferencing Nodes and the Management Node)
- an internal firewall (which must not be NAT) between the enterprise nodes and the public DMZ nodes.

The static route must be applied to the Conferencing Node during its initial deployment phase (otherwise it will not be able to communicate with the Management Node and pick up its configuration).

Externally-facing interface with NAT

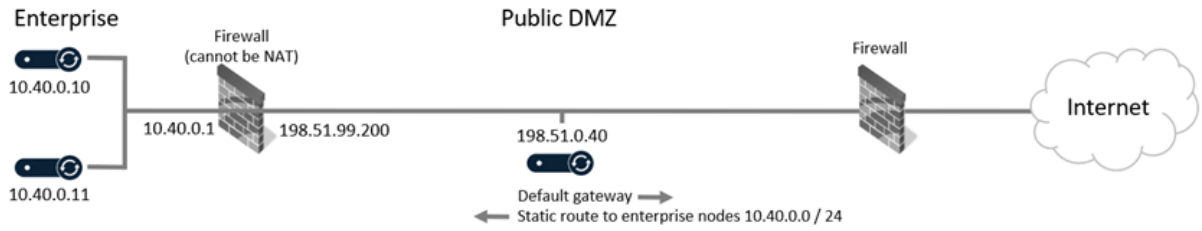
In this example the external firewall performs NAT:



DMZ node has its default gateway at 172.16.0.30 (external firewall)
DMZ node has static route* to 10.40.0.0 / 24 with gateway IP address of 172.16.0.10
* Ensure that the static route is applied to the Conferencing Node during its initial deployment phase

Externally-facing interface without NAT

In this example the external firewall does not perform NAT:



Public DMZ node with its default gateway towards public internet
Public DMZ node has static route* to 10.40.0.0 / 24 with gateway IP address of 198.51.99.200
* Ensure that the static route is applied to the Conferencing Node during its initial deployment phase

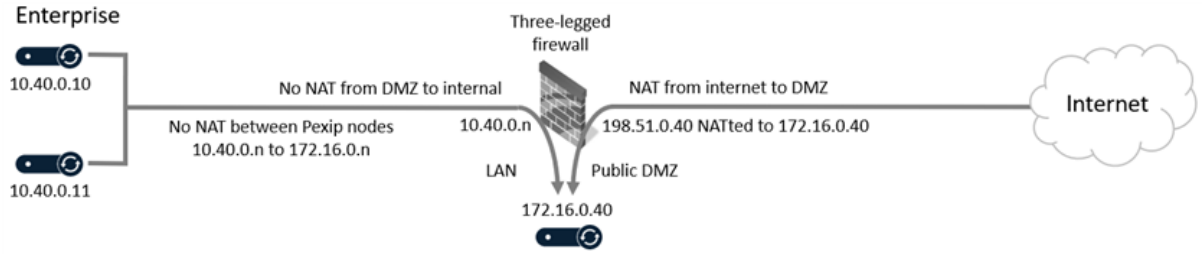
Public DMZ node with single NIC and three-legged firewall (with/without NAT)

These types of deployment have a Conferencing Node in the public DMZ with:

- a single interface — this can be with or without NAT for traffic that is destined for the public internet
- a three-legged firewall that allows bi-directional routing between enterprise nodes in the LAN (any private Conferencing Nodes and the Management Node) — this must always be without NAT.

NAT applied to external traffic

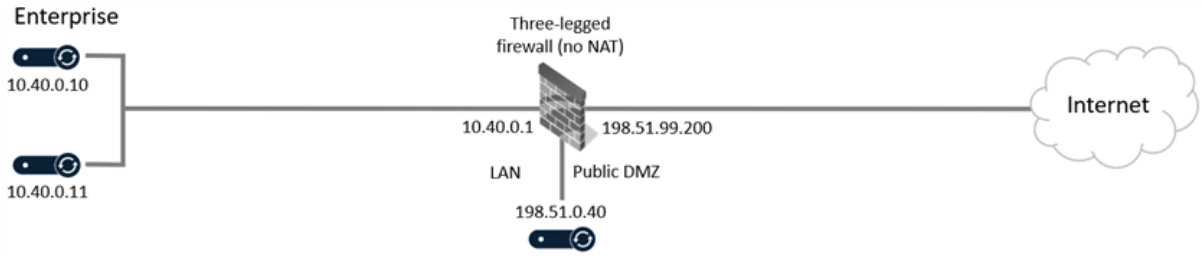
In this example the firewall applies NAT to external traffic (between DMZ and internet):



Firewall allows bi-directional routing between LAN enterprise nodes and public DMZ node. Traffic between DMZ node and internet is NATted.

NAT is not applied to external traffic

In this example the firewall does not apply NAT to external traffic (between DMZ and internet):



Firewall allows bi-directional routing between LAN enterprise nodes and public DMZ node. Traffic between DMZ node and internet is not NATted.

Public DMZ node with dual NICs and externally-facing default gateway (with/without NAT)

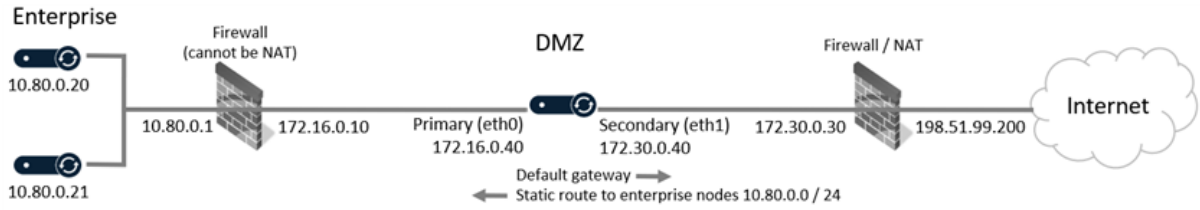
These types of deployment have a Conferencing Node in the public DMZ with:

- dual network interfaces:
  - the primary NIC is internally-facing (eth0 must be used for all inter-node communication)
  - the secondary NIC is externally-facing (eth1 must be used for signaling and media to endpoints and other video devices) — this can be with or without NAT
- the default gateway is out to the public internet (and thus is associated automatically with the secondary NIC)
- a static route back into the enterprise (to communicate with any private Conferencing Nodes and the Management Node)
- an internal firewall (which must not be NAT) between the enterprise nodes and the public DMZ nodes.

The static route must be applied to the Conferencing Node during its initial deployment phase (otherwise it will not be able to communicate with the Management Node and pick up its configuration).

Externally-facing interface with NAT

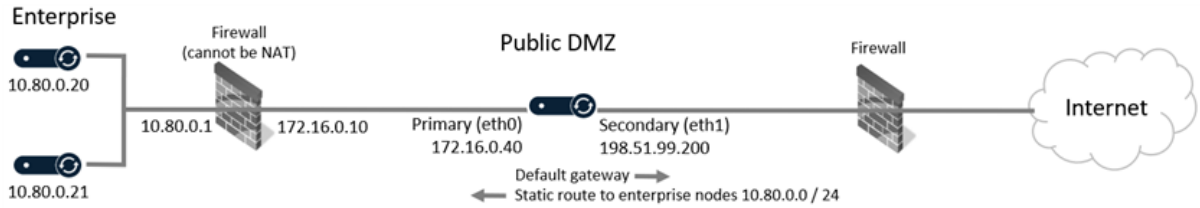
In this example the external firewall performs NAT:



DMZ node has its default gateway at 172.30.0.30 (external firewall) and is associated with secondary interface (eth1)
DMZ node has static route* to 10.80.0.0 / 24 with gateway IP address of 172.16.0.10
* Ensure that the static route is applied to the Conferencing Node during its initial deployment phase

Externally-facing interface without NAT

In this example the external firewall does not perform NAT:



Public DMZ node with its default gateway towards public internet
Public DMZ node has static route* to 10.80.0.0 / 24 with gateway IP address of 172.16.0.10
* Ensure that the static route is applied to the Conferencing Node during its initial deployment phase

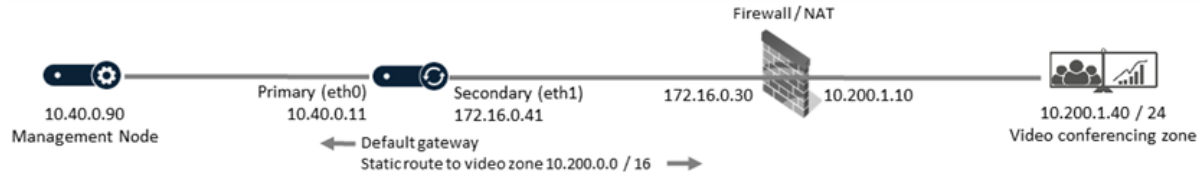
Conferencing Node with dual NICs and routing to a video zone (with/without NAT)

These types of deployment have a Conferencing Node that communicates with a dedicated video conference VLAN. It has:

- dual network interfaces:
  - the primary NIC is internally-facing (eth0 must be used for all inter-node communication)
  - the secondary NIC faces the video zone (eth1 must be used for signaling and media to endpoints and other video devices in the video zone) — this can be with or without NAT
- the default gateway is to the Pexip nodes (and thus is associated automatically with the primary NIC)
- a static route for traffic destined for the video zone.

Video zone routing with NAT

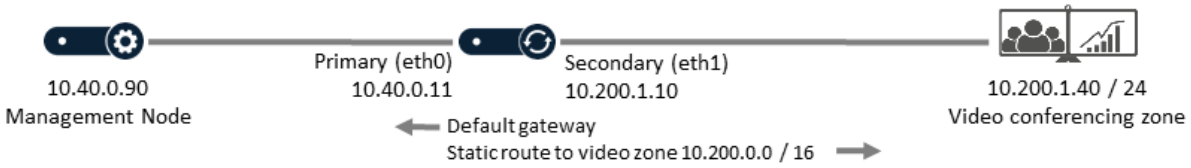
In this example the traffic sent to the video zone has NAT applied:



Conferencing Node default gateway associated with primary interface (eth0) towards Management Node
Conferencing Node static route to 10.200.0.0 / 16 video zone subnet with gateway IP address of 172.16.0.30

Video zone routing without NAT

In this example the traffic sent to the video zone does not have NAT applied:



Conferencing Node default gateway associated with primary interface (eth0) towards Management Node
Conferencing Node static route to 10.200.0.0 / 16 video zone subnet

Dynamic bursting to a cloud service

Pexip Infinity deployments can burst into the Microsoft Azure, Amazon Web Services (AWS) or Google Cloud Platform (GCP) cloud when principal conferencing capabilities are reaching their capacity limits, thus providing additional temporary Transcoding Conferencing Node resources for media processing.

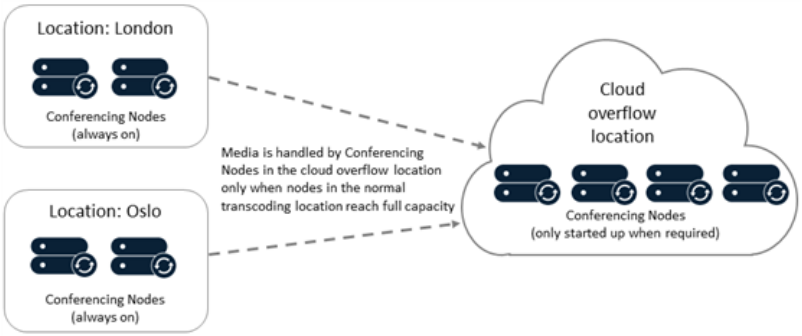
This provides the ability to dynamically expand conferencing capacity whenever scheduled or unplanned usage requires it. The cloud Conferencing Nodes instances are only started up when required and are automatically stopped again when capacity demand normalizes, ensuring that cloud provider costs are minimized.

You can also use the management API to monitor and manually start up your overflow nodes, if for example, you want to configure a third-party scheduling system to start up all of your additional capacity prior to a large conferencing event starting.

Cloud bursting provides additional ad hoc capacity in a cloud service to which you must already have a subscription and for which you will be charged based on time used; you must manage the use of this capacity yourself.

How it works

Dynamic bursting builds upon Pexip Infinity's standard location capacity overflow logic that defines which overflow location's nodes are used when a particular location reaches its media-processing capacity. The dynamic bursting functionality adds to this by automatically starting up those additional cloud nodes when a location is **approaching** full capacity, so that those overflow nodes will be available if required.



After you have deployed your overflow Conferencing Nodes in Azure, AWS or GCP, and enabled cloud bursting and configured your bursting threshold in Pexip Infinity, everything is then controlled automatically by the Management Node:

- Whenever the available capacity in a system location containing your principal (always on) Conferencing Nodes hits or drops below your configured threshold, a Conferencing Node in its overflow location is automatically started up. That new node can then start handling any additional conferencing requirements if the original location reaches full capacity.
- When an overflow Conferencing Node starts to fill up and its location reaches the bursting threshold, a further overflow node is started up, and so on.
- When call levels subside and an overflow Conferencing Node is no longer hosting conferences and is no longer required, the node is automatically shut down again.

  This sequence of events is explained in more detail in a worked example.

Configuring your system for dynamic bursting

Dynamic bursting is supported on Conferencing Nodes hosted in Microsoft Azure, Amazon Web Services (AWS) or Google Cloud Platform (GCP). For specific configuration instructions for your chosen platform, see:

- Dynamic bursting to the AWS cloud
- Dynamic bursting to the Azure cloud
- Dynamic bursting to the Google Cloud Platform

Configuring the bursting threshold

When enabling your platform for cloud bursting the most important decision you must make is the level at which to set the bursting threshold:

- The bursting threshold controls when your dynamic overflow nodes in your cloud service are automatically started up so that they can provide additional conferencing capacity. When the number of additional HD calls that can still be hosted in a location reaches or drops below the threshold, it triggers Pexip Infinity into starting up an overflow node in the overflow location.

  For example, setting the threshold to 5 means that when there are 5 or fewer HD connections still available in a location, an overflow node will be started up.
- When an overflow location reaches the bursting threshold i.e. the number of additional HD calls that can still be hosted on the Conferencing Nodes in the overflow location reaches the threshold, another overflow node in that location is started up, and so on.

  Note that the current number of free HD connections in the original location is ignored when deciding if the overflow location needs to overflow further — however, new calls will automatically use any available media resource that has become available within the original principal location.
- The bursting threshold is a global setting — it applies to every system location in your deployment.
- Note that it takes approximately 5 minutes for a dynamic node instance to start up and become available for conference hosting. If your principal deployment reaches full capacity, and the overflow nodes have not completed initiating, any incoming calls during this period will be rejected with "capacity exceeded" messages. You have to balance the need for having standby capacity started up in time to meet the expected demand, against starting up nodes too early and incurring extra unnecessary costs.

Deployment guidelines

System locations and Conferencing Nodes:

- When configuring your principal "always on" locations, you should normally set the **Primary overflow location** to point at the bursting location containing your overflow nodes, and the **Secondary overflow location** should normally only point at an always-on location.

  Nodes in a bursting location are only automatically started up if that location is configured as a **Primary overflow location** of an always-on location that has reached its capacity threshold. This means that if a bursting location is configured as a **Secondary overflow location** of an always-on location, then those nodes can only be used as overflow nodes if they are already up and running (i.e. they have already been triggered into starting up by another location that is using them as its **Primary overflow location**, or you have used some other external process to start them up manually).
- Typically you should assign all of your bursting Conferencing Nodes to a single overflow system location so that all of your main transcoding locations can make use of the same pool of cloud overflow nodes.

  However, if you expect a single conference to require more than four overflow nodes, then you could configure a second cloud overflow location (containing a different set of bursting nodes) and then configure half of your principal locations to overflow to the first cloud overflow location, and the other half of your principal locations to overflow to the second cloud overflow location. In this case, both overflow locations will act (start up and shut down nodes as appropriate) independently of each other.
- We recommend that you do not mix your "always on" Conferencing Nodes and your bursting nodes in the same system location.
- Cloud bursting nodes must be deployed as Transcoding Conferencing Nodes (not Proxying Edge Nodes).

Additional guidelines:

- An overflow cloud bursting node is automatically stopped when it becomes idle (no longer hosting any conferences). However, you can configure the **Minimum lifetime** for which the bursting node is kept powered on. By default this is set to 50 minutes, which means that a node is never stopped until it has been running for at least 50 minutes. If your service provider charges by the hour, it is more efficient to leave a node running for 50 minutes — even if it is never used — as that capacity can remain on

immediate standby for no extra cost. If your service provider charges by the minute you may want to reduce the **Minimum lifetime**.

- Do not configure your call control system / DNS to route calls directly to your bursting nodes.
- The Management Node requires access to the cloud platform's APIs to start and stop the bursting node instances. These requests can be routed via a web proxy if required.
- A location can trigger a bursting node to start up only if there is some existing media load on that location. This is to ensure that incidents such as a temporary network interruption do not inadvertently trigger bursting.
- No specific licenses are required for your bursting nodes, but you must ensure that your overall system has sufficient licenses to meet peak conference capacity demand.
- Your cloud bursting nodes must all be running on the same cloud platform (Azure, AWS or GCP).
- All log messages that are explicitly related to cloud bursting, such as starting up or shutting down overflow nodes, are tagged with a log module name of administrator.apps.cloudbursting. All other log messages related to those overflow nodes or the conferences they are hosting are reported in the same manner as per standard behavior.

Detailed example of the overflow process

This sequence of actions listed below shows how the process of starting and stopping overflow nodes is managed. It assumes an example scenario where:

- the system is configured with a bursting threshold of 5
- there is a single principal location (London) containing 2 "always on" Conferencing Nodes with a total location capacity of 40 connections
- the principal location is configured to overflow to the cloud overflow location "Bursting Europe"
- the "Bursting Europe" location contains 2 overflow bursting nodes, each with a capacity of 20 connections.

Principal location: London				Cloud overflow location: Bursting Europe		
Action		Call capacity remaining	Dynamic bursting activity	Overflow node A status	Overflow node B status	Call capacity remaining
1.	No conferences in progress	40	*none*	Not running	Not running	
2.	A conference starts and has 33 participants	6*	*none*			
3.	1 more participant joins	5	Principal location threshold reached	Node A starting		
4.		5	Overflow capacity becomes available	Running		20
5.	5 more participants join	0	*none*	↓		20
6.	1 more participant joins	0	Call media handled by overflow node A	↓		18*
7.	12 more participants join	0	All new participants handled by overflow node A	↓		6
8.	1 more participant joins	0	Overflow location threshold reached	↓	Node B starting	5
9.		0	Additional overflow capacity becomes available	↓	Running	25
10.	7 more participants join	0	All new participants handled by overflow nodes A and B	↓	↓	17*

Principal location: London				Cloud overflow location: Bursting Europe		
Action		Call capacity remaining	Dynamic bursting activity	Overflow node A status	Overflow node B status	Call capacity remaining
11.	Conference ends	40	Overflow nodes remain available for further bursting if required	↓	↓	40
12.	A new conference starts with 25 participants	14*	*none*	↓	↓	40
13.			Overflow node A is unused and has been running for 50 minutes	Shutting down	↓	20
14.			Overflow node B is unused and has been running for 50 minutes	Not running	Shutting down	0
15.	Conference ends	40		Not running	Not running	

* A Conferencing Node reserves 1 HD connection for the backplane to other nodes in the same conference.

Note that if other conferences were running on any of the nodes in these locations, they would consume call capacity and bursting would be triggered in exactly the same way when the remaining capacity within the location reached the threshold.

Viewing cloud bursting status

Current bursting status

Go to **Status > Cloud Bursting** to see an overview of the media load of your principal locations (that contain your "always-on" Conferencing Nodes), and whether your overflow nodes and locations are in use.

- Any issues relating to your cloud bursting deployment will also be shown on this page.
- The list of principal locations only includes those locations that are configured with a **Primary overflow location** that contains bursting nodes.
- An **approaching threshold** message is displayed in the **Available HD** connections column for the principal locations when the number of available HD connections is less than or equal to the bursting threshold plus two.

  This message changes to **bursting threshold reached** when the number of available HD connections is less than or equal to the bursting threshold (and therefore overflow nodes are started up).
- You can manually start any overflow nodes by selecting **Start** for the required node (the **Start** option is in the final column of the **Cloud overflow nodes** table).
- The status page dynamically updates every 15 seconds.

Bursting history

Go to **History & Logs > Conferencing Node History** to see all of the events (stop, start or running) that have been applied to overflow Conferencing Nodes and, where appropriate, the reason why the event was applied (for example if a node was shut down as there was no longer a need for the extra capacity).

Handling of media and signaling

Media and signaling for each call to a Pexip Infinity service (VMR, gateway call etc.) may take different paths, depending on the location of the caller, the available capacity of Conferencing Nodes, whether any media overflow locations have been configured, and the role of the Conferencing Node that receives the call signaling.

A Conferencing Node can have either a transcoding or a proxying role:

- Transcoding Conferencing Nodes are required in all deployments; they manage all of the media processing required to host a conference, and can also handle direct connections to/from endpoints if required.
- Proxying Edge Nodes are optional; they handle call signaling and the media connection with the endpoint, but forward the media on to a Transcoding Conferencing Node for processing. For full information, see Deployment guidelines for Proxying Edge Nodes.

Behavior summary

Media and signaling handling for a conference follows these rules:

- Conferencing Nodes handle all conference signaling and media (except for direct media scenarios).
- A single conference instance can span any number of locations and any number of Conferencing Nodes, but with a limit of **4 nodes per location** that are processing media for that conference.
- The Conferencing Node that receives the call signaling always continues to handle the signaling, regardless of where or how the media is routed (which may be through a different node). There is no concept of signaling overflow.
- When a call is received on a Transcoding Conferencing Node, Pexip Infinity selects a transcoding node to process the conference media and perform the lineside media handling. The selection process is based on the **Transcoding location**, and the optional **Primary overflow location** and **Secondary overflow location** configured against the location of the node that is handling the call signaling. In the first case it tries to use a transcoding node in the **Transcoding location**: it will initially select a node that is already transcoding media for that conference (if that node has sufficient capacity to take the new call), otherwise it selects the transcoding node that currently has the most available capacity. If there is not a suitable node in the **Transcoding location** with sufficient capacity to take the media, or it has reached the limit of 4 nodes in that location, it uses a transcoding node in the **Primary overflow location**, or a node in the **Secondary overflow location**.
- When a call is received on a Proxying Edge Node, lineside media handling is allocated to the proxying node with the most available capacity in the location receiving the signaling. That proxying node must then forward the media onto a transcoding node using the same allocation/overflow rules (based on the signaling location's configured transcoding and media overflow locations) as when a call is received on a transcoding node, as described above. A system location should not contain a mixture of proxying nodes and transcoding nodes.
- The transcoding nodes send the call media for the conference to each other over a backplane, with each node sending the media on behalf of all the endpoints connected (or proxied) to it. If two or more transcoding nodes in the same location are processing conference media, a single node per location acts as an intermediary and sends the conference media over a backplane to another intermediary node in the other locations that are also handling that conference.

You should ensure that any locations specified as a **Transcoding location** or as a **Primary** or a **Secondary overflow location** contain only Transcoding Conferencing Nodes.

Note that if you change the media-handling locations for a proxying location (i.e. its transcoding, primary or secondary overflow locations), any proxied calls from that location that are **currently** being handled by the previously configured media locations will be dropped.

More details about all of these rules and some example scenarios are explained in the sections below:

- Locally and globally distributed conferences
- How nodes for signaling and media proxying (if applicable) are determined
- How Pexip Infinity decides which Transcoding Conferencing Node will process the media
- Media overflow locations
- Infinity Gateway calls
- Nominating the outgoing location for outbound calls
- Examples

Locally and globally distributed conferences

Conferences that span more than one Transcoding Conferencing Node can be locally or globally distributed. **Locally distributed conferences** exist across multiple transcoding nodes in the same system location and send conference media between nodes via a local backplane.

Globally distributed conferences exist across several Transcoding Conferencing Nodes, where each node is in a different system location. As system locations are typically used to represent different physical locations, this allows participants in different regions to access the conference from their local Conferencing Node. The nodes send the call media for the conference to each other over a single geo backplane, with each node sending the media on behalf of all the endpoints connected (or proxied) to it, thus minimizing WAN bandwidth usage between locations.

A conference can be **locally and globally distributed** at the same time, if two or more Transcoding Conferencing Nodes in one location and at least one other transcoding node in a different location are involved. In such cases, one transcoding node in each location acts as the intermediary for any other transcoding nodes in the same location that are handling the media for that conference. Call media for each location is sent between the intermediaries only, thus minimizing WAN bandwidth usage between locations.

A single conference instance can span any number of locations and any number of transcoding nodes, but with a maximum limit of **4 nodes per location** that are performing media transcoding for that conference.

How nodes for signaling and media proxying (if applicable) are determined

In all cases, when a call is placed to a Pexip Infinity service, the call (signaling) is received by whichever Conferencing Node was selected by your call control system or routed via DNS.

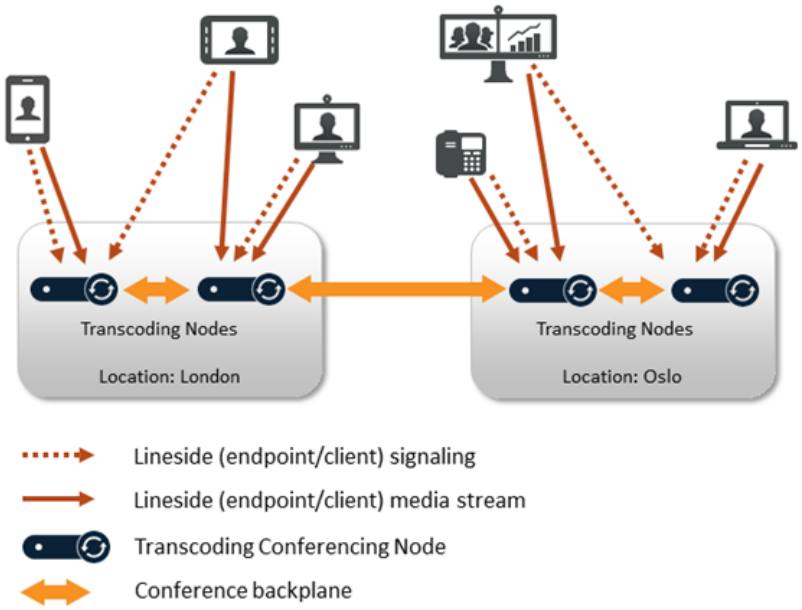
- Signaling always remains routed to the node that received the call. This could be a Proxying Edge Node or a Transcoding Conferencing Node.
- If the signaling is received on a Proxying Edge Node, media proxying is allocated to the proxying node with the most available capacity in the location that received the signaling. The selected proxying node will always handle the media connection with the endpoint, acting as a proxy between the endpoint and a Transcoding Conferencing Node (which should be in a different location).
- If the signaling is received on a Transcoding Conferencing Node, then whichever transcoding node is selected to process the media (which may be a different node to the signaling node) will also directly handle the media connection with the endpoint.

How Pexip Infinity decides which Transcoding Conferencing Node will process the media

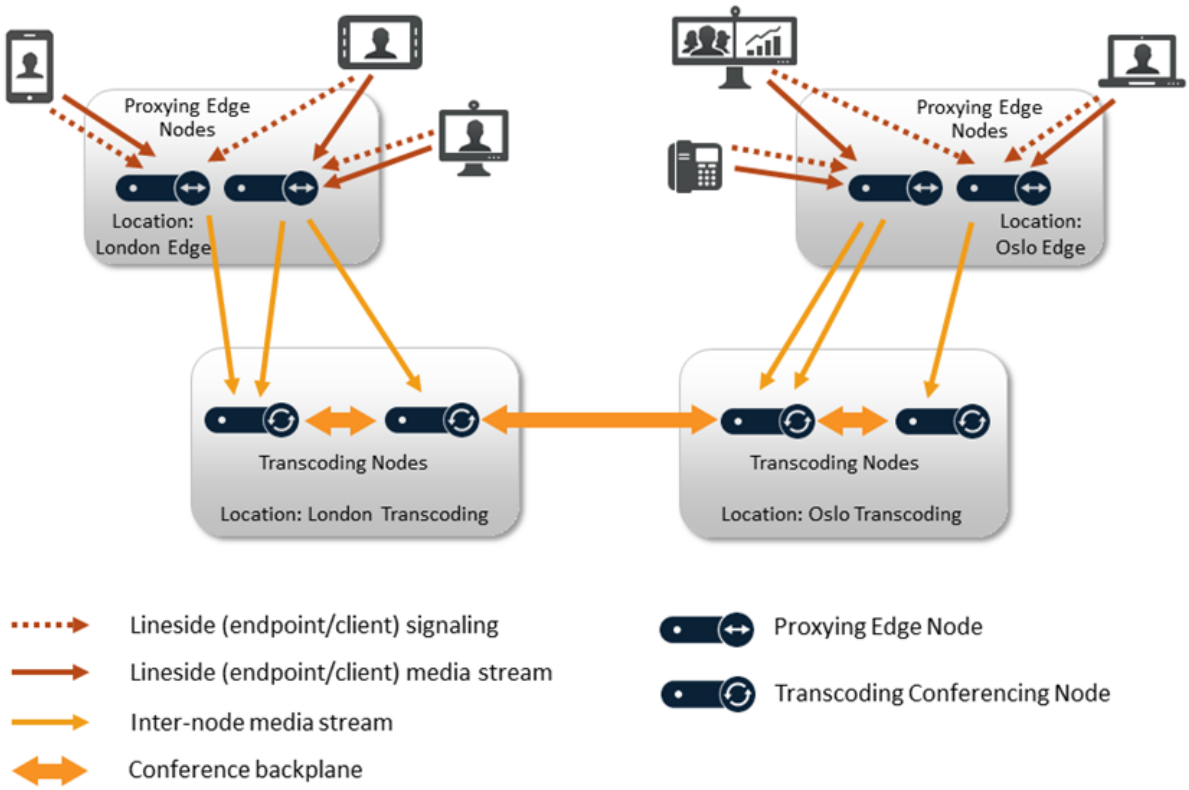
Pexip Infinity intelligently load-balances the call media across all transcoding nodes that are grouped within a system location. It uses the same load-balancing rules to decide which transcoding node will process the media and host the conference, regardless of whether the media is being proxied or not, and whether the signaling is being handled by a proxying node or by a transcoding node:

- The selection process is based on the **Transcoding location**, and the optional **Primary overflow location** and **Secondary overflow location** configured against the location of the node that is handling the call signaling.
- Initially, Pexip Infinity tries to use a Transcoding Conferencing Node that is in the **Transcoding location**:
  - If there is a transcoding node that is already handling media for the conference, and it has spare capacity, then that node will also process the media on the new call.
  - If there are no transcoding nodes that are currently processing media for the conference, or those that are currently processing media for the conference are at full capacity, then the transcoding node in that location that currently has the most available capacity is selected (up to a maximum of four transcoding nodes per location per conference instance).
- If there is **no transcoding resource available in the Transcoding location**, then a transcoding node in a **media overflow location** (if configured) is used. The reasons why there may be no transcoding resource available in the **Transcoding location** are:
  - all of the transcoding nodes in that location are fully-loaded in hosting other conferences
  - there are four fully-loaded transcoding nodes already managing that conference in that location
  - all of the nodes in the **Transcoding location** are proxying nodes (for example, if a location containing proxying nodes has its **Transcoding location** set to ***This location***).

  The selection of the transcoding node in the media overflow location follows the same balancing rules: if a node is currently processing the conference media and has capacity to take the new call then that node is selected, otherwise the node in the overflow location that currently has the most available capacity is selected.
- The call is rejected if there is no transcoding capacity available in the transcoding location or either of the overflow locations associated with the signaling location. Note that endpoints connected (for signaling purposes) to nodes at other locations may still be able to join the conference — this is because each location has its own configurable set of transcoding and overflow locations that it can use for its local endpoints.



Example flows when only using Transcoding Conferencing Nodes and transcoding is performed in the same location as the signaling location



Example flows when using Proxying Edge Nodes and transcoding is performed in a separate location containing Transcoding Conferencing Nodes

Media overflow locations

Media overflow locations are used to tell Pexip Infinity where to process the call media when there is no transcoding resource currently available in the **Transcoding location** associated with the location where the call has been received.

Locations are not explicitly configured as "media overflow" locations — their use depends upon how you have configured your system. For example, you could have one dedicated location that never initially handles any calls, but is used as a common overflow location

for all of your other locations. Alternatively, you could configure your locations to overflow to each other — which may be useful if you have locations around the world, so that a location that is experiencing heavy demand during its working day can make use of idle resources in a location in a different timezone.

The nodes in your overflow location can be "always-on" nodes or you can use dynamic node resources in a cloud service (referred to as "bursting").

To enable overflow, you must configure a location with a **Primary overflow location** and a **Secondary overflow location** (in addition to its standard **Transcoding location**). When overflow resources are required, Pexip Infinity will use any available transcoding resource in the primary overflow location first (using the same rules and limitations as described above), and if there is no suitable resource in the primary overflow location it will use resource in the secondary overflow location.

If you need to nominate more than two overflow locations, you can use local or external policy to specify a list of multiple overflow locations in its response to a media location request.

When determining where to handle the media for any given call, the transcoding location, primary overflow location and secondary overflow location options are all based on what is configured directly against the location containing the Conferencing Node that is handling the signaling for that call — i.e. if an overflow location is configured with its own overflow locations, then those indirectly-configured overflow locations are ignored. For example, if a call is received in location A, and location A is configured to transcode in location B but location B is full, it next uses the overflow location as configured in location A (what is configured in location B is irrelevant to this call). See the examples below for more details and conferencing scenarios.

Note that:

- If you are not using proxying nodes, you must ensure that endpoints can route their media to the transcoding nodes in the overflow locations.
- When each new location is brought into the conference, Pexip Infinity nominates a single transcoding node in each location as the intermediary node for that conference instance, and a geo backplane is established between those intermediary nodes.

Infinity Gateway calls

The same principles described above also apply to calls placed via the Infinity Gateway:

- The call signaling is always handled by the Conferencing Nodes that receive and place the call (note that you can configure Call Routing Rules to call out from a Conferencing Node in a different location to that which received the call).
- The call media is handled by a transcoding node that has the most available capacity in the **Transcoding location** associated with the signaling node's location, or its overflow location. However, see below for additional information about the outgoing location.

Nominating the outgoing location for outbound calls

When specifying the outgoing location for a Call Routing Rule, automatically dialed participant, or when manually dialing out from a conference:

- The call (signaling) is placed from a Conferencing Node in the nominated outgoing location, unless the device being called is registered to Pexip Infinity, in which case the call is always placed from the Conferencing Node where the registration is held.
- If the node placing the call is a Proxying Edge Node, then the proxying node in that signaling node's location with the most available capacity will handle the lineside media connection with the called device and it will proxy the media to a transcoding node in the **Transcoding location** associated with that location (or to an overflow location if there is no transcoding capacity in the **Transcoding location**).
- If the node placing the call is a Transcoding Conferencing Node, then the transcoding node in the **Transcoding location** associated with that signaling node's location (or in an overflow location if there is no transcoding capacity in the **Transcoding location**) with the most available capacity will handle the lineside media connection and the media transcoding.
- Note that if the nominated outgoing location contains a mixture of proxying nodes and transcoding nodes (although we do not recommend this) then Pexip Infinity will choose any node from the location and it will behave as described above according to the selected node's role.

Examples

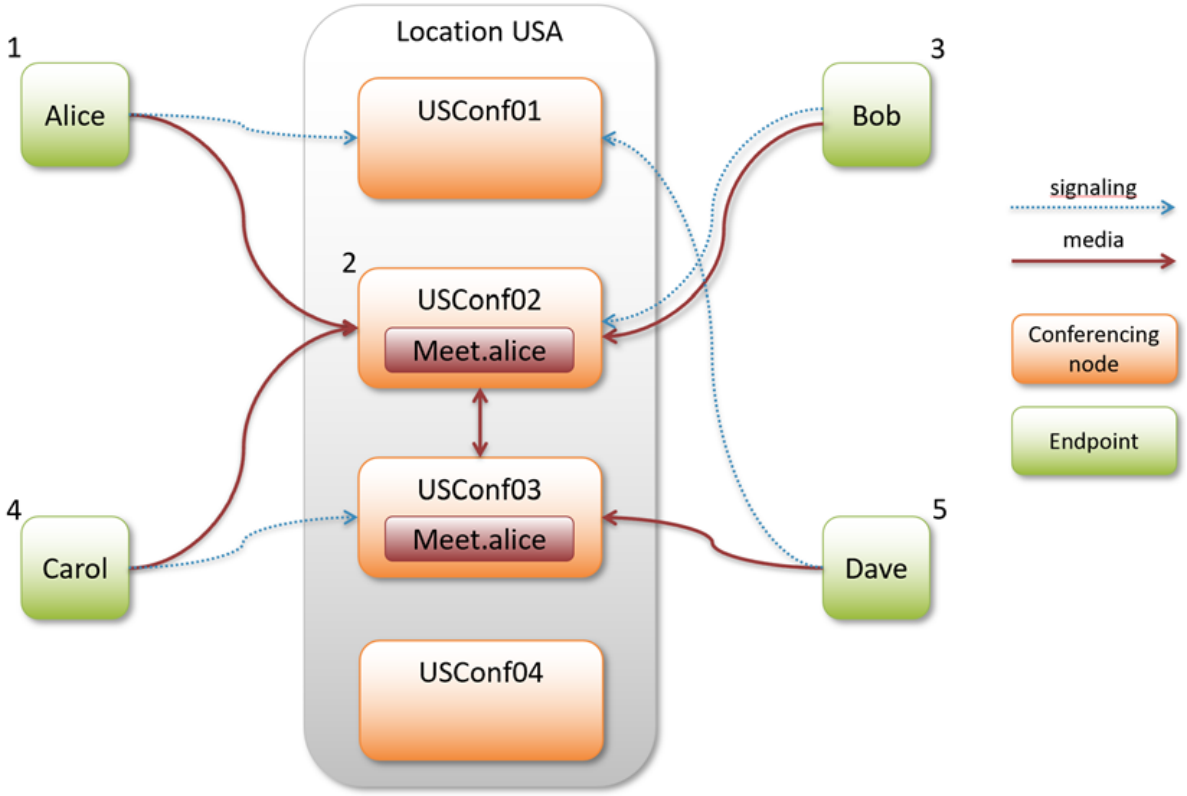
Based on these rules, here are some examples of how signaling and media are routed in different situations:

- Media and signaling flows in a locally distributed conference
- Media and signaling flows in a globally distributed conference

- Media and signaling flows when using Proxying Edge Nodes and Transcoding Conferencing Nodes

Media and signaling flows in a locally distributed conference

This example shows the signaling and media flows for a locally distributed conference (the conference is hosted on nodes all within the same location). In this example, all of the Conferencing Nodes involved are Transcoding Conferencing Nodes.



1. Alice dials **meet.alice@example.com** and is connected via DNS to Conferencing Node **USConf01**.
2. Pexip Infinity determines that another Conferencing Node within that location, **USConf02**, has the most available capacity and so sets up the conference on that node. Alice's signaling is handled by **USConf01** and the media is being sent directly to **USConf02**.
3. Bob dials **meet.alice@example.com**. He is connected to **USConf02**. The conference is already running on that Conferencing Node, so **USConf02** handles both the media and the signaling for Bob.
4. Carol dials **meet.alice@example.com**. She is connected to **USConf03**. Her signaling is handled by **USConf03** but the media is routed directly to **USConf02**, where the conference is running.
5. Dave dials **meet.alice@example.com**. He is connected to **USConf01**. Pexip Infinity determines that **USConf02** is now out of capacity, and **USConf03** has the most available capacity. Dave's media is routed to **USConf03**, but **USConf01** continues to handle his signaling.

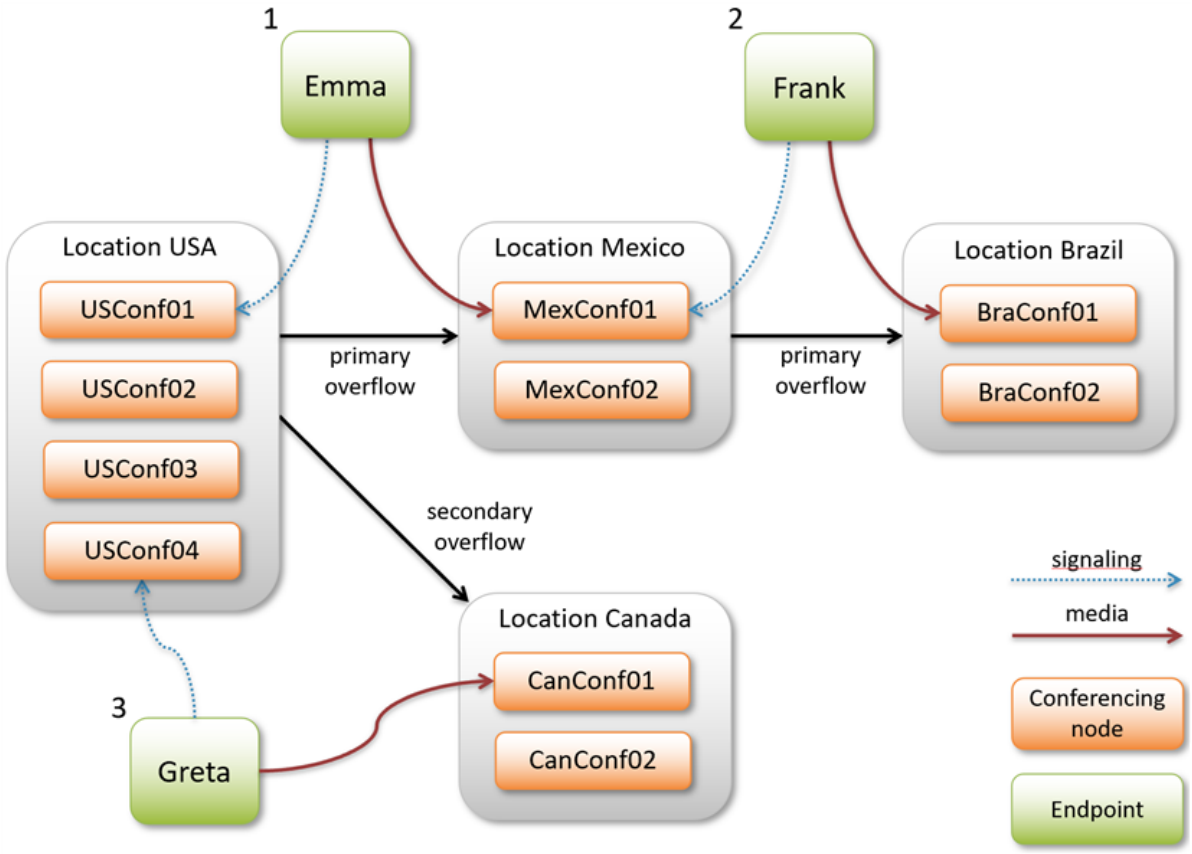
Media and signaling flows in a globally distributed conference

This example builds on the previous example described above. It assumes that the conference **meet.alice@example.com** is in progress and currently has all of its media and signaling in a locally distributed conference in location **USA**. Again, in this example, all of the Conferencing Nodes involved are Transcoding Conferencing Nodes.

This deployment has locations configured for USA, Mexico, Canada and Brazil, and the USA and Mexico locations are configured with transcoding, primary and secondary overflow locations as follows:

Location	Transcoding location	Primary overflow location	Secondary overflow location
USA	***This location*** i.e. USA	Mexico	Canada
Mexico	***This location*** i.e. Mexico	Brazil	-

This example now shows what could happen when three new participants, two from the USA and one from Mexico, join the conference.



1. E
   mma, who lives in USA, dials **meet.alice@example.com** and is connected to **USConf01** in location **USA**.

   Assume that all four Conferencing Nodes in location **USA** have reached their capacity, so Emma's media is routed directly to **MexConf01** in location **Mexico** (because **Mexico** is the primary overflow location for the **USA** location). **USConf01** continues to handle her signaling.
2. Frank, who lives in Mexico, dials **meet.alice@example.com** and is connected to **MexConf01** in location **Mexico**.

   Assume that all of the Conferencing Nodes in location **Mexico** have reached their capacity, so Frank's media is routed directly to **BraConf01** in location **Brazil** (because **Brazil** is the primary overflow location for the **Mexico** location to which Frank is connected). **MexConf01** continues to handle his signaling.
3. Greta, who lives in USA, dials **meet.alice@example.com** and is connected to **USConf04** in location **USA**.

   Assume that all of the Conferencing Nodes in location **USA** have reached their capacity, as have all of the nodes in the USA location's primary overflow location of **Mexico**, so Greta's media is routed directly to **CanConf01** in location **Canada** (because **Canada** is the secondary overflow location for the **USA** location to which Greta is connected; it is irrelevant if location **Brazil** has spare capacity as **Brazil** is not an overflow location for endpoints connected to the **USA** location). **USConf03** continues to handle her signaling.

Note that this example deliberately describes an extreme scenario as its purpose is to demonstrate how location overflow logic is applied.

Conference escalation from locally to globally distributed is handled automatically by Pexip Infinity and is seamless to conference participants.

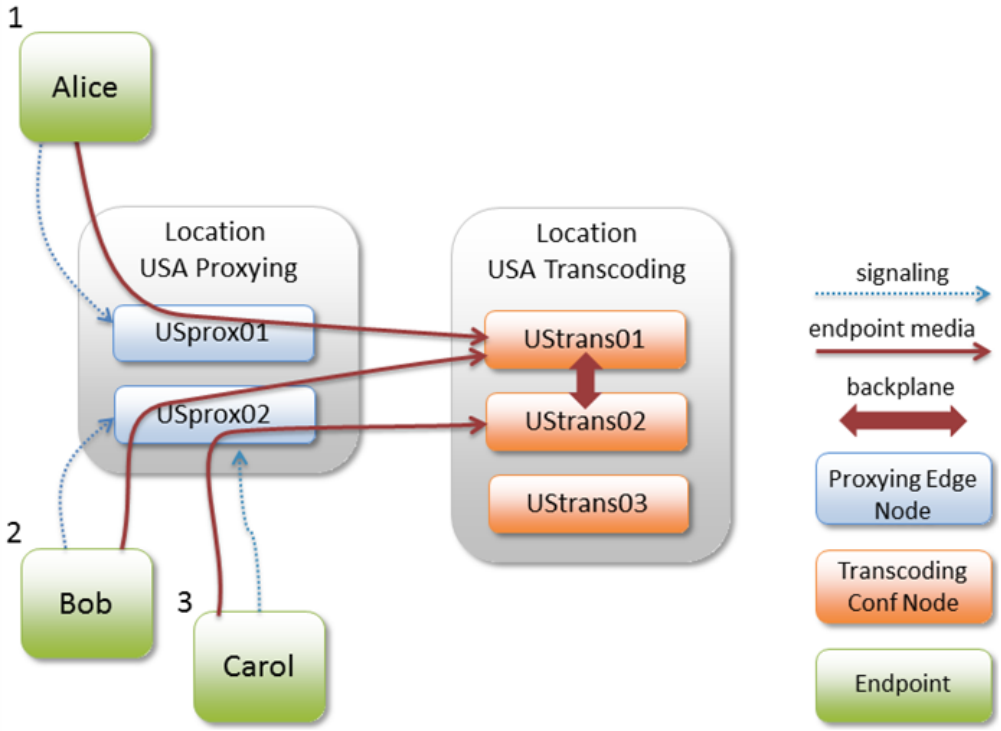
Media and signaling flows when using Proxying Edge Nodes and Transcoding Conferencing Nodes

These examples show the signaling and media flows when you have Proxying Edge Nodes in your deployment. Here, we assume that all callers located in the USA are routed via DNS to a Proxying Edge Node in the **USA Proxying** location, and that location is configured

with a **Transcoding location** of **USA Transcoding** as follows:

Location	Transcoding location	Primary overflow location	Secondary overflow location
USA Proxying	USA Transcoding	Mexico Transcoding	-
Canada Proxying	Canada Transcoding	Mexico Transcoding	-

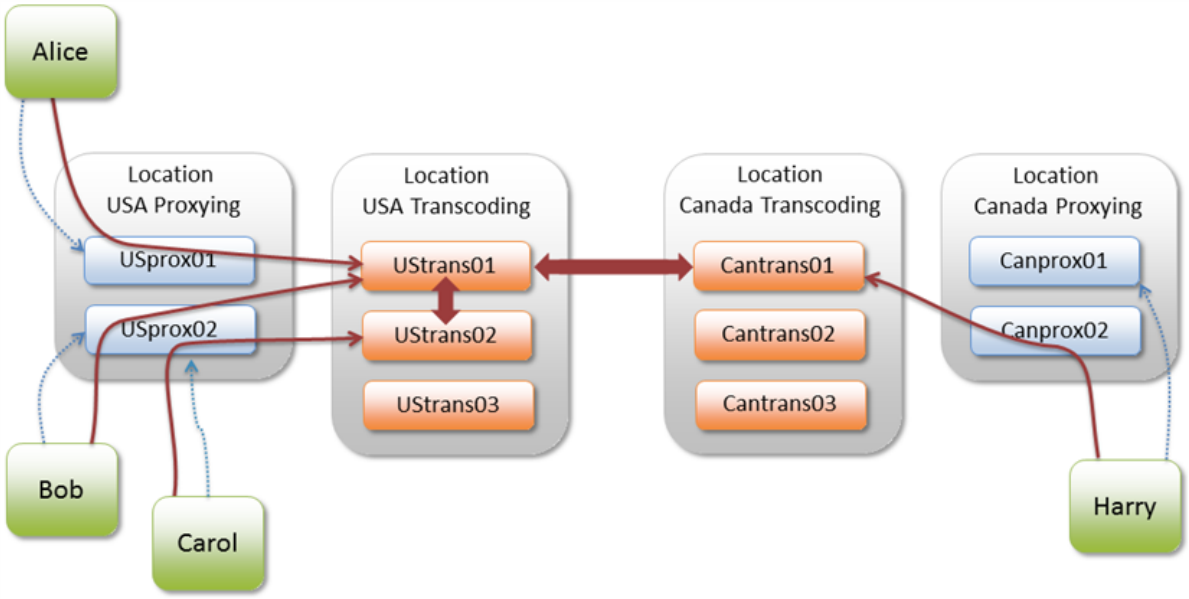
1. Alice dials



   meet.alice@example.com and is connected via DNS to Proxying Edge Node **USprox01**. Her signaling remains routed to **USprox01** throughout the call. **USprox01** is also selected as the proxying node to handle Alice's media, from where it needs to be proxied to a transcoding resource. Pexip Infinity chooses the node with the most available capacity in the transcoding location of **USA Transcoding** — in this case **UStrans01**. Therefore **USprox01** forwards Alice's media onto **UStrans01** which hosts the conference meet.alice.
2. Bob dials **meet.alice@example.com** and he is connected to **USprox02** for signaling purposes. **USprox02** is also chosen to handle his media (as it is the proxying node in that location with the most available capacity) and it needs to proxy his media onto a transcoding resource in the transcoding location. The meet.alice conference is already running on **UStrans01** and has spare capacity, so **USprox02** forwards Bob's media onto **UStrans01**.
3. Carol dials **meet.alice@example.com** and she is connected to **USprox02**. Her media is also allocated to **USprox02**. However, this time **UStrans01** has no spare capacity, so Pexip Infinity chooses another node in the transcoding location as the forwarding destination — **UStrans02** in this case. The conference is now locally distributed within the **USA Transcoding** location and a local backplane is established between **UStrans01** and **UStrans02**.

As before, if the nodes in the transcoding location reach their capacity, the media would be forwarded instead onto any overflow locations that are configured against the location handling the signaling, which is **Mexico Transcoding** in this example.

Escalated to a globally distributed conference



We can now extend this example to show how the conference could become globally distributed when a participant in Canada joins. Here, the **Canada Proxying** location contains Proxying Edge Nodes and it is configured with a transcoding location of **Canada Transcoding**.

Harry dials **meet.alice@example.com** and he is routed via DNS to his local Proxying Edge Node **Canprox01**. His media proxying is allocated to **Canprox02** from where it is forwarded to **Cantrans01** in the transcoding location. The meet.alice conference is now globally distributed and therefore a geo backplane is set up between one of the nodes in location **USA Transcoding** (**UStrans01** in this example) and **Cantrans01**.

Implementing a dial plan

When you deploy the Pexip Infinity platform in your network, you must consider your dial plan and ensure that calls are routed appropriately. This includes:

- aliases that are associated with a conferencing service (Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service)
- person-to-person calls handled by the Infinity Gateway
- calls routed via the Infinity Gateway into an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

If your environment includes a PSTN gateway or uses an ITSP (Internet telephony service provider), consider the potential for toll fraud if you have Call Routing Rules that can route calls to the PSTN gateway or ITSP, or if you allow conference participants to dial out to other participants via the PSTN gateway or ITSP. See PSTN gateways and toll fraud for more information.

Service precedence

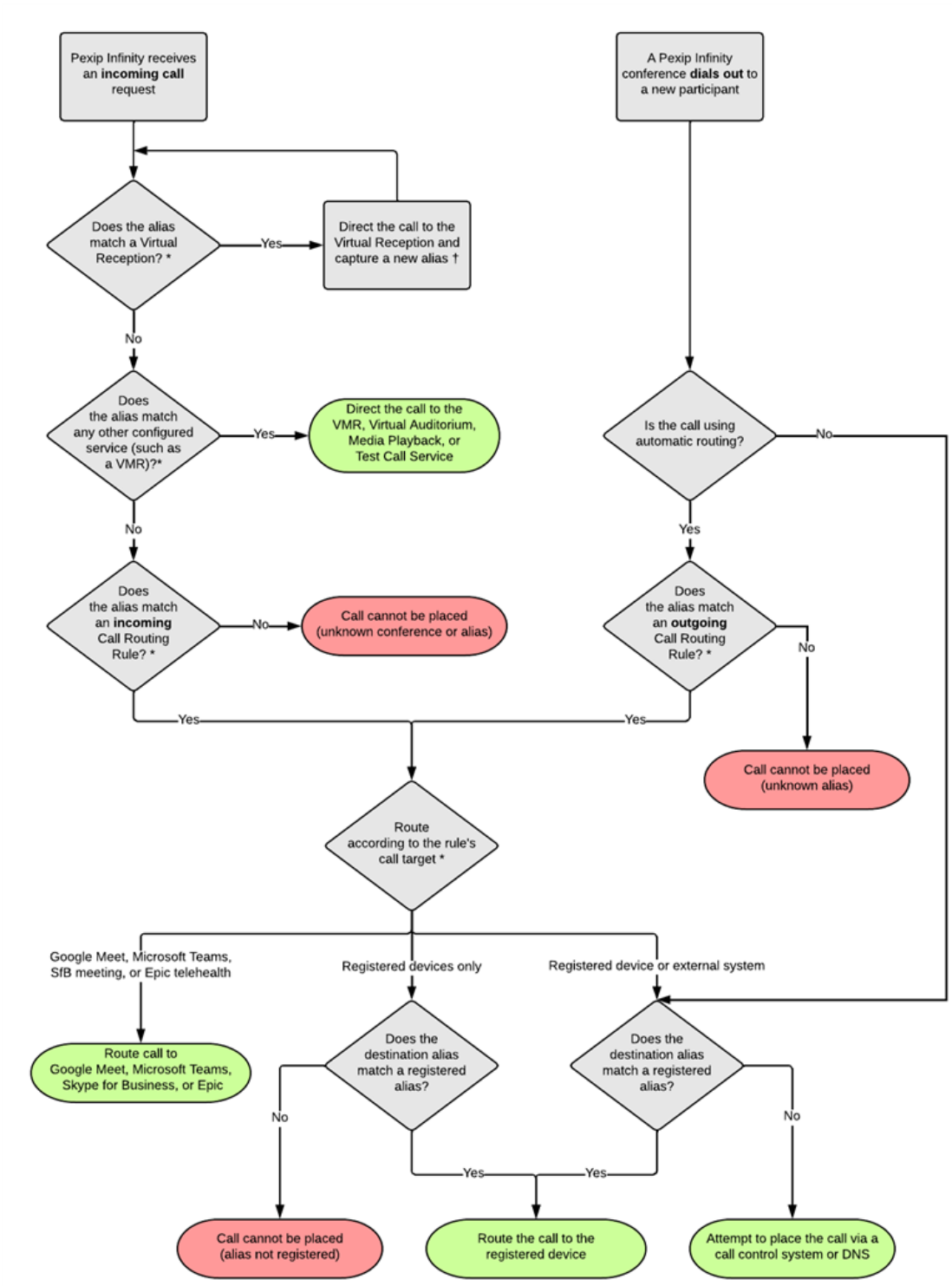
Incoming calls received by Pexip Infinity are routed as follows:

1. Pexip Infinity receives an incoming call via one of its Conferencing Nodes.
2. It checks whether the destination alias belongs to a Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service; if so, it directs the call to that service.
3. If the alias does not belong to any of the above services, Pexip Infinity checks the Call Routing Rules to see if the alias matches any rules specified there for incoming calls. If so, it places an Infinity Gateway call to the destination alias according to the rule's call target settings (which protocol, location and call control system to use, whether to route to registered devices only, etc).

This means that if an alias matches both a Virtual Meeting Room and a Call Routing Rule, the former will always take precedence and the call will be routed to the Virtual Meeting Room. You must therefore ensure that any regular expressions used in a Call Routing Rule

do not unintentionally overlap with any aliases used by a Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service.

Pexip Infinity's call routing logic i.e. the order in which calls are processed, is shown in the following diagram:



* This information can alternatively be provided by using external or local policy.
† If it is a non-Regular type of Virtual Reception (i.e. Google Meet, Microsoft Teams or SfB) the Virtual Reception captures the meeting code for the externally-hosted conference, which then needs to be matched by a suitable Call Routing Rule.

For more information about incoming and outgoing Call Routing Rules and how to specify where calls are routed, see Configuring Call Routing Rules.

Using a standard format for aliases

We recommend that you use a standard format for all Virtual Meeting Room and Virtual Auditorium aliases within your Pexip Infinity deployment. A simple way to do this would be to have a set prefix such as **meet.**, followed by a user name, followed by your domain. For example:

- meet.alice.smith@example.com (for Alice's VMR)
- meet.bob.jones@example.com (for Bob's VMR)
- meet.sales@example.com (for the sales team's Virtual Auditorium)
- meet.reception@example.com (for the Virtual Reception)

Using this standard format will then make it simple to configure your call management system to route calls matching the format to the appropriate Conferencing Nodes.

Routing calls to the local Conferencing Node

When your call management system receives a call to a Virtual Meeting Room, Virtual Auditorium or Virtual Reception alias, it must then decide where to route the call.

A single conference can be hosted across one, two or more Conferencing Nodes with no difference in conference experience from the users' perspective. We recommend that in the first instance, calls are routed to a Conferencing Node that is geographically nearest to the endpoint that placed the call. This reduces the WAN bandwidth used for the conference in two ways: firstly, endpoints are only making local calls; and secondly, if the same conference is hosted in more than one location, only one set of media has to be sent between those locations.

We recommend that you set up rules on your call management system so that calls originating from a particular location (defined by zone, subzone or IP address range) and placed to a Pexip Infinity alias are routed to a specific Conferencing Node. If you are deploying Proxying Edge Nodes, we recommend that you ensure that calls are routed only to those Proxying Edge Nodes and not to Transcoding Conferencing Nodes. To build on the previous example, we have configured subzones on our call management system which group together endpoints in a particular physical location, and we have then set up the following call routing rules:

Source	Alias	Destination zone
Oslo subzone	meet.*@example.com.*	Norway Conferencing Node
New York subzone	meet.*@example.com.*	Americas Conferencing Node
Boston subzone	meet.*@example.com.*	Americas Conferencing Node

In this case, users in Oslo, New York and Boston could all call **meet.sales@example.com** and be routed to the same conference.

See Regular expression (regex) reference for information about writing regular expressions.

Further information

For further information about how to configure your specific call management system to work with Pexip Infinity, see the following documentation:

- Pexip Infinity and Microsoft Skype for Business Deployment Guide
- Pexip Infinity and Cisco VCS Deployment Guide
- Pexip Infinity and Cisco Unified Communications Manager Deployment Guide
- Pexip Infinity and Polycom DMA Deployment Guide

DNS record examples

This section describes the DNS records required to allow endpoints and clients to access Pexip Infinity services.

You can use the tool at http://dns.pexip.com to lookup and check SRV records for a domain.

Enabling endpoints to discover Conferencing Nodes

DNS A-records are required for endpoints and clients to discover Pexip Infinity Conferencing Nodes. Separate, additional records that resolve to these DNS A-records are required for SIP and H.323 endpoints, Pexip apps and federated SfB environments respectively.

The examples below show DNS A and SRV records for the domain **vc.example.com** to allow:

- SIP and H.323 endpoint registration messages to be routed to Pexip Infinity
- calls from SIP and H.323 endpoints to be routed to Pexip Infinity
- calls from remote Skype for Business environments to Pexip Infinity
- Poly/Polycom endpoints to register to One-Touch Join
- Pexip app for Windowss to register to Pexip Infinity
- Pexip apps to make calls to Pexip Infinity.

If you have multiple Conferencing Nodes, you must set up DNS A and SRV records for each node.

Host DNS A-records

This example assumes there are 2 Conferencing Nodes, thus 2 DNS A-records are required.

Hostname	Host IP address
px01.vc.example.com.	198.51.100.40
px02.vc.example.com.	198.51.100.41

In your actual deployment, both the **Hostname** and **Host IP address** should be changed to use the real names and addresses of your Conferencing Nodes.

DNS SRV records for SIP and H.323 endpoints

The following DNS SRV records are required for H.323 and SIP services. For each service there should be one record per host (Conferencing Node). Therefore, in this example, there are 2 records per service — one for each target host — **px01.vc.example.com** and **px02.vc.example.com**.

Name	Service	Protocol	Priority	Weight	Port	Target host
vc.example.com.	h323cs	tcp	10	10	1720	px01.vc.example.com.
vc.example.com.	h323cs	tcp	10	10	1720	px02.vc.example.com.
vc.example.com.	h323ls	udp	10	10	1719	px01.vc.example.com.
vc.example.com.	h323ls	udp	10	10	1719	px02.vc.example.com.
vc.example.com.	h323rs	udp	10	10	1719	px01.vc.example.com.
vc.example.com.	h323rs	udp	10	10	1719	px02.vc.example.com.
vc.example.com.	sip	tcp	10	10	5060	px01.vc.example.com.
vc.example.com.	sip	tcp	10	10	5060	px02.vc.example.com.
vc.example.com.	sips	tcp	10	10	5061	px01.vc.example.com.
vc.example.com.	sips	tcp	10	10	5061	px02.vc.example.com.
vc.example.com.	sip	udp *	10	10	5060	px01.vc.example.com.
vc.example.com.	sip	udp *	10	10	5060	px02.vc.example.com.

* Only required if you intend to enable SIP over UDP.

In your actual deployment, both the **Name** and the **Target host** should be changed to use the real domain name and host names of your Conferencing Nodes.

If you have a mixture of public-facing and privately-addressed Conferencing Nodes, ensure that your public-facing SRV records refer to your public-facing Conferencing Nodes (A-records), and vice versa for your local DNS records and your privately-addressed Conferencing Nodes.

In these examples, the DNS records would be:

```
_h323cs._tcp.vc.example.com. 86400 IN SRV 10 10 1720 px01.vc.example.com.
_h323cs._tcp.vc.example.com. 86400 IN SRV 10 10 1720 px02.vc.example.com.

_h323ls._udp.vc.example.com. 86400 IN SRV 10 10 1719 px01.vc.example.com.
_h323ls._udp.vc.example.com. 86400 IN SRV 10 10 1719 px02.vc.example.com.

_h323rs._udp.vc.example.com. 86400 IN SRV 10 10 1719 px01.vc.example.com.
_h323rs._udp.vc.example.com. 86400 IN SRV 10 10 1719 px02.vc.example.com.

_sip._tcp.vc.example.com.    86400 IN SRV 10 10 5060 px01.vc.example.com.
_sip._tcp.vc.example.com.    86400 IN SRV 10 10 5060 px02.vc.example.com.

_sips._tcp.vc.example.com.   86400 IN SRV 10 10 5061 px01.vc.example.com.
_sips._tcp.vc.example.com.   86400 IN SRV 10 10 5061 px02.vc.example.com.

_sip._udp.vc.example.com.    86400 IN SRV 10 10 5060 px01.vc.example.com.
_sip._udp.vc.example.com.    86400 IN SRV 10 10 5060 px02.vc.example.com.

px01.vc.example.com.         86400 IN A 198.51.100.40
px02.vc.example.com.         86400 IN A 198.51.100.41
```

If the ability to implement intelligent DNS (GeoDNS, GSLB, etc.) exists, then different weights and priorities could be set on the SRV records for different locations. For example, if there is both a European and a North American site, each with separate DNS servers, the European one could return European Pexip nodes at a higher priority than the North American nodes, and vice versa.

DNS pool names

In some scenarios, including Teams Connector integrations, SIP proxies from third-party infrastructure / SBCs, or in SfB integrations, you may need to use a generic DNS "pool name" for handling a pool of Conferencing Nodes.

This is typically a set of round-robin DNS A records, with a common name such as **px.vc.example.com**, that refer to a pool of Conferencing Node resources, for example:

```
px.vc.example.com.   86400 IN A 198.51.100.40
px.vc.example.com.   86400 IN A 198.51.100.41
px.vc.example.com.   86400 IN A 198.51.100.42
```

Note that these A records defined for the **px.vc.example.com** pool are specified in addition to the "standard" A-records that will exist for each Conferencing Node based on their individual hostnames and resolve to the same IP addresses.

Enabling Poly/Polycom endpoints to register to One-Touch Join

If you have a One-Touch Join deployment that includes Poly endpoints in a location with more than one Conferencing Node, you should spread the Poly endpoint registrations across all nodes in the location to maximize performance and provide redundancy. To achieve this, we recommend that all Poly endpoints in a location register to a single FQDN which uses round-robin DNS to resolve to each Conferencing Node in turn. This requires you to set up appropriate DNS records for all Conferencing Nodes in the location, and ensure that your DNS server is configured to round-robin between these records.

These records are specific to One-Touch Join, and are required in addition to the "standard" A-records that will exist for each Conferencing Node (which are based on their individual hostnames and resolve to the same IP addresses).

Note that all Conferencing Nodes in the location that is being used for One-Touch Join must be included in the DNS records.

Example

In this example, we have two locations being used for One-Touch Join that contain Poly endpoints. Each location has three Conferencing Nodes. We set up the following DNS A-records:

```
otj01.vc.example.com.        86400 IN A 10.44.0.10
otj01.vc.example.com.        86400 IN A 10.44.0.11
otj01.vc.example.com.        86400 IN A 10.44.0.12
otj02.vc.example.com.        86400 IN A 10.44.0.20
otj02.vc.example.com.        86400 IN A 10.44.0.21
otj02.vc.example.com.        86400 IN A 10.44.0.22
```

All Poly endpoints in the first location are configured with **otj01.vc.example.com** as their Exchange Server, and those in the second location are configured with **otj02.vc.example.com**.

Enabling access from the Pexip app for Windows and the Pexip mobile apps

You must set up DNS records so that the Pexip app knows which host to contact when placing calls or registering to Pexip Infinity.

The host will typically be a public-facing Conferencing Node (for on-premises deployments where your Transcoding Conferencing Nodes are located within a private network we recommend that you deploy public-facing Proxying Edge Nodes).

To enable access from the Pexip app, each domain used in aliases in your deployment must either have a DNS SRV record for **_pexapp._tcp.<domain>**, or resolve directly to the IP address of a public-facing Conferencing Node.

The SRV records for **_pexapp._tcp.<domain>** should always:

- point to an FQDN which **must** be valid for the TLS certificate on the target Conferencing Nodes
- point to an A/AAAA record
- reference port 443 on the host.

Example

Assume that the following **_pexapp._tcp.vc.example.com** DNS SRV records have been created:

```
_pexapp._tcp.vc.example.com. 86400 IN SRV 10 100 443 px01.vc.example.com.
_pexapp._tcp.vc.example.com. 86400 IN SRV 20 100 443 px02.vc.example.com.
```

These point to the DNS A-records **px01.vc.example.com**, port 443 (HTTPS), with a priority of 10 and a weight of 100, and **px02.vc.example.com**, port 443, with a relatively lower priority of 20 and a weight of 100.

This tells the Pexip app to initially send its HTTP requests to host **px01.vc.example.com** (our primary node) on TCP port 443. The Pexip app will also try to use host **px02.vc.example.com** (our fallback node) if it cannot contact px01.

Enabling access from the Poly Meeting Control App

You must set up DNS records so that the Poly Meeting Control App knows which host to contact when placing calls or registering to Pexip Infinity.

The host will typically be a public-facing Conferencing Node (for on-premises deployments where your Transcoding Conferencing Nodes are located within a private network we recommend that you deploy public-facing Proxying Edge Nodes).

To enable access from the Poly Meeting Control App, each domain used in aliases in your deployment must either have a DNS SRV record for **_pexapp._tcp.<domain>**, or resolve directly to the IP address of a public-facing Conferencing Node.

The SRV records for **_pexapp._tcp.<domain>** should always:

- point to an FQDN which **must** be valid for the TLS certificate on the target Conferencing Nodes
- point to an A/AAAA record
- reference port 443 on the host.

Example

Assume that the following **_pexapp._tcp.vc.example.com** DNS SRV records have been created:

```
_pexapp._tcp.vc.example.com. 86400 IN SRV 10 100 443 px01.vc.example.com.
_pexapp._tcp.vc.example.com. 86400 IN SRV 20 100 443 px02.vc.example.com.
```

These point to the DNS A-records **px01.vc.example.com**, port 443 (HTTPS), with a priority of 10 and a weight of 100, and **px02.vc.example.com**, port 443, with a relatively lower priority of 20 and a weight of 100.

This tells the Poly Meeting Control App to initially send its HTTP requests to host **px01.vc.example.com** (our primary node) on TCP port 443. The Poly Meeting Control App will also try to use host **px02.vc.example.com** (our fallback node) if it cannot contact px01.

Enabling direct federation with remote Skype for Business environments

In public DMZ / hybrid deployments with remote Skype for Business environments, to ensure that calls from those remote SfB environments are routed to your Conferencing Nodes, a DNS SRV record in the format **_sipfederationtls._tcp.<domain>** must be created.

Example

Assume that the following **_sipfederationtls._tcp.vc.example.com** DNS SRV and associated A-records have been created:

```
_sipfederationtls._tcp.vc.example.com. 86400 IN SRV 1 100 5061 px.vc.example.com.
px.vc.example.com.                     86400 IN A 198.51.100.40
px.vc.example.com.                     86400 IN A 198.51.100.41
```

The SRV record points to the DNS A-record **px.vc.example.com**, port 5061, with a priority of 1 and a weight of 100. In other words, it tells Skype for Business to send its TLS requests to **px.vc.example.com** on TCP port 5061.

We then have 2 round-robin DNS A-records for the **px.vc.example.com** pool hostname that resolves px.vc.example.com to the IP addresses of our 2 Conferencing Nodes (198.51.100.40 and 198.51.100.41).

Even if you only intend initially to use a single Conferencing Node to receive incoming SfB calls, this pool-based approach allows you to easily add more nodes in the future. (In your actual deployment, the hostnames (including the pool hostname) and host IP addresses should be changed to use the real hostnames and IP addresses of your Conferencing Nodes.)

Note that these A-records specified for the **px.vc.example.com** pool are required in addition to the "standard" A-records that will exist for each Conferencing Node based on their individual hostnames and resolve to the same IP addresses, as shown in the Host DNS A records section above.

The domain name used in the **_sipfederationtls._tcp.<domain>** SRV record has to match the domain in the corresponding A-record. This is required due to the trust model for SfB federation. For example:

- An SRV record such as _sipfederationtls._tcp.**vc.example.com** must have a corresponding A-record with the same domain, such as px.**vc.example.com**.
- You cannot, for example, configure the _sipfederationtls._tcp.vc.example.com SRV record to point to px.**video.example.com** or px01.**otherdomain.com**.

Testing and next steps after initial installation

After you have completed your installation and initial configuration of Pexip Infinity, you can make a test call to check that your system is working. You can also extend your deployment by integrating it with other call control or third-party systems, or by customizing the user experience. You should also consider how to let your users know about their new video conferencing service.

Making a test call

When you have deployed a Conferencing Node and configured a Virtual Meeting Room and an alias, you can make a test call to check that your system is working.

An easy way to do this is by using the Pexip web app to dial the alias of one of the Virtual Meeting Rooms you've already created.

1. Open a browser (we recommend Chrome or Edge) and type in the IP address (or FQDN, if you've set it up already) of one of the Conferencing Nodes.
   If your browser displays a security warning, this means that it does not trust the Conferencing Node's certificate. This could be because you have not replaced the node's default self-signed certificate, or you have used your own private certificates that have not been signed by an external Certificate Authority.
2. In the **Video address** field, enter the alias of the Virtual Meeting Room you are using for testing.
3. Ensure that you have selected the camera and microphone you wish to use, and they are working correctly:
   - You should see your own image in the video window.
   - The microphone icon shows a green bar to indicate the level of audio being detected. To join without your audio, select the microphone icon; this will change to to indicate that your microphone is off.
4. Select **Join**.
5. From another device, join the conference in the same way.

The two participants should be able to see and hear each other, and share content.

Integrating with a call control system

To integrate Pexip Infinity with your call control system, you must configure a trunk or neighbor zone towards each of the Conferencing Nodes.

For further information about how to configure your specific call management system to work with Pexip Infinity, see the following documentation:

- Pexip Infinity and Microsoft Skype for Business Deployment Guide
- Pexip Infinity and Cisco VCS Deployment Guide
- Pexip Infinity and Cisco Unified Communications Manager Deployment Guide
- Pexip Infinity and Polycom DMA Deployment Guide

Configuring the Pexip Infinity Distributed Gateway

The Pexip Infinity Distributed Gateway ("Infinity Gateway") enables endpoints to make calls to other endpoint devices or systems. This includes calls between devices that use different protocols and media formats, such as SIP and H.323 systems, Skype for Business clients (MS-SIP), and Pexip apps (WebRTC). It also enables you to route calls from VTCs and standards-based endpoints into an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

Registering devices directly to the Pexip Infinity platform

SIP and H.323 endpoints, and some Pexip apps can register directly to Pexip Infinity Conferencing Nodes. This allows Pexip Infinity to route outbound calls to those registered devices without having to go via a SIP proxy or H.323 gatekeeper, or rely on DNS.

Customizing the user experience

You can easily apply your own corporate branding to the Pexip Infinity platform, and produce a personalized user experience for all of your Pexip Infinity services.

Administering Pexip Infinity

For information about how to administer Pexip Infinity, see:

Using the Pexip Infinity Administrator interface 91
Pexip Infinity system configuration 94
Pexip Infinity platform configuration 106
Conferencing Node configuration 153
Pexip Infinity conference types 173
Pexip Infinity conference settings 220
Controlling active conferences 270
Customizing with themes 293
Integrating Google Meet with Pexip Infinity 334
Integrating Microsoft Teams with Pexip Infinity 357
Integrating Epic telehealth with Pexip Infinity 358
Integrating Pexip Infinity with authentication and provisioning services 380
Pexip Infinity maintenance tasks 472
Best practices 500

Using the Pexip Infinity Administrator interface

After you have run the installation wizard, all further configuration should be done using the Pexip Infinity Administrator interface.

This topic covers:

- Accessing the Pexip Infinity Administrator interface
- Page navigation, search and filtering
- Setting the start page
- Setting the session timeout / logging out
- Changing the display language
- Timezones
- Platform information
- Getting help and support

Accessing the Pexip Infinity Administrator interface

The Pexip Infinity Administrator interface is hosted on the Management Node. To access this:

1. Open a web browser and type in the IP address or DNS name that you assigned to the Management Node using the installation wizard (you may need to wait a minute or so after installation is complete before you can access the Administrator interface).
2. Until you have uploaded appropriate TLS certificates to the Management Node, your browser may present you with a warning that the website's security certificate is not trusted. You should proceed, but upload appropriate TLS certificates to the Management Node (and Conferencing Nodes, when they have been created) as soon as possible.

   The **Pexip Infinity Conferencing Platform** login page will appear.
3. Log in using the web administration username and password you set using the installation wizard.

You are now ready to begin configuring the Pexip Infinity platform and deploying Conferencing Nodes.

As a first step, we strongly recommend that you configure at least 2 additional NTP servers or NTP server pools to ensure that log entries from all nodes are properly synchronized.

It may take some time for any configuration changes to take effect across the Conferencing Nodes. In typical deployments, configuration replication is performed approximately once per minute. However, in very large deployments (more than 60 Conferencing Nodes), configuration replication intervals are extended, and it may take longer for configuration changes to be applied to all Conferencing Nodes (the administrator log shows when each node has been updated).

Page navigation, search and filtering

You can view the status of the Pexip Infinity platform, and configure the system's operation, via a series of webpages accessed through the navigation menus (**Status**, **History & Logs**, **System** etc.) at the top of each page of the Administrator interface.

If you know the name of the item you want to configure, you can use the search box (above the menu bar) to search for configuration settings based on their associated menu, field name or field label text.

For example, if you search for "bandwidth" the search results will list all of the settings where bandwidth can be configured, such as the maximum call bandwidth settings at the global level, plus the individual service-level settings for each configured VMR, Call Routing Rule, Media Playback Service and so on. From the list you can then select the relevant item in the **Add or instance** or **Matching field** column to go to straight to the setting for that item.

You can also apply filters to many of the status pages within the Administrator interface by adding appropriate query parameters to the page URL. For example, when viewing participant history you could filter by a specific media node:

https://<manageraddress>/admin/conferencinghistory/participant/?media_node=<IP address>

In most cases you can filter by the same fields as stated in the `filtering` section of the relevant schema which, for participant history for example, is available at:

https://<manageraddress>/api/admin/history/v1/participant/schema/?format=json

Setting the start page

By default, you are taken to the Live view page after logging in, but you can change this to a different default page via the **Global Settings** page (**Platform > Global Settings > Management Web Interface Configuration**) and configuring the **Management start page** setting.

Setting the session timeout / logging out

By default, Administrator interface web sessions time out after 30 minutes of inactivity. You can modify the **Management web interface session timeout** setting via the **Global Settings** page (**Platform > Global Settings > Management Web Interface Configuration**). You can also disable timeouts completely by deselecting the **Enable management web interface session timeout** setting.

When logging out of the Administrator interface (select **Log out** from the top right of the page) you have the option to log out from all of your active sessions.

Changing the display language

The Administrator interface can be displayed in the following languages:

- English (default)
- Simplified Chinese
- Traditional Chinese
- Korean
- Japanese
- Russian

To change the language, you must edit the language preferences within your browser settings and set it to use one of the supported languages. You must then refresh your browser for the changes to take effect. English is displayed if your browser's selected language is not supported.

The following articles provide help in changing your browser's language:

- Chrome: https://support.google.com/chrome/answer/173424
- Firefox: https://support.mozilla.org/en-US/kb/use-firefox-another-language
- Edge: https://support.microsoft.com/en-us/microsoft-edge/use-microsoft-edge-in-another-language-4da8b5e0-11ce-7ea4-81d7-4e332eec551f

Timezones

The Administrator interface displays all times (except for those in the Administrator log and Support log, and usage graphs) in your local time. This is obtained from the timezone configured on the computer you are using to access the Administrator interface. So for example, if you are sitting in an office in London and you log in to a Management Node that is located in New York, call start and end times will be shown in London time.

The timezone being used is shown in brackets after the time, in the format **2024-02-25 22:00:12 (GMT)**.

Logs (and usage graphs, which are based on the information in the logs) are always shown in UTC because of the distributed nature of the Pexip Infinity platform. Logs use the format **2024-02-24T23:16:33.019+00:00**.

Note that the VMs hosting the Management Node and Conferencing Nodes use the UTC timezone. Do not attempt to change the timezone on these systems.

Platform information

From any page, clicking on the **About** link at the top right of the page displays information about the Pexip Infinity platform and software.

Information shown includes:

- Software version and build data
- Free temporary disk space
- FIPS compliance mode status
- Link to the API schema
- OAuth2 endpoints
- Links to patents and third-party software

Getting help and support

There are two types of help integrated into the Pexip Infinity Administrator interface:

- **Field-level help**: by default, most configuration fields show an explanation of the field underneath the input area. You can turn these explanations on or off by clicking on the ? icon on the bottom right of the page.
- **Context-sensitive help**: from any page, clicking on the **Help** link at the top right of the page opens a new browser window showing relevant information from the **Pexip Infinity Administrator Guide**. From there you can use the table of contents on the left-hand side of the help window to navigate and view the entire guide. There is also a search box at the top right of the browser window which you can use to search for individual words or phrases.

More deployment information and PDF downloads of all documentation are available on the Pexip Infinity technical documentation website.

If you cannot find the information you require, contact your Pexip authorized support representative. Technical support for software issues is available while under a valid support contract and running a Pexip Infinity version no more than 2 major software releases behind the current release. Software bug fixes will only be provided in either the current or the next major release of software.

For an in-depth understanding of the Pexip Infinity platform, we recommend that you attend an appropriate training course — for more information, visit the Pexip Academy.

Pexip Infinity system configuration

Configuring DNS servers

Pexip Infinity uses DNS to resolve the hostnames of external system components including NTP servers, syslog servers, SNMP servers and web proxies. It is also used for call routing purposes — SIP proxies, gatekeepers, external call control and conferencing systems and so on. The address of at least one DNS server must be added to your system.

To add, edit or delete the set of available DNS servers, go to **System > DNS Servers**.

The available options are:

Option	Description
Address	The IP address of the DNS server. The address must be an IPv4 address. (Note that IPv6 DNS resolution does not require an IPv6-addressed DNS server.)
Description	An optional description of the DNS server.

After configuring the DNS servers available to your system, you should assign appropriate DNS servers to each location (**Platform > Locations**). Each Conferencing Node in that location will then use those DNS servers.

You can also select the specific DNS servers to be used by the Management Node (**Platform > Management Node**).

While you can assign unlimited DNS servers to a location / Management Node, only three will be used.

Syncing with NTP servers

Pexip Infinity uses NTP servers to obtain accurate system time. This is necessary to ensure correct operation, including configuration replication and log timestamps.

All host servers **must** be synchronized with accurate time before you install the Management Node or Conferencing Nodes on them.

NTP **must** be enabled on the Management Node VM before you deploy any Conferencing Nodes (this is done during installation of the Management Node).

We strongly recommend that you configure at least three distinct NTP servers or NTP server pools on all your host servers and the Management Node itself. This ensures that log entries from all nodes are properly synchronized.

The VMs hosting the Management Node and Conferencing Nodes use the UTC timezone, and all logs are in UTC. Do not attempt to change the timezone on these systems. Note however that the administrator web interface uses your local time.

To add, edit or delete the set of available NTP servers, go to **System > NTP Servers**.

The available options are:

Option	Description
Address	The IP address or FQDN of the NTP server.
Description	An optional description of the NTP server.
Secure NTP key ID	This optional field allows you to configure a key ID which is used in conjunction with the **Secure NTP key** for authenticating access to a secure NTP server.
Secure NTP key	This optional field, used in conjunction with the **Secure NTP key ID**, allows configuration of a SHA1 key for authenticating access to a secure NTP server. Enter the plaintext key; this will be stored as a SHA1 hash.

After configuring the NTP servers available to your system, you should assign appropriate NTP servers to each location (**Platform > Locations**). Each Conferencing Node in that location will then use those NTP servers.

You can also select the specific NTP servers to be used by the Management Node (**Platform > Management Node**).

Using a web proxy

You can specify one or more web proxies to use for outbound web requests from the Management Node, and from the Conferencing Nodes in each system location.

When configured, web proxies are used to route all requests from One-Touch Join, Secure Scheduler for Exchange and Epic telehealth; requests from the Management Node to cloud service providers; and for analytics, incident reporting, and license activation requests.

When a web proxy is configured for the **Management Node**, it is used to route outbound web requests in the following situations:

- when sending usage statistics and incident reports,
- for license activation requests,
- when the Management Node communicates with the service configured for cloud bursting (i.e. Azure, AWS or GCP),
- for OIDC administrator authentication requests,
- (for deployments where One-Touch Join is enabled and is using OAuth) when the Management Node sends requests to the OAuth token endpoint,
- (for deployments where Secure Scheduler for Exchange is enabled) when the Management Node sends requests to the Exchange server / Office 365, and (when OAuth is enabled) the OAuth token endpoint. Note that you can bypass use of the proxy for individual Exchange deployments using the Disable web proxy option.

When a web proxy is configured for a **system location**:

- all Conferencing Nodes in that location will use that web proxy for incident reporting, Epic telehealth REST API requests, participant authentication requests to an OIDC Identity Provider, and
- if the system location is being used for One-Touch Join, by default all One-Touch Join-related outbound requests from Conferencing Nodes in that location will use the web proxy. These requests include:
  - connections to Cisco endpoints
    - You can disable use of the web proxy for Cisco endpoints on a per-Endpoint Group basis.
  - connections to Exchange servers
    - You can disable use of the web proxy when configuring an Exchange Integration (under **Advanced options**).
  - connections to Microsoft Graph
    - You can disable use of the web proxy when configuring a Graph Integration (under **Advanced options**).
  - connections to Google Workspace

To use a web proxy, you must first add its details via **System > Web Proxies**, and then add it to the configuration for the Management Node (**Platform > Management Node**) or the system location (**Platform > Locations**) you wish to use it for.

Option	Description
Name	The name used to refer to this proxy server.
Address	The IP address or FQDN of the proxy server.
Port	The IP port of the proxy server. Default: 8080.
Username Password	The username and password used when accessing the proxy server.

Next steps

You must now add the web proxy to the Management Node or to one or more System Locations.

Monitoring via SNMP

The Management Node and Conferencing Nodes can be monitored using SNMP, and they can also be configured to send SNMP traps to an SNMP Network Management System (NMS). This topic covers:

- SNMP MIB access
- Sending trap notifications to an NMS
- Configuring an SNMP NMS
- Configuring SNMP monitoring on the Management Node
- Configuring SNMP monitoring on Conferencing Nodes

SNMP MIB access

Pexip Infinity supports SNMPv2c (non-secure) and SNMPv3 (secure) access to the basic RFC 1213 MIB-II tree (1.3.6.1.2.1) with read-only functionality. This includes full or partial support for:

- system_mib
- interfaces
- snmp_mib
- at
- ip
- icmp
- udp
- tcp
- RFC 1514/RFC 2790 MIB-II host MIB

Sending trap notifications to an NMS

In addition to enabling read access to the MIB tree, you can also nominate an SNMP Network Management System (NMS) to receive trap notifications from the Management Node and from the Conferencing Nodes within a system location.

The supported SNMP traps are:

Trap	OID	Description
cold start	1.3.6.1.6.3.1.1.5.1	Emitted when the snmpd service running on the node starts or restarts (due to snmp being reconfigured and/or due to the Conferencing Node rebooting).
authentication failure	1.3.6.1.6.3.1.1.5.5	Generated, for example, when any attempt to query SNMP values is made using an incorrect community string.
warm start	1.3.6.1.6.3.1.1.5.2	Generated when any software component fails unexpectedly (and coincides with the generation of a Pexip Incident Report).

The SNMP support in Pexip Infinity is built on top of the popular **net-snmp** open source implementation and therefore inherits some of the same behaviors (for example, generating a coldstart rather than warmstart on reconfiguration). For this reason you may also see some **net-snmp**-specific traps, such as the **nsNotifyShutdown** trap (OID 1.3.6.1.4.1.8072.4.0.2) when the snmpd daemon shuts down.

Pexip Infinity does not currently support traps with SNMPv3. If traps are required, use SNMPv2c.

Configuring an SNMP NMS

If you want to send SNMP notifications from a Management Node or Conferencing Node to a SNMP NMS, you need to configure the details of the NMS. To do this:

1. Go to **System > SNMP NMSs** and select **Add SNMP Network Management System**.
2. Complete the required fields:

Option	Description
Name	The name used to refer to this NMS.

Option	Description
Description	An optional description of the NMS.
Address	The IP address or FQDN of the NMS.
Port	The SNMP port of the NMS. Default: 161.
SNMP trap community	The SNMP trap community name. Default: public

3. Select **Save**.

The NMS can now be selected when configuring the Management Node and system locations, as described below.

Configuring SNMP monitoring on the Management Node

To configure SNMP monitoring on the Management Node:

1. Go to **Platform > Management Node** and select the Management Node.
   You are taken to the **Edit Management Node** page.
2. To enable SNMP read access for the Management Node, configure the following settings:

Option	Description
SNMP mode	Configures the SNMP access mode for the selected node: ***Off***: SNMP is disabled. You cannot use SNMP to query the node for its status. ***SNMPv2c read-only***: enables insecure, read-only access. ***SNMPv3 read-only***: enables secure, read-only access, using the **authPriv** security level with SHA1 authentication and AES 128-bit encryption. When enabled, access is provided to the basic RFC 1213 MIB-II tree (1.3.6.1.2.1). Default: ***Off***.
SNMP community	The SNMP group to which this node belongs. This setting applies to SNMPv2c only. Default: public
SNMPv3 username	The node's SNMPv3 username, used to authenticate SNMPv3 requests.
SNMPv3 privacy password	The node's SNMPv3 privacy password used for encrypting messages between the node and the management station. AES encryption must be used; DES is not supported.
SNMPv3 authentication password	The node's SNMPv3 authentication password, used to authenticate the associated username. The SHA authentication protocol must be used; MD5 is not supported.
SNMP system contact	The contact details (for example, email address) of the person responsible for this particular node.
SNMP system location	A description of the node's location.

3. If you want to send SNMP traps from the Management Node to an NMS, select the NMS from the **SNMP NMS** drop-down menu.
   If you have not already added the NMS, you can do so now by clicking .
4. Select **Save**.

Configuring SNMP monitoring on Conferencing Nodes

To configure SNMP monitoring on a Conferencing Node:

1. Go to **Platform > Conferencing Nodes** and select the Conferencing Node.
   You are taken to the **Edit Conferencing Node** page.
2. To enable SNMP read access for the Conferencing Node, configure the following settings:

Option	Description
SNMP mode	Configures the SNMP access mode for the selected node: ***Off***: SNMP is disabled. You cannot use SNMP to query the node for its status. ***SNMPv2c read-only***: enables insecure, read-only access. ***SNMPv3 read-only***: enables secure, read-only access, using the **authPriv** security level with SHA1 authentication and AES 128-bit encryption. When enabled, access is provided to the basic RFC 1213 MIB-II tree (1.3.6.1.2.1). Default: ***Off***.
SNMP community	The SNMP group to which this node belongs. This setting applies to SNMPv2c only. Default: public
SNMPv3 username	The node's SNMPv3 username, used to authenticate SNMPv3 requests.
SNMPv3 privacy password	The node's SNMPv3 privacy password used for encrypting messages between the node and the management station. AES encryption must be used; DES is not supported.
SNMPv3 authentication password	The node's SNMPv3 authentication password, used to authenticate the associated username. The SHA authentication protocol must be used; MD5 is not supported.
SNMP system contact	The contact details (for example, email address) of the person responsible for this particular node.
SNMP system location	A description of the node's location.

3. If you want to send SNMP traps from the Conferencing Node to an SNMP NMS, you must ensure that the node's system location is configured with the NMS that will receive trap notifications from that node:
   a. Go to **Platform > Locations**.
   b. Select the Location to which the Conferencing Node belongs.
      You are taken to the **Edit System Location** page.
   c. From the **SNMP NMS** drop-down menu, select the NMS to which traps will be sent. Note that this NMS applies to all Conferencing Nodes in this location.
      If you have not already added the SNMP NMS, you can do so now by clicking .
4. Select **Save**.

Using a syslog server

The Management Node acts as a syslog server, collating the logs from itself and each Conferencing Node to produce the Pexip Infinity support log. The support log includes the administrator log.

However, you can also use a remote syslog server to collect copies of each system's logs. The advantage of this is that, should the Management Node become unavailable, logging would still continue on the syslog server.

When a remote syslog server is used, the Management Node and each of the Conferencing Nodes sends its logs directly to the assigned syslog server(s) using the syslog protocol. The Management Node will still continue to collate logs from all the nodes, and it is these unified logs that are shown from the Pexip Infinity Administrator interface. Note that the Management Node does not send the unified logs to the syslog server — it only sends its own logs.

The Management Node shows the **support log** only. If you elect to use a remote syslog server, you can choose to send it **audit logs**, **web server logs** and **system logs**, in addition to, or instead of, the support log.

You can configure more than one remote syslog server, for example if you want to send different combinations of the support, audit and web server logs to different servers.

If more than one syslog server is configured for a Management Node or a system location, logs will be sent to each of them.

To add, edit or delete the remote syslog servers used by Pexip Infinity, go to **System > Syslog Servers**.

The available options are:

Option	Description
Description	An optional description of the syslog server.
Address	The IP address or FQDN of the remote syslog server.
Port	The port on the remote syslog server. Default: 514.
Transport	The IP transport protocol used to communicate with the remote syslog server. Default: UDP.
Syslog protocol format	The format of the syslog protocol used when sending to the remote syslog server. The options are: • RFC3164 with milliseconds • RFC3164 • RFC5424 Default: RFC3164.
Support log	Enables sending of support and administrator log entries to this syslog server.
Audit log	Enables sending of Linux audit log entries to this syslog server. Some security-related certifications may require this log to be recorded.
Web server log	Enables sending of web server log entries to this syslog server. This is a log of all HTTPS requests to the Management Node and Conferencing Nodes.
System log	Enables sending of system log entries to this syslog server.

After configuring the syslog servers available to your system, you should assign those syslog servers for use by the Management Node, and for Conferencing Nodes in each location, as follows:

- For the Management Node, go to **Platform > Management Node > Syslog servers** and add the servers to the ***Chosen Syslog servers*** field.
- For Conferencing Nodes, you must assign the syslog server to the system locations used by the Conferencing Node. To do this, go to **Platform > Locations**, select the location and add the server to the ***Chosen Syslog servers*** field. Each Conferencing Node in that location will then use those syslog servers.

  If you do not assign any syslog servers to a location then there will be no remote logging for any of the nodes in that location.

The following table shows the facility code used for each log type:

Facility code	Log type
local0	Admin log
local2	Support log
local6	Audit log
local7	Web server log
The **system log** includes logs from a variety of system-level applications and may use any facility code that is not listed above.	

Web server logs are provided in the nginx web server format (Module ngx_http_log_module) with the following fields:

```
$remote_addr - $remote_user [$time_local] "$request" $pid $status $body_bytes_sent $request_time "$http_referer" "$http_user_agent"
```

Field	Module
remote_addr	ngx_http_core_module
remote_user	ngx_http_core_module
time_local	ngx_http_log_module
request	ngx_http_core_module
pid	ngx_http_core_module
status	ngx_http_log_module
body_bytes_sent	ngx_http_core_module
request_time	ngx_http_log_module
http_referer/http_user_agent	The values of the Referer/User-Agent headers as sent by the client

Configuring SMTP servers

SMTP servers are used in conjunction with Sending provisioning emails to VMR and device owners when bulk-provisioning VMRs and device aliases from Active Directory via LDAP.

To add, edit or delete the SMTP servers used by Pexip Infinity, go to **System > SMTP Servers**.

The available options are:

Option	Description
Name	The name used to refer to the SMTP server.
Description	A description of the SMTP server.
Address	The IP address or FQDN of the SMTP server.
Port	The port on the SMTP server to connect to. Default: 587.
Username Password	These are optional fields where you can specify the credentials of a valid account on the SMTP server.
From email address	The "from" email address to use when sending emails via this server. This must be an email address that is permitted to be used for sending email using this server and account.
Connection security	The type of connection security to use when connecting to this email server. The default option, ***StartTLS***, uses an encrypted connection.

Managing static routes

Static routes are additional configuration settings that permit routing of traffic to networks that are not accessible through the default gateway.

Typically, static routes are configured on Conferencing Nodes that are deployed in a DMZ, and where the default gateway on those nodes routes traffic out to the internet. The static routes would allow those nodes to communicate with Pexip Infinity nodes or other systems in the local, internal network. (See Network routing and addressing options for Conferencing Nodes for more information.)

- You must configure the static routes you want to use (via **System > Static Routes**) before you can apply them to a Conferencing Node or a Management Node.

- Static routes can be assigned to Conferencing Nodes during the initial deployment process, as well as after it has been deployed.

  In some situations, for example where a Conferencing Node's default gateway is out to the public internet, a static route back to the Pexip Infinity platform must be applied to the Conferencing Node during its initial deployment phase (otherwise it will not be able to communicate with the Management Node and pick up its configuration).
- Static routes can only be assigned to a Management Node after it has been deployed.

The static routes take immediate effect after they have been assigned to a node. You do not have to restart the VM.

Configuring the set of available static routes

To configure the set of static routes that can be applied to a Conferencing Node or Management Node:

1. Go to **System > Static Routes**.
2. Select **Add Static route** and then configure the relevant destination and gateway addresses:

Option	Description
Name	A unique short name or description of the static route.
Destination network address	The IP address used in conjunction with the **Network prefix** to determine the network addresses to which this route applies.
Network prefix	The prefix length used in conjunction with the **Destination network address** to determine the network addresses to which this route applies. The prefix length can be in the range 0–32 for IPv4 addresses and 0–128 for IPv6 addresses. For example, use a **Destination network address** of **10.0.0.0** and a **Network prefix** of **8** to route addresses in the range **10.0.0.0** to **10.255.255.255**.
Gateway IP address	The IP address of the gateway to the network for this route.

3. Select **Save**.

Assigning a static route to an existing node

To assign a static route to an existing Conferencing Node or Management Node:

1. Ensure that the static route that you want to apply has already been configured (**System > Static Routes**).
2. Go to **Platform > Conferencing Node** or **Platform > Management Node** as appropriate:
3. Select the node to which you want to assign the static route.
4. In the **Static routes** section, select from the list of **Available Static routes** the routes to assign to the node, and then use the right arrow to move the selected routes into the **Chosen Static routes** list.
5. Select **Save**.

Assigning a static route to a new Conferencing Node

Before you start the deployment process as described in Deploying new Conferencing Nodes, ensure that the static routes that you want to apply have already been configured (**System > Static Routes**).

Then, while specifying the network configuration parameters to apply to the new Conferencing Node, you can include the static routes that you want to assign to that node.

Using event sinks to monitor conference and participant status

Information about the current status of any conferences that are in progress can normally be obtained using the Management status API. However, in deployments with high levels of live system monitoring activity, such as those managed by service providers, frequent polling of the Management Node via the API can cause performance issues.

To avoid this you can configure the system to automatically send details of every participant and conference management event to an external service known as an **event sink**. When an event occurs, the Conferencing Node sends this information using a simple POST of JSON data to the nominated event sink server.

In particularly busy deployments, you can enable **Bulk support**. This groups and sends events together at a set interval, reducing the number of POSTs and therefore limiting the size of any queue that may result. Note that enabling bulk support will delay the receipt of events by up to the time specified by the Bulk message timeout interval (which is 0.1 seconds by default).

What information is sent?

To view which events are sent to an event sink, go to **System > Event Sinks** and from the bottom right of the page, select **View schema**. The same information can be downloaded as a Swagger or OpenAPI JSON document by selecting **Download Swagger schema** or **Download OpenAPI schema**.

You can also use a test site such as https://webhook.site to view live event data being sent.

Note that:

- Conference start and end event times are from the Conferencing Node's perspective and not the Management Node's perspective.
- You cannot control which events are sent to an event sink.
- You can filter the events sent to an event sink.

Configuring event sinks

Each system location can be configured with one or more event sinks. The same event sink can be used for more than one system location.

Note that conference and participant events are not sent from locations that contain Proxying Edge Nodes — the events for proxied calls are sent from the nodes/locations that perform the transcoding (providing those locations are configured with an event sink).

To add, edit or delete an event sink, go to **System > Event Sinks**. The available options for each event sink are:

Option	Description
Name	The name of this event sink. Each event sink must have a unique name.
Description	An optional description of this event sink.
Version	Select which version of the Event Sink API to use: • ***Version 1: Original*** • ***Version 2: Media streams*** Contains additional participant events: ◦ **participant_media_stream_window** ◦ **participant_media_streams_destroyed** • ***Version 3: Filtered events*** Allows for filtering of events. Default: ***Version 2***
Enable bulk support	When enabled, at the end of each interval specified by the Bulk message timeout, all events received during the interval are batched together and sent as a single POST. When this option is enabled: • There will be a delay between an event occurring and being sent to the event sink of up to the interval specified by the Bulk message timeout. • If only one event is received during the interval, it is sent at the end of the interval as a single event; if more than one event is received, they are sent at the end of the interval in a bulk message event.
URL	The URL of the external server to which events are sent.
Username Password	The username and password used to authenticate to the external server when sending events. The username is case-sensitive. If authentication is not required, leave these fields blank.

Option	Description
Verify TLS	Whether to enable TLS verification when sending events. Only valid if the URL is HTTPS.
Location	The system location(s) in which to use this event sink.
Enable filtered events	(Available when ***Version 3: Filtered events*** is selected) Enable this option to send only a selected subset of events to this event sink.
Events to send	(Available when **Enable filtered events** is selected) Select from the list of **Available events** and then use the right arrow to move the events into the **Chosen events** list. For more information on each event, see Event sink API.
Last attempted restart	This read-only field indicates if and when the event sink was last restarted.

Advanced configuration options

You can configure a range of advanced options that control connectivity and failure retry limits. These settings apply globally across all event sinks and are configured via **Platform > Global Settings > Advanced Event Sink Tuning**:

Option	Description
Event sink HTTP POST timeout	Maximum number of seconds allowed to connect, send, and wait for a response. Default: 7 seconds
Event sink maximum retry backoff	Maximum number of seconds allowed for the retry backoff before raising a "Eventsink Reached Maximum Backoff" alarm and stopping the event publisher. Default: 1800 seconds
Initial retry backoff	Initial time, in seconds, for the first retry attempt when a POST cannot be delivered. Default: 1 second
Internal cache expiration	Internal cache expiration time in seconds. Used to briefly store "participant_disconnected" events in order to gather end-of-call media statistics. Default: 2 seconds
Time to wait for media streams message	Maximum time, in seconds, to wait for an end-of-call media stream message. Default: 1 second
Time between metrics updates	Time between event sink metrics updates. You can specify 0 to disable event sink metrics. Default: 60 seconds
Number of HTTP worker threads	The number of HTTP POST worker threads to start.
Bulk message timeout	When Enable bulk support is enabled, the interval (in seconds) during which events are received, batched, and sent to the event sink in a single POST. Default: 0.1 seconds

Troubleshooting event sink failures

If a POST cannot be delivered to an event sink, the node will try again after 1 second. If it fails again it tries again after 2 seconds, then 4, 8, 16 seconds and so on — it keeps doubling the timeout. In this case, the events may not necessarily be sent in sequence number (**seq** field) order.

If the timeout exceeds 30 minutes it will instead raise an "Eventsink Reached Maximum Backoff" alarm and stop the event sink publisher for that particular event sink.

The retry/timeout parameters are configurable via **Platform > Global Settings > Advanced Event Sink Tuning** as described above.

To resolve event sink related alarms:

1. Check **support.events** logs for the reason for the event sink failures.
2. Take the appropriate action to resolve the failures.
3. Restart the event sink process:
   a. Go to **System > Event Sinks** and select the event sink you want to restart.
   b. Select **Restart**.

   The event sink will restart within approximately 1-2 minutes (after the next Conferencing Node synchronization), and any backlogged events are sent.

Configuring media processing servers

Pexip Infinity can connect to one or more external AI Media Servers (AIMS) to process media, for example to generate live captions.

For full information on AIMS and the Pexip Private AI platform, see Pexip Private AI and AIMS.

Adding a new server

To configure Pexip Infinity with information about an AIMS server to use to process media, go to **Platform > Media Processing Servers** and select **Add media processing server**.

Enter the following information:

Option	Description
FQDN	The FQDN of the AIMS server used to process media.

After you have added the server's FQDN, select **Save**. Pexip Infinity then assigns the server a unique **Application ID** (visible from the main page) and **JWT public key** (visible when the server is selected). These are the values required when configuring AIMS.

Copying the Application ID and JWT public key

When configuring AIMS with details of the Pexip Infinity deployment that it will be receiving requests from, you must provide the **Application ID** and **JWT public key** associated with the AIMS server on Pexip Infinity.

To view and copy these details:

1. Go to **Platform > Media Processing Servers** and select the AIMS server.
2. Click anywhere within the **Application ID** and **JWT public key** fields. The details will be automatically copied to your clipboard in a format that can be uploaded to the AIMS server.

Configuring SSH authorized keys

SSH keys can be used for the admin account for authorized SSH access to your Management Node and Conferencing Nodes.

They enable access authentication via SSH keys (asymmetric key pair) instead of a password.

When you add your keys you can assign them immediately to your Management Node or your existing Conferencing Nodes, or you can assign them later to your Management Node and when deploying or configuring Conferencing Nodes.

When a node is deployed in a cloud environment you can choose any of the SSH keys configured on the Management Node, in addition to the keys configured via the cloud environment (where available). Also, when configuring a node, you can choose to limit SSH access to only allow your own assigned keys.

What's noteworthy?

- The SSH keys only apply for the "admin" user. SSH superuser / sudo still requires password authentication.
- Password login is not allowed when one or more keys are assigned to that node.
- Any modifications to keys that are configured within the cloud service are picked up automatically by the Pexip Infinity nodes within approximately 10 minutes.

Configuring the keys

To add, modify or delete the set of available SSH authorized keys, go to **System > SSH Authorized Keys**.

To add a key:

1. Select **Add SSH Authorized Key**.
2. Enter the **Public key**.
   Do **not** add the private key.
3. You can optionally select the nodes to which you want to assign the key.
4. Select **Save**.
5. You can now optionally enter a **Comment** and select **Save** again. This is a description of the authorized key and is typically a contact email address.

Note that to change the public key you must delete it (and it must not be assigned to any nodes) and then add it again.

Pexip Infinity platform configuration

Configuring the Management Node

The Management Node is the administrative interface of the Pexip Infinity platform, from which administrators can:

- Create and manage Conferencing Nodes.
- Configure Pexip Infinity services (Virtual Meeting Rooms, Virtual Receptions and so on).
- View platform and conference status across all Conferencing Nodes.
- Perform active conference management functions such as adding and disconnecting participants, enabling streaming or recording services, locking a conference, or muting a participant's audio.

The Management Node does not handle any conference media or signaling.

It is deployed using a virtual machine management application such as VMware's vCenter Server, or Microsoft Hyper-V, or on a cloud service such as Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP) or Oracle Cloud Infrastructure.

The initial configuration of the Management Node is provided via the installation wizard. Any subsequent changes to the configuration of the Management Node should be done using the Pexip Infinity Administrator interface or via the management API. Do not make any changes by any other means such as VMware or SSH; doing so may cause the Pexip Infinity service to fail. In particular you must not change the IP address of the Management Node.

To change configuration of the Management Node itself, go to **Platform > Management Node**.

The available options are:

Option	Description
Name	The name used to refer to the Management Node. It comprises the DNS **Hostname** and **Domain suffix** that were assigned to the Management Node during initial installation (when the installation wizard was run).
Description	An optional field to provide more information about the Management Node. This defaults to the **Name** but you can change it here.
DNS servers	Select one or more DNS servers to be used by the Management Node. While you can assign unlimited DNS servers to a Management Node, only three will be used.
NTP servers	Select one or more NTP servers to be used by the Management Node.
Syslog servers	Select one or more Syslog servers to be used by the Management Node.
Web proxy	The web proxy to use for outbound web requests from this Management Node when sending usage statistics, incident reports, and some license activation requests, when communicating with the configured cloud bursting service, and (if One-Touch Join has been enabled and is using OAuth) when sending requests to the OAuth token endpoint.
TLS certificate	The TLS certificate to use on this node.
Configured FQDN	An optional identity for the Management Node. It is used by the web interface when indicating its own identity, for example when it needs to redirect to another page on itself. The name can be the same as its existing **hostname.domain**. If configured, the name must match an identity in the Management Node's TLS certificate. When a **Configured FQDN** is specified, access to the Management Node is limited to anything accessing the node by either the FQDN, or the **IPv4 static NAT address** if it's behind a static NAT.
IPv4 static NAT address	The IPv4 static NAT address for this Management Node. This allows IP address-based access to a Management Node behind a NAT when it has a **Configured FQDN** specified.
IPv6 address	The IPv6 address for the Management Node.

Option	Description
Gateway IPv6 address	The IPv6 address of the default gateway. If this is left blank, the Management Node listens for IPv6 Router Advertisements to obtain a gateway address.
Static routes	From the list of **Available Static routes**, select the routes to assign to the node, and then use the right arrow to move the selected routes into the **Chosen Static routes** list.
MTU	(Maximum Transmission Unit) — the size of the largest packet that can be transmitted via the network interface of the Management Node. It depends on your network topology as to whether you may need to specify an MTU value here. If the Management Node is running in Google Cloud Platform, the MTU must **not** be higher than 1460 bytes.
Enable SSH	Determines whether this node can be accessed over SSH. ***Use Global SSH setting***: SSH access to this node is determined by the global **Enable SSH** setting (**Platform > Global Settings > Connectivity > Enable SSH**). ***Off***: this node cannot be accessed over SSH, regardless of the global **Enable SSH** setting. ***On***: this node can be accessed over SSH, regardless of the global **Enable SSH** setting. Default: ***Use Global SSH setting***.
SSH authorized keys	You can optionally assign one or more SSH authorized keys to use for SSH access. From the list of **Available SSH authorized keys**, select the keys to assign to the node, and then use the right arrow to move the selected keys into the **Chosen SSH authorized keys** list. Note that in cloud environments, this list does **not** include any of the SSH keys configured within that cloud service.
Use SSH authorized keys from cloud service	When a node is deployed in a cloud environment, you can continue to use the SSH keys configured within the cloud service where available, in addition to any of your own assigned keys (as configured in the field above). If you disable this option you can only use your own assigned keys. Default: enabled.

You can also change the Management Node's SNMP settings (see Monitoring via SNMP for more information):

Option	Description
SNMP mode	Configures the SNMP access mode for the selected node: ***Off***: SNMP is disabled. You cannot use SNMP to query the node for its status. ***SNMPv2c read-only***: enables insecure, read-only access. ***SNMPv3 read-only***: enables secure, read-only access, using the **authPriv** security level with SHA1 authentication and AES 128-bit encryption. When enabled, access is provided to the basic RFC 1213 MIB-II tree (1.3.6.1.2.1). Default: ***Off***.
SNMP community	The SNMP group to which this node belongs. This setting applies to SNMPv2c only. Default: public
SNMPv3 username	The node's SNMPv3 username, used to authenticate SNMPv3 requests.

Option	Description
SNMPv3 privacy password	The node's SNMPv3 privacy password used for encrypting messages between the node and the management station. AES encryption must be used; DES is not supported.
SNMPv3 authentication password	The node's SNMPv3 authentication password, used to authenticate the associated username. The SHA authentication protocol must be used; MD5 is not supported.
SNMP system contact	The contact details (for example, email address) of the person responsible for this particular node.
SNMP system location	A description of the node's location.

If you want SNMP traps to be sent from the Management Node to a particular SNMP Network Management System (NMS), select the NMS from the **SNMP NMS** drop-down menu:

Option	Description
SNMP NMS	The Network Management System to which the Management Node sends SNMP traps. If you have not already added the SNMP NMS, you can do so now by clicking . Pexip Infinity does not currently support traps with SNMPv3. If traps are required, use SNMPv2c.

Other details of the Management Node that cannot be changed via the Administrator interface are also shown on this page for your information, as follows:

Option	Description
IPv4 address	The IPv4 address of the Management Node.
Network mask	The IPv4 network mask of the Management Node.
Gateway IPv4 address	The IPv4 address of the default gateway.
Hostname	The DNS hostname of the Management Node.
Domain	The DNS domain of the Management Node.

If you need to change any of the hostname/addressing information, you must do so by Re-running the installation wizard.

To perform other maintenance tasks such as changing the IP address of the Management Node or moving the Management Node to a different host server, see Moving, restoring or changing the IP address of the Management Node.

To configure platform-wide settings, see About global settings.

About global settings

Global settings are system-wide configuration options that affect the entire Pexip Infinity platform. Some of the settings may be overridden at the location, Conferencing Node or VMR level — this is indicated in the table below where applicable.

To configure the global settings, go to **Platform > Global Settings**. The settings are grouped into sections that you can **Show** or **Hide** individually, or by selecting **Expand all** or **Collapse all**, as follows:

You should wait at least 90 seconds for any changes in configuration to be synchronized to all Conferencing Nodes; this may take longer in large deployments. You can go to **Status > Conferencing Nodes** to check when configuration was last updated.

Setting	Description	More information
Service configuration		

Setting	Description	More information
Guests-only timeout	The length of time (in seconds) for which a conference will continue with only Guest participants, after all Host participants have left. Default: 60 seconds	Using PINs to differentiate between Hosts and Guests
Last participant backstop timeout	The length of time (in seconds) for which a conference will continue with only one participant remaining. The type of participant (Host, Guest, automatically dialed, streaming etc) is irrelevant. It also ends any gateway calls into a Microsoft Teams / Google Meet meeting when all of the Teams / Google Meet clients (or other gatewayed participants) have left. The time can be configured to values between 60 seconds and 86400 (1 day), or to 0 (never eject). Default: 0 (never eject)	Automatically ending a conference
PIN entry timeout	The length of time (in seconds) for which a participant is allowed to remain at the PIN entry screen before being disconnected. Default: 120 seconds	Limiting the time a participant can spend at the PIN entry screen
Waiting for Host timeout	The length of time (in seconds) for which a Guest participant can remain at the waiting screen if a Host does not join, before being disconnected. Default: 900 seconds	Limiting how long Guests can wait for a Host
Default theme	The theme to use for services that have no specific theme selected.	Customizing conference images and voice prompts using themes
Maximum inbound call bandwidth (kbps)	Limits the bandwidth of media being received by Pexip Infinity from individual participants, for calls where bandwidth limits have not otherwise been specified.	Managing and restricting call bandwidth
Maximum outbound call bandwidth (kbps)	Limits the bandwidth of media being sent by Pexip Infinity to individual participants, for calls where bandwidth limits have not otherwise been specified.	Managing and restricting call bandwidth
Maximum call quality	Controls the maximum call quality for participants connecting to Pexip Infinity services. You can override this global setting for each individual service (VMR, Call Routing Rule etc). For example, you could use the default option of "HD" for most of your services by default, but enable Full HD on some specific services. The options are: • ***SD***: each participant is limited to SD quality. • ***HD***: each participant is limited to HD (720p) quality. • ***Full HD (1080p)***: allows any endpoint capable of Full HD to send and receive its main video at 1080p. Default: ***HD***	Setting and limiting call quality
Maximum presentation bandwidth ratio	When sending main video and presentation to a standards-based (SIP or H.323) or WebRTC endpoint, this defines the maximum percentage of the call bandwidth to allocate to the presentation content (with the remainder allocated to main video). It must be in the range 25% to 75%. Default: 75% Note that when sending two video streams to a dual-screen endpoint, the call bandwidth is always split 50-50 between video and presentation content (and cannot be changed).	Managing and restricting call bandwidth

Setting	Description	More information
External participant avatar lookup	Determines whether or not avatars for external participants are retrieved using a method appropriate for the external meeting type. Currently this only applies to Microsoft Teams conferences. For all other conference types, and for when this option is not selected, avatars may be retrieved via external policy or user records as per standard behavior. You can also configure this setting on individual Call Routing Rules for Microsoft Teams conferences. Default: enabled.	
Live captions	Select this option to enable live captions by default on all Virtual Meeting Rooms, Virtual Auditoriums and Call Routing Rules. You can override this global setting on individual services. The live captions feature requires integration with a Pexip Private AI platform. For full details, see Pexip Private AI and AIMS. Default: disabled.	Configuring Virtual Meeting Rooms (VMRs) Configuring Call Routing Rules
Minimum PIN length	The minimum PIN length in VMRs which have Host and/or Guest PINs configured. Default: 6 Range: 4-20 digits, including any terminal #.	
Connectivity		
Enable SIP *	Controls support for the SIP protocol over TCP and TLS across all Conferencing Nodes in your Pexip Infinity deployment. Note that disabling SIP will disable support for Skype for Business (MS-SIP). Default: enabled.	Enabling and disabling SIP, H.323, WebRTC and RTMP
Enable SIP UDP	Allows or prevents incoming calls over SIP UDP. Default: disabled.	
Enable H.323 * Enable WebRTC Enable RTMP *	These boxes control support for the selected protocols across all Conferencing Nodes in your Pexip Infinity deployment. Default: all of these settings are enabled by default.	
Enable support for Pexip apps and Client API	Enables support for the Pexip Infinity client API. This is required for integration with the Pexip apps (web, desktop and mobile), and any other third-party applications that use the client API, as well as for integration with Microsoft Teams and Poly OTD endpoints for One-Touch Join. This setting must be enabled if you want to **Enable WebRTC** or **Enable RTMP**. Default: enabled.	
Enable media relay on TCP port 443	This setting enables media relay on TCP port 443 on all Conferencing Nodes. This is intended as a fallback mechanism for use by WebRTC clients that are behind strict firewalls that block RTP media to Pexip's standard ports. This setting should only be enabled when it is impossible to amend the firewall's rules to allow UDP media, as sending media over TCP can result in increased latency and jitter. Enabling this setting may cause disruption to ongoing WebRTC sessions.	

Setting	Description	More information
Enable Far End Camera Control	Enables Pexip apps and SIP/H.323 endpoints to send Far-End Camera Control (FECC) signals to supporting endpoints, in order to pan, tilt and zoom the device's camera. Support for SIP/H.323 endpoints sending FECC signals is for gateway calls only. Default: enabled.	Control another participant's camera
Enable chat	Enables relay of chat messages between conference participants using supported clients such as the Pexip apps. You can override this setting on a per conference basis (for a Virtual Meeting Room or Virtual Auditorium). Default: enabled.	Enabling and disabling chat messages
Enable outbound calls	Controls whether any calls can be made via the Infinity Gateway, and allows dial-out from a conference (via the Pexip apps and the Administrator interface). Default: enabled.	Placing calls via the Pexip Infinity Distributed Gateway Manually dialing out to a participant from a conference
Enable legacy dialout API	This setting controls the system behavior when dialing out via a Pexip app or the client API to a participant from an ongoing conference. When selected (enabled), calls placed via the: • Pexip apps always use automatic routing and thus must match an appropriate Call Routing Rule. • Client API can either use automatic routing or they can specify a dial-out protocol without any need for a Call Routing Rule i.e. it allows end-users to perform arbitrary dial outs (and thus circumvent any administrator-set rules). When not selected (disabled), calls that are placed via: • any Pexip app, or via the client API, always use automatic routing and thus must match an appropriate Call Routing Rule. Note that dial out via the Administrator interface, management API or Automatically Dialed Participants (ADPs) is unaffected by this setting. Default: disabled.	Manually dialing out to a participant from a conference
Pexip Infinity domain	This is an optional field where you can specify the name of the SIP domain that is: • Routed from Skype for Business to Pexip Infinity, either as a static route or via federation. • Used as the default domain in the **From** address for outgoing SIP gateway calls and outbound SIP calls from conferences without a valid SIP URI as an alias. You can also configure the Pexip Infinity domain on a per-location basis, which would override this global setting for Conferencing Nodes in that location.	Pexip Infinity and Microsoft Skype for Business Deployment Guide
Enable Skype for Business auto-escalation	When selected, this automatically escalates a Skype for Business audio call so that it receives video from a conference. Default: disabled.	Controlling media capability
Enable VbSS for Skype for Business	Controls support for Skype for Business Video-based Screen Sharing (VbSS). Note that VbSS is always enabled for Microsoft Teams calls, regardless of this setting. Default: disabled.	For information about enabling VbSS on your Skype for Business infrastructure see https://technet.microsoft.com/en-us/library/mt756736.aspx.

Setting	Description	More information
Enable PowerPoint Live	Enables support for PowerPoint Live content in Microsoft Teams calls. Default: enabled.	
DSCP value for management traffic	The DSCP value for SSH, HTTPS and SNMP management traffic sent from the Management Node and from Conferencing Nodes. This is an optional Quality of Service (QoS) setting used to prioritize different types of traffic in large, complex networks. Also see Configuring system locations.	
Enable SSH	Allows an administrator to log in to the Management Node and all Conferencing Nodes over SSH. This setting can be overridden on individual nodes. Default: enabled.	
Enable directory	When disabled, Pexip apps display aliases from their own call history only. When enabled, registered Pexip apps additionally display the aliases of Virtual Meeting Rooms, Virtual Auditoriums, Virtual Receptions, and devices registered to the Pexip Infinity platform. Default: enabled.	Directory (phone book) of devices and VMRs for registered Pexip apps
Enable restricted routing for Proxying Edge Nodes	When enabled, if a location only contains Proxying Edge Nodes, then those nodes only require IPsec connectivity with other nodes in that location, the transcoding location, the primary and secondary overflow locations, and with the Management Node. When disabled, a full connectivity mesh is required between all nodes in the deployment. Default: enabled.	Deployment guidelines for Proxying Edge Nodes
Media encryption	Controls the media encryption requirements for participants connecting to Pexip Infinity services. You can override this global setting for each individual service (VMR, Call Routing Rule etc). For example, you could use the default option of "best effort" for most of your services, but enforce encryption on some specific services. • ***Best effort***: each participant will use media encryption if their device supports it, otherwise the connection will be unencrypted. • ***Required***: all participants (including RTMP participants) must use media encryption. • ***No encryption***: all H.323, SIP and MS-SIP participants must use unencrypted media. (RTMP participants will use encryption if their device supports it, otherwise the connection will be unencrypted.) Default: ***Best effort***	
Port ranges		
Signaling port range start and end *	The start and end values for the range of ports (UDP and TCP) that all Conferencing Nodes use to send signaling (for H.323, H.245 and SIP). Default: 33000–39999.	
Media port range start and end *	The start and end values for the range of ports (UDP and TCP) that all Conferencing Nodes use to send media for H.323, SIP, Skype for Business, WebRTC and RTMP (note that RTMP uses TCP only). The media port range must contain at least 100 ports. Default: 40000–49999.	
Codecs		

Setting	Description	More information
Codecs	Controls which codecs to offer in audio/video negotiation (SDPs). Some third-party systems can experience issues if they are sent a large SDP from Pexip Infinity. You can reduce the size of the SDP by disabling specific, unwanted codecs. Default: all codecs are enabled except ***AAC-LD128***, ***H.264 High (mode 0)*** and ***H.264 High (mode 1)***. To enable the H.264 High Profile codec, move ***H.264 High (mode 1)*** into the list of **Enabled Codecs**. For optimal interoperability results, only enable ***H.264 High (mode 1)*** — leave ***H.264 High (mode 0)*** in the **Disabled Codecs** list. You can also specify a subset of these codecs to use when Configuring Call Routing Rules.	
Security		
OCSP state	Determines whether OCSP is used to check the status of TLS certificates. ***Off***: OCSP is not used. ***On***: OCSP is used, and the request is sent to the URL specified in the TLS certificate. If no URL is specified in the TLS certificate, the **OCSP responder URL** configured below is used. ***Override***: OCSP is used, and the request is sent to the **OCSP responder URL** specified in the **OCSP responder URL** field, regardless of any URL encoded in the TLS certificate. Default: ***Off***.	Using OCSP to check the status of certificates
OCSP responder URL	The URL to which OCSP requests are sent if either: • the **OCSP state** is set to ***On*** but no URL is present in the TLS certificate, or • the **OCSP state** is set to ***Override*** (in which case any URL present in the certificate is ignored).	Using OCSP to check the status of certificates
SIP TLS certificate verification mode	Determines whether to verify the peer certificate for outbound connections over SIP TLS. The options are: ***Off***: the peer certificate is not verified; all connections are allowed. ***On***: the peer certificate is verified, and the peer's remote identities (according to RFC5922) are compared against the Application Unique String (AUS) identified by Pexip Infinity — the SIP URI — before the connection is allowed. Default: ***Off***.	Verifying SIP TLS connections with peer systems
Maximum log age (days)	The maximum number of days of logs and call history to retain on Pexip nodes. On busy systems, logs may still be rotated before this time due to limited disk space. Enter 0 to have no set limit. Default: 0.	
HTTP Content Security Policy	Determines whether or not HTTP Content-Security-Policy (CSP) headers for Conferencing Nodes are enabled. Default: enabled.	

Setting	Description	More information
HTTP Content Security Policy Header	Defines the contents of the HTTP Content-Security-Policy headers for Conferencing Nodes when CSP is enabled. The default header string contains multiple directives such as **frame-src** and **script-src**, delimited by the **;** character. For more information on CSP, see this MDN article. Default: `upgrade-insecure-requests; default-src 'self'; frame-src 'self' https://telemetryservice.firstpartyapps.oaspapps.com/telemetryservice/telemetryproxy.html https://*.microsoft.com https://*.office.com; style-src 'self' 'unsafe-inline' https://*.microsoft.com https://*.office.com; object-src 'self'; font-src 'self' https://*.microsoft.com https://*.office.com; img-src 'self' https://www.adobe.com data: blob:; script-src 'self' 'unsafe-inline' 'unsafe-eval' https://*.microsoft.com https://*.office.com https://ajax.aspnetcdn.com https://api.keen.io; media-src 'self' blob:; connect-src 'self' https://*.microsoft.com https://*.office.com https://example.com; frame-ancestors 'self';` These defaults are appropriate for typical usage of Pexip Infinity. Note that: • The Microsoft addresses are required for Outlook add-ins. If you are hosting the add-in sources locally, for example in a fully air-gapped environment, you must also add the URL of the web server to the appropriate places. For more information, see Configuring Exchange Scheduler in an air-gapped environment. • The reserved domain **example.com** is included to support a third-party JavaScript library that is used by the legacy Webapp2 for PDF content sharing. You may need to add extra headers if you use custom plugins with the web app. • The '**unsafe-inline**' keyword is only required if you use the legacy Webapp2.	
Break-in resistance		
Enable PIN brute force resistance	Select this option to instruct Pexip Infinity to temporarily block all access to a VMR that receives a significant number of incorrect PIN entry attempts. You can override this setting on a per location basis. Default: enabled.	Break-in resistance settings to mitigate rogue calls
Maximum PIN failures	The maximum number of PIN failures allowed in any 10-minute window before the VMR is blocked. Default: 20.	
Enable VOIP scanner resistance	Select this option to instruct Pexip Infinity to temporarily block service access attempts from any unknown source IP addresses that dial a significant number of incorrect aliases. You can override this setting on a per location basis. Default: enabled.	
Maximum scanner attempts	The maximum number of incorrect dial attempts in any 10-minute window before the source IP address is blocked. Default 20.	
External system integration		
External system username and password	The username and password used by external systems (such as CUCM) when authenticating with Pexip Infinity.	Integrating with external systems

Setting	Description	More information
Management web interface configuration		
Login banner text	Any text entered here is displayed in a message box on the login page. This field supports plain text only.	
Enable management web interface session timeout	Controls whether inactive users are automatically logged out of the Administrator interface after a period of time. If enabled, users are logged out after a number of minutes of inactivity as specified in the **Management web interface session timeout** setting. If disabled, users of the Administrator interface are never timed out. You may want to use this option if, for example, you have an administrator session that permanently monitors the system live view. Default: enabled.	
Management web interface session timeout	The number of minutes a browser session may remain idle before the user is logged out of the Management Node Administrator interface, if **Enable management web interface session timeout** is selected. Default: 30 minutes.	
Show conferences and backplanes in Live View	Controls whether conferences and backplanes are shown in the Live View graph. If you have a very busy deployment, it may be useful to disable conferences and backplanes from the Live View for an improved viewing experience. Note that when conferences and backplanes are removed, the conferences and participants counts in Live View are not shown. Default: enabled.	Viewing live and historical platform status
Management start page	Controls the first page you are directed to after logging into the Administrator interface. Default: ***Live View***	Using the Pexip Infinity Administrator interface
Site banner text Banner background color Banner text color	These options configure the site banner (across the top of every webpage) for the Pexip Infinity Administrator interface. You can define the text to display, and use the color pickers to select the text and background colors.	
Reporting		
Enable incident reporting Incident reporting URL Contact email address	If incident reporting is enabled, reports are sent to the specified URL. The contact email address will be used by Pexip Support when following up on any incident reports received from your deployment. These settings are configured during initial installation of the Management Node (when running the installation wizard).	Automatically reporting errors

Setting	Description	More information
Automatically send deployment and usage statistics to Pexip	Select this option to allow submission of deployment and usage statistics to Pexip. This will help us improve the product. This setting is configured during initial installation of the Management Node (when running the installation wizard).	Automatically sending usage statistics
Advanced event sink tuning		
Event sink settings	A range of advanced options to tune the event sink processes. See Using event sinks to monitor conference and participant status for details.	Using event sinks to monitor conference and participant status
Cloud bursting		
Cloud bursting settings	These options enable and configure the Pexip Infinity platform for dynamic cloud bursting to either Microsoft Azure, Amazon Web Services (AWS) or Google Cloud Platform (GCP).	Dynamic bursting to the AWS cloud Dynamic bursting to the Azure cloud Dynamic bursting to the Google Cloud Platform
Voice activity features		
Enable Voice Focus	Voice Focus is a platform-wide setting that improves the way in which voice activity is detected by better distinguishing between actual speech and background noise. This reduces the probability that people who are not speaking but have audible background noise will be switched into the main speaker position. Note that this does **not** remove any noise from the audio. When enabled it applies to Virtual Meeting Rooms and Virtual Auditoriums, and it uses additional hardware resources (equivalent to an extra 6 audio connections per participant). Default: disabled.	
Breakout rooms		
Enable breakout rooms	Enables the ability to configure breakout rooms on VMRs and Virtual Auditoriums. Default: disabled.	Breakout rooms
Tech preview features		
Enable Softmute	Softmute is advanced speech-aware audio gating which helps to minimize noise coming from a participant who has their microphone turned on in a conference, but is not speaking. If non-voice noise is detected, this feature softens the gain from that participant. It does not entirely suppress noise from an audio signal. When it is enabled here, it can then be individually enabled or disabled at the VMR level (**Services > Virtual Meeting Rooms > Advanced Options > Softmute**). It uses additional hardware resources (equivalent to an extra 6 audio connections per participant). However, if Voice Focus is also enabled, then both features use a combined total of an extra 6 audio connections per participant i.e. there is no extra resource use for enabling Softmute when Voice Focus is also enabled.	Configuring Virtual Meeting Rooms (VMRs)

Setting	Description	More information
Enable Denoise	Denoising is a server-side feature that removes the background noise from somebody who is speaking. In comparison, softmute softens the gain from a noisy, but non-speaking participant. When it is enabled here, it can then be individually enabled or disabled at the VMR level (**Services > Virtual Meeting Rooms > Advanced Options > Denoise**). It uses additional hardware resources (equivalent to an extra 7 audio connections per participant). However, if Voice Focus is also enabled, then both features use a combined total of an extra 7 audio connections per participant i.e. there is no extra resource use for enabling Denoise when Voice Focus is also enabled. If Voice Focus, Softmute and Denoise are all enabled then it still uses a combined total of an extra 7 audio connections per participant. Note that this feature is separate from Google Meet denoising, which is a client-side feature and applies to Google Meet integrations only.	Configuring Virtual Meeting Rooms (VMRs)
Enable clock	Enables support for displaying an in-conference timer or countdown clock.	Displaying an in-conference timer / countdown

* If you change any of these settings, all existing calls will be disconnected and all Conferencing Nodes will be automatically restarted.

About system locations

System locations are typically used to group together Conferencing Nodes that are in the same physical location. System locations serve various purposes:

- They enable Pexip Infinity to make intelligent decisions about routing of media and signaling, including overflowing to another location when a particular location reaches its transcoding capacity, and the separation of Proxying Edge Nodes from Transcoding Conferencing Nodes.
- Services such as DNS, NTP, syslog servers, H.323 gatekeepers, SIP proxies, web proxies, Skype for Business Servers, Teams Connectors, TURN servers, STUN servers, SNMP NMSs, policy profiles, and event sinks are configured on a per-location basis, and are used by all Conferencing Nodes in that location.
- The One-Touch Join service also operates on a per-location basis.
- For deployments integrated with Pexip Private AI, a specific live captions location is used to connect to the AIMS server; other locations are configured to use this location for live captions traffic.

A Conferencing Node's system location is assigned when the node is deployed, but it can be subsequently modified.

If you change the system location of a Conferencing Node, all existing calls will be disconnected and the Conferencing Node will be restarted.

Intelligent and bandwidth efficient routing

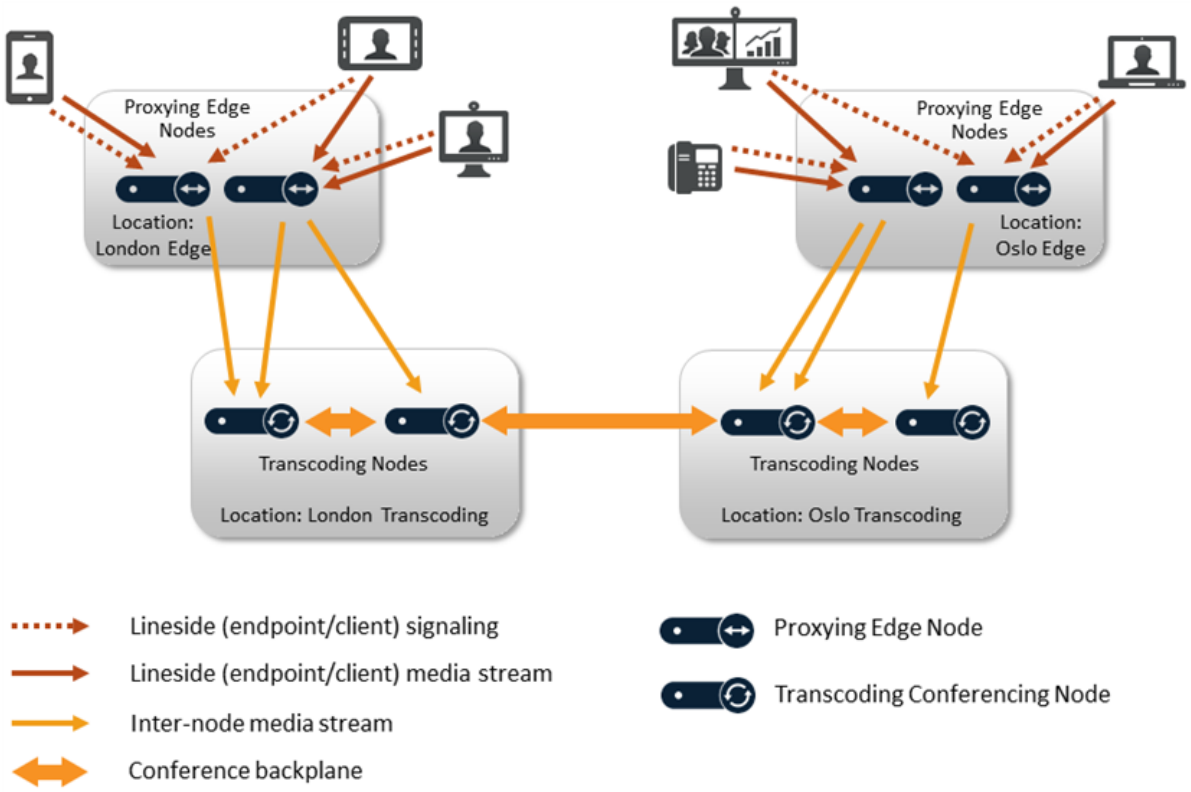
Grouping Conferencing Nodes by location allows Pexip Infinity to make intelligent decisions about routing. For example, if a conference is taking place across many Conferencing Nodes in two different locations, then Pexip Infinity will nominate one node in each location to act as an intermediary for that location. Media streams are sent between each intermediary only, rather than multiple streams being sent between each of the nodes at each of the locations. For more information, see Conference distribution.

Calls that are received (for signaling purposes) can have their media received and transcoded by Conferencing Nodes within the same location as the node that is handling the signaling, or a different location can be nominated to handle the media. If the location handling the media reaches its transcoding capacity for a conference instance, additional "media overflow" locations can be utilized to handle the media for new conference participants that connect to nodes within the original signaling location. For more information, see Media overflow locations.

Locations can also be configured to contain Proxying Edge Nodes that handle the signaling and the media connection with the calling device, but proxy the media onto another location for transcoding purposes (see Deployments including Proxying Edge Nodes below).

When grouping Conferencing Nodes into different locations, you should consider the amount of packet loss within your network. If there is a chance of packet loss due to network congestion between different groups of nodes, then they should be assigned separate system locations even if they are in the same physical location.

Deployments including Proxying Edge Nodes



You can deploy your Pexip Infinity platform as either a mix of Proxying Edge Nodes and Transcoding Conferencing Nodes, or as a system that only contains Transcoding Conferencing Nodes.

A typical deployment scenario is to use Proxying Edge Nodes as a front for many privately-addressed Transcoding Conferencing Nodes. Those outward-facing proxying nodes would receive all the signaling and media from endpoints and other external systems, and then forward that media onto other internally-located transcoding nodes to perform the standard Pexip Infinity transcoding, gatewaying and conferencing hosting functions.

A system location should not contain a mixture of proxying nodes and transcoding nodes. This separation of roles to locations simplifies load-balancing and conference distribution, and makes it easier for you to manage and monitor your externally-facing Proxying Edge Nodes distinctly from your Transcoding Conferencing Nodes. Hence, in the example scenario shown here, the Conferencing Nodes in the two locations "London Edge" and "Oslo Edge" are Proxying Edge Nodes and thus those nodes and locations are not involved in the actual hosting of any conferences. They forward the media onto the Transcoding Conferencing Nodes in the "London Transcoding" and "Oslo Transcoding" locations respectively.

Network deployment considerations

Nodes within a location actively synchronize with each other and may require a relatively high amount of network bandwidth for their communication. Pexip's backplane topology for distributed conferences assumes that there is a high availability, low latency, high-bandwidth network link between nodes within a location.

Therefore, while physical proximity is not a requirement, nodes in the same system location should typically be in the same physical datacenter or in physically proximate datacenters with an excellent link between them. If (as is often the case) you do not have such a network between your datacenters, we recommend that you consider assigning your nodes to different system locations.

There should be no more than 150 ms latency between any of your Conferencing Nodes (both within and between locations).

In addition:

- Conferencing Nodes in a DMZ must not be configured with the same **System location** as nodes in a local network. This is to ensure that load balancing is not performed across nodes in the DMZ and nodes in the local network.
- Any Conferencing Nodes that are configured with a static NAT address must not be configured with the same **System location** as nodes that do not have static NAT enabled. This is to ensure that load balancing is not performed across nodes servicing external clients and nodes that can only service private IP addresses.
- We recommend that a **System location** should not contain a mixture of on-premises and cloud-hosted (Azure, AWS, GCP or Oracle) Conferencing Nodes.

See Network deployment options for more information about the various deployment scenarios.

Configuring services

Many platform and system services are configured on a per-location basis, and are used by all Conferencing Nodes in that location.

DNS and NTP servers

You must select at least one DNS server and at least one NTP server to be used by all of the Conferencing Nodes in that location. This allows you, for example, to assign nodes in a DMZ to a location that uses different DNS and NTP servers from those in a location containing nodes in a local, internal network.

Any public Internet-facing locations should be assigned one or more DNS servers that can correctly resolve SRV and A record queries for any domain.

Syslog servers

You can optionally specify the remote syslog servers to be used by all of the Conferencing Nodes in that location.

H.323 gatekeepers, SIP proxies and Skype for Business servers

You can optionally specify the H.323 gatekeeper, SIP proxy, and Skype for Business server to use to route the outbound H.323/SIP/MSSIP call placed from a node within that location, when adding a new participant to a conference. These are the call control systems that are used when:

- a conference participant uses a Pexip app to add another participant to the call
- the administrator manually dials out to a participant from a Virtual Meeting Room
- a participant is automatically dialed out to from a Virtual Meeting Room
- a third party uses the API to place a call to a participant

For more information, see About H.323 gatekeepers and SIP proxies and About Skype for Business servers.

Web proxies

You can select a web proxy to use for some outbound web requests from all Conferencing Nodes in a location. When selected, the web proxy is used automatically for incident reporting, Epic telehealth requests, and for any One-Touch Join-related requests. For more information, see Using a web proxy.

TURN servers and STUN servers

Pexip Conferencing Nodes can utilize a TURN server and negotiate TURN relays with the following ICE capable clients:

- Skype for Business clients
- WebRTC clients (the web app on the latest browsers, and the desktop and mobile clients)

If these endpoints will be connecting to privately-addressed "on-premises" Conferencing Nodes, you **must** configure Pexip Infinity with the address of at least one TURN server that it can offer to ICE clients.

In some deployment scenarios where the TURN server is not located outside of the enterprise firewall, you may need to configure a separate STUN server so that each Conferencing Node can discover its public NAT address.

You can also configure the STUN servers to be used by Pexip WebRTC clients when they connect to a Conferencing Node in this location.

For more information, see Using TURN servers with Pexip Infinity and Using STUN servers with Pexip Infinity.

SNMP NMSs

If you have enabled SNMP support on one or more Conferencing Nodes in a particular location, you must also select the SNMP Network Management System (NMS) to which SNMP traps will be sent. The selected NMS is used for all Conferencing Nodes in the location that have SNMP support enabled.

Pexip Infinity does not currently support traps with SNMPv3. If traps are required, use SNMPv2c.

For more information, see Monitoring via SNMP.

Policy profiles

Policy profiles specify how Pexip Infinity uses external policy and/or local policy to control its call policy and routing decisions.

You can configure Pexip Infinity to use a different policy profile per system location.

For more information, see Using external and local policy to control Pexip Infinity behavior.

Event sinks

In busy deployments where live event reporting is required, you can configure event sinks for each location. These are external service (s) to which Conferencing Nodes in this location send event information. For more information, see Using event sinks to monitor conference and participant status.

Configuring system locations

To add, edit or delete system locations, go to **Platform > Locations**.

You should wait at least 90 seconds for any changes in configuration to be synchronized to all Conferencing Nodes; this may take longer in large deployments. You can go to **Status > Conferencing Nodes** to check when configuration was last updated.

The available options are:

Option	Description
Name	The name you want to give to this physical location.
Description	An optional field where you can provide more information about the location.
DNS servers	From the list of configured DNS servers, select one or more DNS servers to be used by all Conferencing Nodes in this location. While you can assign unlimited DNS servers to a location, only three will be used.
NTP servers	From the list of configured NTP servers, select one or more NTP servers to be used by all the Conferencing Nodes in this location.
Syslog servers	From the list of configured syslog servers, select one or more syslog servers to be used by all the Conferencing Nodes in this location. If no syslog servers are selected then there will be no remote logging for any of the nodes in this location.
H.323 gatekeeper	The H.323 gatekeeper to use for outbound H.323 calls from this location, when adding an H.323 participant to a conference. For more information, see About H.323 gatekeepers and SIP proxies.
SNMP NMS	The Network Management System to which SNMP traps for all Conferencing Nodes in this location will be sent. For more information, see Monitoring via SNMP.
SIP proxy	The SIP proxy to use for outbound SIP calls from this location, when adding a SIP participant to a conference. For more information, see About H.323 gatekeepers and SIP proxies.
Web proxy	The web proxy to use for some outbound web requests from all Conferencing Nodes in this location. When selected, the web proxy is used automatically for incident reporting, Epic telehealth requests, and for any One-Touch Join-related requests. For more information, see Using a web proxy.

Skype for Business server	The Skype for Business server to use for outbound MS-SIP calls from this location, when adding a SfB participant to a conference. For more information, see About Skype for Business servers.
Microsoft Teams Connector	The Teams Connector to use for outbound calls to Teams meetings from this location, if a Virtual Reception or Call Routing Rule does not explicitly specify the Teams Connector to use.
TURN server	The TURN server to use when ICE clients (such as Skype for Business clients and Pexip WebRTC clients) located outside the firewall connect to a Conferencing Node in this location. For more information, see Using TURN servers with Pexip Infinity.
STUN server	The STUN server to be used by all Conferencing Nodes in this location to determine the public IP address to signal to ICE clients (such as Skype for Business clients and Pexip WebRTC clients) located outside the firewall. For more information, see Using STUN servers with Pexip Infinity.
Client TURN servers	The TURN servers to be provisioned to Pexip WebRTC clients when they connect to a Conferencing Node in this location. For more information, see Using TURN servers with Pexip Infinity.
Enforce media routing via a client TURN server	Select this option to force the WebRTC client to route its media through one of the specified **Client TURN servers**.
Client STUN servers	The STUN servers to be provisioned to Pexip WebRTC clients when they connect to a Conferencing Node in this location. For more information, see Using STUN servers with Pexip Infinity.
MTU	(Maximum Transmission Unit) — the size of the largest packet that can be transmitted via the network interfaces of the nodes in this location. It depends on your network topology as to whether you may need to specify an MTU value here. If any of the Conferencing Nodes in this location are running in Google Cloud Platform, the MTU must **not** be higher than 1460 bytes. On a dual interface node the MTU is only applied to the primary interface since the usual expectation is that MTU applies to media datagrams that are carried in IPsec between nodes and here the fragment size needs to be controlled. For lineside media sent from external-facing nodes (proxying nodes) towards clients, the MTU is applied to media packets at the transcoding location. The MTU should typically align on both the transcoding and proxy locations. Default: 1500
DSCP value for media	An optional setting used to prioritize different types of traffic in large, complex networks. This DSCP value tags the media traffic from Conferencing Nodes in this system location that is sent line side to endpoints and over the IPsec backplanes to other Pexip Conferencing Nodes.
DSCP value for signaling	An optional Quality of Service (QoS) setting used to prioritize different types of traffic in large, complex networks. This DSCP value tags the signaling traffic from Conferencing Nodes in this system location that is sent line side to endpoints and over the IPsec backplanes to other Pexip Conferencing Nodes. Note that some IPsec traffic between nodes — configuration synchronization and other non-realtime traffic — remains untagged. Also see DSCP value for management traffic in **Global Settings**.

Transcoding location	The system location to handle media transcoding for calls (signaling) received in, or sent from, this location. For calls received on a Proxying Edge Node, the media connection with the calling device is handled by a proxying node in this location, and the media is forwarded to a Transcoding Conferencing Node in the nominated **Transcoding location** (or an overflow location if necessary). For calls received on a Transcoding Conferencing Node, the media connection with the calling device and the transcoding is handled by a Transcoding Conferencing Node in the nominated **Transcoding location**. By default the transcoding location is set to ***This location*** i.e. the same location as where the call signaling is being handled, but you can change it to any of the other configured locations. You always need to choose a different transcoding location if this location contains Proxying Edge Nodes. ⓘ All calls have to be transcoded somewhere. You should ensure that the selected location contains Transcoding Conferencing Nodes, otherwise calls will fail due to insufficient resources (unless you have also configured a primary or secondary overflow location). Note that if you change the media-handling locations for a proxying location (i.e. its transcoding, primary or secondary overflow locations), any proxied calls from that location that are **currently** being handled by the previously configured media locations will be dropped. See Handling of media and signaling for more information.
Primary overflow location	An optional field where you can select the system location to handle media when capacity is reached in the **Transcoding location**, for calls (signaling) being handled in this location. For more information, see Media overflow locations.
Secondary overflow location	An optional field where you can select the system location to handle media when capacity is reached in both the **Transcoding location** and the **Primary overflow location**, for calls (signaling) being handled in this location.
Pexip Infinity domain	This is an optional field where you can specify the name of the SIP domain that is: • Routed from Skype for Business to Pexip Infinity, either as a static route or via federation. • Used as the default domain in the **From** address for outgoing SIP gateway calls and outbound SIP calls from conferences without a valid SIP URI as an alias. If configured, it is used instead of the global Pexip Infinity domain in outbound calls from Conferencing Nodes in this location.
Policy profile	The policy profile to be used by Conferencing Nodes in this location. For more information see Using external and local policy to control Pexip Infinity behavior.
Event sinks	The external service(s) to which Conferencing Nodes in this location send event information. For more information, see Using event sinks to monitor conference and participant status.
Enable PIN brute force resistance in this location	Whether PIN brute force resistance is enabled for the Conferencing Nodes in this location: • ***Use Global PIN brute force resistance setting***: as per the global configuration setting. • ***No***: PIN brute force resistance is disabled for nodes in this location. • ***Yes***: PIN brute force resistance is enabled for nodes in this location. When some locations have protection enabled, and other locations do not, the PIN brute force resistance setting is applied according to the location of the node that receives the call signaling. Default: ***Use Global PIN brute force resistance setting***.
Enable VOIP scanner resistance in this location	Whether VOIP scanner resistance is enabled for the Conferencing Nodes in this location: • ***Use Global VOIP scanner resistance setting***: as per the global configuration setting. • ***No***: VOIP scanner resistance is disabled for nodes in this location. • ***Yes***: VOIP scanner resistance is enabled for nodes in this location. When some locations have protection enabled, and other locations do not, the VOIP scanner resistance setting is applied according to the location of the node that receives the call signaling. Default: ***Use Global VOIP scanner resistance setting***.

Live captions outgoing locations	In order of precedence, select the system locations to use to handle live captions for calls received in, or sent from this location. When captions are enabled, a Conferencing Node in the nominated location connects to the **Live captions service API gateway** (AIMS server) configured in the global settings. The system tries to use the first location and will overflow to the second or to the third choice location if the preferred location reaches its transcoding capacity or for any networking issues. Note that: • If no outgoing locations are configured, the live captions feature is disabled for all Conferencing Nodes in this location. • Outgoing locations must only contain Transcoding Conferencing Nodes. • Outgoing locations cannot be cloud bursting locations. Live captions are configured via **Platform > Global Settings > Live Captions**, and enabled globally by default and/or on a per VMR or per Call Routing Rule basis. See Global Settings for more information.

About H.323 gatekeepers and SIP proxies

You can configure the Pexip Infinity platform with the addresses of one or more H.323 gatekeepers and SIP proxies. These are the call control systems that can be used to route outbound H.323 and SIP calls on behalf of Pexip Infinity.

The H.323 gatekeepers and SIP proxies configured here are used for **outbound calls from** Pexip Infinity. They do not determine the systems used to route **inbound calls to** Pexip Infinity. Pexip Infinity **will not** automatically register with any systems configured here. To route inbound calls from a gatekeeper or SIP proxy, you must configure these systems with neighbor zones that direct calls to Pexip Infinity. For more information see Call control.

Outbound calls are made from Pexip Infinity when:

- a conference participant uses a Pexip app to add another participant to the call
- the administrator manually dials out to a participant from a Virtual Meeting Room
- a participant is automatically dialed out to from a Virtual Meeting Room
- a third party uses the API to place a call to a participant
- the Infinity Gateway is used to interwork a call.

Nominating which H.323 gatekeeper and SIP proxy to use

To configure H.323 gatekeepers, go to **Call Control > H.323 Gatekeepers**, and to configure SIP proxies, go to **Call Control > SIP Proxies**.

When configuring the proxy or gatekeeper, the target **Address** can be an IP address or an FQDN, and for a SIP proxy you must also specify the transport protocol.

If the target address is an FQDN, Pexip Infinity looks for DNS SRV records first if the **Port** field is blank. If a port is specified then Pexip Infinity only performs a DNS A / AAAA lookup. When using SRV records, if multiple records are returned they are used in order based on priority and weight, and if the connection to the target host(s) fails then the host(s) associated with the target of the next SRV record are tried (as per RFC3263 for SIP and H.323 Annex O).

Example DNS SRV records for locating H.323 gatekeepers and SIP proxies

This example shows the DNS SRV records required for H.323 gatekeepers and SIP proxies, which in this case are two Cisco VCSs in the vcs.example.com domain. In this example there are two h323ls service records for the two target VCS gatekeepers, and there is a set of two service records for the two target VCS SIP proxies for each transport protocol (TCP, TLS i.e. sips and UDP). Note that if the SIP proxy or gatekeeper is hosted in a remote environment, you (as the administrator for your own domain) will not have any authority over the DNS records for that remote domain — and are not responsible for creating them — you should check that such records exist when troubleshooting any outbound calling issues.

Name	Service	Protocol	Priority	Weight	Port	Target host
vcs.example.com.	h323ls	udp	10	10	1719	vcs01.vcs.example.com.
vcs.example.com.	h323ls	udp	10	10	1719	vcs02.vcs.example.com.

Name	Service	Protocol	Priority	Weight	Port	Target host
vcs.example.com.	sip	tcp	10	10	5060	vcs01.vcs.example.com.
vcs.example.com.	sip	tcp	10	10	5060	vcs02.vcs.example.com.
vcs.example.com.	sips	tcp	10	10	5061	vcs01.vcs.example.com.
vcs.example.com.	sips	tcp	10	10	5061	vcs02.vcs.example.com.
vcs.example.com.	sip	udp	10	10	5060	vcs01.vcs.example.com.
vcs.example.com.	sip	udp	10	10	5060	vcs02.vcs.example.com.

In your actual deployment, both the **Name** and the **Target host** should be changed to use the real domain name and host names of the call control systems.

After adding the details of the H.323 gatekeeper or SIP proxy, you must then associate it with the relevant location (**Platform > Locations**) or Call Routing Rule (**Services > Call Routing**).

- Each System location can have a nominated H.323 gatekeeper and SIP proxy — these are used when adding a new H.323 or SIP participant to a conference and define where to route the outbound H.323/SIP calls placed from nodes within that location.
- Each Call Routing Rule can have a nominated H.323 gatekeeper or SIP proxy — these are used to route the outbound leg of gateway calls for rules that are targeted at external H.323 and SIP systems.

If a Pexip app user adds an H.323 or SIP participant to a conference, the call is routed to the H.323 gatekeeper or SIP proxy that is associated with the system location of the Conferencing Node to which the Pexip app is connected. If the location does not have a configured H.323 gatekeeper or SIP proxy, Pexip app users connected to Conferencing Nodes in that location may still be able to dial in other participants to a conference — to enable this, the user must enter the participant's IP address or FQDN (and for the latter, DNS must be configured properly), thus allowing Pexip Infinity to dial the participant directly.

About Skype for Business servers

Pexip Infinity uses Skype for Business servers to route **outbound** MS-SIP calls.

Outbound calls are made from Pexip Infinity when:

- a conference participant uses a Pexip app to add another participant to the call
- the administrator manually dials out to a participant from a Virtual Meeting Room
- a participant is automatically dialed out to from a Virtual Meeting Room
- a third party uses the API to place a call to a participant
- the Infinity Gateway is used to interwork a call.

You can configure the addresses of one or more SfB servers within Pexip Infinity. These can be Front End Servers or a Director; they cannot be Edge Servers.

Within an **on-prem** Skype for Business environment, you must then associate a specific SfB server with a System location, Call Routing Rule or Virtual Reception to instruct Pexip Infinity to use that SfB server when it places an outbound MS-SIP call from a Conferencing Node in that location, or when it matches that routing rule.

Within a **public DMZ** Skype for Business deployment, you must ensure that a Pexip Infinity location is **not** configured to route calls to a specific SfB server. This is to ensure that each Conferencing Node uses DNS to locate an appropriate SfB system via which to route outbound calls.

To add, edit or delete SfB servers, go to **Call Control > Skype For Business ServerS**.

The available options are:

Option	Description
Name	The name used to refer to this SfB server in the Pexip Infinity Administrator interface.
Description	An optional description of the SfB server.

Option	Description
Address	The IP address or FQDN of the SfB server. This can be a Front End Server or Director; it must **not** be an Edge Server.
Port	The IP port on the SfB server to which the Conferencing Node will connect. Default: 5061.
Transport	The IP transport used to connect to the SfB server. Default: ***TLS***.

To associate a SfB server with a System location, go to **Platform > Locations**, to associate it with a Call Routing Rule, go to **Services > Call Routing**, and to associate it with a Virtual Reception that is acting as an IVR gateway to scheduled and ad hoc SfB meetings, go to **Services > Virtual Receptions**.

For more information on integrating Pexip Infinity with SfB, see Using Microsoft Skype for Business with Pexip Infinity.

Using TURN servers with Pexip Infinity

A TURN server is a media relay/proxy that allows peers to exchange UDP or TCP media traffic whenever one or both parties are behind NAT.

ICE (Interactive Connectivity Establishment) is a framework that allows endpoints to discover paths through which they can exchange media — such as via a TURN server — where direct routing between those devices may not be possible, e.g. when a device is on a private network or is behind a NAT.

Conferencing Nodes

Pexip Conferencing Nodes can utilize a TURN server and negotiate TURN relays with the following ICE capable clients:

- Skype for Business clients
- WebRTC clients (the web app on the latest browsers, and the desktop and mobile clients)

If these endpoints will be connecting to privately-addressed "on-premises" Conferencing Nodes, you **must** configure Pexip Infinity with the address of at least one TURN server that it can offer to ICE clients.

A TURN server is not required if your Conferencing Nodes are publicly-addressable, either directly or via static NAT. For more information, see When is a reverse proxy, TURN server or STUN server required?.

Note that the H.323 specification has no concept of ICE.

The TURN servers that you use with Pexip Infinity:

- Must have a public address (located either on the public internet or in a DMZ).
- Unless a separate STUN server has also been configured, they must be deployed in such a way that traffic from the Conferencing Node towards the TURN server appears as coming from the Conferencing Node's public NAT address (its server reflexive address), otherwise this will prohibit some Skype for Business communication scenarios.
- Must be routable from your Conferencing Nodes.
- Must be standards-based (supporting RFC 5766), for example a Pexip TURN server (see Pexip Reverse Proxy and TURN Server Deployment Guide) or a VCS Expressway.

When using a TURN server with a Conferencing Node:

- Conferencing Nodes only use TURN over UDP (not TCP). However, Conferencing Nodes will perform ICE TCP negotiation.
- Conferencing Nodes always communicate with its configured TURN server over a single UDP port (default UDP/3478). UDP media is multiplexed from the Conferencing Node to that single port on the TURN server. The TURN server will reply back to the same port pair on the Conferencing Node. The TURN server never initiates a connection towards a Conferencing Node.
- As general good practice, we always recommend deploying the TURN server in a suitably secured network segment, such as a DMZ.

Note that Microsoft A/V Edge Server does not support RFC 5766 and therefore cannot be used as a TURN server with Pexip Infinity.

WebRTC clients (Pexip app)

WebRTC clients can also be provisioned with the details of one or more TURN servers that they can use to set up bindings on a TURN server for that client to use to exchange media with other ICE-capable clients.

This may typically be required in direct media scenarios (where media is not sent to, or transcoded by, Conferencing Nodes) where the clients cannot route media directly between each other and must use a TURN server to relay the media instead.

In direct media scenarios we recommend using the Pexip TURN server (see Pexip Reverse Proxy and TURN Server Deployment Guide), and you must configure it to run in **permissive** mode. If you use an alternative TURN server it must be a Coturn implementation.

When using direct media we strongly recommend for enhanced security that you use your own dedicated TURN server that is located in your DMZ.

Nominating the TURN servers used by Pexip Infinity and the Pexip apps

Within Pexip Infinity you can configure one or more TURN servers. You then associate those TURN servers with each System location (with separate configuration for the TURN server used by Conferencing Nodes in that location, and the TURN server details to provision to Pexip apps connected to that Conferencing Node), and with each Call Routing Rule. The same TURN server can be used by more than one location or rule.

Configuring TURN servers

To add, edit or delete TURN server connection details, go to **Call Control > TURN Servers**. The options are:

Option	Description
Name	The name used to refer to this TURN server in the Pexip Infinity Administrator interface.
Description	An optional description of the TURN server.
Address	The IP address or FQDN of the TURN server.
Port	The IP port on the TURN server to which the Conferencing Node or Pexip app will connect. Default: 3478. Conferencing Node behavior: • When a port is specified, Pexip Infinity performs a DNS A-record lookup on the **Address**. • If a port is not specified, then a **_turn._udp** DNS SRV lookup on the **Address** is performed (and the port in the SRV record is used). There is no fallback to an A record lookup if the SRV lookup fails. The behavior of the Pexip app depends on the client web browser (and its implementation of RFC 5928).
TURN server type	Select the type of TURN server and authentication method: • ***Username & Password***: authentication is performed with username and password credentials. • ***Time-limited credentials***: authenticates via a shared secret for limited time period (currently 5 hours). Default: ***Username & Password***
TURN transport type	Select the network transport type for communication with the TURN server: ***UDP***, ***TLS*** or ***TCP***. If you want to support multiple transport types you need to define multiple TURN servers. Default: ***UDP***.
Username and Password	The credentials of an account on the TURN server that can be used to create bindings. (Only applies for a **TURN server type** of ***Username & Password***.)
Shared secret	The shared secret used to authenticate with the TURN server. Maximum length: 100 characters. See https://datatracker.ietf.org/doc/html/draft-uberti-behave-turn-rest-00 for more information about using time-limited credentials. (Only applies for a **TURN server type** of ***Time-limited credentials***.)

After adding the details of the TURN server, you must add it to the relevant locations and Call Routing Rules.

Associating TURN servers with Conferencing Nodes

To associate a TURN server with a Conferencing Node, you must configure the node's system location:

1. Go to **Platform > Locations**.
2. Select the Conferencing Node's location.
3. Select a **TURN server** and select **Save**.

All Conferencing Nodes in that location will use the nominated TURN server for conference calls.

Associating TURN servers with gateway calls

If a gateway call is being placed to a Skype for Business client or Google Meet, the Conferencing Node placing the call will use the TURN server associated with the matching rule. (For gateway calls, the Conferencing Node does not use the TURN sever associated with its system location.)

To associate a TURN server address with a Call Routing Rule:

1. Go to **Services > Call Routing**.
2. Select the relevant rule.
3. Select a **TURN server** and select **Save**.

Configuring the TURN servers provided to Pexip apps

To configure the specific TURN servers that are provisioned to WebRTC clients (i.e. the Pexip app), you must configure the system locations used by the Conferencing Nodes that the clients connect to:

1. Go to **Platform > Locations**.
2. Select the Conferencing Node's location.
3. Select your **Client TURN servers**.
4. Optionally, you can select **Enforce media routing via a client TURN server** if you want to force the WebRTC client to route its media through one of the specified client TURN servers.
5. Select **Save**.
6. Repeat for other locations as necessary.

When a WebRTC client connects to a Conferencing Node in that location, the Conferencing Node will provide it with the details of the nominated TURN servers. These TURN servers may be used by the client to provide a media connectivity path if it cannot make a direct media connection to another client.

How Conferencing Nodes decide which TURN server to offer, and to provision to WebRTC clients

For standard call scenarios, where media is sent to and is transcoded by Conferencing Nodes, the TURN server offered by Conferencing Nodes to ICE clients is determined as follows:

- **Conferences**: uses the **TURN server** associated with the location of the Conferencing Node that is handling the call signaling.
- **Calls placed via the Infinity Gateway**: uses the **TURN server** associated with the Call Routing Rule that matched the call request. If there is no TURN server associated with the rule, then the TURN server associated with the location of the Conferencing Node that is handling the call is used instead. Note that rules can optionally be configured on a per-location basis.

If a TURN server is not configured for the location or rule, a TURN server relay candidate will not be offered.

When a Conferencing Node receives a call from an ICE client, it sends a request to the TURN server to allocate bindings to be used by that client. The client uses these bindings to route its call media through the firewall to the Conferencing Node. The Conferencing Node allocates up to four TURN bindings per call (made up of two TURN bindings per media stream for both audio and video).

Direct media scenarios

In direct media scenarios (where media is not sent to, or transcoded by, Conferencing Nodes) the clients are provisioned with (and should use) the **Client TURN servers** that are associated with the location of the Conferencing Node that is handling the call signaling.

Using STUN servers with Pexip Infinity

A STUN server allows clients, such as Conferencing Nodes or WebRTC clients (such as the Pexip web app), to find out their public NAT address.

When a client is deployed behind a NAT, it can send a STUN request to the STUN server, which responds back to the client and tells it from which IP address it received the STUN request. Using this method, the client can discover its public NAT address, which is important in order for ICE to work between Conferencing Nodes and other ICE-enabled clients (for example, WebRTC and Skype for Business clients). In relation to ICE, this public NAT address is also known as the server reflexive address.

In Microsoft Skype for Business deployments it is essential that a Conferencing Node can discover its public NAT address.

Conferencing Nodes

If a Conferencing Node is deployed on a private network behind a NAT, its system location may already be configured with the details of a TURN server (such as the Pexip TURN server). Often, that TURN server can act as a STUN server and a separate STUN server is not normally required.

By default, Conferencing Nodes send their STUN requests to the TURN server, but if the TURN server is not located outside of the enterprise firewall then the Conferencing Node will not be able to discover its public NAT address. If this is the case in your deployment scenario, you must configure a separate STUN server — the Conferencing Node's STUN requests will then be sent to the STUN server, instead of the TURN server.

A STUN server is not required if:

- your Conferencing Nodes are publicly-addressable, either directly or via static NAT, or
- the STUN requests sent from the Conferencing Nodes to the TURN server will return the public NAT address of the Conferencing Node.

The STUN servers used by Pexip Infinity must be located outside of the enterprise firewall and must be routable from your Conferencing Nodes.

Pexip apps

When connecting to a privately-addressed Conferencing Node, Pexip WebRTC clients that are behind a NAT may also use a STUN server to find out their public NAT address.

When a Pexip app connects to a Conferencing Node, the node will provision it with any **Client STUN server** addresses that are configured against that node's system location. The WebRTC client can then use those STUN servers to discover its public NAT address. This is typically only required if the WebRTC client is communicating via a TURN server.

For more information, see When is a reverse proxy, TURN server or STUN server required?.

How Conferencing Nodes decide which STUN server to use

The STUN server used by a Pexip Infinity Conferencing Node handling a call is determined as follows:

- **Conferences**: uses the STUN server associated with the location of the Conferencing Node that is handling the call signaling.
- **Point-to-point calls via the Infinity Gateway**: uses the STUN server associated with the Call Routing Rule that matched the call request. If there is no STUN server associated with the rule, then the STUN server associated with the location of the Conferencing Node that is handling the call signaling is used instead. Note that rules can optionally be configured on a per-location basis.

If a STUN server is not configured for a location or rule, but a TURN server is configured, the Conferencing Node will send STUN requests to that TURN server.

Nominating the STUN servers used by Pexip Infinity and Pexip WebRTC clients

Within Pexip Infinity you can configure the addresses of one or more STUN servers. You then associate those STUN servers with each System location (with separate configuration for the STUN server used by Conferencing Nodes in that location, and the STUN servers to offer to Pexip apps connected to that Conferencing Node), and with each Call Routing Rule.

Configuring STUN server addresses

To add, edit or delete STUN server connection details, go to **Call Control > STUN Servers**. The options are:

Option	Description
Name	The name used to refer to this STUN server in the Pexip Infinity Administrator interface.
Description	An optional description of the STUN server.
Address	The IP address or FQDN of the STUN server. This should not be the same address as any of your configured TURN servers.
Port	The IP port on the STUN server to which the Conferencing Node will connect. Default: 3478.

Note that Pexip Infinity ships with one STUN server address already configured by default: **stun.l.google.com**. This STUN server uses port 19302 (rather than the common 3478) and can be assigned to system locations for use by Pexip WebRTC clients.

Associating STUN server addresses with Conferencing Nodes

To associate a STUN server address with a Conferencing Node, you must configure the node's system location:

1. Go to **Platform > Locations**.
2. Select the Conferencing Node's location.
3. Select a **STUN server** and select **Save**.

All Conferencing Nodes in that location will use the nominated STUN server for conference calls.

Associating STUN server addresses with gateway calls

If a gateway call is being placed to an ICE-enabled client (such as Skype for Business clients and Pexip WebRTC clients), the Conferencing Node placing the call will use the STUN server associated with the matching rule. (For gateway calls, the Conferencing Node does not use the STUN sever associated with its system location.)

To associate a STUN server address with a Call Routing Rule:

1. Go to **Services > Call Routing**.
2. Select the relevant rule.
3. Select a **STUN server** and select **Save**.

Configuring the STUN server addresses provided to Pexip WebRTC clients

To configure the specific STUN server addresses that are provisioned to Pexip apps, you must configure the system locations used by the Conferencing Nodes that the clients connect to:

1. Go to **Platform > Locations**.
2. Select the Conferencing Node's location.
3. Select one or more **Client STUN servers** and select **Save**.

When a Pexip app connects to a Conferencing Node in that location, the Conferencing Node will provide it with the addresses of the nominated STUN servers. These STUN servers are used by the client to discover its public NAT address.

If no **Client STUN servers** are configured for that node/location, the Pexip app may still be able to communicate by using a TURN relay, if a TURN server is configured on the Conferencing Node, but this may cause delays in setting up media.

For clients on the same network as the Conferencing Nodes, where no NAT is present, users may find that WebRTC call setup time is improved by removing all **Client STUN servers**.

Configuring policy profiles

Policy profiles specify how Pexip Infinity uses external policy and/or local policy to control its call policy and routing decisions.

Each policy profile can be used to:

- control which types of data (e.g. service configuration, participant data, media location, participant avatars etc.) are managed via policy — either by external policy, or by local policy (where local policy is supported for that data type) or by both external and local policy
- nominate the address of an external policy server to which the external policy API requests are sent
- specify the local policy jinja2 script to be executed against the data (service configuration, participant and media location data types only).

You can configure Pexip Infinity to use both external and local policy depending on your requirements. When both external and local policy are enabled, external policy is applied first to retrieve the configuration data from the external system, and then local policy is applied to that retrieved data (which can then conditionally modify that data). See Using external and local policy to control Pexip Infinity behavior for more information.

Each system location is configured with a policy profile and that profile is then used by all of the Conferencing Nodes in that location whenever they need to retrieve configuration data. This means that you could use the same policy profile in all locations (and thus all Conferencing Nodes), or if required you can configure many different profiles with, for example, different local policy scripts or different external policy server URIs, and then assign different policy profiles to different system locations.

ⓘ You must assign policy profiles to locations otherwise they will never be used. If you want to configure just one policy profile to be used globally you need to assign it to all of your locations.

When using external policy within a system location, you must ensure that each Conferencing Node in that location is able to reach the nominated policy server.

To configure policy profiles:

1. Go to **Call Control > Policy Profiles**.
2. Select **Add Policy profile** and then configure that profile. The options are:

Option	Description
Name	The name used to refer to this policy profile in the Pexip Infinity Administrator interface.
Description	An optional description of the policy profile.
External policy server	
URL	The URL of the policy server to use for all external policy API requests from this profile, for example **https://policy.example.com/path**. You can only configure one address URL per policy server. If the request is over HTTPS, Pexip Infinity must trust the certificate presented by the policy server. ⓘ We strongly recommend that you use HTTPS (not HTTP) in production environments.
Username	Optional fields where you can specify the credentials required to access the external policy server. External policy requests support Basic Authentication and basic ASCII-encoded usernames and passwords.
Password	
Avatar policy	
Prefer local avatar configuration	When **Use local avatar configuration** is enabled, requests to fetch avatar images to represent directory contacts and conference participants are sent to the **Avatar URL** associated with the user configured within Pexip Infinity.
Enable external avatar lookup	If enabled, requests are sent to the external policy server to fetch avatar images to represent directory contacts and conference participants. If both **Use local avatar configuration** and **Enable external avatar lookup** are enabled, then the local avatar configuration takes precedence. However, if no matching user record is found, or the user record does not have a configured **Avatar URL** then a request is made to the external policy server instead. If there is an **Avatar URL**, and the request fails for any reason, Pexip Infinity will not fall back to external policy.

Option	Description
Service configuration policy	
Enable external service configuration lookup	If enabled, requests are sent to the external policy server to fetch service configuration data (VMRs, Virtual Receptions, Infinity Gateway calls etc).
Apply local policy	If enabled, the service configuration retrieved from the local database or an external policy server is processed by the local policy script (which may change the service configuration or cause the call to be rejected).
Script	Only applies if **Apply local policy** is selected. Enter a jinja2 script that takes the existing service configuration (if any) and optionally modifies or overrides the service settings.
Participant policy	
Enable external participant lookup	If enabled, requests are sent to the external policy server to allow some of the participant's properties to be overridden, or the call to be rejected.
Apply local policy	If enabled, the original participant's call properties or any override properties returned from external policy are processed by the local policy script (which may itself override the participant properties or cause the call to be rejected).
Script	Only applies if **Apply local policy** is selected. A Jinja2 script that takes the existing participant configuration and optionally overrides the participant settings
Media location policy	
Enable external media location lookup	If enabled, requests are sent to the external policy server to fetch the system location to use for media allocation.
Apply local policy	If enabled, the media location configuration retrieved from the local database or an external policy server is processed by the local policy script (which may change the media location configuration).
Script	Only applies if **Apply local policy** is selected. Enter a jinja2 script that takes the existing media location configuration and optionally modifies or overrides the location settings.
Directory	
Enable external directory lookup	If enabled, requests are sent to the external policy server to fetch directory information (that can be used by some Pexip apps to display a phonebook).
Registration requests	
Enable external registration policy	If enabled, requests are sent to the external policy server to determine whether a device alias is allowed to register to a Conferencing Node.

3. Select **Save**.
4. Go to **Platform > Locations**.
5. Select each location in turn and specify the **Policy profile** that the Conferencing Nodes in that location should use when making policy decisions.

Pexip Infinity license installation and usage

Before you can use the Pexip Infinity platform to make calls, you must install appropriate licenses.

Note that in addition to having sufficient **licenses**, your deployment must also have sufficient **capacity** to support the number of calls being placed. For more information, see Capacity planning.

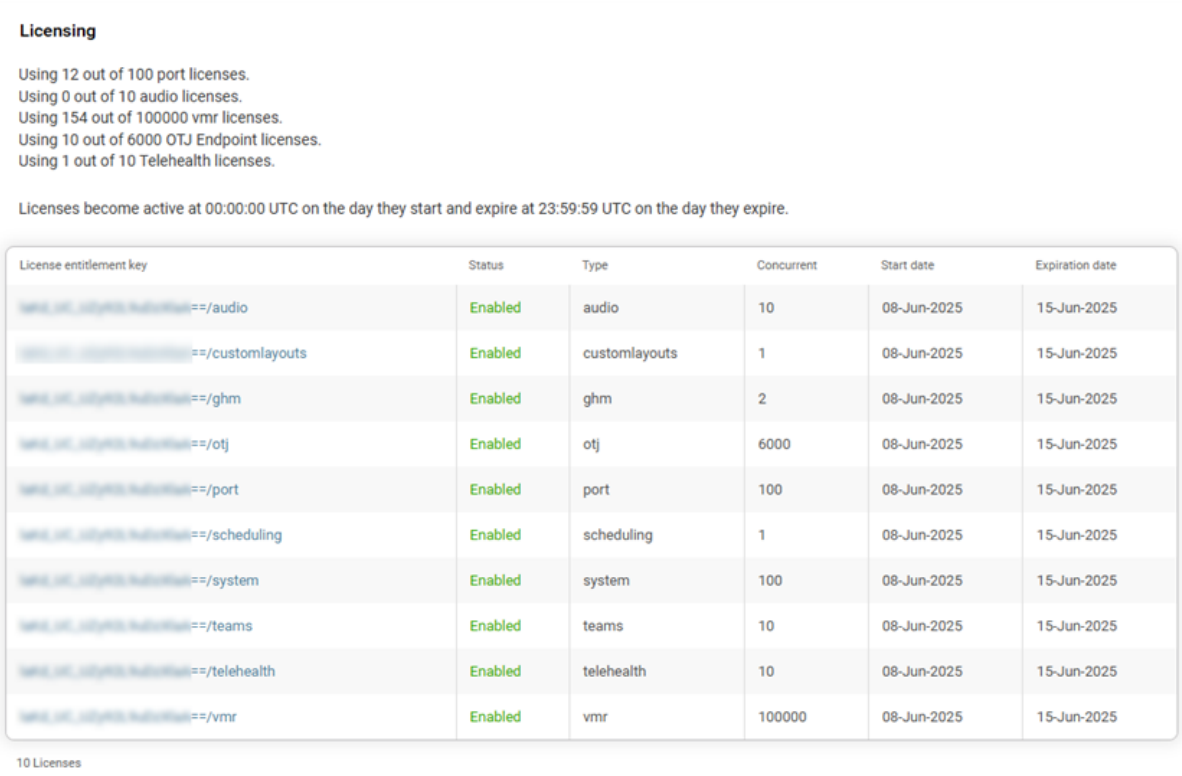
The licenses described in this topic are those that can be installed on Pexip Infinity. Other Pexip products that integrate with the Pexip Infinity platform (such as Pexip Private AI, Scheduler for Web and ERM) require their own separate licensing.

Types of licenses

The terms of your Pexip Infinity license can vary according to your licensing agreement. However, your license ultimately permits the use of the Pexip Infinity software, and additionally specifies:

- the total number of concurrent calls that can be made across the entire Pexip Infinity deployment
- the total number of Virtual Meeting Rooms and Virtual Auditoriums that can be configured at any one time
- whether the Secure Scheduler for Exchange feature is enabled
- the total number of endpoints that can be used for One-Touch Join
- the total number of Google Meet access tokens you can configure for Google Meet integration
- the total number of Microsoft Azure tenants you can configure for Microsoft Teams integration, and whether you are authorized to make and receive 1:1 calls via Microsoft Teams Room devices.
- whether your own custom layouts can be used
- whether Epic telehealth profiles can be configured
- whether Zoom trusted join can be used
- and a system license that permits basic operation of the Pexip Infinity platform.

There is no limit to how many Conferencing Node servers you can deploy, or to the number of locations in which you deploy them. No special licenses are required to deploy to cloud IaaS services such as Microsoft Azure, AWS or Google Cloud Platform. Many other features and services such as registrations, call control, branding, and interoperability with Microsoft Skype for Business also do not require any additional licenses.



Licensing

Using 12 out of 100 port licenses.
Using 0 out of 10 audio licenses.
Using 154 out of 100000 vmr licenses.
Using 10 out of 6000 OTJ Endpoint licenses.
Using 1 out of 10 Telehealth licenses.

Licenses become active at 00:00:00 UTC on the day they start and expire at 23:59:59 UTC on the day they expire.

License entitlement key	Status	Type	Concurrent	Start date	Expiration date
[redacted]==/audio	Enabled	audio	10	08-Jun-2025	15-Jun-2025
[redacted]==/customlayouts	Enabled	customlayouts	1	08-Jun-2025	15-Jun-2025
[redacted]==/ghm	Enabled	ghm	2	08-Jun-2025	15-Jun-2025
[redacted]==/otj	Enabled	otj	6000	08-Jun-2025	15-Jun-2025
[redacted]==/port	Enabled	port	100	08-Jun-2025	15-Jun-2025
[redacted]==/scheduling	Enabled	scheduling	1	08-Jun-2025	15-Jun-2025
[redacted]==/system	Enabled	system	100	08-Jun-2025	15-Jun-2025
[redacted]==/teams	Enabled	teams	10	08-Jun-2025	15-Jun-2025
[redacted]==/telehealth	Enabled	telehealth	10	08-Jun-2025	15-Jun-2025
[redacted]==/vmr	Enabled	vmr	100000	08-Jun-2025	15-Jun-2025

10 Licenses

Examples of the different types of licenses that can be installed

Concurrent calls

Before you can place calls to Pexip Infinity services (Virtual Meeting Room, Virtual Auditoriums, Virtual Receptions and the Infinity Gateway), you must install sufficient call licenses. Two types are available:

- **port**: the total number of concurrent video calls (allows video, audio, and sending and receiving screen share and full-motion HD presentations). Port licenses are mandatory.
- **audio**: the total number of concurrent audio-only calls (the audio-only participant can also send and receive screen share and full-motion HD presentations). Audio licenses are optional.

The Live view page (**Status > Live View**) lets you review current and historic usage charts showing a breakdown of participants by location, protocol, license type and the different conference types being hosted.

VMRs

The **vmr** license specifies the total number of Virtual Meeting Rooms and Virtual Auditoriums that can be configured at any one time.

Scheduling

The **scheduling** license is optional and enables the Secure Scheduler for Exchange feature. If you are using this feature in your deployment, you must also have sufficient VMR licenses installed. See Secure Scheduler for Exchange for more information.

One-Touch Join

The **otj** license is optional and specifies the number of endpoints that can use One-Touch Join. See One-Touch Join for more information.

Google Meet

The **ghm** license is optional and allows you to configure Google Meet access tokens, and to provide gateway services into Google Meet conferences. The **ghm** license specifies the total number of access tokens you can configure. Note that appropriate call licenses are also required for each gateway call that is placed into a Google Meet conference.

Microsoft Teams / Azure tenants and Teams Room calling

The **teams** license is optional. It allows you to configure Microsoft Azure tenants and route calls to Microsoft Teams. The **teams** license controls how many Azure tenants can be configured; there is no limit to the number of Teams Connectors you can install or the number of instances within each Teams Connector. Multiple Teams tenants (tenant IDs) can use the same Teams Connector.

Additionally, the **mtr** license key indicates you are authorized to make and receive 1:1 calls via Microsoft Teams Room devices.

Note that appropriate call licenses are also required for each gateway call that is placed into a Microsoft Teams conference.

See Integrating Microsoft Teams with Pexip Infinity for more information.

Custom layouts

The **customlayouts** license is optional and allows you to upload your own custom layouts as part of a theme. See Custom layouts for more information.

Epic telehealth profiles

The **telehealth** license is optional and allows you to configure Epic telehealth profiles. See Epic telehealth integration with Pexip Infinity for more information.

System license

The **system** license permits basic operation of the Pexip Infinity platform and is always provided with every Pexip Infinity purchase. In the Administrator interface it may appear as a distinct license or it may alternatively be coupled with the **port** license.

Zoom trusted join

The **zoomtrustedjoin** license enables enhanced interoperability for Zoom Rooms calling into Teams meetings.

Call license allocation

When a Transcoding Conferencing Node receives a request from a participant to join a conference or place a call via the Infinity Gateway, it contacts the Management Node to see whether there is an appropriate license available. If so, the call is allowed and one concurrent call license is allocated to that participant for the duration of the call. When the call is terminated, the license is returned to the pool. In general, each participant consumes a single concurrent license, and no participant can consume more than one license at a time. Specifically:

- A **port** license is consumed when making any type of video call (Full HD, HD and SD).
- If your system includes **audio** licenses:
  - a participant making an audio-only call will consume an audio license
  - if a participant attempts to make an audio-only call, but all audio licenses are currently in use, a port license is used instead, if one is available; otherwise the call is rejected due to insufficient call licenses as normal. If a port license has been used, the audio-only call uses the port license for the duration of the call, even if audio licenses become available while that call is still in progress
  - if a participant who is making an audio-only call (and therefore consuming an **audio** license) escalates that call to a video call, a **port** license is used, and the audio license is returned to the pool. Likewise, if a video call becomes an audio-only call, an audio license is used and the port license is returned to the pool.
- If your system **only** includes **port** licenses, then any type of call that normally requires an audio license will consume a port license.
- If you attempt to make a call that requires a **port** license and there are no port licenses currently available, but there is an **audio** license available:
  - the call will go ahead as an audio-only call, and consume an **audio** license
  - the participant sees an "insufficient video licenses" screen, and is seen as an audio-only participant by the other conference participants
  - the call will not automatically escalate to video if a port license becomes available.
- Endpoints do not consume any additional licenses when sending or receiving presentation content or screen sharing.
- Skype for Business participants (connected directly to a VMR) who are only sending or receiving presentations consume an audio license.
- Gateway calls consume two licenses: one for the inbound leg of the call and one for the outbound leg (each license is audio or port as appropriate).
- Any participants who are directly connected to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet, do not consume a call license (they report a license type of "Not required").
- Proxying Edge Nodes do not affect the call licensing requirements for an endpoint connection.

If a Conferencing Node is unable to contact the Management Node, the call is permitted on the assumption that a license is available. After the Management Node has been out of contact for a grace period of 14 days, all calls will be rejected.

If there are no licenses available, or the existing license is invalid, the participant is advised that they cannot join the conference along with the reason why, and a corresponding message is written to the support log.

For more information about license usage, contact your Pexip authorized support representative.

Insufficient licenses

If your Pexip Infinity system reports **insufficient call licenses**, this means that there are valid licenses installed on the system, but at the point at which a participant tried to join a conference, all of the existing call licenses were in use. To remedy this, either wait until one or more existing conferences have completed, thus freeing up some call licenses, or add more call licenses to your system.

In some cases your licenses may include an overdraft. This is intended to cover instances where the number of concurrent calls temporarily exceeds the number of available licenses.

Note that if your system includes port licenses and audio licenses, and you attempt to make a call that requires a **port** license and there are no port licenses currently available, but there is an **audio** license available, then the call will go ahead as an audio-only call and consume an **audio** license. See Call license allocation above for more information.

Invalid license

If your Pexip Infinity reports an **invalid license**, this could mean that:

- the license has not been activated
- the existing license has expired
- the existing license has become corrupt (this could occur, for example, if the Management Node reboots after an upgrade and comes back up on a different physical blade with a new MAC address).

Licenses become valid at 00:00:00 UTC on the day they start and expire at 23:59:59 UTC on the day they expire.

Viewing existing licenses and current usage

To see the licenses that are installed on your system, and the number of licenses currently being consumed, go to **Platform > Licenses**.

The top of the **Licensing** section states, for each type of license installed, how many concurrent licenses are currently being consumed out of the total available. Underneath are listed all the licenses that have been activated successfully and how many concurrent calls/VMRs they specify. You can select a license to see further information about it.

The **Stored License Request** section lists any licenses that were not activated or returned automatically, and are awaiting manual processing.

Network requirements for managing licenses

Before adding, moving, returning or repairing a license:

- You must have an active internet connection from the Management Node.
- License requests are sent to **activation.pexip.com**. You must therefore configure your firewall to allow a connection from the Management Node to **activation.pexip.com** on HTTPS port 443 (unless your Management Node has a web proxy configured, in which case all licensing requests to **activation.pexip.com** are sent via the web proxy).

Adding licenses

Adding a license will not affect the ongoing activity of the platform (such as current calls or registrations) and can therefore be done during normal hours of operation.

To add a new license:

1. On the Pexip Infinity Administrator interface, go to **Platform > Licenses**.
2. Select **Add License**.
3. In the **License entitlement key** field, enter the activation key provided by your Pexip authorized support representative.
4. Leave the **Manually activate** checkbox clear (unless you have been instructed to select this option by your Pexip authorized support representative, or you need to perform a manual activation due to network connectivity issues or for security reasons).
5. Select **Save changes**.

The system will automatically generate a file containing the license request. It then attempts to contact the Pexip licensing server and send it this file to activate the license.

- If the license is activated successfully, you are returned to the **Licensing** page and the new license is shown under the **Licensing** section.
- If the activation attempt is unsuccessful (for example, if the Management Node was unable to establish a connection to the Pexip licensing server), or you selected **Manually activate**, the license is saved as a **Stored license request**. You must then activate it manually.

Moving a license (e.g. when redeploying a Management Node)

You may need to move a license from one Management Node to another. You would typically need to do this if you are redeploying a Management Node, such as when setting up a new test or demonstration environment, or changing the Management Node's IP address (see Moving, restoring or changing the IP address of the Management Node).

To move a license between Management Nodes, you must deactivate the existing license on your 'old' Management Node before reactivating it on the 'new' Management Node.

On the **old** Management Node:

1. Go to **Platform > Licenses**.
2. Select the license you want to deactivate.
3. Make a note of the **License entitlement key**.
4. Select **Return license**.

The system will attempt to contact the Pexip licensing server to automatically deactivate the license:

- If successful, the license is removed from the list of licenses.
- If unsuccessful, you must manually deactivate the license by selecting **Manually Return License** and then following the same steps as described in Manually processing a stored license request (offline activation/return).

On the **new** Management Node:

1. Go to **Platform > Licenses**.
2. Select **Add License**.
3. In the **License entitlement key** field, enter the entitlement key from the 'old' Management Node that you noted previously.
4. Select **Save**.

The system will attempt to activate the license on the 'new' Management Node:

- If the license is activated successfully, you are returned to the **Licensing** page and the new license is shown under the **Licensing** section.
- If the activation attempt is unsuccessful (for example, if the Management Node was unable to establish a connection to the Pexip licensing server), or you selected **Manually activate**, the license is saved as a **Stored license request**. You must then activate it manually.

Returning a license

You may need to return a license, for example if Pexip needs to adjust the licensing. To do this:

1. Go to **Platform > Licenses**.
2. Select the license you want to deactivate.
3. Make a note of the **License entitlement key**.
4. Select **Return license**.

The system will attempt to contact the Pexip licensing server to automatically deactivate the license:

- If successful, the license is removed from the list of licenses.
- If unsuccessful, you must manually deactivate the license by selecting **Manually Return License** and then following the same steps as described in Manually processing a stored license request (offline activation/return).

Manually processing a stored license request (offline activation/return)

To manually process a stored license request:

1. Go to **Platform > Licenses**.

   The **Stored License Request** section lists the licenses that could not be activated or returned automatically, and are awaiting manual activation/return.
2. Select the license request you want to process.
3. Select **Export stored license request**.

   This generates a file containing the request.
4. Download the request file and send it to your Pexip authorized support representative for activation/return. They will respond with a fulfillment file.
5. When you receive the fulfillment file, go back to **Platform > Licenses**, select the stored license request, and select **Complete stored license request**.
6. Browse to the location of the fulfillment file you have received.
7. Select **Save**.

The license should now be activated and appear in the **Licensing** section, or returned as appropriate.

Content of the license request file

The same license request file is used for both automatic and manual license activation/return. The content of this file includes:

- the license request information (i.e. the detail of what is being requested)
- opaque machine identifiers (computed from various information sources, but not exposing the actual values).

Managing TLS and trusted CA certificates

TLS certificates are used by the Management Node and each Conferencing Node to verify their identity to clients connecting to them over HTTPS (web) or SIP TLS. These clients include:

- video endpoints
- web browsers (including the Pexip web app)
- the Pexip app for mobile (certificates are mandatory for these clients)
- third-party video network infrastructure devices
- Outlook clients (if the Secure Scheduler for Exchange service is enabled).

You can use Pexip Infinity's inbuilt Certificate Signing Request (CSR) generator to assist in acquiring a server certificate from a Certificate Authority.

Note that communication between the Management Node and Conferencing Nodes, and between Conferencing Nodes themselves, does not rely on TLS certificates; instead it uses an IPsec transport. For more information see Encryption methodologies.

Certificates overview

The Public Key Infrastructure (PKI) provides a framework for digital certificate management. It provides the following benefits:

- **Authentication**: identities are validated to ensure that only authorized users and devices have access to a server.
- **Encryption**: sessions can be encrypted, so information can be transmitted privately.
- **Data integrity**: ensures that any messages or data transferred to and from devices and servers is not altered.

The primary elements of the PKI are:

- **Public/private key pairs**: public and private keys are used to encrypt and decrypt the information being transferred to a server. Only the private key, which is kept secret by the server, can decrypt the information that is encrypted by the public key. This mechanism is known as asymmetric cryptography (as the encryption is done using non-identical keys); the two keys are mathematically related, and whatever is encrypted with a public key can only be decrypted by its corresponding private key and vice versa.
- **Certificate**: a certificate contains the public key and information about its owner (often referred to as the subject and is typically expressed as a hostname or domain name) and its issuer (typically a trusted, third-party Certificate Authority). This certificate metadata is formatted according to the ITU-T X.509 international standard. A certificate is not considered valid unless it has been directly or indirectly signed by a trusted CA.
- **Certificate Authority (CA)**: a CA is an organization such as Symantec or Comodo that issues digital certificates to corporations after having verified the applicant's identity. The certificate is proof that a certain server or website is owned by a certain organization. A CA can use its own private key to sign the certificates it issues. That signed certificate can then be verified as being signed by a trusted CA by checking the signature against the CA's own root certificate (via its public key). However, many CAs do not sign with their root certificate, but instead with an intermediate certificate — an intermediate authority is a certificate issuer that has itself been issued by a root or another higher level intermediate authority. This method creates a chain of trust.
- **Chain of trust**: when a device validates a certificate, it compares the certificate issuer with its list of trusted CAs. If a match is not found, it checks if the certificate of the issuing CA was issued by a trusted CA, and so on until the end of the certificate chain. The top of the chain, the root certificate, must be issued by a trusted Certificate Authority. Web browsers and other clients typically have a list of CA certificates that they trust, and can thus verify the certificates of individual servers, however, the server is often required to present a full certificate chain along with its server certificate.

TLS certificate usage within Pexip Infinity

The certificates used within Pexip Infinity are standard digital certificates, but they are referred to as TLS certificates as they are used when establishing a TLS connection to a Pexip node.

The clients that connect to Pexip Infinity over TLS must trust the identity of the Certificate Authority (CA) that signed the node's TLS certificate. The Pexip Infinity platform ships with a self-signed certificate for the Management Node, and each Conferencing Node is deployed with a self-signed certificate. These certificates have a 4096 bit public key and are also appended with 2048 bit Diffie-Hellman parameters. Each self-signed certificate will expire after 30 days.

As these certificates are self-signed, they will not be trusted by clients. We therefore recommend that you replace these certificates with your own certificates that have been signed by either an external CA or a trusted internal CA.

You can use a tool such as https://www.sslshopper.com/ssl-checker.html to verify certificates and the chain of trust (specify port 5061 i.e. use the format **<domain>:5061** for the server hostname to ensure that SIP TLS connections are checked).

The Management Node and Conferencing Nodes enable HSTS (HTTP Strict Transport Security) to ensure greater security. This means that if your deployment moves from using a valid TLS certificate to using an invalid certificate (e.g. you redeploy a Conferencing Node, or your certificate expires or is invalidated for some reason) then certain web browsers will stop you from accessing that node via the web when using the DNS name of that node, until you correct the certificate issue. You may browse directly to the IP address of the node in the meantime.

In general, to achieve encrypted communication using TLS the following must happen:

1. The CA issues a signed certificate which is uploaded to the server.
2. When a client needs to communicate with the server, it sends a request to the server asking it to provide identification.
3. The server sends back a copy of its TLS certificate and its public key.
4. The client checks whether the CA that issued the certificate is one that it trusts.
5. If the CA is trusted, and if the certificate is otherwise valid, the client creates a session key encrypted with the server's public key and sends it to the server.
6. The server decrypts the session key. It then uses the session key to encrypt an acknowledgment which it sends to the client in order to initiate the encrypted communication.
7. The server and the client now encrypt all communication using the session key.

Certificate usage guidelines

When requesting/generating certificates for your Pexip Infinity platform:

- The Pexip Infinity platform requires TLS certificates with SHA256 or later signature algorithms. Certificates using legacy algorithms such as SHA1 and MD5 are not supported.
- If browsers (web apps, for example) need to access your Pexip Infinity platform, ensure that your certificates contain at least one subject alternative name (SAN) entry — typically a repeat of the Common Name. (Chrome and Firefox browsers are dropping support for Common Name matching.)
- We recommend using a 4096 bit public key (2048 bit minimum).
- Typically the certificate assigned to a Conferencing Node requires a server authentication EKU (Extended Key Usage). In advanced scenarios it may require a client authentication EKU when responding to SIP/TLS client certificate challenges.

Alarms

An alarm is raised on the Management Node if:

- The Management Node or a Conferencing Node has no associated TLS certificate.
- A TLS certificate has an incomplete chain of trust to the root CA certificate.
- One or more of your trusted CA certificates or TLS certificates is due to expire within the next 30 days.

Managing a node's TLS server certificate

You can upload, view/modify, delete and download the TLS server certificates that are used by the Management Node and by each Conferencing Node, and any client TLS certificate used by a Conferencing Node. You can also generate a certificate signing request for an existing certificate / subject name.

Note that after making any changes to certificates, you need to wait for the files to be synchronized to the relevant Management Node or Conferencing Node (typically after approximately one minute). If changing the certificates in a chain, a reboot of the associated Conferencing Nodes may be required if the changes do not produce the desired effect.

Uploading a TLS server certificate

To upload a new TLS server certificate for the Management Node or a Conferencing Node:

1. From the Pexip Infinity Administrator interface, go to **Certificates > TLS Certificates**.
2. Select **Add TLS certificate**.
3. Complete the following fields:

TLS certificate	Paste the PEM-formatted certificate into the text area or alternatively select the file containing the new TLS certificate. You must upload the certificate file that you have obtained from the Certificate Authority (typically with a .CRT or .PEM extension). Do not upload your certificate signing request (.CSR file). The certificate must be valid for the DNS hostname or FQDN of the Management Node or Conferencing Node to which it will be assigned. You can paste multiple certificates into the text area, but one of those certificates must pair with the associated private key.
Private key	Paste the PEM-formatted private key into the text area or alternatively select the file containing the private key that is associated with the new TLS certificate. Private key files typically have a .KEY or .PEM extension. Pexip Infinity supports RSA and ECDSA keys.
Private key passphrase	If the private key is encrypted, you must also supply the associated passphrase.
TLS parameters	Optionally, paste any additional PEM-formatted parameters into the text area or alternatively select the file containing the parameters that are to be associated with the new TLS certificate. Custom DH parameters and an EC curve name for ephemeral keys can be added. Such parameters can be generated through the OpenSSL toolkit using the commands `openssl dhparam` and `openssl ecparam`. For example, the command `openssl dhparam -2 -outform PEM 2048` generates 2048 bit DH parameters. Note that these parameters can alternatively be added 'as is' to the end of the TLS certificate.
Nodes	Select one or more nodes to which the new TLS server certificate is to be applied. If required, you can upload a certificate and then apply it to a node later.

4. Select **Save**.

If a certificate with the same subject name already exists (e.g. when replacing an expired certificate), the new certificate is uploaded alongside the original certificate (unless the issuer and serial number details are identical, in which case the existing certificate is updated with the new contents from the file). If the original TLS certificate was assigned to one or more Conferencing Nodes you need to move those node assignments over to the new certificate.

Viewing or modifying existing TLS certificates and changing node assignments

To view information about an existing TLS certificate, change a certificate's contents, or change the nodes to which a certificate is applied:

1. Go to **Certificates > TLS Certificates**.

   By default you are shown a list and the current status of **All certificates** that have been uploaded, and whether the certificate has a Server and/or Client authentication EKU applied. Status values include:
   - **Good**: it is a good, valid certificate.
   - **Temporary**: it is a self-signed certificate.
   - **Weak signature**: the certificate is signed with SHA-1 or another old signature. We recommend that you use SHA-256 certificates instead.
   - **Empty subject**: the certificate only has SANs. It is a valid certificate but many browsers will not accept it.
   - **Over <n> days**: all browsers do not trust certificates that are valid for longer than 825 days, and Safari does not trust certificates that are valid for longer than 398 days.

- **<n> days left**: when the certificate is due to expire within the next 60 days.
- **Expired**: when the certificate has expired.

You can alternatively select to view:

- **Certificates by Node** to see which server certificate has been assigned to the Management Node or to a particular Conferencing Node.
- **Client certificates by Node** to see which TLS client certificate has been assigned to a particular Conferencing Node.

2. Select the subject name of the certificate you want to view or modify.

   The decoded certificate data is shown, including any chain of trust information.
3. If required, you can modify the:
   - **Nodes** to which a server certificate is assigned. If you assign a certificate to a node, it will automatically replace any other certificate that was previously assigned to that node.
   - **TLS certificate** data (by expanding the **PEM-formatted data** section).
   - **TLS parameters** associated with the certificate (by expanding the **PEM-formatted data** section).

   You cannot modify the private key.
4. Select **Save**.

Uploading multiple TLS certificate files

To upload a batch of TLS certificates:

1. Go to **Certificates > TLS Certificates**.
2. Select **Import files**.
3. Select **Choose Files** to pick one or more PEM-formatted text files, or PKCS12/PFX files that you want to import.
   - The PEM formatted text files should contain one of the following:
     - trusted CA certificates
     - server TLS certificates with matching private keys.
   - Private keys can be uploaded as separate files or appended to the server TLS certificate file(s).
   - DH or EC parameters may be appended to each server TLS certificate.

   Note that trusted CA certificate files can also be imported via this method if required.
4. Enter the **Private key passphrase** if the private key is encrypted.
5. Select **Import**.

   This adds the certificates in the selected files to the existing list of TLS certificates (or to the list of trusted CA certificates, if appropriate).
6. You can then select each imported TLS certificate in turn and assign it to a Management Node or one or more Conferencing Nodes as appropriate.

If a certificate with the same subject name already exists (e.g. when replacing an expired certificate), the new certificate is uploaded alongside the original certificate (unless the issuer and serial number details are identical, in which case the existing certificate is updated with the new contents from the file). If the original TLS certificate was assigned to one or more Conferencing Nodes you need to move those node assignments over to the new certificate.

Deleting an existing TLS certificate

To delete one or more TLS certificates:

1. Go to **Certificates > TLS Certificates**.
2. Select the boxes next to the certificates to be deleted, and from the **Action** drop-down menu select ***Delete selected TLS certificates*** and select **Go**.

Downloading an existing TLS certificate

To download an existing TLS certificate:

1. Go to **Certificates > TLS Certificates**.
2. Select the subject name of the certificate you want to download.

   The certificate data is shown.

3. Go to the bottom of the page and select **Download**.

   The **Download certificate** page opens.
4. Select what you want to include the download:
   - The **Include certificate** and **Include intermediate certificates** options are selected by default. The leaf certificate is always included if you choose the PKCS12/PFX file format.
   - You can also choose to include the **private key** (in which case you must also supply a **passphrase**).
5. Select the download format, either ***PEM*** or ***PKCS # 12 (PFX)***.
6. Select **Download**.

   A file called **<subject_name>_certificate** or **<subject_name>_keycert** (if the private key was included) with either a **.pem** or **.pfx** extension is downloaded. This file contains the certificate and/or the private key as requested.

   You can also download multiple certificates into one file: select the boxes next to the certificates to be downloaded, and from the **Action** drop-down menu select ***Download*** and select **Go**. In this case the generated file is called **all_certificates** or **all_keycerts** (if private keys were included).

Note that you cannot download a temporary / self-signed certificate.

You can use the Pexip Infinity Management Node to convert PEM certificates to PFX format (or vice versa), by uploading a PEM-formatted certificate and then downloading it again in PFX format. When downloading you can also include the private key and all necessary intermediate certificates in the PFX bundle.

Replacing an existing certificate (by generating a new certificate signing request)

You can generate a certificate signing request (CSR) for an existing certificate / subject name, for example if your current certificate is soon due to expire and you want to replace it. Before generating the CSR you can change the certificate data to be included in the new request, such as adding extra subject alternative names (SANs) to those already present in the existing certificate.

For instructions on how to do this, see Requesting a certificate signing request (CSR) for an existing certificate / subject name.

Client certificates

Typically, for standard Pexip Infinity use, the certificate assigned to a Conferencing Node requires server authentication EKU (Extended Key Usage). However, in some advanced scenarios it may require client authentication capabilities when responding to SIP/TLS client certificate challenges.

Public Certificate Authorities are stopping issuing certificates with client authentication capabilities, therefore if your environment requires your Conferencing Node certificates to have client (and server) EKU then you will need to manage this through your own private Certificate Authority.

Managing trusted CA certificates

Trusted CA certificates are used within Pexip Infinity to:

- verify client certificates presented to Pexip Infinity when SIP TLS verification mode is enabled
- provide a certificate chain of trust when clients connect to a Conferencing Node over SIP TLS
- verify the server certificate on a video network infrastructure device when a Conferencing Node makes a SIP TLS outbound connection to that device, and that device chooses a cipher suite that requires authentication and a certificate is exchanged
- verify connections to an LDAP server
- verify connections to an Exchange on-premises server when using Secure Scheduler for Exchange or One-Touch Join
- verify connections to an endpoint's API, if required, when using One-Touch Join
- verify connections to an AIMS server.

Pexip Infinity ships with an inbuilt list of trusted CA certificates. This list is based on the Mozilla CA Certificate Store and cannot be modified.

In addition, you can upload a customized set of trusted CA certificates to the Pexip Infinity platform.

Chain of trust

For a server's TLS certificate to be trusted by a client, the client must be configured to trust the Certificate Authority (CA) that signed the server's certificate. Many CAs do not sign with their root certificate, but instead with an intermediate certificate. Clients, however,

may only trust the root CA. Therefore the server (in this case the Management Node or Conferencing Node) is often required to present a full certificate chain along with their TLS server certificate.

When this is the case, the chain of intermediate CA certificates must be installed on the Management Node to ensure that the certificate chain of trust is properly established when clients connect to a Conferencing Node over SIP TLS.

To do this you must upload a custom **Trusted CA certificates** file that contains all the required CA certificates, one after each other, in PEM format.

Note that, by default, Pexip Infinity will not use any intermediate CA certificates that have been uploaded to the Management Node for the purposes of verifying certificates for external services. Those external services should present their entire certificate chain. However, some third-party systems such as Azure AD DS (e.g. when used for an LDAPS connection) cannot be configured to present their entire certificate chain; in these cases you can configure Pexip Infinity to use the intermediate certificate in the TLS verification chain as described below.

Uploading and managing additional trusted CA certificates

You can upload a customized set of trusted CA certificates to the Pexip Infinity platform. Any trusted CA certificates uploaded here are used in addition to the default set of trusted CA certificates that ships with Pexip Infinity.

To manage the set of custom trusted CA certificates, go to either **Certificates > Root Trust CA Certificates** (for root-level trusted CA certificates) or **Certificates > Intermediate CA Certificates** (for intermediate CA certificates) . These pages show a list and the current status of all the trusted and intermediate CA certificates that have been uploaded. From here you can:

- **Upload a file of trusted CA certificates**: select **Import**, select **Choose Files** to pick one or more PEM files that you want to import, and then select **Import**.

  This adds the certificates in the selected files to the existing list of trusted root or intermediate CA certificates (or to the list of TLS certificates, depending on the certificate types contained in the file). If a certificate with the same subject name already exists (e.g. when replacing an expired certificate), the new certificate is uploaded alongside the original certificate (unless the issuer and serial number details are identical, in which case the existing certificate is updated with the new contents from the file).
- **View or modify an existing certificate**: select the **Serial number** of the certificate you want to view. The decoded certificate data is shown.

  If required, you can modify the PEM-formatted certificate data, or in the **Advanced** options for an intermediate certificate you can select **Trusted intermediate** (which instructs Pexip Infinity to use that certificate in the TLS verification chain), and select **Save**.
- **Download all certificates**: select **Export**. A **ca-certificates.pem** file containing all of the custom-added certificates in PEM format is created and automatically saved to your local file system.
- **Delete one or more certificates**: select the boxes next to the certificates to be deleted, and from the **Action** drop-down menu select ***Delete selected certificates*** and select **Go**.

Certificate signing requests (CSRs)

To acquire a server certificate from a Certificate Authority (CA), a certificate signing request (CSR) has to be created and submitted to the CA. You can generate a CSR from within Pexip Infinity, and then upload the returned certificate associated with that request.

You can create a new CSR for any given subject name / node, or if you have an existing certificate already installed on a Pexip Infinity node that you need to replace (for example if it is due to expire) you can create a CSR based on the existing certificate data.

By default, CSRs generated via Pexip Infinity request server certificate capabilities only. You can optionally also request client authentication capabilities.

This topic covers:

- Requesting a certificate signing request (CSR) for an existing certificate / subject name
- Creating a new certificate signing request
- Uploading the signed certificate associated with a certificate signing request
- Troubleshooting
- Modifying a CSR

Requesting a certificate signing request (CSR) for an existing certificate / subject name

You can generate a certificate signing request (CSR) for an existing certificate / subject name, for example if your current certificate is soon due to expire and you want to replace it. Before generating the CSR you can change the certificate data to be included in the new request, such as adding extra subject alternative names (SANs) to those already present in the existing certificate.

To generate a CSR for an existing certificate / subject name:

1. Go to **Certificates > TLS Certificates**.
2. Select the subject name of the certificate for which you want to generate a CSR.
   The certificate data is shown.
3. Go to the bottom of the page and select **Create certificate signing request**.
   You are taken to the Add Certificate signing request page, and the CSR data is defaulted to the contents of the certificate you selected.
4. If required you can change the certificate data, such as the subject alternative names (SANs) and subject fields.
   Note that you cannot change the private key — the CSR uses the same private key as the original certificate.
5. Select **Save**.
   The CSR is generated and you are taken to the **Change Certificate signing request** page.
6. Select **Download**.
   This downloads the CSR to your local file system, with a filename in the format **<subject-name>.csr**.
   Note that the private key is not downloaded, or included within the CSR.
7. You can now submit this CSR file to your chosen CA for signing.
   The CA will then send you a signed certificate which you can upload into Pexip Infinity (see Uploading the signed certificate associated with a certificate signing request).

Note that:

- The validity, expiry date etc. of the existing certificate is not affected when you create a CSR to replace it.
- You cannot generate a CSR for an existing temporary / self-signed certificate.
- If the CSR generation fails with a "It was not possible to automatically create a certificate signing request from this certificate" message, then there was a problem with validating the original certificate data, most likely an invalid subject name or an invalid country code. In this case you will have to create the CSR manually.

Creating a new certificate signing request

To generate a CSR within Pexip Infinity:

1. Go to **Certificates > Certificate Signing Requests**.
2. Select **Add Certificate signing request**.
3. Complete the following fields:

TLS Certificate	***Create non-renewal CSR*** is selected by default. This lets you create a new CSR. To create a renewal CSR based on an existing certificate, choose a different subject name / issuer from the list (in which case the subject name and private key fields below are not displayed).
Subject name	Select the name to use as the Common Name field of the requested certificate's subject. This is typically set to the FQDN of the node on which the certificate is to be installed. The available options are prepopulated with the FQDNs (hostname plus domain) of the Management Node and each currently deployed Conferencing Node. The list also includes any **Configured FQDN** names of your Conferencing Nodes, if such names have been configured and are different from the node's FQDN. If you want to specify a custom Common Name instead, select ***User-provided custom Common Name***.
Custom subject name	Enter the name to use as the Common Name field of the requested certificate's subject, if you have selected ***User-provided custom Common Name*** above.

Private key type	Select the type of private key to generate, or select ***Upload user-provided private key*** if you want to provide your own private key. Default: RSA (2048bit)
Private key	Only applies if you have selected ***Upload user-provided private key*** above. Enter the PEM formatted RSA or ECC private key to use when generating your CSR. You can either paste the key into the input field or upload the private key file from your local file system.
Private key passphrase	Only applies if you have selected ***Upload user-provided private key*** above. If the private key is encrypted, you must also supply the associated passphrase.
Subject alternative names	Select the subject alternative names (SANs) to include in the CSR. This allows the certificate to be used to secure a server with multiple names (such as a different DNS name), or to secure multiple servers using the same certificate. You can choose from the same list of names presented in the **Subject name** field. Note that the name you choose as the Common Name is automatically included in the generated CSR's list of SANs (even if you remove it from the **Subject alternative names** list shown here). In some deployments it may be more practical to generate a single CSR in which all of your Conferencing Node FQDNs are included in the list of SANs. This means that the same single server certificate returned by the CA can then be assigned to every Conferencing Node.
Additional subject alternative names	Optionally, enter a comma-separated list of additional subject alternative names to include in the CSR. For example when receiving SIP calls, the certificate on the Conferencing Node receiving the call should include the domain names (e.g. vc.example.com) that are used in any DNS SRV records that are used to route calls to those Conferencing Nodes.
Additional subject fields (if required you can enter the following additional CSR attributes; these are all blank by default)	
Organization name	The name of your organization.
Department	The department within your organization.
City	The city where your organization is located.
State or Province	The state or province where your organization is located.
Country	The 2 letter code of the country where your organization is located.
Advanced (in most scenarios you should leave the advanced options to their default settings)	
Server authentication EKU	Request a certificate with the serverAuth EKU assigned. Default: enabled.
Client authentication EKU	Request a certificate with the clientAuth EKU assigned. Default: disabled.
Include Microsoft certificate template extension	Select this option to specify a (Microsoft-specific) certificate template in the CSR. This is needed when using the Certification Authority MMC snap-in to request a certificate from an enterprise CA. Selecting this option causes the 'WebServer' certificate template to be specified. Default: disabled.

Include Common Name in Subject Alternative Names	Specifies whether to include the requested subject Common Name in the Subject Alternative Name field of the CSR. Default: enabled.

4. Select **Save**.
   You are taken to the **Change Certificate signing request** page.
5. Select **Download**.
   This downloads the CSR to your local file system, with a filename in the format **<subject-name>.csr**.
   Note that the private key is not downloaded, or included within the CSR.
6. You can now submit this CSR file to your chosen CA for signing.
   The CA will then send you a signed certificate which you can upload into Pexip Infinity (see below).

Uploading the signed certificate associated with a certificate signing request

When the Certificate Authority sends you a signed certificate in response to your CSR, you can upload that certificate into Pexip Infinity and assign it to one or more of your nodes. Make sure that you upload it via the **Certificate Signing Requests** page as this ensures that it is linked with the private key associated with your original CSR.

To upload the signed certificate:

1. Go to **Certificates > Certificate Signing Requests**.
2. Select the original CSR that is associated with the signed certificate.
   You are taken to the **Change Certificate signing request** page.
3. In the **Certificate** field either paste the PEM-formatted certificate into the input field or upload the certificate file from your local file system.
   The certificate file that you have obtained from the Certificate Authority typically has a .CRT or .PEM extension. Do not upload your certificate signing request (.CSR file).
4. Select **Complete**.
   Providing it is a valid certificate and is based on the original CSR:
   - the certificate is uploaded and automatically linked with the private key associated with your original CSR.
   - if you are uploading a replacement certificate (same subject name and private key) it will replace the existing certificate and maintain any existing node assignments.
   - the original CSR is deleted.
   - you are taken to the **Change TLS Certificate** page.
5. You can now assign that certificate to the Management Node or one of more Conferencing Nodes as required:
   a. From within the **Change TLS Certificate** page go to the **Nodes** field and from the **Available Nodes** list, select the nodes to which you want to assign the certificate and move them into the **Chosen Nodes** list.
   b. Go to the bottom of the page and select **Save**.

Troubleshooting

This section describes some of the error messages you may see when attempting to upload a signed certificate.

Error message	Possible cause	Resolution
Certificate and private key do not appear to be part of the same key pair	This most likely means that you have tried to upload the certificate against the wrong CSR.	Select the correct CSR and try again.

Modifying a CSR

After a CSR has been created it cannot be modified — the only available actions are to download it (for sending to a CA), or to apply the returned, signed certificate that is associated with that request.

If you need to change the content of a CSR, you should delete the original CSR and create a new CSR with the correct content.

Note that a CSR is automatically deleted when the resulting signed certificate is uploaded.

Verifying SIP TLS connections with peer systems

When a Conferencing Node attempts to connect to a SIP system, such as an endpoint or a video network infrastructure device, the Pexip Infinity platform can be configured to verify whether the certificate presented by the peer system is valid before the connection is allowed.

In addition, if the peer system is also configured to perform TLS verification checking (mutual TLS authentication), then that external system will verify that the client TLS certificate on the Conferencing Node is valid and has the appropriate client authentication capabilities.

Verifying peer certificates for SIP TLS connections

For outbound connections over SIP TLS, you can use the **SIP TLS verification mode** setting to control whether the certificate presented by the peer is verified before the connection from a Conferencing Node is allowed. When this setting is enabled:

- The peer certificate is verified (in date, issued by a trusted authority etc).
- OCSP, if enabled, is used to check that the certificate has not been revoked (see Using OCSP to check the status of certificates below for more information).
- The identity of the peer as presented in the certificate is checked against the identity expected by Pexip Infinity.
- The peer certificate must be a **server** certificate, otherwise you will get "unsupported certificate" errors in the support log.

When this setting is enabled, all peers (including endpoints connecting directly with Pexip Infinity) must have their own certificate.

To enable or disable this setting, go to **Platform > Global Settings > Security**.

Field	Description
SIP TLS verification mode	Determines whether to verify the peer certificate for outbound connections over SIP TLS. The options are: ***Off***: the peer certificate is not verified; all connections are allowed. ***On***: the peer certificate is verified, and the peer's remote identities (according to RFC5922) are compared against the Application Unique String (AUS) identified by Pexip Infinity — the SIP URI — before the connection is allowed. Default: ***Off***.

The following table summarizes the certificate requirements for outbound connections from Conferencing Nodes:

State	Outbound SIP TLS connections
When **SIP TLS verification mode** is ***Off***	Pexip Infinity will authenticate the far end if the far end selects a cipher that requires authentication and a certificate is exchanged. In which case, that certificate must be valid (issued by a trusted authority, in date etc.) and it must be a **server** certificate.
When **SIP TLS verification mode** is ***On***	As above, plus the identity of the peer as presented in the certificate must match the identity expected by Pexip Infinity. This means that Pexip Infinity cannot connect out over TLS to endpoints that don't have a certificate — you must use TCP/UDP instead.
When **OCSP state** is ***On*** or ***Override*** (and **SIP TLS verification mode** is ***On***)	The far end certificate must not be revoked.
If the far end system performs a TLS verification process	The Conferencing Node must have a **Client TLS certificate** assigned, and that certificate must have **client** authentication properties and must **not** be a self-signed certificate (see Mutual TLS authentication and client/server certificates below for more information).

Using OCSP to check the status of certificates

Pexip Infinity allows you to use Online Certificate Status Protocol (OCSP) to check whether a certificate has been revoked. TLS certificates that support OCSP are encoded with the URL of an OCSP responder — a server that will check and respond with the status

of the certificate.

OCSP checking only applies when **SIP TLS verification mode** is ***On***. To enable or disable the use of OCSP, and to configure the URL of the OCSP responder, go to **Platform > Global Settings > Security**. The options available are:

Field	Description
OCSP state	Determines whether OCSP is used to check the status of TLS certificates. ***Off***: OCSP is not used. ***On***: OCSP is used, and the request is sent to the URL specified in the TLS certificate. If no URL is specified in the TLS certificate, the **OCSP responder URL** configured below is used. ***Override***: OCSP is used, and the request is sent to the **OCSP responder URL** specified in the **OCSP responder URL** field, regardless of any URL encoded in the TLS certificate. Default: ***Off***.
OCSP responder URL	The URL to which OCSP requests are sent if either: • the **OCSP state** is set to ***On*** but no URL is present in the TLS certificate, or • the **OCSP state** is set to ***Override*** (in which case any URL present in the certificate is ignored).

Mutual TLS authentication and client/server certificates

Certificates with client authentication properties are typically not required in most Pexip Infinity deployments. However, in advanced scenarios your Conferencing Nodes' certificates may require a client authentication EKU when responding to SIP/TLS client certificate challenges.

If a Conferencing Node makes an outbound connection to, or receives an incoming connection from, a video network infrastructure device (such as a Cisco VCS) that is configured to perform mutual TLS verification checking, then that external system will verify that the certificate presented by the Conferencing Node is valid. In these cases, both the Conferencing Node and the external system can adopt the role of a client as well as acting as a server. Therefore, the certificates presented by the Conferencing Node and by the external system must have **client** and **server** capabilities, as appropriate.

- When a Conferencing Node is acting in a server capacity, the node's configured **TLS certificate** is used.
- When a Conferencing Node is acting in a client capacity, the node's configured **Client TLS certificate** is used.

Availability of client certificates

Public Certificate Authorities are stopping issuing certificates with client authentication EKU (Extended Key Usage) capabilities, due to a change in web browser trusted CA policy. By May 2026, it will be impossible to acquire new certificates with both client and server authentication EKUs from public CAs.

Therefore if your environment requires your Conferencing Node certificates to have client (and server) EKU then you will need to manage this through your own private Certificate Authority.

Requesting client certificates

When requesting certificates for your Conferencing Nodes, if you want the node to be able to verify itself as a client when connecting to an external system, then you must request that the certificate can act as a client certificate (as well as a server certificate). By default, CSRs generated via Pexip Infinity's inbuilt CSR generator only request server certificate capabilities. You can configure the advanced settings in the CSR generator to also request client certificate capabilities (by enabling the **Client authentication EKU** option).

If you create your certificate signing requests via the OpenSSL toolkit, then you can modify the `[v3_req]` section of your openssl request configuration file so that it contains `extendedKeyUsage=serverAuth, clientAuth` . Other Certificate Authorities have different methods of requesting client authentication.

Integrating with external systems

When integrating with external systems, Pexip Infinity may be challenged for authentication credentials, or in some cases communications from an external conferencing system may need to be authenticated by Pexip Infinity.

Managing SIP credentials

When sending a SIP request, Pexip Infinity may be challenged for authentication credentials by the host system or proxy.

The authentication challenge will include a realm that identifies the credentials that Pexip Infinity should use to respond to the challenge. The credentials to use will be provided separately by the administrator of the host system or proxy.

All realms, usernames and passwords are case sensitive.

To configure the SIP credentials used by Pexip Infinity for each realm, go to **Call Control > SIP Credentials**. The options are:

Option	Description
Realm	The realm defines the protection space of the host or proxy (such as its name or domain, e.g. sipproxy.example.com) that is challenging Pexip Infinity for authentication.
Username and Password	The username and password presented by Pexip Infinity when responding to the authentication challenge for the given realm.

CUCM Ad Hoc Conferencing credentials

In many video conferencing deployments, Pexip Infinity will be integrated with other third-party systems and products. An example is Cisco Unified Communications Manager (CUCM), which can be integrated with Pexip Infinity so that when users initiate the **Ad Hoc Conferencing** feature on supported endpoints, the resulting conference will be hosted on Pexip Infinity.

This type of integration requires that communications from the external system are authenticated by Pexip Infinity. (Note that the credentials for this authentication are distinct from those used to access the Pexip Infinity management API.)

Access for external systems is over HTTPS.

To configure Pexip Infinity with the authentication credentials to be used by systems which do not use the management API, go to **Platform > Global Settings > External System Integration**. The options available are:

Field	Description
External system username	The username presented to Pexip Infinity by external systems attempting to authenticate with it.
External system password	The password presented to Pexip Infinity by external systems attempting to authenticate with it.

Enabling and disabling SIP, H.323, WebRTC and RTMP

All call protocols, except for SIP over UDP, are enabled by default in your Pexip Infinity deployment.

If your deployment does not include any endpoints that support a particular protocol, for security reasons you may want to disable support for those protocols across your entire Pexip Infinity deployment.

SIP over UDP is disabled by default so as to reduce the impact of SIP spam which typically uses UDP.

The configurable protocols are:

- SIP over TCP and TLS, including MS-SIP (Skype for Business)
- SIP over UDP (for incoming calls)
- H.323
- WebRTC
- RTMP

WebRTC calls can originate from the Pexip app for Windows, and the Pexip Web App via Google Chrome, Microsoft Edge, Firefox, and Safari (version 11 onwards) browsers.

RTMP and **RTMPS** (for encrypted RTMP) are used to send conference content to streaming and recording services. RTMP authentication is supported; in this case credentials are included in the URI using the syntax **rtmps://username:password@host/...**.

The Live view page (**Status > Live View**) lets you review current and historic usage charts showing a breakdown of participants by location, protocol, license type and the different conference types being hosted.

To enable or disable a particular call protocol across your Pexip Infinity deployment:

1. Go to **Platform > Global Settings > Connectivity**.
2. Select or clear the following checkboxes, as appropriate:
   - **Enable SIP** — this affects both SIP and MS-SIP (Skype for Business) over TCP and TLS
   - **Enable SIP UDP** — this affects incoming calls only over SIP UDP
   - **Enable H.323**
   - **Enable WebRTC**
   - **Enable RTMP**
3. If you have Enabled WebRTC or Enabled RTMP and wish to use the Pexip apps or client API, you must also **Enable support for Pexip apps and Client API**.

Break-in resistance settings to mitigate rogue calls

Common attacks on videoconferencing systems include rogue calls — such as Spam Over Internet Telephony (SPIT) or toll fraud call attempts — that are targeted at an organization's SIP (or, more rarely, H.323) infrastructure. Typically the attacker will place a large volume of calls to numeric aliases to try and gain access to a VoIP to PSTN gateway — and, if successful, use the gateway to commit toll fraud.

To mitigate such attacks, the Pexip Infinity platform enables **PIN brute force resistance** and **VOIP scanner resistance** by default. If required you can disable these settings either at a global platform level, or enable/disable protection for specific locations. You can also specify an allowed set of trusted IP addresses that are exempt from the break-in checks.

These break-in resistance settings form part of a broader strategy for protecting your system; for more information see Security best practices.

This topic covers:

- Alerting the administrator to break-in attempts
- PIN brute force resistance
- VOIP scanner resistance
- Configuring the allow list of IP addresses
- Break-in prevention policy example log messages

Note that any blocks that are applied to a VMR or IP address take immediate effect across the entire Pexip Infinity platform. However, changes to the allow list of IP addresses are subject to the standard replication delay across all Conferencing Nodes.

Alerting the administrator to break-in attempts

When break-in resistance protection has been triggered, an alarm is raised on the Management Node, providing information such as the source IP address of the attack and the associated Conferencing Node. The alarm remains active for the duration of the temporary block, after which time it is lowered automatically.

Note that multiple alarms can be raised for the same remote IP address if a subsequent break-in attempt targets a different node. Secondary alarms raised against additional nodes are not subject to the maximum PIN failures or scanner attempts values as the original block takes immediate effect across the entire platform.

To monitor whether break-in resistance has been triggered in the past, you can review the alarm history, or you can review the administrator log by searching for the relevant break-in policy prevention messages.

PIN brute force resistance

When PIN brute force resistance is enabled, Pexip Infinity will temporarily block all access to a VMR that receives a significant number of incorrect PIN entry attempts (and thus may perhaps be under attack from a malicious actor). It blocks **all** new access attempts to a VMR for up to 10 minutes if more than 20 incorrect PIN entry attempts are made against that VMR in a 10 minute window (you can configure the number of allowed incorrect attempts, but you cannot change the time window). While blocked, it appears to any callers as though the VMR/alias does not exist any longer. There is also a corresponding alarm raised on the Management Node.

Note that this provides a measure of resistance against PIN cracking attacks, but it is not a substitute for having a long PIN (6 digits or longer recommended) and it does not protect against a determined and patient — or lucky — attacker. Also, enabling this feature

could potentially allow a malicious attacker or a legitimate user with incorrect access details to prevent legitimate access to VMRs or other call services for a period.

To configure PIN brute force resistance at the **platform level**:

1. Go to **Platform > Global Settings**.
2. Go to the **Break-in Resistance** section and enable or disable **PIN brute force resistance** as appropriate.

   PIN brute force resistance is enabled by default.
3. Set the **Maximum PIN failures** allowed before the VMR is blocked (the default is 20).

   If you have a lot of meetings with many participants, and with long, complex PINs, you may want to increase the maximum PIN failures limit to tolerate many users mistyping the PIN — however this comes at the expense of reducing resistance to real PIN brute-force attempts.

You can override this setting on a **per location** basis. To do this:

1. Go to **Platform > Locations** and select the required location.
2. Configure **Enable PIN brute force resistance in this location** as appropriate. The options are:
   - ***Use Global PIN brute force resistance setting***: as per the global configuration setting.
   - ***No***: PIN brute force resistance is disabled for nodes in this location.
   - ***Yes***: PIN brute force resistance is enabled for nodes in this location.

   When some locations have protection enabled, and other locations do not, the PIN brute force resistance setting is applied according to the location of the node that receives the call signaling.

   Default: ***Use Global PIN brute force resistance setting***.

VOIP scanner resistance

When VOIP scanner resistance is enabled, Pexip Infinity will temporarily block service access attempts from any unknown source IP address that dials a significant number of incorrect aliases in a short period (and thus may perhaps be attempting to scan your deployment to discover valid aliases to allow the attacker to make improper use of VMRs or gateway rules — such as toll fraud attempts). It blocks **all** new call service access attempts from an IP address if more than 20 incorrect aliases are dialed from that IP address over SIP, H.323 or WebRTC (Pexip app) in a 10 minute window (you can configure the number of allowed incorrect attempts, but you cannot change the time window). There is also a corresponding alarm raised on the Management Node.

Note that this provides a measure of resistance against scanners such as sipvicious which are sometimes used during toll-fraud attempts, but it does not defend against a determined and patient — or lucky — attacker. Also, enabling this feature could potentially allow a malicious attacker or a legitimate user with incorrect access details to prevent legitimate access to VMRs or other call services for a period, if for example, those users are behind the same firewall as other legitimate users.

To configure VOIP scanner resistance at the **platform level**:

1. Go to **Platform > Global Settings**.
2. Go to the **Break-in Resistance** section and enable or disable **VOIP scanner resistance** as appropriate.

   VOIP scanner resistance is enabled by default.
3. Set the **Maximum scanner attempts** i.e. the number of incorrect dial attempts, before the source IP address is blocked (the default is 20).

You can override this setting on a **per location** basis. To do this:

1. Go to **Platform > Locations** and select the required location.
2. Configure **Enable VOIP scanner resistance in this location** as appropriate. The options are:
   - ***Use Global VOIP scanner resistance setting***: as per the global configuration setting.
   - ***No***: VOIP scanner resistance is disabled for nodes in this location.
   - ***Yes***: VOIP scanner resistance is enabled for nodes in this location.

   When some locations have protection enabled, and other locations do not, the VOIP scanner resistance setting is applied according to the location of the node that receives the call signaling.

   Default: ***Use Global VOIP scanner resistance setting***.

Configuring the allow list of IP addresses

You can configure a set of IP addresses that are excluded from the break-in resistance checks. An address can be safelisted for scan attempts and/or incorrect PINs.

- Typically you may want to specify any addresses on a trusted network, where you do not want to penalize genuine mistakes from trusted users.
- You may want to add the address of your reverse proxy (which may itself also employ fail2ban to protect your network at the perimeter) to ignore scan attempts and to ignore incorrect PINs.
- When allowing the address of an upstream SIP or H.323 call control/SBC system, you may want to consider disabling **Ignore incorrect PINs** (i.e. so that Pexip Infinity still performs break-in checks for incorrect PINs from this address) as the call control system cannot police or rate limit incorrect PIN attempts from attackers attempting to brute force a PIN. However, if the call control system has its own VOIP scan resistance behavior — and you trust it — you can enable **Ignore selected scan attempts**.
- For performance reasons, we recommend that you don't add more than a few thousand entries to the allow list table.

To define the allowed addresses:

1. Go to **Call Control > Break-In Attempt Allow List**.
2. Select **Add Allow list address**.
3. Configure the allowed address:

Name	The name of this allow list entry.
Description	A description of the allow list entry.
Network address	The IPv4 or IPv6 address for this allow list IP address range.
Network prefix	The prefix length to use in conjunction with the network address. For example, use a **Network address** of 10.0.0.0 and a **Network prefix** of 8 to specify all addresses in the range 10.0.0.0 to 10.255.255.255. You must specify a prefix.
Ignore selected scan attempts	Select this option to allow unlimited scan attempts (incorrect aliases dialed) that are received from addresses in this range. This should only be enabled in trusted environments. Default: not selected.
Ignore incorrect PINs	Select this option to allow unlimited incorrect PIN attempts that are received from addresses in this range. This should only be enabled in trusted environments. Default: not selected.
Entry type	The type of address. This determines whether Pexip Infinity trusts or ignores any security headers (such as X-Forwarded-For headers) from that source. The options are: ◦ ***Proxy***: use this for a trusted reverse proxy IP address (or address range). Pexip Infinity **trusts** security headers from that source to truthfully reflect the IP address of callers/attackers. ◦ ***User***: use this for an end-user IP address range. Pexip Infinity **ignores** any security headers. Default: ***User***

4. Select **Save**.
5. If required, you can repeat this process to add more addresses.

Break-in prevention policy example log messages

The following examples show messages that may be logged in the **administrator.conference** module of the administrator log (**History & Logs > Administrator Log**) by the break-in prevention policies.

Logged when PIN brute force resistance has temporarily disabled a service, and for all subsequent attempts while the service is blocked:

```
Message="Break-in prevention policy blocking all attempts to join this service." ConferenceAlias="alice" Service="Alice's VMR"
Participant="Crooky McCrookface" Protocol="API" Direction="in" Remote-address="10.44.21.35" Reason="Service appears to be under PIN
break-in attack" remaining_block_duration_seconds="525"
```

Logged when VOIP scanner resistance has temporarily blocked an address:

```
Message="Participant has been quarantined by Break-in prevention policy due to excessive failed join attempts." Participant="Crooky
McCrookface" Protocol="API" Direction="in" Remote-address="10.44.21.35" Reason="Too many attempts to join non-existent aliases"
remaining_block_duration_seconds="488"
```

and then any subsequent attempts generate messages such as:

```
Message="Break-in prevention policy rejecting call attempt from quarantined caller." Protocol="API" Direction="in" Local-alias="
[u'alice']" Remote-address="10.47.250.169" Reason="Suspicious join attempt rejected" remaining_block_duration_seconds="519"
```

Enabling and disabling chat messages

Conference participants who use a chat-enabled client can send messages and share links with each other within a Virtual Meeting Room or Virtual Auditorium, and when calling each other directly via the Infinity Gateway. Supported clients include Skype for Business clients and Pexip's own Pexip apps.

Chat support is enabled by default and is configurable on a platform-wide and per-conference basis. By default, Pexip Infinity does not log or retain the contents of any chat messages.

To configure chat at the platform level:

1. Go to **Platform > Global Settings**.
2. From within the **Connectivity** section, deselect or select **Enable chat**.

You can also override the global setting on a per conference basis if required. To do this:

1. Go to **Services > Virtual Meeting Rooms**, **Services > Virtual Auditoriums** or **Services > Scheduled Conferences**.
2. From within the **Advanced Options** section, choose one of the **Enable chat** options:
   - ***Use global chat setting***: as per the global configuration setting.
   - ***Yes***: chat is enabled.
   - ***No***: chat is disabled.

   Default: ***Use global chat setting***.

When chat is disabled, Pexip apps do not show the chat window.

Providing chat to participants using unsupported clients

Conference participants who are not using one of the supported clients are not able to read or send chat messages.

However, if they have access to a web browser they can use the web app to join the conference without video or audio. This will give them access to the chat room, plus the ability to view and share presentations, view the participant list, and (if they are Host participants) control aspects of the conference.

For more information on using and administering the Pexip apps, see Introduction to Pexip apps.

Conferencing Node configuration

Deploying new Conferencing Nodes

Conferencing Nodes are virtual machines that provide the capacity for conferences. They handle all conference media and signaling.

You can deploy your Pexip Infinity platform as either a mix of Proxying Edge Nodes and Transcoding Conferencing Nodes, or as a system that only contains Transcoding Conferencing Nodes.

A typical deployment scenario is to use Proxying Edge Nodes as a front for many privately-addressed Transcoding Conferencing Nodes. Those outward-facing proxying nodes would receive all the signaling and media from endpoints and other external systems, and then forward that media onto other internally-located transcoding nodes to perform the standard Pexip Infinity transcoding, gatewaying and conferencing hosting functions.

There is no fixed limit on the number of Conferencing Nodes that you can add to the Pexip Infinity platform. Please contact your Pexip authorized support representative or your Pexip Solution Architect for guidance on sizing and capacity planning for your environment.

Each Conferencing Node must have a unique:

- name
- IP address
- hostname
- DNS name (hostname and domain)

For secure deployments, you should also:

- specify a Configured FQDN
- replace the self-signed TLS certificates with your own TLS certificates

Conferencing Nodes can be deployed with dual network interfaces (NICs). Note that you must specify both interface addresses when initially deploying the Conferencing Node; you may also need to assign a static route while deploying the node. For more information, see Conferencing Nodes with dual network interfaces (NICs).

After deploying a new Conferencing Node, it takes approximately 5 minutes before the node is available for conference hosting and for its status to be updated on the Management Node. Until it becomes available, the Management Node reports the status of the Conferencing Node as having a last contacted and last updated date of "Never". "Connectivity lost between nodes" alarms relating to that node may also appear temporarily.

All Conferencing Nodes have identical service configuration, which is obtained automatically from the Management Node.

Do not use VMware, Hyper-V or any other tools to clone instances of existing Conferencing Node virtual machines (VMs). Conferencing Nodes must always be created using the Pexip Infinity Administrator interface or management API.

Prerequisites

- All host servers **must** be synchronized with accurate time before you install the Pexip Infinity Management Node or Conferencing Nodes on them.
- You must enable NTP on the Pexip Infinity Management Node before you deploy any Conferencing Nodes.

Deployment environments

When deploying a new Conferencing Node, a generic instance of a Conferencing Node VM is created, and then it is configured with its specific details such as its IP address and hostname.

This deployment process has been partly automated for on-premises environments using VMware ESXi or KVM hypervisors. When deploying in Microsoft Hyper-V or in a cloud environment such as Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP) or Oracle Cloud Infrastructure, you must also create a suitable VM instance in that environment to host your Conferencing Node before applying a generic configuration file. You can also use a generic VM template to deploy a Conferencing Node in other environments using non-supported hypervisors or orchestration layers.

The different deployment environment options are described in the table below. Your Pexip Infinity platform can contain Conferencing Nodes deployed in any combination of these environments.

Deployment type	Description
Manual (ESXi 8.0 and above) Manual (ESXi 7.0) Manual (ESXi 6.7)	Pexip Infinity generates an **.ova** file that you must then deploy from within VMware on to an ESXi host in order to create the Conferencing Node VM. Choose the appropriate ESXi host version for your environment for deployments that use VMware. For more information, see Deploying a Conferencing Node on an ESXi host.
Manual (KVM)	Choose this option to generate an **.ova** file that is suitable for deploying on a KVM host in order to create the Conferencing Node VM. For more information, see Deploying a Conferencing Node on a KVM host.
Generic (configuration-only)	This option is most typically used when deploying a Conferencing Node in Hyper-V or in a cloud environment. For specific deployment information about each platform, see: • Installing Pexip Infinity in a Microsoft Hyper-V environment • Deploying Pexip Infinity on Amazon Web Services • Deploying Pexip Infinity on Microsoft Azure • Deploying Pexip Infinity on Google Cloud Platform • Deploying Pexip Infinity on Oracle Cloud Infrastructure Pexip Infinity generates a file containing the configuration of the Conferencing Node. You then upload this file to a generic Conferencing Node that has been created from the Pexip-supplied VM template, in order to configure it with the appropriate settings. You can also choose this option for any deployments that do not use ESXi or KVM hypervisors. For more information, see Deploying a Conferencing Node using a generic VM template and configuration file.

Deploying a Conferencing Node on an ESXi host

This process generates an **.ova** file that then must be deployed from within VMware on to an ESXi host.

Note that:

- This file is specific to the Conferencing Node being deployed. It cannot be used to deploy multiple Conferencing Nodes.
- The file is single-use. It cannot be used to re-deploy the same Conferencing Node at a later date. To re-deploy the Conferencing Node, you must first delete it from the Pexip Infinity Management Node and from the host hypervisor, and then deploy a new Conferencing Node with the same configuration as the deleted node.
- Before you start, ensure that you are currently using the same machine that you will subsequently use to upload the generated file on to your host server.

Generating, downloading and deploying the ova file

1. From the Pexip Infinity Administrator interface, go to **Platform > Conferencing Nodes** and select **Add Conferencing Node**.
2. You are now asked to provide the network configuration to be applied to the Conferencing Node, by completing the following fields:

Option	Description
Name	Enter the name to use when referring to this Conferencing Node in the Pexip Infinity Administrator interface.
Description	An optional field where you can provide more information about the Conferencing Node.

Option	Description
Role	This determines the Conferencing Node's role: ○ ***Proxying Edge Node***: a Proxying Edge Node handles all media and signaling connections with an endpoint or external device, but does not host any conferences — instead it forwards the media on to a Transcoding Conferencing Node for processing. ○ ***Transcoding Conferencing Node***: a Transcoding Conferencing Node handles all the media processing, protocol interworking, mixing and so on that is required in hosting Pexip Infinity calls and conferences. When combined with Proxying Edge Nodes, a transcoding node typically only processes the media forwarded on to it by those proxying nodes and has no direct connection with endpoints or external devices. However, a transcoding node can still receive and process the signaling and media directly from an endpoint or external device if required.
Hostname Domain	Enter the hostname and domain to assign to this Conferencing Node. Each Conferencing Node and Management Node must have a unique hostname. The Hostname and Domain together make up the Conferencing Node's DNS name or FQDN. We recommend that you assign valid DNS names to all your Conferencing Nodes.
IPv4 address	Enter the IP address to assign to this Conferencing Node when it is created.
Network mask	Enter the IP network mask to assign to this Conferencing Node. Note that **IPv4 address** and **Network mask** apply to the eth0 interface.
Gateway IPv4 address	Enter the IP address of the default gateway to assign to this Conferencing Node. Note that the **Gateway IPv4 address** is not directly associated with a network interface, except that the address entered here lies in the subnet in which either eth0 or eth1 is configured to use. Thus, if the gateway address lies in the subnet in which eth0 lives, then the gateway will be assigned to eth0, and likewise for eth1.
Secondary interface IPv4 address	The optional secondary interface IPv4 address for this Conferencing Node. If configured, this interface is used for signaling and media communications to clients, and the primary interface is used for communication with the Management Node and other Conferencing Nodes. All other traffic will be routed according to the subnets, static routes, and default gateway.
Secondary interface network mask	The optional secondary interface network mask for this Conferencing Node. Note that **Secondary interface IPv4 address** and **Secondary interface network mask** apply to the eth1 interface.
SNMP mode	Conferencing Nodes can be monitored using SNMP, and they can also be configured to send SNMP traps to an SNMP Network Management System (NMS).
System location	Select the physical location of this Conferencing Node. A system location should not contain a mixture of proxying nodes and transcoding nodes. If the system location does not already exist, you can create a new one here by clicking to the right of the field. This will open up a new window showing the **Add System Location** page.
Configured FQDN	A unique identity for this Conferencing Node, used in signaling SIP TLS Contact addresses.
TLS certificate	The TLS server certificate to use on this node. This must be a certificate that contains the above **Configured FQDN**. Each certificate is shown in the format **<subject name> (<issuer>)**.
Client TLS certificate	The TLS certificate to use on this node when responding to SIP/TLS client certificate challenges. Client TLS certificates are typically not required in most deployments.
IPv6 address	The IPv6 address for this Conferencing Node. Each Conferencing Node must have a unique IPv6 address.
Gateway IPv6 address	The IPv6 address of the default gateway. If this is left blank, the Conferencing Node listens for IPv6 Router Advertisements to obtain a gateway address.

Option	Description
IPv4 static NAT address	The public IPv4 address used by this Conferencing Node when it is located behind a NAT device. Note that if you are using NAT, you must also configure your NAT device to route the Conferencing Node's **IPv4 static NAT address** to its **IPv4 address**.
Static routes	From the list of **Available Static routes**, select the routes to assign to the node, and then use the right arrow to move the selected routes into the **Chosen Static routes** list.
Enable distributed database	This should usually be enabled (checked) for all Conferencing Nodes that are expected to be "always on", and disabled (unchecked) for nodes that are expected to only be powered on some of the time (e.g. nodes that are likely to only be operational during peak times).
Enable SSH	Determines whether this node can be accessed over SSH. ***Use Global SSH setting***: SSH access to this node is determined by the global **Enable SSH** setting (**Platform > Global Settings > Connectivity > Enable SSH**). ***Off***: this node cannot be accessed over SSH, regardless of the global **Enable SSH** setting. ***On***: this node can be accessed over SSH, regardless of the global **Enable SSH** setting. Default: ***Use Global SSH setting***.
SSH authorized keys	You can optionally assign one or more SSH authorized keys to use for SSH access. From the list of **Available SSH authorized keys**, select the keys to assign to the node, and then use the right arrow to move the selected keys into the **Chosen SSH authorized keys** list. Note that in cloud environments, this list does **not** include any of the SSH keys configured within that cloud service.
Use SSH authorized keys from cloud service	When a node is deployed in a cloud environment, you can continue to use the SSH keys configured within the cloud service where available, in addition to any of your own assigned keys (as configured in the field above). If you disable this option you can only use your own assigned keys. Default: enabled.

3. Select **Save**.
4. You are now asked to complete the following fields:

Option	Description
Deployment type	Select ***Manual (ESXi 8.0 and above)***, ***Manual (ESXi 7.0)*** or ***Manual (ESXi 6.7)*** as appropriate.
Number of virtual CPUs to assign	Enter the number of virtual CPUs to assign to the Conferencing Node. We recommend no more than one virtual CPU per physical core, unless you are making use of CPUs that support Hyper-Threading.
System memory (in megabytes) to assign	Enter the amount of RAM (in megabytes) to assign to the Conferencing Node. The number entered must be a multiple of 4. We recommend 1024 MB (1 GB) RAM for each virtual CPU. The field automatically defaults to the recommended amount, based on the number of virtual CPUs you have entered.
SSH password	Enter the password to use when logging in to this Conferencing Node's Linux operating system over SSH. The username is always ***admin***. Logging in to the operating system is required when changing passwords or for diagnostic purposes only, and should generally be done under the guidance of your Pexip authorized support representative. In particular, do not change any configuration using SSH — all changes should be made using the Pexip Infinity Administrator interface.

5. Select **Download**.
   A message appears at the top of the page: "The Conferencing Node image will download shortly or click on the following link".
   After a short while, a file with the name **pexip-<hostname>.<domain>.ova** is generated and downloaded.

Note that the generated file is only available for your current session so you should download it immediately.

6. When you want to deploy the Conferencing Node VM, use a vSphere client to log in to vCenter Server. Select the **VMs and Templates** tab, click on the **Actions** menu and select **Deploy OVF Template....**
7. Follow the on-screen prompts to deploy the **.ova** file; this is similar to the steps you used when deploying the Management Node. You should always deploy the nodes with Thick Provisioned disks.

After deploying a new Conferencing Node, it takes approximately 5 minutes before the node is available for conference hosting and for its status to be updated on the Management Node. Until it becomes available, the Management Node reports the status of the Conferencing Node as having a last contacted and last updated date of "Never". "Connectivity lost between nodes" alarms relating to that node may also appear temporarily.

Enabling automatic startup

After deploying a new Conferencing Node from VMware, you must enable automatic startup of that virtual machine (VM). In VMware, automatic startup is disabled by default for every new VM — which means that if the host server is powered down for any reason, when it restarts the VM will not restart and must be started manually.

You can only enable automatic startup after the Conferencing Node has been deployed.

To enable automatic startup using the vSphere web client (HTML 5):

1. Log in to the VM manager (vCenter Server).
2. From the navigation panel, select the **Hosts And Clusters** tab and navigate to the host server on which the node's VM is installed.
3. From the main panel, select the **Configure** tab.
4. From the left-hand panel, select **Virtual Machines > VM Startup/Shutdown**.
5. At the top right of the page, select **Edit**.
6. In the **System influence** section, select **Automatically start and stop the virtual machines with the system**.
7. Select **OK**.

Disabling EVC

We strongly recommend that you disable EVC (Enhanced vMotion Compatibility) for any ESXi clusters hosting Conferencing Nodes that include a mix of old and new CPUs. If EVC is enabled on such clusters, the Pexip Infinity platform will run more slowly because the Conferencing Nodes assume they are running on older hardware.

To disable EVC:

1. From the vSphere client's navigation panel, select the cluster.
2. From the main panel, select the **Configure** tab.
3. From the left-hand panel, select **Configuration > VMware EVC**.
   The current EVC settings are shown.
4. At the top right of the page, select **Edit** .
5. Select **Disable EVC**.

Deploying a Conferencing Node

This process generates a configuration file that then must be deployed from within Hyper-V.

Note that:

- This file is specific to the Conferencing Node being deployed. It cannot be used to deploy multiple Conferencing Nodes.
- The file is single-use. It cannot be used to re-deploy the same Conferencing Node at a later date. To re-deploy the Conferencing Node, you must first delete it from the Pexip Infinity Management Node and from the host hypervisor, and then deploy a new Conferencing Node with the same configuration as the deleted node.
- Before you start, ensure that you are currently using the same machine that you will subsequently use to upload the generated file on to your host server.
- Hyper-V deployment is only possible via manual deployment from v39 onwards. You must use the Hyper-V generic Conferencing Node image.

Generating and downloading the configuration file

1. From the Pexip Infinity Administrator interface, go to **Platform > Conferencing Nodes** and select **Add Conferencing Node**.
2. You are now asked to provide the network configuration to be applied to the Conferencing Node, by completing the following fields:

Option	Description
Name	Enter the name to use when referring to this Conferencing Node in the Pexip Infinity Administrator interface.
Description	An optional field where you can provide more information about the Conferencing Node.
Role	This determines the Conferencing Node's role: ○ ***Proxying Edge Node***: a Proxying Edge Node handles all media and signaling connections with an endpoint or external device, but does not host any conferences — instead it forwards the media on to a Transcoding Conferencing Node for processing. ○ ***Transcoding Conferencing Node***: a Transcoding Conferencing Node handles all the media processing, protocol interworking, mixing and so on that is required in hosting Pexip Infinity calls and conferences. When combined with Proxying Edge Nodes, a transcoding node typically only processes the media forwarded on to it by those proxying nodes and has no direct connection with endpoints or external devices. However, a transcoding node can still receive and process the signaling and media directly from an endpoint or external device if required.
Hostname Domain	Enter the hostname and domain to assign to this Conferencing Node. Each Conferencing Node and Management Node must have a unique hostname. The Hostname and Domain together make up the Conferencing Node's DNS name or FQDN. We recommend that you assign valid DNS names to all your Conferencing Nodes.
IPv4 address	Enter the IP address to assign to this Conferencing Node when it is created.
Network mask	Enter the IP network mask to assign to this Conferencing Node. Note that **IPv4 address** and **Network mask** apply to the eth0 interface.
Gateway IPv4 address	Enter the IP address of the default gateway to assign to this Conferencing Node. Note that the **Gateway IPv4 address** is not directly associated with a network interface, except that the address entered here lies in the subnet in which either eth0 or eth1 is configured to use. Thus, if the gateway address lies in the subnet in which eth0 lives, then the gateway will be assigned to eth0, and likewise for eth1.
Secondary interface IPv4 address	The optional secondary interface IPv4 address for this Conferencing Node. If configured, this interface is used for signaling and media communications to clients, and the primary interface is used for communication with the Management Node and other Conferencing Nodes. All other traffic will be routed according to the subnets, static routes, and default gateway.
Secondary interface network mask	The optional secondary interface network mask for this Conferencing Node. Note that **Secondary interface IPv4 address** and **Secondary interface network mask** apply to the eth1 interface.
SNMP mode	Conferencing Nodes can be monitored using SNMP, and they can also be configured to send SNMP traps to an SNMP Network Management System (NMS).
System location	Select the physical location of this Conferencing Node. A system location should not contain a mixture of proxying nodes and transcoding nodes. If the system location does not already exist, you can create a new one here by clicking to the right of the field. This will open up a new window showing the **Add System Location** page.
Configured FQDN	A unique identity for this Conferencing Node, used in signaling SIP TLS Contact addresses.

Option	Description
TLS certificate	The TLS server certificate to use on this node. This must be a certificate that contains the above **Configured FQDN**. Each certificate is shown in the format **<subject name> (<issuer>)**.
Client TLS certificate	The TLS certificate to use on this node when responding to SIP/TLS client certificate challenges. Client TLS certificates are typically not required in most deployments.
IPv6 address	The IPv6 address for this Conferencing Node. Each Conferencing Node must have a unique IPv6 address.
Gateway IPv6 address	The IPv6 address of the default gateway. If this is left blank, the Conferencing Node listens for IPv6 Router Advertisements to obtain a gateway address.
IPv4 static NAT address	The public IPv4 address used by this Conferencing Node when it is located behind a NAT device. Note that if you are using NAT, you must also configure your NAT device to route the Conferencing Node's **IPv4 static NAT address** to its **IPv4 address**.
Static routes	From the list of **Available Static routes**, select the routes to assign to the node, and then use the right arrow to move the selected routes into the **Chosen Static routes** list.
Enable distributed database	This should usually be enabled (checked) for all Conferencing Nodes that are expected to be "always on", and disabled (unchecked) for nodes that are expected to only be powered on some of the time (e.g. nodes that are likely to only be operational during peak times).
Enable SSH	Determines whether this node can be accessed over SSH. ***Use Global SSH setting***: SSH access to this node is determined by the global **Enable SSH** setting (**Platform > Global Settings > Connectivity > Enable SSH**). ***Off***: this node cannot be accessed over SSH, regardless of the global **Enable SSH** setting. ***On***: this node can be accessed over SSH, regardless of the global **Enable SSH** setting. Default: ***Use Global SSH setting***.
SSH authorized keys	You can optionally assign one or more SSH authorized keys to use for SSH access. From the list of **Available SSH authorized keys**, select the keys to assign to the node, and then use the right arrow to move the selected keys into the **Chosen SSH authorized keys** list. Note that in cloud environments, this list does **not** include any of the SSH keys configured within that cloud service.
Use SSH authorized keys from cloud service	When a node is deployed in a cloud environment, you can continue to use the SSH keys configured within the cloud service where available, in addition to any of your own assigned keys (as configured in the field above). If you disable this option you can only use your own assigned keys. Default: enabled.

3. Select **Save**.
4. You are now asked to complete the following fields:

Option	Description
Deployment type	Select ***Generic (configuration-only)***.
SSH password	Enter the password to use when logging in to this Conferencing Node's Linux operating system over SSH. The username is always ***admin***. Logging in to the operating system is required when changing passwords or for diagnostic purposes only, and should generally be done under the guidance of your Pexip authorized support representative. In particular, do not change any configuration using SSH — all changes should be made using the Pexip Infinity Administrator interface.

5. Select **Download**.

A message appears at the top of the page: "The Conferencing Node image will download shortly or click on the following link".

After a short while, an XML file with the name **pexip-<hostname>.<domain>.xml** is generated and downloaded.

Note that the generated file is only available for your current session so you should download it immediately.

Deploying the Conferencing Node

1. Download the Hyper-V generic Conferencing Node zip file **Pexip_Infinity_v41_HyperV_ConfNode_<build>.zip** from the Pexip download page, and extract the files.
2. Copy the VHDX disk image to where you store your VM images.

   We recommend a drive separate to the OS drive, for example a D:\ drive.
3. Rename the image to something more meaningful, for example **pexip-conf-node.vhdx**
4. Open a Powershell window and run the following command, changing the name/path of the vhdx file as appropriate:
   ```
   New-VM -Name pexip-conf1 -Generation 2 -VHDPath D:\pexip-conf-node.vhdx -BootDevice VHD
   ```
5. By default, Hyper-V enables secure boot for generation 2 VMs. Pexip Infinity is not compatible with this so it must be disabled. Run the command, setting the VMName parameter as appropriate:
   ```
   Set-VMFirmware -VMName pexip-conf1 -EnableSecureBoot Off
   ```
6. Open Hyper-V manager and configure the CPU and memory as required. (The default values of 1vcpu and 1GB RAM are insufficient.)
7. It may be necessary to enable Virtual LAN identification (via the Hyper-V Manager under Network adapter settings) and provide a VLAN identifier so that the virtual machine can communicate with the network.
8. Start the VM, wait for it to boot and get a DHCP IP address.
9. If you connect to the VM console from Hyper-V manager, it will show the DHCP IP address.
10. Browse to **https://<conferencing-node-ip>:8443/** and use the form provided to upload the XML configuration file to the Conferencing Node VM.
11. The Conferencing Node will apply the configuration and reboot. After rebooting, it will connect to the Management Node in the usual way.

After deploying a new Conferencing Node, it takes approximately 5 minutes before the node is available for conference hosting and for its status to be updated on the Management Node. Until it becomes available, the Management Node reports the status of the Conferencing Node as having a last contacted and last updated date of "Never". "Connectivity lost between nodes" alarms relating to that node may also appear temporarily.

Enabling automatic startup

By default, virtual machines deployed using Hyper-V are configured to restart automatically if they were running when the host server was shut down or powered off. We recommend that you leave this setting as is for the Management Node and all Conferencing Nodes.

Disabling processor compatibility mode

We strongly recommend that you disable processor compatibility mode on all Hyper-V Conferencing Node virtual machines. If processor compatibility mode is enabled, the Conferencing Node may assume it is running on older hardware, and may stop working, with the message **CPU instruction set is not supported; system will be placed in maintenance mode**.

For more information, see https://technet.microsoft.com/en-us/library/dn859550.aspx.

Deploying a Conferencing Node on a KVM host

To deploy a new Conferencing Node onto a KVM host, you must:

1. Use the Pexip Infinity Administrator interface to generate and download the .vmdk image.
2. Convert the .vmdk image for use with KVM.
3. Create a new volume on your KVM server and upload the disk image.
4. Create the Conferencing Node virtual machine.

   Note that we use the libvirt command line tools to perform the import as they provide greater control than Virtual Machine Manager.
5. Enable the virtual machine for automatic startup.

These steps are described in detail below.

Note that:

- This file is specific to the Conferencing Node being deployed. It cannot be used to deploy multiple Conferencing Nodes.
- The file is single-use. It cannot be used to re-deploy the same Conferencing Node at a later date. To re-deploy the Conferencing Node, you must first delete it from the Pexip Infinity Management Node and from the host hypervisor, and then deploy a new Conferencing Node with the same configuration as the deleted node.
- Before you start, ensure that you are currently using the same machine that you will subsequently use to upload the generated file on to your host server.

Generate and download the .vmdk image

1. From the Pexip Infinity Administrator interface, go to **Platform > Conferencing Nodes** and select **Add Conferencing Node**.
2. You are now asked to provide the network configuration to be applied to the Conferencing Node, by completing the following fields:

Option	Description
Name	Enter the name to use when referring to this Conferencing Node in the Pexip Infinity Administrator interface.
Description	An optional field where you can provide more information about the Conferencing Node.
Role	This determines the Conferencing Node's role: ◦ ***Proxying Edge Node***: a Proxying Edge Node handles all media and signaling connections with an endpoint or external device, but does not host any conferences — instead it forwards the media on to a Transcoding Conferencing Node for processing. ◦ ***Transcoding Conferencing Node***: a Transcoding Conferencing Node handles all the media processing, protocol interworking, mixing and so on that is required in hosting Pexip Infinity calls and conferences. When combined with Proxying Edge Nodes, a transcoding node typically only processes the media forwarded on to it by those proxying nodes and has no direct connection with endpoints or external devices. However, a transcoding node can still receive and process the signaling and media directly from an endpoint or external device if required.
Hostname Domain	Enter the hostname and domain to assign to this Conferencing Node. Each Conferencing Node and Management Node must have a unique hostname. The Hostname and Domain together make up the Conferencing Node's DNS name or FQDN. We recommend that you assign valid DNS names to all your Conferencing Nodes.
IPv4 address	Enter the IP address to assign to this Conferencing Node when it is created.
Network mask	Enter the IP network mask to assign to this Conferencing Node. Note that **IPv4 address** and **Network mask** apply to the eth0 interface.
Gateway IPv4 address	Enter the IP address of the default gateway to assign to this Conferencing Node. Note that the **Gateway IPv4 address** is not directly associated with a network interface, except that the address entered here lies in the subnet in which either eth0 or eth1 is configured to use. Thus, if the gateway address lies in the subnet in which eth0 lives, then the gateway will be assigned to eth0, and likewise for eth1.
Secondary interface IPv4 address	The optional secondary interface IPv4 address for this Conferencing Node. If configured, this interface is used for signaling and media communications to clients, and the primary interface is used for communication with the Management Node and other Conferencing Nodes. All other traffic will be routed according to the subnets, static routes, and default gateway.
Secondary interface network mask	The optional secondary interface network mask for this Conferencing Node. Note that **Secondary interface IPv4 address** and **Secondary interface network mask** apply to the eth1 interface.
SNMP mode	Conferencing Nodes can be monitored using SNMP, and they can also be configured to send SNMP traps to an SNMP Network Management System (NMS).

Option	Description
System location	Select the physical location of this Conferencing Node. A system location should not contain a mixture of proxying nodes and transcoding nodes. If the system location does not already exist, you can create a new one here by clicking to the right of the field. This will open up a new window showing the **Add System Location** page.
Configured FQDN	A unique identity for this Conferencing Node, used in signaling SIP TLS Contact addresses.
TLS certificate	The TLS server certificate to use on this node. This must be a certificate that contains the above **Configured FQDN**. Each certificate is shown in the format **<subject name> (<issuer>)**.
Client TLS certificate	The TLS certificate to use on this node when responding to SIP/TLS client certificate challenges. Client TLS certificates are typically not required in most deployments.
IPv6 address	The IPv6 address for this Conferencing Node. Each Conferencing Node must have a unique IPv6 address.
Gateway IPv6 address	The IPv6 address of the default gateway. If this is left blank, the Conferencing Node listens for IPv6 Router Advertisements to obtain a gateway address.
IPv4 static NAT address	The public IPv4 address used by this Conferencing Node when it is located behind a NAT device. Note that if you are using NAT, you must also configure your NAT device to route the Conferencing Node's **IPv4 static NAT address** to its **IPv4 address**.
Static routes	From the list of **Available Static routes**, select the routes to assign to the node, and then use the right arrow to move the selected routes into the **Chosen Static routes** list.
Enable distributed database	This should usually be enabled (checked) for all Conferencing Nodes that are expected to be "always on", and disabled (unchecked) for nodes that are expected to only be powered on some of the time (e.g. nodes that are likely to only be operational during peak times).
Enable SSH	Determines whether this node can be accessed over SSH. ***Use Global SSH setting***: SSH access to this node is determined by the global **Enable SSH** setting (**Platform > Global Settings > Connectivity > Enable SSH**). ***Off***: this node cannot be accessed over SSH, regardless of the global **Enable SSH** setting. ***On***: this node can be accessed over SSH, regardless of the global **Enable SSH** setting. Default: ***Use Global SSH setting***.
SSH authorized keys	You can optionally assign one or more SSH authorized keys to use for SSH access. From the list of **Available SSH authorized keys**, select the keys to assign to the node, and then use the right arrow to move the selected keys into the **Chosen SSH authorized keys** list. Note that in cloud environments, this list does **not** include any of the SSH keys configured within that cloud service.
Use SSH authorized keys from cloud service	When a node is deployed in a cloud environment, you can continue to use the SSH keys configured within the cloud service where available, in addition to any of your own assigned keys (as configured in the field above). If you disable this option you can only use your own assigned keys. Default: enabled.

3. Select **Save**.
4. You are now asked to complete the following fields:

Option	Description
Deployment type	Select ***Manual (KVM)***.

Option	Description
SSH password	Enter the password to use when logging in to this Conferencing Node's Linux operating system over SSH. The username is always ***admin***. Logging in to the operating system is required when changing passwords or for diagnostic purposes only, and should generally be done under the guidance of your Pexip authorized support representative. In particular, do not change any configuration using SSH — all changes should be made using the Pexip Infinity Administrator interface.

5. Select **Download**.
   A message appears at the top of the page: "The Conferencing Node image will download shortly or click on the following link".
   After a short while, a file with the name **pexip-<hostname>.<domain>.vmdk** is generated and downloaded.
   Note that the generated file is only available for your current session so you should download it immediately.

Convert the .vmdk image for use with KVM

To use the Conferencing Node VMDK image file with KVM, you must convert it to raw format:

1. Copy the downloaded VMDK file (named **pexip-<hostname>.<domain>.vmdk**) to the server running KVM.
2. Convert the disk image from VMDK to raw, using the command:
   ```
   qemu-img convert -O raw <downloaded filename> pexip-disk01.raw
   ```
   (This conversion process can take several seconds.)

Create a new volume and upload the disk image

Next, you create a new volume on your KVM server and upload the converted disk image. From within your KVM environment:

1. Use **virsh** to create a new volume on your KVM server:
   ```
   virsh vol-create-as <poolname> <volume_name> 49G --format raw
   ```
   where:
   - `<poolname>` is the name of the storage pool in which to create the volume; typically you would use **default**. (To determine the storage pools available on the target system, use `virsh pool-list`.)
   - `<volume_name>` is the name of your new volume.
   - `49G` is the virtual size of the volume; always use 49G for a Conferencing Node.

   For example:
   ```
   virsh vol-create-as default pexip-conf-01 49G --format raw
   ```
   This example creates a volume named **pexip-conf-01** of size 49 GB and format raw in the storage pool named **default**.
2. Upload the converted disk image to the newly created volume:
   ```
   virsh vol-upload <volume_name> pexip-disk01.raw --pool <poolname>
   ```
   For example:
   ```
   virsh vol-upload pexip-conf-01 pexip-disk01.raw --pool default
   ```
   This example uploads the **pexip-disk01.raw** image to the newly created volume, **pexip-conf-01**, in the storage pool named **default**.

Create the virtual machine

After the disk image has been uploaded, you can create the virtual machine to use it.

Note that we use the libvirt command line tools to perform the import as they provide greater control than Virtual Machine Manager.

1. Identify the filesystem path of the newly uploaded disk image:
   ```
   virsh vol-path <volume_name> --pool <poolname>
   ```
   For example:
   ```
   virsh vol-path pexip-conf-01 --pool default
   ```
   This prints out the absolute path to the disk image file, for example:
   ```
   /var/lib/libvirt/images/pexip-conf-01
   ```
   This path is used in the **disk path** parameter in the **virt-install** command below.
2. Use the **virt-install** command line tool to create the virtual machine:

```
virt-install \
  --import \
  --hvm \
  --name=<vm_name> \
  --arch=x86_64 \
  --vcpus=4 \
  --ram=4096 \
  --cpu host \
  --osinfo=linux2024 \
  --connect=qemu://<hostname>/system \
  --virt-type kvm \
  --disk path=<image_file_path>,bus=virtio,format=raw,cache=none,io=native \
  --network bridge=br0,model=virtio \
  --memballoon virtio \
  --graphics vnc,listen=0.0.0.0,password=<password>
```

This creates a new VM (KVM domain) from the converted disk image.

The command options are described below (items in **bold** may be changed as necessary):

Option	Description
--import	Build guest domain around pre-installed disk image; do not attempt to install a new OS.
--hvm	Create a fully virtualized (i.e. not paravirtualized) VM.
--name=**<vm_name>**	Name of the new VM, where <vm_name> is, for example, **pexip-conf01-vm**.
--arch=x86_64	CPU architecture of new VM (must be x84_64).
--vcpus=**4**	Number of CPUs allocated to new VM. By default, this is **4** for the Conferencing Node.
--ram=**4096**	Memory allocated to new VM (in megabytes).
--cpu host	Expose all host CPU capabilities to new VM (CPUID).
--osinfo=linux**2024**	The guest OS is Linux. Valid year values are **2018** or later. For older osinfo databases, you may need to use **--osinfo=$(virt-install --osinfo list | grep '^linux' | sort -r | head -n 1)**
--connect=qemu://**<hostname>**/system	Connect to KVM on the target system, where <hostname> is the hostname of your KVM server.
--virt-type kvm	Use KVM to host the new VM.
--disk path=**<image_file_path>**, bus=virtio,format=raw,cache=none,io=native	◦ Define the location of the disk image file, where <image_file_path> is as determined in the previous step, for example **/var/lib/libvirt/images/pexip-conf-01**. ◦ Expose it to the guest on the virtio paravirtualized bus (as opposed to IDE/SCSI). ◦ Define the image file as being in raw format. ◦ Instruct the host system not to cache the disk contents in memory. ◦ Use the native IO backend to access the disk device.
--network **bridge=br0,model=virtio**	◦ Create a network interface connected to the **br0** bridge interface on the host. ◦ Expose it to the guest as a **virtio** paravirtualized NIC.
--memballoon virtio	Expose the virtio memory balloon to the guest.
--graphics vnc,listen=**0.0.0.0**, password=**<password>**	Expose the graphical console over VNC, listening on **0.0.0.0** (i.e. all addresses on the target system) and with an access password of **<password>**.

You may receive a warning "Unable to connect to graphical console: virt-viewer not installed"; if so, this message can be safely ignored.

After the VM has been created, it may be managed using the Virtual Machine Manager desktop interface (virt-manager application) or via the command line interface (virsh).

The new node should start automatically. If it does not you can use the Virtual Machine Manager to start the node, or the CLI command:

```
virsh start <vm_name>
```

Note that you can list existing VMs by using:

```
virsh list
```

After deploying a new Conferencing Node, it takes approximately 5 minutes before the node is available for conference hosting and for its status to be updated on the Management Node. Until it becomes available, the Management Node reports the status of the Conferencing Node as having a last contacted and last updated date of "Never". "Connectivity lost between nodes" alarms relating to that node may also appear temporarily.

Enabling automatic startup

After deploying a new Conferencing Node in KVM, you should enable automatic startup of that virtual machine (VM). In KVM, automatic startup is disabled by default for every new VM. This means that if the host server is powered down for any reason, when it restarts the VM will not restart and must be started manually.

You can only enable automatic startup after the Conferencing Node has been deployed.

To enable automatic startup using Virtual Machine Manager:

1. Connect to the Virtual Machine Manager (virt-manager) that is managing the node's VM.
2. Select the node's VM and then, from the toolbar, select the **Show the virtual machine console and details** icon .
   A new window for that VM is opened.
3. If necessary, select **View > Details** to display the VM information.
4. From the sidebar menu, select **Boot Options**.
5. Select the **Start virtual machine on host boot up** check box.
6. Select **Apply**.

Deploying a Conferencing Node using a generic VM template and configuration file

You can also use a generic Pexip Infinity VM template to deploy Conferencing Nodes on other non-supported hypervisors or orchestration layers.

Creating a new generic Conferencing Node is a two-step process:

1. Creating a generic instance of a Conferencing Node VM, using a template provided by Pexip.
2. Configuring the VM with the details of the specific Conferencing Node being deployed, using a file generated from the Pexip Infinity Management Node.

Prerequisites

- The Conferencing Node must be deployed in a VM environment that supports address assignment by DHCP.
- You must know the IP address that will initially be assigned to the Conferencing Node. You will use this IP address to connect to the VM in order to upload the configuration file, but this configuration file may then assign a new IP address to the Conferencing Node.
- Create a system location (**Platform > Locations**) for the Conferencing Node's physical location.

Creating a generic Conferencing Node VM

To create a new generic instance of a Conferencing Node using the VM template:

1. Within your chosen environment, go to https://dl.pexip.com/infinity/index.html, select the appropriate directory for your software version, and then download one of the following files and use it to create a generic instance of a Conferencing Node:
   - **Pexip_Infinity_v41_generic_ConfNode_<build>.ova** in environments that take .ova or .ovf files as input.
   - **Pexip_Infinity_v41_HyperV_ConfNode_<build>.zip** for Hyper-V environments.

   Ensure you are using a VM template with the same Pexip Infinity software version as that which is currently running on the Management Node. This includes dot releases — so for example, for a v27.1 Management Node you must install a v27.1 Conferencing Node rather than a v27 Conferencing Node. If the Management Node has been upgraded, you will need to download

the Conferencing Node VM template corresponding to that software version. For more information, see Upgrading configuration-only deployments.

2. Within your hypervisor, configure the generic Conferencing Node VM with the appropriate number of virtual CPUs and amount of RAM.

Generating, downloading and deploying the configuration file

1. From the Pexip Infinity Administrator interface, go to **Platform > Conferencing Nodes** and select **Add Conferencing Node**.
2. You are now asked to provide the network configuration to be applied to the Conferencing Node, by completing the following fields:

Option	Description
Name	Enter the name to use when referring to this Conferencing Node in the Pexip Infinity Administrator interface.
Description	An optional field where you can provide more information about the Conferencing Node.
Role	This determines the Conferencing Node's role: ○ ***Proxying Edge Node***: a Proxying Edge Node handles all media and signaling connections with an endpoint or external device, but does not host any conferences — instead it forwards the media on to a Transcoding Conferencing Node for processing. ○ ***Transcoding Conferencing Node***: a Transcoding Conferencing Node handles all the media processing, protocol interworking, mixing and so on that is required in hosting Pexip Infinity calls and conferences. When combined with Proxying Edge Nodes, a transcoding node typically only processes the media forwarded on to it by those proxying nodes and has no direct connection with endpoints or external devices. However, a transcoding node can still receive and process the signaling and media directly from an endpoint or external device if required.
Hostname Domain	Enter the hostname and domain to assign to this Conferencing Node. Each Conferencing Node and Management Node must have a unique hostname. The Hostname and Domain together make up the Conferencing Node's DNS name or FQDN. We recommend that you assign valid DNS names to all your Conferencing Nodes.
IPv4 address	Enter the IP address to assign to this Conferencing Node when it is created.
Network mask	Enter the IP network mask to assign to this Conferencing Node. Note that **IPv4 address** and **Network mask** apply to the eth0 interface.
Gateway IPv4 address	Enter the IP address of the default gateway to assign to this Conferencing Node. Note that the **Gateway IPv4 address** is not directly associated with a network interface, except that the address entered here lies in the subnet in which either eth0 or eth1 is configured to use. Thus, if the gateway address lies in the subnet in which eth0 lives, then the gateway will be assigned to eth0, and likewise for eth1.
Secondary interface IPv4 address	The optional secondary interface IPv4 address for this Conferencing Node. If configured, this interface is used for signaling and media communications to clients, and the primary interface is used for communication with the Management Node and other Conferencing Nodes. All other traffic will be routed according to the subnets, static routes, and default gateway.
Secondary interface network mask	The optional secondary interface network mask for this Conferencing Node. Note that **Secondary interface IPv4 address** and **Secondary interface network mask** apply to the eth1 interface.
SNMP mode	Conferencing Nodes can be monitored using SNMP, and they can also be configured to send SNMP traps to an SNMP Network Management System (NMS).

Option	Description
System location	Select the physical location of this Conferencing Node. A system location should not contain a mixture of proxying nodes and transcoding nodes. If the system location does not already exist, you can create a new one here by clicking to the right of the field. This will open up a new window showing the **Add System Location** page.
Configured FQDN	A unique identity for this Conferencing Node, used in signaling SIP TLS Contact addresses.
TLS certificate	The TLS server certificate to use on this node. This must be a certificate that contains the above **Configured FQDN**. Each certificate is shown in the format **<subject name> (<issuer>)**.
Client TLS certificate	The TLS certificate to use on this node when responding to SIP/TLS client certificate challenges. Client TLS certificates are typically not required in most deployments.
IPv6 address	The IPv6 address for this Conferencing Node. Each Conferencing Node must have a unique IPv6 address.
Gateway IPv6 address	The IPv6 address of the default gateway. If this is left blank, the Conferencing Node listens for IPv6 Router Advertisements to obtain a gateway address.
IPv4 static NAT address	The public IPv4 address used by this Conferencing Node when it is located behind a NAT device. Note that if you are using NAT, you must also configure your NAT device to route the Conferencing Node's **IPv4 static NAT address** to its **IPv4 address**.
Static routes	From the list of **Available Static routes**, select the routes to assign to the node, and then use the right arrow to move the selected routes into the **Chosen Static routes** list.
Enable distributed database	This should usually be enabled (checked) for all Conferencing Nodes that are expected to be "always on", and disabled (unchecked) for nodes that are expected to only be powered on some of the time (e.g. nodes that are likely to only be operational during peak times).
Enable SSH	Determines whether this node can be accessed over SSH. ***Use Global SSH setting***: SSH access to this node is determined by the global **Enable SSH** setting (**Platform > Global Settings > Connectivity > Enable SSH**). ***Off***: this node cannot be accessed over SSH, regardless of the global **Enable SSH** setting. ***On***: this node can be accessed over SSH, regardless of the global **Enable SSH** setting. Default: ***Use Global SSH setting***.
SSH authorized keys	You can optionally assign one or more SSH authorized keys to use for SSH access. From the list of **Available SSH authorized keys**, select the keys to assign to the node, and then use the right arrow to move the selected keys into the **Chosen SSH authorized keys** list. Note that in cloud environments, this list does **not** include any of the SSH keys configured within that cloud service.
Use SSH authorized keys from cloud service	When a node is deployed in a cloud environment, you can continue to use the SSH keys configured within the cloud service where available, in addition to any of your own assigned keys (as configured in the field above). If you disable this option you can only use your own assigned keys. Default: enabled.

3. Select **Save**.
4. You are now asked to complete the following fields:

Option	Description
Deployment type	Select ***Generic (configuration-only)***.

Option	Description
SSH password	Enter the password to use when logging in to this Conferencing Node's Linux operating system over SSH. The username is always ***admin***. Logging in to the operating system is required when changing passwords or for diagnostic purposes only, and should generally be done under the guidance of your Pexip authorized support representative. In particular, do not change any configuration using SSH — all changes should be made using the Pexip Infinity Administrator interface.

5. Select **Download**.

   A message appears at the top of the page: "The Conferencing Node image will download shortly or click on the following link".

   After a short while, a file with the name **pexip-<hostname>.<domain>.xml** is generated and downloaded.

   Note that the generated file is only available for your current session so you should download it immediately.
6. Browse to **https://<conferencing-node-ip>:8443/** and use the form provided to upload the configuration file to the Conferencing Node VM.

   If you cannot access the Conferencing Node, check that you have allowed the appropriate source addresses in your ingress firewall rules for management traffic. In public deployments and where there is no virtual private network, you need to use the public address of the node.

   The Conferencing Node will apply the configuration and reboot. After rebooting, it will connect to the Management Node in the usual way.

   You can close the browser window used to upload the file.

After deploying a new Conferencing Node, it takes approximately 5 minutes before the node is available for conference hosting and for its status to be updated on the Management Node. Until it becomes available, the Management Node reports the status of the Conferencing Node as having a last contacted and last updated date of "Never". "Connectivity lost between nodes" alarms relating to that node may also appear temporarily.

Enabling automatic startup

We recommend that all virtual machines are configured to restart automatically if they were running when the host server was shut down or powered off.

Assigning node hostnames and FQDNs

The Management Node and each Conferencing Node within your deployment must have a unique hostname.

Hostnames can be made of a combination of numbers, letters, and the hyphen (-) character, and are case-insensitive. Hostnames cannot be entirely numeric.

A Conferencing Node's assigned hostname/FQDN is not used in any call signaling, and can be different from the actual DNS FQDN you might use to point to that Conferencing Node.

The node's Configured FQDN is used:

- in SIP signaling (if configured)
- MS-SIP signaling (mandatory)
- for any self-referential redirects in web requests.

However, we recommend that the hostname and the domain combination that you assign should match the DNS name or FQDN that you will use to refer to the node for call routing. This makes it easier to manage your deployment, for example by:

- allowing you to use more manageable TLS certificates
- making the system easier to access for the purposes of troubleshooting.

To assign a DNS name to a Conferencing Node, enter a valid **Hostname** and **Domain** combination when first deploying the Conferencing Node.

Configuring existing Conferencing Nodes

To reconfigure or delete an existing Conferencing Node, go to **Platform > Conferencing Nodes** and click on the name of the Conferencing Node.

Any changes to the configuration of the Conferencing Node should be made using the Pexip Infinity Administrator interface. Do not make any changes using other tools (such as VMware or Hyper-V); doing so may cause the Conferencing Node to fail. The only exception is any change to the CPU and RAM allocated to the Conferencing Node — these changes should be made using the hypervisor and should only be done while the Conferencing Node is powered off.

When reconfiguring existing Conferencing Nodes, the following information can be changed:

Option	Description
Name	The name used to refer to this Conferencing Node in the Pexip Infinity Administrator interface. Each Conferencing Node must have a unique name.
Description	An optional field where you can provide more information about the Conferencing Node.
Role	This determines the Conferencing Node's role: • ***Proxying Edge Node***: a Proxying Edge Node handles all media and signaling connections with an endpoint or external device, but does not host any conferences — instead it forwards the media on to a Transcoding Conferencing Node for processing. • ***Transcoding Conferencing Node***: a Transcoding Conferencing Node handles all the media processing, protocol interworking, mixing and so on that is required in hosting Pexip Infinity calls and conferences. When combined with Proxying Edge Nodes, a transcoding node typically only processes the media forwarded on to it by those proxying nodes and has no direct connection with endpoints or external devices. However, a transcoding node can still receive and process the signaling and media directly from an endpoint or external device if required. ⓘ If you change the role of a Conferencing Node, all existing calls will be disconnected and the Conferencing Node will be restarted.
System location	The physical location of this Conferencing Node. A system location should not contain a mixture of proxying nodes and transcoding nodes. ⓘ If you change the system location of a Conferencing Node, all existing calls will be disconnected and the Conferencing Node will be restarted.
Enable maintenance mode	While maintenance mode is enabled, this Conferencing Node will not accept any new conference instances. For more information, see Taking a Conferencing Node out of service. The maintenance mode setting will persist after a reboot.
Configured FQDN	A unique identity for this Conferencing Node, used in signaling SIP TLS Contact addresses.
TLS certificate	The TLS server certificate to use on this node. This must be a certificate that contains the above **Configured FQDN**. Each certificate is shown in the format **<subject name> (<issuer>)**.
Client TLS certificate	The TLS certificate to use on this node when responding to SIP/TLS client certificate challenges. Client TLS certificates are typically not required in most deployments.
IPv6 address	The IPv6 address for this Conferencing Node. Each Conferencing Node must have a unique IPv6 address.
Gateway IPv6 address	The IPv6 address of the default gateway. If this is left blank, the Conferencing Node listens for IPv6 Router Advertisements to obtain a gateway address.
IPv4 static NAT address	The public IPv4 address used by this Conferencing Node when it is located behind a NAT device.
Static routes	From the list of **Available Static routes**, select the routes to assign to the node, and then use the right arrow to move the selected routes into the **Chosen Static routes** list.

Option	Description
Enable SSH	Determines whether this node can be accessed over SSH. ***Use Global SSH setting***: SSH access to this node is determined by the global **Enable SSH** setting (**Platform > Global Settings > Connectivity > Enable SSH**). ***Off***: this node cannot be accessed over SSH, regardless of the global **Enable SSH** setting. ***On***: this node can be accessed over SSH, regardless of the global **Enable SSH** setting. Default: ***Use Global SSH setting***.
SSH authorized keys	You can optionally assign one or more SSH authorized keys to use for SSH access. From the list of **Available SSH authorized keys**, select the keys to assign to the node, and then use the right arrow to move the selected keys into the **Chosen SSH authorized keys** list. Note that in cloud environments, this list does **not** include any of the SSH keys configured within that cloud service.
Use SSH authorized keys from cloud service	When a node is deployed in a cloud environment, you can continue to use the SSH keys configured within the cloud service where available, in addition to any of your own assigned keys (as configured in the field above). If you disable this option you can only use your own assigned keys. Default: enabled.

You can also change the node's SNMP settings (see Monitoring via SNMP for more information):

Option	Description
SNMP mode	Configures the SNMP access mode for the selected node: ***Off***: SNMP is disabled. You cannot use SNMP to query the node for its status. ***SNMPv2c read-only***: enables insecure, read-only access. ***SNMPv3 read-only***: enables secure, read-only access, using the **authPriv** security level with SHA1 authentication and AES 128-bit encryption. When enabled, access is provided to the basic RFC 1213 MIB-II tree (1.3.6.1.2.1). Default: ***Off***.
SNMP community	The SNMP group to which this node belongs. This setting applies to SNMPv2c only. Default: public
SNMPv3 username	The node's SNMPv3 username, used to authenticate SNMPv3 requests.
SNMPv3 privacy password	The node's SNMPv3 privacy password used for encrypting messages between the node and the management station. AES encryption must be used; DES is not supported.
SNMPv3 authentication password	The node's SNMPv3 authentication password, used to authenticate the associated username. The SHA authentication protocol must be used; MD5 is not supported.
SNMP system contact	The contact details (for example, email address) of the person responsible for this particular node.
SNMP system location	A description of the node's location.

Other details of the Conferencing Node that cannot be directly changed (however, see Changing the node's IP address below) are also shown on this page for your information, as follows:

Option	Description
IPv4 address	The IPv4 address of this Conferencing Node.
Network mask	The IP network mask for this Conferencing Node.
Gateway IPv4 address	The IPv4 address of the default gateway.
Secondary interface IPv4 address and network mask	The optional secondary interface IPv4 address and network mask for this Conferencing Node. These fields are only displayed if the Conferencing Node has been deployed with dual network interfaces.
Hostname Domain	The DNS hostname and domain of this Conferencing Node. Together these make up the machine's FQDN, or ***DNS Name*** in VMware. For more information, see Assigning node hostnames and FQDNs.
Enable distributed database	This should usually be enabled (checked) for all Conferencing Nodes that are expected to be "always on", and disabled (unchecked) for nodes that are expected to only be powered on some of the time (e.g. nodes that are likely to only be operational during peak times). You should avoid frequent toggling of this setting. When changing this setting on multiple Conferencing Nodes, update one node at a time, waiting a few minutes before updating the next node.

Assigning a Configured FQDN

If your deployment includes Microsoft Skype for Business, you **must** assign each Conferencing Node with a **Configured FQDN**. However, for security purposes we recommend that all deployments use Configured FQDNs.

In summary, the node's **Configured FQDN** is used:

- in SIP signaling (if configured)
- MS-SIP signaling (mandatory)
- for any self-referential redirects in web requests.

To define the **Configured FQDN**, go to **Platform > Conferencing Nodes** and select each Conferencing Node in turn.

The name in the **Configured FQDN** field is used in SIP signaling over TLS, and specifies the identity that the Conferencing Node service will use when identifying itself to the systems connecting to it. Each Conferencing Node must have a unique **Configured FQDN**.

The **Configured FQDN**:

- must match one of the identities returned in the certificate for the Conferencing Node, and
- must have an entry in DNS

so that the identity of the Conferencing Node can be verified by the systems connecting to it.

The **Configured FQDN** can be the same as the Conferencing Node's FQDN (made up of its Hostname and Domain).

If the **Configured FQDN** field is left blank, the IP address of the Conferencing Node is used in SIP TLS signaling, and, depending on your call control configuration, this may result in calls failing.

For more details on the use of domain certificates in SIP, see section 4 of RFC 5922.

Changing the node's IP address

You can change the IPv4 address of a Conferencing Node by adding a new node with the new address, and deleting the node with the old address.

If you want to preserve the Conferencing Node's existing FQDN, you should:

1. Put the Conferencing Node whose IP address you want to change into maintenance mode.
2. When all calls on that Conferencing Node have terminated, delete the Conferencing Node i.e. shut down and remove it from the hypervisor / cloud platform and then remove its details from Pexip Infinity.
3. Wait at least 90 seconds to allow the deletion to be synchronized across the platform.

4. Add a new Conferencing Node i.e. deploy a new ova file or cloud instance, with the new IP address (but using the same **Name**, **Hostname** and **Domain** etc. as used before, if required).
5. If necessary, make any changes to your call control system so that calls are routed to the Conferencing Node's new IP address.

If you want to maintain full service capacity, and do not need to preserve the node's FQDN, then you can add the new node with the new IP address before deleting the old node with the old IP address.

Do not attempt to change the IP address of a Conferencing Node using utilities available in external tools (such as VMware or Hyper-V) because this will cause Pexip Infinity services to fail.

Deleting Conferencing Nodes

When deleting a Conferencing Node, you should first delete it from the hypervisor/cloud platform and then remove its details from the Pexip Infinity platform using the Pexip Infinity Administrator interface.

To delete a Conferencing Node:

1. Put the Conferencing Node into maintenance mode and wait until all calls on it have terminated.
2. Shut down and remove the Conferencing Node VM.

   For on-premises installations:

   a. Log in to the VM manager, shut down the deleted Conferencing Node and then power it off.
   b. Right-click on the Conferencing Node and select ***Delete from Disk*** (VMware) or ***Delete*** (Hyper-V / KVM).

   For cloud-based installations:

   a. Log in to your cloud provider's management portal.
   b. Terminate the Conferencing Node instance.
   c. If using Azure, delete the resource group and storage containers which hold the instance.
3. From the Pexip Infinity Administrator interface, go to **Platform > Conferencing NodeS**.
4. Select the Conferencing Node to be deleted, and from the **Action** drop-down menu select ***Delete selected Conferencing Nodes***.
5. Select **Go** and on the following page confirm that you want to delete the selected Conferencing Node by selecting **Yes, I'm sure**.

Pexip Infinity conference types

About Pexip Infinity conferences

The Pexip Infinity platform offers a variety of conference types and services:

- Virtual Meeting Rooms and Virtual Auditoriums are used to hold conferences, share presentations, and chat. Participants can join over audio or video from any location using virtually any type of communications tool, such as Skype for Business, a traditional conferencing endpoint, a mobile telephone, or a Pexip app.
- The Virtual Reception IVR service provides a way for conference participants who cannot dial Virtual Meeting Room and Virtual Auditorium aliases directly, to access these services from a central point using DTMF tones. It can also be used to route calls via the Infinity Gateway.
- The Pexip Infinity Distributed Gateway:
  - Enables any type of endpoint, including traditional VTC endpoints, to join externally-hosted meeting services such as Microsoft Teams and Google Meet.
  - Enables endpoints within your deployment to make direct calls to other endpoints. As with calls into VMRs, the gateway can interwork the protocols and media formats used by each type of device (SIP, H.323, WebRTC etc).
  - Can be used with call control systems and other third party services to enable calls from your deployment to external devices including PSTN and mobile phones.
- Pexip's Media Playback Service allows you to play prerecorded video content (such as adverts and informational videos) to consumers. When the media finishes playing, the user can be transferred to another service, such as a VMR conference, or they can be disconnected.
- Secure Scheduler for Exchange integrates Pexip Infinity with Microsoft Exchange. It enables Microsoft Outlook desktop and Pexip web app users (using Office 365, Exchange 2013 or Exchange 2016) to schedule meetings using Pexip VMRs as a meeting resource.
- Pexip's One-Touch Join (OTJ) allows users to schedule a meeting in Microsoft Outlook or Google Calendar and include in the invitation a meeting room with a supported Cisco or Poly videoconferencing endpoint, so that the endpoint in the chosen meeting room displays a **Join** button just before the meeting is scheduled to begin. Participants can then simply walk into the room and select the button, and the endpoint will automatically dial in to the meeting.
- A Test Call Service provides a test loopback service that allows users to check the quality of their video and audio (i.e. that their local camera, microphone and speakers are working properly), and verifies that they can connect to a Conferencing Node.

The Live view page (**Status > Live View**) lets you review current and historic usage charts showing a breakdown of participants by location, protocol, license type and the different conference types being hosted.

About Virtual Meeting Rooms (VMRs) and Virtual Auditoriums

Conferences take place in Virtual Meeting Rooms (VMRs) and Virtual Auditoriums (a type of VMR that is optimized for use by a small number of Hosts and a large number of Guests). They act as a digital space that participants using virtually any type of device can join for reliable, HD group video meetings.

When configuring your VMR and Virtual Auditorium services, you define the aliases that are used by conference participants to access that service. Your services can be secured by the use of PINs or by authentication using SSO — or both, and PINs also allow you to assign differing Host and Guest privileges to conference participants (see About PINs, Hosts and Guests). You can also limit the number of participants that can join a specific meeting.

All Virtual Meeting Rooms and Virtual Auditoriums can be accessed via any Conferencing Node. When a Conferencing Node receives a call to a particular Virtual Meeting Room or Virtual Auditorium alias, it creates a conference instance based on that service's settings. In this way, resources are not used until the first caller actually places a call into the conference. When another endpoint places a call to an alias that belongs to the same Virtual Meeting Room or Virtual Auditorium, the call is placed into the existing conference instance.

You can change the audio and video prompts presented to participants when they are accessing these services by applying themes.

VMRs do not consume resources on your deployment unless they are actually being used to host a conference. The number of VMRs that can be in use at the same time is limited only by the size of your Pexip Infinity deployment (in terms of server capacity and call licenses).

VMRs can be bulk-provisioned from directory information contained in a Windows Active Directory LDAP server, or any other LDAP-accessible database.

- See Configuring Virtual Meeting Rooms (VMRs) for full information about how to configure a VMR.
- See Configuring Virtual Auditoriums for full information about how to configure a Virtual Auditorium.

Allowing end-users to manage their own VMRs

VMR settings such as Host and Guest PIN numbers and whether to display participant names are initially assigned when the VMR is created by the administrator.

The Pexip VMR self-service portal is a separately-installable component that allows end-users to manage their personal Virtual Meeting Room without having to send requests to their administrator to change the configuration or branding of their VMR.

Virtual Meeting Rooms versus Virtual Auditoriums

A Virtual Auditorium is a meeting space that is optimized for use by a small number of Hosts and a large number of Guests, for example:

- A lecture, where the lecturer joins as the Host and all the students join as Guests
- An internal team meeting, where the department heads join as Hosts and all the managers join as Guests.

Similarities

Virtual Auditoriums share many of the same features as Virtual Meeting Rooms:

- Each Virtual Auditorium has one or more aliases associated with it. Participants access the Virtual Auditorium by dialing any one of its aliases — this will route them all to the same conference. For more information, see About aliases and access numbers.
- They use PINs to protect access, and to differentiate between Host and Guest participants. For more information, see About PINs, Hosts and Guests.
- Both support participant authentication, where users must verify their identity before being permitted to join the meeting.
- You can place a limit on the number of participants that can access a Virtual Auditorium at any one time.
- Participants can access Virtual Auditoriums from any video endpoint, or by using one of Pexip's apps — for more information see Introduction to Pexip apps.
- The number of VMRs and Virtual Auditoriums that can be configured on your Pexip Infinity platform is only limited by your **vmr** license. Virtual Auditoriums do not consume resources on your deployment unless they are actually being used to host a conference. Unless you manually restrict access, the number of participants who can access a particular Virtual Auditorium, and the number of Virtual Auditoriums that can be in use at the same time, are limited only by the size of your Pexip Infinity deployment.

Differences

Virtual Auditoriums differ from Virtual Meeting Rooms in that:

- **Raised hands**:
  - **Web app**: the option for Guest participants to raise their hand, to indicate that they wish to speak, is available in both Virtual Auditoriums and Virtual Meeting Rooms. Host participants using a Pexip app can lower guests' hands, as can the Guest themselves.
- Administrators can configure a Virtual Auditorium so that Hosts and Guests see different layouts. In a Virtual Meeting Room, all participants see the same layout.
- In a Virtual Auditorium:
  - **Guests** in a Virtual Auditorium typically only see the Host participant(s), and can hear but not see any of the other Guests, even if a Guest is speaking. However, there are some circumstances where Guests may see other Guests, such as when there are no Hosts left in the conference, or if none of the Hosts are sending video. You can control the precise behavior via the **Guests can see other guests** option when configuring the advanced options for a Virtual Auditorium.
  - **Host** participants see all other Hosts first in the video thumbnails (in order of which Host spoke most recently), followed by Guests (again, in order of who spoke most recently).

  In a Virtual Meeting Room all participants can see all other participants, in order of who spoke most recently, and regardless of whether they are Hosts or Guests.

- Administrators can configure a Virtual Auditorium so that when a presentation is being shown, the person showing the presentation is fixed in the main speaker position. In a Virtual Meeting Room, this option is not available; the main speaker position is always voice-switched, showing the current speaker.

Configuring Virtual Meeting Rooms (VMRs)

A Virtual Meeting Room (VMR) is a virtual meeting space that is always available to hold a Pexip Infinity conference. It can host any number of people from any type of device. Each VMR has one or more aliases associated with it, and participants access the conference by dialing one of these aliases.

Access to VMRs can be protected by the use of PINs or by authentication using SSO — or both. You can also use PINs to separate participants into Hosts and Guests, with different privileges. For more information, see About Pexip Infinity conferences.

To create a new VMR, go to **Services > Virtual Meeting Rooms** and select **Add Virtual Meeting Room**.

To edit an existing VMR, or to view its details (including all the aliases associated with it), go to **Services > Virtual Meeting Rooms** and select the name of the VMR.

As an alternative to manually creating and configuring your VMRs:

- You can import Virtual Meeting Rooms and device aliases from directory information contained in a Windows Active Directory LDAP server, or any other LDAP-accessible database. For more information, see Provisioning VMRs, devices and users from Active Directory via LDAP.
- You can bulk import basic VMR configuration from a CSV file. See Bulk import/export of service configuration data.
- VMRs are also added by the Secure Scheduler for Exchange service; these are shown on a separate page (**Services > Scheduled Conferences**). For more information, see Secure Scheduler for Exchange.

The **vmr** license specifies the total number of Virtual Meeting Rooms and Virtual Auditoriums that can be configured at any one time.

Any configuration changes made to VMRs are replicated to all Conferencing Nodes within around 90 seconds and applied to any subsequent conferences in that VMR. If there are any conferences already in place that use the VMR, any attempts to join it after the configuration has been replicated may be affected by the new configuration settings. Therefore, we do not recommend changing VMR configuration while a conference is in progress in it, as this may lead to an inconsistent user experience.

When adding or editing Virtual Meeting Rooms, the options are:

Option	Description
Name	The name used to refer to this VMR. ⓘ If you can access this VMR via a Virtual Reception then the VMR **Name** entered here is shown to conference participants as they are transferred into the VMR (it is overlaid onto the **virtual_reception_connecting** splash screen of the theme associated with the Virtual Reception that is transferring the call).
Description	A description of the VMR.
Creation time	This read-only field shows the date and time when this record was first configured.
View	The layout controls the maximum number of other participants that each participant will see, and how the participants are arranged on the screen. For more information, see Conference layouts and speaker names. Default: ***Large main speaker and up to 7 other participants (1+7 layout)***.
Show name of active speaker	When active speaker display is enabled, the display name or alias of the current speaker is shown across the bottom of their video image. Default: ***No***.
Show names of participants	If participant name overlays are enabled, the display names or aliases of all participants are shown in a text overlay along the bottom of their video image. Default: ***No***.

Option	Description
Theme	The theme for use with this VMR. For more information, see Customizing conference images and voice prompts using themes. Default: ***<use Default theme>*** (the global default theme is used).
Owner's email address	The email address of the owner of the VMR. • For VMRs created using LDAP synchronization, provisioning messages for this VMR will be sent to this address. • For VMRs created by manual input, CSV import, or via the API, this field is optional.
Participant authentication	
Host PIN	This optional field allows you to set a secure access code that must be entered by participants before they can access the service. If **Allow Guests** is set to ***Yes***, then the **Host PIN** applies to the conference Host(s) only. For more information, see About PINs, Hosts and Guests. • PINs must use the digits **0**-**9** only. • PINs may optionally end with **#**. • PINs must be between 4–20 digits long, including any **#**. • The default minimum length is 6 digits (**Platform > Global Settings > Service Configuration > Minimum PIN Length**).
Allow Guests	***Yes***: the conference can have two types of participants: Hosts and Guests. You must configure a **Host PIN** to be used by the Hosts. You can optionally configure a **Guest PIN**; if you do not configure a **Guest PIN**, Guests can join without a PIN, but the meeting will not start until the first Host has joined. ***No***: all participants have Host privileges. Default: ***No***.
Guest PIN	This optional field allows you to set a secure access code that must be entered by Guests before they can use the service. For more information, see About PINs, Hosts and Guests. • Host PINs and Guest PINs must be different. • PINs must use the digits **0**-**9** only. • PINs may optionally end with **#**. • PINs must be between 4–20 digits long, including any **#**. • If the Host PIN ends in **#** and a Guest PIN is used, the Guest PIN must also end with **#**. • If **#** is not used, Host PINs and Guest PINs must have the same number of digits. • You cannot configure a Guest PIN unless you have already configured a Host PIN.
Host Identity Provider Group	The set of Identity Providers to be offered to Hosts to authenticate with, in order to use the service. If this is blank, Hosts are not required to authenticate. Default: none selected
Guest Identity Provider Group	The set of Identity Providers to be offered to Guests to authenticate with, in order to use the service. If this is blank, Guests are not required to authenticate. Default: none selected

Option	Description
Other participants	(Available when an **Identity Provider Group** has been selected) Determines whether participants joining a SSO-protected service from devices other than the web app (for example SIP or H.323 endpoints) are allowed to dial in to the service. • ***Disallow all***: these devices are placed in a waiting room where they must wait to be admitted by a Host. • ***Allow if trusted***: these devices may join the service if they are locally registered. They must still enter a Host PIN or Guest PIN if either is required. All other devices are placed in a waiting room where they must wait to be admitted by a Host. Default: ***Disallow all***
Advanced options	
Breakout Rooms	Allows this VMR to send its participants into different breakout rooms. See Breakout rooms for more information.
Automatically Dialed Participants	When a conference begins in this VMR, a call will be placed automatically to any participants selected here. To add an Automatically Dialed Participant that is not already on the list, select the icon to the right of the selection fields. For more information, see Automatically dialing out to a participant from a conference.
Guests can present	Controls whether the Guests in the conference are allowed to present content. • ***Yes***: Guests and Hosts can present into the conference • ***No***: only Hosts can present Default: ***Yes***
Enable chat	Whether chat messaging is enabled for the conference: • ***Use global chat setting***: as per the global configuration setting. • ***Yes***: chat is enabled. • ***No***: chat is disabled. Default: ***Use global chat setting***.
Maximum inbound call bandwidth (kbps)	Specifying a maximum inbound call bandwidth will limit the bandwidth of media received by Pexip Infinity from each individual participant dialed in to this VMR. For more information see Managing and restricting call bandwidth.
Maximum outbound call bandwidth (kbps)	Specifying a maximum outbound call bandwidth will limit the bandwidth of media sent from Pexip Infinity to each individual participant dialed in to this VMR. For more information see Managing and restricting call bandwidth.
Conference capabilities	Allows you to limit the media content of the conference. For more information, see Controlling media capability. Default: ***Main video + presentation***.
Maximum call quality	Controls the maximum call quality for participants connecting to this service: • ***Use global setting***: use the global maximum call quality setting. • ***SD***: each participant is limited to SD quality. • ***HD***: each participant is limited to HD (720p) quality. • ***Full HD (1080p)***: allows any endpoint capable of Full HD to send and receive its main video at 1080p. Default: ***Use global setting***

Option	Description
Media encryption	Controls the media encryption requirements for participants connecting to this service. • ***Use global setting***: use the global media encryption setting. • ***Best effort***: each participant will use media encryption if their device supports it, otherwise the connection will be unencrypted. • ***Required***: all participants (including RTMP participants) must use media encryption. • ***No encryption***: all H.323, SIP and MS-SIP participants must use unencrypted media. (RTMP participants will use encryption if their device supports it, otherwise the connection will be unencrypted.) Default: ***Use global setting***
Participant limit	This optional field allows you to limit the number of participants allowed to join this VMR. For more information see Limiting the number of participants.
Service tag	This optional field lets you assign a unique identifier to this VMR, which you can then use to track use of the service.
VMR origin	This read-only field shows the name of the service (if any) used to create this VMR. • For VMRs created using LDAP synchronization, this will be **LDAP Sync template:** followed by the name of the LDAP sync template being used. • For VMRs created by manual input, CSV import, or via the API, this will be blank.
Enable direct media	Allows this VMR to use direct media between participants. When enabled, the VMR provides non-transcoded, encrypted, point-to-point calls between any two WebRTC participants. The options are: • ***Best effort***: use direct media in this VMR where possible (when there are two WebRTC participants only), otherwise use standard, transcoded media connections via a Conferencing Node. • ***Always***: allow direct media calls only — this limits all calls in this VMR to two WebRTC participants. • ***Never***: do not use direct media in this VMR. Default: ***Never*** Note that direct media is not supported if breakout rooms are also enabled for this VMR. See Direct media (end-to-end encrypted calls) for more information.
Direct media notification duration	The number of seconds to show a notification to participants before being escalated into a transcoded call, or de-escalated into a direct media call. Range: 0 to 30 seconds. Default: 0 seconds
Live captions	Determines whether participants using the web app are given the option to turn on the live captions for conferences using this service. • ***Use global live captions setting***: as per global configuration setting. • ***Yes***: live captions may be enabled. • ***No***: live captions cannot be enabled. Default: ***Use global live captions setting*** Live captions are configured via **Platform > Global Settings > Live Captions**, and enabled globally by default and/or on a per VMR or per Call Routing Rule basis. See Global Settings for more information.
Softmute	Controls whether the Softmute* feature is enabled for this VMR. For this option to appear, and for this setting to take effect, the global Softmute setting (**Platform > Global Settings > Tech Preview Features > Enable Softmute**) must be enabled. • ***Enabled***: Softmute is enabled for this VMR. • ***Disabled***: Softmute is disabled for this VMR. Default: ***Disabled***

Option	Description
Denoise	Controls whether the Denoise* feature is enabled for this VMR. For this option to appear, and for this setting to take effect, the global Denoise setting (**Platform > Global Settings > Tech Preview Features > Enable Denoise**) must be enabled. • ***Enabled***: Denoise is enabled for this VMR. • ***Disabled***: Denoise is disabled for this VMR. Default: ***Disabled***
Pinning configuration	Select the pinning configuration to assign to this VMR. The list of available options contains the names of all of the pinning configurations in the theme associated with this VMR.
Aliases	
Alias: #1	
Alias	The alias that, when received by Pexip Infinity, is used to route the call to this service. The alias entered here must match the alias as it is received by Pexip Infinity. Wildcards and regular expressions are not supported. In most cases, the alias received by Pexip Infinity is the same as the alias that the participant used to call the service, but there are some exceptions, described in About aliases and access numbers. You may also want to define multiple aliases for the same service to ensure that it can be accessed by devices and protocols that enforce specific alias formats — for more information, see Using multiple aliases to access the same service.
Description	An optional description of the alias. This is useful if you have more than one alias for a service. Note that this description may be displayed to users of registered Pexip apps who are performing a directory search.
Add another Alias	Select this option if you want the VMR to be accessible by more than one alias. For more information, see Using multiple aliases to access the same service.

Allowing end-users to manage their own VMRs

VMR settings such as Host and Guest PIN numbers and whether to display participant names are initially assigned when the VMR is created by the administrator.

The Pexip VMR self-service portal is a separately-installable component that allows end-users to manage their personal Virtual Meeting Room without having to send requests to their administrator to change the configuration or branding of their VMR.

Configuring Virtual Auditoriums

A Virtual Auditoriums is a virtual meeting space that is optimized for use by a small number of Hosts and a large number of Guests. It can host any number of people from any type of device. Each Virtual Auditorium has one or more aliases associated with it, and participants access the conference by dialing one of these aliases. You use of PINs to assign differing Host and Guest privileges to conference participants (see About PINs, Hosts and Guests). Access to Virtual Auditoriums can be controlled by PINs or by authentication using SSO — or both.

To create a new Virtual Auditorium, go to **Services > Virtual Auditoriums** and select **Add Virtual Auditorium**.

To edit an existing Virtual Auditorium, or to view its details (including all the aliases associated with it), go to **Services > Virtual Auditoriums** and click on the name of the Virtual Auditorium.

You cannot convert an existing Virtual Meeting Room into a Virtual Auditorium. Instead, you must create a new Virtual Auditorium and reassign the existing aliases to it. For more information, see Changing from a Virtual Meeting Room to a Virtual Auditorium and vice versa.

The **vmr** license specifies the total number of Virtual Meeting Rooms and Virtual Auditoriums that can be configured at any one time.

Any changes you make to Virtual Auditorium configuration will be replicated to all Conferencing Nodes within around 90 seconds and will be applied to any subsequent conferences in that Virtual Auditorium. If there are any conferences already in place that use the Virtual Auditorium, any attempts to join it after the configuration has been replicated may be affected by the new configuration

settings. For this reason, we do not recommend changing Virtual Auditorium configuration while a conference is in progress because this will lead to an inconsistent user experience.

When adding or editing Virtual Auditoriums, the options are:

Option	Description
Name	The name used to refer to this Virtual Auditorium. If you can access this Virtual Auditorium via a Virtual Reception then the Virtual Auditorium **Name** entered here is shown to conference participants as they are transferred into the Virtual Auditorium (it is overlaid onto the **virtual_reception_connecting** splash screen of the theme associated with the Virtual Reception that is transferring the call).
Description	A description of the Virtual Auditorium.
Creation time	This read-only field shows the date and time when this record was first configured.
Host view	The maximum number of other participants that Hosts will see, and the layout used to show them. For more information, see Conference layouts and speaker names. Default: ***Large main speaker and up to 7 other participants (1+7 layout)***.
Guest view	The maximum number of Host participants that each Guest will see, and the layout used to show them. For more information, see Conference layouts and speaker names. Default: ***Large Host speaker and up to 7 other Hosts (1+7 layout)***.
Show names of participants	If participant name overlays are enabled, the display names or aliases of all participants are shown in a text overlay along the bottom of their video image. Default: ***No***.
Mute all Guests	When enabled, Guest participants will be muted when they first join the conference. After the conference has started, Hosts can use the Pexip app to unmute Guests, either individually or as a group. Default: ***No***.
Lock presenter as main speaker	When a presentation is being shown, whether the main speaker position shows the presenter or the current speaker. For more information, see Conference layouts and speaker names. ***Yes***: When a presentation is being shown, the main speaker position will always show the video image from the endpoint that is showing the presentation, even if others are speaking. ***No***: When a presentation is being shown the main speaker position will be voice-switched as usual. Default: ***No***.
Theme	The theme for use with this Virtual Auditorium. For more information, see Customizing conference images and voice prompts using themes. Default: ***<use Default theme>*** (the global default theme is used).
Participant authentication	
Host PIN	This optional field allows you to set a secure access code that must be entered by participants before they can access the service. If **Allow Guests** is set to ***Yes***, then the **Host PIN** applies to the conference Host(s) only. For more information, see About PINs, Hosts and Guests. • PINs must use the digits **0**-**9** only. • PINs may optionally end with **#**. • PINs must be between 4–20 digits long, including any **#**. • The default minimum length is 6 digits (**Platform > Global Settings > Service Configuration > Minimum PIN Length**).

Option	Description
Allow Guests	***Yes***: the conference can have two types of participants: Hosts and Guests. You must configure a **Host PIN** to be used by the Hosts. You can optionally configure a **Guest PIN**; if you do not configure a **Guest PIN**, Guests can join without a PIN, but the meeting will not start until the first Host has joined. ***No***: all participants have Host privileges. Default: ***No***.
Guest PIN	This optional field allows you to set a secure access code that must be entered by Guests before they can use the service. For more information, see About PINs, Hosts and Guests. • Host PINs and Guest PINs must be different. • PINs must use the digits **0**-**9** only. • PINs may optionally end with **#**. • PINs must be between 4–20 digits long, including any **#**. • If the Host PIN ends in **#** and a Guest PIN is used, the Guest PIN must also end with **#**. • If **#** is not used, Host PINs and Guest PINs must have the same number of digits. • You cannot configure a Guest PIN unless you have already configured a Host PIN.
Host Identity Provider Group	The set of Identity Providers to be offered to Hosts to authenticate with, in order to use the service. If this is blank, Hosts are not required to authenticate. Default: none selected
Guest Identity Provider Group	The set of Identity Providers to be offered to Guests to authenticate with, in order to use the service. If this is blank, Guests are not required to authenticate. Default: none selected
Other participants	(Available when an **Identity Provider Group** has been selected) Determines whether participants joining a SSO-protected service from devices other than the web app (for example SIP or H.323 endpoints) are allowed to dial in to the service. • ***Disallow all***: these devices are placed in a waiting room where they must wait to be admitted by a Host. • ***Allow if trusted***: these devices may join the service if they are locally registered. They must still enter a Host PIN or Guest PIN if either is required. All other devices are placed in a waiting room where they must wait to be admitted by a Host. Default: ***Disallow all***
Advanced options	
Breakout Rooms	Allows this Virtual Auditorium to send its participants into different breakout rooms. See Breakout rooms for more information.
Automatically dialed participants	When a conference begins in this Virtual Auditorium, a call will be placed automatically to any participants selected here. To add an Automatically Dialed Participant that is not already on the list, select the icon to the right of the selection fields. For more information, see Automatically dialing out to a participant from a conference.
Guests can present	Controls whether the Guests in the conference are allowed to present content. • ***Yes***: Guests and Hosts can present into the conference • ***No***: only Hosts can present Default: ***Yes***

Option	Description
Guests can see other guests	Controls whether Guests see other Guests during a conference. • ***If no hosts are present***: Guests see other guests only if no other Hosts are present. • ***Always***: Guests are always shown to other Guests (if there is space in the layout after Hosts). • ***Never***: Guests never see other Guests. Note that: • When no Hosts are present and Guests do not see other Guests, they see a "Welcome to the lobby" splash screen (the theme screen key is **inlobby**). • This does not effect the behavior at the start of a meeting — Guests always still see the "Waiting for the host" screen. • The **Guests-only timeout** setting still applies and will end the meeting for everybody after the configured timeout period. • However, if Guests are sent to a breakout room and there are no Hosts present in the breakout room then it does apply immediately — Guests are shown the "Welcome to the lobby" splash screen until a Host joins the breakout room, and Guests will only see Hosts (as usual for a Virtual Auditorium). Default: ***If no hosts are present***
Enable chat	Whether chat messaging is enabled for the conference: • ***Use global chat setting***: as per the global configuration setting. • ***Yes***: chat is enabled. • ***No***: chat is disabled. Default: ***Use global chat setting***.
Maximum inbound call bandwidth (kbps)	Enter a value in this field to limit the bandwidth of media being received by Pexip Infinity from each individual participant dialed in to this Virtual Auditorium. For more information see Managing and restricting call bandwidth.
Maximum outbound call bandwidth (kbps)	Enter a value in this field to limit the bandwidth of media being sent from Pexip Infinity to each individual participant dialed in to this Virtual Auditorium. For more information see Managing and restricting call bandwidth.
Conference capabilities	Allows you to limit the media content of the conference. For more information, see Controlling media capability. Default: ***Main video + presentation***.
Maximum call quality	Controls the maximum call quality for participants connecting to this service: • ***Use global setting***: use the global maximum call quality setting. • ***SD***: each participant is limited to SD quality. • ***HD***: each participant is limited to HD (720p) quality. • ***Full HD (1080p)***: allows any endpoint capable of Full HD to send and receive its main video at 1080p. Default: ***Use global setting***
Media encryption	Controls the media encryption requirements for participants connecting to this service. • ***Use global setting***: use the global media encryption setting. • ***Best effort***: each participant will use media encryption if their device supports it, otherwise the connection will be unencrypted. • ***Required***: all participants (including RTMP participants) must use media encryption. • ***No encryption***: all H.323, SIP and MS-SIP participants must use unencrypted media. (RTMP participants will use encryption if their device supports it, otherwise the connection will be unencrypted.) Default: ***Use global setting***
Participant limit	This optional field allows you to limit the number of participants allowed to join this Virtual Auditorium. For more information see Limiting the number of participants.

Option	Description
Service tag	This optional field lets you assign a unique identifier to this Virtual Auditorium, which you can then use to track use of the service.
Live Captions	Determines whether participants using the web app are given the option to turn on the live captions for conferences using this service. • ***Use global live captions setting***: as per global configuration setting. • ***Yes***: live captions may be enabled. • ***No***: live captions cannot be enabled. Default: ***Use global live captions setting*** Live captions are configured via **Platform > Global Settings > Live Captions**, and enabled globally by default and/or on a per VMR or per Call Routing Rule basis. See Global Settings for more information.
Pinning configuration	Select the pinning configuration to assign to this Virtual Auditorium. The list of available options contains the names of all of the pinning configurations in the theme associated with this Virtual Auditorium.
Aliases	
Alias: #1	
Alias	The alias that, when received by Pexip Infinity, is used to route the call to this service. The alias entered here must match the alias as it is received by Pexip Infinity. Wildcards and regular expressions are not supported. In most cases, the alias received by Pexip Infinity is the same as the alias that the participant used to call the service, but there are some exceptions, described in About aliases and access numbers. You may also want to define multiple aliases for the same service to ensure that it can be accessed by devices and protocols that enforce specific alias formats — for more information, see Using multiple aliases to access the same service.
Description	An optional description of the alias. This is useful if you have more than one alias for a service. Note that this description may be displayed to users of registered Pexip apps who are performing a directory search.
Add another Alias	Select this option if you want the Virtual Auditorium to be accessible by more than one alias. For more information, see Using multiple aliases to access the same service.

Changing from a Virtual Meeting Room to a Virtual Auditorium and vice versa

While it is not possible to change an existing Virtual Meeting Room to a Virtual Auditorium (and vice versa), you can change the service to which an alias is routed. For example, if your sales team already have a Virtual Meeting Room with an alias **meet.sales@example.com** and you want these conferences to take advantage of the features available to a Virtual Auditorium instead, then:

1. Set up a new Virtual Auditorium (**Services > Virtual Auditoriums > Add Virtual Auditorium**) with the appropriate configuration, but do not configure any aliases.
2. **Save** the new Virtual Auditorium.
3. Go to the **Aliases** page (**Services > Aliases**) and select the alias of the existing Virtual Meeting Room.
4. From the **Service name** drop-down list, select the name of the new Virtual Auditorium.
5. Repeat for all aliases belonging to the existing Virtual Meeting Room.

All calls made to any of the aliases that previously belonged to the Virtual Meeting Room will now be routed to the Virtual Auditorium. The Virtual Meeting Room will still exist, but will not have any aliases, so no calls can be made to it.

About the Virtual Reception IVR service

The Virtual Reception IVR service provides a way for conference participants who cannot dial Virtual Meeting Room and Virtual Auditorium aliases directly, to access these services from a central lobby using DTMF tones. It can also be used to route calls via the

Pexip Distributed Gateway to join an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

Using the Virtual Reception to access VMRs

Generally, in order to access a Virtual Meeting Room, participants dial one of the Virtual Meeting Room's aliases directly from their endpoint. However, some conference participants may not be able to do this. Reasons might include:

- The participant's endpoint might not support dialing by URI - they can only dial by IP address.
- The participant might be using an audio-only PSTN telephone and can only make calls to direct dial numbers.
- Your enterprise might have a limited number of direct dial numbers that can be used for audio participants, and you do not want to allocate one per VMR.
- Your enterprise uses local, toll-free telephone numbers that audio-only users can dial to access your VMRs and you don't want to have one of these for every VMR in your enterprise.

To allow for such cases, Pexip Infinity enables you to set up a single direct dial number or IP address that participants can dial to access a single, central lobby, known as a Virtual Reception. From here they can use the DTMF tones on their endpoint to enter the number of the specific VMR they want to join.

To implement this you must:

1. Create and configure your Virtual Reception.
2. Ensure that every Virtual Meeting Room that you want to be accessible from the Virtual Reception has at least one alias that consists of digits only. For more information, see Using multiple aliases to access the same service.
3. Provide conference participants with the combination of Virtual Reception alias followed by the Virtual Auditorium or Virtual Meeting Room number that they must dial to access the conference.

Note that when looking at conference status or conference history you will never see a Virtual Reception record for a WebRTC connection (WebRTC clients only send an HTTP request for the initial connection to the Virtual Reception, and no media is started until the participant connects to the VMR), but you will see a record for a video endpoint.

Providing telephone access to Virtual Meeting Rooms

For information on how to set up audio-only PSTN or mobile telephone access to your Virtual Meeting Rooms and Virtual Auditoriums via a Virtual Reception, see Integrating with telephone systems (PSTN).

Including the numeric alias of the VMR in the Virtual Reception dial string

SIP and H.323 endpoints and Skype for Business clients can optionally bypass having to enter the destination alias via DTMF tones.

They can do this by including the numeric alias of the VMR in their dial string when dialing the Virtual Reception. The dial string should be in the format: **<reception_alias>**<destination_alias>@<domain>**.

For example, if the alias of the Virtual Reception is **ivr@example.com** and the numeric alias of the target Virtual Meeting Room is **1234**, then the endpoint can dial **ivr**1234@example.com** to be transferred directly into the VMR.

Note that H.323 devices can also use the dial format **<reception_alias>#<destination_alias>@<domain>**.

Restricting or transforming the aliases entered into a Virtual Reception

To increase the security of your Virtual Reception services you may want to:

- Restrict the aliases or alias patterns that can be entered into a Virtual Reception.
- Optionally transform the entered alias before the Virtual Reception attempts to route the call to that new destination.

You can do this when configuring your Virtual Reception by expanding the **Advanced Options** and specifying the **Destination alias regex match** and **Destination alias regex replace string** fields.

For example, you could:

- Lock down the number ranges that can be reached via the Virtual Reception, e.g. by specifying a regex match of **8(.*)** to only allow calls to numbers starting with "8".
- Restrict and then transform the aliases that are entered, e.g. by specifying a regex match of **(\d{4})** to only allow 4 digit extensions to be entered and then to prepend the entered alias with a prefix of "88" by entering a replace string of **88\1**.

- Append a domain onto any numbers entered (so that you don't need to configure so many alternative VMR aliases), e.g. by specifying a regex match of **(\d{4})** and then appending a domain by using a replace string of **\1@example.com**.
- Only allow access to a specified shortlist of aliases, e.g. by configuring a match string of **(1111|1112|1113)** to reject any other alias than 1111, 1112 or 1113.

Using the Virtual Reception with the Infinity Gateway

You can also use the Virtual Reception to route calls via the Infinity Gateway. This would allow you, for example, to route phone calls towards a Cisco VCS as E.164 numbers, or to join an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

If your environment includes a PSTN gateway or uses an ITSP (Internet telephony service provider), consider the potential for toll fraud if you have Call Routing Rules that can route calls to the PSTN gateway or ITSP, or if you allow conference participants to dial out to other participants via the PSTN gateway or ITSP. See PSTN gateways and toll fraud for more information.

Joining scheduled and ad hoc Microsoft Teams meetings

See Integrating Microsoft Teams with Pexip Infinity for more information.

Joining scheduled and ad hoc Google Meet meetings

Routing phone calls towards a Cisco VCS as E.164 numbers

To implement this you must:

1. Create and configure your Virtual Reception in the normal way.
2. Configure a Call Routing Rule that matches the desired E.164 pattern and then routes the call via the specified Cisco VCS (where the Cisco VCS is the SIP proxy or H.323 gatekeeper as specified in the rule).

   Ensure that the E.164 pattern does not match a Virtual Meeting Room alias (as Virtual Meeting Room routing takes precedence).

Now, if someone dials in to the Virtual Reception and then enters the E.164 number that matches the Call Routing Rule, their call will be routed via the Cisco VCS.

Configuring Virtual Reception IVRs

The Virtual Reception IVR service allows participants to enter a lobby and then use DTMF tones to select the Virtual Meeting Room they want to join. It provides an alternative means to access VMRs for participants who cannot dial VMR aliases directly.

It can also be used in combination with the Infinity Gateway to:

- route calls to an external call control system, for example, to route phone calls towards a Cisco VCS
- join an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

There is no limit to the number of Virtual Receptions you can create.

To create or edit a Virtual Reception IVR service:

1. Go to **Services > Virtual Receptions**.
2. To create a new Virtual Reception, select **Add Virtual Reception**, or to edit an existing Virtual Reception, click on its name.

   The following options are available:

Option	Description
Name	Enter the name you will use to refer to this Virtual Reception.
Creation time	This read-only field shows the date and time when this record was first configured.
Description	An optional field where you can add details about the Virtual Reception.
Theme	Select the theme to apply to this Virtual Reception. For more information, see Customizing conference images and voice prompts using themes. If you leave this blank, the default Pexip theme is used.

Option	Description
Virtual Reception type	The type of Virtual Reception: ○ ***Regular***: the default type of Virtual Reception, used for routing calls to VMRs, or to other devices and call control systems via the Infinity Gateway. ○ ***Skype for Business***: a special type of Virtual Reception, used when you want to provide an IVR gateway to scheduled and ad hoc Skype for Business meetings. ○ ***Google Meet***: a special type of Virtual Reception, used when you want to provide an IVR gateway to scheduled and ad hoc Google Meet meetings. ○ ***Microsoft Teams***: a special type of Virtual Reception, used when you want to provide an IVR gateway to scheduled and ad hoc Microsoft Teams meetings. Default: ***Regular***
Skype for Business options (when a Virtual Reception type of *Skype for Business* is selected)	
Skype for Business server	The SfB server to use to resolve the SfB Conference ID entered by the user. You must then ensure that your Call Routing Rule that routes calls to your SfB environment has **Match against full alias URI** selected and a **Destination alias regex match** in the style **.+@example.com.***
Lookup location	If specified, a Conferencing Node in this system location will perform the SfB Conference ID lookup on the SfB server. If a location is not specified, the transcoding node hosting the Virtual Reception will perform the lookup.
Google Meet options (when a Virtual Reception type of *Google Meet* is selected)	
Access token	Select the name of the access token to use to resolve Google Meet IDs. When configuring a Virtual Reception it does not matter if you use a trusted or untrusted access token.
Lookup location	Specify the location that contains the Conferencing Nodes (typically Proxying Edge Nodes) that will perform the service lookup (meeting ID verification) on Google Meet.
Post-lookup regex match	An optional regular expression used to match against the meeting code entered by the caller into the Virtual Reception. This is typically used in conjunction with the **Post-lookup regex replace string** to transform the meeting code into a distinct alias pattern that will match a Call Routing Rule configured to route calls into Google Meet conferences. For example, you would typically set the regex match to **(.*)** (to match everything) and the replace string pattern to something like **meet.\1** which would prefix the meeting code entered into the Virtual Reception with "**meet.**". You would then configure an associated Call Routing Rule to match calls placed to aliases prefixed with **meet.** which then strips off that prefix (to leave just the meeting code again) before directing the call to Google Meet.
Post-lookup regex replace string	An optional regular expression used in conjunction with the **Post-lookup regex match** field to transform the meeting code into a distinct alias pattern that will match a Call Routing Rule configured to route calls into Google Meet conferences. (Only applies if the post-lookup regex match string is also configured and the entered code matches that regex.)
Microsoft Teams options (when a Virtual Reception type of *Microsoft Teams* is selected)	
Teams Connector	Select the name of the Teams Connector to use to resolve Teams codes.
Lookup location	Specify the location that contains the Conferencing Nodes (typically Proxying Edge Nodes) that will communicate with the Teams Connector to perform the meeting code verification.

Option	Description
Post-lookup regex match	An optional regular expression used to match against the meeting code entered by the caller into the Virtual Reception. This is typically used in conjunction with the **Post-lookup regex replace string** to transform the meeting code into a distinct alias pattern that will match a Call Routing Rule configured to route calls into Microsoft Teams conferences. For example, you would typically set the regex match to **(.*)** (to match everything) and the replace string pattern to something like **meet.\1** which would prefix the meeting code entered into the Virtual Reception with "**meet.**". You would then configure an associated Call Routing Rule to match calls placed to aliases prefixed with **meet.** which then strips off that prefix (to leave just the meeting code again) before directing the call to Microsoft Teams.
Post-lookup regex replace string	An optional regular expression used in conjunction with the **Post-lookup regex match** field to transform the meeting code into a distinct alias pattern that will match a Call Routing Rule configured to route calls into Microsoft Teams conferences. (Only applies if the post-lookup regex match string is also configured and the entered code matches that regex.)
Advanced options	
Destination alias regex match	An optional regular expression used to match against the alias entered by the caller into the Virtual Reception. If the entered alias does not match the expression, the Virtual Reception will not route the call. If this field is left blank, any entered alias is permitted.
Destination alias regex replace string	An optional regular expression used to transform the alias entered by the caller into the Virtual Reception. (Only applies if a regex match string is also configured and the entered alias matches that regex.) Leave this field blank if you do not want to change the alias entered by the caller.
Maximum inbound call bandwidth (kbps)	Enter a value in this field to limit the bandwidth of media being received by Pexip Infinity from each individual participant dialed in to this Virtual Reception. For more information see Managing and restricting call bandwidth.
Maximum outbound call bandwidth (kbps)	Enter a value in this field to limit the bandwidth of media being sent from Pexip Infinity to each individual participant dialed in to this Virtual Reception. For more information see Managing and restricting call bandwidth.
Conference capabilities	Allows you to limit the media content of calls that are connected via the Virtual Reception service. For more information, see Controlling media capability. Default: ***Main video + presentation***.
Maximum call quality	Controls the maximum call quality for participants connecting to this service: ◦ ***Use global setting***: use the global maximum call quality setting. ◦ ***SD***: each participant is limited to SD quality. ◦ ***HD***: each participant is limited to HD (720p) quality. ◦ ***Full HD (1080p)***: allows any endpoint capable of Full HD to send and receive its main video at 1080p. Default: ***Use global setting***
Media encryption	Controls the media encryption requirements for participants connecting to this service. ◦ ***Use global setting***: use the global media encryption setting. ◦ ***Best effort***: each participant will use media encryption if their device supports it, otherwise the connection will be unencrypted. ◦ ***Required***: all participants (including RTMP participants) must use media encryption. ◦ ***No encryption***: all H.323, SIP and MS-SIP participants must use unencrypted media. (RTMP participants will use encryption if their device supports it, otherwise the connection will be unencrypted.) Default: ***Use global setting***

Option	Description
Service tag	This optional field lets you assign a unique identifier to this Virtual Reception, which you can then use to track use of the service.
Aliases	
Alias: #1	
Alias	Enter the alias that participants will dial to access the Virtual Reception.
Description	An optional description of the alias. This is useful if the Virtual Reception has more than one alias.
Add another alias	Select this option if you want the Virtual Reception to be accessible by more than one alias. For more information, see Using multiple aliases to access the same service.

3. Ensure that every Virtual Meeting Room and Virtual Auditorium that you want to be accessible from this Virtual Reception has an alias in the form of a number. To do this:
   a. Go to either **Services > Virtual Meeting Rooms** or **Services > Virtual Auditoriums**.
   b. Select the name of the Virtual Meeting Room or Virtual Auditorium.
   c. In the Aliases section at the bottom of the page, enter the number to be used to access this Virtual Meeting Room or Virtual Auditorium from the Virtual Reception.
   d. Select **Save**.

Configuring a Media Playback Service

Pexip's Media Playback Service allows you to play prerecorded video content (such as adverts and informational videos) to consumers. When the media finishes playing, the user can be transferred to another service, such as a VMR conference, or they can be disconnected.

Third-party integrators with sophisticated Pexip Infinity integrations who use external policy servers, event sinks, and the Pexip Infinity management API can extend this base functionality. For example they could play content in response to events within other systems, such as consumers who are being held in a queuing system, before transferring them into a VMR. Alternatively, consumers could be transferred from a VMR conference into a playback service.

A Media Playback Service can be accessed via any of the Pexip apps, hardware endpoints and audio-only devices. To provide consumers with the most straightforward access to a service via the web app (which removes the need to install a software client) you may want to prepare and share preconfigured links.

Whether being ultimately transferred or disconnected, multiple users can use the same playback service concurrently, but each user views the media independently, and they do not share their own video or audio with anyone while they are connected to the Media Playback Service. After transferring to a VMR, all the usual controls and two-way audio and video features are available.

Configuration and usage overview

To configure a Media Playback Service you need to:

1. Upload MP4 files to your media library.
2. Create a playlist with one or more videos.
3. Configure a Media Playback Service specifying which playlist to run and an alias to access the service.

This topic also covers:

- Participant authentication and roles
- Playing media into a VMR
- Using the Pexip apps with a playback service
- Conference and participant history

Participant authentication and roles

You can control access to a playback service by using PINs and by using Single Sign On (SSO) authentication.

The PIN entered by the user when accessing the playback service also determines the role the user has if they are subsequently transferred into a VMR:

- By default, users keep the same role when transferring to another service afterwards, alternatively you can specify a role. (If you transfer as Guest and the target conference doesn't allow Guests, the participant still keeps the Guest role in that conference.)
- If no PIN was required for the playback service, a Host role is assigned.
- The Host or Guest role makes no difference to the user's experience while they are in the playback service.

The playback service's configuration needs to take into account the security requirements of the service that users are transferred to, if appropriate. This is because users are prompted only once for PINs / authentication details, based on the configuration of the playback service.

To configure the playback service's PINs and authentication, see Participant authentication.

To configure the role a user has when transferring to another service, see Action to perform on playlist completion.

Playing media into a VMR

While the general usage of a Media Playback Service is for a consumer to dial (or be transferred) into an individual playback service, you can also make use of this feature to play prerecorded content into a VMR i.e. all of the participants who are currently in the VMR will see the media content simultaneously. The playback media appears as a standard participant in the conference, so you can use the standard layout controls or spotlighting to highlight the "media playback" participant as required.

To play media into a VMR, you create your playback service with the required media content in the usual way, and then from within an ongoing conference you can add a participant (i.e. dial out) to that playback service, i.e. by specifying the alias of the playback service as the participant you want to add to the conference. Note that:

- The playback service's completion action should be configured as `{"disconnect": true}` to ensure a clean finish and disconnect when the media finishes playing.
- You may need to configure suitable Call Routing Rules as described in Recommendations for handling outgoing calls from a conference.

Uploading media

To upload media for a Media Playback Service, go to **Services > Media Playback Upload**. The options are:

Option	Description
Name	The name used to refer to this Media Library item.
Description	A description of the media.
Media file	The media file you want to upload, it must be in MP4 video format and H.264 encoded. We recommend the file is encoded using the "baseline" profile and "fastdecode", for example: **ffmpeg -i '<media file>' -c:v libx264 -preset slow -profile:v baseline -tune fastdecode 'baseline-fastcpu-720p.mp4'** Individual files up to 2 GB may be uploaded, and the cumulative size of all files together may be up to 10 GB. Later, when you configure the Media Playback Service, you should set the **Maximum call quality** setting to match the resolution of the media files.

Creating a playlist

To configure a Media Playback Service playlist, go to **Services > Media Playback Playlist**. The options are:

Option	Description
Name	The name used to refer to this playlist.
Description	A description of the playlist.

Option	Description
Loop	If enabled, for a given call the whole playlist is repeated until the user disconnects themselves or until the call is terminated programmatically via the management API.
Shuffle	If enabled, the playlist is shuffled so that all media items are played in a random order (media items' specified positions are not used). If the playlist contains an item with a play count greater than one, each instance is shuffled and played once. For example, if an item has a play count of 3, all 3 instances are shuffled among the other media items. If both shuffle and loop are enabled, all iterations of the playlist play in the same order.
Media Playlist Item	
Position of item in playlist	Every item must have a unique position specified. This refers to the order in which in the Media Library Item is played relative to the other items in the playlist. Please note that Position is only used when shuffle is off. When editing an existing playlist, if you want to change the position of multiple playlist items, you need to make each edit and save it individually.
Media Library Item	Media Library Item to be played.
Play count for entry in playlist	The number of times the item is played, the default is 1. If you set the value to 0 the item plays repeatedly until the user disconnects themselves or until the call is terminated programmatically via the management API.

Configuring a Media Playback Service and its aliases

To configure Media Playback Services and their associated aliases, go to **Services > Media Playback Services**. The options are:

Option	Description
Name	The name used to refer to this Media Playback Service.
Description	A description of this Media Playback Service.
Theme	The theme for use with this Media Playback Service. For more information, see Customizing conference images and voice prompts using themes. Note that watermarks are not applied to the playlist videos. Default: ***<use Default theme>*** (the global default theme is used).
Media playlist	The playlist to be played when the Media Playback Service is accessed.

Option	Description
Action to perform on playlist completion	This defines what happens when the playlist finishes (providing **Loop** is not enabled). The user can be disconnected or transferred elsewhere (to a VMR for example). If the action is left blank, the video ends with the final frame remaining in view. The completion action is specified in JSON format. To disconnect the user, enter: `{"disconnect": true}` Alternatively, to transfer the user, the format is: `{ "transfer": {"conference": "<alias>", "role": "<role>"}}` You must provide an **alias** to transfer the user to — this is typically the alias of a VMR. Role is optional. If not supplied or if you specify a role of ***"auto"***, the user is transferred to the next service with the same role they had on accessing the playback service, unless: • no PINs are used to access the playback service, in which case users are transferred as Host by default • you specify ***"host"*** or ***"guest"*** role, in which case all users are transferred using the specified role. For information about how the role for the Media Playback Service is determined see Participant authentication.
Participant authentication	
ℹ The playback service's configuration needs to take into account the security requirements of the service that users are transferred to, if appropriate. This is because users are prompted only once for PINs / authentication details, based on the configuration of the playback service.	
Host PIN	This optional field allows you to set a secure access code that must be entered before the user can access the playback service. This would give the user Host rights if they were subsequently transferred to a VMR, unless a different role is specified in the completion action. • PINs must use the digits **0-9** only. • PINs may optionally end with **#**. • PINs must be between 4–20 digits long, including any **#**. • The default minimum length is 6 digits (**Platform > Global Settings > Service Configuration > Minimum PIN Length**).
Allow Guests	Set this to ***Yes*** if you want to configure a Guest PIN in addition to a Host PIN. Guest/Host roles are treated the same in the playback service, but may be relevant if the user is subsequently transferred to a VMR. Default: ***No***.
Guest PIN	This optional field allows you to set a secure access code that must be entered before the user can access the playback service. This would give the user Guest rights if they were subsequently transferred to a VMR, unless a different role is specified in the completion action. • Host PINs and Guest PINs must be different. • PINs must use the digits **0-9** only. • PINs may optionally end with **#**. • PINs must be between 4–20 digits long, including any **#**. • If the Host PIN ends in **#** and a Guest PIN is used, the Guest PIN must also end with **#**. • If **#** is not used, Host PINs and Guest PINs must have the same number of digits. • You cannot configure a Guest PIN unless you have already configured a Host PIN.

Option	Description
Host Identity Provider Group	The set of Identity Providers to be offered to Hosts to authenticate with, in order to use the service. If this is blank, Hosts are not required to authenticate. Default: none selected
Guest Identity Provider Group	The set of Identity Providers to be offered to Guests to authenticate with, in order to use the service. If this is blank, Guests are not required to authenticate. Default: none selected
Other participants	(Available when an **Identity Provider Group** has been selected) Determines whether participants joining a SSO-protected service from devices other than the web app (for example SIP or H.323 endpoints) are allowed to dial in to the service. • ***Disallow all***: these devices are placed in a waiting room where they must wait to be admitted by a Host. • ***Allow if trusted***: these devices may join the service if they are locally registered. They must still enter a Host PIN or Guest PIN if either is required. All other devices are placed in a waiting room where they must wait to be admitted by a Host. Default: ***Disallow all***
Advanced options	
Service tag	This optional field lets you assign a unique identifier to this service, which you can then use to track use of the service.
Maximum call quality	Controls the maximum call quality for participants connecting to this service. This should be set to match the resolution of the playback video. For example, set it to HD if the playback videos are 720p and set it to Full HD if the videos are 1080p. A failure to match the video resolution with the playback room call quality may degrade performance.
Aliases	
Alias	The alias that, when received by Pexip Infinity, is used to route the call to this service. The alias entered here must match the alias as it is received by Pexip Infinity. Wildcards and regular expressions are not supported. In most cases, the alias received by Pexip Infinity is the same as the alias that the participant used to call the service, but there are some exceptions, described in About aliases and access numbers. You may also want to define multiple aliases for the same service to ensure that it can be accessed by devices and protocols that enforce specific alias formats — for more information, see Using multiple aliases to access the same service.

Using the Pexip apps with a playback service

While a consumer is using a Pexip app to view the playback media, as well as the disconnect button, they have controls for toggling their camera and mic, and they also see their self view, even though they are not sharing their own video or audio with anyone while they are connected to the Media Playback Service. After transferring to a VMR, all the usual VMR controls and two-way audio and video features are available.

Conference and participant history

When viewing conference and participant history it shows both the playback service and the transferred-to VMR if appropriate. The disconnect reasons for the playback service are normally either "Call transferred" or "Call disconnected".

Placing calls via the Pexip Infinity Distributed Gateway

The Pexip Infinity Distributed Gateway ("Infinity Gateway") enables endpoints to make calls to other endpoint devices or systems. This includes calls between devices that use different protocols and media formats, such as SIP and H.323 systems, Skype for Business clients (MS-SIP), and Pexip apps (WebRTC). It also enables you to route calls from VTCs and standards-based endpoints into an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

Traditional hardware gateways are often expensive and are typically centralized in a single location. This means that remote endpoints making gateway calls have to route media over a WAN or over the internet which is costly and uses a lot of bandwidth.

The software-based Infinity Gateway allows for a very cost-efficient deployment of local gateway/transcoding resources in every location. This can result in an improved user experience because of reduced latency as there is no longer a need to hairpin media back to a centralized datacenter. The other benefit is reduced WAN bandwidth usage — again due to not having to hairpin media. Reduced bandwidth usage allows an enterprise to deploy more video systems without having to upgrade the WAN infrastructure.

For example, you can use the Infinity Gateway to enable:

- Users of Pexip apps to place a person-to-person call to a SIP endpoint.
- Skype for Business users in your enterprise to make calls to, and receive calls from, virtually any other type of endpoint.

Calls to externally-hosted conferences

The Infinity Gateway can also be used to route calls from VTC systems and standards-based endpoints into an externally-hosted conference. It allows:

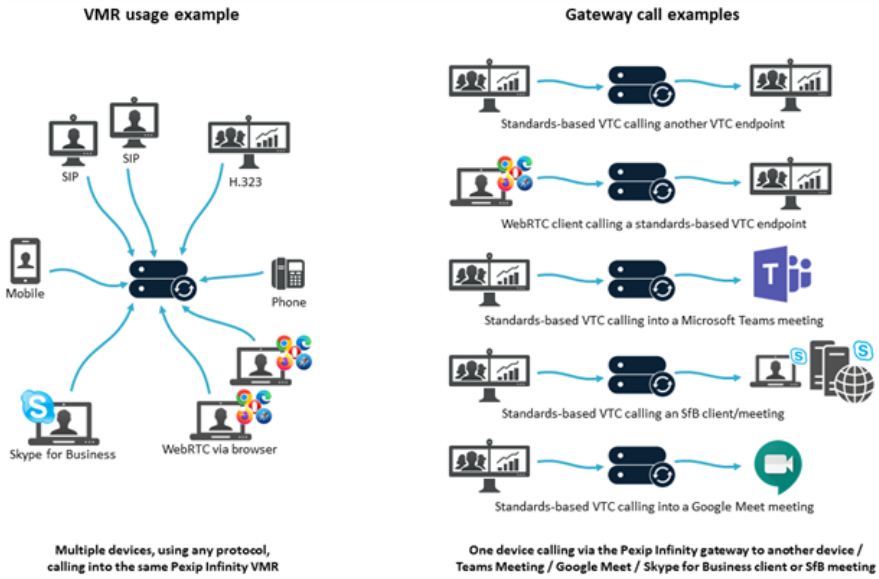
- VTC systems to call directly into a Microsoft Teams or Skype for Business meeting, or a Google Meet conference.
- Participants in a Skype for Business meeting to dial out to (i.e. invite) other non-SfB participants into the conference.

VMRs versus gateway calls — what's the difference?

Pexip Infinity Virtual Meeting Rooms are used to host multiple participants in the same conference. Those participants could be using a range of different endpoint devices to call into that VMR. In large deployments the VMR conference is likely to be distributed across multiple Conferencing Nodes.

In contrast, a gateway call can be considered as either being placed from one device directly to another device (by dialing the alias associated with that device), or as being placed from one device into an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet (by dialing the address associated with that external conference). As with calls into VMRs, the types of devices being connected (SIP, H.323, WebRTC etc.) does not matter. A single Conferencing Node will typically handle the incoming leg of the call and the outgoing leg, however the call could be distributed across two Conferencing Nodes if required for routing purposes.

Note that the person placing the call may not necessarily know if they are calling into a VMR or making a gateway call, as from their perspective they are typically just dialing an alias. Also, in both cases you could use a Virtual Reception to capture the alias of the VMR or device / external conference that the caller wants to connect into.



How it works

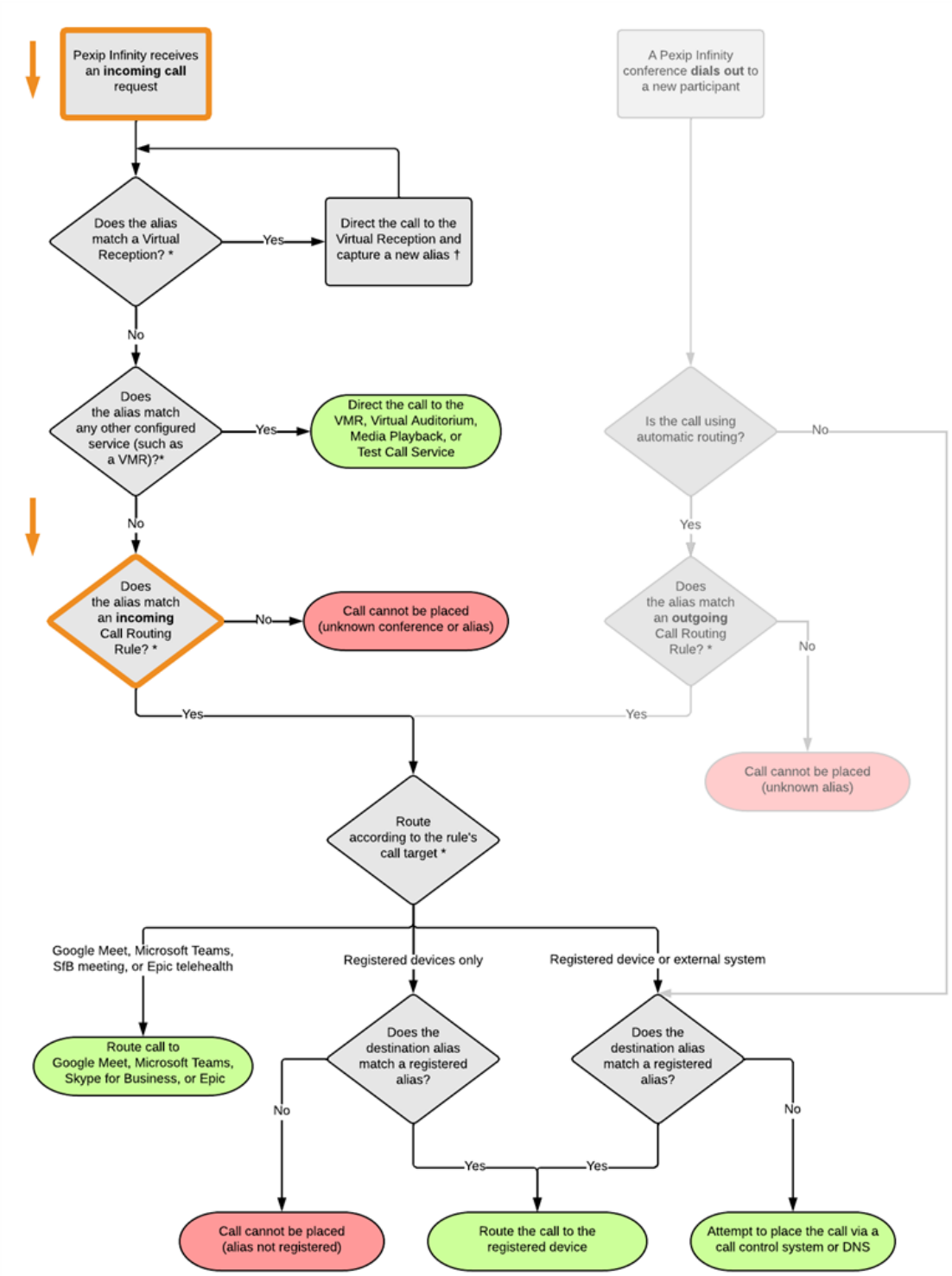
To enable devices to call other devices or systems via the Infinity Gateway, you must configure Call Routing Rules. These rules specify which calls should be interworked, for which protocols, and to where they should be routed.

Incoming calls received by Pexip Infinity are routed as follows:

1. Pexip Infinity receives an incoming call via one of its Conferencing Nodes.
2. It checks whether the destination alias belongs to a Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service; if so, it directs the call to that service.
3. If the alias does not belong to any of the above services, Pexip Infinity checks the Call Routing Rules to see if the alias matches any rules specified there for incoming calls. If so, it places an Infinity Gateway call to the destination alias according to the rule's call target settings (which protocol, location and call control system to use, whether to route to registered devices only, etc).

This means that if an alias matches both a Virtual Meeting Room and a Call Routing Rule, the former will always take precedence and the call will be routed to the Virtual Meeting Room. You must therefore ensure that any regular expressions used in a Call Routing Rule do not unintentionally overlap with any aliases used by a Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service.

The stage where Call Routing Rules are applied in Pexip Infinity's call routing logic for incoming calls is highlighted in the following diagram:



* This information can alternatively be provided by using external or local policy.
† If it is a non-Regular type of Virtual Reception (i.e. Google Meet, Microsoft Teams or SfB) the Virtual Reception captures the meeting code for the externally-hosted conference, which then needs to be matched by a suitable Call Routing Rule.

If your environment includes a PSTN gateway or uses an ITSP (Internet telephony service provider), consider the potential for toll fraud if you have Call Routing Rules that can route calls to the PSTN gateway or ITSP, or if you allow conference participants to dial out to other participants via the PSTN gateway or ITSP. See PSTN gateways and toll fraud for more information.

Note that:

- For additional security you can configure rules so that only registered devices are allowed to make calls via the Infinity Gateway.
- By default, the same Conferencing Node that receives the incoming call is used to place the outgoing call. However, you can configure the matching rule to place the call from a Conferencing Node in a specific location. As with all calls, signaling and media may be handled by different Conferencing Nodes in that location.
- Bandwidth restrictions can be applied to gateway calls; you do this by applying the restriction to the relevant rule.
- If the Infinity Gateway receives DTMF signaling from an inbound call, it will generate similar DTMF on the outbound call.
- In addition to handling gateway calls, rules may also be applied when dialing out from a conference to a new participant (if ***Automatic*** routing is used). When configuring your rules, consider whether the rule is to apply to incoming gateway calls, outgoing calls from a conference or to both incoming and outgoing calls.

Determining the caller's alias

A gateway call consists of two "legs": the incoming call from the caller to Pexip Infinity, and the outgoing call from Pexip Infinity to the endpoint or meeting being called. In most cases, Pexip Infinity is able to determine the alias of the caller and forward this to the endpoint or meeting being called. However, in some cases callers do not include a domain in their alias (for example, H.323 endpoints can be configured with an alias that does not include a domain), so Pexip Infinity will need to construct a valid alias to use for that caller. It does this by appending one of the following (in order of preference):

- the **Pexip Infinity domain (for Skype for Business integration)** for the location from which the outbound call is placed,
- the **Configured FQDN** for the Conferencing Node placing the outbound call,
- the **Static NAT address** for the Conferencing Node placing the outbound call,
- the **Secondary interface IPv4 address**, or if this is not configured, the **IPv4 address**, of the Conferencing Node placing the outbound call.

Note that the above does not apply to gateway calls to or from Skype for Business meetings.

Configuring Call Routing Rules

Call Routing Rules are used to handle two types of calls within Pexip Infinity:

- Calls received by Pexip Infinity that are to be routed via the Infinity Gateway to their ultimate destination (another device or externally-hosted conference).
- Outgoing calls made from within a Pexip Infinity conference (such as when manually adding a participant to a VMR).

This topic explains when Call Routing Rules are used, the main considerations when configuring your rules, our recommendations, how to add or modify rules, and provides an example rule.

When are Call Routing Rules used?

Call Routing Rules may be used for matching incoming calls received by Pexip Infinity, and they may also be used for matching outbound calls made from a Pexip Infinity conference.

Incoming calls

If an incoming call does not match an alias associated with a conferencing service (Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service), it is deemed a gateway call and it must match a Call Routing Rule for the call to be placed to the destination alias / target system.

When using a Virtual Reception to capture an alias, that new alias is treated as a new incoming call i.e. it will be checked against the service aliases first and then matched against the rules.

Call Routing Rules are **not** required to route incoming calls to a conferencing service (Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service). This happens automatically if the alias received by Pexip Infinity matches an alias associated with the conferencing service.

Outgoing calls from a conference

Outgoing calls made from within a Pexip Infinity conference are checked against Call Routing Rules when:

- Automatically Dialed Participants or participants added manually via the Pexip Infinity Administrator interface have their **Route this call** option set to ***Automatically***
- participants are added via a Pexip app.

and then it must match a suitable rule for the call to be placed to the destination alias/target system. See Recommendations for handling outgoing calls from a conference for more information.

Call routing considerations

When configuring a Call Routing Rule, the main points to consider are:

- **Rule priority**: this is the order in which the rules are checked to see if the conditions specified in the rule match the characteristics of the call that Pexip Infinity is trying to route. Rule checking stops when a match is found, even if the call that is placed as a result of that match fails for any reason.
- **Applicability to incoming or outgoing calls**: you must decide whether the rule applies only to incoming gateway calls, or only to outgoing calls placed from within a Pexip Infinity conference (where ***Automatic*** routing has been selected), or to both incoming gateway calls and outgoing calls from conferences. You can also limit the rule so that it only applies if the call is being handled by a Conferencing Node in a specific location.
- **Restricting incoming calls to those from registered devices or specific call protocols**: for incoming gateway calls, whether to limit the rule's applicability to only allow calls to be made from devices that are registered to Pexip Infinity, or to limit them to certain incoming protocols e.g. SIP only.
- **Alias matching and transforms**: the dialed alias or alias patterns that apply to this rule. A rule can be configured to apply to a specific alias e.g. alice@example.com, or to an alias pattern defined by a regex (regular expression) such as .+@example.com (any destination alias with the domain example.com). The destination alias can also, optionally, be transformed before the outgoing call is placed.
- **Call media settings**: whether to limit the media capabilities of the call, or whether to apply a theme.
- **Outgoing call placement**: every rule must define to where and how the call is routed to the destination alias. This includes defining the types of devices or systems to which the call is routed, such as limiting it to registered devices only or by specifying the SIP proxy, H.323 gatekeeper or Skype for Business server to use.
- **Toll fraud**: if you have Call Routing Rules that route calls to an ISDN/PSTN gateway, consider who should be allowed to make those calls (to avoid toll fraud). You could restrict those rules to only apply to incoming calls from registered devices, or to calls that are being handled in an internal location. Also, to restrict the number of people who can dial out from within a conference, we recommend configuring a Host PIN for your conferences (as only Hosts can initiate an outgoing call).

Note that any incoming call that has a destination alias that matches the alias of any Virtual Meeting Room, Virtual Auditorium, Virtual Reception or Test Call Service is always directed automatically to that conferencing service; the call will not be routed via the Infinity Gateway and the Call Routing Rules are not applied.

This diagram shows the call routing logic within Pexip Infinity when handling incoming call requests and when dialing out from within a conference.

* This information can alternatively be provided by using external or local policy.
† If it is a non-Regular type of Virtual Reception (i.e. Google Meet, Microsoft Teams or SfB) the Virtual Reception captures the meeting code for the externally-hosted conference, which then needs to be matched by a suitable Call Routing Rule.

Note that as an alternative to using Pexip Infinity's own routing logic, you can configure Pexip Infinity to instead defer its decision making to an external policy service or to use a local policy script (see policy profiles for more information). This allows Pexip Infinity administrators to implement routing decisions based on their specific requirements.

Recommendations for handling outgoing calls from a conference

In addition to configuring rules to handle person-to-person gateway calls, another consideration when setting up your dial plan is how to deal with outgoing calls placed from within a Pexip Infinity conference, particularly when placed by conference participants using a Pexip app.

The Pexip apps always use ***Automatic*** routing when dialing out from a conference, meaning that the user only has to enter or select the alias of the person they want to invite, and they do not have to select a call protocol.

To support the use of ***Automatic*** routing you **must** configure some appropriate Call Routing Rules otherwise the outbound call will not get placed.

The rules you need to configure depend upon your local requirements, but we recommend that you configure a fallback rule (a rule with a lower priority i.e. a higher number, than all of your other rules that are handling outgoing calls from a conference) that attempts to place the call over SIP and uses either DNS or the local SIP proxy.

Here are some typical rule patterns you may want to use (where all of the rules are configured to handle **Outgoing calls from a conference**):

Priority	Destination alias regex match *	Call target	Protocol and Proxy /Gatekeeper	Purpose
10	.+@example\.com	Registered devices only	n/a	This rule is only required if you are registering endpoints to Pexip Infinity as it will route calls to those registered aliases (devices). You can use a more specific regex if your registered URIs have a more controlled naming convention. You may also want to enable this rule for matching **Incoming gateway calls**.
190	\d+\.\d+\.\d+\.\d+	Registered device or external system	H.323 (to your gatekeeper or via DNS)	This rule matches an alias in the form of an IP address (see Matching an IP address for a more specific regex) and calls out over H.323 either via your local H.323 gatekeeper or via DNS.
200	(?!.*@example\.com$).*	Registered device or external system	SIP (to your proxy or via DNS)	This is your fallback rule. It matches aliases that do not end in @example.com (i.e. calls to domains outside of your environment) and routes it out over SIP either via your local SIP proxy or via DNS. Your fallback rule should ignore all of your domains that are serviced by your Pexip Infinity deployment (for example vc.example.com, vmr.example.com etc). To ignore multiple domains, you could use an expression formatted in the style: `(?!.*@(example\.com|another\.com|vc\.example\.com)$).*` which would ignore the example.com, another.com and vc.example.com domains. Note that these example expressions do not produce any match groups and do not require a replace string (hence the dialed alias is passed on unchanged if the match is successful). You could also tighten the match to only allow certain address formats for the external addresses such as name@domain i.e. `(?!.*@example\.com$).+@.+`

* All of these rules are intended as examples and use a local video domain of **example.com**. They are a basis for your own requirements and they should be adapted accordingly for your local environment.

Note that Call Routing Rules are **not** required to route incoming calls to a conferencing service (Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service).

Creating and modifying Call Routing Rules

To add, edit or delete a Call Routing Rule, go to **Services > Call Routing**. When specifying a Call Routing Rule, the options are:

Option	Description
Name	The name used to refer to this rule. If you are using a Virtual Reception as an IVR gateway to capture a conference ID, and then using this Call Routing Rule to transfer the participant into an external conference such as Google Meet or a Microsoft Teams meeting, then the rule **Name** entered here is shown to conference participants as they are transferred into the meeting (it is overlaid onto the **virtual_reception_connecting** splash screen of the theme associated with the Virtual Reception that is transferring the call).
Service tag	This optional field lets you assign a unique identifier to this rule, which you can then use to track use of the service.
Description	An optional description of the rule.
Priority	Assign a relative priority to this rule, from 1 to 200. Rules are checked in order of priority, starting with 1 and working down the list until a match is found. Rule checking stops when a match is found, even if the call that is placed as a result of that match fails for any reason.
Use this rule for...	
Incoming gateway calls	Applies this rule to incoming calls that have not been routed to a Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service, and thus should be routed via the Infinity Gateway. Default: selected.
Outgoing calls from a conference	Applies this rule to outgoing calls placed from a conferencing service (e.g. when a participant is added to a Virtual Meeting Room) where ***Automatic*** routing has been selected. Default: not selected. Note that the same rule can be applied to both incoming and outgoing calls if required.
Calls being handled in location	You can select a specific location, which means: • for incoming calls, the rule is only applied if the call is being handled (for signaling purposes) by a Conferencing Node in the selected location. • for outgoing calls, the rule is only applied if the call is being initiated from the selected location. This allows you to restrict the types of calls that some users can make. For example, you may use a call control system that routes all calls into Pexip Infinity through Conferencing Nodes in an "internal/trusted" location, whereas calls received from devices in the public internet could all be routed to nodes in a different "public/untrusted" location. You can then, for example, use the **Calls being handled in location** option to allow only calls received in the "trusted" location to place calls via a PSTN gateway. To apply the rule regardless of the location, select ***Any Location***. Default: ***Any Location***.
When matching incoming Gateway calls...	

Option	Description
Match incoming calls from registered devices only	Select this option if you want this rule to only apply to incoming calls from devices, videoconferencing endpoints, soft clients or Pexip apps that are registered to Pexip Infinity. Note that: • The incoming call must be received by the same Conferencing Node as to which the device is registered. This is the normal behavior for most endpoints, however you may need to ensure that the SIP Proxy and Registrar are both configured to resolve to the same node. • The call must also match one of the selected protocols below. Calls placed from non-registered clients or devices, or from the Pexip web app will not be routed by this rule if it is enabled. Default: disabled.
Match Pexip apps (WebRTC/RTMP) Match SIP Match Skype for Business (MS-SIP) Match H.323 Match Teams	Select the source / protocols of the incoming call to which the rule should apply. Note that matching **Skype for Business (MS-SIP)** and **Teams** calls options do not apply if **Match incoming calls from registered devices only** is selected. Default: these options, except for Teams, are all enabled by default.
Alias match and transform	
Match against full alias URI	This setting is for advanced use cases and will not normally be required. By default, Pexip Infinity matches against a parsed version of the destination alias, i.e. it ignores the URI scheme, any other parameters, and any host IP addresses. So, if the original alias received by Pexip Infinity is **sip:alice@example.com;transport=tls** for example, then by default the rule will match against **alice@example.com**. Select this option to match against the full, unparsed alias instead. You must select this option if you are using this rule to support your Skype for Business IVR gateway (where this rule is routing calls handled by a Virtual Reception into the relevant Skype for Business meeting).
Destination alias regex match	The regular expression that the destination alias (the alias that was dialed) is checked against to see if this rule applies to this call. Pexip Infinity supports case-insensitive Perl-style regular expression patterns. Note that the regex must match the entire alias — a partial match is treated as a non-match. See Regular expression (regex) reference for information about writing regular expressions.
Destination alias regex replace string	The regular expression string used to transform the originally dialed alias (if a match was found). If you do not want to change the alias, leave this field blank.
Call media settings	
Maximum inbound call bandwidth (kbps)	Enter a value in this field to limit the bandwidth of media being received by Pexip Infinity from each individual participant dialed in via this rule. For more information see Managing and restricting call bandwidth.
Maximum outbound call bandwidth (kbps)	Enter a value in this field to limit the bandwidth of media being sent from Pexip Infinity to each individual participant dialed out from this rule. For more information see Managing and restricting call bandwidth. When dialing out from a conference, the outbound call bandwidth limit is inherited from the VMR's bandwidth settings. In this case, the rule's outbound bandwidth setting configured here is not used.

Option	Description
Call capability	Allows you to limit the media content of the call. The participant being called will not be able to escalate beyond the selected capability. For more information, see Controlling media capability. Default: ***Main video + presentation***.
Maximum call quality	Controls the maximum call quality for participants connecting to this service: • ***Use global setting***: use the global maximum call quality setting. • ***SD***: each participant is limited to SD quality. • ***HD***: each participant is limited to HD (720p) quality. • ***Full HD (1080p)***: allows any endpoint capable of Full HD to send and receive its main video at 1080p. Default: ***Use global setting***
Media encryption	Controls the media encryption requirements for participants connecting to this service. • ***Use global setting***: use the global media encryption setting. • ***Best effort***: each participant will use media encryption if their device supports it, otherwise the connection will be unencrypted. • ***Required***: all participants (including RTMP participants) must use media encryption. • ***No encryption***: all H.323, SIP and MS-SIP participants must use unencrypted media. (RTMP participants will use encryption if their device supports it, otherwise the connection will be unencrypted.) Default: ***Use global setting***
Theme	The theme to use for calls placed using this rule. See Themes used by Call Routing Rules (gateway calls) for more information. Default: ***<use Default theme>*** (the global default theme is used).
Codecs	Controls which codecs to offer in audio/video negotiation (SDPs). ⓘ The codecs in the **Enabled Codecs** list are offered only if they are also enabled in Global settings. Default: all codecs are selected.
Outgoing call placement (individual fields are only displayed when appropriate)	
Call target	The device or system to which the call is routed. The options are: • ***Registered device or external system***: route the call to a matching registered device if it is currently registered, otherwise attempt to route the call via an external system such as a SIP proxy, Skype for Business server, H.323 gatekeeper or other gateway/ITSP. • ***Registered devices only***: route the call to a matching registered device only (providing it is currently registered). • ***Skype for Business meeting direct (Conference ID in dialed alias)***: route the call via a Skype for Business server to a Skype for Business meeting. Note that the destination alias must be transformed into just a Skype for Business Conference ID. • ***Skype for Business clients, or meetings via a Virtual Reception***: route the call via a Skype for Business server either to a SfB client, or — for calls that have come via a Virtual Reception — to a Skype for Business meeting. For Skype for Business meetings via Virtual Reception routing, ensure that **Match against full alias URI** is selected and that the **Destination alias regex match** ends with **.*** • ***Google Meet meeting***: routes the call to a Google Meet meeting. See Introduction for more information. • ***Microsoft Teams meeting***: routes the call to a Microsoft Teams meeting. See Integrating Microsoft Teams with Pexip Infinity for more information. • ***Microsoft Teams Room***: routes the call to a Microsoft Teams Room. See Integrating Microsoft Teams with Pexip Infinity for more information. • ***Epic Telehealth Profile***: used for telehealth calls. See Epic telehealth integration with Pexip Infinity for more information.

Option	Description
Outgoing location	When calling an external system, this forces the outgoing call to be handled by a Conferencing Node in a specific location. When calling a Skype for Business meeting, a Conferencing Node in this location will handle the outgoing call, and — for ***Skype for Business meeting direct*** targets — perform the Conference ID lookup on the SfB server. Select ***Automatic*** to allow Pexip Infinity to automatically select the Conferencing Node to use to place the outgoing call (which will usually be the node that received the call).
Protocol	The protocol used to place the outgoing call. Note that: • If the call is to a registered device, Pexip Infinity will instead use the protocol that the device used to make the registration. • ***RTMP (streaming)*** always routes to an external system. Default: ***SIP***.
SIP Proxy *	You can optionally specify the SIP Proxy to use to place an outgoing SIP call. Default: ***Use DNS***.
H.323 Gatekeeper *	You can optionally specify the H.323 Gatekeeper to use to place an outgoing H.323 call. Default: ***Use DNS***.
Skype for Business server *	When calling an external system, this is the Skype for Business server to use for outbound SfB (MS-SIP) calls. When calling a Skype for Business meeting, this is the SfB server to use for the Conference ID lookup and to place the call. Default: ***Use DNS*** (note that a server must be selected when the **Call target** is ***Skype for Business meeting direct (Conference ID in dialed alias)***).
TURN server †	You can optionally specify the TURN server to use when placing calls to ICE-enabled clients (such as Skype for Business clients and Pexip WebRTC clients).
STUN server †	You can optionally specify the STUN server to use when placing calls to ICE-enabled clients (such as Skype for Business clients and Pexip WebRTC clients).
Teams Connector	You can optionally select the Teams Connector you want to handle the call. If you do not specify anything, the Teams Connector associated with the outgoing location is used.
Access token	Select the name of the access token to use to resolve Google Meet IDs. You should select either a trusted or untrusted type of token, depending on whether you want to enable the device to be automatically admitted into the Google Meet conference (subject to also being a trusted endpoint from Pexip Infinity's perspective i.e. if the rule also has **Treat as trusted** enabled). Typically, you will use a trusted token if **Treat as trusted** is selected, and an untrusted token if **Treat as trusted** is not selected.
Treat as trusted	This indicates that the target of this Call Routing Rule may treat the caller as part of the target organization for trust purposes. It can be applied to calls placed to: • Microsoft Teams • Google Meet • SIP destinations and registered SIP devices (via a P-Asserted-Identity in the SIP header)

Option	Description
Live captions	Determines whether participants using the web app are given the option to turn on the live captions for conferences using this service. • ***Use global live captions setting***: as per global configuration setting. • ***Yes***: live captions may be enabled. • ***No***: live captions cannot be enabled. Default: ***Use global live captions setting*** Live captions are configured via **Platform > Global Settings > Live Captions**, and enabled globally by default and/or on a per VMR or per Call Routing Rule basis. See Global Settings for more information.
External participant avatar lookup	Applies to Microsoft Teams integrations only. This determines whether or not the Teams Connector requests from Exchange Online an avatar for each participant in the Teams conference. • ***Use global setting***: use the global external participant avatar lookup setting. • ***Yes***: request the participant avatar from Exchange Online via the Teams Connector. • ***No***: use default avatar behavior. Default: ***Use global setting***
Denoise audio	Applies to Google Meet integrations only. Controls whether to remove background noise from audio streams as they pass through the infrastructure. Some endpoints perform their own built-in denoising, in which case, to avoid performing denoising twice, you can configure via the theme's **vendordata.json** file a list of endpoints that perform their own denoising. See Rules and requirements for customized themes for more information. Default: enabled.
Rule state	
Enable this rule	Determines if the rule is enabled or not. Any disabled rules still appear in the rules list but are ignored. Use this setting to test configuration changes, or to temporarily disable specific rules.

* If you do not select a specific **H.323 Gatekeeper**, **SIP Proxy** or **SfB server**, the Conferencing Node will attempt to use DNS to locate an appropriate system via which to route the call (rather than fall back to using the gatekeeper / proxy / server associated with the node's system location).

† If you do not select a specific **TURN server** or **STUN server**, the call will use the TURN / STUN server associated with the node's system location.

Example

In our example organization, every employee has their own video endpoint that is registered to Pexip Infinity with an alias in the format **firstname.lastname@example.com**, and their own Virtual Meeting Room with an alias in the format **meet.firstname.lastname@example.com**.

In most cases, employees will use their standalone SIP or H.323 endpoints to call others within the organization, but sometimes they want to use a Pexip app to make a person-to-person call.

To support this, we set up the following Call Routing Rule (unspecified settings can be left as their default values):

Option	Input	Notes
Name	Pexip app to SIP	
Description	Allow Pexip app users to call registered SIP endpoints directly	
☑ Incoming gateway calls ☐ Outgoing calls from a conference		We want this rule to match incoming gateway calls only.

Option	Input	Notes
Calls being handled in location	***Any location***	We want this rule to apply throughout our deployment.
☑ Match Pexip apps (WebRTC / RTMP) ☐ Match SIP ☐ Match Skype for Business (MS-SIP) ☐ Match H.323		In this example, we only want the rule to apply to calls from Pexip apps.
Destination alias regex match	.+@example.com	This regular expression will match any destination alias with the domain **example.com**.
Destination alias regex replace string	<blank>	We have left this blank because we do not want to amend the alias.
Call target	***Registered devices only***	We want the call to match currently registered devices only.

This rule means that if a Pexip app user dials any alias with the domain **@example.com** (e.g. **alice.jones@example.com**), the call will be routed over SIP. We do not need to worry about this rule applying to VMR aliases with the same domain (e.g. **meet.alice.jones@example.com**) because rules are only applied to incoming calls after checking whether there are any VMRs, Virtual Receptions etc. with that alias.

Configuring the Test Call Service

Pexip Infinity provides a test loopback service that allows users to check the quality of their video and audio (i.e. that their local camera, microphone and speakers are working properly), and verifies that they can connect to a Conferencing Node.

Test Call Services are configured within Pexip Infinity in the same manner as other services such as Virtual Receptions. By default, Pexip Infinity includes a preconfigured Test Call Service (called "Test Call Service") and an associated alias of "test_call" that can be used to dial into the Test Call Service. Additional test call services and aliases can be added if required.

We recommend that you add a **test_call@<yourdomain>** alias to the preconfigured Test Call Service to cater for devices that automatically add a domain to their called alias.

After a call into a Test Call Service (e.g. to the "test_call" alias) is answered, the user receives some instructions and then the user's video and audio of themselves is played back from the Conferencing Node with a 2 second delay. The test call then automatically disconnects after approximately 20 seconds.

The instructions and timeouts can all be customized via themes, and different themes can be applied to different Test Call Services, if required.

As an administrator, you also have the ability to dial out to a user (alias) to initiate a test call from within Pexip Infinity.

Configuring Test Call Services and aliases

To configure Test Call Services and their associated aliases, go to **Services > Test Call Service**. The options are:

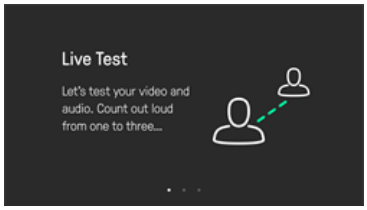
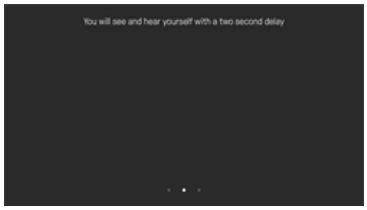
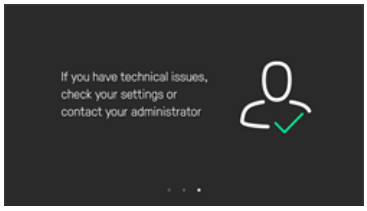
Option	Description
Name	The name used to refer to this Test Call Service.
Description	A description of the Test Call Service.
Theme	The theme for use with this Test Call Service. For more information, see Customizing conference images and voice prompts using themes. Default: ***<use Default theme>*** (the global default theme is used).
Advanced options	

Option	Description
Maximum inbound call bandwidth (kbps)	Enter a value in this field to limit the bandwidth of media being received by Pexip Infinity from the user dialed in to this Test Call Service. For more information see Managing and restricting call bandwidth.
Conference capabilities	Allows you to limit the media content of the conference. For more information, see Controlling media capability. Default: ***Main video + presentation***.
Maximum call quality	Controls the maximum call quality for participants connecting to this service: • ***Use global setting***: use the global maximum call quality setting. • ***SD***: each participant is limited to SD quality. • ***HD***: each participant is limited to HD (720p) quality. • ***Full HD (1080p)***: allows any endpoint capable of Full HD to send and receive its main video at 1080p. Default: ***Use global setting***
Media encryption	Controls the media encryption requirements for participants connecting to this service. • ***Use global setting***: use the global media encryption setting. • ***Best effort***: each participant will use media encryption if their device supports it, otherwise the connection will be unencrypted. • ***Required***: all participants (including RTMP participants) must use media encryption. • ***No encryption***: all H.323, SIP and MS-SIP participants must use unencrypted media. (RTMP participants will use encryption if their device supports it, otherwise the connection will be unencrypted.) Default: ***Use global setting***
Service tag	This optional field lets you assign a unique identifier to this service, which you can then use to track use of the service.
Aliases	
Alias: #1	
Alias	The alias that, when received by Pexip Infinity, is used to route the call to this service. The alias entered here must match the alias as it is received by Pexip Infinity. Wildcards and regular expressions are not supported. In most cases, the alias received by Pexip Infinity is the same as the alias that the participant used to call the service, but there are some exceptions, described in About aliases and access numbers. You may also want to define multiple aliases for the same service to ensure that it can be accessed by devices and protocols that enforce specific alias formats — for more information, see Using multiple aliases to access the same service.
Description	An optional description of the alias. This is useful if you have more than one alias for a service. Note that this description may be displayed to users of registered Pexip apps who are performing a directory search.
Add another Alias	Select this option if you want the Test Call Service to be accessible by more than one alias. For more information, see Using multiple aliases to access the same service.

Customizing the Test Call Service via themes

As with other Pexip Infinity services, the images that are shown, and the audio prompts that the end-user hears, can all be fully customized via themes. You can also control the playback delay and the duration of the test call by configuring settings in the **themeconfig.json** theme file.

The following theme files, splash screens and settings are used during a test call:

Theme file/setting	Default content	Notes
conf-test_call_ 48kHz_mono.wav	"Let's test your video and audio. Count out loud from one to three, now."	Audio file played at the start of a video call to a Test Call Service.
conf-test_call_ audio_only_48kHz_ mono.wav	"Let's test your audio settings. Count out loud from one to three, now."	Audio file played at the start of an audio-only call to a Test Call Service.
conf-test_call_ disconnect_48kHz_ mono.wav	"If you have technical issues, check your settings or contact your administrator."	Audio file played at the end of a call to a Test Call Service.
background_test_ call.jpg		The background image (a black screen) used by default on the Test Call Service splash screens.
test_call_welcome splash screen		Shown at the start of a call to a Test Call Service. Theme elements used: • Icon: icon_test_call_welcome.svg • Label: test_call_welcome_header and test_call_welcome_text • Background: background_test_call.jpg
test_call_in_ progress splash screen		Shown during a call to a Test Call Service. Note that a large, live (with a short delay) video image of the test call participant is shown on top of this screen during a test call. Theme elements used: • Label: test_call_in_progress • Background: background_test_call.jpg
test_call_complete splash screen		Shown briefly prior to automatically disconnecting the participant from a Test Call Service. Theme elements used: • Icon: icon_test_call_complete.svg • Label: test_call_complete • Background: background_test_call.jpg
test_call_service_ media_delay (in **themeconfig.json**)	2	The number of seconds that media is delayed before being looped back to the caller when using a Test Call Service.
test_call_service_ disconnect_timeout (in **themeconfig.json**)	10	The number of seconds that a user can test their media before the disconnect message is played, when using a Test Call Service.

The following sequence describes the test call process and explains when each theme file and setting is used:

1. When a test call is answered, the **test_call_welcome** splash screen is displayed and either the **conf-test_call_48kHz_mono.wav** or **conf-test_call_audio_only_48kHz_mono.wav** audio file is played, depending on whether the caller is connecting with video or just with audio-only.
2. The **test_call_in_progress** splash screen is displayed (after the **conf-test_call_48kHz_mono.wav** audio file has finished).
3. The caller's audio and video media is replayed back to them with a **<test_call_service_media_delay>** seconds delay (2 seconds by default).
4. The media replay stops after **<test_call_service_disconnect_timeout>** seconds (10 seconds by default).
5. The **test_call_complete** splash screen is displayed and the **conf-test_call_disconnect_48kHz_mono.wav** audio file is played.
6. The call automatically disconnects after a further 5 seconds (not configurable).

Dialing out from the Test Call Service

Typically, users will **dial in** to the Test Call Service to check their video and audio, but as an administrator you can **dial out** to a user (alias) to initiate a test call from within Pexip Infinity. When the user answers the call they are taken through the test in the same way as if they had dialed the Test Call Service themself.

To dial a user into a test call:

1. Select the Test Call Service to dial the participant from (go to **Services > Test Call Service** and select a service).
2. At the bottom left of the screen, select **Dial out to participant**.
3. Enter the **Participant alias** you want to dial, and the **Protocol** to use to make the call.
4. Select **Dial out to participant**.

Registering devices to Pexip Infinity

Pexip Infinity can act as a SIP registrar and H.323 gatekeeper, which means that SIP and H.323 endpoints can register directly to Pexip Infinity. This allows Pexip Infinity to route calls to those registered devices without having to go via an external SIP proxy or H.323 gatekeeper, or rely on DNS.

The Pexip app for Windows can also register to Pexip Infinity. Registration is mandatory and allows these apps to receive calls via Pexip Infinity and use directory lookup services.

Devices and apps can only register to Pexip Infinity with a permitted alias (and, if authentication is required, by supplying valid credentials). Allowed aliases and their associated credentials can be configured manually, or they can be bulk provisioned from directory information contained in a Windows Active Directory LDAP server, or any other LDAP-accessible database.

SIP and H.323 devices do not need to be registered in order to make calls to Pexip Infinity. However, you can configure your deployment so that only registered devices can make gateway calls, thus preventing potential for toll fraud.

Pexip Infinity can support up to 1,500 registrations per Conferencing Node, although a large number of registrations could reduce the node's capacity in other areas (proxying, transcoding, One-Touch Join etc).

For help on troubleshooting registration issues, see Troubleshooting registering to Pexip Infinity.

Configuration summary for enabling registrations

To configure Pexip Infinity to accept registrations and route calls to registered devices you must:

1. Ensure that the Pexip Infinity's registrar service is enabled (**Services > Registrar**).
2. Configure the aliases that devices are allowed to register with Pexip Infinity (**Users & Devices > Device Aliases**). For each alias, decide whether authentication is required, and what form that authentication should take.
3. Consider whether to provision individual users with their registration details, and then configure the appropriate services and credentials.
4. Configure appropriate Call Routing Rules (**Services > Call Routing**). Rules are required in all cases except for when using ***Manual*** routing to add a participant to a conference.

To see a list of all device aliases that are currently registered to the Pexip Infinity platform, go to **Status > Registrations**, and to see a history of registered devices go to **History & Logs > Registration History**.

Each configuration step is described in more detail below, after the explanation of how Pexip Infinity manages registered devices.

How it works

Registrar service

The registrar service on Pexip Infinity is enabled by default. However, a device can only register to Pexip Infinity if the alias it wants to register has been added to Pexip Infinity's list of allowed device aliases (**Users & Devices > Device Aliases**).

Authenticating registrations

Device registrations can optionally be subject to authentication, using one of the following methods:

- **Username and password**: the device's registration request is challenged and the device is only allowed to register with that alias if it provides credentials matching those configured on Pexip Infinity for that alias. Pexip Infinity supports Digest authentication only.

  This option is available to SIP and H.323 devices, and the legacy Connect desktop app.
- **SSO using a configured Identity Provider**: the device is only allowed to register if it successfully authenticates with an Identity Provider and the alias it is registering matches the alias configured on the Identity Provider for that user.

  This option is available for Pexip app for Windows. For more information, see Authenticating app registrations using Identity Providers.
- **Using AD FS**: the device is only allowed to register if Pexip Infinity has been configured to use AD FS as an authentication client and the device successfully authenticates using this method.

  This option is being deprecated and is available for the legacy Connect desktop app only.

Registering devices

A device can register to any Conferencing Node. Pexip Infinity supports H.323 alternate gatekeeper registrations for nodes in the same system location, but does not apply any other form of registration load balancing or failover. Multiple devices can register with the same alias. After a device has registered it must periodically refresh that registration.

If your endpoint supports both SIP and H.323, you should register it to Pexip Infinity over just one protocol, either SIP or H.323, but not both.

SIP devices

To register a **SIP device**, use the IP address or FQDN of a local Conferencing Node as the SIP proxy.

For the registration of SIP endpoints, it is important to know how the individual endpoint reacts to different environments. Some devices respect the weight and priority values returned in SRV queries, such that configuring them with the SIP URI and correct SIP domain works very well; other devices may read only the first FQDN returned in an SRV query and never attempt to re-resolve until its SIP settings are changed or the endpoint rebooted. Please contact your your Pexip authorized support representative for further guidance.

H.323 devices

To register an **H.323 device**, use the IP address or FQDN of a local Conferencing Node as the H.323 gatekeeper.

All of the device aliases presented in an H.323 registration request must be in the list of allowed device aliases. If any alias is not present, none of the aliases will be allowed to register. Additionally, if device authentication is being used, all of the device aliases in the request must be configured with the same credentials.

Pexip app for Windows

To register a **Pexip app for Windows** client, users enter the registration alias as the app's **video address**.

- Registration is mandatory for Pexip app for Windows.
- Pexip app for Windows users must authenticate their registration using SSO via an Identity Provider configured on Pexip Infinity.
- Administrators can provision individual users' apps with their registration details. See Provisioning the app for more information.
- When your Pexip for Windows app is registered, as well as being able to receive calls, you can filter and lookup the contact details (phone book / directory) of other devices or VMRs that are set up on the Pexip Infinity platform. See Directory (phone book) of devices and VMRs for registered Pexip apps for more information.

Connect desktop apps

For information on registering a legacy Connect desktop app, see our v39 documentation site.

Calling registered devices

Whenever Pexip Infinity needs to place a call it follows the Call routing logic shown in the diagram below. If the alias it is trying to place the call to is currently registered, then Pexip Infinity will place the call to that registered device instead of attempting to find it via other means such as call control or DNS.

When calling a registered device:

- The Conferencing Node to which the device is registered will place the call to the device (media may flow through other nodes, as per the platform's standard distributed conferencing behavior).
- The outbound call is always placed to the registered device over the same protocol that it used to make the registration.
- If multiple devices are registered with the same alias, Pexip Infinity will place a call to each device, i.e. it will fork the call.

Requirement for Call Routing Rules

In most cases, you must also configure a suitable Call Routing Rule that instructs Pexip Infinity to place the outbound call to the device. A Call Routing Rule **is** required when:

- Making a call to another device or system via the Infinity Gateway.
- Using ***Automatic*** routing when adding a participant to a conference.

A Call Routing Rule **is not** required when:

- Using ***Manual*** routing when adding a participant to a conference.

When configuring rules you must define:

- Which call directions to match — whether the rule applies only to incoming gateway calls, or only to outgoing calls placed from within a Pexip Infinity conference (where ***Automatic*** routing has been selected), or to both incoming gateway calls and outgoing calls from conferences.
- The destination aliases to be matched — this needs to match the pattern of the registered device aliases
- Whether to route the call to ***Registered devices only*** (in which case the call will fail if the device is not currently registered), or to **Registered device or external system** which will route the call to a matching registered device if it is currently registered, otherwise it will attempt to route the call via an external system.

This table shows the rule settings you need to use for the different calling scenarios between, to, and from registered devices:

Purpose	Incoming gateway calls	Outgoing calls from a conference	Match incoming calls from registered devices only	Match <protocol>	Destination alias regex match	Call target	Protocol
Enable calls **between** two registered devices	✓		✓	Limit to the selected source (caller) protocols e.g. SIP, H.323, WebRTC, MS-SIP (Skype for Business) as required	As appropriate for your registered device aliases e.g. **.+@<Infinity domain>**	Registered devices only	n/a
Allow calls **to** registered devices (direct or from a VMR)	✓	✓				Registered devices only	n/a
Allow calls **from** registered devices to an external system	✓		✓		Typically, aliases that are not in the Pexip Infinity domain e.g. **(?!.*@<Infinity domain>$).***	Registered device or external system	Select as required (needs a separate rule per protocol)

Avoiding call looping

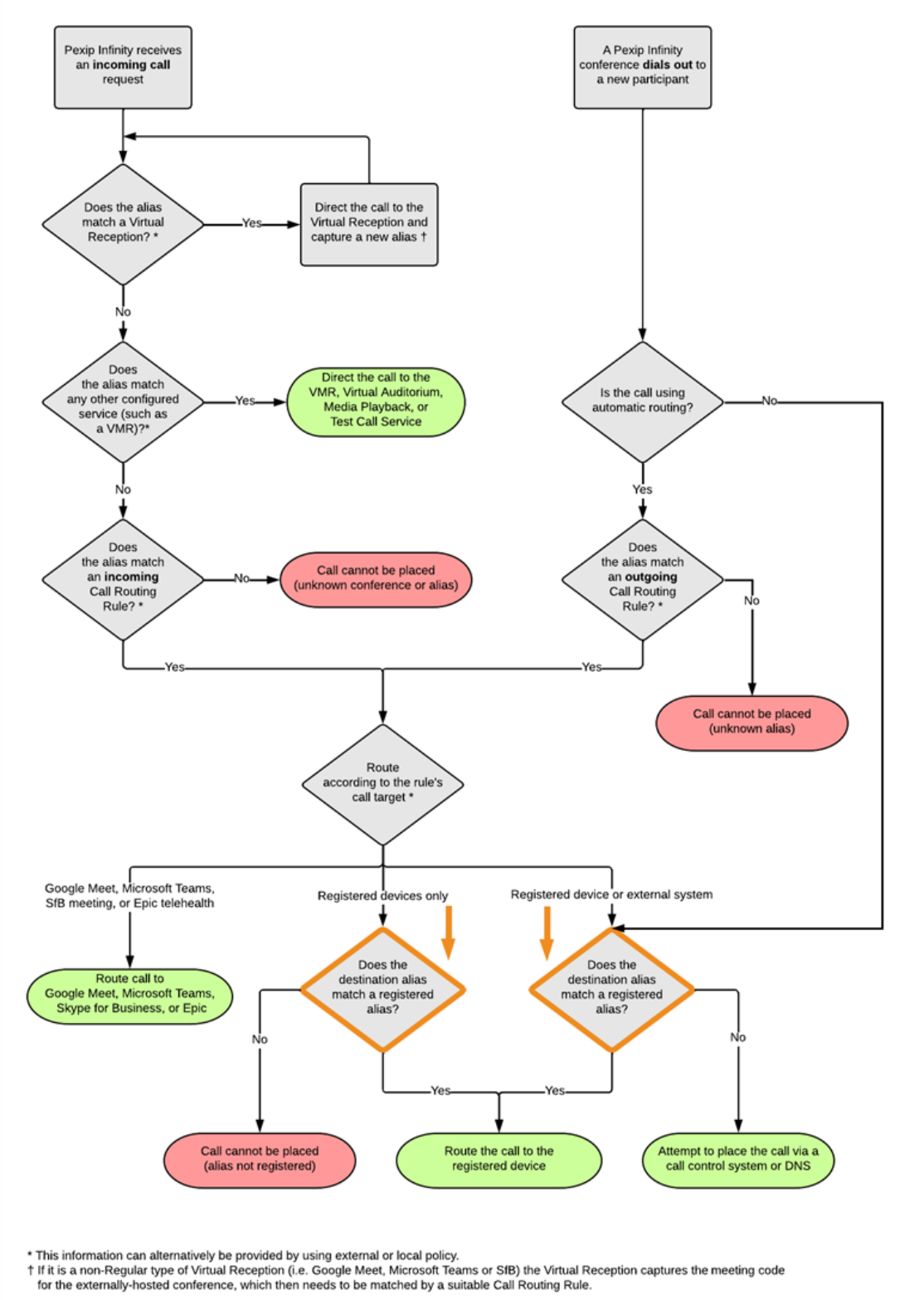
Call Routing Rules that are intended to route calls to devices registered to Pexip Infinity (e.g. Pexip apps) could, depending upon DNS configuration, result in call looping if the destination device alias is not currently registered.

This can occur if the Call Routing Rule handling those calls has its **Call target** set to ***Registered device or external system***. In this case, if the alias is not currently registered, Pexip Infinity will attempt to place the call via DNS or a call control system. This could result in the call being routed back to Pexip Infinity which would then match the same Call Routing Rule and result in a loop.

To avoid this, we recommend that you configure a **Call target** of ***Registered devices only*** for your Call Routing Rules that match those device aliases that you expect to be registered. This ensures that the call will fail cleanly without looping if the device is not currently registered.

Call routing logic

The stage in Pexip Infinity's call routing logic where it checks whether the device that is being called is registered is highlighted in the following diagram:



* This information can alternatively be provided by using external or local policy.
† If it is a non-Regular type of Virtual Reception (i.e. Google Meet, Microsoft Teams or SfB) the Virtual Reception captures the meeting code for the externally-hosted conference, which then needs to be matched by a suitable Call Routing Rule.

Note that you can configure rules so that only registered devices are allowed to make calls via the Infinity Gateway.

Configuring the registrar service

Devices can only register to Pexip Infinity if the registrar service is enabled.

To configure Pexip Infinity's registrar service, go to **Services > Registrar**. The options are:

Option	Description
Enable registrar	Controls whether devices can register to Pexip Infinity. Devices must register with a permitted alias (configured via **Users & Devices > Device Aliases**). The registrar service is enabled by default.
Registration refresh (general)	
Registration refresh strategy	Defines which strategy to use when calculating the expiry time of a SIP or H.323 registration: • ***Adaptive***: Pexip Infinity automatically adjusts the refresh interval depending on the number of current registrations on the Conferencing Node handling the registration request, in order to spread the load of registration refreshes. As the number of devices registered to a Conferencing Node increases, the refresh interval calculated by Pexip Infinity will also typically increase, although it will always be within the bounds of the configured minimum and maximum refresh intervals. • ***Basic***: Pexip Infinity simply uses the configured minimum and maximum settings, along with the requested value, to determine the refresh interval. Default: Adaptive.
Minimum refresh interval (Adaptive strategy)	The minimum interval in seconds before a device's registration must be refreshed, when using the ***Adaptive*** strategy. Default: 60 seconds.
Maximum refresh interval (Adaptive strategy)	The maximum interval in seconds before a device's registration must be refreshed, when using the ***Adaptive*** strategy. Default: 3600 seconds.
Minimum refresh interval (Basic strategy)	The minimum interval in seconds before a device's registration must be refreshed, when using the ***Basic*** strategy. Default: 60 seconds.
Maximum refresh interval (Basic strategy)	The maximum interval in seconds before a device's registration must be refreshed, when using the ***Basic*** strategy. Default: 300 seconds.
Registration refresh intervals for SIP endpoints behind NATs	
Minimum refresh interval (when behind NAT)	The minimum interval in seconds before a device's registration must be refreshed, for a SIP endpoint behind a NAT. Default: 60 seconds.
Maximum refresh interval (when behind NAT)	The maximum interval in seconds before a device's registration must be refreshed, for a SIP endpoint behind a NAT. The refresh interval for SIP endpoints behind a NAT typically has to be shorter than the interval for other endpoints. This is to help keep the NAT pinhole alive. Default: 90 seconds.
Call routing for desktop clients	

Option	Description
Route calls via registrar	(Applies to the legacy Connect desktop app only) When enabled, all calls from registered Pexip Connect desktop apps are routed via the Conferencing Node to which the app is registered, regardless of the domain being called. The app uses the previously-discovered IP address of the Conferencing Node — it does not perform a DNS SRV lookup of the **Registration Host** server address each time a call is placed. When disabled, DNS is used to identify the Conferencing Node to which calls are placed from registered apps. Default: enabled. ⓘ This setting does not affect the new Pexip app for Windows — all calls from the app are routed via the Conferencing Node to which the app is registered.

Note that:

- In all circumstances, requests for a value lower than the relevant **Minimum refresh delay** will result in the registration being rejected.
- The registration refresh strategies apply only to SIP and H.323 endpoints. Pexip apps have a fixed refresh interval of 120 seconds.
- Any changes to the **Route calls via registrar** setting are picked up by the client each time it re-registers or refreshes its registration.

Specifying the aliases that devices are allowed to register with

Devices can only register to Pexip Infinity with a permitted alias, and for specifically-enabled protocols.

You can bulk provision the device aliases from directory information contained in a Windows Active Directory LDAP server, or any other LDAP-accessible database. You can also import device aliases using a CSV file.

To manually configure the aliases that devices are allowed to register to Pexip Infinity, go to **Users & Devices > Device Aliases**. The options are:

Option	Description
Device alias	The alias URI that a device/client can register to Pexip Infinity. The alias must be an exact match; regular expressions are not supported. It cannot be blank.
Creation time	This read-only field shows the date and time when this record was first configured.
Service tag	This optional field lets you assign a unique identifier to this device, which you can then use to track use of the service.
Description	An optional description of the device. Note that this description may be displayed to users of registered Pexip apps who are performing a directory search.
Enable SIP registration *	Allows this device alias to register over the SIP protocol. The registration is optionally authenticated using the specified **Username** and **Password**.
Enable H.323 registration *	Allows this device alias to register over the H.323 protocol. The registration is optionally authenticated using the specified **Username** and **Password**.
Enable legacy Infinity Connect registration (non-SSO) *	Allows a legacy Connect desktop appto register using this alias (not using SSO services). The registration is optionally authenticated using the specified **Username** and **Password**.
Enable Infinity Connect registration using legacy AD FS *	Allows a legacy Connect desktop app to register using this alias, using AD FS to authenticate the registration. ⓘ This method of authentication is being deprecated and is not supported by the new Pexip apps.

Option	Description
Enable registration using IdP SSO *	Allows a Pexip app for Windows to register using this alias, using a SAML or OIDC Identity Provider to authenticate the registration. When this option is enabled, you must also select an **Identity Provider Group** from the drop-down options below.
Username	These credentials apply to aliases that are enabled to register using **SIP**, **H.323** or **non-SSO registration**.
Password	They are used to authenticate the device's registration. Credentials are optional and the device is not challenged if no credentials are entered.
Device origin	The name of the LDAP sync template used to create this device alias (it is blank if the device was created by manual input or via the API). This field is read-only.
Owner's email address	The email address of the owner of the device alias. Provisioning messages for this device alias will be sent to this address. This field is required for devices registering using **Infinity Connect registration using legacy AD FS**.
Identity Provider group	(Available and required when **Enable registration using IdP SSO** is selected) Select the set of Identity Providers to use to authenticate the registration.

* In order for this option to take effect, the Registrar service must be enabled.

Directory (phone book) of devices and VMRs for registered Pexip apps

Pexip Infinity provides a directory search facility to all Pexip apps that are registered to a Conferencing Node.

Upon request from the Pexip app (typically when the user performs a search on their contact list), Pexip Infinity will return a list of device and conference aliases that match the search string entered by the user.

Pexip Infinity checks the supplied search string to see if it is contained within the alias or alias description of any of its configured services (Virtual Meeting Rooms, Virtual Auditoriums or Virtual Receptions), or within the device aliases/descriptions that are allowed to register to Pexip Infinity (it does not matter if that device alias is currently registered or not, and no presence information is supplied).

Pexip Infinity returns up to 5 service aliases and up to 5 device aliases.

The directory service can be controlled by the **Enable directory** global setting (**Platform > Global Settings > Connectivity**). It is enabled by default.

Feature extensions via the external policy API

If you make use of the external policy API, the external system can be used to:

- Determine whether a device alias is allowed to register to a Conferencing Node.
- Obtain extended directory information — this can replace or be used in addition to the list of local device and VMR aliases supplied to registered Pexip apps.

See Using external and local policy to control Pexip Infinity behavior for more information.

Customizing and branding the web app

The branding and styling of the web app can be customized. This changes the look and feel of the web app regardless of which service is being accessed. (However, the theme-based elements of each individual service may also have been customized — a theme changes the look and feel of the actual conference you have joined, or are trying to join.)

Pexip web app customization can be used to control:

- default app settings such as bandwidth, default background image, screen sharing frame rate
- default user settings such as whether microphone and camera are muted on join
- the ability to display an image/logo and accompanying welcome text on a landing page, and to use a custom favicon
- language translations and the default language

- the color scheme for buttons, icons and other graphic indicators; elements can be customized individually or a general color scheme can be applied to all similar items.

To customize the **Pexip web app** you typically create a branding package and then upload it to the Management Node. You can use Pexip's branding portal to quickly and easily create branding packages, or you can create them manually for more advanced customizations.

Customization is not available in the Pexip app for Windows or Pexip app for mobile.

This topic explains the two main steps in applying branding to web apps in your deployment:

1. Creating a branding package:
   - using the Webapp3 branding portal
   - by downloading an existing package for editing (including any of the branding packages that are provided by default).

   For information on creating a branding package manually, see Advanced branding and customization.
2. Uploading the branding package to the Management Node so that it is available to all users.

It also includes information on:

- Updating an existing package
- Removing a package (reverting to default branding)

Creating a branding package using the portal

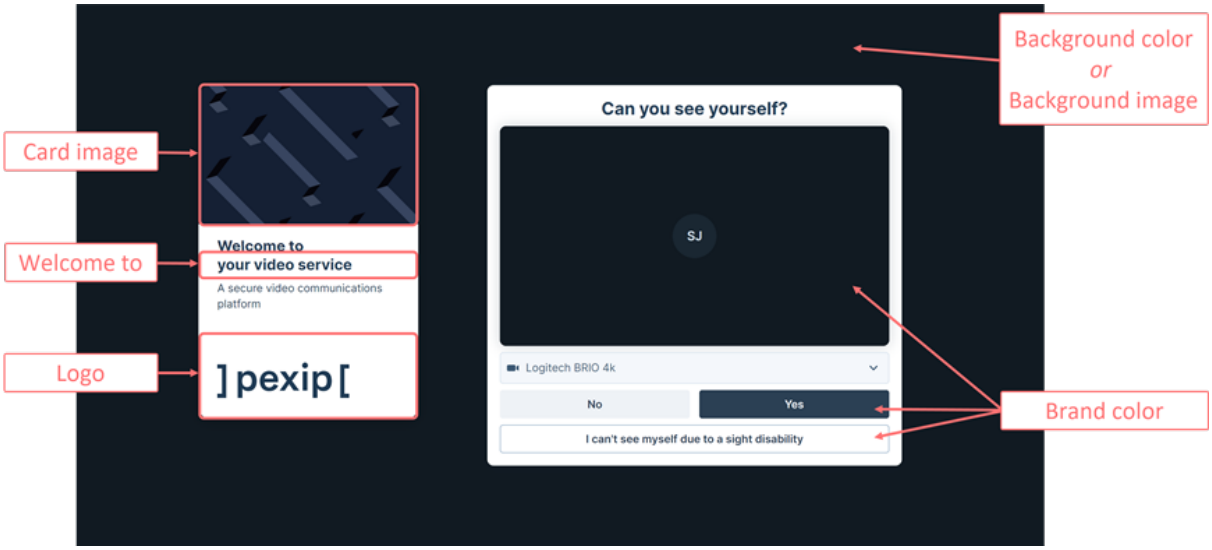
You must create a branding package before you can upload it to the Management Node. Our recommended method for creating a branding package is to use the branding portal for the current web app (Webapp3), although you can also create the required files manually. The branding portal guides you through the selection of your image files and colors (without having to edit individual CSS files etc.), and then generates the customized branding package for you.

For information about the branding portal for the legacy web app (Webapp2), refer to the v36 documentation.

Using the web app branding portal

1. Go to the Pexip branding portal (https://branding.pexip.io/).
2. Select one of the options:
   - to create a completely new package, select **New brand**
   - to use an existing package as the basis of your new package, select **Load existing brand**.
3. From the panel on the left, select each option and edit it to suit your brand's requirements.

   The image below shows what elements of the web app can be customized. The table that follows describes each option in more detail. If you are intending to edit the resulting branding package manually, it also gives the section within the manifest.json file where that element is defined.



Option	Description	Definition in manifest.json
Welcome to	Specifies the text that appears under the card image on the landing page.	"brandName"
Logo	An optional image/logo file that can be displayed at the bottom of the area on the left side of the landing page. This file must meet the following requirements: ◦ transparent ◦ maximum displayed width: 168px ◦ maximum displayed height: 80px ◦ format: we recommend .SVG	"logo"
Brand color	Specifies a color that the branding portal will use as a base to automatically generate the range of colors used for interactive elements of the user interface, such as buttons and text. You can use the color picker tool to select a color from any of the images you have already uploaded.	"baseColor" "colorPalette"
Background color	Specifies the landing page's background color if no background image is specified. You can use the color picker tool to select a color from any of the images you have already uploaded.	"backgroundColor"
Card image	An optional image/logo file that can be displayed at the top of the area on the left side of the landing page. This file must meet the following requirements: ◦ resolution: 640x360 ◦ size: 70-140 kB ◦ format: .JPG, .JPEG, .PNG or .WEBP	"jumbotron"
Background image	An optional image to display behind all other elements on the landing page. Supported file formats are: ◦ .JPG ◦ .JPEG ◦ .PNG ◦ .WEBP Background image files must meet the following requirements: ◦ resolution: 1920x1080 ◦ size: 500-600 kB	"background"
Overlay	Determines the contrast between the background and overlay text.	"overlay"
Overlay opacity	Determines the degree to which the background image is darkened or lightened.	"overlayOpacity"
Application title	Specifies the text that appears on the browser tab when using the Pexip web app.	"appTitle"
Terms of service	Specifies the URL to which the "terms of service" text (which appears on the "video address" step of the join flow, and under **Settings > About**) is directed.	"termsAndConditions"
Disconnect destination	Specifies the URL to which users are directed when a call is completed (instead of returning to the app home page).	"disconnectDestination"

Option	Description	Definition in manifest.json
Handle OAuth 2.0 Redirects	(Supported in Pexip Infinity v34 and later) Enables support for features (such as plugins) that require authentication to a third party using an OAuth 2.0 / OpenID Connect flow, with a redirect destination of **webapp3/oauth-redirect**.	"handleOauthRedirects"

4. When you have finished configuring your branding, from the top right of the page select **Download**.
   This creates and downloads a **brand.zip** file containing your client customizations.
5. If you wish to make further customizations you can unpack the file, edit it, and then re-zip it.
6. Upload the branding package to your Management Node.

Downloading an existing package

As an alternative to using the branding portals to create new branding packages, you can download, edit, and upload an existing branding package. This might be a default branding package downloaded from the Management Node, a package previously created using the branding portals, or a package downloaded from previous versions of Pexip Infinity.

Manually editing an existing package is useful if you have very specific modifications that you want to apply to the branding files, or if you are including plugins. Note that manual configuration requires knowledge of core web-design technologies such as HTML, JavaScript and CSS.

This section covers how to obtain branding packages.

Downloading branding packages from the Management Node

Pexip Infinity includes a number of default branding packages that you can download and edit to create your own customized branding. In addition, if you have existing custom branding files uploaded, you can download and use these as the basis on which to apply your modifications.

You may also wish to download an existing branding package in order to upload it to an external server if you are hosting the web app externally.

To download an existing branding package from the Management Node:

1. Go to **Web App > Web App Branding**.
2. Select the Pexip branding package for the relevant web app version and then from the bottom of the detail page select **Download**.
   A ZIP file containing the selected branding files is downloaded to your local file system.
3. Unpack the downloaded file and apply your modifications to the relevant files.
4. Repackage your branding files into a single ZIP file. This file must have a different name to any packages already uploaded. If you are editing an existing branding package, after uploading simply redirect the current path to point to this new package.
   - The ZIP file must contain the complete set of branding files. You must retain the original file/folder structure in the rebuilt ZIP file.
   - You must include the **manifest.json** file in the **webapp3/branding** folder.
5. Upload the branding package to your Management Node.

Editing an existing package

You can also use as the basis of your new branding packages any other existing branding packages. These might be downloaded from previous versions of Pexip Infinity, from either of the branding portals, or be packages that you created entirely manually. Simply ensure that these packages meet the current branding requirements before zipping and uploading the new package.

Uploading the branding package

Branding packages are uploaded as **.ZIP** files. The files must contain the required branding files in a **webapp3/branding** subfolder.

To upload a branding package to your Management Node:

1. Go to **Web App > Web App Branding**.
2. From the bottom of the page, select **Add Webapp branding package**.
3. On the **Add Webapp branding package** page, enter the following information about the package you wish to upload:

Name	The name for this branding package.
Description	An optional description of this branding package, to help you identify it easily.
Web app version	Select the web app version to which this branding package will apply.
Branding package to upload	Select **Choose File** and select the ZIP file containing your customizations.

4. Select **Save**.
   The branding package is verified and uploaded.
5. After the branding package is uploaded, to use it you must select it for use with one or more web app paths, creating a new path if necessary.

Any changes you make to branding packages and paths must be replicated out to all Conferencing Nodes before being available (typically after approximately one minute).

Updating an existing package

To make changes to a branding package already in use in your deployment:

1. Download the old branding package and edit it with your updates.
2. Save the updated branding package as a ZIP file.
3. Upload the updated branding package (**Web App > Web App Branding**), giving it a different name to the existing version.
4. Edit all paths that use the old version of the branding package so that they use the new version instead.
5. Delete the old branding package.

Don't delete the old branding package before uploading the new package and redirecting to it. If you do, all the paths that used the deleted package will revert to using the default branding, meaning that participants will experience the default branding until you have redirected the relevant paths to use the new package.

Removing a package (reverting to default branding)

Default branding packages cannot be deleted. You can remove customized branding in one of two ways:

- If you wish to keep the branding for possible later use, simply edit the paths that currently use that package, so that the paths either use a different branding package, or use the default branding.
- If you no longer wish to keep the branding package, delete it. Any paths that point to deleted branding packages will revert to using the default branding.

To delete a branding package:

1. On the Management Node, go to to **Web App > Web App Branding**.
2. Select the tick boxes next to the branding packages you wish to remove.
3. From the **Action** drop-down, select ***Delete selected Web app branding packages***.

Wait for the customized branding to be removed from all Conferencing Nodes (typically after approximately one minute). After this time, all new participants accessing the path will see the default branding. Note however that any participants currently using the deleted package will continue to see the deleted branding until they refresh their browser.

Pexip Infinity conference settings

About aliases and access numbers

Every Virtual Meeting Room (VMR), Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service and Test Call Service has one or more aliases associated with it.

When Pexip Infinity receives an incoming call via one of its Conferencing Nodes, it checks whether the destination alias belongs to a Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service; if so, it will direct the call to that service. If the alias does not belong to any of the above services, Pexip Infinity next checks through the Call Routing Rules used by the Infinity Gateway to see if the alias matches any rules specified there — for more information, see Service precedence.

To allow a call that is placed from outside your Pexip Infinity deployment to reach your Conferencing Nodes, the alias that has been dialed by the external participant must usually be in the form of a URI (e.g. **name@domain**), and you must have appropriate DNS records set up so that any such calls to your domain are routed to your Conferencing Nodes. For more information, see DNS record examples.

When Pexip Infinity receives a call to a **Virtual Meeting Room** (including scheduled conferences) or **Virtual Auditorium** alias, it creates a conference instance and routes the call to that conference. Any further calls received to any of the aliases belonging to the same Virtual Meeting Room or Virtual Auditorium are routed to the same conference instance, for the duration of that particular conference. If the service has more than one alias, participants can dial any one of the aliases and be routed to the same conference instance. For more information, see Using multiple aliases to access the same service.

When a participant dials a **Virtual Reception** alias, they are taken to the Virtual Reception IVR service, from which they can use a DTMF keypad to enter the number of the Virtual Meeting Room or Virtual Auditorium they wish to join. This number must correspond to a numeric-only alias of the Virtual Meeting Room or Virtual Auditorium in question. For more information, see About the Virtual Reception IVR service.

When a participant dials a **Media Playback Service** alias, they are taken to the playback service which, after the media playback has completed, can be configured to transfer the participant (via another, different alias) to another service such as a VMR. For more information, see Configuring a Media Playback Service.

Pexip Infinity's **Test Call Service** provide a test loopback service that allows users to check the quality of their video and audio, and verifies that they can connect to a Conferencing Node. See Configuring the Test Call Service for more information.

In most cases, the alias received by Pexip Infinity is the same as the alias that the conference participant dialed from their endpoint, but there are some exceptions, described in Search rules and alias transforms and ENUM.

For simplicity, the following discussion assumes that the alias dialed by the endpoint user is the same as the alias received by Pexip Infinity.

Creating, editing and viewing aliases

Each alias is associated with a single service, but a service can have more than one alias.

To view a list of all existing aliases for all services, go to **Services > Aliases**. From here you can:

- Sort the list of aliases alphabetically by **Alias** or by **Service** name.
- Search for a particular alias.
- Add a new alias.
- Modify an alias, including changing the service with which it is associated.

You can also add or modify aliases while creating or editing a service.

When adding or editing aliases, the options are:

Option	Description
Service	Select the name of the Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service to access when participants dial this alias.

Option	Description
Alias	The alias that, when received by Pexip Infinity, is used to route the call to this service (Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service). The alias entered here must match the alias as it is received by Pexip Infinity. Wildcards and regular expressions are not supported. In most cases, the alias received by Pexip Infinity is the same as the alias that the participant used to call the service, but there are some exceptions, described in About aliases and access numbers. You may also want to define multiple aliases for the same service to ensure that it can be accessed by devices and protocols that enforce specific alias formats — for more information, see Using multiple aliases to access the same service.
Description	An optional description of the alias. This is useful if you have more than one alias for a service. Note that this description may be displayed to users of registered Pexip apps who are performing a directory search.

Guidelines and restrictions

These guidelines apply when configuring aliases:

- An alias can include a domain or subdomain but this is optional (see Domains).
- Aliases configured on Pexip Infinity can contain any unicode character. However, SIP and H.323 devices have limitations on the characters they can dial, so you should ensure that the aliases you configure are dialable from these devices.
- Each alias must be configured separately — wildcards or regular expressions are not supported.
- In certain circumstances an alias can also be in the form of an IP address — for more information see Dialing by IP address.

Case insensitivity

The aliases that you configure on Pexip Infinity are not case-sensitive, and Pexip Infinity treats all incoming aliases as not case-sensitive. For example, this means that:

- a user who dials `meet.alice` will match against a VMR with an alias of `Meet.Alice`
- a user who dials `Meet.Alice` will match against a VMR with an alias of `meet.alice`

Ignoring IP addresses

Pexip Infinity ignores any IP address or IP address and port combination appended to an alias. This is because some SIP endpoints automatically add the IP address of their proxy to the URI that is dialed by the user.

For example, a VMR with a single alias of `meet.alice` will be matched when an endpoint dials any of:

- `meet.alice`
- `meet.alice@<IPaddress>`
- `meet.alice@<IPaddress>:<port>`

Dialing by IP address

We do not recommend dialing by IP address. However, to support endpoints that can only dial by IP address, you can give a service an alias that is the same as the IP address of one of your Conferencing Nodes.

In this case, when a user dials the IP address from that endpoint, the call will be routed directly to the Conferencing Node with that IP address. Pexip Infinity will then perform its standard behavior and check to see if the destination alias of the call (which in this scenario will be the dialed IP address) matches any of its configured aliases; if so, the call will be routed to the service (Virtual Reception, Virtual Meeting Room etc.) that is associated with that alias.

If other endpoints, such as Pexip apps, dial the IP address alias, they will also be connected to the appropriate service (although the Conferencing Node they connect to is determined via other means).

In most cases you would use this feature to assign a Conferencing Node IP address as an alias for a Virtual Reception, so that users could then select which Virtual Meeting Room or Virtual Auditorium they wish to join. For more information, see About the Virtual Reception IVR service.

Ignoring protocol prefixes

Pexip Infinity ignores relevant URI schemes included in an alias, as in the case where a SIP endpoint automatically prefixes the URI dialed by the user with "sip:". The ignored prefixes are "sip:", "sips:" and "h323:". This is because the protocol used for the call does not matter to Pexip Infinity.

For example, a VMR with a single alias of `meet.alice` will be matched when an endpoint dials any of:

- `sip:meet.alice@IPaddress`
- `sips:meet.alice@IPaddress`
- `h323:meet.alice`

Search rules and alias transforms

Some call control systems (for example the Cisco VCS) support the use of search rules and alias transforms. In these cases, the alias that was dialed by the endpoint user and received by the call control system may be changed by the call control system in some way before it is passed on to Pexip Infinity. Often this is done in larger deployments to support complex dial plans.

For example, the call control system might have a rule that replaces the domain `example.com` with `example.net`. This means that when a user dials `meet.alice@example.com`, the call is routed to `meet.alice@example.net`. Therefore it is this latter alias that must be configured on Pexip Infinity in order to match it with a VMR.

ENUM

The ENUM system allows users to dial an E.164 number (for example a telephone number) which is then mapped using DNS to a SIP URI. For more information, see RFC 3761.

For example, your dial plan could be set up so that when a user dials `555 0123`, the call is routed via ENUM to `meet.alice`.

If your dial plan uses ENUM, the resulting SIP URIs must be included as aliases on Pexip Infinity in order to match them with a VMR.

Domains

Domains are optional in aliases. However, when a call is received to an alias in the form of a URI that includes a domain, the domain is not ignored.

For example, if a VMR is configured with an alias of `meet.alice`, users can dial `meet.alice` or `meet.alice@<IPaddress>` to access the meeting room. However, if they dial `meet.alice@example.com` they will not be able to access the VMR because the VMR alias does not include the domain.

When a SIP endpoint user dials an alias that does not include a domain (for example `meet.alice`), the SIP endpoint will automatically add its own domain to the alias (making it for example `meet.alice@example.com`). So even if the VMR is configured with an alias of `meet.alice` and this alias is dialed by a participant from a SIP endpoint, the participant will not be able to join the conference. (H.323 endpoints do not add domains.)

In the absence of a call control system to strip the domain part of the alias, you could instead add to the VMR a second alias of `meet.alice@example.com` so that participants can dial `meet.alice` from either a SIP or H.323 endpoint and access the same conference.

Using multiple aliases to access the same service

You may want conference participants to be able to dial different numbers to access the same Virtual Reception, or to allow different conference participants to dial different aliases but still end up in the same VMR.

Thus, depending on your deployment environment and the types of devices and protocols used to dial into your services, you may want to configure a variety of alias formats to allow access to those services, including:

- **name@domain** (for full URI dialing such as from SIP clients)
- **name** (for systems that strip off or do not require a domain)
- **numeric/E.164** (for endpoints that do not support URI dialing, or can only dial via IP address, or for audio-only callers via PSTN/ISDN gateways that need to be routed via a Virtual Reception).

It is often good practice to create a URI and a numeric alias as a minimum. If numeric/E.164 formatted aliases are used as your primary dial plan, you could also reuse the same number in a number@domain URI-style alias. Some example alias dial plans are given below.

You can add any number of aliases to a single service (Virtual Meeting Room, Virtual Reception etc). Note that the order in which aliases appear in the list of aliases is relevant when Automatically dialing out to a participant from a conference.

Adding additional aliases

To add an additional alias to a service while you are creating it, select **Add another Alias**.

To add another alias to an existing service:

1. Go to **Services > Virtual Meeting Rooms**, **Services > Virtual Auditoriums**, **Services > Virtual Receptions**, **Services > Scheduled Conferences** or **Services > Test Call Service** as appropriate.
2. Select the name of the service you wish to add the alias to.
3. In the **Aliases** section at the bottom of the page, enter the new alias in the empty **Alias** field.
4. Select **Save**.
5. You can change the order in which aliases are listed by selecting the **Move up** ↑ and **Move down** ↓ buttons to the right of the alias panel.

If you have provisioned your VMRs from Active Directory via LDAP and you want to manually change or add aliases to those VMRs, you may want to ensure that **Allow aliases to be overridden** is enabled on the LDAP sync templates used to generate those VMRs, otherwise any changes you make will be overridden the next time that template is synchronized.

Examples

- If some of your endpoints do not support URI dialing, you could set up a VMR for Alice with one alias of **meet.alice.jones** and another alias of **555 25423**. This would give conference participants two different methods of accessing the same conference.
- If your deployment includes a Virtual Reception, every Virtual Meeting Room and Virtual Auditorium that you want to be accessible from the Virtual Reception needs to have a numeric-only alias (so that participants can enter the number using DTMF), in addition to any aliases in the form of a URI. So to continue with our example, you could add a third alias to Alice's VMR of **25423**, which is the number that participants would enter from the Virtual Reception. For more information, see About the Virtual Reception IVR service.
- If you use an ISDN gateway to support telephone participants, you could set up Alice's VMR with aliases of **meet.alice.jones** and an appropriate E.164 number.
- If you do not have or do not wish to use a call control system to transform aliases, you could set up Alice's VMR with aliases of **meet.alice.jones** and **meet.alice.jones@example.com** to support internal and external dialing, and dialing from both SIP and H.323 endpoints.

About PINs, Hosts and Guests

For added security, you can set up your Pexip Infinity Virtual Meeting Rooms (including scheduled conferences) and Virtual Auditoriums with PIN numbers. You can use the same PIN for all participants, or you can use PINs to differentiate between Hosts and Guests.

When a participant dials the alias of a conference that has a PIN, they are presented with an Interactive Voice Response (IVR) screen where they are asked to enter the PIN number. They must enter the correct PIN before they can join the conference.

ⓘ In addition to PINs, you can also require authentication for participants attempting to join a conference held in a Virtual Meeting Room or Virtual Auditorium.

This topic covers:

- Using the same PIN for all participants
- Using PINs to differentiate between Hosts and Guests
- Creating a Hosted Virtual Meeting Room or Virtual Auditorium
- How to combine Host PINs and Guest PINs for differing levels of security
- Setting the default minimum PIN length
- Allowing end-users to manage their PINs
- Incorrect PINs

- Using # at the end of a PIN
- Changing a participant's role from Guest to Host (and vice versa)
- Including the PIN in the dial string to bypass the PIN entry screen
- Limiting the time a participant can spend at the PIN entry screen
- Limiting how long Guests can wait for a Host

Using the same PIN for all participants

All participants in a conference that uses a single PIN have the same Host privileges. (To create a PIN-protected conference where participants have different levels of privileges, see Using PINs to differentiate between Hosts and Guests.)

To set a single PIN number on a conference:

1. Go to **Services > Virtual Meeting Rooms**, **Services > Virtual Auditoriums** or **Services > Scheduled Conferences** and select the service you wish to change.
2. In the **Host PIN** field, enter the PIN number to be used.
3. In the **Allow Guests** drop-down list, select ***No***.

Note that:

- PINs must use the digits **0-9** only.
- PINs may optionally end with **#**.
- PINs must be between 4–20 digits long, including any **#**.
- The default minimum length is 6 digits (**Platform > Global Settings > Service Configuration > Minimum PIN Length**).

Using PINs to differentiate between Hosts and Guests

You can set up your Virtual Meeting Rooms and Virtual Auditoriums so that they are "Hosted". Hosted conferences have two different types of participants: **Hosts** and **Guests**. Hosts and Guests both dial the same alias to join the conference.

You can create two types of Hosted conference:

- the Host must enter a PIN, but Guests do not require a PIN, or
- both Hosts and Guests must enter a PIN.

In both cases:

- Only those participants who enter the **Host PIN** have privileges to control the conference; everyone else is a Guest and can only participate in the conference.
- An administrator can configure individual Virtual Meeting Rooms and Virtual Auditoriums so that Guest participants are not allowed to present into the conference (they can still receive presentation content from other Host participants). By default, Guests are allowed to present content.
- If a conference is locked, Hosts can still join, but Guests are held at a waiting screen.
- For conferences that use Virtual Auditoriums, the administrator can select different layouts for the Host view and Guest view. For more information, see Conference layouts and speaker names.

Guest privileges

- If any Guests join the conference before a Host has arrived, or while the conference is locked, they are shown a holding image and hear a message advising them that they are waiting for the conference Host. The message is repeated every 30 seconds for video participants, and every 15 seconds for audio-only participants. Such participants will be disconnected automatically after 15 minutes waiting, but Administrators can change this time - see Limiting how long Guests can wait for a Host.
- A minute or so after the last Host has left, any remaining Guests are automatically disconnected. Administrators can change the length of time before Guests are disconnected by going to **Platform > Global Settings > Service Configuration > Guests-only Timeout**.
- If the conference is held in a Virtual Auditorium, Guests can only see the Hosts during the conference (they cannot see other Guests). However, after the last Host has left, by default Guests can see any other remaining Guests during the disconnection timeout period, but this can be configured via the Virtual Auditorium's **Guests can see other guests** setting.

Host privileges

- Guests cannot join a conference until the first Host has joined. However, if the Host joins as a presentation and control-only participant via a supported Pexip app, or as a Meeting Manager, that Host has to start the conference manually to allow Guests to join.
- If the first participant to join the conference is a Host, that Host is shown a holding image until another Host or Guest joins the conference.
- If the conference has a Host PIN, participants who enter the Host PIN can join the conference even if it is locked.
- The presence of a Host in a conference may prevent that conference from being automatically terminated.
- A minute or so after the last Host has left, any remaining Guests are automatically disconnected. Administrators can change the length of time before Guests are disconnected by going to **Platform > Global Settings > Service Configuration > Guests-only Timeout**.
- Hosts can use Pexip apps to:
  - disconnect participants from the conference
  - add participants to the conference
  - mute and unmute an individual participant, or all Guest participants simultaneously
  - lock and unlock a conference, and to allow individual participants into a locked conference
  - change the layout and show/hide participant names
  - change the role of other participants in the conference
  - control another participant's camera (if it supports FECC)
  - spotlight a participant in the main video
  - lower the raised hand of another participant
  - send DTMF tones to other participants in the conference.
- The Pexip Infinity administrator may configure a Virtual Auditorium so that Hosts see other participants in a different layout to that seen by Guests. For more information, see Conference layouts and speaker names.

Creating a Hosted Virtual Meeting Room or Virtual Auditorium

To create a Hosted Virtual Meeting Room or Virtual Auditorium:

1. Go to **Services > Virtual Meeting Rooms**, **Services > Virtual Auditoriums** or **Services > Scheduled Conferences** and select the service you wish to change.
2. In the **Host PIN** field, enter the PIN number to be used by the Hosts.
3. In the **Allow Guests** drop-down list, select ***Yes***.
4. If you want to require Guests to enter a PIN, in the **Guest PIN** field, enter the PIN number to be used.

   If you leave this field blank, anyone who dials the alias of the Virtual Meeting Room or Virtual Auditorium will be able to access the conference as a Guest. However, the conference will not start until the first Host has joined and in the meantime all Guests will continue to be shown the holding screen.

Note that:

- Host PINs and Guest PINs must be different.
- PINs must use the digits **0-9** only.
- PINs may optionally end with **#**.
- PINs must be between 4–20 digits long, including any **#**.
- If the Host PIN ends in **#** and a Guest PIN is used, the Guest PIN must also end with **#**.
- If **#** is not used, Host PINs and Guest PINs must have the same number of digits.
- You cannot configure a Guest PIN unless you have already configured a Host PIN.

How to combine Host PINs and Guest PINs for differing levels of security

The table below shows how to use **Host PIN**s (for Host privileges) and **Guest PIN**s (for Guest privileges) to achieve various levels of security when creating hosted Virtual Meeting Rooms and Virtual Auditoriums:

Access	Roles	Host PIN	Allow Guests	Guest PIN
Anyone - no PIN required	All participants have the same Host privileges.	Leave blank	Select ***No***	Leave blank
All participants must enter the same PIN	All participants have the same Host privileges.	Enter the PIN	Select ***No***	Leave blank
Hosts must enter a PIN but Guest do not	Participants who enter the PIN have Host privileges. The conference will not begin until they have joined, and will finish a minute or so after the last Host leaves. All other participants have Guest privileges and do not need to enter a PIN. They will see a holding screen until the first Host joins.	Enter the PIN	Select ***Yes***	Leave blank
Hosts and Guests must enter different PINs	Participants who enter the **Host PIN** have Host privileges. The conference will not begin until they have joined, and will finish a minute or so after the last Host leaves. Participants who enter the **Guest PIN** have Guest privileges and will see a holding screen until the first Host joins.	Enter the PIN (must be different from the **Guest PIN**)	Select ***Yes***	Enter the PIN* (must be different from the **Host PIN**)

* If you configure a **Guest PIN**, you **must** also configure a **Host PIN**.

Setting the default minimum PIN length

The default minimum PIN length is 6 digits and is configured via **Platform > Global Settings > Service Configuration > Minimum PIN Length**.

Allowing end-users to manage their PINs

Host and Guest PIN numbers are initially assigned when the VMR is created by the administrator.

The Pexip VMR self-service portal is a separately-installable component that allows end-users to manage their personal Virtual Meeting Room without having to send requests to their administrator to change the configuration or branding of their VMR.

Incorrect PINs

Participants (other than Pexip apps, and connections via the client API) have three attempts to enter a correct PIN. After the third unsuccessful attempt, they will hear a message advising them that the PIN is invalid, and their call will be disconnected. Further protection against PIN cracking attempts can be achieved using the Break-in resistance settings, which apply to all clients.

All PIN entry attempts, successful and unsuccessful, are logged in the administrator log.

Using # at the end of a PIN

PINs can optionally end with a **#** (known either as the **pound** or **hash** character). The use of a terminal **#**:

- Allows administrators to set different length PINs for Hosts and Guests.
- Allows users who have accidentally entered too many or too few digits (particularly those using audio-only, who have no on-screen indication of the number of digits required) to return to the start of the PIN entry process by entering **#**.
- Adds extra security by disguising the length of the PIN. This is because when a PIN does not end in a **#**, users (except those using Pexip apps) are presented with the "Invalid PIN" message after entering the same number of digits as that of the PIN.

You cannot configure a Host PIN that ends with **#** and a Guest PIN that does not (or vice versa).

For conferences where a terminal **#** is used, Pexip Infinity automatically uses the appropriate audio file, asking participants to "please enter the conference PIN number followed by the pound key". Preconfigured themes allow you to change the use of "pound key" to "hash key".

For Pexip apps, the terminal **#** is optional when entering a PIN.

Examples

- If you want to use a terminal # on a VMR with a Host PIN of 123456 and a Guest PIN of 56789000:
  - in the **Host PIN** field enter **123456#**
  - in the **Guest PIN** field enter **56789000#**.

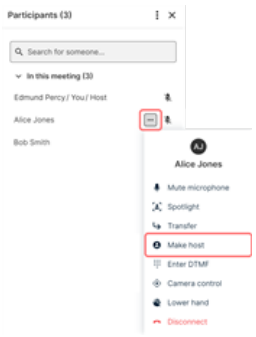
  Now when anyone calls the VMR they will hear the message "please enter the conference PIN number followed by the pound key".
- If you do not want to use a terminal # on a VMR with a Host PIN of 123456 and no Guest PIN:
  - in the **Host PIN** field enter **123456**
  - leave the **Guest PIN** field blank.

  Now when anyone calls the VMR they will hear the message "please enter the conference PIN number".

Changing a participant's role from Guest to Host (and vice versa)

Pexip app users who have a role of Host can change the role of other participants in the conference. This is useful, for example, if a Host needs to leave the conference; they can promote one of the Guests to a role of Host so that the conference will continue after they disconnect.

To change a participant's role: from the **Participants** panel, hover over the participant's name; additional icons appear. Select ··· and then select **Make Host** or **Make Guest**.



You cannot change your own role to Guest.

You can also use this feature to allow a participant who is waiting at the PIN entry screen to join the conference as Host or Guest without entering a PIN. Such users appear in the roster with a role of **Unknown**. (Note that this does not apply to participants using Pexip apps, as they use a different PIN entry process.)

Participants who join via a Skype for Business meeting have a role of **External**; their status cannot be changed.

Including the PIN in the dial string to bypass the PIN entry screen

SIP and H.323 endpoints and Skype for Business clients that dial into PIN-protected conferences can bypass the PIN entry screen by including the PIN in the dialed alias.

This makes it more convenient, for example, when bookmarking PIN-protected conferences.

They can do this by including the relevant PIN (either the Host PIN or the Guest PIN as appropriate) in their dial string when dialing the Virtual Meeting Room or Virtual Auditorium. The dial string should be in the format: **<vmr_alias>**<PIN>@<domain>**.

For example, if the alias of the Virtual Meeting Room is **meet.alice@example.com** and the PIN is **123456**, then the endpoint can dial **meet.alice**123456@e** to go straight into the VMR.

Note that H.323 devices can also use the dial format **<vmr_alias>#<PIN>@<domain>**.

If there is a **#** configured at the end of the PIN e.g. **123456#**, then that **#** is not required in the dial string.

Limiting the time a participant can spend at the PIN entry screen

By default, participants have two minutes to enter a correct PIN before the system disconnects them. The timer starts when the participant gets to the PIN entry screen, and participants will be disconnected at the end of the specified time, whether or not they have entered any PINs during that time (in other words, the timer does not restart after each PIN entry).

You can change the length of time by going to **Platform > Global Settings > Service Configuration** and changing the **PIN entry timeout**.

Note that this limit does not apply to participants using Pexip apps, or to connections via the client API, because they enter PINs via a different process.

Limiting how long Guests can wait for a Host

By default, a Guest participant will be disconnected if they have been waiting for 15 minutes for a Host to either:

- join the conference
- allow the Guest to join a locked conference, either by unlocking the conference or permitting the individual participant to join.

This time applies per-participant, so if Guest A joins a conference five minutes before Guest B and a Host does not join, Guest A will be disconnected five minutes before Guest B.

You can change this length of time by going to time by going to **Platform > Global Settings > Service Configuration** and changing the **Waiting for Host timeout**.

About participant authentication

Participant authentication via single sign-on (SSO) provides an additional, optional layer of security to prevent unauthorized access to meetings. It requires conference participants to authenticate via an Identity Provider service before they can join a Virtual Meeting Room or Virtual Auditorium, or access a Media Playback Service. It can be used instead of, or in combination with, conference PINs.

This topic covers:

- Using SSO-based participant authentication
- Supported Identity Providers
- Supported Pexip apps
- Joining an SSO-protected VMR
- Joining with other devices
- Authentication and PINs
- Transferring participants
- Display names

Using SSO-based participant authentication

You can optionally require participants who are attempting to access a specific Virtual Meeting Room, Virtual Auditorium or Media Playback Service in your deployment to verify their identity using SSO before joining the meeting. The SSO service can also be used to provide a verified display name for meeting participants.

To enable SSO-based authentication, first you set up one or more Identity Providers, which are third-party services (such as ADFS, Microsoft Entra ID or Okta) that store and manage digital identities, and with which users authenticate using SSO. Then, you configure individual services to require that Hosts, Guests, or both authenticate with an Identity Provider to access the service.

SSO-based authentication is supported for participants joining via the latest versions of the Pexip apps.

Participants joining from other devices (such SIP or H.323 endpoints) can optionally be permitted to join the meeting if the device is locally registered (i.e. registered to the same Pexip Infinity deployment that is hosting the VMR they are attempting to access), or manually admitted by a Pexip app user.

You can optionally require participant authentication for meetings scheduled using Pexip's Secure Scheduler for Exchange feature.

When is SSO not required?

In all cases, devices can join a meeting protected by SSO without needing to authenticate if:

- the device is manually dialed out to (either by a conference host or as an Automatically Dialed Participant)
- the device is paired with a participant who is using Webapp2 (and who has successfully authenticated).

Supported Identity Providers

Pexip Infinity can be configured with Identity Providers that support two widely-used standard industry protocols:

- SAML 2.0
- OpenID Connect (OIDC)

For more information about using SSO and Identity Providers with Pexip Infinity, see Using Identity Providers.

Supported Pexip apps

The apps that support participant authentication are:

- Web app from v27 and later
- Pexip app for Windows (any version)

Participants using older versions of the web app cannot join VMRs with participant authentication enabled. Instead they will receive a message saying that their client does not support SSO-protected meetings, and suggesting they upgrade their client.

Participants using other versions of the Pexip web app can still join the meeting if the VMR:

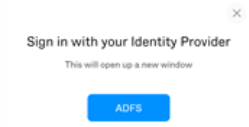
- does not require participant authentication
- requires participant authentication for Hosts only, and the participant is joining as a Guest
- requires participant authentication for Guests only, and the participant is joining as a Host.

Note that all references to Pexip app support also includes bespoke WebRTC clients using the Pexip client APIs.

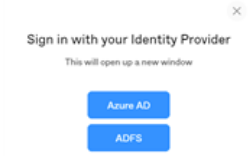
Joining an SSO-protected VMR

To join a Virtual Meeting Room or Virtual Auditorium with participant authentication enabled, using a Pexip app:

1. Access the VMR in the usual way (e.g. by entering the room name, or via a Virtual Reception).
2. If asked, select whether you are a Host or Guest, and enter the PIN (if required).
3. Sign in with your Identity Provider:

   If your organization uses more than one Identity Provider, you see them all listed; choose the one you wish to authenticate with:

   The name that appears on each button in the pop-up is the Name for the Identity Provider that is configured on Pexip Infinity.
4. If this is the first time you have used this Identity Provider, you are redirected to the Identity Provider where you must authenticate. This will usually be done via a new browser tab, which opens automatically. Otherwise, if you've already authenticated with the Identity Provider when accessing this or any other VMR, the authentication happens automatically using your previous credentials.

   Occasionally (at a period determined by the Identity Provider) your authentication session will expire and you will need to re-authenticate.
5. After successfully authenticating, you go straight into the meeting.

Joining with other devices

If authentication is enabled for a VMR or Virtual Auditorium, you can also determine how to treat participants attempting to join from devices other than Pexip apps, such as SIP or H.323 endpoints. You do this via the Other participants setting, where you have options to either put all such devices into a waiting room, or allow these devices to bypass the waiting room, but only if they are locally registered (i.e. registered to a Conferencing Node in the same deployment that is hosting the VMR or VA they are attempting to access).

Note that although some Pexip apps can be registered, the **Other participants** setting does not apply to them.

For each Virtual Meeting Room or Virtual Auditorium:

- if **Other participants** is set to ***Allowed if trusted***:
  - registered devices join the meeting directly (although they still need to enter a Host or Guest PIN if required)
  - unregistered devices are placed in the waiting room;
- if **Other participants** is set to ***Disallow all***:
  - both registered and unregistered devices are placed in the waiting room.

From the waiting room, participants must wait to be manually admitted to the conference by a meeting Host. Upon admission, they are not asked for a PIN and join as a Guest.

Authentication and PINs

VMRs and Virtual Auditoriums that require authentication can optionally use Host and Guest PINs.

For Pexip apps:

- If PINs are configured:
  - Participants need to enter the PIN before authenticating.
  - The PIN that they enter determines whether they are a Host or a Guest, which then determines whether or not they need to authenticate, and if so, the Identity Providers to use.
  - A participant who joins as a Guest can subsequently be escalated to a Host without authentication.
- If PINs are **not** configured, all users are treated as Hosts when authenticating.

For all other devices:

- If PINs are configured, and the device is locally registered, and **Other participants** is set to ***Allowed if trusted***, the participant must enter a PIN before joining the meeting.
- If PINs are **not** configured, and the device is locally registered, and **Other participants** is set to ***Allowed if trusted***, the participant joins the meeting automatically as a Host.
- In all other cases, the participant is held in a waiting room and must wait to be admitted to the meeting by a Host. They are not asked for a PIN and will join as a Guest.

Transferring participants

Participants can be transferred from one VMR into another VMR that requires authentication.

For participants using a supported Pexip app:

- A participant can be transferred into a VMR that requires authentication only if the original VMR also required authentication and used the same Identity Provider as the destination VMR. The participant will not need to re-authenticate to join the destination VMR. If the original VMR did not require authentication, or the destination VMR uses a different identity provider, the transfer is not initiated and the participant remains in the original VMR.
- A participant can be transferred from a VMR that requires authentication into a VMR that does not require authentication. From there, a participant can be transferred into a third VMR that requires authentication only if that VMR uses the same Identity Provider as the original VMR; the participant does not need to re-authenticate in order to join the third VMR. Otherwise, the transfer is not initiated and the participant remains in the second VMR.

For other devices, when a participant using a locally-registered endpoint is transferred, if **Other participants** is set to ***Allowed if trusted*** the participant will join the meeting directly (although they will still need to enter a Host or Guest PIN if required). In all other cases, for both registered and unregistered devices, the transfer will fail and the participant will remain in the original VMR.

Display names

Each Identity Provider offers a given set of attributes for a user (such as their display name, given name, surname and email address).

For Virtual Meeting Rooms and Virtual Auditoriums that require participant authentication, the name shown for each participant (in the participant list, and as a text overlay when **Show names of participants** is enabled) depends on what is configured in the display name setting for the associated Identity Provider, as follows:

For **SAML** Identity Providers, the relevant field is Display Name Attribute Name.

- By default, the **NameId** attribute is used. This is the user's unique ID; what it contains depends on the Identity Provider.
- To use an attribute other than **NameId**, enter it here. This field supports custom attributes.
  Note that the format used varies depending on the Identity Provider.
- To use the name that participants entered themselves in their client, ensure this field is blank.

For **OpenID Connect** Identity Providers, the relevant field is Display Name Claim Name.

- By default, the **Name** attribute is used. This is the user's unique ID; what it contains depends on the Identity Provider.
- To use an attribute other than **Name**, enter it here. This field supports custom attributes.
  Note that the format used varies depending on the Identity Provider.
- To use the name that participants entered themselves in their client, ensure this field is blank.

When the display name is based on information provided by the Identity Provider, it cannot be changed by the participant, the meeting Host, or via the client API. However, it can be changed via .

Viewing a participant's authentication status

Participants and administrators can view whether a participant was required to authenticate in order to join the meeting:

- Meeting participants using the Pexip web app can enable the option to Show authenticated participants.
- Administrators using the Administrator interface can view the participant's status by selecting the participant from **Status > Participants** and scrolling down to the **Authenticated by an Identity Provider** field.

Conference layouts and speaker names

A layout is the view that each participant has of all the other participants in the meeting. The layout for a meeting is configured in advance by the administrator, but can be changed during the meeting by Host participants using SIP/H.323 endpoints that support DTMF keypad controls or by using the Pexip apps.

A wide range of layouts are available, including Pexip's AI-driven Adaptive Composition layout and a variety of classic layouts that have different combinations of speaker-focused and equally-sized participant arrangements. You also have the ability to define your own custom layouts. When joining Microsoft Teams conferences you also have the option of using our Teams-like layout.

By default, the "Large main speaker and up to 7 other participants" (1+7) layout is assigned to each VMR.

You can enable showing the names of active speakers and participants as text overlays, and there are a range of dynamic in-conference options.

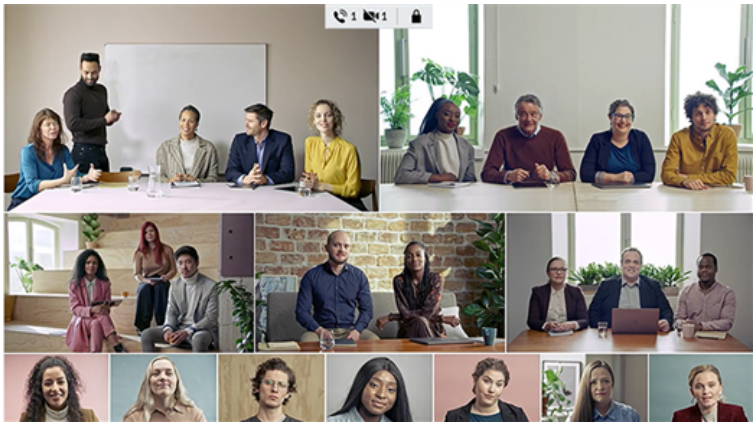
There are also some differences in layout behavior depending on whether the conference is being held in a Virtual Meeting Room (including scheduled conferences) or a Virtual Auditorium.

Adaptive Composition layout

Adaptive Composition is an intelligent meeting layout with real-time automatic face detection and framing:

- Powered by AI and machine learning to give users a more natural, engaging, video-first meeting experience.
- It continuously analyzes each video feed from all participants and uses automatic face detection and framing to create an optimized view of that participant, or group of participants where there are several people in that video feed.
- Layout placement of each participant is based on a combination of face count (number of faces detected within that participant's video feed) and their recent speaking frequency, so as to allocate more space for larger groups of people.
- Independent of where the participants are and which device they are using. No end-user action or configuration is required.

- When using Adaptive Composition you may want to consider disabling speaker tracking camera features in your room systems and endpoints to avoid conflict between the different framing technologies.



When using the Adaptive Composition layout:

- A maximum of 12 video participants are shown, spread across one row of 2 large images, a middle row of 3 slightly smaller images and a bottom row of 7 thumbnail images.
- Single-screen endpoints automatically receive any presentation content as part of the layout mix (replacing some of the other video participants), rather than as a separate stream.
- Any inactive video participants are automatically removed from the video layout. This is triggered if the participant:
  - Walks away from the camera, leaving an empty chair in frame i.e. no face is detected.
  - Points the camera at the ceiling or other place where no movement is detected.
  - Physically closes the camera lid or puts a cover over the camera.

  Note that these participants still continue to **receive** the full video layout.

  The participant is returned to the video layout if video is restored, or if any faces or movement are detected. Face detection is applied periodically on SIP/H.323 endpoints, if audio is detected (all device types), or on mouse activity (Pexip apps).
- An extended Adaptive Composition view is also available* where up to 23 video participants may be shown, initially across three rows (2/3/7 extending to 2/5/7 and then 3/5/7) and then across four rows (3/5/7/8) when required. This is a technology preview feature and it can only be enabled via the transforms functions in the Pexip client APIs.

This layout increases backplane bandwidth and CPU resource usage.

Adaptive Composition is fully supported in all Pexip Infinity conferences, and gateway calls to Microsoft Teams and Google Meet.

Note that Adaptive Composition applies face-detection technology, but does not apply any biometric or facial-recognition technology or store any such related data.

Classic layouts

In addition to Adaptive Composition, you can select from a range of other classic layouts for your Virtual Meeting Rooms and Virtual Auditoriums. These alternative layouts do not apply any face detection or framing technologies.

Speaker-focused layouts

The set of speaker-focused layout options are:

- **1+7** (1 large main speaker and up to 7 other participants) — this is the default layout
- **1+0** (full-screen main speaker only)
- **1+1** (1 large main speaker with one overlaying participant)
- **1+21** (1 main speaker and up to 21 other participants)
- **2+21** (2 main speakers and up to 21 other participants)
- **1+33** (1 main speaker and up to 33 other participants)

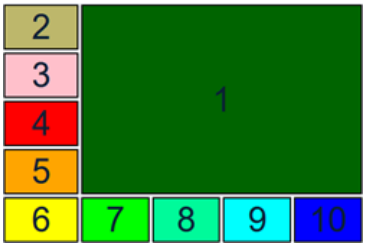
Equal layouts

The set of equal layout options are:

- **2x2** (4 speakers)
- **3x3** (9 speakers)
- **4x4** (16 speakers)
- **5x5** (25 speakers)

For full details on the differences between these layouts, see Virtual Meeting Room layouts and the examples.

Custom layouts



You can design your own layouts and use them in the same way as the standard adaptive composition and classic layouts that are included by default.

Custom layouts are specified through JSON configuration files that are uploaded via themes. The theme can then be assigned as the default theme or applied to specific VMRs and gateway rules as required, to control where your custom layouts may be used.

See the custom layouts section in Rules and requirements for customized themes for instructions on how to create and add custom layouts to your themes.

By using this feature, Pexip may reserve its right to waive at least some of the obligations stated in paragraph 9 of your Service Provider Agreement related to this feature.

Custom layouts support:

- All service types (VMRs, Virtual Auditoriums, gateway calls, Microsoft Teams and Google Meet calls etc).
- Name overlay and active speaker indication, and spotlighting/pinning.
- Receiving the presentation stream as part of the layout mix.
- Multiscreen systems: each screen can have its own layout configuration applied, up to a total of 26 participants across all screens.

Note that:

- Resource usage is similar to that used by the existing classic layouts.
- Face detection or framing technologies are not supported.

Three example custom layouts are included by default in the base theme (see the examples for more details):

- **1+9** (1 large main speaker and up to 9 other participants, with support for receiving the presentation stream as part of the layout mix) — the image, above-right, shows an example of this (the numbers indicate the order in which participants are displayed)
- **1+12** (1 large main speaker and up to 12 other participants)
- **2+8** (2 main speakers and up to 8 other participants)

These example custom layouts are available for use in all service types in the same way as the existing standard layouts.

Conference indicators

All in-conference indicators, such as participant counts, audio participants, recording indicators and locked status are shown in an information bar at the top-center of the layout (this area is referred to as the "notch" in theme customizations).

By default, the classic layouts have a dark information bar with light icons, whereas Adaptive Composition and the 1+33 layout has a light bar with dark icons. The most common indicators are shown below:

36	Total participant count. This is displayed when there are more participants than can be shown in the current layout.
2	Number of participants connected as audio-only. (Not displayed if participant avatars are enabled.)
7	Number of inactive video participants and video-muted Pexip app participants, who are excluded from the video layout. (Not displayed if participant avatars are enabled.)
2	The number of participants with a raised hand.
Alice	The indicator bar also alternates every 5 seconds to show a message containing the name of the first participant in the list of participants with a raised hand.
Paul	An audio or inactive video participant's display name is shown if they start speaking.
Conference locked	Conference locked indicator. Conference locked/unlocked messages are also temporarily displayed when a conference is locked/unlocked.
CC Live captions on	Live captions / transcription indicator. Live captions on/off messages are also temporarily displayed when captions are enabled/disabled.
Streaming enabled	Recording/streaming indicator. Recording/streaming messages are also temporarily displayed when recording/streaming is enabled.

Showing the names of active speakers and participants

Each VMR can be configured to show the names of the participants in a text overlay along the bottom of their video image.

When text overlay is enabled, the name that is shown depends upon whether the participant was invited into the conference, the type of endpoint device, and whether the participant had to authenticate, as follows:

- Automatically dialed participants (regardless of the type of endpoint that was dialed): either the ADP's configured **Participant display name** or, if this field is blank, the **Participant alias**, is shown.

- Manually dialed participants (regardless of the type of endpoint that was dialed): either the participant's configured **Participant display name** or, if this field is blank, the **Participant alias**, is shown.
- Pexip app participants:
  - If participant authentication is enabled for the VMR, the Identity Provider may provide a display name, in which case the user cannot change it. When a display name is provided by the Identity Provider, it will override any other display name provided (e.g. during registration or entered manually by the participant).
  - If the participant is using a registered Pexip app for Windows client, the display name is provided by the Identity Provider during registration and the user cannot change it.
  - For all other Pexip app participants, the display name is the text they entered in the initial **Type your name here**, **Display name**, or **Your display name** field.
- SIP endpoints: the endpoint's display name is shown.
- H.323 endpoints: the endpoint's alias is shown.
- Skype for Business clients: the username portion of the user's sign-in address is used, for example if Alice signs in as **alice@example.com** the text **alice** is shown.

The size of the text overlay varies automatically according to the resolution being received by the endpoint and the type of layout. In low resolutions the text overlays are not shown.

The default font for the in-conference display of participant names is Roboto (which cannot be changed), or if that is not available for the character set, Noto Sans.

Using the Administrator interface

To turn on participant names by default for a service:

1. Go to **Services > Virtual Meeting Rooms**, **Services > Virtual Auditoriums** or **Services > Scheduled Conferences** and select the service you want to change.
2. Next to the **Show names of participants** field select ***Yes***.
3. You can also choose to **Show name of active speaker**, as well as, or instead of, showing the names of all participants.
   - When participant names and active speaker display are both enabled, the name of the person speaking is shown in green instead of white.
   - When only active speaker display is enabled, the name of the person speaking fades in and then fades out again after a few seconds.

Using Pexip apps

Host participants using the Pexip web app can show and hide participant names during a meeting. See Controlling the layout during a conference for more information.

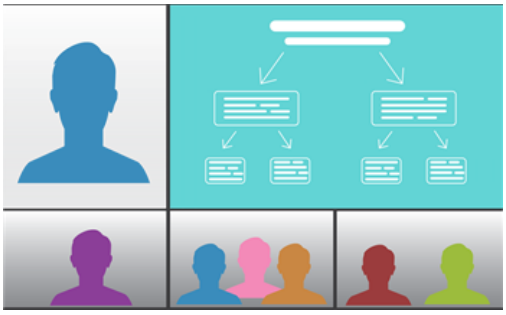
Using local policy

You can use local policy to enable participant names for gateway calls into Microsoft Teams or Google Meet conferences.

Dynamic layout controls, presenting content and single/multiscreen endpoints

Participants have a range of options for dynamically controlling the layout of a conference, how presentation content is displayed, and some additional layout options for multiscreen endpoints.

The options are summarized below, but for more information about the dynamic layout controls, see Controlling the layout during a conference and Using a DTMF keypad to control a conference.



- **Changing the layout during a conference**: Host participants on video endpoints can change the layout currently being used in the conference by sending DTMF/keypad commands to the conference, and participants on Cisco endpoints can also use the Pexip Layout Controls macro. Hosts and Guests may also select a personal layout using the Pexip web app or via DTMF.
- **Receiving the presentation stream as part of the layout mix**: when using Adaptive Composition, single-screen endpoints automatically receive the presentation stream as part of the layout mix (replacing some of the other video participants), but you can choose to switch to receiving it as a separate stream.
- **Multiscreen participant display**: if you have a dual screen endpoint you can display conference participants across both screens, effectively allowing you to show more participants than if you have a single screen. This display mode is currently only available for a subset of meeting and layout types.

Determining if an endpoint has single or multiple screens

Pexip Infinity follows a set of rules to determine whether a specific endpoint is a single-screen or dual-screen device and thus whether it can send presentation content in the layout mix, or make use of multiscreen participant display:

1. In the first instance, if provided by the endpoint, Pexip Infinity uses the display count signaled in the contact header.
2. Otherwise, if the endpoint's user agent string is listed in the theme's **vendordata.json** file, the screen count as defined in that file is used.
3. In either case, the display count can be subsequently set or overridden by participant policy.

If the display count is unknown, then:

- When content is being presented, a separate presentation stream is always sent to the endpoint.
- Multiscreen participant display is not available.

For more information about the **vendordata.json** file, see Base theme and other preconfigured themes.

Allowing end users to change the default layout

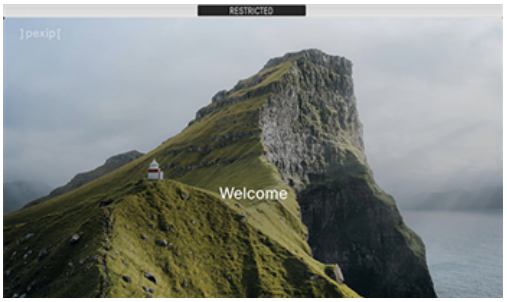
A VMR's settings for its layout and display of participant names are initially assigned when the VMR is created by the administrator. These can be changed dynamically during a meeting by Host participants; any changes will only apply for the duration of that meeting.

The Pexip VMR self-service portal is a separately-installable component that allows end-users to manage their personal Virtual Meeting Room without having to send requests to their administrator to change the configuration or branding of their VMR.

Layout features

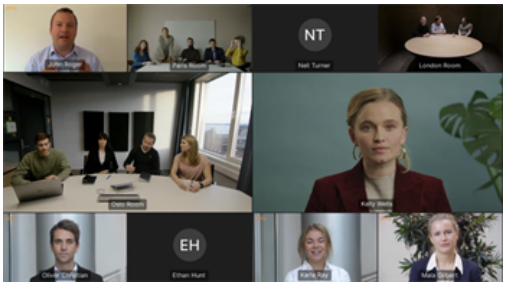
The following layout features and characteristics apply to all layout types (Adaptive Composition and classic layouts) unless stated otherwise:

- Any video-muted Pexip app participant, or a participant who joins without a camera, is removed from the video mix and is included in the inactive/muted indicator count.
- When a participant is the only device that is sending video, that participant sees a holding screen until other video participants join the conference. The holding screen indicates either that the participant is the only participant in the conference, or that all of the other participants are audio-only. The holding images are fully customizable via the theme associated with the service (Virtual Meeting Room, Call Routing Rule etc).
- Watermarks and content classifications can also be enabled/disabled and customized via the theme associated with the service:



  - Video watermarking overlays a small transparent image/logo onto the main speaker video during a conference.

    The default logo watermark is a white "Pexip" logo]pexip[(shown here against a blue background) and is enabled by default.
  - Content classification indicators can be displayed within a conference to, for example, display the current security classification level to meeting participants. By default, there are no content classification indicators configured (and, hence, none are displayed). You also have the option to trigger an audio announcement if the classification level changes during the conference.
- A full list of participants is available using the Pexip apps.
- If you set a timer (countdown clock), or send a banner message to all conference participants, it is displayed at the top of the screen, above any conference indicators. If classification indicators are also in use, they are combined into the same banner:
```
- Timer --- and/or --- Message --- and/or --- Classification -
------------------ In-conference indicators ------------------
```
- The participant avatars setting in the conference theme determines whether any audio-only or video-muted participants are represented as an avatar in the main video layout (excluding Adaptive Composition and Teams-like layouts) instead of those participants being counted in the "notch" indicators.



Features common to classic layouts only

The classic layout types (1+7, 2x2 and so on) have some minor operational differences:

- Video participants who are on hold, or who are experiencing connectivity issues, are represented by a frozen image.
- When sending presentation content to a single-stream endpoint or broadcasting via an RTMP stream (when not dialed out with dual streaming), Pexip Infinity sends the video stream of the active speaker in a small window in the upper right corner of the presentation.

Virtual Meeting Room layouts

Administrators select the layout to use for each VMR, and all video-enabled participants in a Virtual Meeting Room or scheduled conference see that same initial layout regardless of whether they are Hosts or Guests. During the meeting Hosts can change the layout that is seen by all participants, and (when enabled by an administrator) Hosts and Guests can select a layout personal to them alone.

To select the layout to use by default, from the Administrator interface go to **Services > Virtual Meeting Rooms** or **Services > Scheduled Conferences**, select the Virtual Meeting Room, and then select one of the options from the **View** drop-down menu.

The table below lists the layout options that are available. In all cases (except for Adaptive Composition), if one or more participants are spotlighted, they take priority over the current speaker.

VMR layout options

Option	Description
Adaptive Composition	Real-time automatic face detection and framing. See Adaptive Composition layout for more information.
1+7 (1 large main speaker and up to 7 other participants)	Participants see each other in the standard 1+7 layout. • The main video shows the current speaker (or previous speaker, for those currently speaking). • Up to 7 other participants are shown in a single row of live thumbnails at the bottom of the screen. Participants are shown in the thumbnails in order of who spoke most recently, from left to right, regardless of whether they are a Host or Guest.
1+0 (full-screen main speaker)	Participants see a single speaker, with no thumbnails. • The current speaker is shown full-screen. (The current speaker sees the previous speaker.)
1+21 (1 main speaker and up to 21 other participants)	This is the same as the ***1+7 layout*** option, except the main speaker is shown slightly smaller in order to accommodate up to 21 other participants in 3 rows of live thumbnails at the bottom of the screen.
2+21 (2 main speakers and up to 21 other participants)	This is the same as the ***1+21 layout*** option, except the top row contains both the main speaker and most recent speaker side by side.
1+33 (1 main speaker and up to 33 other participants)	 This layout has 1 main speaker, with up to 33 other thumbnail participants arranged around it. However, this layout is equally suitable for smaller numbers of participants as the size of the main speaker, and that of the other speakers, starts relatively large while there are a small number of participants, and then the sizes decrease as more participants join and more thumbnail rows and columns are added to the layout. It displays any watermark at the bottom of the screen. This example image shows the 1+33 layout when there are only 12 participants on screen.

Equal 2x2 (4 speakers)	Participants see up to 4 speakers, with no thumbnails.
	• The current speaker plus the three most recent speakers are shown in a 2x2 format. (The current speaker sees the four most recent speakers.)
	• Participants remain fixed in their position until a participant who is not on screen starts talking, and then the participant who spoke least recently is replaced with the new most recent speaker.
	• If there are fewer than four other participants:
	  ◦ a single participant is shown full screen
	  ◦ two participants are shown side-by-side
	  ◦ three participants are shown in two rows, with one participant in the middle of the top row and the two other participants side-by-side in the bottom row.

Equal 3x3 (9 speakers)	Participants see up to 9 speakers, with no thumbnails. This layout follows the same principles as the 2x2 layout in terms of how the layout is built and arranged when there are fewer than 9 other participants.

Option	Description
Equal 4x4 (16 speakers)	Participants see up to 16 speakers, with no thumbnails. This layout follows the same principles as the 2x2 layout in terms of how the layout is built and arranged when there are fewer than 16 other participants. The watermark displays at 50% (not configurable) of the normal size.
Equal 5x5 (25 speakers)	Participants see up to 25 speakers, with no thumbnails. This layout follows the same principles as the 2x2 layout in terms of how the layout is built and arranged when there are fewer than 25 other participants. The watermark displays at 50% (not configurable) of the normal size.
1+1 (1 large main speaker with one overlaying participant)	This layout has 1 large main speaker with one overlaying participant.
1+9 (1 large main speaker with up to 9 other participants)	This layout has 1 main speaker with up to 9 other participants around it.
1+12 (1 large main speaker and up to 12 other participants)	This layout has 1 main speaker with up to 12 other participants around it.
2+8 (2 main speakers and up to 8 other participants)	This layout has 2 main speakers with up to 8 other participants around it.

Virtual Auditorium layouts

In Virtual Auditoriums, Guest participants typically only see the Host participants, and Hosts see Hosts and Guests (read more). Administrators choose from separate options for the layout shown to Guests and the layout shown to Hosts (unless using Adaptive Composition). Additionally, administrators can configure the Virtual Auditorium so that when a presentation is being shown, the presenter is kept in the main view.

Host participants can change the layout seen by other participants during the course of a meeting — any changes they make apply to both Host and Guest participants, although the Virtual Auditorium's **Guests can see other guests** setting still applies.

To select the layouts to use by default, from the Administrator interface go to **Services > Virtual Auditoriums**, select the Virtual Auditorium, and then select one of the options from the **Host view** and **Guest view** drop-down menus.

Host view

The table below lists the layout options that are available. In all cases, if one or more participants are spotlighted, they take priority over the current speaker.

Virtual Auditorium layout options — Host view

Option	Description
Adaptive Composition	Real-time automatic face detection and framing. See Adaptive Composition layout for more information. If selected, both Host view and Guest view must use Adaptive Composition.

Option	Description

1+7 (1 large main speaker and up to 7 other participants)	Hosts see other Host and Guest participants in the standard 1+7 layout.
	• If there is only one Host, they see the most recent Guest speaker in the main video.
	• If there is more than one Host, they see the currently speaking Host in the main video. (The currently speaking Host sees the previous Host speaker.)
	• Up to 7 other participants are shown in a single row of live thumbnails at the bottom of the screen.
	  ◦ Host participants are shown first in the thumbnails, in order of who spoke most recently, from left to right.
	  ◦ Any remaining thumbnails are used to show Guests (again, in order of who spoke most recently).

1+0 (full-screen main speaker)	Hosts see a single speaker, with no thumbnails. • The current Host speaker is shown full-screen. (The currently speaking Host sees the previous Host speaker, or if there are no other Hosts, they see the most recent Guest speaker.)
1+21 (1 main speaker and up to 21 other participants)	This is the same as the ***1+7 layout*** option, except the main speaker is shown slightly smaller in order to accommodate up to 21 other participants in 3 rows of live thumbnails at the bottom of the screen.
2+21 (2 main speakers and up to 21 other participants)	This is the same as the ***1+21 layout*** option, except the top row contains both the main speaker and most recent speaker side by side. This option is useful in combination with the Lock presenter as main speaker option, so that other participants can see the presenter and the person they are speaking to at the same time.
1+33 (1 main speaker and up to 33 other participants)	This layout has 1 main speaker, with up to 33 other thumbnail participants arranged around it. However, this layout is equally suitable for smaller numbers of participants as the size of the main speaker, and that of the other speakers, starts relatively large while there are a small number of participants, and then the sizes decrease as more participants join and more thumbnail rows and columns are added to the layout. It displays any watermark at the bottom of the screen.

Equal 2x2 (4 speakers)	Hosts see up to four speakers, with no thumbnails.
	• The current speaker plus the three most recent speakers are shown in a 2x2 format. (The current speaker sees the four most recent speakers.)
	• If there are fewer than four other participants:
	  ◦ a single participant is shown full screen
	  ◦ two participants are shown side-by-side
	  ◦ three participants are shown in two rows, with one participant in the middle of the top row and the two other participants side-by-side in the bottom row.

Equal 3x3 (9 speakers)	Participants see up to 9 speakers, with no thumbnails. This layout follows the same principles as the 2x2 layout in terms of how the layout is built and arranged when there are fewer than 9 other participants.
Equal 4x4 (16 speakers)	Participants see up to 16 speakers, with no thumbnails. This layout follows the same principles as the 2x2 layout in terms of how the layout is built and arranged when there are fewer than 16 other participants.
Equal 5x5 (25 speakers)	Participants see up to 25 speakers, with no thumbnails. This layout follows the same principles as the 2x2 layout in terms of how the layout is built and arranged when there are fewer than 25 other participants.
1+1 (1 large main speaker with one overlaying participant)	This layout has 1 large main speaker with one overlaying participant.

Option	Description
1+9 (1 large main speaker with up to 9 other participants)	This layout has 1 main speaker with up to 9 other participants around it.
1+12 (1 large main speaker and up to 12 other participants)	This layout has 1 main speaker with up to 12 other participants around it.
2+8 (2 main speakers and up to 8 other participants)	This layout has 2 main speakers with up to 8 other participants around it.

Guest view

Guests in a Virtual Auditorium typically only see the Host participant(s), and can hear but not see any of the other Guests, even if a Guest is speaking. However, there are some circumstances where Guests may see other Guests, such as when there are no Hosts left in the conference, or if none of the Hosts are sending video. You can control the precise behavior via the **Guests can see other guests** option when configuring the advanced options for a Virtual Auditorium.

The table below lists the layout options that are available. In all cases, if one or more participants are spotlighted, they take priority over the current speaker.

Virtual Auditorium layout options — Guest view

Option	Description
Adaptive Composition	Real-time automatic face detection and framing. If selected, both Host view and Guest view must use Adaptive Composition.
1+7 (1 large main speaker and up to 7 other participants)	Guests see Host participants in the standard 1+7 layout. • If there is only one Host participant, the Host is shown full-screen. • If there is more than one Host participant, the standard 1+7 layout is used, with the main video showing the currently speaking Host and all other Host participants shown in the thumbnails. • Hosts are shown in the thumbnails in order of who spoke most recently, from left to right.
1+0 (full-screen main speaker)	Guests see a single Host speaker, with no thumbnails. • The Host who spoke most recently is shown full-screen.
1+21 (1 main speaker and up to 21 other participants))	This is the same as the ***1+7 layout*** option, except the main Host speaker is shown slightly smaller in order to accommodate up to 21 other Hosts in 3 rows of live thumbnails at the bottom of the screen.
2+21 (2 main speakers and up to 21 other participants)	This is the same as the ***1+21 layout*** option, except the top row contains both the main Host speaker and most recent Host speaker side by side. This option is useful in combination with the Lock presenter as main speaker option, so that other participants can see the presenter and the person they are speaking to at the same time.
1+33 (1 main speaker and up to 33 other participants)	This layout has 1 main speaker, with up to 33 other thumbnail participants arranged around it. However, this layout is equally suitable for smaller numbers of participants as the size of the main speaker, and that of the other speakers, starts relatively large while there are a small number of participants, and then the sizes decrease as more participants join and more thumbnail rows and columns are added to the layout. It displays any watermark at the bottom of the screen.

Option	Description

Equal 2x2 (4 speakers)	Guests see up to four Host speakers, with no thumbnails.
	• The current Host speaker plus up to the three most recent Host speakers are shown in a 2x2 format.
	• If there are fewer than four Hosts:
	  ○ a single Host is shown full screen
	  ○ two Hosts are shown side-by-side
	  ○ three Hosts are shown in two rows, with one participant in the middle of the top row and the two other participants side-by-side in the bottom row.

Equal 3x3 (9 speakers)	Participants see up to 9 speakers, with no thumbnails. This layout follows the same principles as the 2x2 layout in terms of how the layout is built and arranged when there are fewer than 9 other participants.
Equal 4x4 (16 speakers)	Participants see up to 16 speakers, with no thumbnails. This layout follows the same principles as the 2x2 layout in terms of how the layout is built and arranged when there are fewer than 16 other participants.
Equal 5x5 (25 speakers)	Participants see up to 25 speakers, with no thumbnails. This layout follows the same principles as the 2x2 layout in terms of how the layout is built and arranged when there are fewer than 25 other participants.
1+1 (1 large main speaker with one overlaying participant)	This layout has 1 large main speaker with one overlaying participant.
1+9 (1 large main speaker with up to 9 other participants)	This layout has 1 main speaker with up to 9 other participants around it.
1+12 (1 large main speaker and up to 12 other participants)	This layout has 1 main speaker with up to 12 other participants around it.
2+8 (2 main speakers and up to 8 other participants)	This layout has 2 main speakers with up to 8 other participants around it.

Lock presenter as main speaker

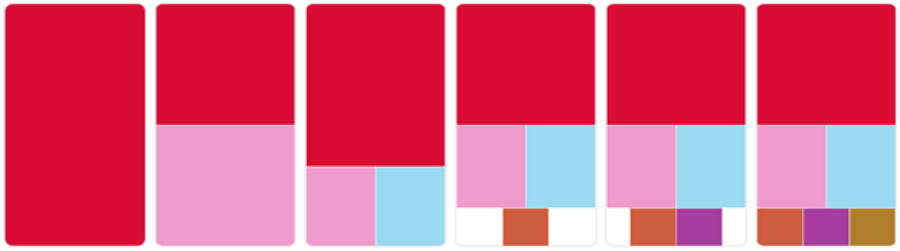
You can configure a Virtual Auditorium so that when a presentation is being shown, the main speaker position always shows the presenter instead of the current speaker.

Option	Description
Yes	When a presentation is being shown, the selected layout rules are overridden so that: • The main speaker position always shows the video image from the endpoint that is showing the presentation, even if others are speaking. • The image that would have been shown in the main view is instead shown in the first available slot/thumbnail.
No	When a presentation is being shown, the main speaker position is voice-switched as usual.

Portrait layouts

Any WebRTC device requesting a preferred aspect ratio of less than 1 (such as 9:16) receives a layout specifically designed for a portrait display.

The portrait layout displays a maximum of 6 people, and builds as shown in the images below when using the Adaptive Composition layout:



It builds like this in classic layouts:

The portrait layout is supported by default in the Pexip web app. It is not supported in custom layouts.

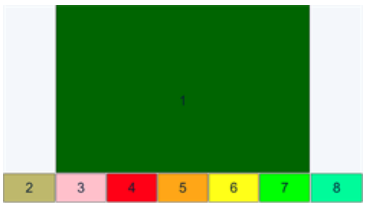
Layout examples

The numbers in the layout examples indicate the order in which participants are displayed as they join the meeting, or become voice switched into view.

Adaptive Composition layout

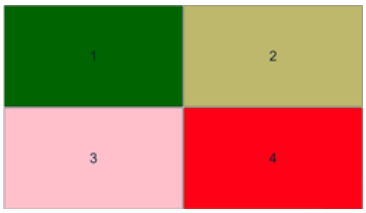
Adaptive Composition layout: maximum of 12 video participants are shown, spread across one row of 2 large images, a middle row of 3 slightly smaller images and a bottom row of 7 thumbnail images.

1+7



Standard 1+7: main video plus a single row of thumbnails.

Equal 2x2



Equal 2x2 main video.

Equal 3x3

Equal 3x3 main video.

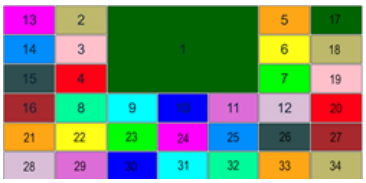
Equal 4x4

Equal 4x4 main video.

Equal 5x5

Equal 5x5 main video.

1+33



1+33: one main video plus up to 33 thumbnails.

1+0

1+0: full-screen main video.

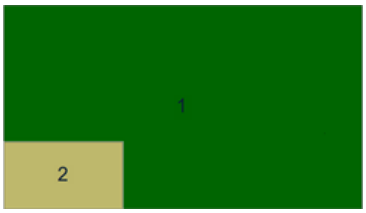
2+21

2+21: two main speakers plus three rows of thumbnails.

1+21

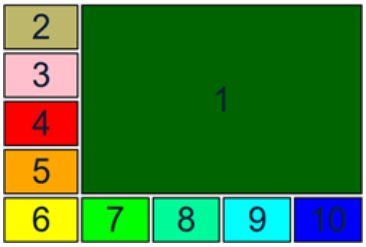
1+21: main video plus three rows of thumbnails.

1+1



1+1: Large main speaker with one overlaying participant.

1+9



1+9: Large main speaker and up to 9 other participants (custom layout).

1+12

1+12: Large main speaker and up to 12 other participants (custom layout).

2+8

2+8: Two main speakers and up to 8 other participants (custom layout).

Changing aspect ratios

Endpoints send and display video images and presentations in various aspect ratios. The most common ratios are 16:9 and 4:3 (i.e. landscape mode), but mobile devices in particular may send video in portrait mode.

If there is a difference between the aspect ratios of the sending and receiving endpoints, then the receiving endpoint and/or Pexip Infinity may crop the image or add black vertical or horizontal borders (also known as "letterboxing" and "pillarboxing"). Pexip Infinity chooses aspect ratios that maximize interoperability and quality. How and when aspect ratios are changed depends on a number of factors, but the general principles are described below.

Main video

- **Pexip Infinity**: If required, Pexip Infinity will change the aspect ratio of main video by cropping the image (to avoid black bars in the thumbnail images). The exception is when video is received in portrait mode; in this case, the image is not cropped and is displayed as is. Pexip Infinity sends a 9:16 portrait-oriented stream to any WebRTC participant that requests a preferred aspect ratio of less than 1.
- **Endpoints**: If the aspect ratio of the main video image being sent from Pexip Infinity does not match the aspect ratio of the receiving endpoint, the endpoint will typically add a border to make it fit. Generally endpoints will not crop the image.

Presentations

- **Pexip Infinity**: In general, Pexip Infinity chooses the resolution and aspect ratio that best matches the presentation content and that will not result in both vertical and horizontal borders (also known as "windowboxing") on a 16:9 endpoint display. It will never crop presentations, but may add either pillar or letterboxing to send a resolution that maximizes interoperability.
- **Endpoints**: If the aspect ratio of the presentation image being sent from Pexip Infinity does not match the aspect ratio of the receiving endpoint, the endpoint will typically add another border to make it fit. Generally endpoints will not crop the image.

Breakout rooms

Breakout rooms are separate sessions that are split off from a main VMR or Virtual Auditorium that allow smaller groups of people to meet together. When participants are in a breakout room they can only see, hear or share content with each other — they are separated from the main room and any other breakout rooms.

Typically, a Host participant using the Pexip web app will manage the breakout rooms — creating rooms, assigning participants to rooms, specifying any time limits, and moving people between rooms or bringing them back to the main room as and when required. Guests in a breakout room can request help from a Host, and any Hosts can join (with full audio and video) any of the breakout rooms at any time.

Breakout rooms can also be managed and joined from the Meeting Manager view.

This topic covers:

- How it works
- What's noteworthy?
- Enabling breakout rooms on a VMR / Virtual Auditorium
- Call and participant live status and history
- Using the web app to control the breakout rooms
- Using breakout rooms as a Guest

How it works

To manage the breakout rooms within a conference (add/remove breakout rooms, assign participants to rooms and so on), you have three options:

- Join the conference as a Host participant using a Pexip web app or Meeting Manager, and use the **Breakout rooms** panel to control the rooms and participants.
- Use to create breakout rooms and assign an incoming caller to a room.
- Use your own bespoke management application in conjunction with the .

Note that participants using other Pexip apps (desktop and mobile apps) or devices such as SIP and H.323 endpoints can participate in breakout rooms, but they cannot manage the participant and breakout room assignments (even if they have Host privileges). Skype for Business endpoints are not supported.

All participants can join the main VMR or Virtual Auditorium in the usual manner, using any of its assigned aliases or other standard joining methods for that room.

When using breakout rooms as a web app Host, you can:

- Assign participants automatically (randomly) to each breakout room, or assign them manually.
- Move participants between rooms, and add or close further rooms during the breakout sessions.
- Specify a time limit, after which all participants are automatically returned to the main room.
- Join any breakout room at any time with full audio and video, presentation and chat capabilities within that room.
- Get notified of requests for help from a breakout room.

What's noteworthy?

When using breakout rooms, note that:

- Any Pexip web app Host can control the breakout rooms, and they can join the VMR or Virtual Auditorium at any time to manage it. You can have multiple Hosts.
- You can create up to 50 breakout rooms and distribute up to 200 participants.
- When a timer is set, all Pexip web app participants see an overlay on their video showing the countdown. SIP and H.323 devices and other Pexip apps do not see the timer.
- Guest participants cannot move themselves between breakout rooms. They can only leave a breakout room when the time set by the Host has ended, when the Host closes the room, or the **Return to main room** button has been enabled for web app users (or they choose to leave the call completely).
- When web app participants are being moved between rooms, they see a modal telling them how long until they are moved (and giving them the option to move immediately). By default this modal lasts for 15 seconds, but administrators can use the transferTimeout branding option to control how long this modal appears, or remove it completely. Participants using other types of endpoints are moved immediately and do not see the modal.
- Breakout rooms can only be configured on Virtual Meeting Rooms (VMRs) and Virtual Auditoriums. They cannot be used with any other types of calls.
- If you use the "Add participant" function from within a breakout room, the dialed participant will join that breakout room.
- Hosts cannot currently send a "broadcast" chat message to all breakout rooms.
- When a breakout room is created, by default:
  - It inherits most of the same properties — such as its layout, theme and chat capabilities — as the main room.
  - The mute all guests and conference lock options are initially disabled (even if they are enabled in the main room).

- Any spotlight settings are reset. For example if a participant in the main room is spotlighted and they are then transferred to a breakout room they will not be spotlighted in the breakout room until a Host performs that action again.

You can override these default settings if the breakout room is created via policy.

- The **Guests-only timeout** setting does not have any effect on breakout rooms (as there is always a control-only API Host assigned to the room). Note that this control-only API Host does not affect the **Guests can see other guests** advanced setting in Virtual Auditoriums.
- In terms of hardware resource allocation, each breakout room is treated in the same way as a standard VMR. Backplane connections are reserved for each breakout room as soon as the room is created (i.e. before any participants have joined the room).

Enabling breakout rooms on a VMR / Virtual Auditorium

Breakout rooms must be enabled at the platform level via **Platform > Global Settings > Breakout Rooms > Enable Breakout Rooms**.

Administrators can then configure breakout rooms on a per-VMR / Virtual Auditorium basis (they are disabled by default). To enable breakout rooms for a VMR or Virtual Auditorium:

1. Go to the required VMR (**Services > Virtual Meeting Rooms**) or Virtual Auditorium (**Services > Virtual Auditoriums**).
2. Expand the **Advanced options** section.
3. Set **Breakout Rooms** to ***Yes***. (Use ***No*** to disable breakout rooms.)
4. Consider the **Participant authentication** settings, so that you can distinguish between Hosts and Guests (so that Host web app participants can control the breakout rooms).

   We recommend that the room has either a **Host and Guest PIN**, or a **Host PIN with Allow Guests** to make the separation between Hosts and Guests. See About PINs, Hosts and Guests for full information about how roles are assigned.
5. When configuring a Virtual Auditorium you may want to consider the **Guests can see other guests** option, which controls whether Guests see other Guests when there is no Host present. See the Virtual Auditorium advanced options for more information.
6. **Save** your changes.

Additional participant-level settings (bypass locked conference)

There is an additional **bypass_lock** configuration option in participant policy which allows a participant to bypass the locked state of a conference and enter the conference as if it was not locked. This setting can only be configured via external or local participant policy.

Call and participant live status and history

Live view (**Status > Live View**) shows each breakout room while it is in progress, and call and participant status and history all show the start and end time of each participant in each breakout room.

The service name is followed by "_breakout_<unique identifier>" to distinguish between separate breakout rooms.

Using the web app to control the breakout rooms

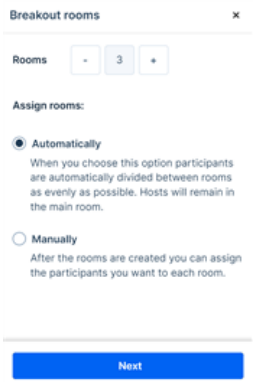
To use the web app to control the breakout rooms, you must join the VMR / Virtual Auditorium as a Host participant.

When breakout rooms are enabled for the VMR / Virtual Auditorium, an additional **Breakout rooms** icon appears in the toolbar. This opens the **Breakout rooms** side panel that you can use to set up and manage the rooms.

Setting up the rooms

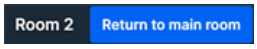
To set up the rooms, from the **Breakout rooms** side panel:

1. Select the initial number of **Rooms**.
2. Choose how you want to **Assign rooms**:
   - **Automatically**: this divides participants randomly and as evenly as possible across the rooms. You can still reallocate participants manually afterwards.
   - **Manually**: this starts with empty rooms and you assign participants manually to the desired rooms.

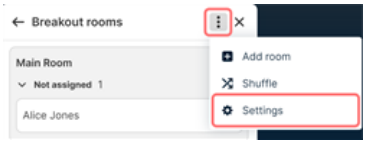


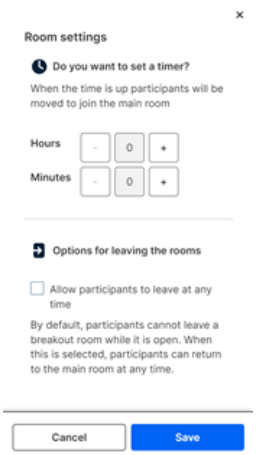
3. Select **Next**.
4. To set a timer and define the options for leaving the room, from the **Breakout rooms** panel select **More** and then **Settings.**

   This opens the **Room settings** dialog.
   - You can optionally set a timer duration. When the time is up, all participants are automatically returned to the main room.
   - You can optionally **Allow participants to leave at any time**. When this is selected, Guest web app participants see a **Return to main room** button at the top left of their screen which allows them to go directly back to the main room:

     Note that leaving this option disabled does not prevent guests leaving a breakout room by disconnecting and then redialing the same conference and rejoining the main room.

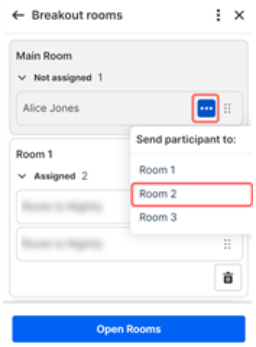




5. Perform any final adjustments:
   - To change the name of a room, select **Edit**.

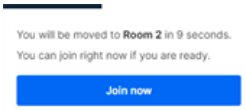
     You can't change a room's name after you have created the rooms.
   - To **move** participants between rooms manually, either select **More** next to the participant's name and select the room you want to move them to, or select **Move** and drag and drop the participant into the room.

     You can also move participants after the rooms have been created.



6. When you are ready, select **Open Rooms**.

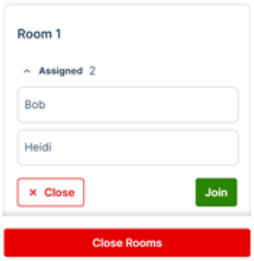
   Web app participants are notified that they are about to be moved, and can elect to move immediately by selecting **Join now** (all other participants will move immediately).



Managing the rooms

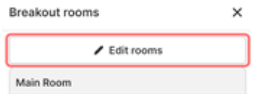
While the rooms are open, the **Breakout rooms** panel shows the current list of rooms and assignments.

From here you can **Close** a specific room, close all rooms or **Join** a room:

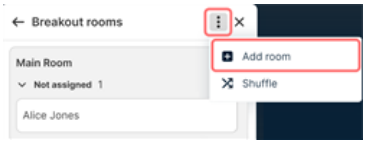


To make other changes, select **Edit rooms**.

You then have a range of options.

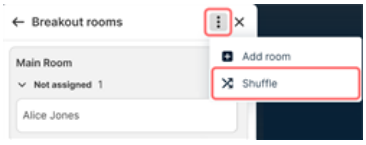


- To **add** more rooms, select **More** and then **Add room**:



- To **close** a room, select **Delete** — any currently assigned participants are moved to the main room:

- To **move** participants between rooms manually, either select **More** next to the participant's name and select the room you want to move them to, or select **Move** and drag and drop the participant into the room.
- To **shuffle** the currently assigned participants (i.e. randomly reassign them across the existing breakout rooms), select **More** and then **Shuffle**:



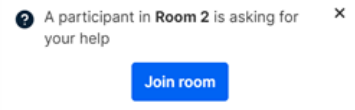
When editing a room, any changes you make do not take effect until you select **Save**.

Participant list (roster), requests for help and raised hands

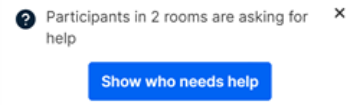
The participant list only shows the other participants who are in the same room (breakout room or main room) as you.

Guest participant/room ask for help

Guest participants in a breakout room can ask for help i.e. summon a Host to the room. When a participant in a breakout room asks for help, all Hosts who are not already in that room see a message:

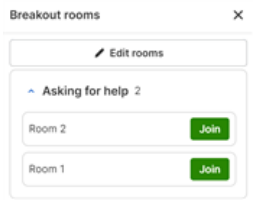


One room requesting help.



Multiple rooms requesting help.

- If a single room is requesting help, Hosts can select **Join room** to join the room immediately.
- If there are multiple rooms, Hosts can select **Show who needs help**. This opens the **Breakout rooms** panel which shows a list of all pending requests for help and provides an option to join that room. The requests are shown in time order (oldest at the top):



The help request is removed from the list as soon as any Host joins the room in response to that request for help.

Note that help requests are for a room, even though they are triggered by an individual participant. If a participant requests help and subsequently disconnects, that request for help remains open until a Host responds to it or another participant in the room removes the request.

Guest participant raises a hand

If a Host is in the same room as the person who raises their hand, then the Host sees a toast notification and an indicator in the participants list (as per standard behavior).

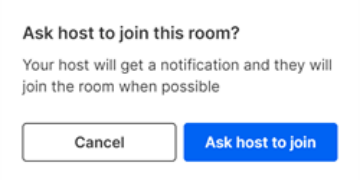
However, if a Host is in a different room from the person raising their hand, then the Host does not see any notifications or indicators.

Using breakout rooms as a Guest

Guest participants (using a Pexip app or a hardware endpoint) cannot control any aspects of the breakout rooms.

Guests in a breakout room can only see, hear, share content or chat (via a Pexip app) with each other. While in a breakout room, the participant list in a Pexip app only shows the other participants who are in the same breakout room.

Ask for help (summon a Host)



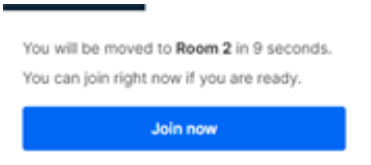
If you are a web app participant and you need to speak to a Host, you can summon a Host to your breakout room by using the **Ask host for Help** option from the toolbar at the left of the screen. You need to confirm that you want to summon the Host and then the Hosts will see a message that somebody in that breakout room is asking for help.

If you no longer need help, select the option again to cancel the request.

Note that:

- The **Raise hand** option from the toolbar at the left of the screen only shows the raised hand toast to the Host if the Host is already in that breakout room.
- Participants on hardware endpoints cannot summon the Host.

Moving between breakout rooms

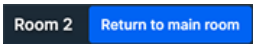


You cannot move yourself between breakout rooms, but if you are a web app participant, when a Host moves you between rooms you see a message that you are about to be moved.

You can optionally select **Join now** to move immediately.

Returning to the main room

If you are a web app participant, depending upon how the breakout room is configured you may see the option to **Return to main room**:



Limiting the number of participants

You can place a limit on the number of concurrent participants that can join a particular Virtual Meeting Room, Virtual Auditorium or scheduled conference at any one time. When such a restriction is in place, when the limit is reached any new participants attempting to join the conference are presented with a **capacity exceeded** message.

If you change this setting while a conference is in progress, the limit that was in place when the conference started is retained until the conference is terminated (i.e. after the last participant has disconnected). The new limit will be applied only when a new conference starts.

If you do not restrict the number of participants, any number of participants can join the service (subject to the available capacity of your Pexip Infinity deployment).

To limit the number of participants in a particular conference:

1. Go to **Services > Virtual Meeting Rooms**, **Services > Virtual Auditoriums** or **Services > Scheduled Conferences** and select the service you wish to change.
2. In the **Advanced options** section, select **Show**.

3. In the **Participant limit** field, enter the maximum number of participants that can use the conference at any one time.
4. Select **Save**.

Note that:

- Presentation streams are not counted separately; such participants are only counted once regardless of whether or not there is a presentation stream in addition to the main video.
- Pexip app users who join as presentation and control-only participants, control-only participants, or Meeting Manager participants count towards the participant limits.

Automatically dialing out to a participant from a conference

You can configure a Virtual Meeting Room, Virtual Auditorium, or scheduled conference to automatically dial out to one or more participants, for example the VMR "owner" or a streaming service, when the first participant joins that conference.

You can also **manually** dial out to participants from a Virtual Meeting Room, Virtual Auditorium, or scheduled conference on an ad hoc basis. For more information, see Manually dialing out to a participant from a conference.

When is the participant called?

When an Automatically Dialed Participant (ADP) has been added to a Virtual Meeting Room, Virtual Auditorium or scheduled conference's configuration, a call is placed from the conference to the ADP's endpoint as follows:

Meeting type	Participant is automatically dialed...
No PIN	when the first participant joins the conference
Guests not allowed; participants must enter a PIN	when the first participant has entered a valid PIN
Guests allowed; Hosts must enter a PIN but Guests do not	when the first participant joins the conference
Guests allowed; Hosts and Guests must enter a PIN	when the first participant has entered a valid PIN

Note that:

- The role of the first participant does not matter — Hosts or Guests can trigger a call to an ADP.
- It takes at least 10 seconds (after the first participant joins) for the call to be placed to the ADP.
- A call is not placed to the ADP if that participant is already in the conference.
- Only one attempt is made to call the ADP — it does not retry.

Host and Guest PINs

When the call is answered, the Automatically Dialed Participant joins the conference as either a Host or Guest, depending on which role was selected when configuring them as an ADP. They are never required to enter a PIN, even if one is required for that role.

Choosing the calling location

Usually, the call to the ADP is placed from the same Conferencing Node that is being used by the participant who initiated the conference.

However, you can optionally configure the call to be placed from a Conferencing Node in a specific location. The behavior varies according to whether it is a manually routed ADP (where **Route this call** is set to ***Manually***) or the ADP is called according to Call Routing Rules (where **Route this call** is set to ***Automatically***).

- **Manually routed ADPs**: the nominated location is the location of the node that will dial the ADP.
- **Automatically routed ADPs**: the nominated location is the notional source location used when considering if a Call Routing Rule applies or not — but the rules themselves determine the location of the node that dials the ADP. In this case, the configured **Outgoing location** for the ADP is deemed to be the location handling the call (when matched against the **Calls being handled in location** setting for the Call Routing Rule) and the rule's **Outgoing location** setting determines the actual location of the node that dials the ADP.

This behavior is useful if, for example, you have configured lowest-cost-routing rules that only allow calls to be placed from "internal" Conferencing Nodes but you have a conference that is triggered by a call to an "external" Conferencing Node. In this case you still want the call to the ADP to be initiated, but you don't want, in general, to allow calls to arbitrary ISDN destinations to be placed from Conferencing Nodes in the external location.

Incoming call alias

When the endpoint is called, the Automatically Dialed Participant sees the call as coming from one of the Virtual Meeting Room or Virtual Auditorium's aliases. This means that if the participant misses the call, they can easily return it by dialing the alias that appears on their endpoint.

When deciding which alias to use to identify the call, Pexip Infinity selects the first in the list that is valid for the selected call protocol. So if your Virtual Meeting Room has two aliases, the first **meet.sales** and the second **meet.sales@example.com**, when you place a call to an ADP over SIP it will show as coming from **meet.sales** because that is the first valid SIP address in the list. However, endpoints may not always be able to return a call to this alias because it is missing the domain. Therefore, if you want ADPs to be able to return calls to Virtual Meeting Rooms and Virtual Auditoriums, we recommend that you list the most routable aliases first when assigning them to these services. Note that you can also use external or local policy (see policy profiles) to specify the source alias for calls to ADPs.

Keeping a conference alive

Automatically Dialed Participants with a role of Host may or may not prevent a conference from being automatically terminated, depending on their **Keep conference alive** setting and whether any other ADPs remain in the conference. For more information, see Automatically ending a conference.

Bandwidth limits

When dialing out from a conference, the outbound call bandwidth limit is inherited from the VMR's bandwidth settings. (If a Call Routing Rule is applied, the rule's bandwidth settings are not used.)

Configuring an ADP

If you have a large number of ADPs to configure, you can import them using a CSV file. For more information, see Bulk import/export of service configuration data.

To configure a Virtual Meeting Room or Virtual Auditorium to automatically dial out to a participant when a conference starts:

1. Go to **Services > Automatically Dialed Participants**.
2. Select **Add Automatically Dialed Participant**.
   You are taken to the **Add Automatically Dialed Participant** page.
3. Complete the following fields:

Field	Description
Participant alias	The alias of the participant to dial when a conference starts. If you select a **Protocol** of ***SIP***, this must be a valid SIP alias.
Creation time	This read-only field shows the date and time when this record was first configured.
Participant display name	An optional user-facing display name for this participant, which may be used in participant lists and as the overlaid participant name (if enabled). If this name is not specified then the **Participant alias** is used as the display name instead.
Description	An optional description of the Automatically Dialed Participant.

Field	Description
Route this call	Select how to route the call: ○ ***Manually***: uses the requested **Protocol** and the defaults for the specified **System location**. ○ ***Automatically***: routes the call according to the configured Call Routing Rules. This means that the dialed alias must match an outgoing Call Routing Rule for the call to be placed (using the protocols, outgoing location and call control systems etc. as configured for that rule). Default: ***Manually***.
Protocol	The signaling protocol to use when dialing the participant. Select either ***SIP***, ***H.323***, or if the endpoint is a Skype for Business client, select ***Skype for Business (MS-SIP)***. The RTMP protocol is typically used when adding a streaming participant. Note that if the call is to a registered device, Pexip Infinity will instead use the protocol that the device used to make the registration. Default: ***SIP***. This field only applies when **Route this call** is set to ***Manually***.
DTMF sequence	An optional DTMF sequence to transmit after the call to the Automatically Dialed Participant starts. A DTMF sequence can include: the digits 0-9, "*" (asterisk), "#" (hash) or "," (comma). The DTMF tones are sent 3 seconds after the call connects, one at a time, every 0.5 seconds. A comma is a special digit that represents a 2 second pause (multiple commas can be used if a longer pause is needed). For example, if you need your Automatically Dialed Participant to dial an audio bridge and then enter conference number 777 followed by #, pause for six seconds and then supply conference PIN 1234 followed by #, you would configure **777#,,,1234#** as your DTMF sequence.
Role	The level of privileges the participant will have in the conference. For more information, see About PINs, Hosts and Guests. Default: ***Guest***.
Conference	Select the names of the Virtual Meeting Rooms, Virtual Auditoriums and scheduled conferences from which this participant will be dialed automatically whenever a conference using that service starts.
Outgoing location	For ***Manually*** routed ADPs, this is the location of the Conferencing Node from which the call to the ADP will be initiated. For ***Automatically*** routed ADPs, this is the notional source location used when considering if a routing rule applies or not — however the routing rule itself determines the location of the node that dials the ADP. For more information, see Choosing the calling location. To allow Pexip Infinity to automatically select the Conferencing Node to use to place the outgoing call, select ***Automatic***. As with all calls, signaling and media may be handled by different Conferencing Nodes in that location.
Streaming	Identifies the dialed participant as a streaming or recording device. When a conference participant is flagged as a streaming/recording participant, it is treated as a receive-only participant and is not included in the video stage layout seen by other participants. See Streaming and recording a conference for more information.
Keep conference alive	Determines whether the conference will continue when all other non-ADP participants have disconnected: ○ ***Yes***: the conference will continue to run until this participant has disconnected (applies to Hosts only). ○ ***If multiple***: the conference will continue to run as long as there are two or more ***If multiple*** participants and at least one of them is a Host. ○ ***No***: the conference will be terminated automatically if this is the only remaining participant. Default: ***If multiple***. For streaming participants, we recommend that this option is set to ***No***. For more information, see Automatically ending a conference.

Field	Description
Call capability	Allows you to limit the media content of the call. The participant being called will not be able to escalate beyond the selected capability. For more information, see Controlling media capability. Default: ***Main video + presentation***. This field only applies when **Route this call** is set to ***Manually***. This field does not apply to RTMP calls.
Dual stream (presentation) URL	When adding a dual streaming RTMP participant, this specifies the RTMP URL for the second (presentation) stream. Leave this field blank when adding a single streaming participant.

4. Select **Save**.

Automatically ending a conference

You may want to ensure that conferences are automatically terminated in certain situations, in order to preserve resources or to ensure that certain types of participants can't keep the conference alive by remaining in a call indefinitely. You can configure Pexip Infinity to automatically end a conference in any of the following situations:

- when only Guests remain
- when there is only one participant present in the conference
- when the only participants remaining are ADPs or participants that have been added by an administrator.

These settings are all independent; a conference needs to meet only one of the above criteria for it to be terminated automatically. In situations where two or more criteria with different timeouts apply, the conference will be terminated when the earliest of the timeouts is reached.

When only Guests remain

Pexip Infinity always terminates a conference a certain period of time after the last Host disconnects and only Guests remain in the call. By default this period is set to one minute, but it can be set to between 0 seconds (i.e. all Guests are disconnected immediately upon the last Host disconnecting), and one day.

The default of 60 seconds should be sufficient time for a Host who has been unintentionally disconnected from a conference to reconnect before the conference is terminated.

To change this setting, go to **Platform > Global Settings > Service Configuration > Guests-only Timeout**.

When there is only one participant present in the conference

You can configure the length of time (in seconds) for which a conference or a Microsoft Teams / Google Meet gateway call will continue with only one participant present. This can be set to between 60 seconds and one day, or to **0** (the default) which means that a conference is never terminated when there is a single participant present.

When this setting is used to terminate a conference, the type of participant (Host, Guest, ADP, administrator-added, streaming, etc.) is irrelevant; if they are the only participant present, the conference / gateway call will be terminated after the specified time.

To change this setting, go to **Platform > Global Settings > Service Configuration > Last Participant Backstop Timeout**.

When the only participants remaining are ADPs and/or administrator-added

In most cases a conference won't be terminated if a Host participant is present. However, if the only remaining Host participants are:

- Automatically Dialed Participants (ADPs), and/or
- participants who have been added to a conference by an administrator (either from the Administrator interface or using the Management API),

then the conference may or may not be terminated, depending on the participants' **Keep conference alive** settings and whether any other ADPs or administrator-added participants remain in the conference.

Keeping a conference alive

There are three options for the **Keep conference alive** setting:

- ***Yes***: if the participant is a Host, the conference will continue to run even when this is the only participant remaining — in other words, they are just like any other Host participant. This is the default used when adding a participant to a conference using the Administrator interface or using the Management API.
- ***No***: the conference will be automatically terminated if this is the only participant remaining. This is the option we recommend for automated systems that are unable to terminate a call themselves, such as streaming participants.
- ***If multiple***: if two or more ADP or administrator-added participants with the ***If multiple*** setting remain in the call and at least one of them is a Host, the conference will not be terminated by Pexip Infinity. However, if all other participants have disconnected and only one ADP or administrator-added participant with the ***If multiple*** setting remains, the conference will be terminated by Pexip Infinity. This is to prevent automated systems (such as recording devices) that are unable to terminate a call themselves from keeping the conference alive indefinitely. For this reason, if you are using this option we recommend that each Virtual Meeting Room or Virtual Auditorium has no more than one such automated system as an ADP. A better alternative is to give automated systems a **Keep conference alive** setting of ***No*** (see above).

  If multiple is the default option when adding an ADP.

If the only remaining participants in a conference are ADPs and/or administrator-dialed participants with a mix of ***Yes***, ***If multiple*** and ***No*** options, the conference will be terminated unless:

- at least one participant has a **Keep conference alive** setting of ***Yes***, or
- there are two or more participants with a setting of ***If multiple*** and at least one of them is a Host.

Controlling media capability

Calls consist of three types of media - audio, video, and presentation - each in their own stream. Pexip Infinity allows you to control which types of media can be used for certain calls. This feature allows better resource management and smaller Session Description Protocols (SDPs) when calling out to known audio-only devices.

You can restrict the media capability either on a per-call basis (the call capability) or a per-conference basis (the conference capability), as described below.

Adding a participant to a conference

When dialing out to a participant from within a conference, either manually or automatically, you can choose whether the call is **audio-only**, **main video only**, or **main video plus presentation**. In such cases, the called participant is not allowed to escalate the call (for example, from an audio-only call to video).

Conference-wide limitations

You can apply limits to individual Virtual Meeting Rooms (including scheduled conferences) or Virtual Auditoriums, restricting them to be **audio-only**, **main video only**, or **main video plus presentation**. In such cases, participants calling in to the conference with a higher media capability have their media limited. For example, when a participant dials in over video to an audio-only VMR, their call is placed as audio-only and they are not subsequently allowed to escalate the call.

Using the **audio-only** setting where appropriate also has the advantage of increasing the overall capacity of your distributed deployment. Each Conferencing Node reserves an extra connection for each Virtual Meeting Rooms and Virtual Auditoriums it is hosting, to use for a backplane should that service become distributed (i.e. hosted on more than one Conferencing Node). By default this connection is equivalent in capacity to an HD call, but if the service is configured as **audio-only** it is instead equivalent to that of an audio-only call.

Virtual Receptions

You can limit the conference capability of a Virtual Reception. Any restrictions are applied to calls while they are accessing the Virtual Reception service (apart from Pexip apps, which do not access a Virtual Reception in the same way as other endpoints). When the call is placed to the destination VMR, the capability of that service will be available instead, but may or may not be used by the endpoint, as follows:

- Pexip apps making a video call via an audio-only Virtual Reception always join a video-capable VMR with video
- Pexip apps making an audio-only call via an audio-only Virtual Reception join a video-capable VMR with audio-only initially, but with the option of enabling video.
- Standards-based endpoints making a video call via an audio-only Virtual Reception always join a video-capable VMR as audio-only initially. Depending on the endpoint, there may be the option to then enable video.

For example, if you limit your Virtual Reception to audio-only, when Alice calls it from her standards-based endpoint she will not receive any video and will only hear the audio prompts. However, when her call is then placed to the destination VMR which has a capability of **main video**, she will initially join it as audio-only but can elect to send and receive video as well.

Infinity Gateway calls

For calls made using the Distributed Gateway, you can configure the relevant Call Routing Rule to limit the media by selecting an appropriate **Call capability** setting:

- You can limit the media capability of the outbound call to **audio-only**, **main video only**, or **main video plus presentation**. For example, if the media capability is set to **audio-only** and Alice makes a video call to Bob, Bob will not be able to answer with video.
- You can match the capability of the outbound call to that of the inbound call by using the **Same as incoming call** option. This means that if, for example, Alice makes an audio-only call to Bob, Bob will not be able to answer with video.

  Note that when using **Same as incoming call** an audio-only call cannot later be escalated to video.

Direct media (end-to-end encrypted calls)

Pexip's direct media feature enables end-to-end encrypted calls. When enabled in a VMR, it provides non-transcoded, encrypted, point-to-point calls between any two WebRTC-based participants.

You can configure the flexibility of the direct media capabilities for each VMR:

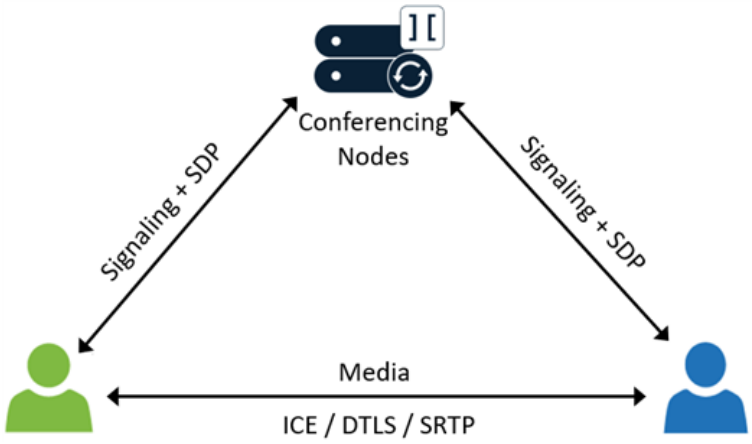
- The same VMR could still be allowed to host conferences with more than two participants, or with different endpoints and call protocols such as SIP and H.323. In these cases, the VMR automatically reverts to Pexip Infinity's standard behavior, which is where each conference participant's call media is transcoded and sent via a Conferencing Node.
- Alternatively you can configure the VMR to restrict it to only ever use direct media calls, and thus always limit each call in that VMR to a maximum of two WebRTC participants.

This topic covers:

- How it works
- What's noteworthy?
- Enabling direct media on a VMR
- Media relay via a TURN server
- Customization and branding

How it works

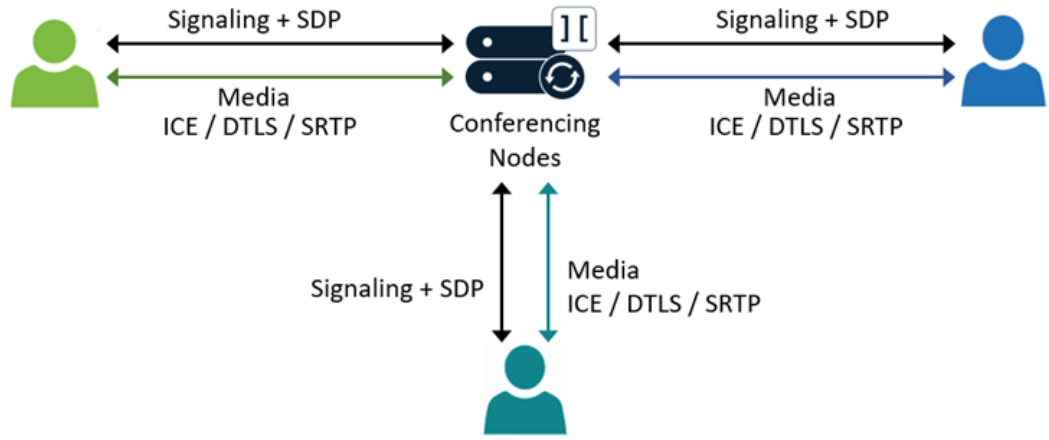
In a direct media call, the two participants negotiate the encryption keys directly with each other. All call signaling is still handled by the Conferencing Node; only the call media is sent directly between the participants:



Direct media call (two WebRTC participants)

Only two WebRTC participants (typically using the Pexip apps) can participate in a direct media call.

If additional participants join the conference (using any call protocol such as WebRTC or SIP for the additional participants) the call is escalated to a standard, transcoded call with media sent via Conferencing Nodes, and is no longer a direct media call:



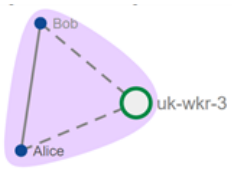
Transcoded media call (escalated for a third participant)

What's noteworthy?

When enabling this feature on a VMR, note that:

- If the first person to join a direct-media-enabled VMR is a WebRTC participant, they join as a direct media participant immediately and do not get the full set of theme-based splash screens and audio notifications that you would normally get when joining a standard, transcoded call. Instead, they see a "Welcome" or "Waiting for the host" screen until the second participant joins and the media connection is established.
- If a direct media conference is escalated to a transcoded call:
  - In conference history, the original participants are shown as being transferred to a new call, and new participant history records are created for the new call.
  - If the call de-escalates back to two WebRTC participants then it becomes a direct media call again (with new participant history records again).
- Presentation / screen sharing, and chat is supported (via the direct media connection). 4k presentation sharing is supported, subject to bandwidth/browser constraints (note that on Chrome and Edge browsers the person sharing has to have their browser active on the 4k monitor to achieve the full outgoing resolution).
- All of the standard authentication methods for joining the VMR are supported.
- If a VMR is also configured with an Automatically Dialed Participant (ADP) then the VMR will always use transcoded calls.
- If a VMR has breakout rooms enabled, the VMR will always use transcoded calls — direct media will not be supported.

- If direct media is set to ***Always*** then a third person who tries to join the VMR, or a non-WebRTC endpoint that attempts to join, will have their call rejected.
- A Conferencing Node can support a maximum of 2,000 concurrent signaling connections for direct media calls.
- Direct media calls always use a port license.
- Within the Pexip apps for web and Windows, each user can view a **Secure check code**. Both users should see the same code, proving that media is not being sent via (and transcoded by) a Conferencing Node. To view the check code, from the **Settings** menu select **Media statistics** — it is at the bottom of the page.
- When using the Administrator interface to monitor calls:
  - When viewing the conference graph, a direct media line is shown between the two participants, plus the signaling connections to the Conferencing Node:



  - Participant media and call quality statistics are available in the current and historic participant status pages.

Enabling direct media on a VMR

Direct media is configurable on a per-VMR basis, and is disabled by default. To enable direct media on a VMR:

1. Go to the required VMR (**Services > Virtual Meeting Rooms**).
2. Expand the **Advanced options** section.
3. Select your required **Enable direct media** option:
   - ***Best effort***: use direct media in this VMR where possible (when there are two WebRTC participants only), otherwise use standard, transcoded media connections via a Conferencing Node.
   - ***Always***: allow direct media calls only — this limits all calls in this VMR to two WebRTC participants.
   - ***Never***: do not use direct media in this VMR.

   Default: ***Never***
4. Optionally, set the **Direct media notification duration** to the number of seconds to display any escalation or de-escalation notifications to participants. The notifications are shown as toasts on top of the main conference video.
5. **Save** your changes.

Media relay via a TURN server

In some network topologies, for example if the Pexip apps are behind a NAT or have firewall restrictions, they may not be able to establish a direct connection between each other. To overcome this, the clients may need to use a TURN server to relay media between each other.

When using direct media we strongly recommend for enhanced security that you use your own dedicated TURN server that is located in your DMZ.

To use a TURN server to relay media:

1. Install your TURN server in your DMZ.

   We recommend using the Pexip TURN server (see Pexip Reverse Proxy and TURN Server Deployment Guide), configured in permissive mode. If you use an alternative TURN server it must be a Coturn implementation.
2. Configure Pexip Infinity to use that TURN server (**Call Control > TURN Servers**).
   - It must have a **TURN server type** of ***Time-limited credentials***.
   - See Using TURN servers with Pexip Infinity for more information.
3. Configure your system locations with the **Client TURN servers** to provision to the WebRTC clients (Pexip app):
   i. Go to **Platform > Locations**.
   ii. Select the Conferencing Node's location.
   iii. Select your **Client TURN servers** (the TURN server you have installed for direct media).

iv. Optionally, you can select **Enforce media routing via a client TURN server** if you want to force the WebRTC client to route its media through one of the specified client TURN servers. This may be required if you want to hide each participant's IP/NAT address from each other.
v. Select **Save**.
vi. Repeat for other locations as necessary.

When a WebRTC client connects to a Conferencing Node in that location, the Conferencing Node will provide it with the details of the nominated TURN servers. These TURN servers may be used by the client to provide a media connectivity path if it cannot make a direct media connection to another client.

Customization and branding

Direct media calls can be customized and branded in the same way as standard transcoded calls.

Customizing the Pexip web app

See Customizing and branding the web app for details about applying branding and styling customizations to the Pexip apps.

Customizing the in-call experience (themes)

The in-call experience provided by the splash screens and messages that are seen by the participants can be branded and customized in the usual way by uploading your own theme and assigning it the relevant services. See Rules and requirements for customized themes for details about how to create and upload your customized theme.

However, in a direct media call, only a limited number of splash screens and message labels are used:

- The following splash screens are used:
  - "direct_media_welcome"
  - "direct_media_waiting_for_host"
  - "direct_media_other_participants_audio_only"
  - "direct_media_escalate"
  - "direct_media_deescalate"
  - "direct_media_host_left"
- The following label keys are used:
  - "welcome": "Welcome"
  - "waiting_for_host": "Waiting for the host..."
  - "other_participants_audio_only": "The other participants are audio only"
  - "direct_media_escalate": "You are about to be transferred into a multi-party conference"
  - "direct_media_deescalate": "You are about to be transferred into a one-to-one call"
  - "host_left": "The host has left the conference. Your call will be disconnected shortly."

Note that:

- The background image file (background.jpg by default) used for each screen can be customized in the theme in the standard way.
- The "direct_media_escalate", "direct_media_deescalate", and "direct_media_host_left" splash screens are provided for bespoke client development via the client APIs. Within the Pexip web app, the notifications are shown as toasts on top of the main conference video.

See Splash screens and Default label keys and usage for details about the specific splash screens and labels, respectively.

Example: changing the notification messages

This shows some example contents of a **themeconfig.json** file that would change the text of the escalation and de-escalation notifications:

```
{
  "theme_version": 2,
  "splash_screens": {
    "labels": {
      "direct_media_escalate": "Additional participants are joining. Transferring to a transcoded call",
      "direct_media_deescalate": "Transferring to a direct media call"
      }
```

```
    }
}
```

Streaming and recording a conference

This topic provides a general overview of Pexip Infinity's support for streaming and recording. For specific step-by-step instructions on how to integrate with some popular streaming services, see Integrating with streaming and recording services.

Pexip VMRs can output a dedicated RTMP/RTMPS multimedia stream to enterprise CDN (Content Delivery Network) streaming and recording services such as Wowza, Quickchannel, Qumu, VideoTool, and Microsoft Stream, and to public streaming services such as YouTube and Facebook. Any Pexip conference can be streamed as a live event to an unlimited number of viewers, and can automatically be recorded and stored for later consumption.

If your conference contains presentation content, you have the option of setting up dual streams so that you can output the main video and presentation content channels separately. This means that viewers can simultaneously access the main video stream of the participants and a separate presentation content stream.

The administrator or meeting Host has to decide whether to set up the conference with a single RTMP stream or with dual RTMP streams. Currently, you cannot set up a single stream purely for presentation content.

When a conference is single streamed:

- Viewers are automatically switched between the main video stream and the presentation stream whenever someone is presenting.
- When someone is presenting, the video stream of the active speaker is shown in a small window in the upper right corner of the presentation. Currently, you cannot set up a single stream purely for presentation content (i.e. without the small video window).
- When someone is presenting in Adaptive Composition layout, the presentation is received in the layout mix and the video stream of the active speaker is shown alongside the presentation.
- The same single stream (with automatic switching between main video and presentation streams) can be made available later as a recording.

When a conference is dual streamed:

- Viewers have to manually switch between the main video stream and the presentation stream, according to their preference. (However, viewers could optionally view the streams in two separate windows — one window showing the main video stream and the other window showing the presentation stream.)
- The main video stream is not embedded as a small window within the presentation stream.
- If nobody is currently presenting, Pexip Infinity sends a placeholder image on the presentation stream.
- If the event is made available later as a recording, it can only be played back as two completely separate streams. There is no synchronization between the streams.

Note that your firewall needs to allow outbound traffic from the Conferencing Node to TCP port 1935 on the RTMP streaming server.

If required, you can set up multiple streams from the same conference to different streaming services.

Streaming/recording/transcribing indicators and announcements

If the conference is being streamed or recorded:

- A icon is displayed at the top center of the layout.
- The participant list on Pexip apps also shows the streaming or recording device to which the stream is being sent. A streaming participant has a streaming badge next to its name (which usually takes the form of the URL to which the stream is being sent).

For administrators looking at a conference graph, streaming participants are identified by a indicator.

Announcements to participants in Google Meet and Microsoft Teams conferences

- If a Google Meet conference is recorded, streamed or transcribed, taking smart notes via Gemini, having its media collected, or AI features are enabled, then an in-conference indicator is displayed and audio prompts indicating that streaming/recording (), transcribing (), note taking (), media collection (), or AI () has been started/stopped are played to callers who are gatewayed via Pexip Infinity into the conference. Distinct messages are used in each case.

- If a Microsoft Teams conference is recorded or transcribed, relevant audio prompts indicating that recording/transcription has been started/stopped are played to VTC participants.

The content of these audio files can also be changed by customizing a theme.

Initiating streaming from a Pexip app

To initiate streaming from a Pexip app, you add the streaming or recording device as a participant in the VMR.

When a Pexip app user adds a new participant to a conference, **Automatic routing** is used, meaning that the dialed alias must match an appropriate Call Routing Rule, and the call is then placed using the protocols and other settings as specified in the rule.

A suitable rule to match rtmp and rtmps aliases would typically include the following settings:

Option	Setting
Name	A suitable description such as "Streaming to <streaming service>".
Priority	This is a very specific rule, so we recommend that you give it a relatively high priority (a low number).
Incoming gateway calls	If you want to use this rule just for calls placed from a conference, then leave this blank (disabled). However, if you also want to use this rule to enable incoming gateway calls to be placed out to streaming or recording devices, then enable this option.
Outgoing calls from a conference	Enabled
Match against full alias URI	Enabled
Destination alias regex match	rtmps?://(.*)
Destination alias regex replace string	Leave this option blank — we want to dial the alias as it was received.
Outgoing location	Ensure that you select a location that is able to place calls to the streaming or recording system, such as a location containing Proxying Edge Nodes when streaming to external RMTP ingest endpoints.
Protocol	RTMP (streaming)

After you have created the above rule, it will also apply to any calls placed using the administrator interface, or to an Automatically dialed participant, where you have selected to route the call automatically.

Alternatively, you can bypass the need for call routing rules by using the administrator interface to dial out to the streaming participant, as follows:

- **Participant alias**: the address of the streaming service (typically prefixed with **rtmp://** or **rtmps://**)
- **Route this call**: ***Manually***
- **Protocol**: ***RTMP***
- **Streaming**: enabled.

In this case, because the call is routed manually, Call Routing Rules are not required.

Streaming when Pexip Infinity enforces media encryption

When streaming over RTMPS for media encryption, you typically need to specify port 443 on the streaming server, if the streaming URL that you created does not already specify a port. For example, a YouTube RTMPS streaming URL would need to be in the format **rtmps**://a.**rtmps**.youtube.com**:443**/live2/dau4-k4z1-5cf3-gg89-0c6x.

Some streaming services support RTMP only, and not RTMPS. Therefore, if Media encryption is set to ***Required*** (either globally or for the specific VMR being used), a call to the RTMP-only streaming service will fail.

In such cases, if appropriate, you may consider changing the **Media encryption** setting for the VMR to ***Best effort***, which will allow outbound calls to be placed over RTMP.

General instructions for setting up streaming

In general, to integrate with a third-party streaming or recording service:

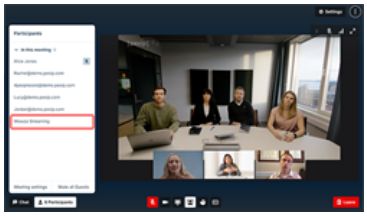
1. From the streaming provider, obtain an address to which the video stream will be sent.
2. Initiate a call from the conference to the streaming address. This is done by adding the streaming address as a conference participant. You can do this either from the Pexip Infinity Administrator interface or from a Pexip app connected to the VMR. Alternatively, for services that offer persistent URLs (such as with YouTube) which therefore can be re-used for subsequent streams, you could set up the URL to be automatically dialed whenever a particular VMR is used.

   When using the Administrator interface, use the following settings:
   - **Participant alias**: the address obtained from the streaming provider.
   - **Protocol**: ***RTMP***
   - **Role**: we recommend selecting ***Guest*** (so that the streaming participant is not shown to other Guests in a Virtual Auditorium layout, and so that it does not keep a conference alive when all other Hosts have left).
   - **Streaming**: select this option.

   When using a Pexip app, use the following settings:
   - **Participant details**: enter your rtmp/rtmps alias e.g. **rtmps://a.rtmps.youtube.com:443/live2/dau4-k4z1-5cf3-gg89-0c6x**

     RTMP authentication is supported; in this case credentials are included in the URI using the syntax **rtmps://username:password@host/...**.

     Note that a suitable Call Routing Rule is required when dialing out to a streaming service via Pexip apps.
   - **Role**: we recommend selecting ***Guest***.
3. When Pexip Infinity has placed the call to the streaming service, the **Streaming enabled** icon is displayed, and for Pexip app users the streaming participant appears in the participant list.



Skype for Business clients can also use the Infinity Gateway service to dial out to an RTMP streaming or recording service from within a Skype for Business meeting.

Streaming address format

When you initiate a call from the conference to the RTMP/RTMPS streaming address, ensure that the streaming address is in the following format:

rtmp(s)://address/application_name/path

Pexip Infinity expects that the address is followed by two parts: the application name and the path. An example of a YouTube streaming address is:

rtmps://a.rtmps.youtube.com/live2/dau4-k4z1-5cf3-gg89-0c6x

Setting and limiting call quality

Many factors can affect the video call quality as seen by individual participants on a call. Some of these factors are external to Pexip Infinity, such as the call protocols used by the endpoints participating in the conference, the compute and camera resources available to the endpoint, and constraints introduced by the network or call control systems. However, other factors that influence quality can be controlled within Pexip Infinity, typically at the platform level but with the ability to override those global settings for specific conferences.

Video call quality typically depends upon the resolution (for sharpness, and often expressed in terms of SD, HD or Full HD) and the frame rate (for smoothness). High resolutions and frame rates require more bandwidth than lower resolutions and frame rates, although the exact requirements vary according to the codec and compression algorithms being used.

The choice of resolution, frame rate and codec also impacts the amount of compute resource required by a Conferencing Node to host a conference, and hence the overall capacity of each Conferencing Node in terms of the number of concurrent participants and conferences it can host.

It is important to note that endpoints ultimately decide what bandwidth and resolution they send to Pexip Infinity, while Pexip Infinity is responsible for deciding what gets sent to the endpoints.

Maximum call quality

While the actual quality of the call as seen by individual participants depends upon the factors discussed above, you can configure within Pexip Infinity a **Maximum call quality** that limits what a Conferencing Node will send to — and request from — each participant in the conference.

The **Maximum call quality** options are SD, HD or Full HD and the associated resulting maximum resolution and frame rate for video are shown below:

Maximum call quality (video resolution)	Maximum resolution	Maximum frame rate (fps)
SD (448p)	768 x 448	30
HD (720p)	1280 x 720	30
Full HD (1080p)	1920 x 1200	30

By default, Pexip Infinity conferences have a maximum call quality of HD. You can configure this at the global platform level and, if required, override it for each individual service (VMR, Call Routing Rule and so on). For example, you could use the default option of "HD" for most of your services by default, but enable Full HD on some specific services.

Note that this is the **maximum** quality that Pexip Infinity will send to conference participants. The configured **Maximum outbound call bandwidth** for a service can cause Pexip Infinity to select a lower quality than the configured **Maximum call quality** (see Managing and restricting call bandwidth for more information).

Impact on resource usage

The **Maximum call quality** setting also controls how much compute resource is allocated and reserved by a Conferencing Node for each participant that joins the conference. This is measured within Pexip Infinity in relation to the amount of resources required by a standard HD connection.

In general, when compared to a single high definition **HD** 720p call:

- a **Full HD** 1080p call uses twice the resource
- an **SD** standard definition call uses half the resource
- an **audio-only** call uses one sixteenth of the resource.

However, note that:

- A WebRTC call using the **VP8** codec uses the same amount of resource as H.264, and the **VP9** codec uses around 25% more resource. Therefore:
  - VP9 at 720p uses the equivalent of 1.25 HD resources
  - VP9 at 1080p uses the equivalent of 2.5 HD resources
  - the maximum number of calls a given Conferencing Node can support will be fewer for VP9 calls, for example a node that supports 39 **H.264**/**VP8** SD calls will support 31 **VP9** SD calls.

  Note that within the same conference some participants may use VP9 (if they are connected to a Conferencing Node using the AVX2 or later instruction set) while other participants may use VP8 (if they are connected to a Conferencing Node on older hardware).
- Conferences or gateway calls that use the **Adaptive Composition** layout, and Teams conferences that use the Teams-like layout, consume additional Conferencing Node resources. The actual amount of additional resource depends on many factors, but as a guide, it uses an additional 1 HD of resource per conference, for up to 3 other video participants, plus approximately another 0.5 HD for each additional (4th, 5th etc.) video participant that is on stage. This is regardless of the call quality / resolution of the conference itself and each individual participant's connection (codec, bandwidth and so on).
- H.323 audio-only calls are treated the same as video calls for resource usage purposes.
- Connections to a Media Playback Service use 1.2 times as much resource as a connection to a VMR.
- When transferring a participant, the transferee can temporarily take two sets of resources while in the process of being transferred. This should not normally last more than a few seconds.

Thus, setting the maximum call quality to a "high" value such as Full HD will result in more resources being reserved than selecting a "low" value such as SD, and the more resources that are used or reserved means a lower capacity in terms of overall concurrent connections (also referred to as ports) for each Conferencing Node.

Considerations for using Full HD (1080p) for main video in calls

Enabling Full HD (1080p) capabilities allows any endpoint capable of Full HD to send and receive its main video at 1080p to those conferences. However, as discussed above, enabling Full HD has implications on bandwidth and capacity across your deployment, specifically:

- Full HD calls require approximately double the Conferencing Node resources and double the bandwidth of an HD call.
- 1 Full HD of capacity will be reserved for backplanes between Conferencing Nodes.

Note that 1080p is automatically used for sharing high-resolution content with HD-capable endpoints if there is sufficient available bandwidth i.e. presentation content may still be sent at 1080p even if Full HD is not allowed for main video.

Setting the maximum call quality for participants in a conference

To set the maximum call quality for all calls across your entire deployment:

1. Go to **Platform > Global Settings**.
2. In the **Service Configuration** section, select the required **Maximum call quality**:
   - ***SD***: each participant is limited to SD quality.
   - ***HD***: each participant is limited to HD (720p) quality.
   - ***Full HD (1080p)***: allows any endpoint capable of Full HD to send and receive its main video at 1080p.
3. Select **Save**.

To override the global default and set the maximum call quality for an individual service or Call Routing Rule:

1. Go to the relevant service or rule:
   - **Services > Virtual Meeting Rooms**
   - **Services > Virtual Auditoriums**
   - **Services > Virtual Receptions**
   - **Services > Scheduled Conferences**
   - **Services > Call Routing**
2. Either select the name of the service or rule you want to edit, or click **Add**.
3. In the **Advanced options** section, or the **Call media settings** section for routing rules, select the required **Maximum call quality**:
   - ***Use global setting***: use the global maximum call quality setting.
   - ***SD***: each participant is limited to SD quality.
   - ***HD***: each participant is limited to HD (720p) quality.
   - ***Full HD (1080p)***: allows any endpoint capable of Full HD to send and receive its main video at 1080p.
4. Select **Save**.

Changes to the maximum call quality take effect for any new conferences initiated after the change has been made.

Managing and restricting call bandwidth

You can restrict the amount of bandwidth used by each participant when dialed in to a Pexip Infinity service (Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service), or for Infinity Gateway calls or other outbound calls that are managed by Call Routing Rules. You can place restrictions on the bandwidth received by participants, sent by participants, or both.

Bandwidth restrictions can be configured on a global basis, or on each individual service or routing rule. Settings applied to an individual service override any global limits that have been applied.

When sending main video and presentation to an endpoint, you can also control the maximum percentage of the call bandwidth to allocate to the presentation content (see presentation bandwidth requirements).

If a Virtual Reception has bandwidth limits, these only apply to participants while they are in the Virtual Reception. When the call is transferred to the selected Virtual Meeting Room or Virtual Auditorium, any bandwidth restrictions for that service will then apply.

When dialing out from a conference, the outbound call bandwidth limit is inherited from the VMR's bandwidth settings. (If a Call Routing Rule is applied, the rule's bandwidth settings are not used.)

Inbound bandwidth restrictions are implemented as requests to the participant endpoints to limit their bandwidth to the specified amount. It is important to note that endpoints ultimately decide what bandwidth and resolution they send — Pexip Infinity cannot actually enforce inbound restrictions. However, Pexip Infinity is responsible for deciding what gets sent to the endpoints.

Restricting video call resolutions

If you want to limit video calls to specific resolutions (and limit the transcoding node resources that are reserved for calls), you should use the **Maximum call quality** setting.

However, bandwidth settings also affect the call resolution as shown below. If there is insufficient bandwidth (the **Maximum bandwidth** settings configured against the service) to support the **Maximum call quality** then the participant's call will use a lower resolution as appropriate for the available bandwidth (and consume less transcoding node resource).

Minimum call bandwidth requirements

This table shows the minimum bandwidth required for Pexip Infinity to be able to send different video resolutions. Note that this **does not** include the audio component, which can add between 8 kbps and 64 kbps.

Video resolution	Minimum required bandwidth
Full HD (1080p)	2400 kbps (1600 kbps for VP9)
HD (720p)	960 kbps (640 kbps for VP9)
SD (448p)	448 kbps
SD (QCIF)	64 kbps

Note that when compared to standards-based endpoints, the WebRTC VP9 codec provides the same resolution at a lower bandwidth, but consumes around 25% more resource on the Conferencing Node. This is why you should use **Maximum call quality** rather than bandwidth restrictions to limit resource consumption.

In most cases you should apply the same bandwidth restrictions to the inbound and outbound calls within a service. However, for example, if you want to allow participants to send their video as SD, but receive the composed layout of all participants (main video and video thumbnails) as HD, you would set the inbound call bandwidth to 960 kbps and the outbound to 1800 kbps (although these settings would not limit clients using VP9).

Applying bandwidth restrictions to an entire deployment

To restrict the bandwidth of calls across your entire deployment:

1. Go to **Platform > Global Settings**.
2. In the **Service Configuration** section, enter the desired values in the following fields:

Option	Description
Maximum inbound call bandwidth (kbps)	Limits the bitrate of media received by Pexip Infinity from a participant. Leave blank if you do not want to apply any restrictions. This can be overridden by any services or rules that have a specific **Maximum inbound call bandwidth** configured.

Option	Description
Maximum outbound call bandwidth (kbps)	Limits the bitrate of media sent from Pexip Infinity to a participant. If left blank, a default limit of 4128 kbps is applied. This can be overridden by any services or rules that have a specific **Maximum outbound call bandwidth** configured.

Applying bandwidth restrictions to a service or Call Routing Rule

To restrict the bandwidth of calls to a particular service or Call Routing Rule:

1. Go to the relevant service or rule:
   - **Services > Virtual Meeting Rooms**
   - **Services > Virtual Auditoriums**
   - **Services > Virtual Receptions**
   - **Services > Scheduled Conferences**
   - **Services > Call Routing**
2. Either select the name of the service or rule you wish to edit, or click **Add**.
3. In the **Advanced options** section, select **Show** (not required for Call Routing Rules).
4. Enter the desired values in the following fields:

Option	Description
Maximum inbound call bandwidth (kbps)	Limits the bitrate of media received by Pexip Infinity from a participant. This overrides any global **Maximum inbound call bandwidth** that may have been configured.
Maximum outbound call bandwidth (kbps)	Limits the bitrate of media sent from Pexip Infinity to a participant. This overrides any global **Maximum outbound call bandwidth** that may have been configured.

5. Select **Save**.

Presentation bandwidth requirements

Bandwidth usage for presentation streams depends on the type of client:

- When sending main video and presentation content to a conference participant on a standards-based (SIP or H.323) or WebRTC endpoint, Pexip Infinity splits the available bandwidth between main video and content.
  - By default up to 75% of the available call bandwidth is allocated to presentation content (and thus 25% is allocated to main video). You can modify this ratio at the global platform level via **Platform > Global Settings > Service Configuration > Maximum Presentation Bandwidth Ratio**.
  - The resolution and frame rate used for presentation content adapts dynamically to match the presented content, and the **Maximum call quality** is used to determine the maximum frame rate that Pexip Infinity will send for a given resolution. For example, if the maximum call quality is HD, then the maximum frame rate for Full HD content is 13 fps, but if an incoming presentation is Full HD at 4 fps then Pexip Infinity will send 4 fps.
- When sending presentation content to Microsoft Teams, Pexip always sends content to Teams at Full HD (1080p) at a frame rate requested by Teams. The resolution and frame rate of content sent from Microsoft Teams is determined by Teams.
- When sending main video and presentation content to a Google Meet meeting, the presentation stream is treated as a separate call and has the same bandwidth limit applied as the main video stream. When sending video or presentation content to Pexip Infinity, Google Meet does not have any fixed bandwidth limitations — it dynamically downspeeds and upspeeds individual calls in response to network conditions, although it typically will not exceed 2 Mbps.

Enabling and disabling chat messages

Conference participants who use a chat-enabled client can send messages and share links with each other within a Virtual Meeting Room or Virtual Auditorium, and when calling each other directly via the Infinity Gateway. Supported clients include Skype for Business clients and Pexip's own Pexip apps.

Chat support is enabled by default and is configurable on a platform-wide and per-conference basis. By default, Pexip Infinity does not log or retain the contents of any chat messages.

To configure chat at the platform level:

1. Go to **Platform > Global Settings**.
2. From within the **Connectivity** section, deselect or select **Enable chat**.

You can also override the global setting on a per conference basis if required. To do this:

1. Go to **Services > Virtual Meeting Rooms**, **Services > Virtual Auditoriums** or **Services > Scheduled Conferences**.
2. From within the **Advanced Options** section, choose one of the **Enable chat** options:
   - ***Use global chat setting***: as per the global configuration setting.
   - ***Yes***: chat is enabled.
   - ***No***: chat is disabled.

   Default: ***Use global chat setting***.

When chat is disabled, Pexip apps do not show the chat window.

Providing chat to participants using unsupported clients

Conference participants who are not using one of the supported clients are not able to read or send chat messages.

However, if they have access to a web browser they can use the web app to join the conference without video or audio. This will give them access to the chat room, plus the ability to view and share presentations, view the participant list, and (if they are Host participants) control aspects of the conference.

For more information on using and administering the Pexip apps, see Introduction to Pexip apps.

Playing notification tones when participants join or leave a conference

Pexip Infinity can automatically play a sound when a participant joins or leaves a conference. These entry and exit tones are disabled by default, but you can enable them on selected services using themes.

Enabling notification tones

Entry and exit tones are applied using themes. To enable entry or exit tones for a particular Virtual Meeting Room or Virtual Auditorium, you must ensure that the theme used for that service includes the following files:

- **conf-participant_entry_tone_48kHz_mono.wav**, containing the sound to play when a participant joins the conference.
- **conf-participant_exit_tone_48kHz_mono.wav**, containing the sound to play when a participant leaves the conference.

By default, entry and exit tones are not played, as the Base theme contains "empty" entry and exit tone files. However, Pexip Infinity also ships with a selection of preconfigured themes, some of which contain entry and exit tones in the **.wav** files listed above.

This means that to enable notification tones you can do any of the following:

- select one of the preconfigured themes containing tones
- use the **.wav** files from those preconfigured themes in your own themes
- produce your own **.wav** files and include those files in your own themes.

Disabling notification tones

The Base theme contains empty entry and exit tone files, and is shipped as the default theme. This means that by default, entry and exit tones will not be played. You can download and copy the empty entry and exit tone files from the Base theme and use them in any custom themes you create for which you don't want to include entry and exit tones. Some of the preconfigured themes also contain empty entry and exit tone files.

If you apply a theme that does not include these files (rather than including empty files), Pexip Infinity will instead use the entry and exit tone files from the theme that has been selected as the default theme.

Therefore, if you want to ensure that entry or exit tones are not played, you must still include the entry tone and exit tone files in the theme, but leave them empty.

Controlling active conferences

Locking a conference and allowing participants to join a locked conference

You can lock a conference if you want to prevent any further participants from joining the meeting after it has started. A conference can be locked and unlocked by Host participants using the Pexip app or using DTMF-enabled endpoints, or by administrators using the Administrator interface.

Locking does not apply to direct media calls.

When a conference is locked:

- A conference locked indicator is displayed.
- Any new participants who attempt to join the conference are held at a waiting screen. They can be allowed in via the Pexip app by Host participants who are already in the conference.

The exact locking behavior depends on whether or not the Virtual Meeting Room or Virtual Auditorium has a Host PIN.

If the service **does not have a Host PIN**:

- Participants can join the conference until it is locked.
- When the conference is locked:
  - Any further participants who attempt to join the conference (including any Automatically Dialed Participants and manually-invited participants who have been given a role of Guest) are held at the **Waiting for the host** screen. However, any ADPs and manually-invited participants with a role of Host will join the conference immediately.
  - All participants who are already in the conference are notified of any participants who are attempting to join the locked conference, and can allow the waiting participants to join. Notifications take the form of an audio message/alert for each participant attempting to join.
- If the conference is unlocked, any participants who are still waiting will automatically join the conference.

If the service **has a Host PIN**:

- Host and Guest participants can join the conference until it is locked.
- When the conference is locked:
  - New participants who enter the Host PIN will join the conference immediately — locking does not apply to them.
  - Any new Guest participants (including any Automatically Dialed Participants and manually-invited participants who have been given a role of Guest) are held at the **Waiting for the host** screen.
  - All Host participants who are already in the conference are notified of any Guest participants who are attempting to join the locked conference, and can allow the waiting Guest participants to join. Notifications take the form of an audio message/alert for each participant attempting to join.
- If the conference is unlocked, any Guest participants who are still waiting will automatically join the conference.

The audio message/alert and the **Waiting for the host** screen can be fully customized via the theme associated with your services.

Locking using the Administrator interface

To lock or unlock a conference from the Administrator interface:

1. Log into the Pexip Infinity Administrator interface.
2. Go to **Status > Conferences**.
3. From the **Service name** column, select the conference you want to lock or unlock.
4. At the bottom left of the page, select **Lock conference** or **Unlock conference** as appropriate.

Locking using Pexip apps

Host participants using the Pexip app can lock and unlock the conference via the in-call toolbar, by toggling the **Lock** icon 🔒.

Locking using DTMF

If DTMF controls are enabled, Host participants using telephones or SIP/H.323 endpoints can lock and unlock the conference using DTMF. The default DTMF entry to do this is ***7** but this may have been customized.

Allowing waiting participants to join a locked conference

When a new participant attempts to join a locked conference, all Host participants (on any endpoint) in the conference are notified via an audio alert that a participant is waiting to join. However, only Host participants who are using Pexip app can admit individual participants into the conference.

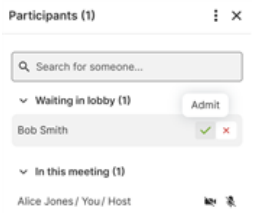
ℹ If a locked conference is then unlocked, all participants who are still waiting will automatically join the conference.

A red dot appears on the **Participants** button whenever any participants are waiting in the meeting lobby because they are:

- waiting to join a locked conference, or
- waiting to join a meeting that requires authentication, but they are using an endpoint that does not support authentication.

Select the button to open the **Participants** panel, and locate the participant in the **Waiting in lobby** section. You then have two options:

- To allow a participant to join the conference, select **Admit**.
- If you do not want them to join, select **Deny**. The participant will be disconnected from the meeting.



Rejecting a request to join a locked conference

If a Host (who is using the Pexip app) does not want a waiting participant to join the conference immediately, they have two options:

- To reject the request completely, the Host participant must click on the **Deny** button / red cross icon next to the waiting participant's name. The waiting participant's call will be disconnected.
- To leave the participant at the waiting for Host screen, the Host participant should do nothing. The waiting participant will remain at the waiting screen until:
  - a Host participant chooses to let the waiting participant join the conference, or
  - the conference is unlocked (after which the waiting participant automatically joins the conference), or
  - the participant has been waiting for longer than the specified waiting time (after which the participant is disconnected), or
  - the conference finishes (after which the waiting participant is disconnected).

Controlling the layout during a conference

Conference participants have a range of options available to them during a conference to control aspects of the conference layout and how they receive presentation content. They can:

- Change the layout of an ongoing conference.
- Apply a personal layout for a specific endpoint or participant.
- Toggle multiscreen participant display for dual-screen endpoints.
- Control how presentation content is displayed when using an Adaptive Composition layout.

- Toggle the display of participant names when using a Pexip app.
- Spotlight a participant when using a Pexip app.

Changing the conference layout

The layout used in a conference can be changed dynamically during the conference by the conference participants. See Conference layouts and speaker names for full details about all of the available layouts and how to configure the initial default layout settings for a conference.

Hosts can apply layout changes to all participants in a Pexip conference, and individual participants can request a personal layout. Not all layouts are supported on dual screen endpoints; these endpoints will revert to displaying the 1+7 layout in place of unsupported layouts.

Hardware endpoint participants (SIP/H.323)

Host participants on video endpoints can change the layout that is seen by all participants by sending DTMF / keypad commands to the conference.

- By default the layout changes every time the endpoint sends ***8** to the conference, and it cycles through most of the available Pexip layout types in this order: 1+7, Adaptive Composition, 1+21, 2+21, 2x2, 3x3, 4x4, 5x5, 1+0, 1+33.
- The ability to change the layout, the layouts that are used, the sequence in which they are displayed, and the DTMF keypad control used to change the layout, are all customizable via themes.

For more information see Using a DTMF keypad to control a conference and Base theme and other preconfigured themes.

In addition, participants on Cisco endpoints can also use the Pexip Layout Controls macro.

Pexip app participants

Host participants using the web app or Pexip app for Windows can dynamically change the layout during a conference. To do this, from the top right of the screen, select **Settings > Meeting layout**.

Bespoke client applications could also make use of the **transformLayout** function in the PexRTC client API.

Applying a personal layout

Host and Guest participants can choose the layout they receive without affecting the conference-wide layout received by other participants.

Enabling personal layouts for a participant

Each participant must be explicitly enabled via participant policy to receive personal layouts before they can request a personal layout. Administrators do this by setting their **can_receive_personal_mix** participant configuration response field to **true**.

A conference can contain a mix of participants who are, and are not, enabled for personal layouts.

Requesting a personal layout

Participants can request a personal layout via the web app or using DTMF tones.

Using the web app

To select a personal layout, from the top right select **Settings > Meeting layout**.

Using DTMF tones

Hardware endpoint participants (SIP/H.323) can request a personal layout by sending a ***1** DTMF tone.

This cycles through the same list of layouts available to the conference as described above.

What's noteworthy?

When using personal layouts, note that:

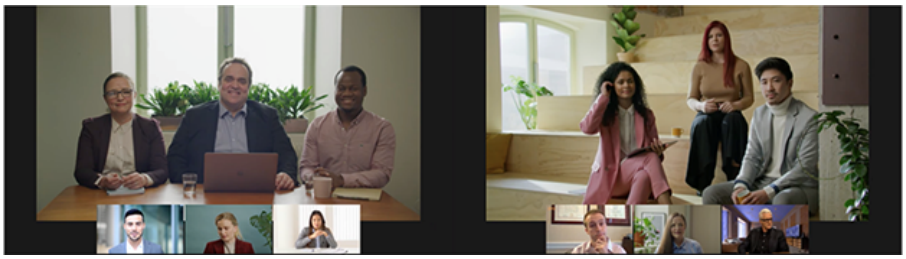
- A personal layout cannot be applied while the main conference layout is either Adaptive Composition or the Teams-like layout. However, if a Host selects a different main layout, participants can then select a personal layout.
- Participants that are enabled for personal layouts have extra connection resources reserved (approximately another ⅓ HD).
- Only the layout and the visibility of the layout notch indicators can be personalized; the conference-wide settings for all other visual elements such as text overlay or spotlight are applied to the personal layout.
- Features such as dual screen and receiving the presentation stream as part of the layout mix are supported — providing they are generally supported in the requested layout.

Multiscreen participant display

If you have a dual screen endpoint you can display conference participants across both screens, allowing you to show more participants than if you have a single screen. This display mode is available for the following meeting, layout and endpoint types:

- Conferences hosted in a VMR or Virtual Auditorium.
- Calls into a Microsoft Teams or Google Meet conference.
- Where the conference layout is 1+7, 1+0, 2x2, 3x3, 4x4, 5x5, custom layouts, Adaptive Composition or Teams-like.
  - If you are using any other layout when multiscreen is enabled then the 1+7 layout is used on both screens.
  - If you are using a 4x4 or 5x5 layout when multiscreen is enabled then each screen will display 3x3 on both screens (this means that when using a 5x5 layout you will see fewer participants in total if multiscreen is enabled).
- SIP and H.323 endpoints.

Multiscreen participant display is used automatically on endpoints that are known to have dual screens — see Determining if an endpoint has single or multiple screens for details.



Note that:

- The same layout format is used on both screens (unless you are using a custom layout).
- As each video participant is displayed they are added alternately to each screen (until the layout on that screen is full).
- If a participant starts presenting, the second screen shows the presentation content.
- In-conference indicators are displayed on the primary screen only.
- When multiscreen is in use, a VTC touch panel will show there is presentation in progress, even if there is no current presentation (this is because the multiscreen feature uses the content channel to display participants on the second screen).
- If there are only two video participants in a meeting (including the VTC) the presentation screen will turn on and remain black (unless you have customized the **unused_screen** theme element) until a third video participant joins.

You can disable multiscreen entirely by adding `"disable_multiscreen": true` to your **themeconfig.json** file.

Toggling multiscreen display

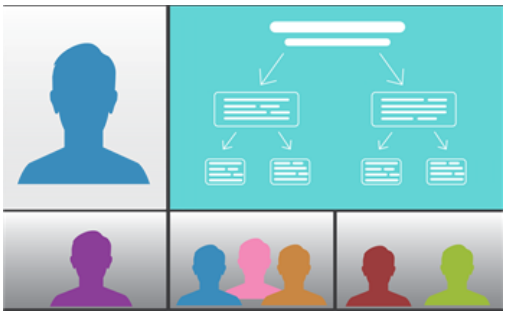
You can toggle multiscreen display mode on and off by a using a DTMF keypad command sent from the endpoint. This command defaults to ***9** but it can be customized via themes. This command can also be used to apply multiscreen display mode to a dual screen endpoint that has dual screens but the additional screen has not been detected.

When toggling the multiscreen mode, it only applies to the endpoint sending the enable/disable command, and not to all participants in the conference.

If you are a meeting Host and want to view an unsupported layout on a dual screen endpoint, you can use this option to disable dual screen on the endpoint, and then use the layout option to cycle through the conference layouts. Note that if you do this, the selected layout will appear on the dual screen endpoint's main screen only.

For more information see Using a DTMF keypad to control a conference.

Receiving the presentation stream as part of the layout mix (Adaptive Composition)



When using Adaptive Composition or the Teams-like layout, most single-screen endpoints will automatically receive the presentation stream as part of the layout mix (replacing some of the other video participants), but you can choose to switch to receiving it as a separate stream.

Note that:

- Cisco endpoints with 2 (or more) HDMI outputs, but that are installed with only a single screen, do not automatically receive the presentation as part of the layout mix. In these cases you need to manually switch to that mode (via the endpoint's keypad / DTMF tones) or customize the default behaviour.
- If the call bandwidth is less than 1.2 Mbps then the presentation is always sent as a separate stream.

When the presentation is received in the layout mix:

- The person presenting is shown alongside the presentation. Note that the "presenter" is the participant that the artificial intelligence driving the layout has decided is the person who is actually presenting (this is based on activity — such as speaking frequency — and not necessarily the device/person that is physically sharing the content).
- A maximum of 12 other video participants are included in the layout, as well as the presentation content.
- If the "presenter" is a group of people, then the presentation occupies the entire top row and the presenter group is shown in the bottom row instead, plus up to 8 other participants.
- The presentation is pillarboxed or letterboxed to fit into its position in the layout.
- The presentation segment has its own framerate regardless of the source framerate of the presentation stream. For example, even though the presentation input stream is 5 fps, the framerate of the presentation in the layout could still be 30 fps.

Dual-screen endpoints automatically receive the presentation stream on the second screen, and the primary screen shows the video participants.

The primary screen shows the person presenting closer to the second screen.

If the "presenter" is a group of people, then the "presenter" occupies the entire top row of the primary screen.

Hardware endpoint participants (SIP/H.323)

The presentation mode can be toggled between receiving it in the layout mix and receiving it as a separate stream by a using a DTMF keypad command sent from the endpoint. This command defaults to ***4** but it can be customized via themes.

- Single-screen endpoints receive the presentation as part of the layout mix by default, and it always resets to this mode when the endpoint joins a new conference. However, within a conference, the current state is remembered if one presentation stops and a new presentation starts.

- When toggling the presentation mode, it only applies to the endpoint sending the enable/disable command, and not to all participants in the conference.
- It does not apply to SIP endpoints with multiple screens — they always receive a separate presentation stream.

See Determining if an endpoint has single or multiple screens for more information.

Pexip app participants

- **Web app**: Participants can elect to receive a presentation stream as part of the layout mix in an Adaptive Composition layout by enabling the Prefer presentation in mix option (**Settings > Advanced**). When disabled, presentation and main video is received as two separate streams in all layouts, and the user can toggle between the two.
- **Pexip app for Windows**: Participants can elect to receive a presentation stream as part of the layout mix in an Adaptive Composition layout by enabling the Prefer presentation in mix option (**Settings > Meeting Layout**). When disabled, presentation and main video is received as two separate streams in all layouts, and the user can toggle between the two.

Enabling and disabling the display of participant names

Each Virtual Meeting Room and Virtual Auditorium can be configured to show the names of the participants in a text overlay along the bottom of their video image. See Showing the names of active speakers and participants for full details about how to enable or disable participant names by default for a conference, and how the display name is determined.

Host participants using the web app can dynamically show and hide participant names during the conference by selecting **Settings > Advanced** and toggling the **Show name labels** option.

Spotlighting a participant (classic layouts only)

Host participants using the Pexip app can "spotlight" themselves or other participants in the meeting.

The spotlight feature locks any spotlighted participants in the primary positions in the stage layout, ahead of any current speakers. When any participants have been spotlighted, the first one to be spotlighted has the main speaker position, the second one has the second position (leftmost small video, for example), and so on. All remaining participants are arranged by most recent voice activity, as usual.

When Lock presenter as main speaker is enabled, a spotlighted participant appears in the main video ahead of any presenters.

To spotlight a participant:

- **Web app / Pexip app for Windows**: From the **Participants** panel, hover over the participant's name, select ⋯ and then select **Spotlight**.
- **Mobile apps**: From the **Participant list**, tap the participant and then select **Spotlight**.

Note that:

- Endpoints that support Far-End Camera Control can also be spotlighted while the FECC dialog is open.
- Spotlighting is not supported in the Adaptive Composition layout.
- In a Virtual Auditorium, where Guests cannot see other Guests, if a Guest participant is spotlighted they only appear in the main video for Host participants; other Guests do not see the spotlighted Guest.

Also see Pinning participants to the layout during a conference.

Controlling breakout rooms

To manage the breakout rooms within a conference (add/remove breakout rooms, assign participants to rooms and so on), you have three options:

- Join the conference as a Host participant using a Pexip web app or Meeting Manager, and use the **Breakout rooms** panel to control the rooms and participants. See Breakout rooms for full details.
- Use to create breakout rooms and assign an incoming caller to a room.
- Use your own bespoke management application in conjunction with the .

Muting a participant's audio

There are several ways the audio being sent from one or more participants can be muted by an administrator or a meeting Host, known as being "administratively muted":

- Using the Administrator interface
- Using the Pexip app
- Using DTMF

Administrative versus local muting

When a participant's audio is administratively muted, Pexip Infinity still receives their audio stream but does not add it to the mix being sent to all other participants. This is different from a participant muting their own microphone locally, when the local client or endpoint does not send any audio to Pexip Infinity.

For the **Pexip web app**, local muting is synchronized with server-side muting so that when a participant:

- mutes or unmutes themself, their mute state is updated on the server (Pexip Infinity)
- is administratively muted:
  - their mute state is updated on their local client
  - they can unmute themselves locally, unless a meeting Host using the web app has selected the option to Prevent guests from unmuting
- is administratively unmuted, their microphone remains muted locally (for privacy reasons) and a dialog appears giving the option whether or not to stay muted locally
- joins a meeting with their microphone unmuted, and where the administrator has previously elected to **Mute all Guests**, the participant will join unmuted.
- views the Participants panel, the color of the mute indicators shows whether participants muted themselves (black) or were muted by a Host (red)

For all other Pexip apps and other standards-based endpoints, local muting is not synchronized with server-side muting. This means that participants must be both locally unmuted and administratively unmuted in order to be heard.

Host participants using telephones or SIP/H.323 endpoints that support DTMF and who have been administratively muted can use DTMF to unmute themselves.

Administrators can customize the Pexip apps so that the microphone is muted locally by default. In these cases, participants may still subsequently unmute and mute themselves locally via the standard controls on their client.

Note that low-level, almost imperceptible background noise is added to the audio mix in all conferences. This creates a similar effect to an open mic and gives reassurance that the conference is alive, even if all participants are muted.

Using the Administrator interface

Administrators can use the Pexip Infinity Administrator interface to either mute individual participants, or mute all Guest participants.

Muting an individual participant

To use the Pexip Infinity Administrator interface to mute a participant's audio:

1. Select the participant. You can do this in two ways:
   - Go to **Status > Participants** and select the participant to mute.
   - Go to **Status > Conferences** and select the Virtual Meeting Room or Virtual Auditorium that the participant is in. From the **Participants** tab, select the participant to mute.
2. At the bottom right of the screen, select **Mute**.

Muting all Guest participants

To use the Pexip Infinity Administrator interface to mute the audio from all Guest participants:

1. Go to **Status > Conferences** and select the Virtual Meeting Room or Virtual Auditorium.
2. At the bottom left of the screen, select **Mute all Guests**.

All Guest participants currently in the conference will be muted. Any Guest participants who subsequently join the conference using clients other than the Pexip web app will be muted (web app participants will join muted or unmuted, depending on their local mute state). Individual Guest participants using clients other than the Pexip web app can still be unmuted and muted by a conference Host or system administrator; web app participants can still unmute and mute themselves locally.

Using the Pexip app

You must have Host privileges to use this feature.

Host participants can mute an individual participant, or mute all Guests simultaneously. Note that it does not mute the participant's speakers, so they will still hear all other unmuted participants, but what that muted participant says will not be heard.

When the Pexip app is used to mute a participant, the **Audio administratively muted?** column of the Conference status page of the Pexip Infinity Administrator interface shows ***YES***.

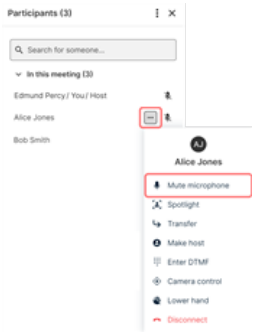
- Participants are not notified that they have been muted or unmuted, although Pexip app participants see a red muted icon next to their name in the participant list.
- Participants can mute and unmute themselves using the Pexip app, but only if they have Host privileges or are using the web app.
- A Pexip app user with Host privileges can unmute a participant previously muted by another Pexip app user.

Muting an individual participant

To use the Pexip app to mute or unmute an individual participant's audio:

From the **Participants** panel, hover over the participant's name, select ··· and then **Mute microphone** or **Unmute microphone**.

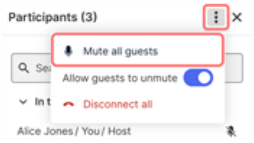
When muted, a icon is shown next to the participant's name.



Muting all Guest participants

To use the Pexip app to mute the audio coming from all Guest participants:

From the top of the **Participants** panel, select ⋮ and then **Mute all guests**.



All Guest participants currently in the conference will be muted (indicated by a icon next to their name). Any Guest participants who subsequently join the conference using clients other than the Pexip web app will be muted (web app participants will join muted or unmuted, depending on their local mute state). Individual Guest participants using clients other than the Pexip web app can still be unmuted and muted by a conference Host or system administrator; web app participants can still unmute and mute themselves locally.

Using DTMF

If DTMF controls have been enabled:

- Host participants can mute and unmute all Guest participants by sending ***5**.
- Host participants that have been muted can unmute themselves via ***6**.

For more information, see Using a DTMF keypad to control a conference.

Manually dialing out to a participant from a conference

The Pexip Infinity platform allows you to dial out to participants from an ongoing conference, on an ad hoc basis. When you dial out to a participant in this way, a call is placed to their endpoint from the Virtual Meeting Room or Virtual Auditorium. If they answer the call, they join the conference as either a Host or Guest, depending on the option that you selected. Participants joining the conference in this way do not go through the IVR screen and do not have to enter a PIN.

The participant could also take the form of a dedicated multimedia stream to enterprise CDN (Content Delivery Network) streaming and recording services such as Wowza, Quickchannel, Qumu, VideoTool, and Microsoft Stream, and to public streaming services such as YouTube and Facebook. Any Pexip conference can be streamed as a live event to an unlimited number of viewers, and can automatically be recorded and stored for later consumption. For more information, see Streaming and recording a conference.

Dialing out from a VMR to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet is not supported.

Administrators and conference hosts can add participants to a conference. Administrators can do so via the Administrator interface, and hosts can do so via the Pexip app.

You can also configure a Virtual Meeting Room or Virtual Auditorium so that one or more participants are dialed out to automatically whenever a conference starts. See Automatically dialing out to a participant from a conference for more information.

If your environment includes a PSTN gateway or uses an ITSP (Internet telephony service provider), consider the potential for toll fraud if you have Call Routing Rules that can route calls to the PSTN gateway or ITSP, or if you allow conference participants to dial out to other participants via the PSTN gateway or ITSP. See PSTN gateways and toll fraud for more information.

When dialing out from a conference, the outbound call bandwidth limit is inherited from the VMR's bandwidth settings. (If a Call Routing Rule is applied, the rule's bandwidth settings are not used.)

Call Routing Rule requirements and controlling dial-out capabilities

The ability to dial out to a participant from a conference is enabled by default. Call Routing Rules may be required when dialing out from a conference to a new participant, depending on the method used:

- **Using the Pexip app or client API**: in most cases, you need to configure appropriate Call Routing Rules to enable end-users using the Pexip app or the client API to place an outbound call. Rules may not be required for the client API if the **Enable legacy dialout API** setting (**Platform > Global Settings > Connectivity**) is selected — see the Connectivity options in global settings for more information.
- **Using administrator tools**: when dialing out via the Administrator interface, management API or when using Automatically Dialed Participants (ADPs), you have the option to choose automatic routing (where a rule is required) or to manually specify the call details (where a rule is not required).

This means that when dialing out from an ongoing conference, any calls made via a Pexip app (including RTMP calls to a streaming or recording service) **always** use automatic routing and thus **must** match an appropriate Call Routing Rule for the call to be placed.

Dialing out via the Administrator interface

You can use the Pexip Infinity Administrator interface to dial out to a participant from a Virtual Meeting Room or Virtual Auditorium. If and when the call is answered, that endpoint will join the conference.

To dial out to a participant using the Administrator interface:

1. Select the Virtual Meeting Room or Virtual Auditorium to dial the participant from. You can do this in any of the following ways:
   - Go to **Services > Virtual Meeting Rooms** and select the name of the Virtual Meeting Room.
   - Go to **Services > Virtual Auditoriums** and select the name of the Virtual Auditorium.
   - If the conference that you want to dial out from is already in progress, go to **Status > Conferences** and select the required Virtual Meeting Room or Virtual Auditorium.
2. At the bottom left of the screen, select **Dial out to participant**.
3. Complete the following fields:

Field	Description
System location	Select the system location from which the call will be placed. If there is more than one Conferencing Node in that location, Pexip Infinity will choose the most appropriate. The system location determines which H.323 gatekeeper or SIP proxy to use to route the call.
Service alias	This lists all the aliases that have been configured for the selected Virtual Meeting Room or Virtual Auditorium. The participant will see the incoming call as coming from the selected alias.
Participant alias	The alias of the endpoint/participant that you want to dial.
Route this call	Select how to route the call: ◦ ***Manually***: uses the requested **Protocol** and the defaults for the specified **System location**. ◦ ***Automatically***: routes the call according to the configured Call Routing Rules. This means that the dialed alias must match an outgoing Call Routing Rule for the call to be placed (using the protocols, outgoing location and call control systems etc. as configured for that rule). Default: ***Manually***.
Participant display name	An optional user-facing display name for this participant, which may be used in participant lists and as the overlaid participant name (if enabled). If this name is not specified then the **Participant alias** is used as the display name instead.
Protocol	The signaling protocol to use when dialing the participant. Select either ***SIP***, ***H.323***, or if the endpoint is a Skype for Business client, select ***Skype for Business (MS-SIP)***. The RTMP protocol is typically used when adding a streaming participant. Note that if the call is to a registered device, Pexip Infinity will instead use the protocol that the device used to make the registration. This field only applies when **Route this call** is set to ***Manually***.
Call capability	Allows you to limit the media content of the call. The participant being called will not be able to escalate beyond the selected capability. For more information, see Controlling media capability. Default: ***Main video + presentation***. This field only applies when **Route this call** is set to ***Manually***.
Role	Select whether you want the participant to join the conference as a Host or Guest.
DTMF sequence	An optional DTMF sequence to transmit after the call to the dialed participant starts. A DTMF sequence can include: the digits 0-9, "*" (asterisk), "#" (hash) or "," (comma). The DTMF tones are sent 3 seconds after the call connects, one at a time, every 0.5 seconds. A comma is a special digit that represents a 2 second pause (multiple commas can be used if a longer pause is needed). For example, if you are dialing an audio bridge and want to enter conference number 777 followed by #, pause for six seconds and then supply conference PIN 1234 followed by #, you would configure **777#,,,1234#** as your DTMF sequence.
Streaming	Identifies the dialed participant as a streaming or recording device. When a conference participant is flagged as a streaming/recording participant, it is treated as a receive-only participant and is not included in the video stage layout seen by other participants. See Streaming and recording a conference for more information.
Dual stream (presentation) URL	When adding a dual streaming RTMP participant, this specifies the RTMP URL for the second (presentation) stream. Leave this field blank when adding a single streaming participant.

Field	Description
Keep conference alive	Determines whether the conference will continue when all other participants have disconnected: ○ ***Yes***: the conference will continue to run until this participant has disconnected (applies to Hosts only). ○ ***If multiple***: the conference will continue to run as long as there are two or more ***If multiple*** participants and at least one of them is a Host. ○ ***No***: the conference will be terminated automatically if this is the only remaining participant. Default: ***Yes***. For streaming participants, we recommend that this option is set to ***No***. For more information, see Automatically ending a conference.

4. Select **Dial out to participant**.

A message **Initiated dial out to participant** appears at the top of the screen.

To confirm whether the participant has joined the conference you can go to **Status > Conferences**, select the conference, and then select the **Participants** tab. The new participant should appear in the list.

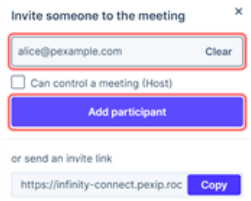
Dialing out via the Pexip app

If you have Host privileges, you can use the Pexip app to dial out to participants from the conference you are in. A call is placed to the participant and if they answer the call they will join the conference directly (they do not go through an IVR screen or have to enter a PIN).

Automatic routing is used when a Pexip app user adds a new participant to a conference. This means that the dialed alias must match an appropriate Call Routing Rule that applies to **Outgoing calls from a conference** for the call to be placed (using the protocols and call control systems etc. as configured for that rule).

To dial out to a participant using the Pexip app, from within a conference:

1. From the bottom of the Participants panel, select **Add participant**.
2. Enter the address of the participant. This must be in the format **name@domain**
3. Select whether you want the participant to have **Host** privileges.
4. Select **Add participant**.



The call is placed from the meeting to the participant and they appear in the participant list. If and when the participant answers the call they will join the meeting; if they do not answer, or do not accept the call, they will disappear from the participant list.

Sending a banner message to all participants

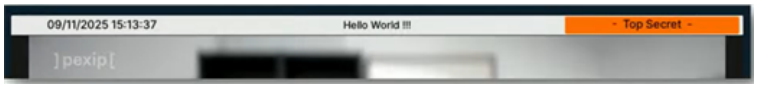
You can send a banner (overlay) message that is displayed at the top of the screen to all conference participants. It is supported in Virtual Meeting Rooms and Virtual Auditoriums, the calling party in Infinity Gateway calls, and in all layouts.

How it works

When a message is displayed, the video layout shrinks slightly to make space for the overlay banner at the top of the layout.

The message is displayed forever, until it is cleared by sending a blank message.

Note that:

- If multiple lines are specified, the banner paginates through the individual lines in the message every 5 seconds.
- The message may appear differently for each participant, depending on the receiving participant's aspect ratio and resolution. We recommend a maximum length of 100 characters per line. If there is not enough space for the entire message it is cropped and ended with "...".
- The overlay message can be combined with a timer and any classification message. For example:


- The banner message is only displayed in main video (including when presentation is part of the layout mix); it is not included in presentation streams.

Setting a message

There are several ways to set the message.

REST client API

If you are using the REST client API:

To set a message:

POST: https://<node_address>/api/client/v2/conferences/<conference_alias>/set_message_text with body: **{ "text": "Your message" }**

You may use **\n** to specify multiple lines.

To retrieve the currently set message:

GET: https://<node_address>/api/client/v2/conferences/<conference_alias>/get_message_text

which returns **{ "text": "Message that was set", "set_time" "UTC+00:00 - time" }**

PexRTC client API

If you are using the PexRTC client API:

Call the **setMessageText(message)** conference control function. You may use **\n** to specify multiple lines.

Web app plugin

It is available as a **Message Overlay** Webapp3 plugin on the Pexip developer portal.

Customizing the banner

The default colors are dark gray text on a light background, although the text, background and border colour of the banner are configurable via **message_box_config** theme settings.

You cannot customize the size or position of the banner, the position of the items within the banner, or the font used for the message text.

Displaying an in-conference timer / countdown

You can set an in-conference timer or countdown clock that is displayed at the top of the screen to all conference participants. It is supported in Virtual Meeting Rooms and Virtual Auditoriums, the calling party in Infinity Gateway calls, and in all layouts.

This is a technology preview feature in v41.

To use this feature you must ensure that **Enable clock** is selected in **Platform > Global settings > Tech preview features**.

How it works

When a timer is displayed, the video layout shrinks slightly to make space for the timer banner at the top-left of the layout.

Timer options

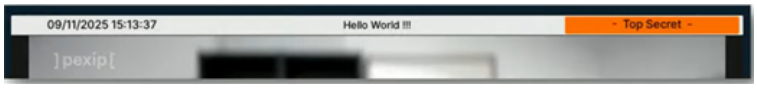
There are 3 types of timer that can be displayed:

- Elapsed time since initiating the timer
- Time remaining countdown (from a provided starting value)
- Current time (UTC in either dd/mm/yyyy or mm/dd/yyyy format)

You can also optionally specify some prefix and suffix text to display either side of the timer.

The timer message takes the format: {PREFIX} {DATE} HH:MM:SS {SUFFIX}

Note that:

- When a countdown timer reaches zero it remains displayed as 00:00:00.
- The in-conference timer can be combined with a banner message and any classification message. For example:


- The message may appear differently for each participant, depending on the receiving participant's aspect ratio and resolution. If there is not enough space for the entire message it is cropped and ended with "...".
- The banner message is only displayed in main video (including when presentation is part of the layout mix); it is not included in presentation streams.

Setting a timer

A conference Host can set the timer via the client APIs.

REST client API

If you are using the REST client API:

To set a timer:

POST: https://<node_address>/api/client/v2/conferences/<conference_alias>/set_clock

Request fields:

type	string	The clock options are: • "elapsed": the clock increments starting from 0. • "remaining": the clock decrements from **starting_value**. • "time": displays the current local time.
starting_value	number	The starting value in seconds for the countdown timer. Only applies for a **type** of ***remaining***. Range: 1-31536000
prefix	string	Optional text to display before the clock.
suffix	string	Optional text to display after the clock.
date	string	The required date format. The options are: • "dd/mm/yyyy" • "mm/dd/yyyy" Only applies for a **type** of ***time***.

Request example to set a countdown clock of 5 minutes:

```
{
    "type": "remaining",
    "starting_value": 300,
    "prefix": "Time remaining: "
}
```

To retrieve the timer configuration that was last assigned:

GET: https://<node_address>/api/client/v2/conferences/<conference_alias>/get_clock

which returns the same configuration as applied via **set_clock**.

To remove the timer:

POST: https://<node_address>/api/client/v2/conferences/<conference_alias>/set_clock without any request fields.

PexRTC client API

If you are using the PexRTC client API:

Call the **setClock(clock_values)** conference control function and specify the same fields as for the REST API.

You can use **setClock()** to remove the timer.

You can use **getClock()** to retrieve the timer configuration that was last assigned.

Customizing the timer

The default colors are dark gray text on a light background, although the text, background and border colour of the banner are configurable via **message_box_config** theme settings.

You cannot customize the size or position of the banner, the position of the items within the banner, or the font used for the message text.

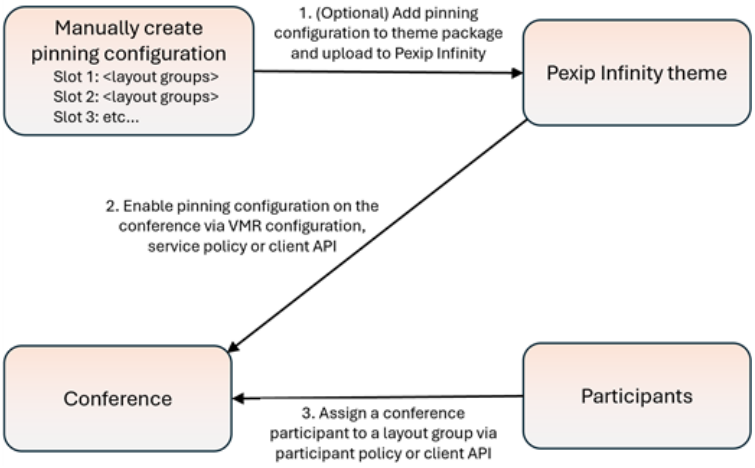
Pinning participants to the layout during a conference

Participant pinning allows you to place and lock specific participants in the most prominent positions in a layout when a conference is in progress.

This topic covers:

- How it works
- What's noteworthy?
- Adding pinning configuration to a theme
- Enabling pinning configuration on a conference
- Assigning conference participants to the layout groups during the conference

How it works



Several steps are involved in the setting up and application of participant pinning. In summary:

1. **(Optional) Add a pinning configuration file to your theme**. This defines the names of the layout groups that control which participants will appear in the primary slots in the layout, and in which order they appear. This must be performed by an administrator before the conference starts.

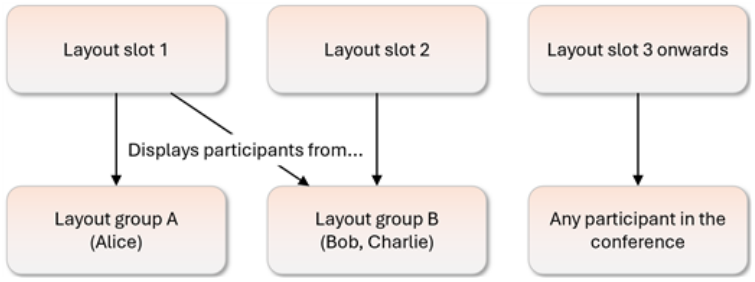
   As an alternative, instead of specifying the pinning configuration in a theme, you can dynamically specify the required pinning configuration via the client REST API while the conference is in progress.
2. **Enable the pinning configuration on a conference**. You can permanently assign a pinning configuration to a VMR, or you can dynamically assign it via service configuration policy, or set it via the client REST API while the conference is in progress.
3. **Assign conference participants to the layout groups**. This occurs while the conference is in progress and is what actually pins specific participants to the slots in the conference layout. You can use participant policy or the client REST API to assign a participant to a layout group.

During the conference the appearance of participants in the layout is then determined according to the pinning configuration, the participant assignments to layout groups, and their voice activity. Note that a pinned participant can remain visible in a slot even if they do not speak. Voice activity is only used to choose between how the participants in a layout group are ordered.

Full details of how to perform these steps is provided further below.

Assignment logic example

Let's assume the following pinning configuration (as defined in the example JSON configuration further below):



- Layout slot 1 is associated with layout group A first and then group B second.
- Layout slot 2 is associated with layout group B only.
- Subsequent layout slots do not have any pinning configuration applied.

Then the following participants are assigned as follows when they join the conference:

- Group A: Alice.
- Group B: Bob and Charlie.

This means that during the conference, the other participants would observe:

- Alice always stays in slot 1 (while she remains present in the conference, and even if she never speaks), as she is the only participant assigned to group A.
- Bob and Charlie alternate in slot 2 as each one of them speaks.
- Slot 3 and later may be occupied by any conference participant except Alice — but including Bob and Charlie even though they are assigned to a layout group — based on standard voice activity logic.
- If Alice leaves the conference, Bob and Charlie would occupy slots 1 and 2.

Alice, Bob and Charlie's own view of the conference would follow a similar logic, but is also dependent on the **remove_self** setting. For example if **remove_self** is true then Alice would always see Bob and Charlie in slots 1 and 2.

Defining and assigning a pinning configuration

You can define and assign a pinning configuration in two ways:

- Define the pinning configuration within a **theme**. That pinning configuration can then be assigned to a conference in any of the following ways:
  - Permanently assign it to a VMR before a conference starts.
  - Dynamically assign it via policy or the client REST API when a conference starts.
- Dynamically **define and assign** the pinning configuration via the client REST API. When using this method note that:
  - The configuration for a reserved slot cannot contain any references to files (typically images for backgrounds or icons).
  - A specific participant can be assigned to a slot using their layout group or alternatively via their UUID.

These options are explained in more detail in the various sections below.

What's noteworthy?

When using participant pinning, note that:

- You do not specify a specific layout as part of the pinning configuration. The participants are pinned to the primary layout slot positions using the sequence in which the conference's current layout is normally filled. See Layout examples for details about the fill order for each layout.
- It is supported in all layouts, including custom layouts, except Adaptive Composition and the Teams-like layout.
- Pinning takes priority over any spotlighting. When pinning configurations are in use, spotlighting only promotes people in the backfilled section of a configuration.
- It can be used in Virtual Meeting Rooms, Virtual Auditoriums and gateway calls.

Adding pinning configuration to a theme

You can add a pinning configuration file to your theme and then subsequently assign that pinning configuration to a VMR (static assignment) or a conference in a VMR that uses that theme (dynamic assignment).

The pinning configuration is specified as a JSON file that defines the names of the layout groups, for example "group1", "group2", that control which participants will appear in the primary slots in the layout, and in which order they appear.

The pinning configuration file must be named with a prefix of "pinning_config_" (must be lower case) and be in a .json format, for example **pinning_config_mygroups.json**.

Ensure that you include the pinning configuration JSON file in your theme package that you upload to Pexip Infinity. You can include multiple pinning configuration files in your theme package.

The JSON file has the following fields:

Field name	Mandatory	Description
name	Yes	This is the name of the pinning configuration and is what is used to refer to the configuration throughout the system, in API calls, policy and in the Administrator interface. It is a string which is limited to a maximum of 50 alphanumeric and "_" characters and no spaces.

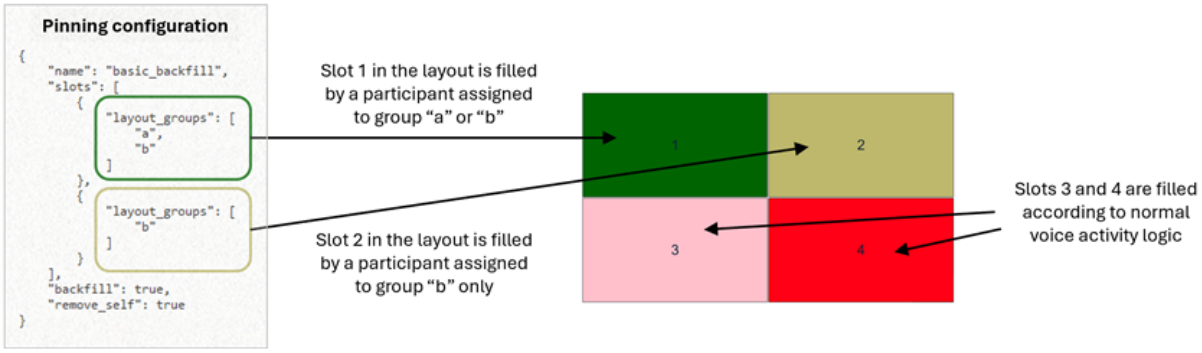
Field name	Mandatory	Description
slots	Yes	This is a list of dictionaries that specify the layout group names which subsequently control who appears in the layout and in which order they appear. A slot dictionary contains: • **layout_groups**: an ordered list of strings that are the group names of the participants you want to appear in this slot. If nobody is assigned to the first group, then a participant from the second (and so on) group is used. The names must be in lower case. When dynamically assigning a pinning configuration to a conference via the client REST API you can also assign a specific participant UUID instead. • **reserved_appearance**: optional configuration for the slot's appearance when there are no participants found for a slot. Any elements specified here override the equivalent settings in the theme's default reserved slot configuration. See Configuring a slot's appearance when there are no participants found for more information. • **show_reserved**: an optional boolean flag that can be set to ***false*** to exclude a reserved slot from the layout when there are no participants found for that slot. This setting is ***true*** by default. Note that Guests in a Virtual Auditorium are never shown in a slot when this is ***false.*** The first layout_groups list determines who may appear in slot 1, the second layout_groups list determines who may appear in slot 2, and so on. Gaps in the slot sequence are not allowed.
backfill	No	Determines whether the rest of the non-pinned slots in the layout should be filled up with other participants (using normal voice-switching rules). This can include participants who are assigned to a layout group but who are not currently appearing in a pinned slot. Default: false
remove_self	No	This controls whether "self view" is removed from the layout mix. Default: false

For example:

```
{
    "name": "basic_backfill",
    "slots": [
        {
            "layout_groups": [
                "a",
                "b"
            ]
        },
        {
            "layout_groups": [
                "b"
            ]
        }
    ],
    "backfill": true,
    "remove_self": true
}
```

The example above defines 2 pinned slots. The first slot in the video layout is filled by a participant assigned to group a, or if nobody in group a is available then somebody from group b. The second slot is filled by somebody from group b only. All of the remaining slots are filled by any participant according to normal voice activity rules.

Thus, if this configuration is used on an **Equal 2x2** layout:



Configuring a slot's appearance when there are no participants found

You can control what is displayed in each slot if there are no participants found to occupy that slot.

Default behavior

You can specify within the main theme configuration file (**themeconfig.json**) the default behavior to apply to all slots.

There is a **reserved_slot** splash screen that represents what is put into a reserved slot when it is not occupied by a participant. The default image is a push pin and it scales to fill the slot.

It uses the following graphics files which you can customize in the usual manner:

- **background_reserved_slot.jpg** (a black screen)
- **icon_reserved_slot.svg** (an image of a push pin)

You can also specify a text overlay:

- **reserved_slot_overlay_text**: an optional field that specifies the overlay text to place in a reserved slot (when the display of participant names is enabled for the layout). Use "" for no overlay text. The same text string is used for each unoccupied slot.

Customized behavior

You can override the default behavior on a per-slot basis within the pinning configuration by defining a **reserved_appearance** dictionary for a slot. This is supported in "version 2" themes only. This dictionary can contain:

- **overlay_text**: the overlay text to display (when the display of participant names is enabled for the layout).
- **config**: a splash screen configuration to display in the slot. This is configured in the same way as a normal splash screen except you cannot use **text** or **pin_entry** elements.

Any elements specified here override the equivalent settings in the theme's default reserved slot configuration. This allows you, for example, to just change the background without having to rewrite all of the other `elements` configuration.

For example, this pinning configuration sets up some different backgrounds and icons on the slots:

```
{
    "name": "custom_appearance",
    "slots": [
        {
            "layout_groups": ["a"],
            "reserved_appearance": {
                "config": {
                    "layout_type": "vertical",
                    "background": {"path": "background-empty-slot.jpg"},
                    "elements": [
                        {"type": "icon", "path": "icon-empty-slot.svg"}
                    ]
                },
                "overlay_text": "Nobody here"
            }
        },
        {"layout_groups": ["b"]},
        {
```

```
            "layout_groups": ["c"],
            "reserved_appearance": {
                "config": {
                    "layout_type": "vertical",
                    "background": {"path": "background-empty-slot.jpg"}
                }
            }
        },
        {
            "layout_groups": ["d"],
            "reserved_appearance": {
                "config": {
                    "layout_type": "vertical",
                    "background": {"path": "background-empty-slot.jpg"},
                    "elements": []
                },
                "overlay_text": ""
            }

        }
    ],
    "remove_self": true
}
```

In the example above the following slot behavior is specified:

- Slot 1: specifies a custom background image, icon and overlay text.
- Slot 2: specifies a new background only, thus the default `elements` i.e. the default pin icon, is still used.
- Slot 3: overrides `elements`, which results in no icon being displayed over a custom background.

Removing unused reserved slots from the layout

You can also set **show_reserved** in the pinning configuration to ***false*** to exclude a reserved slot from the layout when there are no participants found for that slot.

Enabling pinning configuration on a conference

You must enable a pinning configuration on a conference before you can actually pin any participants. You can permanently assign a pinning configuration to a VMR, or you can dynamically assign it via service configuration policy, or set it via the client REST API while the conference is in progress.

Permanently assigning a pinning configuration to a VMR

To permanently assign a pinning configuration to a VMR:

1. Go to the required VMR (**Services > Virtual Meeting Rooms**) or Virtual Auditorium (**Services > Virtual Auditoriums**).
2. Expand the **Advanced options** section.
3. Select the **Pinning configuration** you want to assign. The list of available options contains the names of all of the pinning configurations in the theme associated with that VMR.
4. **Save** your changes.

Using policy to dynamically assign a pinning configuration

You can dynamically assign a pinning configuration to a VMR via service configuration policy. The specified pinning configuration must be available in the theme associated with that conference.

You do this by setting the "**pinning_config**" service configuration response field to the name of the pinning configuration you want to apply.

Using the client REST API to dynamically assign a pinning configuration

You can dynamically assign a pinning configuration to a conference via the client REST API. The specified pinning configuration can either exist in the theme associated with that conference, or it can be dynamically specified within the **set_pinning_config** request.

To assign a pinning configuration, where the configuration is specified within the theme associated with the conference:

POST: https://<node_address>/api/client/v2/conferences/<conference_alias>/set_pinning_config with body: **{"pinning_config": "<name of your pinning configuration>" }**

To dynamically define and assign a pinning configuration to a conference:

POST: https://<node_address>/api/client/v2/conferences/<conference_alias>/set_pinning_config with body: **{"dynamic_pinning_config": <pinning configuration JSON dictionary> }**

where **<pinning configuration JSON dictionary>** follows the same JSON schema as when specifying the pinning configuration within a theme, with the following differences:

- The configuration for a reserved slot cannot contain any references to files (typically images for backgrounds or icons).
- A specific participant can be assigned to a slot using their layout group or alternatively via their UUID prefixed by a !, for example **"slots": [{"layout_groups": ["!abcdabcd-abcd-abcd-abcd-abcdabcdabcd"]}]**.

To clear a pinning configuration:

POST: https://<node_address>/api/client/v2/conferences/<conference_alias>/set_pinning_config with body: **{"pinning_config": "" }**

Assigning conference participants to the layout groups during the conference

You have to assign conference participants to the layout groups while the conference is in progress — this is what actually pins specific participants to the slots in the conference layout. A participant can only be assigned to one layout group at any one time.

You can use participant policy or the client REST API to assign a participant to a layout group.

Using participant policy

You do this by setting their **layout_group** participant configuration response field to the name of the layout group you want to assign.

Using the client REST API

To assign a layout group name:

POST: https://<node_address>/api/client/v2/conferences/<conference_alias>/participants/<participant_uuid>/layout_group with body: **{"layout_group": "<group name>" }**

Note that you can also assign a participant to a layout group when dynamically defining and assigning a pinning configuration via the API.

Disconnecting participants from a conference

There are several ways to disconnect a participant from a conference: Using the Administrator interface, Using Pexip apps and Using DTMF. Each of these options is described below.

Using the Administrator interface

You can use the Administrator interface to disconnect a single participant or all participants.

Disconnecting a single participant

To use the Pexip Infinity Administrator interface to disconnect a single participant from a conference:

1. Select the participant. You can do this in two ways:
   - Go to **Status > Participants** and select the participant to disconnect.
   - Go to **Status > Conferences** and select the Virtual Meeting Room or Virtual Auditorium that the participant is in. From the **Participants** tab, select the participant to disconnect.
2. At the bottom right of the screen, select **Disconnect**.

Disconnecting all participants

To use the Pexip Infinity Administrator interface to disconnect all of the participants from a conference:

1. Go to **Status > Conferences** and select the Virtual Meeting Room or Virtual Auditorium that the participants are in.
2. At the bottom right of the screen, select **Disconnect all**.

3. Select **Yes, I'm sure** to confirm.

All participants will be disconnected and the conference will end.

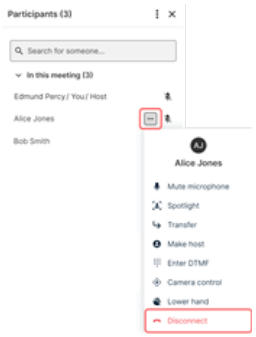
Using Pexip apps

You must be a Host participant in the conference to use this feature.

Disconnecting a single participant

To use a Pexip app to disconnect a participant:

From the **Participants** panel, hover over the participant's name, select and then **Disconnect**. You are asked to confirm that you want to disconnect that participant.

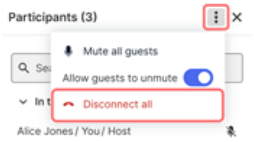


The participant who is disconnected sees a message saying that another participant disconnected them.

Disconnecting all participants

To use a Pexip app to disconnect all participants, including yourself:

From the top of the **Participants** panel select and then **Disconnect all**.



Using DTMF

If DTMF controls are enabled, Host participants can terminate the conference by disconnecting all participants, including themselves. The default DTMF entry to do this is ## but this may have been customized. For more information, see Using a DTMF keypad to control a conference.

Transferring a participant to another conference

You can transfer a participant from one conference to another conference (or to another device or system via the Infinity Gateway). There are several ways to do this: Using the Administrator interface, Using Pexip apps, or Using the APIs.

After the transfer has been successfully initiated, the transferred participant is removed from the original conference. Should the transfer then fail (for example, if it is made to a device that rejects the call), the transferred participant is disconnected i.e. it will not rejoin the original conference.

You can transfer a participant into a VMR that requires participant authentication. For full information, see Transferring participants.

Using the Administrator interface

To use the Pexip Infinity Administrator interface to transfer a single participant from one conference to another conference:

1. Select the participant. You can do this in two ways:
   - Go to **Status > Participants** and select the participant to transfer.
   - Go to **Status > Conferences** and select the Virtual Meeting Room or Virtual Auditorium that the participant is currently in. From the **Participants** tab, select the participant to transfer.
2. At the bottom right of the screen, select **Transfer**.

   The **Transfer Participant** dialog opens.
3. Enter the **Target conference alias**.

   This can be any alias associated with a Virtual Meeting Room, Virtual Auditorium or Call Routing Rule.
4. Select the **Role** the participant will have in the new conference.
5. Select **Transfer**.

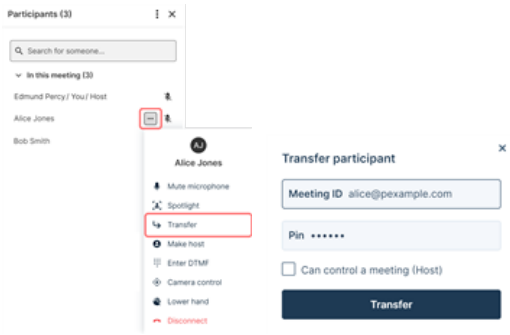
   If the transfer is successful, the participant list is updated to show the participant in the new conference (it may also temporarily show the participant still connected to the previous conference).

Using Pexip apps

To use the Pexip app to transfer a participant from one conference to another, you must have joined as a Host.

From the **Participants** panel, hover over the participant's name, select ··· and then select **Transfer**.

Enter the alias of the conference you wish to transfer the participant to, its PIN (if applicable) and whether the participant should join as a Host, and then select **Transfer**.



You can transfer any participant, except yourself.

Using the APIs

Third-party applications can use the management API and the client APIs to transfer participants to another conference.

For more information, see Management command API and Using the Pexip client APIs.

Using a DTMF keypad to control a conference

Host participants using telephones or SIP/H.323 endpoints that support DTMF can control aspects of a conference by using their keypad.

Default DTMF controls

The default DTMF controls that can be used within a conference (Virtual Meeting Room or Virtual Auditorium) are:

DTMF digits	Control
*1	Request (cycle through) a personal layout †
*3	Toggle raise hand (see Conference indicators)
*4	Toggle presentation in the layout mix †

DTMF digits	Control
*5	Toggle mute and unmute all Guests
*6	Toggle self mute † (see Video watermarking for information about how to enable mute status indicators via themes)
*7	Toggle conference lock and unlock
*8	Cycle through the set of available layouts (this applies to all participants in the conference)
*9	Toggle multiscreen participant display †
##	Terminate the conference (disconnect all participants including yourself)

† This only applies to the endpoint sending the command, and not to all participants in the conference.

Note that with calls made via the Infinity Gateway, any DTMF signals are forwarded to the other party. The only exceptions to this are interop calls to Microsoft Teams and Google Meet where DTMF controls can be used to control aspects of the layout and toggle some participant controls such as self mute and raised hand.

Changing DTMF controls

The DTMF digits for each control can be changed on a per-theme basis by editing the themeconfig.json file.

Each control must be two DTMF digits long, and all the commands must be different. When deciding the digits to use for each control, you should ensure there is no risk of overlap in situations where one of the digits is not successfully received.

For example, if you used:

- **57** to lock the conference
- **75** to mute all Guests
- **77** to end the conference

there is a risk that a user could enter **5775** to lock and mute the conference, but due to packet loss the initial digit was not received. In this case the string that is received would be **775**, which would be interpreted as **77** and thus a command to terminate the conference.

Disabling DTMF controls

DTMF controls are enabled by default. They can be disabled and enabled on a per-theme basis by editing the themeconfig.json file.

Customizing with themes

Customizing conference images and voice prompts using themes

Themes allow you to change the voice prompts and images provided to participants when they use a Pexip Infinity service (Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service) or use the Infinity Gateway to make a person-to-person call or to join an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet. You might change the theme if, for example, you want to use your company's own logo, color scheme or terminology on the screens displayed to conference participants, or you want to change the language used in the voice prompts.

You can also define your own custom conference layouts within a theme and use those layouts in the same way as the standard layouts that are shipped by default in Pexip Infinity.

You have full flexibility when applying themes to services. You apply specific themes to individual services and Call Routing Rules, and you can nominate a global default theme to use for any services that have not had a specific theme applied to them.

Each conference theme is made up of a set of audio and image files and configuration settings. You can replace any or all of these files with your own customized sounds, images or settings in order to create a new theme and apply a theme to a specific service.

Pexip Infinity ships with its own Base theme, which cannot be edited. Files and configuration settings from the Base theme are used when no other theme has been selected, or when the selected theme does not contain a specific file or setting. Pexip Infinity also ships with a number of preconfigured themes — these include alternative themes that contain entry and exit tones and localized audio files — making it easier for you to select and create themes suitable for your deployment.

Pexip branding portal

We recommend using the Pexip branding portal (https://webapp2-branding.pexip.io/) to customize your themes. The portal provides an easy way to change the text, images, splash screen layouts and some of the settings that are used in themes, and it generates a ZIP file that you can upload via the Administrator interface. Note that you cannot change the audio files via the branding portal.

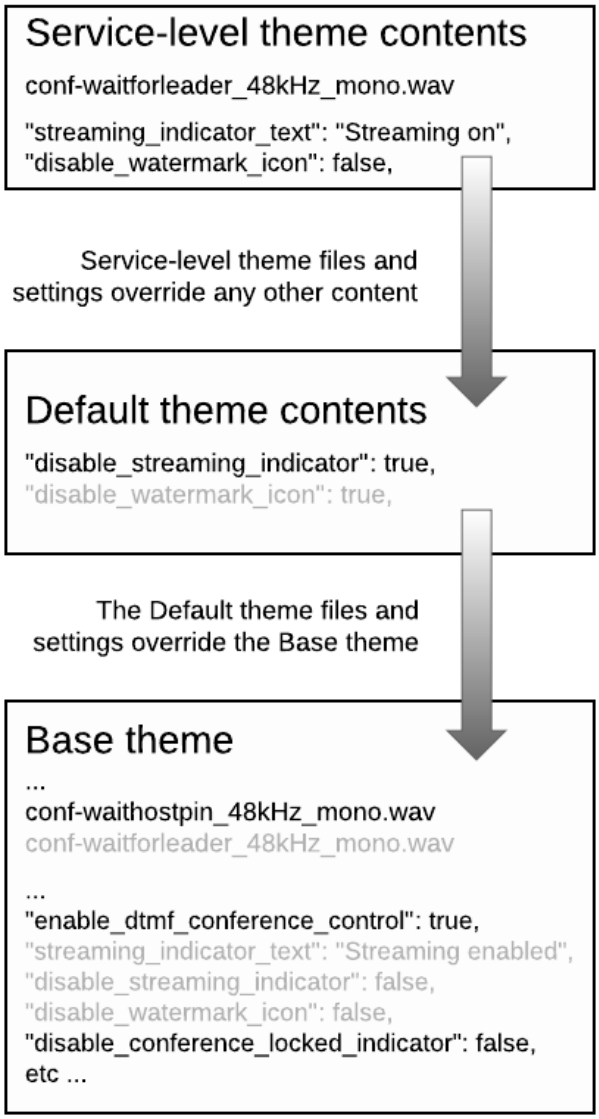
You can also manually configure your own themes as described in this documentation if you have more advanced customization requirements, or want to use your own audio files.

If you use the Pexip apps to access your services, you can also use the branding portal to customize the look and feel of those clients.

How do I know which files and configuration settings will be used in a particular VMR?

When you create a new customized theme, you don't have to upload a complete set of files and configuration settings — you can just upload the files and settings you want to use specifically for that theme, to override the default behavior. When a user accesses a service (such as a VMR) that has that theme applied, Pexip Infinity presents them with the sounds and images from the files included in that theme. If the customized theme does not include a specific file or setting, or the service does not have a theme applied, then Pexip Infinity uses the relevant file or setting from the theme that has been selected as the default theme. If the default theme does not include the specific file or setting either, then the file or setting from the Base theme is used.

For example, this diagram (right) shows how some files and settings in the Base theme (the **conf-waitforleader_48kHz_mono.wav** audio file, and the **streaming_indicator_text**, **disable_streaming_indicator** and **disable_watermark_icon** settings in the **themeconfig.json** file, all shown faded out) are overridden by the equivalent files or settings in the default theme (the **disable_streaming_indicator** setting in this example) and any service-level theme (the **conf-waitforleader_48kHz_mono.wav** file, and the **streaming_indicator_text** and **disable_watermark_icon** settings).



Note that, in this example, even though the default theme includes the **disable_watermark_icon** setting, that setting is itself overridden by the service level theme; however, if a service did not have that service-level theme applied, then the default theme setting of **disable_watermark_icon: true** would apply to that service.

If a theme is only assigned to Call Routing Rules, you only need to customize the relevant subset of the image files (see Themes used by Call Routing Rules (gateway calls) for more information).

If you use the Pexip branding portal to create your theme it will automatically include just the relevant settings and files required to override the default behavior.

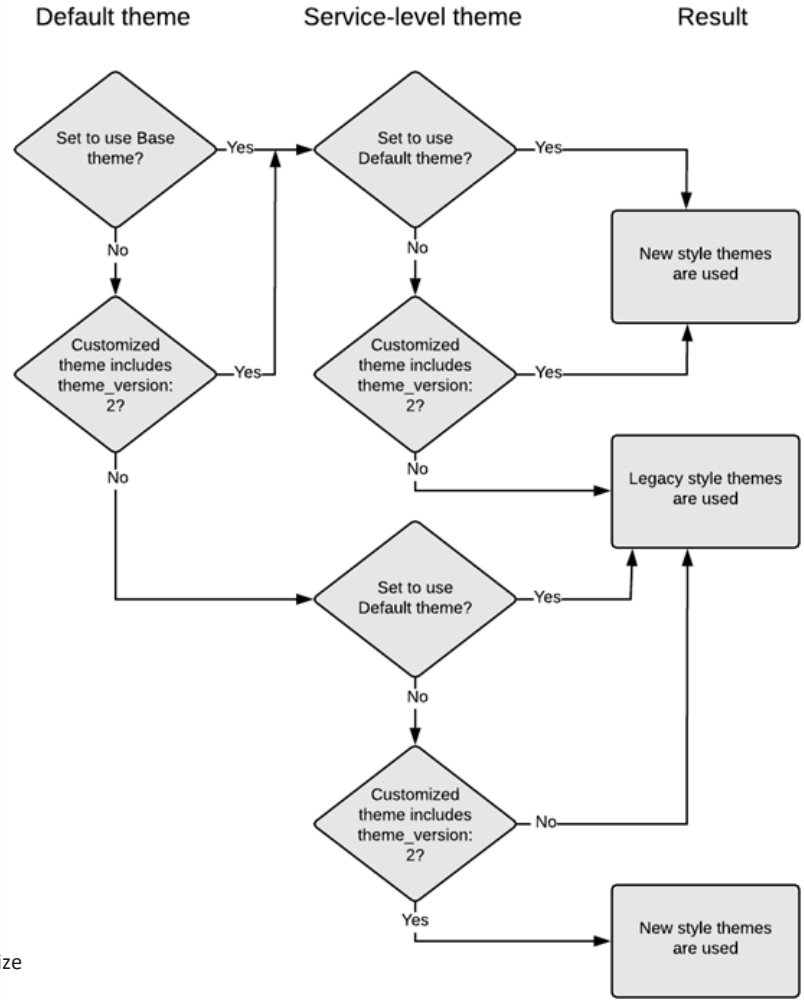
New and legacy style themes

Version 18 of Pexip Infinity introduced a new way to specify the content and layout of your own customized themes that is more efficient than the previous style themes, and offers more flexibility when customizing them for your own requirements.

Legacy style themes (identified as "version 1" themes) and new style themes (identified as "version 2" themes) can both be used on the same platform, with legacy style themes applied to some services and new style themes applied to other services if required. The way in which themes are uploaded to Pexip Infinity and applied to services is the same for both new and legacy style themes. The new style base theme is used by default if you have not applied any customized themes to your services.

Both types of themes behave in exactly the same manner as each other, and look almost the same by default (some of the PIN-entry splash screen graphics used when joining a conference are different). The audio prompts are the same in new and legacy style themes.

The main difference between how legacy and new style themes work and are configured/customized, is that legacy style themes use a JPG file containing a combined background image, graphics and text for each of the splash screens used when joining a conference (such as the Welcome screen and PIN entry screens). The new style themes separate out these elements, allowing you to specify the individual background image, graphics and text elements that are used on each screen, and control where each of those elements is positioned (via new configuration options in the theme's **themeconfig.json** file). See the section on splash screens for more information.



Controlling whether new or legacy theme files are used

A configuration switch in the theme's **themeconfig.json** file is used to control whether the theme is a new or legacy style — if it contains `"theme_version": 2` then new style themes are used. This switch has to be present in a **themeconfig.json** file at the lowest level of the theme file hierarchy where a theme has been applied i.e. if you have applied a theme to a specific service then, for new style themes to be used, that theme must contain a **themeconfig.json** file that includes the `"theme_version": 2` switch.

Therefore, even if you only want to customize a single image or sound file, the pinning configuration, or the vendordata file, you must include in your ZIP upload a **themeconfig.json** file that contains a `"theme_version": 2` switch, if you want to use new-style themes.

The flow chart shows how the inclusion or not of `"theme_version": 2` controls whether a new or legacy-style theme is displayed.

The Pexip branding portal can be used to customize new and legacy-style themes, and any themes created via the branding portal automatically include the relevant new/legacy switches in the theme's **themeconfig.json** file when required.

Creating and applying themes to conferences

This section describes how to create and edit your own themes, change the default theme, and how to apply a theme to a specific service or see which themes you are currently using.

Legacy style themes (identified as "version 1" themes) and new style themes (identified as "version 2" themes) can both be used on the same platform, with legacy style themes applied to some services and new style themes applied to other services if required. The way in which themes are uploaded to Pexip Infinity and applied to services is the same for both new and legacy style themes. The new style base theme is used by default if you have not applied any customized themes to your services. See New and legacy style themes for more information.

Creating a new theme

You can create a new theme by uploading a ZIP file containing the files or configuration settings you want to use instead of the default theme. The ZIP file does not need to contain the complete set of files or settings, just the files or settings that you want to use in place

of, or as an extension to, those from the default theme.

We recommend using the Pexip branding portal (https://webapp2-branding.pexip.io/) to customize your themes. The portal provides an easy way to change the text, images, splash screen layouts and some of the settings that are used in themes, and it generates a ZIP file that you can upload via the Administrator interface. Note that you cannot change the audio files via the branding portal.

To create and upload a new theme:

1. Obtain your theme ZIP file:
   - If you have used the Pexip branding portal to customize your theme:
     i. Build and download the theme (Splash Screens) from the branding portal.
     This creates an **ivr.zip** file.
   - If you are manually creating your own customized theme:
     i. Ensure that any new sound and image files and the **themeconfig.json** file meet the specified file requirements for themes.
     ii. Save all of the files together in a new folder. You can name the folder whatever you like, but within the folder, each file must be saved with the same file name and extension as the default file being replaced. If you are adding extra background files or SVG icon files then you can use any suitable name for those additional files.
     iii. Create a ZIP file of the new folder.
2. Go to **Services > Themes** and select **Add theme**.
3. In the **Name** field, enter the name of the new theme.
4. In the **Theme** field, select **Choose file** and go to the ZIP file you have just downloaded/created.
5. In the **Services** and **Call Routing Rule** sections, select the services (Virtual Meeting Rooms, Virtual Receptions, Call Routing Rules and so on) that will use this new theme.
   Note that if you have more than 10,000 services configured, you cannot select the associated services via this page. Instead you must use another method to assign the theme to a service, such as editing the individual service, using the configuration API, or by using the bulk VMR provisioning facility.
6. Select **Save**.
   If any errors are reported when trying to upload your own manually-created ZIP file, such as "Invalid file in theme: themeconfig.json", you must fix the errors and then try to upload the ZIP file again. Any errors in the themeconfig.json are typically due to invalid JSON structures, such as missing commas between any key value pairs, or having a comma after the last key value pair in any object. Graphics file related errors are typically caused by incorrect image resolutions or using an invalid filename.
7. Wait for the new theme to be replicated out to all Conferencing Nodes (typically after approximately one minute).

You can now test the theme by dialing one of the services to which it has been applied.

Setting the default theme

You can specify a theme to be used by default for all services that have not had a specific theme applied. The default theme does not need to contain a complete set of files or configuration settings, just those that you want to use in place of files from the Base theme. It can be either a legacy or new style theme. If you use the Pexip branding portal to create your theme, it will automatically include just the relevant settings and files.

If you do not select a default theme, the entire Base theme is used as the default. To change the default theme:

1. Go to **Platform > Global Settings > Service Configuration**.
2. From the **Default theme** drop-down, select the theme to use for services where no specific theme has been selected. If you have not already uploaded the theme you want to use, you can do so from here by clicking on the + icon and following the instructions in creating a theme.
3. Select **Save**.

Editing an existing theme

You can edit an existing theme by uploading a new ZIP file containing the new set of files. The ZIP file does not need to contain the complete set of files or settings, just the files or settings that you want to use in place of, or as an extension to, those from the default theme.

Note that when you upload a ZIP file for an existing theme, all of the previously uploaded files for that theme are deleted and replaced with files from the new ZIP file. This means that if you have an existing theme containing **File A** and you want to add **File B** to the

theme, you need to upload a ZIP file containing both **File A** and **File B**. If you use the Pexip branding portal to edit and then rebuild your theme it will automatically include all of the relevant settings and files.

To modify an existing theme:

1. If you originally used the Pexip branding portal to create your theme:
   a. Go to the branding portal https://webapp2-branding.pexip.io/.
   b. Select and then edit your existing theme as required.
   c. Build and download the updated theme from the branding portal as a new ZIP file.
2. If you are manually customizing your theme:
   a. If you want to retain files from the existing theme, first download the current theme.
   b. Make your required changes to the theme files, ensuring that any new sound and image files and the **themeconfig.json** file meet the specified file requirements for themes.
   c. Save all of the new and modified files, along with any other existing files, together in a new folder. You can name the folder whatever you like, but within the folder, each file must be saved with the same file name and extension as the default file being replaced.
   d. Create a ZIP file of the new folder.
3. Go to **Services > Themes** and select the theme you want to update.
4. In the **Theme** field, select **Choose file** and go to the ZIP file you have just downloaded/created.
5. Select **Save**.

   If any errors are reported when trying to upload your own manually-created ZIP file, such as "Invalid file in theme: themeconfig.json", you must fix the errors and then try to upload the ZIP file again. Any errors in the themeconfig.json are typically due to invalid JSON structures, such as missing commas between any key value pairs, or having a comma after the last key value pair in any object. Graphics file related errors are typically caused by incorrect image resolutions or using an invalid filename.
6. Wait for the updated theme to be replicated out to all Conferencing Nodes (typically after approximately one minute).

You can now test the theme by dialing one of the services to which it has been applied.

Downloading an existing theme

You can download a theme to view the files it contains, or to manually edit it. To download a theme:

1. Go to **Services > Themes**.
2. Either:
   - Select **Download Base theme** to download a copy of the default set of files that are shipped with Pexip Infinity.
   - Select **Download Legacy Base theme** to download the base files that applied to legacy-style themes.
   - Select an existing theme and from the **Change Theme** page select **Download theme** to download a copy of the files that make up that customized theme.

A file with a **.ZIP** extension is downloaded.

Changing which themes are associated with which services

There are many ways in which you can change the theme that is associated with a service or Call Routing Rule. You can do this in several ways:

- Go to the **Themes** page (**Services > Themes**) and select the theme you want to assign.
  - You can then select the services/rules you want to apply it to.
  - You can also view the names of any custom layouts or pinning configurations that are available in the theme.
  - Note that you cannot use this method if you have more than 10,000 services configured.
- Edit the individual service or Call Routing Rule: select the service/rule then select the new theme from the **Theme** drop-down menu.
- Use the configuration API.
- Use the bulk VMR provisioning facility.

Viewing the themes currently in use

To find out which themes are currently in use in your deployment, go to **Services > Themes**. The **Service count** column shows you how many services each theme is associated with.

Themes are also shown on the overview page for each type of service (go to **Services** and then select the service). Here you can sort the list by **Theme**.

Base theme and other preconfigured themes

Pexip Infinity ships with its own Base theme. Files and configuration settings from the Base theme are used when no other theme has been selected, or the selected theme does not contain the required file or setting.

You cannot change the contents of the Base theme, but you can create your own themes by uploading customized versions of one or more of the Base files.

When a user accesses a service that has a customized theme applied, Pexip Infinity presents them with the sounds and images from the files included in that theme. If the customized theme does not include a specific file, or the service does not have a theme applied, then Pexip Infinity uses the relevant file from your nominated default theme. If the default theme does not include the specific file either, then the file from the Base theme is used.

The same rules apply when modifying the configuration settings in the **themeconfig.json** file. If any values **within** this file are not specified, Pexip Infinity will use the values from the Base theme's **themeconfig.json** file.

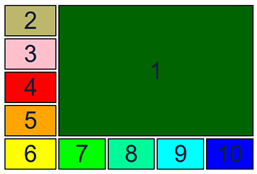
In addition to the files and settings included within the Base theme, there are some advanced settings (such as controlling where screen text is positioned) that are used by default but are not contained with the Base theme's **themeconfig.json** file, but they can be overridden by adding them to your own customized **themeconfig.json** file. See Splash screen elements (to control the size and position of text/graphics) for full information about what can be configured, and Default element configuration for each screen to see the default positions, offsets and sizes used by each splash screen.

Pexip Infinity also ships with a number of preconfigured themes which are similar to the Base theme, but contain alternative audio files with different conference entry and exit tones, and alternative references to the "#" key (as the "**hash key**" or as the "**pound key**").

Custom layouts

The Base theme includes three example custom layouts:

File name	Description
custom_layout_one_main_twelve_around.json	Name: one_main_twelve_around This layout is listed in the Administrator interface as "Large main speaker and up to 12 other participants (Custom Layout)"
custom_layout_two_mains_eight_around.json	Name: two_mains_eight_around This layout is listed in the Administrator interface as "Two main speakers and up to 8 other participants (Custom Layout)"

File name	Description
custom_layout_one_main_nine_around.json	Name: one_main_nine_around  This layout is listed in the Administrator interface as "Large main speaker and up to 9 other participants (Custom Layout)"

Note that:

- These default custom layout files are not included in the Base theme that you can download from the Management Node.
- When you define your own custom layouts the layout name shown in the Administrator interface is the **name** field from the JSON file (which is limited to alphanumeric and "_" characters and no spaces).

Preconfigured themes

Pexip Infinity ships with a number of preconfigured themes. These themes can be copied and edited, making it easier for you to select and create themes suitable for your deployment.

The preconfigured themes are identical to the Base theme with the following exceptions:

- some include entry and exit tones (the Base theme contains "empty" tones files)
- some refer to the "**#**" key as the "**hash key**" (the Base theme refers to this as the "**pound key**").

The files that may differ from those in the Base theme are:

- **Entry tone:** conf-participant_entry_tone_48kHz_mono.wav
- **Exit tone:** conf-participant_exit_tone_48kHz_mono.wav
- **"#" key reference:** 2sd-number-pound-key_48kHz_mono.wav, conf-getpin_pound-key_48kHz_mono.wav, conf-waithostpin_pound-key_48kHz_mono.wav

The content of the above files for each of the preconfigured themes is as follows:

Theme	Entry tone	Exit tone	"#" key references
Pexip theme (English_US) *	<empty file>	<empty file>	"...the pound key"
Pexip theme (English_UK)	<empty file>	<empty file>	"...the hash key"
Pexip theme (English_US) with entry tones	A high tone followed by a low tone	A low tone followed by a high tone	"...the pound key"
Pexip theme (English_UK) with entry tones	A high tone followed by a low tone	A low tone followed by a high tone	"...the hash key"

* This theme is identical to the Base theme, although unlike the Base theme it can be edited.

Rules and requirements for customized themes

We recommend using the Pexip branding portal (https://webapp2-branding.pexip.io/) to customize your themes. The portal provides an easy way to change the text, images, splash screen layouts and some of the settings that are used in themes, and it generates a ZIP file that you can upload via the Administrator interface. Note that you cannot change the audio files via the branding portal.

If you want to manually customize your own themes, or have more advanced customization requirements, the following sections list the complete set of audio, configuration, endpoint data, image and custom layout files that can be used to configure each theme. They include the requirements for each of those files, including how to manage video watermarking, content classifications, and information on which files are relevant to themes used by Call Routing Rules.

Theme ZIP file

When creating or editing a theme, you upload a single ZIP file containing your own customized version of one or more of the files described here. The ZIP file does not need to contain the complete set of files or settings, just the files or settings that you want to use in place of, or as an extension to, those from the default Base theme.

If, for example, you are replacing one of the default background JPGs, icon SVG files or audio files, then you should ensure that you use the same file name as the file you are replacing. If you are using additional background or icon files that you are applying to a subset of screens, then you must also include those new files in your ZIP file.

If `"theme_version": 2` is not present in a **themeconfig.json** file at the lowest level of the theme file hierarchy then it is assumed to be a legacy-style theme (see How do I know which files and configuration settings will be used in a particular VMR? for more information).

Therefore, even if you only want to customize a single image or sound file, the pinning configuration, or the vendordata file, you must include in your ZIP upload a **themeconfig.json** file that contains a `"theme_version": 2` switch, if you want to use new-style themes.

If any errors are reported when trying to upload your own manually-created ZIP file, such as "Invalid file in theme: themeconfig.json", you must fix the errors and then try to upload the ZIP file again. Any errors in the themeconfig.json are typically due to invalid JSON structures, such as missing commas between any key value pairs, or having a comma after the last key value pair in any object. Graphics file related errors are typically caused by incorrect image resolutions or using an invalid filename.

Audio (sound) files

You can replace any of the audio files by using your own version of the file in your customized theme. You must use the same file name as the file you are replacing. All audio files must be:

- .WAV format
- RIFF (little-endian) data, WAVE audio, Microsoft PCM, 16 bit, mono 48000 Hz

Points to note:

- **mono** and **48000Hz** are essential - audio files that do not meet these requirements will fail to upload.
- The volume level of the audio recording is important - use the default Pexip Infinity prompts as a guide.
- Some endpoints may take a few seconds after a call connects before they are able to receive audio. For this reason, we have included a 2-second pause at the start of any audio files that may be played when a user first connects to Pexip Infinity. We recommend that you include a similar pause; use the default Pexip Infinity files as a guide.

If you have used the Pexip branding portal to generate your theme ZIP file and you also want to customize the audio prompts (you cannot currently customize the audio files via the branding portal), then you must add your customized audio (wav) files to the ZIP file generated by the branding portal.

The following table lists the default audio files and their content, which is contained within the Base theme:

Default audio files

These are the default audio files in the Base theme:

File name	Content in Base theme
2sd-invalid-number-three-times-disconnect_ 48kHz_mono.wav	"You have entered an invalid number three times. I will now disconnect the call."
2sd-not-entered-valid-number-disconnect-call_ 48kHz_mono.wav	"You have not entered a valid number. I will now disconnect the call."
2sd-number-not-valid-try-again_48kHz_mono.wav	"That number is not valid. Please try again."
2sd-number-pound-key_48kHz_mono.wav	"Please enter the number you wish to connect to, followed by the pound key." †
2sd-please-hold-connect-you_48kHz_mono.wav	"Please hold while I try to connect you."
conf-ai_started_48kHz_mono.wav ‡‡	"This call is being processed by AI."

File name	Content in Base theme
conf-ai_stopped_48kHz_mono.wav ‡‡	"AI has stopped processing this call."
conf-call-will-be-disconnected_48kHz_mono.wav	"Your call will be disconnected."
conf-classification-changed_48kHz_mono.wav*	<empty file> (For more information, see Content classifications.)
conf-capacity_exceeded_48kHz_mono.wav	"The conferencing system capacity has been exceeded."
conf-external-media-processing-start_48kHz_mono.wav ‡‡	"An external application is collecting media."
conf-external-media-processing-stopped_48kHz_mono.wav ‡‡	"Media is no longer being collected by an external application."
conf-getpin_48kHz_mono.wav	"Please enter the conference PIN number."
conf-getpin_pound-key_48kHz_mono.wav	"Please enter the conference PIN number, followed by the pound key." †
conf-insufficient_licenses_48kHz_mono.wav	"There are insufficient conferencing system licenses available."
conf-invalid_license_48kHz_mono.wav	"The conferencing system license is invalid."
conf-invalidpin_48kHz_mono.wav	"The PIN is invalid for this conference."
conf-leaderhasleft_48kHz_mono.wav	"The Host has left the conference. The conference is about to end."
conf-live_captions_started_48kHz_mono.wav	"Live captions are enabled. Your continued participation in this call will be considered as consent."
conf-live_captions_stopped_48kHz_mono.wav	"Live captions are not in use."
conf-participant_entry_tone_48kHz_mono.wav	<empty file> (For more information, see Playing notification tones when participants join or leave a conference.)
conf-participant_exit_tone_48kHz_mono.wav	<empty file>
conf-participant_is_in_lobby_48kHz_mono.wav ‡	"Welcome to the lobby. Please wait and your meeting host will admit you soon."
conf-participant_locked_out_48Khz_mono.wav	Three knocks. (For more information, see Locking a conference and allowing participants to join a locked conference.)
conf-placeintoconf_48kHz_mono.wav	"Welcome to the conference."
conf-public_streaming_started_48kHz_mono.wav ‡‡	"This call is being streamed publicly."
conf-public_streaming_stopped_48kHz_mono.wav ‡‡	"Public streaming of this call has stopped."
conf-recording_started_48kHz_mono.wav ††	"This call is being recorded."
conf-recording_stopped_48kHz_mono.wav ††	"Recording of this call has stopped."
conf-smart_notes_started_48kHz_mono.wav *	"This call is now taking notes."
conf-smart_notes_stopped_48kHz_mono.wav *	"This call is no longer taking notes."

File name	Content in Base theme
conf-streaming_started_48kHz_mono.wav ‡‡	"This call is being streamed."
conf-streaming_stopped_48kHz_mono.wav ‡‡	"Streaming of this call has stopped."
conf-test_call_48kHz_mono.wav	"Let's test your video and audio. Count out loud from one to three, now." (For more information, see Configuring the Test Call Service.)
conf-test_call_audio_only_48kHz_mono.wav	"Let's test your audio settings. Count out loud from one to three, now."
conf-test_call_disconnect_48kHz_mono.wav	"If you have technical issues, check your settings or contact your administrator."
conf-transcribing_started_48kHz_mono.wav ††	"This call is being transcribed."
conf-transcribing_stopped_48kHz_mono.wav ††	"Transcribing of this call has stopped."
conf-waitforleader_48kHz_mono.wav	"Waiting for the conference Host to join."
conf-waithostpin_48kHz_mono.wav	"Waiting for the conference Host to join. If you are the conference Host, please enter the conference PIN number now."
conf-waithostpin_pound-key_48kHz_mono.wav	"Waiting for the conference Host to join. If you are the conference Host, please enter the conference PIN number followed by the pound key." †
conf-you_are_the_only_participant_48kHz_mono.wav	"You are the only participant in the conference."

* This is new in version 41.

† Alternative recordings using the term "hash key" are also available — see Preconfigured themes.

†† These audio files are only played to callers who are gatewayed via Pexip Infinity into a Google Meet or Microsoft Teams conference.

‡ These audio files are only played to callers who are gatewayed via Pexip Infinity into a Microsoft Teams conference.

‡‡ These audio files are only played to callers who are gatewayed via Pexip Infinity into a Google Meet conference.

Theme configuration file (themeconfig.json)

The **themeconfig.json** file is the theme's primary configuration file. It is a JSON dictionary that specifies the background and layout of the splash screens used when joining a conference (such as the Welcome screen and PIN entry screens), specifies which image files are used, and controls some of the features and overlays that are used within a conference. It does not control any of the audio (sound) elements of the theme.

Whenever you include a **themeconfig.json** file in your customized theme, it must contain: `"theme_version": 2`. If it does not contain this setting, then it is assumed to be a legacy-style theme.

If any settings **within** your customized **themeconfig.json** file are not specified, Pexip Infinity will simply use the equivalent setting from the Base theme or use default behavior. For example, if the only thing you want to customize from the default settings is to disable watermarks, then your **themeconfig.json** file only needs to contain the **disable_watermark_icon** setting (in addition to `"theme_version": 2`). See How do I know which files and configuration settings will be used in a particular VMR? for more information.

Here is the default (Base theme) contents of **themeconfig.json**:

```
{
    "enable_dtmf_conference_control": true,
    "dtmf_conference_control_commands": {
        "*6": "toggle_self_mute",
        "*7": "toggle_lock",
        "*5": "toggle_guest_mute",
        "*4": "toggle_pres_in_mix",
        "*9": "toggle_multiscreen",
        "*8": "cycle_layout",
        "*3": "toggle_raise_hand",
        "*1": "cycle_personal_layout",
        "##": "end_conference"
    },
    "dtmf_allowed_layouts": ["1:7", "ac", "1:21", "2:21", "2x2", "3x3", "4x4", "5x5", "1:0", "1:33"],
    "enable_avatars_in_mix": false,
    "disable_ai_indicator": false,
    "disable_external_media_proc_indicator": false,
```

```
    "disable_streaming_indicator": false,
    "disable_recording_indicator": false,
    "disable_transcribing_indicator": false,
    "disable_smart_notes_indicator": false,
    "disable_watermark_icon": false,
    "disable_watermark_mute_icon": true,
    "disable_conference_locked_indicator": false,
    "ai_indicator_text": "AI",
    "external_media_proc_indicator_text": "External media processing",
    "streaming_indicator_text": "Streaming enabled",
    "public_streaming_indicator_text": "Public streaming enabled",
    "recording_indicator_text": "Recording",
    "transcribing_indicator_text": "Transcribing",
    "smart_notes_indicator_text": "Smart notes enabled",
    "live_captions_on_indicator_text": "Live captions on",
    "live_captions_off_indicator_text": "Live captions off",
    "conference_locked_indicator_text": "Conference locked",
    "conference_unlocked_indicator_text": "Conference unlocked",
    "conference_locked_indicator_n_waiting_text": "{number_of_waiting_participants} waiting for host",
    "test_call_service_media_delay": 2,
    "test_call_service_disconnect_timeout": 10,
    "theme_version": 2
}
```


The default settings within the Base **themeconfig.json** file are described below. Note that the Base file does not contain any text overlay or splash screen layout controls, but you can add your own controls to your customized file — as described in this topic — to override the default behavior.

The **themeconfig.json** file must be UTF-8 encoded, therefore when editing the file you should use a UTF-8 capable editor. We recommend that you do **not** use Notepad on Windows computers.

Description and default values of themeconfig.json settings

All of the colors in these controls are specified using RGB hexadecimal notation (in the format 0xnnnnnn).

Name	Description	Value in Base theme
enable_dtmf_conference_control	Determines whether Host participants can use DTMF to control the conference (**true**) or not (**false**).	`true`

Name	Description	Value in Base theme

dtmf_conference_control_commands	If DTMF controls are enabled, this section specifies the DTMF digits used for each control. Current controls are:
	• **toggle_self_mute (*6)**: toggles the mute status of the participant
	• **toggle_lock (*7)**: toggles the locked/unlocked state of the conference
	• **toggle_guest_mute (*5)**: toggles the muted/unmuted status of all Guest participants
	• **toggle_pres_in_mix (*4)**: toggles whether the presentation stream is sent to that endpoint in the main video mix (replacing some of the other video participants), or as a separate stream
	• **toggle_multiscreen (*9)**: toggles multiscreen participant display
	• **cycle_layout (*8)**: cycles the conference layout through the set of available layouts as defined in **dtmf_allowed_layouts**
	• **toggle_raise_hand (*3)**: toggles the raised hand status of the participant
	• **cycle_personal_layout (*1)**: cycles the personal layout (only received by that requesting endpoint) through the set of available layouts as defined in **dtmf_allowed_layouts**
	• **end_conference (##)**: terminates the conference
	When changing the DTMF controls, you must only edit the keypad digits inside the first pair of quotes on each line (i.e. ***6**, ***7**, **##** etc). Editing anything else will disable DTMF commands.
	If you change the ***6** control for self-mute, you should also change the associated **watermark_mute_icon.png** graphic to reference the revised command.
	Note that another DTMF control command, **toggle_teams_large_gallery**, could be made available for endpoints that are in an interop call to a Microsoft Teams conference. This command toggles between the current layout and Microsoft's Large Gallery view. However, it is not included in the base theme and must be manually added to your custom theme and assigned some appropriate DTMF digits (such as *2).
	Each control must be two DTMF digits long, and all the commands must be different.
	Note that with calls made via the Infinity Gateway, any DTMF signals are forwarded to the other party. The only exceptions to this are interop calls to Microsoft Teams and Google Meet where DTMF controls can be used to control aspects of the layout and toggle some participant controls such as self mute and raised hand.
	See Using a DTMF keypad to control a conference for guidance on best practice.
		`"*6": "toggle_self_mute"`
		`"*7": "toggle_lock"`
		`"*5": "toggle_guest_mute"`
		`"*4": "toggle_pres_in_mix"`
		`"*9": "toggle_multiscreen"`
		`"*8": "cycle_layout"`
		`"*3": "toggle_raise_hand"`
		`"*1": "cycle_personal_layout"`
		`"##": "end_conference"`

dtmf_allowed_layouts	The set of layouts that are cycled through on each press of the **cycle_layout** DTMF command (***8** by default), and by the **cycle_personal_layout** DTMF command (***1** by default). This can also include the names of any custom layouts. The order of the layout identifiers in the list determines the cycle sequence, starting with the layout next in the list after the current layout. However, if the initial (default) layout is not in the list then: • The layout that is first in the list is used in the first instance. • You cannot use DTMF to return to that initial layout during the conference.	["1:7", "ac", "1:21", "2:21", "2x2", "3x3", "4x4", "5x5", "1:0", "1:33"]
enable_avatars_in_mix	Determines whether any audio-only or video-muted participants are represented as an avatar in the main video layout (excluding Adaptive Composition and Teams-like layouts) instead of those participants being counted in the "notch" indicators. See Participant avatars for more information on the additional configuration options that are available when this is enabled.	`false`

Name	Description	Value in Base theme
disable_ai_ indicator	Determines whether the Google Meet AI indicator is disabled (**true**) or enabled (**false**).	`false`
disable_ external_ media_proc_ indicator	Determines whether the Google Meet external media processing indicator is disabled (**true**) or enabled (**false**).	`false`
disable_ streaming_ indicator	Determines whether the streaming indicator icon is disabled (**true**) or enabled (**false**).	`false`
disable_ recording_ indicator*	Determines whether the recording indicator icon is disabled (**true**) or enabled (**false**).	`false`
disable_ transcribing_ indicator*	Determines whether the transcribing indicator icon is disabled (**true**) or enabled (**false**).	`false`
disable_smart_ notes_ indicator*	Determines whether the note taking indicator icon is disabled (**true**) or enabled (**false**).	`false`
disable_ watermark_ icon	Determines whether the watermark icon is overlaid onto the main speaker video (**false**) or is not used (**true**). See Video watermarking for more information.	`false`
disable_ watermark_ mute_icon	Determines whether the "Your mic has been muted" watermark icon is overlaid onto the stream sent to a participant when that participant has been muted (**false**) or is not used (**true**). See Video watermarking for more information.	`true`
disable_ conference_ locked_ indicator	Determines whether the conference locked and conference unlocked indicators are disabled (**true**) or enabled (**false**).	`false`
ai_indicator_ text	The text that is associated with the AI indicator and that is temporarily displayed when Google Meet AI features are enabled.	`AI`
external_ media_proc_ indicator_text	The text that is associated with the external media processing indicator and that is temporarily displayed when Google Meet external media processing features are enabled.	`External media processing`
streaming_ indicator_text	The text that is associated with the streaming indicator and that is temporarily displayed when a conference is being streamed.	`Streaming enabled`
public_ streaming_ indicator_text	The text that is associated with the streaming indicator and that is temporarily displayed when Google Meet public streaming is enabled.	`Public streaming enabled`
recording_ indicator_text	The text that is associated with the recording indicator and that is temporarily displayed when a conference is being recorded.	`Recording`
transcribing_ indicator_text	The text that is associated with the conference transcribing indicator and that is temporarily displayed when a conference is being transcribed.	`Transcribing`

Name	Description	Value in Base theme
smart_notes_indicator_text*	The text that is associated with the note taking indicator and that is temporarily displayed when a Google Meet conference has note taking enabled.	`Smart notes enabled`
live_captions_on_indicator_text	The text that is associated with the live captions indicator and that is temporarily displayed when captions are enabled.	`Live captions on`
live_captions_off_indicator_text	The text that is associated with the live captions indicator and that is temporarily displayed when captions are disabled.	`Live captions off`
conference_locked_indicator_text	The text that is associated with the conference locked indicator and that is temporarily displayed when a conference is locked. This has a maximum limit of 20 characters.	`Conference locked`
conference_unlocked_indicator_text	The text that is associated with the conference unlocked indicator and that is temporarily displayed when a conference is unlocked.	`Conference unlocked`
conference_locked_indicator_n_waiting_text	The text that is associated with the conference locked indicator text, and shows the number of participants waiting in the lobby in Teams CVI calls.	`{number_of_waiting_participants} waiting for host`
test_call_service_media_delay	The number of seconds that media is delayed before being looped back to the caller when using a Test Call Service. (For more information, see Configuring the Test Call Service.)	`2`
test_call_service_disconnect_timeout	The number of seconds that a user can test their media before the disconnect message is played, when using a Test Call Service.	`10`
theme_version	Controls whether legacy (version "1") or new style (version "2") themes are used. If `"theme_version": 2` is not present in a **themeconfig.json** file at the lowest level of the theme file hierarchy then it is assumed to be a legacy-style theme (see How do I know which files and configuration settings will be used in a particular VMR? for more information).	`2`

Additional settings

This table contains any additional settings that are not included in the Base **themeconfig.json** file that you download from the Management Node, but can be used to control theme behavior.

Name	Description	Default behavior
disable_multiscreen	Determines whether the multiscreen feature is available or not.	`false`

Name	Description	Default behavior
enable_solo_streaming_loopback	Normally, the **streaming_in_progress** splash screen is shown if there are no other participants in the conference other than a streaming participant. If you want to show a loopback of the presentation stream instead of this splash screen you need to insert the setting: `"enable_solo_streaming_loopback" : true` into your **themeconfig.json** file.	`false`
disable_comfort_noise	Low-level, almost imperceptible background noise is added to the audio mix in conferences. This can be disabled by updating your theme configuration — either the default theme, or just in themes applied to specific services. Note that comfort noise is not sent towards Google Meet or Microsoft Teams meetings. To disable comfort noise you need to insert the setting: `"disable_comfort_noise": true` into your **themeconfig.json** file.	`false`
message_box_config	Specifies the properties of any timer or message overlay banner text. It contains a dictionary where you can define the text color, the background colour of the box, and the color of the box's border. You can use either: • RGB hexadecimal notation in the format 0xnnnnnn • ARGB hexadecimal notation (0xnnnnnnnn), where the first two digits after the 0x represent the Alpha channel, indicating the level of opacity applied. Example opacity values include: 00 - fully transparent, 80 - 50% opaque, BE - 75% opaque, FF - fully opaque. For example (showing the default colors): `"message_box_config": { "text_color": "0xFF000000", "background_color": "0xE6FFFFFF", "border_color": "0xFF000000" }`	as shown in the example
indicator_color_config	Specifies the colors of the in-conference messages (audio-only speaker names, locked/unlocked and so on) shown in the layout's information bar (referred to in themes as the "notch"). When an indicator message name e.g. "captions", is specified: • The associated `background`, `border` and `text` fields are required. • The `light_mode` field is optional. This lets you define background, border and text colors for a notch that is in light mode (see **notch_color** when Manually creating a custom layout file). If the `light_mode` field is specified then both the associated `background`, `border` and `text` fields for light mode are required. If a message type is not specified, the default values for that message are used.	as shown in the example
reserved_slot_overlay_text	Specifies any overlay text to be placed in a reserved slot when using participant pinning.	null

Endpoint data (vendordata.json)

The **vendordata.json** file contains information about videoconferencing endpoints from different manufacturers.

It is used to determine whether a specific endpoint is a single-screen or two-screen device (unless the endpoint explicitly signals to Pexip Infinity how many screens it has). This information is used when determining if presentation content could be sent as part of the layout mix when using Adaptive Composition or the Teams-like layout, and if multiscreen participant display is supported:

- Single screen endpoints:
  - Presentation content may be sent as part of the layout mix, or as a separate presentation stream.
  - Multiscreen participant display is not available.
- Dual screen endpoints:
  - Presentation content is always sent as a separate presentation stream.
  - Multiscreen participant display is available for the supported layouts.

    For H.323 endpoints, the **vendordata.json** file cannot identify the number of screens (single or dual) and the endpoint cannot explicitly signal how many screens it has, thus they cannot use multiscreen participant display by default. Multiscreen participant display can only be enabled by sending a *9 (toggle to multiscreen) DTMF command to the conference. See Using a DTMF keypad to control a conference and Determining if an endpoint has single or multiple screens for more information.

See Receiving the presentation stream as part of the layout mix (Adaptive Composition) and Multiscreen participant display for more information.

It is also used for Google Meet audio denoising interoperability to configure a list of endpoints that perform their own in-built denoising. See Configuring Pexip Infinity as a Google Meet gateway for more information.

The default **vendordata.json** file is shown below:

```
{
    "endpoints_by_display_count": {
        "1": [
            "Cisco-Board",
            "Cisco-DX",
            "Cisco-Desk",
            "Cisco-EX60",
            "Cisco-EX90",
            "Cisco-MX200",
            "Cisco-MX300",
            "Cisco-RoomKitMini",
            "Cisco-SX10",
            "TANDBERG/257",
            "Yealink VC200",
            "PolycomRealPresenceTrio",
            "PolycomStudioX30",
            "Polycom (HDX 4000",
            "Polycom (HDX 4500",
            "Polycom (HDX 6000"
        ],
        "2": []
    },
    "built_in_denoising": [
    ]
}
```

The JSON dictionary in the file contains:

- An **endpoints_by_display_count** object, which contains two further objects:
  - "**1**": this contains an array of names of single screen endpoints (the absence of a closing bracket for the HDX device names is deliberate).
  - "**2**": this contains an array of names of dual screen endpoints (this array is empty by default).
- A **built_in_denoising** object that contains an array (empty by default) of endpoints that perform their own built-in audio denoising. This information is currently used for Google Meet interoperability only.

The endpoint names used in the arrays are those that are contained in the user agent string used by the endpoint, and can be seen in the **Vendor** field when viewing participant status information.

Pexip intends to maintain and update this file as appropriate in future software versions, however you may amend the contents of the arrays and provide your own **vendordata.json** file as part of a custom theme upload and then apply that theme to your platform, VMRs or rules in the normal way to override the default behavior. Ensure that you do not break the syntax of the JSON file; when adding endpoints to the empty arrays follow the format of the existing "**1**" (single screen endpoints) object.

The **vendordata.json** file must be UTF-8 encoded, therefore when editing the file you should use a UTF-8 capable editor. We recommend that you do **not** use Notepad on Windows computers.

Note that if you submit your own **vendordata.json** file you must also include (as a minimum) in your ZIP upload a **themeconfig.json** file that contains a `"theme_version": 2` switch.

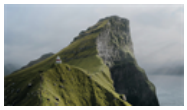
Customizing the splash screens

The splash screens are used when joining a conference (such as to capture a PIN), using a Virtual Reception or the Test Call Service, or to display an error such as insufficient licenses. Each splash screen consists of a background JPG file plus, on most screens, other SVG images or text elements that are overlaid onto the background image.

The **themeconfig.json** file offers a lot of flexibility when customizing the appearance of the splash screens. You can override the default background image, graphics and text elements that are used on each splash screen, and modify where those elements are positioned. Each splash screen can be configured by specifying a `splash_screens` object and associated elements within your **themeconfig.json** file, as described below.

Background image

By default the Base theme contains 3 background files:

File name	Image size (width, height)	Content in Base theme	Notes
background.jpg	1920 x 1080 pixels JPG (RGB mode only)		The background image used by default on all splash screens except for those used by the Test Call Service.
background_test_call.jpg	1920 x 1080 pixels JPG (RGB mode only)		The background image (a dark screen) used by default on the Test Call Service splash screens.
background_reserved_slot.jpg	1920 x 1080 pixels JPG (RGB mode only)		The background image (a black screen) used by default on the **reserved_slot** splash screen when using participant pinning.

You can replace either of these files with your own versions, or you have the flexibility to specify an alternative file to use on a specific splash screen. Ensure that the images display acceptably on both 16:9 endpoints and 4:3 endpoints (if applicable). For example, the original 1920 x 1080 pixel (16:9) images may have approximately 240 pixels cropped from both the left and right sides when displayed as 4:3. You should therefore check that any details on the far left or far right of the image are visible in both formats; for this reason we recommend keeping the image fairly central.

To specify an alternative file, use the `background` element to define the override and the `path` element to define the file to use. The filename must begin with "background". For example, to use an alternative background for the `pin_entry` splash screen:

```
{
  "splash_screens": {
    "screens": {
      "pin_entry": {
        "background": {"path": "background_pin_entry.jpg"},
        ...
```

Make sure that you include the specified background file (background_pin_entry.jpg in this example) in the theme ZIP file that you upload.

Alternatively, instead of specifying a `path` you can define a color, e.g. `"background": {"color": "0x323232"}`

Splash screens

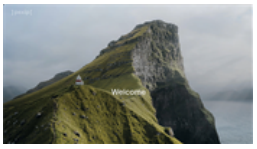
The following table lists the splash screens that you can configure within a theme, and when that screen is shown. It also identifies the screen key to use in a `screens` object if you want to override the default behavior, and any default icon image and text label elements used on that screen.

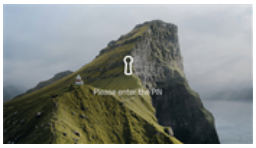
All of the screens used when joining a conference use **background.jpg** as the default background image. The Test Call Service uses **background_test_call.jpg** during the call.

Default splash screens and associated elements

If you are using a theme to customize gateway calls only, then you only need to consider a subset of these screens, as indicated in the table below.

Many of the overlay icon SVG files that are used by the splash screens now contain empty/blank images by default. They are still referenced by each splash screen's elements configuration for backwards compatibility, but as they are blank by default they do not affect each screen's default appearance (from version 37 many screens typically only show a background image and a text label).

Screen key	Default elements used	Usage	Used in gateway calls
welcome	Label: welcome Icon: icon_welcome.svg (blank)	Shown when there are no other participants in the conference. Also shown to a VTC-based participant that is being held in a Skype for Business meeting lobby. 	
connecting	Label: connecting Icon: icon_connecting.svg (blank)	Shown briefly when placing a person-to-person call via the Infinity Gateway.	✓
waiting_for_host	Label: waiting_for_host Icon: icon_waiting_for_host.svg (blank)	Shown to Guests while waiting for a Host to either: • join the conference • allow the Guest to join a locked conference, either by unlocking the conference or permitting the individual participant to join. The length of time Guests can remain at this screen is configurable via the global **Waiting for Host timeout** option (**Platform > Global Settings > Service Configuration**).	
inlobby_status_unknown	Label: inlobby_status_unknown Icon: icon_connecting.svg (blank)	Shown when initially connecting to a Microsoft Teams meeting, while the participant's status is not yet known.	
inlobby	Label: inlobby Icon: icon_is_in_lobby.svg (blank)	This is shown: • while being held in the Microsoft Teams lobby when joining a Microsoft Teams meeting • to Guests in a Virtual Auditorium when the last Host leaves and **Guests can see other guests** is disabled.	
other_participants_audio_only	Label: other_participants_audio_only Icon: icon_other_participants_audio_only.svg (blank)	Shown when all other participants are audio-only or presentation and control-only.	✓

Screen key	Default elements used	Usage	Used in gateway calls
pin_welcome	Label: please_enter_pin Icon: icon_pin_entry.svg	The **pin_** screens are shown to participants† when entering PINs. Participants entering PINs via these screens will be disconnected after three unsuccessful attempts. The length of time participants can remain at these screens is configurable via the global **PIN entry timeout** option (**Platform > Global Settings > Service Configuration**). 	
pin_entry	Icon: icon_pin_entry.svg and icon_pin_entry_digit.svg	An icon_pin_entry_digit.svg icon is displayed as each number is entered, which can go up to 20 icons over 2 lines. See PIN digits element for controlling the layout of the PIN entry digit icons. 	
pin_correct	Label: pin_correct Icon: icon_welcome.svg	Displayed when a correct PIN is entered. 	
pin_invalid	Label: invalid_pin Icon: icon_pin_invalid.svg	Displayed when an incorrect PIN is entered. 	
virtual_reception_welcome	Label: enter_conference_id Icon: icon_virtual_reception_welcome.svg (blank)	Shown to participants† who have connected to a Virtual Reception.	
virtual_reception_conference_id_entry	Label: conference_id_entry Icon: icon_virtual_reception_conference_id_entry.svg (blank)	Shown to participants† as they enter digits ("1234" shown in this example) into a Virtual Reception.	

Screen key	Default elements used	Usage	Used in gateway calls
virtual_ reception_ conference_id_ invalid	Label: conference_id_ invalid Icon: icon_virtual_ reception_ conference_id_ invalid.svg (blank)	Shown to participants† who have connected to a Virtual Reception and have entered an invalid conference number.	
virtual_ reception_ connecting	Label: virtual_ reception_connecting Icon: icon_virtual_ reception_ connecting.svg (blank)	Shown briefly to participants† while being transferred from a Virtual Reception. Note that the **name** of the destination VMR or matching Call Routing Rule is also overlaid onto this image, using the virtual_reception_connecting text label.	
timeout	Label: timeout Icon: icon_timeout.svg (blank)	Shown briefly prior to disconnecting participants† who have connected to a Virtual Reception but did not enter a valid conference number.	
no_incoming_ video	Icon: icon_no_ video.svg	Shown when a participant's video stream is not available. The **no_incoming_video** and **no_main_video** screens both use the same icon image file (**icon_no_video.svg**) and the default background color is #323232 (dark gray), instead of **background.jpg**. 	✓
no_main_video	Icon: icon_no_ video.svg	Used when streaming a conference to indicate that there is currently no main video. The **no_incoming_video** and **no_main_video** screens both use the same icon image file (**icon_no_video.svg**) and the default background color is #323232 (dark gray), instead of **background.jpg**. 	
reserved_slot	Icon: icon_reserved_ slot.svg	Displayed in a reserved slot when it is not occupied by a participant when participant pinning is used. 	✓
unused_screen		Shown on a dual-screen system when the second screen is empty. A black screen is used by default. See Customizing the unused screen in a dual-screen system for more information.	✓

Screen key	Default elements used	Usage	Used in gateway calls
streaming_in_progress	Label: streaming_in_progress Icon: icon_streaming_screen.svg (blank)	Shown if there are no other participants in the conference other than a streaming participant. Alternatively, if you want to show a loopback of the presentation stream instead of this splash screen you need to insert `"enable_solo_streaming_loopback" : true` into your **themeconfig.json** file (see Additional settings).	
stream_waiting	Label: stream_waiting Icon: icon_streaming_screen.svg (blank)	A "holding" splash screen that can be sent to a streaming participant (e.g. YouTube broadcast) while you are waiting for people in the conference to get ready to start.	
no_presentation	Label: no_presentation Icon: icon_no_presentation.svg	Used to indicate that there is currently no presentation stream. 	
error_capacity_exceeded	Label: capacity_exceeded Icon: icon_error.svg (blank)	Shown to participants† when they attempt to join a conference that has reached its maximum number of participants. For more information, see Limiting the number of participants.	
error_insufficient_licenses	Label: insufficient_licenses Icon: icon_error.svg (blank)	Shown to participants† when they are unable to join a conference because all call licenses are currently in use. For more information, see Insufficient licenses.	
error_insufficient_video_licenses	Label: insufficient_video_licenses Icon: icon_error_insufficient_video_licenses.svg (blank)	Shown to participants when they are unable to join a conference because all port (video) licenses are currently in use, but there was an audio license available. For more information, see Insufficient licenses.	✓
error_invalid_license	Label: invalid_license Icon: icon_error.svg (blank)	Shown to participants† when they are unable to join a conference because the Pexip Infinity license is not valid. For more information, see Invalid license.	
direct_media_welcome	Label: welcome Icon: n/a	Shown to a direct media participant who is waiting for the second participant to join the call.	
direct_media_waiting_for_host	Label: waiting_for_host Icon: n/a	Shown to a direct media participant who is waiting for the conference host to join.	
direct_media_other_participants_audio_only	Label: other_participants_audio_only Icon: n/a	Shown to a direct media participant when the other participant is audio only.	

Screen key	Default elements used	Usage	Used in gateway calls
direct_media_escalate	Label: direct_media_escalate Icon: n/a	Shown as a notification message to a direct media participant who is being escalated to a transcoded conference. 	
direct_media_deescalate	Label: direct_media_deescalate Icon: n/a	Shown as a notification message to a direct media participant who is being de-escalated back to a direct media conference. 	
direct_media_host_left	Label: host_left Icon: n/a	Shown as a notification message to a direct media participant when the call is about to be terminated because the Host participant has left.	

* This is new in version 41.

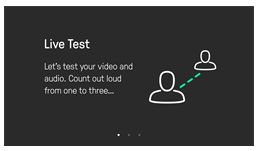
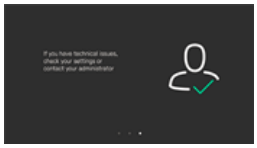
† Except for participants that are using Pexip apps.

†† Only applies to gateway calls into an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

‡ Only applies to gateway calls into Microsoft Teams meetings.

* When using a Virtual Reception in gateway/interop call scenarios.

These are the splash screens used by the Test Call Service:

Screen key	Default elements used	Usage
test_call_welcome	Label: test_call_welcome_header and test_call_welcome_text Icon: icon_test_call_welcome.svg	Shown at the start of a call to a Test Call Service. (For more information, see Configuring the Test Call Service.) 
test_call_in_progress	Label: test_call_in_progress	Shown during a call to a Test Call Service. Note that a large, live (with a short delay) video image of the test call participant is shown on top of this screen during a test call.
test_call_complete	Label: test_call_complete Icon: icon_test_call_complete.svg	Shown briefly prior to automatically disconnecting the participant from a Test Call Service. 

Default element configuration for each screen

If you define `elements` for a splash screen, whatever you specify will completely replace all of the default icon and text elements for that screen. For example, if you want to add a new line of text to a screen, your elements object for that screen must define the original (default) icon and text as well as your new text.

Therefore it is useful to know how the icons and text elements are positioned on each splash screen by default. The default layout of all of the splash screens is available at theme_screen_defaults_v41.json. Note that this element configuration is not contained in the **themeconfig.json** file included within the Base theme that you download from the Management Node.

Text overlays (labels)

The `labels` object is referenced by every text element that is overlaid onto a splash screen and is used to define the actual text displayed to users. For example:

```
"splash_screens": {
  "labels": {
    "welcome": "Welcome",
    "connecting": "Connecting...",
    "timeout": "Time exceeded",
    ...
```

The keys (e.g. "timeout") are used in text elements and the associated values (e.g. "Time exceeded") are the text strings that are shown on the screen. The text content can include \n newlines if you require multi-line capability (newlines must be explicitly used and are never applied automatically). On some splash screens, strings may also reference some predefined variables, such as "{conference_name}" which is used in the virtual_reception_connecting screen.

The default font size is 66px and the default color is 0xffffffff (white, defined in ARGB hexadecimal notation).

Note that the default text `labels` are not contained within the Base theme's **themeconfig.json** file that you can download from the Management Node, but you can override the default settings by adding the relevant labels into your own **themeconfig.json** file. The following table contains the default keys, the default text string for each key, and the screen in which that key is used:

Default label keys and usage

These are the label keys and associated text strings and screens:

Label key	Default text string (to overlay onto the screen background)	Associated screen (by default)
welcome	Welcome	welcome direct_media_welcome
connecting	Connecting...	connecting
timeout	Time exceeded	timeout
waiting_for_host	Waiting for the host...	waiting_for_host direct_media_waiting_for_host
other_participants_audio_only	Other participants\nare not sharing video	other_participants_audio_only direct_media_other_participants_audio_only
please_enter_pin	Please enter the PIN	pin_welcome
pin_correct	Welcome	pin_correct
invalid_pin	Invalid PIN\nPlease try again	pin_invalid
enter_conference_id	Enter conference ID	virtual_reception_welcome
conference_id_entry	{conference_id}	virtual_reception_conference_id_entry
conference_id_invalid	Conference ID invalid	virtual_reception_conference_id_invalid
virtual_reception_connecting	{conference_name}	virtual_reception_connecting

Label key	Default text string (to overlay onto the screen background)	Associated screen (by default)
no_presentation	Waiting for presentation...	no_presentation
streaming_in_progress	Your video is\nbeing streamed	streaming_in_progress
stream_waiting	Streaming will begin shortly...	stream_waiting
inlobby_status_unknown	Call connecting...	inlobby_status_unknown
inlobby	Welcome to the lobby,\nyour meeting host will admit you soon	inlobby
capacity_exceeded	Capacity\nexceeded	error_capacity_exceeded
insufficient_licenses	Insufficient licenses	error_insufficient_licenses
insufficient_video_ licenses	Insufficient video licenses\nYou can still share content and audio	error_insufficient_video_licenses
invalid_license	Invalid license	error_invalid_license
test_call_welcome_ header	Live Test	test_call_welcome
test_call_welcome_text	Let's test your video and\naudio. Count out loud\nfrom one to three...	test_call_welcome
test_call_in_progress	You will see and hear yourself with a two second delay	test_call_in_progress
test_call_complete	If you have technical issues,\ncheck your settings or\ncontact your administrator	test_call_complete
direct_media_escalate	You are about to be transferred\ninto a multi-party conference	direct_media_escalate
direct_media_deescalate	You are about to be transferred\ninto a one-to-one call	direct_media_deescalate
host_left	The host has left the conference.\nYour call will be disconnected shortly.	direct_media_host_left

Adding your own text strings and translations

Adding your own text strings

You can specify your own keys and text strings within the `labels` object and use them on any splash screen. For example, to specify a new label and use the associated text and an alternative graphic on the `welcome` screen:

```
"splash_screens": {
  "labels": {
    "my_label_key": "¡Hola!"
  },
  "screens": {
    "welcome": {
      "elements": [
        {"type": "icon", "path": "icon_my_welcome_image.svg"},
        {"type": "text", "label": "my_label_key"}
      ]
    }
  ...
```

Translations

If you want to use a theme where all of the strings are translated to a different language, you have the ability to change the wording of the text without modifying the layout itself, for example:

```
"splash_screens": {
  "labels": {
    "welcome": "Velkommen",
    "connecting": "Kobler til",
    ...
```

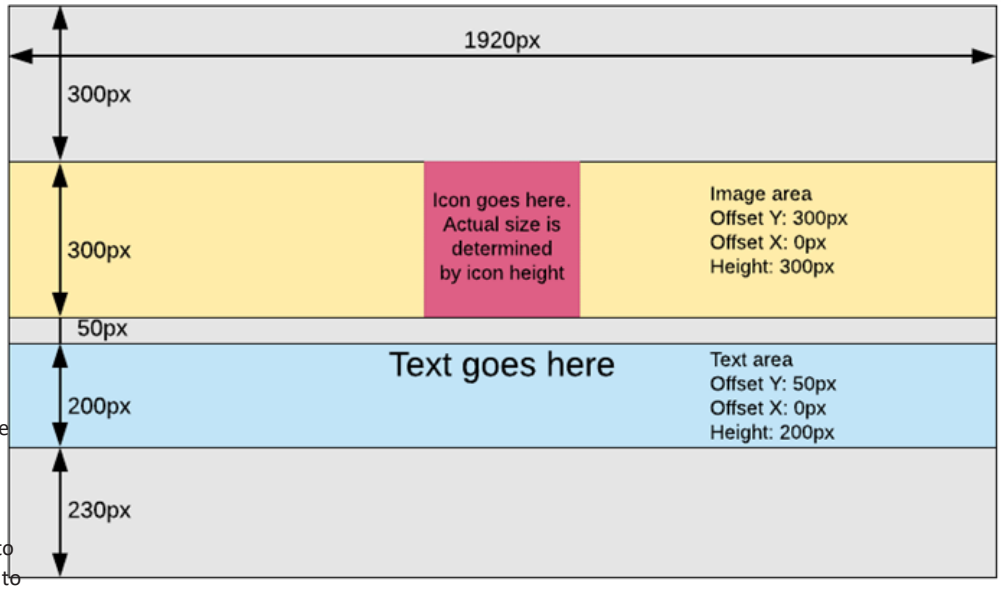
Predefined variables to use in text strings

There is a small set of predefined variables that you can reference within your text strings:

Variable name	Description and usage
{conference_id}	This variable is only supported on the **virtual_reception_conference_id_entry** screen, where it is normally used by the **conference_id_entry** label and it displays the number the user is entering into the Virtual Reception.
{conference_name}	Contains the name of the conference. This variable is used on the **virtual_reception_connecting** screen, where it is included in the **virtual_reception_connecting** label and it refers to the conference into which the user is being transferred. For example: `"labels": {"virtual_reception_connecting": "{conference_name}"}` This variable can also be used in labels on the **welcome**, **unused_screen** and **waiting_for_host** screens.
{number_of_waiting_participants}	Used while a conference is in progress and contains the number of participants waiting to be let into a locked conference. It is normally used with the **conference_locked_indicator_n_waiting_text** setting in the **themeconfig.json** file (note that this is a setting and not a screen label). For example: `"conference_locked_indicator_n_waiting_text": "{number_of_waiting_participants} waiting for host"`

Splash screen elements (to control the size and position of text/graphics)

When customizing a splash screen you can define `elements` to control the appearance and position of images and text on that specific screen. The default screen layout and offsets are shown in this diagram (right).



If you define `elements` for a splash screen, whatever you specify will completely replace all of the default icon and text elements for that screen. For example, if you want to add a new line of text to a screen, your elements object for that screen must define the original (default) icon and text as well as your new text.

The following example shows the JSON structure for how you can define an `icon` and `text` type element for a splash screen (the `welcome` screen in this example). All width, height and offset positions are measured in pixels.

```
{
  "theme_version": 2,
  "splash_screens": {
    "screens": {
      "welcome": {
        "elements": [
          {
            "type": "icon",
            "path": "icon_welcome.svg",
            "offset_y": 300,
            ...
          },
          {
            "type": "text",
```

```
      "label": "welcome",
      "font_size": 66,
      ...
    }
  ],
  ...
```

The full set of keys that you can specify for each element type are described below.

Icon elements (for SVG files used on splash screens)

Icons are the SVG files that are overlaid onto a splash screen. They can be configured by the `icon` element, which controls the position and scale (size) of the SVG file, and which file to use.

To change the color of an SVG image, see Changing the color of the SVG image.

Example icon element usage

This shows how to use the **icon** element and its associated keys.

```
{
  "type": "icon",
  "path": "icon_virtual_reception_welcome.svg",
  "width": 1920,
  "height": 300,
  "offset_y": 300,
  "offset_x": 0
}
```

Key	Description
path	Name of the SVG file (where the filename matches the pattern "icon_[A-Za-z0-9_].svg"). This can be the name of a Base SVG file (where you can also optionally include a replacement file of the same name in the theme ZIP), or a new filename (where you must supply the associated SVG file of that name).
width and height	Icon images are rendered proportionally scaled using the smallest value of **width** or **height**. The icon **width** defaults to 1920, which means that the icon size is normally controlled by the **height**. Width is normally specified only in a free-form layout when you want to restrict icon size based on width to ensure that there is no overflow on the x axis. The icon height is usually the attribute that controls the size of the icon. It defaults to 300px for most screens. If height is set to a larger value than width, then width controls the size of the icon. In vertical layouts, height is also used to calculate the y-axis coordinate of the element below it. Ultimately, the proportions of the image itself control how it is rendered, within a certain width or height constraint. Note that the icon is centered inside the bounding box; for example if you specify `{"width": 1920, "height": 200}` for a 1000x200 icon, it will display centered as expected, but if you then add, for example, `offset_x: 500` the icon will get pushed to the right and cropped.
offset_x	The x-axis coordinate relative to the center in vertical layouts, and absolute in free_form layouts. It can be negative.
offset_y	The y-axis coordinate relative to the element above (or top edge) in vertical layouts, and absolute (from the top of the screen) in free_form layouts. It can be negative.

Text elements

You have full control over the positioning and style of a text string by using a `text` element. All measurements are in pixels. The default font for the in-conference display of participant names is Roboto (which cannot be changed), or if that is not available for the character set, Noto Sans. All other overlay text (splash screen text and in-conference messages) uses Inter and this can be changed by using a customized theme. For example (and showing the default values for non-test-call conferences):

Text element usage

This shows how to use the **text** element and its associated keys.

```
{
  "type": "text",
```

```
  "label": "enter_conference_id",
  "offset_y": 50,
  "offset_x": 0,
  "height": 200,
  "width": 1920,
  "color": "0xffffffff",
  "font_size": 66,
  "font": "Inter",
  "letter_spacing": 0,
  "line_spacing": -13,
  "align": "center"
}
```

Key	Description
label	A reference to an item in the labels object, for example "enter_conference_id", to identify the actual text to display.
offset_x	The x-axis coordinate offset of the bounding box in vertical layouts, and the absolute coordinate of the bounding box in free_form layouts.
offset_y	The y-axis coordinate offset (from the top of the screen) of the bounding box in vertical layouts, and the absolute coordinate of the bounding box in free_form layouts.
width	Width of the bounding box. It usually defaults to 1920 in vertical layouts.
height	Height of the bounding box.
color	Color of the text. You can use either: • RGB hexadecimal notation in the format 0xnnnnnn • ARGB hexadecimal notation (0xnnnnnnnn), where the first two digits after the 0x represent the Alpha channel, indicating the level of opacity applied. Example opacity values include: 00 - fully transparent, 80 - 50% opaque, BE - 75% opaque, FF - fully opaque. Default: 0xffffffff (white, fully opaque). You can specify a default color for all text elements by adding a **text_color** setting to the **themeconfig.json** file, for example adding `"text_color": "0xff0000"` would make all text elements red by default.
font_size	The font size in pixels. Default: 66px.
font	The font name. The supported fonts are: "Noto Sans", "Roboto", "Roboto Condensed", "GT Pressura", "GT Pressura Light", "Inter" and "Inter Medium".
letter_spacing	The spacing between letters. It can be negative. Default: 0.
line_spacing	The spacing between lines. It can be negative. Default: -13.
align	Alignment of text within the bounding box. Possible values: "center", "left", "right". Default: "center".

PIN digits element

The `pin_digits` element is a special element to render up to 20 icons (each 42x42 pixels by default) on 2 lines, and is only used for PIN entry on the `pin_entry` screen. For example (showing the default values):

PIN digits element usage

This shows how to use the **pin_digits** element and its associated keys.

```
{
  "type": "pin_digits",
  "path": "icon_pin_entry_digit.svg",
  "digit_width": 42,
  "digit_height": 42,
```

```
  "digit_spacing": 10,
  "offset_y": 50,
  "offset_x": 0,
  "height": 200,
  "width": 1920,
  "line_spacing": 25
}
```

Key	Description
path	Name of the PIN digit icon SVG file.
digit_ width	The width of each individual PIN digit icon.
digit_ height	The height of each individual PIN digit icon.
digit_ spacing	The space between each PIN digit icon.
offset_x	The x-axis offset of the bounding box.
offset_y	The y-axis offset (from the top of the screen) of the bounding box.
height	Height of the bounding box.
width	Width of the bounding box.
line_ spacing	Spacing between the lines of digits. There are always (up to) 10 digits per line, regardless of the size of the bounding box. You must ensure that the size of the bounding box is big enough to fit the digit icons (based on the maximum length of the PINs used in your deployment).

Vertical and free form layouts

The splash screens support 2 types of layout: vertical (the default) and free form. Layouts are specified by the `layout_type` entry in the screen specification object.

Using vertical and free form layouts

For example, to specify a vertical layout for the `error_invalid_license` screen:

```
{
  "splash_screens": {
    "screens": {
      "error_invalid_license": {
        "layout_type": "vertical"
        ...
```

Vertical layouts

Vertical layouts are centered layouts where the elements (icons and text) stack on top of each other top to bottom in the order they have been specified. The `height` and `offset_y` attributes determine the element's placement on the screen. All elements are centered by default and can be shifted left and right with the `offset_x` variable. For example, to include **icon_welcome.svg** on the **welcome** screen:

```
"splash_screens": {
  "screens": {
    "welcome": {
      "layout_type": "vertical",
      "elements": [
        {"type": "icon", "path": "icon_welcome.svg", "height": 300, "offset_y": 400},
        {"type": "text", "text": "welcome", "height": 200, "offset_y": 50}
      ...
```

In this layout, the icon is centered on the screen, starting 400px from the top of the screen. The text element is placed 50px under the icon, placing it at y=750px (300px height + 400px offset of the icon + 50px offset of the text).

Free form layouts

Free form layouts are, as the name implies, completely free form. This lets you design complex layouts, like the one used for the Test Call Service. In a free form layout, `offset_x` and `offset_y` are absolute coordinates on the screen and you are responsible for placing the elements correctly.

This example shows the layout of the test_call_complete screen (these are the actual values that are used by default):

```
{
  "theme_version": 2,
  "splash_screens": {
    "screens": {
      "test_call_complete": {
        "layout_type": "free_form",
        "background": {"path": "background_test_call.jpg"},
        "elements": [
          {
            "type": "text",
            "label": "test_call_complete",
            "height": 300,
            "offset_y": 361,
            "offset_x": 308,
            "align": "left",
            "font_size": 50
          },
          {
            "type": "icon",
            "path": "icon_test_call_complete.svg",
            "offset_x": 1259,
            "offset_y": 305,
            "height": 386,
            "width": 380
          },
          {
            "type": "icon",
            "path": "icon_inactive_page.svg",
            "offset_x": 872,
            "offset_y": 985,
            "width": 16,
            "height": 16
          },
          {
            "type": "icon",
            "path": "icon_inactive_page.svg",
            "offset_x": 952,
            "offset_y": 985,
            "width": 16,
            "height": 16
          },
          {
            "type": "icon",
            "path": "icon_active_page.svg",
            "offset_x": 1032,
            "offset_y": 985,
            "width": 16,
            "height": 16
          }
        ]
      }
    }
  }
}
```

Helper lines

To aid development, debugging and theme creation, you can display gridlines for a given splash screen by using the `helper_lines` object. Helper lines are added on top of the background and under the icons, text and other elements.

How to add help lines

Example use of helper lines on the error_invalid_license screen:

```
{
  "splash_screens": {
    "screens": {
      "error_invalid_license": {
        "helper_lines": {
          "horizontal": [300, 350, 432, 678],
          "vertical": [660, 960, 1260]
```

```
    }
...
```

Key	Description
horizontal	A list of coordinates (from the top of the screen) to put horizontal helper lines.
vertical	A list of coordinates to put vertical helper lines.

Overlay SVG image files used on splash screens when joining a conference

When creating your own SVG image files for use on the splash screens, do **not** use tools such as Adobe Photoshop or Illustrator to convert JPG or PNG files into SVG format as they typically create complex files/paths that are resource-intensive to render, or in some cases may render incorrectly.

- Use a vector drawing program that uses SVG as its primary format, such as Inkscape.
- Do not create a single complex path — use multiple paths instead.

Default SVG image files

All of the overlay vector images in the Base theme are listed below. All of the images shown in the table below are white, except for icon_inactive_page.svg, and are shown here against a gray background. The table also lists any splash screens on which the overlay images are used by default.

File name	Content in Base theme	Associated default splash screens
icon_active_page.svg		test_call_welcome test_call_in_progress test_call_complete (they are rendered as the progress dots along the bottom of the screen)
icon_inactive_page.svg		test_call_welcome test_call_in_progress test_call_complete (they are rendered as the progress dots along the bottom of the screen)
icon_pin_entry.svg		pin_welcome pin_entry
icon_pin_entry_digit.svg		pin_entry
icon_pin_invalid.svg		pin_invalid
icon_reserved_slot.svg		reserved_slot

File name	Content in Base theme	Associated default splash screens
icon_welcome.svg		pin_correct (also previously used on welcome screen)

The following SVG image files are also included by default in the Base theme but they all contain blank/empty images. In version 36 and earlier they contained images and appeared on various splash screens. From version 37 these SVG files are still referenced by the nominated splash screens for backwards compatibility, but as they are blank by default they do not affect each screen's appearance (the screen only shows the associated background image and any text label). However, if you customize any of these files to include graphical content it will appear on the associated screen.

- **icon_waiting_for_host.svg** (referenced by the waiting_for_host screen)
- **icon_virtual_reception_welcome.svg** (referenced by the virtual_reception_welcome screen)
- **icon_virtual_reception_connecting.svg** (referenced by the virtual_reception_connecting screen)
- **icon_virtual_reception_conference_id_invalid.svg** (referenced by the virtual_reception_conference_id_invalid screen)
- **icon_virtual_reception_conference_id_entry.svg** (referenced by the virtual_reception_conference_id_entry screen)
- **icon_timeout.svg** (referenced by the timeout screen)
- **icon_other_participants_audio_only.svg** (referenced by the other_participants_audio_only screen)
- **icon_error_insufficient_video_licenses.svg** (referenced by the error_insufficient_video_licenses screen)
- **icon_error.svg** (referenced by the error_capacity_exceeded, error_insufficient_licenses and error_invalid_license screens)
- **icon_is_in_lobby.svg** (referenced by the inlobby screen)
- **icon_connecting.svg** (referenced by the connecting and inlobby_status_unknown screens)
- **icon_streaming_screen.svg** (referenced by the streaming_in_progress and stream_waiting screens)

If you want to reinstate the previous graphical images on the splash screens, the SVG icon files used in the previous versions of the default theme (with the original image content) are available for download at theme-svg-files.zip and can be included in a custom theme. Full information on the previous appearance and use of these files and splash screens is contained in our version 36 documentation.

Overlay SVG image files used by the Test Call Service

These overlay SVG image files are used only by the Test Call Service:

File name	Content in Base theme	Associated default splash screens
icon_test_call_complete.svg		test_call_complete
icon_test_call_welcome.svg		test_call_welcome

Changing the color of the SVG image

The color in an SVG file is typically controlled by a **fill** property defined within the file itself. Most of the SVG images used in the Base Pexip theme are white (`fill="#FFFFFF"`).

To change the color of an SVG image you can edit the SVG file in a simple text editor such as Notepad, and change the fill color as required. For example to change the color to red, you would specify `fill="#ff0000"`.

Indicator graphics used within an ongoing conference

The following graphics may be used during a conference. The SVG indicators are rendered on top of a dark gray background (the color is 0x7f323232, where 7f is the alpha channel which makes the background slightly transparent). The gray background is not part of the actual SVG image itself and it cannot be customized. If you use your own customized SVG images, ensure that they look appropriate when displayed on sizes from 20x20px to 48x48px, and are recognizable when displayed as 12x12px for participants that use very low resolutions.

If you are using a theme to customize gateway calls only, then you only need to consider a subset of these images, as indicated in the table below.

Default in-conference graphics files

These are the default graphics files that used during a conference:

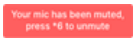
File name	Content in Base theme	Notes	Used in gateway calls
icon_no_ presentation.svg		Used to indicate that there is currently no presentation stream. It is used by default on the **no_presentation** splash screen.	
icon_no_video.svg		Shown when a video stream is not available. It is used by default on the **no_main_video** and **no_incoming_video** splash screens. It should look appropriate when displayed at sizes up to 400 x 400 pixels.	✓
Various icon_ notch57tp_ <item>.svg files	<various>	These icon images are used to support the in-conference indicators, such as participant counts, audio participants, recording indicators and locked status that are shown at the top-center of the layout. We recommend that you do not customize these elements as we cannot guarantee compatibility with future versions of Pexip Infinity.	
presence_avatar_ image.jpg		Shown next to the conference alias when using legacy Pexip apps, and as the contact avatar when using Skype for Business. When the image is used, the corners are cropped so that it can be displayed as a round image. It is a JPG image (RGB mode only) and should be 128 x 128 pixels.	✓
no_powerpoint_ live.jpg		Used in Microsoft Teams interop calls to indicate that a Teams client is performing PowerPoint Live sharing, or is using any other non-supported content share sources, such as Excel or Whiteboard.	
raise_hand.png		Used in the Teams-like layout to indicate that a participant in the Teams conference has raised their hand. This image only applies to Teams integrations, and it should not be customized.	
watermark_ icon.png		Transparent image used for applying a watermark to the main speaker video in a conference. The default image is a white Pexip logo with 40% transparency and is shown here against a blue background. It is a PNG image and should be 200 pixels wide x 100 pixels high. See Video watermarking for more information.	

File name	Content in Base theme	Notes	Used in gateway calls
watermark_ footer_icon.png	]pexip[	A watermark that is only used in the 1 + 33 layout (it appears at the bottom of the layout). It is the same size as the standard **watermark_icon.png** file, but it has inverted colors (so that it shows up on the white background of the layout).	
watermark_mute_ icon.png	Your mic has been muted, press *6 to unmute	An overlay that is displayed to a participant when they have been administratively muted. It is shown under the conference indicators whenever the participant has been muted. It is a PNG image and should be 200 pixels wide x 100 pixels high. Note that: • This overlay is disabled by default, and is controlled by the **disable_ watermark_mute_icon** setting in the **themeconfig.json** file. • It is not always appropriate in all call scenarios to state that the user can unmute themself in this way, such as when the user has dialed in over WebRTC, or for Guest participants in a VMR where all Guests have been muted.	✓

‡ Only applies to gateway calls into Microsoft Teams meetings.

Video watermarking

Video watermarking overlays a small transparent image/logo onto the main speaker video during a conference. The logo watermarks are never added to calls placed via the Infinity Gateway, but the "Your mic has been muted" watermark icon can be overlaid on gateway calls. You can change the watermark images or disable watermarking completely.

- The default logo watermark is a white "Pexip" logo]pexip[(shown here against a blue background) and is enabled by default.
- The default "Your mic has been muted" watermark is 

and is disabled by default.

Enabling, disabling and changing the watermark image

When logo watermarking is enabled:

- The **watermark_icon.png** file is used as the watermark image.
- The top-left corner of the watermark is placed 56 pixels across and 40 pixels down from the top-left corner of the main speaker video, and the watermark area then extends 200 pixels across and 100 pixels down. These sizes are based on Full HD resolution — they will be smaller on lower resolutions.
- The size and position of the watermark area cannot be customized.
- The watermark is shown on the following splash screens: welcome, connecting, streaming_in_progress, stream_waiting, other_ participants_audio_only, no_presentation, no_incoming_video, no_main_video.

When the "Your mic has been muted" watermark is enabled:

- The **watermark_mute_icon.png** file is used as the watermark image.
- It is shown under the conference indicators whenever the participant has been muted.
- The size and position of the watermark area cannot be customized.

Enabling or disabling watermarking

There are 2 settings in the themeconfig.json file that control whether watermarking is enabled or not:

- **disable_watermark_icon**: controls logo watermarking. Set it to ***false*** to enable watermarking, and to ***true*** to disable watermarking. By default, **disable_watermark_icon** is set to ***false***.

- **disable_watermark_mute_icon**: controls the "Your mic has been muted" watermark. Set it to ***false*** to enable watermarking, and to ***true*** to disable watermarking. By default, **disable_watermark_mute_icon** is set to ***true***.

Changing the logo watermark image

When creating your own logo watermark file (**watermark_icon.png**) for inclusion in a theme:

- The file must be PNG format and may include a transparent image (if it is not transparent the overlaid content will be fully opaque).
- We recommend that the watermark image is white with 40% transparency.
- The image should be 200 pixels wide x 100 pixels high. Larger images are automatically scaled to fit to the top left corner of the watermark area (and thus may lose some definition).
- There is an additional watermark file — **watermark_footer_icon.png** — which is used only on the 1 + 33 layout. It is the same size as the standard **watermark_icon.png** file, but it has inverted colors (so that it shows up on the white background of the layout).

Changing the "Your mic has been muted" watermark image

When creating your own "Your mic has been muted" watermark file (**watermark_mute_icon.png**) for inclusion in a theme:

- The file must be PNG format and may include a transparent image.
- The image should be a maximum of 200 pixels wide x 100 pixels high.
- It is not always appropriate in all call scenarios to state that the user can unmute themself in this way, such as when the user has dialed in over WebRTC, or for Guest participants in a VMR where all Guests have been muted.

Customizing the unused screen in a dual-screen system

The second screen in a dual-screen system displays a black screen by default if it is not being used to display conference content.

You can brand this second screen to display a customized image or text by configuring the **unused_screen** splash screen element.

Configuring the unused_screen splash screen element

The **unused_screen** splash screen element is configured by specifying a **splash_screens** object within your **themeconfig.json** file.

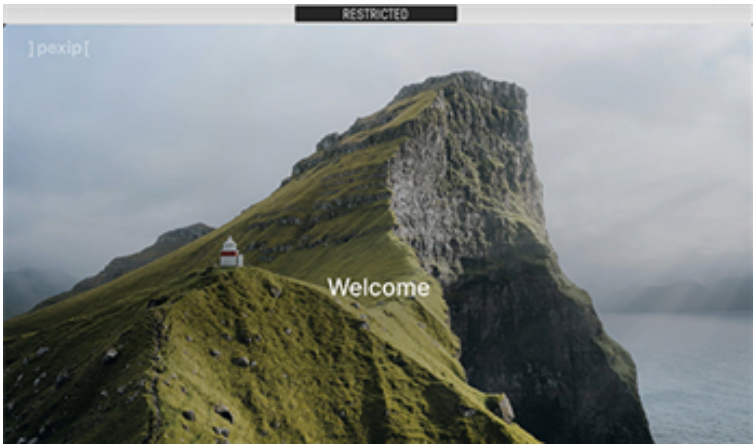
For example, you could create your own overlay svg file (icon_unused_screen.svg) and a background image file (background_unused_screen.jpg) and then define them as follows:

```
"splash_screens": {
  "screens": {
    "unused_screen": {
      "layout_type": "vertical",
      "elements": [
        {
          "type": "icon",
          "path": "icon_unused_screen.svg"
        }
      ],
      "background": {"path": "background_unused_screen.jpg"}
    }
  }
}
```

You can also use the **{conference_name}** variable within any text labels, for example:

```
"splash_screens": {
  "labels": {
    "unused_screen": "{conference_name}"
    },
  "screens": {
    "unused_screen": {
       "layout_type": "vertical",
       "background": {"path": "background_unused_screen.jpg"},
       "elements": [ {"type": "text", "label": "unused_screen"} ]
    }
  }
}
```

Content classifications



When content classifications are configured in a theme, the specified classification text is overlaid (similar to a watermark) onto the video layout and splash screens during a conference (Virtual Meeting Rooms, Virtual Auditoriums, and the calling party in Infinity Gateway calls). It is positioned above any in-conference indicators, and is displayed in dark text on a light background by default.

- Classification indicators are text only.
- They are not included in the base theme. You must create and upload your own theme that includes your classification text strings.
- The colors can be customized, but you cannot configure the position, font or size of the message.

You also have the option to trigger an audio announcement if the classification level changes during a conference.

Configuring and changing the classification message

Configuring the classification messages

To configure your content classification strings you must add a `classification` object to your **themeconfig.json** file, for example:

```
"classification": {
    "levels": {
        "0": "Unclassified",
        "1": "Official",
        "2": "Official Sensitive",
        "3": "RESTRICTED",
        "4": {"text": "Confidential", "color": "0xff000000", "bgcolor": "0xffffffff"},
        "5": "Secret",
        "6": "Top Secret"
    },
    "default": 3
}
```

- The level keys ("0", "1" etc.) must be numeric (but represented as strings). There's no limit to the number of levels, but the keys must be unique. The values must be strings, and can be set to display any message of your choice.
- You can optionally specify colors for the text and background for each classification level. To do this, use the extended form:
  `"0": {"text": "classification level", "color": "0xff000000", "bgcolor": "0xffffffff"}`
  Both color settings are optional. You can use either RGB hexadecimal notation (in the format 0xnnnnnn) or ARGB hexadecimal notation (0xnnnnnnnn).
- The default level is specified explicitly, as shown in the example above, and its numeric value must be a valid key in the levels map.
- Themes without this additional configuration do not show any classification messages.

Dynamically changing the classification message

When a conference is in progress, and its theme contains a `classification` object, the default level message is automatically displayed.

To dynamically display a different message i.e. change the classification level, you must use the Pexip client API. A meeting Host must **POST** a JSON object to:

/api/client/v2/conferences/<conference>/set_classification_level

where the JSON object is as follows:

{"level": 4}

and where 4 — or the level of your choice — is a valid level key from the theme.

Hosts can also use the client API to obtain the set of available levels and the currently active level by sending a **GET** request to:

/api/client/v2/conferences/<conference>/get_classification_level

This returns a JSON document in the usual way, for example:

```
{
    "status": "success",
    "result": {
        "levels": {
            "0": "Unclassified",
            "1": "Official",
            "2": "Official Sensitive",
            "3": "RESTRICTED",
            "4": "Confidential",
            "5": "Secret",
            "6": "Top Secret"
        },
        "current": 1
    }
}
```

Announcing a change to content classification

When there is a change to the content classification level in a conference, the **conf-classification-changed_48kHz_mono.wav** audio file is played out to all conference participants.

However, **conf-classification-changed_48kHz_mono.wav** is an empty file by default (i.e. there is no actual audio message played out).

Therefore, if you want to play out a message then you must supply a substitute **conf-classification-changed_48kHz_mono.wav** file in your customized theme.

We have provided an example audio file at https://dl.pexip.com/resources/themes/index.html which contains the message "The classification level of this call has changed".

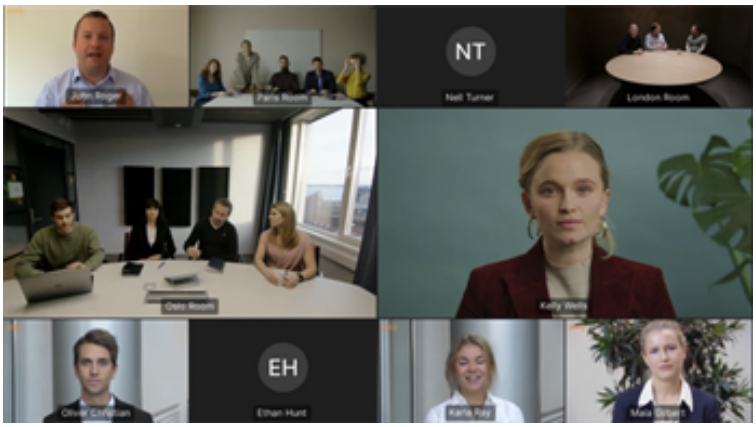
Participant avatars

You can use the **enable_avatars_in_mix** setting in the **themeconfig.json** file to control whether any audio-only or video-muted participants are represented as an avatar in the main video layout (excluding Adaptive Composition and Teams-like layouts) instead of those participants being counted in the "notch" indicators.

The avatar/image used to represent the participant can come from the following sources:

- User records (see Applying user records)
- External policy avatar lookup
- Exchange Online (Microsoft Teams conferences only)

If a participant image is not available then a substitute graphic is generated based on the initials of the participant's name.



When **enable_avatars_in_mix** is enabled you can optionally use the following additional theme settings/files to customize the appearance of the avatar:

Setting / filename	Description	Default
avatar_in_mix_text_color	The text color of the participant's initials (when there is no graphic avatar).	#FFF7F7F7A
avatar_in_mix_shape_color	The background color of the avatar when it contains the participant's initials.	#33DCDCDC
background_avatar_in_mix.jpg	Background image to display in the layout slot occupied by the avatar. Recommended size: 1920x1080	

Custom layouts

You can design your own layouts and use them in the same way as the standard layouts and example custom layouts that are shipped by default.

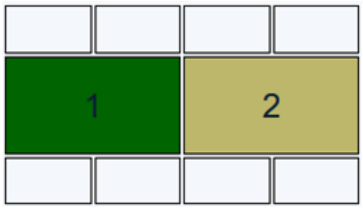
Each layout is specified through a JSON file that must be included in your theme package.

A **customlayouts** license is required to upload your own custom layout files.

By using this feature, Pexip may reserve its right to waive at least some of the obligations stated in paragraph 9 of your Service Provider Agreement related to this feature.

Designing the layout

The layout is designed via a grid of up to 10x10 cells. Within the grid you define up to 26 slots, which are used to display video participants or a presentation stream. A slot can occupy multiple cells i.e. the number of cells in the slot controls the size of the participant/presentation stream within that slot. The NxN grid and each slot must be a **square** dimension i.e. the same number of vertical and horizontal cells, to cater for a 16:9 aspect ratio.



The example image above shows a grid size of 4x4 (thus with 16 grid cells). This layout has 2 slots for video/presentation streams that are defined on the 4x4 grid, where 4 grid cells are used to define a single slot "1" and similarly 4 grid cells for slot "2". The 4 grid cells at the top and the 4 cells at the bottom are unused.

The layout must follow these rules:

- Slots cannot overlap each other.
- The maximum number of slots (i.e. video participants plus a presentation stream) that can be defined is 26.
- Any grid cells that are not used as slots will just be empty space in the layout (and display the background color).
- The grid size can be a maximum of 10x10.
- The grid size must be a square dimension, i.e. number of rows = number of columns.

Using the layout

After loading a layout as part of a theme, you can use them in the same way as the standard layouts that are shipped by default. Note that:

- The VMR (or the relevant conference service / gateway call) must have access to the theme that contains the custom layout. The layouts included in themes follow the standard theme inheritance rules.
- You refer to the custom layout via its specified **name** in the JSON file.
- You may want to consider adding the custom layout name to the **dtmf_allowed_layouts** list in the associated **themeconfig.json** file to enable DTMF cycling for that layout.
- You do not have to import or specify any related SVG file for the custom layout. The system generates the appropriate SVG file automatically and provides it to the web app for use on the **Meeting layout** tab.

Note that the base theme contains three example custom layouts (one_main_twelve_around, two_mains_eight_around and one_main_nine_around).

Layout designer tool

You can use our layout designer tool to create and maintain your own custom layouts, or configure them manually as described below. You can use the tool to:

- Load in existing layout files to edit and view layouts.
- Validate a layout file.
- Design new layouts from scratch, with visual validation.
- Export the designed layouts into a JSON file, for inclusion in your theme package.
- Export an SVG image of the layout (although this is not required by Pexip Infinity or the Pexip web app).

Ensure that you select the appropriate Pexip Infinity version when using the layout designer.

Manually creating a custom layout file

Each custom layout file must be named with a prefix of "custom_layout_" and be in a .json format, for example **custom_layout_mylayout.json**.

The JSON file has the following primary fields:

Field name	Description
name	This is the name of the layout and is what is used to refer to the layout throughout the system, in API calls and in the Administrator interface. It is a string which is limited to a maximum of 50 alphanumeric and "_" characters and no spaces.
modes	This is a list of dictionaries that define the basic layout for displaying video participants. It must contain at least one valid mode. See below for more details.
pres_modes	This is an optional list of dictionaries that define the layout when a presentation is in progress and the layout is including the presentation content in the layout mix. See below for more details.

Field name	Description
dual_screen_enabled	This is an optional boolean (false by default) that controls whether multiscreen is supported. • Must be set to true to use a custom layout as the dual screen layout (otherwise dual screen users will see 1:7 on both screens). • If set to true without specifying **dual_screen_modes** then the main layout is duplicated on both screens. The layouts build out by filling one slot each per screen, alternating until each screen is full.
dual_screen_modes	Each mode pair is defined as per this schema: `{ "primary_modes": list[modes], "secondary_modes": list[modes] }` where **modes** have the same schema as the normal layout modes. Each screen can have its own layout configuration applied, up to a total of 26 participants (slots) across all screens. See the one_main_nine_around custom schema below for example usage.
remove_self	This is an optional boolean that controls whether "self view" is part of the layout mix. It is true by default (i.e. self view is removed from the mix, as per all of the standard layouts).
background_color	This is an optional string that defines the background color for the layout in RGB hexadecimal notation in the format #nnnnnn, for example: `"background_color": "#fffff0"`
watermark_position	Specifies where to place the watermark. Options are: "top-left", "top-right", "bottom-left", or "bottom-right". For example: `"watermark_position": "top-right"`
notch_color	Specifies either a "dark" (the default) or "light" mode choice for the information bar (notch) at the top-center of the layout. For example: `"notch_color": "light"`

This is the basic schema:

```
{
    "name": string,
    "modes": [<basic mode>],
    "pres_modes": [<presentation mode>],
    "dual_screen_enabled": bool,
    "dual_screen_modes":
        {
            "primary_modes": list[modes],
            "secondary_modes": list[modes]
        },
    "remove_self": bool,
    "background_color": string,
    "watermark_position": string,
    "notch_color": string
}
```

Layout modes (basic and presentation)

A layout contains one or more modes. You must define one set of modes (`"modes"`) for the basic layout (where there are only other video participants), and you can optionally define a set of modes for when a presentation is in progress (`"pres_modes"`) and you want to include the presentation in the layout mix.

Each mode is defined for a minimum and maximum number of participants, and is used to determine how the layout is built as video participants are included or removed from the video mix, or when presentation content is received. For example, the basic Adaptive Composition layout has 4 modes: full screen with 1 slot (used when there is just 1 active video participant), side by side 2 slots (2 participants), 2+3 slots (3-5 participants), and finally 2+3+7 slots (6-12 participants).

When defining presentation modes, the presentation stream is always displayed as the 1st participant, and it then follows the same rules as a basic layout.

A mode dictionary has the following fields:

Field name	Description
grid_size	Specifies the size (number of cells) of the grid. It is a tuple of integer values where both integers must be the same and a maximum of 10x10.
constraints	A dictionary containing **min_slots** and **max_slots**, where **min_slots** and **max_slots** are in the range 1-26 inclusive and **min_slots** <= **max_slots**.
slots	A list of slot dictionaries and must have the same number of slots as **max_slots**. The slots fill the layout in the order they are defined in the list. Each slot contains the following fields (mandatory unless stated otherwise):
	start_x: The x coordinate on the grid where to place the top left corner of the slot.
	start_y: The y coordinate on the grid where to place the top left corner of the slot. Note that coordinates start from the top-left of the grid, where the top-leftmost grid cell has coordinates x=0 and y=0.
	width: An integer of the number of grid cells to span this slot across (width and height are the same, as the slots are all 16:9 aspect ratio).
	is_speaker_slot: Optional field. A slot can be defined as a speaker slot which means active speakers based on voice activity will be put in these slots. If multiple slots are defined as speaker slots, the people speaking in those slots won't swap around unless someone outside the speaker slot speaks. By default, the first slot is the speaker slot if this field is not specified. Multiple speaker slots must be defined in consecutive order and the first slot must be a speaker slot.
	is_presentation_slot: This field only applies to presentation modes. The first slot in a presentation mode must contain: `"is_presentation_slot": true` Only one slot (the first slot) can be set to true, and the field is optional (i.e. false) in every other slot.
slot_padding	Optional field. This is a float value between 0 and 0.1 determining the spacing between slots. Default=0.001.

This is the mode dictionary schema:

```
{
    "grid_size": [int, int],
    "constraints": {
        "min_slots": int,
        "max_slots": int
    },
    "slots": [
        {
            "start_x": int,
            "start_y": int,
            "width": int,
            "is_speaker_slot": bool,
            "is_presentation_slot": bool
        }
    ],
    "slot_padding": float
}
```

Themes used by Call Routing Rules (gateway calls)

You can assign a theme to a Call Routing Rule. Only a subset of the theme's image files are used when they are applied to an Infinity Gateway call. Thus, you only need to customize a subset of the files for any themes that are only assigned to one or more Call Routing Rules. The various tables on this page indicate, where appropriate, which image files are used in gateway calls.

Also, other than for Google Meet and Microsoft Teams integration scenarios, the **audio** files are **not** used in themes assigned to Call Routing Rules.

Integrating Google Meet with Pexip Infinity

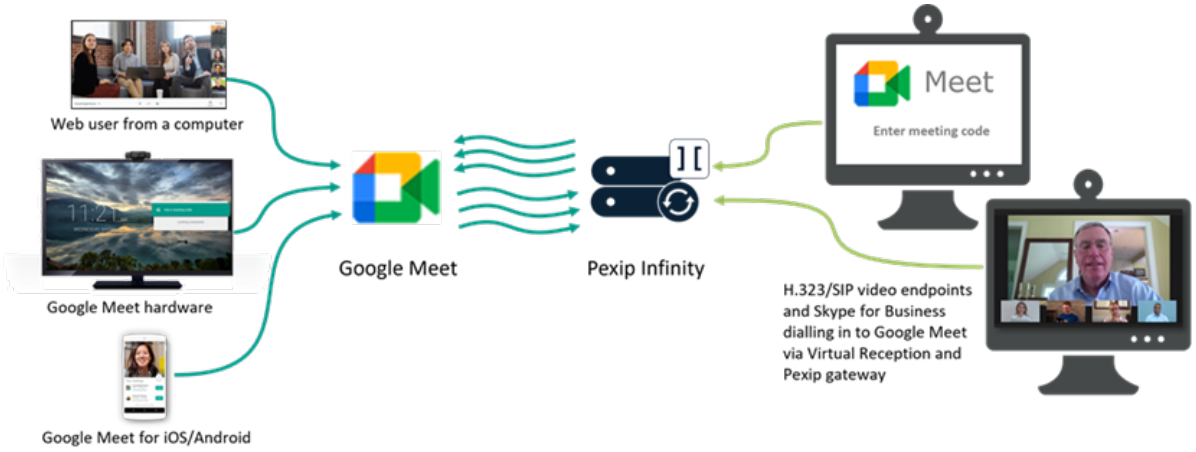
Introduction

The Infinity Gateway provides any-to-any video interoperability with Google Meet. It enables the seamless interoperation of HD video between Google Meet conferences and:

- H.323 & SIP room-based videoconferencing systems, including Cisco, Poly, Lifesize, and others
- Skype for Business
- Browser-based video (WebRTC)

Third-party systems can connect to Google Meet conferences via the Infinity Gateway either by dialing the conference directly or via a Virtual Reception (IVR).

You can also join Google Meet meetings where that meeting is being hosted by an external third-party organization (even if the host's organization has not enabled Pexip interoperability themselves) — this is referred to as "SIP Guest Join". Note that this feature works by routing the call to the external organization via the Pexip Service.



Integration features

The Google Meet in-call features that are supported via the Infinity Gateway include:

- Active speaker switching
- Content sharing (both directions)
- Raise/lower your hand
- Raised hand, recording and AI indicators
- Bandwidth optimizations

Pexip interoperability can be used with all paid Google Workspace licenses.

Note that Google Meet is inherently a dial-in service i.e. you can only dial **from** a third-party video system into Google Meet. You cannot dial **out** from Google Meet to a SIP, H.323 device etc — instead, you have to send the relevant joining instructions/invitation to the user of that device.

Deployment environments

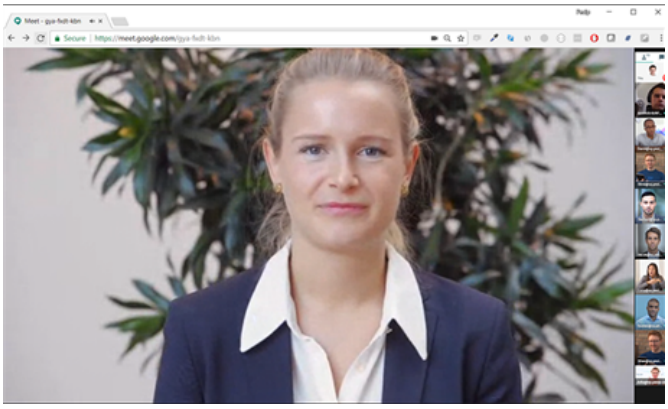
The Pexip Infinity platform can be deployed in any of its supported environments such as on-premises or in a public or hybrid cloud (including Google Cloud Platform).

Pexip Infinity has a close integration with Google Meet and uses Google Meet APIs to provide Infinity's interoperability features. Even though Pexip strives to maintain backwards compatibility between older versions of Pexip Infinity and the latest release of Google Meet, to ensure compatibility with the latest updates to Google Meet we recommend that you aim to keep your Pexip Infinity deployment up-to-date with the latest Pexip Infinity software release. If, for example, you have a large Pexip deployment for non-

Google Meet related services, and you have stringent upgrade procedures meaning that you do not always keep your Infinity software up-to-date with the latest release, you may want to consider deploying a second instance of the Pexip Infinity platform that is dedicated to your Google Meet interoperability requirements, and which can be managed separately and upgraded more frequently.

Native user experience for all participants

All participants receive the appropriate native user experience of the Google Meet conference when there is a mix of direct and gatewayed third-party participants.

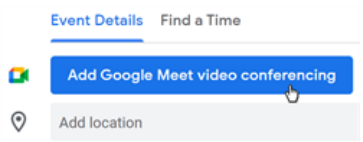


Google Meet experience when third-party VTC systems are connected to the conference

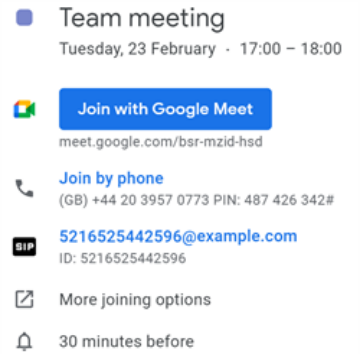
Third-party VTC system experience (Pexip's standard 1+7 layout) when connected to a Google Meet conference

Scheduling and joining conferences

Users in your organization can schedule meetings as normal via their Google Calendar or Outlook (with the Google Outlook add-in for Office 365), and choose to add Google Meet conferencing within their event options. The joining instructions for internal and external standards-based video conferencing systems, and Skype for Business users, are then automatically included in, or linked from, the calendar invitation and event.



All calls are routed into the Google Meet conference by means of the meeting ID that is associated with that conference.



Meeting IDs are generated automatically by Google Meet and are added to calendar events and invites when events are created. Skype for Business joining instructions are available from a "More joining options" link in the calendar event or invite.

For ad hoc conferences, links to the meeting IDs are presented to the host when the meeting is initiated and are also available from the **Meeting details** option while in the conference.

Enabling access and admitting external participants into Google Meet conferences

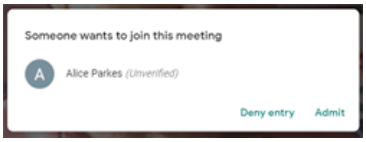
After you have installed and performed the basic configuration of your Pexip Infinity platform, you have to link your Pexip platform to your Google Workspace account, so that it can route calls into your Google Meet conferences. This is handled via **access tokens**, which are private codes that can be used by a third-party system, such as Pexip Infinity, to identify your account.

You can set up two types of access tokens in your Google Workspace account: a **trusted** and an **untrusted** token. You can use these two token types to control whether an endpoint that is routed via Pexip Infinity into a Google Meet conference is automatically admitted into the conference, or whether an existing conference participant has to explicitly admit it into the conference. When you configure Pexip Infinity, you decide which type of token to associate with the access rules and dial patterns that allow devices to be routed into Google Meet conferences.

Pexip Infinity also adds an additional layer of trust control by including an explicit setting on each Call Routing Rule to indicate whether or not the devices that are routed via that rule into Google Meet are trusted endpoints from Pexip Infinity's perspective (for example, you could treat the device as trusted if the caller is coming from a specific location, or if the device is registered to Pexip Infinity).

In essence, when Pexip Infinity routes a call to Google Meet, it provides three pieces of information:

- the meeting ID (so that the endpoint joins the correct conference)
- the access token, which can be either a "trusted" or "untrusted" token
- a "domain member" flag, which indicates if the calling endpoint is a trusted endpoint from Pexip Infinity's perspective.



If the access token is a trusted token **and** the endpoint is trusted by Pexip Infinity, then the device is automatically admitted into the conference.

In all other cases, the device has to be explicitly admitted into the conference (this takes the form of a popup dialog as shown right, which is displayed to all participants who are connected directly to the conference). Any of those participants can then choose to allow (admit) or deny access. Guest devices must be admitted within 60 seconds.

Since version 31 of Pexip Infinity, to enable Pexip's gateway interoperability, you must also request and upload a gateway token (provided by Pexip) that is used to authenticate your Pexip Infinity system with Google Meet.

See Configuring Google Workspace for Google Meet integration and Configuring Pexip Infinity as a Google Meet gateway for details about configuring access tokens, requesting and uploading a gateway token, and Registering devices to Pexip Infinity for registration information.

Using SIP Guest Join

Pexip's standard Google Meet interoperability solution allows your own video conferencing endpoints (and your guests) to join Google Meet meetings that you are hosting. The "SIP Guest Join" (SGJ) feature lets you join Google Meet meetings with your own video conferencing endpoints where that meeting is being hosted by an external third-party organization who has not enabled Pexip interoperability themselves.

Note that this is a guest join experience, therefore someone has to be in the Google Meeting to admit you from the lobby. If nobody admits you within 30 seconds, you will have to dial again.

If you are invited to an externally-hosted Google Meet meeting, you can join it via One-Touch Join on your video conferencing endpoint.

Joining from a meeting room

1. Invite the room (as a room resource) to the meeting. Either:
   - Copy the contents of the meeting invite, and use that to book the room (in this case the accept/reject message from the room is **sent to you**); or
   - Forward the meeting invite to the room's mailbox (in this case the accept/reject message is **sent to the meeting organizer**).
2. At the start time of the meeting: enter the room and press the **Join** button.

Joining from a personal video endpoint (that is set up for One-Touch Join)

1. Accept the meeting invite as usual (OTJ only looks for "accepted" meetings).
2. Press **Join** at the start of the meeting.

See Enabling SIP Guest Join for configuration and setup details.

Raising and lowering hands

Participants connecting to Google Meet via Pexip Infinity can raise and lower their hand in meetings, and see that other participants have their hand raised.

Raise and lower your hand

VTC participants can raise or lower their hand via DTMF/keypad controls.

This is via ***3** by default but can be customized via themes.

Raised hand indicators

Notifications are shown to any VTC participants if a Google Meet participant raises their hand:

- It indicates the number of participants with a raised hand.
- The indicator bar alternates every 5 seconds to show a message containing the name of the first participant in the list of participants with a raised hand.

Note that you cannot use the Pexip apps to lower the hand of a Google Meet participant.

Presentation / content sharing

If a participant who is using the native Google Meet web client starts presenting, any VTC participants can see both the presentation content and the presenter's standard video stream. Similarly, if a VTC participant starts presenting, the other participants in the conference will see both the presentation stream and the video from that participant.

Recording, streaming, transcribing, note taking, media collection and AI

If a Google Meet conference is recorded, streamed or transcribed, taking smart notes via Gemini, having its media collected, or AI features are enabled, then an in-conference indicator is displayed and audio prompts indicating that streaming/recording (), transcribing (), note taking (), media collection (), or AI () has been started/stopped are played to callers who are gatewayed via Pexip Infinity into the conference. Distinct messages are used in each case.

Configuring Google Workspace for Google Meet integration

This topic explains the Google Workspace configuration steps that are required when integrating Pexip Infinity with Google Meet.

It covers the following processes:

- Generating your interoperability access tokens
- Google Meet interoperability settings
- Host management and video lock

You can configure your Google Meet settings from the Google Admin console via **Apps > Google Workspace > Google Meet**.

For more information about setting up Google interoperability, see https://support.google.com/a/answer/7673980.

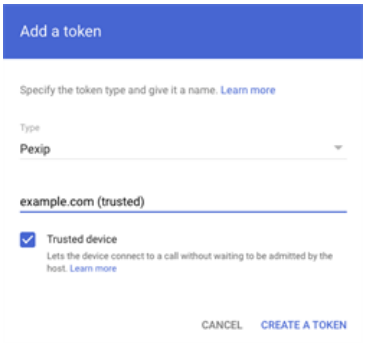
Generating your interoperability access tokens

Interoperability access tokens are the private codes assigned to your Workspace account that are used by Pexip Infinity when it routes calls into your Google Meet conferences. Tokens can be defined as "trusted" or "untrusted", see Enabling access and admitting external participants into Google Meet conferences for details.

The generated tokens are only displayed once in Google Workspace, at the time of creating them. These tokens need to be configured on your Pexip Infinity system. Therefore you should have your Pexip Infinity Administrator interface open and available at the same time as you are generating the tokens within Google Workspace.

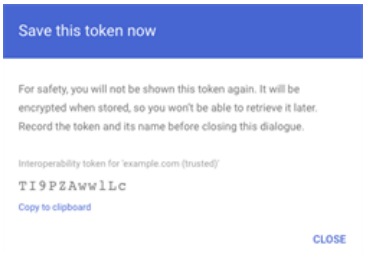
To set up your trusted and untrusted interoperability access tokens from the Google Workspace administrator console:

1. Go to **Apps > Google Workspace > Google Meet > Interoperability tokens**.
2. Select **Add Token**.
3. Select a **Type** of ***Pexip***.
4. Enter a token name, for example the domain of your Pexip Infinity deployment plus we recommend adding a "trusted" or "untrusted" label, for example "example.com (trusted)".
5. Select whether it is a **Trusted** token or not (to be used with trusted devices).



6. Select **Create a token**.

   You are shown the generated interoperability token.



   This is the token you must enter into Pexip Infinity.
7. Use the option to **copy** the token to your clipboard.

   This is the only time you will be able to see the token before it is stored and encrypted in Google Workspace.
8. Switch to your Pexip Infinity Administrator interface and configure this interoperability token in Pexip Infinity:

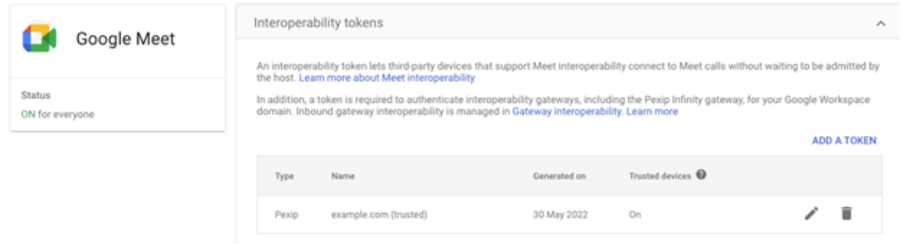
a. Go to **Call Control > Google Meet Access Tokens** and select **Add Google Meet access token**.
b. Add the details of your token:

Name	The name that you specify here is used elsewhere in the Pexip Infinity interface when associating the token with a Call Routing Rule or Virtual Reception, so we recommend including an indication if it is your trusted or untrusted token.
Access token	Paste in the access token from your clipboard.

c. Select **Save**.

9. You can now return to the Google Workspace console and **Close** the token dialog.
10. If you want to create a trusted and an untrusted token, repeat the above steps to generate the second interoperability token.

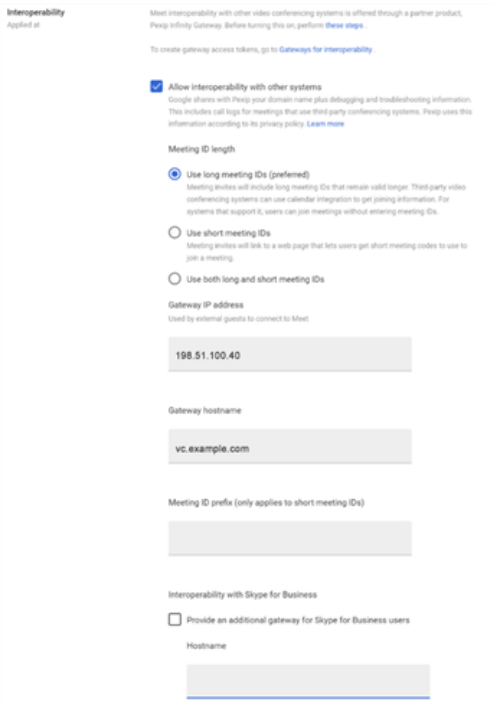
    You can create as many trusted and untrusted tokens as required, although one of each type is normally sufficient. Service providers may need to apply multiple pairs of access tokens for each tenant they are managing.



See Configuring Pexip Infinity as a Google Meet gateway for more information about Pexip Infinity configuration requirements.

Google Meet interoperability settings

You also need to enable Google Meet interoperability to allow other systems to dial into your Google Meet calls. You do this via **Apps > Google Workspace > Google Meet** and then configure the **Meet video settings** and select the **Interoperability** settings.



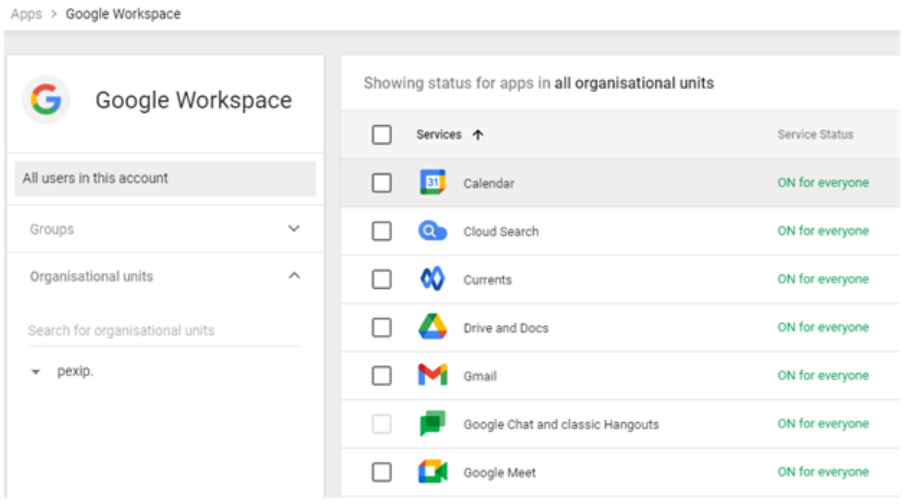
You should configure the following **Interoperability** options:

Name	Description
Allow interoperability with other systems	Select this option to enable gateway interoperability and to configure the other settings. By default, interoperability is made available to everyone in the organization; see Controlling access to gateway interoperability for details about how to limit access.
Meeting ID length	Select ***Long meeting IDs***. (Short IDs are the previous, deprecated method.)
Gateway IP address and Gateway hostname	The IP address and hostname address you specify here are used when Google Meet generates the addresses that allow third-party systems to access the conference. They are used to direct callers to your Pexip Infinity Conferencing Nodes that will then connect them into the Google Meet conference. • **Gateway IP address**: set this to the IP address of one of your Conferencing Nodes. • **Gateway hostname**: set this to the name of the DNS SRV record that you have set up for your Conferencing Nodes (e.g. vc.example.com). Alternatively you can use the FQDN of one of your nodes (e.g. px01.vc.example.com). Ensure that you refer to Conferencing Nodes that will be routable from those systems and devices that will be using those dial-in addresses. See DNS record examples for more information about enabling endpoints to route their calls to Conferencing Nodes.
Meeting ID prefix	Leave blank.
Provide an additional gateway for Skype for Business users Hostname	In most cases this option is not required and should not be selected — normally the address shown via the "More joining options" link in the calendar event or invite is suitable for both SIP devices and Skype for Business users to join your meetings (as they will typically use the same **Gateway hostname** domain). Only select this option if you use a different domain for Skype for Business and you need to generate a separate joining address for Skype for Business users. In which case, select this option and specify the domain used for Skype for Business calls as the **Hostname**.

Controlling access to gateway interoperability

When you enable gateway interoperability (by selecting **Allow interoperability with other systems**), it is made available by default to everyone in the organization.

However, you can restrict access to the interoperability functionality to specific Organizational Units (OUs) or groups, although it is not currently possible to enable it for individual users. See https://support.google.com/a/answer/9493952 for more details about how to do this.

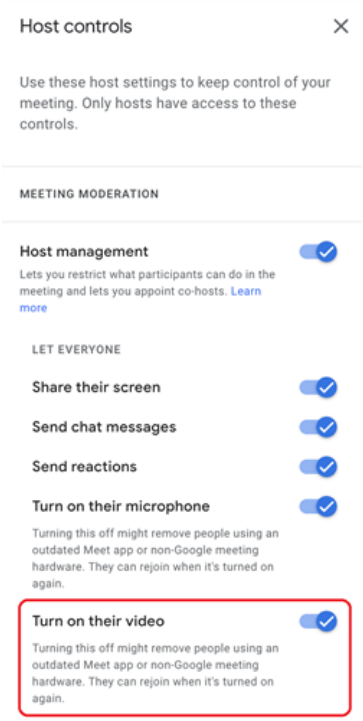


Note that:

- You can configure a subset of specific OUs/groups for interop access **before** turning on the main **Allow interoperability with other systems** to avoid temporarily making it available to everybody in the organization. If you subsequently want to broaden the access you can add other OUs/groups or just enable it for everybody in the organization.
- The access tokens apply to the entire Google Workspace tenant (i.e. to all specified OUs, groups or the entire organization, as appropriate).

Host management and video lock

Within a meeting, a meeting host must ensure that the **Turn on their video** setting (within the **Host management** options for that meeting) is enabled, otherwise VTCs will not be able to join the meeting.



Configuring Pexip Infinity as a Google Meet gateway

The Infinity Gateway provides any-to-any video interoperability with Google Meet.

Third-party systems can connect to Google Meet conferences via the Infinity Gateway either by dialing the conference directly or via a Virtual Reception (IVR).

You can also join Google Meet meetings where that meeting is being hosted by an external third-party organization (even if the host's organization has not enabled Pexip interoperability themselves) — this is referred to as "SIP Guest Join". Note that this feature works by routing the call to the external organization via the Pexip Service.

This topic describes the configuration and administrative steps that are required to use Pexip Infinity as a Google Meet gateway. It assumes that you have already performed a basic installation and configuration of Pexip Infinity. This topic covers:

- Ensuring a Google Meet (ghm) license is installed
- Configuring your access tokens
- Requesting and uploading your gateway token
- Configuring Virtual Receptions and Call Routing Rules
  - Routing directly via the Infinity Gateway
  - Routing indirectly via a Virtual Reception (IVR gateway)
- Enabling SIP Guest Join
- Configuring the layout seen by VTC participants (and other participant controls)

- DNS and ports requirements
- Call and participant status
- Additional deployment information

See Introduction for more information about the user experience within Google Meet conferences.

Ensuring a Google Meet (ghm) license is installed

You must have a **ghm** license enabled on your platform (**Platform > Licenses**). This allows you to configure access tokens and route calls to Google Meet.

The ghm license specifies the total number of access tokens you can configure, and is required in addition to the standard Pexip Infinity call licenses. If necessary, contact your Pexip authorized support representative to purchase the required license.

Configuring your access tokens

All communication between Pexip Infinity and Google Meet is authenticated by access tokens that identify your Google Workspace account (see Generating your interoperability access tokens).

To configure your trusted and untrusted access tokens in Pexip Infinity:

1. Go to **Call Control > Google Meet Access Tokens** and select **Add Google Meet access token**.
2. Add the details of your **trusted** token:

Name	The name that you specify here is used elsewhere in the Pexip Infinity interface when associating the token with a Call Routing Rule or Virtual Reception, so we recommend including an indication if it is your trusted or untrusted token.
Access token	The access token value as shown in your Google Workspace account when you generated it.

3. Select **Save**.
4. Repeat the above steps to add the details of your **untrusted** token.

These tokens will now be available to associate with any Virtual Receptions and Call Routing Rules that you configure (as described below) to handle Google Meet conferences.

Note that:

- The access tokens apply to the entire Google Workspace tenant, but you can enable interoperability on a per-OU (organizational unit) basis within Google Workspace.
- Service providers may need to apply multiple pairs of access tokens for each tenant they are managing.

Requesting and uploading your gateway token

Google Meet is transitioning to a new method of authenticating third-party systems that join a Google Meet conference via Pexip's interoperability gateway.

The new method uses the JWT (JSON Web Token) standard. To use JWT as your authentication method you need to upload to your Pexip Infinity system a token that has been signed by Pexip. This token then provides the authentication between your Pexip Infinity system and Google Meet whenever a third-party device uses Pexip's interoperability gateway to join a Google Meet conference.

If you do not upload a gateway token you will still be able to use Pexip's interoperability gateway in the short term, but we expect Google Meet to enforce JWT authentication in the near future. You must obtain and install your gateway token as soon as practicable after upgrading to version 31 or later.

Note that you must still create and configure your access tokens (as described above), in addition to your gateway token. The access tokens continue to identify your account and determine whether an endpoint is trusted or not.

To obtain and upload your gateway token to Pexip Infinity:

1. Go to **Call Control > Google Meet Gateway Token**.
2. In the **Gateway token request** section, select **Download**.

   This generates and downloads a token request file called **gms_gateway_token_request.csr**.

3. Send the **gms_gateway_token_request.csr** file and the **License entitlement key** from your ghm license (**Platform > Licenses**) to your Pexip authorized support representative and request your gateway token for Google Meet interoperability.
4. When you receive your gateway token, return to the **Call Control > Google Meet Gateway Token** page.
5. Upload your token into the **Gateway token upload** section. You can either paste the token contents into the text area, drag and drop the gateway token file into the text area, or use the **Browse...** button to select your file.

   You must use the file generated by Pexip (typically with a .CRT or .PEM extension). Do **not** upload your original token request file (with a .CSR extension).
6. Select **Complete**.

   Your gateway token will now be used to authenticate your Pexip Infinity system with Google Meet.

Token status and requesting a new token

The **Gateway token status** section at the bottom of the page tells you when your gateway token will expire. A "Google Meet Gateway Token expiring" alarm is also raised when the gateway token is due to expire within the next 30 days.

You can only have one gateway token uploaded at any one time. In the future, when your token's expiry date is approaching, you'll need to request a new token and then upload it (replacing the existing token) using the same procedure as above.

Configuring Virtual Receptions and Call Routing Rules

There are two ways in which you can configure Pexip Infinity to route calls into Google Meet conferences:

- **Routing directly via the Infinity Gateway**: here you use a Call Routing Rule to route incoming calls for specific alias patterns — that will typically include the meeting ID — directly into the relevant Google Meet conference. This means that the endpoint can dial an alias, such as **1234567890123@example.com** and be taken directly into the conference.
- **Routing indirectly via a Virtual Reception**: here you configure Pexip Infinity to act as an IVR gateway or "lobby" by configuring a Virtual Reception to prompt the caller to enter the meeting ID of the required conference, and then use a Call Routing Rule (typically the same rule as used for direct routing) to route the call into the Google Meet conference. This means that the endpoint can dial a general alias, such as **gmeet@example.com** and then enter the specific meeting ID, such as **1234567890123**, and then be transferred into the conference.

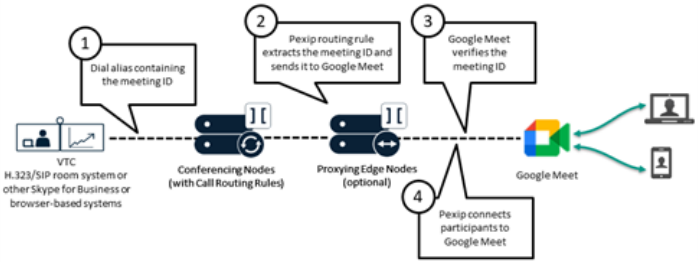
You can use either or both of these two methods, depending upon your requirements. The configuration required for direct and indirect routing is explained below.

Depending on your dial plan requirements, you may want to use multiple Call Routing Rules, where some rules use a trusted token and other rules use an untrusted token. For example, if you want to associate calls received via a particular location as trusted, and all calls received in other locations as untrusted then you will need to configure two rules — one rule for calls received in the trusted location that is associated with the trusted token, and then another lower priority rule that is associated with the untrusted token.

Routing directly via the Infinity Gateway

To route calls to Google Meet conferences directly via the Infinity Gateway you need to consider the alias pattern that participants will dial to access the Google Meet conferences. This will typically include the meeting ID element, for example the pattern could be just **<meeting ID>** or **<meeting ID>@<domain>** i.e. the meeting ID and then, optionally, the domain of your Pexip Infinity platform, for example **1234567890123@example.com** to access a Google Meet conference with a meeting ID of 1234567890123.

You also need one or more Call Routing Rules that match that alias pattern and, if necessary, transform it such that it contains just the Google Meet meeting ID which it can then use to connect to the conference. You can use multiple rules to differentiate between devices that are to be treated as trusted or not from Pexip Infinity's perspective, and hence which type of **Access token** is selected and whether **Treat as trusted** is selected or not:



- If devices register to Pexip Infinity we recommend using two gateway rules: one higher-priority rule to specifically handle registered devices, and one lower-priority rule to handle any device (registered and non-registered).

- If you are using third-party call control systems you also may want to use different rules to distinguish between calls arriving at Conferencing Nodes in different locations.

To configure each rule:

1. Go to **Services > Call Routing** and select **Add Call Routing Rule**.
2. Configure the following fields (leave all other fields with default values or as required for your specific deployment):

Option	Description
Name	The name to use to refer to this rule.
Priority	Assign the priority for this rule. If you are creating multiple rules where one rule has **Match incoming calls from registered devices only** selected, and other rules do not have this option selected, then ensure that the rules that do have **Match incoming calls from registered devices only** selected have a higher priority (lower number) than those rules where it is not selected (to avoid the call matching first against the "not selected" rule if all of the other rule settings are the same).
Incoming gateway calls	Ensure this option is selected.
Outgoing calls from a conference	Leave unselected.
Calls being handled in location	Applies the rule only if the incoming call is being handled by a Conferencing Node in the selected location. To apply the rule regardless of the location, select ***Any Location***.
Match incoming calls from registered devices only	If devices register to Pexip Infinity, we recommend using two gateway rules: one higher-priority rule to specifically handle registered devices, and one lower-priority rule to handle all devices (registered and non-registered). ◦ Select this option if you want the rule to apply only to calls received from devices that are registered to Pexip Infinity. ◦ Leave this option unselected if you want the rule to apply regardless of whether the device is registered to Pexip Infinity.
Match Pexip apps (WebRTC / RTMP) Match SIP Match Skype for Business (MS-SIP) Match H.323 Match Teams	Select one or more of the match options as appropriate, depending on which types of systems/protocols you want to offer interoperability into Google Meet.
Match against full alias URI	Leave unselected.
Destination alias regex match	Enter a regular expression that will match the calls that are sent to a Google Meet conference. For example, to match an alias in the form **<meeting ID>** or **<meeting ID>@example.com**, you could use: **(^\d{12,13})($\|(@example\.com$))**
Destination alias regex replace string	If required, enter the regular expression string to transform the originally dialed (matched) alias into the meeting ID to use to place the call into the Google Meet conference. If you do not need to change the alias, leave this field blank. When used with the example for **Destination alias regex match** shown above you would use: **\1** which would extract just the <meeting ID> element of the alias.

Option	Description
Call capability	To support incoming calls via SIP, H.323 and WebRTC/RTMP, ***Same as incoming call*** is recommended as this makes the most efficient use of Pexip Infinity resources. If you also want to support calls from Skype for Business, then you should select ***Main video + presentation*** instead to ensure that any SfB participants are able to escalate from audio-only to a video call after joining the conference (alternatively you can configure a separate rule just for matching incoming calls from Skype for Business and set that rule to use ***Main video + presentation***).
Theme	If required, assign a customized theme to this rule (which will then be used for callers that use this rule to gateway into Google Meet). For example, the theme could use alternative labels on some of the splash screens that are displayed when connecting to a conference. You can also customize the in-conference DTMF/keypad layout control and raise/lower hand options, or the endpoints that have their own in-built audio denoising.
Call target	Select ***Google Meet meeting***.
Outgoing location	If required, you can ensure that the outgoing call to the Google Meet conference is handled by a Conferencing Node in a specific location. If an outgoing location is not specified, the call is placed from a Conferencing Node in the same location as the Conferencing Node that is handling the incoming call.
Access token	Select the name of the access token to use to resolve Google Meet IDs. You should select either a trusted or untrusted type of token, depending on whether you want to enable the device to be automatically admitted into the Google Meet conference (subject to also being a trusted endpoint from Pexip Infinity's perspective i.e. if the rule also has **Treat as trusted** enabled). Typically, you will use a trusted token if **Treat as trusted** is selected, and an untrusted token if **Treat as trusted** is not selected.
Treat as trusted	Select **Treat as trusted** if you want Google Meet to treat the devices routed via this rule as part of the target organization for trust purposes. Typically you will select this option if the rule is handling devices that are trusted from Pexip Infinity's perspective, for example, you could treat the device as trusted if the caller is coming from a specific location, or if the device is registered to Pexip Infinity. This setting is used in conjunction with the **Access token** setting to control whether the device is automatically admitted into the Google Meet conference.
Denoise audio	Applies to Google Meet integrations only. Controls whether to remove background noise from audio streams as they pass through the infrastructure. Some endpoints perform their own built-in denoising, in which case, to avoid performing denoising twice, you can configure via the theme's **vendordata.json** file a list of endpoints that perform their own denoising. See Rules and requirements for customized themes for more information. Default: enabled.

3. Select **Save**.
4. If you are creating multiple rules, for example when handling whether a device is registered to Pexip Infinity or not, return to step 1 and create the next rule.

Add Call Routing Rule

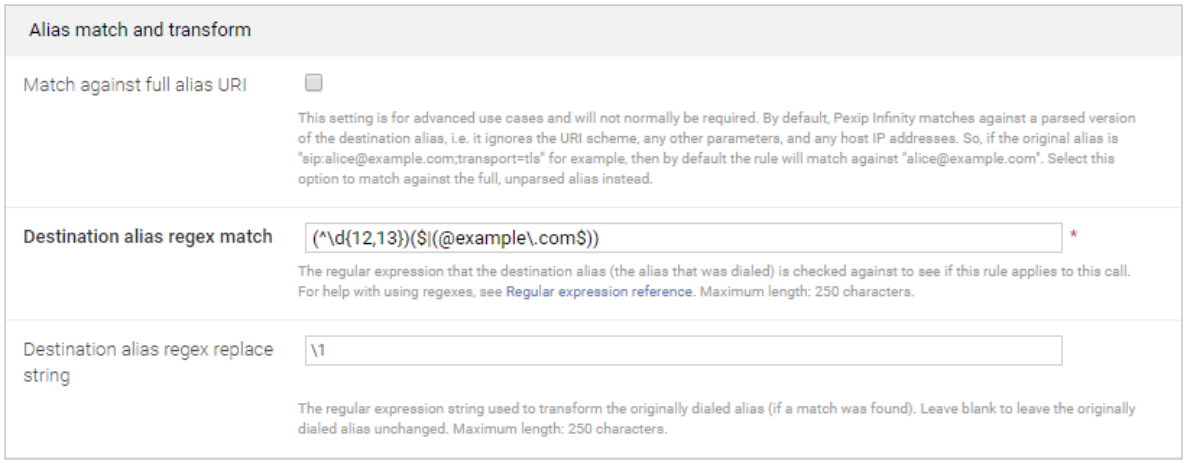
Name	Joining Google Meet * The name used to refer to this Call Routing Rule. Maximum length: 250 characters.
Service tag	 A unique identifier used to track usage of this Call Routing Rule. For more information, see Tracking usage with a service tag. Maximum length: 250 characters.
Description	 A description of the Call Routing Rule. Maximum length: 250 characters.
Priority	30 * The priority of this rule. Rules are checked in ascending priority order until the first matching rule is found, and it is then applied. Range: 1 to 200.

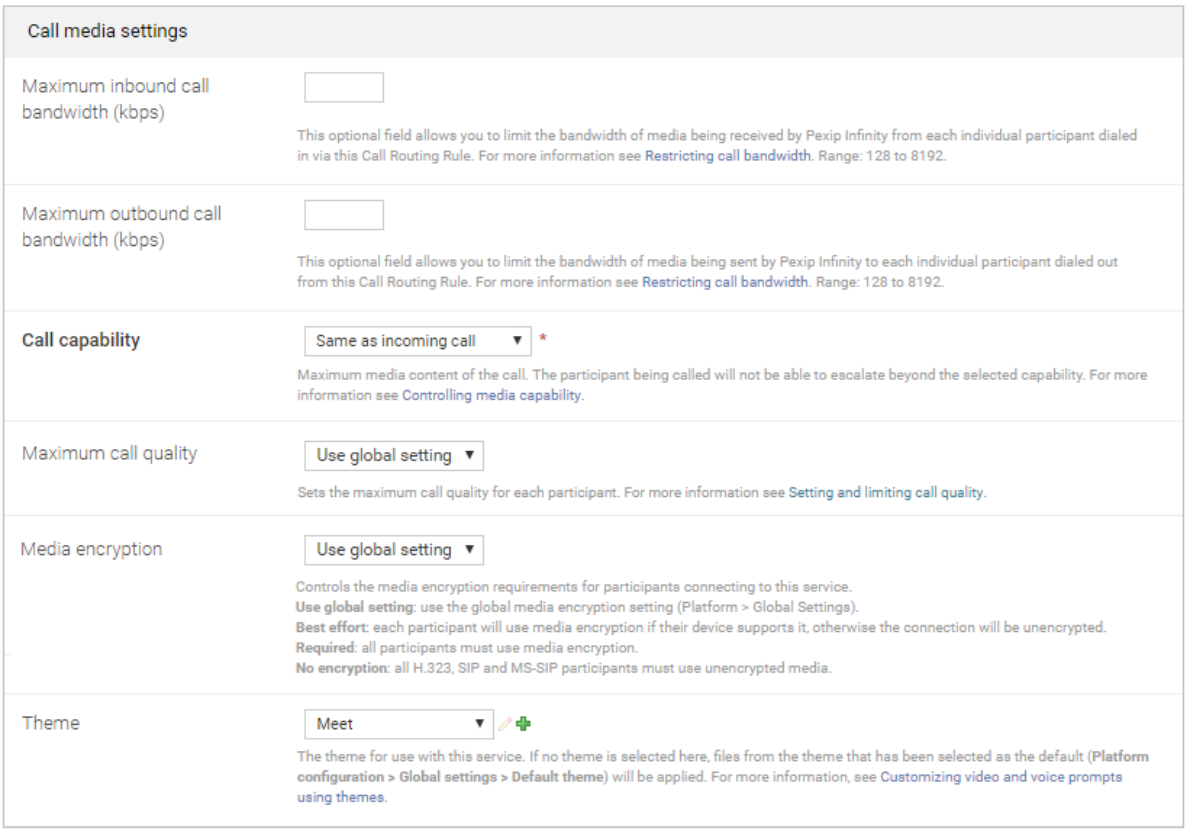
Use this rule for...

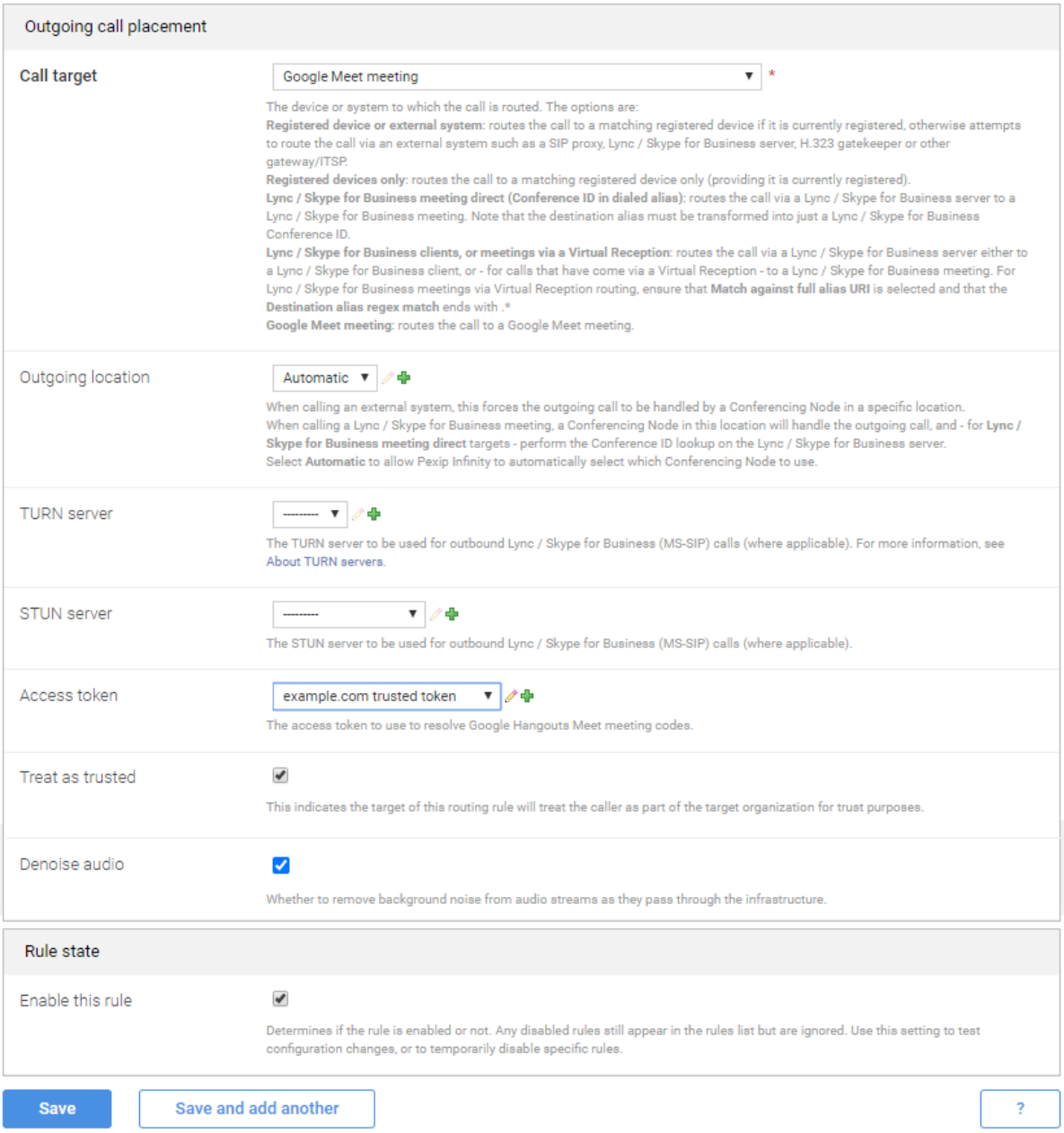
Incoming gateway calls	☑ Applies this rule to incoming calls that have not been routed to a Virtual Meeting Room or Virtual Reception, and should be routed via the Pexip Distributed Gateway service.
Outgoing calls from a conference	☐ Applies this rule to outgoing calls placed from a conference service (e.g. when adding a participant to a Virtual Meeting Room) where **Automatic** routing has been selected. For more information see Configuring Call Routing Rules.
Calls being handled in location	Any Location ▼ Applies the rule only if the incoming call is being handled by a Conferencing Node in the selected location or the outgoing call is being initiated from the selected location. To apply the rule regardless of the location, select **Any Location**.

When matching incoming Gateway calls...

Match incoming calls from registered devices only	☑ Only apply this rule to incoming calls from devices, videoconferencing endpoints, soft clients or Infinity Connect clients that are registered to Pexip Infinity. Note that the call must also match one of the selected protocols below. Calls placed from non-registered clients or devices, or from the Infinity Connect Web App will not be routed by this rule if it is enabled.
Match Infinity Connect (WebRTC / RTMP)	☑ Select whether this rule should apply to incoming calls from Infinity Connect clients (WebRTC / RTMP).
Match SIP	☑ Select whether this rule should apply to incoming SIP calls.
Match Lync / Skype for Business (MS-SIP)	☐ Select whether this rule should apply to incoming Lync / Skype for Business (MS-SIP) calls.
Match H.323	☐ Select whether this rule should apply to incoming H.323 calls.


Alias match and transform
Match against full alias URI
This setting is for advanced use cases and will not normally be required. By default, Pexip Infinity matches against a parsed version of the destination alias, i.e. it ignores the URI scheme, any other parameters, and any host IP addresses. So, if the original alias is "sip:alice@example.com;transport=tls" for example, then by default the rule will match against "alice@example.com". Select this option to match against the full, unparsed alias instead.
Destination alias regex match
(^\d{12,13})($|(@example\.com$))
The regular expression that the destination alias (the alias that was dialed) is checked against to see if this rule applies to this call. For help with using regexes, see Regular expression reference. Maximum length: 250 characters.
Destination alias regex replace string
\1
The regular expression string used to transform the originally dialed alias (if a match was found). Leave blank to leave the originally dialed alias unchanged. Maximum length: 250 characters.


Call media settings
Maximum inbound call bandwidth (kbps)
This optional field allows you to limit the bandwidth of media being received by Pexip Infinity from each individual participant dialed in via this Call Routing Rule. For more information see Restricting call bandwidth. Range: 128 to 8192.
Maximum outbound call bandwidth (kbps)
This optional field allows you to limit the bandwidth of media being sent by Pexip Infinity to each individual participant dialed out from this Call Routing Rule. For more information see Restricting call bandwidth. Range: 128 to 8192.
Call capability
Same as incoming call
Maximum media content of the call. The participant being called will not be able to escalate beyond the selected capability. For more information see Controlling media capability.
Maximum call quality
Use global setting
Sets the maximum call quality for each participant. For more information see Setting and limiting call quality.
Media encryption
Use global setting
Controls the media encryption requirements for participants connecting to this service.
Use global setting: use the global media encryption setting (Platform > Global Settings).
Best effort: each participant will use media encryption if their device supports it, otherwise the connection will be unencrypted.
Required: all participants must use media encryption.
No encryption: all H.323, SIP and MS-SIP participants must use unencrypted media.
Theme
Meet
The theme for use with this service. If no theme is selected here, files from the theme that has been selected as the default (Platform configuration > Global settings > Default theme) will be applied. For more information, see Customizing video and voice prompts using themes.



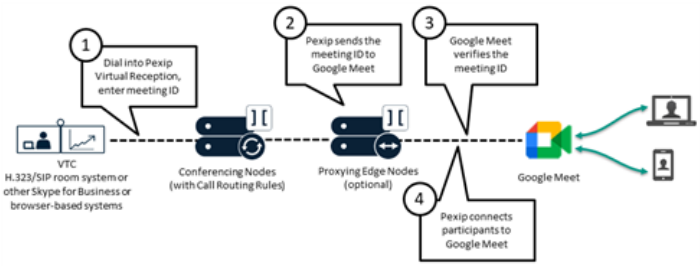
Using the direct gateway service

After the Call Routing Rule has been configured, third-party systems and devices can now dial an alias that matches your specified pattern (e.g. **1234567890123** or **1234567890123@example.com**) to be routed directly into the appropriate Google Meet conference (in this example the conference with a meeting ID of 1234567890123).

Routing indirectly via a Virtual Reception (IVR gateway)

To route calls to Google Meet conferences via a Virtual Reception (IVR gateway) you need:

- A Virtual Reception configured specifically to handle Google Meet conferences.
- A Call Routing Rule to route the calls handled by the Virtual Reception into the relevant Google Meet conference. Typically you would configure the Virtual Reception and Call Routing Rule patterns so that the same rule can also support direct routing as described above.



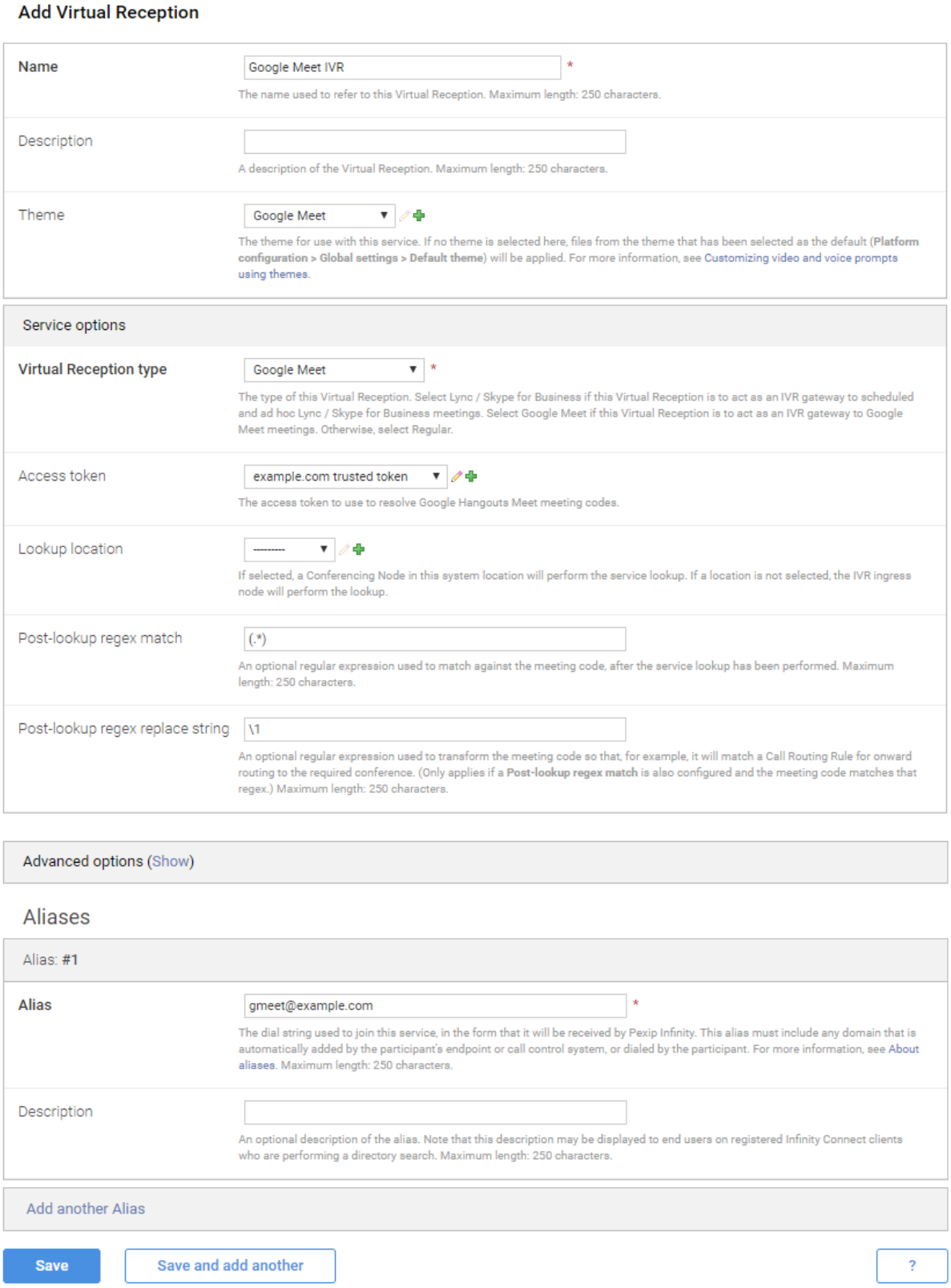
The Virtual Reception requests the caller to enter the Google Meet meeting ID which it then sends to Google Meet for verification. You can then optionally transform the meeting ID to meet any dial plan requirements, before the Infinity Gateway then matches the (optionally transformed) meeting ID and routes the caller to the appropriate Google Meet conference.

To configure the Virtual Reception:

1. Go to **Services > Virtual Receptions** and select **Add Virtual Reception**.
2. Configure the following fields (leave all other fields with default values or as required for your specific deployment):

Option	Description
Name	The name to use to refer to this Virtual Reception, for example "Google Meet IVR gateway".
Theme	If required, assign a customized theme to this Virtual Reception to brand it as the gateway to Google Meet conferences, for example by customizing the voice prompts or changing the appearance of the Virtual Reception splash screen.
Virtual Reception type	Select ***Google Meet***.
Access token	Select the name of the access token to use to resolve Google Meet IDs. When configuring a Virtual Reception it does not matter if you use a trusted or untrusted access token.
Lookup location	Specify the location that contains the Conferencing Nodes (typically Proxying Edge Nodes) that will perform the service lookup (meeting ID verification) on Google Meet.
Post-lookup regex match	This is an optional field. It is typically used in conjunction with the **Post-lookup regex replace string** to transform the meeting ID entered into the Virtual Reception into a distinct alias pattern that will match a Call Routing Rule that is configured to route calls into Google Meet conferences. Note that this match and transform occurs after the meeting ID entered by the user has been sent to Google Meet for verification. In most cases, you would typically set the regex match to: **(.*)** which matches everything entered into the Virtual Reception.
Post lookup regex replace string	This may be used in conjunction with the **Post-lookup regex match** field to transform the meeting ID entered by the caller. In our example, setting it to: **\1** will simply pass on the entered meeting ID unchanged. This will then typically work in conjunction with the Call Routing Rule that you configured above for direct routing and then directs the call to Google Meet.
Alias	Enter the alias that users will dial to use this Virtual Reception to place calls into Google Meet conferences, for example **gmeet@example.com**.

3. Select **Save**.


Add Virtual Reception
Name
Google Meet IVR
The name used to refer to this Virtual Reception. Maximum length: 250 characters.
Description
A description of the Virtual Reception. Maximum length: 250 characters.
Theme
Google Meet
The theme for use with this service. If no theme is selected here, files from the theme that has been selected as the default (Platform configuration > Global settings > Default theme) will be applied. For more information, see Customizing video and voice prompts using themes.
Service options
Virtual Reception type
Google Meet
The type of this Virtual Reception. Select Lync / Skype for Business if this Virtual Reception is to act as an IVR gateway to scheduled and ad hoc Lync / Skype for Business meetings. Select Google Meet if this Virtual Reception is to act as an IVR gateway to Google Meet meetings. Otherwise, select Regular.
Access token
example.com trusted token
The access token to use to resolve Google Hangouts Meet meeting codes.
Lookup location

If selected, a Conferencing Node in this system location will perform the service lookup. If a location is not selected, the IVR ingress node will perform the lookup.
Post-lookup regex match
(.*)
An optional regular expression used to match against the meeting code, after the service lookup has been performed. Maximum length: 250 characters.
Post-lookup regex replace string
\1
An optional regular expression used to transform the meeting code so that, for example, it will match a Call Routing Rule for onward routing to the required conference. (Only applies if a Post-lookup regex match is also configured and the meeting code matches that regex.) Maximum length: 250 characters.
Advanced options (Show)
Aliases
Alias: #1
Alias
gmeet@example.com
The dial string used to join this service, in the form that it will be received by Pexip Infinity. This alias must include any domain that is automatically added by the participant's endpoint or call control system, or dialed by the participant. For more information, see About aliases. Maximum length: 250 characters.
Description
An optional description of the alias. Note that this description may be displayed to end users on registered Infinity Connect clients who are performing a directory search. Maximum length: 250 characters.
Add another Alias
Save
Save and add another
?

To configure the associated Call Routing Rule:

- Configure the Call Routing Rule as described above for direct routing.
- If you want to use a different rule for routing via a Virtual Reception than the rule you are using for direct routing (e.g. because you want to limit the supported incoming call protocols, or use a different outgoing location for calls placed via the Virtual Reception), then follow the same principles as the direct routing rule, but use a different alias pattern in your Virtual Reception's **Post-lookup regex replace string** and your rule's **Destination alias regex match string**.

Using the Google Meet IVR gateway service

After the Virtual Reception and Call Routing Rule have been configured, third-party systems and devices can now dial the alias of the Virtual Reception (e.g. **gmeet@example.com**) and then, when prompted by the IVR service, enter the meeting ID of the Google Meet conference they want to join.

The Infinity Gateway will then route the call into the appropriate Google Meet conference.

Enabling SIP Guest Join

Pexip's standard Google Meet interoperability solution allows your own video conferencing endpoints (and your guests) to join Google Meet meetings that you are hosting. The "SIP Guest Join" (SGJ) feature lets you join Google Meet meetings with your own video conferencing endpoints where that meeting is being hosted by an external third-party organization who has not enabled Pexip interoperability themselves.

When using SIP Guest Join, your calls are routed via the Pexip Service, therefore you need to contact your Pexip authorized support representative to perform the necessary steps to enable the guest join process on your behalf. This includes:

- Purchasing a **Trusted Devices** license for the number of endpoints you own. Note that the Enterprise Room Connector (ERC) premium licensing package for Infinity includes SIP Guest Join.
- Providing the domains and IP addresses of **all** of the systems that will send SIP call signaling to the Pexip Service (typically these are the addresses of your externally-facing Conferencing Nodes, or VCS Expressway if relevant).

Note that you do not need to register your endpoints to the Pexip Service, or route any of your non-guest Google Meet interop calls via the Pexip Service.

Enabling guest join in Pexip Infinity

As a prerequisite to using the guest join feature you must be using One-Touch Join on the endpoints that you will use to join the Google Meet conferences.

To enable guest join you must create a suitable OTJ Meeting Processing Rule to handle invitations to Google Meet meetings, and ensure that your guest join calls are routed to the Pexip Service:

1. Add an OTJ Meeting Processing Rule (**One-Touch Join > OTJ Meeting Processing Rules**) with a **Meeting type** of ***Google Meet SIP Guest Join*** and a **Domain** of ***google.pexip.me***.
2. Enter your **Pexip Service Customer ID**. This will ensure that these calls (which are from endpoints not registered to the Pexip Service) are tracked within Pexip Control Center.

   If you do not know your Pexip Service Customer ID, your your Pexip authorized support representative can provide it. Alternatively, see Finding your Pexip Service Customer ID.
3. Ensure that the SIP Guest Join calls placed from your endpoint are routed to the Pexip Service.

   Typically you can do this via DNS as follows:
   a. Add a Call Routing Rule (**Services > Call Routing**).
   b. Configure the following fields (leave all other fields with default values or as required for your specific deployment):

Option	Description
Priority	Choose a suitable priority based on your existing rules.
Incoming gateway calls	Ensure this option is selected.

Option	Description
Outgoing calls from a conference	Leave unselected.
Match incoming calls from registered devices only	Select this option if you want the rule to apply only to calls received from devices that are registered to Pexip Infinity.
Match Pexip apps (WebRTC / RTMP) Match SIP Match Skype for Business (MS-SIP) Match H.323 Match Teams	Select the relevant types of system/protocols you want to support, typically **Match SIP** or **Match H.323**.
Match against full alias URI	Leave unselected
Destination alias regex match	Enter a regular expression that will match the Google Meet calls that are sent to the Pexip Service. We recommend: **[a-z\-]{10,12}@google\.pexip\.me**
Destination alias regex replace string	Leave blank.
Call target	Select ***Registered device or external system***.
Protocol	Select ***SIP***.
SIP proxy	Select ***Use DNS***.

Controlling the layout seen by VTC participants

VTC participants see Pexip's standard 1+7 layout by default, but this can be changed to use other layouts. Meeting participants can also use DTMF/keypad controls to control the meeting layout during an ongoing conference, or participants using Cisco endpoints could also use the Pexip Layout Controls macro. DTMF/keypad controls can also be used to control the participant's mute status and to raise/lower the participant's hand.

Setting the default layout

Pexip's 1+7 layout is used by default for VTC participants within a Google Meet meeting, however this default can be changed to any of the other supported layouts.

The easiest way to change the default layout for all Google Meet gateway calls is to use local policy. This method also lets you control other layout options, such as the display of participant names, if required. You could, for example, use local policy to set the default layout to Adaptive Composition and then use one of the methods described below to dynamically change the layout to something else when and if required.

This example local policy script enables the display of participant names and selects the Adaptive Composition layout for all Google Meet gateway calls.

```
{
    {% if service_config and service_config.service_type == "gateway" and service_config.called_device_type == "gms_conference" %}
        "action" : "continue",
        "result" : {{service_config|pex_update({"enable_overlay_text": true, "view":"five_mains_seven_pips"})|pex_to_json}}
    {% elif service_config %}
        "action" : "continue",
        "result" : {{service_config|pex_to_json}}
    {% else %}
        "action" : "reject",
        "result" : {}
    {% endif %}
}
```

If you want to apply the policy to a specific Call Routing Rule you can test against **service_config.name** (the **Name** of the rule), for example: `service_config.name == "name of routing rule"`

Dynamically changing the layout during a conference

Cisco endpoint macros

Cisco endpoints can use the Pexip Layout Controls macro that allows you to dynamically change the layout.

DTMF/keypad commands

All VTC participants can control the layout during a conference by using DTMF/keypad commands directly from their endpoint. This allows the user to select which layout they receive, and to control how presentation content is displayed.

By default the layout changes every time the endpoint sends ***8** to the conference, and it cycles through most of the available Pexip layout types in this order: 1+7, Adaptive Composition, 1+21, 2+21, 2x2, 3x3, 4x4, 5x5, 1+0, 1+33.

You may want to remove some layouts from the cycle list. You can do this by customizing the behavior via themes: the DTMF/keypad controls that are suitable for use in a Google Meet conference, and the sequence in which the layouts are displayed (by pressing *8), can all be customized by setting up your own theme in Pexip Infinity.

Here is an example theme configuration file (**themeconfig.json**) that offers a customized list of DTMF controls and layout options:

```
{
    "theme_version": 2,
    "dtmf_conference_control_commands": {
        "*3": "toggle_raise_hand",
        "*6": "toggle_self_mute",
        "*8": "cycle_layout",
        "*9": "toggle_multiscreen"
    },
    "dtmf_allowed_layouts": ["1:7", "ac", "2x2", "3x3", "4x4", "5x5", "1:0"]
}
```

This theme would override the default behavior and limit the DTMF controls to the following options that are relevant to a VTC connected to a Google Meet conference:

- ***3** command: toggles the raised/lowered hand of the participant
- ***6** command: toggles the mute status of the participant
- ***8** command: cycles the conference layout through the set of available layouts as defined in **dtmf_allowed_layouts**
- ***9** command: toggles multiscreen participant display

It also removes the 1+21 and 2+21 layouts from the cycle list. If required, you can adapt the list to change the order, or to further reduce the available layouts by removing the layout types you do not want to use.

You can then apply this theme to your Call Routing Rule(s) used to route calls into Google Meet (or you can set it as your default theme if appropriate for your deployment).

Note that:

- You cannot use the Pexip branding portal to create your theme configuration file (**themeconfig.json**). You have to manually create your own theme configuration file (using the example above as the basis for this) and package it into a theme ZIP file.
- If multiple endpoints are gatewayed into the same Google Meet conference then they will all start with the same default layout, but any commands to change the layout received by the endpoint only apply to the endpoint issuing the commands.
- If you switch from a less resource-intensive layout (such as "1+7") to a more resource-intensive layout (such as "ac") this will consume additional Conferencing Node hardware resources.

Other layout control options

You can also use the Pexip APIs to control the layout:

- **Client API**: you can use the client API (typically via the web app) to dynamically change the layout applied to a Google Meet gateway call. This involves using the transform_layout function with the client REST API or the transformLayout function with the

PexRTC JavaScript client API.

- **Management API**: you can use the transform_layout management API command.

DNS and ports requirements

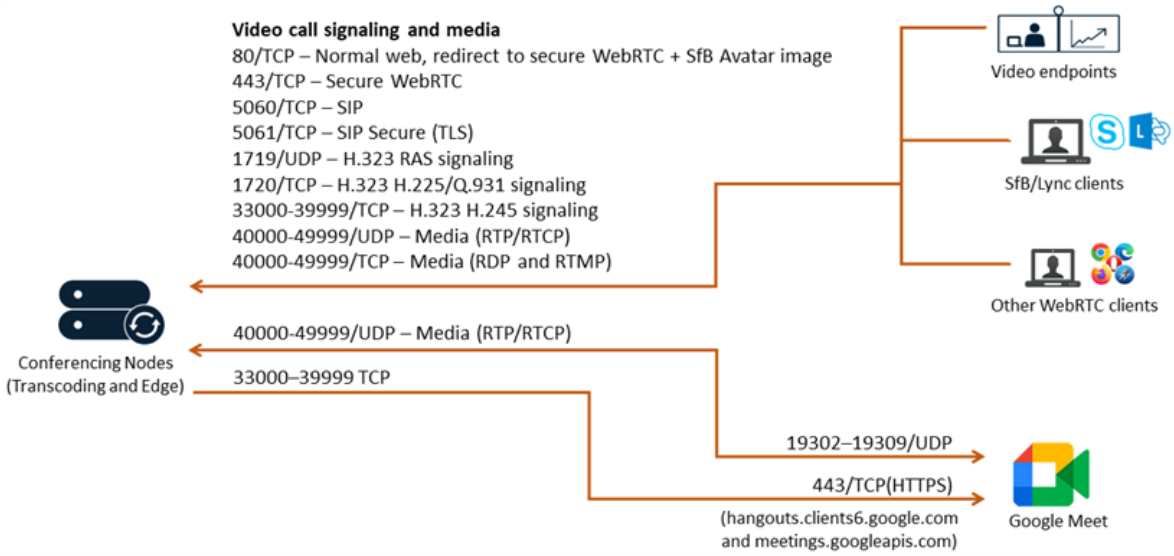
You need to ensure that the endpoints and systems you want to gateway into Google Meet can call into Pexip Infinity Conferencing Nodes, and that Conferencing Nodes can call out to Google Meet.

These are the port usage rules for call signaling and media between Google Meet and Conferencing Nodes (Proxying Edge Nodes and Transcoding Conferencing Nodes):

Source address	Source port	Destination address	Dest. port	Protocol	Notes
Google Meet ‡	19302–19309	Conferencing Node	40000–49999 **	UDP	SRTP/SRTCP
Conferencing Node	33000–39999 **	Google Meet (**hangouts.clients6.google.com** and **meetings.googleapis.com**)	443	TCP (HTTPS)	
Conferencing Node	40000–49999 **	Google Meet	19302–19309	UDP	SRTP/SRTCP

‡ See https://support.google.com/a/answer/1279090 for the actual IP ranges that need to be allowed.

** Configurable via the Media port range start/end, and Signaling port range start/end options (see About global settings).



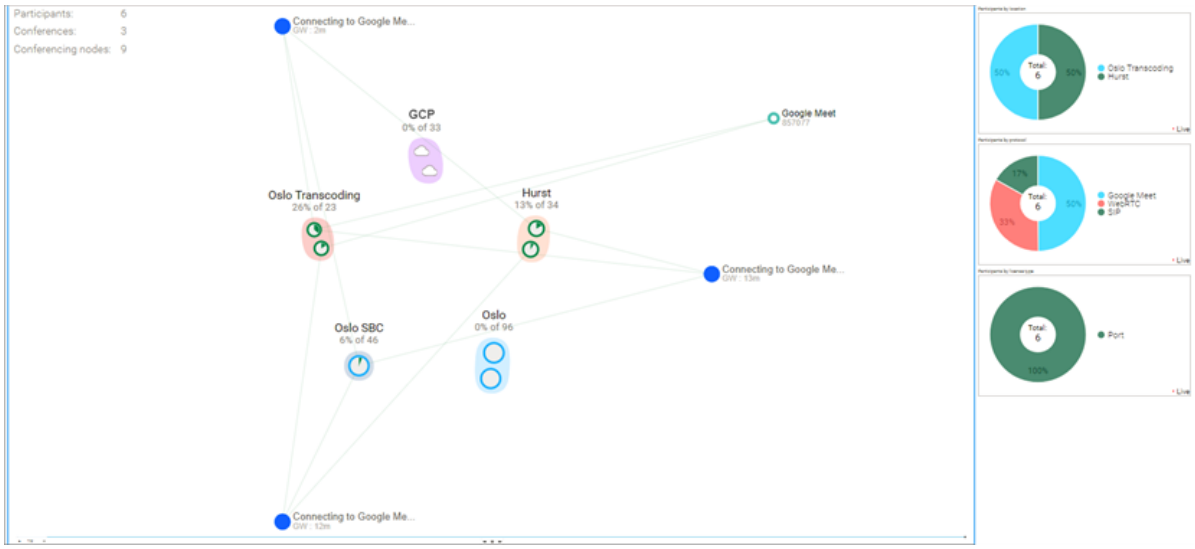
Call signaling and media ports

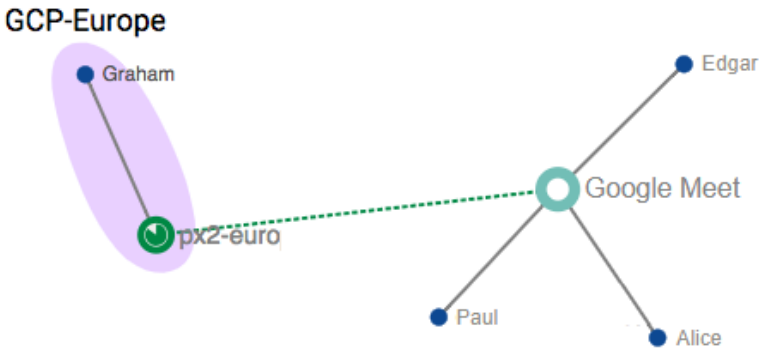
- See DNS record examples for information about enabling endpoints to route their calls to Conferencing Nodes.
- See Pexip Infinity port usage and firewall guidance for complete information about the ports used when Conferencing Nodes connect to Google Meet and other devices.

Call and participant status

When using the Pexip Infinity Administrator interface to monitor calls that are placed into Google Meet conferences, you should note that:

- Each participant who is gatewayed into a Google Meet conference is listed as a separate gateway call. However, if multiple participants are connected to the same Google Meet conference, the Live View (**Status > Live View**) shows them as connected to the same external conference.





- When viewing the participant status for a gateway call, the meeting ID, such as 1234567890123, is shown as a participant alias. This participant represents the gateway call leg into Google Meet. If you look at the media streams associated with that participant you see that:
  - Pexip Infinity sends (subject to bandwidth) three VP8 video streams (each at different resolutions) and one 1 audio stream to Google Meet for that participant.
  - Pexip Infinity receives one video and one audio stream for each external participant in the conference, up to a maximum of 8 video streams (to support Pexip's standard 1+7 layout). If there are more than 8 other participants then only an audio stream is received for those extra participants.

Media streams

Type	Start time	Node	Tx codec	Tx bitrate (kbps)	Tx resolution	Tx framerate	Tx packets sent	Tx packets lost	Tx current packet loss	Tx jitter (ms)	Rx codec	Rx bitrate (kbps)	Rx resolution	Rx framerate	Rx packets received	Rx packets lost	Rx current packet loss	Rx jitter (ms)
Video	2018-06-25 13:43:32 (BST)	10.47.2.21	VP8	129	320x180	30.0	2403	1	0.0%	33.3	Off	0		0.0	0	0	0.0%	0.0
Video	2018-06-25 13:43:32 (BST)	10.47.2.21	Off	0		0.0	0	0	0.0%	0.0	VP8	2516	1280x720	30.0	18911	0	0.0%	0.8
Video	2018-06-25 13:43:32 (BST)	10.47.2.21	Off	0		0.0	0	0	0.0%	0.0	VP8	128	320x180	25.0	1222	0	0.0%	2.3
Video	2018-06-25 13:43:32 (BST)	10.47.2.21	VP8	303	640x360	30.0	3474	2	0.0%	33.3	Off	0		0.0	0	0	0.0%	0.0
Video	2018-06-25 13:43:32 (BST)	10.47.2.21	VP8	1266	1280x720	30.0	10612	0	0.0%	33.3	Off	0		0.0	0	0	0.0%	0.0
Audio	2018-06-25 13:43:32 (BST)	10.47.2.21	Opus	64			3987	5	0.0%	0.9	Off	0			0	0	0.0%	0.0
Audio	2018-06-25 13:43:32 (BST)	10.47.2.21	Off	0			0	0	0.0%	0.0	Opus	1			2118	1	0.0%	1.2
Audio	2018-06-25 13:43:32 (BST)	10.47.2.21	Off	0			0	0	0.0%	0.0	Opus	0			446	0	0.0%	3.1

8 Media streams

Other participant aliases that are displayed for that call include the device that placed the call (such as name@example.com) and one or more aliases in the format spaces/<id>/devices/<id> which represent the other participants in the Google Meet conference.

- You cannot control (e.g. disconnect, mute or transfer) any of the other participants connected to the Google Meet conference.
- VTCs may be unable to connect to the meeting, or get disconnected from the meeting, if the meeting host disables the **Turn on their video** option in the meeting settings. These types of disconnections are reported as SIP 480 errors (with a cause of "Call

rejected" and type of "RemoteDisconnect").

Additional deployment information

- Each participant who is gatewayed via Pexip Infinity into a Google Meet conference consumes two call licenses (one for the inbound leg of the call and one for the outbound leg, as is standard for calls via the Infinity Gateway calls). Any external participants who are connected directly to the Google Meet conference do not consume a license. See Pexip Infinity license installation and usage for more information.
- You cannot limit the **Maximum outbound call bandwidth** (the call leg towards Google Meet) — it is fixed at 2 Mbps.
- If the Google Meet conference is recorded, "streaming enabled" indicators are included in the video stream sent to gatewayed participants.
- Chat messages are supported in both directions between Google Meet and other chat-enabled clients. However, the name of the sender from the Google Meet side is not identified on messages received by Skype for Business clients.

Integrating Microsoft Teams with Pexip Infinity

For the most recent information, see Integrating Microsoft Teams with Pexip Infinity.

Integrating Epic telehealth with Pexip Infinity

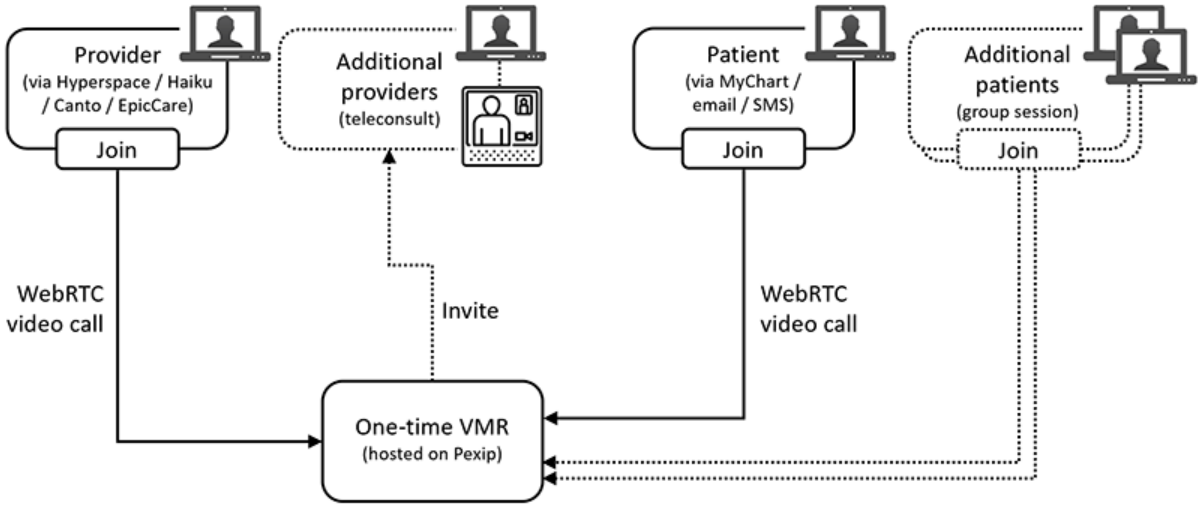
Epic telehealth integration with Pexip Infinity

Pexip's Epic telehealth integration enables healthcare organizations to hold video-based visits in Pexip Virtual Meeting Room (VMRs). The Pexip solution:

- Provides secure, easy-to-join telehealth visits from any location.
- Integrates seamlessly with Epic's standard video visit workflow.
- Uses context-aware linking within an Epic appointment to enable providers and patients to meet together over video.
- Allows providers (physicians/doctors) to directly launch their video visit from Hyperspace, Haiku, Canto or EpicCare.
- Allows patients to launch the video visit directly from MyChart, or via join links sent by email or SMS text messages.
- Supports clinic-to-clinic and teleconsult workflows (remote provider).
- Requires no downloads or plugins.
- Enables HIPAA compliance.

Epic Electronic Health Record (EHR) customers include hospitals, health systems, and physician practices. For more information, see Pexip's listing in the Epic Vendor Services.

Each video visit is held in a one-time VMR within Pexip Infinity via the WebRTC-based web app. Typically the visit will involve a single provider and patient, but multiple providers could join the same video visit — invited either by Epic-based workflows, or by calling out directly from the Pexip VMR. Multiple patients can also join the same video visit if it is a group session.



How it works

When it is time to go to their appointment, the patient clicks a button in MyChart, or clicks a join link sent to them by email or SMS text message, and this launches a video visit browser session (using the Pexip app).

The provider (doctor) can see the patient's appointment in Epic and may also be notified that the patient has connected and is ready to be seen. The provider clicks a button in their Epic system to launch their video visit and join the session.

The Pexip Infinity implementation and join process works as follows:

- Each Epic appointment is held in a one-time VMR within Pexip Infinity:
  - When the provider presses the Join button in their Epic app, a web app call is placed via the Infinity Gateway into the one-time VMR. The join URL is configured to take them straight into the VMR.
  - Similarly when the patient presses their Join button or clicks their join link, they also launch a web app call directly into the same one-time VMR.

  - Epic generates a unique, one-time-use, join URL for each participant (provider and patient) for every join attempt. Each URL contains a unique Pexip Infinity alias, which is derived from the appointment information.
  - If a call fails to connect for whatever reason, or the user gets disconnected, they can rejoin the same appointment but they must not try to re-use the same join URL (as it will always fail on subsequent re-use). They must close the browser tab, go back to their Epic healthcare application (typically their particular hospital's Mychart portal) and re-launch the call, or click again on their email/SMS join link. They will end up in the same meeting as before, just via a different alias.
  - Within Pexip Infinity, each alias for the same appointment is associated with the same Pexip Infinity service name (which is also derived from the appointment information). This ensures that each join attempt for the same appointment is taken to the same one-time VMR.
  - Thus, multiple providers and patients all meet in the same VMR if they all share the same appointment.
- Providers are treated as Hosts and patients are treated as Guests. The one-time VMR has a Host PIN, but no Guest PIN.
- Providers (Hosts) can dial out from the VMR and invite other participants if required.
- If patients (Guests) join the VMR before a provider (Host) has connected, they are held at the **Waiting for the host** screen until a provider joins (who automatically opens the conference with the Host PIN included within their join URL).
- Each video visit launches into an external browser session so as to allow the user continued access to either Hyperspace or MyChart.
- The launching of the external web app from the various Epic platforms uses SMART on FHIR OAuth 2.0 authentication (a set of open specifications to integrate apps with Electronic Health Records, portals, Health Information Exchanges, and other health IT systems). When a provider/patient clicks "join" to launch the Pexip video session they may get challenged by OAuth to re-enter their Epic sign-on credentials. This is purely down to timing and is not in Pexip's control.

Outline of the integration process

Here is an overview of the integration process, including the steps and interactions taken between the customer, Pexip and Epic:

1. The customer deploys a standard Pexip Infinity platform in their self-hosted environment.
2. The customer obtains and applies a telehealth license from Pexip (in addition to any other Pexip Infinity licenses they need).
3. The customer performs a basic (non-telehealth) test call to ensure that at least one Conferencing Node is reachable from the Internet, has proper certificates, and that call connectivity is working correctly (e.g. by calling two people into a test VMR via the web app).
4. The customer formally requests the Pexip integration app via Epic Vendor Services.
5. A Pexip administrator approves the customer request and securely stores the production client secret and non-production client secret.
6. Pexip informs the customer of the production and non-production client secrets via secure means. The customer is then responsible for storing them securely and entering them into the Pexip Infinity Administrator interface.
7. Pexip also informs the customer of the patient and provider application client IDs, and the backend OAuth2 application client ID that are appropriate to their environment.
8. The customer configures their Epic FDI record and Epic telehealth profile to their Pexip Infinity deployment. This is best performed simultaneously as there is some data to be shared between the two systems. The customer should:
   - Create an Epic FDI record, and generate their encryption key and securely share it with Epic.
   - Add an Epic telehealth profile to their Pexip Infinity deployment, using the appropriate client secrets, application client IDs and encryption key settings. The profile's uuid identifier should be included in the corresponding CRYPTURL in the Epic FDI record (see Creating an Epic telehealth profile).
9. The customer sets up the private/public keypairs to support OAuth2 authentication to Epic for patients joining via email/SMS.
10. Epic November 2024 or later may require a JWK Set URL (JKU) at configuration time on the Epic side.

    The value to use is specific to your local Pexip deployment and takes the form:

    https://<Infinity web application base URL>/api/telehealth/v1/patient/static/<Unique uuid identifier for this telehealth profile>/publickeys.json

    For example, if on your telehealth profile your **Infinity web application base URL** is **https://edge.example.com** and your **Unique uuid identifier for this telehealth profile** is 8e9dfd66-e3b5-4332-89c9-450fdf1103c5 then your JWK Set URL (JKU) will be:

    https://edge.example.com/api/telehealth/v1/patient/static/8e9dfd66-e3b5-4332-89c9-450fdf1103c5/publickeys.json
11. The customer provides Pexip with:

- The patient OAuth2 redirect URL and provider OAuth2 redirect URL that they have configured on their deployment.
- The public key files they created to support OAuth2 authentication for patients joining via email/SMS.

12. Pexip asks Epic to add the two OAuth2 redirect URLs to the patient and provider applications on the Epic side, to add the public key files, and to synchronize the client secrets and encryption keys.
13. When the changes are made and have propagated on the Epic backend and on Pexip Infinity, the system is ready for testing and validation.
14. Pexip permanently and securely destroys any customer keys in our possession — secure storage of these (e.g. for backup or restore purposes) is now the customer's responsibility.

More information

For full details on the mandatory and optional integration configuration steps, and further reference information, see:

- Configuring Pexip Infinity to integrate with Epic telehealth
- Optional features and customizations for Epic telehealth integrations
- Monitoring, maintenance and reference information for Epic telehealth integrations
- Troubleshooting and call setup information for Epic telehealth integrations

Configuring Pexip Infinity to integrate with Epic telehealth

This topic describes the mandatory configuration and administrative steps that are required to integrate Pexip Infinity with an Epic telehealth system. This topic covers:

- Prerequisites and integration basics
- Platform routing and firewall requirements
- Ensuring a telehealth integration license is installed
- Creating an Epic telehealth profile
- Creating a Call Routing Rule for incoming calls
- Configuring Pexip Infinity to integrate with Epic telehealth
- Optional features

Prerequisites and integration basics

Before you perform the telehealth integration steps described here, you should:

- Perform a basic installation and configuration of Pexip Infinity. Any of Pexip Infinity's deployment models may be used (on-premises, cloud-hosted etc).
- Confirm that (non-telehealth) call connectivity is working correctly e.g. by calling two people into a test VMR via the Pexip web app.
- Ensure that you are running Pexip Infinity version 28 or later if you are using Epic August 2022 or later.

Summary of Pexip Infinity configuration steps

After you have completed your standard implementation of the Pexip Infinity platform, these are the specific Epic telehealth integration configuration steps that you must perform within Pexip Infinity:

1. Install a telehealth integration license (**Platform > Licenses**).
2. Create an Epic telehealth profile (**Call Control > Epic Telehealth Profile**).
3. Create a Call Routing Rule to manage and route the incoming telehealth calls from providers and patients (**Services > Call Routing**).

All of these steps are described in detail below.

You can also perform some optional configuration to customize the integration or use additional features.

Recommended Epic crypto algorithms and compatibility

The recommended crypto algorithms for different Epic versions are:

- AES-256-CBC is recommended for Epic versions from August 2019 onwards
- AES-128-CBC may be used for older Epic versions (August 2018 onwards)

The oldest Epic version that works with Pexip Infinity is August 2018.

Epic configuration requirements

As well as the Pexip Infinity configuration guidelines contained within this documentation, you must also refer to Epic's *Pexip Implementation Guide*.

The configuration of the Epic and the Pexip Infinity installations need to align precisely (with certain shared secret configuration values) for the integration to work correctly. Please note the following important configuration requirements that must be implemented within Epic:

- FDI records must be created in Epic to integrate Epic with the Pexip Infinity installation. These ensure that both Epic and Pexip Infinity use the same algorithm and key to pass information between themselves.
- The uuid identifier of the **Epic telehealth profile** on Pexip Infinity must be included in the corresponding Launch URL (CRYPTURL) in the Epic FDI record.
- The OAuth2 redirect URLs defined in the **Epic telehealth profile** on Pexip Infinity must be registered within the Epic configuration.
- Refer to Epic/Citrix support if there is a need to launch a browser in a Hyperspace/Citrix environment to ensure that Chrome is used for the video call instead of Internet Explorer.

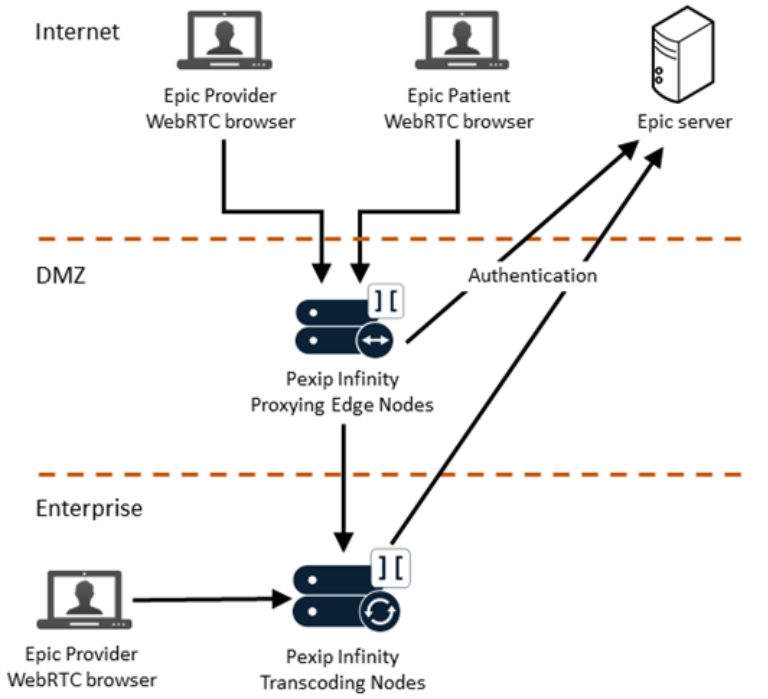
Platform routing and firewall requirements

Each Epic customer requires a self-hosted Pexip Infinity platform with the following routing/firewall requirements:

- At least one Conferencing Node (which could be a Proxying Edge Node) must be accessible from the computers/devices running the Epic apps, which typically means that the node needs to be publicly reachable. Alternatively, access could be routed via a reverse proxy. If required, the jinja templates can be used to load balance traffic across multiple Conferencing Nodes or to split provider traffic and patient traffic between different pools of Conferencing Nodes. For large environments (100+ concurrent video calls) we recommend that you contact your Pexip authorized support representative to discuss your architectural design.
- Firewall and DNS requirements are as for a standard Pexip Infinity public deployment. However, note that:
  - All inbound calls are placed via the Pexip web app (WebRTC), so no SIP or H.323 ports are required for incoming calls.
  - All Conferencing Nodes that may be involved in handling telehealth calls, including all "internal" transcoding nodes that are processing conference media, must be able to connect out to port 443/TCP (HTTPS) on the Epic provider and patient systems.
  - If you want the ability to call out from the VMR to invite other participants into the meeting, then you need to ensure that you allow SIP or H.323 call signaling and media from your Conferencing Nodes to those video endpoints.
- There is nothing from the Epic/public side of the integration that needs to access the Pexip Infinity Management Node.

The web entry point used for the OAuth2 redirect URLs (typically a Proxying Edge Node or reverse proxy) must be stable. Subsequent changes to DNS names will require hard-to-coordinate changes on Epic's side. Ensure that you use a DNS name that is suitable over the long term.

A typical deployment environment may look like this:



Ensuring a telehealth integration license is installed

You must have a **telehealth** integration license enabled on your Pexip Infinity platform (**Platform > Licenses**).

The **telehealth** license enables you to configure an Epic telehealth profile, and is required in addition to the standard Pexip Infinity call licenses. If necessary, contact your Pexip authorized support representative to purchase the required license.

Creating an Epic telehealth profile

The Epic telehealth profile configured on Pexip Infinity defines how the join URLs for providers and patients are constructed, and includes the shared encryption settings and keys that allow Pexip Infinity and the Epic servers to communicate securely.

At the start of the integration process, your Pexip authorized support representative will have given you various configuration items, including customer-specific secret credentials. Be very careful about copying and pasting the Epic encryption key and Application client secrets into your Epic telehealth profile. If there are any extra (or missing) characters in these fields, telehealth calls will fail to launch. Furthermore, as these fields are stored encrypted and are not decryptable by Pexip support, this can make it difficult to discover any typos when troubleshooting.

Go to **Call Control > Epic Telehealth Profile**, select **Add Telehealth Profile** and configure the profile as described:

Option	Description
Name	The name of the telehealth profile. This name appears in the web app participant list (roster) and in the Pexip Infinity administrator interface as part of the service name, so we recommend setting it to the name of the hospital or healthcare organization. Example: **Healthcare Org**
Description	A description of the telehealth profile.

Option	Description
Unique uuid identifier for this telehealth profile	A unique identifier for the telehealth profile. A value is automatically assigned and there is normally no need to modify it. Example: **12daaebc-ea75-43c1-a6c7-c044be66a36e** The corresponding Launch URL (CRYPTURL) in the Epic FDI record should include this uuid, for example: https://yourpexipinfinityserver.example.com/api/telehealth/v1/patient/oauth2smartlaunch/**12daaebc-ea75-43c1-a6c7-c044be66a36e**?<parameters> with the <parameters> replaced with appropriate parameters for your deployment as per Epic's own Pexip integration guide documentation. Note that the CRYPTURL configured in the Epic FDI record determines which telehealth profile is used for a particular call.
Domain name	The name of the domain to use for telehealth aliases associated with this telehealth profile location.
Base URL of the Epic server	The base HTTPS URL of the Epic server. Example: **https://epic.example.com/**
Epic OAuth2 base URL	The base HTTPS URL of the Epic server's OAuth2 APIs. Example: **https://epic.example.com/interconnect-aocurprd-oauth**
Patient OAuth2 redirect URL	The OAuth2 redirect URL for the patient telehealth application. Format: **https://[infinity deployment]/api/telehealth/v1/patient/oauth2authorized/[uuid]** where: • **[infinity deployment]** is the address of your public Conferencing Node, Proxying Edge Node, reverse proxy or load balancer. • **[uuid]** is a unique uuid identifier for this telehealth profile (typically this should be the same as the uuid identifier entered above for this profile). Example: **https://yourinfinitydeployment.example.com/api/telehealth/v1/patient/oauth2authorized/12daaebc-ea75-43c1-a6c7-c044be66a36e**
Provider OAuth2 redirect URL	The OAuth2 redirect URL for the provider telehealth application. Format: **https://[infinity deployment]/api/telehealth/v1/provider/oauth2authorized/[uuid]** where: • **[infinity deployment]** is the address of your public Conferencing Node, Proxying Edge Node, reverse proxy or load balancer. • **[uuid]** is a unique uuid identifier for this telehealth profile (typically this should be the same as the uuid identifier entered above for this profile). Example: **https://yourinfinitydeployment.example.com/api/telehealth/v1/provider/oauth2authorized/12daaebc-ea75-43c1-a6c7-c044be66a36e**
Infinity web application base URL	The public base URL for the Pexip Infinity web app. Example: **https://[infinity deployment]/** where: • **[infinity deployment]** is the address of your public Conferencing Node, Proxying Edge Node, reverse proxy or load balancer. Example: **https://yourinfinitydeployment.example.com/**
Patient application Client ID	The unique OAuth2 application Client ID assigned by Epic for the Pexip telehealth patient application to use when contacting clients.
Provider application Client ID	The unique OAuth2 application Client ID assigned by Epic for the Pexip telehealth provider application to use when contacting clients.

Option	Description
Template for provider aliases	The jinja2 template used for generating provider aliases used by providers (e.g. doctors) when joining a telehealth conference. This must include the value of **{{base_telehealth_alias}}** somewhere in the alias, although you can use your own choice of prefix and/or suffix. Default: **telehealth.{{base_telehealth_alias}}@{{telehealth_integration.telehealth_call_domain}}** See Supported jinja2 template variables for the full set of template variables that can be used in provider alias templates.
Template for patient aliases	The jinja2 template used for generating patient aliases used by patients when joining a telehealth conference. This must include the value of **{{base_telehealth_alias}}** somewhere in the alias, although you can use your own choice of prefix and/or suffix. Default: **telehealth.{{base_telehealth_alias}}@{{telehealth_integration.telehealth_call_domain}}**
Template for provider display names	The jinja2 template used for generating display names used by providers (e.g. doctors) when joining a telehealth conference. Default: **{{telehealth_integration.name}}**
Template for patient display names	The jinja2 template used for generating display names used by patients when joining a telehealth conference. By default this uses the first, middle and last name for the patient. Thus, for some group consultations, the hospital may want to construct their patient display name template so that it hides the surname. Default: **{{" ".join([launch_information.fname or "", launch_information.mname or "", launch_information.lname or ""]).replace(" ", " ")}}**
Provider WebRTC join link template	The jinja2 template used for generating HTTPS web join links used by providers (e.g. doctors) when joining a telehealth conference. It must include: • **{{telehealth_alias}}** variable somewhere in the link (usually as a conference=parameter) • **pin** parameter (so that the provider automatically opens the conference when they join) and you may customize other aspects of the URI if required, including the branding path. Default: **{{telehealth_integration.infinity_webapp_server_base_url}}/webapp/#/?conference={{telehealth_alias}}&pin={{pin}}&name={{display_name}}** See Supported jinja2 template variables for the full set of template variables that can be used in join link templates.
Patient WebRTC join link template	The jinja2 template used for generating HTTPS web join links used by patients when joining a telehealth conference. It must include: • **{{telehealth_alias}}** variable somewhere in the link (usually as a conference=parameter) and you may customize other aspects of the URI if required, including the branding path, although you should not include the **pin** parameter. Default: **{{telehealth_integration.infinity_webapp_server_base_url}}/webapp/#/?conference={{telehealth_alias}}&name={{display_name}}&role=guest**

Option	Description
Service name template	The jinja2 template used for generating the telehealth conference/appointment name. It must include: • **{{unique_encounter_id}}** variable somewhere in the generated name; however you may add your own prefix or suffix to the ID if required. The same service name is used for all participants who join the conference. Therefore if this integration is used for group appointments, the template **must not** include **{{launch_information.fname}}** or **{{launch_information.lname}}** as these will differ from patient to patient — and in which case you must modify the default template content. Default: **Appointment for {{launch_information.fname}} {{launch_information.lname}} {{telehealth_integration.name}}:{{unique_encounter_id}}{% if launch_information.encfacnpiname %} in {{launch_information.encfacnpiname}}{% endif %}** This default template generates a name of "Appointment for <first name> <last name> <profile name>:<Epic appointment ID" and also adds "in <encounter department>" if the department information has been provided by Epic. See Supported jinja2 template variables for the full set of template variables that can be used in the service name template.
Error page template for launch failures	The jinja2 template used for generating the error page shown to users if the telehealth call launch fails. You may adapt the template styles and text as appropriate for your environment. See Error page template for launch failures for more information.
Epic encryption key	The encryption key used to encrypt and decrypt telehealth parameters passed from Epic to Pexip Infinity when launching calls associated with this telehealth Integration. • If the encryption algorithm is AES-256-CBC then the key type must be base64-encoded key (**not** password), and the key itself must be a base64-encoded 32 byte random value. • If the encryption algorithm is AES-128-CBC then the key type may be base64-encoded key or password. If a base 64 key is used the key itself must be a base64-encoded 16 byte random value. A password may be any length. Both Epic and Pexip Infinity must be configured to use exactly the same algorithm and key in order for information to be passed correctly. See Epic encryption keys, client secrets and OAuth2 keypairs for more information.
Encryption key type	The type of the encryption key. It may be a simple text password from which a key will be derived, or it may be a predefined base64-encoded key. If AES-256-CBC is the encryption algorithm used, a base64-encoded key is mandated. Options are: • Simple password • Base 64 encoded key Default: ***Simple password***
Epic encryption algorithm	The encryption algorithm used by Epic when encrypting telehealth call parameters during telehealth call launch. For Epic releases from August 2019 onwards AES-256-CBC is recommended. Options are: • AES-128-CBC • AES-256-CBC Default: ***AES-128-CBC***
Patient application Client Secret	The unique OAuth2 application Client ID assigned by Epic for the Pexip telehealth patient application. Currently, the same value is used for both the **Patient** and **Provider** secrets on a single telehealth profile. See Epic encryption keys, client secrets and OAuth2 keypairs for more information.
Provider application Client Secret	The unique OAuth2 application Client ID assigned by Epic for the Pexip telehealth provider application. This should be set to the same value as the **Patient application Client Secret**.
Email and SMS helper application	

Option	Description
Backend OAuth2 application Client ID	The unique OAuth2 application Client ID assigned by Epic for the Pexip backend application to use when contacting the Epic server. See Backend OAuth2 client ID and private/public keypairs for more information.
Epic OAuth2 token URL for the backend application	The HTTPS URL of the Epic server's OAuth2 token URL. Example: **https://epic.example.com/interconnect-aocurprd-oauth/oauth2/token**
Backend OAuth2 private key	The private key for the Pexip backend OAuth2 application to use when authenticating to Epic. See Backend OAuth2 client ID and private/public keypairs for more information.

Supported jinja2 template variables

The following table shows which variables may be used in which templates. A ✓ indicates that the variable **may** be included in the template, whereas indicates that the variable **must** be present in the template.

		Allowed in template?				
Variable name	**Description**	**Alias**	**Display name**	**Join link**	**Service name**	**Error page**
base_telehealth_alias	A unique one-time alias for this join attempt, that must be included in the alias templates. It forms part of the overall conference alias that is specified in the WebRTC join link. This is the element that the associated Call Routing Rule should extract from the dialed conference alias when routing the incoming call.					
detailed_debug_information	Contains debug information about the failure to launch a call that can be used to help diagnose the problem.					✓
display_name	The name of the provider/patient displayed in the VMR. This variable is generated from the **Template for provider display names** and **Template for patient display names** options on the telehealth profile. By default this means: • Providers: the **telehealth_integration.name** (e.g. "Healthcare Org"). • Patients: the first name / middle name / last name of the patient e.g. "Beverley P Spinder".			✓		
error_name	The error name (returned from Epic) that can be used to decide which messages to display to the user on the error page: • An error_name of "BorvoGetTelehealthDirectLaunchTokenError" is returned if the launch link was not valid. This error could be expected to occur in a working environment. It might be because the user has selected a link for an old appointment, or for an appointment that isn't ready to start yet. • Any other error names that might be returned are the result of an unexpected error, and we recommend that these are all treated in the same manner.					✓

Variable name	Description	Allowed in template? Alias	Display name	Join link	Service name	Error page
launch_information	This variable contains appointment information provided by Epic. It contains the following fields: • **fname**: patient first name • **lname**: patient last name • **encfacnpiname**: encounter department Example: "Appointment for {{launch_information.fname}} {{launch_information.lname}} in {{launch_information.encfacnpiname}}"		✓		✓	
pin	A randomly-generated 8-digit Host PIN. This is mandatory for provider join links, and optional for patient join links.					
telehealth_alias	The provider/patient alias (generated from the provider and patient alias templates).					
telehealth_integration	This variable contains various properties of the telehealth profile configured in Pexip Infinity. It contains the following fields: • **telehealth_call_domain**: the profile's **Domain name** • **name**: the profile's **Name** • **description**: the profile's **Description** • **integration_uuid**: the profile's **uuid identifier** • **infinity_webapp_server_base_url**: The profile's **Infinity web application base URL** Example: "telehealth.{{base_telehealth_alias}}@{{telehealth_integration.telehealth_call_domain}}"	✓	✓	✓	✓	
telehealth_request_id	The telehealth call identifier. This is unique for every attempt to join an appointment.	✓	✓	✓		✓
unique_encounter_id	A unique appointment ID provided by Epic. This is the same for all participants (providers and patients) joining a given appointment (or "encounter" in Epic terminology). It **must** be referenced in the service name template.	✓	✓	✓		
utc_time	The current time in UTC, used to indicate when a call launch error occurred.					✓

* Mandatory.

See Jinja2 templates and filters for more information about using jinja templates.

Error page template for launch failures

An error page is displayed to users If a telehealth call fails to launch correctly.

The contents of the page is determined by the **Error page template for launch failures** field in the telehealth profile. This field is an HTML/jinja2 template that typically contains a mixture of HTML markup and styles, literal text and jinja2 variables. It can also contain jinja2 control structures that allow you to vary the content of the page based upon certain conditions.

The default template is shown below but you may adapt the content as appropriate for your environment:

```
<!DOCTYPE html>
```

```
<html lang="en">
  <head>
      <title>Something went wrong</title>
      <style>
        .pexip-cell {
        padding: 20px;
        }
        .main-title {
        font-size: 22px;
        color: #4a4a4a;
        }
        .pexip-heading {
        font-weight: bold;
        font-size: 14px;
        color: #4a4a4a;
        }
        .pexip-info {
        font-size: 12px;
        color: #4990e2;
        }
          </style>
  </head>
  <body>
    <br>
    <table style=
           "width: 100%; border-collapse: collapse; border: 1px solid #e9e9e9; font-family: Calibri, sans-serif, serif;">
      <tbody>
        <tr>
          <td colspan="2" class="pexip-cell" style="vertical-align: top;">
            {% if error_name == "BorvoGetTelehealthDirectLaunchTokenError" %}
            <p><span class="main-title">That link isn't working at the moment. It might be a link for an old appointment, or for an appointment
that isn't ready to start yet.</span></p>
            <p><span class="pexip-heading">Please check that you are using the correct link for your appointment today.</span></p>
            <p>
              <span class="pexip-heading">Extra information:</span><br>
                <span class="pexip-info">telehealth_request_id: {{telehealth_request_id}}</span><br>
                {% if detailed_debug_information %}
                <span class="pexip-info">Detailed debugging information: {{detailed_debug_information}}</span>
                {% endif %}
            </p>
            {% else %}
            <p><span class="main-title">Something went wrong - looks like an error on our end or on the network.</span></p>
            <p><span class="pexip-heading">Please close this window and try reconnecting.</span></p>
            <p>
              <span class="pexip-heading">If the error persists, please share the information below with your provider for
troubleshooting:</span><br>
                <span class="pexip-info">utc_time: {{utc_time}}</span><br>
                <span class="pexip-info">telehealth_request_id: {{telehealth_request_id}}</span><br>
                {% if detailed_debug_information %}
                <span class="pexip-info">Detailed debugging information: {{detailed_debug_information}}</span>
                {% endif %}
            </p>
            {% endif %}
          </td>
        </tr>
      </tbody>
    </table>
  </body>
</html>
```

The default template has the following logic:

- First, It defines some styles for use within the page.
- It tests `if error_name == "BorvoGetTelehealthDirectLaunchTokenError"`:
  - If true, it displays a message that the user may have used a link that is out of date or is not ready yet.
  - Otherwise (error_name is not "BorvoGetTelehealthDirectLaunchTokenError") this means an unexpected error has occurred and an appropriate message is displayed ("Something went wrong - looks like an error on our end or on the network").
- In both cases, the telehealth_request_id is displayed, and if detailed_debug_information is available then that is also displayed on the page.

The following variables are available in the error page template for launch failures: detailed_debug_information, error_name, telehealth_request_id and utc_time (full details are included above).

Email and SMS helper application and "direct join links"

"Direct join links" allow invitees, including guests, to join video visits directly. The links can be shared with end users via, for example, email or SMS and so on, using the Epic "Send Video Visit Link" activity. To enable direct join links:

- The Email and SMS helper application section of the Telehealth Profile on Pexip Infinity must be populated.
- You must enable the **Pexip Direct Launch Backend** app (this is a hidden app, not publicly listed on the Epic Vendor Services).
- Work with your Epic technical support representative to create appropriate Integration Records for Guest and Provider Direct Join Links records in your Epic environment. The records created must align with the settings on the Pexip Infinity Telehealth Profile configuration.

See Backend OAuth2 client ID and private/public keypairs for more information.

Creating a Call Routing Rule for incoming calls

You must create a Call Routing Rule within Pexip Infinity to manage and route the incoming telehealth calls appropriately.

1. Go to **Services > Call Routing** and select **Add Call Routing Rule**.
2. Configure the following fields (leave all other fields with default values or as required for your specific deployment):

Option	Description
Name	The name of this rule e.g. "Epic telehealth".
Priority	Assign the priority for this rule. If you are creating multiple rules you need to ensure that any other rules with a higher priority (lower number) will not process your telehealth calls.
Incoming gateway calls	Ensure this option is selected.
Outgoing calls from a conference	Leave this option unselected.
Calls being handled in location	Applies the rule only if the incoming call is being handled by a Conferencing Node in the selected location. To apply the rule regardless of the location, select ***Any Location***.
Match incoming calls from registered devices only	Leave this option unselected.
Match Pexip apps (WebRTC / RTMP) Match SIP Match Skype for Business (MS-SIP) Match H.323 Match Teams	Select **Match Pexip apps (WebRTC / RTMP)**. Do not select any of the other protocols.
Match against full alias URI	Leave this option unselected.
Destination alias regex match	The objective of the match/replace strings is to extract the **base_telehealth_alias** from the dialed alias in the incoming call received by Pexip Infinity from Epic. The match string should map to the provider/patient alias templates in the telehealth profile. The example regex match of **telehealth\.(.*)@(.*)** works perfectly with the default telehealth profile alias templates. Example: **telehealth\.(.*)@(.*)**

Option	Description
Destination alias regex replace string	When used with the example **Destination alias regex match** shown above you would use: **\1** which would extract the **base_telehealth_alias** from the dialed alias.
Theme	If required, assign a customized theme to this rule. For example, the theme could use alternative labels on some of the splash screens that are displayed when connecting the call. See Optional features and customizations for Epic telehealth integrations for more information.
Call target	Select ***Epic Telehealth meeting***.

3. Select **Save**.

Optional features

In addition to the basic configuration requirements described here, there are extra optional features you may want to consider that require additional configuration, including:

- **Branding and customization**: you can apply a range of branding or customization options:
  - Create your own theme to customize the splash screens (such as the **Waiting for the host** screen).
  - Create a customized web app to control the web app's appearance and behavior on call completion.
  - Use local policy to change the default layout and other properties of the VMR.
- **Dial out (teleconsult)**: enable dialing out to another provider from within the conference.

See Optional features and customizations for Epic telehealth integrations for more information.

Optional features and customizations for Epic telehealth integrations

This topic describes some optional features, customization and branding options for Epic telehealth integrations with Pexip Infinity:

- Customizing the audio prompts and splash screens via themes
- Changing the branding and styling of the Pexip web app
- Advanced web app customization via plugins
- Customizing the behavior at the end of a call
- Changing the disconnect timeout
- Overriding the one-time VMR layout and configuration via local policy scripts
- Calling out from a VMR to invite other consultants
- Using the Connect desktop app

Customizing the audio prompts and splash screens via themes

The audio prompts that are played out to patients and providers, and the splash screens that are displayed prior to joining the meeting, can all be customized by creating and uploading an alternative theme.

In particular you may want to customize the **Waiting for the host** screen that is displayed to patients while they are waiting for the provider to join.

We have also provided an alternative **conf-waithostpin_48kHz_mono.wav** file where instead of the standard "Waiting for the conference Host to join. If you are the conference Host, please enter the conference PIN number now" message, the alternative file says "Waiting for your healthcare provider to start the appointment". You can download this alternative theme file as **healthcare.zip** from https://dl.pexip.com/resources/themes/index.html.

You can either set a new default theme (for all calls on the Pexip Infinity platform), or apply the theme to the Call Routing Rule handling the incoming telehealth calls.

See Customizing conference images and voice prompts using themes for more information.

Changing the branding and styling of the Pexip web app

The look and feel of the web app can be customized, for example to apply language translations or to change the color scheme for buttons, icons and other graphic indicators on the home page of the web app. (Note that the in-call elements, such as the splash screens, are customized via themes instead).

You can use the Pexip branding portal to apply a whole range of customizations — see Customizing and branding the web app for more information.

Advanced web app customization via plugins

If you want to offer dynamic control of the one-time VMR to the provider (VMR Host) so that, for example, they can change the layout for group sessions then you can develop a bespoke Pexip app plugin.

Other potential uses for plugins could be to facilitate access to a translation call center service.

Customizing the behavior at the end of a call

By default, when a user finishes a call placed via the web app they are returned to the app home page. However, you can change this behavior so that they are sent to a different URL/webpage instead, such as:

- The healthcare organization's home page.
- A custom webpage that says something like "This call has ended. Please close your browser page."

We recommend doing this as it ensures that the patient is taken away from the web app when the call disconnects or if the call drops. This is particularly important as the URLs generated when a patient or provider first joins a meeting are single use. Thus it's not possible to rejoin a meeting using the old alias — the user has to go back into the Epic healthcare app (e.g. MyChart) and rejoin from there.

Note that you cannot apply this type of customization via the branding portal. This redirect behavior can only be configured via manual customization of the web app, by defining a **disconnectDestination** in the **settings.json** file. However, you can first use the branding portal to apply your other customization requirements and then add the **disconnectDestination** option to the files generated by the branding portal before uploading the branding package to the Management Node.

Changing the disconnect timeout

By default, guests (e.g. patients) who are waiting for hosts (e.g. healthcare providers) are automatically disconnected after 900 seconds (15 minutes). If you'd like a longer or shorter timeout, configure **Waiting for Host timeout** as appropriate in **Platform > Global Settings > Service Configuration**.

Overriding the one-time VMR layout and configuration via local policy scripts

The one-time VMR that is generated for each telehealth call has the following properties:

- 1+7 layout
- Display names enabled
- Host PIN only
- Guests can present
- Chat: as per global settings (**Platform > Global Settings > Connectivity**, then deselect or select **Enable chat**.)
- Uses the theme defined in the Call Routing Rule handling the incoming call.

If you want to use a different layout or disable overlay names for every telehealth call, you can configure a local policy script to override the default configuration. Local policy uses the jinja2 scripting language and allows you to control Pexip Infinity's call behavior.

When using local policy, the call information available to policy includes a **telehealth_request_id** field. This field is only present in telehealth calls and identifies each individual call or join attempt. You could also identify telehealth calls based on the **local_alias** (dialed alias) pattern.

Calling out from a VMR to invite other consultants

A provider who is already within a video call can use the Pexip app to call out directly to other providers (if appropriate, instead of using Epic's remote consultation order procedures) by manually entering the video address (alias) of the person they want to invite

into the conference.

This process requires an "outgoing" Call Routing Rule to be configured within Pexip Infinity to handle the outgoing call and ensure it is routed correctly. You should ensure that the **Destination alias regex match** is appropriate to match the video addresses of the remote providers you may want to invite into the video visit, so that you only need one rule that will work for all providers.

To configure the rule:

1. Go to **Services > Call Routing** and select **Add Call Routing Rule**.
2. Configure the following fields (leave all other fields with default values or as required for your specific deployment):

Option	Description
Name	The name used to refer to this rule e.g. "Telehealth dial out".
Priority	Assign the priority for this rule. If you are creating multiple rules you need to ensure that any other rules with a higher priority (lower number) will not process these outgoing calls.
Incoming gateway calls	Leave this option unselected.
Outgoing calls from a conference	Ensure this option is selected.
Destination alias regex match	This is a regular expression that must match the destination alias i.e. the video address of the participant being dialed. Note that the regex must match the entire alias — a partial match is treated as a non-match. See Regular expression (regex) reference for information about writing regular expressions.
Call target	The device or system to which the call is routed. The suitable options are: ○ ***Registered device or external system***: route the call to a matching registered device if it is currently registered, otherwise attempt to route the call via an external system such as a SIP proxy or H.323 gatekeeper. ○ ***Registered devices only***: route the call to a matching registered device only (providing it is currently registered).
Protocol	The protocol used to place the outgoing call. Select ***SIP*** or ***H.323*** as appropriate for the type of device you need to call. Note that if the call is to a registered device, Pexip Infinity will instead use the protocol that the device used to make the registration.
SIP Proxy	You can optionally specify the SIP Proxy to use to place an outgoing SIP call, otherwise select ***Use DNS***.
H.323 Gatekeeper	You can optionally specify the H.323 Gatekeeper to use to place an outgoing H.323 call, otherwise select ***Use DNS***.

3. Select **Save**.

Using the Connect desktop app

Some providers may want to use the Connect desktop app instead of the web app.

In this case the **Provider WebRTC join link template** in the Epic telehealth profile should be set to: **pexip://{{telehealth_alias}}?pin={{pin}}**

This URL structure causes the telehealth call to be placed via the Connect desktop app (which must already be installed on the provider's computer).

We recommend that patients continue to use the web app.

Monitoring, maintenance and reference information for Epic telehealth integrations

This topic describes some monitoring options and additional reference information for Epic telehealth integrations with Pexip Infinity:

- Monitoring appointments via Pexip Infinity
- Appointment and participant names
- Epic encryption keys, client secrets and OAuth2 keypairs
- Access and use of data shared between Epic and Pexip Infinity
- Removing a Pexip/Epic integration

For troubleshooting information, see Troubleshooting and call setup information for Epic telehealth integrations.

Monitoring appointments via Pexip Infinity

Active and historic appointments (conferences) can be viewed via the Pexip Infinity Administrator interface.

Each appointment is represented as a single conference within the **Status** pages on Pexip Infinity, and each conference typically has two participants — the provider and the patient. However, there could be additional participants if it is a group session or an additional provider is added to the call.

The appointment and participant names are constructed as described below.

Appointment and participant names

Appointment and participant names are displayed within the call itself (in the Pexip app roster) to all of the participants, and they are also shown within the Pexip Infinity Administrator interface status pages.

By default the appointment name (referred to as the service or conference name within Pexip Infinity) name takes the form:
"Appointment for <patient name> <telehealth profile name>:<Epic unique appointment id> in <encounter department>"
but you can control this by configuring the **Service name template** in the Pexip Infinity telehealth profile.

Note that "in <encounter department>" is only included if the department is known. For example:

- "Appointment for Beverley P Spinder Healthcare Org:6ea6d1427f684721abd21c1634386f7b6 in Radiology" or
- "Appointment for Kai Oosman Picardo Healthcare Org:5ec8f1667f531781afe19c1774322f7b8"

The display name in the participants list is configured by the **Template for provider display names** and **Template for patient display names** options on the telehealth profile. By default this uses:

- The **Name** field of the telehealth profile for the provider e.g. "Healthcare Org".
- The first name / middle name / last name for the patient e.g. "Beverley P Spinder".

Epic encryption keys, client secrets and OAuth2 keypairs

This section contains more information about how to complete some of the Pexip Infinity telehealth profile settings concerning encryption keys, client secrets and backend OAuth2 keypairs.

Epic encryption key

The **Epic encryption key** is a secret used for encrypting some of the information passed from Epic to Pexip. It should be a cryptographically strong random value. It is used in the **Epic telehealth profile** within Pexip Infinity and FDI records within Epic.

To generate a secret for use with Epic AES-256-CBC crypto (used with Epic version August 2019 onwards), you must generate a random 32 byte, base64 encoded key. (If you are using an older version of Epic, you can use either a 16 byte key generated using a similar method to that described below, or a simple password.)

You can generate the key using any tools that Epic may provide, or by using a Pexip command line tool as described below:

1. Connect over ssh into the Management Node as user admin with the appropriate password.
2. Issue the following command:
   ```
   pextelehealthkeygen 32
   ```

Example output:

```
value: fTW2KwlJZ6JEpRS+RlN2fruKPgggBe+Y9txLIk3QpA0=
```

The **value** output to the screen is a random 32 byte key, generated by OpenSSL, that is encoded into human-readable text using base64 encoding.

We recommend that you store this value securely (for example, using your corporate password database) in case you need to recover or reinstall your Pexip system without a suitable backup being available.

Client Secrets

The **Provider application Client Secret** and **Patient application Client Secret** that must be configured in the **Epic telehealth profile** are generated by Epic during the signup process and passed to Pexip, who will relay them securely to the end customer. Currently, the same value is used for both the provider and patient secrets on a single Epic telehealth profile.

A production and non-production variant of each secret is generated. These should be treated as being as sensitive as passwords, and you should ensure that the production and non-production variants are noted distinguishably.

Backend OAuth2 client ID and private/public keypairs

This section explains the procedure for configuring the **Email and SMS helper application** settings in an Epic telehealth profile, to support patients joining via links sent by email or SMS text messages.

Your Pexip representative will share the **Backend OAuth2 application Client ID** with you. In common with other application secrets that make up an integration, your production and non-production Epic environments use different Client IDs.

Each client ID needs its own associated unique public/private keypair, so you need to perform the following procedure twice to generate a keypair:

1. Log in to the Pexip Infinity Management Node over SSH.
2. Run the following commands:

   ```
   cd /dev/shm
   openssl genrsa -out ./privatekey.pem 3072
   openssl req -new -x509 -key ./privatekey.pem -out ./publickey509.pem -subj '/CN=PexipBackendDirectLaunchApp'
   ```

   This generates two files: **privatekey.pem** and **publickey509.pem**.

   Make sure that the key length is at least 2048 bits (our instructions above specify 3072 bits).
3. Copy **privatekey.pem** and **publickey509.pem** off the Management Node using an SCP (Secure Copy) client, for example WinSCP.
4. Delete the files from the Management Node by running the following commands:

   ```
   rm ./privatekey.pem
   rm ./publickey509.pem
   ```
5. Make a secure backup of the **privatekey.pem** and **publickey509.pem** files (you will need them if you need to reinstall your Pexip system).

   Make sure to indicate clearly if the keys are intended for your production environment or your non-production environment. You may want to rename the generated files to distinguish between the files you want to use for your production and non-production environments, and to ensure you do not overwrite the first set of files with the second set of files when you repeat this process.

   privatekey.pem is sensitive — so ensure you store it in a manner that complies with your organization's secret storage security policy (e.g. in a secure location such as your corporate password database).
6. Repeat the process to generate a second pair of keys (one pair for your production environment and one pair for your non-production environment).

You can now complete your configuration:

1. Configure your Pexip Infinity Epic telehealth profile for your production environment:
   a. In a text editor, open the **privatekey.pem** file you want to use for your production environment.
   b. Copy/paste the contents into the **Backend OAuth2 private key** field in the telehealth profile.
   c. Check you have also entered the production **Backend OAuth2 application Client ID** that Pexip shared with you, and select **Save**.
2. Repeat the step above for your non-production Pexip Infinity environment, but this time using the non-production **Backend OAuth2 application Client ID** and the other **privatekey.pem** file.
3. Send your two **publickey509.pem** files via email to your Pexip authorized support representative who will relay them to Epic so they can add the keys to their central system. Make sure to indicate clearly which key is for your production environment, and which is for your non-production environment.

   Only send the public keys. Do not send the private keys.

Access and use of data shared between Epic and Pexip Infinity

Please refer to Epic's documentation ("Integration Record Setup" in their *Pexip Implementation Guide*), which is available from your Epic support representative, for information about the data passed from Epic to Pexip Infinity.

The Pexip Infinity deployment obtains data from Epic that is necessary to launch the appointment in Pexip Infinity. This typically includes the first, middle and last name of the patient, the username of the launching user, the department encounter name e.g. radiology, and the CSN (contact serial number — an encrypted ID that identifies an appointment). The same data is also used for providers.

Pexip Infinity writes data to Epic. In particular it informs Epic when telehealth appointment calls terminate.

Privacy and security

The Pexip Infinity self-hosted solution supports the industry standards for communication encryption for end-user devices, ensuring that all video calls and shared media content is secure and kept private even if it crosses the internet. The entire deployment and all its data, including call status, diagnostic logs and call history, is completely under the ownership and control of the enterprise, even when deployed in the cloud.

Removing a Pexip/Epic integration

If you no longer require Epic telehealth integration, you must remove the specific Epic-related configuration from your Pexip Infinity deployment:

- The Epic telehealth profile (**Call Control > Epic Telehealth Profile**).
- The Call Routing Rule that manages the incoming telehealth calls (**Services > Call Routing**).
- Remove the telehealth integration license from your platform (**Platform > Licenses**).

You should also remove any optional customizations you may have applied, including:

- Any branding or customization overrides:
  - Any themes that customized the splash screens (**Services > Themes**).
  - Any customized Pexip web app settings (**Services > Web App Customization**).
  - Any local policy to customize the VMR (**Call Control > Policy Profiles**).
- Any additional Call Routing Rules that manage dial out (teleconsult) from the conference (**Services > Call Routing**).

Please talk to your Epic provider for information about how to remove your Pexip integration from the Epic system itself.

Troubleshooting and call setup information for Epic telehealth integrations

This topic provides some guidance on troubleshooting installation/integration and call launching/completing issues with telehealth integrations with Pexip Infinity, and it also contains some detailed call setup information.

Installation and integration issues

During initial integration it is possible that various misconfigurations may cause authentication with Epic to fail when trying to make any telehealth calls. If authentication fails, the system will usually display a screen containing a short textual error message, and the telehealth call will not attempt to launch. As the failures here are potentially security related, this error page intentionally does not give too much away, but there is usually a short error reason presented, as described below.

Symptom	Possible cause	Resolution
"TelehealthIntegrationNotFoundException Error"	The Epic server is configured with an incorrect launch URL.	Check that the Epic server is configured to use a launch URL that is correctly formatted and that contains the exact same (unique to your deployment) uuid as configured in the telehealth profile's **Unique uuid identifier for this telehealth profile** field.

Symptom	Possible cause	Resolution
"MetadataFetchError"	This error generally indicates some sort of network related problem (such as connectivity or DNS resolution issues). It occurs when Pexip Infinity was unable to access an unauthenticated (open) API exposed by the Epic server that provides Pexip Infinity with metadata pertaining to the Epic deployment.	Verify that the Epic server configured in the telehealth profile's **Base URL of the Epic server** and **Epic OAUTH2 base URL** fields is operational, reachable, and that it can be resolved using the DNS settings configured in all system locations (including both edge and transcoding locations as appropriate), and is not being blocked by a firewall. If a web proxy is configured in any location, verify that it too is reachable (and, if applicable, its DNS name is resolvable using the DNS server in the relevant location).
"Oauth2 Launch Error"	This could be caused by a range of issues including misconfiguration or calls being blocked by a firewall.	If this error occurs, check that: • The Epic server is passing non-blank "launch=" and "iss=" parameters as part of the initial launch URL. • The **Epic encryption key** was entered correctly in the telehealth profile, and that **Encryption key type** and **Epic encryption algorithm** are both configured appropriately to match what is configured on the Epic side. • The **Patient application Client ID** and **Provider application Client ID** are appropriately configured in the telehealth profile for this deployment, and that you are using the production or non-production values as appropriate to match what is configured on the Epic side. • The **Patient application Client Secret** and **Provider application Client Secret** were both entered correctly in the telehealth profile for this deployment, and that you are using the production or non-production values as appropriate. • Pexip Infinity's access to the Epic server is not being blocked by a firewall.
"Epic OAuth2 Error"	This Epic-generated message can occur if the **OAuth2 redirect URIs** are not configured within Epic or there is a mismatch between what is configured on the Epic and Pexip sides.	Check that the **OAuth2 redirect URIs** are the same on the Epic and Pexip sides.
"BorvoGetTelehealthDirectJoinTokenError"	Incorrect backend client id	Set **Backend OAuth2 application Client ID** on the relevant telehealth profile to the value appropriate for your environment (the correct value differs between production and non-production Epic environments)
	Invalid private key	Ensure **Backend OAuth2 private key** is correctly configured on the relevant telehealth profile.
	Poorly synced NTP clock	Fix any NTP sync alarms raised on Pexip Infinity.

Symptom	Possible cause	Resolution
"Malformed or missing state cookie"	The appointment launch URI is misconfigured on Epic.	Ensure that the launch URI from MyChart or Hyperspace is pointing to the launch URI e.g. /api/telehealth/v1/patient/oauth2smartlaunch/11111111-2222-3333-4444-555555555555 and **not** the (similar, but different) redirect uri /api/telehealth/v1/patient/oauth2authorized/11111111-2222-3333-4444-555555555555 (where 11111111-2222-3333-4444-555555555555 is replaced by the "Unique uuid identifier for this telehealth profile" appropriate for your deployment's Epic integration). This correction needs to be applied to the FDI records for the Pexip integration within the Epic solution.

If none of the above steps helps solve your initial launch integration problems, your Pexip authorized support representative may ask you to enable extra debug tracing, reproduce the failing call scenario and then afterwards to download a diagnostic snapshot.

Call launching/completion issues

This section describes some of the occasional issues that can occur when launching or completing a telehealth call.

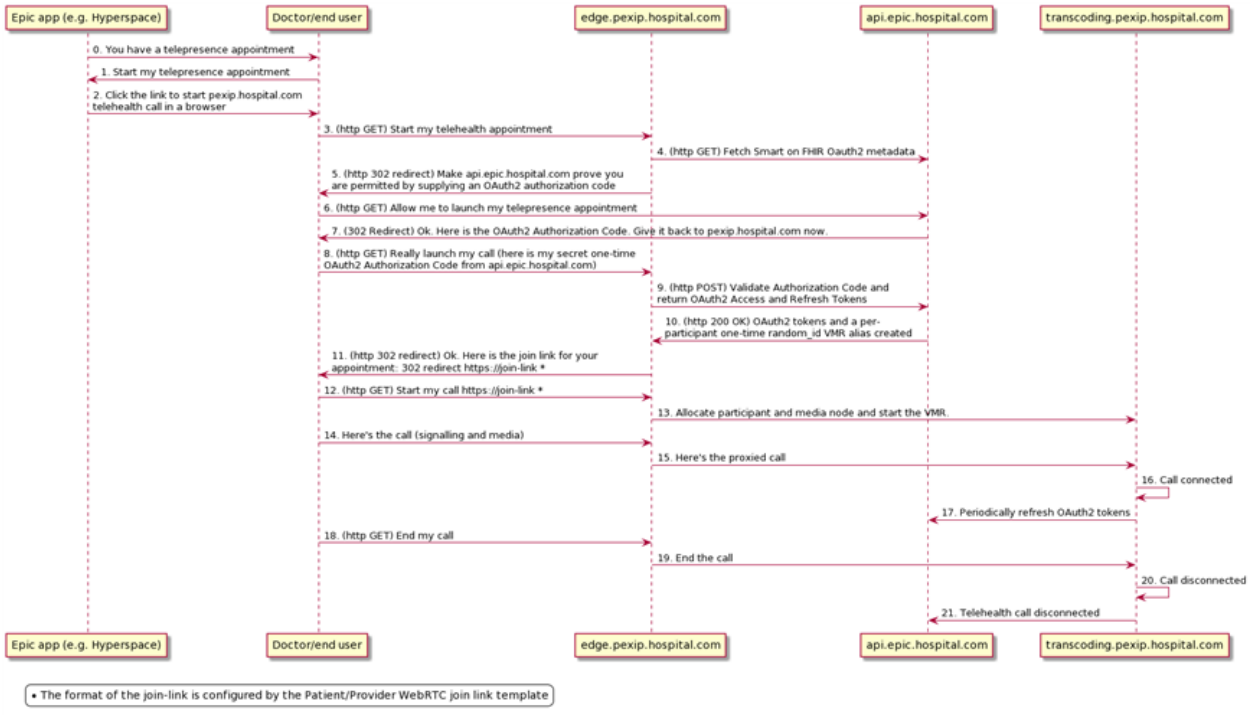
Symptom	Possible cause	Resolution
The telehealth call launches, but provider or patient audio/video is not being sent to the other meeting participants	Healthcare and hospital networks often have multiple network segments that are independently and very tightly/securely locked down. If they are very locked down, that may prevent audio and video being transmitted to and from an end user on a particular network segment to the Pexip Infinity server.	Verify that the network configuration allows audio and video to be sent to Pexip Infinity. Check that all the users can join a regular non-telehealth Virtual Meeting Room from their respective networks and that they can see and hear each other. Troubleshoot/fix the network configuration for simple VMRs by working with your video and networking teams, before re-testing telehealth calls.
	Transient USB webcam driver issues preventing camera/microphone selection.	Disconnect and reconnect your USB webcam, then place the call again.
	Web browser issues preventing camera/microphone selection.	Close all tabs on all web browsers and restart the browser, then place the call again.
	Third party video applications such as MS Teams can prevent other applications (such as Pexip) from using the webcam. (See this Microsoft community article.)	Exit any third party application (such as MS Teams) which may be attempting to use your webcam.
When a telehealth call terminates, including accidental termination or termination due to unexpected network stability issues, the appointment is automatically checked out within Epic	This is an Epic configuration issue.	This can be worked around on the Epic side from Epic May 2021 onwards. Please contact your Epic support representative and refer them to https://galaxy.epic.com/Redirect.aspx?DocumentID=100106247

Detailed call setup information

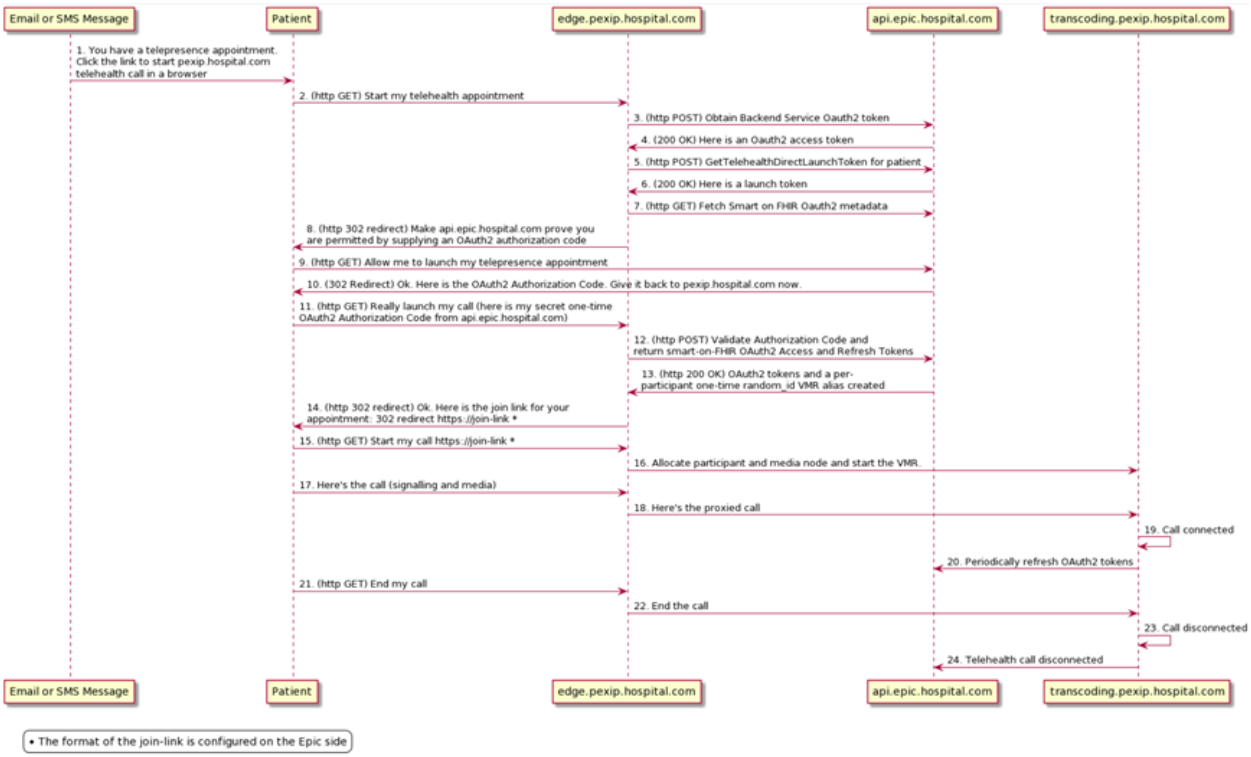
Pexip Infinity needs to access Epic at several points while setting up a telehealth call:

1. During OAuth authentication before the call starts (after a user has proven their identity to Epic's satisfaction they get redirected back to Pexip Infinity with a one-time-use ID which Pexip needs to take and exchange for Epic OAuth2 tokens).
2. When the call is connected and while it is ongoing, Pexip may need to periodically renew its OAuth2 tokens as they typically have a finite lifetime of approximately 1 hour.
3. When the call ends, Pexip uses its OAuth2 token to make an API call to let Epic know that the call has ended.

The following sequence diagram shows the call flow when establishing and finishing a telehealth call via the Epic apps. It shows the touch points between the Epic app, the patient/provider, the Pexip proxying and transcoding nodes and the API calls to the Epic server.



This diagram shows the flow when a patient joins via an SMS or email notification (step 7 onwards in this flow is the same as step 4 onwards in the Epic apps flow above):



Note that:

- The Epic API (at api.epic.hospital.com in our example) is publicly reachable.
- All transcoding nodes (and proxying nodes) need to be able to reach out to api.epic.hospital.com. There are no inbound messages to transcoding nodes.
- The api.epic.hospital.com entity in the sequence diagrams is sometimes referred to (in Epic terminology) as the "interconnect" server.
- You can use a reverse proxy instead of a proxying node (the addresses are configured in the Epic telehealth profile).

Integrating Pexip Infinity with authentication and provisioning services

Pexip Infinity can be configured to connect to various external services in order to authenticate, authorize, and provision accounts that are allowed to connect to Pexip Infinity and its clients and services. It can also use local accounts and databases for authentication.

For information about when and how to deploy Pexip Infinity with these authentication and provisioning services, see these individual topics:

Authentication overview

Pexip Infinity can be configured to connect to various external services in order to authenticate, authorize, and provision accounts that are allowed to connect to Pexip Infinity and its clients and services. It can also use local accounts and databases for authentication.

The table below shows which authentication services can be used for which Pexip features and services.

	Local database	LDAP	OpenID Connect	OAuth2	SAML/OIDC via IdP	AD FS
Pexip Infinity Administrator interface	✓	✓	✓			
Pexip Infinity management API	✓	✓		✓		
VMR / device bulk provisioning		✓				
Participant authentication to access VMRs					✓	
Pexip app for Windows registration					✓	
Legacy Connect desktop app registration (deprecated feature)						✓

- By default, Pexip Infinity has a single local administrator account for accessing the Pexip Infinity Administrator interface and the Pexip Infinity management API, authenticated with a username and password. For more information, see Managing local administrator authentication.
- Pexip Infinity can be configured to connect to a Windows Active Directory LDAP server, or any other LDAP-accessible database, in order to authenticate and authorize the login accounts that are allowed to connect to the Pexip Infinity Administrator interface or the Pexip Infinity management API, and to bulk-provision individual Virtual Meeting Rooms or devices for every member of the directory. For more information, see Managing administrator access via LDAP and Provisioning VMRs, devices and users from Active Directory via LDAP.
- Access to the Pexip Infinity Administrator interface can also be controlled via an OpenID Connect (OIDC) Identity Provider, allowing you to make use of the provider's single sign-on (SSO) and multi-factor authentication (MFA) capabilities. For more information see Managing administrator access via OIDC.
- Access to the Pexip Infinity management API can also be controlled via OAuth2. This is an alternative to LDAP for environments that make frequent API requests, as it can significantly reduce the number of authentication requests sent to the LDAP server. For more information, see Managing API access via OAuth2.
- Conference participants can be required to use single sign-on (SSO) to authenticate with a SAML 2.0 or OpenID Connect Identity Provider, in order to access conferences. For more information, see About participant authentication.
- Pexip app for Windows must authenticate with an Identity Provider in order to register with Pexip Infinity. For more information, see Authenticating app registrations using Identity Providers.

- Pexip Infinity can integrate with Active Directory Federation Services (AD FS) to provide the legacy Connect desktop app with single sign-on (SSO) access to allow users to register their clients using their AD credentials.

  This method of authentication is being deprecated and is not supported by the new Pexip apps.

  For more information, see our v39 documentation site.

Managing local administrator authentication

By default, Pexip Infinity only has a single local administrator account for accessing the Pexip Infinity Administrator interface and the Pexip Infinity management API. The username and password for this account are set during the installation process, when running the installation wizard. The default username is **admin**.

When logging in using this account, the credentials are authenticated against Pexip Infinity's own local (internal) database.

This local administrator account can be used on its own, or in addition to other administrator authentication methods such as LDAP, OpenID Connect, and OAuth.

Administrator roles are not relevant to the local admin account. The account always has full read-write access, and this is not configurable.

Configuring Pexip Infinity

To allow administrators to use the local administrator account when they log in to the Pexip Infinity Administrator interface or management API, go to **Users & Devices > Administrator Authentication**. The relevant options are:

Option	Description
Authentication source	The database to query for administrator authentication and authorization. ***Local database***: uses the Pexip Infinity local on-box database. Administrators can only log in via the default account (typically **admin**) and will have full administrator privileges. ***LDAP database***: administrators can only log in using an account configured on the LDAP database and obtain privileges according to the groups and roles associated with that account. Note that if this option is selected and the LDAP server is inaccessible for any reason, administrators will not be able to log in to the Pexip Infinity web-based Administrator interface or management API. ***LDAP database and local database***: administrators can log in using either the default local **admin** account or via an account configured on the LDAP database. ***Open ID Connect service***: administrators can only log in using an account configured on the OpenID Connect provider and obtain privileges according to the groups and roles associated with that account. Note that if this option is selected and the OpenID Connect provider is inaccessible for any reason, administrators will not be able to log in to the Pexip Infinity web-based Administrator interface or management API. ***Open ID Connect service and local database***: administrators can log in using either the default local **admin** account or via an account configured on the OpenID Connect provider. ***LDAP database, Open ID Connect service and local database***: administrators can log in using the default local **admin** account, via an account configured on the LDAP database, or via an account configured on the OpenID connect provider. To use the local admin account, ensure that ***Local database*** is selected, either on its own or in combination with another authentication source. Default: ***Local database***.
Require client certificate	This option is only applicable if using LDAP as an authentication source. For more information, see Certificate-based authentication.
Management API Oauth2 settings	
The settings in this section apply to management API access via OAuth only. They do not affect local, LDAP or OpenID Connect access. For more information, see Managing API access via OAuth2.	

Option	Description
LDAP configuration and Advanced LDAP configuration	
The settings in these sections apply to LDAP access only. They do not affect local or OpenID Connect access. For more information, see Managing administrator access via LDAP.	
OpenID Connect configuration	
The settings in this section apply to OpenID Connect access only. They do not affect local or LDAP access. For more information, see Managing administrator access via OIDC.	

Reinstating the local admin account

If necessary you can reinstate access via the Pexip Infinity local on-box database, so that administrators can log in via the default admin account with full administrator privileges. You may need to do this if, for example, the **Authentication source** is configured as ***LDAP database*** only and your connectivity to the LDAP server goes down or your credentials become invalid.

To reactivate your local admin account, log in to the Management Node over SSH and run one of the following commands:

Command	Enables access via...
authset LDAP LOCAL	local admin account only
authset LDAP REMOTE	LDAP accounts only
authset LDAP BOTH	local admin account and LDAP accounts
authset LDAP ALL	local admin account, LDAP accounts and OIDC accounts
authset OIDC LOCAL	local admin account only
authset OIDC REMOTE	OIDC accounts only
authset OIDC BOTH	local admin account and OIDC accounts
authset OIDC ALL	local admin account, LDAP accounts and OIDC accounts

If you forget the password for the Pexip Infinity Administrator interface, you can re-run the installation wizard, being sure to change only the **Web administration password** setting.

Managing administrator access via LDAP

You can configure the Pexip Infinity platform to authenticate and authorize administrator login accounts via a centrally managed LDAP-accessible server. Integration with LDAP provides increased security, better auditing of changes, and more control and flexibility as you can assign different privileges to specific groups of users.

By default, Pexip Infinity only has a single local administrator account. Integration with an LDAP directory service allows multiple users to administer the platform. These users log in with their directory credentials, which is generally a Windows AD domain. When using LDAP:

- Instead of authenticating the supplied username and password credentials against its own internal database, Pexip Infinity contacts the LDAP server to authenticate the administrator's user account.
- It uses the account's LDAP group attributes in combination with role mappings defined in Pexip Infinity to determine which Pexip Infinity features the administrator is authorized to access.
- Every management API request is preceded by an authentication request to the LDAP server. If the frequency of API requests in your environment means that this is causing a significant load on your LDAP server, you may wish to consider using OAuth2 as an alternative authentication method for the management API. (The LDAP server will still receive requests for Administrator interface authentication.)

You can also enable client certificate authentication when using LDAP.

The configuration described here applies to all administrator accounts connecting to the **Pexip Infinity Administrator interface** or the **Pexip Infinity management API**. It does not apply to SSH connections. When using LDAP authentication, Pexip Infinity is configured by default to work with a Windows Active Directory LDAP server, but it can also be configured to work with other LDAP-accessible databases.

In addition to LDAP, you can authenticate and authorize administrator accounts connecting to the **Pexip Infinity Administrator interface** using Open ID Connect. For more information, see Managing administrator access via OIDC.

You can also allow administrator accounts connecting to the **Pexip Infinity management API** to use OAuth instead of, or in addition to, LDAP. For more information, see Managing API access via OAuth2.

The following sections describe:

- Configuration summary for LDAP authentication
- Configuring Pexip Infinity
- Configuring administrator roles
- Configuring role mappings
- Examples: configuring permissions for an AD group
- Reinstating the local admin account

Configuration summary for LDAP authentication

To enable authentication and authorization via LDAP, you need to configure both the LDAP database (if it is not already configured with user details) and the Pexip Infinity platform.

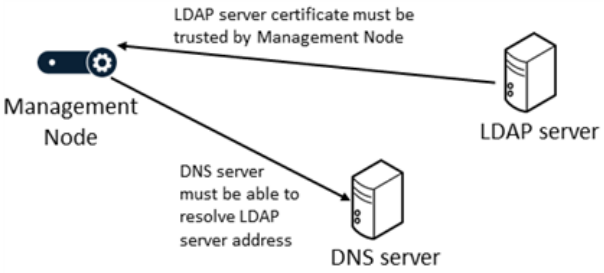
The LDAP database must be configured with:

- administrators' user credentials
- groups that define the capabilities of the users.

The Pexip Infinity platform must be configured with:

- an authentication source setting that uses an LDAP database
- connection details for the LDAP server; if the server address is an FQDN, ensure that it is resolvable over the DNS server configured for the Management Node
- administrator roles to control the actions that administrators can perform
- LDAP role mappings that map LDAP groups to administrator roles.

If a secure TLS connection between the LDAP server and the Management Node is required, ensure that:



- The LDAP server address is specified as an FQDN (so that it matches the name on the certificate presented by the LDAP server, which is typically created for the host name rather than the IP address).
- The Management Node trusts the certificate presented by the LDAP server; typically this means that the LDAP server certificate has to be uploaded to the Management Node as a trusted CA certificate (as the LDAP server's certificate is often generated by an internal authority which would not be included in Pexip's inbuilt list of trusted CA certificates).

  Note that the Management Node's server certificate does not have to be trusted by the LDAP server (unless the LDAP server has been explicitly configured to demand a client certificate).

The Pexip Infinity platform configuration steps for specifying an LDAP authentication source, and configuring administrator and LDAP role mappings are described in more detail in the following sections, and there is an example that shows how to configure permissions for an AD group. For information about installing server and trusted CA certificates, see Managing TLS and trusted CA certificates.

Configuring Pexip Infinity

To allow administrators to use LDAP authentication when they log in to the Pexip Infinity Administrator interface or management API, go to **Users & Devices > Administrator Authentication**. The options are:

Option	Description
Authentication source	The database to query for administrator authentication and authorization. ***Local database***: uses the Pexip Infinity local on-box database. Administrators can only log in via the default account (typically **admin**) and will have full administrator privileges. ***LDAP database***: administrators can only log in using an account configured on the LDAP database and obtain privileges according to the groups and roles associated with that account. Note that if this option is selected and the LDAP server is inaccessible for any reason, administrators will not be able to log in to the Pexip Infinity web-based Administrator interface or management API. ***LDAP database and local database***: administrators can log in using either the default local **admin** account or via an account configured on the LDAP database. ***Open ID Connect service***: administrators can only log in using an account configured on the OpenID Connect provider and obtain privileges according to the groups and roles associated with that account. Note that if this option is selected and the OpenID Connect provider is inaccessible for any reason, administrators will not be able to log in to the Pexip Infinity web-based Administrator interface or management API. ***Open ID Connect service and local database***: administrators can log in using either the default local **admin** account or via an account configured on the OpenID Connect provider. ***LDAP database, Open ID Connect service and local database***: administrators can log in using the default local **admin** account, via an account configured on the LDAP database, or via an account configured on the OpenID connect provider. When using an LDAP database either on its own or in combination with another authentication source, you must configure the items in the LDAP configuration section. By default, Pexip Infinity checks the entered username against the Active Directory **sAMAccountName** attribute (as configured in the **LDAP user search filter** advanced setting below). Default: ***Local database***.
Require client certificate	Controls whether administrators are authenticated via a client certificate. By default, administrators log in to the Pexip Infinity Administrator interface via the standard login page, or provide an authorization header when accessing the management API. Instead, users can be required to present (via their browser) a client certificate containing their user identification details. The options are: ***Not required***: Client certificates are not required. Administrators log in via the standard login page and provide a password which is authenticated against the selected **Authentication source**. Management API requests require an authorization header. ***Required (user identity in subject CN)***: administrators identify themselves via the identity contained in the subject CN (common name) of the client certificate presented by their browser. ***Required (user identity in subjectAltName userPrincipalName)***: administrators identify themselves via the identity contained in the **subjectAltName userPrincipalName** attribute of the client certificate presented by their browser. Default: ***Not required***. ⓘ When a client certificate is required, the standard login page is no longer presented. Administrators will not be able to access the Pexip Infinity Administrator interface or the management API if their browser does not present a valid certificate that contains a user identity which exists in the selected **Authentication source**. To reinstate access via the Pexip Infinity Administrator interface or management API, see Disabling certificate-based authentication.
Management API Oauth2 settings	

Option	Description
The settings in this section apply to management API access via OAuth only. They do not affect local, LDAP or OpenID Connect access. For more information, see Managing API access via OAuth2.	
LDAP configuration	
LDAP server address	The domain name (for DNS SRV lookup), FQDN (for DNS A/AAAA lookup) or IP address of the LDAP server. If using a domain or an FQDN, ensure that it is resolvable over DNS. You must also ensure that Pexip Infinity has trusted CA certificates for the authority that signed the LDAP server's certificate (if a TLS connection is required). We strongly recommend that you do not use an IP address. If an IP address is used, and a TLS connection is required, this will only work if the IP address is specified as the common name in the LDAP server's certificate. See Troubleshooting LDAP server connections for more information about how the system establishes a connection to the LDAP server and how to troubleshoot connection issues.
LDAP bind username and password	The username and password of the bind account on the LDAP server. This should be a domain user service account, not the Administrator account.
LDAP base DN	The base DN (distinguished name) of the LDAP forest to query (e.g. **dc=example,dc=com**).
OpenID Connect configuration	
The settings in this section apply to OpenID Connect access only. They do not affect local or LDAP access. For more information, see Managing administrator access via OIDC.	
Advanced LDAP configuration	
By default the advanced LDAP configuration settings are preconfigured for Windows Active Directory, and may also be appropriate for other LDAP databases such as OpenLDAP.	
Search global catalog	Select this option to expand the scope of the search to the entire Active Directory Global Catalog instead of traditional LDAP. Note that this uses ports 3268 (TCP) and 3269 (TLS).
LDAP user search DN	The DN relative to the **LDAP base DN** to query for user records (e.g. **ou=people**). If blank, the **LDAP base DN** is used. In deployments with large user bases, you may want to configure this to optimize the LDAP user queries.
LDAP user filter	The LDAP filter used to match user records in the directory. Default: (&(objectclass=person)(!(objectclass=computer)))
LDAP user search filter	The LDAP filter used to find user records when given the user name. The filter must contain {username} to indicate locations into which the username is substituted. This filter is applied in conjunction with the **LDAP user filter** and must contain at least one substitution. If client certificate-based authentication is used, this filter usually must include 'userPrincipalName={username})' either in addition to, or instead of, the default value; for example '(|(uid={username})(sAMAccountName={username})(userPrincipalName={username}))'. To log in using an email address, you can use '(|(uid={username})(sAMAccountName={username})(mail={username}))' — note that this requires the use of LDAPS. Default: (|(uid={username})(sAMAccountName={username}))
LDAP group attributes	A comma-separated list of attributes in the LDAP user record to examine for group membership information. The attribute value must contain the DN of each group the user is a member of. If no attributes are specified, or none of the specified attributes are present in the LDAP user record, an LDAP group search (using the remaining advanced configuration options below) is performed instead. Default: memberOf

Option	Description
LDAP group search DN	The DN relative to the **LDAP base DN** to query for group records (e.g. **ou=groups**) when no group attributes are present in the LDAP user record. If blank, the **LDAP base DN** is used. In deployments with large user bases, you may want to configure this to optimize the LDAP group queries.
LDAP group filter	The LDAP filter used to match group records in the directory. Default: (|(objectclass=group)(objectclass=groupOfNames)(objectclass=groupOfUniqueNames)(objectclass=posixGroup))
LDAP group membership filter	The LDAP filter used to search for group membership of a user. The filter may contain **{userdn}** to indicate locations into which the user DN is substituted. The filter may contain **{useruid}** to indicate locations into which the user UID is substituted. This filter is applied in conjunction with the LDAP group filter and must contain at least one substitution. Default: (|(member={userdn})(uniquemember={userdn})(memberuid={useruid}))

If authentication against an LDAP database is configured, you can save the settings only if Pexip Infinity can successfully contact the specified LDAP server.

Note that all LDAP distinguished names must be entered as per the LDAP standard (RFC 4514). LDAP configuration is case insensitive.

Certificate-based authentication

When using LDAP as an authentication source, you can configure the Pexip Infinity platform for client certificate authentication. This means that instead of logging in to the Pexip Infinity Administrator interface via the standard login page, or providing an authorization header when accessing the management API, administrators present (via their browser) a client certificate containing their user identification details. The validation of the presented certificate acts as the authentication phase and the username attributes in the certificate are used to determine which features the administrator is authorized to access.

To enable client certificate authentication, from the Require client certificate field select either of the ***Required...***options.

Disabling certificate-based authentication

When a client certificate is required, the standard login page is no longer presented. Administrators will not be able to access the Pexip Infinity Administrator interface or the management API if their browser does not present a valid certificate that contains a user identity which exists in the selected **Authentication source**.

To disable client certificate authentication so that you can log in to the Pexip Infinity Administrator interface via the standard login page:

1. Log in to the Management Node over SSH.
2. Run the command:
   ```
   authset CBA OFF
   ```

Supporting nested security groups in Windows Active Directory

The default LDAP configuration does not support nested security groups in Windows Active Directory. For example, if group A is allowed to log in via LDAP, and if group B is a member of group A, then any user who is only a member of group B will not be allowed to log in.

To allow members of a nested Active Directory security group to log in over LDAP:

1. Go to **Users & Devices > User Authentication** and expand the **Advanced LDAP configuration** section.
2. Ensure that **LDAP group attributes** is empty (i.e. remove the default "memberOf" content).
3. Change **LDAP group membership filter** to "(member:1.2.840.113556.1.4.1941:={userdn})"
4. Select **Save**.

(This configuration uses the LDAP_MATCHING_RULE_IN_CHAIN OID. More information on this can be found at https://msdn.microsoft.com/en-us/library/aa746475%28VS.85%29.aspx.)

Configuring administrator roles

Administrator roles control the actions that administrators can perform after they have been authenticated. You create one or more administrator roles, and then map them to your LDAP groups, allowing you to grant different privileges to different sets of administrator users.

To add, edit or delete administrator roles, go to **Users & Devices > Administrator Roles**. For full details of the options that are available, see Managing administrator roles.

Configuring role mappings

LDAP role mappings are used to map the LDAP groups associated with LDAP user records to the Pexip Infinity administrator roles. You must configure a separate LDAP role mapping for each LDAP group for which you want to map one or more Pexip Infinity administrator roles.

To add, edit or delete LDAP role mappings, go to **Users & Devices > Role Mapping**. When configuring LDAP role mappings, the options are:

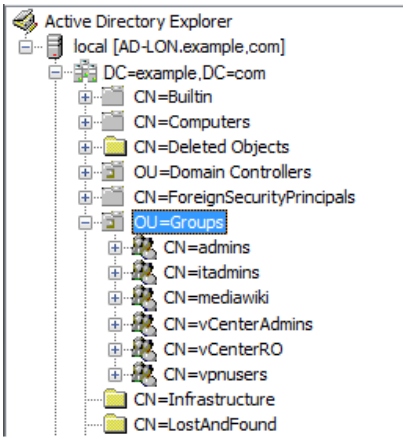
Option	Description
Name	A descriptive name of the role mapping, e.g. "domain administrator with full privileges".
Source	The authentication source to which this role mapping applies. Select ***LDAP***.
LDAP group DN	Select the LDAP group against which you want to map one or more administrator roles. The list of LDAP groups is only populated when there is an active connection to an LDAP server (**Users & Devices > Administrator Authentication**). Note that the LDAP groups used for role mappings cannot be the pre-defined AD groups such as **Domain Users** etc. but need to be explicitly configured custom groups.
Roles	Select from the list of **Available roles** the administrator roles to associate with the LDAP group and then use the right arrow to move the selected roles into the **Chosen Roles** list. All of the underlying permissions within a role are "positive" permissions, i.e. they allow the administrator to do something. If more than one role is selected, all of the permissions associated with each role are combined and granted to the relevant administrator. Note that you can select which opens a new window from where you can configure a new administrator role. When you save the role it is automatically added to the set of **Chosen Roles**.

Examples: configuring permissions for an AD group

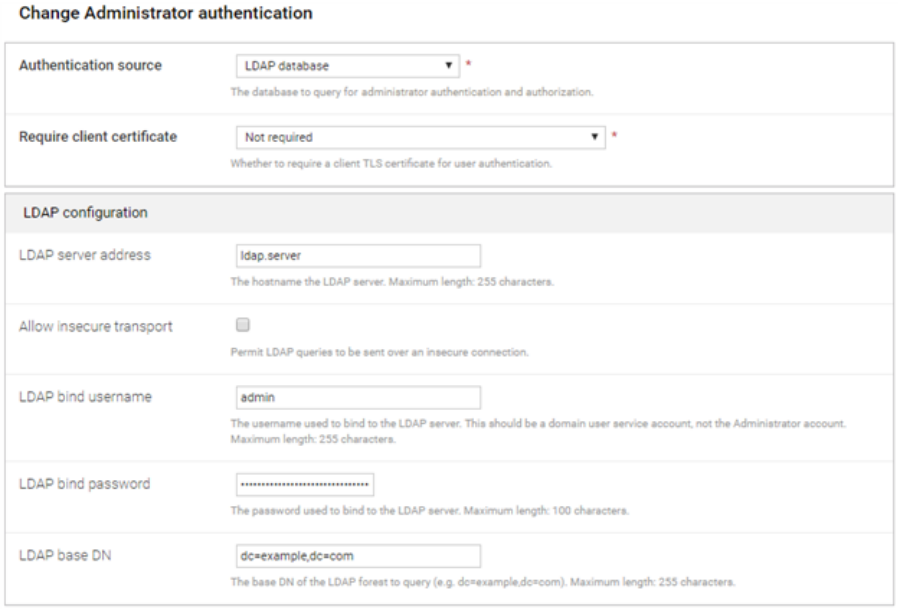
These examples show how you can configure the specific actions (permissions) that all members of an AD group are allowed to perform when administering Pexip Infinity, and provide methods to filter the groups that are displayed.

The filtering options are not mandatory but they do make it easier to select the appropriate LDAP groups, and can optimize system performance.

Let's assume that you have the following set of groups already configured in Windows Active Directory:



In both of the examples below you need to ensure that you have configured an LDAP authentication source (**Users & Devices > Administrator Authentication**) that can access your AD server, for example:



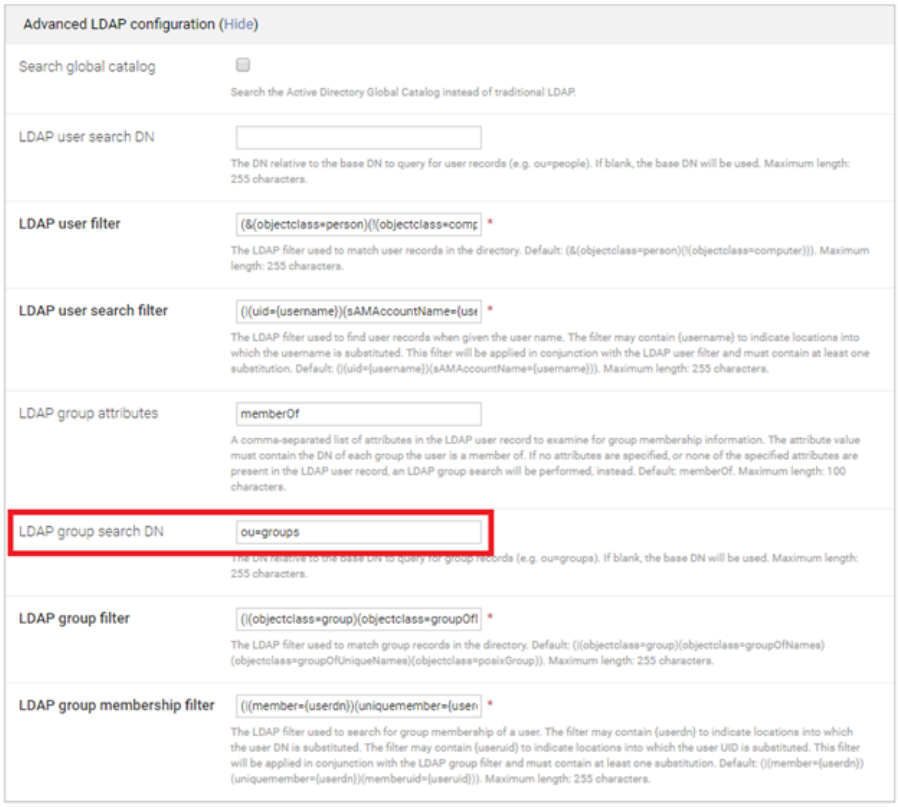
Example: Filtering by specifying an LDAP group search DN

This example shows how to configure all AD users who are members of the "itadmins" group to be able to add, modify and delete VMR/conference related settings, but only be able to view other configuration aspects of Pexip Infinity (system settings, logs etc).

To make it easier to select the "itadmins" group we have specified an **LDAP group search DN** to limit the number of LDAP groups that are presented when configuring your LDAP roles:

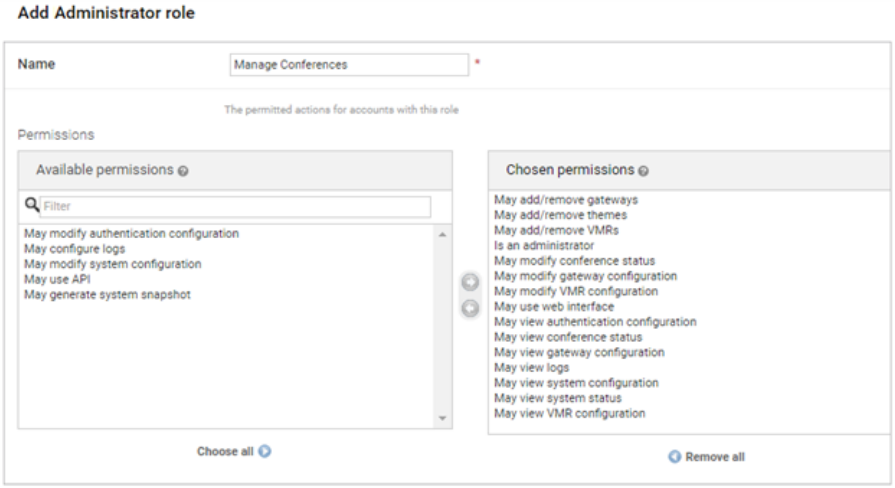
1. Go to **Users & Devices > Administrator Authentication** where your LDAP configuration has been completed, as shown above, and open the **Advanced LDAP Configuration** section.
2. In this case, we want to define permissions based upon membership of specific AD groups, therefore we have configured the **LDAP group search DN** setting to **ou=groups**.

   This means that when we configure the LDAP roles, the set of LDAP groups that is presented is filtered to include only those in the **groups** organizational unit (ou).



3. We now need to configure an administrator role (**Users & Devices > Administrator Roles**) that defines the set of actions that can be performed by administrators who have been assigned that role.

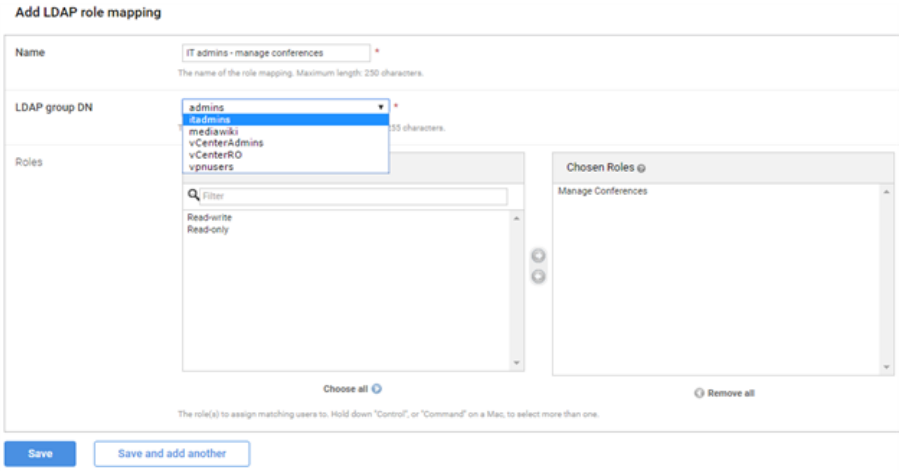
   Here, a "Manage Conferences" role has been created. The **Chosen permissions** allow an administrator to use the web interface to configure all service-related items such as VMRs, themes, gateway rules, but to only be able to view (and not modify) all other configuration.



4. The final step is to associate this administrator role with an LDAP role/group (**Users & Devices > Role Mapping**).

   Here, we have configured an "IT admins - manage conferences" role with a **Source** of ***LDAP***. The **LDAP group DN** drop-down presents a list of LDAP groups from AD. In our case this list is filtered to only show those groups in the **ou=groups** organizational unit (due to the **LDAP group search DN** configuration in step 2).

   We have selected the **itadmins** group and associated it with the **Manage Conferences** role we created in step 3. (Note that you can associate the LDAP role with more than one administrator role if required.)



This means that AD users who are in the itadmins group can now sign in to the Pexip Infinity Administrator interface, using their AD credentials, and configure service-related settings only.

To set up different permissions for members of other AD groups, repeat steps 3 and 4 to create different administrator role and LDAP role associations.

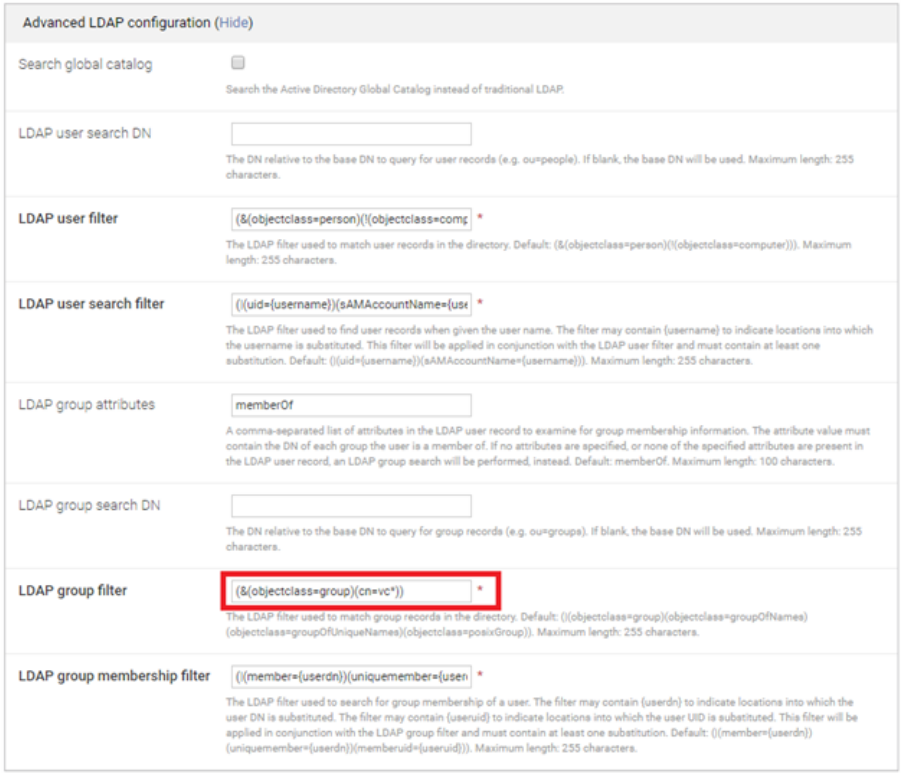
Example: Limiting groups by specifying an LDAP group filter

This example is similar to the example above, but shows an alternative method of limiting the number of LDAP groups that are presented when configuring your LDAP roles.

In this case we show how to specify an **LDAP group filter** to limit the groups that are displayed.

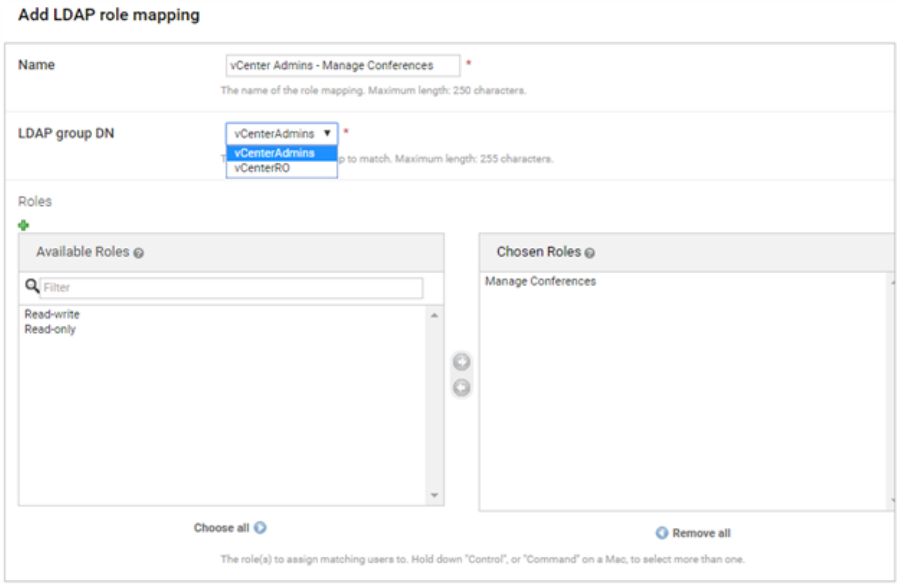
1. Go to **Users & Devices > Administrator Authentication** where your LDAP configuration has been completed, as shown above, and open the **Advanced LDAP Configuration** section.
2. In this example, we are specifying a group filter so that when we configure the LDAP roles, the set of LDAP groups that is presented is filtered to only show those whose name starts with "vc".

   We do this by configuring the **LDAP group filter** to **(&(objectclass=group)(cn=vc*))**



3. As with the previous example, you need to ensure that you have configured an administrator role (**Users & Devices > Administrator Roles**), such as "Manage Conferences", that defines the set of actions that can be performed by administrators who have been assigned that role.
4. The final step is to associate this administrator role with an LDAP role/group (**Users & Devices > Role Mapping**).

   Here, we have configured a "vCenter Admins - Manage Conferences" role. The **LDAP group DN** drop-down presents a list of LDAP groups from AD. In our case this list is filtered to only show those groups whose name starts with "vc".



This means that AD users who are in the vCenterAdmins group can now sign in to the Pexip Infinity Administrator interface.

Reinstating the local admin account

If necessary you can reinstate access via the Pexip Infinity local on-box database, so that administrators can log in via the default admin account with full administrator privileges. You may need to do this if, for example, the **Authentication source** is configured as ***LDAP database*** only and your connectivity to the LDAP server goes down or your credentials become invalid.

For full instructions, see Reinstating the local admin account.

Managing administrator access via OIDC

You can configure the Pexip Infinity platform to authenticate and authorize administrator login accounts for the Pexip Infinity Administrator interface via the OpenID Connect (OIDC) protocol. Integration with an OIDC Identity Provider allows you to make use of the provider's single sign-on (SSO) and multi-factor authentication (MFA) capabilities and provides increased security, better auditing of changes, and more control and flexibility as you can assign different privileges to specific groups of users.

By default, Pexip Infinity only has a single local administrator account. Integration with OIDC allows multiple users to administer the platform. These users log in with their OIDC credentials to an OpenID provider such as Microsoft Azure. When using OIDC:

- Instead of authenticating the supplied username and password credentials against its own internal database, Pexip Infinity contacts the OpenID provider to authenticate the administrator's user account.
- It uses the account's OIDC group attributes in combination with role mappings defined in Pexip Infinity to determine which Pexip Infinity features the administrator is authorized to access.

In addition to OIDC, you can authenticate and authorize administrator login accounts using a centrally managed LDAP-accessible server. For more information, see Managing administrator access via LDAP.

The configuration described here applies to all administrator accounts connecting to the **Pexip Infinity Administrator interface**. It does not apply to connections to the **Pexip Infinity management API**, or SSH connections.

The following sections describe:

- Configuration summary for OIDC authentication
- Configuring Azure as an OIDC provider
- Configuring Pexip Infinity
- Configuring administrator roles
- Configuring OIDC role mappings
- Reinstating the local admin account

Configuration summary for OIDC authentication

To enable authentication and authorization via OIDC, you need to configure both the OIDC provider (if it is not already configured with user details) and the Pexip Infinity platform.

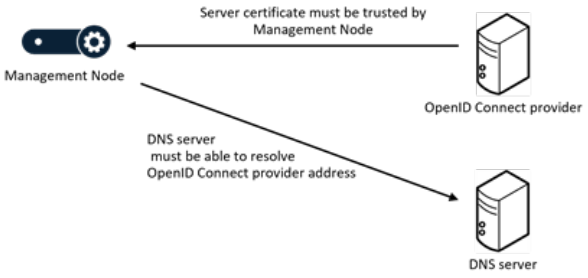
The OpenID provider must already be configured with:

- administrators' user credentials
- groups that define the capabilities of the users.

The Pexip Infinity platform must then be configured with:

- an Administrator **Authentication source** setting that uses an OIDC provider (i.e. includes ***OpenID Connect service***)
- connection details for the OIDC provider; if the server address is an FQDN, ensure that it is resolvable over the DNS server configured for the Management Node
- administrator roles to control the actions that administrators can perform
- role mappings that map OIDC groups to administrator roles.

If a secure TLS connection between the OIDC provider and the Management Node is required, ensure that:



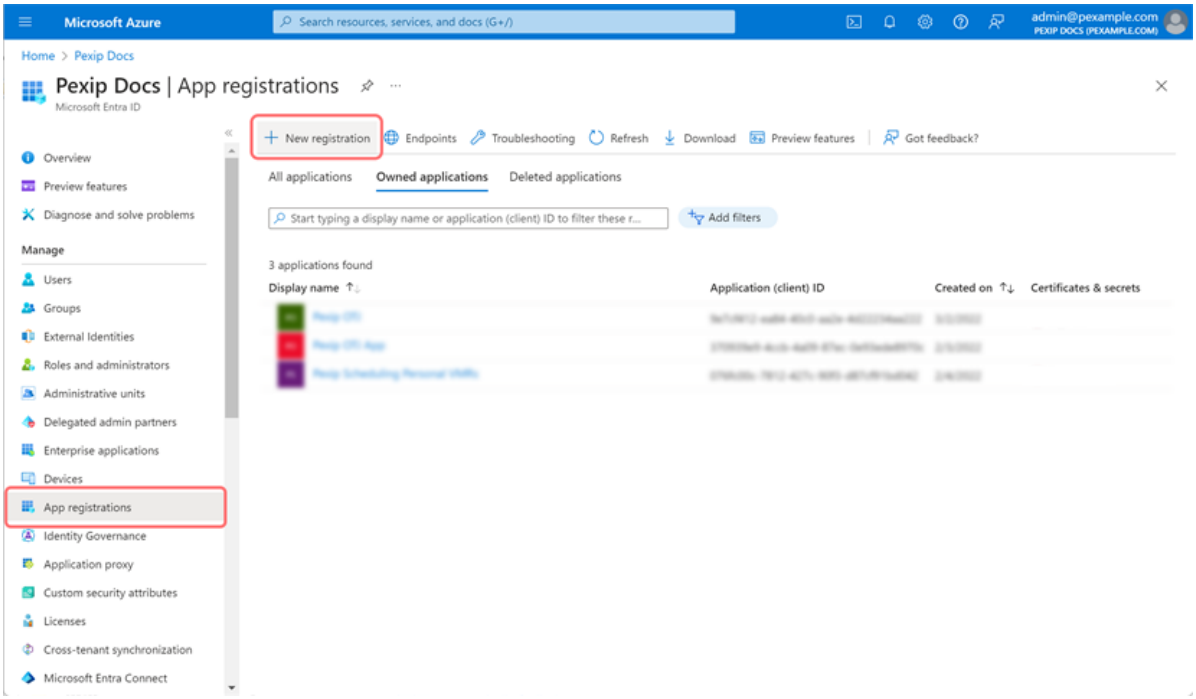
- The OIDC provider address is specified as an FQDN (so that it matches the name on the certificate presented by the OIDC provider, which is typically created for the host name rather than the IP address).
- The Management Node trusts the certificate presented by the OIDC provider; typically this means that the OIDC provider server certificate has to be uploaded to the Management Node as a trusted CA certificate (as the OIDC provider server's certificate is often generated by an internal authority which would not be included in Pexip's inbuilt list of trusted CA certificates).

  Note that the Management Node's server certificate does not have to be trusted by the OIDC provider (unless the OIDC provider has been explicitly configured to demand a client certificate).

The steps for configuring Azure as an OIDC provider, and configuring the Pexip Infinity platform to use Azure as an OIDC authentication source, are described in more detail in the following sections. For information about installing server and trusted CA certificates, see Managing TLS and trusted CA certificates.

Configuring Azure as an OIDC provider

1. Log in to the Azure portal and select **Microsoft Entra ID**. From the panel on the left select **App registrations** and then select **New registration**:

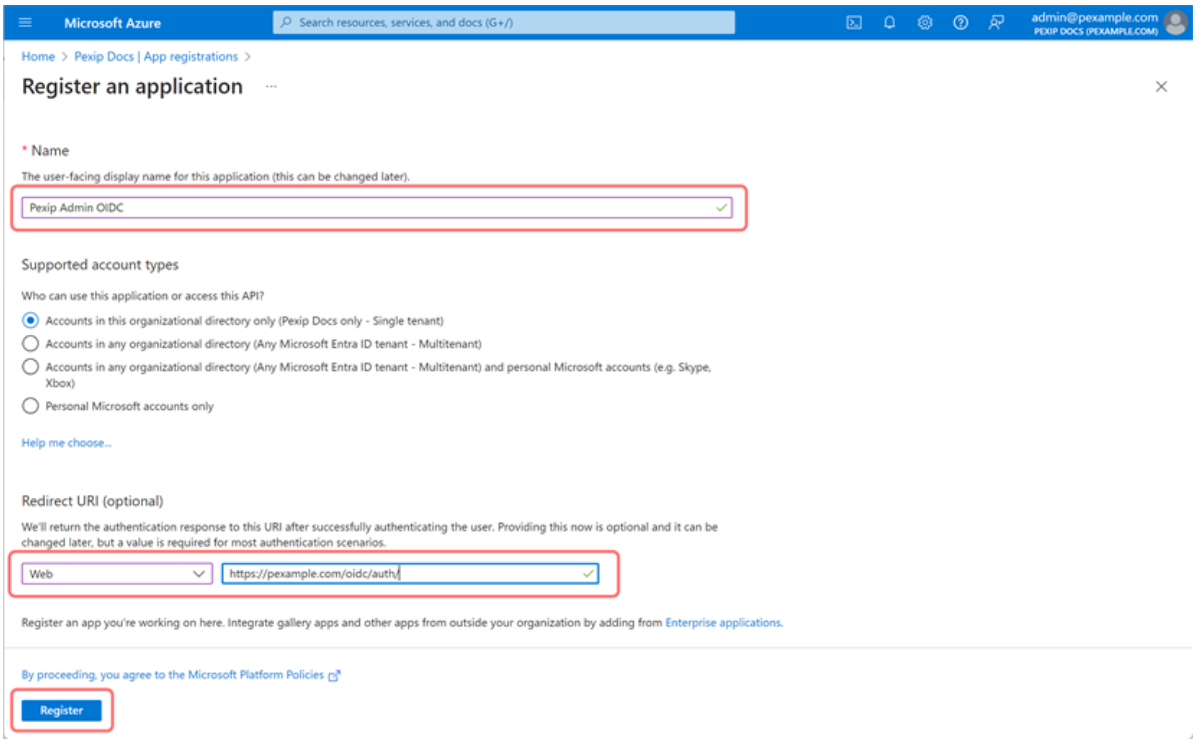


2. On the **Register an application** page:
   - Enter an appropriate **Name**
   - In the **Redirect URI** section, from the drop-down options select ***Web*** (if you are using an Authentication method of ***Client secret***) or ***Single-page application*** (if you are using ***PKCE***) and enter a redirect URI in the format **https://<Management Node FQDN>/oidc/auth/**

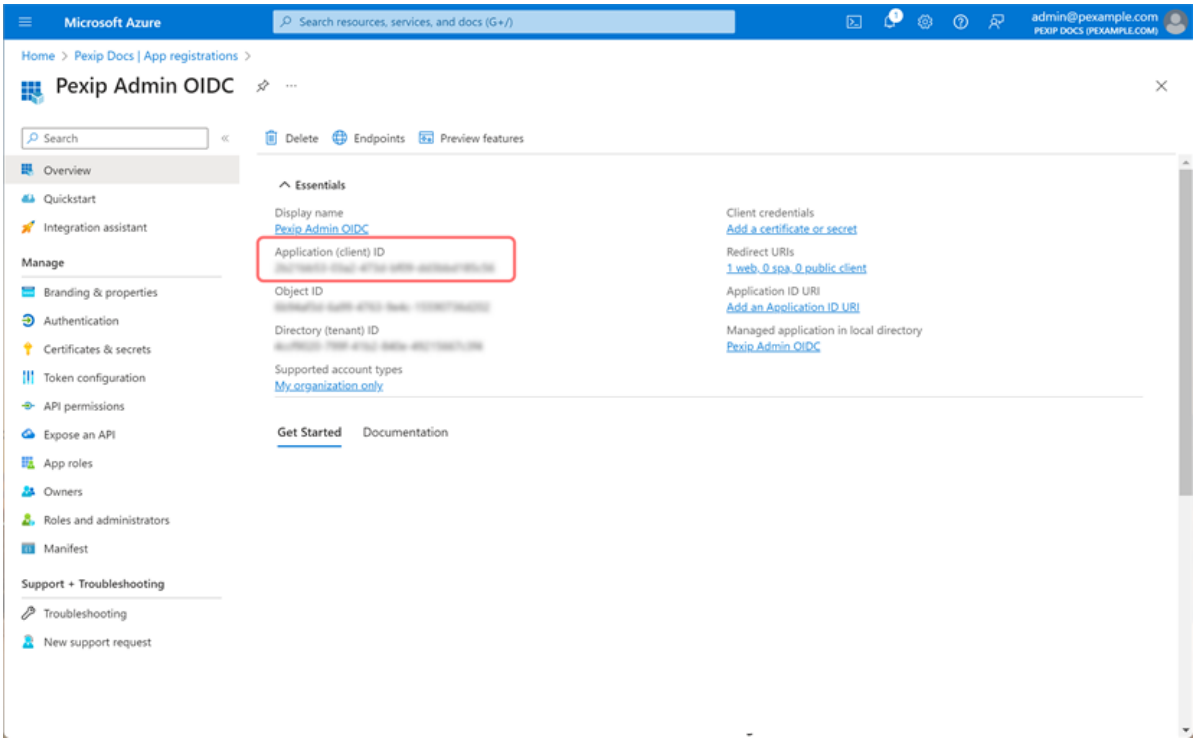
     The **Redirect URI** is the page on the Administrator interface to which the Pexip Infinity administrator will be returned after they have successfully signed in to the OIDC provider. Because it is a page on the Management Node, this URI is internal to your deployment and only needs to be accessible from the administrator's web browser; you do not need to

make it externally accessible.

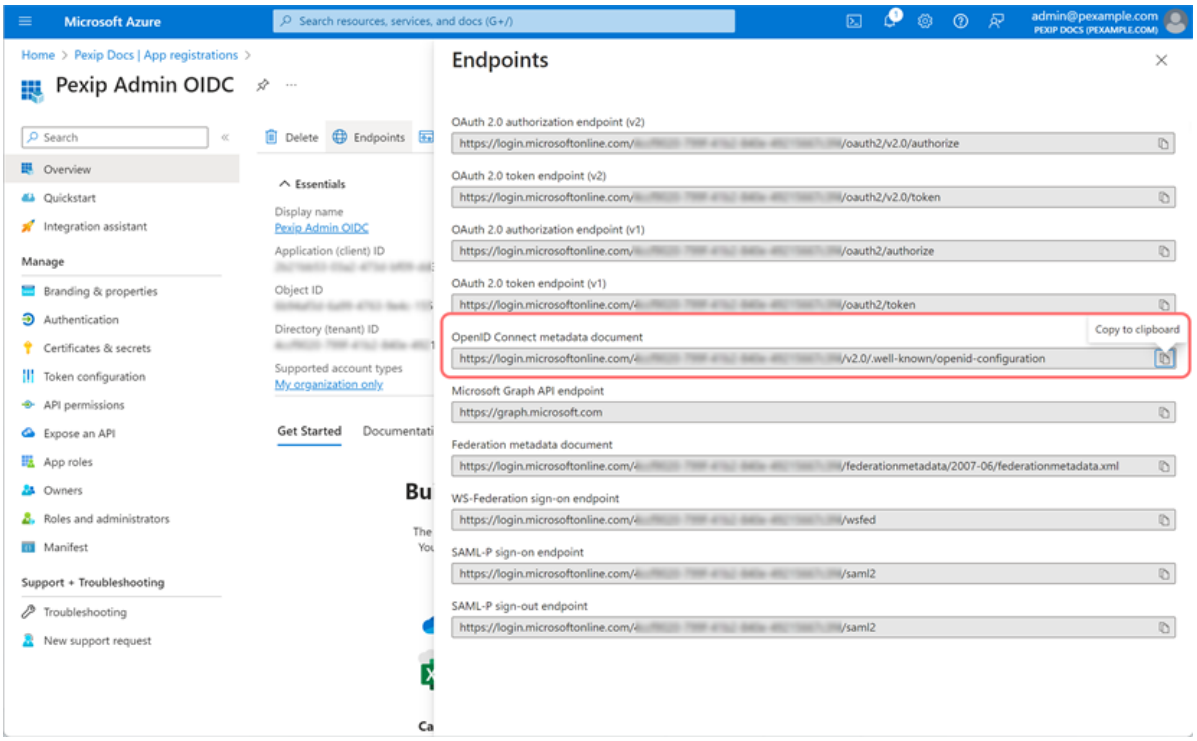
- Select **Register**:



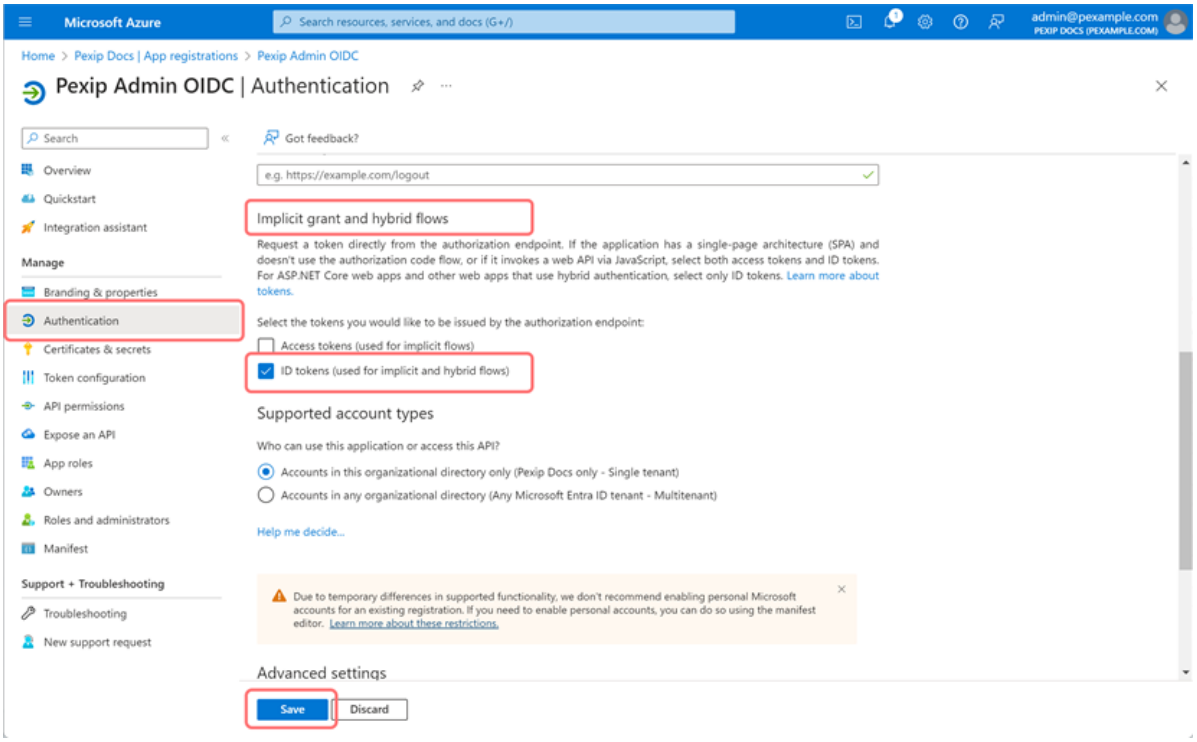
3. Note the Application (client) ID. You must enter this as the Client ID when configuring Pexip Infinity:



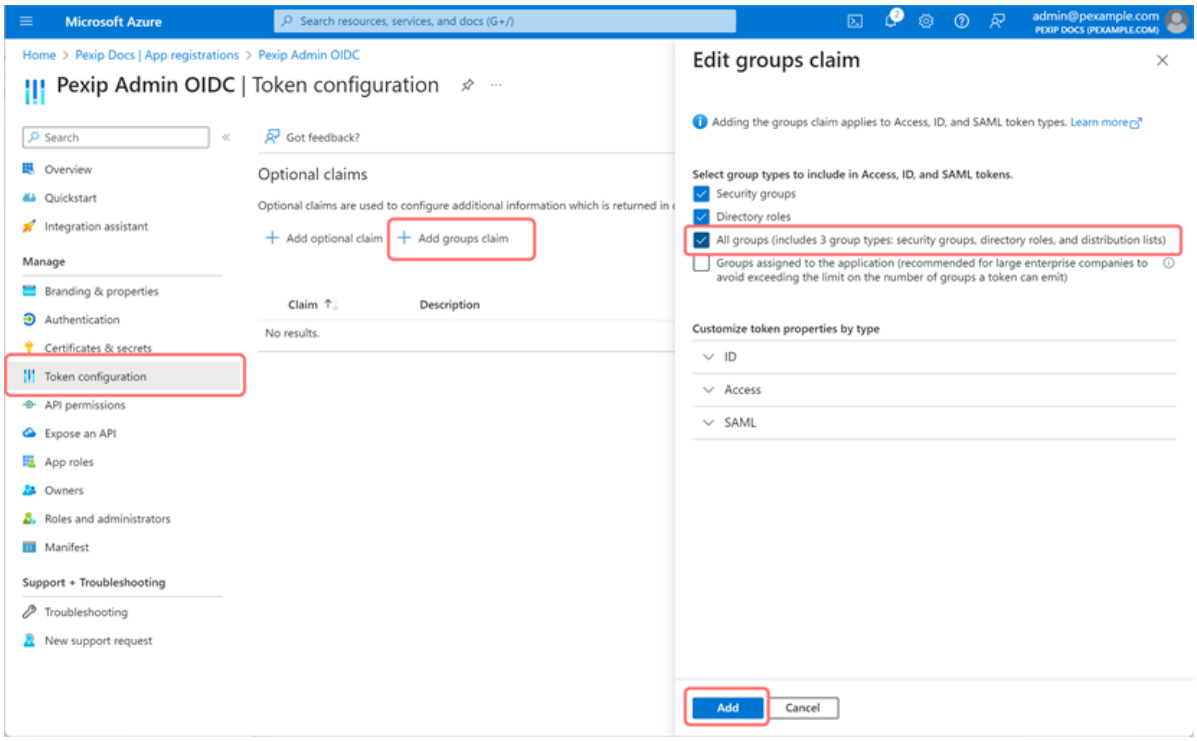
4. Select the **Endpoints** tab, and copy the **OpenID Connect metadata document** (this ends in **openid-configuration**). You must enter this as the Metadata URL when configuring Pexip Infinity:



5. From the panel on the left, select **Authentication** and scroll down to **Implicit grant and hybrid flows**. Select **ID tokens** and then select **Save**:

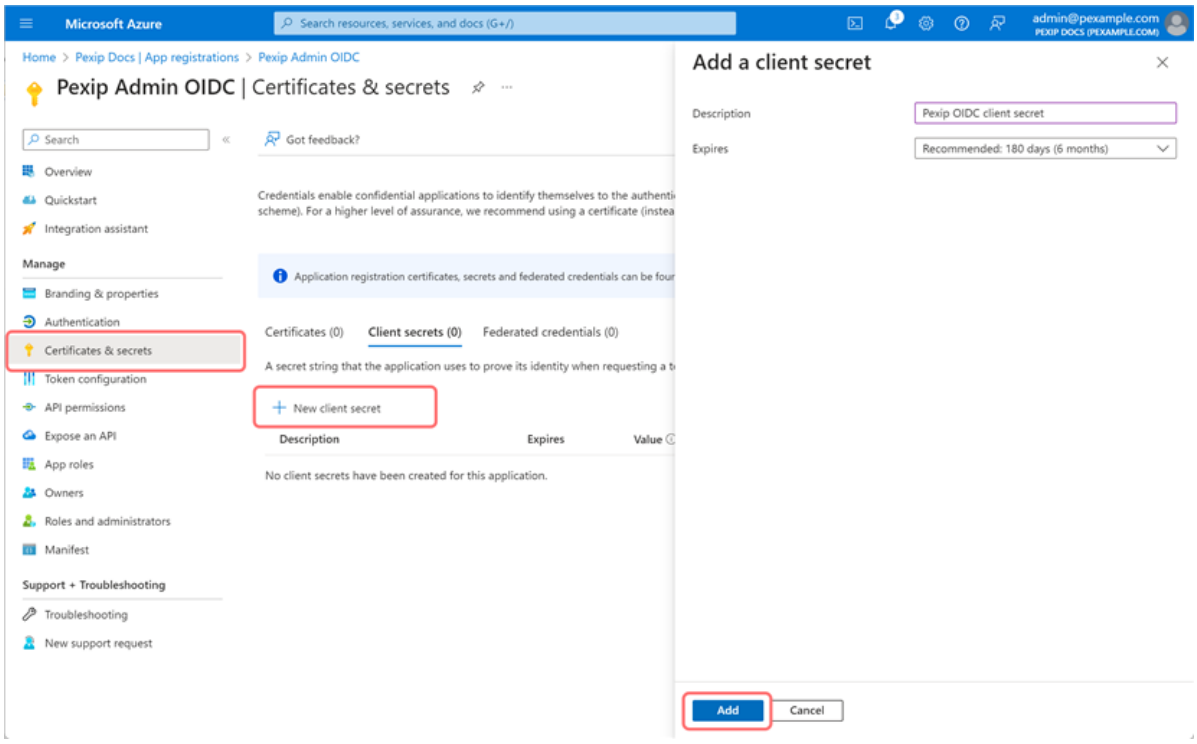


6. From the panel on the left, select **Token configuration**. Select **Add groups claim**, select **All groups...**, and then select **Add**:



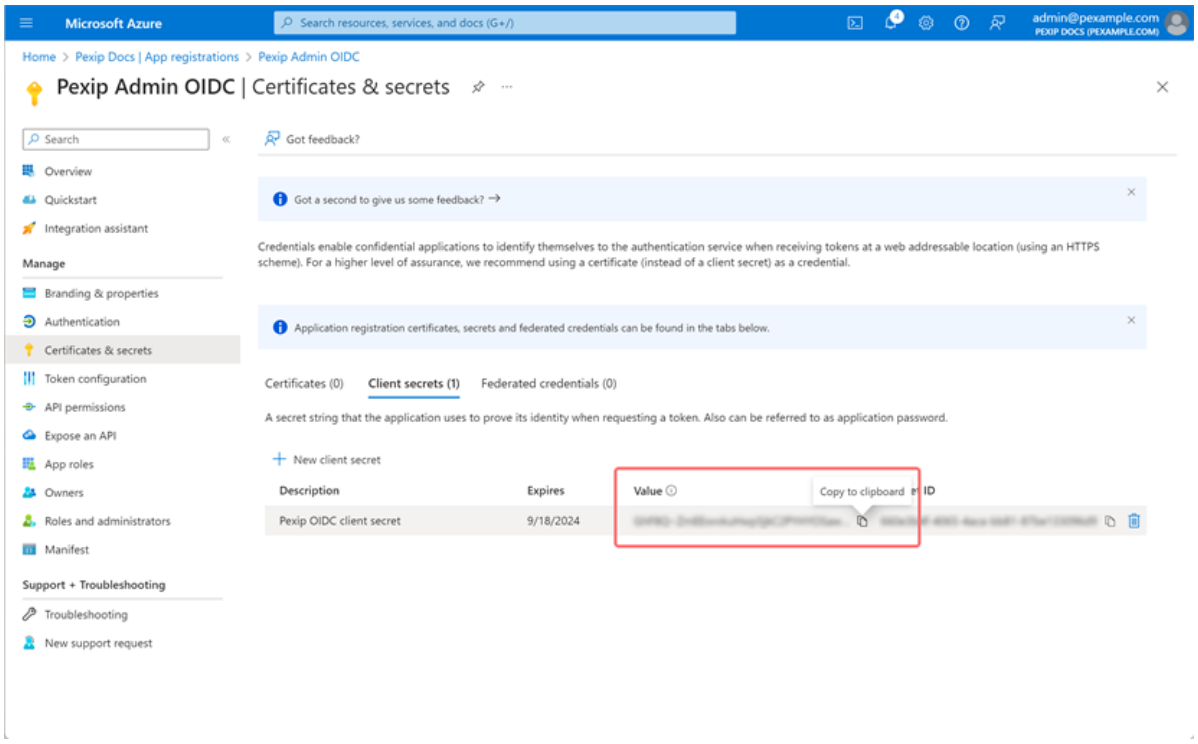
7. (Not required if using an Authentication method of ***PKCE***)

   From the panel on the left, select **Certificates & secrets**. Select **New client secret**, complete the fields, and then select **Add**:



8. (Not required if using an Authentication method of ***PKCE***)

   Before you navigate away from the page you must copy the string in the **Value** field (do **not** copy the string in the **Secret ID** field). You must enter this as the Client secret when configuring Pexip Infinity:


Microsoft Azure
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admin@pexample.com
PEXIP DOCS (PEXAMPLE.COM)
Home > Pexip Docs | App registrations > Pexip Admin OIDC
Pexip Admin OIDC | Certificates & secrets
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Branding & properties
Authentication
Certificates & secrets
Token configuration
API permissions
Expose an API
App roles
Owners
Roles and administrators
Manifest
Support + Troubleshooting
Troubleshooting
New support request
Got feedback?
Got a second to give us some feedback?
Credentials enable confidential applications to identify themselves to the authentication service when receiving tokens at a web addressable location (using an HTTPS scheme). For a higher level of assurance, we recommend using a certificate (instead of a client secret) as a credential.
Application registration certificates, secrets and federated credentials can be found in the tabs below.
Certificates (0)
Client secrets (1)
Federated credentials (0)
A secret string that the application uses to prove its identity when requesting a token. Also can be referred to as application password.
New client secret
Description
Expires
Value
Copy to clipboard
ID
Pexip OIDC client secret
9/18/2024

Configuring Pexip Infinity

The next step to allow administrators to use OIDC authentication when they log in to the Pexip Infinity Administrator interface is to configure Pexip Infinity with information about the OIDC provider. To do this, go to **Users & Devices > Administrator Authentication**. The relevant options are:

Option	Description
Authentication source	The database to query for administrator authentication and authorization. ***Local database***: uses the Pexip Infinity local on-box database. Administrators can only log in via the default account (typically **admin**) and will have full administrator privileges. ***LDAP database***: administrators can only log in using an account configured on the LDAP database and obtain privileges according to the groups and roles associated with that account. Note that if this option is selected and the LDAP server is inaccessible for any reason, administrators will not be able to log in to the Pexip Infinity web-based Administrator interface or management API. ***LDAP database and local database***: administrators can log in using either the default local **admin** account or via an account configured on the LDAP database. ***Open ID Connect service***: administrators can only log in using an account configured on the OpenID Connect provider and obtain privileges according to the groups and roles associated with that account. Note that if this option is selected and the OpenID Connect provider is inaccessible for any reason, administrators will not be able to log in to the Pexip Infinity web-based Administrator interface or management API. ***Open ID Connect service and local database***: administrators can log in using either the default local **admin** account or via an account configured on the OpenID Connect provider. ***LDAP database, Open ID Connect service and local database***: administrators can log in using the default local **admin** account, via an account configured on the LDAP database, or via an account configured on the OpenID connect provider. When using OIDC either on its own or in combination with another authentication source, you must configure the items in the OpenID Connect configuration section. Default: ***Local database***.
Require client certificate	This option is only applicable if using LDAP as an authentication source. For more information, see Certificate-based authentication.
Management API Oauth2 settings	
The settings in this section apply to management API access via OAuth only. They do not affect local, LDAP or OpenID Connect access. For more information, see Managing API access via OAuth2.	
LDAP configuration and Advanced LDAP configuration	
The settings in these sections apply to LDAP access only. They do not affect local or OpenID Connect access. For more information, see Managing administrator access via LDAP.	
OpenID Connect configuration	
Metadata URL	Enter the URL of the **OpenID Connect metadata document**, copied from your OIDC provider. This ends with **openid-configuration** To obtain the metadata information, the Management Node must have a connection to the OpenID Connect provider. Alternatively, you can leave this field blank; the **Configuration metadata**, **Authorization URL** and **Token endpoint URL** fields appear and you can instead paste the information from Azure into the relevant fields.
Configuration metadata	(Available if **Metadata URL** is blank) The OpenID Connect configuration metadata, copied from your OIDC provider.

Option	Description
Authorization URL	(Available if **Metadata URL** is blank) The OpenID Connect authorization URL, copied from your OIDC provider.
Token endpoint URL	(Available if **Metadata URL** is blank) The OpenID Connect token endpoint URL, copied from your OIDC provider.
Client ID	The **Application (client) ID**, copied from your OIDC provider.
Authentication method	The OpenID Connect authentication method. For Azure, select one of the following, depending on how you configured the Azure app registration: • ***Client secret***: for web applications using a client secret • ***PKCE***: for single-page applications
Client secret	(Only available if an **Authentication method** of ***Client secret*** is selected) Enter the string from the **Value** field of the client secret, copied from your OIDC provider.
Scope	The OpenID Connect OAuth2 scope to request.
Username field	The field in the authentication token response to use as the username.
Groups field	The field in the authentication token response to use as the list of groups. You apply role mappings to one or more of these groups by referencing the group in the Value field. If this field is left blank, all users authenticated using OIDC will have the combined access rights of all role mappings that specify a Authentication source of ***OpenID Connect***.
Required key	If there is a field in the authentication token response which must be present, enter the name of the field here.
Required value	If you have specified that there is a **Required key**, enter the value of the required key here.
Domain hint	A domain to pass to the OpenID Connect service as a hint to the expected login account. If you are logged into multiple accounts with different domains you should be presented with the most suitable domain. For example, if the domain hint is "pexample.com", and you are logged into pexample.com and pexample.org, it will send the domain hint "pexample.com" and the OIDC provider should automatically select your pexample.com login.
Login button text	The text to use for the OpenID Connect button on the login page of the Pexip Infinity Administrator interface. Default: **OpenID Connect**

If authentication against an OpenID Connect provider is configured, you can save the settings only if Pexip Infinity can successfully contact the specified OpenID Connect provider.

Configuring administrator roles

Administrator roles control the actions that administrators can perform after they have been authenticated. You create one or more administrator roles, and then map them to your OIDC groups, allowing you to grant different privileges to different sets of administrator users.

To add, edit or delete administrator roles, go to **Users & Devices > Administrator Roles**. For full details of the options that are available, see Managing administrator roles.

Configuring OIDC role mappings

Role mappings are used to map the list of groups obtained from Azure to the Pexip Infinity administrator roles. You must configure a separate role mapping for each group for which you want to map one or more Pexip Infinity administrator roles.

To add, edit or delete OIDC role mappings, go to **Users & Devices > Role Mappings**. When configuring OIDC role mappings, the options are:

Option	Description
Name	A descriptive name of the role mapping, e.g. "domain administrator with full privileges".
Source	The authentication source to which this role mapping applies. Select ***OpenID Connect***.
Value	The group to which this mapping applies. For Azure, enter the Object ID; for Okta, enter the group name. This group must be included in the list that is returned in the specified Groups field field of the authentication token response. If the **Groups field** is blank, this **Value** will be ignored.
Roles	Select from the list of **Available roles** the administrator roles to associate with the OIDC group and then use the right arrow to move the selected roles into the **Chosen Roles** list. All of the underlying permissions within a role are "positive" permissions, i.e. they allow the administrator to do something. If more than one role is selected, all of the permissions associated with each role are combined and granted to the relevant administrator. Note that you can select which opens a new window from where you can configure a new administrator role. When you save the role it is automatically added to the set of **Chosen Roles**.

Reinstating the local admin account

If necessary you can reinstate access via the Pexip Infinity local on-box database, so that administrators can log in via the default admin account with full administrator privileges. You may need to do this if, for example, the **Authentication source** is configured as ***OpenID Connect service*** only and your connectivity to the Azure goes down or your credentials become invalid.

For full instructions, see Reinstating the local admin account.

Managing API access via OAuth2

You can configure the Pexip Infinity platform to use OAuth2 to authenticate and authorize administrator accounts connecting to the Pexip Infinity management API.

Using OAuth2 for management API authentication is an alternative to access using LDAP for environments that make frequent API requests, as it can significantly reduce the number of requests sent to the LDAP server. Using LDAP, every management API request is preceded by an authentication request to the LDAP server. Using OAuth2, the API client authenticates once with the Management Node to obtain a token that is then valid for a specified amount of time (1 hour by default).

When configured, OAuth2 applies to management API access only. It can not be used to access the Pexip Infinity web-based Administrator interface; access to this will always use the specified administrator **Authentication source** (as configured under **Users & Devices > Administrator Authentication**).

Management API authentication via OAuth supports OAuth2 with JWT bearer assertion signed with ES256 with private key.

The steps required to enable management API access via OAuth2 are:

1. Creating an appropriate administrator role.
2. Creating a management API OAuth2 client and assigning the new administrator role to it.
3. Configuring management API OAuth2 settings.

This topic also includes information on:

- Using OAuth2 to access the management API
- Viewing active management API clients
- Postman API access via OAuth2

All usernames and passwords are case sensitive.

Creating an administrator role

Administrator roles control the actions that administrators can perform after they have been authenticated. You create one or more administrator roles, and then associate each with an OAuth2 client, allowing you to grant different privileges to different sets of

administrator users.

To add, edit or delete administrator roles, go to **Users & Devices > Administrator Roles**. For full details of the options that are available, see Managing administrator roles.

Creating an OAuth2 client

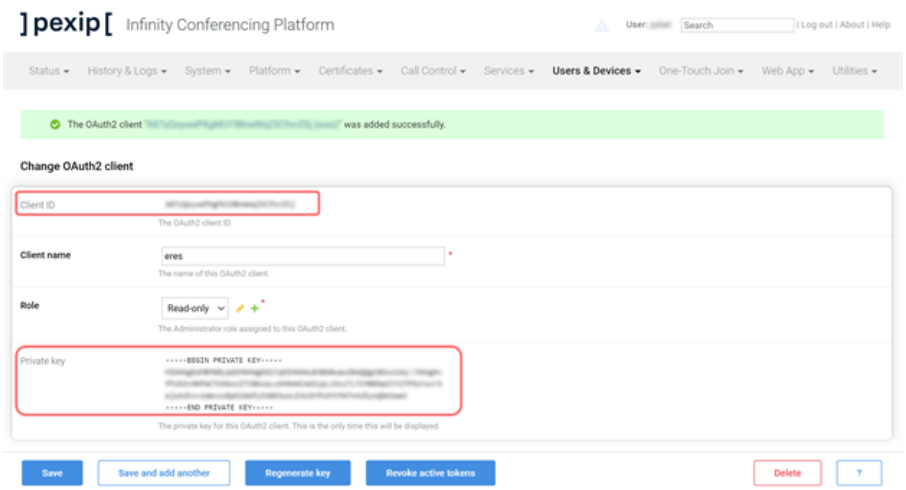
To create OAuth2 client credentials that can be used to access the management API, go to **Users & Devices > OAuth2 Clients** and select **Add OAuth2 client**. Enter the following:

This process will generate a private key that you must copy before navigating away from the page.

Option	Description
Client name	Enter a name for this particular OAuth client and role combination.
Role	Select the Administrator role to assign to the management API when it is authenticated using this client.

Select **Save**. Additional **Client ID** and **Private key** information appears. These are the credentials that the OAuth2 client must use in order to access the management API.

You must copy the private key before you navigate away from the page as it will not be available afterwards.



Configuring management API authentication

To configure management API access using OAuth2, go to **Users & Devices > Administrator Authentication** and scroll down to the **Management API Oauth2 settings** section. The options are:

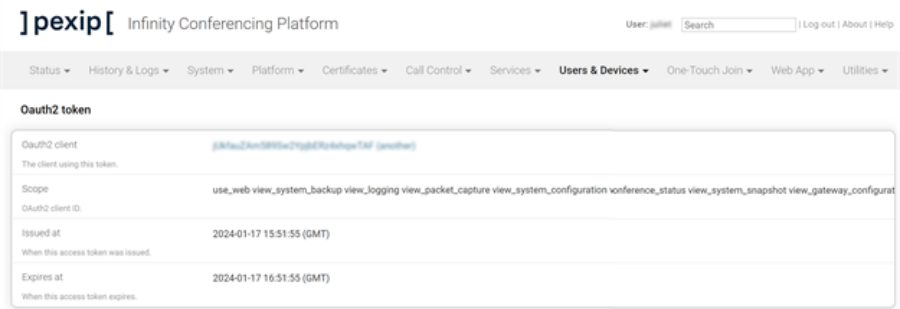
Option	Description
Management API Oauth2 settings	
Access token expiration	The length of time (in seconds) after which the management API OAuth2 client must re-authenticate. Default: 3600 (1 hour)
Disable Basic authentication	When selected, basic authentication is not available to management API users; they can only authenticate using OAuth2. When this option is not selected, management API users can authenticate using either OAuth2 (if configured) or basic authentication — which includes the local admin username and password, or a valid LDAP username and password.
Allow all permissions	When enabled, management API clients authenticated using OAuth2 can use all permissions specified in the assigned role. When this option is disabled, these API clients cannot modify authentication configuration even if "may modify authentication configuration" is specified in the role.

Using OAuth2 to access the management API

When management API access using OAuth2 has been enabled as described above, users can access the management API by configuring their OAuth2 client with the **Client ID** and **Private key** credentials, and then directing the client to **https://<management_node_fqdn>/oauth/token/**

Viewing active management API clients

To view all OAuth2 clients with a currently active token, go to **Users & Devices > OAuth2 Tokens**. Select a client to view details of the permissions assigned to the client, and its token expiry.



Postman API access via OAuth2

Accessing the API via OAuth2 requires the following flow:

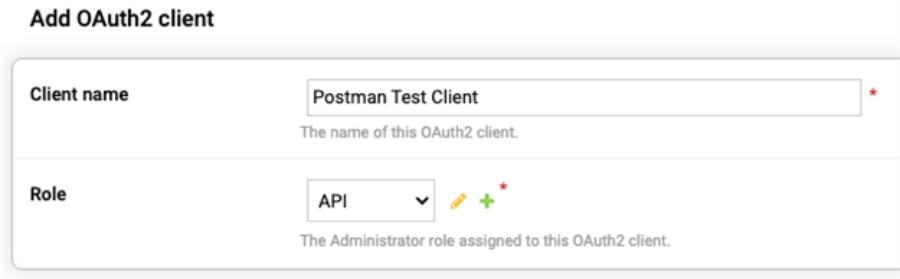
- A bearer token must be requested via the **/oauth/token/** endpoint
- This bearer token is then used to access the API, i.e. **GET /api/admin/status/v1/alarm/**

You can configure Postman to do this by using a pre-request script; the details are below.

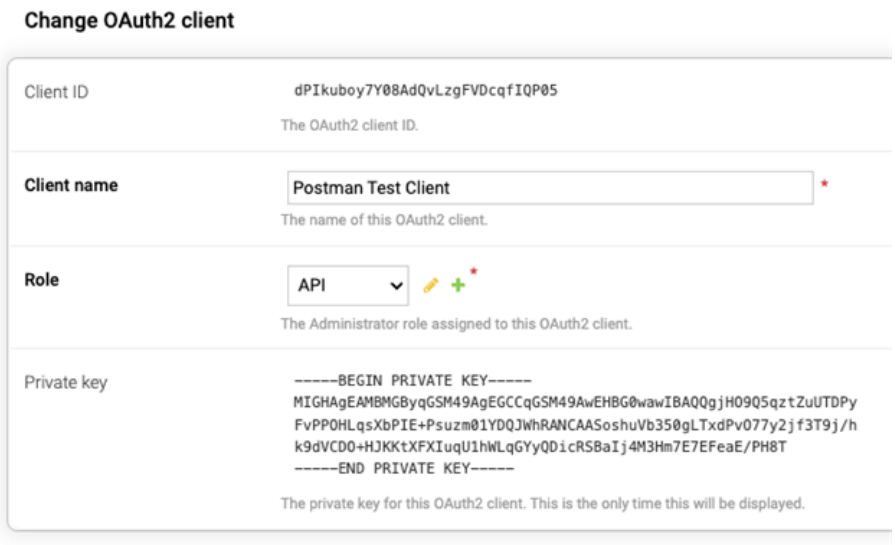
- Create an API client
- Set up a Postman environment
- Set up a Postman collection
- Configure the collection
- Create a request
- Troubleshooting

Create an API client

Log in to the Management Node and create an API client.



Make a note of the **Client ID** and the **Private Key**.



Set up a Postman environment

1. In Postman select **Environments > New > Environment**.
2. Give it a name.
3. Add the following variables:

Variable	Type	Value
host	default	mgmt.domain.com
pmlib_code	default	See postman-util-lib
client_id	default	<client_id>
client_private_key	secret	<client_private_key>
jwt_signed	secret	
access_token	secret	qwerty
expires	default	qwerty
scope	default	

The **host** value must match the host value of a valid OAuth2 endpoint listed under **OAuth2 Endpoints** on the **About** page on the Management Node.

Set up a Postman collection

1. In Postman select **Collections > New > Collection**.
2. Give it a name.
3. Set the environment you created above to this collection.

Configure the collection

1. Select the collection you created above.
2. Enter the following in the **Authorization** tab.
   - Auth Type: Bearer Token
   - Token: {{access_token}}

3. Open **Scripts > Pre-request**.
4. Input the following script:

```
// ]pexip[
                //
                // Pre-request postman script for requesting a bearer token to access the API via OAuth2
                //
                // Documentation: https://docs.pexip.com/admin/managing_API_oauth.htm
                //

                // Imports
                const crypto = require("crypto-js");
                eval(pm.environment.get('pmlib_code'));

                // Set Access token expiration (in seconds)
                // Set this to match the configuration on Pexip;
                //  - Administrator Authentication > Management API OAuth2 settings > Access token expiration
                const token_expiration = 3600

                // Get environment variables
                var mgmt = pm.environment.get('host');
                var clientId = pm.environment.get('client_id');
                var clientPrivateKey = pm.environment.get('client_private_key');

                // Set JWT header
                var header = {
                'alg': 'ES256',
                'typ': 'JWT'
                };

                // Prepare timestamp in seconds
                const timestampCurrent = Math.floor(Date.now() / 1000)
                const timestampExpiry = timestampCurrent + token_expiration

                // Set JWT payload
                var payload = {
                'sub': clientId,
                'iss': clientId,
                'aud': 'https://'+mgmt+'/oauth/token/',
                'iat': timestampCurrent,
                'exp': timestampExpiry,
                'jti': crypto.lib.WordArray.random(36).toString()
                };

                // Sign the JWT using the pmlib.rs* library (https://joolfe.github.io/postman-util-lib/)
                var JWS = pmlib.rs.jws.JWS;
                signedJWS = JWS.sign(header.alg, header, payload, clientPrivateKey);
                pm.environment.set('jwt_signed', signedJWS);

                // Check if we have a bearer token
                var token_exists = pm.environment.get('access_token') || '';

                var token_expiry = pm.environment.get('expires') || '';
                const token_expiry_date = new Date(token_expiry * 1000);

                // Check if the token is valid
                const diff = token_expiry - timestampCurrent;
                if (diff > 0) {
                // Bearer token is still valid
                console.log('Bearer token still valid (Expires: '+token_expiry_date.toLocaleString('en-GB', { timeZone: 'UTC' })+' (UTC))');
                } else {
                // Bearer token has expired
                console.log('Bearer token expired ('+token_expiry_date.toLocaleString()+' (UTC)), requesting new token');
                // Request a bearer token
                const postRequest = {
                url: 'https://'+mgmt+'/oauth/token/',
                method: 'POST',
                header: {
                'Content-Type': 'application/x-www-form-urlencoded; charset=UTF-8',
                },
                body: {
```

```
            mode: 'urlencoded',
            urlencoded : [
            { key: 'grant_type', value: 'client_credentials' },
            { key: 'client_assertion_type', value: 'urn:ietf:params:oauth:client-assertion-type:jwt-bearer' },
            { key: 'client_assertion', value: signedJWS },
            { key: 'scope', value: 'is_admin use_api' },
            ]
            },
            };
            pm.sendRequest(postRequest, (error, response) => {
            console.log(error ? error : response.json());

            // Set/update environment variables
            jsonData = response.json()
            if (!jsonData.error){
            pm.environment.set('access_token', jsonData.access_token);
            pm.environment.set('expires', timestampExpiry);
            pm.environment.set('scope', jsonData.scope);
            console.log('New access token: '+jsonData.access_token)
            }
            });
        }
```

Create a request

For this example request, we'll use the alarm status endpoint:

1. Select the collection you created above.
2. Select **New > HTTP**.
3. Give it a name.
4. Enter the following for the host: **https://{{host}}/api/admin/status/v1/alarm/**
5. Submit the request.

Troubleshooting

400 Response

```
{error: "invalid_client"}
```

Things to check:

- Confirm that the **client_id** is correct.
- Confirm that the **client_private_key** is correct.
- Confirm that the OAuth2 client has the correct permissions for the endpoint.

  You can read more about the roles at Creating an administrator role. The roles map to the scope. The minimum roles for a client are **is_admin use_api**, but this only permits base access.
- Confirm that the host value matches a value listed on the about page on the management node (**https://mgmt.domain.com/admin/platform/about/**)

Here is an example collection and an environment file to import into Postman.

- Postman collection
- Postman environment

Managing administrator roles

Administrator roles control the actions that administrators can perform in the web-based Administrator interface or management API after they have been authenticated. You can map different administrator roles to different sets of users, based on their group membership within your chosen authentication provider, giving you greater control over the actions that individual administrators can perform. For example, you can configure a role that allows an administrator to only view (and not modify) specific items of configuration data or status information via the Administrator interface.

For information on using administrator roles in conjunction with your authentication provider, see Managing administrator access via LDAP, Managing administrator access via OIDC, or Managing API access via OAuth2 as appropriate.

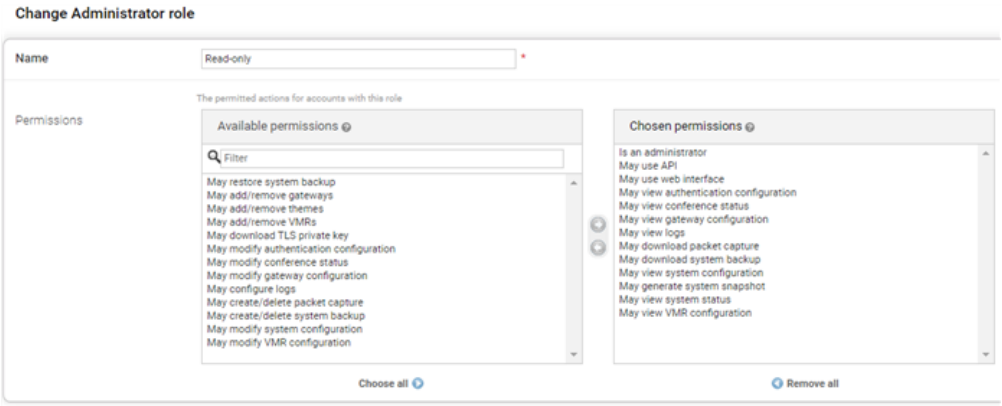
Roles are not relevant to the local admin account, which always has full read-write access.

When an administrator has restricted permissions, all navigation menu options are still displayed, but they are given an **Access denied** message if they try to select a menu option that they are not authorized to use. For read-only restrictions, the relevant **Add <item>** options are not displayed.

Two roles are present by default:

- ***Read-only***: allows read-only access to all configuration settings and status information when accessing the system via the web-based Administrator interface or the management API. An administrator with this role can also take diagnostic snapshots and backups, view logs, and make packet captures. Note that this role has full read access to sensitive information — you can create more restricted roles if necessary.
- ***Read-write***: allows full administrative access when accessing the system via the web-based Administrator interface or the management API.

To add, edit or delete administrator roles, go to **Users & Devices > Administrator Roles**. When configuring roles, the options are:

Option	Description
Name	A descriptive name of the role, e.g. "auditor" or "management system".
Permissions	Select from the list of **Available permissions** the set of permitted actions for the role and then use the right arrow to move the selected actions into the **Chosen permissions** list. For more information on each permission, see Description of all available administrator permissions. All roles must include the ***Is an administrator*** permission for access to the system. In addition, the ***May use web interface*** and ***May use API*** permissions must be included for access via the web-based Administrator interface and management API respectively. You must then also add all of the other permissions, such as ***May modify system configuration*** and so on, that you want to apply to the role — if a role has, for example, only the ***Is an administrator*** and ***May use web interface*** permissions, an administrator with that role is able to log in via the web-based Administrator interface but cannot perform any actions. When creating an OAuth2 client for management API access, the permissions ***Is an administrator*** and ***May use API*** are assumed and do not need to be set explicitly. The permissions that are applied to the default ***Read-only*** role are shown below: 

Using Identity Providers

Identity Providers (IdPs) are third-party services (such as AD FS, Microsoft Entra ID or Okta) that store and manage digital identities, and with which users authenticate using SSO. Pexip Infinity supports Identity Provider SSO authentication for participant access to VMRs, and for client application registration.

This topic covers:

- About Single Sign-on
- Using SSO with Pexip Infinity
- Supported Identity Providers

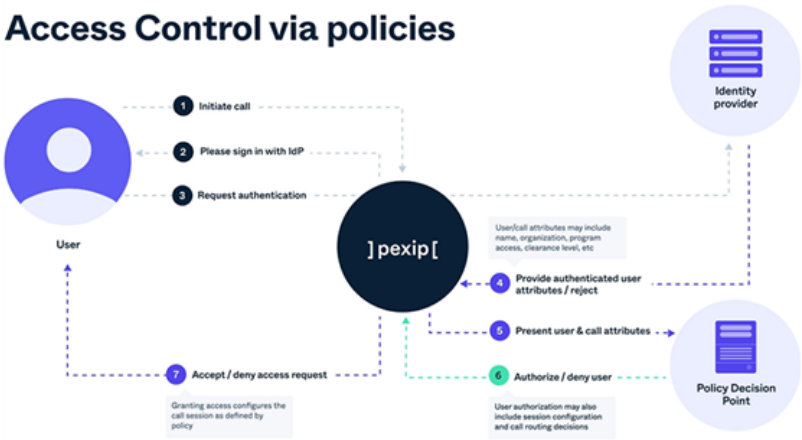
- Process for enabling Identity Providers
- Adding Identity Providers to Pexip Infinity

About Single Sign-on

Single Sign-On (SSO) is a common feature of web services and business tools that enables users to authenticate and log into a third-party service or web page, using the same login credentials as the other systems they already use.

The following animation demonstrates the conference participant authentication process with an Identity Provider:

This can be extended, as shown in the next animation, to include local or external participant policy in the decision making process for finer control. This allows you, for example, to override a participant's role, alias and display name before they join a conference.



Using SSO with Pexip Infinity

You can require Identity Provider SSO authentication for either or both of the following:

- For participants who are attempting to access a specific VMR in your deployment, to verify their identity before joining the meeting. The SSO service can also be used to provide a verified display name for meeting participants. For more information, see About participant authentication.
- For client applications (such as the Pexip app for Windows) to register their device to Pexip Infinity. For more information, see Authenticating app registrations using Identity Providers.

Supported Identity Providers

Pexip Infinity supports two widely-used standard industry protocols:

- SAML 2.0
- OpenID Connect (OIDC)

We have provided step-by-step guides for configuring the following SAML Identity Providers:

- ADFS
- Microsoft Entra ID (formerly called Azure AD)
- Okta
- Keycloak (an open-source option, which may be useful if you only require an IdP for Pexip app registrations).)

For guidance on configuring other Identity Providers, please contact your Pexip authorized support representative.

Supported signature algorithms

Pexip Infinity supports the following:

- SAML Identity Providers: RSA/SHA signature and digest algorithms.
- OIDC Identity Providers: RS256 and HS256 JWT signature schemes.

Process for enabling Identity Providers

When setting up an Identity Provider, some configuration needs to be generated on Pexip Infinity and then added to the Identity Provider, and vice versa.

To make the setup easier, for SAML Identity Providers we have provided the ability to download a configuration file from Pexip Infinity which can be uploaded to the Identity Provider. You can also subsequently import some configuration from SAML and OIDC Identity Providers to Pexip Infinity.

In summary, the process is as follows:

1. On Pexip Infinity, create the Identity Provider record and download the configuration. For full details see Adding Pexip Infinity service configuration.
2. On the Identity Provider, create a new service and upload the Pexip Infinity configuration to it. For full details see Configuring individual SAML Identity Providers.
3. Return to Pexip Infinity and either import the configuration file from the Identity Provider, or complete the configuration manually. For full details, see Adding the Identity Provider configuration to Pexip Infinity.
4. On Pexip Infinity, add the Identity Provider to one or more Identity Provider Groups. For full details see Creating Identity Provider groups.

After the Identity Provider configuration is complete, you can then:

- Configure individual Virtual Meeting Rooms and Virtual Auditoriums to use one of the Identity Provider Groups for participant authentication. For details, see Configuring Virtual Meeting Rooms (VMRs).
- If your deployment uses Pexip's Secure Scheduler for Exchange feature, you can also require participant authentication for single-use VMRs.
- Configure individual device aliases to use one of the Identity Provider Groups to authenticate the alias registration. For more information, see Authenticating registrations.

Adding Identity Providers to Pexip Infinity

About redirect URLs

When configuring an Identity Provider, you must list one or more **Redirect URLs** that are valid for that provider. Redirect URLs are used in the authentication process to identify the source of a request, and also as an address to which the response to the request is returned. Redirect URLs must include the FQDN of the webapp as part of the URL.

If users in your deployment are able to access the web app from more than one FQDN, then you must configure each Identity Provider (both on Pexip Infinity and on the Identity Provider itself) with the same number of corresponding **Redirect URLs**. To help you, we have included an option to automatically add redirect URLs for every node FQDN used in your deployment. (Note that if you use this option, you still need to add the individual redirect URLs to your Identity Provider configuration.) You can also add additional redirect URLs manually from within the Advanced options section (available for both SAML and OpenID Connect).

About pop-up windows

By default, the authentication process for users attempting to join a Virtual Meeting Room takes place via a separate pop-up window. If your environment does not allow or support pop-ups, you can Disable pop-up windows on a per-Identity Provider basis. When this

option is selected, a web app user's current browser session is redirected away from the web app in order to perform the authentication process, and then automatically returned to the web app.

If users have browser close confirmation enabled (via either their user settings, or as a default setting), they may be shown a confirmation pop-up when redirected to the IdP. In these cases, the user must select the option to **Leave** the site in order to proceed.

If you are using the pop-up-free authentication flow for users attempting to join a meeting and you are either:

- hosting the web app externally, (i.e. on something other than a Conferencing Node), or
- accessing the web app from a URL that is not a Configured FQDN for any Conferencing Node (for example, when using DNS A records to access the web app)

you must add the web app's host address / URL (as applicable) to the "allow list" of External Web App Hosts (**Web App > External Web App Hosts**).

Adding Pexip Infinity service configuration

The configuration for the Identity Provider is in two sections on Pexip Infinity.

Firstly, go to **Users & Devices > Identity Providers** and select **Add Identity Provider**. Complete the **Service configuration** section, as follows:

Option	Description
Service configuration	
Name	The name used to refer to this Identity Provider. This name will be visible to end users, so you should use a name that will help users differentiate between Identity Providers without compromising security.
Description	An optional description of the Identity Provider.
UUID	A unique identifier for this Identity Provider configuration. A value is automatically assigned and there is normally no need to modify it.
Identity Provider protocol	Select the protocol used by this Identity Provider: • ***SAML***: the IdP uses SAML 2.0 • ***OpenID Connect***: the IdP uses OpenID Connect The remaining configuration fields depend on the option selected here.
SAML options (available when an Identity Provider type of *SAML* is selected)	
Infinity public key (PEM format)	Paste the contents of the public key used by Pexip Infinity when communicating with the Identity Provider. This must be in PEM (certificate) format, must be unencrypted, and must include all the text in the file between (and including) -----**BEGIN CERTIFICATE**----- and -----**END CERTIFICATE**----. If you do not already have a public / private keypair, you can use a third-party tool such as PuTTYgen to generate one for use here.
Infinity private key	Paste the contents of the private key used by Pexip Infinity when communicating with the Identity Provider. This must be unencrypted, and must include all the text in the file between (and including) -----**BEGIN PRIVATE KEY**----- and -----**END PRIVATE KEY**-----.

Option	Description
Redirect URL	An Assertion Consumer Service (ACS) URL that can be used in the authentication process with this Identity Provider. This should be in the format: **https://<webapp_fqdn>/api/v1/samlconsumer/<uuid>** where **<webapp_fqdn>** is the FQDN from which the web app is accessed, and **<uuid>** is the UUID shown in the field above. The entire URL must be in lower-case. You should provide one **Redirect URL** for every web app FQDN used in your deployment; Additional redirect URLs can be added from the Advanced options section below. For more information, see About redirect URLs.
SAML 2.0 Entity ID for this service	An identifier for the service on Pexip Infinity. We recommend that you use the FQDN from which the web app is accessed, for consistency.
Signature algorithm	Signature algorithm used to sign SAML authentication request messages and service metadata
Digest algorithm	Digest algorithm used to sign SAML authentication request messages and service metadata
OpenID Connect options (available when an Identity Provider type of *OpenID Connect* is selected)	
OpenID Connect flow	Select the flow used by the OpenID Connect Identity Provider: • ***Implicit*** • ***Authorization Code*** (default) • ***Authorization Code with PKCE***
OpenID Connect client identifier	The client identifier provided by the OpenID Connect Identity Provider.
OpenID Connect client secret	(Not required for the ***Authorization Code with PKCE*** flow) The client secret provided by the OpenID Connect Identity Provider.
Redirect URL	A URL that can be used in the authentication process with this Identity Provider. This should be in the format: **https://<webapp_fqdn>/api/v1/oidcconsumer/<uuid>** where **<webapp_fqdn>** is the FQDN from which the web app is accessed, and **<uuid>** is the UUID shown in the field above. The entire URL must be in lower-case. You should provide one **Redirect URL** for every web app FQDN used in your deployment; Additional redirect URLs can be added from the Advanced options section below. For more information, see About redirect URLs.
Advanced options (available for both *SAML* and *OpenID Connect*):	
Create redirect URLs from Conferencing Node FQDNs	Automatically generate allowed redirect URLs from the configured FQDNs for each Conferencing Node.
Additional redirect URL	Enter any additional redirect URLs valid for use with this Identity Provider.

Option	Description
Attributes	An Identity Provider attribute is a custom attribute that is associated with an Identity Provider and that has been created to support a specific business need (for example whether somebody is a member of a particular group of people). The attributes you assign here are made available to any , when the participant uses this provider for authentication. These attributes can then be used in the participant policy decision-making process (for example whether to accept or reject a call, or assign a specific role). Custom attributes can also be used to provide an alternative display name. **Managing the available attributes** To create a new attribute, select + and then enter the **Name** and **Description** of the attribute in the popup form that appears. The **Name** entered here must match the associated name in your Identity Provider configuration. You can also go to **Users & Devices > Identity Provider Attributes** to directly create, modify or delete any attributes. **Assigning an attribute** Select from the list of **Available Attributes** the set of attributes to associated with this Identity Provider and then use the right arrow to move the selected attribute into the **Chosen Attributes** list.
Disable pop-up windows	By default, the authentication process for users attempting to join a Virtual Meeting Room takes place via a separate pop-up window. If your environment does not allow or support pop-ups, select this option. For more information, see About pop-up windows.
Advanced options available for *SAML* only:	
Download service metadata	(Available when saved) This option allows you to download the configuration in a format that can be imported by the Identity Provider.
Advanced options available for *OpenID Connect* only:	
OpenID Connect Token Endpoint authentication type	The authentication method used by Pexip Infinity to authenticate when using the token endpoint.
OpenID Connect JWT signature scheme	The algorithm used by the Identity Provider to sign the contents of the token.
OpenID Connect UserInfo Endpoint URL	You can optionally enter here the URL of an OpenID Connect UserInfo Endpoint if you wish to use the UserInfo Endpoint (rather than the JWT) to retrieve information about the user such as their display name.
eIDAS authentication	The eIDAS level to use in requests and responses. This should not be changed from the default ***Disabled*** unless advised by Pexip support.
Additional OpenID Connect scopes	Space-separated list of additional scopes to request from the OpenID Connect Identity Provider.

Configuring the Identity Provider

The next step is to create a new service on the Identity Provider and configure it with details of Pexip Infinity.

All Identity Providers can be configured manually, but some SAML Identity Providers also allow you to upload configuration that has been exported from Pexip Infinity. To do this:

1. On Pexip Infinity go to **Users & Devices > Identity Providers** and select the SAML Identity Provider you have just created.
2. At the bottom of the page select **Download service metadata**.
3. Download the file and import it to the Identity Provider.

For full step-by-step instructions on configuring the main supported Identity Providers, see Configuring individual SAML Identity Providers.

Adding the Identity Provider configuration to Pexip Infinity

After you have configured the Identity Provider, you must add its configuration to Pexip Infinity. Some Identity Providers will have provided the option to download a configuration file in XML format (for SAML) or JSON (for OpenID Connect); you have the option to upload this to Pexip Infinity. Note that this will not include the configuration for the display name; this must be done manually.

To complete the configuration, on Pexip Infinity, go back to **Users & Devices > Identity Providers** and select the Identity Provider. Under the **Identity Provider configuration** section, complete the following fields (the individual Identity Provider configuration instructions explain where to find this information for each Identity Provider).

Note that the available fields differ between SAML Identity Providers and OpenID Connect Identity Providers:

SAML Identity Provider configuration

Option	Description
Identity Provider configuration	
Upload file	If you have downloaded configuration from the Identity Provider in XML format, this allows you to import it to Pexip Infinity in order to automatically populate the fields below.
Identity Provider public key	The public key used to verify assertions signed by this Identity Provider.
Identity Provider SSO URL	The URL to which users are sent when authenticating with this Identity Provider. Custom query string parameters may be appended, e.g. **https://<url>?foo=bar**.
Identity Provider identifier	The identifier for this Identity Provider integration. For SAML IdPs this is the Entity ID.
Display Name Attribute Name	The SAML 2.0 attribute name from which the user's display name will be extracted. By default the **NameID** value is used. You can also use a custom attribute. If this field is blank, participants will be able to enter their own display name. Note that the format used will vary depending on the Identity Provider.
Registration Alias Attribute Name	The SAML 2.0 attribute name from which the user's registration alias will be extracted. This field is required if the Identity Provider is being used for device registration. The alias returned must match the alias being registered, otherwise registration will not be permitted.

OpenID Connect Identity Provider configuration

Option	Description
Identity Provider configuration	
Upload file	If you have downloaded configuration from the Identity Provider as a JSON discovery document, this allows you to import it to Pexip Infinity in order to automatically populate the fields below.
Identity Provider SSO URL	The URL to which users are sent when authenticating with this Identity Provider. Custom query string parameters may be appended, e.g. **https://<url>?foo=bar**.
OpenID Connect Token Endpoint URL	OpenID Connect Token Endpoint URL used for exchanging codes for tokens in the Authorization Code Flow. Not required when using the Implicit Flow.

Option	Description
OpenID Connect JSON Web Key Set URL	Download location for your Identity Provider's JSON Web Key Set (JWKS) to enable signature verification. Not required when using HS256 signatures.
Identity Provider identifier	The identifier for this Identity Provider integration. For OpenID Connect IdPs this is the Issuer for returned JWTs.
Display Name Claim Name	The claim name from which the user's display name will be extracted. This can come from either the JWT, or data from the UserInfo endpoint (if one is configured). By default the **name** value is used. You can also use a custom attribute. If this field is blank, participants will be able to enter their own display name. Note that the format used will vary depending on the Identity Provider.
Registration Alias Claim Name	The claim name from which the user's registration alias will be extracted. This field is required if the Identity Provider is being used for device registration. The alias returned must match the alias being registered, otherwise registration will not be permitted.

Creating Identity Provider groups

Each Identity Provider must belong to at least one Identity Provider group in order to be used.

- An Identity Provider group can contain just a single Identity Provider — for example, if you use only one Identity Provider in your deployment, or if you wish to restrict access to certain VMRs to participants who have authenticated with a particular Identity Provider.
- If you are using an Identity Provider to authenticate Pexip app registrations, we recommend that you create a new Identity Provider group containing a single Identity Provider and use this for the sole purpose of authenticating Pexip app for Windows registrations.
- A group can contain more than one Identity Provider — for example, you may have more than one Identity Provider in use within your enterprise, and you wish users from some or all of them to be able to access the same SSO-protected VMRs.
- An Identity Provider can belong to more than one Identity Provider group. For example, you might have one Identity Provider group that contains all the Identity Providers in your enterprise, and other Identity Provider groups that contain subsets of those Identity Providers, or just a single Identity Provider.

To create an Identity Provider group, go to **Users & Devices > Identity Providers** and complete the following fields:

Option	Description
Name	The name used to refer to this Identity Provider Group.
Description	An optional description of the Identity Provider Group.
Identity Providers	From the list of configured Identity Providers, select one or more Identity Providers to add to this Identity Provider Group.

Configuring individual SAML Identity Providers

This topic provides guidance on how to configure the following Identity Providers using SAML, in order to support Participant authentication and device registration authentication:

- Microsoft Entra ID (formerly known as Azure AD)
- AD FS
- Okta
- Keycloak

For guidance configuring other Identity Providers, please contact your Pexip authorized support representative.

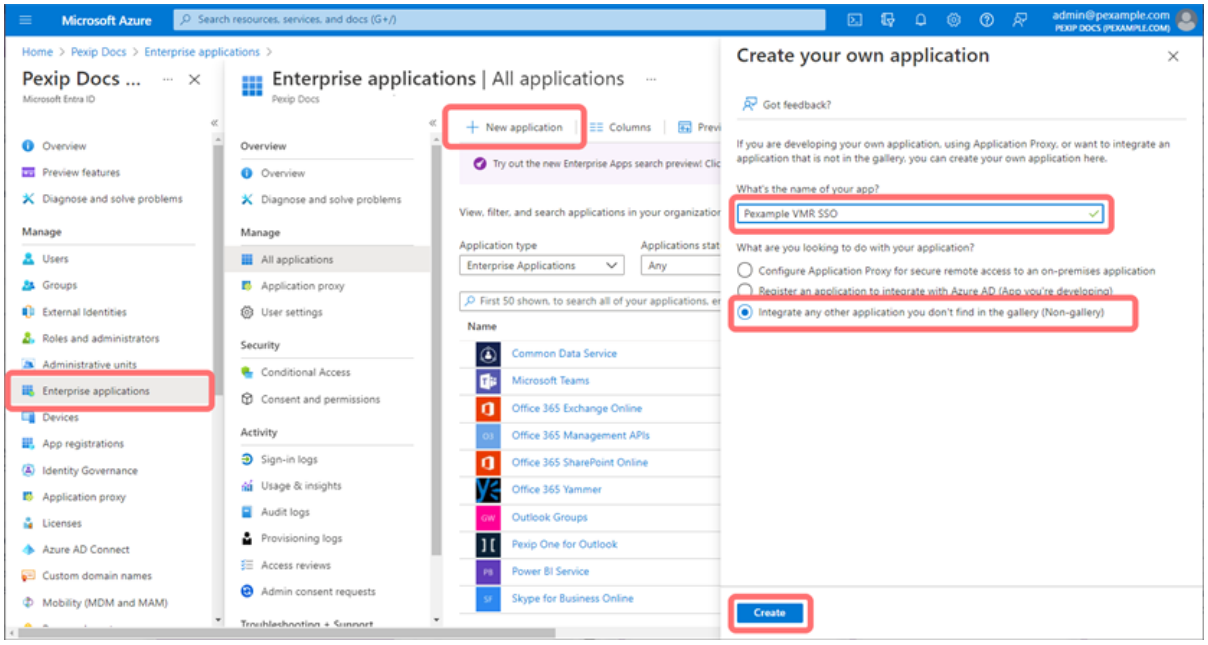
For general information on the use of Identity Providers with Pexip Infinity, see Using Identity Providers.

Prerequisites

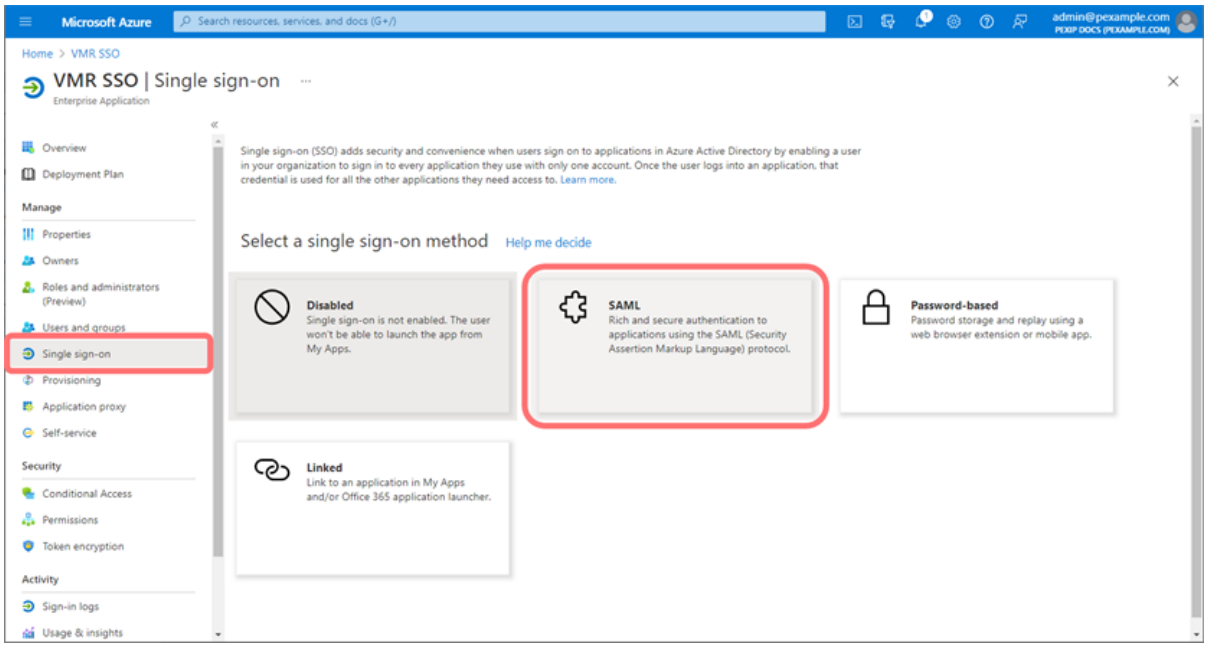
You should have already created a record for the Identity Provider on Pexip Infinity — for more information on the steps involved, see Process for enabling Identity Providers.

Microsoft Entra ID (formerly known as Azure AD)

1. From the Azure portal select **Microsoft Entra ID > Enterprise Applications > New application > Create your own application**. Give the application a name, select the option to **Integrate any other application you don't find in the gallery** and then select **Create**:



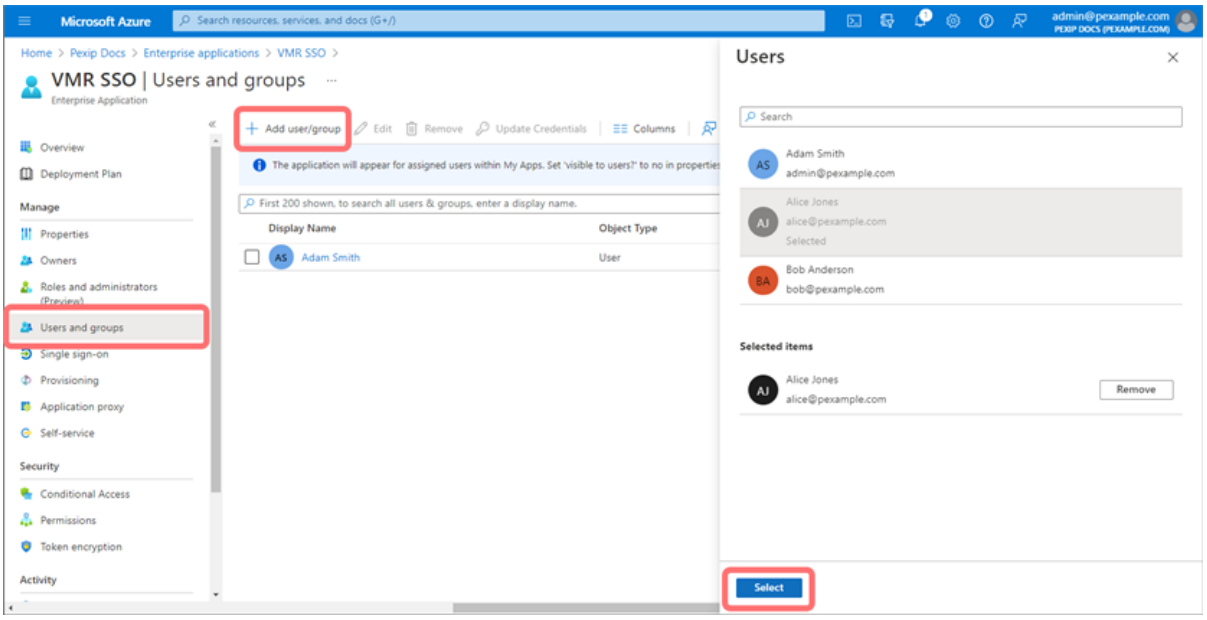
2. When the application has been created, select **Single sign-on > SAML**:



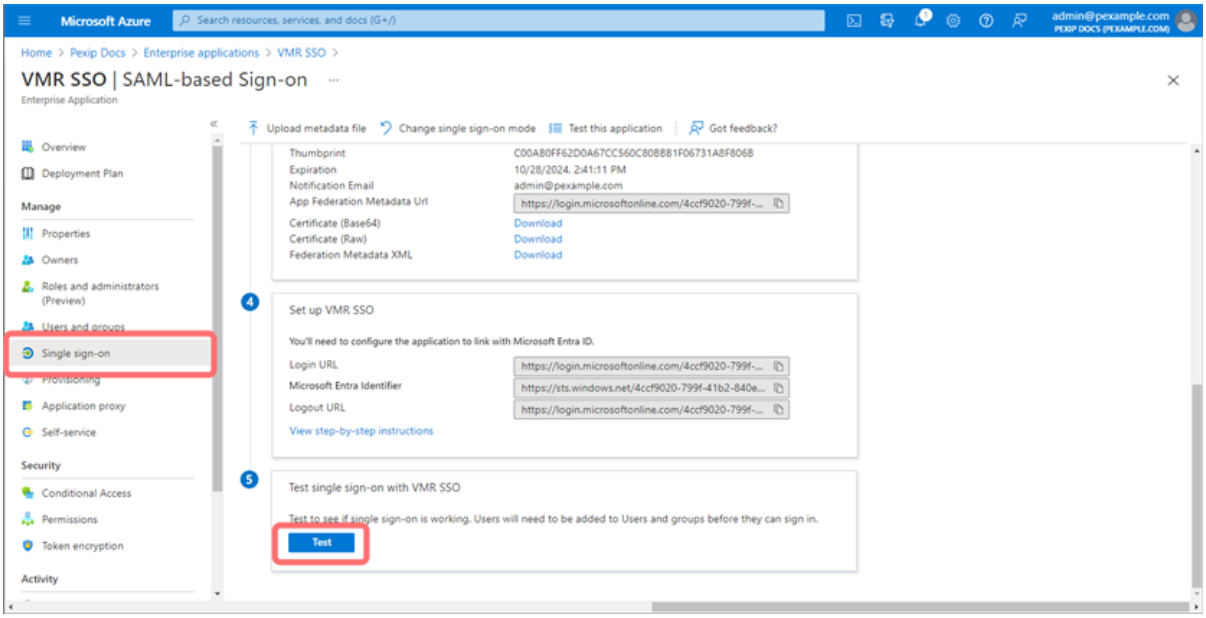
3. Configure the application with the details of the Identity Provider you configured earlier on Pexip Infinity:

Azure option	Infinity Identity Provider option	Notes
Basic SAML Configuration		
Identifier (Entity ID)	SAML 2.0 Entity ID for this service	We recommend that you use the same FQDN from which the web app is accessed.
Reply URL (Assertion Consumer Service URL)	Redirect URL	This should be in the format: **https://<webapp_fqdn>/api/v1/samlconsumer/<uuid>** where **<webapp_fqdn>** is the FQDN from which the web app is accessed, and **<uuid>** is the UUID generated by Pexip Infinity. The entire URL must be in lower-case.
Sign on URL Relay State Logout URL		Leave blank
Attributes and claims		
	Display Name Attribute Name	This page lists the options available for users' display names. Select the format you wish to use (or create a new claim) and enter the corresponding **Claim name** from AD into the Display Name Attribute Name field on Pexip Infinity. For example, you may wish to hide surnames in Virtual Meeting Rooms that are used for B2C purposes. To do this, on Pexip Infinity enter: **http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname** If you don't specify a claim in Pexip Infinity, participants must enter their own display name.
SAML signing certificate		
Signing Option		Select either ***Sign SAML assertion*** or ***Sign SAML response and assertion***.
Signing Algorithm		This feature supports any of the following: SHA1, SHA256, SHA384 or SHA512 with RSA certificates
Certificate (Base64)	Identity Provider public key	Download the certificate from Azure and paste into Pexip Infinity.
Set up <application name>		
Login URL	Identity Provider SSO URL	Copy this value from Azure and paste into Pexip Infinity.
Microsoft Entra Identifier	Identity Provider identifier	Copy this value from Azure and paste into Pexip Infinity.
Logout URL		This is not used by Pexip Infinity.

4. Select **Users and Groups > Add user/group** and select the users you wish to allow:



5. Go back to **Single sign-on** and at the bottom of the page, select **Test**:



6. Because this test is being initiated from Entra and Pexip Infinity does not support Identity Provider-initiated flows, you won't see the SSO page that users will see when attempting to join the VMR. Instead, you should be able to reach Pexip Infinity and see the following page:

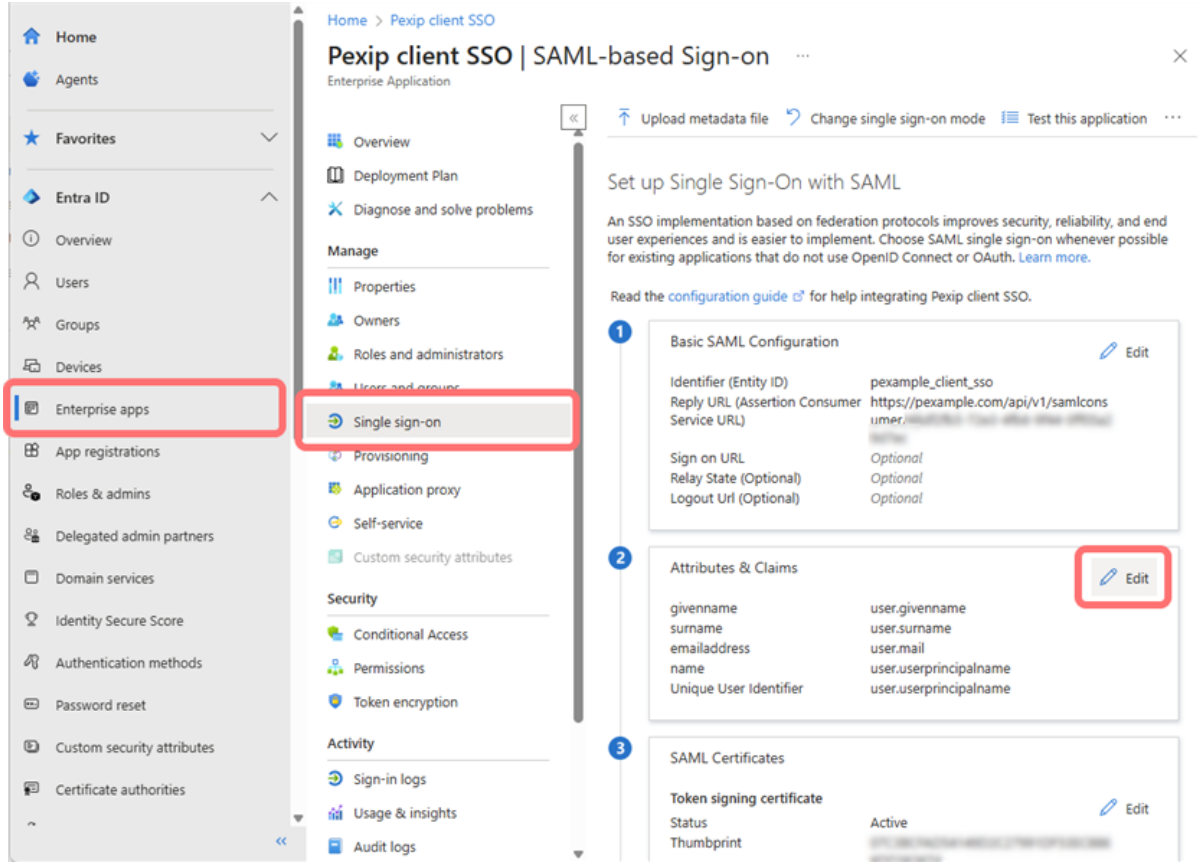
Now that you have set up Microsoft Entra ID as an Identity Provider, you can configure your VMRs and Virtual Auditoriums to use SSO to authenticate participants, and require devices to use SSO when registering an alias.

Setting a custom session timeout for registrations

To prevent a user from authenticating with your Identity Provider and staying authorized indefinitely, the Pexip app for Windows periodically invalidates the session and requires users to re-authenticate their registration. The session will be invalidated 24 hours after successful authentication unless you customize the session timeout duration.

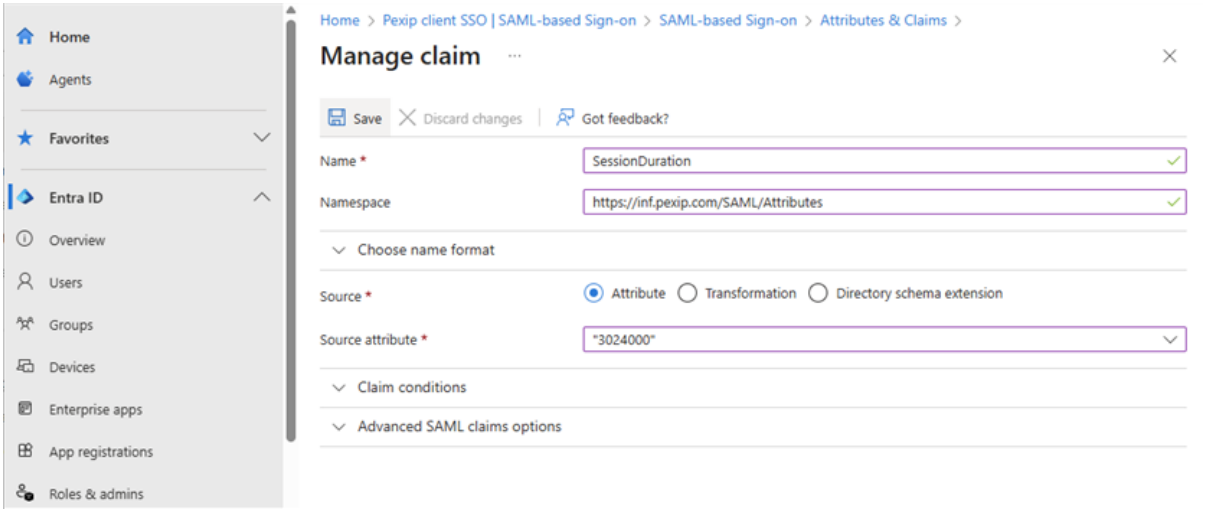
For Microsoft Entra ID, you customize the session timeout duration by configuring a custom `SessionDuration` attribute and value, as follows:

1. From the Azure portal select **Microsoft Entra ID > Enterprise Applications** and select your application.
2. Select **Single sign-on** and from the **Attributes & Claims** panel select **Edit**:

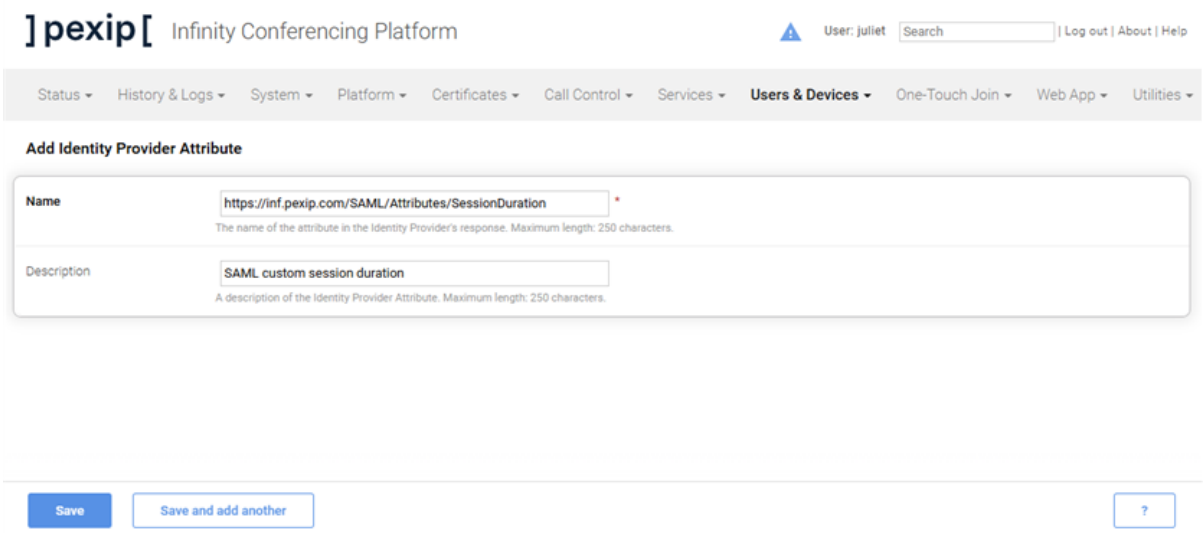


3. From the **Attributes & Claims** page, select **Add new claim**.
4. From the **Manage claim** page, enter the following:

Name	SessionDuration
Namespace	https://inf.pexip.com/SAML/Attributes
Source	Attribute
Source attribute	Enter the session duration, in seconds. The maximum is 3024000 (35 days).



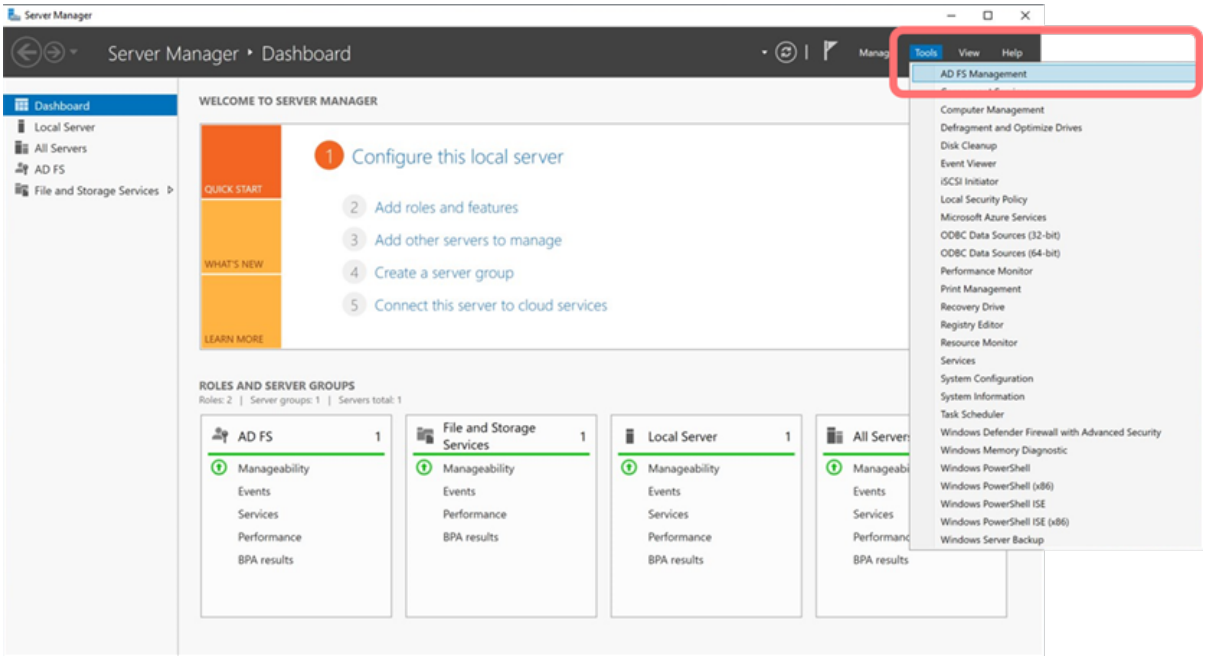
5. From the Pexip Infinity Administrator interface, go to **Users & Devices > Identity Provider Attributes** and add an attribute with a **Name** of **https://inf.pexip.com/SAML/Attributes/SessionDuration**:



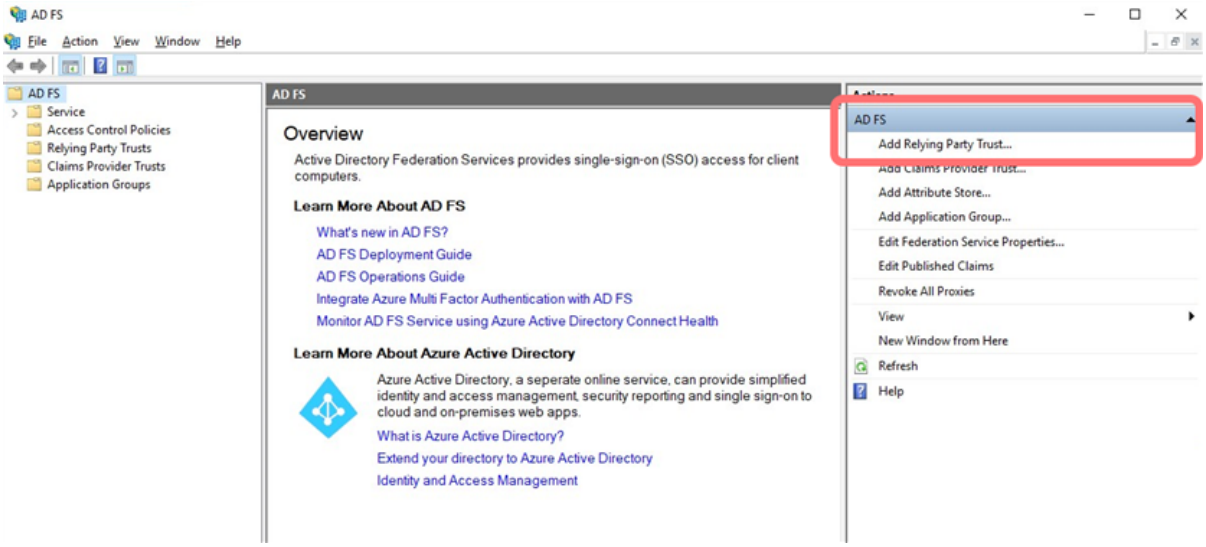
6. From **Users & Devices > Identity ProviderS**, select the Identity Provider. Scroll down to the **Advanced options** section, and from the **Attributes** field, select and apply the **https://inf.pexip.com/SAML/Attributes/SessionDuration** attribute.

AD FS

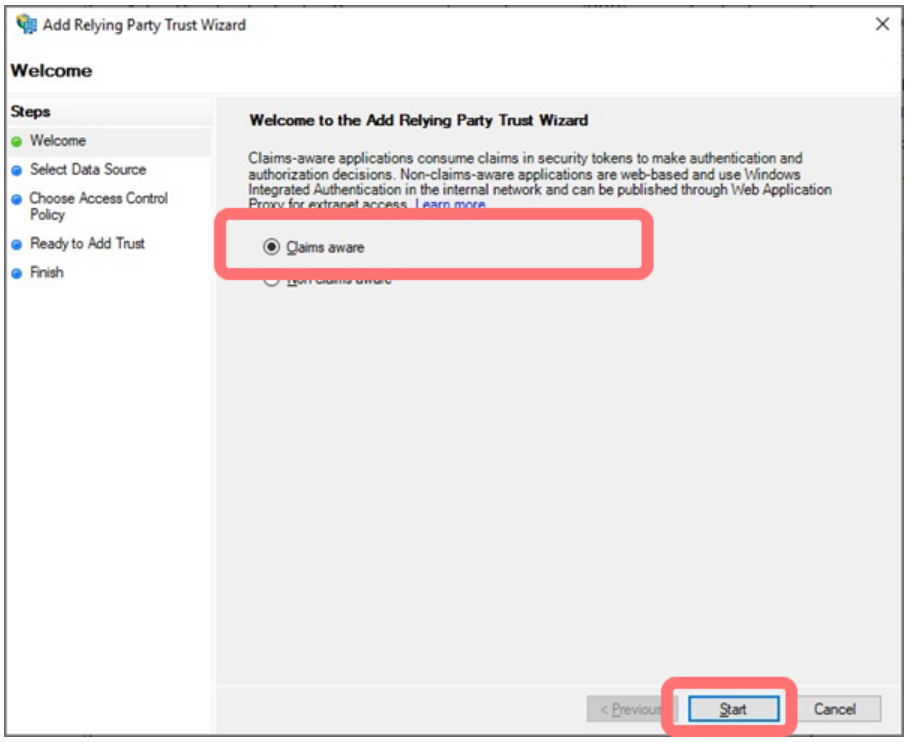
1. From the AD FS server, open the **Server Manager** application. From the top right, select **Tools > AD FS Management**:



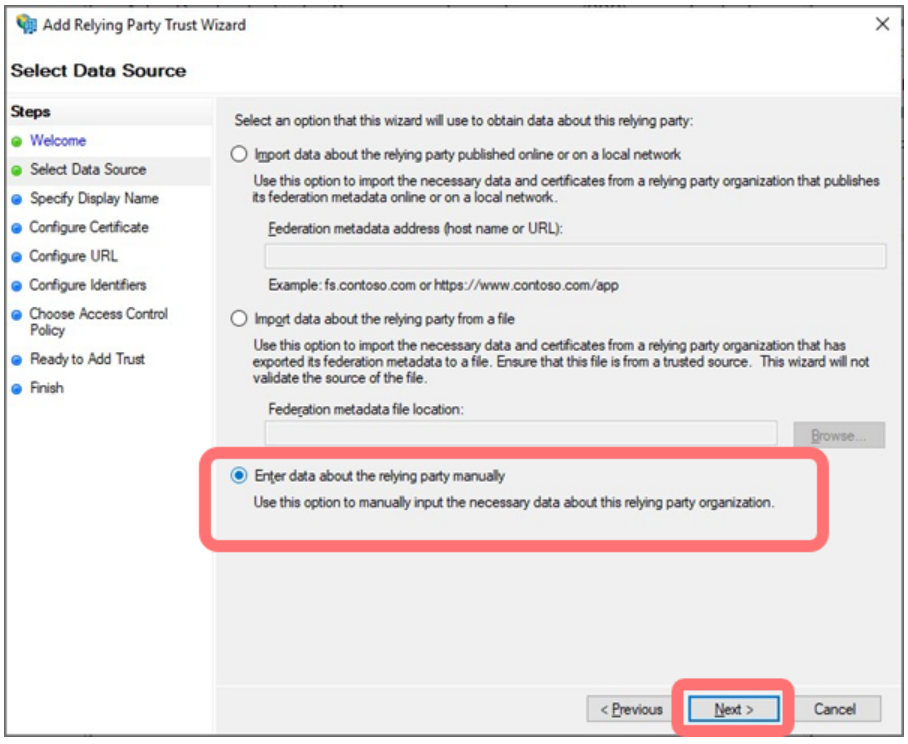
2. From the **Actions** panel, select **Add Relying Party Trust**:



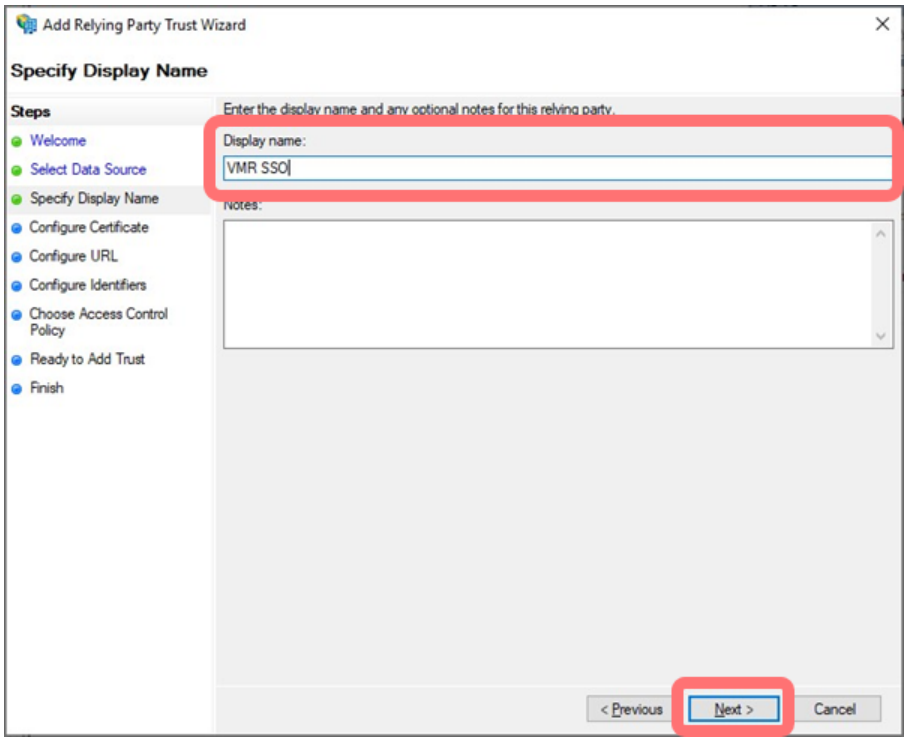
3. At the **Add Relying Party Trust Wizard**, select **Claims aware** and then **Start**:



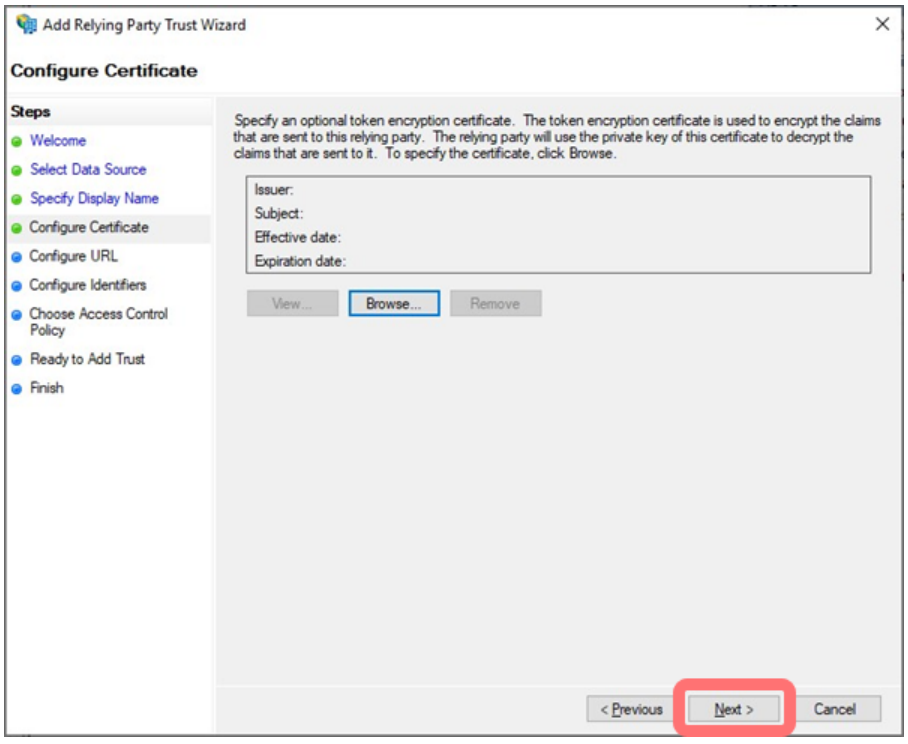
4. Select **Enter data about the relying party manually**, and then **Next**:



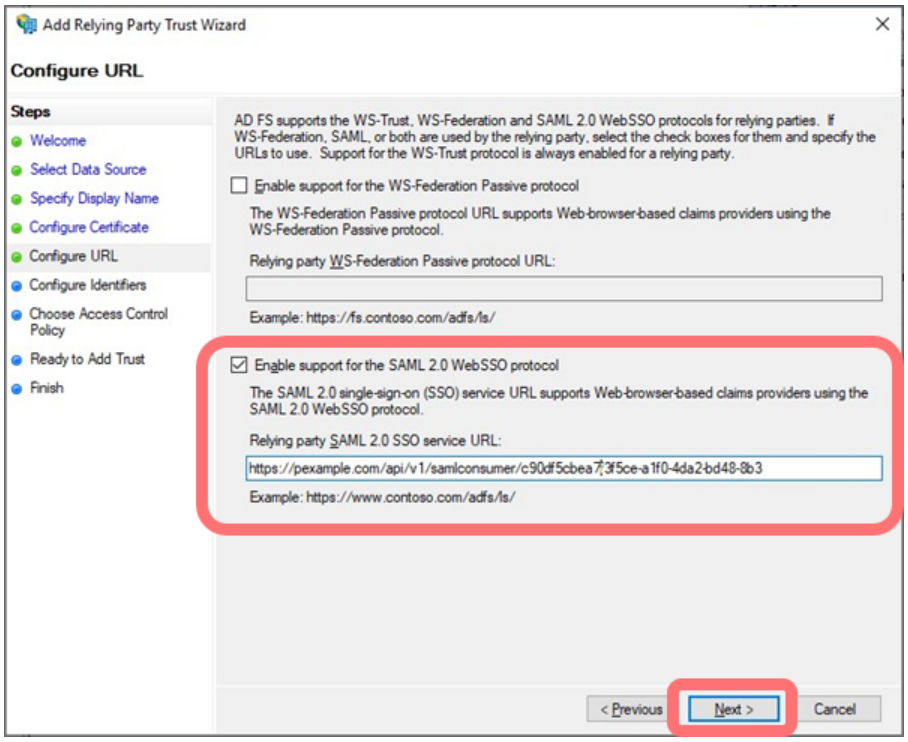
5. Enter a display name for your own reference, and select **Next**:



6. Upload Pexip Infinity's token encryption certificate and select **Next**:

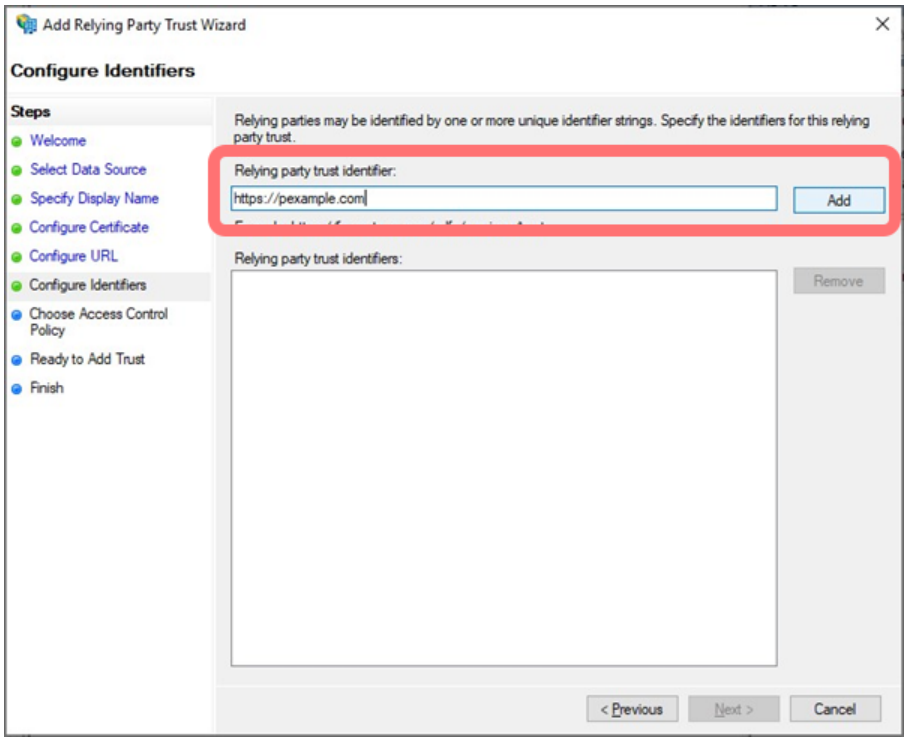


7. Select **Enable support for the SAML 2.0 WebSSO protocol**, and then enter the Pexip Infinity Redirect URL:



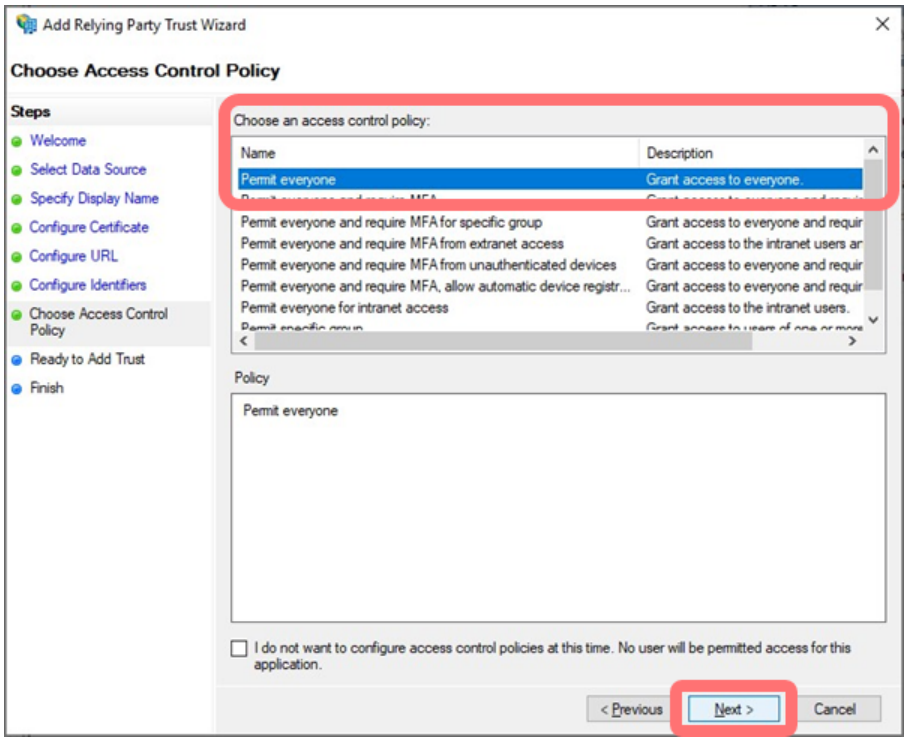
AD FS option	Infinity Identity Provider option	Notes
Relying party SAML 2.0 SSO service URL	Redirect URL	This should be in the format: **https://<webapp_fqdn>/api/v1/samlconsumer/<uuid>** where **<webapp_fqdn>** is the FQDN from which the web app is accessed, and **<uuid>** is the UUID generated by Pexip Infinity. The entire URL must be in lower-case.

8. Enter the **Relying party trust identifier**:

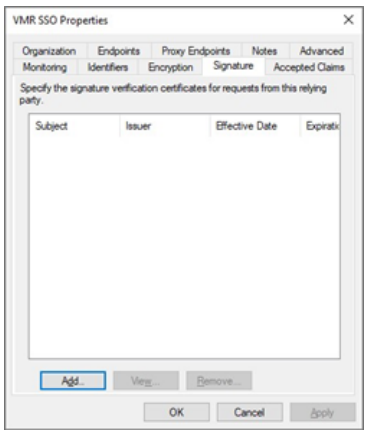


AD FS option	Infinity Identity Provider option	Notes
Relying party trust identifier	SAML 2.0 Entity ID for this service	We recommend that you use the same FQDN from which the web app is accessed.

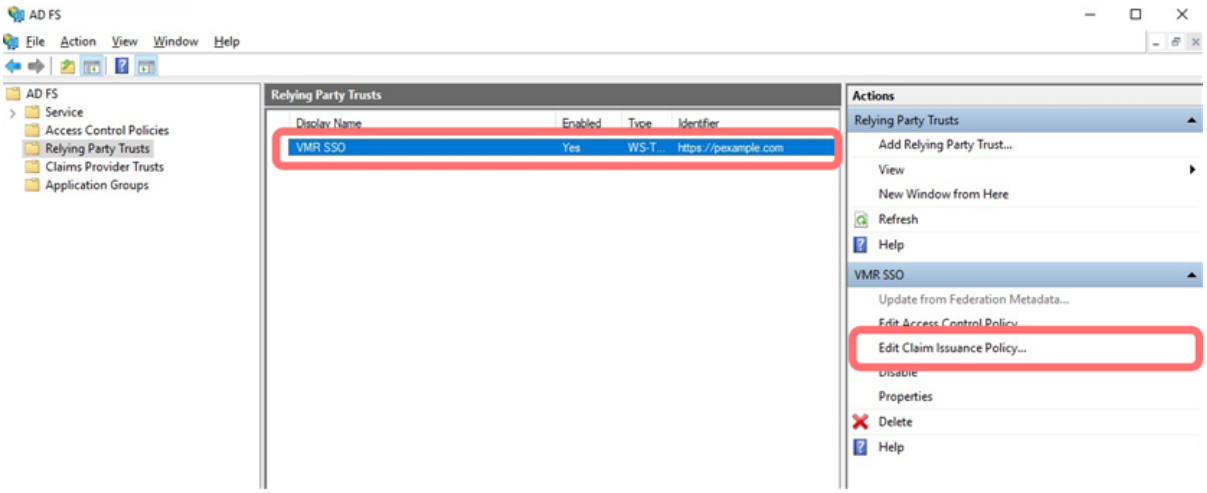
9. Select **Permit everyone**:



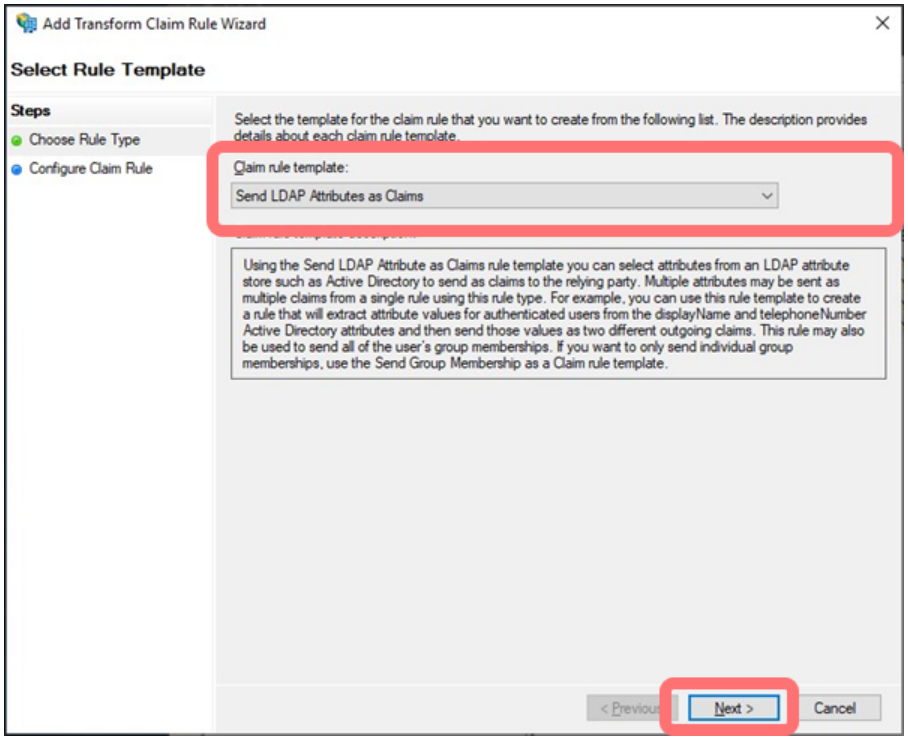
10. Close the wizard, and then select the relying party trust you have just created. Open its **Properties**, and select the **Signature** tab. **Add** the same certificate as you added earlier:



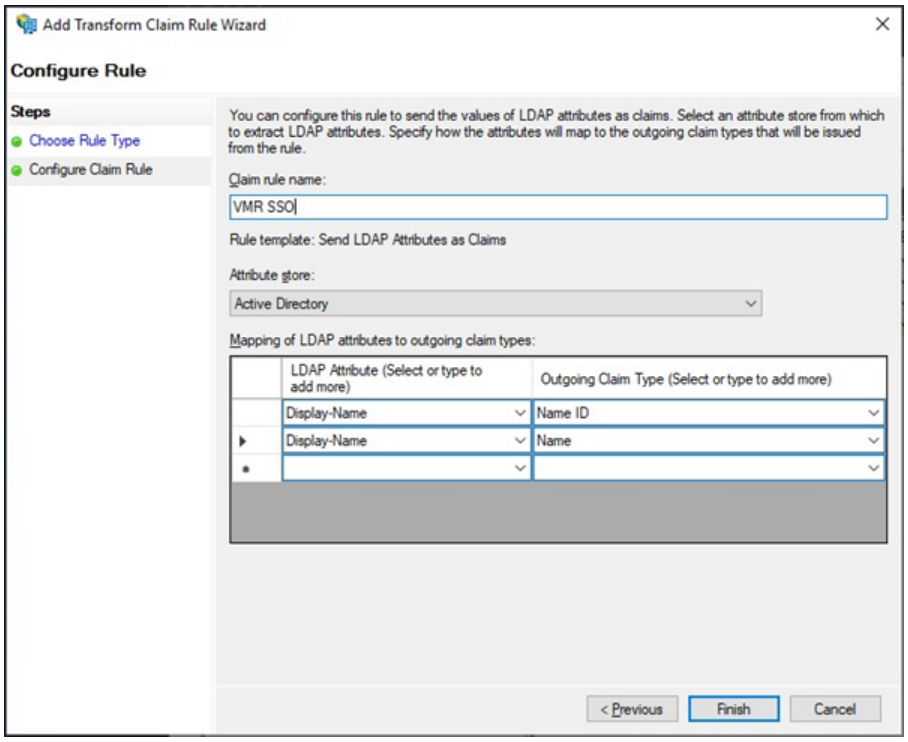
11. Go back and select the relying party trust you have just created, and from the **Actions** panel select **Edit Claim Issuance Policy**:



12. Select **Add rule**, and then at the wizard select a **Claim rule template** of ***Send LDAP Attributes as Claims***:

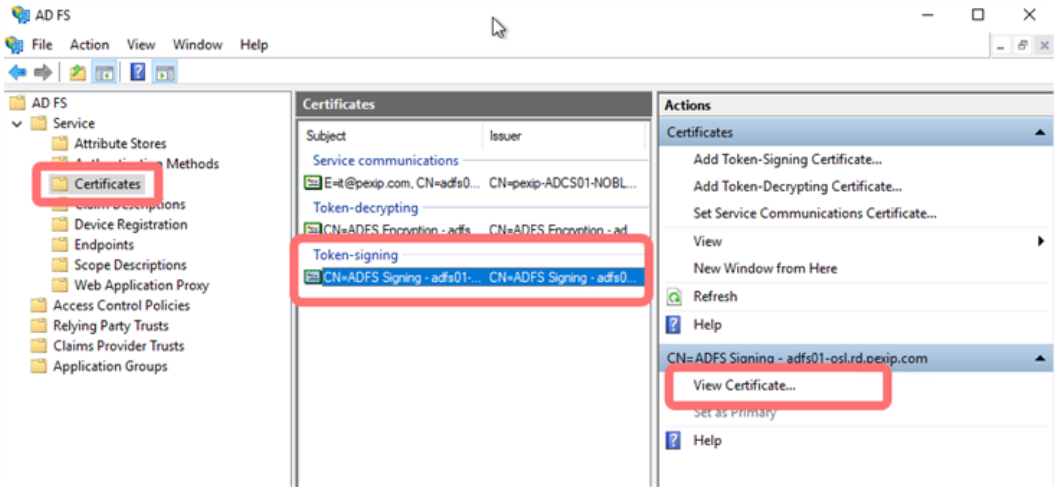


13. Enter a name for your own reference. From the **Attribute store** option, select ***Active Directory***. From the lists that then appear, map the following attributes to claim types:
    - Display-Name > Name ID
    - Display-Name > Name



AD FS option	Infinity Identity Provider option	Notes
Outgoing claim type	Display Name Attribute Name	http://schemas.xmlsoap.org/ws/2005/05/identity/claims/name Confirm that the default (**NameID**) is appropriate for this VMR's use. For example, you may wish to hide surnames in Virtual Meeting Rooms that are used for B2C purposes. If you don't specify a claim in Pexip Infinity, participants must enter their own display name.

14. Next you must get the AD FS signing certificate. To do this, from the left panel select **AD FS > Service > Certificates**. From the **Certificates** panel select **Token-signing** and then from the **Actions** panel select **View Certificate...**:



AD FS option	Infinity Identity Provider option	Notes
Certificate	Identity Provider public key	Download the certificate from AD FS and paste it into Pexip Infinity. By default, certificates created in AD FS expire after 365 days and AD FS will then automatically generate a new certificate. You should therefore set an annual reminder to download this new certificate from AD FS and update the Identity Provider configuration on Pexip Infinity with the new certificate.

Okta

1. From Okta, select **Applications > Applications > Create App Integration**. At the **Create a new app integration** screen, select **SAML 2.0**:

2. Give the app a name and select **Next**:

3. Under the SAML settings section:

enter the following:

Okta option	Infinity Identity Provider option	Notes
Single sign on URL	Redirect URL	This should be in the format: **https://<webapp_fqdn>/api/v1/samlconsumer/<uuid>** where **<webapp_fqdn>** is the FQDN from which the web app is accessed, and **<uuid>** is the UUID generated by Pexip Infinity. The entire URL must be in lower-case.
Audience URI (SP Entity ID)	SAML 2.0 Entity ID for this service	We recommend that you use the same FQDN from which the web app is accessed.

All other fields can be left as the defaults, or edited according to your deployment.

4. If you wish to enable assertion encryption, or change the signing algorithms, open the **Advanced Settings**:

5. In the **Attribute Statements** section, , add a custom attribute with a **Name** of ***email*** and a **Value** of ***user.email***:

Okta option	Infinity Identity Provider option	Notes
Attribute statements name	Display Name Attribute Name	

6. Select **Next** and then **Finish**.
7. From the **Settings** page, select **View Setup Instructions**:

8. Enter the information provided by Okta into the Pexip Infinity Identity Provider configuration:

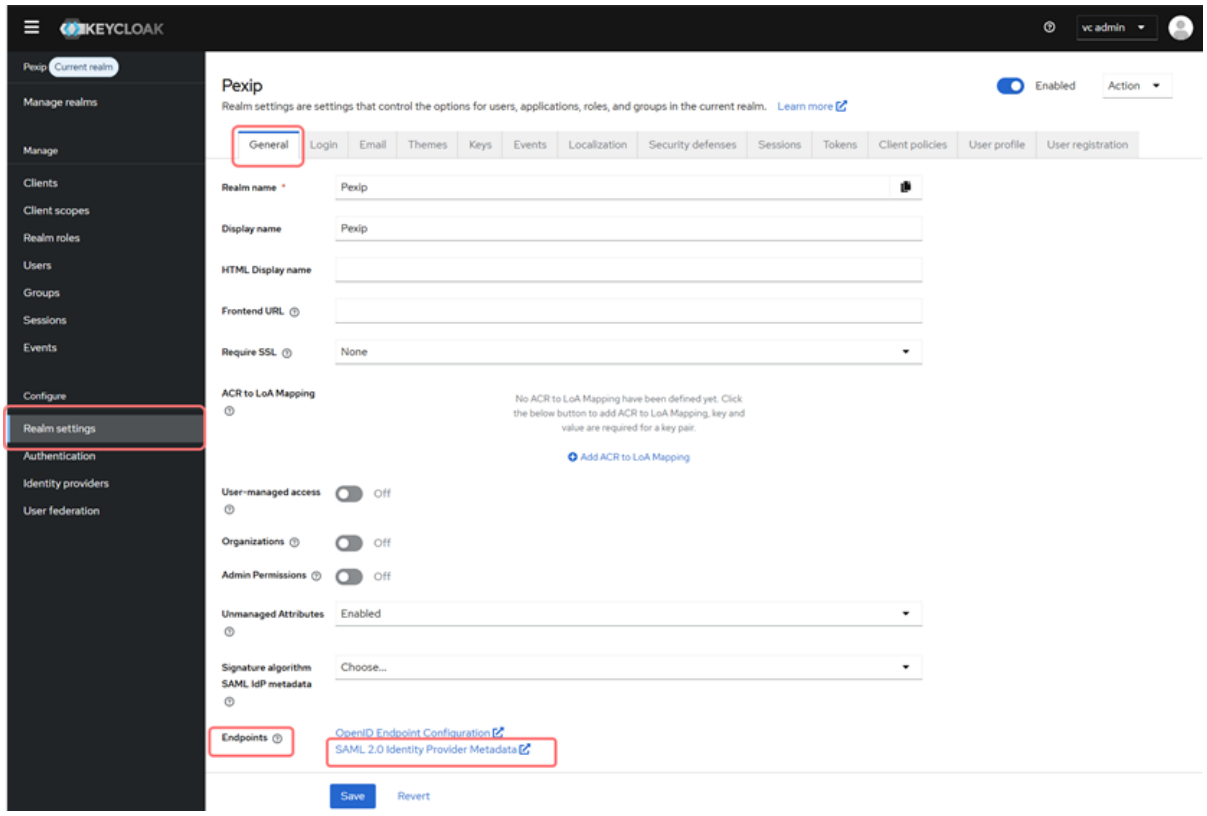
Okta option	Infinity Identity Provider option	Notes
Sign-On URL	Identity Provider SSO URL	
Issuer	Identity Provider identifier	
Signing Certificate	Identity Provider public key	

9. From the **Assignments** tab, select **Assign > Assign to People** and select the users you wish to allow access:

Keycloak

Downloading the Keycloak metadata

1. From your Keycloak admin console, select **Realm Settings** and then select the **General** tab. Scroll down to the **Endpoints** field, select **SAML 2.0 Identity Provider Metadata** and save the .XML file:



Configuring Keycloak as an IdP on Pexip Infinity

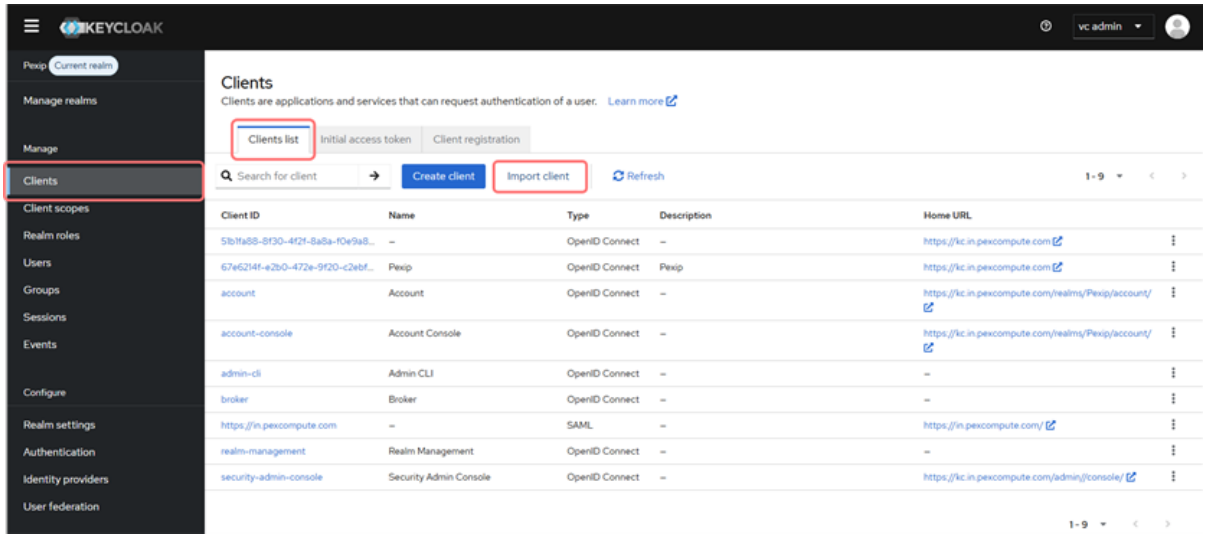
1. From the Pexip Infinity Administrator interface (**Users & Devices > Identity ProviderS > Add Identity Provider**), create a record for the Keycloak Identity Provider, using the following information (all other fields can be left as the defaults, or edited according to your deployment):

Option	Description
Service configuration	
Identity Provider protocol	Select ***SAML***
SAML options	
Certificate	Paste the certificate used by Pexip Infinity when communicating with the Identity Provider.
Private key	Paste the contents of the private key used by Pexip Infinity when communicating with the Identity Provider. This must be unencrypted, and include all the text in the file between (and including) -----**BEGIN PRIVATE KEY**----- and -----**END PRIVATE KEY**-----.
Redirect URL	This should be in the format: **https://<webapp_fqdn>/api/v1/samlconsumer/<uuid>** where **<webapp_fqdn>** is the FQDN from which the web app is accessed, and **<uuid>** is the UUID generated by Pexip Infinity. The entire URL must be in lower-case.
Identity Provider configuration	
Upload file	Upload the .XML file you downloaded from Keycloak in the previous step.

2. **Save** the configuration. You are returned to the **Select Identity Provider to change** page.
3. Select the Keycloak Identity Provider you have just created. From the bottom of the page, select **Download service metadata** and save the .XML file.

Importing the Pexip Infinity configuration into Keycloak

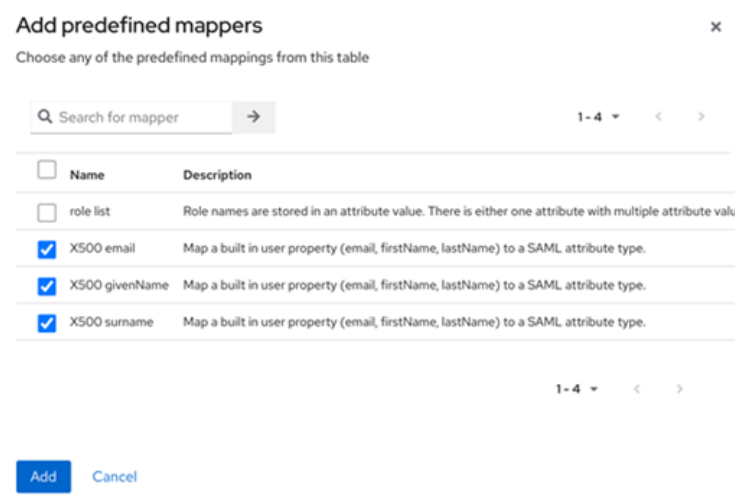
1. From the Keycloak admin console, select **Clients** > **Clients list** > **Import client**:



2. From the **Import client** page, in the **Resource file** section select **Browse** and upload the .XML file you downloaded from Pexip Infinity in the previous step, and select **Save**.
3. From the **Clients** page, select the client you have just imported. From the **Settings** tab, enter the following information, ensuring it matches the configuration on Pexip Infinity:

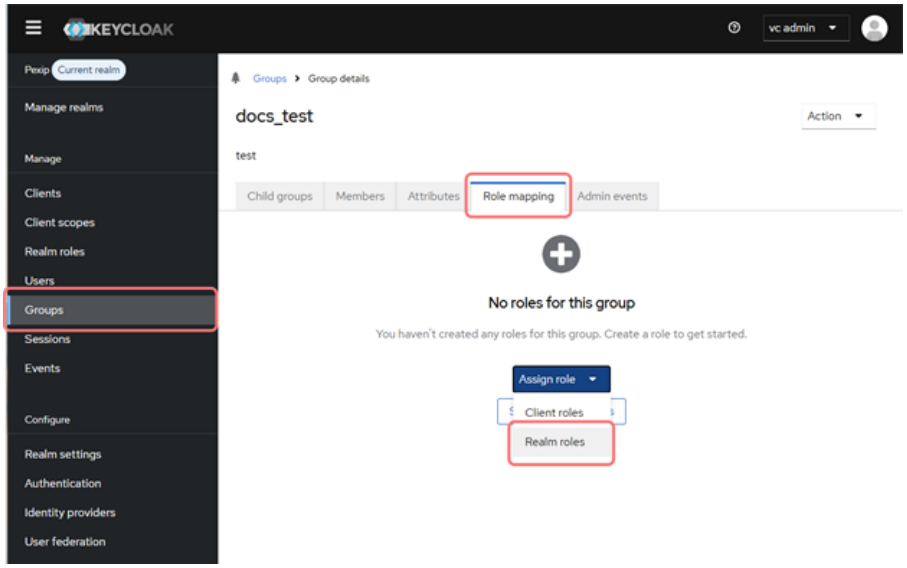
Keycloak option	Infinity Identity Provider option	Notes
Valid Redirect URIs	Redirect URL	This should be in the format: **https://<webapp_fqdn>/api/v1/samlconsumer/<uuid>** where **<webapp_fqdn>** is the FQDN from which the web app is accessed, and **<uuid>** is the UUID generated by Pexip Infinity. The entire URL must be in lower-case.

4. From the **Client scopes** tab:
   a. Select **https://<your_domain>-dedicated**.
   b. From the **Mappers** tab, select **Add predefined mapper**.
   c. Select the **X500 email**, **X500 givenName**, and **X500 surname** attributes:

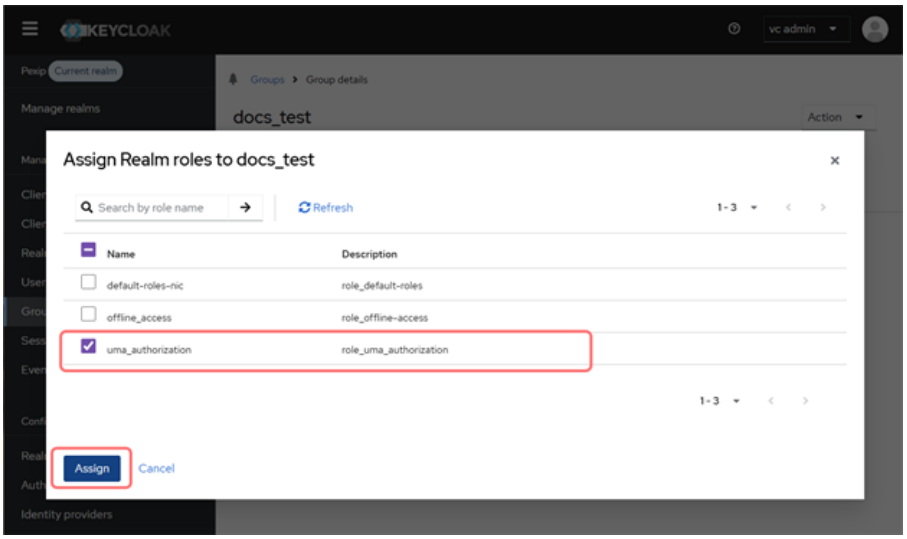


d. Select **Add**.

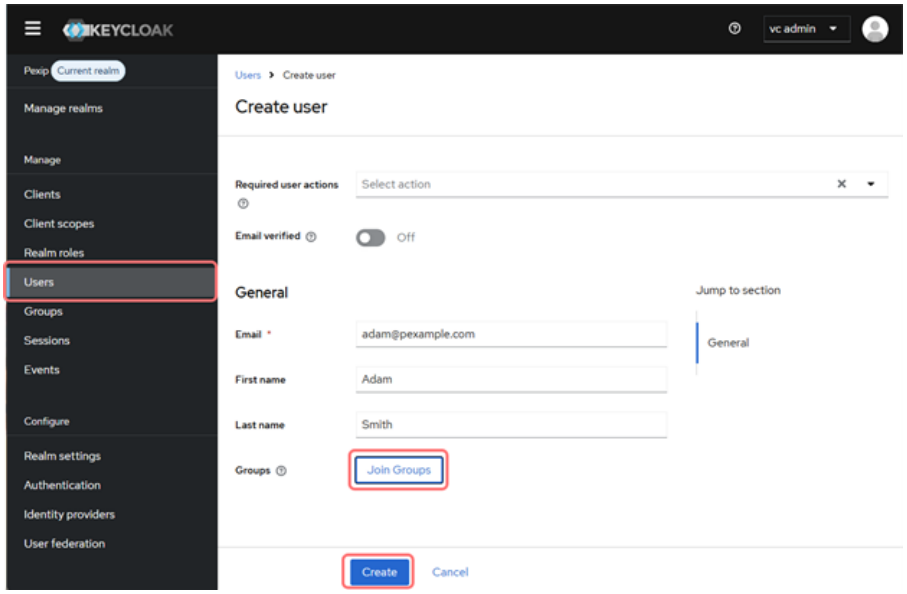
5. From **Groups**, create a new group. Then select the new group and from the **Role mapping** tab select **Assign role > Realm roles**:



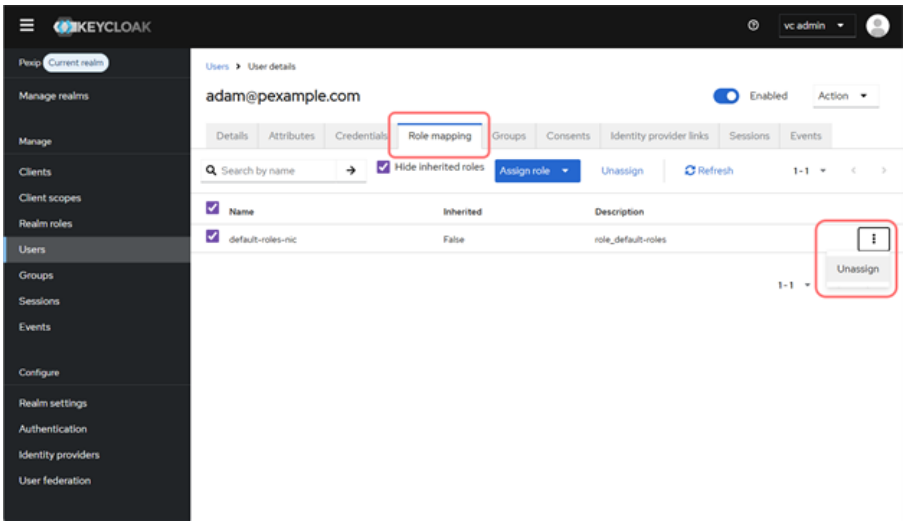
6. From the **Assign Realm roles** page, enable **uma_authorization**:



7. From **Users**, create a new user. Enter their details and select **Join Groups** to assign them to the **Group** created above:



8. Select the new user and from the **Role mapping** tab, select the menu to the right of the **default-roles-nic** and select **Unassign**:



Configuring devices to use Keycloak for registration authentication

1. From Pexip Infinity (**Users & Devices > Identity Provider Groups**), create an Identity Provider Group and add the Keycloak Identity Provider to the group.
2. From Pexip Infinity, for each device that will use Keycloak to authenticate, select the device alias (**Users & Devices > Device Aliases**) and:
   a. ensure that **Enable registration using IdP SSO** is selected
   b. from the **Identity Provider Group** drop-down, select the Keycloak Identity Provider group you created earlier.

Authenticating app registrations using Identity Providers

The Pexip app for Windows must be registered in order to make or receive calls, and the registration must be authenticated with an Identity Provider using SSO. This topic explains how to configure Pexip Infinity and your chosen Identity Provider to enable users to register their Pexip app for Windows using their SSO credentials.

To enable registration authentication you must:

- Configure Pexip Infinity with the aliases to be registered — see Creating the registration alias.
- Configure Pexip Infinity with details of an Identity Provider to use to authenticate registrations — see Configuring Pexip Infinity with Identity Providers.
- Configure the Identity Provider to support authentication of registrations — see Configuring the Identity Provider.

  We have provided step-by-step guides for configuring common SAML Identity Providers, including the open-source Keycloak (which may be an option if you only require an IdP for Pexip app registrations).

Registration process

The process of the app registering to Pexip Infinity is outlined below.

1. The user enters the alias they wish to register in the app's **Your video address** field, for example **alice@pexample.com**.
2. The app performs a DNS SRV lookup on **_pexapp._tcp.<domain>**, for example **_pexapp._tcp.pexample.com**, to locate a Conferencing Node where it can send its registration request.
3. When the Conferencing Node receives the request, it checks whether the alias is configured under **Users & Devices > Device Aliases**.
4. If the alias is configured and has **Enable registration using IdP SSO** enabled, the Conferencing Node checks the configured **Identity Provider group** and obtains the configured **Identity Provider SSO URL**.
5. The Conferencing Node provides the SSO URL to the app; the app then sends its registration request to this URL, including the alias being registered.
6. Upon receipt of the registration request, the Identity Provider requests authentication information from the app.
7. The user provides the Identity Provider with their SSO username and password via their default browser.

8. If the user successfully authenticates with the Identity Provider, the Identity Provider returns the registration alias and display name for that user.
9. If the registration alias that is returned matches the alias being registered, the registration will be permitted and the display name that was returned is used as the app's display name (and cannot be changed by the user).

Creating the registration alias

When registering a Pexip app to Pexip Infinity, the alias (video address) being registered by the Pexip app must match one of the device alias entries on the Management Node. You can either create device registration aliases manually, or import them from a LDAP source.

For full information about registrations, see Registering devices to Pexip Infinity. The sections below describe the configuration that is specifically required for device aliases being registered by Pexip app for Windows.

Configuring the alias manually

To configure a device alias for Pexip app registrations, on the Management Node under **Users & Devices > Device Aliases** select **Add Device alias**. The following fields are required (other fields can be configured as appropriate for your deployment):

Device alias	The alias that app users will enter in the **video address** field when registering their app.
Description	The description of this device; this appears to other app users when searching the directory.
Enable registration using IdP SSO	Select this option. The **Identity Provider group** option appears.
Identity Provider group	From the drop-down list, select the Identity Provider group containing the Identity Provider to use to authenticate the registration.

Importing device aliases from LDAP

To import device aliases from a LDAP source using a LDAP synchronization template, from the Management Node go to **Utilities > LDAP Sync Templates** and select **Add LDAP sync template**. The following fields are required (other fields can be configured as appropriate for your deployment):

Sync devices	Select this option. The **Device settings** and **Device registration settings** panels appear.
Device alias pattern	The pattern for the alias that app users will enter in the **video address** field when registering their app. If users' video addresses are the same as their email addresses, use the default **{{mail}}**.
Device description pattern	The pattern for the description of this device; this appears to other app users when searching the directory.
Enable registration using IdP SSO	Select this option. The **Identity Provider group** option appears.
Identity Provider group	From the drop-down list, select the Identity Provider group containing the Identity Provider to use to authenticate the registration.

There is a known limitation in Pexip Infinity version 37 and earlier when using an LDAP sync template for device alias registrations. The options to **Enable registration using IdP SSO** and enter a corresponding **Identity Provider group** are not included, so you must manually set these fields after importing the new aliases or use an update script. Contact your Pexip authorized support representative with the reference **42719** to obtain the update script.

Configuring Pexip Infinity with Identity Providers

Pexip app registrations must be authenticated with an Identity Provider using SSO. You can use an existing Identity Provider, or create a new one.

For full information about creating and configuring Identity Provider, see Using Identity Providers. The sections below describe the configuration that is specifically required for Pexip app registration.

Creating Identity Providers

To configure Pexip Infinity with an Identity Provider to use for registration, go to **Users & Devices > Identity Providers > Identity Provider Configuration**. The following fields relate specifically to registration authentication (other fields can be configured as appropriate for your deployment):

Registration Alias Attribute Name (SAML)	You must enter a value in this field.
Registration Alias Claim Name (OpenID Connect)	The alias returned by the Identity Provider must match the alias being registered, otherwise the registration is not permitted.
Display Name Attribute Name (SAML)	This field is optional.
Display Name Claim Name (OpenID Connect)	The name returned is used as the user's display name. If the field is blank, the user's alias is used as their display name. Users cannot change the display name provided during registration.

Creating Identity Provider groups

We recommend that you create a new Identity Provider group containing a single Identity Provider and use this group for the sole purpose of authenticating Pexip app for Windows registrations. This is the Identity Provider group you select when configuring device aliases.

The Identity Provider in this group can be used for other purposes as part of another Identity Provider group.

Configuring the Identity Provider

When authenticating with the IdP using SSO, users enter their email address as their SSO username. If your users' aliases (video addresses) are different from their email addresses, we recommend you use a SAML-based IdP. This allows you to create a custom attribute in the IdP to validate that the video address being registered matches the user's email address.

The Pexip app requires specific IdP attribute entries in your configuration for validating user authentication. The video address that a user signs in and registers with must match an IdP attribute to validate that the correct person is using the video address to sign in.

- If users' video addresses are the same as their email, use the **email** attribute for validation.
- If users' video addresses are not the same as their email, you must create a **custom** attribute. This is a top-level SAML configuration and does not require a new entry for every user.

Session duration and timeout

To prevent a user from authenticating with your Identity Provider and staying authorized indefinitely, the Pexip app for Windows periodically invalidates the session and requires users to re-authenticate their registration. The session will be invalidated 24 hours after successful authentication unless you customize the session timeout duration.

For OIDC IdPs, the session timeout duration is controlled via the required `exp` field of the JWT provided by the IdP.

For SAML IdPs, you customize the session timeout duration either:

- via the `SessionNotOnOrAfter` attribute and value (if this is supported by your IdP), or
- by configuring a `SessionDuration` custom attribute and value. For an example of how to do this for Microsoft Entra ID, see Microsoft Entra ID.

Registering and provisioning Pexip apps

When your Identity Provider integration is complete, Pexip app for Windows users can use their SSO credentials to register their Pexip apps to Pexip Infinity. For information on the registration experience for end users, see Registering.

As an administrator you can provision individual users with pre-configured settings for their app, including their registration alias. See Provisioning the app for instructions about how to do this and for information on the associated end-user experience.

Authenticating registrations using AD FS

Pexip Infinity can integrate with Active Directory Federation Services (AD FS) to provide the legacy Connect desktop app with single sign-on (SSO) access to allow users to register their clients using their AD credentials.

This method of authentication is being deprecated and is not supported by the new Pexip apps.

For information on how to configure AD FS and Pexip Infinity to enable users to register their legacy Connect clients using their AD FS credentials, see our v39 documentation site.

About participant authentication

Participant authentication via single sign-on (SSO) provides an additional, optional layer of security to prevent unauthorized access to meetings. It requires conference participants to authenticate via an Identity Provider service before they can join a Virtual Meeting Room or Virtual Auditorium, or access a Media Playback Service. It can be used instead of, or in combination with, conference PINs.

This topic covers:

- Using SSO-based participant authentication
- Supported Identity Providers
- Supported Pexip apps
- Joining an SSO-protected VMR
- Joining with other devices
- Authentication and PINs
- Transferring participants
- Display names

Using SSO-based participant authentication

You can optionally require participants who are attempting to access a specific Virtual Meeting Room, Virtual Auditorium or Media Playback Service in your deployment to verify their identity using SSO before joining the meeting. The SSO service can also be used to provide a verified display name for meeting participants.

To enable SSO-based authentication, first you set up one or more Identity Providers, which are third-party services (such as ADFS, Microsoft Entra ID or Okta) that store and manage digital identities, and with which users authenticate using SSO. Then, you configure individual services to require that Hosts, Guests, or both authenticate with an Identity Provider to access the service.

SSO-based authentication is supported for participants joining via the latest versions of the Pexip apps.

Participants joining from other devices (such SIP or H.323 endpoints) can optionally be permitted to join the meeting if the device is locally registered (i.e. registered to the same Pexip Infinity deployment that is hosting the VMR they are attempting to access), or manually admitted by a Pexip app user.

You can optionally require participant authentication for meetings scheduled using Pexip's Secure Scheduler for Exchange feature.

When is SSO not required?

In all cases, devices can join a meeting protected by SSO without needing to authenticate if:

- the device is manually dialed out to (either by a conference host or as an Automatically Dialed Participant)
- the device is paired with a participant who is using Webapp2 (and who has successfully authenticated).

Supported Identity Providers

Pexip Infinity can be configured with Identity Providers that support two widely-used standard industry protocols:

- SAML 2.0
- OpenID Connect (OIDC)

For more information about using SSO and Identity Providers with Pexip Infinity, see Using Identity Providers.

Supported Pexip apps

The apps that support participant authentication are:

- Web app from v27 and later
- Pexip app for Windows (any version)

Participants using older versions of the web app cannot join VMRs with participant authentication enabled. Instead they will receive a message saying that their client does not support SSO-protected meetings, and suggesting they upgrade their client.

Participants using other versions of the Pexip web app can still join the meeting if the VMR:

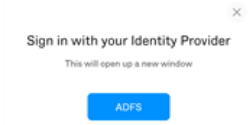
- does not require participant authentication
- requires participant authentication for Hosts only, and the participant is joining as a Guest
- requires participant authentication for Guests only, and the participant is joining as a Host.

Note that all references to Pexip app support also includes bespoke WebRTC clients using the Pexip client APIs.

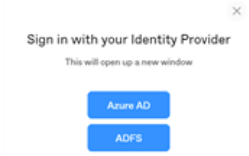
Joining an SSO-protected VMR

To join a Virtual Meeting Room or Virtual Auditorium with participant authentication enabled, using a Pexip app:

1. Access the VMR in the usual way (e.g. by entering the room name, or via a Virtual Reception).
2. If asked, select whether you are a Host or Guest, and enter the PIN (if required).
3. Sign in with your Identity Provider:

   If your organization uses more than one Identity Provider, you see them all listed; choose the one you wish to authenticate with:

   The name that appears on each button in the pop-up is the Name for the Identity Provider that is configured on Pexip Infinity.
4. If this is the first time you have used this Identity Provider, you are redirected to the Identity Provider where you must authenticate. This will usually be done via a new browser tab, which opens automatically. Otherwise, if you've already authenticated with the Identity Provider when accessing this or any other VMR, the authentication happens automatically using your previous credentials.

   Occasionally (at a period determined by the Identity Provider) your authentication session will expire and you will need to re-authenticate.
5. After successfully authenticating, you go straight into the meeting.

Joining with other devices

If authentication is enabled for a VMR or Virtual Auditorium, you can also determine how to treat participants attempting to join from devices other than Pexip apps, such as SIP or H.323 endpoints. You do this via the Other participants setting, where you have options to either put all such devices into a waiting room, or allow these devices to bypass the waiting room, but only if they are locally registered (i.e. registered to a Conferencing Node in the same deployment that is hosting the VMR or VA they are attempting to access).

Note that although some Pexip apps can be registered, the **Other participants** setting does not apply to them.

For each Virtual Meeting Room or Virtual Auditorium:

- if **Other participants** is set to ***Allowed if trusted***:
  - registered devices join the meeting directly (although they still need to enter a Host or Guest PIN if required)
  - unregistered devices are placed in the waiting room;
- if **Other participants** is set to ***Disallow all***:
  - both registered and unregistered devices are placed in the waiting room.

From the waiting room, participants must wait to be manually admitted to the conference by a meeting Host. Upon admission, they are not asked for a PIN and join as a Guest.

Authentication and PINs

VMRs and Virtual Auditoriums that require authentication can optionally use Host and Guest PINs.

For Pexip apps:

- If PINs are configured:
  - Participants need to enter the PIN before authenticating.
  - The PIN that they enter determines whether they are a Host or a Guest, which then determines whether or not they need to authenticate, and if so, the Identity Providers to use.
  - A participant who joins as a Guest can subsequently be escalated to a Host without authentication.
- If PINs are **not** configured, all users are treated as Hosts when authenticating.

For all other devices:

- If PINs are configured, and the device is locally registered, and **Other participants** is set to ***Allowed if trusted***, the participant must enter a PIN before joining the meeting.
- If PINs are **not** configured, and the device is locally registered, and **Other participants** is set to ***Allowed if trusted***, the participant joins the meeting automatically as a Host.
- In all other cases, the participant is held in a waiting room and must wait to be admitted to the meeting by a Host. They are not asked for a PIN and will join as a Guest.

Transferring participants

Participants can be transferred from one VMR into another VMR that requires authentication.

For participants using a supported Pexip app:

- A participant can be transferred into a VMR that requires authentication only if the original VMR also required authentication and used the same Identity Provider as the destination VMR. The participant will not need to re-authenticate to join the destination VMR. If the original VMR did not require authentication, or the destination VMR uses a different identity provider, the transfer is not initiated and the participant remains in the original VMR.
- A participant can be transferred from a VMR that requires authentication into a VMR that does not require authentication. From there, a participant can be transferred into a third VMR that requires authentication only if that VMR uses the same Identity Provider as the original VMR; the participant does not need to re-authenticate in order to join the third VMR. Otherwise, the transfer is not initiated and the participant remains in the second VMR.

For other devices, when a participant using a locally-registered endpoint is transferred, if **Other participants** is set to ***Allowed if trusted*** the participant will join the meeting directly (although they will still need to enter a Host or Guest PIN if required). In all other cases, for both registered and unregistered devices, the transfer will fail and the participant will remain in the original VMR.

Display names

Each Identity Provider offers a given set of attributes for a user (such as their display name, given name, surname and email address).

For Virtual Meeting Rooms and Virtual Auditoriums that require participant authentication, the name shown for each participant (in the participant list, and as a text overlay when **Show names of participants** is enabled) depends on what is configured in the display name setting for the associated Identity Provider, as follows:

For **SAML** Identity Providers, the relevant field is Display Name Attribute Name.

- By default, the **NameId** attribute is used. This is the user's unique ID; what it contains depends on the Identity Provider.
- To use an attribute other than **NameId**, enter it here. This field supports custom attributes.
  Note that the format used varies depending on the Identity Provider.
- To use the name that participants entered themselves in their client, ensure this field is blank.

For **OpenID Connect** Identity Providers, the relevant field is Display Name Claim Name.

- By default, the **Name** attribute is used. This is the user's unique ID; what it contains depends on the Identity Provider.
- To use an attribute other than **Name**, enter it here. This field supports custom attributes.
  Note that the format used varies depending on the Identity Provider.
- To use the name that participants entered themselves in their client, ensure this field is blank.

When the display name is based on information provided by the Identity Provider, it cannot be changed by the participant, the meeting Host, or via the client API. However, it can be changed via .

Viewing a participant's authentication status

Participants and administrators can view whether a participant was required to authenticate in order to join the meeting:

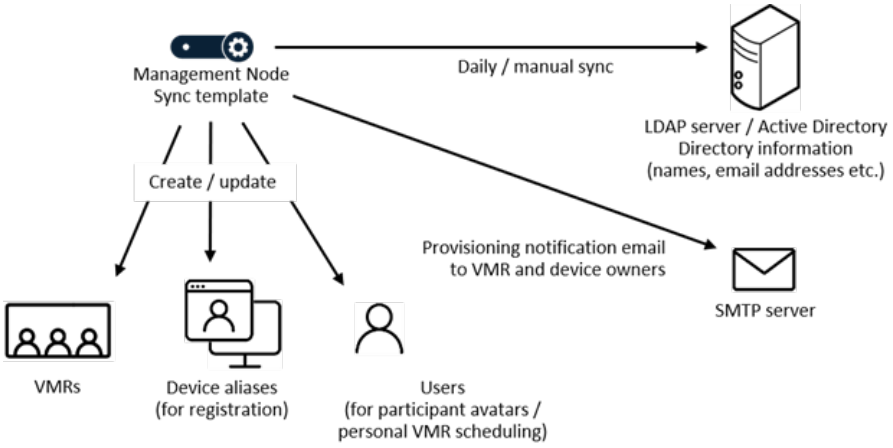
- Meeting participants using the Pexip web app can enable the option to Show authenticated participants.
- Administrators using the Administrator interface can view the participant's status by selecting the participant from **Status > Participants** and scrolling down to the **Authenticated by an Identity Provider** field.

Provisioning VMRs, devices and users from Active Directory via LDAP

In large organizations with many employees and users of video conferencing, you may need to configure lots of Virtual Meeting Rooms (VMRs) and associated records to support those employees. Much of this data can be bulk-provisioned from directory information contained in a Windows Active Directory LDAP server, or any other LDAP-accessible database.

The following Pexip Infinity configuration settings can be provisioned directly from an LDAP data source:

- **Virtual Meeting Rooms (VMRs)**: this provides a simple way to automatically provide a personal VMR for every employee in an organization. Data such as employee names can be imported from the directory and used to generate a unique name and alias for each VMR, following a pattern such as **meet.<username>@example.com**. Other VMR settings such as PINs, the layout and theme can also be assigned, depending upon your VMR template configuration.
- **Device aliases**: these are the aliases that people can use to register their endpoint or Pexip app to a Conferencing Node, and how those devices can be authenticated.
- **User records**: these are required for two optional features:
  - Scheduling meetings in personal VMRs — part of the Secure Scheduler for Exchange feature.
  - Participant avatars — conference participants and directory contacts within Pexip Infinity can be represented by an avatar or image. You can configure user records to represent those participants/contacts and associate each user with an avatar URL that points to an external service (such as Gravatar) which can be used to retrieve that user's avatar/image. The user's **Email address** is used as the primary identifier of each user record, and this must match the **Owner's email address** associated with the device alias of the conference participant. To use your avatar URLs, you must set up a policy profile that has **Use local avatar configuration** enabled. See Applying user records for more information.



You control which data is provisioned by setting up one or more **sync templates** in Pexip Infinity. Each sync template can be configured to automatically synchronize against its LDAP **sync source** once per day. This ensures that the Pexip Infinity VMR, device and user configuration stays in step with all additions and removals that have occurred in the source directory. Many of the VMR/device/user settings can be tagged as "overridable" which means that if that setting (the VMR PIN for example) is manually overridden after the VMR, device or user record has been created, a subsequent resynchronization will not reset it back to another value.

If Pexip Infinity is configured with the details of an SMTP server, you can send an email to the VMR owner whenever a synchronization creates a new VMR or modifies an existing VMR, or to inform a user of the device alias and credentials that they need to use to register their device. Emails are not sent for any user records that are created.

Configuration summary

To provision your Virtual Meeting Rooms, device aliases or user records from an LDAP database you must:

1. Configure the connection details of an LDAP data source. You must also ensure that Pexip Infinity has trusted CA certificates for the authority that signed the LDAP server's certificate (if a TLS connection is required).
2. Configure an LDAP synchronization template that controls what to synchronize — VMRs, device aliases or users — and how individual settings such as the VMR name, dialable alias and PINs are generated, which of those values can be manually overridden, and whether to synchronize automatically against the LDAP data source.

   If required, you can configure multiple synchronization templates that use the same LDAP data source (or different templates that use different LDAP sources). For example, you may want to use a different template for VMR syncing than the template used for device alias syncing if you need to use different LDAP filter criteria for those two user bases.
3. Configure an SMTP server and construct an email template, if you want to send a provisioning notification email to the VMR owner or intended user of a device alias.
4. Generate the VMRs, device aliases and user records using your template and LDAP data source. Set the template to sync automatically when you are happy that the template is configured correctly.

These configuration steps are described in more detail in the following sections.

In addition, you can also:

- View the status of ongoing or completed LDAP template synchronization processes via **Status > LDAP Sync**.
- Delete all of the VMRs, devices and users that have been created from a specific template.

Configuring an LDAP data source

To configure Pexip Infinity with the connection details of an LDAP data source, such as a Windows Active Directory LDAP server:

1. Go to **Utilities > LDAP Sync Sources**.
2. Select **Add LDAP sync source**, and then complete the following fields:

Option	Description
Name	The name used to identify this LDAP sync source.
Description	An optional description of the sync source.
LDAP server address	The domain name (for DNS SRV lookup), FQDN (for DNS A/AAAA lookup) or IP address of the LDAP server. If using a domain or an FQDN, ensure that it is resolvable over DNS. You must also ensure that Pexip Infinity has trusted CA certificates for the authority that signed the LDAP server's certificate (if a TLS connection is required). We strongly recommend that you do not use an IP address. If an IP address is used, and a TLS connection is required, this will only work if the IP address is specified as the common name in the LDAP server's certificate. See Troubleshooting LDAP server connections for more information about how the system establishes a connection to the LDAP server and how to troubleshoot connection issues.
Search global catalog	Select this option to expand the scope of the search to the entire Active Directory Global Catalog instead of traditional LDAP. Note that this uses ports 3268 (TCP) and 3269 (TLS).
LDAP bind username and password	The username and password of the bind account on the LDAP server. This should be a domain user service account, not the Administrator account.
LDAP base DN	The base DN (distinguished name) of the LDAP forest to query (e.g. **dc=example,dc=com**).

3. Select **Save**.
   You can save the sync source details only if Pexip Infinity can successfully contact the specified LDAP server.

Note that all LDAP distinguished names must be entered as per the LDAP standard (RFC 4514). LDAP configuration is case insensitive.

Configuring an LDAP synchronization template

LDAP synchronization templates control what to synchronize — VMRs, device aliases and/or users — and how individual settings such as the VMR name, dialable alias and PINs are generated, and what to include in provisioning notification emails sent to the VMR or device owner. You must also tell the template which LDAP data source to use.

Many of the VMR, device and user settings are based on **patterns** in the template that define how that setting is generated (such as using an email address as the device alias). See Using templates, variables and filters when provisioning VMRs, devices and users for more information.

To configure Pexip Infinity with an LDAP synchronization template:

1. Go to **Utilities > LDAP Sync Templates**.
2. Select **Add LDAP sync template**, and then complete the following fields:

Option	Description
Name	The name used to identify this synchronization template.
Description	An optional description of the synchronization template.
LDAP user search DN	The DN relative to the **LDAP base DN** of the sync source to query for user records (e.g. **ou=people**). If blank, the **LDAP base DN** is used. In deployments with large user bases, you may want to configure this to optimize the LDAP user queries.
LDAP user filter	The LDAP filter used to match user records in the directory. Default: (&(objectclass=person)(!(objectclass=computer))) For more information, see LDAP search and filter examples.

Option	Description
LDAP sync source	Select the LDAP data source to use when syncing records. Note that you can select which will open a new window from where you can configure a new sync source.
Sync VMRs	Enables VMR synchronization for this template. Default: enabled.
Sync devices	Enables device alias synchronization for this template. Default: disabled.
Sync users	Enables user record synchronization for this template. Default: disabled.
Enable automatic daily sync	Select this option to instruct Pexip Infinity to automatically synchronize this template against its LDAP sync source once per day. This ensures that VMRs, devices and users are regularly updated, deleted or created as appropriate based on the latest data in the sync source. Therefore, for example, if a user is removed from Active Directory or their account is disabled, their corresponding VMR, device or user record in Pexip Infinity will be deleted via the next daily sync. As template synchronization can result in the automatic creation, modification or deletion of large numbers of VMRs, devices and users, we recommend that you only enable automatic syncing after you have manually synced at least once and have verified that you are satisfied with your sync template configuration. All automatic synchronizations are initiated at 01:00 UTC (this start time cannot be configured). After an initial sync, which can take several minutes in a large organization, an ongoing daily sync is typically much faster as it only processes changes since the previous sync. Default: disabled.
Device settings (shown when Sync devices is selected)	
Device alias pattern	The pattern for the alias that will be registered by the device and be used by people trying to call the device. For more information, see Using templates, variables and filters when provisioning VMRs, devices and users. Default: {{mail}} If a device with the generated alias already exists, that existing device configuration is left unchanged.
Device tag pattern Allow device tag to be manually overridden	The pattern for the unique identifier used to track usage of the device. You can also allow the auto-generated device tag to be manually overridden for each device alias.
Device description pattern Allow device description to be manually overridden	The pattern to use when generating the optional description of this device alias. You can also allow the auto-generated device description to be manually overridden for each device alias.
Device username pattern Allow username to be manually overridden	The pattern for the device username. Note that you can use the same username for different device aliases (although we recommend you only do this for multiple devices used by the same person). Default: {{mail}} You can also allow the auto-generated username to be manually overridden for each device alias.

Option	Description
Device password pattern	The pattern to use when generating the password for this device. A password pattern must be specified if a username pattern is configured. Example: **{{ ("some random string"+(mail|pex_reverse))|pex_hash|pex_tail(16) }}**
Allow password to be manually overridden	You can also allow the auto-generated password to be manually overridden (by the administrator) for each device alias.
Sync disabled accounts	Syncs all device aliases, even if the corresponding LDAP account is disabled. By default, device aliases are only provisioned if the corresponding LDAP account is enabled in the LDAP directory. If **Sync disabled accounts** is selected, a device alias will be created for LDAP accounts that are marked as disabled. This may be useful, for example, if you have a disabled machine account in LDAP corresponding to a SIP or H.323 room system — however it is not generally useful for staff accounts because if an employee leaves an organization you usually want their device record to be deleted automatically after their account is disabled in the corporate LDAP directory. Default: disabled.
Device registration settings (shown when Sync devices is selected)	
Enable SIP registration	Allows this device alias to register over the SIP protocol.
Enable H.323 registration	Allows this device alias to register over the H.323 protocol.
Enable legacy Infinity Connect registration (non-SSO)	Allows a legacy Connect desktop appto register using this alias (not using SSO services).
Enable Infinity Connect registration using legacy AD FS	Allows a legacy Connect desktop app to register using this alias, using AD FS to authenticate the registration. ⓘ This method of authentication is being deprecated and is not supported by the new Pexip apps.
Allow registration settings to be manually overridden	Allows all of the auto-generated device registration settings to be manually overridden for each device.
Enable registration using IdP SSO Identity Provider group	Allows a Pexip app for Windows to register using this alias, using a SAML or OIDC Identity Provider to authenticate the registration. When this option is enabled, you must also select an **Identity Provider Group** from the drop-down options below.
VMR name and description (shown when Sync VMRs is selected)	
VMR name pattern	The pattern to use to generate the name of the VMR. You should structure this pattern to generate a unique VMR name. For more information, see Using templates, variables and filters when provisioning VMRs, devices and users. Example: **{{givenName}} {{sn}} VMR** If a VMR with the generated name already exists, that existing VMR configuration is left unchanged.

Option	Description
VMR description pattern Allow VMR description to be manually overridden	The pattern to use to generate the VMR description. This field is optional. Examples: **{{givenName}} {{sn}}'s meeting room** You could use a more advanced pattern such as: **{{givenName}} {{sn}}'s {%if department %} ({{department}}){% endif %} meeting room** which will insert "(<department name>)" before "meeting room", but only if a value exists in the LDAP **department** field. You can also allow the auto-generated VMR description to be manually overridden for each VMR.
VMR Participant Authentication settings (shown when Sync VMRs is selected)	
Host PIN pattern	The pattern to use to generate the Host PIN (if required). For example, if you want a VMR to keep the same random PIN after every resync you could use a pattern like this: **{% set pl=6 %}{% set hp=(("an ungu3ssable Pa$$phr4s3" ~mail)\|pex_hash\|pex_tail(pl)) %}{{hp}}** where you: ◦ set **pl** to the number of digits required in the PIN (it is set to 6 in this example, but can be from 4-20), and ◦ replace **"an ungu3ssable Pa$$phr4s3"** with your own passphrase string (we recommend at least 14 characters long). This will generate a random number that is **pl** digits long, but it will always generate the same number per VMR (assuming the VMR owner's LDAP **mail** field remains unchanged). If you want to do a periodic update of all of the PINs associated with VMRs generated from this template, then just change the passphrase string and they will be updated to a new random PIN (assuming **Allow PIN settings to be manually overridden** is not selected). Note that the `~` operator is similar to a `+` and converts arguments, nulls etc. safely to strings.
Allow Guests	***Yes***: the conference can have two types of participants: Hosts and Guests. You must configure a **Host PIN** to be used by the Hosts. You can optionally configure a **Guest PIN**; if you do not configure a **Guest PIN**, Guests can join without a PIN, but the meeting will not start until the first Host has joined. ***No***: all participants have Host privileges. Default: ***No***.
Guest PIN pattern	The pattern to use to generate the Guest PIN (if required). Note that if you set a Host PIN and a Guest PIN for a VMR, those two PINs must be different. Therefore we recommend that you generate Host and Guest PINs of different lengths, or ensure that some aspect of each PIN will be different, for example: **{% set pl=6 %}{% set hp=(("an ungu3ssable Pa$$phr4s3" ~mail)\|pex_hash\|pex_tail(pl)) %}{% set gp=(("a different Pa$$phr4s3" ~mail)|pex_hash|pex_tail(pl)) %}{% if hp == gp %}{{("000000000000000000000"~((gp|int)+123))|pex_tail(pl)}}{% else %}{{gp}}{% endif %}** This works in conjunction with the example above that is used to generate the Host PIN. The pattern includes the recipe used to generate the Host PIN (**hp**) and uses the same technique to generate a random Guest PIN (**gp**). It then checks if the Guest PIN accidentally clashes with the Host PIN. If it does clash, the Guest PIN is adjusted (it is incremented by 123 in this example). As with the Host PIN pattern, you should: ◦ set **pl** to the number of digits required in the Host and Guest PINs, and ◦ replace **"an ungu3ssable Pa$$phr4s3"** and **"a different Pa$$phr4s3"** with your own passphrase strings (both strings must be different).

Option	Description
Allow PIN settings to be manually overridden	Allows the auto-generated Host and Guest PIN settings to be manually overridden for each VMR. Note that this would also preserve the generated PIN value, even if it does not actually get manually overridden.
Host Identity Provider Group	The set of Identity Providers to be offered to Hosts to authenticate with, in order to use the service. If this is blank, Hosts are not required to authenticate. Default: none selected
Guest Identity Provider Group	The set of Identity Providers to be offered to Guests to authenticate with, in order to use the service. If this is blank, Guests are not required to authenticate. Default: none selected
Allow IdP settings to be manually overridden	Allows the Host and Guest Identity Provider Group settings to be manually overridden for each VMR. Default: disabled.
Other Participants Allow Other Participants setting to be manually overridden	Determines whether participants joining a SSO-protected service from devices other than the web app (for example SIP or H.323 endpoints) are allowed to dial in to the service. ○ ***Disallow all***: these devices are placed in a waiting room where they must wait to be admitted by a Host. ○ ***Allow if trusted***: these devices may join the service if they are locally registered. They must still enter a Host PIN or Guest PIN if either is required. All other devices are placed in a waiting room where they must wait to be admitted by a Host. Default: ***Disallow all*** You can also allow the **Other Participants** setting to be manually overridden for each VMR.
VMR layout and theme (shown when Sync VMRs is selected)	
Layout Show name of active speaker	The layout controls the maximum number of other participants that each participant will see, and how the participants are arranged on the screen. For more information, see Conference layouts and speaker names. Default: ***Large main speaker and up to 7 other participants (1+7 layout)***.
Show names of participants	If participant name overlays are enabled, the display names or aliases of all participants are shown in a text overlay along the bottom of their video image.
Allow layout settings to be manually overridden	You can also allow the type of layout and whether to show names of participants to be manually overridden for each VMR.
Theme Allow theme to be manually overridden	The theme for use with this VMR. For more information, see Customizing conference images and voice prompts using themes. Default: ***<use Default theme>*** (the global default theme is used). You can also allow the auto-generated theme to be manually overridden for each VMR.
VMR aliases (shown when Sync VMRs is selected)	
VMR alias 1 pattern	The pattern to use to generate the alias (string) that participants can dial to join this VMR. You should structure this pattern to generate a unique alias for the VMR. Example: **meet.{{givenName|lower}}.{{sn|lower}}@example.com** or **meet.{{mail}}** If the generated alias is already assigned to another VMR, that alias is left assigned to that existing VMR.
Description 1 pattern	The pattern to use to generate a description of the alias. This field is optional. Example: **The alphanumeric URI alias**

Option	Description
VMR alias 2...8 pattern and descriptions	You can optionally enter patterns for a further 7 alternative aliases, such as an E.164 alias, to associate with this VMR. (Aliases 5-8 are configured by expanding the **Advanced VMR alias options** section.) Do not use the **pex_random_pin()** filter to generate numeric aliases as this is likely to generate duplicate aliases. Where possible, it is **always** best to use a phone number, employee ID or other typically unique value from the user's LDAP directory record as the basis of a numeric alias. If a unique value does not exist, a potential workaround (at small scale) is to use a **pex_hash()** expression such as the following to generate a 6-digit random E164-type alias from a hash of each user's email address: **{{ ("my alias passphrase"+(mail|pex_reverse))|pex_hash|pex_tail(6) }}** However it is important to note that these truncated hashes may not be unique between different users, which is why it is preferable to use to use a unique numeric value from the user's LDAP directory record instead.
Allow aliases to be overridden	(This option is configured by expanding the **Advanced VMR alias options** section.) Allows aliases and alias descriptions for a VMR to be added, removed or modified in any way. When enabled this means that after the initial creation of a VMR and its aliases, subsequent syncs will not change any of the aliases or their descriptions (including any which were created by **VMR alias 1..8 pattern** and **Description 1..8 pattern** even if those patterns are modified in this template).
Advanced options (shown when Sync VMRs is selected)	
Guests can present Allow Guests can present to be manually overridden	Controls whether the Guests in the conference are allowed to present content. ○ ***Yes***: Guests and Hosts can present into the conference ○ ***No***: only Hosts can present Default: ***Yes*** You can also allow the auto-generated Guests can present setting to be manually overridden for each VMR.
Direct media enabled Allow direct media to be manually overridden	Allows this VMR to use direct media between participants. When enabled, the VMR provides non-transcoded, encrypted, point-to-point calls between any two WebRTC participants. The options are: ○ ***Best effort***: use direct media in this VMR where possible (when there are two WebRTC participants only), otherwise use standard, transcoded media connections via a Conferencing Node. ○ ***Always***: allow direct media calls only — this limits all calls in this VMR to two WebRTC participants. ○ ***Never***: do not use direct media in this VMR. Default: ***Never*** Note that direct media is not supported if breakout rooms are also enabled for this VMR. See Direct media (end-to-end encrypted calls) for more information. You can also allow the direct media setting to be manually overridden for each VMR.
Direct media notification duration Allow the direct media notification duration to be manually overridden	The number of seconds to show a notification to participants before being escalated into a transcoded call, or de-escalated into a direct media call. Range: 0 to 30 seconds. Default: 0 seconds You can also allow the direct media notification duration setting to be manually overridden for each VMR.
Enable chat Allow enable chat to be manually overridden	Whether chat messaging is enabled for the conference: ○ ***Use global chat setting***: as per the global configuration setting. ○ ***Yes***: chat is enabled. ○ ***No***: chat is disabled. Default: ***Use global chat setting***. You can also allow the auto-generated enable chat setting to be manually overridden for each VMR.

Option	Description
Maximum inbound call bandwidth (kbps) Maximum outbound call bandwidth (kbps) Allow maximum bandwidth settings to be manually overridden	Specifying a maximum inbound call bandwidth will limit the bandwidth of media received by Pexip Infinity from each individual participant dialed in to this VMR. For more information see Managing and restricting call bandwidth. Specifying a maximum outbound call bandwidth will limit the bandwidth of media sent from Pexip Infinity to each individual participant dialed in to this VMR. For more information see Managing and restricting call bandwidth. You can also allow the auto-generated maximum inbound and outbound call bandwidth to be manually overridden for each VMR.
Participant limit Allow participant limit to be manually overridden	This optional field allows you to limit the number of participants allowed to join this VMR. For more information see Limiting the number of participants. You can also allow the auto-generated participant limit to be manually overridden for each VMR.
Service tag pattern Allow service tag to be manually overridden	This optional field lets you assign a unique identifier to this VMR, which you can then use to track use of the service. You can also allow the auto-generated service tag to be manually overridden for each VMR.
Conference capabilities Allow conference capabilities to be manually overridden	Allows you to limit the media content of the conference. For more information, see Controlling media capability. Default: ***Main video + presentation***. You can also allow the auto-generated conference capabilities to be manually overridden for each VMR.
Maximum call quality Allow maximum call quality to be manually overridden	Controls the maximum call quality for participants connecting to this service: ○ ***Use global setting***: use the global maximum call quality setting. ○ ***SD***: each participant is limited to SD quality. ○ ***HD***: each participant is limited to HD (720p) quality. ○ ***Full HD (1080p)***: allows any endpoint capable of Full HD to send and receive its main video at 1080p. Default: ***Use global setting***
Media encryption Allow media encryption to be manually overridden	Controls the media encryption requirements for participants connecting to this service. ○ ***Use global setting***: use the global media encryption setting. ○ ***Best effort***: each participant will use media encryption if their device supports it, otherwise the connection will be unencrypted. ○ ***Required***: all participants (including RTMP participants) must use media encryption. ○ ***No encryption***: all H.323, SIP and MS-SIP participants must use unencrypted media. (RTMP participants will use encryption if their device supports it, otherwise the connection will be unencrypted.) Default: ***Use global setting*** You can also allow the auto-generated media encryption setting to be manually overridden for each VMR.
User basic options (shown when Sync Users is selected)	

Option	Description
User description pattern Allow user description to be manually overridden	The pattern to use to generate the user description. This field is optional. Examples: **The user for {{displayName}}** You could use a more advanced pattern such as: **The user for {{displayName}} {% if title or department%}({% if title %}{{title}}, {%endif%} {%if department %}{{department}}{% endif %}){% endif %}** which will include the user's job title and department name if they are specified in the LDAP source. You can also allow the auto-generated user description to be manually overridden for each user.
User name options (shown when Sync Users is selected)	
First name pattern	The pattern to use to generate the first name of the user. Default: **{% if givenName %}{{givenName}}{% endif %}**
Last name pattern	The pattern to use to generate the last name of the user. Default: **{% if sn %}{{sn}}{% endif %}**
Display name pattern	The pattern to use to generate the display name of the user. Example: **{% if displayName %}{{displayName}}{% endif %}** Note that the display name is not currently used in Pexip Infinity (for example, it does not affect participant name overlays and cannot be used when provisioning Pexip apps).
Allow user names to be manually overridden	Allows the auto-generated user names to be manually overridden for each user.
User contact options (shown when Sync Users is selected)	
Telephone number pattern	The pattern to use to generate the telephone number of the user. Default: **{{telephoneNumber|pex_clean_phone_number}}**
Mobile number pattern	The pattern to use to generate the mobile number of the user. Default: **{{mobile|pex_clean_phone_number}}**
Allow user contacts to be manually overridden	Allows the auto-generated user contact information (telephone and mobile numbers) to be manually overridden for each user.
User other options (shown when Sync Users is selected)	
Title pattern	The pattern to use to generate the title of the user. Default: **{% if title %}{{title}}{% endif %}**
Department pattern	The pattern to use to generate the department of the user. Default: **{% if department %}{{department}}{% endif %}**

Option	Description
Avatar URL pattern	The pattern to use to generate the avatar URL of the user. Note that: ○ The avatar URL must be an unprotected resource (username and password credentials cannot be supplied with the request), and it must be reachable from Conferencing Nodes. ○ The image retrieved from the avatar URL must be a JPEG. ○ When a Conferencing Node sends an image request to the avatar URL, Pexip Infinity adds on extra URL parameters that specify the required dimensions, for example **?width=100&height=100&s=100** (the **s** is a size parameter used by Gravatar). Pexip Infinity only ever requests square images. ○ The Pexip app requests a 460x460 image size. ○ All JPEG images must use the RGB or RGBA color space (CMYK is not supported), and be of the requested size (width, height). See Applying user records for more information. Default: **https://www.gravatar.com/avatar/{{mail|trim|lower|pex_md5}}?d=404**
Allow the user's other personal details to be manually overridden	Allows the auto-generated user personal information (title, department and avatar URL) to be manually overridden for each user.
User advanced options (shown when Sync users is selected)	
UUID pattern	The pattern to use to generate the UUID of the user. This field is required and must be unique for each user and must be in a UUID format. Therefore it is strongly recommended to use {{objectGUID|pex_to_uuid}} as the value of this pattern. Default: **{{objectGUID|pex_to_uuid}}**
Exchange mailbox UUID pattern	The pattern to use to generate the Exchange Mailbox UUID of the user. This field is not required but if included it must be in a UUID format and be unique for each user. Default: **{% if msExchMailboxGuid %}{{msExchMailboxGuid|pex_to_uuid}}{% endif %}**
Allow user advanced options to be manually overridden	Allows the auto-generated user advanced options to be manually overridden for each user.
Email options	
Send emails VMR / device owner's email address Allow email address to be manually overridden SMTP server	These email options allow you to send an email to the VMR owner whenever a synchronization creates a new VMR or modifies an existing VMR, or when a synchronization creates a new device alias or modifies an existing device alias. The **VMR/Device owner's email address**: ○ Defaults to **{{mail}}** ○ Is used as the email address of the user records that are created when syncing users. ○ When using the VMR self-service portal, it determines which VMRs can be viewed and edited (the LDAP **mail** attribute of the user logged into the VMR portal must match the **VMR owner's email address**). See Sending provisioning emails to VMR and device owners for more information about these options.
VMR email options	
VMR email subject VMR email template	Templates for the subject line and body of the email to be sent when a VMR is created or updated. See Sending provisioning emails to VMR and device owners for more information about these options.
Device email options	
Device email subject Device email template	Templates for the subject line and body of the email to be sent when a device alias is created or updated. See Sending provisioning emails to VMR and device owners for more information about these options.

3. Select **Save**.

   You must save the template before you can use it to generate VMRs.

Allowing generated fields to be manually overridden

Many of the settings can be tagged as "overridable" which means if that setting is manually overridden after the VMR, device or user has been created, a subsequent resynchronization of that template will not reset it back to another value.

For example, you could configure the template with **Allow PIN settings to be manually overridden** selected. Then, after generating a VMR with a random Host PIN, the administrator could manually change the PIN to another specific value. This "overridable" setting ensures that the new value is protected and is not reset to another random value when that template is next used to synchronize the VMRs.

If you make a field overridable you do not have to wait for another sync before making your override changes to the relevant field. To make a field modifiable via the template again, you must clear the relevant **Allow <field> to be manually overridden** template setting and then re-sync with that template. Note that a setting tagged as overridable via one template could still be modified by a different template.

All of the override options are disabled by default.

LDAP search and filter examples

You can configure multiple synchronization templates that all use the same LDAP data source (or different LDAP sources, if required), but that apply different filters to that LDAP source. This means that you could apply different VMR configuration patterns based on different LDAP organizational groups. For example, you could configure the VMRs for all members of the European sales team to have a Guest PIN and a different theme to those VMRs generated for the American accounting department.

If you want to apply different VMR patterns to different types of LDAP directory users, you can use the **LDAP user search DN** and **LDAP user filter** template fields to filter the records from the LDAP sync source that are used to generate VMRs via that template.

This section contains some examples that you could use in a Windows Active Directory environment. Note that all LDAP user search and user filter contents are not case sensitive.

Filtering based on group membership

To import users that belong to a specific group, you can filter on the **memberOf** AD attribute. For example, to only import users who are members of the cn=sales,ou=groups,dc=example,dc=com group, you would set the **LDAP user filter** on your template to:
(&(objectCategory=person)(objectClass=user) (memberOf=cn=sales,ou=groups,dc=example,dc=com))

You could then configure a second template for the accounting department that uses the same **LDAP sync source** but with the **LDAP user filter** set to:
(&(objectCategory=person)(objectClass=user) (memberOf=cn=accounting,ou=groups,dc=example,dc=com))

To only import users who are **not** a member of the sales group, you would set the **LDAP user filter** to:
(&(objectCategory=person)(objectClass=user) (!(memberOf=cn=sales,ou=groups,dc=example,dc=com)))

To import users that belong to a specific group and any other group **nested** within that group, you can use the memberOf:1.2.840.113556.1.4.1941 filter for users. For example, to import users who are members of the cn=sales,ou=groups,dc=example,dc=com group or are members of a group nested within the sales group, you would set the **LDAP user filter** on your template to:
(&(ObjectCategory=user)(memberOf:1.2.840.113556.1.4.1941:=cn=sales,ou=groups,dc=example,dc=com))

Filtering by LDAP field names

To import only those users who have an email address, you can set the **LDAP user filter** to:
mail=*

To import all users **except** for some named individuals, you could use the following **LDAP user filter** model:
(&(objectCategory=person)(objectClass=user) (!(cn=Alice Parkes))(!(cn=Bob Lee))(!(cn=Carol Jones)))

Filtering by organizational unit (e.g. region/country)

To import users that are located in a specific organizational unit such as a region or country, you can restrict the user search to the appropriate 'ou' objects. For example, to only import users who are based in 'Europe', you would set the **LDAP user search DN** to:
ou=europe,ou=people
(this is the DN relative to the base DN defined in the LDAP sync source)

Escaping special characters in search/filter strings

If your search/filter strings include the following special characters: `()*\` you must use a single `\` to escape those characters.

For example if you have a group **cn=group-with-parenthese(s)-in-name,ou=groups,dc=example,dc=com** then you would set the **memberOf** filter to:
(memberOf=CN=group-with-parenthese\(s\)-in-name,ou=groups,dc=example,dc=com))

More information on Active Directory LDAP filtering can be found at https://social.technet.microsoft.com/wiki/contents/articles/5392.active-directory-ldap-syntax-filters.aspx.

Generating / syncing VMRs, devices and users with LDAP

After you have configured a sync source and a template you can generate your VMRs, devices and users for the first time, or you can synchronize them to any changes that have occurred in the LDAP database since the last time the VMRs, devices and users were updated.

When synchronizing a template:

- Pexip Infinity checks all of the VMRs, devices and user records previously created by that template and it updates, deletes or creates new records as appropriate based on the current data in the sync source. Therefore, for example, if a user is removed from Active Directory or their account is disabled, then their corresponding VMR, device alias and user record will be deleted.
- If a generated VMR name, device alias or user email address already exists (either created manually or via a different template), or if a generated alias for a VMR is already assigned to another VMR, those existing VMR properties, VMR-to-alias assignments, device aliases or user records are not overwritten.

Any configuration changes made to Virtual Meeting Rooms are replicated to all Conferencing Nodes (typically after approximately one minute) and applied to any subsequent conferences in that Virtual Meeting Room. If there are any conferences already running that use the Virtual Meeting Room, any attempts to join it after the configuration has been replicated may be affected by the new configuration settings.

Changes to device aliases are also replicated to all Conferencing Nodes (again, typically after approximately one minute) and are applied to any subsequent registration requests.

The template synchronization process itself can also take several minutes if you have a large number of VMRs, devices or users. Therefore you must wait for at least one minute after the entire synchronization process has completed to ensure that all synced data has been replicated to all Conferencing Nodes.

For these reasons, we do not recommend changing Virtual Meeting Room configuration while a conference is in progress as this may lead to an inconsistent user experience.

Automatic synchronization

To automatically synchronize a template on a daily basis:

1. Go to **Utilities > LDAP Sync Templates**.
2. Select the template you want to use (to open the **Change LDAP Sync Template** page).
3. Select **Enable automatic daily sync**.
4. Select **Save**.

As template synchronization can result in the automatic creation, modification or deletion of large numbers of VMRs, devices and users, we recommend that you only enable automatic syncing after you have manually synced at least once and have verified that you are satisfied with your sync template configuration.

All automatic synchronizations are initiated at 01:00 UTC (this start time cannot be configured). After an initial sync, which can take several minutes in a large organization, an ongoing daily sync is typically much faster as it only processes changes since the previous sync.

Manual synchronization

To manually generate (for the first time) or synchronize the VMRs, devices and aliases in Pexip Infinity with the LDAP data source:

1. Go to **Utilities > LDAP Sync Templates**.
2. Select the template you want to use (to open the **Change LDAP Sync Template** page).
3. Scroll down to the bottom of the page and select **Sync now**.
4. A status page is displayed which shows the progress of the synchronization.
   You can safely navigate away from this page without affecting the synchronization.
5. For large imports that take a long time to complete, you can monitor the ongoing status of the import at any time via **Status > LDAP Sync**.

Alternatively, you can sync one or more templates by going to **Utilities > LDAP Sync Templates**, selecting the check box next to the templates that you want to use, and then from the **Action** drop-down menu, select ***Sync selected LDAP sync templates*** and then select **Go**.

Deleting all VMRs, devices and users associated with a template

You can easily delete all of the VMRs, devices and users that have been created from a specific template, for example if you have incorrectly specified the template's **LDAP user filter**.

To delete all VMRs, devices and user records associated with a synchronization template:

1. Go to **Utilities > LDAP Sync Templates**.
2. Select the template whose associated VMRs, devices and users you want to delete (to open the **Change LDAP Sync Template** page).
3. Scroll down to the bottom of the page and select **Delete all VMRs, devices and users created from this template**.
4. Confirm that you want to proceed.
5. The VMRs, devices and users will be deleted and a status message is displayed confirming the number of records that were deleted.

Deleting a template

If you delete a synchronization template, all of the VMRs, devices and users associated with that template will also be deleted.

Applying user records

User records can be used to support the use of the following features within Pexip Infinity:

- Participant avatars
- Scheduling meetings in personal VMRs
- Controlling permissions in the VMR portal

User records can be created manually (**Users & Devices > Users**), or they can be generated from directory information contained in an AD/LDAP server via Pexip's provisioning mechanism (**Utilities > LDAP Sync Templates**).

User records and participant avatars

Conference participants and directory contacts within Pexip Infinity can be represented by an avatar or image.

You can configure user records to represent those participants/contacts and associate each user with an avatar URL that points to an external service (such as Gravatar) which can be used to retrieve that user's avatar/image. Then, when that user is participating in a conference, their avatar can be shown to any of the other conference participants who are using a Pexip app, or displayed to all participants within the main video layout if avatars are enabled in the conference theme.

When using avatars, in addition to creating the user record, you must also set up and enable an **Avatar policy** (**Call Control > Policy Profiles**).

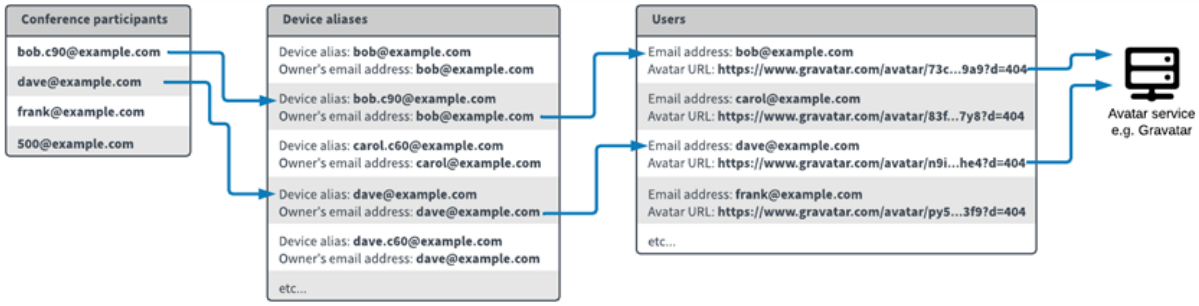
For a participant's avatar to be requested and available for display in the Pexip app (in the participant list tab), that participant's device must be registered to Pexip Infinity (any device type or protocol).

Setting up user records and avatar URLs

When configuring end users and their associated avatars, there are two main attributes of each user record to consider:

- **Email address**: the user's **Email address** is used as the primary identifier of each user record. When attempting to retrieve a user's avatar, the system locates the relevant user record (and thus the user's avatar URL) by finding the user **Email address** that matches the **Owner's email address** associated with the device alias of the conference participant. We recommend using LDAP sync templates to provision the device and user records as this will ensure that a matching email address is used.
- **Avatar URL**: this is the link to where the avatar can be requested:
  - The avatar URL must be an unprotected resource (username and password credentials cannot be supplied with the request), and it must be reachable from Conferencing Nodes.
  - The image retrieved from the avatar URL must be a JPEG.
  - When a Conferencing Node sends an image request to the avatar URL, Pexip Infinity adds on extra URL parameters that specify the required dimensions, for example **?width=100&height=100&s=100** (the **s** is a size parameter used by Gravatar). Pexip Infinity only ever requests square images.
  - The Pexip app requests a 460x460 image size.
  - All JPEG images must use the RGB or RGBA color space (CMYK is not supported), and be of the requested size (width, height).

The following example diagram shows the relationship between participant aliases, device aliases and user records when obtaining an avatar URL:



To configure user records and avatar URLs:

1. Set up a policy profile with local avatar configuration enabled:
   a. Go to **Call Control > Policy Profiles** and create a profile (or modify an existing profile).
   b. In the **Avatar policy** section, enable **Use local avatar configuration**.
   c. Save the policy profile.
   d. Assign the policy profile to your locations (**Platform > Locations**).
2. Configure user records and their associated avatar URLs:
   - You can bulk provision the user records from directory information contained in a Windows Active Directory LDAP server, or any other LDAP-accessible database, or
   - You can manually configure user records (see below).

Note that other user attributes can also be configured (such as names and contact numbers) but these are not currently used within Pexip Infinity.

User records and personal VMR scheduling

Pexip Infinity's Secure Scheduler for Exchange feature can offer users the ability to schedule meetings in their own personal VMRs. This feature requires users to sign in to an Outlook add-in using their SSO email address, and requires Pexip Infinity to be configured with both:

- a user record with an **Email address** that matches the user's SSO email address (usually the user record will already exist, but you can configure this feature to create the required user record if required, in which case the User origin displays the name of the associated Secure Scheduler for Exchange Integration), and
- at least one VMR with an **Owner's email address** that matches the user's SSO email address.

Controlling permissions in the VMR portal

User records are also used to control web app branding and VMR management permissions in the VMR portal.

Manually configuring user records

To manually configure the user records, go to **Users & Devices > Users**. The full set of options are:

Option	Description
Email address	The email address of the user. If you wish to use Personal VMRs, this must be the user's SSO email address.
Description	The description of the user.
First name	The first name of the user.
Last name	The last name of the user.
Display name	The display name of the user. Note that the display name is not currently used in Pexip Infinity (for example, it does not affect participant name overlays and cannot be used when provisioning Pexip apps).
Telephone number	The telephone number of the user.
Mobile number	The mobile number of the user.
Title	The job title of the user.
Department	The department of the user.
Avatar URL	The avatar URL of the user.
Advanced options	
UUID	The unique identifier (UUID) of the user. This field is required and must be unique for each user and must be in a UUID format. Therefore it is strongly recommended to use the generated default value.
Exchange mailbox ID	The unique identifier of the user's Exchange Mailbox. This field is not required but if included it must be in a UUID format and be unique for each user. Normally, this field is only populated when LDAP sync is enabled for customers with Exchange on-premises environments.
Exchange User ID	The unique identifier of the user on Exchange on-premises. This field is read-only and appears for Secure Scheduler for Exchange users who have scheduled a meeting in their personal VMR.
Microsoft 365 User ID	The user's Microsoft 365 Object ID, which is the unique identifier of the user on Microsoft 365 services. This field is read-only and appears for Secure Scheduler for Exchange users who have scheduled a meeting in their personal VMR.
User origin	• If the user was provisioned from Active Directory, this is the name of the LDAP sync template used to create this user. • If the user was created as part of scheduling meetings in personal VMRs, this is the name of the Secure Scheduler for Exchange Integration that created the user. • If the user was created by manual input or via the API, this will be blank. This field is read-only.

Using templates, variables and filters when provisioning VMRs, devices and users

When you configure an LDAP synchronization template to provision VMRs, devices and users, you define patterns for how some of those properties — such as the VMR name or a device alias — are generated from the data in the LDAP sync source.

Pexip Infinity's LDAP sync templates use a subset of the jinja2 templating language (see Template Designer Documentation).

These templates consist of the following elements:

- **literal text** that you want to add to the output or result, such as prefixing every generated VMR name with **meet.**
- **variables** that are substituted with values from the LDAP sync source, such as **givenName**
- **filters** that can manipulate text or modify the content of variables or text strings, such as **join** or **lower**
- **delimiters** such as **{{...}}** and pipes **|** which are used to enclose variables and define filter expressions
- **jinja statements and control structures** (see Jinja Control Structures)

An example pattern would be **meet.{{givenName|lower}}.{{sn|lower}}**. This concatenates the literal text **meet.** with the content of the **givenName** variable that has been converted to lower case by the **lower** filter. This is then concatenated with a period **.** and then the **sn** (surname) variable, also piped through the **lower** filter.

Therefore if Alice Parkes had an entry in the LDAP directory with **givenName=Alice** and **sn=Parkes**, the example pattern above would produce **meet.alice.parkes**.

The following sections provide more information about the supported variables, jinja2 filters and custom Pexip filters, with each section including example expressions that demonstrate how to format your patterns.

Supported variables

When syncing each VMR, device or user record, Pexip Infinity extracts LDAP fields from the directory records in the LDAP data source and makes them available as variables (with the same name) for use in the sync template. Pexip Infinity offers several standard fields, and administrators can also add their own custom set of fields. Fields with multiple values are also supported.

When using variables:

- You must enclose the variable name within **{{...}}** delimiters, for example {{givenName}}.
- Any combination of the variables can be used in any synchronization template pattern field.
- Fields that are not present in the LDAP source will appear as "None".
- Capitalization of the variable name is important. The variable **must** be spelled exactly as shown below.

Standard LDAP fields

Pexip Infinity automatically extracts and makes available the following standard LDAP fields:

Variable name	Description
company	Company/organization name
department	Department name
displayName	User's preferred display name
employeeID *	Employee reference number
givenName	First name
mail	Email address
mailNickname *	Typically used to store an alternative email address (e.g. based on a maiden name); see Change of name below
mobile	Mobile phone number
objectGUID	Globally unique identifier for the directory object (user entry)
sAMAccountName	Logon name (domainname\username format)

Variable name	Description
sn	Last name or surname
telephoneNumber	Telephone number
title	Job title
userPrincipalName	Logon name (username@domainname format)

* Not present in all AD schemas.

Custom LDAP fields

In addition to the standard set of fields made available automatically by Pexip Infinity, administrators can configure their own custom set of fields to support additional attributes in their LDAP/AD schemas.

To add a custom LDAP field:

1. Go to **Utilities > LDAP Sync Fields**.
2. Select **Add LDAP sync field**, and then complete the following fields:

LDAP field name	The name of LDAP field to be read from the LDAP server. The name is case sensitive. If the name does not match exactly against a field in the LDAP data source its value will appear (when used in a template) as "None".
Template variable name	The name of the variable to use in sync templates that will contain the value of the LDAP field. We recommend that you set this name to something similar to the LDAP field name, but note that this name cannot contain hyphens.
Description	An optional description of the LDAP attribute.
Is binary (advanced options)	In advanced scenarios, some binary LDAP fields such as GUIDs require special encoding. In such cases, expand the advanced options and select **Is binary**. Do not select this option for ordinary textual or numeric LDAP fields.

3. Select **Save**.

   The **Template variable name** that represents the custom LDAP field can now be used in a template along with the standard LDAP fields.

LDAP fields with multiple values

In some scenarios, an LDAP field could contain multiple values for a given user. For example the LDAP **proxyAddresses** field (which you would have to configure as a custom LDAP field as described above) could contain many addresses.

In these cases, a special **multi_valued_attrs** variable is automatically created. This variable will contain **all** of the LDAP fields that are found to have multiple values for a given user. This means if two fields have multiple values, the **multi_valued_attrs** variable will contain all of the values for those two fields. It will not contain any of the fields that have just a single value.

You can then use the **pex_find_first_match** filter to extract values from the variable. This filter can be used to extract from a string list the first value that matches a specified regex.

Example usage: **{{pex_find_first_match(multi_valued_attrs.proxyAddresses, "^sip:.*")}}**

This would extract the first **proxyAddresses** value that starts with "sip:" i.e. the first proxy that has a SIP-style address.

However, as the **multi_valued_attrs** variable is only populated with LDAP fields that contain multiple values, a better expression in this case would be: **{{pex_find_first_match(multi_valued_attrs.proxyAddresses, "^sip:.*") or proxyAddresses}}** which will return the first/only value in the standard **proxyAddresses** variable if only one value is available or the regex does not find a match.

More complex matching is possible by iterating over the **multi_valued_attrs** collection in other ways.

Change of name

A typical scenario encountered by IT administrators is when someone changes their name (e.g. after getting married) and wants to preserve their original contact address alongside their new contact details.

One way in which you could support this in Pexip Infinity is by defining multiple VMR alias patterns in your template. This example approach assumes that the LDAP **mail** field holds the user's contact address, and a **mailNickname** field is used to hold an alternative address. Thus, you could define the following patterns:

VMR alias 1 pattern:	meet.{{mail}}
VMR alias 2 pattern:	{%if mailNickname %}{#Add alias based on user's maiden name mailNickname#}meet.{{mailNickname}}{% else %}{#intentionally leave the alias blank#}{% endif %}

For most users, this will generate a single VMR alias in the format **meet.<email address>**, and the VMR's second alias will remain blank as the LDAP **mailNickname** field for those users will be empty.

For example, user Ann Jones has her LDAP **mail** field set to ann.jones@example.com and her LDAP **mailNickname** field is blank. Her VMR will have a single alias of **meet.ann.jones@example.com**.

When a user changes their name, the administrator would update the user's LDAP **mail** field to match their new name and populate the **mailNickname** field with the previous value of the **mail** field.

Now, the next time the administrator performs a template synchronization, the user's VMR aliases will be updated. The first alias will be changed to a new string based on the user's new email address, and a second alias will also be generated based upon the **mailNickname** field. This means that the user can be contacted via either their previous VMR alias or the new alias.

In our example, let's assume that Ann Jones changes her name to Ann Smith. The administrator changes her LDAP **mail** field to ann.smith@example.com and sets her LDAP **mailNickname** field to ann.jones@example.com. Her VMR will now have two aliases: **meet.ann.smith@example.com** and **meet.ann.jones@example.com**.

Supported jinja2 filters

Pexip Infinity supports a subset of filters from the jinja2 templating language. Any jinja filters that are not listed below have been disabled in Pexip Infinity. See List of Builtin Filters for more information about these filters.

abs	float	last	replace	truncate
capitalize	format	length	round	upper
default	int	lower	striptags	
first	join	range	trim	

To use a filter you would typically follow the syntax **{{<source_value>|<filter_name>}}**.

In most cases the <source_value> is likely to be a variable, for example **{{givenName|upper}}**, although it could be one or more literal values, for example **{{ [1, 2, 3, 4]|join }}**.

Some filters take parameters, for example **{{sn|truncate(5)}}**. You can also use multiple filters in the same expression, for example **{{sn|truncate(5)|upper}}**.

The **trim** filter is often used. This trims leading and trailing whitespace from the string held in the <source_value>.

Example usage: **{{title|trim}}**

If the **title** field contained " Project Manager ", this would be converted to "Project Manager".

The **replace** filter is also often used. This replaces one string with another string. The first argument of the filter is the substring that should be replaced, the second is the replacement string. If the optional third argument count is given, only the first count occurrences are replaced.

Example usage: **{{ department|replace("Personnel", "HR") }}**

If the **department** field contained "Personnel Department Personnel", this would be converted to "HR Department HR".

Example usage: **{{ department|replace("Personnel", "HR", 1) }}**

In this case a count of 1 is specified, thus if the **department** field contained "Personnel Department Personnel", it would be converted to "HR Department Personnel".

For more complicated search and replace patterns, use the custom Pexip **pex_regex_replace** filter described below.

Custom Pexip filters

In addition to the jinja filters, Pexip also provides the following custom filters, which are typically used to manipulate data:

Filter	Description and example usage
pex_base64	Performs Base64 encoding on the input field. Example usage: **{{provisiondata|pex_base64}}** In this example, a local variable named **provisiondata** is encoded as Base64.
pex_clean_phone_number	This extracts only +0123456789 characters (and removes ()&%#@|"':;, A-Z,a-z etc). Example usage: **{{ telephoneNumber|pex_clean_phone_number }}** In this example, if telephoneNumber is '+44 (20) 12345678', this expression would return '+442012345678'.
pex_debug_log (message)	The **pex_debug_log** filter can be used to help debug your script. It writes debug messages to the Pexip Infinity support log. You can include literal text and variables. To avoid filling the support log and causing it to rotate, remove all **pex_debug_log** filters from your scripts as soon as they are working correctly.
pex_find_first_match (string_list, 'find_regex')	This extracts from the list the first value that matches the specified regex. It is typically used to extract a value from the **multi_valued_attrs**variable where an LDAP field, such as **proxyAddresses**, contains multiple values. Example usage: **{{pex_find_first_match(multi_valued_attrs.proxyAddresses, "^sip:.*")}}** This would extract the first **proxyAddresses** value that starts with "sip:" i.e. the first proxy that has a SIP-style address.
pex_hash	Performs a hash of a field. You could use this, for example, to generate random-looking PINs based on a hash of one or more fields from the directory (you would also need to apply another filter such as pex_tail to set the PIN to an appropriate valid length). Example usage: **{{ sAMAccountName|pex_hash|pex_tail(6) }}**
pex_head (maxlength)	Returns, at most, the first maxlength characters from the input field. Example usage: **{{ givenName|pex_head(4) }}** In this example, for a givenName of 'Alice' this expression would return 'Alic', and a givenName of 'Bob' would return 'Bob'.
pex_in_subnet	Tests whether a given address is within one or more subnets. It takes as input the address you want to test, and one or more subnet ranges, and returns either True or False.
pex_md5	Applies an MD5 hash to the input field. Example usage: **https://www.gravatar.com/avatar/{{mail|trim|lower|pex_md5}}?d=404** This example generates a URL that conforms with how the Gravatar service constructs its URLs based on users' email addresses.

Filter	Description and example usage
pex_now (timezone)	The **pex_now** filter takes an optional parameter of a timezone description e.g. 'UTC', 'Asia/Tokyo', or 'US/Eastern' and returns the current date and time for that timezone. UTC is assumed if a timezone is not provided. The resulting available attributes are **year**, **month**, **day**, **hour**, **minute**, **second** and **microsecond**. Example usage: **{% set now = pex_now("Europe/London") %}** **{% if now.month == 2 and now.day == 29 %}**
pex_random_pin (length)	Generates a random PIN of the given length. Note that this filter does not take any input. Example usage: **{{ [9,pex_random_pin(5)]|join }}** for the Host PIN, and **{{ [2,pex_random_pin(5)]|join }}** for the Guest PIN. A usage pattern such as this (where the Host PIN starts with a 9, and the Guest PIN starts with a 2) ensures that a VMR can never be configured with a Host PIN that is the same as the Guest PIN. You would typically use this filter in conjunction with the **Allow PIN settings to be manually overridden** option to ensure that the PIN is not reset every time a template resync is performed. Do not use pex_random_pin() to generate aliases. This filter generates a truly random number, with each number generated independently of any previous output. Therefore, with many thousands of users, a 5 digit numeric alias (e.g. pex_random_pin (5)) is statistically quite likely to clash with a random alias generated using pex_random_pin(5) for another user. With 10,001 or more users and a 4 digit random alias, a clash is guaranteed (with multiple clashes being likely).
pex_regex_replace ('find_regex', 'replace_string')	This performs a regex find and replace. Example usage: **{{mail|pex_regex_replace('@.+','@otherdomain.com')}}** This example takes as input an email address contained in the **mail** variable and changes the domain portion of the address to @otherdomain.com. For example, it will transform **user1@domainA.com** to **user1@otherdomain.com**, and **user2@domainB.com** to **user2@otherdomain.com** etc.
pex_regex_search ('regex pattern', 'string_to_search')	This performs a regex search for the first location that matches the pattern and returns the regex groups. Example usage: **{% set groups = pex_regex_search("([a-z0-9.-]+)@([a-z0-9.-]+.com)", "example string with someone@example.com") %}** **{% if groups %}** **{{ groups[0] }}@{{ groups[1] }}** **{% endif %}** This example takes as input a string containing an email address and extracts the email using two regex groups.
pex_require_min_length (length)	This validates that the input string field has the specified minimum length. Syntax: {{ some_string|pex_require_min_length(2) }} You can use this filter to control if VMRs are created or not, based on the length of a string or variable. For example, you could set the VMR description pattern as **{{ mail|pex_require_min_length(1) }}**. This means that the VMR will not be created if the **mail** variable is empty. (When subsequently syncing, the VMR would be deleted if the condition is not met.) Example usage: **{{ telephoneNumber|pex_require_min_length(4) }}** This example ensures that a VMR is created only if **telephoneNumber** is at least 4 characters long.
pex_reverse	This reverses the characters in the input field. Example usage: **{{ givenName|pex_reverse|lower }}** In this example, if givenName is 'Alice', this expression would return 'ecila'.

Filter	Description and example usage
pex_safelinks_decode	This filter converts a string that has been rewritten by Safe Links in Microsoft Defender for Office 365 into the original URL. The syntax is: {{ pex_safelinks_decode('URL_to_decode') }} Example usage: **{{ pex_safelinks_decode ('https://nam10.safelinks.protection.outlook.com/?url=https%3A%2F%2Fpexip.me%2Fmeet%2F16571366&data=05%7C01%7Ctest%40pexip.com%7CUnknown%7C3000%7C%7C%7C;reserved=0') }}** would produce: **https://pexip.me/meet/16571366&data=05|01|test@pexip.com|Unknown|3000|||;reserved=0**

pex_strlen	Returns the length of string. The basic usage syntax is:
	`{% set some_length = "Example"|pex_strlen %}`
	`{# sets a variable named some_length to the value 7 #}`
	Example usage: **{%set a = telephoneNumber|pex_clean_phone_number|pex_head(8)%}{%set b = mobile|pex_clean_phone_number|pex_head(8)%}{%if a|pex_strlen== 8%}{{a}}{%else%}{{b}}{%endif%}**
	In this example, the expression returns the value of the telephoneNumber field if it is present and at least 8 characters long after cleaning and truncation, otherwise it returns the value of the mobile field that has been cleaned and truncated to at most 8 characters.
	Example usage: **{%if telephoneNumber|pex_strlen == 5 %}{#return user's phone number#}{{telephoneNumber|pex_clean_phone_number}}{% else %}{#intentionally leave blank#}{% endif %}**
	In this example, the expression returns a cleaned value of the telephoneNumber field if it is 5 characters long, otherwise it returns an empty string.

pex_tail (maxlength)	Returns, at most, the last maxlength characters from the input field. Example usage: **{{ telephoneNumber|pex_tail(4) }}** In this example, if telephoneNumber is '+44 (20) 12345678', this expression would return '5678'.
pex_to_json	Converts a Python dictionary variable into JSON format.
pex_to_uuid	Converts a base64 string to a UUID. GUIDs that are retrieved from LDAP are encoded as base64 and typically need converting to a human readable UUID such as a59da36d-9b16-430a-80f7-cb1b01d4bd45 for subsequent use within Pexip Infinity. Example usage: **{{objectGUID|pex_to_uuid}}**
pex_update	Updates Python dictionary variables.

Filter	Description and example usage
pex_url_decode	This filter decodes a previously URL-encoded string. For example, it can be used with OTJ custom rules to simplify the regular expressions required. It converts any percent-encoded characters within a string into the reserved characters they represent. The syntax is: {{ some_val_to_decode | pex_url_decode }} Example usage: **{{https%3A%2F%2Fteams.microsoft.com%2FmeetingOptions%2F%3ForganizerId%3Db9080d25-fc86-49da-bdfc-8ade02c06d5c%26amp%3BtenantId%3Dd08b6073-7208-49d3-8155-fc16f147990a%26amp%3BthreadId%3D19_meeting_YWI3MTB1NmQtZAJmOS00ZjEyLTlhNjUtMTJmMmU4MTVjMTVk%40thread.v2%26amp%3BmessageId%3D0%26amp%3Blanguage%3Den-GB | pex_url_decode}}** would produce: **https://teams.microsoft.com/meetingOptions/?organizerId=b9080d25-fc86-49da-bdfc-8ade02c06d5c&tenantId=d08b6073-7208-49d3-8155-fc16f147990a&threadId=19_meeting_YWI3MTB1NmQtZAJmOS00ZjEyLTlhNjUtMTJmMmU4MTVjMTVk@thread.v2&messageId=0&language=en-GB**
pex_url_encode	This filter creates URL parameters that are safely URL-encoded. It takes any number of two-element tuples and converts them into a percent-encoded string of key=value pairs. It does not take any input; the syntax is: {{ pex_url_encode(('key_1', 'data_1'), ('key_2', 'data_2'), ..., ('key_n', 'value_n')) }} Example usage: **{{ pex_url_encode(('data', '8J+UpcKvXF8o44OEKV8vwq/wn5Sl'), ('message', 'A test message')) }}** would produce: **data=8J%2BUpcKvXF8o44OEKV8vwq%2Fwn5Sl&message=A+test+message**
pex_urldefense_decode	This filter converts a string that has been rewritten by Proofpoint's URL Defense into the original URL. The syntax is: {{ pex_urldefense_decode('URL_to_decode') }} Example usage: **{{ pex_urldefense_decode('https://urldefense.proofpoint.com/v2/url?u=https-3A__www.microsoft.com_microsoft-2Dteams_join-2Da-2Dmeeting&d=DwMFAw&c=euGZstcaTDllvimEN8b7jXrwqOf-v5A_CdpgnVfiiMM&r=sSO2li5AaOA0GiOJlvT_QWecc2Am_scbeG3PEUJ41Ws&m=0XEC91GMwTHvPxUsjSw-B5Q0Sd686oZ7HwAeEHjKMfg&s=G0pifb_3at7c5yOf4aFOElh8NybdOgM4400EXtjFAM0&e=') }}** would produce: **https://www.microsoft.com/microsoft-teams/join-a-meeting**
pex_uuid4()	This generates a uuid (universally unique identifier). Note that this filter does not take any input. Example usage: **{{ pex_uuid4() }}** In a similar manner to the **pex_random_pin** filter, you would typically use this in conjunction with the appropriate **Allow <field> to be manually overridden** template setting, to ensure that after a uuid has been assigned to a field, another different uuid is not assigned every time a template resync is performed.

Sending provisioning emails to VMR and device owners

When bulk-provisioning VMRs and device aliases from Active Directory via LDAP, an email can be sent out to the VMR owner or device owner telling them the:

- VMR properties, such as its aliases and security PINs
- device alias properties, such as its associated username and password for registration purposes.

Email generation is specified on a per sync template basis. The content of the email is free-form and can be customized for each generated VMR or device alias by using variables — such as {{pin}} to include the Host PIN in VMR-related emails — that are substituted with the relevant value when each individual email is generated.

This topic covers:

- Guidelines and limitations
- Configuring an LDAP sync template to generate emails
- Constructing the VMR provisioning email
- Example VMR email templates
- Constructing the device alias provisioning email
- Example device email templates
- Sending reminder emails

Guidelines and limitations

When constructing your email template:

- Ensure that the subject line of the generated email is a single line.
- You can use HTML markup (UTF-8 characters only).

If Pexip Infinity experiences connectivity issues with the SMTP server, it will retry sending each individual email up to 3 times before moving onto the next email, and it will attempt to re-establish dropped SMTP server connections up to 100 times per bulk-send operation.

Note that an email is triggered when a synchronization process results in changes to the VMR's or device alias's properties, regardless of whether that property is available as an email variable, and whether that variable is actually used in an email template. Changes to the email template do not trigger the sending of an email.

If you want to update all VMRs, for example with a new layout, but do not want to trigger emails, then disable the **Send emails** option for the sync template, make the change to the template, synchronize, and when completed, turn **Send emails** back on again.

Configuring an LDAP sync template to generate emails

When configuring a sync template (**Utilities > LDAP Sync Templates**) you can specify the following email-related options:

Option	Description
Email options	
Send emails	When selected, the system generates and sends an email to the: • VMR owner when a synchronization creates a new VMR or modifies an existing VMR (when **Sync VMRs** is enabled for the template) • device owner when a synchronization creates a new device alias or modifies an existing device alias (when **Sync devices** is enabled for the template). Separate emails are sent for VMR provisioning and for device provisioning when both **Sync VMRs** and **Sync devices** are enabled. Note that you can also manually send reminder emails.
VMR / device owner's email address	The email address of the owner of this VMR or device alias. The generated email(s) will be sent to this address. This field is also used as the email address of the user records that are created when **Sync users** is enabled. When using the VMR self-service portal, it determines which VMRs can be viewed and edited (the LDAP **mail** attribute of the user logged into the VMR portal must match the **VMR owner's email address**). Example: **{{mail}}**
Allow email address to be manually overridden	Allows the auto-generated email address to be manually overridden for each VMR or device alias.
SMTP server	Select the SMTP server to use for sending provisioning emails (see Configuring SMTP servers).

Option	Description
VMR email options	
VMR email subject	A template for the subject line of the email to be sent when a VMR is created or updated.
VMR email template	A template for the body of the email to be sent when a VMR is created or updated. See Constructing the VMR provisioning email below for more information on how complete this field.
Device email options	
Device email subject	A template for the subject line of the email to be sent when a device alias is created or updated.
Device email template	A template for the body of the email to be sent when a device alias is created or updated. See Constructing the device alias provisioning email below for more information on how complete this field.

Constructing the VMR provisioning email

The **VMR email subject template** and **VMR email body template** fields contain the pattern for the email to be sent to the VMR owner when that VMR is first created (provisioned) or is updated.

The templates will typically contain a mixture of literal text and variables, and it can also contain control structures that allow you to vary the content of the email based upon certain conditions.

Supported variables

The **VMR email subject template** and **VMR email body template** fields support a different set of variables from the other synchronization template pattern fields (because it is based on the generated VMR properties, rather than the source data used to build those properties).

The following variables are available:

Variable name	Description
primary_owner_email_address	The email address of the owner of the VMR.
description	The description of the VMR.

Variable name	Description

aliases	A list of tuples containing the VMR alias and its corresponding description e.g. **[("alice", "The short alias"), ("meet.alice@example.com", "The alphanumeric URI alias"), ("123456", "The numeric alias")]**
	
	Thus, in your email template, `aliases[0][0]` will extract the first alias in the **aliases** list, `aliases[0][1]` extracts the description of the first alias in the list, and `aliases[1][0]` extracts the next alias in the list, and so on.
	
	Note that:
	• The order of the aliases contained in the **aliases** variable may not reflect the order of the alias templates in the sync template, or the order that the aliases appear when viewing the VMR via the Administrator interface or via the management API i.e. the alias generated by **VMR alias 1 pattern** may not appear before the alias generated by **VMR alias 2 pattern** and so on.
	• As a VMR alias pattern template could generate a blank alias (which will be discarded when syncing) there is no guarantee that you will have 4 aliases configured even if a sync template has VMR alias 1, 2, 3 and 4 patterns configured.
	
	Extracting a specific alias
	
	There are several ways in which you can extract a specific alias from the **aliases** variable. Let's assume that **aliases** contains the following (in any order):
	
	[("alice", "The short alias"), ("meet.alice@example.com", "The alphanumeric URI alias"), ("123456", "The numeric alias")]
	
	To extract an alias with a particular **description**, you can use this jinja code and text in your VMR email template:
	
	`{% macro _aliases_with_description(description) -%}`
	`{%for alias in aliases%}{%if alias[1]==description%}{{alias[0]}}{%endif%}{%endfor%}`
	`{%- endmacro %}`
	`The telephone number is {{_aliases_with_description("The numeric alias")}}`
	`The alphanumeric URI is {{_aliases_with_description("The alphanumeric URI alias")}}`
	
	and this will result in an email containing:
	
	`The telephone number is 123456`
	`The alphanumeric URI is meet.alice@example.com`
	
	Alternatively, if you want to look at a **start and end pattern** of the alias itself to see if it is the alias you want to use, you can use this jinja code and text in your VMR email template:
	
	`{% macro _aliases_with_start_and_end(aliasstart, aliasend) -%}`
	`{%for alias in aliases%}{%if alias[0].startswith(aliasstart) and alias[0].endswith(aliasend)%}{{alias[0]}}{%endif%}`
	`{%endfor%}`
	`{%- endmacro %}`
	`The URI is {{_aliases_with_start_and_end("meet.", "@example.com")}}`
	
	which in this example only lists those aliases that start with **meet.** and end with **@example.com** i.e. it will result in an email containing:
	
	`The URI is meet.alice@example.com`
	
	Alternatively, you can use the system-generated **uri_alias** and **numeric_alias** variables as described below.

allow_guests	Whether to distinguish between Host and Guest participants: • true — the conference has two types of participants: Hosts and Guests. • false — all participants have Host privileges
pin	The Host PIN.
guest_pin	The Guest PIN.
max_callrate_in	The maximum inbound call bandwidth (kbps).
max_callrate_out	The maximum outbound call bandwidth (kbps).

Variable name	Description
participant_limit	The maximum number of participants allowed to join this VMR.
tag	The VMR's service tag.
Additional generated variables	
action	Possible values are: • "created" — when the VMR has just been created • "updated" — when the existing VMR has been updated • "reminder" — when sending reminder emails.
uri_alias	Set to any one of the configured URI aliases of a VMR. For example, **meet.alice@example.com** is a possible URI alias, as is **123456@example.com**. Thus if a VMR has both of those aliases then either might appear in the **uri_alias** variable. (Note that **123456** is not a URI alias.) If your VMRs have several URI-style aliases, you may want to programmatically select the alias that meets your required pattern from the **aliases** variable as described above. It is blank if there is no URI-style alias.
numeric_alias	Set to any one of the VMR's all-numeric aliases i.e. an alias consisting entirely of digits 0-9. It is blank if there is no numeric alias.

Example VMR email templates

Here are some example email body templates that you can use as the basis for your own emails.

Basic VMR email template

This is a basic email template that can be used to show joining instructions for VMRs that do not have a PIN, and where the default branding path is used.

It makes use of the `{{numeric_alias}}` and `{{uri_alias}}` variables which, when the email is generated, will be substituted with the appropriate alias values for that specific VMR.

You can remove the instructions for any of the joining methods that are not appropriate for your deployment, such as the instructions for joining via a Pexip app or via telephone. If you use this example, remember to replace `node.example.com` with the address of your Conferencing Node, and to use your own direct dial telephone number instead of `+1 555 0123`.

```
Here are the details of your Pexip Virtual Meeting Room (VMR). You can copy and paste these instructions when inviting people to join your
meetings.

From a web browser, go to: https://node.example.com/webapp/m/?conference={{numeric_alias}} and follow the prompts

From a video conferencing endpoint dial {{uri_alias}}

To join from the Pexip for Windows app, click pexip://{{uri_alias}} (you can download it from https://www.pexip.com/apps)

To join from a telephone, dial +1 555 0123 then, when prompted, enter the conference number {{numeric_alias}} followed by "#"

Instructions for using your VMR can be found at https://docs.pexip.com/admin/using_vmr.htm

You can check your connection and the quality of your video and audio - from a web browser, go to:
https://node.example.com/webapp/conference/test_call
```

Constructing the device alias provisioning email

The **Device email subject template** and **Device email body template** fields contains the pattern for the email to be sent to the device owner when that device alias is first created (provisioned) or is updated.

The templates will typically contain a mixture of literal text and variables, and it can also contain control structures that allow you to vary the content of the email based upon certain conditions.

Supported variables

The **Device email subject template** and **Device email body template** fields support a different set of variables from the other synchronization template pattern fields (because it is based on the generated device properties, rather than the source data used to build those properties).

The following variables are available:

Variable name	Description
primary_owner_email_address	The email address of the owner of the device alias.
device_alias	The alias of the device that can be registered to Pexip Infinity.
device_description	The description of the device alias.
device_username	The username associated with the device alias. This should be used in association with the **device_password** when registering the **device_alias** to Pexip Infinity.
device_password	The password associated with the device alias.
device_tag	The device alias's service tag.
Additional generated variables	
action	Possible values are: • "created" — when the device alias has just been created • "updated" — when the existing device alias has been updated • "reminder" — when sending reminder emails.
standard_sso_registration	Whether to include a more detailed description of how to configure a specific type of endpoint for registration. Possible values are: • True — include information for this type of endpoint, e.g. SIP • False — do not include information for this type of endpoint, e.g. H.323
sip_registration	
h323_registration	
infinity_connect_non_sso_registration	

Example device email templates

Here is an example email body template that you can use as the basis for your own emails. It is shown in plain text and in HTML format.

Basic device email template in plain text

This is a basic email template that can be used to provide the information required for registering a device.

It supplies the **device_alias** to register to Pexip Infinity, along with the credentials (**device_username** and **device_password**) that must be used when registering devices other than the Pexip app for Windows.

If you use this example, remember to replace `confnode.example.com` with the address of your Conferencing Node.

```
Hi {{primary_owner_email_address}},
```

```
{%if action=="created" %}Your Pexip app account has just been created.{% else %}Service update: your Pexip app account has been updated with
new settings. {%endif%}

This email contains the settings to use when registering your hardware-based video endpoint.

Video address (alias): {{device_alias}}

Registration server address: confnode.example.com

User name: {{device_username}}

Password: {{device_password}}

When your device is registered, other users will be able to call you using your personal video alias: {{device_alias}}

Instructions for registering your Pexip app can be found at https://docs.pexip.com/apps/using_windows.htm#register
```

Basic device email template in HTML format

This is a repeat of the basic email template above, but illustrates how HTML formatting may be applied.

```
<html>
<p>Hi {{primary_owner_email_address}},</p>
<p>{%if action=="created" %}Your Pexip app account has just been created.{% else %}Service update: your Pexip app account has been updated with
new settings. {%endif%}</p>
<p>This email contains the settings to use for your hardware-based video endpoint.</p>
<p>To configure your Pexip app (or hardware device) you will need to enter the following settings:</p>
<p>Video address (alias): <b>{{device_alias}}</b></p>
<p>Registration server address: <b>confnode.example.com</b></p>
<p>User name: <b>{{device_username}}</b></p>
<p>Password: <b>{{device_password}}</b></p>
<p>When your device is registered, other users will be able to call you using your personal video alias: {{device_alias}}</p>
<p>Instructions for registering your Pexip app can be found on the <a href="https://docs.pexip.com/apps/using_windows.htm#register" target="_
blank">Pexip documentation website</a></p>
</html>
```

Sending reminder emails

You can manually send reminder emails for a specific VMR or device alias, or for all VMRs and devices associated with a specific sync template. Note that:

- No status information is shown on the **LDAP Sync Status** page when sending email reminders (as the reminder process does not resync the content from the LDAP data source). However, all related activity is recorded in the administrator log as usual.
- You can only resend emails for VMRs or device aliases that have been created by an LDAP sync template.

Sending a reminder email to a VMR owner

To send a reminder email to the owner of a VMR:

1. Go to **Services > Virtual Meeting Rooms**.
2. Select the Virtual Meeting Room(s) for which you want to send reminders (select the checkbox next to the VMR name).
3. From the **Action** drop-down list, select ***Send reminder emails to VMR owners***.
4. Select **Go**.

   This will generate and send an email to the owner of each selected VMR.

To see or change the "owner" of a VMR, go into the VMR details page and show the **Advanced options**. The **Owner's email address** field is at the bottom of that section.

Sending a reminder email to a device owner

To send a reminder email to the owner of a device alias:

1. Go to **Users & Devices > Device Aliases**.
2. Select the device aliases for which you want to send reminders (select the checkbox next to the device alias name).
3. From the **Action** drop-down list, select ***Send reminder emails to device owners***.
4. Select **Go**.
   This will generate and send an email to the owner of each selected device.

To see or change the "owner" of a device, go into the device alias details page and review the **Owner's email address** field.

Sending reminder emails for VMRs and devices associated with a sync template

This option is useful if you have initially configured your system not to send emails — for example while you are ensuring that the VMRs and devices are being created as expected — and then want to send emails to all of those generated VMRs and devices associated with that template (as the synchronization process only automatically generates emails for an existing VMR or device if its configuration changes in some way).

To send reminder emails for all VMRs and devices associated with a specific sync template:

1. Go to **Utilities > LDAP Sync Templates**.
2. Select the sync template(s) for which you want to send reminders (select the checkbox next to the template name).
3. From the **Action** drop-down list, select ***Send VMR and device reminder emails for selected LDAP sync templates***.
4. Select **Go**.
   This will generate and send an email to the owner of each VMR and device that is associated with that LDAP sync template.

Troubleshooting LDAP server connections

Pexip Infinity can be configured to connect to a Windows Active Directory LDAP server, or any other LDAP-accessible database, in order to:

- bulk-provision individual Virtual Meeting Rooms or devices for every member of the directory
- authenticate and authorize the login accounts that are allowed to connect to the Pexip Infinity Administrator interface or the Pexip Infinity API.

This section explains how Pexip Infinity connects to the LDAP server, and provides guidance on how to troubleshoot connection problems.

Note that all LDAP distinguished names must be entered as per the LDAP standard (RFC 4514). LDAP configuration is case insensitive.

Connecting to the LDAP server

When resolving the **LDAP server address**, the system supports DNS SRV and DNS A/AAAA lookups. The system always tries in the first instance to set up a TLS connection with the LDAP server.

To establish a TLS connection, the Pexip Infinity platform must trust the certificate presented by the LDAP server i.e. the LDAP server's certificate must be signed by an authority within the Pexip Infinity trusted CA certificates store. In addition, the resolved LDAP server address must match the CN (common name) contained within the certificate presented by the LDAP server.

The system will connect to the port returned by an SRV lookup, otherwise it will connect to 389 (TCP) or 636 (TLS). Requests to search the Active Directory Global Catalog use ports 3268 (TCP) and 3269 (TLS).

Connection process

The system treats the given **LDAP server address** as a domain or FQDN and attempts to resolve the address via DNS lookups in the following sequence:

1. Perform a DNS SRV lookup against _ldaps._tcp.<**LDAP server address**>
   (or _ldaps._tcp.gc._msdcs.<**LDAP server address**> if searching the AD global catalog).
2. Perform a DNS SRV lookup against _ldap._tcp.<**LDAP server address**>
   (or _ldap._tcp.gc._msdcs.<**LDAP server address**> if searching the AD global catalog).
3. Perform a DNS A/AAAA lookup against <**LDAP server address**>.

When a DNS lookup is successful, the system will attempt to establish a TLS connection with the server at the returned address. Only TLS connections are attempted as a result of _ldaps lookups.

If multiple addresses are returned by SRV lookups, the system will attempt to connect to each address in priority order.

Connectivity error messages and using the support log

Diagnostic information is also recorded in the support log (**Status > Support Log**).

When Pexip Infinity connects successfully to the LDAP server, the support log will contain an entry similar to this:

2015-06-05T11:15:00.550+00:00 mgmt 2015-06-05 11:15:00,550 Level="INFO" Name="support.ldap" Message="Successfully connected to LDAP server" Address="server.example.com" Uri="ldaps://server.example.com"

Unable to contact the LDAP server

If Pexip Infinity cannot contact the configured LDAP server, the support log will contain an entry similar to this:

2015-06-05T08:40:29.707+00:00 mgmt 2015-06-05 08:40:29,704 Level="INFO" Name="support.ldap" Message="Failed connecting to LDAP server" Address="server.example.com" Reasons="
ldaps://server.example.com : Can't contact LDAP server
ldap://server.example.com : Can't contact LDAP server"

Ensure that the server is available at the configured address and, if the server address is specified by domain name or FQDN, ensure that DNS records exist and resolve to the correct address.

Connection errors: TLS certificate issues

If Pexip Infinity can reach the configured LDAP server, but cannot connect to it due to TLS certificate issues, the support log will contain an entry similar to this:

2015-06-05T08:55:49.042+00:00 mgmt 2015-06-05 08:55:49,042 Level="INFO" Name="support.ldap" Message="Failed connecting to LDAP server" Address="server.example.com" Reasons="
ldaps://server.example.com : Can't contact LDAP server
ldap://server.example.com : Connect error"

The reason "**Connect error**" means that Pexip Infinity cannot verify the LDAP server's certificate.

Ensure that the LDAP server's TLS certificate (or the CA certificate that signed it, if it is not self-signed) is in the Pexip Infinity trust store (**Certificates > Root Trust CA Certificates**).

Note that, by default, Pexip Infinity will not use any intermediate CA certificates that have been uploaded to the Management Node for the purposes of verifying certificates for external services. Those external services should present their entire certificate chain. However, some third-party systems such as Azure AD DS (e.g. when used for an LDAPS connection) cannot be configured to present their entire certificate chain; in these cases you can configure Pexip Infinity to use the intermediate certificate in the TLS verification chain (see Uploading and managing additional trusted CA certificates).

Connection errors: binding to the server fails e.g. invalid credentials

If Pexip Infinity can reach the configured LDAP server, but cannot connect to it due to binding errors, such as invalid credentials, the support log will contain an entry similar to this:

2015-06-05T09:11:03.765+00:00 mgmt 2015-06-05 09:11:03,765 Level="INFO" Name="support.ldap" Message="Failed connecting to LDAP server" Address="server.example.com" Reasons="
ldaps://server.example.com : Invalid credentials
ldap://server.example.com : Invalid credentials"

Ensure that you have entered the correct credentials. They should be for an enabled, non-expired, domain user service account (not the Administrator account), which has a password set to never expire. All usernames and passwords are case sensitive.

If you are certain that the account you are trying to bind with is configured correctly, try to bind using the:

- bare username of the service account (e.g. **ldapuser**)
- full DN of the service account (e.g. **CN=ldapuser,CN=Users,DC=example,DC=com**)
- Windows logon of the service account (e.g. **EXAMPLE\ldapuser**).

Connection errors: Error syncing with LDAP

You can receive an "Error syncing with LDAP" error message when attempting to perform a template synchronization when provisioning VMRs, devices or users.

This can be caused by invalid syntax in the template's **LDAP user filter** or **LDAP user search DN** fields. Check that all parentheses are balanced and are in the correct places, and that all operators are correctly positioned.

This message can also be received if you have not selected an LDAP sync source when configuring your sync template.

Cannot log in to Pexip Infinity despite using correct credentials

If users receive a "Please enter the correct username and password for an administrator account" message when trying to log in to Pexip Infinity, but they are using the correct username and password, this typically means that either:

- The LDAP server cannot be contacted:
  - These errors are recorded in the Support log; see the connectivity troubleshooting guidelines above for more information.
- The LDAP server can be contacted but the correct user records are not being searched:
  - Check the Pexip Infinity LDAP configuration settings (**Users & Devices > Administrator Authentication**) to ensure that all objectClass and LDAP field names have been spelled correctly, and that the base DN and user search DN fields contain the correct domain and organizational unit settings.
  - If you are using nested AD security groups, see Supporting nested security groups in Windows Active Directory.
- The LDAP server can be contacted and the user records can be found and authenticated, but the user is not authorized to access Pexip Infinity:
  - Check that administrator roles and role mappings have been configured on Pexip Infinity (**Users & Devices > Administrator Roles** and **Users & Devices > LDAP Role Mappings**).
  - Ensure that the user's LDAP account is associated with the LDAP group DNs / role combinations that are configured on Pexip Infinity.
- Note that usernames and passwords are case sensitive — ensure that you are using the correct case for your credentials.

Recovering local access

If necessary you can reinstate access via the Pexip Infinity local on-box database, so that administrators can log in via the default admin account with full administrator privileges. You may need to do this if, for example, the **Authentication source** is configured as ***LDAP database*** only and your connectivity to the LDAP server goes down or your credentials become invalid.

For full instructions, see Reinstating the local admin account.

VMRs, devices or user records not created as expected by a sync template

User search or user filters not being applied

If more or fewer VMRs, devices or users than expected (or no VMRs/devices/users at all) were created after performing a template synchronization, it is likely that the **LDAP base DN**, **LDAP user search DN** and **LDAP user filter** fields have been misconfigured.

Check that all objectCategory, objectClass and LDAP field names have been spelled correctly. Note that all LDAP user search and user filter contents are not case sensitive.

More information on Active Directory LDAP filtering can be found at https://social.technet.microsoft.com/wiki/contents/articles/5392.active-directory-ldap-syntax-filters.aspx.

Using ldapsearch or AD Explorer to view the LDAP database

Mac and Linux systems

You can use a command line tool such as **ldapsearch**, which is available for Mac and Linux systems, to help test and diagnose connectivity issues with the LDAP server. Note that **ldapsearch** is installed by default on all Pexip Infinity nodes.

Here are some example **ldapsearch** queries you could use (after adapting the parameters as appropriate for your environment):

```
$ ldapsearch -v -W -H ldaps://dc01.example.com -D 'admin123' -b OU=people,DC=example,DC=com
```

This fetches the contents of OU (org unit) **people** from the LDAP server at **dc01.example.com**, binding as user **admin123**.

If your LDAP server requires the full DN for a user — rather than just a username — you can use:

```
$ ldapsearch -v -W -H ldaps://dc01.example.com -D 'CN=admin123,OU=people,DC=example,DC=com' -b OU=people,DC=example,DC=com
```

making sure that the parameter to `-D` is the full DN of your user.

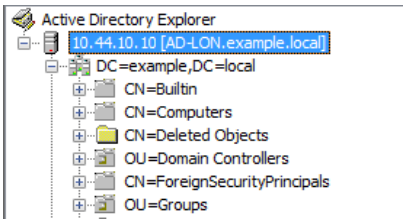
If your server does not support LDAPS, but only LDAP+STARTTLS, you can use:

```
$ ldapsearch -v -W -ZZ -H ldap://dc01.example.com -D 'admin123' -b OU=people,DC=example,DC=com
```

Windows

Windows users can use Active Directory Explorer (AdExplorer) to navigate around and view AD structures and entries. See https://technet.microsoft.com/en-us/sysinternals/adexplorer for more information and links to download the software.

The example below shows how you can discover your AD server's actual hostname (**AD-LON.example.local** in this case) if you use AdExplorer to connect to your server via its IP address (**10.44.10.10** in this case):



Pexip Infinity maintenance tasks

Typical maintenance tasks include upgrading the Pexip Infinity software, creating and restoring backups of configuration data, and managing administrator account passwords.

Backing up and restoring configuration

You should take regular backups of the configuration data on your Management Node, particularly before and after Upgrading the Pexip Infinity platform. This allows you to restore your configuration to a specific point in time, or to restore your configuration if you have to deploy a new Management Node.

There are two ways to maintain copies of your Management Node configuration data:

- use your hypervisor's tools to create a full backup or snapshot of the Management Node VM
- use the backup and restore mechanism built into the Pexip Infinity Administrator interface (automatic and manual backup options are available)

For **on-premises deployments**, we recommend that you use both the hypervisor and Pexip's inbuilt methods to preserve your configuration data. A VM snapshot should be your primary mechanism prior to an upgrade, as this allows you to easily restore your system back to its state at the time the snapshot was taken. The Pexip Infinity backup and restore mechanism is your fallback mechanism, as this allows you to preserve a copy of your data in an alternative location, in case you lose your VM environment. For **cloud-based deployments** (Azure, AWS, GCP or Oracle) if you want to take a VM snapshot we recommend that you perform a graceful OS shutdown (not a hard power-off) before taking the snapshot; alternatively, you may use the Pexip Infinity backup and restore mechanism.

You can also separately backup and restore just your Virtual Meeting Room and Virtual Auditorium configuration, see Bulk import/export of service configuration data.

If you are using Secure Scheduler for Exchange, we recommend that you run the scheduling recovery script after restoring the Management Node to ensure that any meetings that were scheduled after the backup was taken are reinstated.

See Resilience strategies — redundancy, backing up and restoring data for more general guidance on resilience strategies.

Backing up Conferencing Nodes

Conferencing Nodes do not need to be backed up. They receive all their configuration information from the Management Node and can simply be redeployed if necessary (i.e. delete and recreate). However, if your Conferencing Nodes are geographically spread out and redeploying them would consume significant bandwidth or take a significant length of time, they can also be backed up with your hypervisor's backup tools.

Using your hypervisor's tools to take and restore backups

You can use your hypervisor's own backup tools, or any other third-party tool that supports VM backups, to create a full backup of the Management Node. This backup can then be re-deployed at a later date if required.

Be aware that snapshots are not backups. Snapshots are a tool to roll back to a given time. Therefore, we recommend taking snapshots only when necessary (such as prior to an upgrade) and deleting the snapshot as soon as possible after the upgrade is confirmed to be successful. You should only create and delete VMware snapshots at a time of minimal usage. Taking or removing snapshots can cause virtual machines to become unresponsive.

VM backups should use a proper hypervisor VM backup tool (e.g. VMware VDP — vSphere Data Protection) or similar, and restoration should be tested and verified (preferably after the inbuilt backup methods have been set up, to ensure that you have another way of recovering if your restoration fails).

Consult your hypervisor documentation for more information about taking and restoring backups and snapshots.

Using backup and restore via the Pexip Infinity Administrator interface

You can use Pexip Infinity's inbuilt backup and restore mechanism to backup and restore the configuration data on the Management Node.

You can enable regular automatic backups, and you can also take a manual backup whenever it is appropriate, for example, before and after you make any configuration changes or perform a software upgrade.

- All backup files are encrypted — the administrator supplies a passphrase and must remember this for any subsequent restoration.
- Restoration can only occur on exactly the same version that the backup was taken from.
- The data contained in the backup contains all configuration data, including IP addresses, custom themes, certificates and call history.
- The backup data does not contain licenses, the administrator log, the support log, usage statistics or the operating system password.
- The system keeps on the host VM only the 5 most recent manually-taken backups and the 5 most recent automatic backups. Older backup files are deleted when a new file is created.

Note that this function can only be used to restore configuration or to replicate the configuration of a previous Management Node onto a new Management Node. It cannot be used to redeploy Conferencing Nodes.

Managing backup files

The system keeps on the host VM only the 5 most recent manually-taken backups and the 5 most recent automatic backups. Older backup files are deleted when a new file is created.

- To see the backup files that are currently stored on the host VM, go to **Utilities > Backup/Restore** and look at the list of **Existing Backup Files**.
- If you want to manually download a backup file from the host VM to another machine, go to **Utilities > Backup/Restore** and select the **Download backup** option. You can also configure the automatic backup to upload each backup file to an external FTP server.
- **SHA-256 Checksums** are presented in the list of backup files and should be stored securely. The checksum of a backup file is presented when the file is uploaded, and you should use it to verify the file before the backup is restored.

A backup file is a collection of configuration and history, and therefore contains information such as IP addresses, conference names, aliases, keys and certificates. This information will be sensitive to your organization, so the file must be saved and handled securely.

Enabling automatic backups

You can enable Pexip Infinity to automatically backup the Management Node configuration data on a regular basis.

When automatic backups are enabled:

- By default each backup is taken at 01:00 UTC every day, but you can vary the start hour and perform multiple backups per day.
- Successful backup operations are recorded in the administrator log (with a "Created automatic system backup" message).
- Automatic backup filenames take the format:
  pexip_auto_backup_<hostname>_<version>_<build>_<date>_<time>.pexbak

To enable automatic backups:

1. Go to **Utilities > Automatic Backups**.
2. Select the **Enable automatic backups** checkbox (this is disabled by default).
3. Enter a **Backup passphrase**.
   The text entered here is used to encrypt the backup file. You must remember this text as it will be required if you need to subsequently restore the data from the file.
4. Enter the **start hour** and **interval**:
   - **Automatic backup start hour** is the hour of the day in UTC to run the backup. Range is 0-23. Default: 1.
   - **Automatic backup interval** is the interval in hours between each automatic backup for that day. Range is 1-24. Default: 24 (i.e. daily).

   Examples:
   - Start 6, interval 24 = 6am UTC every day
   - Start 0, interval 6 = 0:00/6:00/12:00/18:00 every day
   - Start 0, interval 1 = hourly
   - Start 6, interval 10 = 6:00 and 16:00 every day

5. The system always keeps the 5 most recent automatic backups on the host VM. In addition, you can upload each automatic backup file to an external FTP server. To do this, you must specify the **Upload URL** (supported schemes are FTPS and FTP) and the **Username** and **Password** credentials of the FTP server.
6. Select **Save**.

Manually creating a backup file

To manually create and download a backup file:

1. Go to **Utilities > Backup/Restore**.
2. In the **Create backup** section, enter a **Passphrase** and then enter it again in the **Re-enter passphrase** field.

   The text entered here is used to encrypt the backup file. You must remember this text as it will be required if you need to subsequently restore the data from the file.
3. Select **Create backup**.

   After a few seconds you see a message: "Successfully created the backup file: <file_name>" where <file_name> takes the format: **pexip_backup_<hostname>_<version>_<build>_<date>_<time>.pexbak**
4. Download the file from the host VM:
   a. From the **Existing backup files** section at the bottom of the page, select **Download backup** for the file you have just created.
   b. Follow your browser's prompts to save or download the file to your local file system.
   c. If required, you can delete unwanted backup files from your host VM by selecting **Delete backup**.

      The system keeps on the host VM only the 5 most recent manually-taken backups and the 5 most recent automatic backups. Older backup files are deleted when a new file is created.

Restoring data to the Management Node

You can restore configuration data to the Management Node. This could be restored to the original Management Node from which the backup was taken, or it could be restored to a newly deployed Management Node (if the original node was lost due to, for example, issues with its VM environment).

Any in-progress calls will not be affected while data is being restored to the Management Node.

To restore configuration data:

1. If required, deploy a new Management Node (see Installation overview for links to the appropriate instructions for your hypervisor):
   - Complete the installation wizard as normal.
   - The IP address you enter at this stage is temporary (it can be the same as the previous Management Node).
   - All the configuration data you enter, including the IP address, will be subsequently replaced.
2. If any new Conferencing Nodes have been deployed since the backup was taken:
   - Power these Conferencing Nodes off and delete them before restoring the Management Node data.
   - These additional Conferencing Nodes will not be recognized by the restored Management Node. You will have to create them again after the restore has completed.
3. If previously configured Conferencing Nodes have been deleted since the backup was taken:
   - There is no need to do anything at this stage; the restore will complete successfully.
   - Simply delete the Conferencing Nodes from the restored configuration, after the restore has completed.
4. Restore your previous backup file:
   a. Go to **Utilities > Backup/Restore**.
   b. In the **Restore backup** section, enter the **Passphrase**.

      The text entered here must be identical to the text that was used to create the backup file.
   c. Select **Choose File** and then choose the backup file that you want to restore.

      The file must be chosen from your local file system (you cannot select a file from the list of **Existing backup files**).
   d. Select **Restore backup**.

      You are taken to the **Restore Backup** confirmation page.

If instead, you see "Failed to restore the system from the backup file" with a "Decryption Error: decryption failed" message, the most likely reason for this is that you have entered an incorrect passphrase.

e. The confirmation page shows the date that the backup was taken, and the Management Node IP address that will be restored to this system.

   Select **Restore backup** to confirm the restoration.

5. If the restore is successful, after a few seconds you will see a "Successfully restored the backup file" message and the Management Node will reboot.

   You need to wait for the node to return from rebooting before you can access it again.

   If you have restored a file where the IP address of the Management Node in the backup file is different from the node's current address (prior to the restore), you need to manually enter this IP address into the web browser to access the restored Management Node's web interface login page.

6. If you have restored your configuration onto a new VM (and were unable to return the licenses from your previous VM), you must contact your Pexip authorized support representative to get your licenses reapplied on your new VM.

Upgrading the Pexip Infinity platform

ℹ We have designed the upgrade process to minimize disruption. However, there may be some temporary loss of functionality or unpredictable system behavior due to Conferencing Nodes running conflicting software versions, or some Conferencing Nodes being placed in maintenance mode. For this reason, we recommend upgrading at a time of minimal usage.

ℹ There is a known limitation with upgrades from v39.x and v40.x whereby Conferencing Nodes are put into maintenance mode but do not wait until all calls have cleared before upgrading and rebooting. You must therefore upgrade from these versions in a maintenance window to minimize the impact on any ongoing calls.

The upgrade process

When you initiate an upgrade of the Pexip Infinity software, the following steps occur automatically:

1. The Management Node software is upgraded, after which the Management Node automatically reboots.
2. The first 10 Conferencing Nodes are put into maintenance mode. This means that all further incoming calls to those nodes will be rejected.

   (In Pexip Infinity deployments of 10 Conferencing Nodes or fewer, all of the nodes are placed into maintenance mode.)
3. When all calls have cleared from a Conferencing Node that is in maintenance mode, its software is upgraded and the node is rebooted. This process should take around 2 minutes to complete, but may take longer depending on the connection speed between the Management Node and the Conferencing Node (as the upgrade files have to be transferred between the nodes).

   If all of the calls on a Conferencing Node that is in maintenance mode have not cleared after 1 hour, the node is taken out of maintenance mode and put at the back of the queue of nodes to be upgraded. A further attempt to upgrade that node will be made after all other nodes have been upgraded (or had upgrade attempts made).
4. After a Conferencing Node has been upgraded successfully (or has been put back in the queue for a later upgrade attempt) and is again available, another Conferencing Node is selected and put into maintenance mode.
5. The process continues with each subsequent Conferencing Node being put into maintenance mode, and, after all calls have cleared, being upgraded and then rebooted. Up to 10 Conferencing Nodes may simultaneously be in maintenance mode or in the process of being upgraded at any one time.

   Any Conferencing Nodes used for dynamic cloud bursting will be automatically started up and upgraded. Bursting nodes are normally selected for upgrade after the system has upgraded (or attempted to upgrade) the "always-on" nodes.
6. If the upgrade of the Management Node and all Conferencing Nodes has not completed successfully after 24 hours, the process will stop and all nodes will be left in their existing upgrade state. This is designed to prevent situations where some Conferencing Nodes cannot be upgraded, which would otherwise leave the system in a permanent state of upgrading.

If the upgrade process does not complete successfully and stops after 24 hours, you may have a mix of upgraded and non-upgraded nodes. You will then need to repeat the upgrade process. During a repeat upgrade, only those nodes that have not already been upgraded will be included in the upgrade process.

During the 24-hour period from when an upgrade has been initiated, you cannot re-initiate an upgrade using the Administrator interface. If you must re-initiate an upgrade during this time, you must reboot the Management Node and then start the process again.

When to upgrade

While the upgrade is in progress, some Conferencing Nodes will be running the newer version of the software and some will be running the older version. These Conferencing Nodes will be incompatible until they are all again running the same version. This means that there may be instances where two endpoints dial the same Virtual Meeting Room alias but if they are routed to different Conferencing Nodes that are running different versions of the software, the two endpoints will be in different conferences. It may also mean that, if an endpoint is in an ongoing call on a node that is put into maintenance mode in preparation for an upgrade, some call functionality such as presentation sharing may be limited. For this reason, **we recommend upgrading at a time of minimal usage**.

Alternatively, to avoid unpredictable system behavior due to Conferencing Nodes running conflicting software versions, you may want to manually put **all** of your Conferencing Nodes into maintenance mode before initiating the upgrade process. This will allow all existing calls to finish, but will not admit **any** new calls. You should then actively monitor your Conferencing Nodes' status and manually take each node out of maintenance mode after it has been upgraded to the new software version, so that the system can start taking new calls again on those upgraded nodes.

Upgrading from version 35 or later to version 41

ℹ If you are running a software version between v31.4 and v34.2 inclusive, you must first upgrade to version **35.1**, and then upgrade again to version 41; see Upgrading from versions 31.4-34.2 to version 41.

ℹ If you are upgrading from versions prior to v32, you must remove any existing MD5/SHA1 certificates, except root certificates, and replace them with new certificates before upgrading to v32 or later.

How to upgrade

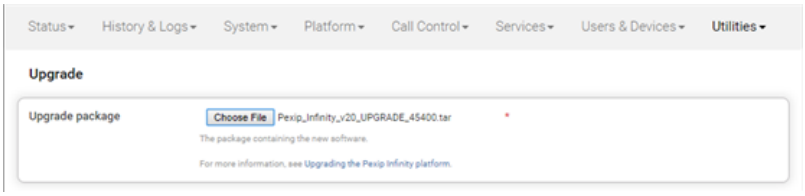
When upgrading, please note that:

- A Management Node upgrade may take a long time, potentially up to 1 hour. Do **not** reboot the Management Node under any circumstances. If you are concerned about the time the upgrade is taking, please contact your Pexip authorized support representative.
- There is normally no need to reboot a Conferencing Node. If a Conferencing Node appears to be stuck for over 1 hour then contact your Pexip authorized support representative — do **not** reboot the node.
- If you experience issues when upgrading, contact your Pexip authorized support representative — do **not** attempt to roll back to VM snapshots without first contacting them.
- When upgrading from version 38.x or earlier, **all** Conferencing Nodes must have completed upgrade before **any** nodes become available. If a Conferencing Node appears to have failed upgrade then contact your Pexip authorized support representative.

To upgrade Pexip Infinity software from v35.0 or later to v41:

1. Before upgrading an on-premises deployment, we recommend that you use your hypervisor's snapshot functionality to take a full VMware/Hyper-V snapshot of the Management Node. We recommend that you do **not** take a snapshot of your Conferencing Nodes — you can simply redeploy them from the Management Node (after it has been rolled back) in the unlikely event that this is required.

   Before upgrading a cloud-based deployment (Azure, AWS, GCP or Oracle), you should backup the Management Node via Pexip Infinity's inbuilt mechanism (**Utilities > Backup/Restore**).
2. Download the Pexip Infinity upgrade package for v41 from the Pexip download page.
3. Before upgrading, ensure that all "always-on" Conferencing Nodes are powered on and are reachable (i.e. no Connectivity Loss errors), and are all running the same version from which you are upgrading. You do not need to power on any cloud bursting nodes.
4. From the Pexip Infinity Administrator interface, go to **Utilities > Upgrade**.
5. Select **Choose File** and browse to the location of the upgrade package.



6. Select **Continue**.

- When upgrading from versions prior to v39.0, there is a short delay while the upgrade package is uploaded.

  After the upgrade package has been uploaded, you are presented with a confirmation page showing details of the existing software version and the upgrade version.
- When upgrading from v39.0 or later, you are presented with a page showing details of the existing software version and the upgrade version, along with a progress bar indicating how long is left until upload of the new upgrade package is complete.

7. To proceed, select **Start upgrade**.

   You are taken to the **Upgrade Status** page, showing the current upgrade status of the Management Node and all Conferencing Nodes. This page automatically refreshes every 5 seconds.
8. When the upgrade completes, all nodes will show a status of **No upgrade in progress** and have the new **Installed version**.

   If the upgrade process completes and there are some nodes that have failed to upgrade, you can restart the upgrade process by uploading the upgrade package to the Management Node again via **Utilities > Upgrade**. This will skip over any nodes that have already been upgraded.
9. If you have Pexip CVI for Microsoft Teams you must also upgrade your associated Teams Connector deployment in Azure to the same version as your Pexip Infinity deployment (including minor/"dot" releases).

No additional actions are required to upgrade the Management Node or Conferencing Nodes individually.

We recommend that you take a fresh backup of your system after upgrading.

Error messages during upgrade

The following Error and Warning messages are expected during an upgrade and do not require any action:

```
2017-03-03T12:12:02.400+00:00 <manager_hostname> 2017-03-03 12:12:02,456 Level="ERROR"
Name="administrator.system.connectivity" Message="Unable to contact node." Src-Node="<node_
ip_fqdn>" Node="<node_ip_fqdn>"
```

```
2017-03-09T07:04:08.460+00:00 <manager_hostname> 2017-03-09 07:04:08,460 Level="WARNING"
Name="administrator.alarm" Message="Alarm raised" Node="<node_ip_fqdn>"
Alarm="connectivity_lost" Time="2017-03-09 07:04:08,455" Instance="Source=<node_ip_fqdn>,
Destination=<node_ip_fqdn>" Detail=""
```

Upgrade prevented due to invalid licenses:

If you are upgrading from v27, in some cases you may be prevented from running the upgrade process and are shown a "This system includes active licenses that will not be valid after upgrade" message. This may occur if you still have legacy-style licenses. In this case contact your Pexip authorized support representative to obtain replacement license keys so that you can continue with the upgrade.

Upgrading from versions 31.4-34.2 to version 41

If you are running a software version between v31.4 and v34.2 inclusive, you must first upgrade to version **35.1**, and then upgrade again to version 41. To do this:

1. Before upgrading, ensure that all "always-on" Conferencing Nodes are powered on and are reachable (i.e. no Connectivity Loss errors), and are all running the same version from which you are upgrading. You do not need to power on any cloud bursting nodes.
2. Download the Pexip Infinity **v35.1** upgrade file.
3. Follow the steps outlined in Upgrading from version 35 or later to version 41, but when asked to **Choose File** browse to the location of the **v35.1** upgrade file.
4. Verify that the upgrade has completed successfully.
5. Download the Pexip Infinity **v41** upgrade file.
6. Follow the steps outlined in Upgrading from version 35 or later to version 41, and when asked to **Choose File** browse to the location of the **v41** upgrade file.

Upgrading configuration-only deployments

The automatic upgrade process described above updates the Management Node and all Conferencing Nodes, including Conferencing Nodes that have been created using the configuration only deployment type.

However, if after the upgrade you wish to deploy new configuration-only Conferencing Nodes, you must download and use a version of the Conferencing Node VM template that matches the version of the Management Node that you have upgraded to. Creating a new

Conferencing Node from a VM template containing a different version of the Pexip Infinity software than that which is running on the Management Node is not supported and will not work. For more information, see Deploying a Conferencing Node using a generic VM template and configuration file.

Definition of upgrade statuses

During an upgrade, the **Upgrade Status** page reports the status of each Conferencing Node as follows:

Status	Description
No upgrade in progress	During a platform upgrade, this is the default state that occurs before a Conferencing Node is upgraded, or after the node has rebooted.
Upgrade pending	An upgrade of the platform is in progress and this Conferencing Node is in the queue to be upgraded.
Preparing to upgrade	The Conferencing Node is preparing to upgrade. During this time the node is put into maintenance mode, and other services are stopped.
Waiting for calls to clear	The Conferencing Node is in maintenance mode and is waiting for existing conferences to complete.
Timeout waiting for calls to clear	Not all conferences had cleared after 1 hour. This Conferencing Node has been removed from maintenance mode and the upgrade will be attempted again later.
Upgrade in progress	The new software is being unpackaged and installed on the Conferencing Node. During this time the Conferencing Node will not synchronize configuration with the Management Node.
Rebooting	The upgrade has completed and the Conferencing Node is rebooting.
Could not communicate with conferencing node	This error is reported if the Conferencing Node cannot be contacted.

Downgrading or recovering from a failed upgrade

If you experience issues when upgrading, contact your Pexip authorized support representative before attempting to roll back or downgrade your deployment. Downgrading should only be done following their advice.

To downgrade your system to the previous version, or to recover your system after a failed upgrade:

1. Restore the Management Node from the VM snapshot that you took using your hypervisor at the start of the upgrade process.
2. Restore the individual Conferencing Nodes by either:
   - redeploying each Conferencing Node from the Management Node - see Deploying new Conferencing Nodes, or
   - restoring the hypervisor snapshot of each Conferencing Node that was taken at the start of the upgrade process.

Checking the installed version

Management Node

To view which software version is running on the Management Node, click on the **About** link at the top right corner of the Pexip Infinity Administrator interface.

Conferencing Nodes

To view which software version is running on individual Conferencing Nodes, go to **Status > Conferencing Nodes**. The software version number is shown in the **Installed version** column.

Installing and updating software bundles

The software bundles feature enables administrators to update individual components of the Pexip Infinity platform (for example the Pexip apps), independently of the platform itself.

The use of software bundles allows Pexip to provide our customers with more frequent and timely access to features and bug fixes for individual components, and allows our customers to install, test and deploy the components that they are most interested in without having to wait for the next major Pexip Infinity software release.

Software bundles available in v41

Each new major release of Pexip Infinity includes the latest available set of bundled software components, which are available for your use without any further action required. Between major releases, we may release one or more updates to each of these components which you can optionally download from our website and install on the Pexip Infinity Management Node.

Baseline software bundles

This release of Pexip Infinity ships with the following components that can subsequently be updated:

Software bundle	Description	Original revision
PexRTC software bundle	Core WebRTC library	41-41.0
Webapp2 software bundle	Webapp2	41-1.14
Webapp3 software bundle	Webapp3	41-13.0

Viewing and changing installed software bundles

To view all bundled software components currently installed and available on your Pexip Infinity platform, from the Pexip Infinity Administrator interface, go to **Utilities > Software Bundles**.

The **Bundle Type** column lists each of the bundled software components that are currently installed; the **Revision** column shows the version of that component that is currently in use.

To change which version of a particular software bundle is currently in use on your on your Pexip Infinity platform, select the bundle and then select the **Selected revision** drop-down. This lists all the available versions of this software bundle; select the version you wish to use.

Installing new software bundles

Each software bundle is compatible with a specific major version of Pexip Infinity, indicated by the first part of the bundle's revision number. Thus, for example, you cannot install a 39-n-n bundle on a Pexip Infinity deployment running v38. Upon upgrade to a new major version of Pexip Infinity, all existing software bundles are upgraded automatically to the latest available version.

Any web app customizations apply to all revisions of the web app.

To install or update a software bundle:

1. Contact your Pexip authorized support representative for information on how to download the latest software bundles, which you should save to your local machine.
2. From the Pexip Infinity Administrator interface, go to **Utilities > Software Bundles**.
3. From the bottom of the page, select **Upload**.
4. Select **Choose File** and browse to the location of the new software bundle.
5. Select **Upload**.
6. After a short delay you are taken to the **Change Software bundle revision** page (alternatively, you can go to this page by selecting **Utilities > Software Bundles** and selecting the software component you have just updated).
7. From the **Selected revision** drop-down, select the version you have just installed.
8. Select **Save**.

The updated software bundles are replicated to each Conferencing Node, and the updated web app is available to end users as soon as the Conferencing Node they are connected to has updated, although they may need to refresh their browser to receive the changes.

Setting and changing usernames and passwords

There are a number of different components of the Pexip Infinity platform that require a username and password:

- Pexip Infinity Administrator interface
- Management Node operating system
- Conferencing Node operating system

Information on how to set and change these is given below.

Some external systems such as CUCM also use usernames and passwords to authenticate with Pexip Infinity. For more information on these, see Integrating with external systems.

Pexip Infinity Administrator interface

A username and password are required for logging in to the Pexip Infinity Administrator interface. By default these credentials are verified against the local database and are the **web administration username** and **web administration password** that were set during initial installation (via the installation wizard). You can also configure the system to authenticate users against an LDAP-accessible database and/or an OpenID Connect (OIDC) provider.

Local database authentication

The username (typically **admin**) cannot be changed after being set, but the password can.

To change the password used to log in to the Pexip Infinity Administrator interface:

1. Log in to the Pexip Infinity Administrator interface.
2. At the top right of the screen, select **Change password**.
3. Enter the existing password, and then enter the new password twice.
4. Select **Change my password**.

You will see a message:

Password change successful

Your password was changed.

Note that the **Change password** option is not available if the system has its **Authentication source** configured as either ***LDAP database*** only, or ***Open ID Connect service*** only.

LDAP and OIDC authentication

Usernames and passwords for any LDAP-accessible database and OpenID Connect provider cannot be managed via the Pexip Infinity Administrator interface.

Changing the Management Node operating system password

The Management Node virtual machine (VM) runs on a Linux operating system. The username is **admin** and cannot be changed.

The password for the Management Node operating system was set during initial installation of the Management Node (when running the installation wizard).

The password can be changed by logging in to the Management Node operating system over SSH (providing SSH access has not been disabled) and running the standard UNIX command `passwd`.

Conferencing Node operating system password

Conferencing Nodes run on a Linux operating system. The username for each Conferencing Node operating system is **admin** and cannot be changed.

The password was set during Deploying new Conferencing Nodes when configuring the **SSH password** field.

The password can be changed by logging in to the Conferencing Node operating system over SSH (providing SSH access has not been disabled) and running the standard UNIX command `passwd`.

Lost or forgotten passwords

If you forget the password for the Pexip Infinity Administrator interface, you can re-run the installation wizard, being sure to change only the **Web administration password** setting.

If you re-run the installation wizard to only reset the web administration password then you will not lose your configuration data and you will not need to delete and redeploy your Conferencing Nodes, providing you re-enter exactly the same hostname/addressing information as before. However, after completing the wizard you must then edit every system location (**Platform > Locations**) and reselect the DNS and NTP servers for that location.

If you forget the password used to log in to a Conferencing Node or Management Node operating system, contact your Pexip authorized support representative for assistance.

Re-running the installation wizard

You can rerun the installation wizard if you previously exited the wizard before completing all the steps, or you entered the wrong hostname/addressing information (for server-based deployments) during the initial configuration.

Re-running the installation wizard will overwrite any existing Management Node configuration and lose all connections to existing Conferencing Nodes. Therefore, if you re-run the installation wizard after deploying any Conferencing Nodes, you must delete any existing Conferencing Nodes and then redeploy them from the updated Management Node.

If you re-run the installation wizard to only reset the web administration password then you will not lose your configuration data and you will not need to delete and redeploy your Conferencing Nodes, providing you re-enter exactly the same hostname/addressing information as before. However, after completing the wizard you must then edit every system location (**Platform > Locations**) and reselect the DNS and NTP servers for that location.

If you interrupt the installation wizard, or it does not complete properly for any reason, you must reboot the Management Node and then rerun the installation wizard.

If you want to change the IP address of the Management Node you must first return your licenses from your current Management Node. If you do not return your licenses before re-running the installation wizard you will lose your licenses from your new Management Node (with the new IP address) and you will not be able to add them back to that new Management Node.

Do not re-run the installation wizard on cloud-based deployments (Azure, AWS, GCP or Oracle) in order to change Management Node configuration data such as its IP address or hostname. To change such data you must terminate the existing instance and deploy a new Management Node instance. You should only re-run the installation wizard on cloud-based deployments if you need to reset the web administration password (and then you should not change any of the other configuration data).

To re-run the installation wizard:

1. Open a console window on the Management Node VM.

   If your Management Node is deployed on Azure, AWS, GCP or Oracle, use an SSH client to access the Management Node, supplying your private key file if appropriate.
2. At the login prompt, log in as **admin**.
3. If requested, enter the operating system password created when the installation wizard was run. If you did not get as far as creating a new password, you are asked to create one now.
4. Type `installwizard`.

   If your Management Node is deployed on Azure, AWS, GCP or Oracle, you are prompted again for the admin password.

   The Pexip installation wizard begins.

   (If you interrupt the installation wizard, or it does not complete properly for any reason, you must reboot the Management Node and then rerun the installation wizard.)

(Note that the steps described above can vary slightly depending on whether it is a cloud or server-based deployment, and whether you previously completed all of the installation wizard steps.)

Follow the prompts to set the following configuration for the Management Node.

If you press enter, the default value is applied:

Setting	Default value	Multiple entries allowed?	Can be changed via Pexip Infinity Administrator interface?
IP address	As assigned by DHCP, otherwise 192.168.0.100 *	No	No ‡

Setting	Default value	Multiple entries allowed?	Can be changed via Pexip Infinity Administrator interface?
Network mask	As assigned by DHCP, otherwise 255.255.255.0 *	No	No ‡
Gateway	As assigned by DHCP, otherwise 192.168.0.1 *	No	No ‡
Hostname	As assigned by DHCP, otherwise <no default>	No	No ‡
Domain suffix	As assigned by DHCP, otherwise <no default>	No	No ‡
DNS servers	As assigned by DHCP, otherwise 8.8.8.8	Yes, if separated by a space or comma	Yes
NTP servers †	As assigned by DHCP, otherwise two of the following: • 0.pexip.pool.ntp.org • 1.pexip.pool.ntp.org • 2.pexip.pool.ntp.org • 3.pexip.pool.ntp.org	Yes, if separated by a space or comma	Yes
Web administration username	admin	No	No ‡
Web administration password	<no default>	No	Yes
Enable incident reporting (yes/no)	<no default>		Yes
Contact email address **	<no default>	No	Yes
Send deployment and usage statistics to Pexip (yes/no)	<no default>		Yes

* The addresses entered here are assigned as static IP addresses. When deploying in a cloud service, these values are replaced with the IP address and network settings for your instance.

** Shown and required if incident reporting is enabled.

† The NTP server must be accessible by the Management Node at the time the startup wizard is run. Installation will fail if the Management Node is unable to synchronize its time with an NTP server.

‡ After they have been configured, do not attempt to change these settings by any other means. To change these settings on server-based deployments, you must re-run the installation wizard. To change these settings on cloud-based deployments, you must terminate the existing instance and deploy a new Management Node instance.

Migrating Conferencing Nodes between host VMware servers

Using the tools available in VMware, you can move Conferencing Nodes from one host server to another.

You may want to do this if the existing host server is running low on capacity or is experiencing performance issues.

Pexip Infinity supports VMware's vMotion for the live migration of Conferencing Nodes.

You must put the Conferencing Node into maintenance mode and wait until all conferences on that node have finished before migrating it to another host server.

Prerequisites

The source and target servers must be enabled for vMotion and use the same shared storage.

Manual migration via VMware's vMotion

1. Ensure that the Conferencing Node is in maintenance mode and that all conferences on that node have finished.
2. In the VMware inventory view, right-click on the Conferencing Node and select **Migrate**. The **Migrate Virtual Machine** window will appear.
3. Select **Change host** then select **Next**.
4. Select the destination host server or cluster and then select **Next**.
5. Select the destination resource pool and then select **Next**.
6. Select the priority and then select **Next**.
7. Check the details of the migration, and if they are all correct, select **Finish**.

Taking a Conferencing Node out of service

If you need to take a Conferencing Node out of service, you must first put it into maintenance mode to ensure that it is not hosting any conferences.

When a Conferencing Node goes into maintenance mode, it will not accept any new calls or registrations. Any existing calls on that Conferencing Node will not be affected (this applies to all call processing i.e. handling signaling, media proxying and transcoding). After all existing calls have terminated, the Conferencing Node will still be live and contactable but will not be handling any calls.

A Conferencing Node will go into maintenance mode if:

- The system administrator elects to put the Conferencing Node into maintenance mode via the Pexip Infinity Administrator interface (see Manually placing a Conferencing Node into maintenance mode below).

  This setting will persist after a reboot.
- The Pexip Infinity platform is being upgraded, during which time each Conferencing Node in turn is automatically put into maintenance mode and upgraded. For more information, see Upgrading the Pexip Infinity platform.
- The Conferencing Node has been installed on a system that does not meet the CPU instruction set requirements. In such a case:
  - after initial deployment, the following message will appear in the admin log: `CPU instruction set is not supported; system will be placed in maintenance mode`
  - each time the Conferencing Node rejects a call, the following message will appear in the admin log: `Message="Participant failed to join conference." Reason="System in maintenance mode"`
  - any manual changes to the **Enable maintenance mode** setting will have no impact - the Conferencing Node will remain in maintenance mode regardless of this setting.

  For more information on how to resolve this, see Troubleshooting the Pexip Infinity platform.
- The Conferencing Node is running on a VM in Azure which has a scheduled maintenance event.

Note that when a Conferencing Node is in maintenance mode, it reports a media load of 100%. This is to indicate that there is no current capacity available.

Manually placing a Conferencing Node into maintenance mode

When putting multiple nodes into maintenance mode, to ensure maximum availability for registration data and other data that may have replicas stored in the distributed database on that node, and to ensure maximum service availability, we recommend that you:

- Activate maintenance mode on one node at a time.
- Wait a couple of minutes before putting the next node into maintenance mode.

This allows registrations and any relevant distributed data to gradually be migrated to other nodes.

To manually put a Conferencing Node into maintenance mode:

1. Go to **Platform > Conferencing Nodes**.
2. Select the Conferencing Node(s).
3. From the **Action** menu at the top left of the screen, select **Enable maintenance mode** and then select **Go**.

   While maintenance mode is enabled, this Conferencing Node will not accept any new conference instances.
4. Wait until any existing conferences on that Conferencing Node have finished. To check, go to **Status > Live View**.

When all conferences on the Conferencing Node have finished you can take it out of service.

Long term maintenance

If you intend to take one or more nodes offline for a long period of time (days or weeks) then you should consider temporarily deleting those nodes altogether — and redeploying them afresh when they are next needed. This will avoid distracting connectivity alarms being raised for those nodes that might make it harder for the administrator to spot other "unexpected" alarms in the meantime.

Detecting maintenance mode

Load balancers can use the **https://<node_address>/api/client/v2/status** REST API command to check whether a Conferencing Node is in maintenance mode.

Rebooting and shutting down a Conferencing Node

Occasionally, it may be necessary to reboot or shut down a Conferencing Node.

Rebooting a Conferencing Node

In many cases, Conferencing Nodes will reboot automatically as a result of certain configuration updates:

- Changing some specific **Global Settings** such as the media and signaling port ranges, or when you have enabled or disabled call protocols.
- Changing the system location of a Conferencing Node.
- Changing a Conferencing Node's role from a transcoding node to a proxying node or vice versa.

However, sometimes a manual reboot may be required, for example if a Conferencing Node fails to upgrade and it remains on a **Waiting for calls to clear** status.

There are two main ways to reboot a Conferencing Node:

- Log in to the hypervisor or your cloud service that is managing the host server and reboot the virtual machine using the relevant processes according to the management system.
- Connect to the Conferencing Node via SSH, log in as admin, and issue the command `sudo reboot`.

Shutting down a Conferencing Node

Sometimes you may need to shut down a Conferencing Node for maintenance, or if you are deleting it permanently. There are two main ways to do this:

- Log in to the hypervisor that is managing the host server and power off the virtual machine using the relevant processes according to the management system.
- Connect to the Conferencing Node via SSH, log in as admin, and issue the command `sudo poweroff`.

If your nodes are running in a cloud service, you should power off via SSH / serial console for a clean shutdown, and then use the cloud service's management portal to shutdown/stop the virtual machine to ensure that its associated resources are released.

Moving, restoring or changing the IP address of the Management Node

These procedures explain how to change the IP address of your Management Node, or physically move your Management Node to a new host server, without losing your Pexip Infinity platform configuration or invalidating your Pexip Infinity licenses.

Before performing any of these procedures we recommend that you follow standard IT practices, as per your company policy. For example, ensure that you take a snapshot of the Management Node prior to making any changes, and if possible take offline backups of the VMs.

In all cases you will have a loss of service while you perform the maintenance procedure.

Note that these instructions only apply to on-premises deployments. For cloud-based deployments, if you need to change the IP address of the Management Node you must decommission your entire deployment, terminating all of the existing instances, and then redeploy your system.

Changing the IP address of the Management Node (same physical host)

To change the IP address of your Management Node, where you want to keep the same physical host for the Management Node:

1. We recommend that you backup your current configuration:
   a. Go to **Utilities > Backup/Restore**.
   b. In the **Create backup** section, enter a **Passphrase** and then enter it again in the **Re-enter passphrase** field.
      The text entered here is used to encrypt the backup file. You must remember this text as it will be required if you need to subsequently restore the data from the file.
   c. Select **Create backup**.
      After a few seconds you see a message: "Successfully created the backup file: <file_name>" where <file_name> takes the format:
      pexip_backup_<hostname>_<version>_<build>_<date>_<time>.pexbak
   d. Download the file from the host VM:
      i. From the **Existing backup files** section at the bottom of the page, select **Download backup** for the file you have just created.
      ii. Follow your browser's prompts to save or download the file to your local file system.
      iii. If required, you can delete unwanted backup files from your host VM by selecting **Delete backup**.
      The system keeps on the host VM only the 5 most recent manually-taken backups and the 5 most recent automatic backups. Older backup files are deleted when a new file is created.
2. Return your Pexip Infinity license.
   You must return your license before decommissioning your current host installation.
   a. Go to **Platform > Licenses**.
   b. Select the license you want to deactivate.
   c. Make a note of the **License entitlement key**.
   d. Select **Return license**.
3. Decommission your Conferencing Nodes i.e. delete the Conferencing Nodes and remove the associated VMs. Follow this procedure for each Conferencing Node:
   a. Put the Conferencing Node into maintenance mode and wait until all calls on it have terminated.
   b. Shut down and remove the Conferencing Node VM.
      i. Log in to the VM manager, shut down the deleted Conferencing Node and then power it off.
      ii. Right-click on the Conferencing Node and select ***Delete from Disk*** (VMware) or ***Delete*** (Hyper-V / KVM).
   c. From the Pexip Infinity Administrator interface, go to **Platform > Conferencing NodeS**.
   d. Select the Conferencing Node to be deleted, and from the **Action** drop-down menu select ***Delete selected Conferencing Nodes***.
   e. Select **Go** and on the following page confirm that you want to delete the selected Conferencing Node by selecting **Yes, I'm sure**.
4. Re-run the installation wizard and supply the new IP address of the Management Node:
   a. Open a console window on the Management Node VM.
   b. At the login prompt, log in as **admin**.
   c. Enter the operating system password.
   d. Type `installwizard`.
      The Pexip installation wizard begins.
      At the **IP address** prompt, enter your new Management Node IP address.
      (If you interrupt the installation wizard, or it does not complete properly for any reason, you must reboot the Management Node and then rerun the installation wizard.)
5. Reapply your previous license:
   a. Go to **Platform > Licenses**.
   b. Select **Add License**.

c. In the **License entitlement key** field, enter the entitlement key from the 'old' Management Node that you noted previously.
d. Select **Save**.

The system will attempt to activate the license on the 'new' Management Node:

- If the license is activated successfully, you are returned to the **Licensing** page and the new license is shown under the **Licensing** section.
- If the activation attempt is unsuccessful (for example, if the Management Node was unable to establish a connection to the Pexip licensing server), or you selected **Manually activate**, the license is saved as a **Stored license request**. You must then activate it manually.

6. Re-provision your Conferencing Nodes.

   See Deploying new Conferencing Nodes for links to the appropriate instructions for your hypervisor.

Moving/restoring a Management Node to a new physical host (same IP address)

Note that if your Management Node is installed on VMware and you do not need to perform a fundamental redeployment of the Management Node, i.e. you only want to move it across host servers, then you should use vMotion to perform the move.

To move or restore a Management Node to a new physical host, where you are keeping the same IP address for the Management Node on the new host server:

1. Backup your current configuration:
   a. Go to **Utilities > Backup/Restore**.
   b. In the **Create backup** section, enter a **Passphrase** and then enter it again in the **Re-enter passphrase** field.

      The text entered here is used to encrypt the backup file. You must remember this text as it will be required if you need to subsequently restore the data from the file.
   c. Select **Create backup**.

      After a few seconds you see a message: "Successfully created the backup file: <file_name>" where <file_name> takes the format:

      pexip_backup_<hostname>_<version>_<build>_<date>_<time>.pexbak
   d. Download the file from the host VM:
      i. From the **Existing backup files** section at the bottom of the page, select **Download backup** for the file you have just created.
      ii. Follow your browser's prompts to save or download the file to your local file system.
      iii. If required, you can delete unwanted backup files from your host VM by selecting **Delete backup**.

      The system keeps on the host VM only the 5 most recent manually-taken backups and the 5 most recent automatic backups. Older backup files are deleted when a new file is created.
2. Return your Pexip Infinity license.

   You must return your license before decommissioning your current host installation.
   a. Go to **Platform > Licenses**.
   b. Select the license you want to deactivate.
   c. Make a note of the **License entitlement key**.
   d. Select **Return license**.
3. Install the Management Node on the new host server.

   Note that all of the configuration data that you enter via the installation wizard, including the IP address, will be subsequently replaced when you restore your backup.

   See Installation overview for links to the appropriate instructions for your hypervisor.
4. Restore the previous configuration to the new Management Node from your backup.
   a. Go to **Utilities > Backup/Restore**.
   b. In the **Restore backup** section, enter the **Passphrase**.

      The text entered here must be identical to the text that was used to create the backup file.
   c. Select **Choose File** and then choose the backup file that you want to restore.

      The file must be chosen from your local file system (you cannot select a file from the list of **Existing backup files**).

d. Select **Restore backup**.

   You are taken to the **Restore Backup** confirmation page.

   If instead, you see "Failed to restore the system from the backup file" with a "Decryption Error: decryption failed" message, the most likely reason for this is that you have entered an incorrect passphrase.

e. The confirmation page shows the date that the backup was taken, and the Management Node IP address that will be restored to this system.

   Select **Restore backup** to confirm the restoration.

If the restore is successful, after a few seconds you will see a "Successfully restored the backup file" message and the Management Node will reboot.

You need to wait for the node to return from rebooting before you can access it again.

5. Reapply your previous license:
   a. Go to **Platform > Licenses**.
   b. Select **Add License**.
   c. In the **License entitlement key** field, enter the entitlement key from the 'old' Management Node that you noted previously.
   d. Select **Save**.

   The system will attempt to activate the license on the 'new' Management Node:
   - If the license is activated successfully, you are returned to the **Licensing** page and the new license is shown under the **Licensing** section.
   - If the activation attempt is unsuccessful (for example, if the Management Node was unable to establish a connection to the Pexip licensing server), or you selected **Manually activate**, the license is saved as a **Stored license request**. You must then activate it manually.

Moving/restoring a Management Node to a new physical host (different IP address)

To manually move or restore a Management Node to a new physical host server, where you also want to use a different IP address for the Management Node:

1. Decommission your Conferencing Nodes i.e. delete the Conferencing Nodes and remove the associated VMs. Follow this procedure for each Conferencing Node:
   a. Put the Conferencing Node into maintenance mode and wait until all calls on it have terminated.
   b. Shut down and remove the Conferencing Node VM.
      i. Log in to the VM manager, shut down the deleted Conferencing Node and then power it off.
      ii. Right-click on the Conferencing Node and select ***Delete from Disk*** (VMware) or ***Delete*** (Hyper-V / KVM).
   c. From the Pexip Infinity Administrator interface, go to **Platform > Conferencing NodeS**.
   d. Select the Conferencing Node to be deleted, and from the **Action** drop-down menu select ***Delete selected Conferencing Nodes***.
   e. Select **Go** and on the following page confirm that you want to delete the selected Conferencing Node by selecting **Yes, I'm sure**.
2. Backup your current configuration:
   a. Go to **Utilities > Backup/Restore**.
   b. In the **Create backup** section, enter a **Passphrase** and then enter it again in the **Re-enter passphrase** field.

      The text entered here is used to encrypt the backup file. You must remember this text as it will be required if you need to subsequently restore the data from the file.
   c. Select **Create backup**.

      After a few seconds you see a message: "Successfully created the backup file: <file_name>" where <file_name> takes the format:

      pexip_backup_<hostname>_<version>_<build>_<date>_<time>.pexbak
   d. Download the file from the host VM:
      i. From the **Existing backup files** section at the bottom of the page, select **Download backup** for the file you have just created.

ii. Follow your browser's prompts to save or download the file to your local file system.

iii. If required, you can delete unwanted backup files from your host VM by selecting **Delete backup**.

The system keeps on the host VM only the 5 most recent manually-taken backups and the 5 most recent automatic backups. Older backup files are deleted when a new file is created.

3. Return your Pexip Infinity license.

You must return your license before decommissioning your current host installation.

a. Go to **Platform > Licenses**.
b. Select the license you want to deactivate.
c. Make a note of the **License entitlement key**.
d. Select **Return license**.

4. Install the Management Node on the new host server.

All the configuration data you enter via the installation wizard, including the IP address, will be subsequently replaced when you restore your backup. Thus, at this stage you can use the same IP address as you used for your previous Management Node installation.

See Installation overview for links to the appropriate instructions for your hypervisor.

5. Restore the previous configuration to the new Management Node from your backup (this also restores the original Management Node IP address — you set the new address in the next step):

a. Go to **Utilities > Backup/Restore**.
b. In the **Restore backup** section, enter the **Passphrase**.

The text entered here must be identical to the text that was used to create the backup file.

c. Select **Choose File** and then choose the backup file that you want to restore.

The file must be chosen from your local file system (you cannot select a file from the list of **Existing backup files**).

d. Select **Restore backup**.

You are taken to the **Restore Backup** confirmation page.

If instead, you see "Failed to restore the system from the backup file" with a "Decryption Error: decryption failed" message, the most likely reason for this is that you have entered an incorrect passphrase.

e. The confirmation page shows the date that the backup was taken, and the Management Node IP address that will be restored to this system.

Select **Restore backup** to confirm the restoration.

If the restore is successful, after a few seconds you will see a "Successfully restored the backup file" message and the Management Node will reboot.

You need to wait for the node to return from rebooting before you can access it again.

6. Re-run the installation wizard and supply the new IP address of the Management Node:

a. Open a console window on the Management Node VM.
b. At the login prompt, log in as **admin**.
c. Enter the operating system password.
d. Type `installwizard`.

The Pexip installation wizard begins.

At the **IP address** prompt, enter your new Management Node IP address.

(If you interrupt the installation wizard, or it does not complete properly for any reason, you must reboot the Management Node and then rerun the installation wizard.)

7. Reapply your previous license:

a. Go to **Platform > Licenses**.
b. Select **Add License**.
c. In the **License entitlement key** field, enter the entitlement key from the 'old' Management Node that you noted previously.
d. Select **Save**.

The system will attempt to activate the license on the 'new' Management Node:

- If the license is activated successfully, you are returned to the **Licensing** page and the new license is shown under the **Licensing** section.
- If the activation attempt is unsuccessful (for example, if the Management Node was unable to establish a connection to the Pexip licensing server), or you selected **Manually activate**, the license is saved as a **Stored license request**. You must then activate it manually.

8. Re-provision your Conferencing Nodes.

   See Deploying new Conferencing Nodes for links to the appropriate instructions for your hypervisor.

Bulk import/export of service configuration data

If you want to configure your Pexip Infinity platform with a large number of Virtual Meeting Rooms, Virtual Auditoriums, Virtual Receptions, Automatically Dialed Participants (APDs) or device aliases, you can import the configuration for each of these items from a CSV file.

You can also export all of your existing service configuration data to a CSV file. You may want to do this for backup purposes or to transfer configuration between, for example, a test system and a production system.

You can also use a CSV file to import multiple endpoints for One-Touch Join.

You can import Virtual Meeting Rooms and device aliases from directory information contained in a Windows Active Directory LDAP server, or any other LDAP-accessible database. For more information, see Provisioning VMRs, devices and users from Active Directory via LDAP.

Preparing the CSV file for import

When formatting your import file:

- A header row in the CSV file is optional. If included, it must use the same field names as shown in the following sections, but you may change the order of the fields. If a header row is not used, fields must be in the same order as shown for each import file type.
- All non-blank fields must contain valid data, for example the **pin** field must only contain digits and the **dtmf_sequence** field must only contain digits or commas.
- If non-ASCII characters are used, the file must be encoded as UTF-8 text.
- Fields with values of **true** or **false** are not case-sensitive. All other fields are case-sensitive.
- Values may optionally be enclosed in double quotation marks; any strings containing commas must be enclosed in double quotation marks e.g. "description for x, y and z".

Note that you can perform an export of existing data to produce an example file in the correct format.

The sections below contain the specific field requirements for your service, device alias or ADP import file.

Services (Virtual Meeting Rooms, Virtual Auditoriums, Virtual Receptions and Media Playback Services)

This section explains how to format a CSV import file of Virtual Meeting Rooms, Virtual Auditoriums, Virtual Receptions or Media Playback Services, and how to perform an import or export.

This allows you to import all configuration apart from the Automatically Dialed Participants (ADPs) to be associated with each service. These can be imported separately; see Automatically dialed participants (ADPs) below.

You must prepare separate CSV files for Virtual Meeting Rooms, Virtual Auditoriums, Virtual Receptions and Media Playback Services, using the fields as shown below.

Virtual Meeting Room file format

```
name,description,pin,allow_guests,guest_pin,alias_alias,alias_description,tag,max_callrate_in,max_callrate_out,call_type,host_
view,enable_overlay_text,ivr_theme_name,participant_limit,primary_owner_email_address,guests_can_present,enable_chat,crypto_
mode,max_pixels_per_second,enable_active_speaker_indication,host_identity_provider_group_name,guest_identity_provider_group_
name,non_idp_participants,live_captions_enabled,softmute_enabled,denoise_enabled,direct_media,direct_media_notification_
duration,breakout_rooms,pinning_config
```

Virtual Auditorium file format

```
name,description,pin,allow_guests,guest_pin,alias_alias,alias_description,tag,max_callrate_in,max_callrate_out,call_type,host_
view,guest_view,force_presenter_into_main,enable_overlay_text,ivr_theme_name,participant_limit,mute_all_guests,guests_can_
present,enable_chat,crypto_mode,max_pixels_per_second,host_identity_provider_group_name,guest_identity_provider_group_name,non_idp_
participants,live_captions_enabled,softmute_enabled,denoise_enabled,breakout_rooms,guests_can_see_guests,pinning_config
```

Virtual Reception file format

```
name,description,alias_alias,alias_description,tag,max_callrate_in,max_callrate_out,call_type,ivr_theme_name,mssip_proxy_
name,teams_proxy_name,match_string,replace_string,system_location_name,post_match_string,post_replace_string,two_stage_dial_
type,gms_access_token_name,crypto_mode,max_pixels_per_second
```

Media Playback Service file format

```
name,description,pin,allow_guests,guest_pin,media_playlist_name,on_completion,alias_alias,alias_description,tag,ivr_theme_
name,host_identity_provider_group_name,guest_identity_provider_group_name,non_idp_participants
```

where (as appropriate for the relevant service type):

Field name	Content
name	The name used to refer to this Virtual Meeting Room, Virtual Auditorium, Virtual Reception or Media Playback Service. This field cannot be left blank. If you can access this service via a Virtual Reception then the service **Name** entered here is shown to conference participants as they are transferred into the service (it is overlaid onto the **virtual_reception_connecting** splash screen of the theme associated with the Virtual Reception that is transferring the call).
description	An optional description of the Virtual Meeting Room, Virtual Auditorium, Virtual Reception or Media Playback Service.
pin	This optional field allows you to set a secure access code that must be entered by participants before they can join the conference. If a **Guest PIN** has also been set, then the **PIN** will apply to the conference Host(s) only. For more information, see About PINs, Hosts and Guests.
allow_guests *	Determines whether the conference will allow participants with Guest privileges. For more information, see About PINs, Hosts and Guests. • true: the conference has two types of participants: Hosts and Guests. The **pin** to be used by Hosts must be specified. A **guest_pin** can optionally be specified; if a **guest_pin** is not specified, Guests can join without a PIN. • false: all participants have Host privileges Default: false
guest_pin	This optional field allows you to set a secure access code that must be entered by Guests before they can join the conference. For more information, see About PINs, Hosts and Guests.
media_playlist_name	The media playlist for the Media Playback Service.
on_completion	The action to perform on playlist completion for the Media Playback Service.

Field name	Content
alias_alias	The alias that, when received by Pexip Infinity, is used to route the call to this service (Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service). The alias entered here must match the alias as it is received by Pexip Infinity. Wildcards and regular expressions are not supported. In most cases, the alias received by Pexip Infinity is the same as the alias that the participant used to call the service, but there are some exceptions, described in About aliases and access numbers. You may also want to define multiple aliases for the same service to ensure that it can be accessed by devices and protocols that enforce specific alias formats — for more information, see Using multiple aliases to access the same service.
alias_ description	An optional description of the alias. This is useful if you have more than one alias for a service. Note that this description may be displayed to users of registered Pexip apps who are performing a directory search.
tag	This optional field lets you assign a unique identifier to this service, which you can then use to track use of the service.
max_callrate_ in	The maximum media bandwidth in kbps that Pexip Infinity will receive from each individual participant dialed in to the service. Range 128 to 4096 kbps. For more information see Managing and restricting call bandwidth.
max_callrate_ out	The maximum media bandwidth in kbps that Pexip Infinity will send to each individual participant dialed in to the service. Range 128 to 4096 kbps. For more information see Managing and restricting call bandwidth.
call_type *	Allows you to limit the media content of the conference. For more information, see Controlling media capability. Valid values are: • "video": main video plus presentation • "video-only": main video only • "audio": audio-only Default: "video"
host_view *	The maximum number of other participants that each host participant will see, and the layout used to show them. For more information, see Conference layouts and speaker names. Valid values are: • "one_main_zero_pips": full-screen main speaker only • "one_main_seven_pips": large main speaker and up to 7 other participants • "one_main_twentyone_pips": main speaker and up to 21 other participants • "two_mains_twentyone_pips": 2 main speakers and up to 21 other participants • "one_main_thirtythree_pips": 1 small main speaker and up to 33 other participants • "four_mains_zero_pips": 2 x 2 layout, up to a maximum of 4 speakers • "nine_mains_zero_pips": 3 x 3 layout, up to a maximum of 9 speakers • "sixteen_mains_zero_pips": 4 x 4 layout, up to a maximum of 16 speakers • "twentyfive_mains_zero_pips": 5 x 5 layout, up to a maximum of 25 speakers • "five_mains_seven_pips": Adaptive Composition layout • "one_main_one_pip": large main speaker with one overlaying participant • "one_main_nine_around": one main speaker and up to 9 other participants • "one_main_twelve_around": large main speaker and up to 12 other participants • "two_mains_eight_around": two main speakers and up to 8 other participants Default: "one_main_seven_pips"

Field name	Content
guest_view *	The maximum number of Host participants that each Guest participant will see in a Virtual Auditorium, and the layout used to show them. (Guests will only see Hosts; they can hear but not see any of the other Guests.) For more information, see Conference layouts and speaker names. Valid values are: • "one_main_zero_pips": full-screen main speaker only • "one_main_seven_pips": large main speaker and up to 7 other participants • "one_main_twentyone_pips": main speaker and up to 21 other participants • "two_mains_twentyone_pips": 2 main speakers and up to 21 other participants • "one_main_thirtythree_pips": 1 small main speaker and up to 33 other participants • "four_mains_zero_pips": 2 x 2 layout, up to a maximum of 4 speakers • "nine_mains_zero_pips": 3 x 3 layout, up to a maximum of 9 speakers • "sixteen_mains_zero_pips": 4 x 4 layout, up to a maximum of 16 speakers • "twentyfive_mains_zero_pips": 5 x 5 layout, up to a maximum of 25 speakers • "five_mains_seven_pips": Adaptive Composition layout (you must also set Host view to Adaptive Composition) • "one_main_one_pip": large main speaker with one overlaying participant • "one_main_nine_around": one main speaker and up to 9 other participants • "one_main_twelve_around": large main speaker and up to 12 other participants • "two_mains_eight_around": two main speakers and up to 8 other participants Default: "one_main_seven_pips"
force_presenter_into_main *	When a presentation is being shown in a Virtual Auditorium, this option controls whether the main speaker position shows the presenter or the current speaker. For more information, see Conference layouts and speaker names. • true: the Host sending the presentation stream will always hold the main video position • false: the main video position is voice-switched Default: false
enable_active_speaker_indication *	When active speaker display is enabled, the display name or alias of the current speaker is shown across the bottom of their video image. • true: active speaker is indicated • false: active speaker is not indicated Default: false
enable_overlay_text *	If participant name overlays are enabled, the display names or aliases of all participants are shown in a text overlay along the bottom of their video image. • true: participant names are shown • false: participant names are not shown Default: false
ivr_theme_name	The name of the theme to use with the service. If no theme is specified, the default Pexip theme is used. For more information, see Customizing conference images and voice prompts using themes.
participant_limit	This optional field allows you to limit the number of participants allowed to join this VMR. For more information see Limiting the number of participants.
primary_owner_email_address	The email address of the owner of the VMR.

Field name	Content
mute_all_guests *	Determines whether Guest participants are muted when they first join the conference in a Virtual Auditorium. • true: mute Guests when they first join the conference • false: do not mute Guests when they first join the conference Default: false
guests_can_present *	Controls whether the Guests in the conference are allowed to present content. • true: Guests and Hosts can present into the conference • false: only Hosts can present Default: true
enable_chat *	Determines whether chat message support is enabled in the conference. Valid values are: • "default": as per the global configuration setting • "yes": chat is enabled • "no": chat is disabled Default: "default"
crypto_mode *	Controls the media encryption requirements for participants connecting to this service. • <null>: use the global media encryption setting. • "besteffort": each participant will use media encryption if their device supports it, otherwise the connection will be unencrypted. • "on": all participants (including RTMP participants) must use media encryption. • "off": all H.323, SIP and MS-SIP participants must use unencrypted media. (RTMP participants will use encryption if their device supports it, otherwise the connection will be unencrypted.) Default: <null> (use global setting)
max_pixels_per_second *	Controls the maximum call quality for participants connecting to this service: • <null>: use the global maximum call quality setting. • "sd": each participant is limited to SD quality. • "hd": each participant is limited to HD (720p) quality. • "fullhd": allows any endpoint capable of Full HD to send and receive its main video at 1080p. Default: <null> (use global setting)
host_identity_provider_group_name	The set of Identity Providers to be offered to Hosts to authenticate with, in order to use the service. If this is blank, Hosts are not required to authenticate. Default: <null>
guest_identity_provider_group_name	The set of Identity Providers to be offered to Guests to authenticate with, in order to use the service. If this is blank, Guests are not required to authenticate. Default: <null>
non_idp_participants *	Determines whether participants joining a SSO-protected service from devices other than the web app (for example SIP or H.323 endpoints) are allowed to dial in to the service. • "disallow_all": these devices are placed in a waiting room where they must wait to be admitted by a Host. • "allow_if_trusted": these devices may join the service if they are locally registered. They must still enter a Host PIN or Guest PIN if either is required. All other devices are placed in a waiting room where they must wait to be admitted by a Host. Default: "disallow_all"

Field name	Content
live_captions_enabled	Determines whether participants using the web app are given the option to turn on the live captions for conferences using this service. • "default": as per global configuration setting. • "yes": live captions may be enabled. • "no": live captions cannot be enabled. Default: "default"
softmute_enabled	Controls whether the Softmute* feature is enabled for this VMR. • true: Softmute is enabled for this VMR. • false: Softmute is disabled for this VMR. Default: false
denoise_enabled	Controls whether the Denoise* feature is enabled for this VMR. • true: Denoise is enabled for this VMR. • false: Denoise is disabled for this VMR. Default: false
direct_media	Allows this VMR to use direct media between participants. When enabled, the VMR provides non-transcoded, encrypted, point-to-point calls between any two WebRTC participants. The options are: • "best_effort": use direct media in this VMR where possible (when there are two WebRTC participants only), otherwise use standard, transcoded media connections via a Conferencing Node. • "always": allow direct media calls only — this limits all calls in this VMR to two WebRTC participants. • "never": do not use direct media in this VMR. Default: "never" Note that direct media is not supported if breakout rooms are also enabled for this VMR.
direct_media_notification_duration	The number of seconds to show a notification to participants before being escalated into a transcoded call, or de-escalated into a direct media call. Range: 0 to 30 seconds. Default: 0 seconds
breakout_rooms	Whether Breakout Rooms are enabled for the conference. Default: false
guests_can_see_guests	Controls whether Guests see other Guests during a conference. This applies to Virtual Auditoriums only. Default: no_hosts
mssip_proxy_name	The name of the Skype for Business server to use to resolve the SfB Conference ID entered by the user in the Virtual Reception.
pinning_config	The name of the pinning configuration to apply (from the theme associated with the conference).
teams_proxy_name	The name of the Teams Connector to use to resolve Microsoft Teams meeting codes entered in a Virtual Reception. If you do not specify anything, the Teams Connector associated with the outgoing location is used.
match_string	An optional regular expression used to match against the alias entered by the caller into the Virtual Reception. If the entered alias does not match the expression, the Virtual Reception will not route the call. If this field is left blank, any entered alias is permitted.

Field name	Content
replace_string	An optional regular expression used to transform the alias entered by the caller into the Virtual Reception. (Only applies if a regex match string is also configured and the entered alias matches that regex.) Leave this field blank if you do not want to change the alias entered by the caller.
system_ location_name	This is an optional field used in conjunction with the **two_stage_dial_type** setting, when a type other than ***regular*** is selected. If specified, a Conferencing Node in this system location will perform the SfB Conference ID lookup on the SfB server, or the Microsoft Teams or Google Meet code lookup, as appropriate. We recommend that a location is specified here, otherwise the transcoding node hosting the Virtual Reception will perform the lookup (which may lead to routability issues).
two_stage_ dial_type *	The type of Virtual Reception: • "regular": the default type of Virtual Reception, used for routing calls to VMRs, or to other devices and call control systems via the Infinity Gateway. • "mssip": a special type of Virtual Reception, used when you want to provide an IVR gateway to scheduled and ad hoc Skype for Business meetings. • "gms": a special type of Virtual Reception, used when you want to provide an IVR gateway to scheduled and ad hoc Google Meet meetings. • "teams": a special type of Virtual Reception, used when you want to provide an IVR gateway to scheduled and ad hoc Microsoft Teams meetings. Default: "regular"

* If this field is left blank, the default value is used.

Multiple aliases

For services with more than one alias, you must add an additional record for each additional alias that repeats all fields except the **alias_alias** and **alias_description**:

```
name1,description1,pin1,allow_guests,guest_pin1,alias_alias1,alias_description1,...
name1,description1,pin1,allow_guests,guest_pin1,alias_alias2,alias_description2,...
```

Duplicates

If any records in the CSV file have the same **name** field, and any of the other fields apart from **alias_alias** and **alias_description** are different, only one service with that name will be created. This service will use the last record that was imported.

If any records in the CSV file have the same **name** as an existing service, the existing configuration will be overwritten by the imported service's configuration.

Examples

To import a Virtual Meeting Room called **alice** with a single alias of **meet.alice**, and a second Virtual Meeting Room called **bob** with aliases **meet.bob** and **meet.bobby**, you would create the following CSV file:

```
alice,,,,,meet.alice
bob,,,,,meet.bob
bob,,,,,meet.bobby
```

To import Virtual Meeting Rooms for Alice, Bob and Charlie that each have different Host and Guest PINs you would create the following CSV file:

```
alice,,123456,True,678900,meet.alice
bob,,456789,True,987654,meet.bob
charlie,,5432145,True,5556789,meet.charlie
```

Importing the CSV file

Virtual Meeting Rooms, Virtual Auditoriums and Virtual Receptions require separate CSV files.

To import the data in the CSV file to Pexip Infinity:

1. On the Pexip Infinity Administrator interface, go to either:
   - **Services > Virtual Meeting Rooms** or
   - **Services > Virtual Auditoriums** or
   - **Services > Virtual Receptions**.
2. Select **Import**.
3. Choose the CSV file to import and select **Save**.

Exporting existing service configuration

You can export all of your existing Virtual Meeting Room, Virtual Auditorium or Virtual Reception configuration data to a CSV file. This produces a CSV file in the same format as that used for importing configuration data (as described above). The file includes a header row.

This feature exports all configuration except the Automatically Dialed Participants associated with each service.

To export the data:

1. On the Pexip Infinity Administrator interface, go to either:
   - **Services > Virtual Meeting Rooms** or
   - **Services > Virtual Auditoriums** or
   - **Services > Virtual Receptions**.
2. Select **Export**. This takes you to the **Export Configuration** page.
3. Select **Download**.
4. Follow your browser's prompts to save or open the file.

Device aliases

This section explains how to format a CSV import file of device aliases, and how to perform an import or export.

You must prepare the CSV file using the fields as shown below:

```
alias,tag,description,username,password,primary_owner_email_address,enable_sip,enable_h323,enable_infinity_connect_non_sso,enable_infinity_connect_sso,enable_standard_sso
```

where:

Field name	Content
alias	The alias URI that a device/client can register to Pexip Infinity. The alias must be an exact match; regular expressions are not supported. It cannot be blank.
tag	This optional field lets you assign a unique identifier to this device, which you can then use to track use of the service.
description	An optional description of the device.
username	The username with which to authenticate the device. The username and password credentials are optional and the device is not challenged if no credentials are entered. These credentials do not apply when using Pexip app registration via SSO services.
password	The password with which to authenticate the device, in association with the username. When passwords are exported to a CSV file, they are encrypted. You can re-import the encrypted form of the password and it will be saved as the original, providing you use the same Management Node for the import and export.
primary_owner_email_address	The email address of the owner of the device alias.
enable_sip	Allows this device alias to register over the SIP protocol.

Field name	Content
enable_h323	Allows this device alias to register over the H.323 protocol.
enable_infinity_connect_non_sso	Allows a legacy Connect desktop appto register using this alias (not using SSO services).
enable_infinity_connect_sso	Allows a legacy Connect desktop app to register using this alias, using AD FS to authenticate the registration. This method of authentication is being deprecated and is not supported by the new Pexip apps.
enable_standard_sso	Allows a Pexip app for Windows to register using this alias, using a SAML or OIDC Identity Provider to authenticate the registration.

Duplicates

If any records in the CSV file have the same **alias** field, only one device alias with that name will be created. This device alias will use the last record that was imported.

If any records in the CSV file have the same **alias** as an existing device alias, the existing configuration will be overwritten by the imported device alias's configuration.

Examples

```
alias,tag,description,username,password,sync_tag,primary_owner_email_address
```

To create aliases that will allow Alice and Bob to register their devices to Pexip Infinity with a username and password, you would create the following CSV file:

```
alice@example.com,,,alice,1234
bob@example.com,,,bob,5678
```

Importing the CSV file

To import the data in the CSV file to Pexip Infinity:

1. On the Pexip Infinity Administrator interface, go to **Users & Devices > Device Aliases**.
2. From the bottom of the page, select **Import**.
3. Choose the CSV file to import and select **Save**.

If you are importing device data that contains encrypted passwords i.e. a CSV file of device data that you have previously exported, then you can only re-import the file if you are using the same Management Node. It may work across different software versions but we do not ensure that it is forwards/backwards compatible across software versions.

Exporting existing device alias configuration

You can export all of your existing device alias configuration data to a CSV file. This produces a CSV file in the same format as that used for importing configuration data. The file includes a header row.

To export the data:

1. On the Pexip Infinity Administrator interface, go to **Users & Devices > Device Aliases**.
2. From the bottom of the page, select **Export**. This takes you to the **Export Device Alias Configuration** page.
3. Select **Download**.
4. Follow your browser's prompts to save or open the file.

Automatically dialed participants (ADPs)

This section explains how to format a CSV import file of Automatically dialed participants (ADPs), and how to perform an import or export.

You must prepare the CSV file using the fields as shown below:

```
alias,remote_display_name,description,protocol,dtmf_sequence,role,streaming,keep_conference_alive,call_type,routing,presentation_
url,system_location_name,conference_name
```

where:

Field name	Content
alias	The alias of the participant to dial when a conference starts. It cannot be blank.
remote_ display_name	An optional friendly name for this participant. This may be used instead of the participant's alias in participant lists and as a text overlay in some layout configurations.
description	An optional description of the Automatically Dialed Participant.
protocol *	The protocol to use to place the outgoing call: • "sip" • "h323" • "rtmp" • "mssip" (for calls to Microsoft Skype for Business) • "auto" (to use Call Routing Rules) Default: sip ⓘ The protocol is ignored if the **routing** field is set to ***routing_rule***.
dtmf_sequence	An optional DTMF sequence to transmit after the call to the dialed participant starts. ⓘ If one or more commas are used in the DTMF sequence (as a 2-second pause), the entire string within the field must be contained in double quotes.
role *	The level of privileges the participant has in the conference: • "chair": the participant has Host privileges • "guest": the participant has Guest privileges Default: guest
streaming *	Identifies the dialed participant as a streaming or recording device. • true: the participant is a streaming or recording device • false: the participant is not a streaming or recording device Default: false
keep_ conference_ alive *	Determines whether the conference continues when all other non-ADP participants have disconnected: • "keep_conference_alive": the conference continues to run until this participant disconnects (applies to Hosts only). • "keep_conference_alive_if_multiple": the conference continues to run as long as there are two or more "keep_ conference_alive_if_multiple" participants and at least one of them is a Host. • "keep_conference_alive_never": the conference terminates automatically if this is the only remaining participant. Default: keep_conference_alive_if_multiple
call_type *	Allows you to limit the media content of the call. The participant being called will not be able to escalate beyond the selected capability. For more information, see Controlling media capability. Valid values are: • "video": main video plus presentation • "video-only": main video only • "audio": audio-only Default: "video"

Field name	Content
routing *	Specifies how to route the call: • "manual": uses the requested **protocol** and the defaults for the specified **system_location_name**. • "routing_rule": routes the call according to the configured Call Routing Rules. This means that the dialed alias must match an outgoing Call Routing Rule for the call to be placed (using the protocols, outgoing location and call control systems etc. as configured for that rule). Default: "manual"
presentation_url	This additional parameter can be specified for RTMP calls to send the presentation stream to a separate RTMP destination.
system_location_name	The location of the Conferencing Node from which to place the call.
conference_name	The name of the Virtual Meeting Room or Virtual Auditorium from which this participant will be dialed automatically whenever a conference using that service starts.
* If this field is left blank, the default value is used.	

Duplicates

You can upload any number of records with the same **alias** field, as long as the **conference_name** field is different for each.

If any records in the CSV file have the same **alias** and **conference_name** fields, only one ADP for that conference name will be created. This ADP will use the last record that was imported.

If any records in the CSV file have the same **alias** and **conference_name** as an existing ADP, the existing configuration will be overwritten by the imported ADP's configuration.

Examples

To import an ADP for Alice's endpoint (**alice@example.com**) that will be dialed in as a Host when her VMR (**meet Alice**) is called, and as a Guest when the sales team's Virtual Auditorium (**meet Sales**) is called, you would create the following CSV file:

```
alice@example.com,Alice Jones,,,,chair,,,,,,,meet Alice
alice@example.com,Alice Jones,,,,guest,,,,,,,meet Sales
```

Importing the CSV file

To import the data in the CSV file to Pexip Infinity:

1. On the Pexip Infinity Administrator interface, go to **Services > Automatically Dialed Participants**.
2. From the bottom of the page, select **Import**.
3. Choose the CSV file to import and select **Save**.

Exporting existing ADP configuration

You can export all of your existing ADP configuration data to a CSV file. This produces a CSV file in the same format as that used for importing configuration data. The file includes a header row.

To export the data:

1. On the Pexip Infinity Administrator interface, go to **Services > Automatically Dialed Participants**.
2. From the bottom of the page, select **Export**. This takes you to the **Export Automatically Dialed Participant Configuration** page.
3. Select **Download**.
4. Follow your browser's prompts to save or open the file.

Best practices

Performing routine checks

Here is a list of tasks that you should perform on a regular basis to ensure that your Pexip Infinity deployment is running optimally.

Check live status

The Live view page gives a real-time overview of your deployment, and should be viewed regularly. In particular, check for the following:

- Whether there are any alarms on the Management Node or Conferencing Nodes. Any alarms are listed in the top right of the Live view page, and any Conferencing Nodes that have an alarm are shown on the graph with an orange or red outline. For more information, see Viewing alarms.
- The current usage and capacity of your deployment.
- Any conferences or conference participants that are experiencing call quality issues.

For more information, see Viewing live and historical platform status.

Check alarm history

Active alarms are cleared as soon as the issue is resolved, so you may not always be aware of temporary issues such as lack of capacity or licenses. However, the **History & Logs > Alarm History** page shows the details of all alarms that have been raised, so it is useful to review this information to see if there are any trends that need to be addressed.

For more information, see Viewing alarms.

Check LDAP sync status

If you are Provisioning VMRs, devices and users from Active Directory via LDAP in your deployment, periodically go to **Status > LDAP Sync** and check each configured sync template for any warnings or errors. If there are any, you should investigate, remedy and re-sync. For more information, see Viewing LDAP sync template results and Troubleshooting LDAP server connections.

Check usage history

For each location, go to the historical usage graph and review the level of usage. If usage is nearing existing capacity, you may need to consider adding more capacity.

Check for upcoming software releases

Make sure you are aware of the features that are coming in future releases of the Pexip Infinity software, and ensure you have a plan in place for upgrading your current deployment.

- Sign up for the Pexip newsletter at https://www.pexip.com (enter your address in the subscription section at the bottom of the page).
- Review what's in the current release at https://docs.pexip.com/admin/whats_new.htm.
- Review whether any fixes and new features are relevant for your environment at https://docs.pexip.com/admin/release_notes.htm.

Perform regular backups

You should take regular backups of the configuration data on your Management Node. Conferencing Nodes do not generally require backups, as they receive all their configuration information from the Management Node.

For more information, see Backing up and restoring configuration.

Advanced checks

The following checks require an in-depth understanding of the Pexip Infinity platform. We recommend that you attend an appropriate training course — for more information, visit the Pexip Academy.

- Query participant history and filter on `tx_packet_loss__gte=4` and `rx_packet_loss__gte=4` to see any calls that have more than 4% packet loss. If so, investigate the cause and consider whether further action is required.
- Search administrator and support logs for `irregular_pulse`. This is usually indicative of over-committed hardware. For more information, see Hardware instability detected.
- Search administrator and support logs for `incident`. If there are any, contact your Pexip authorized support representative. If your deployment is Automatically reporting errors, you may have already been contacted about the incident.
- Search administrator log for `error` and `warning`.

Security best practices

The Pexip Infinity platform uses industry-standard encryption and security protocols to control access and to prevent unwanted audiences from listening in and stealing communications. It is also designed to comply with the strictest US Federal requirements.

This topic describes how you can help secure your Pexip Infinity deployment from network-based attacks and integrate Pexip Infinity into your existing network security architecture. For a full discussion of general best practices to prevent your videoconferencing deployment from being compromised, read this Pexip white paper.

Attacks on the operating system and management interfaces

Pexip Infinity uses a customized, cut-down version of Linux which has been designed to avoid exposing unnecessary network services and thus naturally limits the "attack surface" available to an attacker. Pexip regularly releases new software versions which incorporate the very latest operating system security patches (see Pexip security bulletins for more information). In addition, all Pexip Infinity APIs and management interfaces are password or PIN protected.

To mitigate attacks on the operating system and management interfaces:

- Disable the local management account and instead authenticate and authorize login accounts via a centrally managed Windows Active Directory / LDAP-accessible server or an OpenID Connect provider. You can also limit what certain administrator groups can do (for example, support teams may not need the ability to deploy Conferencing Nodes). You can also configure client-certificate-based access if required.
- Configure Global settings to define an administration session timeout and login banner text.
- Configure Global settings to disable access over SSH if it is not required.
- If you use Simple Network Management Protocol (SNMP), use SNMPv3 to both encrypt and authenticate incoming SNMP discovery and monitoring between the Management Node and the SNMP manager (see Monitoring via SNMP).
- Use secure NTP to obtain accurate system time (see Syncing with NTP servers).
- Use a firewall to prevent unauthorized network traffic from reaching your devices, and to block unauthorized access to services and network ports that are not required to be exposed for video communications to work correctly. For example, the management user interface HTTPS, SNMP and SSH services do not usually need to be accessible to anyone other than your network administrator. See Pexip Infinity port usage and firewall guidance for more information.
- Use secure remote logging via the industry-standard syslog protocol (see Using a syslog server).
- Use the latest release of Pexip Infinity software.

DOS/DDOS attacks

Pexip Infinity, as with many network services, can be vulnerable to Denial of Service (DOS) or a Distributed Denial of Service (DDOS) attacks. To mitigate such attacks:

- Use a firewall to block unauthorized access to services and network ports that are not required to be exposed for video communications (see Pexip Infinity port usage and firewall guidance).
- Disable unneeded services altogether. You can do this by configuring services and protocols via Global settings.

Eavesdropping and rogue calls

Pexip Infinity supports the latest industry standards for encryption for communication with end-user devices and employs IPsec security to provide strong protection of all inter-node communications (see Encryption methodologies). Inter-node traffic is restricted to only protocols that are expected; any unexpected traffic/protocols are dropped. Pexip Infinity also works with all popular video call control systems in the market today, and can connect legacy devices in the corporate network (which may not themselves support encryption) and encrypt on behalf of those devices when connecting to external devices that do support encryption.

Common attacks on videoconferencing systems include rogue calls — such as Spam Over Internet Telephony (SPIT) or toll fraud call attempts — that are targeted at an organization's SIP (or, more rarely, H.323) infrastructure. Typically the attacker will place a large volume of calls to numeric aliases to try and gain access to a VoIP to PSTN gateway — and, if successful, use the gateway to commit toll fraud.

To mitigate eavesdropping and rogue calls:

- Use proper TLS/SSL certificates from a respectable source on all Pexip Infinity nodes so that clients and other servers can verify that they have genuinely connected to the correct Pexip Infinity server and not an impostor or "man-in-the-middle" (see Managing TLS and trusted CA certificates). You can also:
  - Use OCSP to check the status of certificates.
  - If appropriate, enable SIP TLS verification / mutual authentication to require that peer certificates are verified. Typically, this is recommended for Microsoft Skype for Business integrations, but most other videoconferencing deployments are **not** equipped to provide a proper SIP TLS certificate so SIP TLS verification is not recommended unless you only expect calls from a closed circle of users.

  Note that all Pexip Infinity nodes use HSTS (HTTP Strict Transport Security) to ensure greater security.
- Enable PIN protection on your Virtual Meeting Rooms:
  - use a long (at least 6 digits), unique, randomly-generated PIN for each Virtual Meeting Room (you can automate this if you are Provisioning VMRs, devices and users from Active Directory via LDAP),
  - use a trailing # at the end of each PIN to disguise the length,
  - regularly change the PIN on each VMR.
- When using numeric aliases for your VMRs, make them at least 6 digits long.
- If you are using the Pexip Reverse Proxy and TURN Server, you should enable fail2ban on the reverse proxy.
- Ensure that **PIN brute force resistance** and **VOIP scanner resistance** are enabled, either globally or in specific locations.
- When using the Infinity Gateway, consider:
  - Limiting your Call Routing Rules applicability to only allow calls to be made from devices that are registered to Pexip Infinity, or are received in certain locations.
  - Limiting calls to certain incoming protocols e.g. SIP only.
  - Only allowing calls to be placed to devices that are registered to Pexip Infinity.
  - Using precise regular expressions in your Call Routing Rules for the domains, dial plans and alias patterns that you want to support.

  See Configuring Call Routing Rules for more information about routing calls via the Infinity Gateway service.
- Pexip Infinity disables SIP UDP by default (as SIP UDP is the most commonly targeted signaling service). However, consider disabling other unused protocols (see Enabling and disabling SIP, H.323, WebRTC and RTMP).
- Disable any other features that are not required in your deployment, such as the ability to make outbound calls, and support for the Pexip app / API clients (see Global settings).
- Protect against toll fraud by ensuring that access to your VoIP gateway or your VoIP provider's SIP trunk and other important resources is carefully restricted – especially for unauthenticated external SIP/H.323 callers (see PSTN gateways and toll fraud).
- Implement a local policy script such as our example Reject specified user agents script that rejects calls from known bad User Agents. This script will deflect the majority of "casual" scan attacks.
- Monitor the administrator log and use features of your firewall to block offenders.

Resilience strategies — redundancy, backing up and restoring data

This topic discusses the resilience, redundancy, backup and restore strategies to consider when deploying and maintaining your Pexip Infinity platform.

Resilience and redundancy

Pexip is designed for multiple layers of resilience and redundancy. Companies and service providers should consider which situations they want to protect against. All options can be combined, and this is typically a consideration of cost versus benefit, and how much downtime can be tolerated in a worst case scenario.

The main levels of resilience redundancy and our associated recommendations are described below.

Hardware and physical considerations

- Dual hot swap power supplies for each server connected to different power circuits, optionally with UPS or backup power.
- Dual network cards in each server, connected to dual switches (VMware NIC Teaming). Switches are connected to redundant routers, allowing for any component in the network path to fail. Consider if the data center is robust if the fiber cable to the data center is cut.
- Redundant storage, either by adding a hardware RAID controller and operating two disks in RAID 1 (mirror) or by using redundant external SAN solutions.
- Redundant servers — we recommend that service providers deploy n+1 to always allow for one physical server to be unavailable.
- Redundant datacenters — consider providing Pexip Infinity from multiple data centers (either multiple data centers in one region or in various international regions).

VMware considerations

Note that some of the options listed below are not relevant for all node types in a Pexip deployment. Typically a Pexip Management Node, Reverse Proxy and the Conferencing Nodes responsible for signaling should be kept up by using technologies such as vMotion or DRS.

- For Conferencing Nodes with a consistently high media load, we recommend deploying n+1 (or more) Conferencing Nodes, so that you always have spare capacity if a node becomes unavailable.
- vMotion: proactively move the node to another server if maintenance/changes are to be conducted on the server where it is currently running. Do not vMotion a Conferencing Node that has active conferences.
- DRS: automatically move VMs around depending on load for servers. This should not be used with Conferencing Nodes that handle media.

Call control or DNS considerations

- If using Pexip Infinity to provide registration and call control services, deploy multiple Conferencing Nodes to provide resiliency options.
- If using DNS SRV records for call control server discovery, ensure that each SRV record points to multiple A records (the names of multiple servers) so that call control can still be provided by other Conferencing Nodes if one fails.
- Consider DNS SRV records with different priorities that fail over to an alternative datacenter if available.
- Be prepared to modify DNS configurations to direct traffic to alternative datacenters in case of a major disaster in one datacenter.
- Ensure that call control systems outside of Pexip (other SIP registrars, H.323 gatekeepers, Skype for Business servers, or other components) are integrated with multiple Pexip Conferencing Nodes (often via the DNS strategies mentioned above) to allow service continuity should any node in the Pexip deployment be unavailable.

See DNS record examples for more information.

Physical locations of Pexip nodes

As the Pexip Infinity platform can be deployed with multiple nodes in multiple locations, make use of this flexibility and design the setup as required for the spread of customers being served.

Backup mechanisms

It is mandatory for any critical installation to have a proper backup of the Management Node.

The Pexip Infinity platform can be backed up in many ways — with different advantages to each option. In some scenarios, for fast recovery of various situations, using multiple (or all) options is possible, and in most cases we recommended using at least two backup strategies.

For **on-premises deployments**, we recommend that you use both the hypervisor and Pexip's inbuilt methods to preserve your configuration data. A VM snapshot should be your primary mechanism prior to an upgrade, as this allows you to easily restore your system back to its state at the time the snapshot was taken. The Pexip Infinity backup and restore mechanism is your fallback mechanism, as this allows you to preserve a copy of your data in an alternative location, in case you lose your VM environment. For **cloud-based deployments** (Azure, AWS, GCP or Oracle) if you want to take a VM snapshot we recommend that you perform a graceful OS shutdown (not a hard power-off) before taking the snapshot; alternatively, you may use the Pexip Infinity backup and restore mechanism.

In all cases ensure that you take regular backups, refreshing your backups after making configuration changes to your Management Node.

VMware backup

VMware backups should be performed in the same datacenter (with an offsite replication of the backup in case of a local disaster) to allow fast restoration of the data.

Be aware that snapshots are not backups. Snapshots are a tool to roll back to a given time. Therefore, we recommend taking snapshots only when necessary (such as prior to an upgrade) and deleting the snapshot as soon as possible after the upgrade is confirmed to be successful. You should only create and delete VMware snapshots at a time of minimal usage. Taking or removing snapshots can cause virtual machines to become unresponsive.

VM backups should use a proper hypervisor VM backup tool (e.g. VMware VDP — vSphere Data Protection) or similar, and restoration should be tested and verified (preferably after the inbuilt backup methods have been set up, to ensure that you have another way of recovering if your restoration fails).

Conferencing Nodes do not need to be backed up. They receive all their configuration information from the Management Node and can simply be redeployed if necessary (i.e. delete and recreate). However, if your Conferencing Nodes are geographically spread out and redeploying them would consume significant bandwidth or take a significant length of time, they can also be backed up with your hypervisor's backup tools.

Pexip's inbuilt Management Node configuration backup process

You can use Pexip Infinity's inbuilt backup and restore mechanism to backup and restore the configuration data on the Management Node.

You can enable regular automatic backups, and you can also take a manual backup whenever it is appropriate, for example, before and after you make any configuration changes or perform a software upgrade.

- All backup files are encrypted — the administrator supplies a passphrase and must remember this for any subsequent restoration.
- Restoration can only occur on exactly the same version that the backup was taken from.
- The data contained in the backup contains all configuration data, including IP addresses, custom themes, certificates and call history.
- The backup data does not contain licenses, the administrator log, the support log, usage statistics or the operating system password.
- The system keeps on the host VM only the 5 most recent manually-taken backups and the 5 most recent automatic backups. Older backup files are deleted when a new file is created.

As the restore does not contain the license key, if you use this method to restore your configuration data onto a fresh Management Node you will need to contact your Pexip authorized support representative to obtain a new license key.

We recommended that you configure Pexip Infinity to schedule regular backups and send the backup file to an external FTP server. See Backing up and restoring configuration for instructions.

Pexip import/export of service configuration data

This is your "last resort" if you do not have the ability to restore configuration from a VM backup or from a Pexip backup. You can export and import your service configuration data (Virtual Meeting Rooms, Virtual Auditoriums, Virtual Receptions, device aliases and Automatically Dialed Participants) to and from CSV files.

You can apply this data to a fresh (or existing) Pexip installation if, for example, you had to redeploy the entire Pexip platform. Note that Call Routing Rule and other platform configuration must be documented for manual restoration in this scenario. See Bulk import/export of service configuration data for instructions.

If you use provisioning you can restore your VMRs and device aliases from your LDAP/AD source.

Impact of lost connectivity to the Management Node

If the Management Node disappears from the network (due to network outage, server outage, lack of VMware HA, or during HA restoration etc.) there will be some impact on your Pexip Infinity platform:

- **No configuration updates pushed to Conferencing Nodes**: a Conferencing Node typically checks in with the Management Node once every 60 seconds for configuration updates. This will fail, hence no new VMRs, Call Routing Rules, or platform configuration updates such as DNS servers etc. will be added to the Conferencing Node's local replicated database.
- **No CDR/log data sent back to the Management Node**: the Conferencing Nodes will try to push syslogs back to the Management Node with log data. This will also fail, and the Conferencing Nodes will buffer the log data until the Management Node is operational (there may be some limitations here — if you leave the Management Node down for a very long time and there is a lot of traffic, at some point the Conferencing Node logs will rotate to avoid filling up the disk, but you should not normally have such long outages).
- **No visibility in Pexip Management Node interface**: new calls will not appear in the Management Node's Administrator interface (assuming the Administrator interface is accessible, but some Conferencing Nodes cannot reach the Management Node due to a network split, for example).
- **Licensing**: if a Conferencing Node is unable to contact the Management Node, the call is permitted on the assumption that a license is available. After the Management Node has been out of contact for a grace period of 14 days, all calls will be rejected.
- **Rebooting/restarting Conferencing Nodes**: do **not** restart or reboot a Conferencing Node while the Management Node is unavailable. The 14-day licensing grace period only works if the Conferencing Node has had a valid sync with the Management Node after the Conferencing Node's most recent reboot.
- **Syslog to external syslog server (TCP/TLS)**: for reliable syslog output, each Conferencing Node can send its syslog data directly your own external syslog server, for quick realtime analysis with tools like Splunk etc. By using TCP/TLS based syslog you will use a reliable data channel so you know that the traffic is received in the other end. As the log data is sent directly from the Conferencing Nodes to your configured syslog server, it is not affected by the Management Node not being available.

Some features are not affected or are only partly affected:

- **New and existing calls**: Conferencing Nodes will continue to handle both new and existing calls as before — this is because no conference media or signaling passes through the Management Node.
- **One-Touch Join**: if the Management Node is offline the One-Touch Join service will continue to function, although you will not be able to make any changes to the configuration of the service during this time.
- **Secure Scheduler for Exchange**: when a single-use VMR is requested, the meeting invitation is still sent to all participants (including the equipment resource), but it will include placeholder text instead of join instructions — these will be added and the invitation re-sent when the Management Node comes back online. Additionally, cancellations and updates to existing meetings will not be processed until the Management Node is back online.

Considerations when restoring the Management Node from backup

Your procedures to restore the Management Node from a backup depend upon the type of backup available and the problem that has occurred:

Restoring from VMware

This is probably the quickest and most reliable way of restoring the Management Node.

- You can perform a Management Node restore from VMware during production hours.
- When the restore is complete the configuration replication will start as soon as the IPSec tunnels between the Management Node and the Conferencing Nodes are back up. Due to the nature of IPsec, some Conferencing Nodes might take a while to sync back up. Rebooting a Conferencing Node will solve this, but that will impact operations and will drop any active calls that are being handled on that node — and note that if the node is not in sync with the Management Node you cannot use the Administrator interface to check if that node has any running calls.
- Any configuration on the Conferencing Nodes will be refreshed with the data that has been restored to the Management Node.
- Any logs / call data records (CDRs) that have already been sent to the old Management Node will be lost as Conferencing Nodes do not check what is in the Management Node database — it only knows they had been delivered. To recover these CDRs, use external syslog data and parse it as an additional import to your billing system. New CDRs will be queued awaiting the Management Node to become operational again.

Restoring a file created by Pexip's inbuilt Management Node configuration backup process

We recommend using this approach if the Management Node needs to be reinstalled or rebuilt.

When you perform a restore of Management Node configuration data, the node's IP address is restored (as this is used by the IPSec tunnels to all of the Conferencing Nodes). To be able to restore a Management Node into another datacenter (in case of local disaster

recovery), the Management Node needs an IP address from your ISP that can be re-routed to another site. For high-availability deployments, this is something worth discussing with the network provider as you cannot just move nodes around after being deployed.

- You can perform a Management Node restore from a Pexip backup during production hours.
- If you are restoring onto a fresh Management Node:
  - As the IP address of the new Management Node will be the same as the old node, they should not both be powered on in the same VLAN to avoid IP conflict.
  - It will not replicate configuration to the Conferencing Nodes until the backup file is restored (as the new Management Node does not contain the right certificates to communicate with the Conferencing Nodes). When the Pexip backup is restored, it will connect with the Conferencing Nodes and replicate configuration. Any configuration on the Conferencing Nodes will be refreshed with the data that has been restored to the Management Node.
  - Any logs / call data records (CDRs) that have already been sent to the old Management Node will be lost as Conferencing Nodes do not check what is in the Management Node database — it only knows they had been delivered. To recover these CDRs, use external syslog data and parse it as an additional import to your billing system. New CDRs will be queued awaiting the Management Node to become operational again.

Restoring service configuration data from CSV files

An import of service configuration data can be performed at any time.

- The primary key for the Virtual Meeting Room / Virtual Auditorium / Virtual Reception database is the name of that service. So, for example, if you import VMRs on top of an existing VMR database, the existing configuration for those VMR names will be overwritten by the imported VMR's configuration. Any other existing VMR records will be left unchanged.
- The primary key of device alias data is the alias itself, and for Automatically Dialed Participants it is the combination of the conference name and the alias to be dialed.
- Any new or modified service configuration will take effect after approximately 60 seconds.
- CDR data is not affected by a restore of service configuration data.

PSTN gateways and toll fraud

If your environment includes a PSTN gateway or uses an ITSP (Internet telephony service provider), consider the potential for toll fraud if you have Call Routing Rules that can route calls to the PSTN gateway or ITSP, or if you allow conference participants to dial out to other participants via the PSTN gateway or ITSP.

As you might intentionally want to allow users to route calls via the PSTN gateway or ITSP (and thus incur toll charges), we recommend that you use a suitable call control solution such as a Cisco VCS to configure an appropriate dial plan and authentication mechanism for your network.

The ways in which Pexip Infinity calls may be routed via a PSTN gateway or ITSP are described below.

Infinity Gateway calls

If your dial plan allows Infinity Gateway calls and has a Call Routing Rule which, for example, matched 9.* and routed the call via the PSTN gateway or ITSP, this would allow anyone who could route a call to the Pexip Infinity platform to then send a call via the PSTN gateway or ITSP.

You could restrict those rules to only apply to incoming calls from registered devices, or to calls that are being handled in an internal location.

Virtual Receptions

If you have configured Virtual Receptions and also have Call Routing Rules that match numeric aliases (such as 9.*), then anyone that can reach the Virtual Reception could match the Call Routing Rule and potentially route their call via the PSTN gateway or ITSP.

Thus, if any call control rules are in place to restrict who may dial numbers which correspond with numeric Call Routing Rules, then the same restrictions should also be placed on who may call any Virtual Receptions.

Manually dialing out to a participant from a conference

Pexip Infinity allows you to manually dial out to participants from a conference, on an ad hoc basis.

This means that conference participants using Pexip apps could place outbound calls via the PSTN gateway or ITSP. Note that these types of calls may dial out directly to the destination alias or they may use Call Routing Rules.

In these circumstances, to reduce (but not eliminate) the risk of toll fraud, we recommend that you use PIN-protected conferences (so that only Hosts can dial out).

Example emails for sending to new users

Based on feedback from our customers, we know that issues can occur when end users who are new to Pexip Infinity don't understand what clients to use, or don't know what aliases to dial in order to join a meeting. So we've put together some best practice templates that you as an administrator can use as a basis for emails you'll send to your users, to let them know how they can access their Virtual Meeting Rooms and let others join them for meetings, and which Pexip apps to use and how to use them.

All deployments will differ, but some general recommendations and considerations are:

- Your email to new users should be very clear about how they can access their own VMR, and also provide clear instructions for them to forward on to anyone they want to invite to their VMR as a guest. You should also make it clear which devices or clients they should use when calling in to your deployment.
- Users connecting from outside your organization and who do not have their own video device should generally use the web app to access meetings hosted on Pexip Infinity. This means that they won't need to download or install anything in order to join meetings, but will still have the same high-quality user experience and functionality of participants using the Pexip app for Windows.

  You'll need to make sure that at least one Conferencing Node is accessible externally, and you'll also need to set up appropriate DNS records for connections from both inside and outside your network.
- Users connecting from inside your organization should also use the web app, unless you want them to be able to register to receive incoming calls — in which case they need to use the Pexip app for Windows.
- The Pexip app for Windows should be used if you want to take advantage of the additional registration (to receive incoming calls) and internal directory service features. Administrators can also set up Call Routing Rules that apply to registered devices only, meaning that you can permit registered Pexip app for Windows users to make calls that web app users cannot.

  If you are deploying the Pexip app for Windows in your environment, we recommend that you make use of provisioning, and you'll also need to set up appropriate DNS records.
- If you have a Skype for Business environment, consider whether you want users to continue to use those familiar clients to access their VMRs.
- Consider whether you want users to be able to join or control meetings from a mobile device using the Pexip app for Android or iOS. If so you'll need to set up appropriate DNS records.
- Consider whether you want to allow users to call in to meetings from a telephone. If so, you will need to set up appropriate telephone numbers and Virtual Receptions.
- Where possible, simplify the join experience for users by providing URLs that they can click to join meetings directly.

We already provide example joining instructions for use with the Secure Scheduler for Exchange feature, and provisioning, but here we also give some examples for Accessing a VMR and Using Outlook to schedule meetings in a VMR.

Accessing a VMR

Below is some example text you can send to new users to let them know how they and their guests can access their VMRs. You should edit this as appropriate for your own deployment and for each individual user by:

- adding the URLs
- specifying the VMR aliases and Host PINs for each user's VMR
- specifying the Guest PIN (or deleting references to Guests PINs if you don't use these in your deployment)
- adding download links for the Pexip app for Windows
- deleting any content not relevant to your deployment
- adding formatting and hyperlinks to the text.

You can include URLs in the text that, when clicked, will open a specific client and pre-fill some of the required fields, such as VMR alias and PIN. For guidance on creating these URLs, see Creating preconfigured links to launch conferences via Pexip app.

> Welcome to your own personal Virtual Meeting Room (VMR).

You will usually be the Host for any meetings in this VMR. This means you can control when the meeting starts and ends, and you can also mute and unmute noisy participants, lock the virtual meeting room (to prevent anyone else from joining part way through the meeting), and eject unwanted participants.

How to join your VMR as a host

- Using a web browser from within <organization>, go to <internal URL>, enter the address of your VMR which is <alias>, and follow the prompts to join the meeting.
- From a web browser outside <organization>, go to <external URL>, enter the address of your VMR which is <alias>, and follow the prompts to join the meeting.
- From a video conferencing endpoint inside <organization> dial <internal alias> then enter the PIN: <Host PIN>.
- From a video conferencing endpoint outside <organization> dial <external alias> then enter the PIN: <Host PIN>.
- From Skype for Business, dial <alias> then enter the PIN: <Host PIN>.
- To use a Pexip app for Windows, first download the app from <link>. After it is installed, click this link <URL>, or dial <alias> then enter the PIN: <Host PIN>.
- From a telephone, dial <telephone no.> then, when prompted, enter the conference number <numeric alias> followed by "#". When asked, enter the host PIN: <Host PIN>.
- To control the meeting from an Android or iOS device, first install the Pexip app for mobile from the Google Play store or the Apple store. After it is installed, click this link <URL> or dial <alias> then enter the PIN: <Host PIN>.

How your guests can join your VMR

You can copy and paste the text below when inviting guests to join your VMR.

To join my VMR, you can simply use a web browser if you don't already have a video device you can use:

- Using a web browser from within <organization>, go to <internal URL> and in the "Video address" field enter <alias>. Follow the prompts to join the meeting as a Guest using the PIN <Guest PIN>.
- Using a web browser from outside <organization>, go to <external URL> and in the "Video address" field enter <alias>. Follow the prompts to join the meeting as a Guest using the PIN <Guest PIN>.
- From a video conferencing endpoint inside <organization> dial <internal alias> then enter the PIN: <Guest PIN>.
- From a video conferencing endpoint outside <organization> dial <external alias> then enter the PIN: <Guest PIN>.
- From Skype for Business, dial <alias> then enter the PIN: <Guest PIN>.
- From a Pexip app for Windows within <organization>, click this link <URL>, or dial <alias> then enter the PIN: <Guest PIN>.
- From a telephone, dial <telephone no.> then, when prompted, enter the conference number <numeric alias> followed by "#". When asked, enter the Guest PIN: <Guest PIN>.

If you arrive before the meeting host, you will be placed in a virtual waiting room. You will be automatically admitted to the meeting when the host arrives.

Using Outlook to schedule meetings in a VMR

Below is some example text you can send to new users to let them know about the Secure Scheduler for Exchange feature. An end user guide for this feature is available (see Using Outlook to schedule meetings in VMRs); you can either copy some information from there, or send your users the direct link to the guide.

The way in which users locate and activate the add-in will depend on the Outlook client being used, so you will need to replace **<Pexip Scheduling Service option>** with the relevant details for your deployment.

Using Outlook to schedule video meetings

As part of the roll-out of Virtual Meeting Rooms (VMRs) in our organization, you now have the ability to schedule any of your meetings as video meetings, using a one-off VMR created just for that meeting.

To do this, just create your meeting invitation in the usual way and then select the <Pexip Scheduling Service option>.

This will automatically add joining instructions to the body of the meeting invitation.

Note that for these sorts of meetings, everyone joins in the same way using the same links, and everyone has Host privileges.

For full instructions, see www.docs.pexip.com/admin/scheduling_user_guide.htm

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Glossary of Pexip Infinity terms

Here is a list and description of many of the terms used in this documentation.

Adaptive composition

An intelligent conference layout with real-time automatic face detection and framing.

AIMS

Pexip's AI Media Server (AIMS) is a self-hosted, standalone virtual machine (VM) that allows you to access Pexip's AI-powered features (such as live captions) in a secure environment.

Alarm

An alert that is raised when there is an issue on your Pexip Infinity deployment that requires attention. For more information, see Viewing alarms.

Alias

An alias is the string that is typically dialed by a participant when placing a call, and causes the call to be routed to the appropriate conference. It either triggers the creation of a conference instance, or if a conference associated with that alias already exists, causes the call to be routed to that conference.

Each alias is associated with a conferencing service (Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service), which defines the type and settings (such as PIN) for the conference that is created.

In most cases, the alias received by the Conferencing Node is the same as the alias that the conference participant dialed from their endpoint, but there are some exceptions (for more information see About aliases).

Depending on the dial plan, multiple aliases can be used throughout a network to access the same service.

Authentication

Pexip Infinity integrates with a range of external services to authenticate, authorize and provision access to clients and services — for a full summary see Authentication overview.

Automatically dialed participant (ADP)

A participant/device that will have a call placed to it from a Virtual Meeting Room or Virtual Auditorium whenever a conference using that service starts.

For more information, see Automatically dialing out to a participant from a conference.

Backplane

A link between the Management Node and a Conferencing Node, or between two Conferencing Nodes, used to transmit Pexip control messages. Backplanes between Conferencing Nodes also transmit conference audio, video, and data packets.

All packets are secured through authentication and encryption designed to protect the privacy of the data. For more information, see Encryption methodologies.

Local backplanes exist between Conferencing Nodes in a single location.

Geo backplanes exist between Conferencing Nodes in different locations.

Branding and customization

You can easily apply your own corporate branding to the Pexip Infinity platform, and produce a personalized user experience for all of your Pexip Infinity services.

You can use themes to customize the voice prompts and images used in your conferences, and you can apply branding customizations to change the look and feel of the Pexip apps (Pexip's free video apps, which can also be used to access those services).

Branding packages and paths

Branding packages and paths allow you to provide a variety of differently-branded web app experiences within your environment:

- Branding packages define the look and feel of the web app used to join a conference.
- Branding paths are the URL paths (such as /webapp, /webapp2, and /webapp3) that are used to access the web app, from where people can join a conference. This means that you can link a different branding package to each of the available paths; then depending on which URL is used to access the web app, you can provide a different branding for each path.

Breakout rooms

Breakout rooms are separate sessions that are split off from a main VMR or Virtual Auditorium that allow smaller groups of people to meet together. When participants are in a breakout room they can only see, hear or share content with each other — they are separated from the main room and any other breakout rooms.

Typically, a Host participant using the Pexip web app will manage the breakout rooms — creating rooms, assigning participants to rooms, specifying any time limits, and moving people between rooms or bringing them back to the main room as and when required. Guests in a breakout room can request help from a Host, and any Hosts can join (with full audio and video) any of the breakout rooms at any time.

See Breakout rooms for more information.

Bursting

Pexip Infinity deployments can burst into cloud-hosted services when primary conferencing capabilities are reaching their capacity limits, thus providing additional temporary Transcoding Conferencing Node resources. See Dynamic bursting to a cloud service for more information.

Call tag

An optional identifier for each participant in a call that can be specified in client API requests and then used by app developers to correlate other API requests. For more information, see Tracking usage via service and participant call tags.

Cloud service

As an alternative to deploying your Pexip Infinity platform on premises (in your own datacenters), you can deploy nodes on a hosted cloud platform. See Deploying as a cloud service via Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP) or Oracle Cloud Infrastructure for more information.

Cloud Video Interop (CVI)

Cloud Video Interop (CVI) is a Microsoft Qualified third-party solution that enables third-party meeting rooms (telepresence) and personal video devices (VTCs) to join Microsoft Teams meetings.

Pexip Infinity is a Microsoft-certified video interoperability platform for Microsoft Teams. See Integrating Microsoft Teams with Pexip Infinity.

Conference

A general term that can be used to refer to a specific instance of a meeting being held in a Virtual Meeting Room or Virtual Auditorium.

Conference instance

A conference with active participants that exists on one or more Conferencing Nodes. A unique conference instance is created when the first participant dials an alias associated with a Pexip Infinity service alias, and lasts until the last participant disconnects.

Conferencing Node

A virtual machine (VM) that provides the capacity for conferences, handling the media processing and call signaling.

A Conferencing Node can have either a transcoding or a proxying role:

- Transcoding Conferencing Nodes are required in all deployments; they manage all of the media processing required to host a conference, and can also handle direct connections to/from endpoints if required.
- Proxying Edge Nodes are optional; they handle call signaling and the media connection with the endpoint, but forward the media on to a Transcoding Conferencing Node for processing.

For more information, see Conferencing Nodes.

Connection (to a conference)

A connection can be a call or presentation from an endpoint to a Virtual Meeting Room or Virtual Auditorium, a backplane between Transcoding Conferencing Nodes, or a call into or out of the Infinity Gateway. In this context, a **connection** is analogous to a **port**. Note that a Skype for Business client may require two connections (one for the video call, and one for presentation content).

Direct media

Pexip's direct media feature enables end-to-end encrypted calls. When enabled in a VMR, it provides non-transcoded, encrypted, point-to-point calls between any two WebRTC-based participants.

See Direct media (end-to-end encrypted calls) for more information.

Distributed conference

A conference instance that exists across two or more Conferencing Nodes. It can be **locally distributed**, **globally distributed**, or both:

Locally distributed conferences exist across two or more Conferencing Nodes in the same location.

Globally distributed conferences exist across two or more Conferencing Nodes in physically different locations.

Locally and globally distributed conferences exist across two or more Conferencing Nodes in one location and at least one other Conferencing Node in a different location.

Endpoint

A hardware device or soft client capable of participating in a conference. The endpoint's capabilities can vary from audio-only to full audio, video, and data sharing support.

Enhanced Room Management (ERM)

Enhanced Room Management (ERM) is a separately-installable Pexip product that provides management of your Cisco and Poly devices and room systems. From system monitoring to bulk provisioning of software upgrades, address books, and branding profiles, it provides everything you need to manage your systems from one single management interface.

Event sink

An external service to which Pexip Infinity can send details of every participant and conference management event. For more information, Using event sinks to monitor conference and participant status.

Host participant

A conference participant who has privileges to control aspects of the conference. For more information, see About PINs, Hosts and Guests.

Host server

The physical hardware on which the virtual **Management Node** and **Conferencing Node**s reside. For more information, see Host servers.

Hypervisor

An application that is used to create and manage virtual machines. For more information on the hypervisors supported by Pexip Infinity, see Supported hypervisors.

Identity Provider

Identity Providers (IdPs) are third-party services (such as AD FS, Microsoft Entra ID or Okta) that store and manage digital identities, and with which users authenticate using SSO. Pexip Infinity supports Identity Provider SSO authentication for participant access to VMRs, and for client application registration.

IVR (Interactive Voice Response)

IVR technology allows participants to use a DTMF keypad to interact with Pexip Infinity services to:

- select which Virtual Meeting Room or Virtual Auditorium they wish to join. For more information, see About the IVR service.
- enter a PIN, for those Virtual Meeting Rooms that have restricted access. For more information, see About PINs, Hosts and Guests.

Justice

The Pexip Secure Meetings for Justice ("Pexip Justice") solution enables judiciaries to hold virtual or hybrid court hearings, video arraignments and virtual interactions in corrections systems, in a Pexip Virtual Meeting Room (VMR).

Layouts

The layout used in a conference determines how meeting participants are displayed when they are connected over video, and how other participants — such as those with an audio-only connection — are indicated.

The layout for a meeting is configured in advance by the administrator, but can be changed during the meeting by Host participants using SIP/H.323 endpoints that support DTMF keypad controls (*8 by default), or by using a Pexip app, or on Cisco endpoints via the Pexip Layout Controls macro.

See Conference layouts and speaker names for more information.

Management Node

A virtual machine (VM) on which the **Pexip Infinity** software is installed. This machine hosts the Pexip Infinity Administrator interface. It is used to create one or more Conferencing Nodes and configure information about the conferences that can exist on those Conferencing Nodes.

Media Playback Service

Pexip's Media Playback Service allows you to play prerecorded video content (such as adverts and informational videos) to consumers. When the media finishes playing, the user can be transferred to another service, such as a VMR conference, or they can be disconnected.

MTU (Maximum Transmission Unit)

The size of the largest packet of data that can be transmitted via the network interface of a device (endpoint, Conferencing Node or Management Node).

Multiscreen participant display

If you have a dual screen endpoint you can display conference participants across both screens, allowing you to show more participants than if you have a single screen. This display mode is available for the following meeting, layout and endpoint types:

- Conferences hosted in a VMR or Virtual Auditorium.
- Calls into a Microsoft Teams or Google Meet conference.
- Where the conference layout is 1+7, 1+0, 2x2, 3x3, 4x4, 5x5, custom layouts, Adaptive Composition or Teams-like.
  - If you are using any other layout when multiscreen is enabled then the 1+7 layout is used on both screens.
  - If you are using a 4x4 or 5x5 layout when multiscreen is enabled then each screen will display 3x3 on both screens (this means that when using a 5x5 layout you will see fewer participants in total if multiscreen is enabled).
- SIP and H.323 endpoints.

Multiscreen participant display is used automatically on endpoints that are known to have dual screens — see Determining if an endpoint has single or multiple screens for details.

See Controlling the layout during a conference.

NUMA

NUMA is an architecture that divides the computer into a number of nodes, each containing one or more processor cores and associated memory. A core can access its local memory faster than it can access the rest of the memory on that machine.

OAuth 2.0

An authorization framework that enables users to grant third-party access to an HTTP service / web resource without sharing their passwords.

One-Touch Join (OTJ)

Pexip's One-Touch Join (OTJ) allows users to schedule a meeting in Microsoft Outlook or Google Calendar and include in the invitation a meeting room with a supported Cisco or Poly videoconferencing endpoint, so that the endpoint in the chosen meeting room displays a **Join** button just before the meeting is scheduled to begin. Participants can then simply walk into the room and select the button, and the endpoint will automatically dial in to the meeting.

Participant

A conference participant typically refers to the endpoint device or system that is connected to a conference — this could be a personal device or a room system (where the room might contain multiple people). A participant could be a video participant or an audio-only participant. Such participants are only counted once regardless of whether or not there is a presentation stream in addition to the main video, and they only consume a single call license. A Pexip app can also join a conference without audio or video, for example as a presentation and control-only participant or a Meeting Manager participant.

Personal layout

Personal layouts is a feature that allows Host and Guest participants to choose the layout they receive without affecting the conference-wide layout received by other participants.

See Applying a personal layout for more information.

Pexip app

A suite of software clients that allows end users to connect to Pexip Infinity services from a web browser, an installable desktop client, or a mobile client. For more information, see Introduction to Pexip apps.

Pexip Infinity

A virtualized and distributed multipoint conferencing platform that comprises a single Management Node and one or more Conferencing Node(s).

See Introduction to .

Pexip Infinity software files

The files that are used to deploy the Management Node and Conferencing Nodes onto the physical virtual machine (VM) infrastructure or cloud service.

Pexip Private AI

Pexip Private AI is a stand-alone platform powered by Pexip's AI Media Server (AIMS) which integrates with Pexip Infinity to provide access to AI-powered features (such as live captions) in a secure environment.

Port

The term "port" may be used to describe a connection (such as a call or presentation) from an endpoint to a Conferencing Node or a backplane between Transcoding Conferencing Nodes. For more information, see license installation and usage.

Port can also refer to the virtual data connections used for network traffic between devices. For more information on the ports used by the Management Node and Conferencing Nodes to connect to other devices, see port usage and firewall guidance.

Provisioning room systems via ERM

ERM supports the provisioning of configurations, commands, firmware, branding, macros/panels and call settings to one or more room systems.

Provisioning VMRs, devices and users via LDAP

In large organizations with many employees and users of video conferencing, you may need to configure lots of Virtual Meeting Rooms (VMRs) and associated records to support those employees. Much of this data can be bulk-provisioned from directory information contained in a Windows Active Directory LDAP server, or any other LDAP-accessible database.

Proxying Edge Node

A Proxying Edge Node handles all media and signaling connections with an endpoint or external device, but does not host any conferences — instead it forwards the media on to a Transcoding Conferencing Node for processing. For more information, see Distributed media-proxying Edge nodes.

Registering

Pexip Infinity can act as a SIP registrar and H.323 gatekeeper, which means that SIP and H.323 endpoints can register directly to Pexip Infinity. This allows Pexip Infinity to route calls to those registered devices without having to go via an external SIP proxy or H.323 gatekeeper, or rely on DNS.

The Pexip app for Windows can also register to Pexip Infinity. Registration is mandatory and allows these apps to receive calls via Pexip Infinity and use directory lookup services.

Reverse proxy

A reverse proxy is an application that can proxy HTTP and HTTPS traffic from an externally-located client to a web service application located on the internal network — in our case a Pexip Conferencing Node. A reverse proxy can also be referred to as a load balancer.

Scheduling a conference

Pexip scheduling solutions allow you to create and manage conferences, and enable users to add a Pexip VMR to their meeting invitations. For more information about the different scheduling solutions available, see:

- Secure Scheduler for Exchange (previously known as VMR Scheduling for Exchange): provides an integration for Outlook users in Office 365 and Exchange environments to schedule meetings
- Pexip Secure Scheduler for Web: a web-based solution for scheduling and managing meetings
- Pexip Secure Meetings for Justice: a specific solution for judiciaries to schedule, host and manage court hearings.

Secure meetings

The Pexip Infinity secure video collaboration platform offers secure meetings via:

- a range of deployment environments to suit your physical security requirements
- integration with internal and external services to authenticate, authorize and provision access to clients and services — for a full summary see Authentication overview
- the optional PexipSecure Scheduler for Exchange feature, which integrates with Outlook to allow users to add a secure Pexip VMR to their meeting invitations
- the Pexip Secure Scheduler for Web product, a web-based tool for creating and managing secure virtual meetings
- the Pexip Secure Meetings for Justice product, specifically tailored to the unique requirements of judicial VMRs.

Secure Scheduler for Exchange

Pexip scheduling solutions allow you to create and manage conferences, and enable users to add a Pexip VMR to their meeting invitations. For more information about the different scheduling solutions available, see:

- Secure Scheduler for Exchange (previously known as VMR Scheduling for Exchange): provides an integration for Outlook users in Office 365 and Exchange environments to schedule meetings
- Pexip Secure Scheduler for Web: a web-based solution for scheduling and managing meetings
- Pexip Secure Meetings for Justice: a specific solution for judiciaries to schedule, host and manage court hearings.

Service

In the context of this documentation, a Pexip Infinity service is one of the following: Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service.

Note that, separate to the Pexip Infinity self-hosted platform described in this documentation, Pexip also has a managed video conferencing-as-a-service (VCaaS) offering, referred to as the Pexip Service.

Service tag

An optional identifier that an administrator can assign to a service, allowing them to track usage of the service via the administrator log. For more information, see Tracking usage with a service tag.

Single Sign On (SSO)

Single Sign-On (SSO) is a common feature of web services and business tools that enables users to authenticate and log into a third-party service or web page, using the same login credentials as the other systems they already use.

Identity Providers (IdPs) are third-party services (such as AD FS, Microsoft Entra ID or Okta) that store and manage digital identities, and with which users authenticate using SSO. Pexip Infinity supports Identity Provider SSO authentication for participant access to VMRs, and for client application registration.

SIP Guest Join (SGJ)

Pexip's standard Microsoft Teams and Google Meet interoperability solutions allow your own video conferencing endpoints (and your guests) to join Microsoft Teams / Google Meet meetings that you are hosting. The "SIP Guest Join" (SGJ) feature lets you join Microsoft Teams and Google Meet meetings with your own video conferencing endpoints where that meeting is being hosted by an external third-party organization who has not enabled Pexip interoperability themselves.

Note that this is a guest join experience, therefore someone has to be in the Microsoft Teams / Google Meet meeting to admit you from the lobby.

Socket

The socket on the host server's motherboard where one processor is installed.

System location

A label that allows you to group Conferencing Nodes together, typically according to where they are physically located. For more information, see About system locations.

Teams Connector

The Pexip Teams Connector is a Pexip application that is deployed in Microsoft Azure and is used to enable Cloud Video Interoperability (CVI) with Microsoft Teams. It handles all Teams communications and meeting requests from the Pexip Infinity platform and passes them on to the Microsoft Teams environment.

Test Call Service

A Pexip Infinity loopback service that allows users to check the quality of their video and audio prior to joining a conference, and verifies that they can connect to a Conferencing Node.

Theme

Themes allow you to change the voice prompts and images provided to participants when they use a Pexip Infinity service (Virtual Meeting Room, Virtual Auditorium, Virtual Reception, scheduled conference, Media Playback Service, or Test Call Service) or use the Infinity Gateway to make a person-to-person call or to join an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet. You might change the theme if, for example, you want to use your company's own logo, color scheme or terminology on the screens displayed to conference participants, or you want to change the language used in the voice prompts.

You can also define your own custom conference layouts within a theme and use those layouts in the same way as the standard layouts that are shipped by default in Pexip Infinity.

See Customizing conference images and voice prompts using themes for more information.

Thumbnail

A smaller window at the bottom of the main picture which displays the live video stream from a conference participant. Sometimes referred to as a PiP (Picture in Picture).

Transcoding Conferencing Node

In large Pexip Infinity deployments, a Transcoding Conferencing Node handles all the media processing, protocol interworking, mixing and so on that is required in hosting Pexip Infinity calls and conferences. When combined with Proxying Edge Nodes, a transcoding node typically only processes the media forwarded on to it by those proxying nodes and has no direct connection with endpoints or external devices. However, a transcoding node can still receive and process the signaling and media directly from an endpoint or external device if required.

See Distributed s for more information.

TURN server

A TURN server is a media relay/proxy that allows peers to exchange UDP or TCP media traffic whenever one or both parties are behind NAT.

Virtual Auditorium

A meeting space that is optimized for use by a small number of Host participants and a large number of Guests. For more information, see About services.

Virtual machine (VM)

A software implementation of a computer, which runs on a host server and is implemented and managed using a hypervisor. The Management Node and Conferencing Nodes are virtual machines. In a cloud environment it is typically referred to as a VM instance.

Virtual Meeting Room (VMR)

A Virtual Meeting Room (VMR) is a virtual meeting space that is always available to hold a Pexip Infinity conference. It can host any number of people from any type of device. Each VMR has one or more aliases associated with it, and participants access the conference by dialing one of these aliases.

Access to VMRs can be protected by the use of PINs or by authentication using SSO — or both. You can also use PINs to separate participants into Hosts and Guests, with different privileges. For more information, see About services.

Virtual Reception

The Virtual Reception IVR service provides a way for conference participants who cannot dial Virtual Meeting Room and Virtual Auditorium aliases directly, to access these services from a central lobby using DTMF tones. It can also be used to route calls via the Pexip Distributed Gateway to join an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

For more information, see About the IVR service.

VMR self-service portal

The Pexip VMR self-service portalis a separately-installable component that allows end-users to manage their personal Virtual Meeting Room without having to send requests to their administrator to change the configuration or branding of their VMR.

Voice Focus

Voice Focus is a platform-wide setting that improves the way in which voice activity is detected by better distinguishing between actual speech and background noise. This reduces the probability that people who are not speaking but have audible background noise will be switched into the main speaker position. Note that this does **not** remove any noise from the audio.

When enabled it applies to Virtual Meeting Rooms and Virtual Auditoriums, and it uses additional hardware resources (equivalent to an extra 6 audio connections per participant).

Regular expression (regex) reference

Regular expressions can be used in conjunction with Pexip Infinity features including Call Routing Rules, Provisioning VMRs, devices and users from Active Directory via LDAP, configuring Regex-based meeting processing rules for One-Touch Join, and searching the support and admin logs.

This topic contains information on:

- Regex testing tool
- Regex syntax
- Pattern matching examples
- Search and replace examples

Regex testing tool

The Pexip Infinity Administrator interface contains an inbuilt regex testing tool:

1. Go to **Utilities > Regular Expression Tester**.
2. Enter your test input and regex:

Option	Description
Input	The test input to match against the regular expression, such as a dialed alias.
Regex match	The regular expression that the **Input** is checked against (see Regex syntax below).
Regex replace string	The optional regular expression string used to transform the **Input** (if a match was found). Leave this field blank to leave the original input unchanged.

3. Select **Test Regular Expression** (at the bottom of the page).

   The **Result** field shows the result of transforming the **Input** using the **Regex match** and **Regex replace string**. If a **Regex replace string** was not provided then the **Result** will be the same as the **Input**, providing it was a full match.

   The **Debug information** shows extra information such as whether it was a successful, partial or failed match, and lists any match groups.

Regex syntax

BRE (basic regular expression) syntax is used when searching the support and admin logs.

Otherwise (i.e. for call routing and provisioning), Pexip Infinity supports case-insensitive Perl-style regular expression patterns as described in the rest of this topic. The table below describes some of the special characters that are commonly used in regular expressions, and provides examples of how they can be used.

Character	Description	Example
Basic syntax		
non-special character	Matches the specified characters literally (providing they are not any of the regex special characters).	**abc** matches **abc**, **ABC**, **aBC** etc
(...)	Groups a set of matching characters together. Multiple groups can be specified. Each group can then be referenced in order (from left to right) using the characters **\1**, **\2**, etc. as part of a replace string.	See Search and replace examples below
\	Escapes a regular expression special character: **{ } [ ] ( ) ^ $. | * + ? ** Also used to reference a group in a replace expression.	**ab\+** matches **ab+**
|	Matches either one expression or an alternative expression.	**.*@example\.(net|com)** matches against any URI for the domain example.com or the domain example.net

Character	Description	Example
Wildcard and character matching		
.	Matches any single character.	**a.c** matches **aac**, **abc**, **azc**, **a2c**, **a$c** etc.
\d	Matches a decimal digit character (i.e. 0-9).	**a\d** matches **a1**, **a2**, **a3** etc. but not **aa**, **ab** etc.
\D	Matches a non-digit character.	**a\D** matches **ab** but not **a1**, **a2**, **a3** etc.
\s	Matches any whitespace character (space, tab, newline).	**ab\sd** matches **ab d** but not **abcd**, **abxd** etc.
\S	Matches any non-whitespace character.	**ab\Sd** matches **abcd**, **abxd** etc. but not **ab d**
\w	Shorthand for **[a-zA-Z0-9_]**. Matches any alphabetical or digit character, or underscore.	**\w+** matches **bob** and **bob_jones** but not **bob.jones**. **[\w.-]+** matches **bob**, **bob.jones**, **bob_jones**, and **bob-jones**.
[...]	Matches the characters specified in the brackets. You can specify a range of characters by specifying the first and last characters in the range, separated by a hyphen. You cannot use other special characters within the [ ] — they will be taken literally.	**9[aeiou]** matches **9a**, **9e**, **9i** etc. and **9A**, **9E**, **9I** etc, but not **9b**, **9c**, **9B**, **9C** etc **9[a-z]** matches **9a**, **9b**, **9z** etc. and **9A**, **9B**, **9Z** etc. but not strings such as **91**, **99** or **9(** **[0-9#*]** matches any single E.164 character (digits 0-9, hash key or asterisk)
[^...]	Matches anything except the set of specified characters.	**[^a-z]** matches any non-alphabetical character **[^0-9#*]** matches anything other than an E.164 character
Repetition factors		
*	Matches 0 or more repetitions of the previous character or expression.	**ab*c** matches **ac**, **abc**, **abbc**, but not **ab** or **abd**
+	Matches 1 or more repetitions of the previous character or expression.	**ab+c** matches **abc** and **abbc**, but not **ab**, **ac** or **abd** **[a-z.]+** matches any string containing the letters a to z, A to Z, or dots (periods)
?	Matches 0 or 1 repetitions of the previous character or expression.	**ab?c** matches **ac** and **abc**, but not **ab**, **abbc** or **abd**
.*	Matches against any sequence of characters.	**a.*** matches everything beginning with **a**
{n}	Matches n repetitions of the previous character or expression.	**ab{2}c** matches **abbc**, but not **abc** or **abbbc** **a\d{3}** matches **a123**and **a789**, but not **a12** or **456**
{n,m}	Matches n to m repetitions of the previous character or expression.	**ab{2,4}c** matches **abbc**, **abbbc** and **abbbbc**, but not **abc** or **abbbbbc**
{n,}	Matches at least n repetitions of the previous character or expression.	**ab{2,}c** matches **abbc**, **abbbc**, **abbbbc** etc, but not **abc**
Position matching		
^	Matches the beginning of a line.	**^meet\.(.*)** matches any string that starts with **meet.** and places the rest of the string into a group (which could be used in a replace string).
$	Matches the end of the line.	**.*\.com$** matches any string that ends in **.com**

Character	Description	Example
(?!...)	(Not supported in OTJ Regex RE2 meeting processing rules) Negative lookahead. Defines a subexpression that must not be present immediately after the current match position, for example **regex1(?!regex2)** where a match is found if **regex1** matches and **regex2** does not match.	**(?!.*@example\.com$).*** matches any string that does not end with **@example.com** **(?!meet).*** matches any string that does not start with **meet** **meet(?!\.).*** matches any string that starts with **meet** providing it is not followed by a period
(?<!...)	(Not supported in OTJ Regex RE2 meeting processing rules) Negative lookbehind. Defines a subexpression that must not be present immediately before the current match position, for example **(?<!regex1)regex2** where a match is found if **regex1** does not match and **regex2** matches.	**.*(?<!@)example\.com.*** matches any string containing **example.com** providing it is not preceded by **@**

Note that this is only a subset of the full range of expressions. For a full description of regular expression syntax see http://perldoc.perl.org/perlre.html.

Pattern matching examples

VMR naming patterns

If your videoconferencing rooms have an alias naming convention in the form oslo1@example.com, oslo2@example.com, newyork1@example.com etc, this could be matched with an expression such as:

[a-z]+\d@example\.com

Or, if the rooms are named 555 followed by exactly three digits, e.g. 555100@example.com and 555234@example.com, you could use the expression:

555\d{3}@example\.com

If your room aliases could follow either pattern, you could merge the two expressions like this:

([a-z]+\d|555\d{3})@example\.com

Matching an IP address

To match against an IP address, use:

^([1-9][0-9]?|1[0-9]{2}|2[01][0-9]|22[0-3])(\.([0-9]|[1-9][0-9]|1[0-9]{2}|2[0-4][0-9]|25[0-5])){3}$

To match against somebody dialing an IP address (e.g. from an H.323 device or the web app), **or** someone dialing a SIP URI in the format IP_address@example.com, use a **Destination alias regex match** of:

^(([1-9][0-9]?|1[0-9]{2}|2[01][0-9]|22[0-3])(\.([0-9]|[1-9][0-9]|1[0-9]{2}|2[0-4][0-9]|25[0-5])){3})(@example\.com)?$

with a **Destination alias regex replace string** of: **\1**

and this combination of settings will ignore any @example.com element and pass just the IP address.

Search and replace examples

Replacing an alias domain

This example transforms any alias that ends in example.net into an alias that ends in example.com:

Match string:	**(.+)(@example\.net$)**
Replace string:	**\1@example.com**

This example builds on the previous example by transforming any alias that ends in example.com, example.net or example.co.uk into a common alias that ends in example.com:

Match string:	**(.+)(@example\.(com|net|co\.uk)$)**
Replace string:	**\1@example.com**

Strip leading 9

This example strips a leading 9 from any all-numeric (0-9) alias (e.g. for integrating with an ITSP phone gateway service):

Match string:	**9([0-9]+$)**
Replace string:	**\1**

If your environment includes a PSTN gateway or uses an ITSP (Internet telephony service provider), consider the potential for toll fraud if you have Call Routing Rules that can route calls to the PSTN gateway or ITSP, or if you allow conference participants to dial out to other participants via the PSTN gateway or ITSP. See PSTN gateways and toll fraud for more information.

Jinja2 templates and filters

Pexip Infinity uses a subset of the jinja2 templating language (see the Jinja Template Designer Documentation) to assist in creating content or deciding on processing logic when configuring various features such as One-Touch Join, VMR scheduling, LDAP provisioning or writing local policy scripts.

Template content

Jinja2 templates consist of the following elements:

- **literal text** that you want to add to the output or result, such as prefixing every generated VMR name with **meet.**
- **variables** that are either locally defined within the script or template, or that are provided automatically according to the context in which the template is being used (for example there are **call_info** and **participant** variables when configuring local policy), or a **givenName** variable when syncing VMRs via LDAP.
- **filters** that can manipulate text or modify the content of variables or text strings, such as **join**, **pex_update** or **pex_to_json**
- **delimiters** such as {{...}} and pipes | which are used to enclose variables and define filter expressions
- **jinja statements and control structures** (see Jinja Control Structures)

Supported jinja2 filters

Pexip Infinity supports a subset of filters from the jinja2 templating language. Any jinja filters that are not listed below have been disabled in Pexip Infinity. See List of Builtin Filters for more information about these filters.

abs	float	last	replace	truncate
capitalize	format	length	round	upper
default	int	lower	striptags	
first	join	range	trim	

To use a filter you would typically follow the syntax **{{<source_value>|<filter_name>}}**.

In most cases the <source_value> is likely to be a variable, for example **{{givenName|upper}}**, although it could be one or more literal values, for example **{{ [1, 2, 3, 4]|join }}**.

Some filters take parameters, for example **{{sn|truncate(5)}}**. You can also use multiple filters in the same expression, for example **{{sn|truncate(5)|upper}}**.

The **trim** filter is often used. This trims leading and trailing whitespace from the string held in the <source_value>.

Example usage: **{{title|trim}}**

If the **title** field contained " Project Manager ", this would be converted to "Project Manager".

The **replace** filter is also often used. This replaces one string with another string. The first argument of the filter is the substring that should be replaced, the second is the replacement string. If the optional third argument count is given, only the first count occurrences are replaced.

Example usage: **{{ department|replace("Personnel", "HR") }}**

If the **department** field contained "Personnel Department Personnel", this would be converted to "HR Department HR".

Example usage: **{{ department|replace("Personnel", "HR", 1) }}**

In this case a count of 1 is specified, thus if the **department** field contained "Personnel Department Personnel", it would be converted to "HR Department Personnel".

For more complicated search and replace patterns, use the custom Pexip **pex_regex_replace** filter described below.

Custom Pexip filters

In addition to the jinja filters, Pexip also provides the following custom filters, which are typically used to manipulate data:

Filter	Description and example usage
pex_base64	Performs Base64 encoding on the input field.
pex_clean_phone_number	This extracts only +0123456789 characters (and removes ()&%#@|"':;, A-Z,a-z etc).
pex_debug_log (message)	The **pex_debug_log** filter can be used to help debug your script. It writes debug messages to the Pexip Infinity support log. You can include literal text and variables. To avoid filling the support log and causing it to rotate, remove all **pex_debug_log** filters from your scripts as soon as they are working correctly.
pex_find_first_match(string_list, 'find_regex')	This extracts from the list the first value that matches the specified regex.
pex_hash	Performs a hash of a field.
pex_head (maxlength)	Returns, at most, the first maxlength characters from the input field.
pex_in_subnet	Tests whether a given address is within one or more subnets. It takes as input the address you want to test, and one or more subnet ranges, and returns either True or False.
pex_md5	Applies an MD5 hash to the input field.
pex_now(timezone)	The **pex_now** filter takes an optional parameter of a timezone description e.g. 'UTC', 'Asia/Tokyo', or 'US/Eastern' and returns the current date and time for that timezone. UTC is assumed if a timezone is not provided. The resulting available attributes are **year**, **month**, **day**, **hour**, **minute**, **second** and **microsecond**. Example usage: **{% set now = pex_now("Europe/London") %}** **{% if now.month == 2 and now.day == 29 %}**
pex_random_pin (length)	Generates a random PIN of the given length. Note that this filter does not take any input.
pex_regex_replace ('find_regex', 'replace_string')	This performs a regex find and replace. Example usage: **{{mail|pex_regex_replace('@.+','@otherdomain.com')}}** This example takes as input an email address contained in the **mail** variable and changes the domain portion of the address to @otherdomain.com. For example, it will transform **user1@domainA.com** to **user1@otherdomain.com**, and **user2@domainB.com** to **user2@otherdomain.com** etc.
pex_regex_search ('regex pattern', 'string_to_search')	This performs a regex search for the first location that matches the pattern and returns the regex groups. Example usage: **{% set groups = pex_regex_search("([a-z0-9.-]+)@([a-z0-9.-]+.com)", "example string with someone@example.com") %}** **{% if groups %}** **{{ groups[0] }}@{{ groups[1] }}** **{% endif %}** This example takes as input a string containing an email address and extracts the email using two regex groups.
pex_require_min_length(length)	This validates that the input string field has the specified minimum length. Syntax: {{ some_string|pex_require_min_length(2) }}

Filter	Description and example usage
pex_reverse	This reverses the characters in the input field.
pex_safelinks_decode	This filter converts a string that has been rewritten by Safe Links in Microsoft Defender for Office 365 into the original URL.

pex_strlen	Returns the length of string. The basic usage syntax is:
	`{% set some_length = "Example"|pex_strlen %}`
	`{# sets a variable named some_length to the value 7 #}`

pex_tail(maxlength)	Returns, at most, the last maxlength characters from the input field.
pex_to_json	Converts a Python dictionary variable into JSON format.
pex_to_uuid	Converts a base64 string to a UUID.
pex_update	Updates Python dictionary variables.
pex_url_decode	This filter decodes a previously URL-encoded string. For example, it can be used with OTJ custom rules to simplify the regular expressions required.
pex_url_encode	This filter creates URL parameters that are safely URL-encoded.
pex_urldefense_decode	This filter converts a string that has been rewritten by Proofpoint's URL Defense into the original URL.
pex_uuid4()	This generates a uuid (universally unique identifier). Note that this filter does not take any input.

Encryption methodologies

This page describes the encryption methods that are used for Pexip Infinity. It covers:

- Pexip nodes
- Endpoints
- Pexip upload portal

Pexip nodes

All communication links between the Management Node and Conferencing Nodes, and between Conferencing Nodes, use an IPsec transport with the following settings:

- 256-bit AES-GCM for encryption
- a 4096 bit Diffie-Hellman modulus for key exchange.

No other ciphers, hashes or moduli are permitted.

These settings apply to both the initial channel set up for key exchange (ISAKMP) and the secondary channel over which application data is transported (ESP).

Inter-node traffic is restricted to only protocols that are expected for the successful operation of Pexip Infinity, including but not necessarily limited to call signaling, media, status, and configuration information; any unexpected traffic/protocols are dropped.

Endpoints

Encrypted connections between Pexip Infinity and endpoints (including SIP trunks) use:

- AES 128-bit or 256-bit encryption for media
- TLS for SIP call control (for more information, see Managing TLS and trusted CA certificates)
- SRTP for SIP media
- H.235 for H.323 media

Pexip apps (web/desktop/mobile) use:

- HTTPS TLS for signaling
- DTLS-SRTP for WebRTC media

Pexip upload portal

The logs and snapshots uploaded to the Pexip upload portal use the following settings:

- 256-bit AES-GCM for encryption
- Salt parameter for additional security
- The encryption key is derived from 128-bit character password

Interoperability

We endeavor to make Pexip Infinity interoperable with all relevant standards-based devices. For the most up-to-date information on the devices and software versions with which Pexip Infinity is known to work, along with any known issues, see https://docs.pexip.com/admin/interoperability.htm.

We welcome your feedback on any known issues with any devices, and we would also like to hear of any other devices that you have used with Pexip Infinity. Please send your comments to your Pexip authorized support representative.

Supported RFCs

Pexip Infinity supports the following RFCs:

- RFC 1889 RTP: A Transport Protocol for Real-time Applications
- RFC 2190 RTP Payload Format for H.263 Video Streams
- RFC 2429 RTP Payload Format for the 1998 Version of ITU-T Rec. H.263 Video (H.263+)
- RFC 2782 DNS RR for specifying the location of services (DNS SRV)
- RFC 2790 Host Resources MIB
- RFC 2976 The SIP INFO Method
- RFC 3261 SIP: Session Initiation Protocol
- RFC 3263 Locating SIP Servers
- RFC 3264 An Offer/Answer Model with SDP
- RFC 3550 RTP: A Transport Protocol for Real-Time Applications
- RFC 3581 Symmetric Response Routing
- RFC 3605 RTCP attribute in SDP
- RFC 3711 The Secure Real-time Transport Protocol (SRTP)
- RFC 3840 Indicating User Agent Capabilities in SIP
- RFC 3890 A Transport Independent Bandwidth Modifier for SDP
- RFC 3891 SIP "Replaces" Header
- RFC 3984 RTP Payload Format for H.264 Video
- RFC 3986 Uniform Resource Identifier (URI): Generic Syntax
- RFC 4320 Actions Addressing Identified Issues with the Session Initiation Protocol's (SIP) Non-INVITE Transaction
- RFC 4321 Problems Identified Associated with the Session Initiation Protocol's (SIP) Non-INVITE Transaction
- RFC 4566 SDP: Session Description Protocol
- RFC 4568 SDP: Security Descriptions for Media Streams
- RFC 4574 The Session Description Protocol (SDP) Label Attribute
- RFC 4585 Extended RTP Profile for RTCP-Based Feedback
- RFC 4587 RTP Payload Format for H.261 Video Streams
- RFC 4629 RTP Payload Format for ITU-T Rec. H.263 Video
- RFC 4733 RTP Payload for DTMF Digits, Telephony Tones and Telephony Signals
- RFC 4796 The SDP Content Attribute
- RFC 5168 XML Schema for Media Control
- RFC 5245 Interactive Connectivity Establishment (ICE)
- RFC 5389 Session Traversal Utilities for NAT (STUN)
- RFC 5577 RTP Payload Format for ITU-T Recommendation G.722.1
- RFC 5763 Framework for Establishing a Secure Real-time Transport Protocol (SRTP) Security Context Using Datagram Transport Layer Security (DTLS)
- RFC 5764 Datagram Transport Layer Security (DTLS) Extension to Establish Keys for the Secure Real-time Transport Protocol (SRTP)
- RFC 5766 Traversal Using Relays around NAT (TURN)
- RFC 6026 Correct Transaction Handling for 2xx Responses to Session Initiation Protocol (SIP) INVITE Requests
- RFC 6125 Representation and Verification of Domain-Based Application Service Identity
- RFC 6416 RTP Payload Format for MPEG-4 Audio/Visual Streams
- RFC 7714 AES-GCM Authenticated Encryption in the Secure Real-time Transport Protocol (SRTP)
- RFC 8446 The Transport Layer Security (TLS) Protocol Version 1.3
- RFC 8855 The Binary Floor Control Protocol
- RFC 8856 SDP Format for BFCP Streams
- draft-ietf-avt-rtp-h264-params-01.txt
- draft-ietf-payload-rtp-opus-01.txt

- draft-ietf-payload-rtp-vp8-10.txt
- draft-ietf-mmusic-sdp-g723-g729-04.txt

Patents

This table is intended to serve as notice under 35 U.S.C. § 287(a).

Product	Patent
Pexip Infinity	US 8971407
Pexip Infinity	US 8976225
Pexip Infinity	US 9118808
Pexip Infinity	US 9407933
Pexip Infinity	US 10382337
Pexip Infinity	US 10873745
Pexip Infinity	US 10972702
Pexip Infinity	US 11088860
Pexip Infinity	US 11538169
Pexip Infinity	US 11729238
Pexip Infinity	US 11909800
Pexip Infinity	US 12015875
Pexip Infinity	US 12137053

Accessibility

This page summarizes the accessibility features of the Pexip Infinity platform.

Web app

The current Pexip Web App (Webapp3) is certified as fully compliant with the Web Content Accessibility Guidelines (WCAG) version 2.2 to AA standard. For more information, please contact your Pexip authorized support representative.

Other products

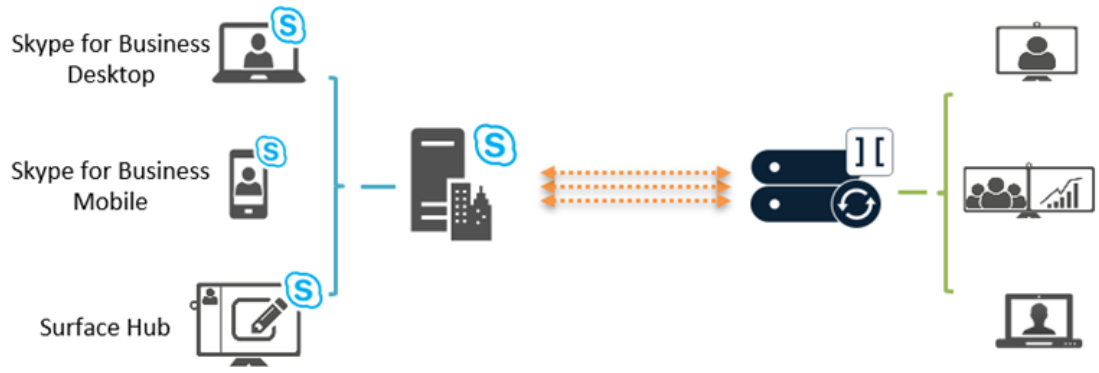
The following features are designed to improve accessibility of the Pexip Infinity user experience, including the Pexip apps, in support of AAA WCAG 2.0 compliance.

Feature	More information
Progress animations and customizable spinners	Colors and contrast can be customized through the Pexip branding portal.
Resizing text	Text can be resized to 200% without loss of content or function.
Events stream	All significant events and chats are displayed textually in the events panel.
Screen reader support	Screen readers can access the Events stream live.
Navigation and focus	Navigation has been designed to streamline keyboard input.
Help	Tooltips are used throughout the client to create context and provide help.
Audio prompts	Audio prompts complement splash screen messaging. These are customizable.
Text overlay	Text overlays displaying participant names can be enabled.
Alarms	The presence of active alarms is indicated by a flashing blue icon on the Management Node Administrator interface.
Choice of colors	Colors used in the Administrator interface have been chosen to provide maximum accessibility and contrast. For example, in the Administrator Logs and Support logs, warnings and errors are highlighted with an orange or blue background respectively; Live View search is yellow and blue.

Using Microsoft Skype for Business with Pexip Infinity

Pexip Infinity allows Microsoft Skype for Business users to meet with other people regardless of the system they are using – Skype for Business, web browsers or traditional video conferencing systems. All participants can enjoy wideband audio, high definition video and cross-platform presentation sharing.

It can be integrated with SfB as part of an existing, on-premises SfB environment inside an enterprise network, or as a standalone Pexip environment deployed in a public DMZ that enables direct federation with remote SfB environments, or as a hybrid deployment where SfB users may be homed either on-premises or in Office 365.



Pexip Infinity enables full interoperability between Microsoft's H.264 SVC/RDP and H.263, H.264, VP8 (WebRTC) and BFCP/H.239 for truly seamless video and content sharing in any-to-any configurations, such as multiparty conferences.

In addition to enabling SfB participants to join conferences hosted on Pexip Infinity, Pexip Infinity can act as a gateway between SfB and standards-based endpoints. This enables SfB clients to receive and initiate point-to-point calls with H.323/SIP endpoints and registered Pexip apps, and invite those devices into a SfB meeting while retaining the native meeting experience on each device.

Architecture options

Pexip Infinity can be integrated with Microsoft Skype for Business in three different ways:

- As part of an existing, on-premises SfB environment inside an enterprise network, referred to as **on-premises deployment**.

  To integrate Pexip Infinity with an existing, on-premises SfB environment, one or more SIP domains are statically routed from the SfB environment towards one or more Pexip Infinity Conferencing Nodes. Then, when a SfB user dials a conference alias, such as meet.john@vc.example.com, or the alias of a standards-based endpoint, the user is placed into the appropriate Pexip-hosted conference. The SfB user can also pin one or more such aliases to their contact list for easy access later.
- As a standalone Pexip environment deployed in a public DMZ, referred to as **public DMZ deployment**.

  As Pexip Infinity supports SfB natively, it can be deployed to enable SfB interoperability without having any existing, on-premises SfB infrastructure. In such a deployment, Pexip Infinity can federate directly with remote SfB environments (on-premises environments as well as SfB Online/Office 365), without the need for a local SfB environment.
- As a **hybrid deployment** which is a mix of on-premises and Office 365 deployments where users may be homed in either environment. A hybrid deployment has the same configuration requirements as a **public DMZ deployment**.

You will typically choose one of these methods, depending on requirements and preference.

Pexip Infinity as a SfB gateway

Pexip Infinity can act as a gateway between SfB and standards-based endpoints. This enables SfB clients to:

- invite H.323/SIP endpoints and registered Pexip apps into a SfB meeting via manual dialout or drag and drop from the contacts list
- use the Infinity Gateway to route incoming calls directly into an ad hoc or scheduled SfB meeting
- when dialed into a Pexip VMR conference, invite other SfB or external contacts into that same Pexip VMR (this creates a new SfB meeting which is merged with the existing Pexip VMR)
- receive and initiate person-to-person calls with standards-based devices, and then optionally add other participants (to escalate to a multipoint SfB meeting).

More information

For full information on using Microsoft Skype for Business with Pexip Infinity, see Integrating Microsoft Skype for Business with Pexip Infinity.

When is a reverse proxy, TURN server or STUN server required?

Since version 16 of Pexip Infinity, we recommend that you deploy Proxying Edge Nodes instead of a reverse proxy and TURN server if you want to allow externally-located clients to communicate with internally-located Conferencing Nodes.

If you do not want to deploy Proxying Edge Nodes, and all of your Conferencing Nodes are privately addressed, you will need to use a reverse proxy and a TURN server to allow external endpoints such as Pexip apps to access your Pexip Infinity services, and you may need to use a TURN server for Skype for Business clients. A TURN server can also act as a STUN server, however, in some Pexip Infinity deployment scenarios where the TURN server is deployed inside your enterprise firewall, you may need to configure a separate, external STUN server.

When connecting to a privately-addressed Conferencing Node, Pexip WebRTC clients that are behind a NAT may also use a STUN server to find out their public NAT address.

If you are using direct media then you may also want to provision details of a client TURN server to the WebRTC clients.

When using direct media we strongly recommend for enhanced security that you use your own dedicated TURN server that is located in your DMZ.

The following table shows when a reverse proxy, TURN server or STUN server needs to be deployed (if you are not using Proxying Edge Nodes). When used, they must be publicly accessible, and routable from your on-premises Conferencing Nodes.

External endpoint / client	Conferencing Node addresses	Reverse proxy	TURN server	STUN server (for Conferencing Nodes)	STUN server (for WebRTC clients behind NAT)
Pexip WebRTC clients	Private (on-premises)	✓	✓	✓ (if the TURN server is inside the firewall)	✓
Skype for Business clients *	Private (on-premises)	-	✓ (only required if internal Conferencing Node cannot route to the public-facing interface of the SfB Edge server)	✓ (if the TURN server is inside the firewall)	✓
Any endpoint / client	Publicly reachable — either directly or via static NAT	-	-	-	-
Pexip WebRTC clients using direct media	-	-	✓ (provisioned as a Client TURN server)	-	✓

* Also requires a Skype for Business Edge Server when Conferencing Nodes are privately addressed.

Note that you may still want to deploy a reverse proxy in front of your Proxying Edge Nodes if, for example, you want to:

- host customized web app content
- use it as a load balancer for Pexip's Secure Scheduler for Exchange service, to proxy requests from Outlook clients to Conferencing Nodes.

Integrating with streaming and recording services

This section provides step-by-step instructions on how to integrate with popular streaming services. For an overview of streaming with Pexip Infinity, see Streaming and recording a conference.

Streaming a conference to YouTube

This guide explains how to stream a conference being held in a Pexip Virtual Meeting Room or Virtual Auditorium to YouTube.

YouTube prerequisites

Before you can obtain an RTMPS streaming URL from YouTube, you must ensure that you have a verified YouTube account, and that the account is enabled for live events.

1. From your YouTube account settings page, select **Channel status and features** (www.youtube.com/features).
2. If your **Account status** is not verified, select **Verify** and follow the YouTube instructions.
3. Ensure that your account is enabled for **Live streaming**.

Note that enabling your first live stream may take up to 24 hours.

Setting up streaming

To stream a conference to YouTube, you must first obtain a live streaming URL via YouTube. You then initiate a call from the VMR to YouTube, by adding the YouTube URL as a conference participant. There are two ways in which you can obtain an RTMPS streaming URL from YouTube:

- Use Pexip's own utility at https://yt.pexip.com. This method simplifies the generation process and automatically uses the appropriate settings. See Using Pexip's own utility to obtain a YouTube streaming URL.
- Generate your URL directly from within your YouTube account at https://studio.youtube.com/channel/. See Manually generating your own streaming URL within YouTube Studio.

Note that the live stream will have a 20-30 second delay. This is because YouTube buffers the stream so that it can tolerate brief connection losses and to ensure a good consistent experience. This is standard streaming behavior.

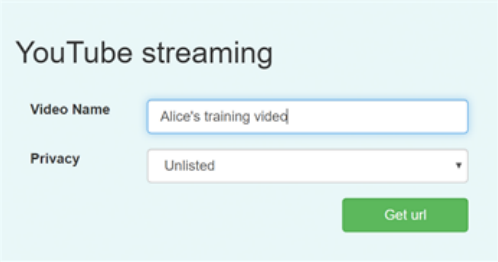
The appropriate procedures for obtaining your streaming URL, starting streaming, and streaming at Full HD (1080p) are described below.

Using Pexip's own utility to obtain a YouTube streaming URL

This procedure explains how to use Pexip's own utility to request a URL for a YouTube RTMPS stream on your behalf.

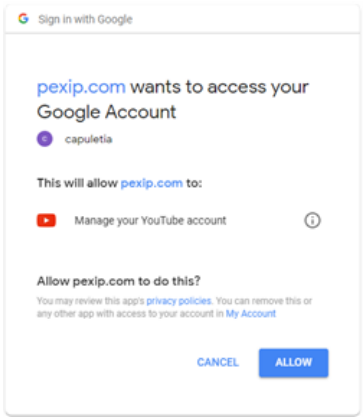
1. Go to https://yt.pexip.com.
2. Enter a **Video Name** — this is the name that will appear in YouTube.
3. Select a **Privacy** level:
   - ***Unlisted***: viewers must know the streaming URL to see the stream.
   - ***Public***: anybody can find the stream on YouTube. This is not recommended unless you are streaming very public content.
   - ***Private***: restricts access to only people that you have explicitly allowed to view the stream.

   Default: ***Unlisted***.
4. Select **Get url**.

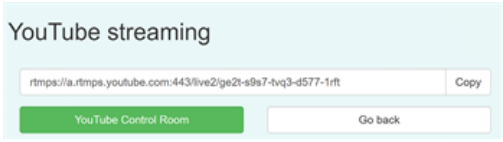


5. If you are not already signed in to a Google Account, you must either sign in or select an account.

6. At the prompt, **Allow** pexip.com to access your Google account:



7. The streaming URL will be generated and displayed.



   If you receive a "The user is not enabled for live streaming" error message, this means that you either do not have a verified YouTube account, or that the account is not enabled for live events.
8. Copy the rtmps:// address. Note that the **Copy** button does not always work; you must select the address and copy it manually. Leave this browser window open.
9. You can now go to Adding the participant URL and enabling streaming.

Manually generating your own streaming URL within YouTube Studio

This procedure explains how to use YouTube to generate a streaming URL.

Note that the YouTube **Privacy** setting is ***Public*** by default, so we recommend that you change this to ***Unlisted***.

1. In your YouTube account go to your Studio channel dashboard https://studio.youtube.com/channel/.
2. Select **Go live**.
3. Select **Schedule Stream** (at the top right of the page).
4. If this is your first event, select **Create New**, or, if appropriate, you can use the settings from a previous event.
5. Enter the event **Details**:

Title	The title of the video.
Description	An optional description of the event.
Broadcast type	Select ***Streaming software***.
Category, Thumbnail, Playlists, Audience	Select the options as appropriate for your event.

6. Select **Next**.
7. Select the **Customisation** options as appropriate for your event, and select **Next**.
8. Enter the **Visibility** settings:

Privacy	We recommend ***Unlisted***, which means that viewers must know the streaming URL to see the stream.
Schedule	Leave the default values (the current date/time) if you are going live "now" or amend them as appropriate.

9. Select **Done**.
10. YouTube generates the stream and takes you to a page showing the stream settings and displaying a stream preview window

(which is waiting for your connection from the conference).

11. You can now go to Adding the participant URL and enabling streaming.

Adding the participant URL and enabling streaming

Now that you have set up your stream, you can initiate a call from the Virtual Meeting Room to YouTube, and then begin streaming.

Creating the participant URL

Regardless of whether you used Pexip's utility or manually generated your own streaming URL, you may need to do some manipulation to it before you add it as a participant URL to your conference:

- **Pexip utility**: you should have copied an address in the style **rtmps://a.rtmps.youtube.com:443/live2/ge2t-s9s7-tvq3-d577-1rft** from the utility webpage.
- **Manual generation**: from within the YouTube studio stream settings, use the **Copy** buttons to produce the participant URL for your video stream as **Stream URL/Stream Key**:
  a. Take the **Stream URL**, for example **rtmps://a.rtmps.youtube.com:443/live2**
  b. Append a **/** (slash).
  c. Append the **Stream key**, for example **ge2t-s9s7-tvq3-d577-1rft**

  In this example, the final RTMPS URL is **rtmps://a.rtmps.youtube.com:443/live2/ge2t-s9s7-tvq3-d577-1rft**

Start streaming

1. Initiate a call from the Virtual Meeting Room to the streaming address. This is done by adding the participant URL you have generated as a conference participant. You can do this either from the Pexip Infinity Administrator interface or from a Pexip app connected to the VMR.

   When using the Administrator interface, use the following settings:
   - **Participant alias**: the YouTube participant URL.
   - **Protocol**: ***RTMP***
   - **Role**: we recommend selecting ***Guest*** (so that the streaming participant is not shown to other Guests in a Virtual Auditorium layout, and so that it does not keep a conference alive when all other Hosts have left).
   - **Streaming**: select this option.

   When using a Pexip app, use the following settings:
   - **Participant details**: enter your YouTube participant URL e.g. **rtmps://a.rtmps.youtube.com:443/live2/dau4-k4z1-5cf3-gg89-0c6x**

     RTMP authentication is supported; in this case credentials are included in the URI using the syntax **rtmps://username:password@host/...**.

     Note that a suitable Call Routing Rule is required when dialing out to a streaming service via Pexip apps.
   - **Role**: we recommend selecting ***Guest***.

   When Pexip Infinity has placed the call to the streaming service, the **Streaming enabled** icon is displayed, and for Pexip app users the streaming participant appears in the participant list.
2. Go to YouTube Studio (if you used Pexip's URL generator you can select the green link below the rtmps:// address that you copied) and wait for the stream to appear (this can take up to 1 minute) in the Preview window.

   Note that this is your preview only — at this stage the stream is not being broadcast. The stream has a 20-30 second delay.
3. When you are ready, select **Go Live** to start broadcasting.
4. You are now streaming to anyone who is allowed to access or find your streams (according to your **Privacy** settings).
5. When you have finished streaming you can select **End Stream** in YouTube Studio, and disconnect the RTMPS participant from the conference.

Streaming at Full HD (1080p)

If you want to stream at Full HD (1080p) resolution you must:

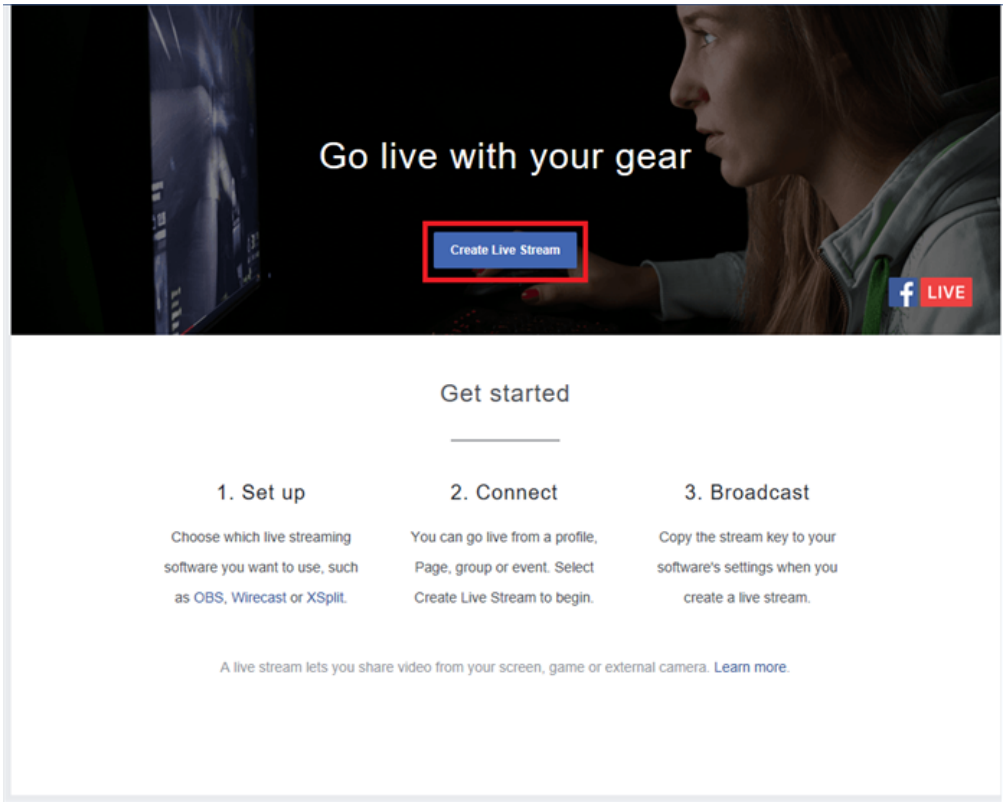
1. Ensure that the VMR you want to stream is configured with a **Maximum call rate** of ***Full HD*** (or uses a global default of Full HD).
2. Configure the outbound bandwidth on the VMR you want to stream to be 4096 kbps (**Maximum outbound call bandwidth** in the **Advanced options** of the VMR settings).
3. Ensure that the stream is capable of receiving 1080p. Within YouTube you must manually configure a suitable **Stream key** (in the stream settings) with a resolution of 1080p (with a bit rate of 3000 Kbps - 6000 Kbps).

Streaming a conference to Facebook

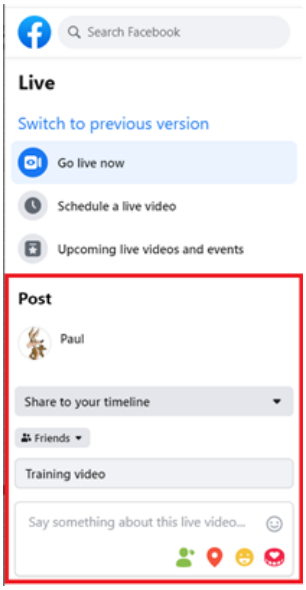
This guide explains how to stream a conference being held in a Pexip Virtual Meeting Room or Virtual Auditorium to Facebook.

To stream a conference to Facebook:

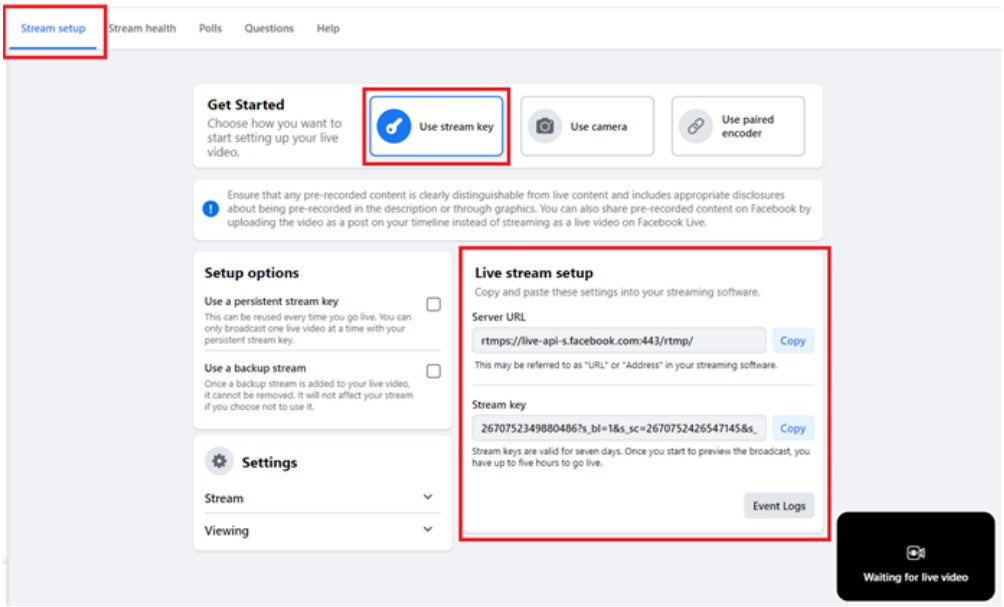
1. Go to facebook.com/live/create and select **Create Live Stream**:



2. From within the **Post** panel to the left of the page:
   - Select where to post your live broadcast (***Share to your timeline***, for example), and who should see it.
   - You can also add a **Live video title** and an associated message.



3. In the main body of the page, on the **Stream setup** tab, ensure that **Use stream key** is selected and that you can see the **Server URL** and **Stream key** settings. You will need these on the next step.



4. You must now initiate a call from the Virtual Meeting Room to Facebook, by adding the Facebook streaming address as a conference participant. You can do this either from the Pexip Infinity Administrator interface or from a Pexip app connected to the VMR.

   When using the Administrator interface, use the following settings:

   - **Participant alias**: the **Server URL** followed directly by the **Stream key**
   - **Protocol**: ***RTMP***
   - **Role**: we recommend selecting ***Guest*** (so that the streaming participant is not shown to other Guests in a Virtual Auditorium layout, and so that it does not keep a conference alive when all other Hosts have left).

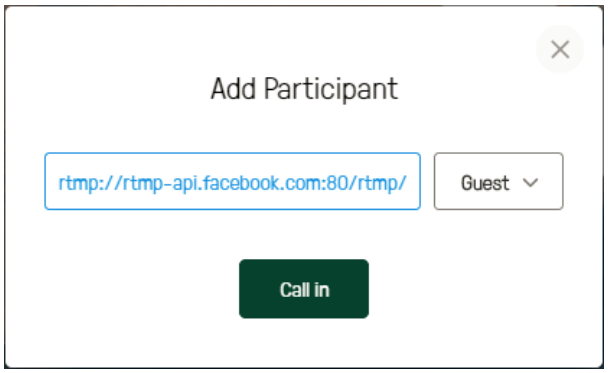
   When using a Pexip app, use the following settings:

   - **Participant details**: the **Server URL** followed directly by the **Stream key**, for example:

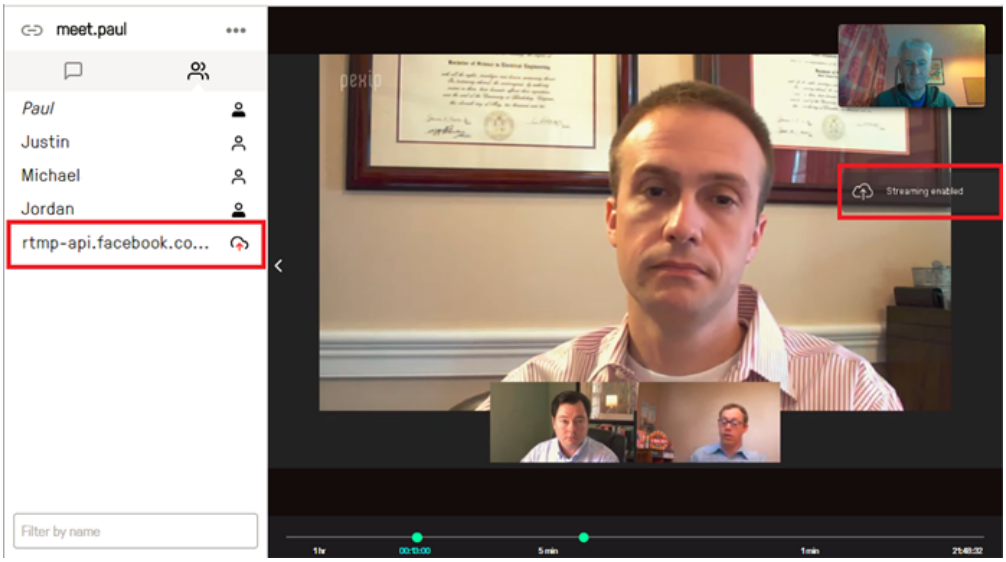
     rtmps://live-api-s.facebook.com:443/rtmp/2670752349880486?s_bl=1&s_sc=26707145&s_sw=0&s_vt=api-s&a=Abx7ECwitZM

Note that a suitable Call Routing Rule is required when dialing out to a streaming service via Pexip apps.

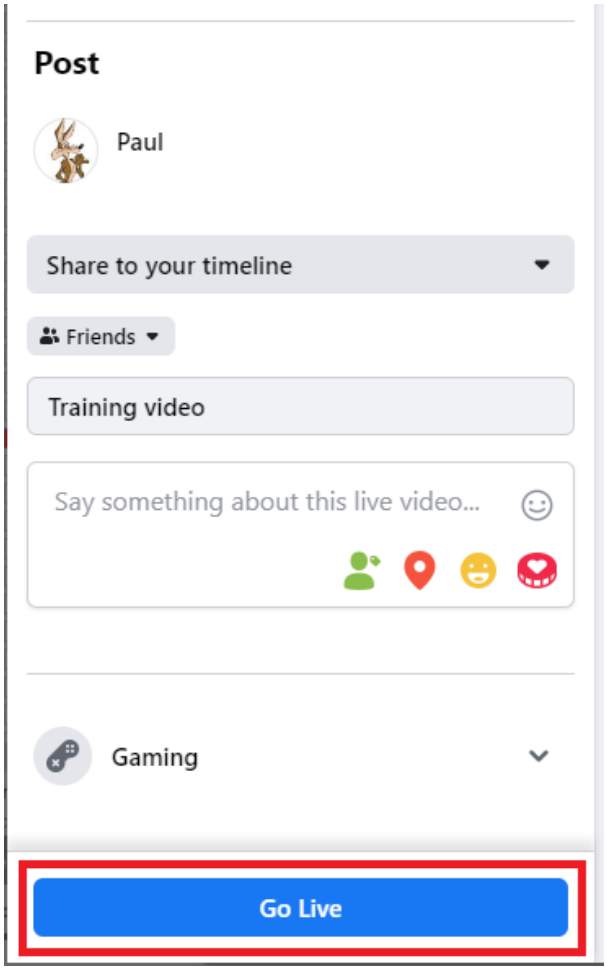
- **Role**: we recommend selecting ***Guest***.



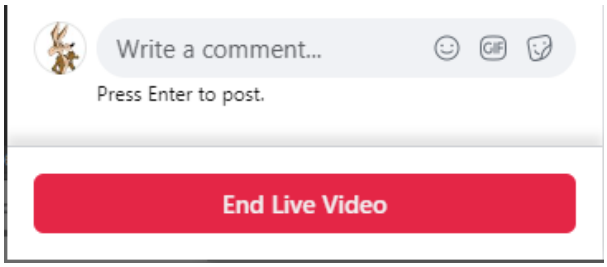
5. When Pexip Infinity has placed the call to the streaming service, the **Streaming enabled** icon is displayed, and for Pexip app users the streaming participant appears in the participant list.



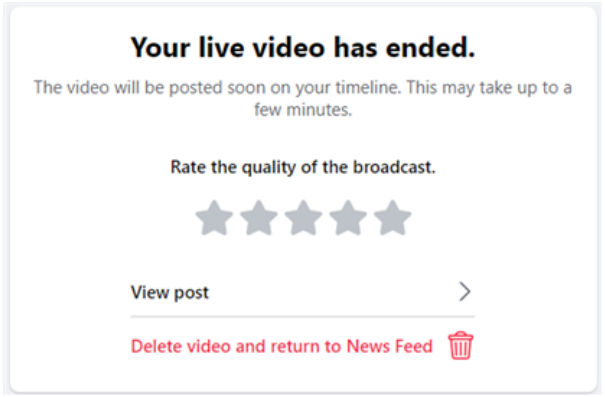
6. Return to Facebook. After a short delay you should see a preview of the video to be streamed. When you are ready, select **Go Live**:



7. You can see the live video and any comments as they are posted. When you have finished, select **End Live Video**:



8. Your video will be available on your timeline:



Streaming a conference to Wowza Video

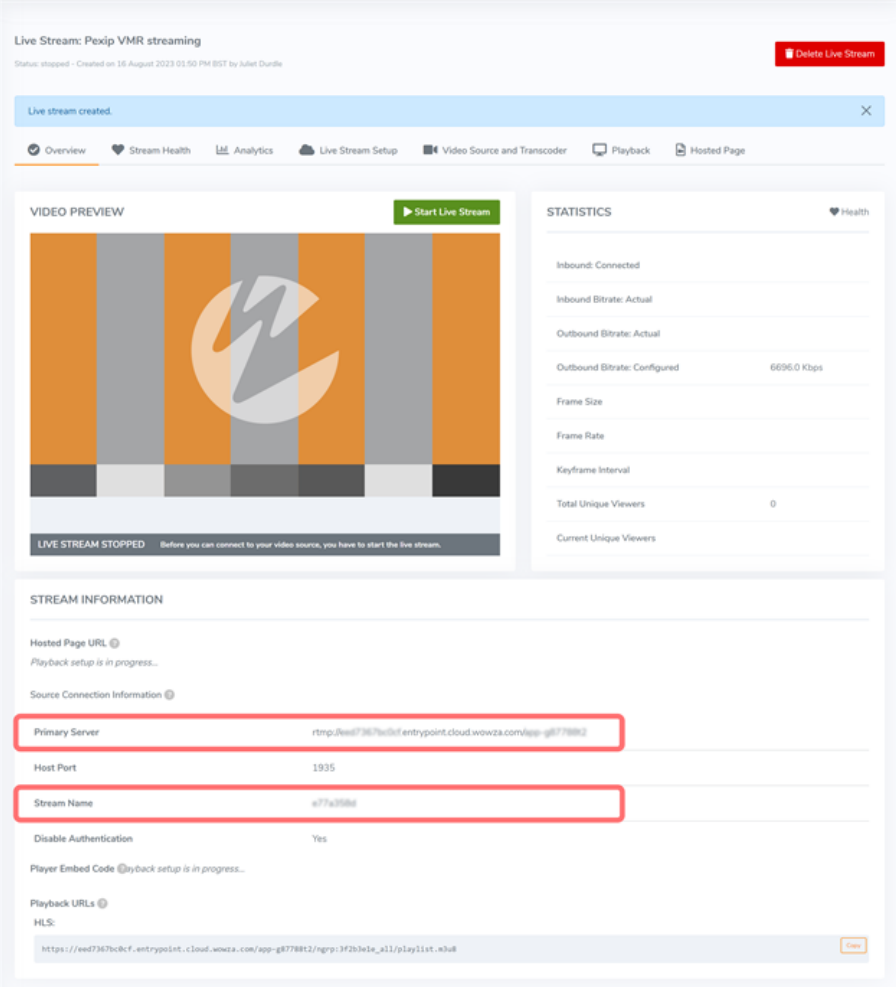
This guide explains how to stream a conference being held in a Pexip Virtual Meeting Room or Virtual Auditorium to Wowza Video.

To stream a conference to Wowza Video:

1. Log in to your Wowza Video account at https://video.wowza.com/.
2. Go to **Live Streams > Add Live Stream**. You are taken through a series of pages asking you to provide information about the stream you are creating. Below are the settings you **must** select; for all other settings, choose the option most suited to your requirements:

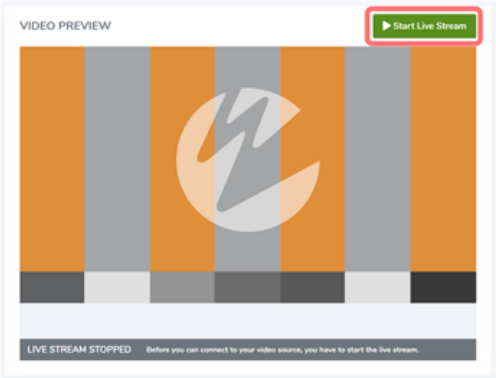
Video Source and Transcoder settings	Option to select
What camera or encoder will you use to connect to Wowza Video?	RTMP
Do you want to push or pull your stream?	Push Stream
Source Security	Disable authentication on the video source

3. Select **Finish**. You are taken to the **Overview** page for your new stream. In the **Source Connection Information** section you can see details of the stream's **Primary Server** and **Stream Name**:

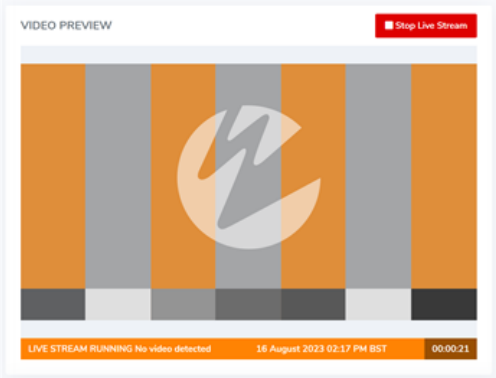


4. When you are ready to begin streaming, select **Start Live Stream**.

   You must begin the stream **before** you initiate the call from the VMR.



When streaming has begun, you will see **No Video Detected** underneath the video thumbnail:



5. Initiate a call from the Virtual Meeting Room to Wowza Video by adding the Wowza stream as a conference participant. You can do this either from the Pexip Infinity Administrator interface or from a Pexip app connected to the VMR. Alternatively, since Wowza streams can be re-used for subsequent events, you could set it up to be automatically dialed whenever a particular VMR is used.

   When using the Administrator interface, use the following settings:

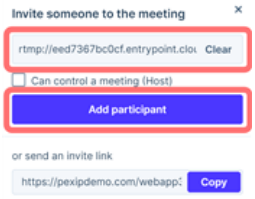
   - **Participant alias**: the **Primary Server**, followed by **/**, followed by the **Stream Name**
   - **Protocol**: ***RTMP***
   - **Role**: we recommend selecting ***Guest*** (so that the streaming participant is not shown to other Guests in a Virtual Auditorium layout, and so that it does not keep a conference alive when all other Hosts have left).

   When using a Pexip app, use the following settings:

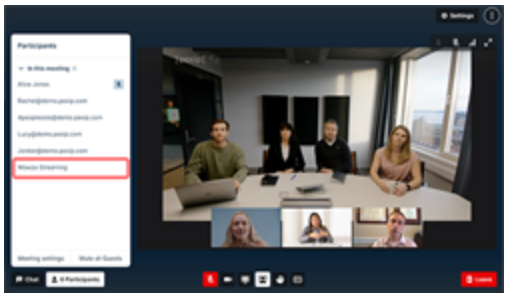
   - **Participant details**: the **Primary Server**, followed by **/**, followed by the **Stream Name**, for example:

     rtmp://48338d.entrypoint.cloud.wowza.com/app-33a3/ed7d6d5b

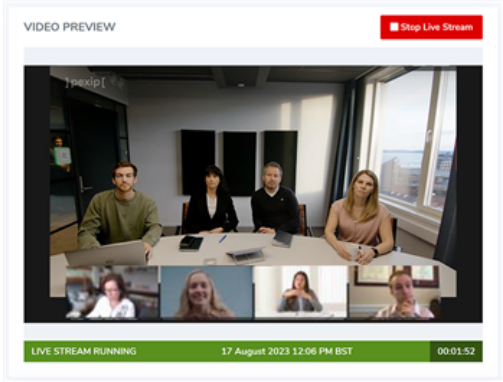
     Note that a suitable Call Routing Rule is required when dialing out to a streaming service via Pexip apps.
   - **Role**: we recommend that you do not add the participant as a Host.



6. When Pexip Infinity has placed the call to the streaming service, the **Streaming enabled** icon is displayed, and for Pexip app users the streaming participant appears in the participant list.



7. Return to Wowza Video. After a short delay you should see a preview of the video being streamed:



8. The stream is now available for viewing. If you chose the option to have Wowza Video host a webpage that plays back your video, anyone with the **Hosted Page URL** can view the live stream.
9. When you have finished, select **Stop Live Stream**.
10. If you have chosen to record the stream, after a short delay you can view and download this and any previous recordings of the stream by selecting the stream from the left-hand panel and then selecting **Recordings.**

Integrating with telephone systems (PSTN)

Pexip Infinity diagnostics

For information about how to monitor and troubleshoot Pexip Infinity, see:

Viewing live and historical platform status 547
Conference status 555
Diagnostics tools and reporting 578
Viewing Conferencing Nodes 586
Viewing system location status 588
Viewing cloud bursting status 589
Viewing alarms 590
Viewing login history 599
About the support log 600
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Creating and viewing diagnostic graphs 616
Disconnection reasons 618
Pexip Infinity port usage and firewall guidance 622
Troubleshooting the Pexip Infinity platform 631
Troubleshooting registering to Pexip Infinity 640

Viewing live and historical platform status

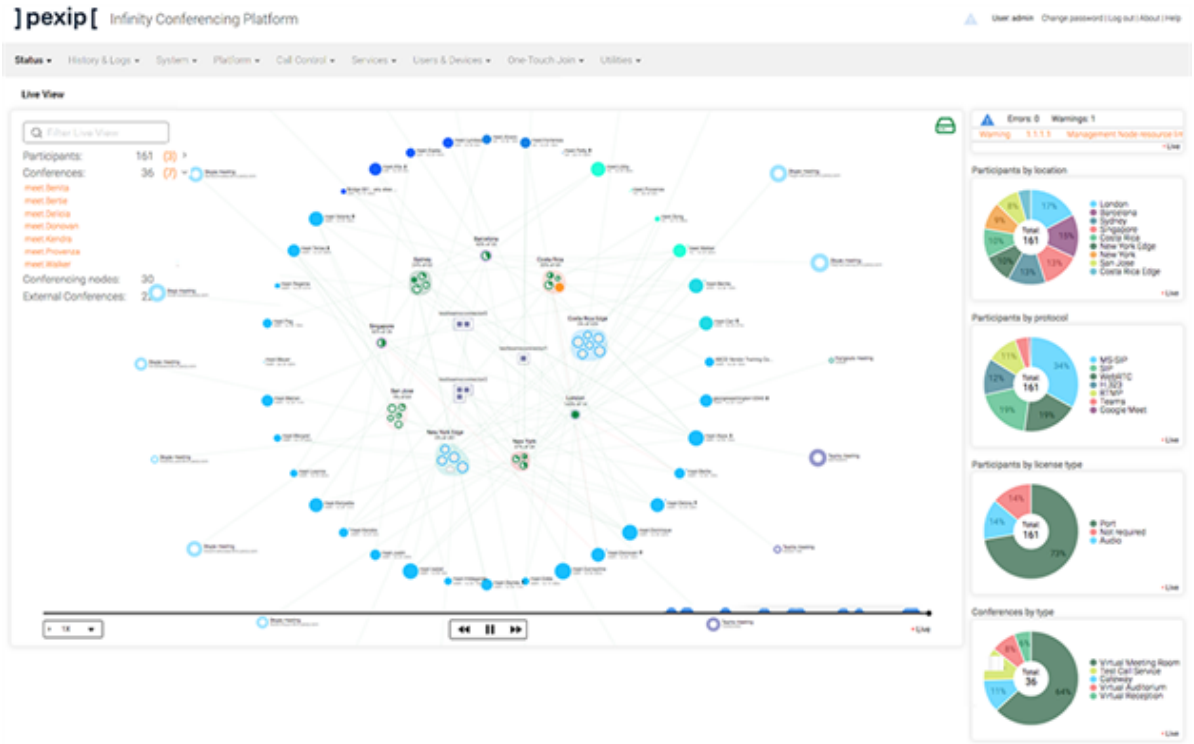
For a real-time overview of your Pexip Infinity deployment, go to **Status > Live View**. The interactive graphs and charts on this page show at a glance:

- the locations and Conferencing Nodes that are currently deployed and the available capacity and current load on each node
- all conferences that are currently taking place, and the nodes on which they are being hosted
- any conferences or conference participants that are experiencing call quality issues
- any error or warning alarms
- (during an upgrade process) which nodes are currently being upgraded, which nodes are still waiting to be upgraded, and which are in maintenance mode
- pie charts showing a breakdown of participants by location, protocol, license type and the different conference types being hosted.

You can interact with the graphs and charts by hovering your mouse pointer over each item to view specific information on each location, node and conference. You can also drill down to view more detailed current and historical information.

You can filter the view to show specific conferences or participants. You can also choose to remove conferences and backplanes entirely from the view if they take up too much space.

By using the timeline controls, you can rewind and replay the graph to view full node status and conference activities during the previous seven days. For a complete history of all conferences, see Viewing historical information about conferences.



Key to icons and symbols

The status graphs use the following symbols:

Icon	Meaning	Icon	Meaning
Management Node			

Icon	Meaning	Icon	Meaning
	The Management Node in normal operation.	(Pulsating)	The Management Node with an error-level alarm.
(Pulsating)	The Management Node with a warning-level alarm.		
Conferencing Nodes *			
	A Transcoding Conferencing Node during normal operation. The amount of green fill within the circle indicates the current media load (in terms of percentage of estimated HD ports in use), so an unused node is white and a fully loaded node is filled entirely green.		A Proxying Edge Node during normal operation. The amount of green fill within the circle indicates the current media load (in terms of percentage of proxying capacity in use), so an unused node is white and a fully loaded node is filled entirely green.
	A cloud bursting node that is currently on standby i.e. stopped.		A node that is in maintenance mode.
(Rotating)	A cloud bursting node that is currently starting (green circle spinning clockwise) or stopping (spinning counter-clockwise).		A node that is waiting to be upgraded. When you begin an upgrade of your platform, all nodes will have this status but will still be able to handle calls. After each node is upgraded in turn, it will return to normal operation.
(Pulsating)	A node that has an error-level alarm.	(Pulsating)	A node that is currently upgrading.
(Pulsating)	A node that has a warning-level alarm.		
Teams Connectors			
	When Azure Event Hub is not enabled, Teams Connector instances are only displayed when they are currently involved in a Teams call. Each purple square represents a Teams Connector instance.		When the Azure Event Hub is enabled, all Teams Connectors that are integrated with Pexip Infinity are shown. Each purple square represents a Teams Connector instance and the fill level of the square represents the current media load. A filled square in lighter purple represents an instance that is draining.
Conferences †			
	Conference being held in a Virtual Meeting Room.		Gateway call.
	Conference being held in a Virtual Auditorium.		Call to the Test Call Service.

Icon	Meaning	Icon	Meaning
	Call to a Virtual Reception.		A Skype for Business meeting.
	Call to a Media Playback Service.		A Microsoft Teams meeting.
	A conference that is experiencing call quality issues displays a blue asterisk next to the conference icon.		A Google Meet meeting.
	Conference is locked.		

The labels shown against each conference are based on the name of the VMR, Virtual Auditorium and so on, or the Call Routing Rule name for gateway calls.

Participants			
	The participant is currently presenting content.		API / Meeting Manager participant.
	Streaming participant.		

* For Conferencing Nodes, the size of each icon is relative to its total capacity.

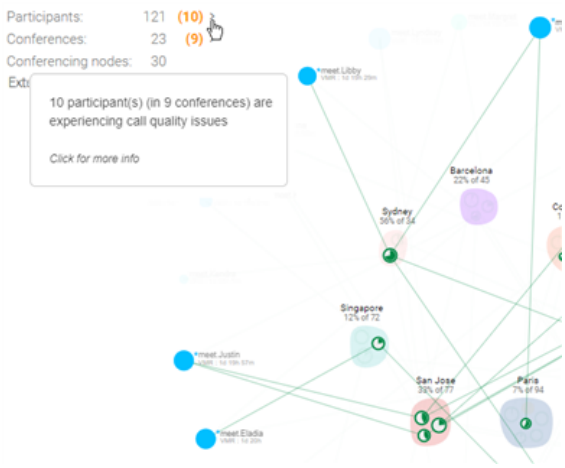
† For conferences, the size of each icon is relative to the number of participants.

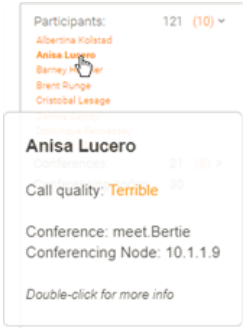
Platform summary status, call quality issues and alarms

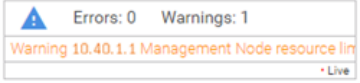
The statistics at the top of the graph summarize the platform status. They show the number of participants currently connected to the platform, the number of concurrent conferences, the number of Conferencing Nodes, and the number of externally-hosted conferences (i.e. Microsoft Teams or Skype for Business meetings, or Google Meet).

The issues statistics (in orange) are only shown if any of the participants or individual conferences are experiencing call quality issues. If you hover over this area, the affected participants and conferences are indicated.

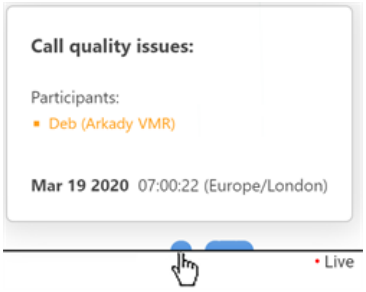
You can click to expand the list of issues and you can also review the details of individual participants and conferences.







The alerts area shows any error and warning alarms. You can click on the alarm for more details, to acknowledge a warning-level alarm (which hides it from Live View only), and to access troubleshooting information.



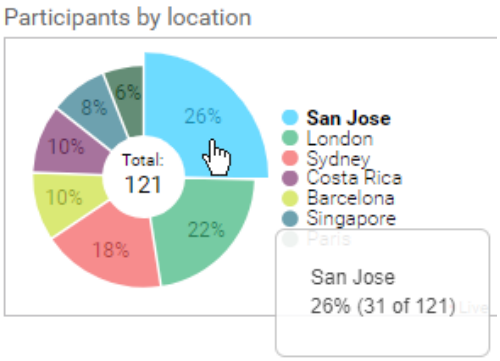
The interactive timeline indicates in blue any times in the past when a participant or backplane had call quality issues. You can hover over each blue bubble to get details of the issues that occurred at that time. In very large/busy systems with poor networks this is limited to the last 10,000 events over the last 7 days.

Pie charts and detailed participant usage graphs

The pie charts show a breakdown of:

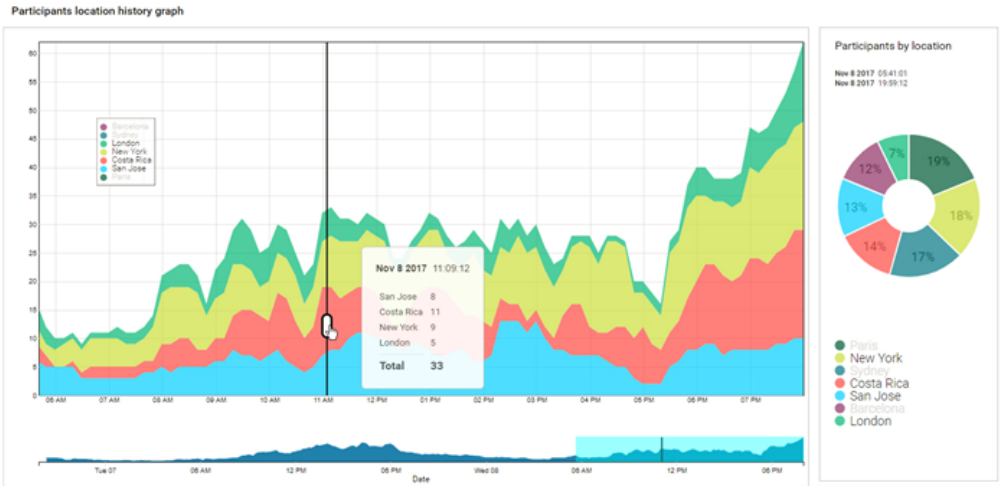
- Participants by location
- Participants by protocol
- Participants by call license type (port, audio, or not required) — note that this shows total participants; some types of participant calls consume 2 licenses
- Conference types (VMRs, Virtual Receptions, gateway calls etc.)

You can hover over an area of the chart to see more information.



If you click on a pie chart you are taken to a more detailed interactive graph showing historical information about those locations, protocols, licenses or conference types, where you can filter out or select specific elements e.g. you could choose to only view participants using the SIP protocol, or all participants except those in the London location.

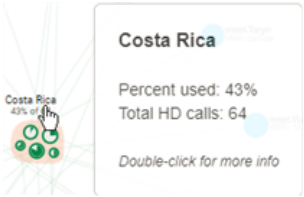
This example here shows the participants by location history graph, but all of the other graphs (protocol, license, conference type) work in the same way. You can:



- Click on individual locations (or protocols, licenses, conference types) to include/exclude them from the chart.
- Grab and move the vertical timebar up or down to adjust the time period that is displayed, or move the bar left or right to show the breakdown of participants for that point in time, or just click anywhere on the graph to move the vertical timebar to that point. The historical data is captured at 10 minute intervals.
- Adjust, or grab and move the timebox area in the bottom graph to quickly change the date and time period that is displayed in detail in the main chart. You can double-click on the bottom graph to reset the main chart to zoom out to show all available data.
- Grab and move the legend to another area of the chart.

Viewing location status

Each location is represented by a group of one or more Conferencing Nodes against a colored background. A different color is used for each location. The label gives the name of the location and summarizes the amount of available HD port capacity across all Conferencing Nodes in that location, and shows how much of that capacity is currently in use.

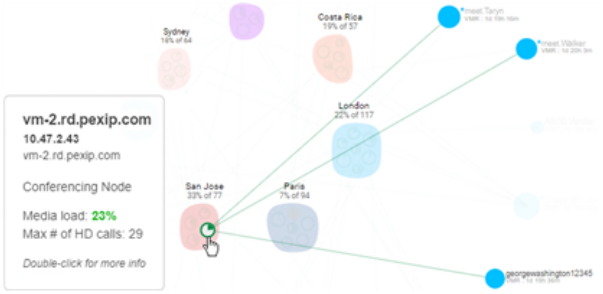


You can double-click to get more detailed location information, including an interactive status chart showing the load history for that location.

The example here shows the **Costa Rica** location, which is current using 43% of the 64 available HD ports.

Viewing Conferencing Node status

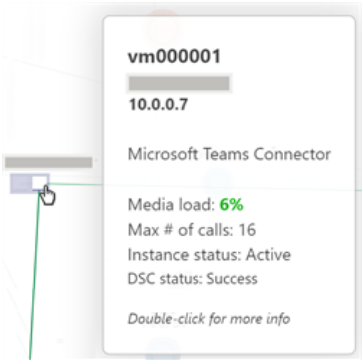
Each Conferencing Node is represented by a node icon . Nodes in the same location are grouped together. Hovering over each node's icon provides information about that node's hostname, IP address, current load and estimated capacity (in terms of the estimated maximum number of calls the node can handle). Hovering also highlights all the conferences for which that node is handling the media.



You can double-click to get more detailed node information, including an interactive status chart showing the load history for that node.

The example here shows the information that is available for one of the four Transcoding Conferencing Nodes in the **San Jose** location.

Viewing Teams Connector and Teams meeting status



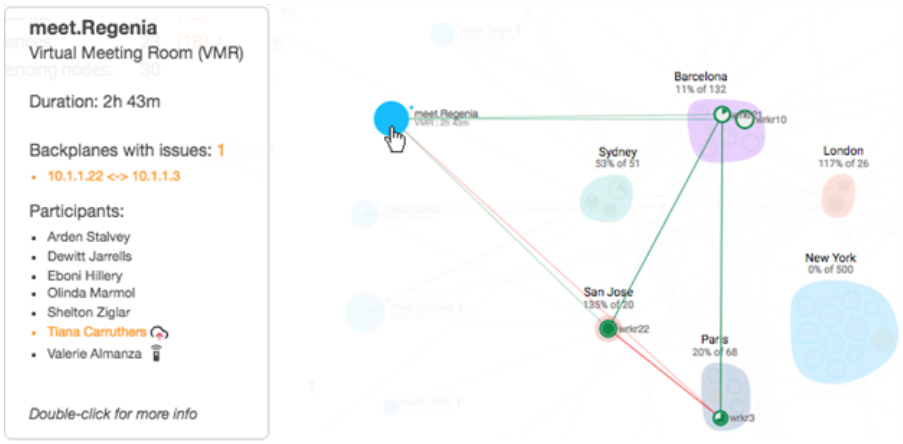
You can view the status of each Teams Connector instance, such as call capacity and current media load, when you hover over an instance , providing enhanced Teams status information is enabled.

If enhanced status information is not enabled then Live View only displays Teams Connector instances when a Teams call is in progress.

Each purple square represents a Teams Connector instance. A single instance can handle more than one Teams meeting; if you hover over an instance you are shown all of the Teams meetings that it is currently running.

You can also hover over an individual Teams meeting to see a list of all the participants, and highlight those who are gatewayed into the meeting.

Viewing conference status



Conferences are represented by ● icons. Hovering over the icon provides details of the conference, including its duration and a list of participants.

Hovering also highlights all the Conferencing Nodes that are handling media for that conference. Any backplane issues are highlighted — if a backplane link has poor quality, it is shown in red, and if a participant has poor quality, the link between their node and the conference is also shown in red.

You can double-click to get full conference details and to view an interactive conference graph.

The example here shows the information that is available for the conference being hosted in the **meet.Regenia** Virtual Meeting Room. The conference is experiencing one backplane issue.

Gateway calls to externally-hosted conferences

When viewing a Gateway call to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet, all of the externally-connected participants are listed (grayed out).

When you hover over the external conference it also shows you which Conferencing Node it's connected to, and a link to the associated Pexip gateway call.

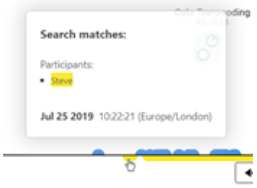
Filtering by participant or conference

The box at the top of the graph allows you to filter the view by conference or participant name.

When you enter text in the filter box, you will see only:

- Conferences with the text in their name,
- Conferences with the text in their service tag,
- Conferences which contain a participant with the text in their name, and
- Conferences which contain a participant with the text in their alias.

You can also apply a filter to an individual conference graph.



When a filter is applied in live view, the timeline at the bottom of the page indicates in yellow all the times in the past when there was a conference or participant matching the filter. If you hover over these yellow indicators, information about the match will appear.

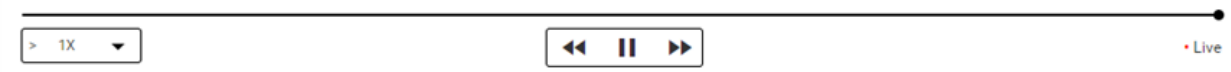
If you select a conference to view while the filter is applied, or you apply a filter to conference graph, any participants whose name or alias matches the filter text are highlighted in the conference graph. When viewing a conference graph with a filter applied, the timeline indicates in yellow whenever there was a participant in the conference who matched the filter. Again, if you hover over these yellow indicators, information about the match will appear.

Rewinding and replaying status

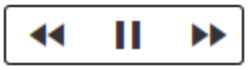
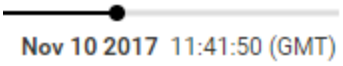
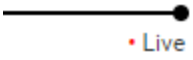
You can use the controls at the bottom of the graph to review the platform status during the previous seven days (data persists across reboots and upgrades). Any times when a participant or backplane had call quality issues are indicated in blue. You can use the timeline to:

- View historical Conferencing Node activities such as nodes being added or placed into maintenance mode.
- View conferences and the nodes on which they are hosted.
- Drill down into individual conferences to review conference activities such as participants joining, leaving or presenting, and examine a participant's media statistics.
- Pause, rewind and replay the graph at a variety of speeds.

The pie charts and alarms list also reflect the selected time period.



The following controls are available:

Control	Description
	Use the control at the bottom left of the timeline to select the speed at which to rewind or replay the graph.
	Use the control below the middle of the timeline to rewind, fast-forward, play or pause the graph.
	Drag the black dot to view a particular point in the timeline. The selected date and time are shown above the timeline.
	When viewing historical information, the date and time of the graph currently being displayed is shown at the bottom right of the timeline.
	To return to the live view, drag the black dot to the far right of the timeline, or fast-forward to the end. When you are viewing the live graph, "Live" will be shown at the bottom right of the timeline.

To maintain optimum system performance, the timeline is not available for large deployments of more than 40 Conferencing Nodes.

Remove conferences from Live View

You can control whether conferences and backplanes are shown in the Live View graph. If you have a very busy deployment, it may be useful to clear conferences and backplanes for an improved viewing experience.

To remove conferences and backplanes from the Live View:

1. Go to **Platform > Global Settings > Management Web Interface Configuration**
2. Clear the **Show conferences and backplanes in Live View** option.
3. Select **Save**.

Note that when conferences and backplanes are removed, the conferences and participants counts in Live View are not shown.

Conference status

Viewing current conference status

To see a list of all the conferences currently in progress, go to **Status > Conferences**. You can select individual conferences to see more detailed information, including an interactive conference graph, and perform some conference control functions such as muting all Guests or disconnecting all participants.

To view historical information on conferences after they have finished, including the ability to rewind and replay the conference graph, see Viewing historical information about conferences.

The pages showing conference status information do not refresh automatically — you must refresh them manually — except for the conference graph which does dynamically refresh. A "**Conference instance no longer active**" message indicates that the conference whose details you were viewing has now finished.

The Administrator interface uses color coding when reporting media statistics, such as perceived call quality, packet loss and jitter. In general, statistics that are shown in green text represent good quality, orange represents intermediate quality, and red is used for bad quality. See media statistics and perceived call quality for more information.

Each conference has the following information available:

Field	Description
Service name	The name of the Virtual Meeting Room, Virtual Auditorium, Virtual Reception, Media Playback Service, Test Call Service or Call Routing Rule. For Infinity Gateway calls, the rule name is followed by a unique identifier to distinguish between separate calls. For Virtual Receptions, Media Playback Services and Test Call Services, if there are multiple concurrent users of that service you will see a single instance of that service (rather than one instance per participant, as all participants are using the same service even though they cannot see or hear each other). Click on the service name to view more information.
Duration	For Virtual Meeting Rooms and Virtual Auditoriums: the length of time since the first participant joined the conference. For Virtual Receptions, Media Playback Services and Test Call Services: the length of time that this instance of the service has been in continuous use (note that this could involve more than one participant if their usage overlapped). For Infinity Gateway calls: the length of time since the call was received by the Infinity Gateway.
Participant count	The number of participants currently in the conference or using the service.
Service type *	The type of conference, e.g. Virtual Meeting Room.
Service tag *	The unique identifier that an administrator has assigned to this service. If this field is blank, no tag has been assigned. For more information, see Tracking usage via service and participant call tags.
Is locked *	Indicates whether a conference has been locked to prevent further participants from joining. For more information, see Locking a conference and allowing participants to join a locked conference.
Guests muted *	Indicates whether all Guest participants are muted.

* Only displayed when you have selected an individual conference to view.

To view more information about the conference, click on the service name. This gives you access to some conference control functions, and 3 tabs: Participants, Backplanes and Graph.

Conference control

When viewing conference details you can use the controls at the bottom of the page to:

- dial out to a new participant
- mute all Guests

- lock the conference
- disconnect all participants from the conference.

Participants

The **Participants** section lists all the participants that are currently in the conference.

For more details about a particular participant, including media stream statistics, click on the **Participant alias**. This takes you to the Participant details page.

Field	Description
Participant alias	The name of the user or the registered alias of the endpoint. When viewing a Google Meet conference, you will see additional aliases (typically in the form "spaces/<id>/devices/<id>") for each external participant.
Duration	The length of time since this participant joined the conference or accessed the service.
Display name	The name that has been configured on the participant's endpoint.
System location	The system location of the Conferencing Node to which the endpoint is connected. However, when the participant is connected to a Proxying Edge Node, this is the location of the Transcoding Conferencing Node that is processing the conference media for this participant.
Signaling node	The IP address and name of the Conferencing Node to which the endpoint is connected. This node is handling the call signaling but may or may not be handling the call media (for more information, see Handling of media and signaling in locally distributed conferences).
Media node	The IP address and name of the Transcoding Conferencing Node that is processing the call media for this participant (for more information, see Handling of media and signaling in locally distributed conferences).
Role	**Host** indicates that either: • the conference has no PINs configured (in which case all participants have a role of **Host**) • the participant accessed the conference using the **Host PIN**. **Guest** indicates that the participant accessed the conference using the **Guest PIN**. **Unknown** indicates one of the following: • the participant is at the PIN entry screen and has not yet successfully entered a PIN • the participant is at the Waiting for Host screen but their role has not yet been determined • the participant is connected to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet (these participants appear in the Pexip app participant list with a role of **External Guest**). For more information, see About PINs, Hosts and Guests.
Is presenting	Indicates whether the endpoint is currently sending a presentation stream.
Is administratively muted	Indicates whether the endpoint's audio has been muted (using a Pexip app, the Administrator interface, or by a third party using the Pexip API).
Is locally muted	Indicates whether the endpoint has been muted via the client.
Streaming or recording device	Indicates if the participant is a streaming or recording device.

Backplanes

The **Backplanes** section provides information about the media streams being transmitted between Transcoding Conferencing Nodes for the selected conference. Backplane links between Conferencing Nodes are unidirectional, so for a conference involving two transcoding nodes there will be two backplane links: one from node A to node B, and another from node B to node A. Note that a bidirectional backplane is created when a Conferencing Node connects to a Teams Connector or to a Skype for Business meeting.

Field	Description
Media node	The IP address and name of the Conferencing Node that is transmitting media. For details about the media streams being sent over a particular backplane link, click on the media node's IP address.
Remote media node	The IP address and name of the Conferencing Node or remote system e.g. a Teams Connector, that is receiving media.
Remote conference name *	The name of the conference on the remote node. For external backplanes, this identifies the conference on the other platform, such as a Microsoft Teams conference ID.
System location *	The system location of the Conferencing Node that is transmitting media.
Duration	The length of time since the connection was established.
Backplane type	**Geographic** indicates that the two Conferencing Nodes are in different system locations. **Local** indicates that the two Conferencing Nodes are in the same system location. **External** indicates a link between a Conferencing Node and an external node, such as a Teams Connector.

* Only displayed when you have selected an individual media node to view.

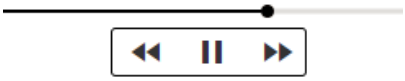
Backplane media streams

Media stream details are displayed when you have selected an individual node to view.

Field	Description
Type	Indicates whether the information is for an **Audio**, **Video**, or **Presentation** stream.
Start time	The time that the media stream started.
Tx codec	The format used by the transmitting Conferencing Node to encode and decode the media stream being transmitted.
Tx bitrate (kbps)	The quantity of data currently being sent from the transmitting Conferencing Node to the recipient Conferencing Node for this particular media stream.
Tx resolution	The display resolution of the image being sent from the transmitting Conferencing Node.
Tx framerate	The video frame rate per second being sent from the transmitting Conferencing Node.
Tx packets sent	The total quantity of packets sent from the transmitting Conferencing Node to the recipient Conferencing Node since the start of the conference.
Tx packets lost	The total quantity of packets sent from the transmitting Conferencing Node but not received by the recipient Conferencing Node.
Tx jitter (ms)	The variation in the expected periodic arrival of packets being sent from the transmitting Conferencing Node to the recipient Conferencing Node, in milliseconds.

Graph

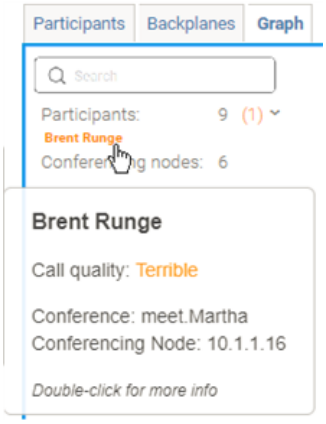
This section displays a dynamic graphical view of the connections for this conference, as follows:



You can use the timeline controls at the bottom of the graph to rewind and replay the graph at a variety of speeds. When viewing or replaying the graph you can:

- See when participants and Conferencing Nodes joined or disconnected from the conference.
- See when participants started and stopped presenting.
- View participant packet loss statistics during the conference by hovering over a connection.
- View summary details of individual participants, such as the protocol they are using and their bandwidth usage, by hovering over a participant. You can double-click on a participant to see more information.
- View summary details of individual nodes, such as its media load or any alarms, by hovering over a node. You can double-click on a node to see more information.
- Click within the graph to use your mouse to pan and zoom.

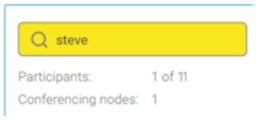
(See Rewinding and replaying status for more information about how to use the controls.)



Conference statistics and issues: the number of Conferencing Nodes and participants that are involved in the conference is displayed at the top left of the graph.

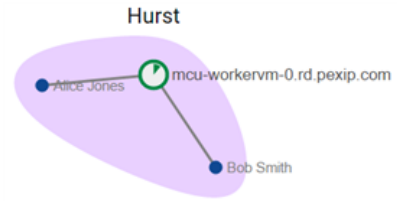
If any participants are experiencing call quality issues then the number of affected participants is displayed (in orange). You can click on this number to reveal the affected participants and also drill down to view more details about each of those participants.

The timeline indicates in blue any times when a participant or backplane had call quality issues. You can hover over these blue indicators to see more details of the issue.

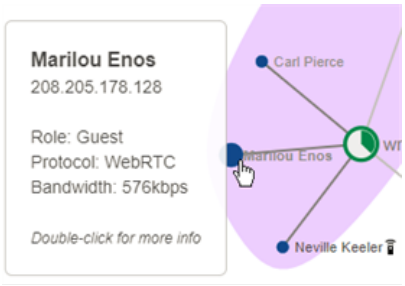


Filtering: the Search box at the top left of the graph allows you to search for participants by name or alias.

When a filter is applied, any participants who match the filter text are highlighted in yellow. The timeline also indicates in yellow when there was a participant who matched the filter. You can hover over these yellow indicators to see more information about the match.



Colored areas: each colored area highlights a system location and shows the Conferencing Nodes and endpoint connections within that location. A different color is used for each location.



Small dark blue dots: all participant endpoints. Some may have an icon next to their name, as follows:

- for Pexip app presentation and control-only / Meeting Manager participants
- for participants who are currently presenting content
- for streaming participants.

You can hover over an endpoint to view participant information.

Large green circles: the Transcoding Conferencing Nodes to which the endpoints are connected, or are processing conference media. The amount of green fill within the circle indicates the current media load (in terms of percentage of estimated HD ports in use), so an unused node is white and a fully loaded node is filled entirely green.

Large blue circles: the Proxying Edge Nodes to which the endpoints are connected. The amount of green fill within the circle indicates the current media load (in terms of percentage of proxying capacity in use), so an unused node is white and a fully loaded node is filled entirely green.

Large pale blue circles: an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

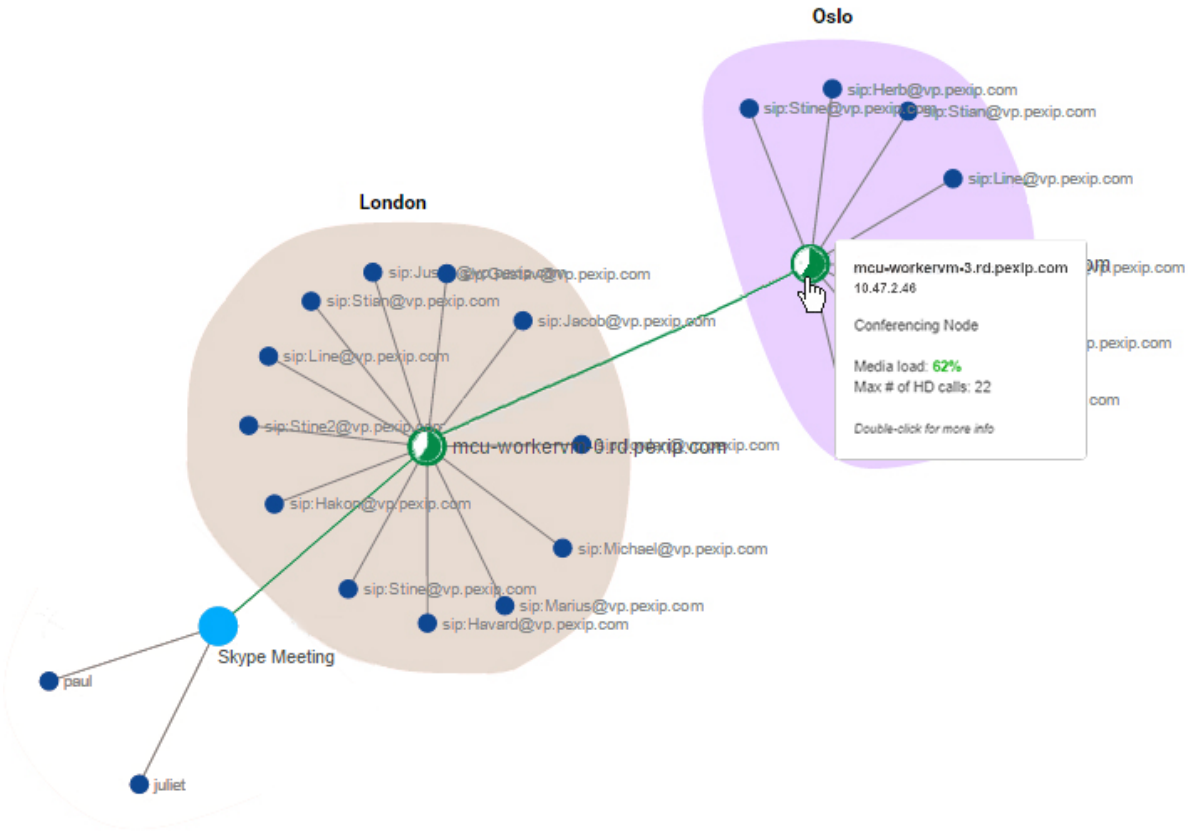
Green lines: backplane links between Conferencing Nodes, or links to external nodes. These become **dashed green lines** if total packet loss is greater than 1%.

Gray lines: connections between an endpoint and a Conferencing Node. These become **dashed gray lines** if total packet loss is greater than 1%.

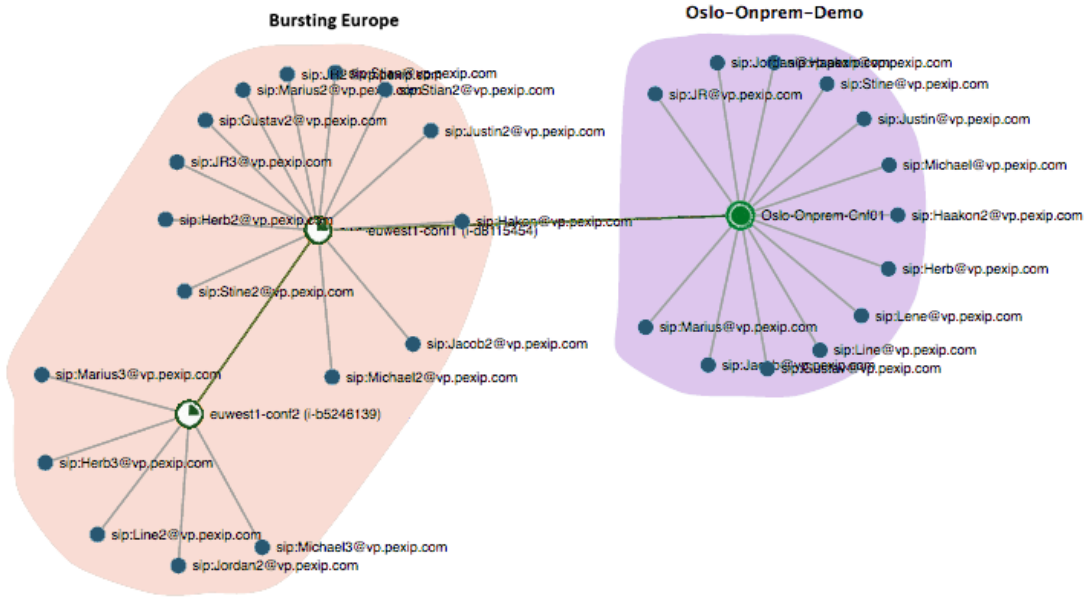
Red dashed lines: any connections with total packet loss greater than 2%.

Blue lines: a media-forwarding link between a Proxying Edge Node and a Transcoding Conferencing Node. Only one link is shown regardless of how many connections/streams are being proxied. Packet loss information is not available on media-forwarding links.

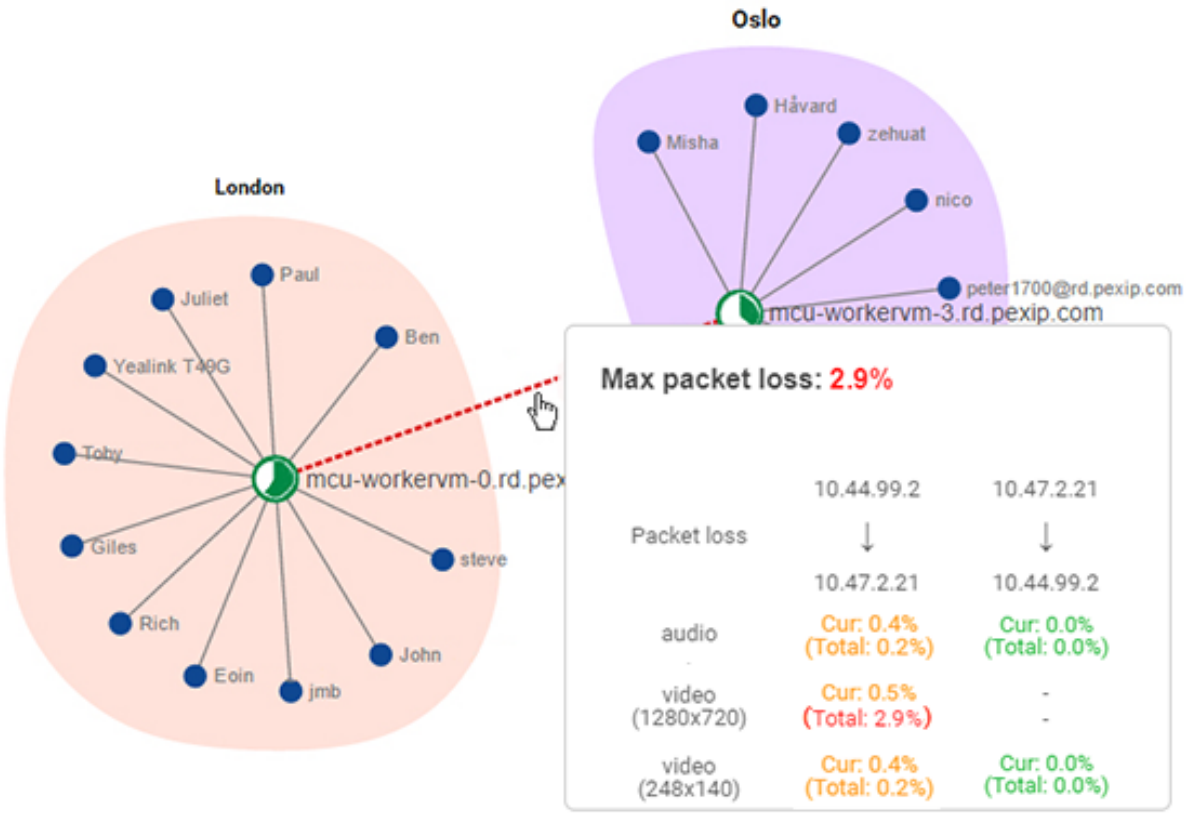
Keyhole: a keyhole in the top right of the screen indicates that the conference is locked.



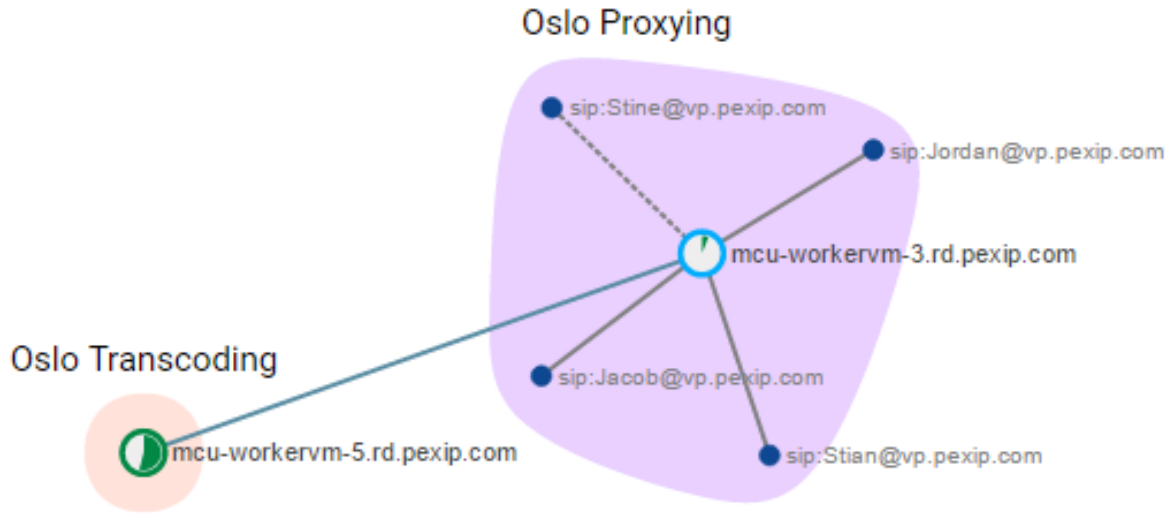
Example graph showing endpoints and a Skype for Business meeting connected to two Conferencing Nodes, where each Conferencing Node is in a separate location.



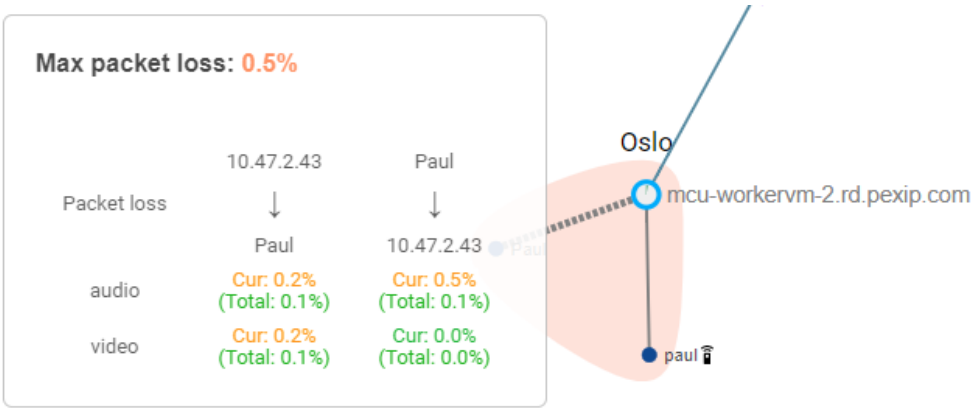
Example graph showing a system location containing two Conferencing Nodes that are hosted in AWS and are being used as an overflow location from an on-premises location containing one Conferencing Node.



Example graph showing packet loss on a backplane between two Conferencing Nodes in separate locations.



Example graph showing endpoints connected via a Proxying Edge Node to a VMR hosted on a Transcoding Conferencing Node in the "Oslo Transcoding" location.



Example graph showing packet loss for a connection.

Viewing participant status

To see a list of all the current conference participants across all Pexip Infinity services, go to **Status > Participants**. This shows a list of all participants; to view a particular participant's details, click on the **Participant alias**.

When viewing a participant's details you can also use the controls at the bottom of the page to:

- **Mute** the participant
- **Disconnect** the participant from the conference
- **Transfer** the participant to another conference

Note that when viewing participants in a gateway call, these controls cannot be used on any remote participants that are connected to an externally-hosted conference.

To view historical information on participants after a conference has finished, see Viewing historical information about participants.

Participant details

The following table lists the information shown for each participant. Note that some information, such as the Call ID, is not always available for participants who are directly connected to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

Field	Description
Perceived call quality *	A graphical representation of the participant's call quality over time. A blue line at the top of the graph indicates Good, down to a red line at the bottom which indicates Terrible. The percentage number indicates the amount of the call where the quality is perceived as Good or OK (above the line in blue). For example: Note that: • A call quality of Unknown is reported for all calls of less than 20 seconds duration. • All calls over RTMP (of more than 20 seconds duration) always report a call quality of 100% Good, as they are placed over TCP. • The perceived call quality graph is only available for calls with a duration up to 24 hours. See media statistics and perceived call quality for more information.
Participant alias	The name of the user or the registered alias of the endpoint. Click on the participant alias to view detailed information about the call.

Field	Description
Service name	The name of the Virtual Meeting Room, Virtual Auditorium, Virtual Reception, Media Playback Service, Test Call Service or Call Routing Rule. For Infinity Gateway calls, the rule name is followed by a unique identifier to distinguish between separate calls. Select **View status** to view the current status of the conference.
Call quality	The current quality of the call based on packet loss over the 3 most recent 20 second time windows. See media statistics and perceived call quality for more information.
Connect time *	The date and time that signaling was established between the participant's endpoint and Pexip Infinity.
Duration	The length of time since this participant joined the conference or accessed the service.
Display name	The name that has been configured on the participant's endpoint.
Destination alias *	For participants that have dialed in to the conference or service themselves, this is the alias that they dialed. For participants that have been dialed out to manually or automatically from a Virtual Meeting Room or Virtual Auditorium, this is the alias of the endpoint that was dialed.
System location	The system location of the Conferencing Node to which the endpoint is connected. However, when the participant is connected to a Proxying Edge Node, this is the location of the Transcoding Conferencing Node that is processing the conference media for this participant.
Proxying system location *	The system location of the Proxying Edge Node that is handling the call, if applicable.
Signaling node	The IP address and name of the Conferencing Node to which the endpoint is connected. This node is handling the call signaling but may or may not be handling the call media (for more information, see Handling of media and signaling in locally distributed conferences).
Media node	The IP address and name of the Transcoding Conferencing Node that is processing the call media for this participant (for more information, see Handling of media and signaling in locally distributed conferences).
Media proxying node *	The IP address and name of the Proxying Edge Node that is proxying the call media for this participant, if applicable.
Service type *	**Virtual Meeting Room**, **Virtual Auditorium**, **Virtual Reception**, **Test Call Service** or **Gateway**: the participant has successfully accessed the service indicated. **PIN collection IVR**: the participant is currently accessing the Interactive Voice Response screens (where they are asked to enter a valid PIN). **Waiting for Host**: the participant is being shown a holding screen while they wait for a conference Host to join. **Insufficient Capacity Screen**: the participant is being shown a holding screen indicating that they cannot join the conference due to a lack of capacity on Pexip Infinity, or because the service's participant limit has been reached. **Insufficient Licenses Screen**: the participant is being shown a holding screen indicating that they cannot join the conference due to a lack of available call licenses on Pexip Infinity. For more information, see Insufficient licenses. **Invalid License Screen**: the participant is being shown a holding screen indicating that they cannot join the conference because there are no valid licenses available on Pexip Infinity. For more information, see Invalid license.
Protocol	The communication protocol used by the endpoint.

Field	Description
Role	**Host** indicates that either: • the conference has no PINs configured (in which case all participants have a role of **Host**) • the participant accessed the conference using the **Host PIN**. **Guest** indicates that the participant accessed the conference using the **Guest PIN**. **Unknown** indicates one of the following: • the participant is at the PIN entry screen and has not yet successfully entered a PIN • the participant is at the Waiting for Host screen but their role has not yet been determined • the participant is connected to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet (these participants appear in the Pexip app participant list with a role of **External Guest**). For more information, see About PINs, Hosts and Guests.
License count *	The number of licenses consumed by this participant. Media participants consume 1 license but API-only participants (e.g. Pexip app users who are not sending media, or Meeting Manager participants) do not consume a license. Participants who are directly connected to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet, do not consume a license.
License type *	The type of license used, either: • **Port**: audio/video participant. • **Audio**: audio-only participant — only applies if the system has audio licenses installed, otherwise port licenses are used for audio-only calls) • **Not required**: a presentation and control-only participant; a Meeting Manager participant; or a participant directly connected to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.
Is presentation stream supported *	Indicates whether the endpoint is able to support a separate media stream for presentations negotiated by H.239 or BFCP.
Is presenting	Indicates whether the endpoint is currently sending a presentation stream.
Is muted	Indicates whether the endpoint's audio has been muted (using a Pexip app, the Administrator interface, or by a third party using the Pexip API).
Client muted*	Indicates whether the endpoint has been muted via the client.
Call direction *	**In**: the call was placed by an external endpoint and received by Pexip Infinity. **Out**: the call was placed by Pexip Infinity to an endpoint or other device.
Bandwidth (kbps) *	The maximum bandwidth, in kbps, negotiated for use between the Conferencing Node and the endpoint. Actual bandwidth used is shown in the Media streams section (**Tx bitrate** and **Rx bitrate**).
Streaming or recording device *	Indicates if the participant is a streaming or recording device.
Encryption *	Indicates whether the media stream being sent to and from the Conferencing Node towards the endpoint is encrypted.
Vendor *	Information about the endpoint's manufacturer and software.
Remote IP address *	The IP address of the system from which signaling from this endpoint is being sent and received. This may be the endpoint itself, or it may be a call control system if one is in use in your network.
Remote port *	The port on the system from which signaling from this endpoint is being sent and received.

Field	Description
Call ID *	A unique identifier that can be used to trace the call in the administrator log and support log. Select **View call logs** to see a filtered view of the support log showing only events containing this Call ID. Select **View log summary** to see a condensed view of the call signaling messages in the support log for this Call ID. Calls made via the Virtual Reception generate two separate participant calls but these both have the same Call ID. Calls made via the Infinity Gateway generate separate participant calls with different Call IDs.
Is disconnect / transfer / mute supported *	Indicates whether the participant can be disconnected, transferred or muted via the Management Node.
Authenticated by an Identity Provider *	Indicates whether the participant was required to authenticate in order to join the conference. For more information, see About participant authentication.
Identity Provider *	The name of the Identity Provider with which the participant successfully authenticated.
Direct media *	Indicates whether this participant used direct media when connected to this VMR. See Direct media (end-to-end encrypted calls) for more information.
Transcoding enabled *	Indicates whether transcoding is enabled for this participant.

* Only displayed when you have selected an individual participant to view.

Media streams

Media stream details are displayed when you view the details of an individual participant. Note that media streams are not displayed for any participants who are directly connected to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

For standard transcoded calls this section is labeled "Conferencing Node media streams" and the Tx and Rx statistics are shown from the perspective of the Conferencing Node (and the descriptions in the table below are written from that perspective). For direct media calls this section is labeled "Client media streams" and the statistics are instead shown from the perspective of the client participant.

Type	Indicates whether the information is for the **Audio**, **Video**, or **Presentation** stream. When viewing the stream details of a completed VMR call you may see multiple instances of each stream type (for example, if the participant had started presenting, stopped and then started presenting again). For in-progress VMR calls you only see a maximum of one instance of each stream type (reflecting what the participant is currently doing). In gateway calls to an externally-hosted conference you may see a separate stream for every resolution/frame rate being sent. Note that a presentation stream is not shown for Pexip apps that are sending or receiving still images or PDF pages (as opposed to screen sharing, or receiving full motion presentation).
Start time	The date and time that the media stream started.
Node	The address of the Transcoding Conferencing Node handling the media.
Tx codec	The format used by the Conferencing Node to encode the media stream being sent to the endpoint.
Tx bitrate (kbps)	The quantity of data currently being sent from the Conferencing Node to the endpoint, in kilobits per second.
Tx resolution	The display resolution of the image being sent from the Conferencing Node.

Tx framerate	The video frame rate per second being sent from the Conferencing Node.
Tx packets sent	The total quantity of packets sent from the Conferencing Node to the endpoint since the start of the conference.
Tx packets lost	The total quantity of packets sent from the Conferencing Node but not received by the endpoint. This value is reported to the Conferencing Node by the endpoint. Endpoints that do not support RTCP are not able to supply this information, so the value will always be **0**.
Tx jitter (ms)	The variation in the expected periodic arrival of packets being sent from the Conferencing Node to the endpoint, in milliseconds. This value is reported to the Conferencing Node by the endpoint. Endpoints that do not support RTCP cannot supply this information, so the value will always be **0**.
Rx codec	The format used by the Conferencing Node to decode the media stream being sent from the endpoint. ***Off*** indicates that no decodable media for this stream has been received in the last 10 seconds or so from the codec. For example, for a Pexip app being used for presentation and control-only, there will be no video transmitted. ***Off stage*** indicates that the participant is currently not being shown in the main video or thumbnails, so Pexip Infinity is not attempting to decode the media stream. ***Telephone event*** may be displayed if an audio codec that use silence suppression (such as G729B) is muted and sends DTMF. If this field is blank, this may mean that Pexip Infinity has negotiated which codec to use but it is yet to receive any media from the endpoint to determine which codec it is actually sending.
Rx bitrate (kbps)	The quantity of data currently being received by the Conferencing Node from the endpoint, in kilobits per second.
Rx resolution	The display resolution of the image being received by the Conferencing Node.
Rx framerate	The video frame rate per second being received by the Conferencing Node. This can fluctuate over time as it is measured by the Conferencing Node.
Rx packets received	The total quantity of packets received by the Conferencing Node from the endpoint since the start of the conference.
Rx packets lost	The total quantity of packets sent from the endpoint but not received by the Conferencing Node.
Rx jitter (ms)	The variation in the expected periodic arrival of packets being received by the Conferencing Node from the endpoint, in milliseconds.

Viewing current and historic registrations

To see a list of all device aliases that are currently registered to the Pexip Infinity platform, go to **Status > Registrations**, and to see a history of registered devices go to **History & Logs > Registration History**.

Devices can only register to Pexip Infinity if the alias it wants to register has been added to Pexip Infinity's list of allowed device aliases (**Users & Devices > Device Aliases**). For more information, see Registering devices to Pexip Infinity.

The following information about each registration is available:

Field	Description
Alias	The alias that has been registered, including its URI scheme (e.g. sip:). Click on the alias to view detailed status information.
Username	The username associated with the alias that was used to authenticate the registration.
Remote IP address	The device's remote IP address that is used for signaling.
Node	The address of the node to which the device is registered.

Field	Description
Protocol	The protocol over which the device alias is registered, such as SIP, H.323 or WebRTC.
Vendor *	Vendor information for the device.
Start time	The date and time the registration started.
End time	The date and time the registration ended (registration history only).
Is natted *	Indicates if the registered device is probably behind a NAT (its contact address is different from its source IP address).
Registration ID *	The UUID associated with this registration (registration history only). Select **View registration logs** to see the log history for this registration.

* Only displayed when you have selected an individual alias to view.

Viewing historical information about conferences

To see a list of all the completed conferences on the Pexip Infinity platform, go to **History & Logs > Conference History**. This shows a list of the most recent completed conferences (up to a limit of 10,000). To view details of a particular conference, including the ability to rewind and replay the conference graph, click on the **Service name**.

To view information on conferences currently running on Pexip Infinity, see Viewing current conference status.

The Administrator interface uses color coding when reporting media statistics, such as perceived call quality, packet loss and jitter. In general, statistics that are shown in green text represent good quality, orange represents intermediate quality, and red is used for bad quality. See media statistics and perceived call quality for more information.

The following information is available for each completed conference:

Field	Description
Service name	The name of the Virtual Meeting Room, Virtual Auditorium, Virtual Reception, Media Playback Service, Test Call Service or Call Routing Rule. For Infinity Gateway calls, the rule name is followed by a unique identifier to distinguish between separate calls. For Virtual Receptions, Media Playback Services and Test Call Services, if there were multiple concurrent users of that service you will see a single instance of that service (rather than one instance per participant, as all participants are using the same service even though they cannot see or hear each other). Click on the service name to view more information.
Start time	The date and time that the first participant connected to the service.
End time	The date and time that the last participant's call ended.
Duration	The length of time that the conference or service was in use.
Participant count	The total number of participant calls made to this conference. Note that if a single participant disconnects from the conference and then reconnects to it, this will be counted as two participant calls.
Service type *	The type of conference, e.g. Virtual Meeting Room.
Service tag *	The unique identifier that an administrator has assigned to this service. If this field is blank, no tag has been assigned. For more information, see Tracking usage via service and participant call tags.
Instant message count *	The total number of instant messages sent during the conference.

* Only displayed when you have selected an individual conference to view.

To view more information about the conference, click on the service name. You will then see 3 tabs for the selected conference: Participants, Backplanes and Graph.

Participants

The **Participants** section lists all the participants that were in the conference.

For more details about a particular participant, including media stream statistics, click on the **Participant alias**. This takes you to the Participant history page.

Field	Description
Participant alias	The name of the user or the registered alias of the endpoint.
Start time	The date and time that the participant's call reached Pexip Infinity.
End time	The date and time that the participant's call ended.
Duration	The length of time that the participant was connected to Pexip Infinity. This includes time connected to the Virtual Meeting Room or Virtual Auditorium, and any time spent at the Virtual Reception or PIN entry screens.
Display name	The name that has been configured on the participant's endpoint.
System location	The system location of the Conferencing Node to which the endpoint is connected. However, when the participant is connected to a Proxying Edge Node, this is the location of the Transcoding Conferencing Node that is processing the conference media for this participant.
Role	**Host** indicates that either: • the conference had no PINs configured (in which case all participants had a role of **Host**) • the participant accessed the conference using the **Host PIN**. **Guest** indicates that the participant accessed the conference using the **Guest PIN**. **Unknown** indicates one of the following: • the participant reached the Virtual Reception but did not proceed to a Virtual Meeting Room or Virtual Auditorium • the participant reached the PIN entry screen but did not successfully enter a PIN • the participant reached the Waiting for Host screen but their role was not determined • the call was via the Infinity Gateway. For more information, see About PINs, Hosts and Guests.

Backplanes

The **Backplanes** section provides information about the media streams being transmitted between Transcoding Conferencing Nodes for the selected conference. Backplane links between Conferencing Nodes are unidirectional, so for a conference involving two transcoding nodes there will be two backplane links: one from node A to node B, and another from node B to node A. Note that a bidirectional backplane is created when a Conferencing Node connects to a Teams Connector or to a Skype for Business meeting.

Field	Description
Media node	The IP address and name of the Conferencing Node that is transmitting media. For details about the media streams being sent over a particular backplane link, click on the media node's IP address.
Remote media node	The IP address and name of the Conferencing Node or remote system e.g. a Teams Connector, that is receiving media.
Remote conference name *	The name of the conference on the remote node. For external backplanes, this identifies the conference on the other platform, such as a Microsoft Teams conference ID.
System location *	The system location of the Conferencing Node that is transmitting media.

Field	Description
Start time	The date and time that the connection was established.
End time	The date and time that the connection was brought down.
Duration	The length of time since the connection was established.
Backplane type	**Geographic** indicates that the two Conferencing Nodes are in different system locations. **Local** indicates that the two Conferencing Nodes are in the same system location. **External** indicates a link between a Conferencing Node and an external node, such as a Teams Connector.
Disconnect reason *	The reason that the backplane link was disconnected.

* Only displayed when you have selected an individual media node to view.

Backplane media streams

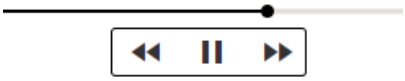
Media stream details are displayed when you have selected an individual node to view.

Field	Description
Type	Indicates whether the information is for an **Audio**, **Video**, or **Presentation** stream.
Start time	The time that the media stream started.
End time	The time that the media stream ended.
Tx codec	The format used by the transmitting Conferencing Node to encode and decode the media stream being transmitted.
Tx bitrate (kbps)	The quantity of data currently being sent from the transmitting Conferencing Node to the recipient Conferencing Node for this particular media stream.
Tx resolution	The display resolution of the image being sent from the transmitting Conferencing Node.
Tx framerate	The video frame rate per second being sent from the transmitting Conferencing Node.
Tx packets sent	The total quantity of packets sent from the transmitting Conferencing Node to the recipient Conferencing Node since the start of the conference.
Tx packets lost	The total quantity of packets sent from the transmitting Conferencing Node but not received by the recipient Conferencing Node.

Graph

This section displays a dynamic graphical view of the connections for this conference, as described below.

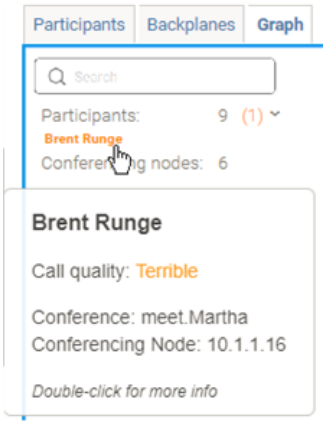
Initially the graph shows the midpoint state of the conference. You can use the interactive timeline and controls to go forwards or backwards to replay the graph and review the entire conference.



You can use the timeline controls at the bottom of the graph to rewind and replay the graph at a variety of speeds. When viewing or replaying the graph you can:

- See when participants and Conferencing Nodes joined or disconnected from the conference.
- See when participants started and stopped presenting.
- View participant packet loss statistics during the conference by hovering over a connection.
- View summary details of individual participants, such as the protocol they are using and their bandwidth usage, by hovering over a participant. You can double-click on a participant to see more information.
- View summary details of individual nodes, such as its media load or any alarms, by hovering over a node. You can double-click on a node to see more information.
- Click within the graph to use your mouse to pan and zoom.

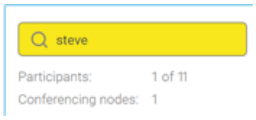
(See Rewinding and replaying status for more information about how to use the controls.)



Conference statistics and issues: the number of Conferencing Nodes and participants that are involved in the conference is displayed at the top left of the graph.

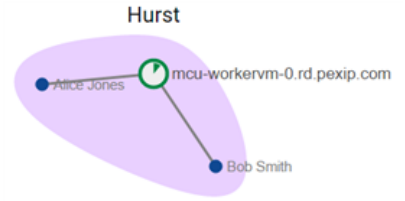
If any participants are experiencing call quality issues then the number of affected participants is displayed (in orange). You can click on this number to reveal the affected participants and also drill down to view more details about each of those participants.

The timeline indicates in blue any times when a participant or backplane had call quality issues. You can hover over these blue indicators to see more details of the issue.

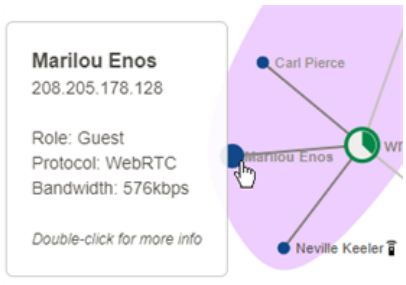


Filtering: the **Search** box at the top left of the graph allows you to search for participants by name or alias.

When a filter is applied, any participants who match the filter text are highlighted in yellow. The timeline also indicates in yellow when there was a participant who matched the filter. You can hover over these yellow indicators to see more information about the match.



Colored areas: each colored area highlights a system location and shows the Conferencing Nodes and endpoint connections within that location. A different color is used for each location.



Small dark blue dots: all participant endpoints. Some may have an icon next to their name, as follows:

- for Pexip app presentation and control-only / Meeting Manager participants
- for participants who are currently presenting content
- for streaming participants.

You can hover over an endpoint to view participant information.

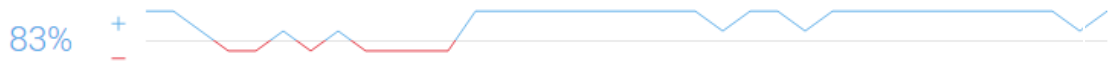
	Large green circles: the Transcoding Conferencing Nodes to which the endpoints are connected, or are processing conference media. The amount of green fill within the circle indicates the current media load (in terms of percentage of estimated HD ports in use), so an unused node is white and a fully loaded node is filled entirely green.
	Large blue circles: the Proxying Edge Nodes to which the endpoints are connected. The amount of green fill within the circle indicates the current media load (in terms of percentage of proxying capacity in use), so an unused node is white and a fully loaded node is filled entirely green.
	Large pale blue circles: an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.
	Green lines: backplane links between Conferencing Nodes, or links to external nodes. These become **dashed green lines** if total packet loss is greater than 1%.
	Gray lines: connections between an endpoint and a Conferencing Node. These become **dashed gray lines** if total packet loss is greater than 1%.
	Red dashed lines: any connections with total packet loss greater than 2%.
	Blue lines: a media-forwarding link between a Proxying Edge Node and a Transcoding Conferencing Node. Only one link is shown regardless of how many connections/streams are being proxied. Packet loss information is not available on media-forwarding links.
	Keyhole: a keyhole in the top right of the screen indicates that the conference is locked.

Viewing historical information about participants

To see a list of all the calls made to the Pexip Infinity platform, go to **History & Logs > Participant History**. This shows a list of all calls made to the 10,000 most recent completed conferences. To view details of a particular call, click on the **Participant alias**.

To view information on participants currently connected to Pexip Infinity, see Viewing participant status.

The following table lists the information shown for each participant. Note that some information, such as the Call ID, is not always available for participants who are directly connected to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.

Field	Description
Perceived call quality *	A graphical representation of the participant's call quality over time. A blue line at the top of the graph indicates Good, down to a red line at the bottom which indicates Terrible. The percentage number indicates the amount of the call where the quality is perceived as Good or OK (above the line in blue). For example:  Note that: • A call quality of Unknown is reported for all calls of less than 20 seconds duration. • All calls over RTMP (of more than 20 seconds duration) always report a call quality of 100% Good, as they are placed over TCP. • The perceived call quality graph is only available for calls with a duration up to 24 hours. See media statistics and perceived call quality for more information.
Participant alias	The name of the user or the registered alias of the endpoint. Click on the participant alias to view detailed information about the call.
Service name	The name of the Virtual Meeting Room, Virtual Auditorium, Virtual Reception, Media Playback Service, Test Call Service or Call Routing Rule. For Infinity Gateway calls, the rule name is followed by a unique identifier to distinguish between separate calls. Select **View conference** to view the historical status of the conference, and optionally to replay the conference graph.
Call quality	The perceived overall quality of the call. See media statistics and perceived call quality for more information.
Start time	The date and time that the participant's call reached Pexip Infinity.
End time	The date and time that the participant's call ended.
Duration	The length of time that the participant was connected to Pexip Infinity. This includes time connected to the Virtual Meeting Room or Virtual Auditorium and any time spent at the Virtual Reception or PIN entry screens.
Display name	The name that has been configured on the participant's endpoint.
Conference alias *	The alias that the participant dialed to access the service. If the participant included a PIN number in the dial string, this would also be included in the alias shown here.
System location	The system location of the Conferencing Node to which the endpoint is connected. However, when the participant is connected to a Proxying Edge Node, this is the location of the Transcoding Conferencing Node that is processing the conference media for this participant.
Proxying system location *	The system location of the Proxying Edge Node that is handling the call, if applicable.
Signaling node *	The IP address and name of the Conferencing Node to which the endpoint is connected. This node is handling the call signaling but may or may not be handling the call media (for more information, see Handling of media and signaling in locally distributed conferences).
Media node	The IP address and name of the Transcoding Conferencing Node that is processing the call media for this participant (for more information, see Handling of media and signaling in locally distributed conferences).
Media proxying node *	The IP address and name of the Proxying Edge Node that is proxying the call media for this participant, if applicable.

Field	Description
Service type *	Indicates whether the participant was connected to a Virtual Meeting Room, Virtual Auditorium, Virtual Reception, Test Call Service or the Infinity Gateway.
Protocol	The communication protocol used by the endpoint.
Role	**Host** indicates that either: • the conference had no PINs configured (in which case all participants had a role of **Host**) • the participant accessed the conference using the **Host PIN**. **Guest** indicates that the participant accessed the conference using the **Guest PIN**. **Unknown** indicates one of the following: • the participant reached the Virtual Reception but did not proceed to a Virtual Meeting Room or Virtual Auditorium • the participant reached the PIN entry screen but did not successfully enter a PIN • the participant reached the Waiting for Host screen but their role was not determined • the call was via the Infinity Gateway. For more information, see About PINs, Hosts and Guests.
License count *	The number of licenses consumed by this participant. Media participants consume 1 license but API-only participants (e.g. Pexip app users who are not sending media, or Meeting Manager participants) do not consume a license. Participants who are directly connected to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet, do not consume a license.
License type *	The type of license used, either: • **Port**: audio/video participant. • **Audio**: audio-only participant — only applies if the system has audio licenses installed, otherwise port licenses are used for audio-only calls) • **Not required**: a presentation and control-only participant; a Meeting Manager participant; or a participant directly connected to an externally-hosted conference, such as a Microsoft Teams or Skype for Business meeting, or Google Meet.
Call direction *	**In**: the call was placed by an external endpoint and received by Pexip Infinity. **Out**: the call was placed by Pexip Infinity to an endpoint or other device.
Bandwidth (kbps) *	The maximum bandwidth, in kbps, negotiated for use between the Conferencing Node and the endpoint. Actual bandwidth used is shown in the Media streams section (**Tx bitrate** and **Rx bitrate**).
Streaming or recording device *	Indicates if the participant is a streaming or recording device.
Encryption *	Indicates whether the media stream being sent to and from the Conferencing Node towards the endpoint is encrypted.
Vendor *	Information about the endpoint's manufacturer and software.
Remote IP address *	The IP address of the system from which signaling from this endpoint is being sent and received. This may be the endpoint itself, or it may be a call control system if one is in use in your network.
Remote port *	The port on the system from which signaling from this endpoint is being sent and received.
Call ID *	A unique identifier that can be used to trace the call in the administrator log and support log. Select **View call logs** to see a filtered view of the support log showing only events containing this Call ID. Select **View log summary** to see a condensed view of the call signaling messages in the support log for this Call ID. Calls made via the Virtual Reception generate two separate participant calls but these both have the same Call ID. Calls made via the Infinity Gateway generate separate participant calls with different Call IDs.

Field	Description
Disconnect reason	The reason that the call was disconnected. This is provided by either the endpoint or Pexip Infinity, depending on how the call was terminated. For more information, see Disconnection reasons.
Authenticated by an Identity Provider *	Indicates whether the participant was required to authenticate in order to join the conference. For more information, see About participant authentication.
Identity Provider *	The name of the Identity Provider with which the participant successfully authenticated.
Direct media *	Indicates whether this participant used direct media when connected to this VMR. See Direct media (end-to-end encrypted calls) for more information.
Transcoding enabled *	Indicates whether transcoding is enabled for this participant.
* Only displayed when you have selected an individual participant to view.	

Media streams

Historical media stream details are displayed when you view the details of an individual participant.

Reporting of media statistics and perceived call quality

When using the Administrator interface to look at the current and historic status of calls and of individual participants within a call, various media statistics and guidance as to perceived call quality are displayed.

Perceived call quality

The current and historic participant status screens provide guidance to the perceived quality of a call by looking at packet loss over multiple time windows.

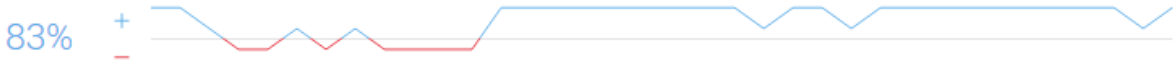
For an ongoing call, the system looks at packet loss in both directions (Tx and Rx) over the 3 most recent 20 second time windows. Packet loss is used to calculate call quality for a time window: < 1% packet loss is perceived as Good quality; < 3% is OK; < 10% is Bad; otherwise it is Terrible.

The system then reports the current call quality as the most frequent call quality in those last 3 windows. If all time windows report a different quality, then any of the three qualities may be indicated.

For completed calls, the system looks at packet loss in both directions (Tx and Rx) over multiple 20 second time windows throughout the call and calculates the call quality per window. It then reports the overall call quality as the average call quality of those windows.

The system also provides a graphical representation of the participant's call quality over the duration of the call.

A blue line at the top of the graph indicates Good, down to a red line at the bottom which indicates Terrible. The percentage number indicates the amount of the call where the quality is perceived as Good or OK (above the line in blue). For example:



Note that:

- A call quality of Unknown is reported for all calls of less than 20 seconds duration.
- All calls over RTMP (of more than 20 seconds duration) always report a call quality of 100% Good, as they are placed over TCP.
- The perceived call quality graph is only available for calls with a duration up to 24 hours.

Color coding for media statistics and quality

The Administrator interface uses color coding when reporting media statistics, such as perceived call quality, packet loss and jitter. In general, statistics that are shown in green text represent good quality, orange represents intermediate quality, and red is used for bad quality.

- **Perceived call quality** is calculated as described above and is shown in green if the quality is Good or OK, or in red if the quality is Bad or Terrible.
- **Packet loss** is shown in green when it is < 0.2%, in orange when it is between 0.2% and 2%, and in red when it is >= 2%.
- **Jitter** is shown in green when it is < 10ms, in orange when it is 10–50ms, and in red when it is >= 50ms.

Viewing usage statistics

To view the daily usage statistics, go to **History & Logs > Usage Statistics**. This shows:

- the total number of minutes used (the sum of all participant durations) per location, and
- the highest number of concurrent call licenses used.

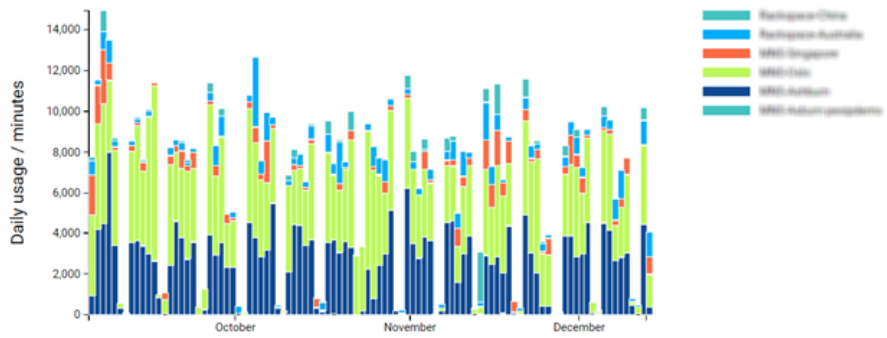
Each day represents usage for the 24-hour period from midnight UTC. The participant minutes information is also broken down by ingress location within each day. The number of days for which statistics are shown here depends on how quickly your administrator log file is rotated, but information will persist over reboots and upgrades.

You can view specific information on each location and day by hovering over each item.

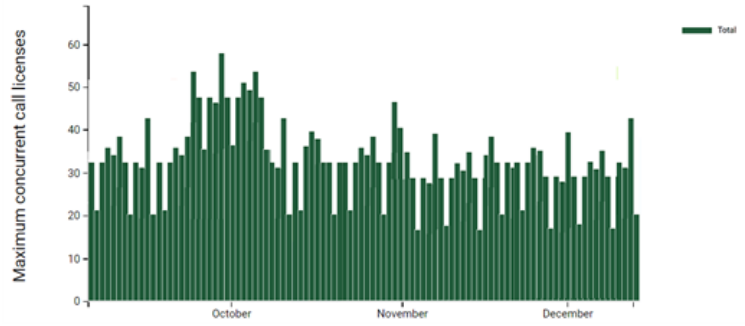
If there is no information available, for example if your system has been deployed in the last 24 hours, you will see the following message: **Information not yet available. This can take up to 1 hour.**

The Live view page (**Status > Live View**) lets you review current and historic usage charts showing a breakdown of participants by location, protocol, license type and the different conference types being hosted.

Example graphs



Graph showing usage in participant minutes per day for a deployment spread across six locations



Graph showing the maximum number of concurrent call licenses used each day

Viewing source data

To view the source data used to create each graph, click on the **View source data** link below the graph. This takes you to a page showing the source data in CSV format. All dates are in Unix timestamp format.

The source data goes back for the same amount of time as the administrator log, from which they are extracted. For more information, see About the administrator log.

Viewing LDAP sync template results

To view the status of ongoing or completed LDAP template synchronization processes, go to **Status > LDAP Sync**.

You are shown the details of the most recent or in-progress run of each LDAP synchronization template. Information shown includes:

- run status e.g. Initializing sync: connecting to LDAP, Syncing, Sync succeeded
- number of VMRs, devices and users that were created, deleted, updated and unchanged
- number of warnings
- last updated date and time.

If necessary, you can select a template to view the details of the last error/warning message it generated. Typical messages include:

Warning	Comment
Error syncing with LDAP	This can occur if Pexip Infinity fails to connect to, or authenticate with, the LDAP server. In which case the Pexip Infinity support log will provide more information. Alternatively, this can be caused by invalid syntax in the template's **LDAP user filter** or **LDAP user search DN** fields.
Alias clash: alias held by an existing conference has not been assigned to the newly synced conference	This occurs when a template generates an alias that is identical to an existing alias that is already assigned to another VMR (either created manually or via another mechanism). In this case, the alias is not reassigned to the VMR currently being created/updated.
Duplicate alias: attempting to create the same alias more than once for the same conference	This occurs when the template attempts to create the same alias more than once for the same conference. Check the alias patterns for your template.
Conference clash: not overwriting manually created conference	This occurs when a template generates a VMR name that already exists, and that VMR was created manually. In this case, the existing VMR with that name is left unchanged.
Sync template clash: not overwriting conference created by another mechanism	This occurs when a template generates a VMR name that already exists, and that VMR was created via another mechanism such as a different template or Secure Scheduler for Exchange. In this case, the existing VMR with that name is left unchanged.
Validation error when creating conference	This occurs if the synchronization tries to set a field to an invalid value (e.g. a PIN containing letters rather than numbers, or an empty VMR name). The error message will include some addition parameters such as Conference, Error and Data which will help to identify the VMR, the field and source LDAP data that is causing the problem.
Validation error when creating device	This occurs if the synchronization tries to set a field to an invalid value (such as an empty alias). The error message will include some addition parameters such as Device-Alias, Error and Data which will help to identify the device alias, field and source LDAP data that is causing the problem.
Sync template clash: not overwriting device created by another template	This occurs when a template generates a device alias that already exists, and that alias was created via another template. In this case, the existing device alias is left unchanged.

Warning	Comment
Device clash: not overwriting manually created device	This occurs when a template generates a device alias that already exists, and that alias was created manually. In this case, the existing device alias is left unchanged.
Sync template clash: not overwriting User created by another template	This occurs when a template generates a user email address that already exists, and that user record was created via another template. In this case, the existing user record is left unchanged.
User clash: not overwriting manually created User	This occurs when a template generates a user email address that already exists, and that user record was created manually. In this case, the existing user record is left unchanged.

All of the warning messages generated from a sync run can be viewed in the Administrator log (**Status > Administrator Log**) as described below.

Viewing all historical sync results via the Administrator log

To view all historic details of completed LDAP synchronization processes:

1. Go to the Administrator log (**Status > Administrator Log**).
2. Search for **ldap.sync** to see details of all log entries relating to all VMR, device and user synchronization processes.

   Note that you will see "Beginning" and "Completed" log entries for VMR, device and user syncing even if the template is not configured to perform both types of syncing. This is because all sync processes are always performed in case they need to delete VMRs, devices or users from previous synchronizations.

More information

For more information about bulk-provisioning VMRs and devices from an LDAP-accessible database, see:

- Provisioning VMRs, devices and users from Active Directory via LDAP
- Troubleshooting LDAP server connections

Diagnostics tools and reporting

Automatically reporting errors

In order to allow continual monitoring and improvement of the Pexip Infinity platform, we encourage customers to send us details of any incidents that occur in their products. When this feature is enabled, incident reports are sent automatically to a specified URL. By default this is the address of a secure web server owned and managed by Pexip, but you have the option to change the URL.

We collate and analyze this information in order to identify issues with the software that can be resolved in future product releases. The information we receive will not be used for any other purpose.

The feature is enabled and disabled on a Pexip Infinity platform-wide basis, but when enabled each Management Node and Conferencing Node will send their individual incident reports directly to the incident reporting server.

When incident reporting is enabled:

- If you are Using a web proxy for the Management Node, or for the Location to which the Conferencing Node belongs, all incident reports will be sent via the specified web proxy.
- If you are not using a web proxy, you must ensure that your IP routing and firewall configuration allows outbound communications from each of these systems to the nominated URL. By default, reports are sent to **https://acr.pexip.com** over TLS to port 443.

Content of incident reports

The exact content of each incident report will vary depending on the nature of the problem, but all reports **always** include:

- the date and time that the incident occurred
- the FQDN of the system on which the incident occurred
- the IP address of the system on which the incident occurred
- information about the Pexip Infinity software version
- details of the internal software processes running at the time the incident occurred.

The incident reporting server will also annotate each report with the IP address from which it was received (generally this will be the public IP address of the sender's network). This information is not contained in the original incident report.

The incident reports **may** also include:

- system configuration information
- stack traces.

The incident reports **never** include:

- system logs.

Enabling and disabling automatic sending of incident reports

Automatic sending of incident reports is enabled or disabled during initial installation of the Management Node (when running the installation wizard).

To enable or disable the automatic submission of incident reports, or to change the URL to which they are sent:

1. On the Pexip Infinity Administrator interface, go to **Platform > Global Settings**.
2. From the **Reporting** section, select or deselect the **Enable incident reporting** box as required.
3. If appropriate, in the **Incident reporting URL** field enter the URL to which reports will be sent.
4. Optionally, enter a **Contact email address** of somebody who can be contacted by Pexip for further information. If specified, this email address is added to incident reports.
5. Select **Save**.

Tracking usage via service and participant call tags

To assist in tracking usage of the Pexip Infinity platform you can assign **service tags** to conferencing services such as VMRs, Virtual Auditoriums, Virtual Receptions and Call Routing Rules, and **participant call tags** to each participant within a conference.

You can also use the service and participant call tags for decision-making purposes in local and external policy.

Service tags

Each Virtual Meeting Room (including those used for scheduled conferences), Virtual Auditorium, Virtual Reception, Call Routing Rule and registered device can be assigned a unique identifier, known as a service tag. You can then use this tag to track usage of the service or device, for example for billing purposes.

Every conference event and participant event associated with the conference will include the service tag.

In most cases, if you are using service tags in your deployment you would obtain the information you require from the logs either from the Pexip Infinity Management API (for more information, see Log output), or from an event sink. However, you can also search the Administrator log directly from the web-based Administrator interface (**History & Logs > Administrator Log**) using the service tag to track usage of the service.

Example

A service provider is using the Pexip Infinity Management API to create Virtual Meeting Rooms automatically on behalf of customers. They assign each new VMR a user-friendly **Service name** - this is the name that is presented to participants accessing the conference and appears in Skype for Business users' contact lists. They also assign each VMR a randomly generated UUID as the **Service tag**. The service provider then uses the Service tag UUID to track usage of each VMR in order to bill each customer appropriately for using their services.

Participant call tags

Every participant event associated with a conference can include a call tag that is specific to that participant.

The tag can be specified in client API requests and participant policy responses, and then used by app developers to correlate other API requests. The participant call tag is present in policy requests, participant updates to other participants, event sinks and the status/history databases for management API requests.

Note that the call tag is not displayed in the Administrator interface.

Automatically sending usage statistics

As part of our desire to continuously improve operation of the Pexip Infinity platform and Pexip apps based on real-life customer usage data, we have enabled the ability for customers, at their own discretion, to automatically provide us with statistics relating to the usage and configuration of their deployment.

When enabled at a platform level, relevant conference information (such as conference duration and number of participants, call duration and protocol) and selected configuration settings (such as version, license information, location and number of conferencing nodes) is sent automatically every 30 minutes to a third party service provider on behalf of Pexip.

We will aggregate and analyze the information for each customer who has enabled these options, in order to better understand the ways our products are being used in production deployments. This in turn helps us determine the features and requirements of future software releases, which will ultimately provide a product that is of maximum benefit to you.

The analytics information we receive will not contain any identifying information (such as user names or passwords), and will not be shared with any other organizations. A full list of the data that will be sent to us at a platform level is given in Information sent when usage reporting is enabled. For any more information, please contact your Pexip authorized support representative.

When this option is enabled, either your Management Node must have a web proxy configured, or your firewall must be configured to allow the Management Node to send outbound traffic over HTTPS.

Enabling and disabling automatic sending of platform usage statistics

The option to automatically send usage statistics is enabled or disabled during initial installation of the Management Node (when running the installation wizard).

To enable or disable the automatic submission of usage statistics after initial installation or upgrade:

1. On the Pexip Infinity Administrator interface, go to **Platform > Global Settings**.
2. From the **Reporting** section, select or deselect the **Automatically send deployment and usage statistics to Pexip** box as required.
3. Select **Save**.

Information sent when usage reporting is enabled

Platform

When **Automatically send deployment and usage statistics to Pexip** is enabled, the following platform usage statistics are sent by the Management Node to Pexip's service provider:

Information sent
Common
The version of Pexip Infinity software running on the Management Node.
The Pexip Infinity software license ID.
Public IP address of the Management Node from which the report has been sent.
The name of the Management Node.
Call
Reason the call was disconnected.
Whether the call was to a Virtual Meeting Room, Virtual Auditorium, Virtual Reception, or Infinity Gateway.
Number of seconds this call leg was alive.
Vendor of the endpoint that placed the call.
Protocol used by the endpoint (SIP, H.323, WebRTC, etc).
Maximum bandwidth signaled.
Whether the participant was a Guest or Host.
IP address of the Conferencing Node that handled the media for the call.
IP address of Conferencing Node that handled the signaling for the call.
Whether the media stream was encrypted.
Unique identifier of this call (UUID).
Unique identifier of the conference instance (UUID).
The time at which the call was connected.
The time at which the call was disconnected.
Identifier of the system location of the Conferencing Node to which the call was connected.
Whether the call was incoming or outgoing.
The perceived overall quality of the call.
The type of license used by the call.
The number of licenses consumed by the call.
Whether the call was end-to-end encrypted.
Audio/Video/Presentation (statistics for each type of stream, where present)
Whether the stream is audio, video or presentation.

Information sent
The format used by the Conferencing Node to encode the media stream being sent to the endpoint.
The quantity of data sent to the endpoint.
The display resolution of the image sent to the endpoint.
The total quantity of packets sent to the endpoint.
The total quantity of packets sent from the Conferencing Node but not received by the endpoint.
The format used by the Conferencing Node to decode the media stream being sent from the endpoint.
The quantity of data received from the endpoint.
The display resolution of the image received from the endpoint.
The total quantity of packets received by the Conferencing Node from the endpoint.
The total quantity of packets sent from the endpoint but not received by the Conferencing Node.
Conference
Cryptographic hash of the name of the Virtual Meeting Room, Virtual Auditorium, Virtual Reception or Infinity Gateway that was used.
The date and time that the first participant connected to the service.
The date and time that the last participant's call ended.
Number of seconds conference was active.
Whether the call was to a Virtual Meeting Room, Virtual Auditorium, Virtual Reception, or Infinity Gateway.
Total number of participants who connected to this conference instance.
Whether direct media was enabled for the service.
The number of chat messages exchanged on this conference.
Platform (sent once a day for each system location)
Numeric identifier of the system location.
Number of nodes in that system location.
Configuration
Whether RTMP is enabled.
Whether SIP (TCP and TLS) is enabled.
Whether SIP TLS certificate verification mode is set to ***On*** or ***Off***.
Whether WebRTC is enabled.
The URL to which OCSP requests will be sent.
The start value for the range of ports that all Conferencing Nodes use to send media.
The end value for the range of ports that all Conferencing Nodes use to send media.
The start value for the range of ports that all Conferencing Nodes use to send signaling.
The end value for the range of ports that all Conferencing Nodes use to send signaling.

Information sent
Whether outbound calls are enabled.
Whether SSH is enabled.
Whether OCSP will be used when checking the validity of TLS certificates.
The DSCP value for management traffic sent from the Management Node and Conferencing Nodes.
The number of minutes a browser session may remain idle before the user is logged out of the Pexip Infinity Administrator interface.
The URL to which incident reports will be sent (if enabled).
Whether H.323 is enabled.
Whether incident reporting is enabled.
Whether support for Pexip apps is enabled.
The number of configured devices (including those that are not registered).
The number of configured services (Virtual Meeting Rooms + Virtual Auditoriums + Virtual Receptions).
Whether chat messages are enabled.
Whether SIP UDP is enabled.
Registration
The number of users registered over SIP, H.323 and WebRTC.

Downloading a diagnostic snapshot

If you are experiencing issues with your Pexip Infinity service, you may be asked by your Pexip authorized support representative to provide them with a snapshot of your system to assist them in diagnosis of the issue.

To download the diagnostic snapshot:

1. On the Pexip Infinity Administrator interface, go to **Utilities > Diagnostic Snapshot**.
2. To download all available data, select **Generate full snapshot**.

   or

   To download a subset of the snapshot, containing a specified time period:
   a. Type in, or adjust the sliders to the start and end times (in terms of how many hours ago from the current time) of the diagnostic data to be downloaded. The number of hours available will vary depending on the logs available on the system.
   b. If requested by your Pexip authorized support representative, select **Include diagnostic metrics from all Conferencing Nodes**.

      If this is selected, then metrics from the Management Node and all Conferencing Nodes are included with the snapshot.

      If this is not selected, then metrics from the Management Node only are included with the snapshot.
   c. Select **Generate limited duration snapshot**.
3. A **Snapshot request** page appears showing the status of your snapshot.

   While the snapshot generation is in progress you can optionally navigate away from this page and return to it later.
4. The status of any ongoing/available snapshots is shown in an **Existing Snapshot Requests** section at the bottom of the **Utilities > Diagnostic Snapshot** page. To download a completed snapshot:
   a. Select **View Snapshot Request** to return to the relevant **Snapshot request** page.
   b. Select **Download** and follow your browser's prompts to save the file.

You can generate multiple snapshots but when you generate a new snapshot, any existing snapshots older than an hour will be deleted.

File contents

The diagnostic snapshot is a collection of logs, incident reports, and metrics, and therefore contains information such as IP addresses, conference names, aliases, Pexip Infinity configuration and system logs. Some of this information may be sensitive to your organization, so the snapshot should be saved and handled securely.

Note that Management Node metrics are always included in a snapshot. Conferencing Node metrics are included if you download a full snapshot or if you download a limited duration snapshot and have selected **Include diagnostic metrics from all Conferencing Nodes**.

File size

When each log reaches its maximum size, the oldest data is overwritten.

The maximum file size of each log in the snapshot is as follows:

- Administrator: 1 GB
- Support: 2 GB
- Internal developer-specific logs: 2 GB

The diagnostic snapshot also includes all available incident reports:

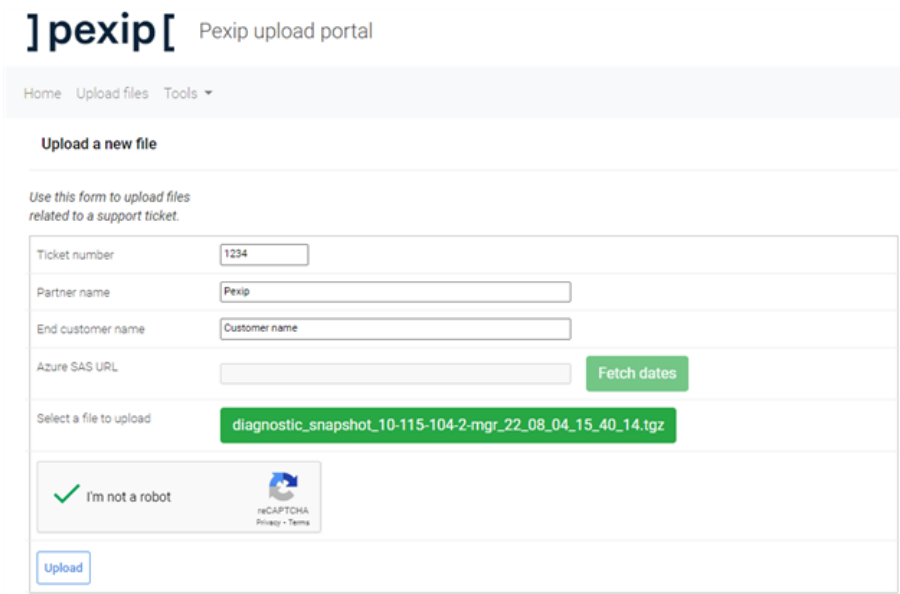
- the full snapshot contains up to 28 days of incident reports, with a cap at 5 GB
- the limited duration snapshot contains incident reports from the selected number of hours.

Uploading to the Pexip upload portal

The Pexip upload portal is a web tool used by Pexip Infinity administrators to upload snapshots or logs when requested by your Pexip authorized support representative. When the data is uploaded, they are encrypted (see Encryption methodologies) and your Pexip authorized support representative receives the encrypted file along with the encryption key.

1. Go to https://upload.pexip.com/ and sign in.
2. Select **Upload Files**.
3. Fill in the page details:

Ticket number	Enter your support ticket number.
Azure SAS URL	This field is only applicable for Teams Connector integrations.
Select a file to upload	Select **Choose file** and select the diagnostic snapshot file you downloaded from the Management Node.



4. Select **Upload**.

   You see a message that the that the snapshot file has been uploaded.

You can now exit the Pexip upload portal.

Performing a network packet capture

The packet capture utility allows you to capture network traffic to and from Conferencing Nodes in one or more locations, and the Management Node. Packet captures can be helpful when diagnosing various network issues, for example problems with DNS, NTP and firewalls.

If you are experiencing issues with your Pexip Infinity service, you may be asked by your Pexip authorized support representative to provide them with one or more packet captures of your system to assist them in diagnosis of the issue.

To perform a packet capture:

1. From the Administrator interface, go to **Utilities > Packet Capture**.
2. Select the **System locations** where you want to run the packet capture and move them into the **Chosen System locations** list.

   The packet capture will run against all of the Conferencing Nodes in the chosen locations. It will capture both interfaces if a Conferencing Node has dual network interfaces.

   You can also choose the Management Node.
3. Select the **Traffic types** you want to include in the capture. The options are:
   - All traffic
   - All traffic excluding IPSec (backplane IPsec traffic between Conferencing Nodes)
   - UDP only
   - TCP only
4. Encryption keys for SIP, H.323 and WebRTC traffic, and IPsec traffic if that is being captured, are captured by default.

   Clear the **Log encryption keys** check box if you do not want to capture encryption keys.

   The keys are captured in the log files so you'll typically need to download a diagnostic snapshot as well.
5. Select the **Duration** of the packet capture. It can be in the range 10-600 seconds.
6. You can optionally specify an **Address filter list** by entering a comma-separated list of subnet(s) or IP address(es) to filter the captured traffic.
7. Select **Start packet capture**.

   All previous packet captures are deleted when a new packet capture is started. Make sure that you have downloaded any previous packet capture files that you want to keep.

   The capture will run for the specified duration. You cannot stop a packet capture after it has started.

   A **Time remaining** countdown will display while the packet capture is running.
8. When the capture completes, the captured "pcap" files are listed at the bottom of the page.

   The page may take a few seconds to sync. If all of the expected files aren't present after one minute then refresh the page.
9. You can **Download** or **Delete** any or all of the individual packet capture files.

Taking a manual packet capture from non-communicating nodes

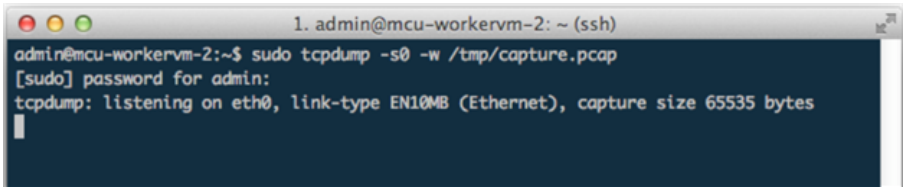
If a Conferencing Node has lost communication with the Management Node, or if otherwise instructed by your Pexip authorized support representative, you may have to perform a manual capture via a direct SSH connection with that node:

1. Connect to the Pexip node over SSH (using Putty, SecureCRT or another SSH client), logging in as user **admin**.
2. At the SSH command line, issue the following command, supplying the admin password when prompted.:

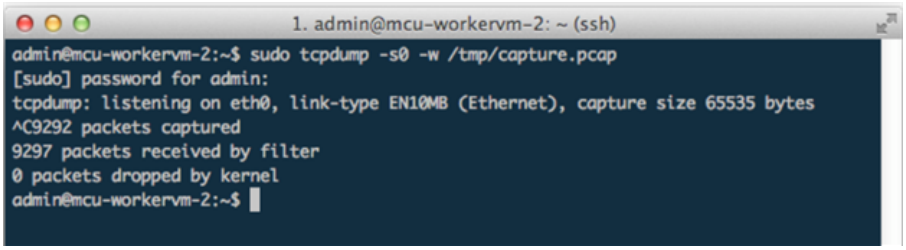
   ```
   sudo tcpdump -s0 -w /tmp/capture.pcap
   ```

   If the node has dual interfaces you need to add either the `-i nic0` or `-i nic1` switch to the tcpdump command, depending on which interface you want to capture.

   This starts a packet capture, storing the captured data in file **/tmp/capture.pcap** on the Pexip node.
3. While the capture is running, the following output can be observed:

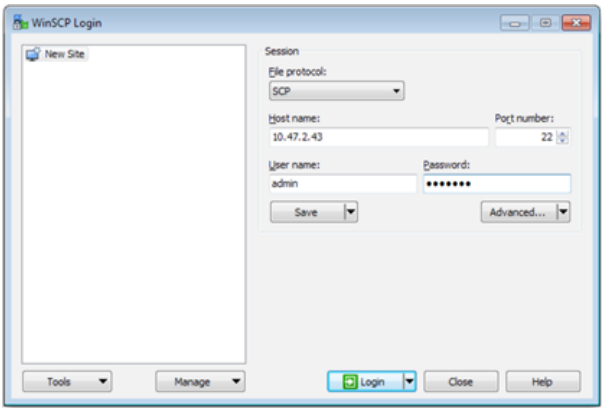


4. To stop the capture, press **Control + C** in the SSH window:

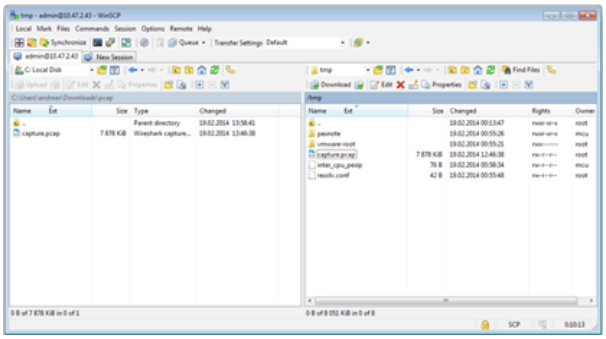


   You can now close the SSH session.

5. To download the captured file **capture.pcap**:
   a. Connect to the Pexip node with an SCP (Secure Copy) client, for example WinSCP — make sure to use SCP as the transfer protocol.
   b. Log in as user **admin** and supply the admin password.



   c. Navigate to folder **/tmp** and download the capture file, named **capture.pcap**.



Viewing Conferencing Nodes

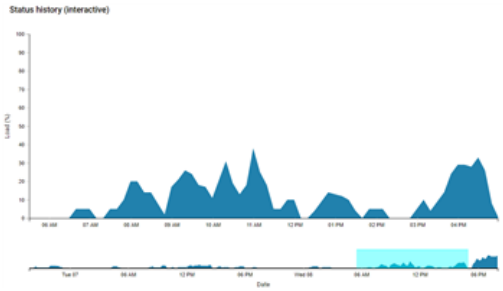
You can view details of all of your current Conferencing Nodes, including an interactive Conferencing Node status chart showing the load history for that node. You can also review a list of historic events (such as starting or stopping cloud bursting nodes) that have occurred on your Conferencing Nodes.

Viewing current Conferencing Nodes

To see a list of all Conferencing Nodes, go to **Status > Conferencing NodeS**.

Each Conferencing Node has the following information available:

Field	Description
Name	The name of the Conferencing Node. Click on the name to view detailed status information.
System location	The physical location of the Conferencing Node.
Role	Indicates the node's role — either a Transcoding Conferencing Node or a Proxying Edge Node.
IPv4 Address	The primary IP address of the Conferencing Node.
Secondary address *	The optional secondary interface IPv4 address of the Conferencing Node.
Static NAT address *	The optional static NAT address of the Conferencing Node.
Version	The version number of the Pexip Infinity software that is currently installed on this Conferencing Node.
Number of vCPUs	The configured number of virtual CPUs.
System memory	The amount of RAM configured on this Virtual Machine.
Maintenance mode	Indicates whether the node is currently in maintenance mode.
Config sync status	The configuration synchronization status: • **Synchronized**: the node's configuration is synchronized (as at the **Last updated** date/time). • **Suspended**: synchronization is paused — this applies to nodes that are configured for dynamic bursting but are not currently active. • **Not synchronized**: the node can be reached and is responding, however it is not currently synchronized. If this situation persists (for more than an hour) then you should contact your Pexip authorized support representative. When the Management Node is restarted, it is expected that Conferencing Nodes will show as not synchronized for several minutes. • **Unreachable**: the node cannot be reached (does not respond) — this could be due to network issues or it may be powered off. A "Configuration not synchronized" alarm is raised if the status is "Not synchronized" or "Unreachable".
Last contacted	The date and time that the Conferencing Node last responded to the Management Node. In typical deployments, configuration replication is performed approximately once per minute. However, in very large deployments (more than 60 Conferencing Nodes), configuration replication intervals are extended, and it may take longer for configuration changes to be applied to all Conferencing Nodes (the administrator log shows when each node has been updated). • **Green** text indicates that there are no issues. • **Orange** text indicates that no responses have been received within the last 2 expected replication intervals (typically within the last 3–5 minutes) and the node is not synchronized. • **Red** text indicates that no contact has been made for some time — typically none within the last 5 minutes, but note that this replication interval is extended in very large deployments. • **Light gray** text indicates that the node is suspended (a dynamic bursting node that is not currently active).
Last updated	The last time that the Conferencing Node's configuration was successfully updated.

Field	Description
Maximum connections Maximum direct media connections Audio connections * SD connections * HD connections * Full HD connections * Proxy audio connections * Proxy video connections *	For a Transcoding Conferencing Node, the initial table shows an estimate of the total number of simultaneous high-definition (HD) 720p video calls this node can handle.† After you have selected an individual transcoding node to view, you can also review the estimated SD, HD, Full HD and audio call capacity. For a Proxying Edge Node, the initial table shows an estimate of the total number of proxied video calls this node can handle. After you have selected an individual proxying node to view, you can also review the estimated proxying capacity for audio calls.
Media load	An estimate of how much of its total capacity the Conferencing Node is currently using.† Note that when a Conferencing Node is in maintenance mode, it reports a media load of 100%. This is to indicate that there is no current capacity available.
Call count *	The number of signaling connections that the node is currently handling.
CPU model * CPU instruction set * Hypervisor *	Extra information about the Virtual Machine hosting this Conferencing Node, including: • The underlying CPU model. • The highest CPU instruction set supported by this Virtual Machine. • The hypervisor on which this Virtual Machine is running.
Usage graph *	Shows media usage % statistics for this Conferencing Node for the last 4 weeks.  You can click on the graph and then scroll to zoom in and out, and you can drag the timebox to the left or right to move forwards or backwards in time.

* Only displayed when you have selected an individual Conferencing Node to view.

† Capacity estimates assume that only calls of that media type are being handled. For more information on server capacity, see Capacity planning.

Viewing historic Conferencing Node events

Go to **History & Logs > Conferencing Node History** to see all of the events that have occurred on your Conferencing Nodes. These events include:

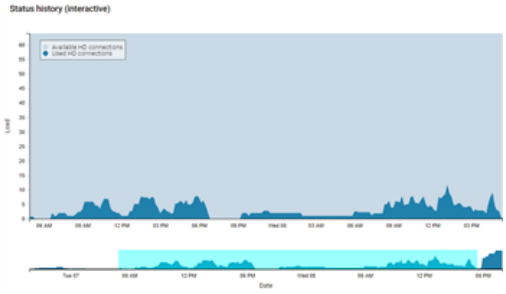
- Adding or removing a node.
- Placing a node into or out of maintenance mode.
- Starting up or stopping a cloud bursting node.

Viewing system location status

To see a list of all system locations currently configured, and their media usage statistics, go to **Status > Locations**.

The Live view page (**Status > Live View**) lets you review current and historic usage charts showing a breakdown of participants by location, protocol, license type and the different conference types being hosted.

The following information about each location is available:

Field	Description
Name	The name of the location.
Maximum connections Audio connections * SD connections * HD connections * Full HD connections * Proxy audio connections * Proxy video connections *	The initial table shows an estimate of the maximum number of connections (in terms of **HD** calls for a location containing transcoding nodes, and **video** calls for a location containing proxying nodes) that can be supported by all of the Conferencing Nodes assigned to this location. After you have selected an individual location to view, you can also review the estimated SD, HD, Full HD and audio call capacity for a location containing transcoding nodes, or the estimated proxying capacity for audio and video calls for a location containing proxying nodes.
Media load	An estimate of how much of the location's total capacity is currently being used.
Transcoding resources	The system location to handle media transcoding for calls (signaling) received in, or sent from, this location.
Usage graph *	Shows media usage statistics for this location for the last 7 days. It indicates the resources that are actually being consumed (dark blue area) over time, relative to the maximum possible media load (light blue area). The load is shown in terms of HD calls or proxied video calls as appropriate for the nodes in that location (see Call types and resource requirements for more information). The maximum possible media load typically stays consistent over time, but can vary if Conferencing Nodes are added to, or removed from, the location.  You can click on the graph and then scroll to zoom in and out, and you can drag the timebox to the left or right to move forwards or backwards in time.

* Only displayed when you have selected an individual location to view.

If a location has no associated Conferencing Nodes (for example, after the location has just been created), the location reports a media load of 100%. This indicates that there is no current capacity available in that location.

Viewing cloud bursting status

You can view the current status of your overflow nodes and locations, and view a history of all events that have been applied to overflow nodes.

Viewing current status

Go to **Status > Cloud Bursting** to see an overview of the media load of your principal locations (that contain your "always-on" Conferencing Nodes), and whether your overflow nodes and locations are in use.

- Any issues relating to your cloud bursting deployment will also be shown on this page.
- The list of principal locations only includes those locations that are configured with a **Primary overflow location** that contains bursting nodes.
- An **approaching threshold** message is displayed in the **Available HD** connections column for the principal locations when the number of available HD connections is less than or equal to the bursting threshold plus two.

  This message changes to **bursting threshold reached** when the number of available HD connections is less than or equal to the bursting threshold (and therefore overflow nodes are started up).
- You can manually start any overflow nodes by selecting **Start** for the required node (the **Start** option is in the final column of the **Cloud overflow nodes** table).
- The status page dynamically updates every 15 seconds.

Viewing historic events

Go to **History & Logs > Conferencing Node History** to see all of the events (stop, start or running) that have been applied to overflow Conferencing Nodes and, where appropriate, the reason why the event was applied (for example if a node was shut down as there was no longer a need for the extra capacity).

Viewing alarms

When there are active alarms on your Pexip Infinity deployment, a flashing blue triangle appears at the top right of each page of the Administrator interface. To view details of the current alarms, click on this icon or go to the **Alarms** page (**Status > Alarms**).

- Alarms remain in place for as long as the issue exists. After the issue has been resolved (for example, if a conference ends, therefore freeing up licenses) the associated alarm will automatically disappear from the **Alarms** page.
- Multiple instances of the same type of alarm can be raised. For example if two Conferencing Nodes are not correctly synchronized to an NTP server, you will see an alarm for each node.
- You can select an individual warning-level alarm and **Acknowledge** it. Any acknowledged alarms are still shown on the **Alarms** page but are no longer displayed at **Status > Live View**.
- You can select an individual alarm and view the associated documentation (this guide) for suggested causes and resolutions.

The **History & Logs > Alarm History** page shows the details of all historic alarms including the severity level, and the time the alarm was raised and lowered.

An alarm is raised in each of the following situations:

Alarm	ID	Level	Cause	Suggested resolutions
The Management Node does not have a TLS certificate	20	Critical	The Management Node has no associated TLS certificate.	Upload a TLS certificate and associate it with the Management Node.
A Conferencing Node does not have a TLS certificate	9	Critical	A Conferencing Node has no associated TLS certificate.	Upload a TLS certificate and associate it with the Conferencing Node. Alternatively, and if appropriate for your deployment, associate an existing certificate with your Conferencing Node. When doing this, the existing certificate should already contain a SAN (Subject Alternative Name) that matches your Conferencing Node's FQDN. See Managing a node's TLS server certificate for more information.
CPU instruction set not supported	10	Critical	A Conferencing Node has gone into maintenance mode because it was deployed on a server with an unsupported processor instruction set (e.g. SSE4.1). This could also be caused by setting the EVC mode on a VMware cluster to too low a level, such as Westmere.	Deploy the Conferencing Node on a server with AVX or later.
Eventsink Reached Maximum Backoff	36	Critical	An event cannot be delivered to an event sink, and the system has reached its retry timeout limit.	See Troubleshooting event sink failures.
Eventsink Reached Maximum Concurrent POSTs	37	Critical	More than the configured **Maximum number of background POSTs** events (default 1000) are queued for an event sink but have not been sent.	See Troubleshooting event sink failures.

Alarm	ID	Level	Cause	Suggested resolutions
Azure key vault certificate has expired	48	Critical	A Teams Connector certificate (in the Azure key vault) has expired.	Update the Teams Connector certificate.
NTP not synchronized	11	Error	A node has failed to synchronize with the configured NTP servers. A virtual machine on VMware has been migrated.	Ensure that NTP is enabled on the Management Node, and that NTP servers are assigned to, and accessible from, each location. See Syncing with NTP servers for more information. See https://kb.vmware.com/s/article/2108828 for VMware migration issues.
Configuration not synchronized	18	Error	This alarm is raised if the Conferencing Node status "Last contacted" time has not been updated within the last 2 expected replication intervals (typically no contact within the last 3 minutes).	In typical deployments, configuration replication is performed approximately once per minute. However, in very large deployments (more than 60 Conferencing Nodes), configuration replication intervals are extended, and it may take longer for configuration changes to be applied to all Conferencing Nodes (the administrator log shows when each node has been updated). If configuration synchronization fails this may indicate network connectivity or routing issues between the Management Node and the Conferencing Node, which could be due to a malfunction or misconfiguration of devices such as routers or firewalls etc. Ensure that all of the appropriate Pexip nodes are fully routable to each other in both directions. See General network requirements.
MS Exchange Connection Failure	22	Error	The Management Node cannot connect to the Exchange server.	Check that the details entered in the EWS URL (**System > Secure Scheduler For Exchange IntegrationS**) are correct and the Exchange server is online.
Automatic backup upload failed	25	Error	The Management Node cannot connect to the FTP server to upload a backup file.	Check that the **Upload URL** (supported schemes are FTPS and FTP) and the **Username** and **Password** credentials of the FTP server are correct (**Utilities > Automatic Backups**) and that the Management Node can reach the FTP server.
LDAP sync failed	28	Error	An LDAP template synchronization process has failed. This alarm duplicates the information shown for the error listed at **Status > LDAP Sync**.	See Troubleshooting LDAP server connections for help with resolving LDAP connection issues. The alarm is lowered when you resync the template (although it will get re-raised if the issue has not been resolved).
Scheduled scaling: cannot allocate some or all of the requested Teams Connector instances	42	Error	This is raised if Pexip Infinity requests more instances than Azure will allow (above the limit set by the instance count (slider) configuration in the Azure portal for the **Virtual machine scale set** in your Teams Connector resource group). Note that Azure will still create as many instances as it can up to the maximum.	You should review your maximum instances setting (slider) in Azure and your scheduled scaling policies to ensure you do not request scaling up beyond your maximum limit.

Alarm	ID	Level	Cause	Suggested resolutions
Scheduled scaling: some or all of the requested Teams Connector instances are not operational	43	Error	This is raised if the number of required instances (**Minimum number of instances** plus the policy's **Number of instances to add**) are not running at the policy's activation date/time. This can occur if Azure failed to start the instances, running instances have failed, or there is some other problem with Azure's scaling/provisioning processes. It can also occur if the **Azure Event Hub connection string** field is not configured correctly. This alarm can also be raised temporarily if the **Minimum number of instances** is increased. It will last for a few minutes until the new instances are up and running. This is expected behavior and can occur with or without any scheduled scaling policies.	This requires investigation of the VM scale set in the Azure portal as to the cause of failure, and manual intervention to resolve the issue. Problem scenarios could include: • The instance may be running but not sending heartbeat events. • The instance failed to start. In most cases the resolution is to restart the instance via the Azure portal. Also check that the **Azure Event Hub connection string** field in Pexip Infinity is configured correctly.
Teams scheduled scaling: Event hub for management events does not exist	44	Error	This is raised if "Enable Azure Event Hub" is enabled but Pexip Infinity cannot connect to the Azure Event Hub queue for scheduled scaling. The most likely reason for this is that you have not created the Teams Connector API app.	Disabling the "Enable Azure Event Hub" setting will lower the alarm. However, to use scheduled scaling you must follow the instructions to create the Teams Connector API app and redeploy your Teams Connector.
System integrity is compromised	46	Error	This means that one or more of the Pexip Infinity system files have been modified by an external party/event and thus the integrity of the system has been compromised. (These are internal files, not Management Node configuration data.)	You need to perform a full redeploy of your platform. You should also notify your security administrator if the source of the event is unknown.
License limit reached	2	Warning	A Conferencing Node is unable to accept a call because there are not enough concurrent licenses available on the system at this time. For more information, see Pexip Infinity license installation and usage.	• Wait until one or more of the existing conferences have finished and the licenses have been returned to the pool. • Contact your Pexip authorized support representative to purchase more licenses. Note that when a license subsequently becomes available (e.g. because a participant leaves a conference, or because the administrator adds more licenses), the alarm is not cleared immediately; the alarm is cleared after the next participant successfully joins a conference.
Licenses expiring	3	Warning	One or more of your licenses is due to expire within the next 60 days.	Contact your Pexip authorized support representative to renew your licenses.

Alarm	ID	Level	Cause	Suggested resolutions
Call capacity limit reached	1	Warning	A call has not been accepted because all Conferencing Nodes that are able to take the media for this call are at capacity. It could be either Proxying Edge Nodes or Transcoding Conferencing Nodes that are out of capacity. Note: to understand how often this issue is occurring in your deployment, search the Administrator log for "out of proxying resource" or "out of transcoding resource". This alarm clears either when an existing call is disconnected or the next time a new call is successfully placed.	• Deploy more Conferencing Nodes in either the proxying or transcoding location as appropriate. • Move existing Conferencing Nodes onto more powerful servers. • Allocate more virtual CPUs for Conferencing Nodes on existing servers (if there are sufficient CPU cores). Note that the Conferencing Node will have to be rebooted for this to take effect. • Configure each location with a primary and secondary overflow location. • If a call is received in a location that contains Proxying Edge Nodes, that location must be configured with a **Transcoding location** that contains your Transcoding Conferencing Nodes. Note that some types of call consume more resources than other calls. Thus, for example, if you are at full capacity and an audio-only call disconnects, there may still not be sufficient free resource to connect a new HD video call.
Management Node limit reached	5	Warning	The Management Node does not have sufficient resources for the current deployment size (number of Conferencing Nodes).	Increase the amount of RAM and the number of virtual CPUs assigned to the Management Node. See the recommended hardware requirements in Server design recommendations.
Trusted CA certificates expiring	6	Warning	One or more of your trusted CA certificates is due to expire within the next 30 days, or has already expired.	Obtain and upload an updated certificate for the certificate authority.
TLS certificates expiring	7	Warning	One or more of your TLS certificates is due to expire within the next 30 days, or has already expired.	Obtain and upload an updated TLS certificate. You may also need to delete the old certificate.
Incomplete TLS certificate chains	8	Warning	A TLS certificate has an incomplete chain of trust to the root CA certificate.	Obtain and upload the appropriate chain of intermediate CA certificates to the Management Node (the certificate provider normally provides the relevant bundle of intermediate CA certificates).
Syslog server inaccessible	4	Warning	A syslog server has been configured to use TCP or TLS but either is not responding to contact requests, or the connection has dropped.	• Check your network connectivity. • Check that the syslog server is running.
Connectivity lost between nodes	19	Warning	Communication to a Pexip Infinity node has been lost.	Check network connectivity and routing as for "Configuration not synchronized" above, or in the case of a software upgrade, wait for the upgrade process to complete.

Alarm	ID	Level	Cause	Suggested resolutions
Hardware instability detected	21	Warning	Pexip Infinity has detected that the underlying VM infrastructure has paused the Pexip virtual machine. This is usually indicative of over-committed hardware, which we do not support. Pexip Infinity is a real time system and requires dedicated access to the underlying CPU and RAM resources of the hardware host.	Ensure that the Management Node and all Conferencing Nodes have dedicated access to their own RAM and CPU cores. See the recommended hardware requirements in Server design recommendations.
CPU instruction set is deprecated	23	Warning	The node is deployed on a server that is not using the AVX or later CPU instruction set (e.g. if it uses SSE4.2). This alarm is raised when a Conferencing Node restarts and is automatically cleared after 48 hours.	Deploy the Conferencing Node on a server with AVX or later.
Hardware IO (input/output) instability detected	24	Warning	Pexip Infinity has recently detected consistent read latency greater than 100ms or write latency greater than 400ms.	• Avoid having multiple VMs using the same physical hard drive. • Check the hard drive for failures.
VOIP scanner resistance has detected excessive incorrect aliases being dialed in a short period	26	Warning	Pexip Infinity's VOIP scanner resistance has detected excessive incorrect aliases being dialed in a short period, and has temporarily blocked access attempts from the suspected VOIP scanner IP addresses.	See the administrator log for details of the calls.
PIN brute force resistance has detected excessive incorrect PIN entry attempts in a short period	27	Warning	Pexip Infinity's PIN brute force resistance has detected excessive incorrect PIN entry attempts in a short period, and has temporarily blocked access attempts to one or more conferencing services.	See the administrator log for details of the calls.
Scheduled maintenance event (freeze)	39	Warning	A scheduled maintenance event in Microsoft Azure has been detected.	No action is required. Any Conferencing Node running on the affected VM is automatically placed into maintenance mode until the event completes.
Scheduled maintenance event (redeploy)	40	Warning		
Scheduled maintenance event (preemption)	41	Warning		
Azure key vault certificate expiring	47	Warning	A Teams Connector certificate (in the Azure key vault) is due to expire within the next 30 days.	Update the Teams Connector certificate.
Google Meet Gateway Token expiring	49	Warning	The Google Meet gateway token is due to expire within the next 30 days.	Update the .

Cloud bursting alarms

The following alarms may be raised in relation to issues with dynamic cloud bursting. See Dynamic bursting to a cloud service for more information about resolving these alarms.

Alarm	ID	Level	Cause	Suggested resolutions
Not authorized to perform this operation	15 & 16	Error	Pexip Infinity is not authorized to view instance data or to start and stop instances in the cloud service.	For AWS, ensure that an appropriate policy document is configured in AWS and is attached to the user that is being used by the Pexip platform. For Azure, check your Active Directory (AD) application and its associated role/permissions. For GCP, check your service account and its associated role/permissions.
Authentication failure while trying to communicate with the cloud provider	17	Error	Pexip Infinity cannot sign in to the cloud service.	Check your cloud bursting settings in **Platform > Global Settings > Cloud Bursting**: • For AWS, check that the **Access Key ID** and **Secret Access Key** match the **User Security Credentials** for the user you added within **Identity And Access Management** in the AWS dashboard. • For Azure, check that your **subscription**, **client** and **tenant** IDs and **secret key** are correct for your Active Directory application. • For GCP, check that your configured **GCP project ID**, **service account ID** and **private key** are correct for your GCP service account.
Cloud bursting process encountered an unexpected error	12	Error	Pexip Infinity encountered an unexpected error while managing the cloud overflow nodes.	Check the status of your cloud bursting nodes within Pexip Infinity (**Status > Cloud Bursting**) and of your instances within your cloud provider. Also check administrator and support log messages that are tagged with a log module name of **administrator.alarm** to see additional error message information.
Cloud-bursting node found, but no corresponding Conferencing Node has been configured	13	Warning	This occurs when Pexip Infinity detects a bursting instance with a tag matching your system's hostname but there is no corresponding Conferencing Node configured within Pexip Infinity.	This message can occur temporarily in a normal scenario when deploying a new Conferencing Node and you have set up the VM instance in your cloud provider but you have not yet deployed the Conferencing Node in Pexip Infinity. In this case, the issue will disappear as soon as the Conferencing Node is deployed.
A location contains cloud bursting nodes, but no other locations are using it for overflow	14	Warning	A location contains some cloud overflow nodes, but no other locations are using it as an overflow location.	Set the location containing the cloud overflow nodes as the **Primary overflow location** of the locations containing your "always on" Conferencing Nodes.

One-Touch Join alarms

The following alarms may be raised in relation to issues with One-Touch Join:

Alarm	ID	Level	Instance	Cause	Suggested resolutions
OTJ Google Gatherer Error	29	Error	Google Connection Test Failure	The connection test to Google Workspace has failed. This could be because your service account credentials are incorrect.	Check your service account details, specifically the service account email and private key.
			Google Room Connection Failure	OTJ cannot connect to one of the rooms you have specified. This could be because the room is misconfigured within Google Workspace.	Check the steps to set up a new room. Is the room resource email correct? Has it been shared with the service account?
OTJ Exchange Gatherer Error	30	Error	Exchange Connection Test Error	The connection test to Exchange has failed. This could be because your service account credentials are incorrect.	Check your Exchange service account username and password.
			Exchange OAuth Error	OTJ cannot use OAuth to sign into Exchange.	Check your OAuth credentials.
			Exchange Room Connection Error	OTJ cannot connect to the room specified in the alarm description. This could be because the room is misconfigured.	Check the room has been correctly set up.
OTJ Endpoint Configurator Error	31	Error	Endpoint Misconfigured Error	The OTJ endpoint does not have a username and password configured, and there is no default username and password.	Provide a username and password for the OTJ endpoint or for the associated OTJ profile.
			Endpoint Request Error	OTJ cannot connect to the endpoint.	Check that the endpoint is configured correctly.
			Endpoint Non-200 Status Code	The endpoint returns a non-200 status code.	Check the status code that is given in the logs. This is likely a configuration error with the endpoint.
OTJ Meeting Processor Failure	32	Error	Meeting Processor Rendering Error, Template Error or Runtime Error	The meeting processing rule could not extract a meeting alias.	Check and edit the rule using the test tool.
OTJ Poly Endpoint Error	35	Error	Poly Endpoint Not Polled	A Poly endpoint that has Raise alarms enabled has not made contact with the OTJ calendaring service within the last 10 minutes.	Ensure that Enable support for Pexip apps and Client API is enabled. Ensure that the configuration for endpoint on Pexip Infinity and on the endpoint itself is correct, in particular that the username and password configured on both match. Ensure that the endpoint is showing as registered to the calendaring service. Restart the endpoint.

Alarm	ID	Level	Instance	Cause	Suggested resolutions
OTJ Webex Failure	38	Error	Webex Endpoint failed with <Webex error message>	One-Touch Join has received an error from Webex when attempting to send the request. Examples of these messages include: • Cloud Calendar is configured • Device has not registered as an XAPI provider • Webex Request Error • No response received from request • The server, while acting as a gateway or proxy, received an invalid response from the upstream server it accessed in attempting to fulfill the request	The resolution will depend on the issue, for example: • Disable the cloud calendar. • Check that the Device ID is correct. • Confirm that the correct ports are open, and that Pexip Infinity can reach Webex. • Confirm that the endpoint is switched on and connected to the internet, and that Webex can reach it.
			Webex OAuth Error	The system cannot get an access token for your Webex integration.	Sign in with the Webex service account again.
			Webex Configuration Error	A Webex endpoint is configured but there is no Webex integration configured on the OTJ profile.	Make sure you configure a Webex integration on the OTJ profile.

Alarm	ID	Level	Instance	Cause	Suggested resolutions
OTJ Graph Gatherer Error	45	Error	Graph Connection Test Error	A successful connection to the Graph API could not be made for any room. Possible causes: • A network connection issue between the Conferencing Node and Graph API (**https://graph.microsoft.com**). • The room email addresses (for the OTJ Endpoints used by this O365 Graph Integration) are incorrect. • The application permission (the Application set up in Azure for OAuth) has not been granted for the rooms.	• Check network connectivity. • Check the email addresses used by the rooms. • Check the Azure App registration configuration.
			Graph OAuth Error	There was a problem getting an OAuth access token that could be used to connect to the Graph API. Possible causes: • There is a network connection issue between the Conferencing Node and the token endpoint URI (which is **https://login.microsoftonline.com** by default). • The OAuth Client ID, Client secret, or OAuth 2.0 token endpoint URL are not entered correctly in the O365 Graph Integration configuration on the Pexip Infinity Management Node. • The Azure application is not set up correctly (for example, the correct API permissions were not added, or admin consent was not provided).	• Check network connectivity. • Check the O365 Graph integration configuration. • Check the Azure App registration configuration.
			Graph Room Connection Error	There was a problem connecting to one or more rooms (but the OTJ service has successfully connected to other rooms).	• Check that the room email address is correct. • Check the application permission is configured correctly to allow OTJ to read this room's calendar.

Viewing login history

To see a record of attempts to log in to the Pexip Infinity Administrator interface, go to **History & Logs > Login History**.

The system displays the 100 most recent login attempts. It displays successful and failed attempts, and for each attempt shows the time of the attempt, the username that was submitted, and the address and port of the remote client.

Note that you are taken directly to the **Login History** page after logging in to the Pexip Infinity Administrator interface if the system has been configured with a login banner and certificate-based authentication has been enabled.

About the support log

The Management Node collates the logs from itself and all Conferencing Nodes and compiles them into the support log.

The support log records all events occurring across the Pexip Infinity deployment, including:

- configuration changes on the system
- conferences starting and ending
- participants joining and leaving conferences
- participants presenting in conferences
- SIP, H.323 and BFCP signaling
- device registration requests
- DNS lookups.

Sensitive information, such as the content of chat messages, is not logged.

Log timestamps always use UTC.

The information in the support log is retained during reboot and upgrade.

ⓘ You can also configure Pexip Infinity to use an external syslog server. This allows you to use the syslog server's associated tools to collate and process the log entries to obtain specific information about real-time events, tailored to your requirements.

For more information on the content of the support log, see Log output.

Viewing the support log

To view the support log, go to **History & Logs > Support Log**. If required, you can export the support log to a text file by selecting **Download**.

The log is presented in the order the messages were received by the Management Node.

The log appears in the format:

```
syslog_time system originating_time level name details
```

where:

Field	Description
`syslog_time`	In the format: `year-month-dayThour:minute:second.millisecondUTC_offset` The time that the event was logged by syslog on the originating system.
`system`	The IP address or host name of the system that sent the log message.
`originating_time`	In the format: `year-month-day hour:minute:second,millisecond` The time at which the event occurred on the originating system.
`level`	The severity of the event. The levels, in order of increasing severity, are: • DEBUG • INFO • WARNING • ERROR ⓘ In the Pexip Infinity Administrator interface, warnings and errors are highlighted with an orange or blue background respectively.
`name`	The system module producing the log output.
`details`	Information about the event, as a series of name=value pairs.

Searching the support log

You can filter the entire support log to only show messages that contain a particular string (for example, an IP address or alias) using the search box at the top right of the web page.

The search box also supports BRE syntax regular expressions (regex), so for example, you could search for:

- **^2017-10-07T09:[34][[:digit:]]:** to limit the displayed log messages to those emitted between 09:30:00 and 09:49:59 UTC on 7 October 2017
- **.*"Registration added".*"sip:alice@example.com"** to find all instances of "Registration added" messages for the SIP alias alice@example.com
- **.*\("error"\|administrator.alarm\).*** to review error and warning alarms that have been raised or lowered.

Note that the results are not paginated.

Summarizing support log messages

You can select **Log summary** to generate a condensed view of the messages in the support log, showing a summary of the call signaling — similar to traditional ladder diagrams, but text based — for the current search/filtered log messages, such as a filter on a specific call ID.

File size

The support log has a maximum size of 2 GB, after which it will be overwritten, starting with the oldest entries. Up to 10 MB is available in a single page view on the Pexip Infinity Administrator interface; you can navigate through additional log pages using the buttons at the bottom left of the page. (Note that each page is a separate file, so the first page, which is the end of the file, may not be full.)

About the administrator log

The administrator log is a subset of the support log. It contains information about events occurring during the normal use of the system which may be of interest to a system administrator, notably:

- Conferencing Node deployment and communication status
- configuration changes on the system
- conferences starting and ending
- participants joining and leaving conferences
- participants presenting in conferences
- device registrations
- VMR imports
- system backups
- break-in prevention policy triggers.

Sensitive information, such as the content of chat messages, is not logged.

Log timestamps always use UTC.

The information in the administrator log is retained during reboot and upgrade.

Viewing the administrator log

To view the administrator log, go to **History & Logs > Administrator Log**. If required, you can export the administrator log to a text file by selecting **Download**.

The log is presented in the order the messages were received by the Management Node.

The log appears in the format:

```
syslog_time system originating_time level name details
```

where:

Field	Description
`syslog_time`	In the format: `year-month-dayThour:minute:second.millisecondUTC_offset` The time that the event was logged by syslog on the originating system.
`system`	The IP address or host name of the system that sent the log message.
`originating_time`	In the format: `year-month-day hour:minute:second,millisecond` The time at which the event occurred on the originating system.
`level`	The severity of the event. The levels, in order of increasing severity, are: • DEBUG • INFO • WARNING • ERROR In the Pexip Infinity Administrator interface, warnings and errors are highlighted with an orange or blue background respectively.
`name`	The system module producing the log output.
`details`	Information about the event, as a series of name=value pairs.

Searching the administrator log

You can filter the entire administrator log to only show messages that contain a particular string (for example, an IP address, alias or service tag) using the search box at the top left of the web page.

The search box also supports BRE syntax regular expressions (regex), so for example, you could search for:

- **^2017-10-07T09:[34][[:digit:]]:** to limit the displayed log messages to those emitted between 09:30:00 and 09:49:59 UTC on 7 October 2017
- **.*"Registration added".*"sip:alice@example.com"** to find all instances of "Registration added" messages for the SIP alias alice@example.com
- **.*\("error"\|administrator.alarm\).*** to review error and warning alarms that have been raised or lowered.

Note that the results are not paginated.

For more information on the content of the administrator log, see Log output.

File size

The administrator log has a maximum size of 1 GB, after which it will be overwritten, starting with the oldest entries. Up to 10 MB is available in a single page view on the Pexip Infinity Administrator interface; you can navigate through additional log pages using the buttons at the bottom left of the page. (Note that each page is a separate file, so the first page, which is the end of the file, may not be full.)

Log output

The Pexip Infinity Management Node collates the logs from itself and all Conferencing Nodes and compiles these into the support log and the administrator log (which is a subset of the events in the support log).

To view these logs from the Pexip Infinity Administrator interface, go to **History & Logs > Support Log** or **History & Logs > Administrator Log**. If required, you can export the logs to a text file by selecting **Download**.

You can also use a syslog server to collate logs remotely.

Log timestamps always use UTC.

The log is presented in the order the messages were received by the Management Node.

The log appears in the format:

```
syslog_time system originating_time level name details
```

where:

Field	Description
`syslog_time`	In the format: `year-month-dayThour:minute:second.millisecondUTC_offset` The time that the event was logged by syslog on the originating system.
`system`	The IP address or host name of the system that sent the log message.
`originating_time`	In the format: `year-month-day hour:minute:second,millisecond` The time at which the event occurred on the originating system.
`level`	The severity of the event. The levels, in order of increasing severity, are: • DEBUG • INFO • WARNING • ERROR In the Pexip Infinity Administrator interface, warnings and errors are highlighted with an orange or blue background respectively.
`name`	The system module producing the log output.
`details`	Information about the event, as a series of name=value pairs.

Administrator log system modules

The administrator log contains information about events occurring during the normal use of the system which may be of interest to a system administrator.

The following list shows typical events that are shown in the administrator log. In addition to these, logs may also contain one-off events, such as bulk configuration import and incident reports.

For the sake of brevity, the examples shown below feature only the `name` and `details` sections of the log message.

administrator.system

Logs when the Pexip Infinity application has started on this Conferencing Node. Also logs web-based activities such as:

- import of service configuration
- deployment of a Conferencing Node
- configuration synchronization towards Conferencing Nodes
- liveness status of Conferencing Nodes and connectivity paths between nodes (either a direct path, or via other nodes in some cases for Proxying Edge Nodes)

Examples

```
Name="administrator.system" Message="Starting Infinity application." Pid="6361" Version="15 (34498.0.0)"
Name="administrator.system" Message="Deploying conferencing node virtual machine." Address="192.168.0.2"
Name="administrator.system.configuration" Message="Conferencing node configuration updated." Node="conferencingnode1"
Name="administrator.system.connectivity" Message="Unable to contact node." Src-Node="192.168.0.5" Node="192.168.0.2"
Name="administrator.system.connectivity" Message="Connectivity re-established." Src-Node="192.168.0.5" Node="192.168.0.2" Path="Direct" Last-Reported="Wed Mar 22 23:13:50 2017"
Name="administrator.system.connectivity" Message="Connectivity established." Src-Node="10.47.2.17" Node="10.47.3.46" Path="10.47.3.4, 10.47.3.7"
```

(in this last example there is an indirect connectivity path between the source node and the target node, which is 10.47.2.17 > 10.47.3.4 > 10.47.3.7 > 10.47.3.46)

administrator.configuration

This module logs when:

- configuration is added or changed
- a user logs in or logs out of the system, or fails to log in
- a participant is manually dialed into a conference
- license requests are processed

Messages

```
Configuration added
Configuration changed
Configuration deleted
User logged in
User logged out
User login failed
Initiated dial command
License activation request stored for offline processing
Processed license request
```

Parameters

Parameter	Definition
User	The user who made the change.
Type	The resource that was changed/added.
ID	The ID of the resource that was changed/added.
Fields	The fields in the resource that were changed. Applies to "Configuration changed" logs only.
Remote-Address	The IP address of the system from which the request was received.
Remote-Port	The port on the system from which the request was received.

Examples

```
Name="administrator.configuration" Message="User login failed" User="admins" Remote-Address="192.0.2.0" Remote-Port="53991"
Name="administrator.configuration" Message="Configuration added" User="schedulingservice" Type="Scheduled Alias" ID="123456@example.com" Remote-Address="192.0.2.0" Remote-Port="53991"
Name="administrator.configuration" Message="Configuration changed" User="admin" Type="Virtual Meeting Room" ID="Alice's VMR" Fields="Description, Show names of participants" Remote-Address="192.0.2.0" Remote-Port="53991"
```

administrator.conference

This module logs conference activity and break-in prevention policy triggers.

Parameters for messages related to conference activity

Each log entry includes information in the form of a series of name=value pairs. These are defined as follows:

Parameter	Definition
Call-ID	An identifier that allows correlation of messages from the same call.
Conference	The name of the service being connected to.
ConferenceAlias	The alias that was dialed in order to connect to the service.
Conversation-Id	An identifier that allows correlation of messages across separate "calls" for video+audio, RDP, or chat.
Detail	The reason the call was disconnected (for disconnect events). For more information, see Disconnection reasons.
Direction	Either: • **in**: a call into Pexip Infinity • **out**: a call dialed out from Pexip Infinity.
DisplayName	The display name of the participant.
Duration	The duration of the call (for disconnect events).
License-Type	The type of license used in the call ("port", "audio" or "None").
Licenses	The number of licenses allocated for the call.
Location	The system location of the Conferencing Node that is handling the media processing for the call.
Media-Node	The IP address of the Conferencing Node handling the media processing for the call.
Participant	The name or alias of the conference participant.
Participant-Id	Allows correlation of messages from the same participant URI per connection type, for example an API connection will have a different Participant-Id from a WebRTC connection.
Protocol	The protocol of the call (may also indicate "BACKPLANE" when a backplane between Conferencing Nodes is established or removed).
Proxy-Location	The system location of the Proxying Edge Node.
Proxy-Node	The IP address of the Proxying Edge Node.
Remote-Address	The source IP address for the signaling for this call.
Requester	The Call ID of the participant requesting this operation.

Parameter	Definition
Role	The role of this participant: • **chair**: the participant is a Host • **guest**: the participant is a Guest
Service-Tag	A unique identifier used to track usage of this service.
Service-Type	The service type of the conference. This is one of the following: • **conference**: a Virtual Meeting Room • **lecture** : a Virtual Auditorium • **two_stage_dialing**: a Virtual Reception • **media_playback**: a Media Playback Service • **test_call**: a Test Call Service call • **gateway**: an Infinity Gateway call.
Signaling-Location	The name of the Pexip Infinity location handling the signaling for this call.
Signaling-Node	The IP address of the Conferencing Node handling the signaling for the call.
Vendor	System details about the endpoint of the participant, such as manufacturer name and version number for hard endpoints, or browser and operating system details for soft clients.

Examples

Trigger for the log message	Example log entry
A participant tries to connect to a conference (at this stage it is the conference alias that is logged).	`Message="Participant attempting to join conference." ConferenceAlias="sip:meet.alice@example.com" Participant="sip:alice@example.com" Protocol="SIP" Direction="in" Remote-Address="10.47.2.55" Participant-Id="44be63a0-...c52" Registered="False" Location="Europe"`
A conference instance is created on a Conferencing Node. This occurs when the first participant attempts to join this conference on that Conferencing Node. The **Conference** parameter indicates the name of the service, not the alias being dialed.	`Message="Conference has been created." Conference="Meet Alice" Service-Tag="" Service-Type="conference" Conference-ID="cc7f6255-0986-447d-8ee9-561c5bbf4a24"`
A participant joins the conference. This log message is issued before a role is assigned.	`Message="Participant has joined." Conference="Meet Alice" Service-Tag="" Service-Type="conference" ConferenceAlias="sip:meet.alice@example.com" Participant="sip:alice@example.com" DisplayName="Alice Smith" Protocol="SIP" Direction="in" Vendor="TANDBERG/257 (TE4.1.1.273710)" Call-Id="fe7107ba-...16c" Conversation-Id="fe7107ba-...16c" Participant-Id="44be63a0-...c52" Remote-Address="10.47.2.55" Location="London" Licenses="1" Signaling-Node="192.168.0.1" Signaling-Location="London" Media-Node="192.168.0.2" Conference-ID="cc7f6255-0986-447d-8ee9-561c5bbf4a24" Proxy-Node="10.47.2.46" Proxy-Location="London Proxy"`
Conference creation fails.	`Message="Failed creating conference." Conference="meet" Service-Tag=""`

Trigger for the log message	Example log entry

A participant cannot join the conference. The **Reason** may vary as appropriate.
 	
If the reason is "No direct route between Edge and Transcoding", the most likely cause is that local or external media policy is in place and it has nominated a Transcoding Conferencing Node that cannot be directly reached from the Proxying Edge Node that received the call.
	`Message="Participant failed to join conference." ConferenceAlias="sip:meet.invalid@example.com" Participant="sip:bob@example.com" Protocol="SIP" Direction="in" Remote-Address="10.44.100.75" Participant-Id="6696b7fe-...2cb" Reason="Neither conference nor gateway found"`
	
	other reasons include:
	
	`Reason="No direct route between Edge and Transcoding"`
	`Reason="Out of proxying resource"`
	`Reason="Out of transcoding resource"`
	`Reason="System in maintenance mode"`

A participant leaves the conference. Includes the **Duration** (in seconds) for which they were present in the conference. This message provides the definitive record of the length of time a call was connected to a conference. For more information on the reasons given in the **Detail** field, see Disconnection reasons.	`Message="Participant has disconnected." Conference="Meet Alice" Service-Tag="" Service-Type="conference" ConferenceAlias="sip:meet.alice@example.com" Participant="sip:alice@example.com" DisplayName="Alice Smith" Protocol="SIP" Direction="in" Call-Id="fe7107ba-...16c" Conversation-Id="fe7107ba-...16c" Participant-Id="44be63a0-...c52" Remote-Address="10.47.2.237" Location="London" Licenses="1" License-Type="port" Signaling-Node="192.168.0.1" Signaling-Location="London" Media-Node="192.168.0.1" Conference-ID="cc7f6255-0986-447d-8ee9-561c5bbf4a24" Proxy-Node="10.47.2.46" Proxy-Location="London Proxy" Duration="63.123" Detail="Remote disconnect"`
A conference stops on a particular Conferencing Node. The **Duration** of this conference is given in seconds.	`Message="Conference has been stopped." Conference="Meet Alice" Service-Tag="" Service-Type="conference" Duration="62.234" Conference-ID="cc7f6255-0986-447d-8ee9-561c5bbf4a24"`

A participant starts/stops presenting.	`Message="Participant is presenting." Conference="meet" Service-Tag="" Participant="sip:chuck@example.com" Call-Id="e5dbd44b...243" Conversation-Id="e5dbd44b...243"`
	
	`Message="Participant has stopped presenting." Conference="meet" Service-Tag="" Participant="sip:chuck@example.com" Call-Id="e5dbd44b...243" Conversation-Id="e5dbd44b...243"`

A participant using a Pexip app, or any third party using the Pexip Client API, controls participants in a conference.	`Message="Participant hold requested by API." Conference="meet" ...`
	
	`Message="Participant resume requested by API." Conference="meet" ...`
	
	`Message="Participant disconnect requested by API." Conference="meet" ..."`
	
	`Message="Participant mute requested by API." Conference="meet" ...`
	
	`Message="Participant unmute requested by API." Conference="meet" ...`
	
	`Message="Participant presentation receipt enabled by API." Conference="meet"...`
	
	`Message="Participant presentation receipt disabled by API." Conference="meet" ...`
	
	`Message="Outgoing call requested" Conference="meet" Participant="sip:carol@example.com" Protocol="SIP" Role="chair"`
	
	`Message="Sending DTMF to participant." Conference="meet" Service-Tag="" Participant="sip:alice@example.com" Call-Id="e5dbd44b...243" Conversation-Id="e5dbd44b...243" Digits="4"`
	
	`Message="Participant transfer requested by API." Conference="meet.alice" Service-Tag="" Requester="66eb0873-...8ce" Participant="sip:bob@example.com" Call-Id="d583e3a1...@10.44.26.36" Conversation-Id="2a8ea679-...8a8" To-Conference="3259oknfaoi"`

A participant uses a Pexip app or DTMF commands to control the conference.	`Message="Conference terminated by participant" Conference="meet" ...`
	
	`Message="Conference lock requested by participant" Conference="meet" ...`
	
	`Message="Conference unlock requested by participant" Conference="meet" ...`
	
	`Message="Mute all guests requested by participant" Conference="meet" ...`
	
	`Message="Unmute all guests requested by participant" Conference="meet" ...`

Trigger for the log message	Example log entry

Participants attempt to join a PIN-protected conference.	`Message="PIN entry correct." Participant="sip:bob@example.com" ConferenceAlias="sip:meet@example.com" Call-Id="27efdd36...6be" Role="host" Detail="Participant entered correct conference PIN and will be placed into the conference."`
	
	`Message="Participant entered incorrect conference PIN" RetriesRemaining="2" Participant="sip:bob@example.com" ConferenceAlias="sip:meet@example.com" Call-Id="27efdd36...6be"`
	
	`Message="Participant exceeded PIN entry retries. Call will be disconnected." Participant="sip:bob@example.com" ConferenceAlias="sip:meet@example.com" Call-Id="27efdd36...6be"`
	
	`Message="PIN entry timed out." Participant="sip:bob@example.com" ConferenceAlias="sip:meet@example.com" Call-Id="a15dc445-...dfc" Detail="The correct PIN has not been received. Call will be disconnected."`

Participants are involved in a PIN-protected conference.	`Message="Host participant joining." Participant="sip:alice@example.com" ConferenceAlias="sip:meet@example.com" Call-Id="4d3ad9a1-...833" Role="chair" Detail="Participant will be automatically placed into the conference."`
	
	`Message="Conference host timeout." Participant="sip:bob@example.com" ConferenceAlias="sip:meet@example.com" Call-Id="4d3ad9a1-...833" Detail="The conference host has not joined. Call will be disconnected."`
	
	`Message="Guest participant waiting." Participant="sip:bob@example.com" ConferenceAlias="sip:meet@example.com" Call-Id="50a86594-...797" Role="guest" Detail="Waiting until participant is allowed to join the conference."`
	
	`Message="Guest participant joining." Participant="sip:bob@example.com" ConferenceAlias="sip:meet@example.com" Call-Id="50a86594-...797" Role="guest" Detail="Participant will be automatically placed into the conference."`

A conference is automatically ended.	`Message="Last host left the conference and has not re-joined; conference will be terminated." Conference="Meet Alice" Service-Tag=""`
	
	`Message="Removing participant due to conference termination" Participant="sip:bob@example.com" Conference="Meet Alice" Service-Tag=""`

Break-in prevention policy example log messages

The following examples show messages that may be logged by the break-in prevention policies.

Logged when PIN brute force resistance has temporarily disabled a service, and for all subsequent attempts while the service is blocked:

```
Message="Break-in prevention policy blocking all attempts to join this service." ConferenceAlias="alice" Service="Alice's VMR" Participant="Crooky McCrookface" Protocol="API" Direction="in" Remote-address="10.44.21.35" Reason="Service appears to be under PIN break-in attack" remaining_block_duration_seconds="525"
```

Logged when VOIP scanner resistance has temporarily blocked an address:

```
Message="Participant has been quarantined by Break-in prevention policy due to excessive failed join attempts." Participant="Crooky McCrookface" Protocol="API" Direction="in" Remote-address="10.44.21.35" Reason="Too many attempts to join non-existent aliases" remaining_block_duration_seconds="488"
```

and then any subsequent attempts generate messages such as:

```
Message="Break-in prevention policy rejecting call attempt from quarantined caller." Protocol="API" Direction="in" Local-alias="[u'alice']" Remote-address="10.47.250.169" Reason="Suspicious join attempt rejected" remaining_block_duration_seconds="519"
```

administrator.participantdialer

This module logs outbound calls placed from a conference or via the Infinity Gateway.

Examples

Logged when dialing out from a conference:

```
Message="Placing outbound call" ConferenceAlias="meet.alice@example.com" DisplayName="meet.alice" Participant-alias="sip:bob@example.com" Protocol="SIP" Role="host"
```

Logged when making a call via the Infinity Gateway:

```
Message="Placing outbound call" Conference="Route to London:423a18c5-b06c-4654-9d34-e6b82f308b64" ConferenceAlias="alice@example.com" DisplayName="Alice" Participant-alias="sip:carol@example.com" Protocol="SIP" Role="guest"
```

administrator.registration

This module logs endpoint registration activities.

Examples

Logged when a registration is added:

```
Message="Registration added" Alias="alice@example.com" Protocol="SIP" Registration-Id="8695ea23-66fb-41d2-9266-d16598f5d133"
Natted="False" Location="Europe"
```

Logged when a registration is removed:

```
Message="Registration deleted" Alias="bob@example.com" Protocol="WebRTC" Registration-Id="6936f540-6dbc-40ac-9d55-5469bbf4e561"
Location="Europe" Reason="Registration expired"
```

administrator.ldap.sync

This module logs VMR, device and user synchronization template activities.

Examples

Logged when a VMR template synchronization is initiated:

```
Message="Beginning VMR Sync" Template="All employees" Creating="219" Deleting="0" Modifying="0" Unchanged="0"
```

Logged when a VMR template synchronization finishes:

```
Message="Completed VMR Sync" Template="Europe" Status="sync_succeeded" Created="18" Deleted="10" Updated="5" Unchanged="34"
Warnings="3" Last_warning="Conference clash: not overwriting manually created conference"
```

Logged when an alias clash occurs during VMR template synchronization:

```
Message="Sync detected alias clash: alias held by an existing conference has not been assigned to the newly synced conference."
NewConference="Alice's VMR" ExistingConference="Bob's VMR" Alias="meet@example.com"
```

administrator.apps.cloudbursting

This module logs cloud bursting activities.

Examples

Logged when an overflow Conferencing Node is started up:

```
Message="On-prem location 'Oslo' is overloaded! Starting instance i-0043088a9073b57e6 (stopped) in location 'AWS eu-west-1'"
```

Logged when an overflow Conferencing Node is shut down:

```
Message="Shutting down idle worker i-597394d2 (running) (uptime: 50 minutes)"
```

administrator.alarm

This module logs all **Alarm raised**, **Alarm acknowledged**, and **Alarm lowered** events.

Examples

Logged when a "connectivity lost" alarm is lowered:

```
"Message="Alarm lowered" Node="10.47.2.46" Alarm="connectivity_lost" Instance="Source=10.47.2.46, Destination=10.47.2.17"
```

administrator.web

This module logs activity relating to conference and participant control by an administrator using the Administrator interface.

Examples

```
Message="Initiated mute command." User="admin" Participant-alias="Alice" Participant-
Id="25fca17e-...374"
Message="Initiated unmute command." User="admin" Participant-alias="Alice" Participant-
Id="25fca17e-...374"
Message="Initiated transfer command." User="admin" Participant-alias="Alice" Participant-
Id="25fca17e-...374" Conference-alias="meet.alice2" Role="chair"
```

```
Message="Initiated conference lock." User="admin" Conference="meet.alice2"
Message="Initiated conference unlock." User="admin" Conference="meet.alice2"
Message="Initiated disconnect command." User="admin" Participant-alias="Alice" Participant-Id="21300b40-...557"
Message="Initiated conference disconnect." User="admin" Conference="meet.alice"
```

administrator.scheduling

This module logs the activities of the Secure Scheduler for Exchange service.

Parameters

Parameter	Definition
ExchangeConnectorID	The database ID of the Pexip Exchange Integration which is the cause of this log activity.
CorrelationID	A UUID which identities which specific item this log refers to. The logs can be searched for an item's CorrelationID to easily find all logs pertaining to that item.
Name	The name of the VMR associated with the scheduled conference.
VmsID	The database ID of the VMR associated with the scheduled conference.
ScheduledAliasID	The database ID of the scheduled alias associated with the scheduled conference.
UUID	The UUID which identifies the Alias associated with the scheduled conference (this is the UUID used for the PXPS tag).
Alias	The full alias of the VMR associated with the scheduled conference, including the domain part
NumericAlias	The alias of the VMR associated with the scheduled conference, without the domain part
ScheduledConferenceID	The database ID of the single conference.
RecurringConferenceID	The database ID of the recurring conference.
OccurrenceScheduledConferenceIDs	The database IDs of the individual conference occurrences associated with the recurring conference. This is a list of values separated by commas.

Examples

Trigger for the log message	Example
New scheduled alias created.	`Message="Scheduled Alias added" ScheduledAliasID="4983" UUID="5b195574-d8e0-4f5a-b07e-7826f419db7f" Alias="8359689@rd.pexip.com" NumericAlias="8359689" ExchangeConnectorID="2"`
New single meeting created.	`Message="Scheduled Conference added" VmsID="35111" Name="Example new meeting (Toby)" ScheduledConferenceID="5001" ScheduledAliasID="4983" ExchangeConnectorID="2" CorrelationID="4d566414-a1d3-5c3f-aa08-67fda6737fa7"`
Recurring meeting changed to a single meeting. The `RecurringConferenceID` is deleted and the new `ScheduledConferenceID` assigned.	`Message="Recurring Conference changed to Scheduled Conference" VmsID="35000" Name="Example meeting (Jim Bob)" RecurringConferenceID="1" ScheduledConferenceID="10" ExchangeConnectorID="1" CorrelationID="e55af8b7-ec66-4376-8521-16a41de918eb"`
Single meeting updated.	`Message="Scheduled Conference changed" VmsID="35111" Name="Example edited meeting (Toby)" ScheduledConferenceID="5001" ExchangeConnectorID="2" CorrelationID="4d566414-a1d3-5c3f-aa08-67fda6737fa7"`
New recurring meeting created.	`Message="Recurring Conference added" VmsID="1234" Name="Example new recurring meeting (John Smith)" RecurringConferenceID="1000" OccurrenceScheduledConferenceIDs="501, 502" ScheduledAliasID="2000" ExchangeConnectorID="1" CorrelationID="5a304f92-6705-4f45-87b3-428d8a7261ef"`

Trigger for the log message	Example
Single meeting changed to a recurring meeting. The `ScheduledConferenceID` is deleted and the new `RecurringConferenceID` assigned.	`Message="Scheduled Conference changed to Recurring Conference" VmsID="10" Name="Another example meeting (Darth Vader)" ScheduledConferenceID="12" RecurringConferenceID="14" OccurrenceScheduledConferenceIDs="13, 14, 15" ExchangeConnectorID="1" CorrelationID="ab40e1f4-5a46-494a-a39f-8156eded538e"`
Recurring meeting updated. The `OccurrenceScheduledConferenceIDs` replace any occurrences which already exist.	`Message="Recurring Conference changed" VmsID="1234" Name="Example edited recurring meeting (John Smith)" RecurringConferenceID="1000" OccurrenceScheduledConferenceIDs="503" ExchangeConnectorID="1" CorrelationID="5a304f92-6705-4f45-87b3-428d8a7261ef"`
An individual occurrence of an existing recurring conference is updated but the occurrence is not currently in the database. Modified occurrences are always added to the database.	`Message="Occurrence Scheduled Conference added" VmsID="1234" Name="Example edited recurring meeting (John Smith)" OccurrenceScheduledConferenceID="504" RecurringConferenceID="1000" ExchangeConnectorID="1" CorrelationID="5a304f92-6705-4f45-87b3-428d8a7261ef"`
An individual occurrence of an existing recurring conference is updated and the occurrence is already in the database.	`Message="Occurrence Scheduled Conference changed" VmsID="1234" Name="Example edited recurring meeting (John Smith)" OccurrenceScheduledConferenceID="504" ExchangeConnectorID="1" CorrelationID="5a304f92-6705-4f45-87b3-428d8a7261ef"`
Meeting canceled — this normally happens when a single meeting or entire recurring meeting is canceled, but it may also happen if an item is rejected because an update to it was invalid. Note that deleting the conference also causes the associated single conference or recurring conference and any related aliases to be deleted at the same time.	`Message="Conference deleted" VmsID="35111" ExchangeConnectorID="2" CorrelationID="4d566414-a1d3-5c3f-aa08-67fda6737fa7"`
Occurrence canceled — this normally happens when an individual occurrence in a recurring series is canceled, but it may also happen if the occurrence is rejected because an update to it was invalid. Note that the `VmsID` is left unchanged.	`Message="Occurrence Scheduled Conference deleted" VmsID="35111" OccurrenceScheduledConferenceID="504" ExchangeConnectorID="1" CorrelationID="5a304f92-6705-4f45-87b3-428d8a7261ef"`
Single meeting recovered while running the Scheduling Recovery tool. This means a single meeting was added back to the database with a new alias.	`Message="Scheduled Conference recovered" VmsID="10" Name="Example recovered meeting (Fred)" ScheduledConferenceID="12" ScheduledAliasID="12" ExchangeConnectorID="1" CorrelationID="1b770df6-d27f-4c32-b499-1e3104c2f45b"`
Recurring meeting recovered while running the Scheduling Recovery tool. This means a recurring meeting was added back to the database with a new alias.	`Message="Recurring Conference recovered" VmsID="2" Name="Example recovered recurring meeting" RecurringConferenceID="2" OccurrenceScheduledConferenceIDs="2, 3" ScheduledAliasID="3" ExchangeConnectorID="2" CorrelationID="65dc5d19-c8fb-4040-9dd5-447ef196eda2"`

Trigger for the log message	Example
The scheduling background tasks successfully updated a recurring conference. `NewOccurrenceScheduledConferenceIDs` can be empty if no new occurrences were added at this point. `IsDepleted` indicates whether there are any more occurrences left in this recurring series.	`Message="Background tasks updated Recurring Conference" VmsID="42" Name="Meaning of life" RecurringConferenceID="42" NewOccurrenceScheduledConferenceIDs="" CurrentIndex="3" IsDepleted="False" ExchangeConnectorID="1" CorrelationID="79e4ee44-3a05-41b2-b988-4e6d98c15072"`
The scheduling background tasks encountered an error when updating a recurring conference. This can happen if we can no longer update a recurring series. For example, the room/equipment resource could have been changed and this was a recurring series stored on the old resource.	`Message="Recurring Conference deleted because it can no longer be processed" VmsID="13" RecurringConferenceID="13" ExchangeConnectorID="2" CorrelationID="e86b9fca-c0ba-4ef2-8b56-8f75fcb60d12"`
The scheduling background tasks successfully deleted an expired single meeting.	`Message="Background tasks deleted expired Scheduled Conference" VmsID="101" ScheduledConferenceID="201" ExchangeConnectorID="1" CorrelationID="359803da-f34f-4515-b505-313b3c9f53fc"`
The scheduling background tasks successfully deleted an expired meeting occurrence. Note that when an individual occurrence expires, it is only the corresponding `OccurrenceScheduledConferenceID` that gets deleted. There may still be other occurrences which are not expired.	`Message="Background tasks deleted expired Occurrence Scheduled Conference" VmsID="14" OccurrenceScheduledConferenceID="15" ExchangeConnectorID="1" CorrelationID="375c005e-5edc-4a44-876c-43c8cedf6657"`
The scheduling background tasks successfully deleted an expired recurring conference.	`Message="Background tasks deleted expired Recurring Conference" VmsID="53" RecurringConferenceID="33" ExchangeConnectorID="3" CorrelationID="a874129f-da34-4bfb-8422-211711af52d0"`
The scheduling background tasks successfully deleted a used scheduled alias.	`Message="Background tasks deleted expired used Scheduled Alias" ScheduledAliasID="5" ExchangeConnectorID="2"`
The scheduling background tasks successfully deleted an unused scheduled alias.	`Message="Background tasks deleted expired unused Scheduled Alias" ScheduledAliasID="7" ExchangeConnectorID="2"`

administrator.otj

This module logs the activities of the One-Touch Join service when:

- a One-Touch Join meeting is created
- a One-Touch Join meeting is deleted
- a One-Touch Join meeting is changed.

Parameters

Parameter	Definition
OTJProfileName	The name of the OTJ Profile associated with this meeting.
OTJProfileID	The ID of this OTJ Profile.
Room	The email address of the room resource in whose calendar the meeting has been scheduled.
Subject	The text that appears in the subject line of the meeting invitation.
OrganizerEmail	The email address of the person who created the meeting invitation.
StartTime	The scheduled start time of the meeting.
EndTime	The scheduled end time of the meeting.
Alias	The alias that the endpoint will use to dial in to the meeting.
OTJRuleName	The name of the meeting processing rule that was matched and used to process this meeting.

Examples

Trigger for the log message	Example
New OTJ meeting created	`Message="OTJ Meeting Created" OTJProfileName="Test MJX integration0" OTJProfileID="1" Room="roomresource@resource.calendar.google.com" Subject="Test Meeting" OrganizerEmail="jack@pexip.com" StartTime="2020-02-05 15:00:00" EndTime="2020-02-05 16:00:00" Alias="1234567890@pexip.com" OTJRuleName="Pexip Rule"`
Existing OTJ meeting changed	`Message="OTJ Meeting Changed" OTJProfileName="Test MJX integration0" OTJProfileID="1" Room="roomresource@resource.calendar.google.com" Subject="Test Meeting" OrganizerEmail="jack@pexip.com" StartTime="2020-02-05 15:30:00" EndTime="2020-02-05 16:30:00" Alias="1234567890@pexip.com" OTJRuleName="Pexip Rule"`
Existing OTJ meeting deleted	`Message="OTJ Meeting Deleted" OTJProfileName="Test MJX integration0" OTJProfileID="1" Room="roomresource@resource.calendar.google.com" Subject="Test Meeting" OrganizerEmail="jack@pexip.com" StartTime="2020-02-05 15:30:00" EndTime="2020-02-05 16:30:00" Alias="1234567890@pexip.com" OTJRuleName="Pexip Rule"`

Support log system modules

Support logs provide a more detailed record of activity on the Pexip Infinity system:

Module	Contains
support.bfcp	BFCP signaling.
support.conference	Logs the aliases that are matched against Call Routing Rules. The parsed or unparsed alias match type depends upon the rule's **Match against full alias URI** setting. For example: `Message="Alias matched gateway rule" Rule="Route calls from London" Description="" Service-Tag="" Unparsed-alias="alice@example.com" Parsed-alias="alice@example.com" Matched-alias-Type="Parsed-alias"`
support.conferenceservice	Logs when a participant has sent a chat message (but does not include the message contents).
support.dns	DNS lookups associated with SIP signaling.
support.events	Errors when attempting to connect to an external event sink server.
support.externalpolicy	Requests sent to an external policy server and the associated response.
support.gms	Gateway calls into Google Meet.

Module	Contains
support.h323.q931, support.h323.ras, support.h323.h245	H.323 signaling.
support.jinja2	Messages generated by the pex_debug_log filter when inserted into local policy jinja2 scripts.
support.otj	Messages related to One-Touch Join.

support.participant	Details about conference participants, including presentation stream activation, loss of incoming video, and call media statistics when a call is completed.
	An example of end-of-call media statistics is as follows:
	`Message="Media Stream destroyed" Participant="sip:alice@example.com" Call-Id="890b38520c27737c@192.168.0.1" Conversation-Id="1238035a...a2de" Detail="Stream 0 (audio)`
	`RX: rate 54kbps loss 0.77% jitter 6ms`
	`TX: rate 61kbps loss 0.00% jitter 2ms`
	`codec MP4A-LATM rate 64kbps"`
	`Message="Media Stream destroyed" Participant="sip:alice@example.com" Call-Id="890b38520c27737c@192.168.0.1" Conversation-Id="1238035a...a2de" Detail="Stream 1 (video)`
	`RX: rate 1085kbps loss 0.00% jitter 3ms`
	`TX: rate 699kbps loss 0.00% jitter 5ms`
	`codec H264 rate 700kbps resolution 768x448 fps 30"`
	`Message="Current participant is the presenter" Participant="sip:bob@example.com" Call-Id="...`
	`Message="Presentation started" Presenter="sip:bob@example.com" Participant="sip:alice@example.com" Call-Id="...`
	`Message="Stopped sending presentation to participant" Last-presenter="sip:bob@example.com" Participant="sip:alice@example.com" Call-Id="...`
	`Message="Call is on hold" Participant="sip:alice@example.com" Call-Id=...`
	`Message="Call has resumed" Participant="sip:alice@example.com" Call-Id="...`
	`Message="Lost incoming video" Participant="sip:alice@example.com" Call-Id="...`

support.pulse	Logs scheduled maintenance events in Azure. For example: `New scheduled maintenance event: {'Description': 'The system will Reboot', 'EventId': '2262d851-75e0-11eb-92d9-000d3abdf5e0', 'EventSource': 'Platform', 'EventStatus': 'Scheduled', 'EventType': 'Reboot', 'NotBefore': '2021-01-06T11:10:18+00:00', 'ResourceType': 'VirtualMachine', 'Resources': ['azure-mcu001']}`
support.rest	Messages to and from the Pexip Client API (used by the Pexip apps, and other third parties).
support.rtmp	Calls (streaming) to an RTMP device.

support.scheduled_ maintenance_events	Logs actions taken by Pexip Infinity resulting from detecting scheduled maintenance events in Azure. For example:
	`Message="Scheduled maintenance event affecting MCU node has been detected. System will be placed into maintenance mode."`
	`Message="Maintenance event completed. Maintenance mode will be unset."`

support.sip	SIP signaling.
support.teams	Gateway calls into Microsoft Teams.

Content of signaling messages

All log lines for signaling-related messages contain source and destination IP addresses and ports.

Multi-line messages are encoded on a single line with the character sequence `^M` representing a linebreak.

In addition to observing remote IP addresses, for H323 logging, a `Uuid` is present for H.225 (RAS) and H.245 signaling messages. These come from the same IP address but different ports, and the Uuid allows you to identify that they relate to the same call.

For BFCP signaling, all log lines are tied to a SIP `Call-Id`.

Most of the content of the support log is the signaling messages themselves; however it also includes logging such as TCP and TLS connection attempts, successes, and failures.

Creating and viewing diagnostic graphs

The Pexip Infinity Administrator interface includes a series of graphs that can be used to monitor the status and performance of the platform. You can view and edit the default graphs, create your own graphs, and change the order in which the various graphs appear.

To view and manage the graphs, go to **Status > Diagnostic Graphs**.

Information shown in the graphs

The graphs show data for the selected metrics starting from either when the relevant node was created, or when it was first rebooted following a platform upgrade to version 20 or later. Further metrics may be added in subsequent releases of Pexip Infinity; when you upgrade a node, data for any newly-added metrics will be available starting from the point at which the node was rebooted following the upgrade.

If a node has been temporarily out of service at any point, there will be a corresponding gap in the graphs. This also applies to cloud bursting nodes for the times that they were not in use.

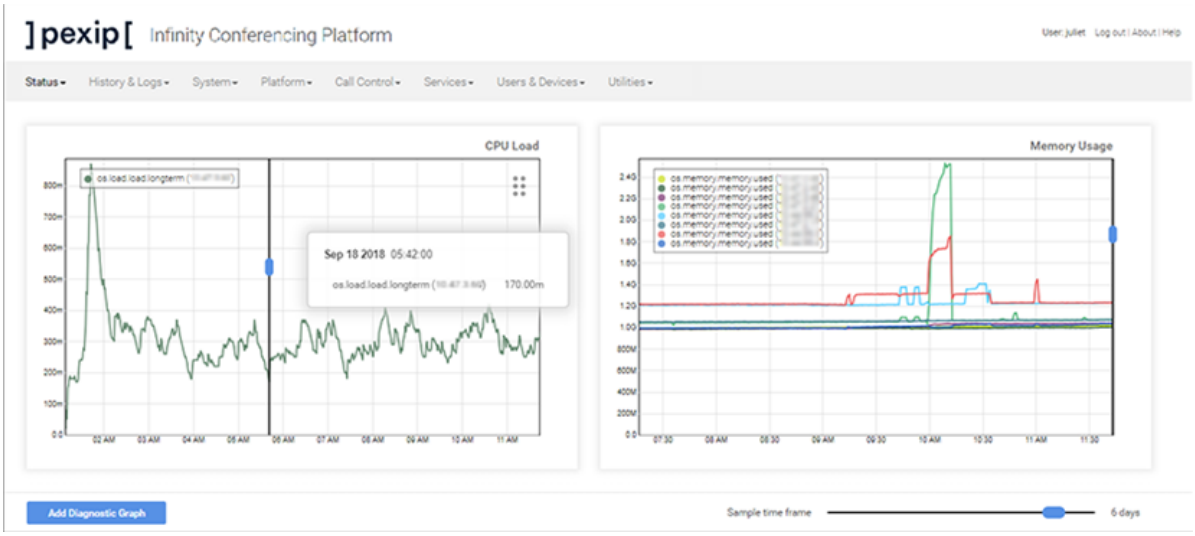
Diagnostic graphs are generated dynamically from information obtained from the relevant nodes, so the node must be currently available in order for its measurements to be shown in the graph. This means that any nodes that are currently out of service are not shown; nor are any cloud bursting nodes, unless they are currently in use.

Default graphs

Two graphs are included by default:

- **Memory usage** shows the memory being used by the Management Node's application and OS processes
- **CPU load** shows the system load on the Management Node.

Viewing and controlling graphs



- To change the time period covered by all graphs shown on the page, drag the **Sample time frame** control at the bottom right of the window to the left to zoom out, or right to zoom in.
- To change the time period covered by an individual graph, select the blue handle on the vertical bar at the right of the graph and drag it downwards to zoom out, or upwards to zoom in. Note that in some cases after zooming in or out the lines may be cropped. To fix this, click on the line twice.
- To view the values at a particular point in the graph, drag the vertical bar to the point in time you're interested in. The values at that point will appear in a pop-up next to the control.
- To toggle individual lines of the graph on or off, click on them from within the legend. Those that are available but not currently showing will appear grayed out.

- To move a graph, select the icon at the top right of the graph, and drag it to the desired location.

Creating new graphs

You can create new custom graphs to add to the **Diagnostic Graphs** page. Newly-created graphs will still show retrospective information, starting from when the selected node was upgraded to version 20 or later.

1. From the bottom left of the window, select **Add Diagnostic Graph**.
2. In the field above the new graph, enter an appropriate title.
3. From the **Available nodes** and **Available metrics** boxes below the graph, select the combination(s) you wish to appear in the graph.
   - For more information on a particular metric, hover over it — the description will appear in a tooltip.
   - To select multiple items, hold down the **Ctrl** button (Windows) or **Command** button (Mac).
   - The metrics that are available depend on whether the selected node is a Management Node, Transcoding Conferencing Node, or Proxying Edge Node.
   - If you have selected multiple nodes, only those metrics that are common to all selected node types are available.

   When you have made your selection, select **Add selected metric**.
4. The chosen combinations will appear in the **Selected metrics** table. From here you can refine the appearance of the graph, such as changing the color of individual lines or changing the magnification.

   If a selected metric does not yet have a value (for example, if you have selected the count of node allocations and none have yet taken place), the **Minimum**, **Average**, **Maximum** and **Last** values will be blank.
5. From the bottom left of the window, select **Save**.

The new graph will be added to the add to the **Diagnostic Graphs** page. Click on the graph to drag it to the desired location.

Editing and deleting graphs

To **edit** an existing graph, double-click it. You can then change the name, add and remove metrics, and change the appearance of the graph in the same way as when creating a new graph. When you have finished, from the bottom left of the window select **Save**.

To **delete** a graph, double-click it. From the bottom right of the window, select **Delete**.

Disconnection reasons

When a `Participant has disconnected` message appears in the admin log, further information about the reason will be given in the `Details` field of the message.

This **Disconnect reason** is also shown when viewing historical information about a participant.

The table below lists reasons for disconnections that may be given, along with an explanation of each. Note that this is not an exhaustive list, because in some cases the disconnect reason is provided by the endpoint.

Any **Abnormal** events are not usually seen during the expected functioning of the system and may require further investigation or intervention.

Reason for disconnection	Normal or Abnormal event	Message code	Meaning/resolution
Conference host ended the conference with a DTMF command	Normal	#pex120	A Host participant ended the call using a DTMF command.
Conference terminated by a Host participant	Normal	#pex121	A Pexip app Host participant has selected "disconnect all", or a client API command was used to terminate the conference.
Conference terminated by an administrator	Normal	#pex122	An administrator using the Pexip Infinity Administrator interface has selected “disconnect all”, or a management API command was used to end the conference.
Disconnected by an administrator	Normal	#pex123	An administrator using the Pexip Infinity Administrator interface has disconnected this particular participant.
Disconnected by another participant	Normal	#pex124	A Host using a Pexip app has disconnected a specific participant.
Conference terminated by another participant	Normal	#pex125	A Pexip app Host participant has selected "disconnect all", or a client API command was used to terminate the conference.
Timeout waiting for conference host to join or permit access to locked conference	Normal	#pex126	The participant timed out because the conference Host either did not join the conference, or did not permit the participant to join a locked conference.
Signaling node disconnected	Abnormal	#pex129	The media node lost connectivity to the signaling node.
Media process disconnected	Abnormal	#pex130	The Conferencing Node hosting the media has encountered an unexpected behavior.
Media node disconnected	Abnormal	#pex131	The signaling node lost connectivity to the media node.
Proxied participant disconnected	Abnormal	#pex132	The proxying node lost connectivity to the transcoding node.
No participants can keep conference alive	Normal	#pex140	This was the only remaining participant, and they were an ADP that was not configured to keep the conference alive.
All conference hosts departed hosted conference	Normal	#pex141	There are no Host participants remaining in the conference.
Last remaining participant removed from conference after timeout	Normal	#pex142	This was the only participant remaining, and they were disconnected after the configured amount of time.
Test call finished	Normal	#pex143	This was a call to the Test Call Service that was automatically disconnected after the specified time.
Call rejected	Normal	#pex150	The person being called did not answer or could not be reached.
Call disconnected	Normal	#pex151	A Pexip app has been disconnected by themselves or another system other than Pexip Infinity.

Reason for disconnection	Normal or Abnormal event	Message code	Meaning/resolution
Call failed - please contact your administrator		#pex158	A call routing rule with an invalid regex replace string has been configured.
Failed to gather IP addresses.	Abnormal	#pex170	The browser cannot find the local IP address. This may be due to ad blockers. A Pexip app could not determine its IP address. This may because there are privacy extensions installed.
Call Failed: Error: Could not get access to camera/microphone. Have you allowed access? Has any other application locked the camera?	Abnormal	#pex171	A Pexip WebRTC participant has not allowed their camera or microphone to be shared, or has no camera or microphone available.
Timer expired awaiting token refresh	Abnormal	#pex190	A Pexip app was unable to refresh its token after 2 minutes. This is likely due to network issues. On mobile devices, if you switch to a different browser tab or app and there is significant load on the device, it may suspend the browser tab that is running the Pexip app, thus preventing it from refreshing the token.
Resource unavailable	Abnormal	#pex191	There was insufficient transcoding or proxying capacity on the Transcoding Conferencing Node or the Proxying Edge Node on which the call landed.
Participant exceeded PIN entry retries	Normal	#pex192	The participant exceeded the allowed number of PIN entry attempts (3).
AVMCU cannot connect to PIN-protected or locked conference	Normal		There are some limitations with merging and escalating Skype for Business meetings with PIN-protected Pexip Infinity conferences.
Backplane disconnected	Abnormal		A connection between two Conferencing Nodes was lost, and the participant was connected to one of the nodes.
Browser closed	Normal		A Pexip app participant closed their web app or desktop client.
Call transferred	Normal		The participant was transferred to another conference.
CCCP call disconnected	Normal		The call from Pexip Infinity to the SfB server has been disconnected by a system other than Pexip Infinity. The reasons include: • **Conference Terminated – Organizer Ended Session**: the normal meeting ended scenario (e.g. the organizer pressed "End Meeting"). • **Participant Removed**: the gatewayed participant (VTC/VMR) was removed from the meeting by the organizer. • **Conference Terminated – Enterprise User Absent**: all of the SfB clients have left the meeting and the gateway participants have been timed out. • **Conference Terminated – Inactivity**: the meeting timed out due to inactivity (no new users have joined in the last 24 hours). • **Conference Terminated**: the organizer has ended the meeting. • **RFC3263 lookup failure**: the call could not be connected because DNS lookup failed.

Reason for disconnection	Normal or Abnormal event	Message code	Meaning/resolution
Conference number entry attempts exceeded	Normal		A participant using a Virtual Reception has exceed the allowed number of attempts (3) to enter a conference number. Does not apply to Pexip apps.
Connection to the other side was lost in a non-clean fashion	Normal		An H.323 participant was disconnected by themselves or another system (other than Pexip Infinity).
Connection was closed cleanly	Normal		An H.323 participant was disconnected by themselves, another system, or Pexip Infinity.
Detected AVMCU call failure	Abnormal		Call to the Skype for Business server failed.
Dialog has failed	Abnormal		(SIP) A reverse connection could not be established. This is likely due to a TCP connection issue.
Insufficient capacity	Abnormal		There was insufficient transcoding capacity on the Conferencing Nodes within the location the call landed on, and any configured media overflow locations, or a participant limit was reached.
Local disconnect (No response to Round Trip Delay Request)	Normal		Pexip Infinity disconnected the participant because their endpoint did not respond to the request sent by Pexip Infinity.
No conference number entered	Normal		The participant exceeded the allowed time at the Virtual Reception (120 seconds). Does not apply to Pexip apps.
Participant did not connect in time	Normal		The outgoing call was not answered.
Participant failed to start audio channel			An ACK was not received from the far side. This may be because the far side never received, or failed to process, the 200 OK from Pexip Infinity, and therefore never set up media.
PIN entry timed out	Normal		The participant exceeded the allowed time at the PIN entry prompt. Does not apply to Pexip apps.
Presentation stopped	Normal		A Skype for Business participant stopped presenting.
RFC3263 lookup failure	Abnormal		The call could not be connected because DNS lookup failed.
Re-INVITE timer has expired	Abnormal		Failed to establish a connection to the far end in order to send a SIP INVITE message.
Remote disconnect	Normal		A SIP/H.323 participant was disconnected by themselves or another system (other than Pexip Infinity).
Session renegotiation failed	Abnormal		A reverse connection could not be established. This is likely due to a TLS (certificate) issue.
Timeout expired waiting for ACK	Abnormal		SIP: no ACK message was received within the default timeout period. This often occurs due to issues with DNS (such as unresolvable DNS records, or resolving to the wrong host) as referenced in the `Record-Route` or `Contact` SIP signaling headers.
Transaction failed INVITE <random string of characters>	Abnormal		SIP: INVITE message was not responded to within the default timeout period.

Reason for disconnection	Normal or Abnormal event	Message code	Meaning/resolution
Transaction failed UPDATE <random string of characters>	Abnormal		SIP: UPDATE message was not responded to within the default SIP timeout period (32 seconds).
Undefined reason	Normal		An H.323 participant has been disconnected by themselves or another system (other than Pexip Infinity).
User initiated disconnect	Normal		A Pexip WebRTC participant manually disconnected themselves.

Pexip Infinity port usage and firewall guidance

The diagrams and tables below show the ports used when the Management Node and Conferencing Nodes connect to other devices.

Firewall, routing and NAT guidance

Note that in all Pexip Infinity deployment scenarios:

- The Management Node must be able to reach all Conferencing Nodes (Proxying Edge Nodes and Transcoding Conferencing Nodes) and vice versa.
- Each Conferencing Node must be able to reach every other Conferencing Node (Proxying Edge Nodes and Transcoding Conferencing Nodes), except:
  - When a location contains Proxying Edge Nodes, those nodes only require IPsec connectivity with:
    - any other proxying nodes in that location
    - all nodes in the transcoding location, and the primary and secondary overflow locations that are associated with that location
    - the Management Node.

    This means that the proxying nodes in one location do not need to have a direct network connection to other proxying nodes in other locations.
- Any internal firewalls must be configured to allow UDP port 500 and traffic using IP protocol 50 (ESP) in both directions between all Pexip nodes.
- There cannot be a NAT between any Pexip nodes.

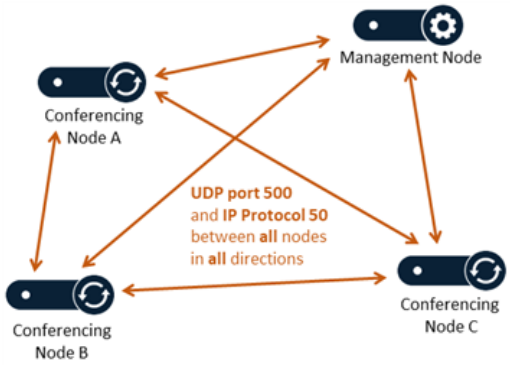
When a secondary network address is configured on a Conferencing Node:

- The primary address is always used for inter-node communication to the Management Node and to other Conferencing Nodes.
- SSH connections can be made only to the primary interface.
- The secondary address is always used for signaling and media (to endpoints and other video devices).
- Connections to DNS, SNMP, NTP, syslog and so on, go out from whichever interface is appropriate, based on routing.
- You can have a mixture of any number of single-interfaced and dual-interfaced Conferencing Nodes, providing all nodes can communicate with each other via their primary interfaces.

Inter-node communication (Conferencing Nodes and Management Node)

These are the port usage rules for all inter-node communication (local and remote) — between Conferencing Nodes, and between the Management Node and Conferencing Nodes:

Source address	Source port	Destination address	Dest. port	Protocol	Notes
Management Node	500	Conferencing Node	500	UDP	ISAKMP (IPsec) inter-node communication
Management Node	n/a	Conferencing Node	n/a	ESP	IPsec / IP Protocol 50 inter-node communication
Conferencing Node	500	Management Node / Conferencing Node	500	UDP	ISAKMP (IPsec) inter-node communication
Conferencing Node	n/a	Management Node / Conferencing Node	n/a	ESP	IPsec / IP Protocol 50 inter-node communication



- **No NAT** between nodes – path must be directly routable
- **UDP port 500** (IKE), and **IP Protocol 50** (IPsec ESP) to pass in **both** directions.
- If a location only contains Proxying Edge Nodes, then those proxying nodes in that location only require IPsec connectivity with any other proxying nodes in that location, the transcoding location, and the primary and secondary overflow locations, and with the Management Node.

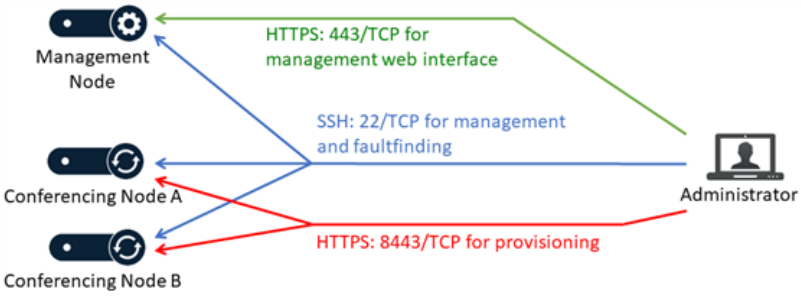
Port requirements for inter-node communication

Administration access

These are the port usage rules for administrative access to the Management Node and Conferencing Nodes:

Source address	Source port	Destination address	Dest. port	Protocol	Notes
SSH client	<any>	Management Node / Conferencing Node	22	TCP	SSH *
Web browser / API workstation	<any>	Management Node	80*/443	TCP (HTTP/HTTPS)	Management web and API administration
Web browser / API workstation	<any>	Conferencing Node	8443	TCP (HTTPS)	Provisioning a Conferencing Node (primarily for Azure/GCP/AWS deployments)

* Only required if you want to allow administrative access via this port.



Port requirements for administrative access

Peripheral services

These are the port usage rules for the mandatory and optional peripheral services used by the Management Node and Conferencing Nodes:

Source address	Source port	Destination address	Dest. port	Protocol	Notes
Standard features					

Source address	Source port	Destination address	Dest. port	Protocol	Notes
Management Node / Conferencing Node	55000–65535	DNS server	53	TCP/UDP	DNS
Management Node / Conferencing Node	123, 55000–65535	NTP server	123	UDP	NTP
Management Node	55000–65535	Pexip Licensing server (**activation.pexip.com** ◊)	443	TCP (HTTPS)	Platform licensing requests
Secure Scheduler for Exchange only					
Management Node	<any>	Exchange server (for Exchange on-premises)	443	TCP (HTTPS)	
Management Node	<any>	**login.microsoftonline.com** (for Microsoft 365)	443	TCP (HTTPS)	
Management Node	<any>	Kerberos Key Distribution Center (KDC) (if Kerberos Authentication is enabled)	88	TCP/UDP	
Management Node	<any>	KDC Proxy (if Kerberos Authentication and Kerberos KDC HTTPS proxy is enabled)	443	TCP (HTTPS)	
Management Node	<any>	Load balancer (if in use)	443	TCP (HTTPS)	
Conferencing Node	<any>	User OAuth token URI (if personal VMRs are enabled)	443	TCP (HTTPS)	
Outlook client/add-in	<any>	Add-in server FQDN (Conferencing Node / reverse proxy)	443	TCP (HTTPS)	Must be reachable either directly, or by using split DNS.
Outlook client/add-in	<any>	Either **https://appsforoffice.microsoft.com** or, for locally-hosted add-in sources, the address of the web server that is being used.	443	TCP (HTTPS)	
One-Touch Join only					
Management Node / Conferencing Node	55000–65535	Web proxy (optional)	8080 †	TCP	HTTP web proxy
Management Node	55000–65535	OAuth token endpoint • for Exchange integrations connecting to O365 using OAuth for the service account: **login.microsoftonline.com** • for Google Workspace domain user authorization: **oauth2.googleapis.com/token** • for Webex-registered endpoints: **webexapis.com**	443 †	TCP (HTTPS)	Not required if a web proxy is used

Source address	Source port	Destination address	Dest. port	Protocol	Notes
Conferencing Node	55000–65535	OAuth token endpoint • for Exchange integrations connecting to O365, or O365 Graph integrations: **login.microsoftonline.com** • for Google Workspace service account authorization: **googleapis.com/oauth2/v4/token** • for Google Workspace domain user authorization: **oauth2.googleapis.com/token** • for Webex-registered endpoints: **webexapis.com**	443 †	TCP (HTTPS)	Not required if a web proxy is used
Conferencing Node	55000–65535	**graph.microsoft.com** (for O365 Graph Integrations) ◊	443 †	TCP (HTTPS)	Not required if a web proxy is used
Conferencing Node	55000–65535	Exchange on-premises or Office 365 (for Exchange Integrations or O365 EWS Integrations)	80/443 †‡	TCP (HTTP/HTTPS)	Not required if a web proxy is used
Conferencing Node	55000–65535	Kerberos Key Distribution Center (KDC) (if Kerberos Authentication is enabled)	88	UDP and TCP	Conferencing Node
Conferencing Node	55000–65535	KDC Proxy (if both Kerberos Authentication and Kerberos KDC HTTPS proxy is enabled)	443	TCP (HTTPS)	Conferencing Node
Conferencing Node	55000–65535	**googleapis.com** (for Google Workspace Integrations) ◊	443	TCP (HTTPS)	Not required if a web proxy is used
Conferencing Node	55000–65535	Cisco endpoint API	80/443 †	TCP (HTTP/HTTPS)	Not required if a web proxy is used
Conferencing Node	55000–65535	Cisco Webex cloud (**webexapis.com**)	443 †	TCP (HTTPS)	Not required if a web proxy is used
Poly endpoint	<any>	Conferencing Node	443	TCP (HTTPS)	
Other additional features (ports only required if the relevant feature is configured)					
SNMP server	<any>	Management Node / Conferencing Node	161	UDP	SNMP
Management Node	55000–65535	FTP server	21 + server's FTP port range	TCP	FTP server for daily backup files
Management Node	55000–65535	LDAP server	389 / 636 3268 / 3269	TCP	LDAP / LDAPS AD global catalog searches
Management Node / Conferencing Node	55000–65535	Web proxy	8080 †	TCP	HTTP web proxy

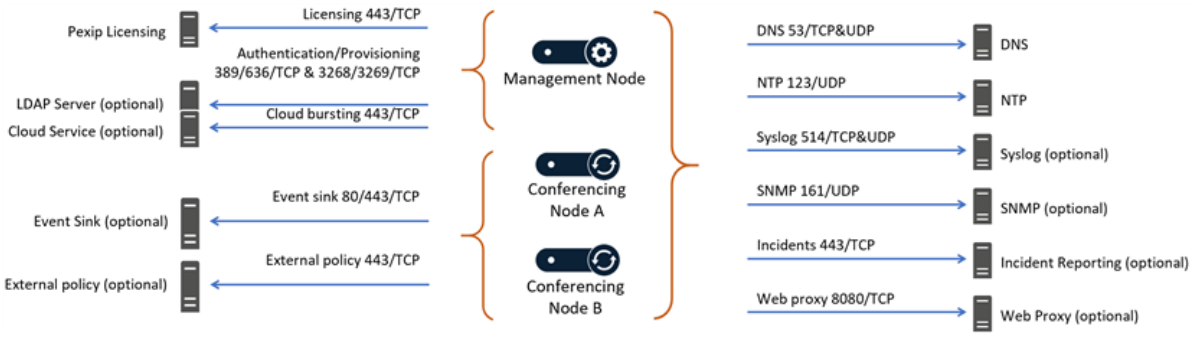
Source address	Source port	Destination address	Dest. port	Protocol	Notes
Management Node / Conferencing Node	55000–65535	Incident reporting server (**acr.pexip.com**)	443	TCP (HTTPS)	Incident reporting ◊
Management Node	55000–65535	Usage statistics server (**api.keen.io**)	443	TCP (HTTPS)	Usage statistics ◊
Management Node	55000–65535	Cloud service provider	443	TCP (HTTPS)	Dynamic bursting to cloud service ◊
Management Node	ephemeral	Teams Connector Azure Event Hub	5671	AMQPS	When the Event Hub is enabled for advanced status reporting in Teams integrations *
Management Node	55000–65535	SMTP server	587	TCP	SMTP (provisioning emails)
Management Node / Conferencing Node	<any>	SNMP NMS	161 †	UDP	SNMP Network Management System (NMS)
Management Node / Conferencing Node	55000–65535	Syslog server	514 †	UDP †	Syslog
Management Node / Conferencing Node	55000–65535	OIDC server	443	TCP (HTTPS)	Single Sign-On (SSO) with OpenID Connect ◊
Conferencing Node	55000–65535	AD FS server	443	TCP (HTTPS)	Single Sign-On (SSO) with AD FS
Conferencing Node	55000–65535	Event sink server	80/443	TCP (HTTP/HTTPS)	Event sink
Conferencing Node	55000–65535	Epic server	443	TCP (HTTPS)	Epic telehealth REST API requests ◊
Conferencing Node	55000–65535	External policy server	443	TCP (HTTPS)	External policy server REST API requests
Conferencing Node	<any>	AI Media Server (AIMS)	443	TCP (HTTPS)	Live captions service
Cisco endpoint	<any>	Conferencing Node	443	TCP (HTTPS)	Meeting Controls macro

† Configurable by the administrator.

‡ Determined by Exchange.

◊ Does not apply if a web proxy has been configured.

Note also that the ephemeral port range (55000–65535) is subject to change.



Port requirements for peripheral services (only the most commonly-used services are shown)

Conferencing Node call signaling and media

These port usage rules for call signaling and media apply to Proxying Edge Nodes and Transcoding Conferencing Nodes:

Source address	Source port	Destination address	Dest. port	Protocol	Notes
Standard call signaling and media					
Endpoint	<any>	Conferencing Node	80	TCP (HTTP)	Redirects to HTTPS for web/API access, and for Skype for Business conference avatars (if SfB is in use)
Endpoint	<any>	Conferencing Node	443	TCP (HTTPS)	Web browser/ API interface / Pexip app for mobile
Endpoint / call control system	<any>	Conferencing Node	5060	TCP	SIP
Endpoint / call control system	<any>	Conferencing Node	5061	TCP	SIP/TLS
Endpoint / call control system	<any>	Conferencing Node	40000–49999 **	TCP/UDP	Endpoint / call control system / Skype for Business system / Pexip app†† RTP / RTCP / RDP / VbSS / DTLS / STUN / TURN
Conferencing Node	33000–39999 **	Endpoint / call control system	5060	TCP/UDP	SIP
Conferencing Node	33000–39999 **	Endpoint / call control system	5061	TCP	SIP/TLS
Conferencing Node	40000–49999 **	Endpoint / call control system	<any>	TCP/UDP	RTP / RTCP / RDP / VbSS / DTLS / STUN / TURN Endpoint / call control system / Skype for Business system / Pexip app††
Conferencing Node	40000–49999 **	STUN / TURN server	3478 †	UDP	STUN / TURN
Conferencing Node	40000–49999 **	RTMP streaming server	1935	TCP	RTMP streaming
Conferencing Node	55000–65535	SfB Web Conferencing service	443 / 8057 ‡‡	TCP (TLS)	PSOM (PowerPoint presentation from SfB)

Source address	Source port	Destination address	Dest. port	Protocol	Notes
Conferencing Node	55000–65535	SfB Front End Server or Edge Server and WAC/OWA/OOS server	443	TCP (TLS/HTTPS)	PowerPoint presentation from SfB
Endpoint / call control system	<any>	Conferencing Node	5060	UDP	SIP UDP (if enabled)
H.323 only					
Endpoint / call control system	<any>	Conferencing Node	1719	UDP	H.323 (RAS signaling)
Endpoint / call control system	<any>	Conferencing Node	1720	TCP	H.323 (H.225/Q.931 signaling)
Endpoint / call control system	<any>	Conferencing Node	33000–39999 **	TCP	H.323 (H.245 signaling)
Conferencing Node	33000–39999 **	Endpoint / call control system	1719	UDP	H.323 (RAS signaling)
Conferencing Node	33000–39999 **	Endpoint / call control system	1720 / <any>	TCP	H.323 (H.225/Q.931 signaling) (to <any> if the device is registered to Pexip Infinity)
Conferencing Node	33000–39999 **	Endpoint / call control system	<any>	TCP	H.323 (H.245 signaling)
Microsoft Teams only					
Client application	<any>	Conferencing Node	443	TCP (HTTPS)	Microsoft Teams (client application viewing the meeting invitation Alternative Dial Instructions) *
Teams Connector	ephemeral	Conferencing Node	443 4277	TCP	Microsoft Teams signaling *
Teams Connector	50000-54999	Conferencing Node	40000–49999 **	UDP	Microsoft Teams SRTP/SRTCP *
Conferencing Node	33000–39999 ** & ephemeral	Teams Connector load balancer & Teams Connector instance	443	TCP (HTTPS)	Microsoft Teams signaling *
Conferencing Node	40000–49999 **	Teams Connector instance	50000–54999	UDP	Microsoft Teams SRTP/SRTCP *
Google Meet only					
Google Meet ‡	19302–19309	Conferencing Node	40000–49999 **	UDP	Google Meet SRTP/SRTCP
Conferencing Node	33000–39999 **	Google Meet (**hangouts.clients6.google.com** and **meetings.googleapis.com**)	443	TCP (HTTPS)	Google Meet

Source address	Source port	Destination address	Dest. port	Protocol	Notes
Conferencing Node	40000–49999 **	Google Meet	19302–19309	UDP	Google Meet SRTP/SRTCP

† Configurable by the administrator.

** Configurable via the Media port range start/end, and Signaling port range start/end options (see About global settings).

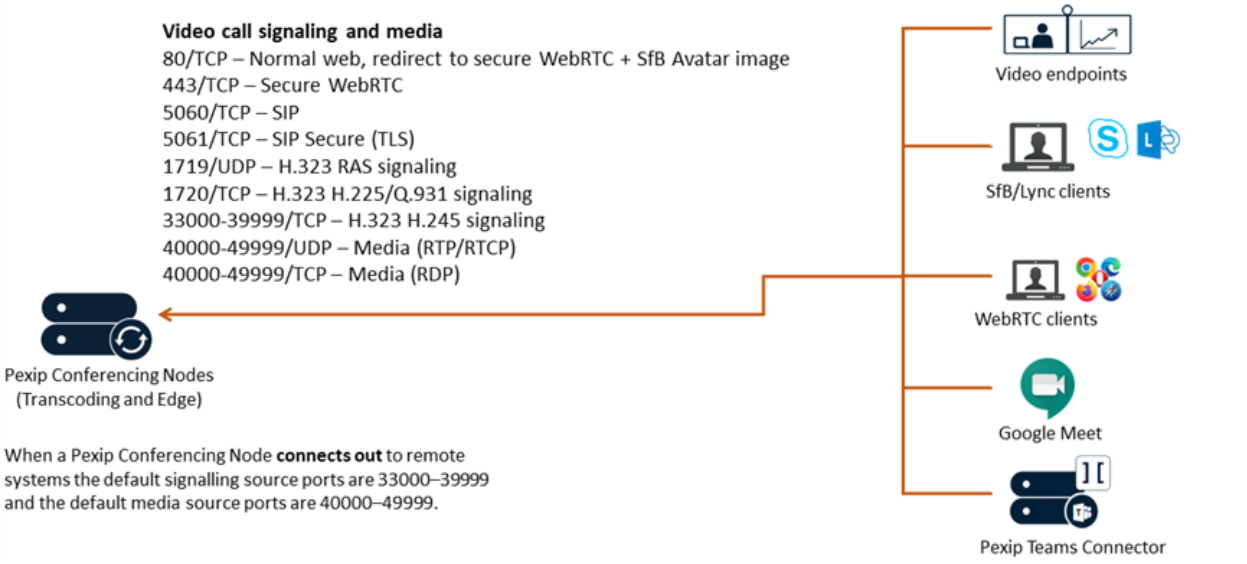
†† Pexip app web, mobile and desktop (installable) clients.

‡ See https://support.google.com/a/answer/1279090 for the actual IP ranges that need to be allowed.

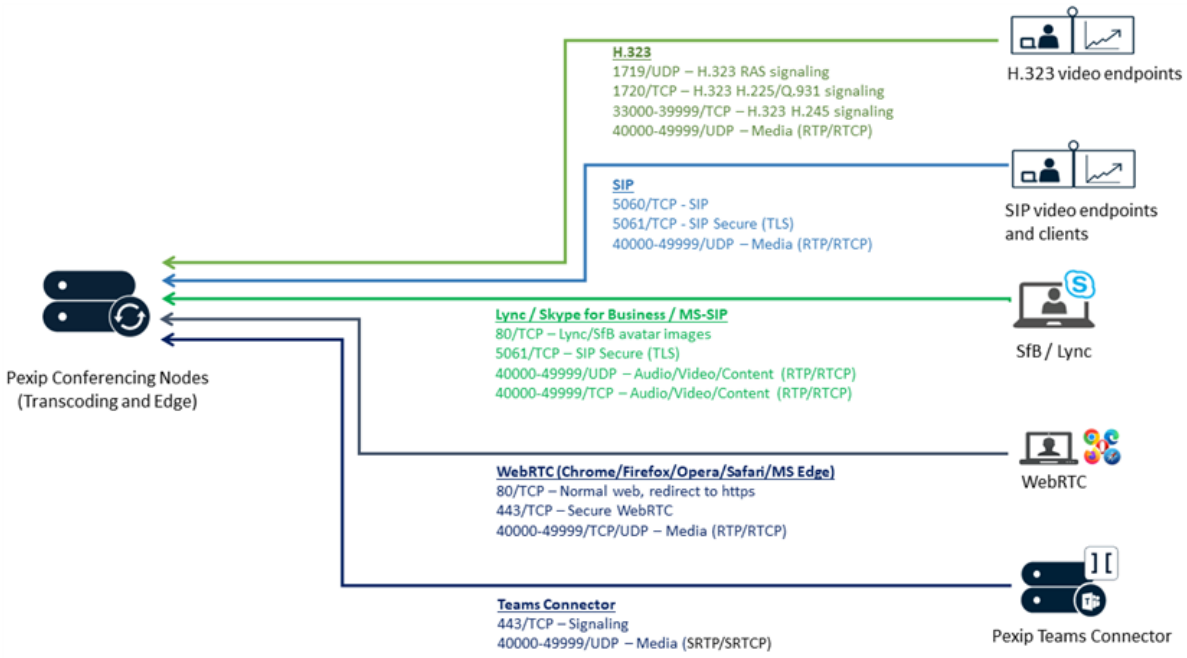
‡‡ Typically 443 for Web Conferencing Edge and 8057 for a SfB Front End Server / FEP.

Note also that:

- ICE calls allocate 4 ports per media line/stream.
- The ephemeral port range (55000–65535) is subject to change.



Call signaling and media ports overview (inbound)



Call signaling and media ports details (inbound)

Troubleshooting the Pexip Infinity platform

This topic contains a list of symptoms, possible causes and suggested resolutions for some issues you may experience when using the Pexip Infinity platform.

It includes the following sections:

- Pexip Infinity deployment and upgrading
- Dynamic bursting to a cloud service
- Joining a conference and viewing content
- Conference connectivity and TLS issues
- Pexip Infinity administration
- Pexip apps

For an up-to-date list of devices that are supported by Pexip Infinity, including any known issues, see Interoperability.

Contacting support

If you cannot find the information you require, contact your Pexip authorized support representative. Technical support for software issues is available while under a valid support contract and running a Pexip Infinity version no more than 2 major software releases behind the current release. Software bug fixes will only be provided in either the current or the next major release of software.

Pexip Infinity deployment and upgrading

Symptom	Possible cause	Resolution
During upgrade, one or more Conferencing Nodes are stuck with a status of "waiting for calls to clear", but there are no active calls reported on the Management Node.	A completed call has not cleared properly from the Conferencing Node.	Reboot the Conferencing Node.
A Conferencing Node does not accept calls even though it is powered on and is contactable on the network.	If time is not properly synchronized between the Management Node and the host server, certificates issued by the Management Node may be invalidated by Conferencing Nodes within the same Pexip Infinity deployment. As a result, the Conferencing Nodes will not communicate properly with the Management Node, causing calls to fail.	Ensure all virtual machines (i.e. the Management Node and all Conferencing Nodes) within the Pexip Infinity platform, and the host servers on which they are running, are using accurate times according to the public or private standard NTP clock. We strongly recommend that you configure at least 3 distinct NTP servers or NTP server pools in each instance to ensure proper synchronization. To synchronize time on Pexip Infinity: 1. Synchronize time on the host servers (for instructions, see the relevant hypervisor installation guide). 2. Enable NTP on Management Node. 3. Reboot all VMs.

Symptom	Possible cause	Resolution
A newly deployed Conferencing Node does not accept calls and its last contacted status on the Management Node shows "Never", even though it is powered on and is contactable on the network.	After deploying a new Conferencing Node, it takes approximately 5 minutes before the node is available for conference hosting and for its status to be updated on the Management Node. Until it becomes available, the Management Node reports the status of the Conferencing Node as having a last contacted and last updated date of "Never". "Connectivity lost between nodes" alarms relating to that node may also appear temporarily.	Wait for the Conferencing Node to finish initializing.
The wrong information was entered while running the installation wizard.		On server-based deployments you can re-run the installation wizard by following the instructions in Re-running the installation wizard. Do not re-run the installation wizard on cloud-based deployments (Azure, AWS, GCP or Oracle) in order to change Management Node configuration data such as its IP address or hostname. To change such data you must terminate the existing instance and deploy a new Management Node instance. You should only re-run the installation wizard on cloud-based deployments if you need to reset the web administration password (and then you should not change any of the other configuration data).
A new Management Node or Conferencing Node does not work. It was created by cloning it through VMware.	You cannot use cloning to create Management Nodes or Conferencing Nodes.	Create the Management Node according to our instructions (see Installation overview). Create all Conferencing Nodes by following the instructions in Deploying new Conferencing Nodes.
A newly-deployed Conferencing Node has gone into maintenance mode with the message "CPU instruction set is not supported; system will be placed in maintenance mode".	The Conferencing Node has been installed on a system that does not meet the CPU instruction set requirements.	Do one of the following: • Migrate the Conferencing Node to another server. • Re-install the Conferencing Node on another server. • If the instruction set is limited because you are using Enhanced vMotion Compatibility (EVC) (or equivalent), increase the minimum EVC level to L4 (Sandy Bridge).
When deploying a Management Node or Conferencing Node VM in Azure with SSH keys provisioned at boot time, the VM cyclically reboots.	This is a known limitation in Pexip Infinity versions 37.1 and earlier. It is the result of an underlying change in Azure meaning that you may not be able to successfully deploy any new VM instances (even in existing Azure environments).	Either: • Upgrade to v37.2 or later. • Deploy the VM with password authentication only, then connect over SSH to run the install wizard. After installation, use the Management Node to configure SSH keys.

Dynamic bursting to a cloud service

Symptom	Possible cause	Resolution
No bursting nodes appear in the **Cloud overflow nodes** area of the **Status > Cloud Bursting** page	The cloud node instances are not tagged correctly.	Check that the node instances running in your service provider have been assigned the **pexip-cloud** tag and that the tag value is set to the Management Node **hostname**.
You see a status issue "Instance <name> (with IP <address>) was found, but no corresponding Conferencing Node has been configured".	This occurs when Pexip Infinity detects a bursting instance with a tag matching your system's hostname but there is no corresponding Conferencing Node configured within Pexip Infinity.	This message can occur temporarily in a normal scenario when deploying a new Conferencing Node and you have set up the VM instance in your cloud provider but you have not yet deployed the Conferencing Node in Pexip Infinity. In this case, the issue will disappear as soon as the Conferencing Node is deployed.
You see connectivity errors in the administrator log while overflow nodes are being started/stopped.	This is normal behavior.	No action required.
"Not authorized to perform an operation on <instance ID or region name>. Check the policy created for the AWS user." error.	This means that there is a problem with the AWS policy document, or the AWS user is not attached to the policy.	See Deploying Pexip Infinity on Amazon Web Services for more information.
You see a status issue "Cloud bursting process encountered the following error: unsupported operand type(s) for -: 'datetime.datetime' and 'NoneType'"	A system location's configured **Transcoding location** only contains bursting nodes.	Ensure that the system location's configured **Transcoding location** contains "always-on" nodes.

Joining a conference and viewing content

Symptom	Possible cause	Resolution
Participants cannot join a conference due to insufficient capacity. • When users attempt to join a conference they get a message saying "Participants cannot join a conference due to insufficient capacity." • There is an alarm "Call capacity limit reached". • The administrator log is reporting "Participant failed to join conference" and "out of proxying resource" or "out of transcoding resource".	A call has not been accepted because all Conferencing Nodes that are able to take the media for this call are at capacity. It could be either Proxying Edge Nodes or Transcoding Conferencing Nodes that are out of capacity.	• Deploy more Conferencing Nodes in either the proxying or transcoding location as appropriate. • Move existing Conferencing Nodes onto more powerful servers. • Allocate more virtual CPUs for Conferencing Nodes on existing servers (if there are sufficient CPU cores). Note that the Conferencing Node will have to be rebooted for this to take effect. • Configure each location with a primary and secondary overflow location. • If a call is received in a location that contains Proxying Edge Nodes, that location must be configured with a **Transcoding location** that contains your Transcoding Conferencing Nodes. Note that some types of call consume more resources than other calls. Thus, for example, if you are at full capacity and an audio-only call disconnects, there may still not be sufficient free resource to connect a new HD video call.
	The participant has dialed in to the Conferencing Node while it is still starting up and an internal capacity-checking tool is running.	Wait for one minute and then attempt to join the conference.
	The Virtual Meeting Room or Virtual Auditorium has a participant limit applied, and this limit has been reached.	Increase the participant limit, if appropriate.
Participants cannot join a conference due to an **invalid** license. • When users attempt to join a conference they get a message saying "Participants cannot join a conference due to an invalid license." • The admin log is reporting "Participant failed to join conference" and "no valid license available".	If your Pexip Infinity reports an **invalid license**, this could mean that: • the license has not been activated • the existing license has expired • the existing license has become corrupt (this could occur, for example, if the Management Node reboots after an upgrade and comes back up on a different physical blade with a new MAC address).	• Check the status of your licenses from the **Licensing** page (**Platform > Licenses**). • Contact your Pexip authorized support representative for assistance. For more information, see Pexip Infinity license installation and usage.

Symptom	Possible cause	Resolution
Participants cannot join a conference due to **insufficient** licenses. • When users attempt to join a conference they get a message saying "Participants cannot join a conference due to insufficient licenses." • There is an alarm saying "License limit reached". • The admin log is reporting "Participant failed to join conference" and "license limit reached".	There are not enough call licenses available on the system at this time. For more information, see Pexip Infinity license installation and usage.	• Wait until one or more of the existing conferences have finished and the call licenses have been returned to the pool. • Contact your Pexip authorized support representative to purchase more call licenses.
An H.323 endpoint has its bandwidth restricted when joining a conference via a Virtual Reception, or when placing a call using the Distributed Gateway after first connecting to a Virtual Reception.	If there has been a bandwidth restriction placed on the Virtual Reception, any H.323 endpoints using that service will not be able to subsequently increase their bandwidth, even after being transferred to a Virtual Meeting Room or using a Call Routing Rule that has a higher (or no) limit.	• Make the call using a SIP endpoint or Pexip app. • Do not place a bandwidth restriction on the Virtual Reception.
Presentations do not display full screen.	If the presentation being shared is either: • an application that is not in full-screen mode • a full screen image that is being sent from a non-standard aspect ratio screen then the image being sent may have a non-standard aspect ratio. To send the image inside a standard resolution window (for example 640x480 [4:3]) or 1280x720 [16:9]), the endpoint may add horizontal or vertical mattes (known as letterboxing or pillarboxing respectively).	Ensure that presenters always either: • share their entire screen, or • share individual applications when they are in full-screen mode only.
Images are not displaying as expected: • they are being cropped • they have black bars at the top or sides	Endpoints send and display video images and presentations in various aspect ratios, most commonly 16:9 and 4:3. If there is a difference between the aspect ratios of the sending and receiving endpoints, then the endpoint and/or Pexip Infinity may crop the image or add vertical or horizontal mattes.	For more information, see Changing aspect ratios.
Occasional video freezes are seen in media received from SIP or H323 endpoints, where Conferencing Nodes in GCP are in use.	System locations that include Conferencing Nodes running in Google Cloud Platform must be configured with a maximum MTU of 1460 bytes. Some older endpoints default to a MTU higher than this.	Reduce the MTU on the endpoint to 1350.

Symptom	Possible cause	Resolution
Cisco endpoints running TC version 5.x or earlier software do not receive video when dialed out to from Pexip Infinity. Video is still sent as normal.	This may occur when additional video codecs (e.g. H.264 High Profile) are enabled on the Pexip Infinity deployment, and the TC5.x or older is called from a VMR. This occurs because more codecs are offered to the TC endpoint than it can cope with.	• Upgrade the TC5.x endpoint to a more recent software version. • Disable all video codecs (i.e. H.264 High Profile) not required in the deployment (**Platform > Global Settings > Codecs**).
A Cisco endpoint running TC7.1.4 might not receive video from Pexip in H.323 calls.	This is due to a bug in TC7.1.4 software.	• Disconnect and reconnect the call. • Use SIP instead of H.323. • Upgrade the TC software.
Cisco endpoints running TC7.0.x or earlier may crash when connecting to or from Pexip Infinity v24 or later over SIP.	This happens when AES-256 cipher is included in the SDP.	Upgrade the TC software to the latest version.
Poly endpoint is displaying its SIP address instead of its System Name in calls.	The endpoint is configured to use its SIP address.	To configure your Polycom device to display its system name in calls instead of its SIP address: 1. Go to **General Settings > System Settings** and set a **Device name** (remember to press **Save** at the bottom). 2. Go to **Call Configuration > Call Settings** and select the **Display System Name Instead of SIP Address** option. Note that some older software versions do not support these options — the minimum requirements are: • RealPresence Group requires 6.2.2 or later. • Studio X Series requires 3.13.0 or later.
Participants keep hearing themselves repeated back after a short delay. This happens in a conference with one or more other participants connected using the web app via Edge, Safari or Firefox, and who are using their computer's microphone and speakers.	Edge, Safari and Firefox do not have adequate echo cancellation, and in certain circumstances may experience a delay in playing audio. When this happens, sounds played through the computer's speakers are picked up by the computer's microphone and replayed back to other participants.	Participants using Edge, Safari or Firefox should: • use a headset • mute themselves when not speaking • consider using Chrome or a Pexip app instead.
In-band DTMF tones may not be detected if they are input too quickly.	False detections can be caused, for example, by poor line quality, line noise and echo.	This is best resolved through using out-of-band DTMF tones.
Participants are disconnected from conferences and "Backplane disconnected" messages are recorded in the administrator log.	VMware snapshots were being taken or deleted while conferences were in progress. Taking or removing snapshots can cause virtual machines to become unresponsive. For more information, see this VMware knowledge base article.	Only create and delete VMware snapshots at a time of minimal usage.
Participants cannot join a VMR that requires authentication. Pexip Infinity logs report: Message="Failed to verify SAML Assertion" Reason="Signature validation failed. SAML Response rejected"	The SAML Identity Provider's certificate has expired. By default, certificates created in AD FS expire after 365 days.	Download a new certificate from the Identity Provider and paste it into the Identity Provider public key field for the Identity Provider on Pexip Infinity.

Conference connectivity and TLS issues

Symptom	Possible cause	Resolution
Calls fail when dialing out to, or receiving calls from, video network infrastructure devices, such as a Cisco VCS or CUCM. Calls may fail immediately, or after a period of time, and SSL alerts are raised in the support log with an "unsupported certificate" description.	Either the external system or the Conferencing Node is verifying the other party's certificate and it is rejecting the connection because the certificate does not have **client** authentication properties.	Ensure that the certificates on your external systems and on your Conferencing Nodes contain "TLS Web Client Authentication" Enhanced Key Usage properties (see Mutual TLS authentication and client/server certificates). From Pexip Infinity version 39 or later you must assign a **Client TLS certificate** to your Conferencing Nodes (**Platform > Conferencing Nodes > Client TLS Certificate**) for mutual TLS purposes.

TLS certificate administration

Error message	Possible cause	Resolution
Certificate and private key do not appear to be part of the same key pair	This most likely means that you have tried to upload the certificate against the wrong CSR.	Select the correct CSR and try again.

Pexip Infinity administration

Symptom	Possible cause	Resolution
A policy profile is configured but it is being ignored.	The policy profile has not been assigned to a system location.	Assign the policy profile to your locations (**Platform > Locations**).
Log timestamps appear to be inaccurate or log entries appear to be out of sequence.	Time is not properly synchronized between the Management Node, Conferencing Nodes and their host servers, causing different systems to use different timestamps. This could be because: • Insufficient NTP servers or NTP pools have been configured on a host server or the Management Node (we recommend a minimum of 3). • One or more NTP servers are unreachable or have inaccurate time themselves. • NTP is otherwise not configured according to our recommendations.	Ensure all virtual machines (i.e. the Management Node and all Conferencing Nodes) within the Pexip Infinity platform, and the host servers on which they are running, are using accurate times according to the public or private standard NTP clock. We strongly recommend that you configure at least 3 distinct NTP servers or NTP server pools in each instance to ensure proper synchronization. To synchronize time on Pexip Infinity: 1. Synchronize time on the host servers (for instructions, see the relevant hypervisor installation guide). 2. Enable NTP on Management Node. 3. Reboot all VMs.
How do I register Pexip Infinity to a gatekeeper?		You don't register Pexip Infinity to a gatekeeper. Instead, configure your call control system to route calls to Pexip Infinity. See Call control.

Symptom	Possible cause	Resolution
Oracle Acme Packet SBC has the error "Message Too Large".		Increase the following configuration parameters on the Acme Packet SBC: • `sip-message-len 16000` • `option +max-udp-length=0`
VMware datastore is showing disk I/O alarms.		Enabling SIOC on your datastores might help. For more information, see this VMware knowledge base article.
A participant's **Display name** has some characters replaced with asterisks.	For H.323 endpoints, the display name should use IA5 characters, but some endpoints use UTF-8 or other encodings. If Pexip Infinity receives any non-IA5 characters, it replaces them with an asterisk.	Ensure the display name used by such endpoints uses a string that contains valid IA5 characters only.
Unable to take snapshots on deployments hosted in Microsoft Azure.	The Azure Load Balancer is timing out.	Increase the TCP idle timeout setting in Azure for the Azure Load Balancer. See https://docs.microsoft.com/en-us/azure/load-balancer/load-balancer-tcp-idle-timeout for configuration instructions.

Pexip apps

The table below describes general issues that may occur when using the Pexip apps within your deployment. For a list of specific error messages that may be presented to Pexip app users, along with their meaning and suggested resolution (where appropriate), see Troubleshooting Pexip app error messages.

Symptom	Possible cause	Resolution
Participants cannot use Pexip apps. Web, desktop, and mobile app users are presented with a message "This feature has been disabled."	Support for these Pexip apps has been disabled.	Enable support as follows: 1. Go to **Platform > Global Settings > Connectivity**. 2. Select the **Enable support for Pexip apps and Client API** checkbox.
Participants attempting to connect to a conference using the Pexip mobile app for iOS get a message "Certificate Error. Please contact your system administrator."	Your deployment does not use valid, trusted certificates. Pexip mobile apps require deployments with HTTPS and valid certificates.	Install valid certificates. For more information, see Managing TLS and trusted CA certificates.
Participants using a Pexip app cannot write or see chat messages.	Chat has been globally disabled.	Enable chat.
Guest participants using a Pexip app cannot see the chat window, participant list, or presentation.	Guest participants are not allowed in to the conference if the conference is locked, or until the first Host participant has joined.	• Allow an individual Guest to join the locked conference. • Unlock the conference, allowing all Guests to join. • Connect at least one endpoint to the conference as a Host participant.
Users of the web app see "An error occurred. The page you are looking for is currently unavailable." error message when connecting via a reverse proxy.	You are running version 3 or earlier of the Pexip Reverse Proxy and TURN Server and the web browser cannot make a TLS connection.	Upgrade to latest version of the Pexip Reverse Proxy and TURN Server.

Symptom	Possible cause	Resolution
Users see a **Connection Alert** message at the top of their window	There is packet loss of 3% or more between the Pexip app and Pexip Infinity.	Users who experience this issue regularly should reduce their bandwidth by selecting a lower Connection Quality.

Troubleshooting registering to Pexip Infinity

This topic explains registration behavior and provides assistance in troubleshooting registration issues.

To see a list of all device aliases that are currently registered to the Pexip Infinity platform, go to **Status > Registrations**, and to see a history of registered devices go to **History & Logs > Registration History**.

Each device alias configured in Pexip Infinity has to be enabled for the individual protocols over which that alias can be registered. If the alias is configured but the device cannot register, you should check which protocols have been enabled for registration for that alias.

This topic covers:

- Checking the administrator log
- Maintenance mode
- SIP registration behavior
- H.323 registration behavior
- Pexip app for Windows SSO registration behavior

Checking the administrator log

Successful registration requests and expired registrations are recorded in the administrator log, for example:

```
Message="Registration added" Alias="bob@pexample.com" Protocol="SIP" Registration-id="958e52ba-4328-424b-8d77-c18a59f4c1da"
Natted="False" Location="Europe"
```

```
Message="Registration deleted" Alias="55170" Protocol="H323" Registration-id="302fd156-85bd-497d-85dc-c5b29a16d612"
Location="Europe" Reason="Registration expired"
```

Failed registration requests (due to, for example, an unknown device alias or authorization failure) and refreshed registrations are recorded in the support log only.

If a call fails to be placed to a registered device, check that an appropriate Call Routing Rule has been configured that has suitable destination alias matching strings for the registered alias (and ensure that call looping cannot occur).

Note that registrations are enabled by default when Pexip Infinity is installed. Therefore, until device aliases are configured, every registration request will be rejected.

Maintenance mode

Devices cannot register to a Conferencing Node that is in maintenance mode:

- Requests from **SIP devices** and **Pexip apps** are rejected with "503 Service Unavailable".
- Requests from **H.323 endpoints** receive no response.
  - If the H.323 endpoint has previously registered and received a list of alternate gatekeepers, it should then attempt to register to one of the alternates instead.
  - If the H.323 endpoint has not previously registered (and therefore has not received a list of alternate gatekeepers) it will typically be unable to register. However, it may attempt to register to another Conferencing Node if multiple h323rs DNS SRV records have been configured and the endpoint can handle multiple records.

SIP registration behavior

The alias registered by a SIP device alias is normally a URI, for example **alice@example.com**, and thus the full device alias (including any domain) must match an entry in the **Device aliases** list.

When troubleshooting failed registrations, you need to look in the **support log**. Typical messages related to various successful and unsuccessful SIP registration attempts are described in the following table:

Registration scenario	Behavior and example support log messages
No authentication is required and the alias exists in the list of device aliases	Register request is accepted with 200 response `Message="Summarised received SIP request" Src-address="10.44.2.77" Src-port="42576" Dst-address="10.44.155.21" Dst-port="5061" Transport="TLS" Method="REGISTER" From="sip:bob@pexample.com" To="sip:bob@pexample.com" Contacts="<sip:bob@10.44.2.77>;+sip.instance="<urn:uuid:f0b7095d-5ee0-548a-80a8-c522aafdf94b>"" Call-ID="58078da9a56dbcee@10.44.2.77" CSeq="62195" Request-URI="sip:pexample.com" Received-time="2017-03-23T16:11:59,795756"` `Message="Registration added" Alias="bob@pexample.com" Protocol="SIP" Registration-id="cacef4f2-dc7b-4cbe-9a6f-6c69aea640fb" Natted="False" Location="Europe"` `Message="Summarised sending SIP response" Src-address="10.44.155.21" Src-port="5061" Dst-address="10.44.2.77" Dst-port="42576" Transport="TLS" Method="REGISTER" From="sip:bob@pexample.com" To="sip:bob@pexample.com" Contacts="<sip:bob@10.44.2.77>;+sip.instance="<urn:uuid:f0b7095d-5ee0-548a-80a8-c522aafdf94b>";expires=74" Call-ID="58078da9a56dbcee@10.44.2.77" CSeq="62195" Status-code="200"`
Requested alias does not exist in the list of device aliases	Register request is rejected with 403 response `Message="Summarised received SIP request" Src-address="10.44.2.77" Src-port="42576" Dst-address="10.44.155.21" Dst-port="5061" Transport="TLS" Method="REGISTER" From="sip:bob@pexample.com" To="sip:bob@pexample.com" Contacts="<sip:bob@10.44.2.77>;+sip.instance="<urn:uuid:f0b7095d-5ee0-548a-80a8-c522aafdf94b>"" Call-ID="58078da9a56dbcee@10.44.2.77" CSeq="62191" Request-URI="sip:pexample.com" Received-time="2017-03-23T16:02:29,795311"` `Message="Summarised sending SIP response" Src-address="10.44.155.21" Src-port="5061" Dst-address="10.44.2.77" Dst-port="42576" Transport="TLS" Method="REGISTER" From="sip:bob@pexample.com" To="sip:bob@pexample.com" Contacts="" Call-ID="58078da9a56dbcee@10.44.2.77" CSeq="62191" Status-code="403"`
Authentication is required but the wrong credentials are supplied	Register request is rejected with 401 response (two requests and two 401 responses) `Message="Summarised received SIP request" Src-address="10.44.2.77" Src-port="42576" Dst-address="10.44.155.21" Dst-port="5061" Transport="TLS" Method="REGISTER" From="sip:bob@pexample.com" To="sip:bob@pexample.com" Contacts="<sip:bob@10.44.2.77>;+sip.instance="<urn:uuid:f0b7095d-5ee0-548a-80a8-c522aafdf94b>"" Call-ID="58078da9a56dbcee@10.44.2.77" CSeq="62171" Request-URI="sip:pexample.com" Received-time="2017-03-23T15:44:17,530041"` `Message="Summarised sending SIP response" Src-address="10.44.155.21" Src-port="5061" Dst-address="10.44.2.77" Dst-port="42576" Transport="TLS" Method="REGISTER" From="sip:bob@pexample.com" To="sip:bob@pexample.com" Contacts="" Call-ID="58078da9a56dbcee@10.44.2.77" CSeq="62171" Status-code="401"` (this pair of messages is then repeated)
Authentication is required and correct credentials are supplied	The first register request is rejected with 401 response, then the second request (when the device will supply the requested credentials) is accepted with 200 `Message="Summarised received SIP request" Src-address="10.44.2.77" Src-port="42576" Dst-address="10.44.155.21" Dst-port="5061" Transport="TLS" Method="REGISTER" From="sip:bob@pexample.com" To="sip:bob@pexample.com" Contacts="<sip:bob@10.44.2.77>;+sip.instance="<urn:uuid:f0b7095d-5ee0-548a-80a8-c522aafdf94b>"" Call-ID="58078da9a56dbcee@10.44.2.77" CSeq="62177" Request-URI="sip:pexample.com" Received-time="2017-03-23T15:48:39,817656"` `Message="Summarised sending SIP response" Src-address="10.44.155.21" Src-port="5061" Dst-address="10.44.2.77" Dst-port="42576" Transport="TLS" Method="REGISTER" From="sip:bob@pexample.com" To="sip:bob@pexample.com" Contacts="" Call-ID="58078da9a56dbcee@10.44.2.77" CSeq="62177" Status-code="401"` `Message="Summarised received SIP request" Src-address="10.44.2.77" Src-port="42576" Dst-address="10.44.155.21" Dst-port="5061" Transport="TLS" Method="REGISTER" From="sip:bob@pexample.com" To="sip:bob@pexample.com" Contacts="<sip:bob@10.44.2.77>;+sip.instance="<urn:uuid:f0b7095d-5ee0-548a-80a8-c522aafdf94b>"" Call-ID="58078da9a56dbcee@10.44.2.77" CSeq="62178" Request-URI="sip:pexample.com" Received-time="2017-03-23T15:48:39,873782"` `Message="Registration added" Alias="bob@pexample.com" Protocol="SIP" Registration-id="aecae4f2-da7b-4dbe-9a6f-6c69feb689ea" Natted="False" Location="Europe"` `Message="Summarised sending SIP response" Src-address="10.44.155.21" Src-port="5061" Dst-address="10.44.2.77" Dst-port="42576" Transport="TLS" Method="REGISTER" From="sip:bob@pexample.com" To="sip:bob@pexample.com" Contacts="<sip:bob@10.44.2.77>;+sip.instance="<urn:uuid:f0b7095d-5ee0-548a-80a8-c522aafdf94b>";expires=111" Call-ID="58078da9a56dbcee@10.44.2.77" CSeq="62178" Status-code="200"`

H.323 registration behavior

An H.323 device often registers multiple aliases such as an H.323 ID, an E.164 number, and in some cases the system name. H.323 ID aliases can take the form of a full URI.

All of the device aliases presented in an H.323 registration request must be in the list of allowed device aliases. If any alias is not present, none of the aliases will be allowed to register. Additionally, if device authentication is being used, all of the device aliases in the request must be configured with the same credentials.

When an alias does not exist in the list of device aliases, a message is written to the support log in the form:

```
Name="support.registration" Message="H.323 device authentication failed" Reason="Alias not found" Alias="(u'Bob Jones', 'h323_ID')"
```

To check all the aliases that are being presented from an endpoint, you can filter the support log by "registrationRequest", and then search for one of the aliases that you know is being presented. When you have found a suitable message, the `terminalAlias` section lists all of the aliases that are being presented, for example:

```
Name="support.h323.ras" Message="Received RAS message" Src-address="10.44.2.77" Src-port="1719" Dst-address="10.44.157.21" Dst-
port="1719" Detail="registrationRequest:
...
terminalAlias: [
h323_ID:
 Bob Jones,
h323_ID:
 bob@pexample.com
dialedDigits:
 123456
]
```

Pexip app for Windows SSO registration behavior

The following message is a normal part of the registration authentication flow. It is written to the support log when registration is initiated, prior to authentication with the IdP:

```
Name="support.registration" Message="REST device authentication failed" Alias="alice@pexample.com" Reason="Redirect to IdP"
```

When the selected IdP does not have a username that matches the alias being registered, a message is written to the support log in the form:

```
Name="support.registration" Message="REST device authentication failed" Alias="alice@pexample.com" Reason="Alias mismatch"
```

A message is written to the support log in the following form when:

- an alias does not exist in the list of device aliases
- the alias exists but **Enable registration using IdP SSO** is not selected

```
Name="support.registration" Message="REST device authentication failed" Alias="alice@pexample.com" Reason="No checker for
twisted.cred.credentials.IAnonymous"
```